**NOT MEASUREMENT SENSITIVE**

**MIL-STD-40051E**
**25 March 2025**

**SUPERSEDING**
**MIL-STD-40051-1D/2D**
**22 November 2021**

# DEPARTMENT OF DEFENSE STANDARD PRACTICE

PREPARATION OF DIGITAL TECHNICAL INFORMATION
FOR
TECHNICAL MANUALS (TMs)

**AREA TMSS**

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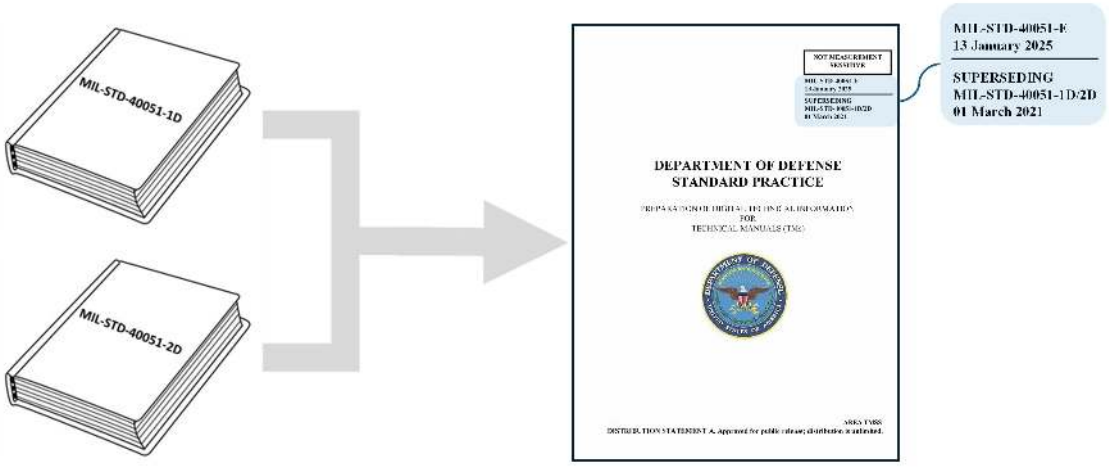
# FOREWORD

1. This standard is approved for use by the Department of the Army and the United States Marine Corps and is available for use by all Departments and Agencies of the Department of Defense (DoD).
2. This standard establishes the technical content requirements and mandatory style and format requirements for the preparation of Technical Manuals (TMs) and other types of equipment publications specified herein and subsequent revisions required to support the various types of equipment and weapon systems within the Department of the Army and the Marine Corps. The requirements contained in this standard cover operation (except aviation) and maintenance at all levels through depot, including Depot Maintenance Work Requirements (DMWRs) and National Maintenance Work Requirements (NMWRs). The requirements also cover destruction to prevent enemy use, Battle Damage Assessment and Repair (BDAR), Lubrication Orders (LOs), Preventive Maintenance Checklists (PMCs), general maintenance manuals, software manuals, hand receipts (HRs), and Technical Bulletins (TBs).
3. This standard shall supersede both MIL-STD-40051-1D and MIL-STD-40051-2D. Therefore, the information contained herein shall serve as content requirements for both page-based and electronic publications.




**Note**

This revision is comprehensive, and many things have changed. Some existing content has been rearranged, while other content is new. Also, numerous new graphics have been integrated to improve usability and understanding. Therefore, changes are not denoted with vertical bars or any other change markers. This standard is intended to be reviewed and utilized as a new document guiding the creation of publications.

4. Comments, suggestions, or questions should be addressed to USAMC Logistics Data Analysis Center (LDAC), ATTN: AMLD-LST Bldg 4400, Martin Rd SW, Redstone Arsenal, AL 35898 or emailed to or e-mailed to usarmy.redstone.ldac.mbx.tmss@army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/.

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# 1 SCOPE

## 1.1 Scope.

This standard establishes the technical content, style, and format requirements for all Technical Manuals (TMs) and publications for major weapon systems and their related systems, subsystems, equipment, assemblies, components, Shop Replaceable Units (SRUs), and Line Replaceable Units (LRUs).

This standard provides requirements for operator and maintenance technical manuals, Depot Maintenance Work Requirements (DMWRs), National Maintenance Work Requirements (NMWRs), Preventive Maintenance Daily (PMD), Preventive Maintenance Services (PMS), Phased Maintenance Inspection (PMI), destruction of equipment to prevent enemy use manuals, Battle Damage Assessment and Repair (BDAR) manuals, Preventive Maintenance Checklists (PMCs), Lubrication Orders (LOs), ammunition DMWRs, software manuals, and general maintenance manuals, Hand Receipts (HRs), ammunition data sheets, and Technical Bulletins (TBs). The requirements are applicable for all maintenance levels, including DMWRs and NMWRs.

The requirements in MIL-STD-40051E are used to develop maintenance technical and equipment publications regardless of output. The terms "IETM" and "ETM" to refer to output-specific requirements when necessary. When the text does not specify whether a requirement applies specifically either to ETMs or IETMs, the requirement applies to both outputs.

## 1.2 Approach to this revision.

While MIL-STD-40051E is technically a revision, its updates are so pervasive that to many users, it will feel like a new standard. Due to the comprehensive nature of this update, change bars would be visually disruptive and are not used, nor is a revision summary provided. Users are encouraged to approach this document as though it were a wholly new standard.

## 1.3 Paragraphs with limited applicability.

This standard contains paragraphs and specific requirements that are not applicable to all services. Such paragraphs or requirements are prefixed to indicate the Services to which they pertain: (A) Army, (N) Navy, (MC) Marine Corps, (F) Air Force, and (SF) Space Force. Portions not prefixed are applicable to all services. Paragraphs prefixed with "MC" pertain to manuals that are for the Marine Corps only and unless otherwise stated do not apply to multi-service manuals involving the Marine Corps.

## 1.4 Use of technical content.

In addition to using the technical content requirements provided herein for the development of technical publications, the technical information developed in accordance with this standard can be used to provide the necessary input to other external systems that are designed to collect and report operations, maintenance, historical, and parts requisition data required for efficient management and support of aviation and non-aviation weapon systems and their related systems, equipment, and components/modules.

## 1.5 Examples/figures; figure and table titles.

The figures used in this standard are only examples. While they are intended to clarify and illustrate concepts, in the event of a conflict between the text and figures, the text of this standard takes precedence over the figures. The figures in this standard represent standard-compliant material, but they may not represent all variations of standard-compliant material. Figure appearance may vary slightly based on version of DTD/stylesheets utilized.

Figure and table titles in this standard do not use title case. Technical publications prepared in accordance with MIL-STD-40051, however, use title case (see 4.17.5.10).

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## 1.6 Materiel developer decisions.

For requirements that involve a MATDEV decision (those preceded by "as specified by the MATDEV" or "as required by the MATDEV"), determination should be made based on the requirements of the applicable weapon system/equipment.

## 1.7 Single viewer for IETMs: IADS.

Since the beginning of FY24, Interactive Authoring and Display Software (IADS) is now the only authorized viewer for all IETMs. PMs will use IADS to simplify and expedite maintenance of IETMs. The IADS viewer will be used to support the high functionality requirements of simple and complex systems such as aircraft, ground combat systems, and all other Army systems/equipment utilizing IETMs. IADS is available for download at https://iads.redstone.army.mil/.

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# 2 APPLICABLE DOCUMENTS

## 2.1 General.

The documents listed in this section are specified in sections 3, 4, and 5 of this standard. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples.

While every effort has been made to ensure the completeness of this list, users are cautioned that they must meet all specified requirements documents cited in Sections 3, 4, and 5 of this standard regardless of whether they are listed here.

## 2.2 Government documents.

The following other Government documents and publications form a part of this document to the extent specified herein. Unless specified otherwise, the issues are those cited in the solicitation or contract.

### 2.2.1 Specifications, standards, and handbooks.

The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

SPECIFICATIONS

| DEPARTMENT OF DEFENSE | |
|---|---|
| MIL-PRF-2104 | Lubricating Oil, Internal Combustion Engine, Combat Tactical Service |
| MIL-PRF-63002 | Technical Manuals: Requirements for Preparation of Modification Work Orders (MWOs) |
| MIL-PRF 63033 | Performance Specification Technical Manuals, Technical: Demilitarization of Surplus Military Items, Preparation of |
| MIL-PRF-63029 | Manuals, Technical: Requirements for Operator's Manuals and Checklists for Manned and Unmanned Aircraft Systems |
| MIL-PRF-63049 | Manuals, Technical: List of Applicable Publications (LOAP) |

STANDARDS

| DEPARTMENT OF DEFENSE | |
|---|---|
| MIL-STD-129 | Military Marking for Shipment and Storage |
| MIL-STD-882 | System Safety |
| MIL-STD-1309 | Definitions of Terms for Testing, Measurement and Diagnostics |
| MIL-STD-3003 | Vehicles, Wheeled: Preparation for Shipment and Storage |

HANDBOOKS

| DEPARTMENT OF DEFENSE | |
|---|---|
| MIL-HDBK-113 | Guide for the Selection of Lubricants, Functional Fluids, Preservatives and Specialty Products for use in Ground Equipment Systems |

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| | |
|---|---|
| MIL-HDBK-275 | Guide for Selection of Lubricants, Fluids, and Compounds for Use in Flight Vehicles and Components |
| MIL-HDBK-1222 | Guide to the General Style, Format, and Digital Development of U.S. Army WP Technical Manuals |

Copies of these documents are available from the Document Automation and Production Service online at https://quicksearch.dla.mil.

H6 Federal Item Name Directory: Copies of Handbook H6 are available on CD-ROM from the Commander, Defense Logistics Services Center, Battle Creek, MI 49017-3084, or H6 searches can be done at https://www.dla.mil/Working-With-DLA/Federal-and-International-Cataloging/H6/.

### 2.2.2 Other government documents and publications.

The following other Government documents and publications form a part of this document to the extent specified herein. Unless specified otherwise, the issues are those cited in the solicitation or contract.

| | |
|---|---|
| ABCAs Directory | Abbreviations, Brevity Codes, and Acronyms Directory (https://armypubs.army.mil/abca/SearchABCA.aspx) |
| AR 25-30 | Army Publishing Program |
| AR 25-38 | Army Printing and Distribution Program |
| AR 71-32 | Force Development and Documentation Consolidated Policies |
| AR 75-1 | Malfunctions Involving Ammunition and Explosives |
| AR 95-1 | Flight Regulations |
| AR 385-10 | The Army Safety and Occupational Health Program |
| AR 700-20 | Ammunition Peculiar Equipment |
| AR 700-82 | Joint Regulation Governing the Use and Application of Uniform Source, Maintenance, and Recoverability Codes |
| AR 700-138 | Army Logistics Readiness and Sustainability |
| AR 750-1 | Army Materiel Maintenance Policy |
| AR 750-10 | Army Modification Program |
| DA Pam 25-38 | Army Printing and Distribution Procedures |
| DA Pam 25-40 | Army Publishing Program Procedures |
| DA Pam 95-9 | Management of Aviation Critical Safety Items |
| DA Pam 385-63 | Range Safety |
| DA Pam 700-27 | Instructions for Basis of Issue Plan Feeder Data Submissions |
| DA Pam 385-64 | Ammunition and Explosives Safety Standards |
| DA Pam 738-751 | Functional Users Manual for the Army Maintenance Management System-Aviation (TAMMS-A) |

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| | |
|---|---|
| DA Pam 742-1 | Ammunition Surveillance Procedures |
| DA Pam 750-8 | The Army Maintenance Management System (TAMMS) Users Manual |
| (Application for copies of the above should be addressed to Commander, U. S. Army Publishing Directorate, Media Distribution Division, ATTN: JDHQSV-PAS, 1655 Woodson Road, St. Louis, MO 63114-6128 or online at https://www.apd.army.mil/). | |
| DOD Terminology Program | https://jdeis.js.mil |
| DoDM 5200.01, Volume 1 | DoD Information Security Program: Overview, Classification, and Declassification |
| DoDM 5200.01, Volume 2 | DoD Information Security Program: Marking of Information |
| DoDM 5200.01, Volume 3 | DoD Information Security Program: Protection of Classified Information |
| DoDI 5200.48 | Controlled Unclassified Information (CUI) |
| DoDI 4140.01 | DoD Supply Chain Materiel Management Policy |
| DoDI 5230.24 | Distribution Statements on DOD Technical Information |
| Copies of DoD documents are available online at https://www.esd.whs.mil/DD/DoD-Issuances/. | |
| TC 4-02.1 | First Aid |
| ATP 4-31 | Recovery and Battle Damage Assessment and Repair |
| TB 43-0118 | Field Instructions for Painting and Preserving Communications-Electronics Equipment |
| TB 43-0213 | Corrosion Prevention and Control (CPC) for Army Ground Equipment |
| TB 750-93-1 | Department of the Army Technical Bulletin Maintenance of Supplies and Equipment Functional Grouping Codes: Combat, Tactical, Support Vehicles, Medical and Special Purpose Equipment |
| ATP 3-04.7 | Army Aviation Maintenance |
| TM 1-1500-204-23 (vol 1–11) | Aviation Unit Maintenance (AVUM) and Aviation Intermediate Maintenance (AVIM) Manual for General Aircraft Maintenance (General Maintenance and Practices) |
| TM 1-1500-328-23 | Technical Manual Aeronautical Equipment Maintenance Management Procedures |
| TM 1-1500-335-23 | Nondestructive Inspection Methods, Basic Theory |
| TM 1-1500-344-23 (vol 1-4) | Cleaning and Corrosion Control<br>Volume I<br>Corrosion Program and Corrosion Theory |
| TM 1-1500-345-23 | Painting and Marking of Army Aircraft |
| TM 4-33.31 | Cold Weather Maintenance Operations |

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| | |
|---|---|
| TM 43-0139 | Chemical Agent Resistant Coating (CARC) Application and Quality Assurance on Army Ground Equipment |
| TM 55-1500-342-23 | Joint Service Technical Manual Organizational, Intermediate, and Depot Maintenance Aircraft Weight and Balance |
| (Copies of these publications are available from the U.S. Army Publishing Directorate, Media Distribution Division, 1655 Woodson Road, St. Louis, MO 63114-6128. Copies of TMs and TBs may be obtained from ETMs online on the LDAC website, https://www.ldac.army.mil/. Copies of FM 4-25.11, ATP 3-04.7, and TM 4-33.31 may be obtained from the TRADOC website, https://www.tradoc.army.mil/. | |
| Public Law 91-956 | OSHA Act of 1970 |
| Copies of this document may be obtained at https://www.osha.gov. | |
| EO 12196 | Occupational Safety and Health Programs for Federal Employees |
| EO 13423 | Strengthening Federal Environmental, Energy, and Transportation Management |
| EO 13526 | Classified National Security Information |
| Copies of these documents may be obtained at https://www.federalregister.gov/presidential-documents/executive-orders. | |
| H6 Item Name Directory | https://flisancillaryu.dla.mil/H6/H6home.aspx |
| U.S. Government Publishing Office (GPO) Style Manual | https://www.govinfo.gov/content/pkg/GPO-STYLEMANUAL-2016/pdf/GPO-STYLEMANUAL-2016.pdf |

## 2.3 Non-Government publications.

The following documents form a part of this document to the extent specified therein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

| | |
|---|---|
| AMERICAN INSTITUTE OF AERONAUTICS AND ASTRONAUTICS (AIAA) | |
| ANSI/AIAA S-102.2.4-2015 | Capability-Based Product Failure Mode, Effects and Criticality Analysis (FMECA) Requirements |
| AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME) | |
| ASME Y14.38 | Abbreviations and Acronyms for Use on Drawings and Related Documents |
| ASME Y14.100 | Engineering Drawing Practices |
| Copies may be obtained by visiting the American Society of Mechanical Engineers' website at https://www.asme.org/. DoD users may obtain copies via LDAC's Technical Publications Branch by submitting a request at https://armyeitaas.sharepoint-mil.us/teams/LDACTechnicalPublications. | |
| AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) | |
| ASTM-F856 | Standard Practice for Mechanical Symbols, Shipboard— Heating,Ventilation, and Air Conditioning (HVAC) |

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| | |
|---|---|
| Copies may be obtained by visiting American Society for Testing Material's website at https://www.astm.org/. DoD users may obtain copies via LDAC's Technical Publications Branch by submitting a request at https://armyei-taas.sharepoint-mil.us/teams/LDACTechnicalPublications. | |
| INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE) | |
| IEEE Std 91 | IEEE Standard Graphic Symbols for Logic Functions |
| IEEE Std 260.1 | IEEE Standard Letter Symbols for Units of Measurement (SI Units, Customary Inch-Pound Units, and Certain Other Units) |
| INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO) | |
| ISO 1219-1 | Fluid Power Systems and Components - Graphic Symbols and Circuit Diagrams – Part 1: Graphic Symbols for Conventional Use and Data Processing Systems |
| ISO 1219-2 | Fluid Power Systems and Components - Graphic Symbols and Circuit Diagrams – Part 2: Circuit Diagrams |
| ISO/IEC 8632 series | Information Technology – Computer Graphics – Metafile for the Storage and Transfer of Picture Description Information |
| ISO 9000 Series | Quality Management |
| ISO 10303 Series | Standard for the Exchange of Product Model Data (STEP) |
| Copies may be obtained by visiting the International Organization for Standardization's website at https://www.iso.org/. DoD users may obtain copies via LDAC's Technical Publications Branch by submitting a request at https://armyeitaas.sharepoint-mil.us/teams/LDACTechnicalPublications. | |
| IEEE Std 280 | IEEE Standard Letter Symbols for Quantities Used in Electrical Science and Electrical Engineering |
| IEEE Std 945 | IEEE Recommended Practice for Preferred Metric Units for Use in Electrical and Electronics Science and Technology |
| Copies may be obtained by visiting the Institute of Electrical and Electronics Engineers' website at https://www.ieee.org/. DoD users may obtain copies via LDAC's Technical Publications Branch by submitting a request at https://armyeitaas.sharepoint-mil.us/teams/LDACTechnicalPublications. | |

## 2.4 Order of Precedence.

In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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# 3 DEFINITIONS

## 3.1 Acronyms.

| | |
|---|---|
| AAL | Additional Authorization List |
| ABCA | American, British, Canadian, Australian |
| | Abbreviations, Brevity Codes, and Acronyms |
| ABN | Authentication Block Number |
| AFI | Air Force Instruction |
| AIAA | American Institute of Aeronautics and Astronautics |
| AMC | Aviation Maintenance Company |
| | Army Materiel Command |
| AMCOM | U.S. Army Aviation and Missile Command |
| Ao | Operational Availability |
| AOAP | Army Oil Analysis Program |
| APD | Army Publishing Directorate |
| APE | Ammunition Peculiar Equipment |
| API | Application Programming Interface |
| ANSI | American National Standards Institute |
| AR | Army Regulation |
| ASB | Aviation Support Battalion |
| ASC | Aviation Support Company |
| ASCC | Air Standardization Coordination Committee |
| ASME | American Society of Mechanical Engineers |
| ASTM | American Society for Testing and Materials |
| ATE | Automatic Test Equipment |
| ATO | Authority to Operate |
| ATP | Army Techniques Publication |
| AVIM | Aviation Intermediate Maintenance |
| AVUM | Aviation Unit Maintenance |
| AVMAC | Aviation Maintenance Allocation Chart |
| BDAR | Battle Damage Assessment and Repair |
| BII | Basic Issue Items |
| BIT | Built-in Test |
| BITE | Built-in Test Equipment |
| BOI | Basis of Issue |

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| | |
|---|---|
| CAD | Computer Assisted Design |
| CAGE | Commercial and Government Entity |
| CAGEC | Commercial and Government Entity Code |
| CARC | Chemical Agent Resistant Coating |
| CATT | Computer Automated Transportation Tool |
| CBRNE | Chemical, Biological, Radiological, Nuclear, and Explosives |
| CBT | Computer-Based Training |
| CECOM | U.S. Army Communications-Electronics Command |
| CFR | Code of Federal Regulations |
| CG | Coast Guard |
| CGM | Computer Graphics Metafile |
| CIC | Custom Industrial Coatings |
| CIS | Communication and Information System |
| CL | Component List |
| CM | Collateral Material |
| COAI | Combination of Adopted Items |
| COEI | Components of End Item |
| COMSEC | Communications Security |
| CONUS | Continental United States |
| CPC | Corrosion Prevention and Control |
| CSI | Critical Safety Item |
| CSP | Cyber Security Protection |
| CTA | Common Table(s) of Allowance |
| CUI | Controlled Unclassified Information |
| DA | Department of the Army |
| DD | Department of Defense (forms designation) |
| DMWR | Depot Maintenance Work Requirement |
| DoD | Department of Defense |
| DoDAC | Department of Defense Ammunition Code |
| DoDI | Department of Defense Instruction |
| DoDIC | Department of Defense Identification Code |
| DoDM | Department of Defense Manual |
| DOT | Department of Transportation |
| DR | Deficiency Report |



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| | |
|---|---|
| DSN | Defense Switching Network |
| DTC | Diagnostic Trouble Code |
| DTD | Document Type Definition |
| DTR | Defense Transportation Regulation |
| ECM | Electronic Countermeasure |
| ECP | Engineering Change Proposal |
| EIC | End Item Code |
| EIR | Equipment Improvement Recommendation |
| EM | Electronic Media |
| EMP | Electromagnetic Pulse |
| EMO | Electronic Media Only |
| EO | Executive Order |
| EPCO | Equipment Publications Control Officer/Office |
| ESD | Electrostatic Discharge |
| ETM | Electronic Technical Manual |
| FEDLOG | Federal Logistics Database |
| FGC | Functional Group Code |
| FM | Field Manual |
| FMC | Fully mission capable |
| FMECA | Failure Mode, Effects and Criticality Analysis |
| FO | Foldout |
| FP | Foldout Page |
| FRC | Final Reproducible Copy |
| FSC | Federal Supply Classification |
| GPO | Government Publishing Office |
| GCSS-A | Global Combat Support System - Army |
| HAP | Hazardous Air Pollutant |
| HAZMAT | Hazardous Material |
| HCI | Hardness Critical Item |
| HCP | Hardness Critical Process |
| HDBK | Handbook |
| HCSDS | Hazardous Component Safety Data Statement |
| HR | Hand Receipt |
| HVAC | Heating, Ventilation, and Air Conditioning |
| IADS | Interactive Authoring and Display Software |



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| | |
|---|---|
| IAW | in accordance with |
| IDN | initial distribution number |
| IEC | International Electrotechnical Commission |
| IEEE | Institute of Electrical and Electronics Engineers |
| IETM | Interactive Electronic Technical Manual |
| INSCOM | Information Systems Command |
| IR | Infrared |
| IRRD | Issue Receipt Release Document |
| ISO | International Organization for Standardization |
| ISPM | International Standard for Phytosanitary Measures |
| IUID | Item Unique Identification |
| JDRS | Joint Deficiency Reporting System |
| JP | Joint Publication |
| JPEG | Joint Photographic Experts Group |
| JTA | Joint Table of Allowances |
| LCMC | Lifecycle Management Command |
| LCN | Logistics Control Number |
| LDAC | Logistics Data Analysis Center |
| LIN | Line Item Number |
| LO | Lubrication Order |
| LOAP | List of Applicable Publications |
| LOEP | List of Effective Pages |
| LPD | Logistics Product Data |
| LRU | Line Replaceable Unit |
| MAC | Maintenance Allocation Chart |
| MAP | Minor Alteration Procedure |
| MATDEV | Materiel Developer |
| MC | Marine Corps |
| MEL | Maintenance Expenditure Limit |
| M/H | Man-hour |
| M&O | Maintenance and Overhaul |
| MHP | Munitions History Program |
| MIL | Military |
| MIL-PRF | Military Performance Specification |
| MIL-HDBK | Military Handbook |



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| | |
|---|---|
| MIL-STD | Military Standard |
| MOC | Maintenance Operational Checks |
| MOS | Military Occupational Specialty |
| MRP | Mandatory Replacement Part |
| MRPL | Mandatory Replacement Parts List |
| MSL | Military Shipping Label |
| MTBF | Mean Time Between Failures |
| MTF | Maintenance Test Flight |
| MTMC | Military Traffic Management Command |
| MTOE | Modified Table of Organization and Equipment |
| MTTR | Mean Time to Repair |
| MUX | Multiplex |
| MWO | Modification Work Order |
| N/A | Not Available/Applicable |
| NATO | North Atlantic Treaty Organization |
| NAVAIR | Naval Air Systems Command |
| NAVSEA | Naval Sea Systems Command |
| NDI | Nondestructive Inspection |
| NEPR | Naval Environmental Production Research |
| NETR | Nationwide Environmental Title Research |
| NHA | Next Higher Assembly |
| NIIN | National Item Identification Number |
| NMCM | Not Mission Capable Maintenance |
| NMCS | Not Mission Capable Supply |
| NMWR | National Maintenance Work Requirement |
| NSA | National Security Agency |
| NSN | National Stock Number |
| MSDS | Materials Safety Data Sheet |
| PA-BoM | Property Accountability - Bill of Materials |
| OCONUS | Outside the Continental United States |
| ODC | Ozone Depleting Chemicals |
| ODS | Ozone Depleting Substances |
| OIP | Overhaul Inspection Procedure |
| OJCS | Organization of the Joint Chiefs of Staff |



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| | |
|---|---|
| OSD | Office of the Secretary of Defense |
| OSHA | Occupational Safety and Health Act |
| Pam | Pamphlet |
| PCN | Publication Control Number |
| PDF | Portable Document Format |
| PDREP | Product Data Reporting and Evaluation Program |
| PEO | Program Executive Office |
| PID | Personal Identification |
| PIN | Publication Identification Number |
| PM | Phased Maintenance |
| PMAC | Preliminary Maintenance Allocation Chart |
| PMC | Preventive Maintenance Checklist |
| PMCS | Preventive Maintenance Checks and Services |
| PMD | Preventive Maintenance Daily |
| PMI | Phased Maintenance Inspection |
| PMS | Preventive Maintenance Services |
| P/N | Part Number |
| POC | Point of Contact |
| POL | Petroleum, Oils, and Lubricants |
| PQDR | Product Quality Deficiency Report |
| PRF | Performance |
| QA | Quality Assurance |
| QDR | Quality Deficiency Report |
| QTY | Quantity |
| RAM | Reliability, Availability, Maintainability |
| RCM | Reliability Centered Maintenance |
| REFDES | Reference Designator |
| RGL | Reading Grade Level |
| RMS | Reliability, Maintainability, and Supportability |
| RPSTL | Repair Parts and Special Tools List |
| SAE | Society of Automotive Engineers |
| SAM | Software Administrators Manual |
| SDDC | Surface Deployment and Distribution Command |
| SB | Supply Bulletin |
| SC | Supply Catalog |



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| | |
|---|---|
| SF | Standard Form |
| SKO | Sets, Kits, and Outfits |
| SMR | Source, Maintenance, and Recoverability |
| SOFM | Safety of Flight Message |
| SOUM | Safety of Use Message |
| SPI | special packaging instruction |
| SRA | Specialized Repair Activity |
| SRU | Shop Replaceable Unit |
| SSR | Supply System Responsibility |
| STANAG | Standardization Agreement |
| STD | Standard |
| STEP | Standard for Exchange of Product Model Data |
| SUM | Software Users Manual |
| SVC CO | Service company |
| SVG | Scalable Vector Graphic |
| SVGF | Tagged Image File Format |
| TACOM | U.S. Army Tank-automotive and Armaments Command |
| TAMMS | The Army Maintenance Management System |
| TAMMS-A | The Army Maintenance Management System-Aviation |
| TASMG | Theater Aviation Sustainment Maintenance Group |
| TB | Technical Bulletin |
| TBO | Time Between Overhaul |
| TC | Training Circular |
| TDA | Tables of Distribution and Allowances |
| TIL | Tools Identification List |
| ROD | Report of Discrepancy |
| TEA | Transportation Engineering Agency |
| TM | Technical Manual |
| TMDE | Test, Measurement, and Diagnostic Equipment |
| TMSS | Technical Manual Specifications and Standards |
| TO | Technical Order |
| TOC | Table of Contents |
| TOE | Table of Organization and Equipment |
| TPS | Test Program Set |
| TRADOC | Training and Doctrine Command |

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| | |
|---|---|
| U/I | Unit of Issue |
| UOC | Usable On Code |
| U.S. | United States |
| USA | United States Army |
| USAF | United States Air Force |
| USAMC | United States Army Materiel Command |
| USMC | United States Marine Corps |
| USN | United States Navy |
| UURI | Using Unit Responsibility Items |
| UUT | Unit Under Test |
| W3C | World Wide Web Consortium |
| WP | Work Package |
| WPM | Wood Packaging Materials |
| WS | Warning Summary |
| WTB | Warranty Technical Bulletin |
| XML | eXtensible Markup Language |
| XSL | eXtensible Stylesheet Language |
| XSL-FO | eXtensible Stylesheet Language – Formatting Object |
| XSLT | eXtensible Stylesheet Language Transformation |

## 3.2 Definitions

**TABLE I. Definitions**

| TERM | DEFINITION |
|---|---|
| **Abbreviation** | A shortened or contracted form of a word or phrase used to represent the whole word or phrase (for example, "U.S." for "United States"). Abbreviations may contain punctuation. |
| **Acronym** | A word formed from the initial letters or groups of letters of words in a set phrase or series of words, such as "PMCS" for "Preventive Maintenance Checks and Services." Acronyms contain no punctuation. |
| **Additional Authorization List (AAL) items** | Items that are optional (discretionary), are not essential to operate the end item, and are not listed on engineering drawings. Items are not turned in with the end item. |
| **Application** | Software usually designed to handle specific tasks for users. Used interchangeably with " software application" and often abbreviated to "app." |
| **Army Oil Analysis Program (AOAP)** | HQDA directed program to enhance crew safety, improve equipment readiness, and reduce the consumption of resources through application of advanced diagnostic technology to detect premature materiel degradation and support failure trend analysis. |

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**TABLE I. Definitions (continued)**

| TERM | DEFINITION |
|---|---|
| **Assembled item** | An item that has an "A" as the first letter of the Source, Maintenance, and Recoverability code (SMR). This indicates the item is not stocked as an assembly but is assembled from its constituent repair parts. |
| **Assembly** | A combination of components and/or modules and parts used as a portion of, and intended for, further installation in an equipment end item (for example, engine, transmission, rotor head, electronic chassis, rack, and/or cabinet). |
| **Assess-Only** | The Assess-Only process facilitates incorporation of new capabilities into existing approved environments while minimizing the need for additional Authorizations to Operate. Information Technology (IT) products (hardware, software), IT services, and Platform IT not authorized for operation through the full Risk Management Framework process must be securely configured in accordance with applicable DoD policies and security controls, and they must undergo special assessment of their functional and security-related capabilities and deficiencies. |
| **Attribute** | Declares an attribute name, specifies the form and XML-specific aspects of possible values, and specifies the action (such as providing a default value) to be taken if an attribute's value is not specified (**<element_name attribute= "value">**). Attributes appear inside the element start tag. |
| **Auxiliary equipment** | Equipment, accessories, or devices that, when used with basic equipment, extend or increase its capability (for example, Modified Table of Organization and Equipment (MTOE) items). |
| **Basic Issue Items (BII)** | The minimum essential items not listed in the drawings but required to place the equipment in operation, to operate it, and to perform emergency repairs. Although shipped separately packaged, BII must be with the equipment during operation and whenever they are transferred between property accounts. BII may be packed with Communications Security (COMSEC) equipment. |
| **Basis of Issue (BOI)** | The quantity of an item (special tool/kit) authorized for the end item density spread or for the unit level specified. |
| **Basic** | The original version of a new, authenticated publication that exists before any changes are published. |
| **Block diagram** | A modified schematic diagram in which each group of maintenance significant components that together performs one or more functions is represented by a single symbol or block. The block or symbol representing the group of components shows simplified relevant input and output signals pertinent to the subject diagram. |
| **Boilerplate** | Portions of verbatim text intended, and in many cases required, to be customized when reproduced in a publication. Customizations are made by supplying information usually indicated in italics within parentheses. |
| **Bug (Software)** | An error, flaw, failure, or fault in a computer program, application, or system that causes it to produce an incorrect or unexpected result, or to behave in unintended ways. |
| **Built-In Test (BIT)** | A test approach using Built-In Test Equipment (BITE) or other integral hardware designed into equipment or components under test to self-test and fault diagnose all and/or part of the equipment or component under test. |

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**TABLE I. Definitions (continued)**

| TERM | DEFINITION |
|---|---|
| **Built-In Test Equipment (BITE)** | Any identifiable, removable device that is part of equipment or components under test that is used for the express purpose of testing. |
| **Bulk material** | Material issued in bulk for manufacture or fabrication of support items (for example, sheet metal, pipe tubing, bar stock, or gasket material); excludes expendable items. |
| **Callout** | Anything placed on an illustration to aid in identifying the objects being illustrated, such as numeric callouts, nomenclature callouts, leader lines, and arrows. |
| **Change** | A change is composed of corrected pages/work packages to the basic manual (or previous editions). It consists of information that improves or clarifies the manual without requiring rewriting or reorganizing the technical content of the manual. |
| **Change bars** | Vertical bars that appear near changed material in a publication. |
| **Change levels** | The level of the change to the basic version of the publication (for example, Change 1, Change 2, etc.). |
| **Change-level attribute** | Identifies the change type and level in the XML. |
| **Change notation** | In a change package, the notation in the footer that identifies the change and its number (Change 1, Change 2, etc.). |
| **Change package** | An abbreviated version of a changed ETM containing only changed work packages prefaced by a change transmittal, authentication for the change, LOEP (only for classified publications and ammunition data sheets), cover (only if changed), and DA Form 2028 (only if changed). Sometimes called the "change only" file. Change packages allow for quick distribution of updated material and reduce costs related to labor and printing. |
| **Change transmittal** | The page or pages that serve as the primary historical record of a changed ETM. Sometimes called the "change sheet." |
| **Chemical, Biological, Radiological, Nuclear, and Explosives (CBRNE)** | Reference to warfare involving the use of chemical, biological, nuclear, or explosive weapons or where any of these hazards are present. Also includes the decontamination procedures performed on equipment and/or personnel exposed to chemical, biological, radiological, and nuclear weapons. The term "explosives" was added to account for improvised devices. |
| **Commercial and Government Entity Code (CAGEC)** | A five-character code assigned to commercial entities that manufacture or supply items used by the Federal Government and to Government activities that control design or are responsible for the development of certain specifications, standards, or drawings that control the design of Government items. CAGEC can be found at https://cage.dla.mil/. |
| **Complete repair** | Maintenance capacity, capability, and authority to perform all the corrective maintenance tasks in order to restore serviceability to a failed item. Excludes the prescriptive maintenance tasks, overhaul, and rebuild. |
| **Component** | A combination of parts mounted together during manufacturing that may be tested, replaced as a unit, or repaired (for example, starter, generator fuel pump, and printed circuit board (PCB)). |

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**TABLE I. Definitions (continued)**

| TERM | DEFINITION |
|---|---|
| **Components of End Item (COEI)** | Items identified on the engineering drawing tree that are physically separated and distinct from the end item. |
| **Comprehensibility** | The completeness with which a user in the target audience understands the information in the publication. |
| **Computer Graphics Metafile (CGM)** | Provides a means of graphics data interchange for computer representation of 2D graphical information independent from any particular application, system, platform, or device. CGM contains a metafile that describes the content and additional function as in the standard. Essentially, CGM is a wrapper for the data and the data are explained in the metafile. Refer to ISO/IEC 8632. |
| **Condition Based Maintenance (Predictive Logistics)** | A set of maintenance processes and capabilities derived primarily from real-time assessment of weapon system condition obtained from embedded sensors and/or external test and measurements using portable equipment. |
| **Consolidated file** | The version of an authenticated "change" publication that incorporates changed material into the original publication. Sometimes called the "full-up" or "included file." |
| **Corrosion Prevention and Control (CPC)** | Systematic maintenance steps/procedures taken to prevent, retard, repair, or otherwise mitigate the gradual rusting, oxidation, pitting, embrittlement, or other degradation/destruction of metallic and non-metallic surfaces, such as rubber and plastic, due to exposure to corrosive environments. CPC examples include spot painting, application of corrosion inhibiting compounds, cleaning, abrasive blasting, etc. |
| **Crew (operator) maintenance** | The first and most critical operation of the Army Maintenance System. Begins with the operator and/or crew performing Preventive Maintenance Checks and Services (PMCS) using the applicable operator manuals.<br><br>The before-and-during-PMCS concentrates on ensuring equipment is fully mission capable (FMC). Maintenance operations normally assigned to operator and/or crew include the following:<br><br>**a.** Performance of PMCS.<br><br>**b.** Inspections by sight and touch of accessible components per the operator manual and condition based maintenance indicators or instrumentation.<br><br>**c.** Lubrication, cleaning (including corrective actions to repair corrosive damage), preserving (including Corrosion Inhibiting Compound application and spot painting), tightening, replacement, and minor adjustments authorized by the Maintenance Allocation Chart (MAC).<br><br>**d.** Replacement of combat spares (unserviceable parts, modules, and assemblies) as authorized by the MAC and carried on board the equipment or system. |
| **Critical Safety Item (CSI)** | Any part, assembly, subassembly, installation procedure, or production process that would have hazard probability level A, B, C, or D chance of resulting in an unsafe condition if not in accordance with design data or quality requirements. |
| **Critical Torque** | The dual engine torque value that, when exceeded, may not allow the aircraft to maintain % RPM R within normal limits under single-engine operations in the same flight conditions. |

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**TABLE I. Definitions (continued)**

| TERM | DEFINITION |
|---|---|
| **Debug** | To find and reduce the number of errors, flaws, faults, failures, or defects in software. |
| **Degradation** | The reduction in system, subsystem, or component performance capability. |
| **Department of Defense (DoD)** | The Office of the Secretary of Defense (OSD) (including all boards and councils), the Military Departments (Army, Navy, Marine Corps, Air Force, and Space Force), the Organization of the Joint Chiefs of Staff (OJCS), the Unified and Specified Commands, the National Security Agency (NSA), and the Defense Agencies. |
| **Department of Defense Ammunition Code (DoDAC)** | An eight-character code developed to indicate interchangeability of ammunition and explosive items in Federal Supply Classification (FSC) Group 13 and Group 14. This eight-character code is divided into two parts. The two parts are separated by a hyphen. The first four digits represent the Federal Supply Class (FSC) and the last four are alphanumeric characters representing the DoD Identification Code that is assigned to items that are interchangeable in function and use. The DoDAC is used for ammunition operations such as worldwide stock status reporting and requisitioning. |
| **Depot-level maintenance** | Materiel maintenance requiring major overhaul or a complete rebuilding of parts, assemblies, subassemblies, and end items, including the manufacture of parts, modifications, testing, and reclamation as required. Depot maintenance serves to support lower categories of maintenance by providing technical assistance and performing that maintenance beyond their responsibility. Depot maintenance provides stocks of serviceable equipment because it has more extensive facilities available for repair than are available in lower maintenance activities. Depot maintenance includes all aspects of software maintenance. |
| **Depot Maintenance Work Requirement (DMWR)** | A maintenance serviceability standard for depot-level repairables designated for repair and return to Army Working Capital Fund stock. It prescribes the scope of work to be performed on an item by organic depot maintenance facilities or contractors, and/or qualified below-depot sources of repair; types and kinds of materiel to be used, and quality of workmanship. The DMWR also addresses repair methods, procedures and techniques, modification requirements, fits and tolerances, equipment performance parameters to be achieved, quality assurance discipline, and other essential factors to ensure that an acceptable and cost-effective product is obtained. |
| **Digital graphics form** | Graphic file types for use in electronic publications include 2D vector and raster formats. Vector file format is based on mathematical shapes on an X-Y grid and is scalable without loss of detail. Scalable Vector Graphics file (SVG) and Computer Graphics Metafile (CGM) vector file formats are preferred for line drawings, charts, and diagrams. Vector format has a smaller file size than raster format. Raster, or bitmap, file format is based on an array of pixels and cannot scale up without loss of detail (pixelization). Joint Photographic Experts Group (JPEG) and Tagged Image File Format (SVGF) raster file format are acceptable for photographs and software screenshot illustrations. |
| **Disc** | Removable and portable storage medium used to house data (for example, CD or DVD. |
| **Distribution medium** | The method of distribution for a technical publication (for example, paper, CD, DVD, etc.). |

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**TABLE I. Definitions (continued)**

| TERM | DEFINITION |
|---|---|
| **Document instance** | The instance is the actual document text and its accompanying eXtensible Markup Language (XML) tags conforming to the specifications and restrictions set forth in the Document Type Definition (DTD). |
| **Document Type Definition (DTD)** | The definition of the markup rules for a given class of documents. A DTD or reference to one should be contained in any XML conforming document. |
| **Durable items** | Supplies not consumed in use that retain their original identity during the period of use, such as weapons, machines, tools, and equipment. |
| **Effectivity** | The act or process of identifying weapon systems or end-items and their hardware and software system and subsystems by their associated Usable On Code (UOC), serial number, model number, Part Number (P/N)/CAGEC, National Stock Number (NSN), End Item Code (EIC), software version or Modification Work Order (MWO). Effectivity is included to signify that certain configurations or modifications apply to a given weapon system/equipment. |
| **Electronic Countermeasure (ECM)** | An electrical or electronic device designed to trick or deceive radar, sonar, or other detection systems, like Infrared (IR) or lasers. It may be used both offensively and defensively to deny targeting information to an enemy. The system may make many separate targets appear to the enemy, or make the real target appear to disappear or move about randomly. It is used effectively to protect aircraft from guided missiles. Most air forces use ECM to protect their aircraft from attack. It has also been deployed by military ships and recently on some advanced tanks to fool laser/IR guided missiles. It is frequently coupled with stealth advances so the ECM systems have an easier job. Offensive ECM often takes the form of jamming. Defensive ECM includes using blip enhancement and jamming of missile terminal homers. |
| **Electronic Media (EM) Number** | A chronologically numbered four-digit number (assigned by Army Publishing Directorate (APD)), using zeros when necessary to maintain four digits, following the letters "EM" (for example, EM 0001, EM 0002, EM 0003). The EM number functions as the disc nomenclature assigned to an Electronic ETM/ IETM composed of one or more discs (for example, the same EM number applies to all discs distributed as a set if the series and size of related equipment/weapon system publications dictate use of more than one disc). EMs have an additional "EM PIN" in addition to a regular TM PIN. |
| **Electronic Media Only (EMO) publications** | Publications that, while they may be printed at some time in the future, will not be printed at the time of authentication. |
| **Electronic Technical Manual (ETM)** | A page-oriented file usually based on a paper original that may or may not be prepared from a digital database. An ETM usually has hyperlinks added. ETMs can be distributed as digital media (for example, Portable Document Format (PDF)) or printed on paper. |
| **Electrostatic Discharge (ESD)** | Static electricity. A transfer of electrostatic charge between objects of different potentials caused by direct contact or induced by an electrostatic field. Devices such as integrated circuits and discrete devices (for example, resistors, transistors, and other semiconductor devices) are susceptible to damage from electrostatic discharge. |
| **Element** | A component of the hierarchical structure defined by a document type declaration (DTD) or Schema. It is identified in a document instance by |

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**TABLE I. Definitions (continued)**

| TERM | DEFINITION |
|---|---|
| | descriptive markup, a start-tag and end-tag, shown as: **<element_type_name attribute = "value" attribute= "value">**. |
| **Em space** | A space equal to the type size in points. |
| **Embedded** | Describes hardware and/or software that forms an integral part/component of some larger system and that is expected to function without human intervention. An embedded system usually does not include peripherals (for example, keyboard, monitor, storage, etc.). Embedded systems most often will provide real-time response. |
| **En space** | Half an em, and thus a space equal to half the type size in points. |
| **End Item Code (EIC)** | A 3-position alphanumeric code assigned to each end item managed or used by the Army, which meets all the following criteria:<br>**a.** End items with an NSN recorded in the Army master data file.<br>**b.** Type classified standard, low-rate production, or limited procurement-urgent in accordance with AR 700–142.<br>**c.** Assigned appropriation/budget activity account code A through Q inclusive. |
| **Equipment** | Assemblies, components, modules, and parts that are designed to perform an operational function and is ready for intended use. These end items may require other end items to perform a mission (for example, tank, truck, radio, generators, and machine guns). |
| **Equipment conditions** | Any conditions that must be met before an operator or maintenance work package can be started. Equipment conditions are listed in the initial setup and linked to appropriate work packages. |
| **Equipment Improvement Recommendation (EIR)** | Written reports on an SF 368 to report equipment faults in design operations and manufacturing of new equipment received that is below standard quality in workmanship under AR 702–7 and AR 702–7–1. |
| **Equipment nomenclature** | The official name of the equipment as shown in the FEDLOG H6 listing (https://www. dla. mil/Working-With-DLA/Federal-and-International-Cataloging/H6/), except for software. |
| **Essential** | Those systems/subsystems/components that are required for a designated mission or system operation. |
| **Evacuation** | A combat service support function that involves the movement of recovered materiel from a main supply route; maintenance collection materiel may be returned to the user, to the supply system for reissue, or to property disposal activities. |
| **Expendable items** | Items other than repair parts that are consumed in use (for example, paint, lubricants, wiping rags, tape, cleaning compounds, sandpaper). |
| **eXtensible Markup Language (XML)** | A set of rules for encoding documents electronically through the use of markup. Its primary purpose is to facilitate the sharing of structured data across different information systems. It is a product of the World Wide Web Consortium (W3C). |

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**TABLE I. Definitions (continued)**

| TERM | DEFINITION |
|---|---|
| **eXtensible Stylesheet Language (XSL)** | A stylesheet language that can be used for rendering XML documents. |
| **eXtensible Stylesheet Language Formatting Objects (XSL-FO)** | A subset of eXtensible Stylesheet Language Transformation (XSLT) that is used to format valid and well-formed XML into a page-oriented output. This output may be a direct print to paper or it may be to an electronic page-oriented presentation such as a Portable Document Format (PDF) file. |
| **eXtensible Stylesheet Language Transformation (XSLT)** | A declarative, XML-based language used to transform XML documents into other XML documents. XSLT is supported by the World Wide Web Consortium (W3C). |
| **Field-level maintenance** | The first operation of the Army Maintenance System characterized by the performance of maintenance tasks "on system" in a tactical environment using trained personnel, tools, and test, measurement, and diagnostic equipment (TMDE). Field-level maintenance is typically operator and/or crew maintenance and repair and return to user maintenance operations. |
| **Final Reproducible Copy (FRC)** | The final document ready for reproduction and publication as an authenticated publication, including all necessary changes made as a result of validation/ verification and acquisition activity conditions of acceptance or approval. |
| **Follow-on maintenance** | A maintenance task or tasks that must be accomplished after the completion of a maintenance task or tasks. Follow-on maintenance is used to clean up or undo actions performed prior to or during a maintenance task and may be done directly after the task or after several tasks. For example, if a panel is removed to perform maintenance, it must be installed when the maintenance task is complete or may be installed after several tasks requiring removal of the panel are completed. |
| **Footer** | One or more lines of standard text that appear at the bottom of each page. |
| **Functional diagram** | A type of illustration in which symbols are connected by lines to show relationships among the symbols. The symbols may be rectangles or other shapes, standard electronic symbols representing components or functions, or pictorials representing equipment or components. Where appropriate, voltage readings are shown. The lines may represent procedures or processes, such as signal or logic flow, and physical items, such as wires. Functional diagrams include schematics, wiring and piping diagrams, flow charts, and block diagrams. |
| **Functional Group Code (FGC)** | A numeric or alphanumeric code assigned to identify major components, assemblies, and subassemblies to a functional system. Subordinate subfunctional groups/subassemblies are coded to relate back to the basic (top position) FGC in a sequential, Next Higher Assembly (NHA) relationship. For aviation systems, FGCs are prescribed by DA Pam 738-751. TB 750-93-1 may be used for some tactical vehicles as specified by the acquiring activity. Software used on the equipment is considered a major component and would be assigned an FGC accordingly. |
| **General maintenance** | Procedures that can be applied to multiple types of equipment and can be provided in a maintenance manual for the equipment or in a separate general maintenance manual. Examples include painting, lubrication, preservation, cleaning, marking, and corrosion prevention. |

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**TABLE I. Definitions (continued)**

| TERM | DEFINITION |
|---|---|
| **Graphic** | Any type of presentation or representation that gives a clear visual impression; often used interchangeably with "illustration," "figure," "drawing," "diagram," and "artwork." |
| **Hazardous Air Pollutant (HAP)-free** | A material that contains no more than 0.1 percent by mass of any individual HAP that is an Occupational Safety and Health Act (OSHA) defined carcinogen as specified in 29 Code of Federal Regulations (CFR) 1910.1200(d)(4) and no more than 1.0 percent by mass for any other individual HAP, as demonstrated by a specification or a standard, or a manufacturer's representation, such as in a Safety Data Sheet (SDS). |
| **Hardness Critical Item (HCI)** | A support item that provides the equipment with special protection from Electromagnetic Pulse (EMP) damage during a nuclear attack. |
| **Hardness Critical Process (HCP)** | A process affecting a mission critical item that could degrade system survivability in a nuclear, biological, or chemical hostile environment if hardness were not considered. Nuclear HCPs are processes, finishes, specifications, manufacturing techniques, and/or procedures that are hardness critical, and that, if changed, could degrade nuclear hardness. |
| **Hardtime interval maintenance** | Scheduled maintenance conducted at predetermined fixed intervals because of age, calendar, or use such as operating time, flying hours, miles driven, or rounds fired. |
| **Header** | One or more lines of standard text that appear at the top of each page. |
| **Horizontal (landscape) format** | Positioning of technical manual so that page horizontal (width) dimensions are greater than vertical (height) dimensions. |
| **Hover** | To move a cursor over a clickable object, such as a link, without clicking on that object. |
| **Icon** | Pictorial representation; visual image to give immediate recognition of a hazard or to provide essential information. |
| **Illustration** | A general term for graphic presentations of all types, to include pictorials, functional diagrams, line graphs, and photographs. Often used interchangeably with "figure," "graphic," "drawing," "diagram," and "artwork." |
| **Inline** | Data such as dialog boxes, figures, graphics, and icons that are placed in the manual so that they are embedded with the text. |
| **Institute of Electrical and Electronics Engineers (IEEE)** | Membership organization that includes engineers, scientists, and students in electronics and allied fields. Founded in 1963, it has over 300,000 members and is involved with setting standards for computers and communications. |
| **Interactive Electronic Technical Manual (IETM)** | A technical manual designed for electronic window display and possessing the following three characteristics:<br>**a.** Format and style are optimized for window presentation; that is, the presentation format is frame-oriented, not page-oriented.<br>**b.** The elements of technical data constituting an IETM are so interrelated that a user's access is facilitated and is achievable by a variety of paths.<br>**c.** An IETM can function interactively (as a result of user requests and information input). |

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**TABLE I. Definitions (continued)**

| TERM | DEFINITION |
|---|---|
| **Interchangeability** | The ability to use a part/component on more than one end item or assembly due to similar fit, form, and function. |
| **International Organization for Standardization (ISO)** | Organization that sets international standards, founded in 1946 and headquartered in Geneva. It deals with all fields except electrical and electronics, which is governed by the older International Electrotechnical Commission (IEC), also in Geneva. With regard to information processing, ISO and IEC created the Joint Technical Committee (JTC 1) for Information Technology. |
| **(Item Unique Identification (IUID)** | A system of marking items delivered to the DoD with Unique Item Identifiers (UII) that have machine-readable data elements to distinguish an item from all other like and unlike items. UIIs assigned to items are registered in the DoD IUID Registry. |
| **Landscape mode** | A horizontal orientation of a document or image, commonly used to print charts, images, and/or text that may not fit on a normal (vertical) page. |
| **Leak rate** | The speed or rate of flow of accidental escape of fluid or gas from a system, which is caused by damage. The leak rate is influenced by such factors as the hole size, internal/external pressures, and fluid level. |
| **Legend** | A tabular listing and explanation of the numbers or symbols on a figure or an illustration. |
| **Limited repair** | Scope of corrective repair authorized to be performed by a level of maintenance lower than the level of authorized complete repair. |
| **Line Replaceable Unit (LRU)** | A combination of components and/or modules installed in an item of equipment or system that is replaceable in the operational environment (that is, under field or combat conditions). An LRU may be a PCB, black box, component, major component, alternator, carburetor, avionics, tank engine, and road wheel assembly installed weapons. This repair by replacement is normally accomplished as far forward as possible by unit (organizational) maintenance personnel. |
| **Linear Data** | Linear data are technical data that are displayed in a sequential or document-oriented manner. The sequence of the data presentation is largely determined by the data author. It is an organization of technical data that often replicates the order of information found in a page-based document. There is generally a default "path" through the technical data. Linear data should be used for information/tasks that must be done or must follow a certain order such as operating instructions, general information, some maintenance tasks, and simple troubleshooting tasks. |
| **List of Applicable Publications (LOAP)** | A separate listing of publications related to a specific piece of equipment, group of equipment, or system. For additional information, refer to MIL-PRF-63049. |
| **Logistics Product Data (LPD)** | The LPD comprises the support and support-related engineering and logistics data acquired for use in materiel management processes such as those for initial provisioning, cataloging, and item management. Depending upon specific program requirements, these data may be in the form of summary reports, a set of specific data products, or both. |

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**TABLE I. Definitions (continued)**

| TERM | DEFINITION |
|---|---|
| **Maintainer Maintenance** | Consists of the following:<br>**a.** Performance of PMCS.<br>**b.** Inspections.<br>**c.** Lubrication, cleaning, preserving, tightening, replacement, and minor adjustments authorized by the MAC.<br>**d.** Diagnosis and fault isolation as authorized by the MAC.<br>**e.** Replacement of unserviceable parts, modules, and assemblies as authorized by the MAC.<br>**f.** Requisition, receipt, storage, and issue of repair parts.<br>**g.** Verification of faults and level of repair of unserviceable materiel prior to evacuation.<br>**h.** If beyond the MAC authorization, evacuate to next higher level (sustainment) or turn in to the appropriate supply support activity.<br>**i.** Recovery or coordination for transportation of equipment to and from the support unit of action.<br>**j.** Accomplishment of all actions directed by the AOAP.<br>**k.** Materiel readiness reporting.<br>**l.** Coordination and annotation of field-level MWOs.<br>**m.** Providing maintenance support to sustainment maintenance activities.<br>**n.** Diagnosis and isolation of materiel or module malfunctions, adjustment, and alignment of modules that can be readily completed with assigned tools and TMDE.<br>**o.** Performance of light body repair.<br>**p.** Turn-in of components with maintenance repair codes of F, D, H, and L recoverable components to the support activity.<br>**q.** Fabrication as identified by the appropriate publication.<br>**r.** Operation of cannibalization points, when authorized.<br>**s.** Identify and annotate corrosion and take corrective actions within the organization's capability to prevent or repair corrosion damage to Army materiel.<br>**t.** Apply Corrosion Inhibiting Compounds IAW TB 43-0213, Corrosion Prevention and Control (CPC) for Army Ground Equipment.<br>**u.** Performance of Spot-Painting IAW TM 43-0242, Chemical Agent Resistant Coating Spot Painting for Army Ground Equipment.<br>**v.** Performance of Repainting IAW TM 43-0139, Chemical Agent Resistant Coating (CARC) Application and Quality Assurance on Army Ground Equipment.<br>**w.** Ensure that inspection, maintenance requests, and deficiency reports involving corrosion are submitted using the Corrosion/Rotted cause code |

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**TABLE I. Definitions (continued)**

| TERM | DEFINITION |
|---|---|
| | (170). This includes failures where corrosion contributed to system or equipment failure or for maintenance performed as either a preventive measure or as a result of corrosion. Reports include:<br>**1.** DA Form 2404 (Equipment Inspection and Maintenance Worksheet).<br>**2.** DA Form 5988-E (Equipment Maintenance and Inspection Worksheet).<br>**3.** DA Form 2407 (Maintenance Request).<br>**4.** DD Form 1225 (Storage Quality Control Report).<br>**5.** SF 364 (Report of Discrepancy (ROD)).<br>**6.** SF 368 (Product Quality Deficiency Report). |
| **Maintenance Allocation Chart (MAC)** | A list of equipment maintenance tasks showing maintenance level and corresponding man-hours required for each task. The MAC is arranged in either functional group code sequence or top-down breakdown sequence and uses the same sequence as used in the RPSTL. |
| **Maintenance level** | The primary division of maintenance activities. The U.S. Army uses a two-level maintenance concept of "field" and "sustainment."<br>Maintenance sublevels identify and implement the specific activity, identified by the MAC, to perform the maintenance. The maintenance levels of both the field and sustainment maintenance are further separated by aviation and non-aviation and the corresponding levels are shown below:<br>**a.** Field-level:<br>**1.** Aviation:<br>(a) Theater Aviation Sustainment Maintenance Group (TASMG) – MAC code – L.<br>(b) Depot – corresponds to MAC code – D.<br>**2.** Non-aviation:<br>**a.** Below Depot – MAC code – H.<br>**b.** Depot – corresponds to MAC code – D.<br>**b.** Sustainment-level:<br>**1.** Aviation:<br>**a.** Aviation Maintenance Company (AMC) – MAC code – O.<br>**b.** Aviation Support Battalion (ASB) – MAC code – F.<br>**2.** Non-aviation:<br>**a.** Theater Aviation Sustainment Maintenance Group (TASMG) – MAC code – L.<br>**b.** Depot – MAC code – D. |
| **Maintenance significant item** | An end item, assemblage, component, or system intended for issue to the Army in the field that will require corrective maintenance services on a recurring basis. End items, assemblages, components, or systems that require |

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**TABLE I. Definitions (continued)**

| TERM | DEFINITION |
|---|---|
| | maintenance while in storage; generally, these are Line Item Numbers (LINs) contained within the Army Maintenance Master Data File (MMDF). |
| **Maintenance task** | A group of instructions with supporting illustrations on how to perform a maintenance action, such as remove or install. Each task has a definite beginning and end. A task is made up of one or more procedures. |
| **Man-hours** | Man-hour time is only shown in the Maintenance Allocation Chart and is determined by multiplying Task Time by the number of maintenance personnel required to perform the task. Man-hours are used for vehicle classification inspections and when determining Maintenance Expenditure Limits (MEL). |
| **MATDEV (Materiel Developer)** | A command or agency responsible for research and development, production, and fielding of a new materiel system (primarily Army; however, also used in various DoD-level publications as a descriptive term for acquisition commands, agencies, and Program Offices (POs)). The principal Army MATDEVs are the Army Program Executive Offices (PEOs)and/or PMs. For non-PEO and/or PM-managed systems, other MATDEVs include AMC, U. S. Army Information Systems Command (INSCOM), Chief of Engineers, the Surgeon General, and Strategic Defense Command. |
| **Mean Time Between Failure (MTBF)** | A basic measure of reliability. The average number of failures of a specific item occurring during a specified time interval. |
| **Mean Time To Repair (MTTR)** | A basic measure of maintainability. The sum of corrective maintenance times at any specific level of repair, divided by the total number of failures within an item repaired at that level, during a interval under stated conditions. |
| **Modified Table of Organization and Equipment (MTOE)** | A modified version of a TOE that authorizes the unit organization, personnel, and equipment needed to perform an assigned mission in a specific geographical or operational environment. |
| **Modification Work Order (MWO)** | Detailed instructions (including text and graphics) for making changes/ improvements to a particular system in order to bring the system up to date and/ or to improve its overall efficiency. Refer to MIL-PRF-63002. |
| **Module** | **1.** Formerly "chapter." A section of a publication that groups together related work packages.<br>**2.** An assembly containing a complete self-contained circuit or sub-circuit. It may consist of a single printed circuit board (PCB); in which case it is synonymous with a PCB or may comprise two or more PCBs mechanically attached to one another and removable from the next high assembly as a single unit. |
| **Mouse-over** | A software program element that triggers a change on an item (typically a graphic change, such as making an image or hyperlink appear) in a viewer when the pointer passes over it. The change usually signifies that the item is a link to related or additional information. Mouse-overs are used in navigation bars, pop-up dialog boxes, windows, and/or in form submissions. |
| **Multi-service publication** | A publication containing principles, terms, tactics, techniques, and procedures used and approved by the forces of two or more Services to perform a common military function consistent with approved joint doctrine. |

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**TABLE I. Definitions (continued)**

| TERM | DEFINITION |
|---|---|
| **Multi-sheet** | A series of related illustrations in a series that are broken into "sheets" (for example, "Figure 17, Sheet 1"). |
| **Multi-view** | Illustrations that identify significant features with views oriented and enlarged as necessary and text clarifying how these views relate to the main illustration (for example, "Rotated 180 degrees" ). |
| **National Item Identification Number (NIIN)** | The last nine digits of the National Stock Number (NSN). The first two digits of the NIIN identify the country assigning the number and the remaining seven digits are a serially assigned number. |
| **National Maintenance Work Requirement (NMWR)** | A maintenance serviceability standard for field-level reparables designated for repair and return to Army working capital fund stock. It prescribes the statement of work to be performed on an item by organic depot maintenance facilities, certified non-depot national providers, or contractors; types and kinds of materiel to be used; and quality of workmanship. The NMWR also addresses repair method, procedures and techniques, modification requirements, fits and tolerances, equipment performance parameters to be achieved, quality assurance discipline, and other essential factors to ensure that an acceptable and cost-effective product is obtained. |
| **National Stock Number (NSN)** | A 13-digit number assigned to a repair part to be used for requisitioning purposes. It consists of the four-digit Federal Supply Classification (FSC) and the nine-digit National Item Identification Number (NIIN) (for example, 5940-23-333-1234). |
| **Next Higher Assembly (NHA)** | The assembly/subassembly to which a part/component is attached. In a hierarchical parent/child part structure, next higher assembly refers to the "parent" of the part in question. |
| **Nomenclature** | The approved name or alphanumeric identifier assigned to an item, piece of equipment, or component in agreement with an organized designation system except for software. |
| **Non-linear data** | Technical data that are not displayed in a sequential fashion. There are high levels of interactivity between the data and the user. The order of presentation is dictated by inputs from the user, from external sources, or from events (as in diagnostics). Non-linear organization of content does not follow a document or page-based paradigm. Non-linear data may be used for areas of the publication that do not require a specific order, such as complex troubleshooting or maintenance tasks. |
| **Operating instructions** | Explicit step-by-step actions for the user/operator to operate a piece of equipment. |
| **Operator and/or crew maintenance** | The cornerstone of Army maintenance, the first and most critical operation of the Army Maintenance System. Operator and/or crew maintenance starts with the operator and/or crew performing PMCS using the applicable TM–10 series. The before- and during-PMCS concentrate on ensuring equipment is FMC. Maintenance operations normally assigned to operator and/or crew include the following:<br>**a.** Performance of PMCS.<br>**b.** Inspections by sight and touch of accessible components in accordance with the TM–10 series and CBM indicators or instrumentation. |

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**TABLE I. Definitions (continued)**

| TERM | DEFINITION |
|---|---|
| | **c.** Lubrication, cleaning (including corrective actions to repair corrosive damage), preserving (including spot painting), tightening, replacement, and minor adjustments authorized by the MAC.<br>**d.** Limited diagnosis and fault isolation as authorized by the MAC. This requires appropriate resources onboard the equipment or system to perform these tasks.<br>**e.** Replacement of combat spares (unserviceable parts, modules, and assemblies) as authorized by the MAC and carried on board the equipment or system. |
| **Orphan** | Last line of a paragraph pushed to a new page, stranded alone (orphaned) at the top of the page without the rest of its paragraph. |
| **Outfit** | An assemblage of tools or equipment, type classified, assigned a LIN, with a unit of issue of outfit; it may include separately type classified items as a component, such as pneumatic tool and compressor outfit, water purification outfit, tool outfit hydraulic systems repair, and tool outfit pioneer portable electric tools. |
| **Overhaul Inspection Procedure (OIP)** | Routine maintenance inspection conducted just before a period specified for removal of aircraft for overhaul or retirement. |
| **Pane** | Any of the rectangular frames within the main content area of the inner shell into which a computer display can be divided and in which text/graphics/ multimedia output can be displayed. |
| **Part Number (P/N)** | A primary number used to identify an item used by the manufacturer (individual, company, firm, corporation, or Government activity) that controls the design, characteristics, and production of the item by means of its engineering drawings, specifications, and inspection requirements. |
| **Parts information module** | Parts information includes all data for major equipment systems, their subsystems, assemblies, components, Shop Replaceable Units (SRUs), and Line Replaceable Units (LRUs) and a subset of data (within parts information) known as a Repair Parts and Special Tools List (RPSTL) which authorizes for the requisitioning, issue, and disposition of spares, repair parts, and special tools in accordance with Source Maintenance and Recoverability (SMR) codes (see full "RPSTL" definition below). Parts information for an equipment system can stand alone as its own technical publication or be included with other operations and maintenance data as a module in a publication (see "parts information module" definition below).<br><br>Often the terms "parts information" and "RPSTL" are used interchangeably in common discourse; however, a RPSTL is a subset of parts information. |
| **Parts information module** | A parts information module is composed of an introduction work package (WP) and one or more accompanying WPs that authorize repair to system components. Parts information modules can appear in one of three ways:<br>**a.** a stand-alone technical publication comprised of only a parts information module and contains a subset RPSTL section (see full "RPSTL" definition below); |

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**TABLE I. Definitions (continued)**

| TERM | DEFINITION |
|---|---|
| | **b.** a module containing a RPSTL subsection and included in a larger Depot Maintenance Work Requirement (DMWR) / National Maintenance Work Requirement (NMWR) publication;<br>**c.** or a module containing an RPSTL subsection and included technical publication for an equipment system. |
| **Phased Maintenance Inspection (PMI) (aircraft)** | A thorough and searching examination of the aircraft and associated equipment. Removal of access plates, panels, screens, and some partial disassembly of the aircraft is required to complete the inspection. Inspections are due after an appointed number of flight hours since new or from the completion of the last inspection. |
| **Pictorial** | A type of diagram used to show the physical view of components and to show relative location and size. |
| **Portrait mode** | A vertical orientation of a document or image; the most commonly used format for reading/printing publications. |
| **Pre-shop analysis** | To determine, before beginning maintenance activities, the extent of maintenance required to return the end item, assembly, subassembly, or component to a serviceable condition as specified by the depot-level maintenance instructions. |
| **Preventive maintenance (scheduled maintenance)** | The performance of scheduled inspections and maintenance tasks necessary to keep the equipment in serviceable condition and ready for its primary mission. |
| **Preventive Maintenance Checklist (PMC)** | A stand-alone publication that contains PMCS information for any or all maintenance levels and intervals performed by the operator and/or maintainer to ensure the equipment is mission capable and in good operating condition. The information in the PMC is extracted from the associated operator and/or maintenance manuals. |
| **Preventive maintenance daily (aircraft)** | Inspection of aircraft and associated equipment after the last flight of the mission day or before the first flight of the next day. Some operational checks and removal of screens, panels, and inspection plates may be required to accomplish the inspection. |
| **Preventive maintenance services inspection (aircraft)** | Special recurring inspection of aircraft and associated equipment after an appointed number of flight hours or days, whichever occurs first (for example, 10 flight hours or 14 days). Some operational checks and removal of screens, panels, and inspection plates may be required to accomplish the inspection. |
| **Preventive Maintenance Checks and Services (PMCS)** | Preventive maintenance checks and service is the care, servicing, inspection, detection, and correction of minor faults before these faults cause serious damage, failure, or injury. The procedures and the category of maintenance to perform PMCS are found in the LO and TM operator and maintainer manuals. PMCS is for non- aviation systems only. Scheduled PMCS is usually performed by field maintainers and takes the equipment out of service for a period of time. |
| **Procedure** | A set of step-by-step actions required to accomplish a task or a portion thereof. There are one or more procedures in a task. |
| **Proponent** | The agency or command responsible for initiating, developing, coordinating, approving content, and issuing a publication, as well as identifying a |

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**TABLE I. Definitions (continued)**

| TERM | DEFINITION |
|---|---|
| | publication for removal. Each publication has only one proponent. Only HQDA principal officials can be proponents for DA policy publications. |
| **Publication Control Number (PCN)** | A unique 11–digit identification number assigned to Marine Corps publications. |
| **Publication Identification Number (PIN)** | A number (for ETMs) or numbers (for IETMs) assigned by Army Publishing Directorate (APD) to each publication. PINs are composed of nine numbers: six numbers followed by a hyphen and a three-digit "change number" that permits ordering a specific change to the publication (for example, -001 for change 1,-023 for change 23).<br><br>A change retains the basic publication's PIN and adds the number of the current change (for example, 067394-001, 067394-002, etc.).<br><br>In addition to the regular PIN described above, IETMs have a second EM ("Electronic Media" ) PIN. It is presented in the same format as the PIN. |
| **Publication number** | The number shown on the cover of each publication as constructed per DA Pam 25-40 (for example, TM 1-1520-238-10). It is also displayed in the header of every page of page-based publications. |
| **Quality Assurance (QA)** | A planned and systematic pattern of all actions necessary to provide adequate confidence the item or product conforms to established technical requirements. |
| **Raster graphics** | Raster graphics use a dot matrix data structure (bitmap) representing a generally rectangular grid of pixels, or points of color. Raster graphics are resolution dependent and cannot scale up to an arbitrary resolution without loss of apparent quality, and they are not easily editable. File format examples of raster graphics include SVGF and JPEG. |
| **Reading Grade Level (RGL)** | A measurement of reading difficulty of text related to grade levels (such as ninth-grade level, fourteenth-grade level, etc.). |
| **Reference Designator (REFDES)** | Letters, numbers, or both, used to identify and locate discrete units, portions thereof, and basic parts of a specific equipment, assembly, or subassembly. |
| **Reliability, Maintainability and Supportability (RMS) and Operational Availability (Ao)** | Requirements imposed on materiel systems to ensure that they are operationally ready for use when needed, will successfully perform assigned functions, and can be economically operated and maintained within the scope of logistic concepts and policies. |
| **Reliability Centered Maintenance (RCM)** | A systematic approach for identifying preventive maintenance tasks for an equipment end item in accordance with a specified set of procedures and for establishing intervals between maintenance tasks. |
| **Repairable item** | An item that can be restored to perform all its required functions by corrective maintenance. |
| **Repair part** | Consumable items or material required for the maintenance, overhaul, or repair of a system, equipment, or end item. |
| **Repair Parts and Special Tools List (RPSTL)** | Provides authorized spares and repair parts; special tools; special Test, Measurement, and Diagnostic Equipment (TMDE); and other special support equipment required for performance of all levels of maintenance of the weapon system/equipment, subsystems, assemblies, and components. The RPSTL |

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**TABLE I. Definitions (continued)**

| TERM | DEFINITION |
|---|---|
| | authorizes the requisitioning, issue and disposition of spares, repair parts and special tools in accordance with the SMR codes. |
| **Revision** | A revision is composed of corrected, updated, or additional pages or work packages to the current edition of a publication. It consists of replacement work packages that contain new or updated technical information or improves, clarifies, or corrects existing information in the current edition of the manual. |
| **Schematic diagram** | A graphic representation showing the interrelationship of each component or group of components in the system/ equipment. The essential characteristic of these diagrams is that every maintenance-significant functional component is separately represented. Also, where appropriate, voltage readings, hydraulic values, and pneumatic values are shown. |
| **Service** | Operations required periodically to keep an item in proper operating condition, such as replenishing fuel, lubricants, chemical fluids, or gases. |
| **Set** | A unit and necessary assemblies, subassemblies, and parts connected together or used in association to perform an operational function (for example, radio receiving set, measuring set, radar, or homing set that includes parts, assemblies, and units such as cables, microphones, and measuring instruments). |
| **Source, Maintenance, and Recoverability (SMR) code** | This six digit code is composed of four parts consisting of a two-position source code, a two-position maintenance code, a one-position recoverability code, and a one-position Service option code. The first two positions of the SMR code indicate the source for acquiring the item for replacement purposes. The third position represents who can install, replace, or use the item. The fourth position dictates whether the item is to be repaired and identifies the lowest maintenance level with the capability to perform a complete repair action. The fifth position indicates the recoverability intention/disposition intent and the maintenance level authorized to carry that out for unserviceable support items and for repairable items. The sixth position is unique to each Service and is used to disseminate specific instructions to that Service's logistics business process. |
| **Spare part** | Repairable component or assembly used for maintenance replacement purposes in the end items of equipment. They are articles identical to or interchangeable with the components of end items on contract that are procured over and above the quantity needed for initial installation for support of a system. |
| **Special tools/kits** | Those tools/kits that have single or peculiar application to a specific end item/ system. |
| **Specialized repair activity** | A level of maintenance usually characterized by the capability to perform maintenance tasks requiring specialized skills, disciplined quality control, highly sophisticated and expensive special tools/kits, and Test, Measurement, and Diagnostic Equipment (TMDE). Its phases normally consist of adjustments, calibration, alignment, testing, troubleshooting, assembly, disassembly, fault isolation, and repair of unserviceable parts, modules, and Printed Circuit Boards (PCBs). |
| **Standard format tables** | Tables that have a structure and format pre-defined by this standard. LDAC controls table output format with the stylesheet. Refer to 4.8 for a list of tables that contain standard format information. |

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**TABLE I. Definitions (continued)**

| TERM | DEFINITION |
|---|---|
| **Standard for the Exchange of Product model Data (STEP)** | STEP is a file format which defines a vendor neutral data format that allows the digital exchange of information among Computer-Aided Design (CAD) systems. STEP is defined in the ISO 10303 series. |
| **Standardization Agreement (STANAG)** | A North Atlantic Treaty Organization (NATO) standardization document that specifies the agreement of member nations to implement a standard, in whole or in part, with or without reservation, in order to meet an interoperability requirement. |
| **Subassembly** | Two or more parts that form a portion of an assembly or a component replaceable as a whole, but having a part or parts that are individually replaceable (for example, gun mount stand, window recoil mechanism, floating piston, intermediate frequency strip, mounting board with mounted parts). |
| **Supply Catalog (SC)** | The DA publication that serves as the configuration control document providing the user identification of Sets, Kits, and Outfits (SKO) and its components. For Army, there is only one official SC with multiple Component Lists (CL). |
| **Sustainment maintenance** | The second operation of the Army Maintenance System. Sustainment maintenance is characterized by the performance of maintenance tasks, "off system" in a secure environment using trained personnel, tools, and TMDE. Sustainment maintenance is typically repair and return to stock and depot-maintenance operations. |
| **System** | A combination of equipment end items, assemblies, major components, components, modules, and parts assembled as a single functional unit to perform a task or mission. |
| **System of systems** | A set or arrangement of systems that results when independent and useful systems that are integrated into a larger system that delivers unique capabilities. |
| **Table of Contents (TOC)** | A sequential list of chapter and WP titles and sometimes figure and table titles with corresponding WP sequence numbers, if applicable, for information within the publication. Within an IETM, it is a sequential list of module and WP titles and sometimes figure and table titles; it may contain WP sequence number references if they are used within an IETM. May contain page number references for ETMs. |
| **Tags** | Descriptive markup in XML, as in "start tag" and "end tag." For example, the start tag for a paragraph is **<para>**, and the end tag is **</para>**. |
| **Tailoring** | The process of evaluating individual potential requirements to determine their pertinence and cost effectiveness. The tailoring of data requirements is limited to the exclusion of information requirement provisions and selecting or specifying applicable requirements. |
| **Task** | A procedure or set of procedures. Refer to definitions of operating instructions and maintenance task. |
| **Task time** | Time noted in the WP initial setup represents time-to-complete from start to finish, including equipment conditions and follow-on maintenance. |
| **Technical Bulletin (TB)** | A publication used to provide supplementary information to TMs. TBs may contain such things as maintenance expenditure limits, equipment improvement recommendation material, functional group code information, general information such as medical data, procedures for kit installation or special |

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**TABLE I. Definitions (continued)**

| TERM | DEFINITION |
|---|---|
| | mission changes to materiel, warranty information, schematics and/or diagrams to supplement a TM, ammunition information, and so forth. TBs do not contain operator or maintenance instructions, are not used for quick reference guides or other similarly named publications, are not used in lieu of operator or maintenance manuals or to change them, and do not contain parts information. |
| **Technical Manual (TM)** | A manual that contains instructions for the installation, operation, maintenance, and support of weapon systems, weapon system components, and support equipment. TM information may be presented, according to prior agreement between the contractor and the Government, in any form or characteristic, including hard printed copy, audio and visual displays, electronic embedded media, discs, other electronic devices, or other media. They normally include operational and maintenance instructions, parts lists, and related technical information or procedures exclusive of administrative procedures. |
| **Test, Measurement, and Diagnostic Equipment (TMDE)** | Any system or device used to evaluate the operational condition of an end item or subsystem thereof to identify and/or isolate any actual or potential malfunction. This TMDE includes diagnostic and prognostic equipment; semiautomatic and automatic test equipment, to include test program sets (with issued software); and calibration test or measurement equipment. When the term TMDE is used, it refers to both general-purpose TMDE and special-purpose TMDE. |
| **Test Program Set (TPS)** | The combination of interface devices, software test programs (such as those residing in logic storage media or in permanent digital memory), and documentation (for example, technical manuals and technical data packages) that together allows the automatic test equipment operator to perform the testing/diagnostic action on the Unit Under Test (UUT). |
| **Time Between Overhaul (TBO) items** | Those items having a definite retirement schedule within a defined overhaul interval (for example, those items that must be replaced within a system assembly, subassembly, or component between scheduled overhauls). |
| **Tool kit** | An assembly of tools/components in a small pouch or box, designed for use of and carried by an individual or crew, type classified with a unit of issue of kit. |
| **Tool set** | A collection of tools/components used by a group, section, squad, platoon or unit usually supplemented by tool kits to perform an organizational mission, type classified, assigned a LIN, with a unit of issue of set. |
| **Top-down breakdown** | The pyramidal breakdown of an end item, with the top item being the complete end item. The process of breakdown is established from the engineering drawing structure in an NHA progression until the lowest repairable in each family tree group is identified. All non-repairables (spare parts) can be identified in like manner to establish their NHA relationships. |
| **Usable On Code (UOC)** | A three-position alphanumeric code representing the applicable configuration in which an item is used. |
| **User** | A person using the publication. |
| **Vector graphics** | Vector graphics are made up of paths rather than individual pixels. These paths can be used to represent lines and shapes within the image. Vector images are made up of many individual, scalable objects. These objects are defined by mathematical equations rather than pixels, so they always render at the highest |

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**TABLE I. Definitions (continued)**

| TERM | DEFINITION |
|---|---|
| | quality. Since vector graphics store image data as paths, they can be enlarged without losing quality, which makes them a good choice for line drawings, diagrams, charts and graphs, and other types of drawings. Vector (for example, SVG and CGM) is the preferred format for graphics used in Army technical publications. |
| **Verbatim text** | Text that must be reproduced in a publication precisely as written in this standard. |
| **Viewer** | A program that allows a file to be displayed but not changed. Viewers are often freely distributable and platform independent, even when the editor application is not. This characteristic allows authors to create IETMs with an editor application and make the viewer, which displays the IETM, available to other users. The single viewer for IETMs is Interactive Authoring and Display Software (IADS). |
| **Waiver** | A written request submitted to and approved by LDAC seeking an exception to requirements. Waivers granted are valid for one year. |
| **Work package** | Presentation of information functionally divided into tasks in the logical order of work sequence. These work packages (WPs) should stand alone and may contain one or more tasks. If capture of individual times and/or personnel information is required for a program such as Condition Based Maintenance (CBM), maintenance work packages must contain only one task.The following work package types are covered in this standard: general information, operator instructions, troubleshooting tasks, maintenance tasks, RPSTL, supporting information, destruction of equipment to prevent enemy use, Battle Damage Assessment and Repair (BDAR), preventive maintenance checklist, and lubrication orders. A work package should contain all information or references required to support the work package type. |
| **Wiring diagram** | A diagram illustrating signal flow or wiring connections. Where appropriate, voltage readings should be shown. |

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# 4 GENERAL REQUIREMENTS

## 4.1 General.

This standard provides the technical content requirements and mandatory style and format requirements for the preparation of technical publications and subsequent revisions covering operation and maintenance, at all levels of maintenance through sustainment, Depot Maintenance Work Requirements (DMWRs) and National Maintenance Work Requirements (NMWRs).

This standard also provides format and content requirements for aircraft maintenance manuals, PMDs, PMSs, PMIs, parts information manuals (includes RPSTLs), troubleshooting manuals, destruction to prevent enemy use manuals, BDAR manuals, LOs, PMCs, software manuals, and general maintenance manuals. All requirements throughout this standard for DMWRs shall be followed for NMWRs.

General style and format requirements are provided in 4.15 through 4.17.5.14. Specific technical content requirements are provided in the following appendices.

| | |
|---|---|
| APPENDIX A | Technical Manual Requirements with Content Selection Matrices |
| APPENDIX B | General Information, Equipment Description, and Theory of Operation |
| APPENDIX C | Operator Instructions (Except Aviation) |
| APPENDIX D | Troubleshooting Procedures |
| APPENDIX E | Maintenance Instructions |
| APPENDIX F | Parts Information |
| APPENDIX G | Supporting Information |
| APPENDIX H | Destruction of Equipment to Prevent Enemy Use |
| APPENDIX I | Battle Damage Assessment and Repair (BDAR) |
| APPENDIX J | Preventive Maintenance Checklist |
| APPENDIX K | Lubrication Orders |
| APPENDIX L | Ammunition Depot Maintenance Work Requirements |
| APPENDIX M | Software Users Manuals (SUMs) and Software Administrators Manuals (SAMs) |
| APPENDIX N | General Maintenance Manuals |
| APPENDIX O | Hand Receipts (HRs) |
| APPENDIX P | Ammunition and Munitions Equipment Data Sheets Technical Manuals |
| APPENDIX Q | Technical Bulletins (TBs) |

## 4.2 Types of technical publications.

APPENDIX A lists specific technical content requirements for each type of publication covered by this standard. Each publication type listed below shall provide in detail the maintenance coverage prescribed for the applicable maintenance level(s) by the Maintenance Allocation Chart (MAC) and Source, Maintenance, and Recoverability (SMR) coded items. The following types of technical publications shall be prepared as stand-alone manuals.

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| PUB TYPE | TITLES |
|---|---|
| -10 | Operator Manual for *insert system* |
| -12 | Operator and Field Maintenance Manual for *insert system* [1] |
| -12&P | Operator and Field Maintenance Manual including Repair Parts and Special Tools List (RPSTL) for *insert system* [1] |
| -20 | Field Maintenance Manual for *insert system*[1] |
| -20&P | Field Maintenance Manual including Repair Parts and Special Tools List (RPSTL) for *insert system* [1] |
| -20P | Field Maintenance Manual<br>Repair Parts and Special Tools List (RPSTL) for *insert system* [1] |
| DMWR or NMWR<br>with RPSTL | Depot Maintenance Work Requirements (DMWR) including Repair Parts and Special Tools List (RPSTL) for *insert system* |
| | National Maintenance Work Requirements (NMWR) including Repair Parts and Special Tools List (RPSTL) for *insert system* |
| DMWR / NMWR | Depot Maintenance Work Requirements (DMWR) for *insert system* |
| | National Maintenance Work Requirements (NMWR) for *insert system* |
| | Depot Maintenance Work Requirement Containing National Overhaul Standards *insert system* |
| SUM/SAM | Software Users Manual (SUM) for *insert system* |
| | Software Administrators Manual (SAM) for *insert system* |
| | Software Administrators Manual (SUM/SAM) for *insert system* |
| Ammunition DMWR | Depot Maintenance Work Requirements (DMWR) for Demilitarization of *insert system* |
| | Depot Maintenance Work Requirements (DMWR) for Maintenance of *insert system* |
| Ammunition Data Sheets | Army Ammunition Data Sheets for *insert system* |
| | Army Munitions Equipment Data Sheets for *insert system* |
| (Aviation)<br>-20, -20&P | Field Maintenance Manual for *insert system* [1] |
| | Field Maintenance Manual including RPSTL for *insert system* [1] |
| (Aviation)<br>Preventive Maintenance Daily (PMD)<br>Preventive Maintenance Services (PMS) | Preventive Maintenance Daily (PMD) Manual for *insert system* |
| | Preventive Maintenance Services (PMS) Manual for *insert system* |
| (Aviation)<br>Phased Maintenance (PMI) | Phased Maintenance Inspection (PMI) Checklist for *insert system* |
| BDAR | *Insert maintenance level* Battle Damage Assessment and Repair for *insert system* |
| Destruction | Destruction of Equipment to Prevent Enemy Use for *insert system* |

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| PUB TYPE | TITLES |
|---|---|
| Lubrication Order (LO) | Lubrication Orders for *insert system* |
| Preventive Maintenance Checklists (PMC) | Preventive Maintenance Checklists (PMC) for *insert system* |
| General Maintenance | General Maintenance Manual for *insert system* |
| Hand Receipt (HR) | Hand Receipt Covering Contents of Components of End Item (COEI), Basic Issue Items (BII), and Additional Authorization List (AAL) for *short end item title* |
| Technical Bulletin (TB) | Technical Bulletin for *insert system* |

[1] Army Training and Doctrine Command (TRADOC) maintainer representative approval is required for any inclusion of sustainment-level maintenance. If approval is granted, "sustainment" may be included in title. See also A.6.2 for publication titling guidance.

### 4.2.1 Applications ("apps").

Software applications, or "apps," will meet the requirements of and go through the steps of the publications process (for example, scheduling, coordination, verification, authentication, having XML data compliant with this standard, and so forth). Contact the Equipment Publications Control Office (EPCO) with any specific “app” questions.

## 4.3 Publication numbering conventions.

Publication numbers reveal details about what a technical publication contains, such as maintenance designations, parts information, and volume count. Refer to A.6.2 for general publication numbering details.

### 4.3.1 Maintenance designations.

Publication maintenance levels are designated at the end of the publication numbers and signal which of two levels of Army maintenance the publication falls under: field and sustainment.

**a. Field-level only publications**: provide instructions for operators and maintainers (-10 TMs for operators, -20 TMs for maintainers, and -12 TMs for operator/maintainers combined).

**b. Field and Sustainment publications**: provide, when approved by an Army Training and Doctrine Command (TRADOC) maintainer representative, additional sustainment information for -12 and -20 technical publications.

### 4.3.2 Parts information.

The inclusion of parts information is signified with the presence of a "P" in publication numbers. The possibilities are listed below.

**a.** Publications numbers that *do not* include a "P" *do not* contain parts information. See FIGURE 1.

**b.** Publications numbers that include "&P" after the maintenance level *do* contain parts information as well as operator, maintainer, and, when approved by a TRADOC maintainer representative, additional sustainment information. See FIGURE 2.

**c.** Publications that include a "P" (without "&") in the number contain *only* parts information. See FIGURE 3.

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### 4.3.3 Volume count.

When a publication consists of multiple volumes, volume numbers shall appear at the end of the publication number with a dash. For example, a two-volume TM that includes a RPSTL shall be represented by TM XX-XXXX-XXX-XX&P-1 and TM XX-XXXX-XXX-XX&P-2, with "Xs" signifying the TM number.

**Note**

Figure appearance may vary slightly based on version of DTD/stylesheets utilized.

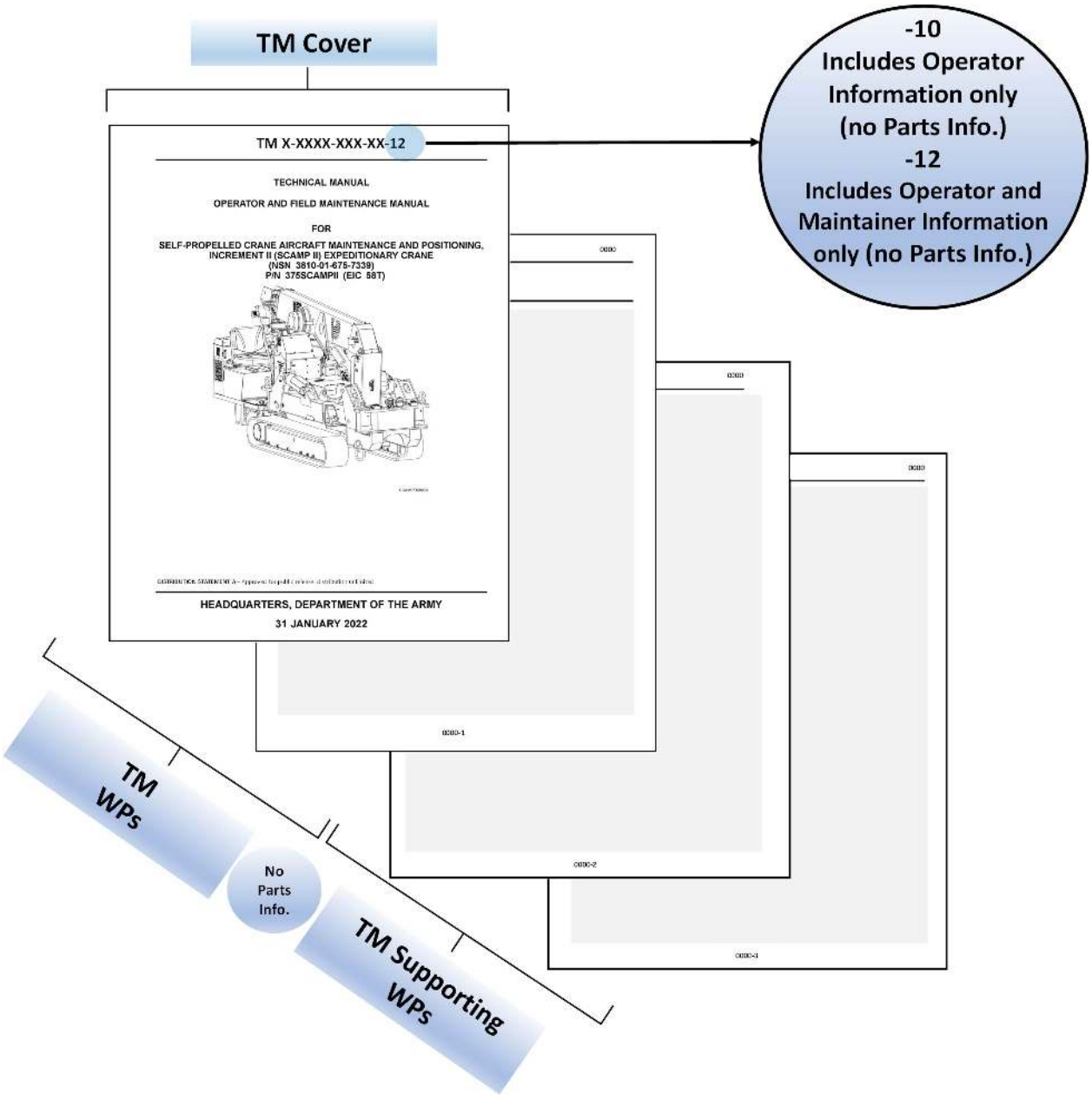


**FIGURE 1. Naming conventions for maintenance designation (sheet 1 of 3).**

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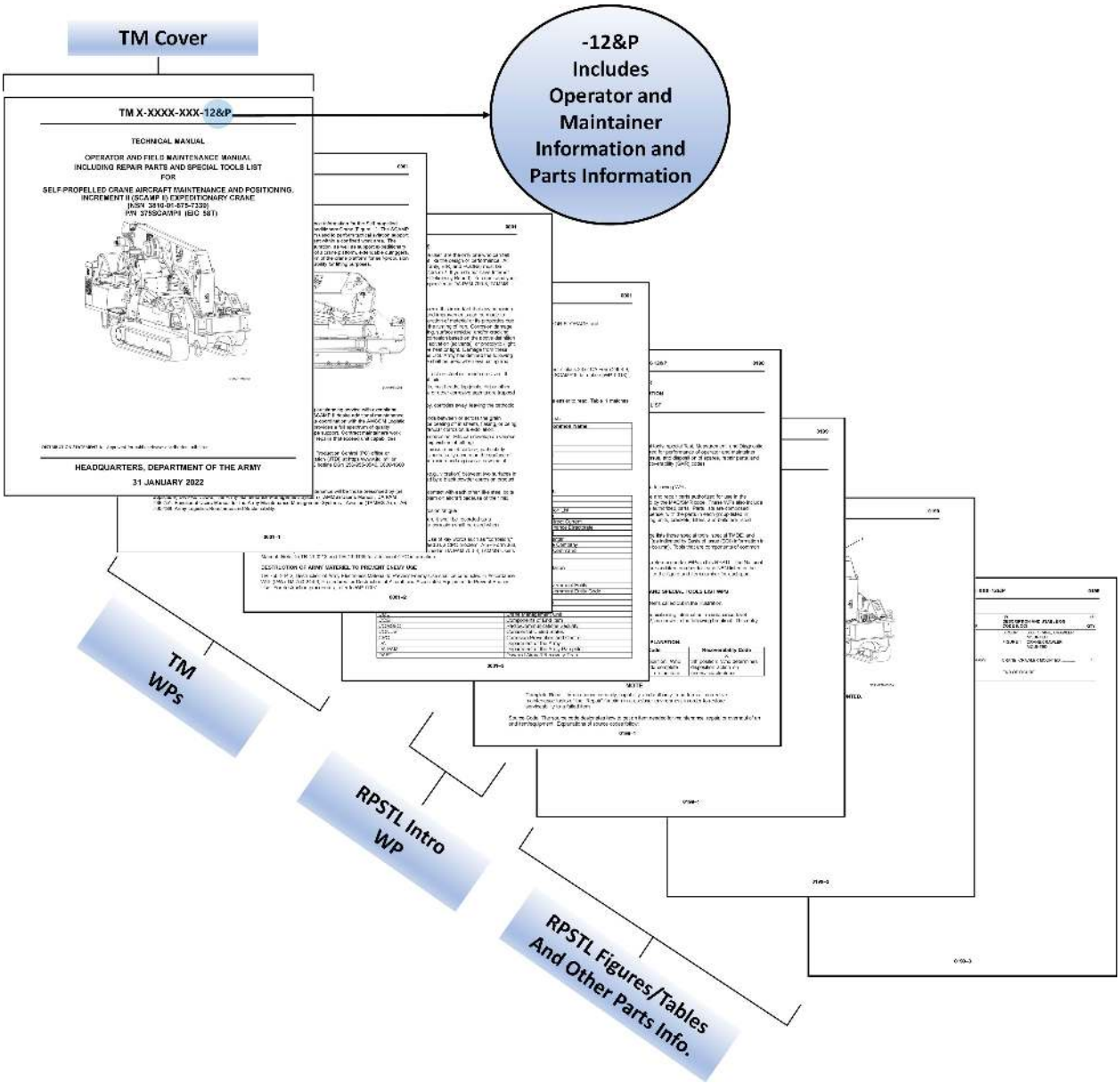


**FIGURE 2. Naming conventions for maintenance designation (sheet 2 of 3).**

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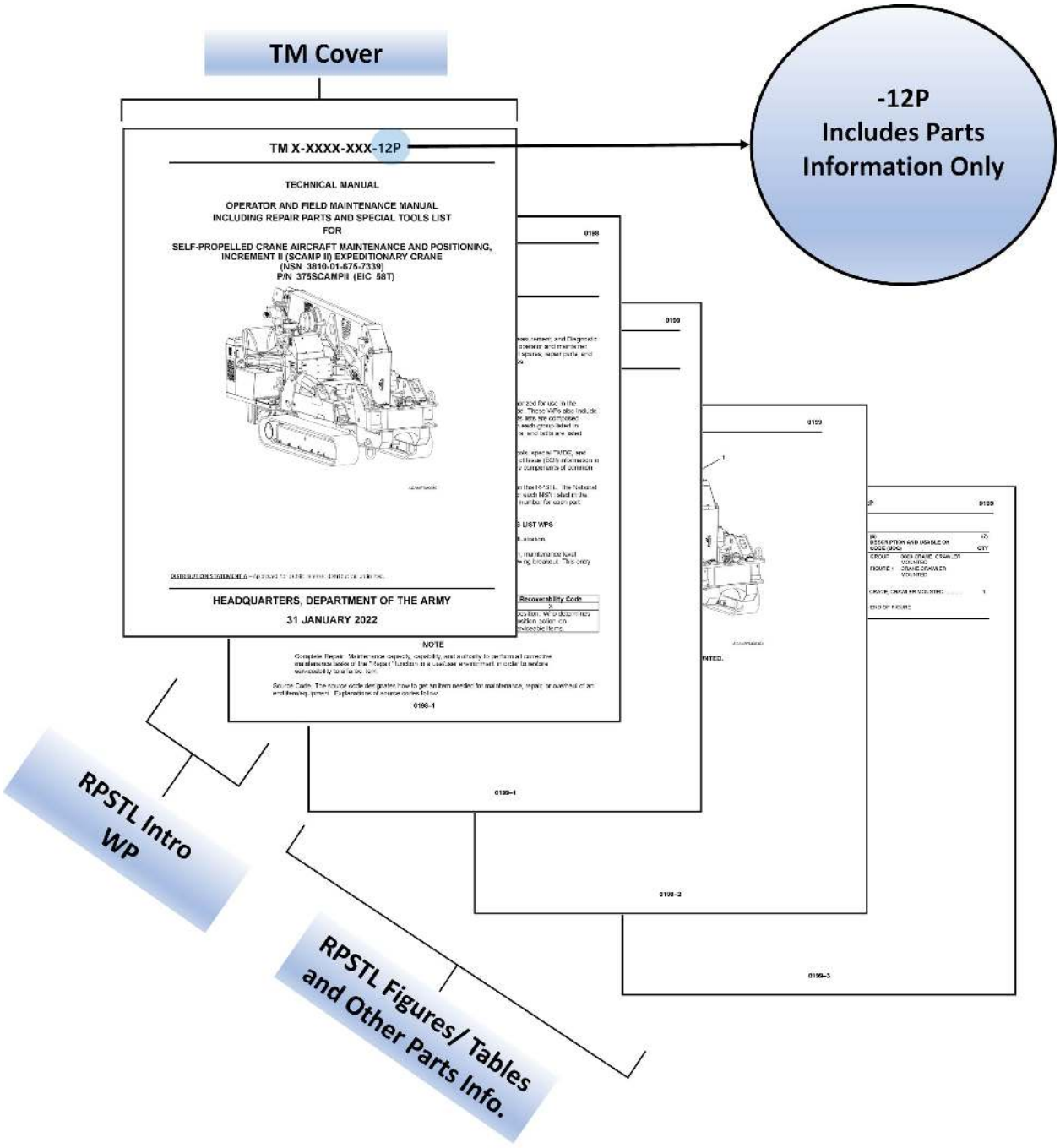


**FIGURE 3. Naming conventions for maintenance designation (sheet 3 of 3).**

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## 4.4 Selective application and tailoring of content.

This standard contains some requirements that may not be applicable to the preparation of all technical publications. Selective application and tailoring of requirements contained in this standard are the responsibility of the materiel developer (MATDEV) and shall be accomplished using APPENDIX A.

The applicability of some requirements is also designated by one of the following statements: "unless specified otherwise by the MATDEV"; "as specified by the MATDEV"; or "when specified by the MATDEV."

Some guidance is not applicable to all Services. Such guidance is prefixed to indicate the Services to which the guidance pertains: (A) Army, (N) Navy, (MC) Marine Corps, (F) Air Force, (CG) Coast Guard, and (SF) Space Force. Portions not prefixed are applicable to all Services. Paragraphs prefixed with MC pertain to manuals that are for the Marine Corps only, and unless otherwise stated, do not apply to multi-service manuals involving the Marine Corps.

The content of this standard can also be tailored based on the following designations:

**a.** Required. Content items that are designated as required by the use of the word "shall" must be included in the publication and shall not be tailored out. If a required item is not applicable to a publication's system, the item shall be listed in the "Scope" section of the General Information work package (WP) as "Not Applicable."

**b.** Optional. The content items usually are caveated by "as applicable," "as specified by the MATDEV," or "unless specified by the contracting activity." They may contain use of the word "may" or "should," and they are areas that are flexible and can be tailored in or out.

**c.** Prohibited. In some cases, this standard uses the phrase "shall not" or states that content is prohibited. These items must not be included or used for the publication covered.

**d.** Conditionally Required. Items marked with "CR" are handled slightly different than items marked with just an "R." These items are "conditionally" required, meaning they are only required if the "parent" of that item is included in a publication. For example, all TMs are not required to include a "Software Information" module. Thus, this module is shaded ("grayed out") in the APPENDIX A matrices. If a MATDEV decides that this module shall be included in a TM, however, then this module becomes the "parent" and every individual WP (or "child") marked with a "CR" shall be included as part of the software module.

**e.** Shaded cells with plus (+) sign.Shaded cells with a plus (+) sign designate content that contains information that most systems typically include. For these cells, technical writers shall research to ensure a specific equipment system does not require these sections. If content is not included in a technical publication, the "SCOPE" section of WP 0001 General Information shall include a statement that content is not applicable (see B.7.1.1). APPENDIX A, FIGURE A-5 shows example Mandatory Replacement Parts (MRP) or Critical Safety Items (CSI) entries. If these WPs are not included in a TM, WP 0001 General Information WP, Scope shall list these items as not applicable (see B.7.1.1).

The standard cannot address every possible content or format issue and state whether it is required, optional, or prohibited. If the standard does not specifically say that a content item or a particular format can be used, coordination shall take place between the materiel developer, the LCMC Equipment Publications Control Officer (EPCO), and the Logistics Data Analysis Center (LDAC) to determine if a waiver request or other documentation will be required. This coordination should take place as early as possible in the development process.

## 4.5 Preparation of digital data for electronic delivery with the Army Document Type Definition (DTD) and stylesheets.

Data shall be prepared and delivered digitally in accordance with this standard. Data shall be eXtensible Markup Language (XML) tagged using the Army Document Type Definition (DTD), which interprets the technical content and structure for the functional requirements contained in this standard. The Army-developed stylesheets shall be used.

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XML tags used in the Army DTD are noted throughout the text of this standard in bracketed, bold characters (for example,**<descwp>**) as a convenience for the author and to ensure that the tags are used correctly when developing a document instance.

The Army DTD and stylesheets may be obtained from LDAC as follows:

| | |
|---|---|
| LDAC website (URL): | https://www.ldac.army.mil |
| Telephone: | Commercial: (256) 955-0855, DSN: 645-0855 |
| Email: | usarmy.redstone.ldac.mbx.tmss@army.mil |

## 4.6 IETM functionality requirements.

The specific level of functionality and user interaction to be provided in IETMs shall be in accordance with the functionality matrix contained in APPENDIX A. The IETM functionality matrix shall be tailored according to equipment diagnostic/prognostic technology and user requirements towards the goal of reducing maintenance down time. The mandatory functionality requirements for IETMs provided in APPENDIX A supplement the technical content requirements provided in APPENDIX B through APPENDIX N. These requirements shall apply for the presentation/display of IETM information.

## 4.7 IETM installation data/access.

Instructions for installing the disc on the computer and launching the IETM in Interactive Authoring and Display Software (IADS) shall be prepared. This information shall be placed in a "readme" file and on the mailer rather than inside the IETM itself.

The disc containing the IETM may include third party software (Adobe Acrobat, for example) and may only be used on Army networks if that software is approved via Assess-Only and documentation is provided during the authentication process.

## 4.8 Standard format tables.

Standard format tables have a structure and format pre-defined by this standard and are controlled by the stylesheet.

The standard format tables' specified data shall have the content requirements prescribed in APPENDIX B through APPENDIX G. They shall have no deviations to the number of columns, the titles in the column headings, and the required format.

A list of standard format tables is provided below:

**a.** Description and use of controls and indicators (C.7.1).

**b.** Checking unpacked equipment (E.7.3.1.3.2).

**c.** Preventive Maintenance Checks and Services (PMCS) (E.7.2).

**d.** Classification of Materiel Defects (E.7.4.4.11.3).

**e.** Overhaul and retirement schedule (E.7.16).

**f.** Overhaul inspection procedures (OIPs) (E.7.21).

**g.** Depot mobilization requirements (E.7.22.2).

**h.** Special instructions (E.7.12.3).

**i.** Repair parts list (F.8.2.1).

**j.** Kit parts list (F.8.4).

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**k.** Bulk items (F.8.5).

**l.** Special tools/kits list (F.8.6).

**m.** Maintenance Allocation Chart (Aviation, Non-Aviation, Marine Corps) (G.7.2, G.7.3).

**n.** Tools and test equipment requirements (MACs) (G.7.3.2.4).

**o.** Remarks (MACs) (G.7.3.2.5).

**p.** Component of End Items (COEI) list (G.7.4).

**q.** (MC) Supply System responsibility (SSR) List (G.7.4).

**r.** Basic Issue Items (BII) List (G.7.4.6).

**s.** Additional Authorization List (AAL) (G.7.5).

**t.** (MC) Using Unit Responsibility Items (UURI) List (G.7.5).

**u.** (MC) Collateral Material (CM) List (G.7.6).

**v.** Expendable and durable items list (G.7.7.1).

**w.** Tool identification list (G.7.8.2).

**x.** Mandatory replacement parts list (G.7.9.1).

**y.** Critical safety items (CSI) (G.7.10).

## 4.9 Final Reproducible Copy (FRC).

FRC shall be the final manuscript, reproducible copy, or electronic media delivery, with all necessary changes and corrections incorporated and including final resolution of all comments and recommendations made as a result of validation, verification, testing, and user review. The primary copy of any publication is a set of digital files, not the hard-copy results.

The FRC of ETMs shall be a Portable Document Format (PDF) file. The PDF shall be capable of being edited and linked, and it shall have intelligent text (searchable text that can be copied and pasted) with all fonts embedded.

### 4.9.1 XML requirement.

A publication's FRC package shall deliver all XML files necessary for validation with the DTD. For IETMs, the files shall be capable of fully compiling to an .iads dataset equivalent to the FRC delivery. For ETMs, the files shall be capable of fully compiling a PDF equivalent to the PDF delivered in 4.9.

## 4.10 Waiver requests.

When MATDEVs seek an exception to this standard's requirements, they shall submit a waiver request through their designated EPCO office to LDAC as early as possible when preparing the publication, and definitely prior to submitting the FRC. If approved by LDAC, waiver requests shall be included with the final submission package for all publications.

## 4.11 Boilerplate text.

"Boilerplate" text is verbatim text that shall be used with applicable customizations usually indicated by italicized text in parentheses.

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## 4.12 Structure of content.

### 4.12.1 IETM data and organization.

Most IETMs will contain both linear and non-linear data depending on the type of publication.

Linear data shall be used for data that require a certain order, such as operating instructions. Non-linear data shall be used for data where order is dictated by dialog box responses or other user input, such as complex troubleshooting and most maintenance procedures. Non-linear data are utilized when information is presented in a narrative form meant to enlighten a user about a subject (for example, the system-level theory behind how components of a system work together to accomplish a mission).

For non-linear data, alphabetization is not used in the data or TOC entries for Preventive Maintenance Checks and Services, General Information, or Equipment Description and Data. Alphabetization may be used in the troubleshooting index if it aids usability.

The content selection matrices in Appendix A (A.4.4) as well as the DTD associated with this standard will aid in selection of data prepared as linear data versus data prepared as non-linear data.

### 4.12.2 Technical publication sections.

Technical publications are composed of the following possible sections:

**a.** Front Matter.

**b.** General Information, Equipment Description and Data, and Theory of Operation.

**c.** Operator Instructions.

**d.** Troubleshooting Information.

**e.** Maintenance Instructions (operator and maintainer maintenance tasks).

**f.** Destruction of Equipment (depending on type of publication).

**g.** Software Information (depending on type of publication).

**h.** Parts Information (includes RPSTL).

**i.** Supporting Information.

**j.** Rear Matter.

### 4.12.3 Technical publication data hierarchy.

The hierarchical breakdown of a publication is:

1. Volumes (if required, and only for ETMs).
2. Modules (formerly chapters; refer to 4.12.5).
3. Work Packages.
4. Paragraphs.
5. Subparagraphs.
6. Steps.

Volumes and steps (and any subordinate steps) shall have at least two occurrences (except for procedural steps, which shall be tagged in accordance with 4.12.8.1). For example, where there is a Volume 1, there shall be a Volume

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2, etc. Work packages may have only one occurrence within a module depending on publication type and equipment system requirements.

Volumes shall be partitioned only between modules. Stand-alone parts information manuals may be volumized between parts lists **<plwp>**, special tools/kits and parts list **<stl_partswp>**, kits **<kitswp>**, bulk items list **<bulk_ itemswp>**, and special tools/kits list **<stlwp>**.

### 4.12.4 Volume requirements for ETMs.

Division into volumes shall occur when the number of printed pages (excluding pocket-sized TMs) exceeds 1,500 pages or 750 sheets. Each volume shall not exceed 1,500 pages or 750 sheets. A pocket-sized manual (4 x 5-1/2 inches) shall not exceed 200 pages or 100 sheets. Pocket-sized manuals shall not be divided into volumes.

Each volume of a series shall display the publication number on its cover and all pages that make up the volume. Front matter for each volume of a series shall include a warning summary(if applicable), change transmittal (as applicable per volume), and a Table of Contents (TOC). The first volume shall contain a complete TOC (including information for all volumes) and, if applicable, a warning summary that covers all volumes of the publication. Refer to 5.2.7.

Rear matter for each volume of a series shall contain as a minimum one blank DA Form 2028 (Recommended Changes to Publications and Blank Forms) and a back cover. Foldouts are included as applicable.

Separate volumes shall not be used to distinguish between models of equipment (for example, -10 for basic model, -10- 1 for model A, -10-2 for model B, etc.).

When a change causes a manual to exceed page limits above, a revision shall be required.

### 4.12.5 Modules.

Modules (formerly "chapters") shall be used to divide publication data into specific functional information. Module types include, but are not limited to, the following: General Information, Operating Instructions, Troubleshooting Information, and Maintenance Information. Refer to A.4.4 for content selection matrices, which list all available module types depending on publication type.

Each module shall be made up of one or more WPs. For simple equipment, a single module may suffice for each functional element (for example, maintenance at all levels for a rifle may be placed in one combined module). For complex weapon systems/equipment, it may be necessary to create several modules for each of the functional elements. Maintenance of an aircraft's electrical system, fuel system, and flight control system may be placed in separate modules, for example.

Module titles shall be based on the titles given in the matrices in APPENDIX A and may be augmented based on equipment system nomenclature.

#### 4.12.5.1 Module title page <titlepg>.

For standard-sized technical publications, each module shall begin with a module title page (see FIGURE 4). Some exceptions may apply when multiple operating instruction modules are utilized in an IETM (see C.5.2.1.2).

Module title pages for standard-sized technical publications shall contain:

**a.** Publication number.

**b.** Module number.

**c.** Module title.

**d.** Equipment system identification.

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Module title pages shall be unnumbered.

For ETMs, a module title page shall always be a right-hand page.

A separate module title page is not required for pocket-sized manuals. For pocket-sized manuals, the module number and title may be placed on the top of the first page of the first WP of the module.

**Note**

Figure appearance may vary slightly based on version of DTD/stylesheets utilized.

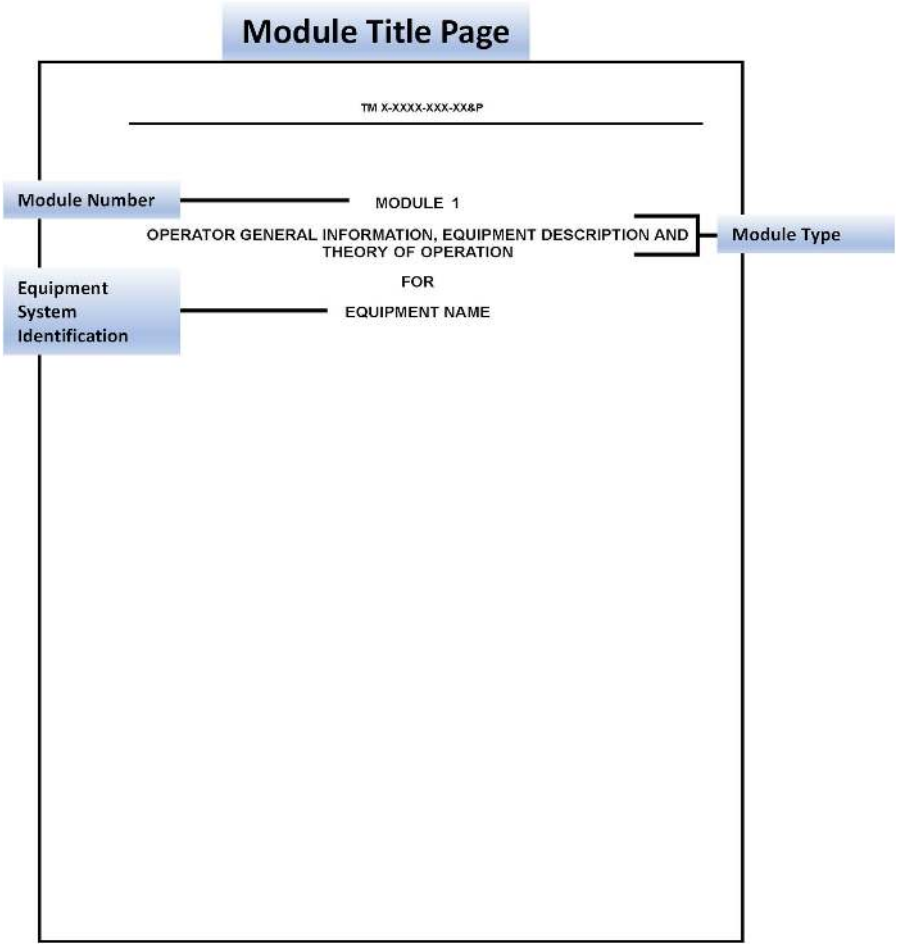


**FIGURE 4. Module title page for standard-sized technical publication.**

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### 4.12.5.2 Module numbering.

Modules shall be numbered in sequential order throughout the publication using Arabic numerals. Additionally, modules shall be numbered sequentially across volumes.

### 4.12.5.3 Module indexes.

Module indexes are optional. When utilized, module indexes shall include WP titles and WP sequence numbers, and they shall be used on all modules. Module indexes are not listed on the matrices, but they are included in the DTD and can be "switched off" to disappear in the publication at the MATDEV's discretion. See FIGURE 5.

**Note**

Figure appearance may vary slightly based on version of DTD/stylesheets utilized.

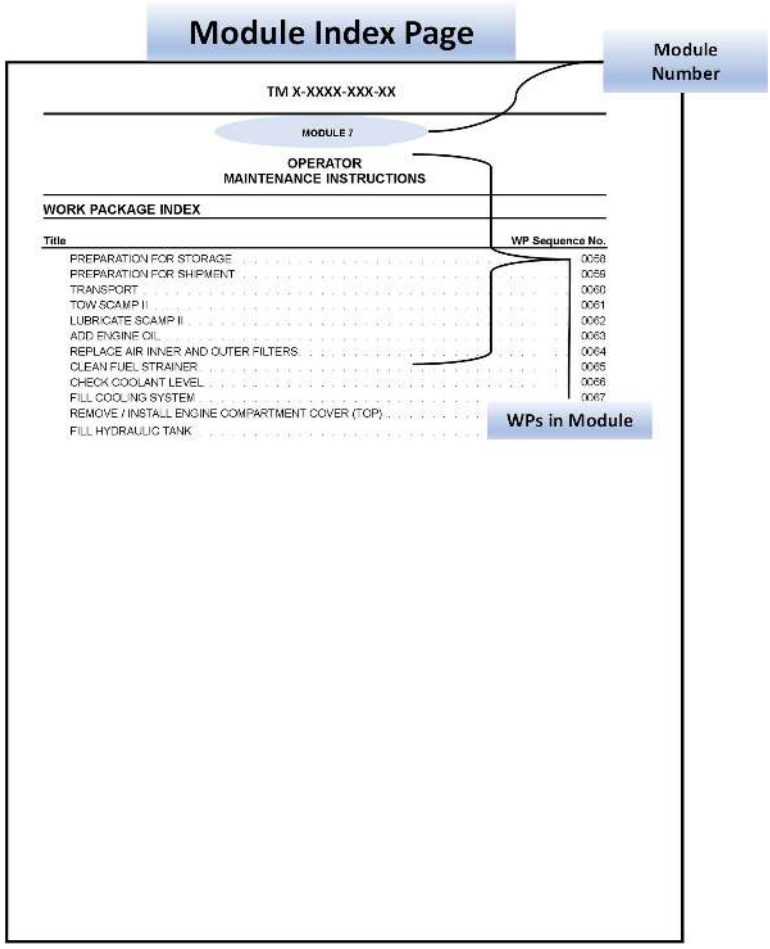


**FIGURE 5. Module index.**

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### 4.12.6 Work packages.

WPs shall be used to logically divide publication data into narrative or task-oriented information (refer to FIGURE 6, FIGURE 7, and FIGURE 8). Each WP shall be developed as an independent, stand-alone data unit. WPs shall be prepared with the following considerations:

**a.** To facilitate the change/revision process, all WPs including the Parts Information WPs shall be kept small. IETMs should limit the size of WPs to keep them from becoming overwhelming.

**b.** Each WP may contain one or more tasks.

**(1)** Multiple tasks shall only be grouped into one WP if all the tasks have the same personnel required and equipment condition (see FIGURE 6) sections. In other words, tasks shall not be grouped into one WP if the personnel required (also known as the Military Occupational Specialty (MOS)) and equipment condition is not applicable to every task in that WP.

**(2)** The content in the remaining initial setup sections should be consolidated, but be applicable to at least one task in the WP. Liberal use of references to individual WPs is encouraged to prevent duplication of information and/or procedures. With the exceptions of RPSTL figures and foldouts, references shall not be made to figures in separate WPs.

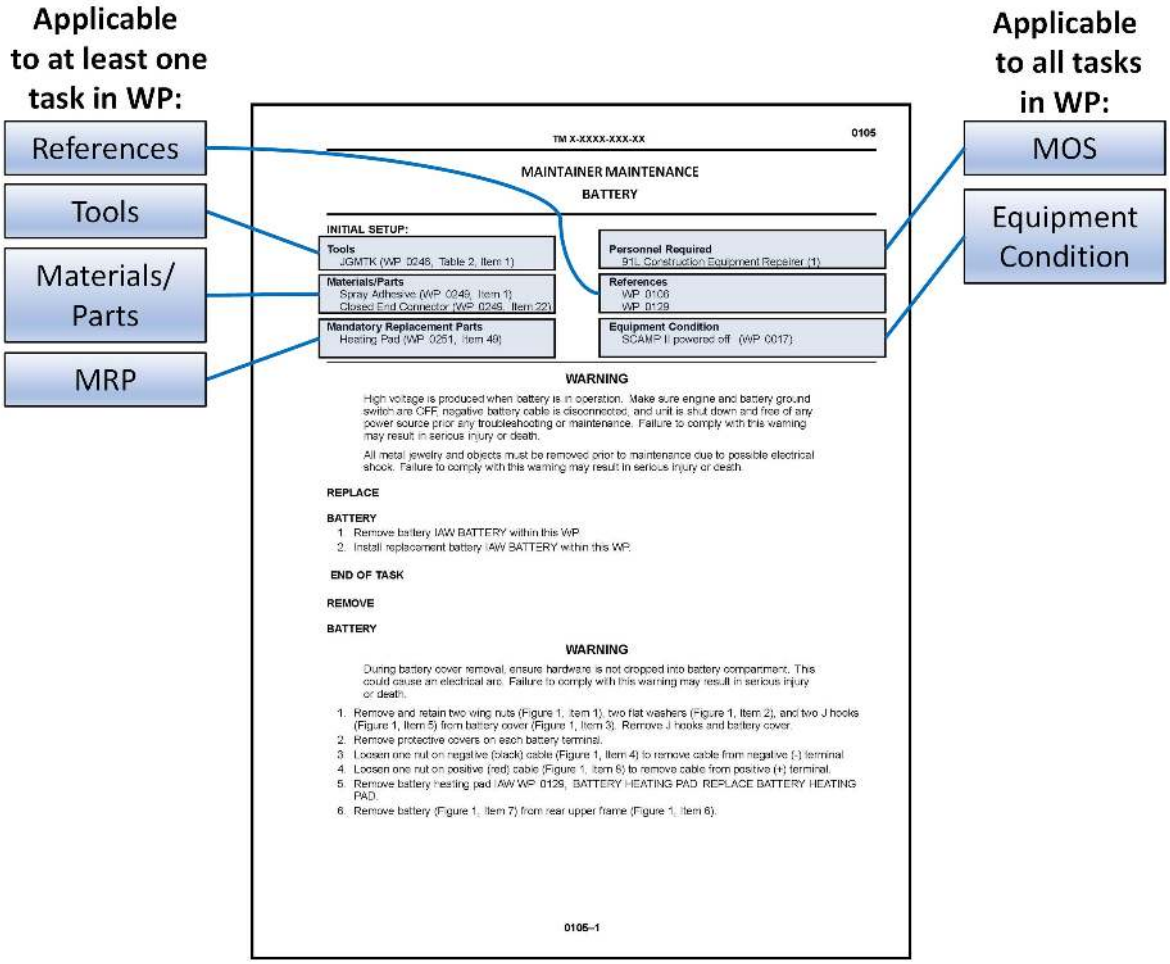


**FIGURE 6. Initial setup example.**

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**c.** WPs shall begin on a right-hand page in ETMs.

Refer to APPENDIX B through APPENDIX Q for more information on each module and for guidance on building individual WPs in these modules.

Refer to FIGURE 7 for an example of a narrative WP and FIGURE 8 for a task-oriented example of a WP with multiple tasks. Division or selection of coverage will depend on various factors outlined in 4.12.6.1.

**Note**

Figure appearance may vary slightly based on version of DTD/stylesheets utilized.

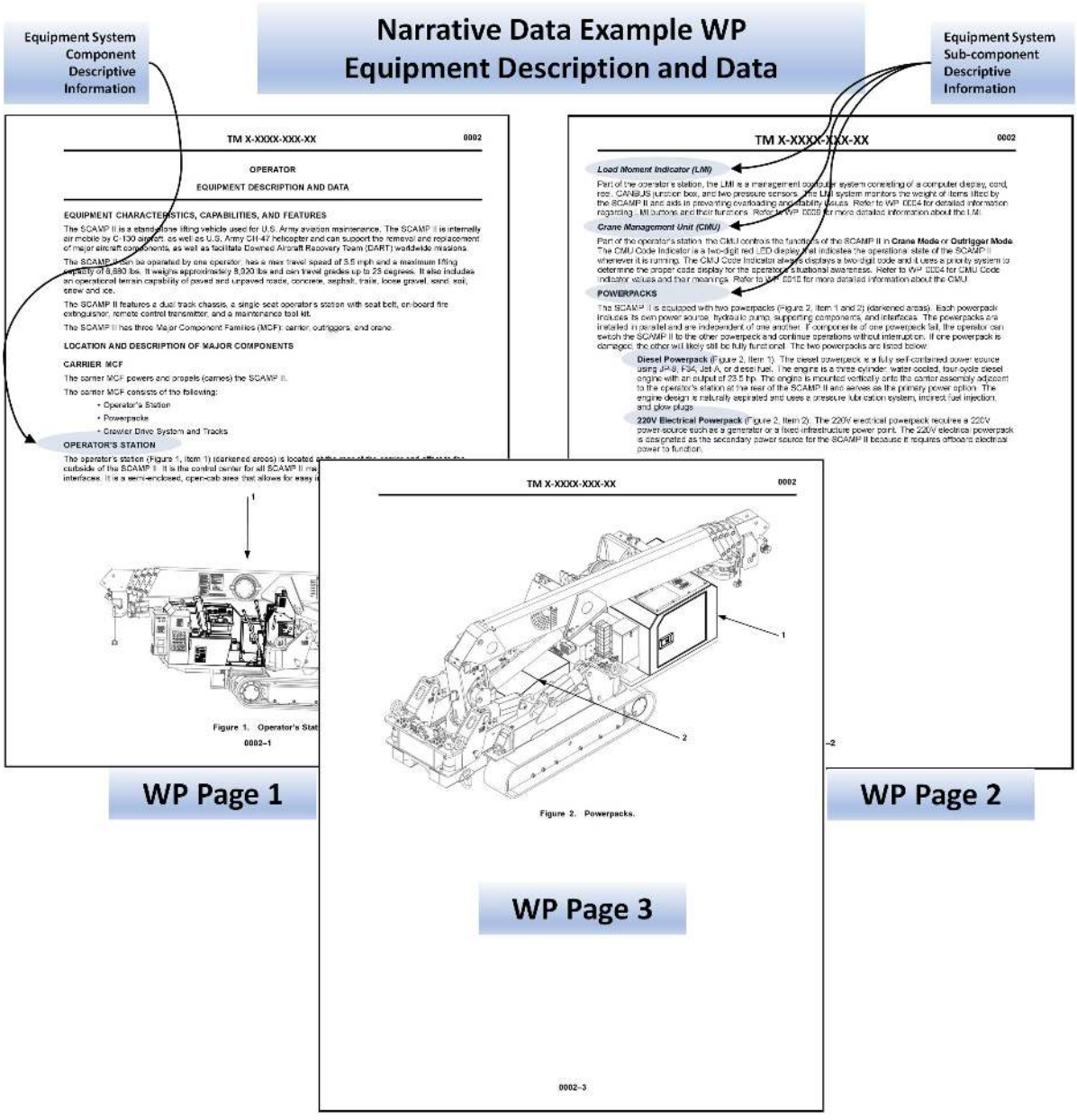


**FIGURE 7. Narrative example WP.**

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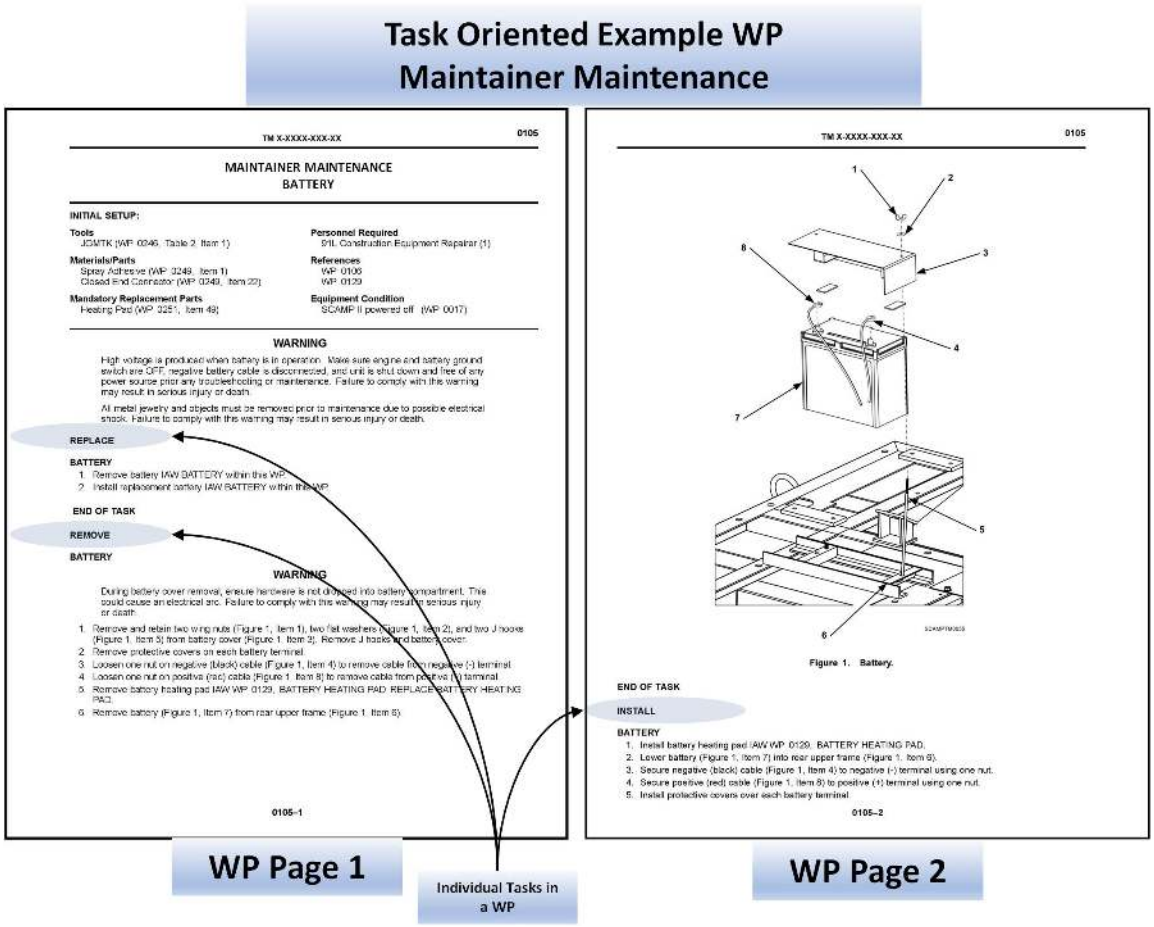


**FIGURE 8. Task-oriented example WP.**

### 4.12.6.1 Development of individual WPs.

Inclusion of a WP in a publication will depend on various factors, some of which are outlined below.

**a.** Specific WPs are required by this standard. Refer to the matrices for each publication type in APPENDIX A for the specific content requirements by publication type. Required WPs are denoted in the matrices with an "R."

**b.** Based on system analysis by the MATDEV, specific WPs may be required regardless of publication type. In these instances, content selection matrices shall be filled out in accordance with A.4.3.2. Prohibited WPs, indicated by "P" in matrices, shall not be included. The system data listed below may determine the need to require specific WPs.

**(1)** Operational modes – based on system complexity, information system operation data may all be included in one WP or separated out into multiple WPs (or even multiple modules). Refer to APPENDIX C for more information about the organization of operational information.

**(2)** Troubleshooting complexity – some more complex systems may benefit from organizing each troubleshooting malfunction into its own dedicated WP to prevent confusion and improve usability. Refer to APPENDIX D.

**(3)** Maintenance actions – maintenance actions may benefit from separate WPs for each maintenance task or grouping maintenance tasks together based on the types of maintenance tasks included in a publication.

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Refer to APPENDIX E for more information about maintenance task types and maintenance WP organization methodology.

**(4)** Levels of maintenance for required maintenance tasks – separate WPs may be developed for the same equipment or component for different maintenance levels (for example, a WP for operator maintenance and a WP for maintainer maintenance for the same item of equipment).

**(5)** System configurations – two or more WPs for an identical maintenance task may be required when the task is performed differently due to differences in configurations.

**(6)** Initial setup information – separate WPs may be necessary due to different initial setup information or operating procedurefor a set of maintenance tasks for a repairable component (see FIGURE 6).

**(7)** WP and/or task referencing – separate WPs may be preferable if a task will be referenced in other tasks (especially relevant to maintenance WPs). Refer to APPENDIX E for more information about maintenance task types and maintenance WP organization methodology.

**(8)** Usability – At times, confining information to one WP may be necessary because dividing the information into several WPs would degrade usability. Conversely, development of more than one WP may be necessary because the reduction in the size of the WP would make it more usable (4.12.6).

### 4.12.6.2 Work package title <title>.

The title of an individual WP shall be derived from the titles listed in content selection matrices (A.4.4). To illustrate, FIGURE 9 shows that the first WP of a publication shall be titled "GENERAL INFORMATION" based on the content selection matrix entry for the first WP in a publication (the stylesheet automatically populates the "OPERATOR" portion of the title). Similarly, FIGURE 10 demonstrates that operation under usual conditions WPs shall contain the text "OPERATION UNDER USUAL CONDITIONS–" followed by a system-specific task title (the stylesheet automatically populates the "OPERATOR INSTRUCTIONS" portion of the title). In some cases where multiple operator instruction modules are utilized in a publication, the "OPERATION UNDER USUAL CONDITIONS–" portion of a title may be omitted for usability purposes in IETMs. These cases are explained further in C.5.2.1.2.

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**Note**

Figure appearance may vary slightly based on version of DTD/stylesheets utilized.

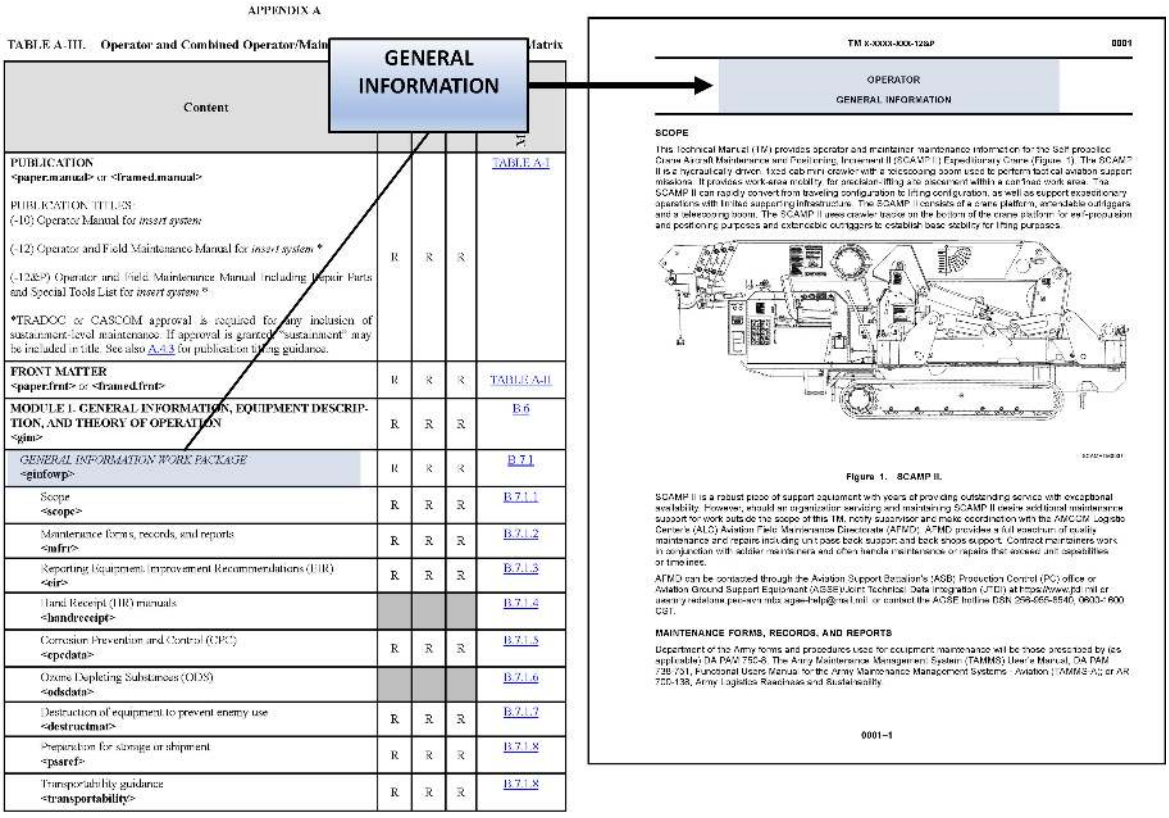


**FIGURE 9. Work package titles (sheet 1 of 2).**



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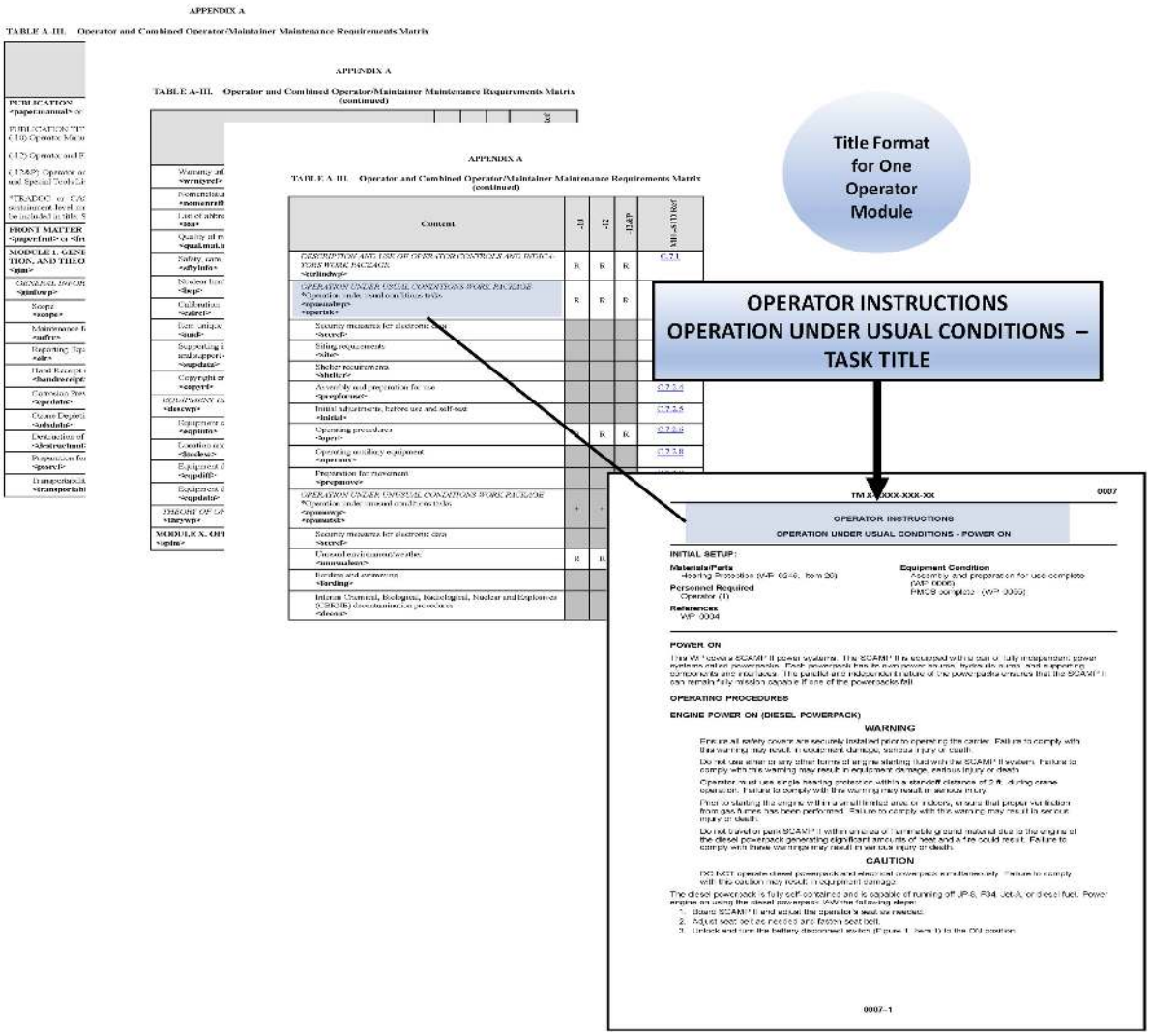


**FIGURE 10. Work package titles (sheet 2 of 2).**

### 4.12.6.3 Work package identification number.

A unique number shall be assigned to each work package. This identifier may be used for database retrieval purposes.

The WP identification number (wpno) shall not appear on the printed page or user's viewer and should not be confused with the WP sequence number in 4.12.6.4. It shall be assigned when preparing the document instance in accordance with the DTD and shall not be changed throughout the life of the work package. The "wpno" attribute of the top-level tag and the file name should match.

The WP identification number shall consist of an alpha designation for the type of information contained in the work package, a five-digit block number assigned by the MATDEV, and the publication number. The publication number is used only to provide uniqueness and avoid duplication of a WP identification number. Other than that purpose, it shall not have significance. To aid with configuration management of publications, it is recommended that WP identification numbering include the maintenance level of publications (further discussed in ). Additionally, use the

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letter "n" instead of an ampersand (for example, X-XXXX-XXXX-20nP) to avoid an error within the file (formatting also shown in FIGURE A-7).

When reusing a work package, the WP identification number shall remain the same throughout the life of the WP and shall not be changed in the event the WP is reused in another manual.

The following alpha designators shall be assigned to the specific types of information contained within the WPs.

| | |
|---|---|
| A | Ammunition information |
| B | BDAR information |
| C | PMC checklist information |
| D | Destruction information |
| G | General information |
| L | LO information |
| M | Maintenance instructions |
| O | Operator instructions |
| R | Parts Information |
| S | Supporting information |
| T | Troubleshooting procedures |
| W | Software information |

The following examples explain WP database identification numbering:

**M00432-9-1425-646-10**

| | |
|---|---|
| M | Identifies a WP containing maintenance instructions. |
| 00432 | Identifies the 432nd WP containing specific maintenance instructions. |
| 9-1425-646-10 | The operator TM under which this WP was initially developed. |

**T02000-1-1520-238-20nP**

| | |
|---|---|
| T | Identifies a WP containing troubleshooting procedures. |
| 02000 | Identifies the 2000th WP containing specific troubleshooting procedures. |
| 1-1520-238-20nP | The maintainer TM with a RPSTL under which this WP was initially developed. |

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### 4.12.6.4 Work package sequence numbers.

Work package sequence numbers are different from WP identification numbers (discussed in 4.12.6.3), which do not appear within WPs when displayed—regardless of media output. WP sequence numbers are automatically generated by stylesheets. They usually consist of a four-digit number beginning with "0001" placed in the top right-hand corner of the WP (in ETMs) and run consecutively throughout a publication (see FIGURE 11). Further, WP numbering between modules follows the same organizational parameters: WP numbering that starts in Module 2 follows the last WP number in Module 1 (for example, if 0010 is the last WP in Module 1, 0011 will be the first WP in Module 2). Refer to through for more information about WP sequencing for change and revision efforts.

**Note**

Figure appearance may vary slightly based on version of DTD/stylesheets utilized.

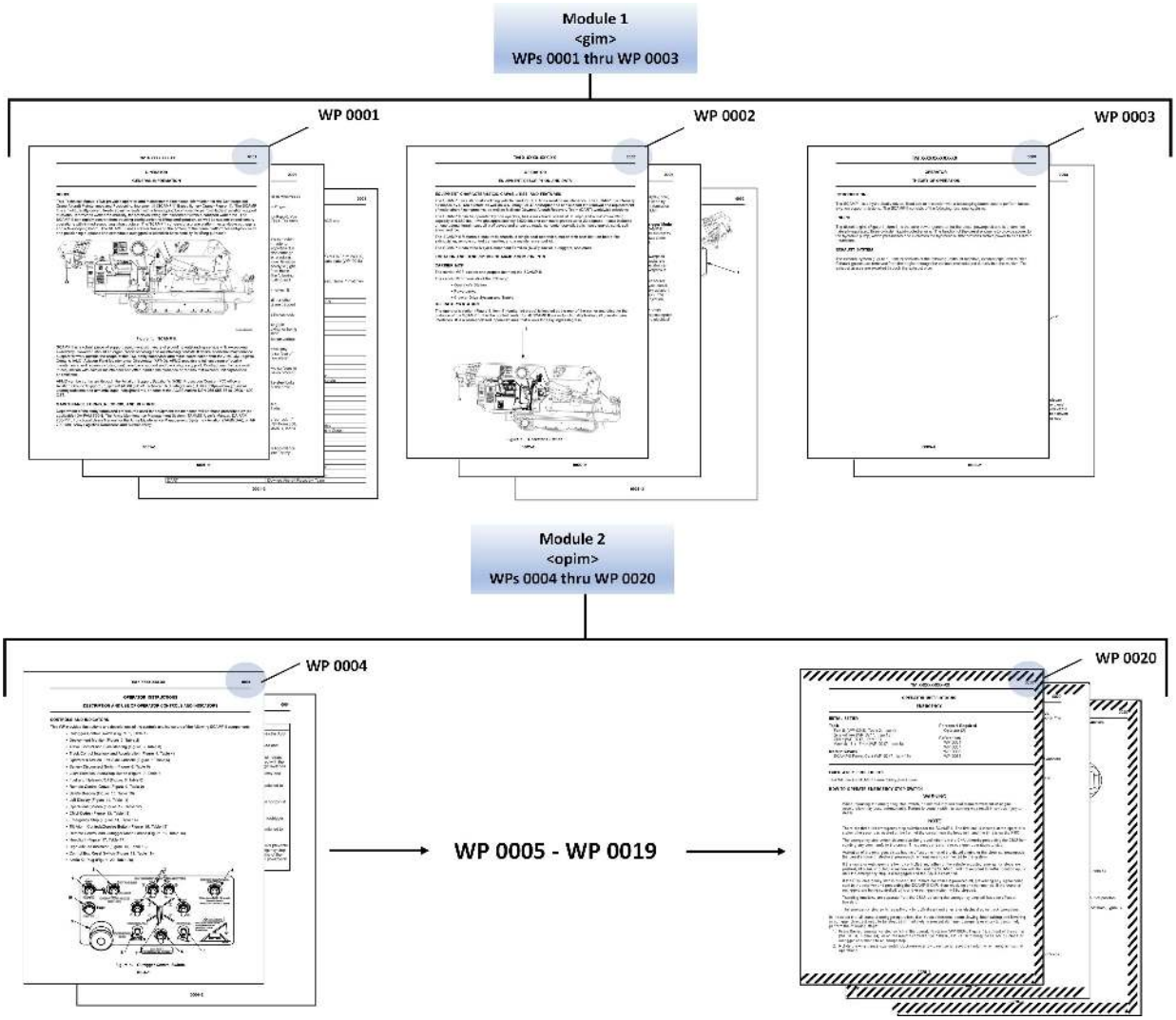


**FIGURE 11. WP sequence number example.**

### 4.12.6.5 Assignment of WP sequence numbers in volumized ETMs.

When an ETM is divided into two or more volumes, the WP sequence number shall continue consecutively. The first volume shall contain as many WPs as necessary beginning with 0001. The WPs contained in the second and

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subsequent volumes shall be numbered consecutively, beginning with the number immediately following the last WP sequence number in the preceding volume.

#### 4.12.6.6 Work package page numbering in ETMs.

Each WP shall be page numbered consecutively using the four-digit WP sequence number followed by -1, -2, -3, and so on (for example, 0001-1, 0001-2, etc.). In ETMs, page numbers shall be centered at the bottom of the page. Even numbers shall be assigned to the left-hand pages and odd numbers to right-hand pages.

### 4.12.7 Paragraphs.

Multiple primary paragraphs may be included in a work package. Primary paragraphs shall contain no more than four subparagraph levels.

#### 4.12.7.1 Paragraphs and subparagraph titles.

**a.** In narrative text, paragraphs **<para0>** and subparagraphs **<subpara1>** shall have titles. The title shall begin at the left margin.

**b.** Multiple blocks of text under a title are allowed.

#### 4.12.7.2 Paragraph and subparagraph numbering (ETMs).

Paragraphs and subparagraphs within a WP may be unnumbered in ETMs.

#### 4.12.7.3 Paragraph and subparagraph format.

TABLE II shows paragraph and subparagraph formatting requirements.

**TABLE II. Paragraph and subparagraph format.**

| Paragraph level | Starting position | Title format | Block text format |
|---|---|---|---|
| Primary Paragraph (para0) | Flush left | Bold, all caps | Start on a separate line; include a blank line between title and text block |
| Subparagraph Level 1 (subpara1) | Indent first line 2.5 em spaces; remaining text indented 2.5 em spaces from left | Bold, mixed case | Start on a separate line; include a blank line between title and text block |
| Subparagraph Level 2 (subpara2) | Indent first line 5 em spaces; remaining text indented 5 em spaces from left | Bold, mixed case and end with a period | Start after title separated by 2 en spaces |



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**TABLE II. Paragraph and subparagraph format. (continued)**

| Paragraph level | Starting position | Title format | Block text format |
|---|---|---|---|
| Subparagraph Level 3 (subpara3) | Indent first line 7.5 em spaces; remaining text indented 7.5 em spaces from left | Bold, mixed case and end with a period | Start after title separated by 2 en spaces |
| Subparagraph Level 4 (subpara4) | Indent first line 10 em spaces; remaining text indented 10 em spaces from left | Bold, mixed case and end with a period | Start after title separated by 2 en spaces |

### 4.12.8 Steps.

#### 4.12.8.1 Procedural steps.

**a.** Procedural steps shall be used to present detailed step-by-step instructions for performing operational or maintenance tasks.

**b.** Procedural steps shall start with an action verb. An introductory phrase citing the tool to use may precede the action verb (for example, "Using torque wrench, tighten bolts...").

**c.** When required to denote situations in which an "if/then" scenario exists, conditional steps (which are a type of procedural step) may be used. For example, if maintenance shall only be performed when certain observed conditions are present, conditional steps direct a user how to proceed. Conditional steps shall be presented as steps rather than included as a note. See FIGURE 12.

**d.** Procedural steps shall not have titles.

**e.** With the exception of steps in troubleshooting work packages, a minimum of two steps shall be required to use the tag **<step1>** . If only one step exists, use a **<para>** tag. Single steps shall be unnumbered.

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**Note**

Figure appearance may vary slightly based on version of DTD/stylesheets utilized.

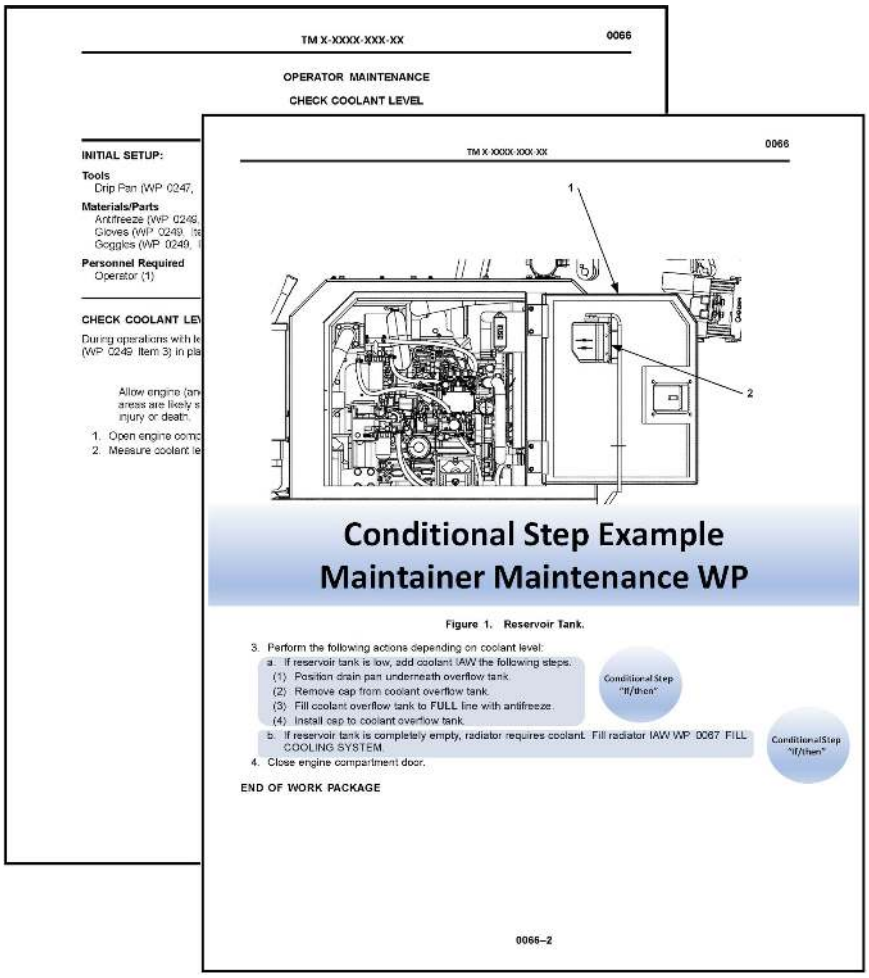


**FIGURE 12. Conditional step example.**

## 4.12.8.2 Procedural step levels.

Procedural steps shall be divided into no more than six levels (see FIGURE 13). The following example demonstrates how procedural steps and subordinate step levels shall be formatted and numbered.

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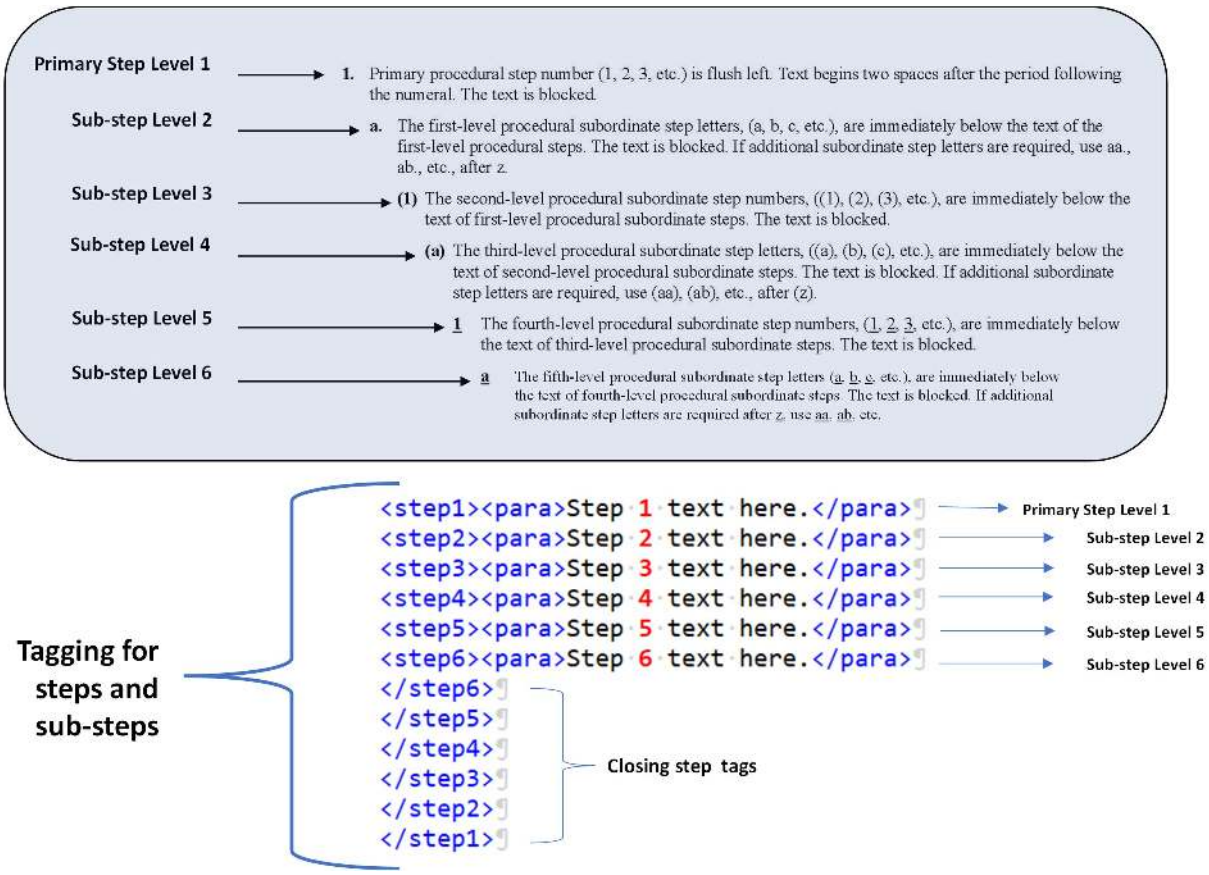


**FIGURE 13. Procedural step formatting.**

## 4.13 Publication updates.

When technical publications require new information, either changes or revisions are prepared. Section 4.13.1 discusses guidelines for changes, and section 4.13.2 discusses guidelines for revisions.

### 4.13.1 Changes.

A change is used to update, emphasize, correct, expand upon, improve, or clarify existing information in the most recent version of an authenticated publication. Changes allow for the quick distribution of updated information affecting fielded publications and may contain updated procedures, steps, and tasks, etc. For example, changes may be the result of minor updates to a Maintenance Allocation Chart (MAC) or minor changes mandated by an Engineering Change Proposal (ECP), a Modification Work Order (MWO), or the integration of DA Form 2028 corrections.

The amount of affected content in a change shall not affect more than half (**50%**) of publication content. In other words, the number of changed pages in a publication change package shall be less than half of the total page count of the authenticated publication. If greater than **50%** of content is updated in a publication, the effort shall be a revision (4.13.2).This section includes guidelines for changes to technical publications. 4.13.1.1 covers instructions applicable to all technical publications while section 4.13.1.2 covers instructions applicable only to ETMs.

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#### 4.13.1.1 Change instructions for all technical publications.

TABLE III serves as a guide for publication change requirements. Section 4.13.1.2 includes the tagging information necessary to comply with requirements listed in TABLE III.

**TABLE III. Change requirements.**

| CHANGE REQUIREMENTS | |
|---|---|
| Publication type | IETMs and ETMs. |
| Compliance | Changes shall be completed in accordance with (IAW) the same version of MIL-STD-40051 as the original publication. |
| Dates | Changes shall be identified by a change date, which shall appear on the change transmittal (not the publication cover). |
| Issuance | The FRC of the changes shall include the following two items:<br>**1.** The issuance of a change only file.<br>**2.** The re-issuance of the consolidated TM (incorporating all updates in the change package). |
| Markings | Changes shall include the following two items:<br>**1.** Change bars that identify updated data within a publication.<br>**2.** Change number identification information in the footer of each changed page. |
| Summary | Changes shall include a change transmittal listing all affected publication WPs. |

##### 4.13.1.1.1 Tagging changes.

Changes include alterations to an original text including insertions, modifications, deletions, etc. Changes shall be integrated into XML utilizing two available options: change elements (tags) and change attributes. The type of change being integrated into a publication shall determine whether elements and/or attributes shall be used to integrate the change.

##### 4.13.1.1.2 Change elements.

For changes involving only text (only a few words), the **<change>** tag (or element) may be used. The scope and type of changed information shall be identified by assigning attributes to the **<change>** tag. Change tags shall include comment data as detailed in section . See FIGURE 14 for an example.



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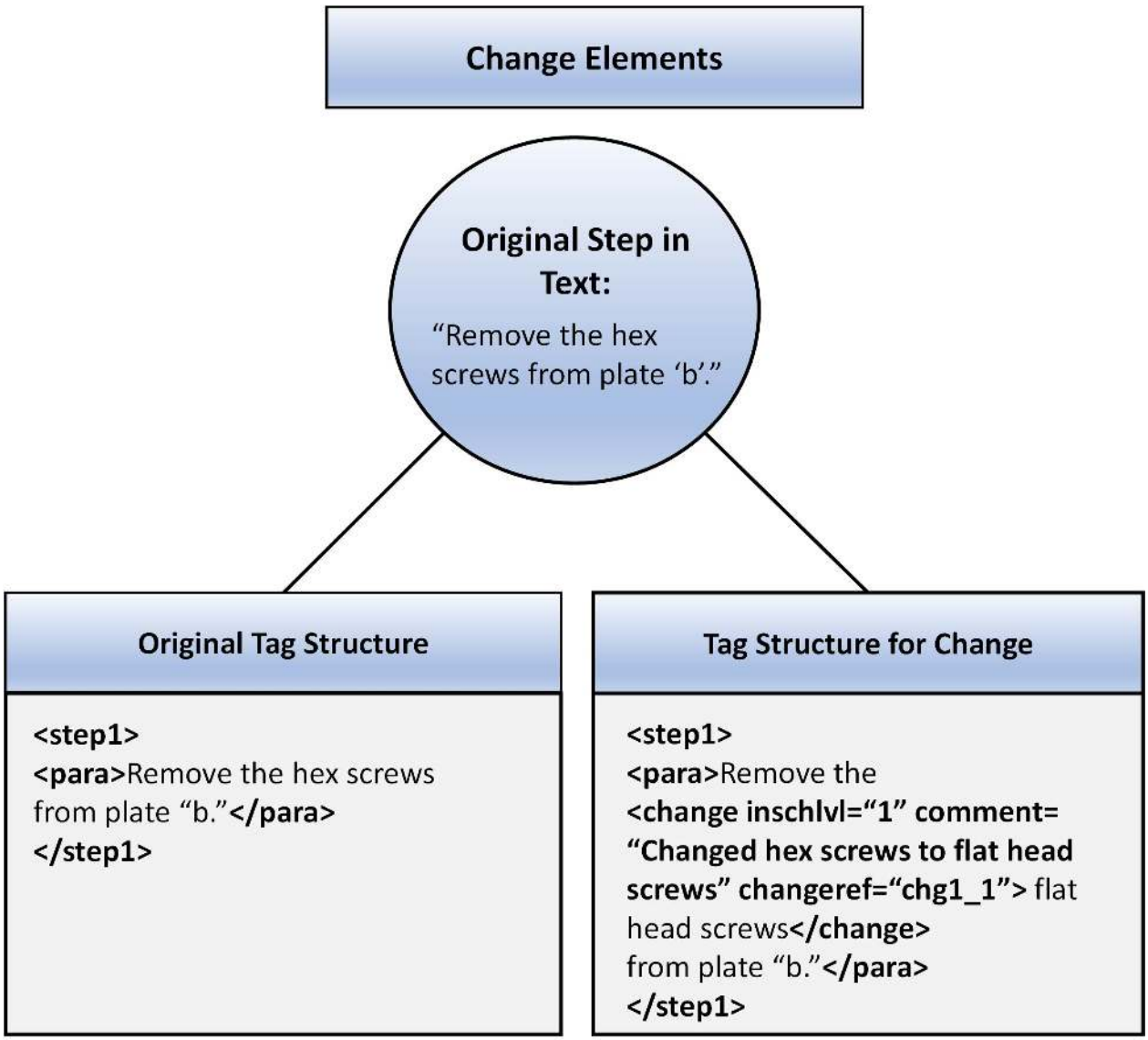


**FIGURE 14. Change elements.**

### 4.13.1.1.3 Change attributes.

For changes involving larger amounts of text (steps or paragraphs), changes shall be integrated into XML by updating the attribute of the step/paragraph tag (element) itself (rather than inserting **<change>** tags). To visualize how tag attributes shall be updated, see FIGURE 15. Section 4.13.1.1.4 describes the functions of available change attributes.

### 4.13.1.1.4 Change attribute functions.

There are five attributes related to change documentation:

**a. inschlvl** – contains numeric change designator; for use on any change or addition.

**b. delchlvl**– contains numeric change designator; for use on a deletion.

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**c. comment** – very brief explanation for publications manager/technical writer about the change.

**d. changeref** – link to applicable **<change.history>** giving authority for the change.

**e. id** – (if available on the element) allows revision summary to link directly to the change.

#### 4.13.1.1.5 Comment data in change attributes.

Change attribute comment data shall be integrated into the attributes of not only **<change>** tags but also any other element (tag) containing change data. Change attribute comments are defined in 4.13.1.1.4, and an example of comment data is highlighted in FIGURE 15 and FIGURE 16 (right-side boxes). Comment data is important because it provides context for changes and provides change detail that would not be readily apparent in the updated version of the text. In IETMs, the content of the comment attribute will be displayed when the pointer is hovered over the change bar.

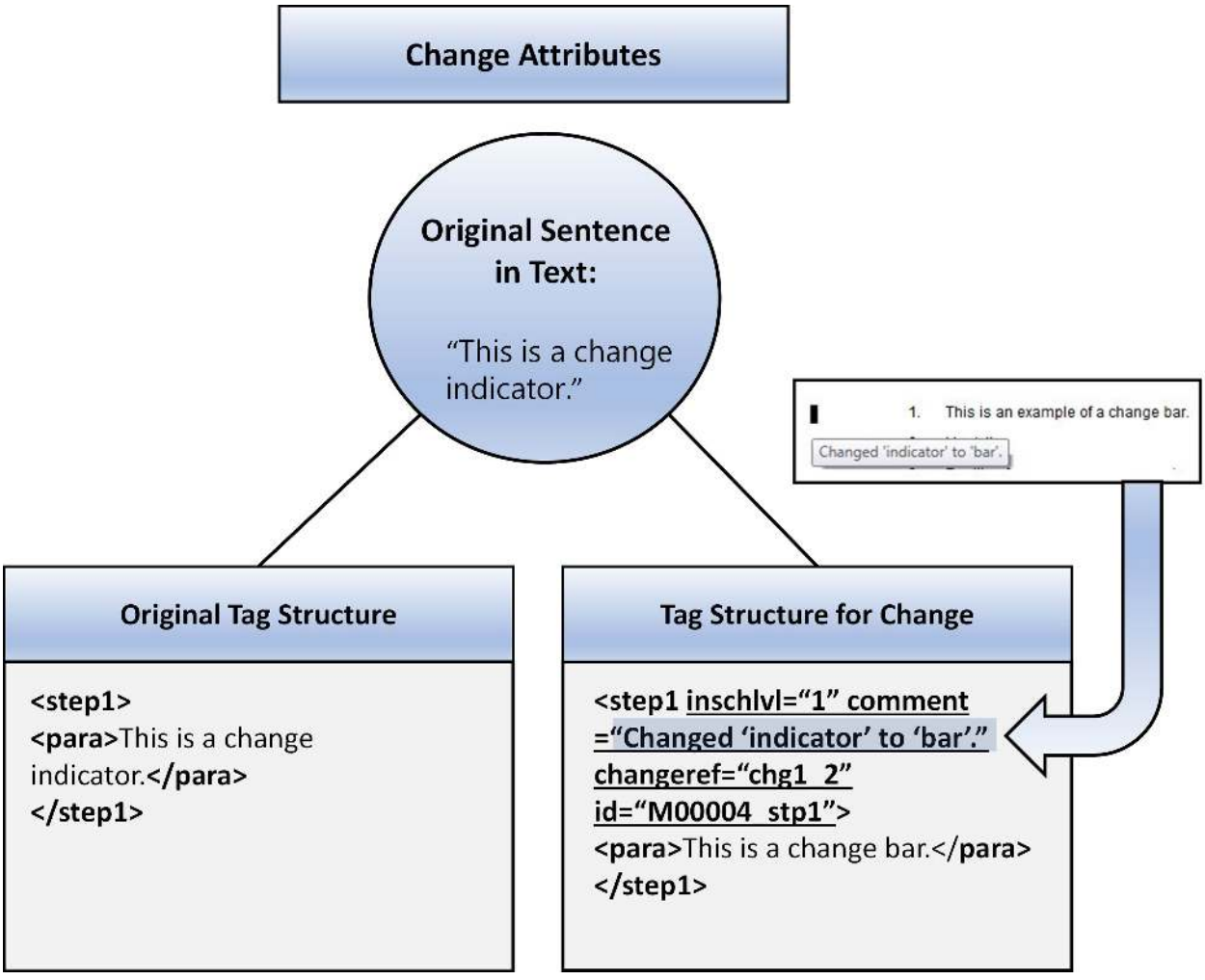


**FIGURE 15. Change attributes (sheet 1 of 2).**

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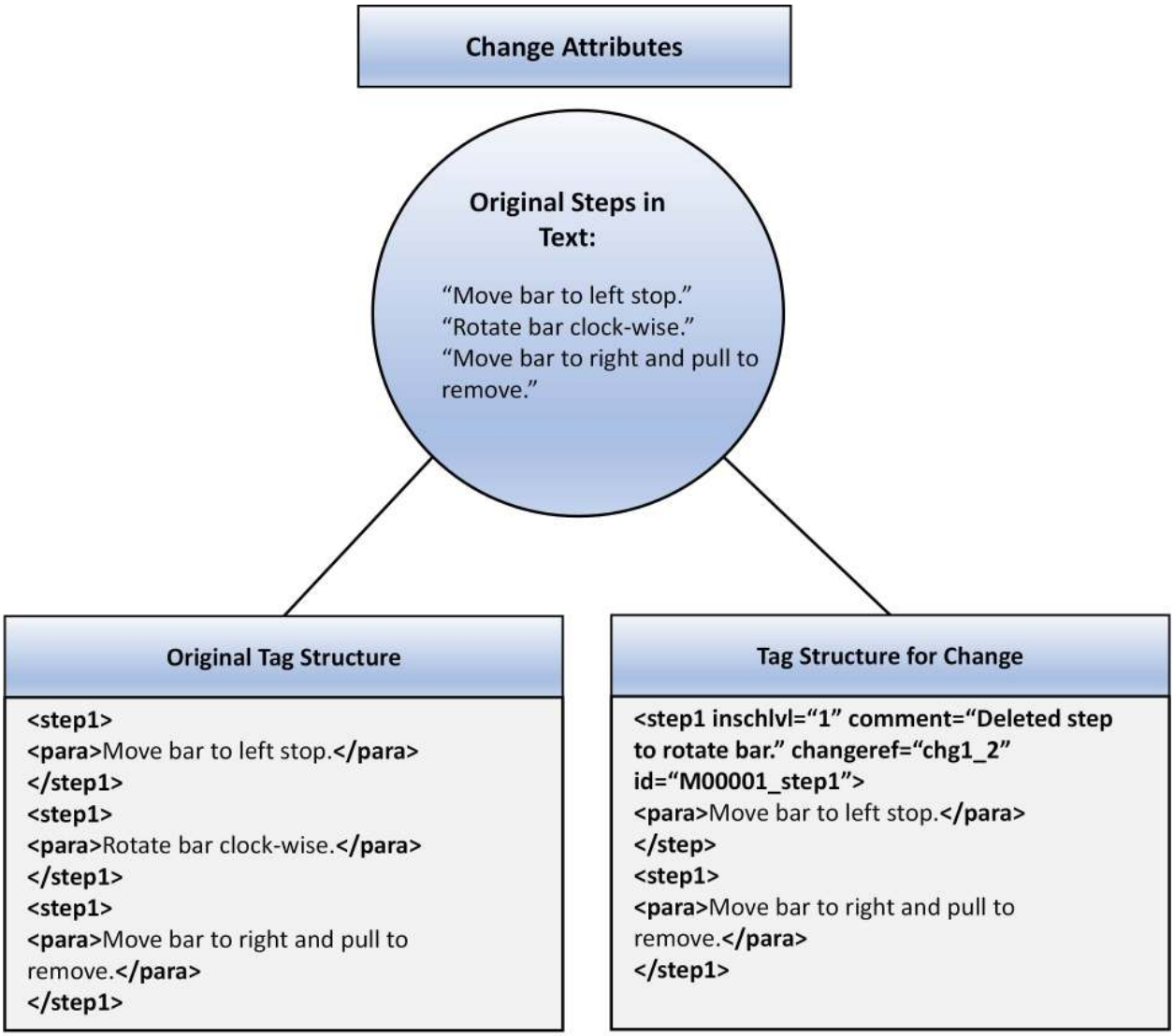


**FIGURE 16. Change attributes (sheet 2 of 2).**

### 4.13.1.1.6 Change markings.

Change bars shall be inserted to identify technical updates in text, tables, and figures as discussed below. FIGURE 17 through FIGURE 21 demonstrate how change bars shall appear within technical publications. Change bars shall display current modifications only. Change bars from a previous change shall not be displayed when a page/file is subsequently modified; however, change tag attributes shall record change history in accordance with 4.13.1.1.3.

Additionally, change bars are not required for correction of minor inaccuracies, such as spelling, punctuation, renumbering, etc., unless such correction alters the meaning of the information. Procedural steps or list items whose number/letter changed after adding or deleting material shall not be marked with a change bar and shall not be considered changed material.

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**a. Changes to content within WPs.**

**(1)** Change bars shall appear at the outer margin opposite updated, deleted, or added text, tables, figures (including cover figures) and table/figure titles. With the exception of TOC content, margins alternate (right side for odd pages, left side for even pages. TOC change bars shall appear on the right side only).

**(2)** Change bars shall span the entire side of a changed figure. If only the figure title has changed, the change bar shall appear adjacent to the title only. For numbering guidance with changed figures, see 4.17.5.9.

**(3)** For changes to the content of a table, change bars shall appear adjacent to the table title and all changed table rows. If only the table title has changed or a new table has been added, the change bar shall appear adjacent to the title only.

**(4)** Change bars shall appear to the left of changed initial setup information.

**(5)** If a callout is deleted from an illustration, the word "DELETED" may be placed after the appropriate number in the legend, if applicable. If a callout is deleted from an illustration without a legend, such as those used to supplement a RPSTL, the word "DELETED" may be placed on the illustration at the end of the leader line.

**b. Changes to WPs and modules.**

**(1) Deleted WP.** Refer to 4.13.1.2.1.2 for requirements related to deletion of WPs in a change.

**(2) Inserted WP.** Refer to 4.13.1.2.1.1 for requirements related to addition of WPs in a change.

**(3) Inserted module.** Module insertion simply includes the addition of module tags as well as at least one WP (4.13.1.2.1.1).

**c. Changes to RPSTLs.**

**(1) Table item.** When a RPSTL item is updated, deleted, or added a change bar shall be placed in the appropriate margin next to the item number column.

**(2) Deleted WP.** When a RPSTL WP is deleted from a publication, remaining RPSTL figure numbers shall not be renumbered to reflect the deleted WP and associated figure until the next revision. Refer to 4.13.1.2.1.2 for further requirements related to deletion of WPs in a change.

**(3) Inserted WP.** When a RPSTL WP is inserted between two RPSTL work packages, the RPSTL figure number may be modified to add alpha characters between existing numbers (22A, 22B, for example). The remaining RPSTL figure numbers shall not be changed until the next revision. The WP numbering shall follow guidelines in 4.13.1.2.1.1.

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**Note**

Figure appearance may vary slightly based on version of DTD/stylesheets utilized.

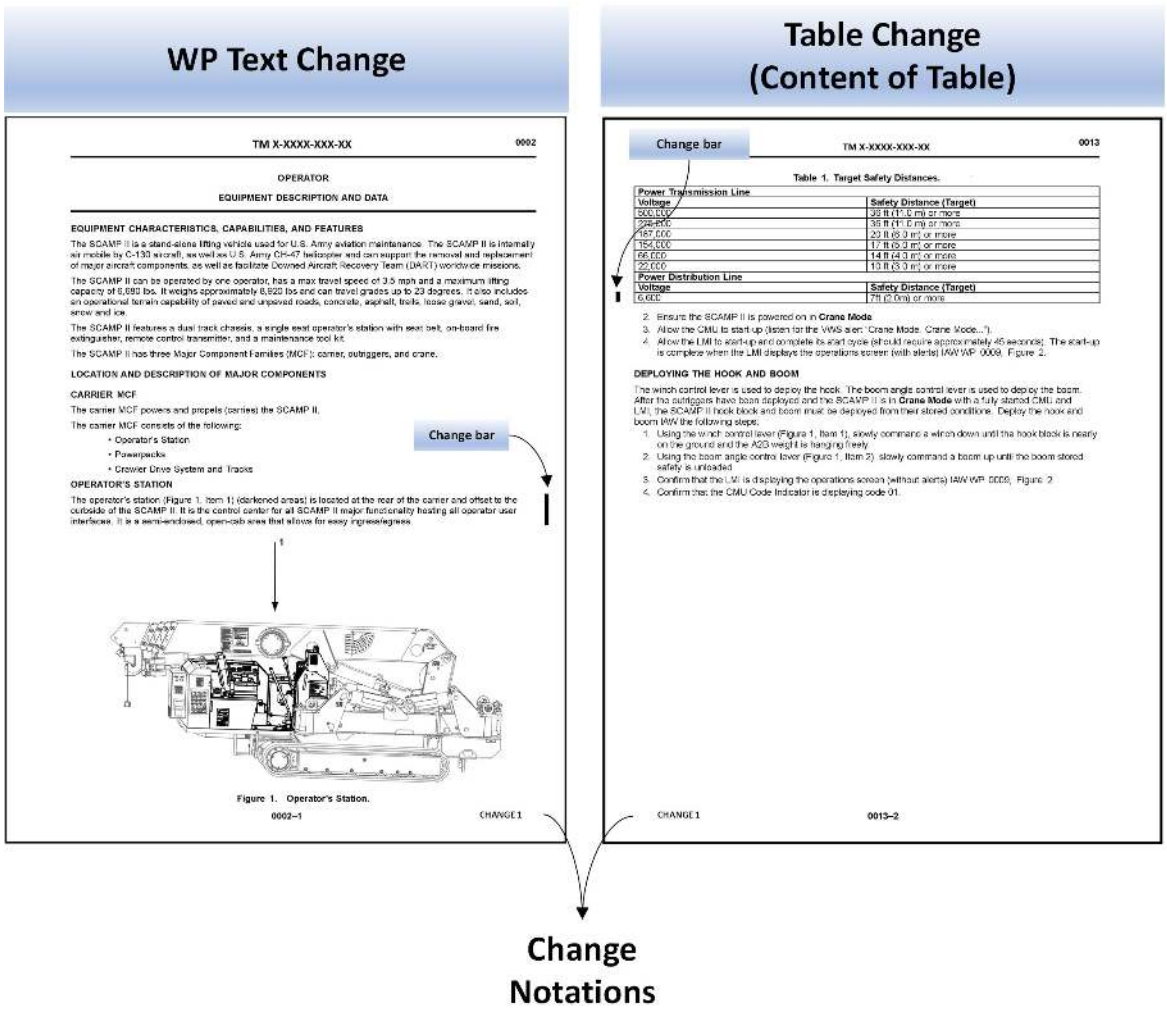


**FIGURE 17. Change markings for changes to WP text and table content.**

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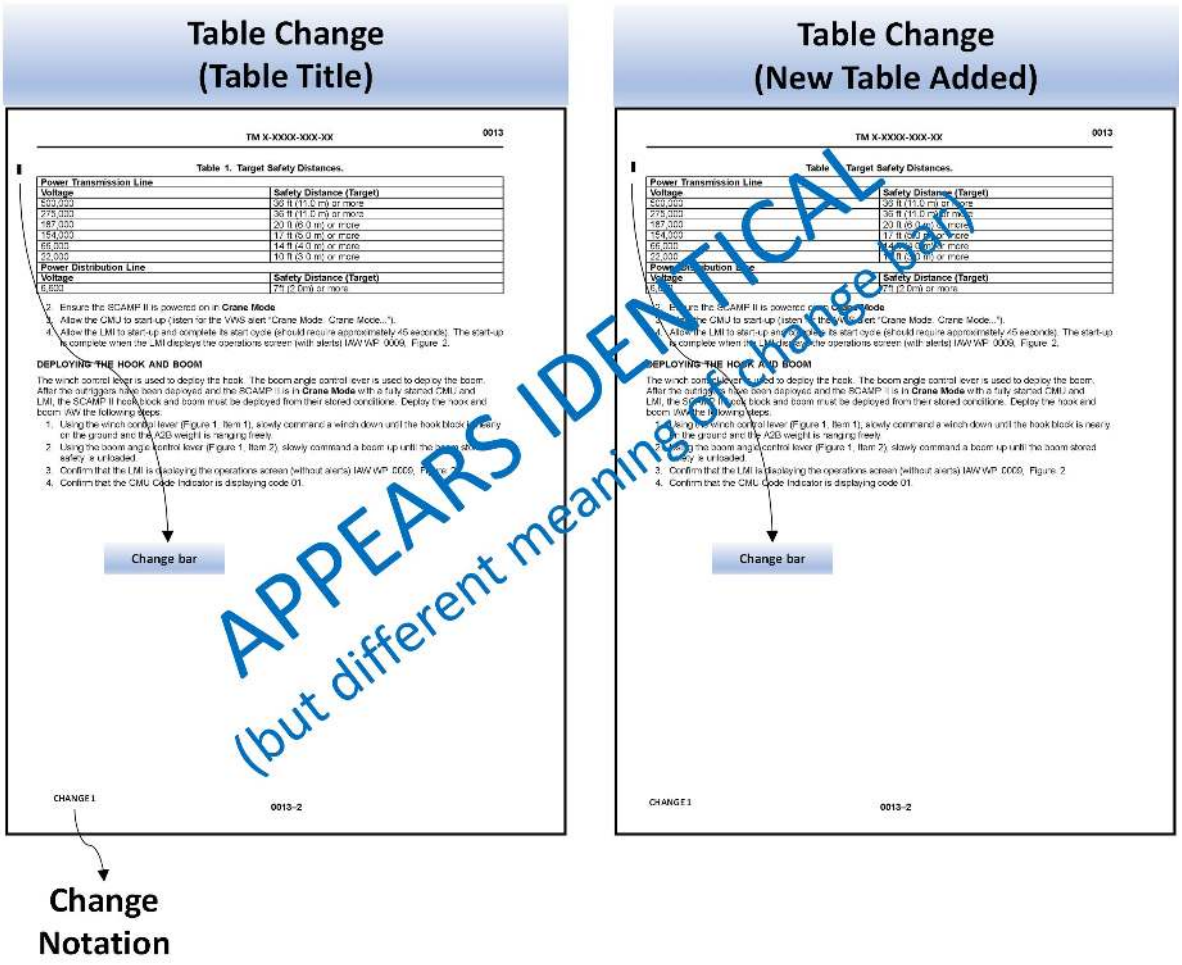


**FIGURE 18. Change markings for changes to table title and new table added.**

APPEARS IDENTICAL
(but different meaning)



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Check the source to verify that this is the current version before use.

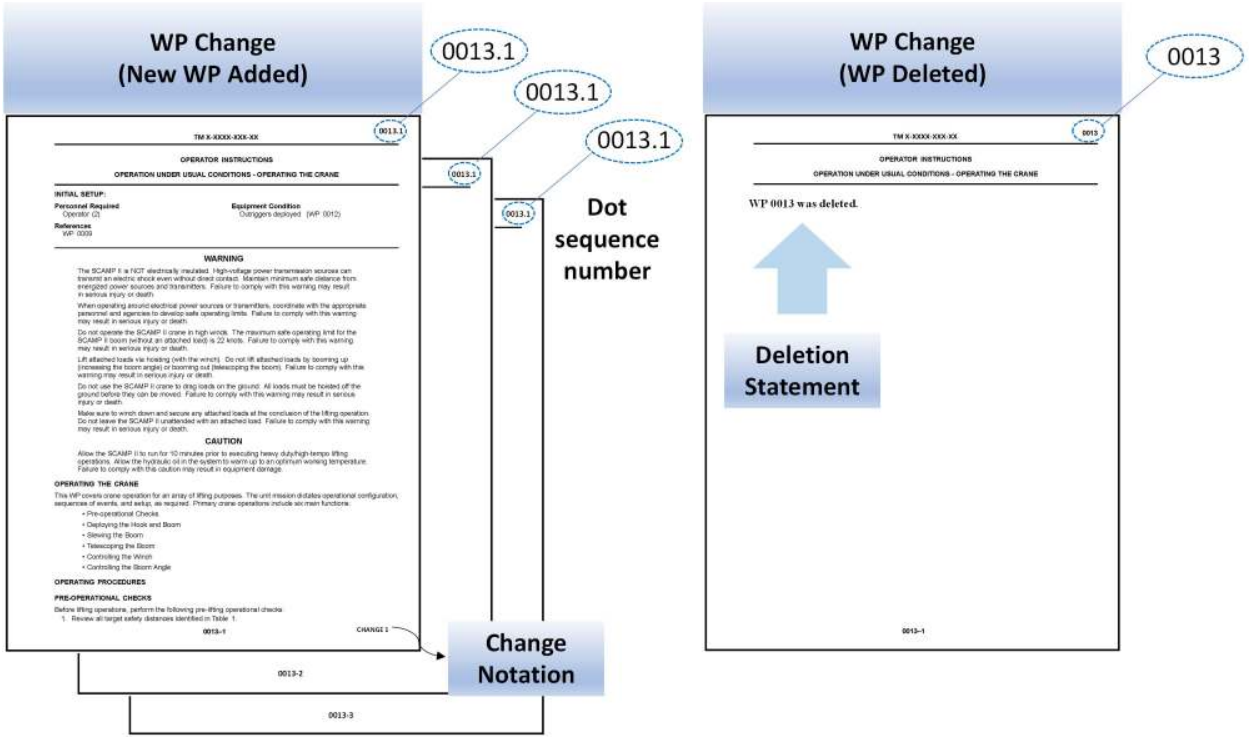


**FIGURE 19. Change markings for added or deleted WPs.**



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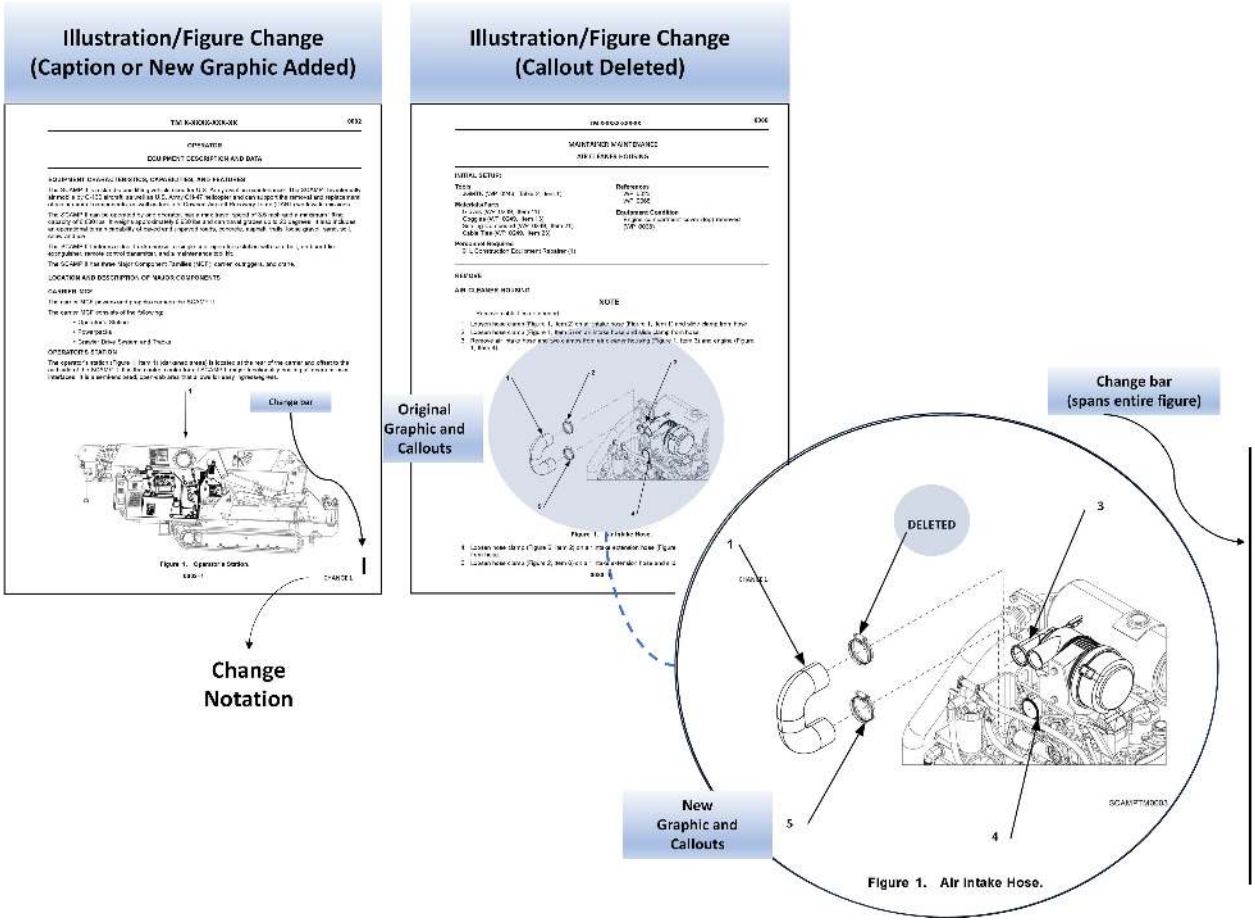


**FIGURE 20. Change markings for changes to figure title and deleted callout.**

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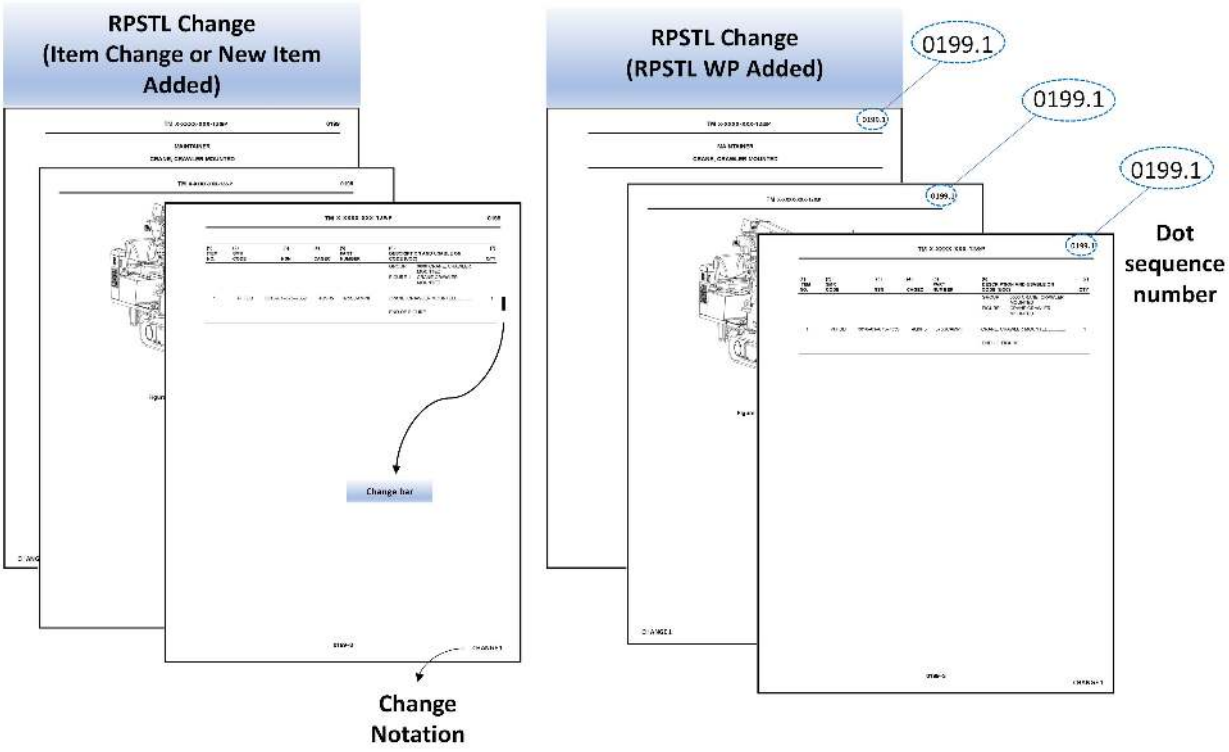


**FIGURE 21. Change markings for a changed or added RPSTL table item and added RPSTL WP.**

### 4.13.1.2 Change package requirements for ETMs.

A change package is a subset of a consolidated file for an ETM change (4.13.1.2.1). General change package requirements are listed below.

**a. Change bars.** All change materials including front matter shall be marked with change bars (excludes cover, revision summary (if applicable), and LOEP (if applicable)).

**b. Change numbers.** All pages of a changed work package, including front and rear matter, shall include the applicable change number located on the outer edge of the page opposite the binding side at the bottom of the page (excludes cover).

**c. Change PIN.** Change PIN shall appear on the last page of the change package on the bottom right-hand side of the page (content may be present on the page, or it may be blank other than the change PIN). The change retains the basic publication's PIN and adds the number of the current change (for example, 067394-001, 067394-002, etc.).

**d. Change pages.** All like-numbered pages in a publication change package must be changed as a unit. For example, if a change appears on page ii of the publication, all pages with small Roman numerals must be included in the change package and identified on the change transmittal page. If any single page in a WP includes a change, however minor, the entire WP must be included in the change package and identified on the transmittal page.

**e. Change transmittal.** A change package shall only contain the change transmittal for the current change (see 5.2.5 for change transmittal guidance). All previous change transmittals shall not be included.

**f. Change package file order.** Change packages shall consist of the items listed below.

**(1)** Change transmittal page (Item 1, FIGURE 22).

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**(2)** Authentication for the change (Item 2, FIGURE 22).

**(3)** Affected front matter pages (some front matter shown in Item 3, FIGURE 22).

**(4)** Applicable WPs (Items 4-6, FIGURE 23).

**(5)** 2028 (only if required, such as when the MATDEV's contact info changes).

**(6)** Foldout pages (only if affected).

**(7)** Cover (only if affected. Changes to the publication do not change the cover's date).

**(8)** PIN page for the change (Item 6, FIGURE 23).

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**Note**

Figure appearance may vary slightly based on version of DTD/stylesheets utilized.

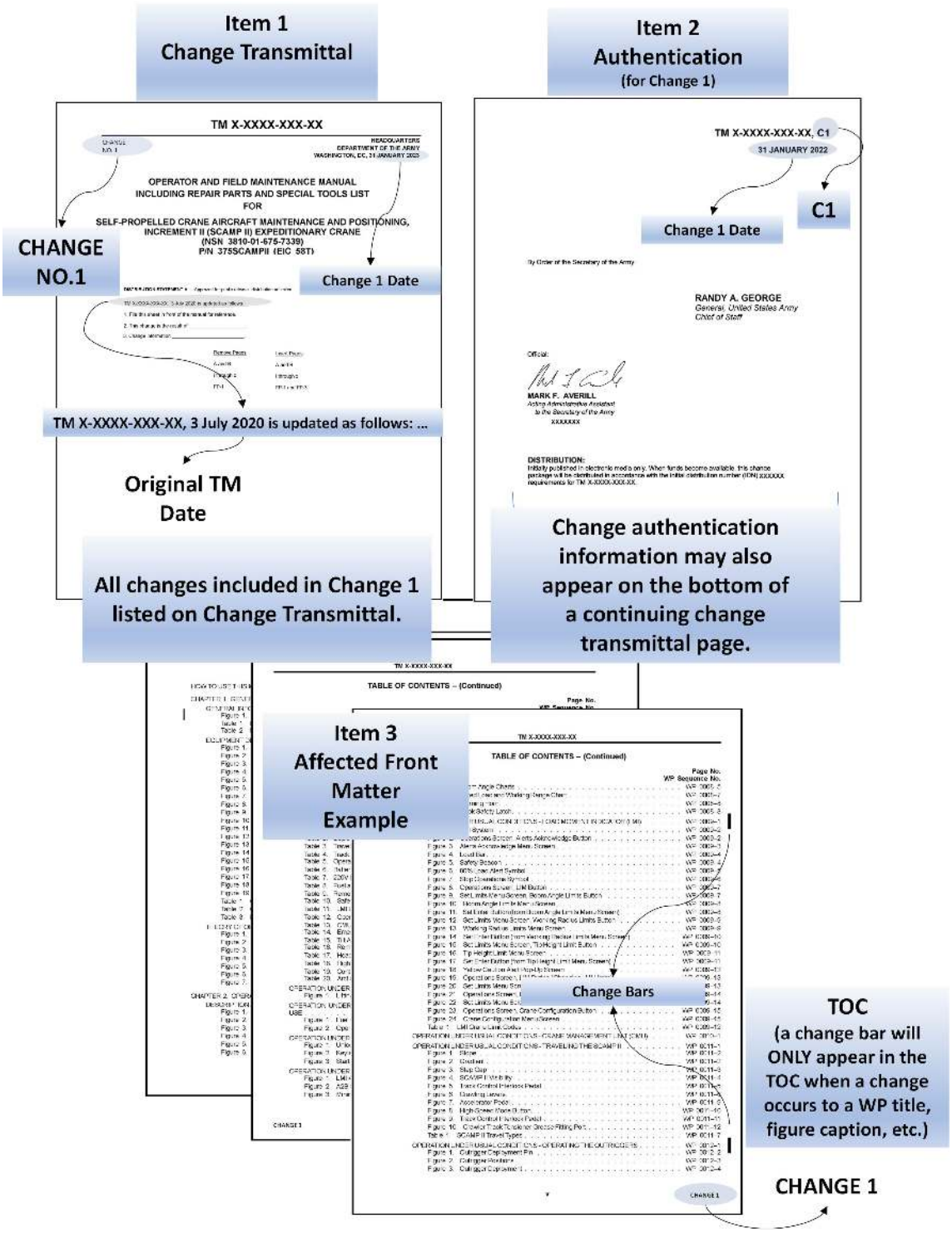


**FIGURE 22. Change package - change 1 example (sheet 1 of 2).**

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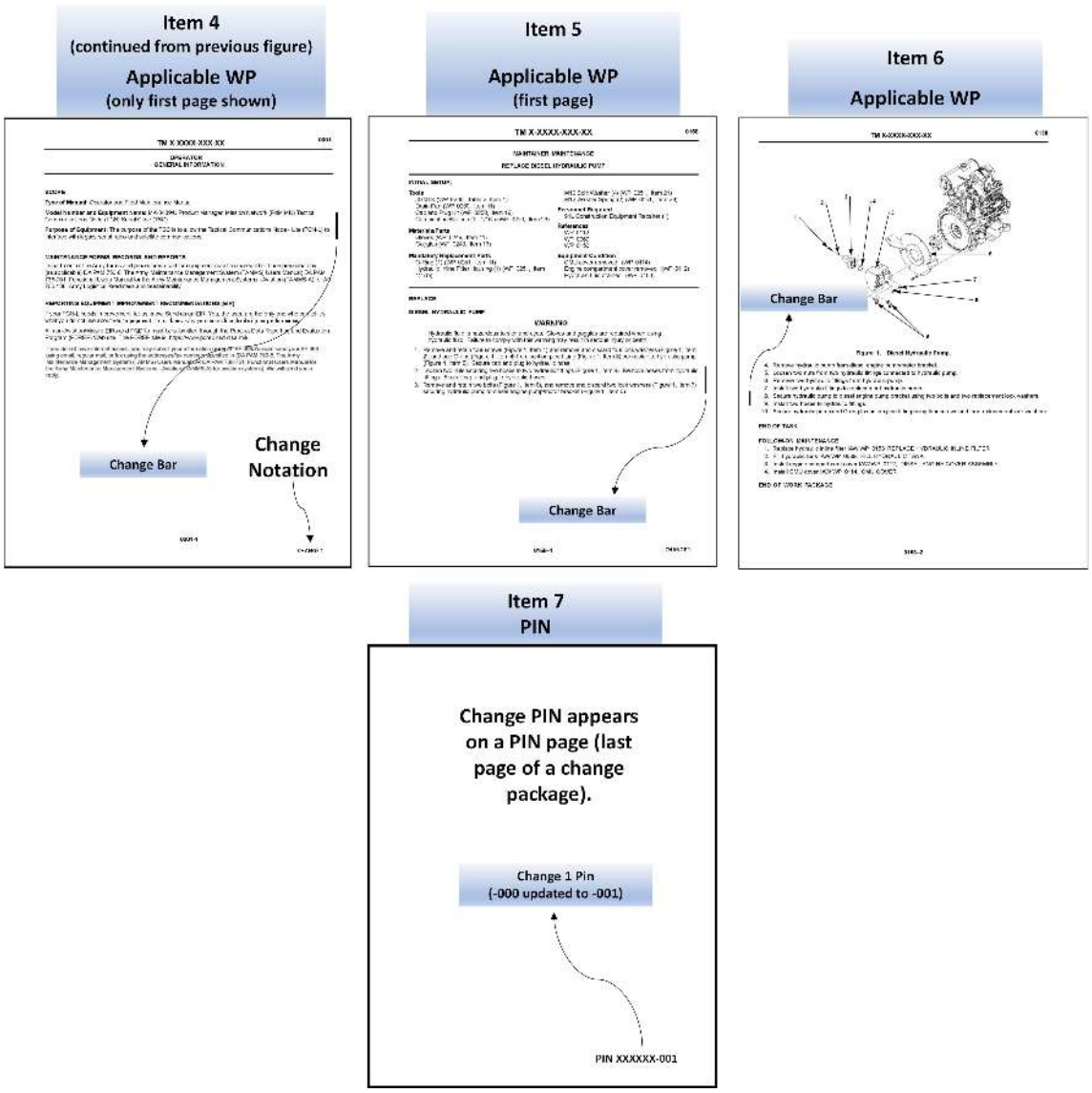


**FIGURE 23. Change package - change 1 example (sheet 2 of 2).**

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#### 4.13.1.2.1 Preparing the consolidated file for an ETM change.

A change package (4.13.1.2) is inserted into the authenticated publication to create a consolidated file. Items included in a consolidated file (sometimes called the "full up" or the "included file" ) for an ETM change are listed below. FIGURE 24 through FIGURE 26 illustrate a consolidated file for a change 3 effort. Section 4.13.1.1.3 discusses change markings that shall appear in a change, while sections 4.13.1.2.1.1 and 4.13.1.2.1.2 discuss guidelines for inserting and deleting WPs in an ETM change.

a. **Item # 1: Cover.** The cover appears first with its inside blank page unless items have "rolled over."

b. **Item # 2: (not shown in figures) Warning summary (if included).** The warning summary shall appear next in a consolidated file.

c. **Items # 2 through 4: Change transmittal page(s).** All change transmittal pages shall appear as the third, fourth, etc. item(s) in a consolidated file. For change second, third, and fourth, etc. changes, the change transmittal pages shall appear in descending order (Change, 4, Change 3, Change 2, Change 1, for example).

d. **Items # 5 through 8: Publication with change pages inserted.** The rest of the publication appears as specified throughout this standard, but with the changed WPs inserted in place of the previous WPs and any updated rear matter added as specified by the change transmittal. The consolidated file retains the TM's original PIN (for example, ######-000). The change PIN only appears in the change package file.

**Note**

Figure appearance may vary slightly based on version of DTD/stylesheets utilized.

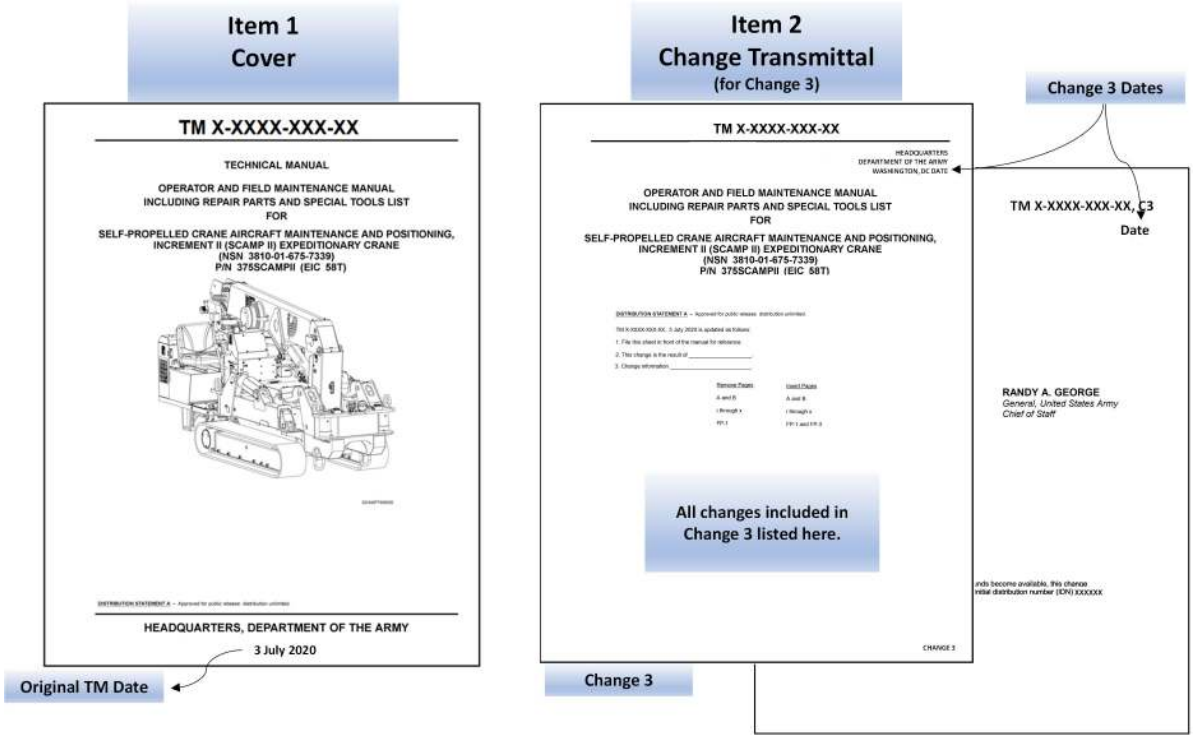


**FIGURE 24. Consolidated file - change 3 example (sheet 1 of 3).**

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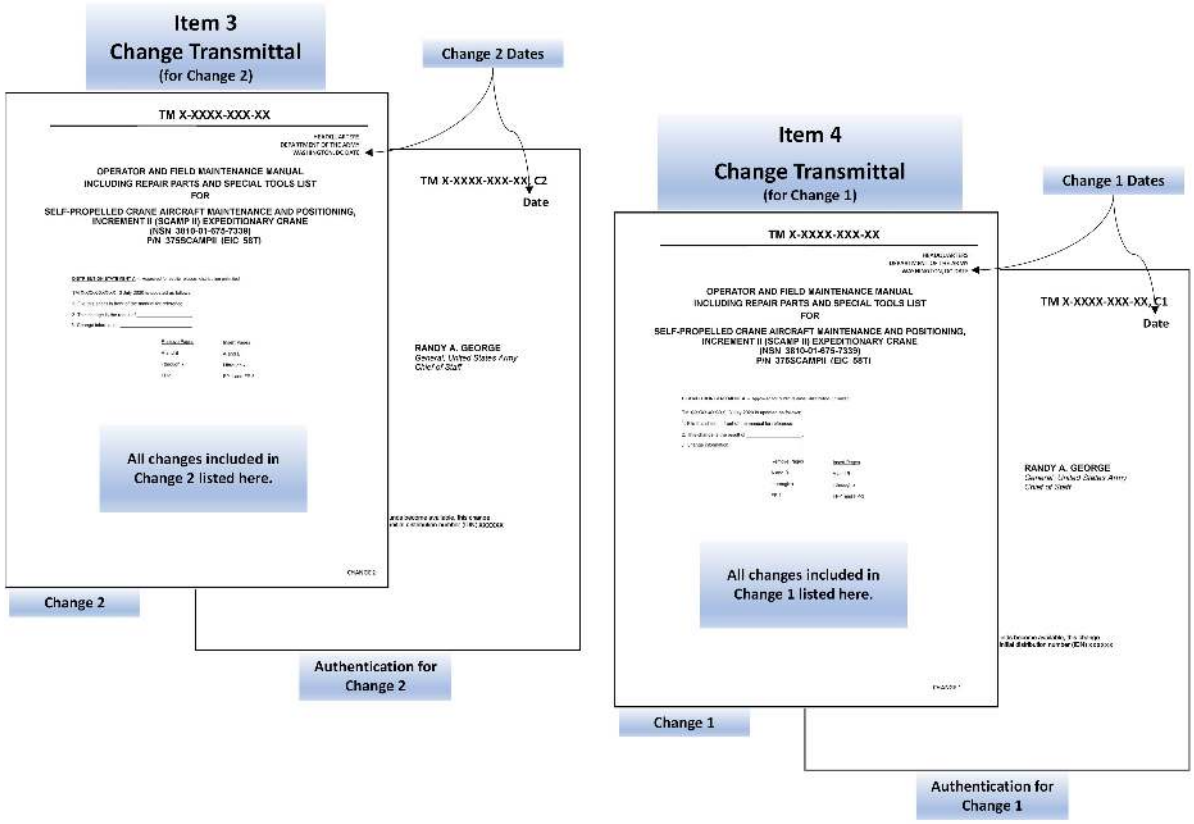


**FIGURE 25. Consolidated file - change 3 example (sheet 2 of 3).**



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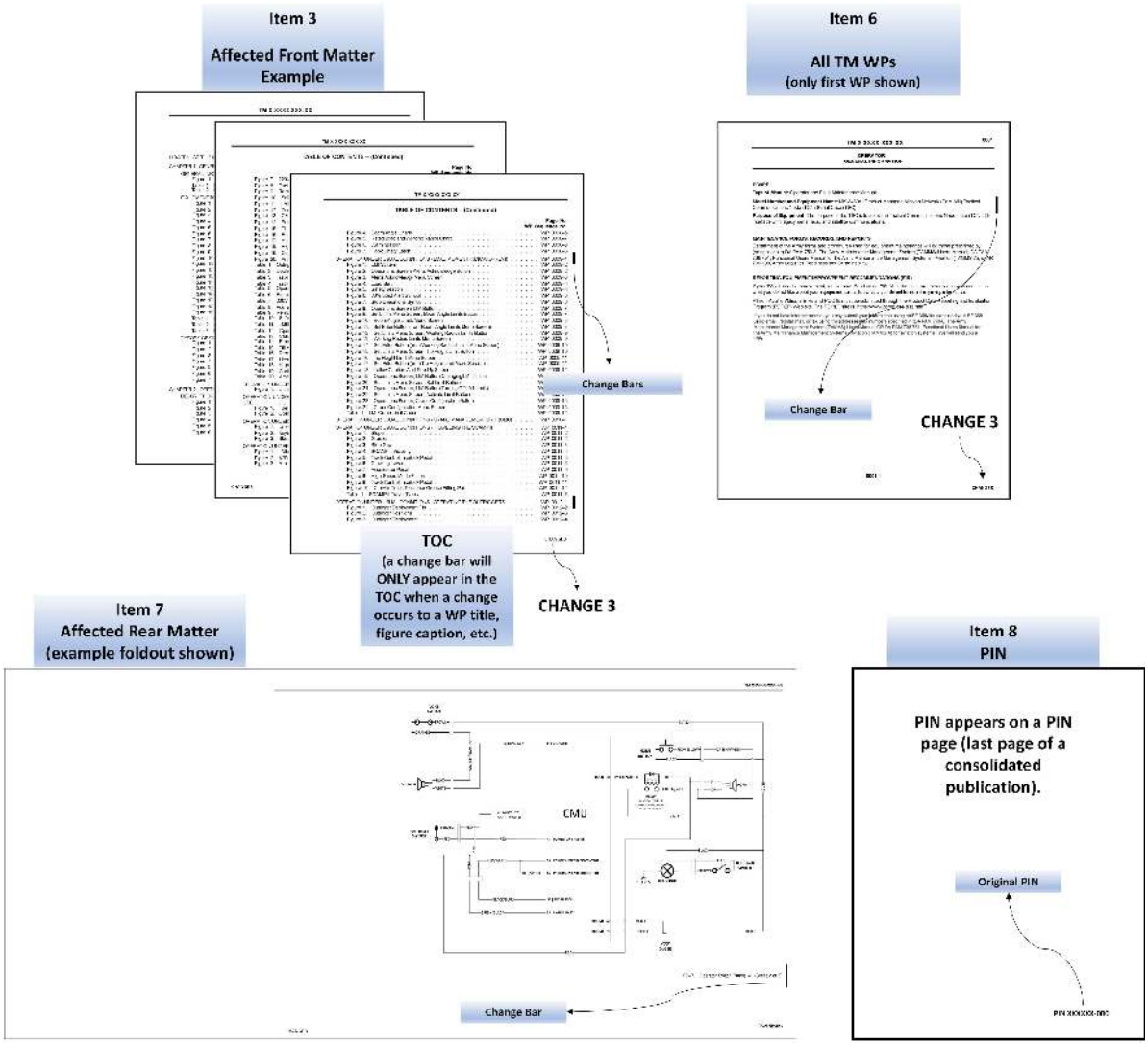


**FIGURE 26. Consolidated file - change 3 example (sheet 2 of 3).**

### 4.13.1.2.1.1 WP insertion in ETM publication changes.

A new WP that is added to the end of a non-volumized ETM or to the end of the last volume of a multi-volume ETM shall use the next available four-digit WP number. For example, if 0098 is the number of the last WP in the ETM, 0099 shall be the number of the new work package.

A new WP that is inserted between two WPs shall use a point numbering scheme to create a new sequence number that logically fits between the two existing WP numbers. Point numbers shall start with "1" and continue in numerical sequence as needed. See the example case below:

**EXAMPLE CASE**

Lubrication WP added to an existing publication as part of a change 1 effort. WP tag shall include the following attributes:

**<lubewp chngno="1" inschlvl="1" insertwp = "1" wpno="M0000101–X-XXXX-XXX-XX">**

In the production file, a change number, for example "1," is also required for the <production> tag.

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Refer to TABLE IV for examples of sequence numbers for inserted WPs.

**TABLE IV. New WP sequence numbers for a change (ETMs).**

| Existing WPs | Number of new WPs inserted between existing WPs | Sequence numbers for new WPs |
|---|---|---|
| 0010, 0011 | 3 | 0010.1, 0010.2, and 0010.3 |
| 0010.3, 0011 | 1 | 0010.4 |
| 0011, 0012 | 1 | 0011.1 |

### 4.13.1.2.1.2 WP deletion in ETM publication changes.

When a WP is deleted in an ETM change, the WP shall be removed and a page inserted with a statement that says, "WP was deleted" and a change bar shall be placed next to this statement (FIGURE 27 and FIGURE 28). The deleted WP shall be listed on the change transmittal with the word "DELETED" next to it. All work packages following the deleted WP shall retain their original WP sequence number.

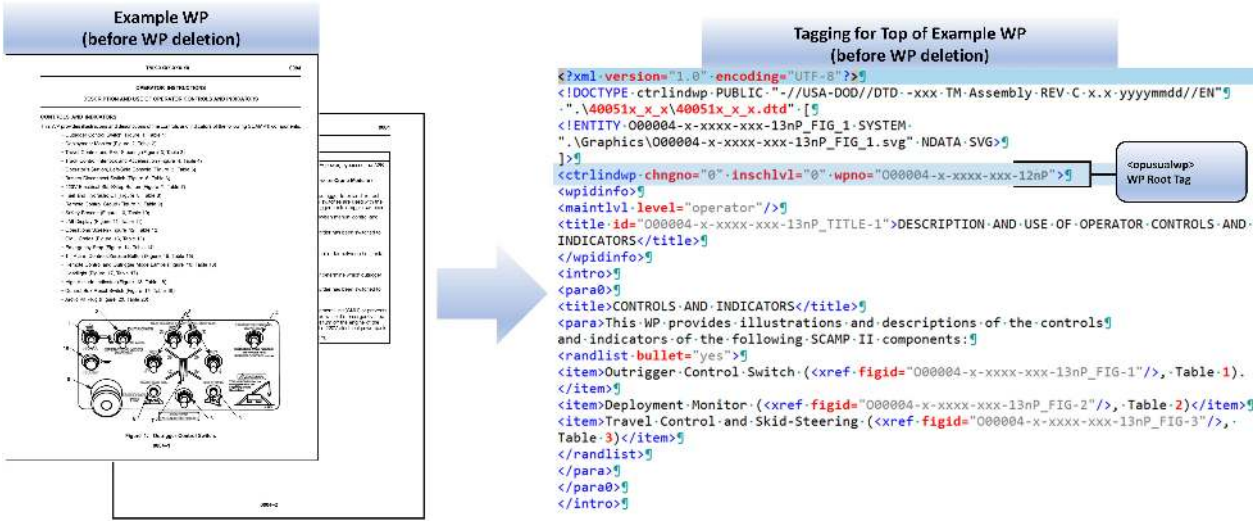


**FIGURE 27. Deletion of WP in an ETM change (before deletion).**

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Check the source to verify that this is the current version before use.

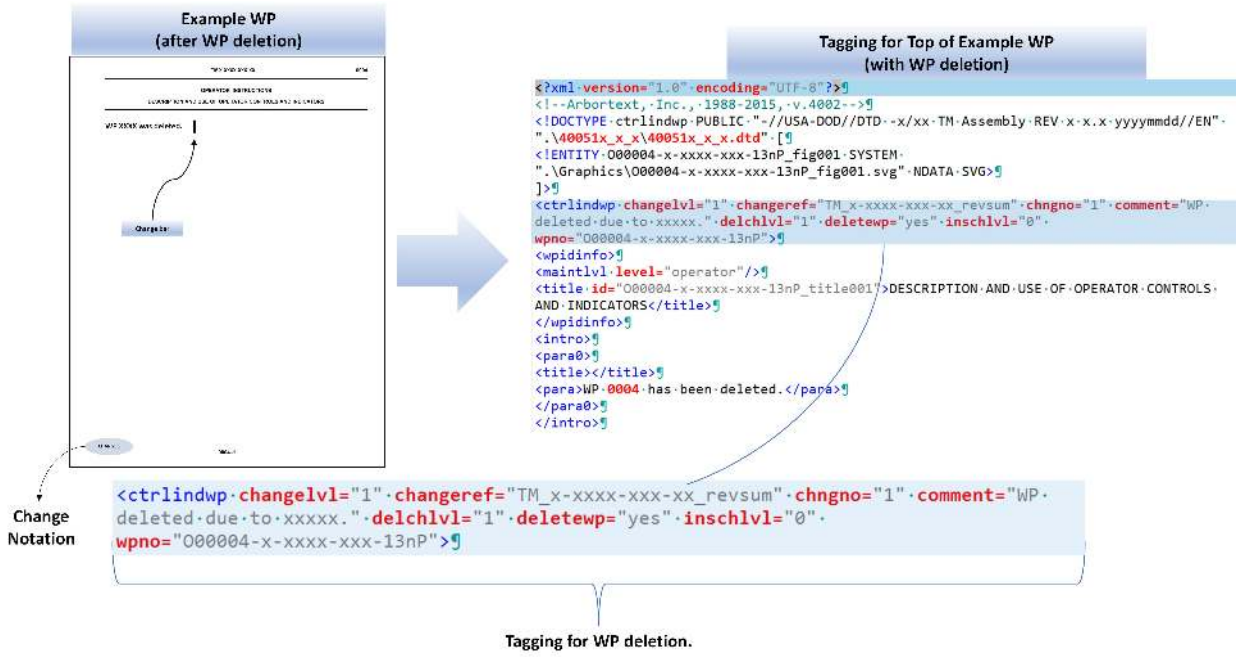


**FIGURE 28. Deletion of WP in an ETM change (after deletion).**

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### 4.13.2 Revisions.

A revision shall be implemented when more than half (**50%**) of original publication content is affected by updates. Revisions may be the result of minor content updates that impact many areas of a publication (such as tool changes, environmental warnings, CUI markings, reference updates, MIL-STD conversions, etc.). Revisions, however, may also be the result of significant LPD updates (such as equipment system upgrades or configuration changes) that are often documented in ECPs, MWOs, or other system alteration documents. Types of LPD-driven updates include, but are not limited to, the examples listed below:

**a.** Updated operator instructions to power on, assemble, operate, and power off new equipment associated with a system.

**b.** The integration of additional troubleshooting procedures and/or WPs pertaining to the new equipment.

**c.** Updated MAC entries that may rearrange existing FGCs or incorporate new FGCs and associated maintenance tasks.

**d.** The integration of additional RPSTL WPs for new equipment changed or added to a system.

TABLE V shall serve as a guide for revision requirements. Contact the MATDEV with any equipment change questions that may be incorporated into a revised publication. Contact the EPCO with any questions about revised publication requirements.

**TABLE V. Revision requirements.**

| **REVISION REQUIREMENTS** | |
|---|---|
| Publication type | ETMs and IETMs. |
| Alignment | All impacted sections of a revised TM shall be assessed to ensure alignment among instructions, tasks, MAC, and RPSTL. |
| Compliance | Revisions shall be completed IAW the most recent version of MIL-STD-40051 at the time of the revision or what was put on contract. |
| Dates | Revisions shall be identified by a revision date on the front cover. A supersession notice also shall be included on the front cover. |
| Issuance | Revisions include the re-issuance of the entire publication. |
| Markings | Any markings from a previous change shall be removed. |
| Summary | May contain a revision summary page. |

#### 4.13.2.1 WP sequence numbers for revisions (IETMs).

When an IETM is revised, all work packages are renumbered to account for any new or deleted WPs; thus, the WP sequence numbers will be updated to absorb added or deleted WPs.

#### 4.13.2.2 Supersession notices for revisions.

Supersession notices shall be used whenever a revision is done to provide the user with information about what the revised manual supersedes. The LOEP (if applicable) shall contain the same supersession notice as the cover. Refer to 5.2.1.11.2 for examples of supersession notices.

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## 4.14 Safety information: warnings, cautions, and notes.

In general, warnings identify the risk of bodily injury or death, cautions identify risk of damage to equipment, and notes highlight other pertinent information.

The implications of procedures can be nuanced, however, particularly as technology evolves. For example, FIGURE 30 below offers a scenario in which an equipment failure should be a warning rather than a caution.

Warnings, cautions, and notes should be used judiciously to preserve their value and to avoid overuse.

Safety office approval for warnings is not meant to guarantee that they are compliant with this standard. The guidance in 4.14 through 4.14.11.3 is mandatory and intended for compliance.

Examples in 4.14.1 through 4.14.3 are shown in ETM format. For IETM format, refer to 4.14.7.

### 4.14.1 Warning <warning>.

A warning is a short message that calls attention to an essential operating or maintenance procedure, practice, condition, statement, etc.; if not strictly observed, a warning could result in injury to or death of personnel (see FIGURE 29).

Warnings should concisely identify a hazard, explain how to avoid the hazard, and explain what to do if coming in contact with the hazard. While the warning summary does cite the first aid training circular, any immediate actions that must be taken to preserve the safety of injured individuals shall be identified in the warning.

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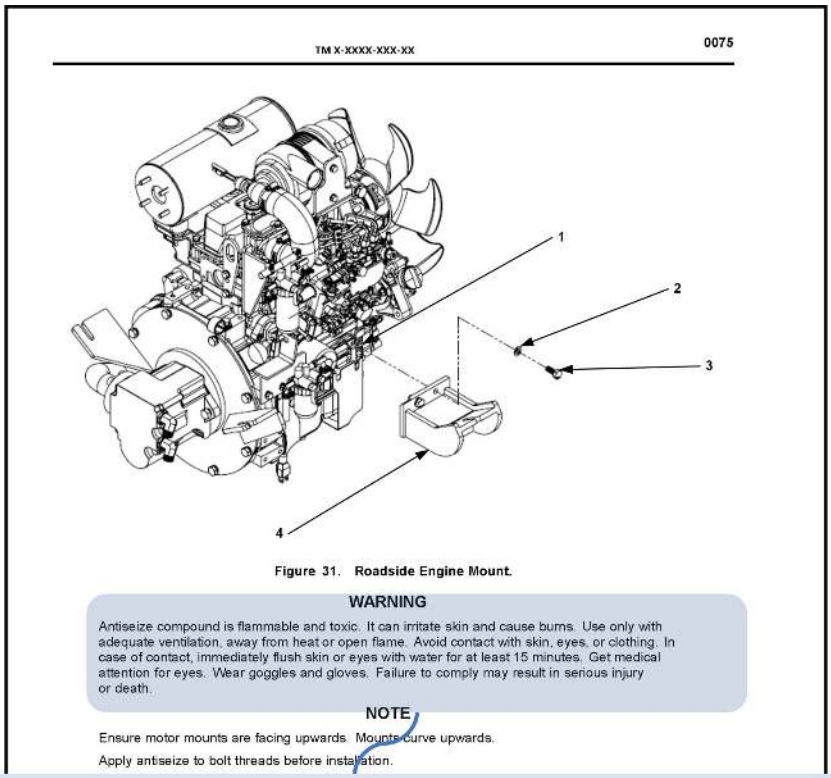
Check the source to verify that this is the current version before use.

**Note**

Figure appearance may vary slightly based on version of DTD/stylesheets utilized.




**WARNING**

Antiseize compound is flammable and toxic. It can irritate skin and cause burns. Use only with adequate ventilation, away from heat or open flame. Avoid contact with skin, eyes, or clothing. In case of contact, immediately flush skin or eyes with water for at least 15 minutes. Get medical attention for eyes. Wear goggles and gloves. Failure to comply may result in death or serious injury.

**FIGURE 29. Basic warning, ETM format.**

A warning also shall be used when there is both danger of injury or death and danger of damage to the equipment or environment during that procedure. In the example below, a warning is used to identify the risk of injury or death that could result from what is otherwise just an equipment failure (see FIGURE 30).

**WARNING**

Prior to operation, ensure all ice, snow, and moisture are removed from aircraft surfaces. Failure to comply may result in death or serious injury to personnel and potential damage to or loss of aircraft.

**FIGURE 30. Example of a warning also containing equipment damage, ETM format.**

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### 4.14.2 Caution <caution>.

A caution is a short message that calls attention to an essential operating or maintenance procedure, practice, condition, statement, etc.; if not strictly observed, a caution could result in damage to or destruction of equipment, or loss of mission effectiveness. Cautions should concisely identify a hazard, explain how to avoid the hazard, and explain what to do if coming in contact with the hazard. See FIGURE 31.

**FIGURE 31. Equipment caution, ETM format.**

### 4.14.3 Note <note>.

A note is a short message that describes an unusual procedure or condition to which special attention must be paid for any reason. Notes shall be brief and shall not be used to add steps. See FIGURE 32.

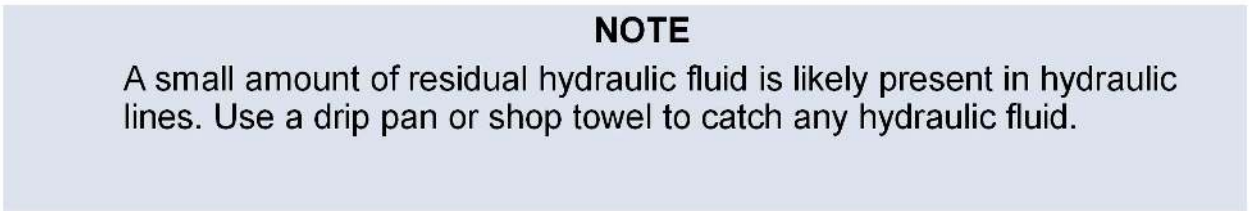


**FIGURE 32. Note, ETM format.**

### 4.14.4 Prohibited features for warnings, cautions, and notes.

Warnings, cautions, and notes:

**a.** Shall not be used for filtering.

**b.** Shall not contain references to figures, tables, or numbered lists (this does not apply to essential figures in the warning summary).

**c.** Shall not contain references to steps, except to indicate applicability when multiple steps are present or different models of equipment are used.

**d.** Shall not be used to insert steps and thus avoid renumbering.

**e.** Shall not use all caps for the text of the warning, caution, or note.

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### 4.14.5 Warning, caution, and note placement.

Warnings, cautions, and notes shall be placed according to the guidance in TABLE VI.

**TABLE VI. Warning, caution, and note placement.**

| When applicable to | Placement |
|---|---|
| One step, one paragraph, etc, of the warning, caution, or note | Immediately before the step, paragraph, etc. |
| Entire WP | Immediately after the initial setup |
| Entire task/procedure | Immediately following the title of the associated procedure |
| Single step | Immediately preceding the applicable step |
| Multiple steps | Immediately before the first step to which they apply |
| The same text (for multiple warnings, cautions, and notes) | Warnings first, cautions second, notes last |

### 4.14.6 ETM formatting requirements for warnings, cautions, and notes.

**a.** For ETMs, warning, caution, and note text shall be indented on the right and left.

**b.** The text shall be left justified for warnings, cautions, and notes.

**c.** The layout shall not result in warnings, cautions, and notes being divided so first lines of text or groups of icons appear on one page and remaining lines or groups of icons on another page.

**d.** The layout shall avoid warnings, cautions, and notes being placed on a different page than the paragraph, step, graphic, etc. to which they apply.

**e.** The headers WARNING, CAUTION, and NOTE shall be in all caps.

**f.** (MC) Single line warnings, cautions, and notes shall be centered.

### 4.14.7 Header formatting requirements for warnings, cautions, and notes (IETMs).

**a.** The warning header shall have the word WARNING in white text preceded with a white exclamation point surrounded with a black triangle. The warning header shall be inside a red rectangular box with a black border.

**b.** The warning header shall have the word WARNING in white text preceded with a white exclamation point surrounded with a black triangle. The warning header shall be inside a red rectangular box with a black border.

**c.** Warnings may have a safety or hazard icon or icons and these shall appear below the warning header. The warning header, icons, and text shall be enclosed within a larger white box with a red and black border. The warning text shall be left justified. See FIGURE 33.

**d.** For IETMs, the caution header shall have the word CAUTION in black text preceded with a white exclamation point surrounded with a black triangle. The caution header shall be inside a yellow rectangular box with a black and yellow border. Cautions may have icons depicting equipment damage and these shall appear below the caution header. The caution text shall be left justified. See FIGURE 34.

**e.** Notes shall have the word NOTE in white and shall be inside a blue rectangular box with a black and blue border as shown in FIGURE 35.

**f.** When a warning, caution, or note consists of two or more paragraphs, the header shall not be repeated above each paragraph.

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**Note**

Figure appearance may vary slightly based on version of DTD/stylesheets utilized.

**FIGURE 33. Example of a warning, IETM format.**

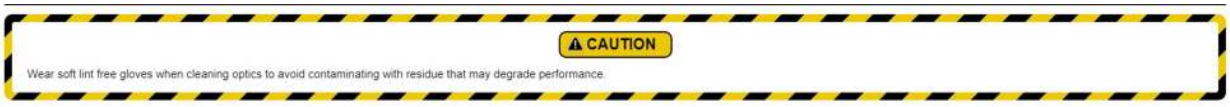


**FIGURE 34. Example of a caution, IETM format.**

**FIGURE 35. Example of a note, IETM format.**

## 4.14.8 Header formatting requirements for warnings, cautions, and notes (ETMs).

**a.** The header WARNING, CAUTION, or NOTE shall be bold and centered above the appropriate text.

**b.** Headers shall not be numbered or bulleted.

**c.** When a warning, caution, or note consists of two or more paragraphs, the header shall not be repeated above each paragraph.

**d.** If safety, hazard, or equipment damage icons are included, they shall appear below the header.

## 4.14.9 Grouping warnings, cautions, and notes.

**a.** Warnings, cautions, and notes shall not be mixed together under one header (warnings grouped with cautions, etc.).

**b.** Within each category, however, unrelated topics that pertain to the same task, procedure, or step(s) may be grouped under one heading. Thus, warnings may be grouped with warnings, cautions with cautions, and notes with notes.

**c.** When grouping warnings, cautions, or notes, each shall be separated by at least one line and may be bulleted but shall not be numbered.

## 4.14.10 Icons <icon-set>.

Refer to FIGURE 36 for an example of icons and signal words.

**a.** Icons are not required for every warning. Icons shall be used only if they clarify the notice, clearly depict the relevant hazard, and can be reproduced clearly.

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**b.** Icons and signal words shall be used consistently to depict the same message throughout the publication. For this reason, the use of standardized icons to improve readers' recognition of hazards is encouraged. A full list of approved icons for use in publication warnings also is available online at https://armyeitaas.sharepoint-mil.us/teams/LDACTechnicalPublications.

**c.** Additional non-standard warning icons shall be approved by the proponent's safety office and shall be forwarded by the relevant EPCO office to LDAC for review and approval.

**d.** Equipment damage caution icons shall be approved by the proponent's safety office.

**e.** Icons used shall be defined in the warning summary 5.2.4. If icons are not used in the text, they shall not appear in the warning summary.

**Note**

Figure appearance may vary slightly based on version of DTD/stylesheets utilized. Placeholder icons are included since icons are subject to change.

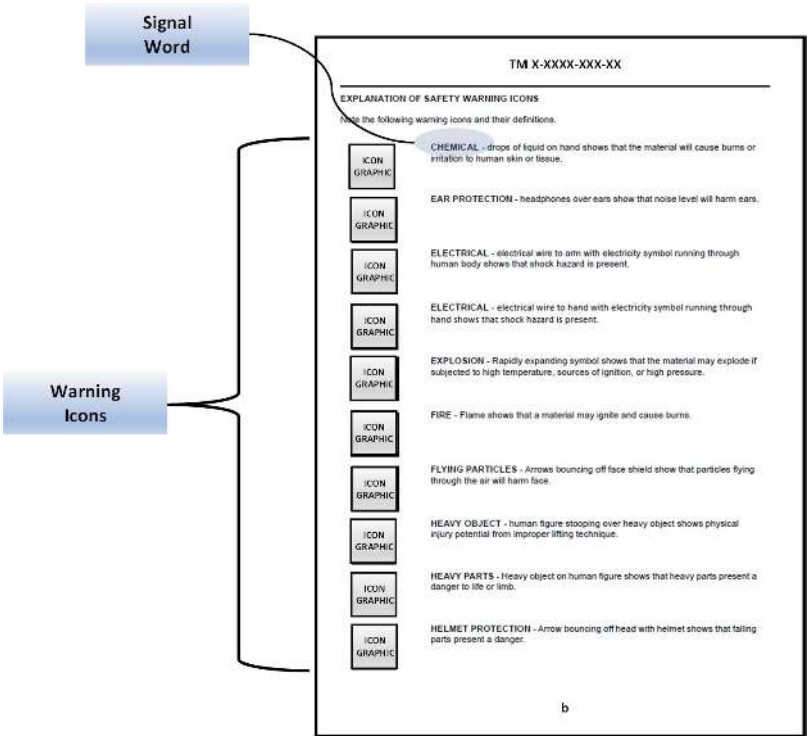


**FIGURE 36. Warning icon with signal word.**

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#### 4.14.10.1 Safety warnings with icons <icon-set>.

The approved safety warning icons are available on the LDAC website (https://armyeitaas.sharepoint-mil.us/teams/LDACTechnicalPublications) and can be used in conjunction with the WARNING header and description of the hazard.

### 4.14.11 Hazardous materials warnings <warning>.

**a.** Procedures prescribed for the operation and maintenance of equipment shall be consistent with the safety standards established by the Occupational Safety and Health Act (OSHA) Public Law 91-596 and Executive Order 12196.

**b.** When exposure to hazardous chemicals or other adverse health factors or use of equipment cannot be eliminated, guidance pertaining to the exposure shall be included in the publication. A list of personnel protective devices also shall be included.

**c.** Hazardous materials warnings may be presented in the standard warning format without an icon, in conjunction with an icon, or with a combination of icons as described in 4.14.11.2. OSHA icons are approved for use.

The proponent's safety office shall approve the use of icons other than those presented on the LDAC website at https://armyeitaas.sharepoint-mil.us/teams/LDACTechnicalPublications.

Hazards that result from a combination of materials shall clearly be identified to indicate that mixing or combining the materials creates the hazard.

#### 4.14.11.1 Format for hazardous materials warnings with icons <icon-set>.

Hazardous materials warnings with icons consist of a WARNING header, the icon(s), and a full description of the hazardous material and the precautions to be taken.

#### 4.14.11.2 Abbreviated format for hazardous materials warnings with icons <icon-set>.

For commonly used substances only (for example, dry cleaning solvent, hydraulic fluids, paints, etc.), an abbreviated format may be used for hazardous materials warnings. The abbreviated format consists of the WARNING header, the icon(s), and the signal word(s) (for example, ISOPROPYL ALCOHOL, TT-I-735) of the hazardous material.

The signal word or words for warning icons appear outside a box centered below the icon(s). The full description of the warning shall be placed in the warning summary. Icons may be used in publication warnings either individually or in combination.

#### 4.14.11.3 Equipment damage caution icons <icon-set>.

The equipment damage caution icons may be used in conjunction with the CAUTION header and description of the equipment damage.

### 4.15 General style and format of textual elements.

This standard provides style and format requirements for technical content requirements. These requirements are considered mandatory and are intended for compliance.

The examples provided are an accurate interpretation of the technical content, style, and format requirements contained in this standard and are provided to permit the effective use of the DTD. Any conflicts between examples and the text of the standard shall be resolved in favor of the text (1.5).

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### 4.15.1 Style and format examples.

The U.S. Government Publishing Office (GPO) Style Manual shall be used as a general guide for capitalization, punctuation, compounding of words, numerals in text, and spelling of nontechnical words.

### 4.15.2 Font size and style.

For ETMs, font sizes and styles are controlled by XSL-FO and thus should not be changed.

### 4.15.3 Page size and orientation for ETMs.

The publication shall be prepared in a size selected from TABLE VII as specified by the proponent. Orientation of pages, either vertical (portrait) or horizontal (landscape), shall be consistent throughout a given manual for ease of use. The growing prevalence of publications used in electronic display mode instead of paper makes this consistency extremely important. All pages shall have the same orientation (with the exception of foldouts grouped together at the end of the publication).

Ensure to take into account the binding edge when determining margins.

**TABLE VII. Manual styles and trim sizes.**

| Style | Trim Size | Orientation | Maximum Printing Area |
|---|---|---|---|
| Pocket-sized | 5½ x 4 | Horizontal | 5 x 3 1/8 |
| | 4 x 5½ | Vertical | 3 1/8 x 5 |
| Logbook | 6½ x 9½ | Vertical | 5½ x 8½ |
| | 9½ x 6½ | Horizontal | 8½ x 5½ |
| Standard | 8½ x 11 | Vertical | 7 x 10 |
| Double Standard | 17 x 11 | Horizontal | 15 3/4 x 9 |

### 4.15.4 Margin data for ETMs.

**a.** Margin data (usually headers and footers) shall be placed outside the area of the page used for either text, full-page tabular data, or full-page illustrations, but within the printing area dimensions of the page (TABLE VII).

**b.** Complete headers shall be prepared for all pages except for the following: front and back covers, Marine Corps signature pages, blank pages, DA Form 2028, and pocket-sized TMs (where the TM numbers shall only be on the front cover and back cover along with the PIN).

**c.** Complete footers shall be prepared for all TM pages except for the following: front and back covers, change transmittal pages, Marine Corps signature pages, blank pages, authentication pages, and DA Form 2028.

#### 4.15.4.1 Headers for ETMs.

**a.** Headers shall consist of the security classification markings (4.15.10.6.1) if any, the TM number (5.2.1.2, 5.2.1.3) and the WP sequence number (4.12.6.4).

**b.** The security classification shall be centered at the top of the page. The publication number shall be centered beneath the security classification. The WP sequence number shall be on the same line as the publication number but shall be right justified.

**c.** All pages of the change transmittal shall have the change number (upper left), publication number (centered) , and change date (upper right).

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**d.** The authentication shall have only the security classification, publication number, publication date, and change number (if applicable) in the header.

**e.** For multi-service publications, all services' numbers shall appear on the front cover with the proponent service listed first. For Army and Marine Corps proponent pubs, all other pages shall have only the proponent service's number in the header. Refer to 5.2.1.3.

**f.** TM numbers for pocket-sized publications are required on front and back covers only. For pocket-sized manuals only, the WP sequence number may be placed only on the first page of the WP providing it is included as part of the page number on all pages of the work package.

**g.** For foldout pages, the security classification and publication number shall be right-justified at the top of the foldout page. See FIGURE 37 .

**Note**

Figure appearance may vary slightly based on version of DTD/stylesheets utilized.

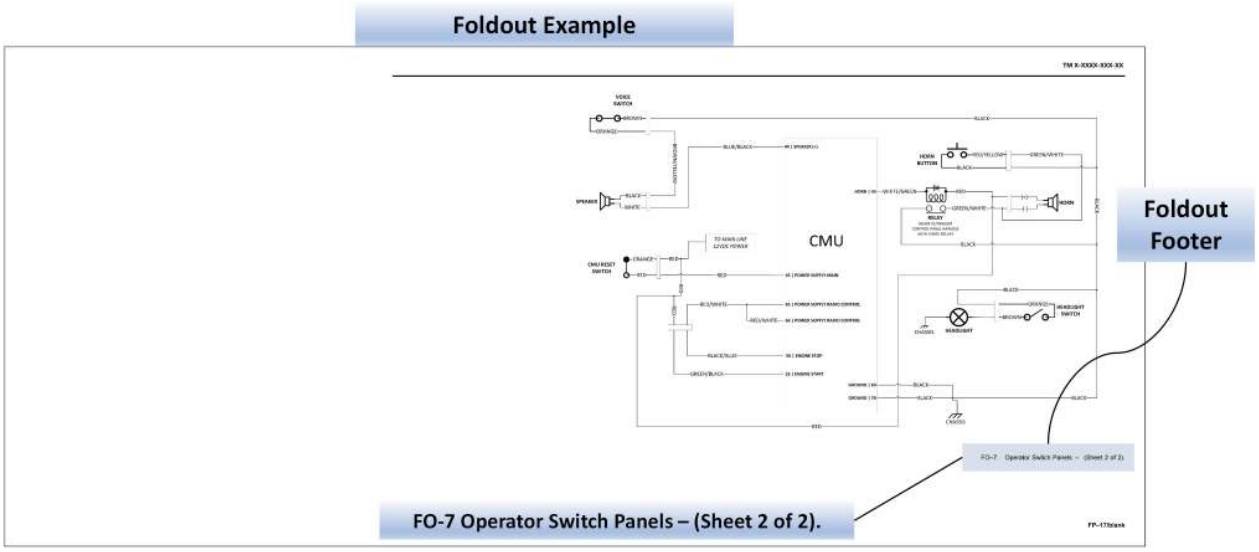


**FIGURE 37. ETM headers and footers.**

### 4.15.4.2 Footers for ETMs

Footers shall include the page numbers (4.12.6.6) and any of the following that are applicable: security classification markings (4.15.10.6.2 ) change designator, and footnotes (4.15.11.3.14).

## 4.15.5 Foldout page size and orientation for ETMs.

**a.** Foldout pages, if needed, shall be the same height as regular pages in the standard manual only, and shall be folded 2, 4, or 6 times, depending on the width necessary.

**b.** If the publication will be printed, each foldout shall have a blank apron wide enough for the user to look at the data while reading text elsewhere in the publication. For EMO publications, the apron is not required. The foldout shall fold up narrower than the apron to accommodate holes. Foldouts shall not be used in RPSTLs or operator-only publications. TABLE VIII lists the foldout maximum trim sizes and foldout maximum printing area for foldout pages. The minimum margin is ½-inch top and bottom and ½ inch on the side opposite the binding edge.

**c.** Foldout pages shall be the last printed material in the publication or volume.

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**TABLE VIII. Foldout maximum trim sizes (in inches).**

| Manual Trim Size | Foldout Maximum Page Trim Size (Including Apron) | Foldout Maximum Printing Area |
|---|---|---|
| 8½ x 11 | 45 x 11 | 36 x 10 |
| 11 x 8½ | 11 x 45 | 10 x 36 |

### 4.15.6 Page numbering and position for ETMs.

Page numbering and position shall be as follows in TABLE IX:

**TABLE IX. Page numbering and position for ETMs.**

| Type of page | Numbering method | Must start on a right-hand page? |
|---|---|---|
| Front cover | Unnumbered | Yes |
| Revision summary | Unnumbered | Yes |
| Marine Corps signature page | Unnumbered | Yes |
| Warning summary | Numbered consecutively using lower case letters (a, b, c, etc.) | Yes |
| Change transmittal (if applicable) | Unnumbered | Yes |
| LOEP (if applicable) | Numbered consecutively using upper case letters (A, B,C, etc.). | Yes |
| Table of contents | Numbered consecutively using lower case Roman numerals beginning with "i" (i, ii, iii, etc.). | Yes |
| "About this manual" | Numbered consecutively using lower case Roman numerals after TOC; does not start over with "i" for "About" information. | Yes |
| Blank page | Unnumbered with the relevant page denoted as a blank. For example, if page 0001-10 of a WP is blank, page 0001-9 is numbered 0001-9/blank. Blank pages contain no text (even "This page intentionally left blank") or any other information. | No |
| Module title pages | Unnumbered | Yes |
| Module index pages | Unnumbered | No |
| DA Form 2028 | Unnumbered | Yes |
| Authentication | Unnumbered | Yes |
| Foldout pages | Numbered consecutively using Arabic numbers prefixed by the letters "FP." The reverse side of foldout pages are blank. Each foldout page number includes a blank page notation (for example, FP-1/blank, FP-3/blank, etc.). Refer to FIGURE 37. | Yes |
| Back covers | Unnumbered | No |



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### 4.15.7 Equations.

The use of equations shall be held to the minimum use required by the needs of the user.

### 4.15.8 Tables.

Tables are discussed in this section. For details on standard format tables, see 4.8.

#### 4.15.8.1 Placement of tables for ETMs.

For ETMs, tables shall be placed in the publication on the same page or as soon after the first reference in the text as possible. The table number and title shall be placed at the top of the table.

#### 4.15.8.2 Table numbering.

Tables shall be numbered, even if only one table exists in a work package. Table numbers shall be consecutive within each WP in the order of their reference starting with Arabic number 1.

#### 4.15.8.3 Table titles.

Tables shall have unique titles that identify the contents or purpose of the table and distinguish it from others in the publication. Table titles shall be formatted as follows:

**a.** The table title consists of the word "Table" followed by the table number, a period, two spaces, and the title. A period follows the last word of the title. Capitalize the first letter of the first and each major word of the title. Table titles are in boldface type.

**b.** Center table titles above the table. If the title is too long to fit on one line, the text wraps and remains centered.

**c.** Identify tables applicable to one Service only in a multi-service publication. (For example, "Table 3. Fuel Indicator Correction Factors (Army Only)").

**d.** Identify tables applicable to more than one service in a multi-service publication. (For example, "Table 1. UHF Radio Controls (Army and Navy Only)").

**e.** When a table is continued on more than one page, the closing rule is omitted at the foot of a continued table, and the opening rule is continued at the head of the continued portion along with the column heading data.

#### 4.15.8.4 Lists

Lists may be used when appropriate. Lists may be unnumbered, numbered sequentially, bulleted, or lettered alphabetically. They may have an optional title. The three types of lists are:

**a.** Definition list **<deflist>**. The definition list shall consist of the term **<term>** and the definition **<def>**. The definition list may have headers such as "Term" and "Definition" above the appropriate sections of the list (the abbreviation/acronym list in general information work package, for example).

**b.** Random list **<randlist>**. The random list shall consist of one or more items in a random order.

**c.** Sequential list **<seqlist>**. The sequential list shall consist of two or more items that occur in a specified order, such as alphabetic, numeric, or alphanumeric. A sequential list shall not be used for procedural steps.

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### 4.15.9 Abbreviations and acronyms.

Use of abbreviations and acronyms shall use the following criteria:

**a.** Common abbreviations and acronyms shall be taken from ASME Y14.38.

**b.** DoD unique abbreviations and acronyms shall be taken from the DOD Terminology Program website (https://jdeis.js.mil).

**c.** Army abbreviations and acronyms shall be taken from the Abbreviations, Brevity Codes, and Acronyms (ABCA) tool at https://armypubs.army.mil/ .

**d.** Any new abbreviations and acronyms shall be developed in accordance with AR 25-30.

**e.** Acronyms, abbreviations, and unusual terms may be used in any WP text.

**f.** The first use of abbreviations and acronyms in each WP shall have the word(s) spelled out completely with the abbreviation or acronym in parentheses immediately after the word(s). Generally, it is not advisable to define acronyms in titles; instead, define them in WP text. At the MATDEV's discretion, abbreviations/acronyms may instead be defined in titles assuming titles do not become needlessly long and show consistency among WPs.

**g.** Acronyms such as "PMCS" shall be in all capital letters and shall contain no spaces or periods, but abbreviations such as "U.S." may contain periods.

**h.** All abbreviations and acronyms used in the manual, including those in tables or figures, shall be defined in the "list of abbreviations/acronyms" paragraph of the General Information WP (B.7.1.11). Acronyms that are accepted as words (radar, sonar, laser, etc.) by the general public need not be included, nor do extremely common abbreviations and acronyms (lb, oz, AC/DC, CD, DVD, USB, LED, IP address, etc.). Terms that are commonly used within technical manuals (TM, TB, WP, IAW) shall be included in the list of abbreviations/acronyms (B.7.1.11), but parenthetical identification within any WP is optional. Contact your EPCO when questions arise.

**i.** When abbreviations or acronyms are used as markings on the equipment (placarding), the same abbreviations or acronyms shall be used in the publication.

### 4.15.10 Symbols.

All nonstandard symbols shall be defined in the list of abbreviations and acronyms contained in the General Information WP (B.7.1.11). New symbols shall not duplicate those presently listed in ASTM-F856 where possible.

#### 4.15.10.1 Metric symbols.

Metric symbols shall be in accordance with IEEE Std 945.

#### 4.15.10.2 Nuclear hardness (hardness critical processes) marking.

All Hardness Critical Processes shall be preceded with the acronym "HCP." The acronym shall be prepared in boldface type and in the same style and size as the adjacent text. The acronym shall not be shown with the titles in the TOC. Use of the acronym is as follows:

**a.** When the entire task and all subordinate paragraphs and steps relate to establishing nuclear hardness, the acronym "HCP" shall precede the task title (for example, **HCP** DISASSEMBLY).

**b.** When the entire task and all subordinate paragraphs and steps do not contribute to establishing nuclear hardness, only those that do contribute shall be annotated with the acronym "HCP."

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For example:
"SERVICING

**(1)** _____

**(2) HCP** _____"

**c.** Operating or maintenance actions that could degrade hardness but are not directly involved in establishing nuclear hardness shall not be annotated with the acronym. Instead, they shall be preceded by a caution statement.

### 4.15.10.3 Electrostatic Discharge Sensitive (ESD) marking.

All paragraphs addressing handling or maintenance that could damage ESD sensitive parts shall be marked with the acronym "ESD" as shown in the following. The acronym shall be prepared in boldface type and in the same style and size as the adjacent text. The acronym shall not be shown with the titles in the TOC. Use of the acronym is described in the following list:

**a.** When the entire task and all subordinate paragraphs and steps relate to ESD sensitive parts, the acronym "ESD" shall precede the task title (for example, "**ESD** DISASSEMBLY").

**b.** When the entire task and subordinate paragraphs and steps are not directly related to ESD sensitive parts, only those that do apply shall be annotated with the acronym "ESD." For example:

For example:
"REMOVAL

**(1)** _____

**(2) ESD** _____"

**c.** Handling or maintenance actions that could damage ESD sensitive parts but are not directly related to handling or maintenance of ESD sensitive parts shall not be annotated with the acronym "ESD." Instead, they shall be preceded by a caution statement.

**d.** Mark figures, drawings, and schematics with the "ESD" acronym in accordance with ASME Y14-100.

### 4.15.10.4 Quality Assurance (QA) marking.

Depot and aviation maintenance procedures that have a QA impact shall be identified by the acronym "QA" in boldface letters preceding the text. Only procedures at the step level shall be labeled with "QA." For example:

"1. **QA** _____"

### 4.15.10.5 Cybersecurity Protection Symbol (CSP).

Technical publications shall be written to ensure Cybersecurity Protection (CSP) measures are taken during maintenance. Procedures shall be written in such a way as to alleviate or eliminate cyber security risks, as directed by DoDI 8500.01.

Unless otherwise specified by the MATDEV, all cybersecurity procedures shall be marked with the symbol. When approved by the MATDEV, "**CSP**" may be used in lieu of the box CSP symbol. The symbol shall be prepared in the same style and size as the applicable paragraph sidehead. The symbol shall not be included in the paragraph title in the TOC. Use of the symbol is as follows:

**a.** When the entire procedure and all subordinate paragraphs/steps describe process/procedures that could be a Cybersecurity threat, the CSP symbol shall be inserted as an attribute before the paragraph (for example, "**CSP**1.2 LRU REPAIR").

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**b.** When all subordinate paragraph and steps are not related to process/procedures that could provide a Cybersecurity threat, only those related will be annotated.

**c.** Illustrations, drawings, and schematics related to Cybersecurity threat process/procedures shall be marked with the CSP symbol.

**"CAUTION**

Cyber Security Protection (CSP) Measures. Certain procedures within this technical manual may present a risk of attack from malicious logic, such as malware, viruses, and Trojan horses. Equipment items used in this technical manual, such as support equipment software programmers, test equipment, and laptops (eTools) may provide a point of entry for malware when connected to the weapon system. Contact local cybersecurity POC for guidance."

### 4.15.10.6 Special page markings.

This section covers page markings for classified manuals, those with other restrictive markings, and emergency page markings.

#### 4.15.10.6.1 Security classification guidelines.

**a.** When the MATDEV requires the development of a classified publication, it shall be properly marked as cited in 4.15.10.6.2 and the current security directives. To ensure proper protection of classified markings, if there is a conflict between the text contained herein and the current security directives, the current security directives shall take precedence.

**b.** The security classification markings for classified publications, titles of parts, modules, WPs, paragraphs, illustrations, tables, and their contents shall be identified in accordance with Part 117 of Title 32, Code of Federal Regulations.

**c.** Downgrading/declassification shall be done in accordance with DoDM 5200.01 volumes 1-3.

**d.** For guidance on classification and handling restrictive markings on discs, refer to DoDM 5200.01 volumes 1-3.

**e.** The overall security classification assigned to a publication shall agree with the highest security classification assigned to any portion within.

#### 4.15.10.6.2 Protective markings.

When specified by the MATDEV, a CONTROLLED UNCLASSIFIED INFORMATION (CUI) protectively marked publication shall be prepared. Publications with distributions other than " A" contain CUI information and shall contain the required protective markings. Refer to DoDI 5200.48 for specific requirements on using the CUI protective marking.

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### 4.15.10.6.3 Emergency page markings.

When specified by the MATDEV, emergency pages shall be prepared. Pages containing emergency information shall have a dark border that indicates to the user that they are emergency pages. The border should go to the edge of the page if the composition system allows it, and it should be made up of characters such as large Xs, large asterisks, or large slashes. See FIGURE 38 for examples of emergency page markings.

**Note**

Figure appearance may vary slightly based on version of DTD/stylesheets utilized.

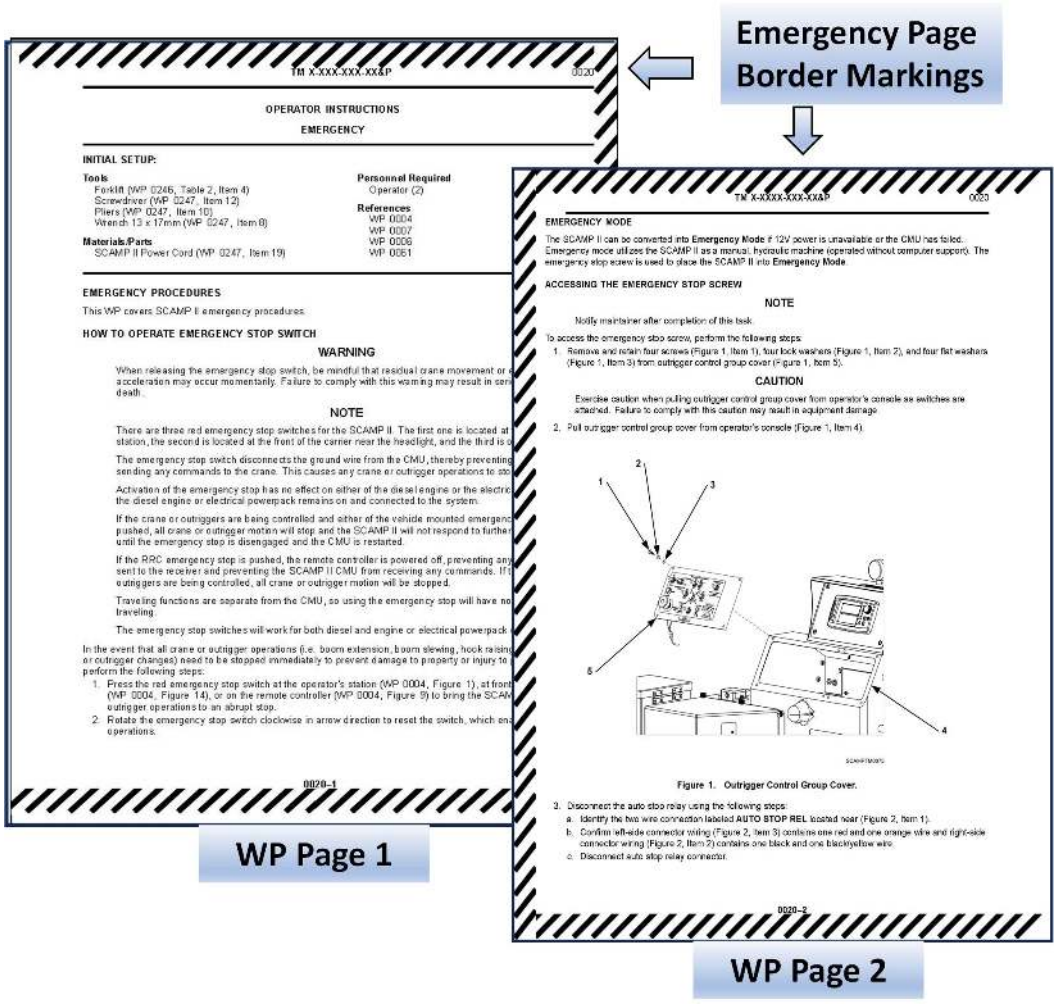


**FIGURE 38. Emergency page border markings.**

## 4.15.11 Referencing.

### 4.15.11.1 External documents <extref> or <link>.

Reference shall be made only to other documents that are authorized to the user and that are available through normal publications channels such as Army Publishing Directorate, Media Distribution Division, St. Louis, MO, or a post library.

**a.** For Government specifications and standards, reference shall be made to the basic publication number.

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**b.** For non-Government documents, reference shall be made by the publication number if one exists, or the document title if there is no publication number.

**c.** Reference to unauthenticated draft DA/DoD publication actions shall not be made.

#### 4.15.11.2 Websites <internet>

**a.** As needed, technical publications may reference external government websites. Contact the Equipment Control Publications Office (EPCO) with any questions regarding approved websites for publication referencing.

**b.** TMs shall not reference outside, non-government websites.

#### 4.15.11.3 Internal (within a publication) <xref> or <link>.

When it becomes necessary to reference to other WPs, descriptive information, maintenance tasks, or other data within the same publication, those items shall be linked to the top of the relevant WP. If referenced by title, the title shall be the same as the title of the work package. Links shall be functional across disc sets regardless of installation type.

##### 4.15.11.3.1 Publication modules.

References to any major division of the manual shall be made by module number and WP sequence number.

##### 4.15.11.3.2 Volumes.

References to information in another volume within the publication shall include the volume number.

##### 4.15.11.3.3 Maintenance tasks, procedures, and paragraphs.

Reference to maintenance tasks, procedures, and paragraphs in another WP shall be by WP sequence number and reference to title, as necessary (for example, "WP 0025, Disassembly" or "WP 0012, Equipment Data").

##### 4.15.11.3.4 Equipment, components, and parts.

References to parts of the equipment and to equipment components may be made by nomenclature, model, type, reference designator, or figure and item number, as applicable. References shall be made only to models or types of equipment covered by the publication. The referenced items may be linked to a graphic for identification and location.

##### 4.15.11.3.5 National Stock Numbers (NSNs) and Part Numbers (P/Ns).

**a.** Reference to NSNs or P/Ns shall be made only in tables, other tabular material, and lists.

**b.** Reference to NSNs or P/Ns shall not be made in the narrative portions of the publication such as procedural steps, initial setups, illustrations, or legends, except when essential for identification or otherwise allowed by the standard.

**c.** NSN and P/N information for all equipment, components, and parts shall be accessible at any point in the presentation of WP text, tables, and illustrations via links or references to WPs containing the NSN and P/N information.

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### 4.15.11.3.6 Equipment panel markings (placarding.

Reference shall be made to panel markings and switch positions exactly as marked on the equipment. Symbols on panel markings, however, shall be spelled out when they cannot be produced by the software, composing equipment, or printers used in producing the manual, such as the symbol for ohm, infinity, etc.

### 4.15.11.3.7 Metric and United States (U.S. standard measurements.

Unless specified otherwise by the MATDEV, all measurements shall be expressed in both U.S. standard units (for example, ounces, pounds, gallons, inches, feet, knots, miles, etc.) and metric units (for example, grams, kilograms, liters, centimeters, kilometers per hour, kilometers, etc.). U.S. standard measurements shall be followed by the metric conversion in parentheses unless the equipment, instrument, or tool, etc., is calibrated in metric units. In that case, metric units shall be first, followed by the U.S. standard units (for example, "169.5 N•m (125 lb-ft)").

### 4.15.11.3.8 Temperature.

Reference shall be made to temperature readings as calibrated on the equipment. If other than Fahrenheit, the equivalent in Fahrenheit shall follow in parentheses. General temperature references, such as room temperature, shall be given in degrees Fahrenheit (for example, "78 °F").

### 4.15.11.3.9 Tables.

**a.** References/links shall be made to tables within a WP by table number (for example, "Table 2").

**b.** References shall refer in the text to tables in a different WP by WP sequence number and table number (for example, "WP 0012, Table 2"), but these links should direct users to the top of the relevant WP.

**c.** References/links shall be made only to tables within the same manual or another volume of the same manual.

### 4.15.11.3.10 Figures and multi-sheet figures.

**a.** References shall be made only to figures within the same manual or another volume of the same manual.

**b.** References shall be made to figures within a WP by figure number (for example, "Figure 2"); for multi-sheet figures, add the sheet number (for example, "Figure 17, Sheet 1").

**c.** With the exceptions of RPSTL figures and foldouts, references shall not be made to figures in separate WPs. For example, a reference such as "WP 0003, Figure 3" would not be allowed. Figures may be repeated in different WPs.

**d.** Each parts information work package shall have only one figure. Multi-sheet figures are allowed.

**e.** With the exceptions of wiring diagrams and schematics, figures shall not contain embedded references to other illustrations or tables.

**f.** In IETMs, figure item numbers in procedural steps shall be linked to figures. The link shall display the figure so the user can instantly read the procedural steps while simultaneously viewing the supporting illustration without the need to relocate the figure.

### 4.15.11.3.11 Numeric callouts.

Where numeric callouts are used to identify parts, figures and numbers shall be used in text to identify items and parts on illustrations. For example:

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"Remove safety disc (Figure 1, Item 3) and safety disc washer (Figure 1, Item 4) from valve body (Figure 1, Item 2)."

#### 4.15.11.3.12 Nomenclature callouts.

For figures where nomenclature callouts are used, reference to those callouts with the text shall be by figure only, and the nomenclature in the text shall match the nomenclature in the figure. For example:

"Remove safety disc (Figure 1) and safety disc washer (Figure 1) from valve body (Figure 1)"

#### 4.15.11.3.13 Items on diagrams.

References shall be made to parts on diagrams by sufficient description or reference designator to identify the item (for example, "resistor A6R11").

#### 4.15.11.3.14 Footnotes.

For ETMs, references shall be made to footnotes when essential for explanation, comments, or other information. Testing procedures shall not contain footnotes.

#### 4.15.11.3.15 Controls and Indicators.

Figure and table references are not required in this section when the tabular option is used. If using the narrative option, figure and table references shall be used.

#### 4.15.11.3.16 Repeating information.

Repeating information shall be allowed to ensure the WP information is complete. Information consisting of two pages or less (or the digital equivalent) may be repeated; information consisting of more than two pages (or the digital equivalent) shall be referenced.

### 4.16 Word choice.

### 4.16.1 Comprehensibility.

Publications shall be written for the target audience. Reading Grade Level (RGL) shall be as specified by the MATDEV.

### 4.16.2 Nomenclature: official/approved, consistency, applicability.

**a.** Nomenclature, other terms, and names shall be consistent within a manual and throughout the RPSTL, MAC, and other directly related manuals. Statements that explain applicability for individual items of equipment shall use specific serial numbers, block designations, model designations, or similar identification. Such terms as "on later equipment" and "on early serial numbers" shall not be used.

**b.** Unless specified otherwise by the MATDEV only approved names and official nomenclature shall be used. Official nomenclature shall be the nomenclature used in the FEDLOG H6 listing (https://www.dla.mil/Working-With-DLA/Federal-and-International-Cataloging/H6/) except for software.

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**c.** If the MATDEV approves unofficial nomenclature (common name), an appropriate nomenclature cross-reference list shall be prepared for the publication and placed in the general information work package.

**d.** When necessary for clarity and when the LCN nomenclature is used in the MAC, the LCN nomenclature may be used in place of the nomenclature used in the FEDLOG H6 listing in other parts of the manual except for the RPSTL.

### 4.16.3 Military terms.

Military terms shall be in accordance with the DOD Terminology Program (https://jdeis.js.mil) or any approved dictionary or glossary of Army military terms.

### 4.16.4 Automatic electronic test and checkout terminology.

Terms used for automatic electronic test and checkout shall be in accordance with MIL-STD-1309.

## 4.17 General style and format of graphic elements.

The term "graphic" is used interchangeably with "illustration" and " figure."

### 4.17.1 Placement of graphics.

Illustrations shall be placed as close to their reference in text as possible. For ETM procedures, locate illustrations (except foldouts) as close as possible to the applicable text step(s). Link to the illustration in the applicable step(s).

In ETMs:

**a.** Whenever possible, place illustrations on the same or facing page of associated text.

**b.** Foldout illustrations shall not be included in WPs; instead, they shall be placed before the back cover of the publication.

**c.** When an illustration is wider than the page, the illustration may be placed sideways on a page (rotated 90 degrees counterclockwise). Foldouts, however, shall not be rotated.

### 4.17.2 Graphic format.

Vector graphics (such as .svg) are preferred. Format of graphics (.svg, .png, .jpg, etc.) shall be as specified by the proponent. Graphics shall be:

**a.** In an editable format.

**b.** Sized as small as possible (as long as usability and clarity do not suffer) to enable shorter load times.

**c.** Resizable to accommodate zooming and different size requirements.

**d.** Obtained in the format in which they were created (the source/raw file), as well as any exported formats for use in the publication.

**e.** Delivered separately as source/raw files and with any exported formats such as .svg.

### 4.17.3 Types of graphics.

As applicable, the following types of graphics shall be used in the preparation of publications:

**a.** Line drawings (preferred type of graphic).

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**b.** Photographs.

**c.** Engineering drawings (DMWR/NMWR only).

**d.** Diagrams.

**e.** Charts and graphs.

**f.** Tools and test equipment illustrations.

### 4.17.3.1 Line drawings.

Line drawings, including exploded views, locator views, and detailed views shall be used to support the operational, troubleshooting, and maintenance procedures.

**a.** When numbers are used to locate and identify equipment components or parts, they shall be used as specified in 4.17.5.5.

**b.** To assist the maintenance technician or operator in locating major components, controls and indicators, etc., locator views may be included.

**c.** When the illustration does not adequately or clearly depict the subject matter or part(s), specific detailed views may be included to support the main illustration.

**d.** (NMWR/DMWR only) Engineering drawings or parts of engineering drawings shall not be used in place of conventional line drawings in any technical equipment publication except for NMWR/DMWR as permitted by 4.17.3.3.

### 4.17.3.2 Photographs.

Photographs may be used for illustrations when a photograph provides for better clarity than a line drawing. The proponent shall determine acceptability of photographs and usage of line drawings.

**a.** All photographs, regardless of source, shall be delivered as digital photographs. Photographs shall be of sufficient resolution to allow a reasonable level of user zooming into IETMs without loss of detail.

**b.** Photographs shall be detailed and sharp, free of heavy shadows, distorted objects, cluttered foregrounds and backgrounds, and of good contrast. Photographs shall provide sufficient detail for the user to clearly identify all components.

**c.** Photographic retouching shall be held to a minimum. Retouching shall be used only to emphasize detail, exclude unwanted detail, correct slight photographic defects, and eliminate undesirable shadow from that portion of the photograph related to the text only.

**d.** For photographs that cannot meet these requirements, line drawings shall be prepared and used. Photographs and line art may appear in the same figure.

### 4.17.3.3 Engineering drawings (NMWR/DMWR only).

Engineering drawings may be used with the approval of the MATDEV.

**a.** Engineering drawings are controlled documents, and when used, they shall be used in their entirety without modification. All copyright requirements in 5.2.1.11.6.1 also shall be met.

**b.** They shall be reduced or redrawn to meet page size/viewer restrictions.

**c.** When the controlled elements of an engineering drawing (for example, title block, sources of supply, revision data, etc.) are removed, leaving only the drawing itself, it shall be treated as a typical line drawing.

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### 4.17.3.4 Diagrams.

When authorized by the MATDEV, specific types of diagrams such as schematic and wiring diagrams may be provided in an authenticated TB that supplements a publication. The following types of diagrams may be included in publications:

**a.** Block diagrams.

**b.** Schematic diagrams.

**c.** Pictorial diagrams.

**d.** Cutaway diagrams.

**e.** Wiring diagrams/wire lists.

**f.** Cable diagrams.

**g.** Piping diagrams.

**h.** Test setup diagrams.

Diagrams shall be prepared in accordance with the following specifications and shall include legends or keys that explain symbols used.

**TABLE X. Diagram specifications.**

| Subject | Equipment Covered | Specification |
|---|---|---|
| Abbreviations | All | ASME Y14.38, DOD Terminology Program (https://jdeis.js.mil), and Abbreviations, Brevity Codes, and Acronyms Directory (https://armypubs.army.mil/abca/SearchABCA.aspx) |
| Engineering Drawing | All | ASME Y14.100 |
| Graphic Symbols | Electrical and Electronic | IEEE Std 280 |
| | Mechanical | ASTM-F856 |
| | Digital (Logic) | IEEE Std 91 |
| | Fluid Power | ISO 1219-1, ISO 1219-2 |
| Unit Symbols | All | IEEE Std 260.1 |
| Logic | All | IEEE Std 91 |

### 4.17.3.5 Charts and graphs.

Charts and graphs may be prepared as illustrations or may be created as interactive graphics. Instructions shall be provided for the use and interpretation of complex graphs.

### 4.17.3.6 Tools and test equipment illustrations.

**a.** Uncommon or unusual uses and connections for test purposes shall be illustrated only if they are essential to avoid misunderstanding.

**b.** Unusual operations shall also be illustrated.

**c.** Special tools/kits and test equipment shall be illustrated, as applicable.

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**d.** Standard tools and test equipment shall not be illustrated nor shall self-evident or generally known uses be shown.

**e.** Connections for test purposes shall be illustrated to prevent misunderstanding.

### 4.17.4 Special considerations for figures.

**a.** Multi-view and multi-sheet illustrations may be used to clarify, identify significant features, or further detail equipment assemblies, subassemblies, and detailed parts.

**Note**

Figure titles for repeated illustrations must be unique in each work package.

**b.** Illustrations shall not be repeated in WPs containing multi-page tasks, but they may be repeated to support a different requirement in another part of the publication. Use discretion to prevent graphic overuse.

### 4.17.5 Elements of graphics/illustrations.

Graphic techniques and guidance are provided in 4.17.5.1 through 4.17.5.14.

#### 4.17.5.1 Border rules and boxes.

Border rules and boxes shall not be used for single illustrations; they are used to separate multi-section illustrations on the same page or for locator/detail views.

#### 4.17.5.2 Use of the human figure.

**a.** When necessary, illustrations may include parts of the human body such as a hand, arm, leg or foot.

**b.** The human figure shall not contain any information that can identify the person including a face, rank insignia, identification numbers, unit/company/brigade patches, medals/wings, etc.

**c.** Jewelry shall not appear in any illustration.

**d.** The human figure shall not be permitted to obscure details of the equipment necessary for a complete understanding of its operation.

**e.** The human figure shall be clothed as designated by the MATDEV.

**f.** A cross-section of races and sexes shall be used.

#### 4.17.5.3 Credit lines.

No credit lines shall be included on graphics.

#### 4.17.5.4 Callouts.

Callouts, whether numeric callouts, reference designators, or nomenclature callouts, shall be used to identify specific parts of an illustration. See FIGURE 39 for examples of callout guidance listed below:

**a.** When an item is first illustrated and its location has not yet been specified, a simplified general locator illustration may be used to identify the location of the equipment item within the system.

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**b.** Both numeric and nomenclature callouts can be used in the same document. They shall not, however, be used together in the same illustration. For example, a graphic should not use text to describe one component of equipment and the number "2" to denote another component on that same piece of equipment.

**c.** Callouts and their leader lines should be easily distinguishable from components and other lines of the illustration.

**d.** When practical, all callouts should be placed outside the boundaries of the parts illustrated so that the parts are not obscured.

**e.** Use leader lines or sweep arrows to help readers orient themselves with respect to the illustration and to provide directional movement in tasks (4.17.5.6).

**f.** Use a type size no smaller than 8 points and no larger than 10 points in ETMs. Particularly with nomenclature callouts, type smaller than 8 points degrades readability.

**g.** So that graphics remain clear and readable, nomenclature callouts should only be used for graphics with a small number of callouts, generally no more than ten. If more than ten callouts are required, use numeric callouts to avoid overcrowding.

**h.** When hotspot techniques are used in conjunction with callouts, an explanation shall be provided in the "About" portion of IETMs.

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**Note**

Figure appearance may vary slightly based on version of DTD/stylesheets utilized.

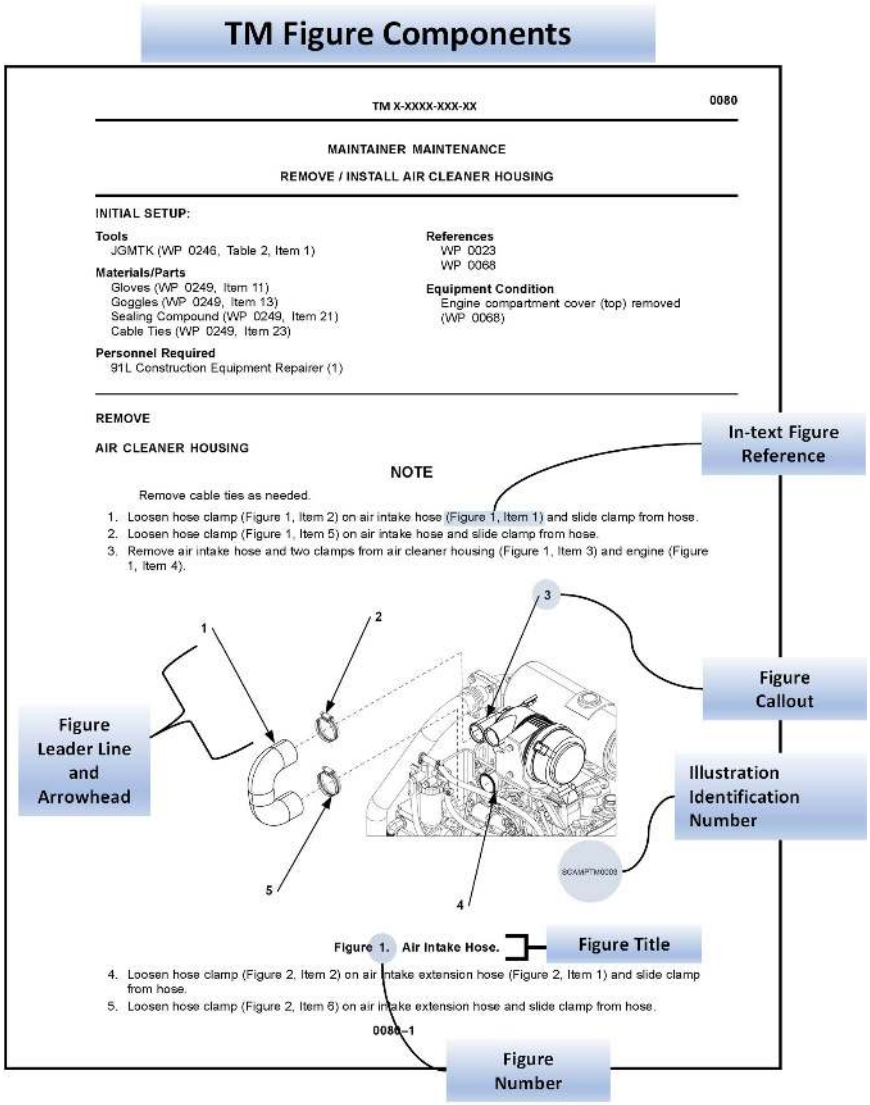


**FIGURE 39. Figure components.**

## 4.17.5.5 Numeric callouts.

**a.** Numbers shall start with Arabic numeral "1" and continue consecutively within an illustration.

**b.** For multi-sheet illustrations, numeric callouts shall continue in sequence from one sheet to another; once an item in a multi-sheet illustration has been assigned a number, it shall retain that number throughout that multi-sheet illustration.

**c.** All items shown as exploded shall be identified. Items drawn in phantom need not be identified.

**d.** Numbers shall not be contained within a shape of any kind (for example, circle, square, triangle, etc.).

**e.** Numbers shall be presented in one of the following orders: in clockwise sequence, beginning at 11 o'clock (FIGURE 40; preferred method), or in inspection or disassembly/assembly order.

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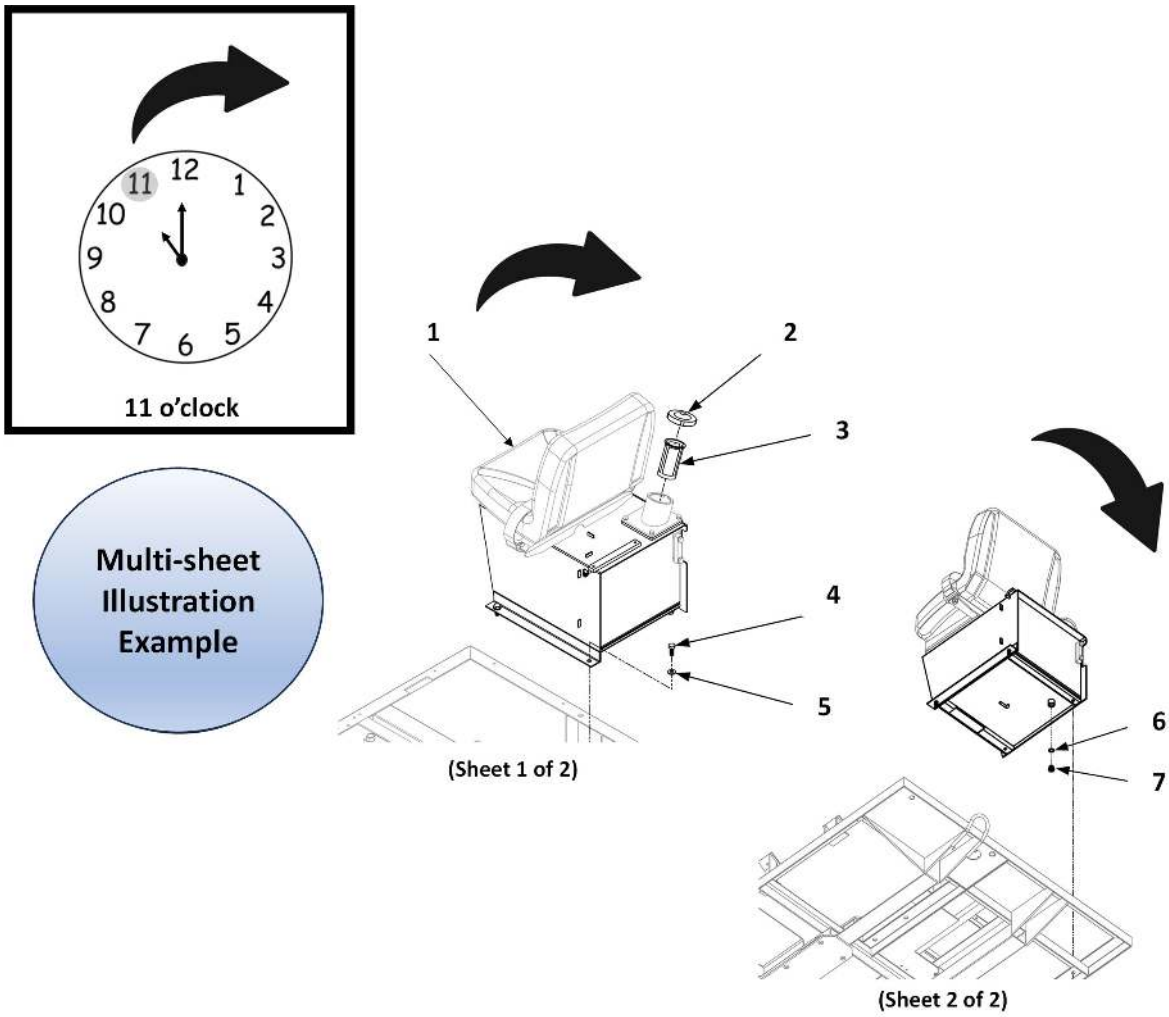


**FIGURE 40. Numeric callouts on multi-sheet illustrations.**



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### 4.17.5.6 Leader lines and arrowheads.

**a.** Leader lines shall be uniform, clearly visible, short, and as straight as possible.

**b.** Leader lines shall be placed at an angle from the item they identify.

**c.** Do not allow leader lines to cross each other or touch the callout.

**d.** If a leader line needs to enter an object, a halo may be placed around the arrowhead or a breakoff symbol may be used in lieu of an arrowhead.

**e.** Arrowheads may be added for clarity. Arrowheads shall touch the outside edge of the item to which they apply (see arrowheads in FIGURE 39). Do not allow arrowheads to enter the object to which they apply.

**f.** Leader lines and arrowheads should never obscure essential details.

### 4.17.5.7 Illustration legends.

When necessary for clarity, legends shall be prepared to identify numeric callouts on illustrations. Legends shall not be a part of the illustration and shall be placed in the text area.

Illustration legends may be used in IETMs, but their use is discouraged. Hovers are preferred.

### 4.17.5.8 Procedures on illustrations.

Procedural steps shall not be placed on illustrations.

### 4.17.5.9 Figure numbers.

Figure numbers shall conform to the requirements listed below. Refer to TABLE XI for figure numbering for special cases.

**a.** Figure numbers shall be included for all illustrations except inline graphics (for example, equations).

**b.** Figures shall be numbered using Arabic numbers sequentially within each WP starting with the Arabic numeral 1.

**c.** The figure number shall precede the title.

**d.** The figure number and title shall not be an integral part of the figure.

**e.** The figure number and title shall be separated from the graphic so the text can have the capability of being searched.

**f.** When an illustration is changed, numeric callouts added between existing numbers may be the same as the preceding callouts with added alpha characters (for example, 22A, 22B). This system may also be used in basic publications when errors are discovered so late in preparation that renumbering all following callouts would delay submittal. In complete revisions, index numbers with added alpha characters should be eliminated as soon as financially feasible.

**g.** Special cases shall be treated as specified in TABLE XI.

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**TABLE XI. Figure numbering special cases.**

| Type of Figure | Numbering Guidance |
|---|---|
| RPSTL figure | Number sequentially within the RPSTL using Arabic numerals beginning with 1. Figure numbering does not start over with each work package. |
| RPSTL revision figures | Renumber all figures when adding or deleting figures. |
| Foldout figures | Number in consecutive ascending numerical sequence within each publication, beginning with Arabic number 1 (FO-1, FO-2, etc.), in the order the figures are referenced in the text. Place foldout figure numbers before the figure title under the illustration. |
| Multi-sheet figures (including RPSTL) | Number consecutively by sheet. Include the total number of sheets following the title ("Figure 2. Wing Hydraulic Assembly (Sheet 1 of 3)."(Sheet 1 of 2)." Number remaining sheets in order (Sheet 2 of x, Sheet 3 of x, etc.). Multi-sheet RPSTL figures are used as specified by the MATDEV. |

### 4.17.5.10 Figure title format.

Each figure shall have a unique figure title within each work package, with the exception of inline graphics (for example, an equation). Figure title format shall adhere to the following:

**a.** Include the word "Figure" with the "F" capitalized, followed by the figure number, a period, two spaces, and the title (for example, "Figure 3. Fuel Indicator.").

**b.** Capitalize the first letter of the first word of the title after the period, as well as each major word of the title (title case).

**c.** End with a period following the last word.

**d.** Identify illustrations applicable to one service in a multi-service TM (for example, "Figure 3. Fuel Indicator (Army Only).").

**e.** Identify illustrations applicable to more than one service in a multi-service TM (for example, "Figure 3. Fuel Indicator (Army and Air Force Only).").

### 4.17.5.11 Figure title placement.

Figure title placement shall:

**a.** Center the figure title below the graphic.

**b.** Begin the title on the same line with the figure number.

**c.** Wrap and remain centered when the title is too long to fit on one line.

**d.** For figures rotated 90 degrees, place the figure number and title where the bottom of the page would have been before rotation so that the angle of the equipment is apparent.

### 4.17.5.12 Illustration identification numbers.

**a.** Each illustration shall be assigned a unique identification number provided by the MATDEV.

**b.** The contractor's identification number may be used when approved by the MATDEV.

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**c.** The identification number shall be placed in the lower right-hand corner of the illustration (within the graphics area) sufficiently removed to avoid being confused as part of the illustration. When the identification number is to be printed in the TM, that number shall be approximately 4- to 6-point type.

### 4.17.5.13 Portraying signal flow.

Signal flow, especially for electrical and electronic equipment, critically affects the understandability of diagrams. To assist the TM user in following the diagram, major signal or pressure flow shall be from left to right, and feedback or return flow shall be from right to left, if possible. Animation or color may be used to indicate signal flow.

### 4.17.5.14 Use of color.

Unless specified otherwise by the MATDEV, backgrounds shall be white and text shall be black for publications. Prior approval for color in ETMs will be obtained by the MATDEV from the Logistics Data Analysis Center (LDAC). Written approval will be provided in the FRC package for printed publications. No approval for color is required for EMO publications.

The use of some colors may not be appropriate for certain environmental conditions, such as night operations.

## 4.18 Multimedia presentation in IETMs.

Multimedia shall never be the primary means of presenting information. Audio, video, and animation techniques shall only be used in an IETM when it results in enhancing the presentation of information or makes the procedures more effective. Every instance of use must be discussed with and approved by the MATDEV before any audio, video, or animation presentation is included in an IETM.

Multimedia standards to be used for presentation techniques shall be as specified by the MATDEV in accordance with AR 25-30.

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# 5 DETAILED REQUIREMENTS

## 5.1 Technical content preparation.

TM data developed in accordance with this standard shall be task-oriented and fully consistent with the maintenance concepts derived from the baseline documents described in the following:

**a.** Logistics Product Data (LPD). The technical data and instructions developed by the requirements of LPD, along with the DoD Requirements for LPD (including the MAC), shall be used as the baseline to prepare publications.

**b.** Maintenance Allocation Chart (MAC).

**c.** Additional source data. The following source data shall be used in the preparation of specific instructions and the development of specific supporting illustrations: engineering drawings, sound engineering principles and techniques, available engineering analyses, service experience, performance data on the item and on similar items, and all other Reliability, Maintainability, Supportability (RMS) and Operational Availability (Ao) data.

## 5.2 Preparation of front matter content <paper.frnt> (ETMs) <framed.frnt> (IETMs).

Requirements for the preparation of the front matter necessary to supplement the technical content modules and associated content in APPENDIX B through APPENDIX N are provided in 5.2.1 through 5.3. APPENDIX A provides detailed content and assembly requirements for all TMs covering operation, maintenance, and parts information at all maintenance levels through sustainment that shall be tailored to meet system and user requirements.

For manuals with multiple maintenance levels, the front matter items shall be combined. There shall be only one instance of each front matter item rather than multiple items divided by maintenance level.

As applicable, the front matter shall consist of the following WPs in the order specified:

1. Front cover **<frntcover>** or **<frontcover_abbreviated>** (5.2.1).
2. Revision summary **<revisionsummary>** (if applicable, except for initial release) (5.2.2).
3. (MC) Signature **<promulgation>** (5.2.3).
4. Warning summary **<warnsum>** (5.2.4).
5. Change transmittal **<chgsheet>**, if applicable (ETMs only; 5.2.5).
6. LOEP **<loep>**, if applicable (classified ETMs and ammunition data sheets ETMs only; 5.2.6). Examples are shown in FIGURE 50, FIGURE 51, and FIGURE 52).
7. Table of contents **<contents>** (ETMs only) (5.2.7).
8. "About this manual" **<about>** (5.2.8).

### 5.2.1 Front cover <frntcover>.

A front cover, which refers either to the first page in an ETM or the front cover information frame in an IETM, shall be prepared for each publication. Unless otherwise specified, the front cover shall contain the following content information:

**a.** Security classification (when required).

**b.** Publication number, single service **<tmno>** (5.2.1.2) or multi-service **<tminfono>** (5.2.1.3). While **<tmno>** is a child of **<tmtitle>**, the publication number appears before the title on the page.

**c.** National overhaul standards statement (TMs/DMWRs/NMWRs with national overhaul standards only; 5.2.1.4).

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**d.** Publication title **<tmtitle>** includes but is not limited to the following components:

**(1)** System or item nomenclature **<sysnomen>**. This information shall be presented in the order specified below. This group may be repeated to present all variations covered in the publication.

**(a)** Model number **<modelno>** (5.2.1.5.1.1).

**(b)** National Stock Number (NSN) **<nsn>** (5.2.1.5.1.2) (Required for equipment publications if available, but not for general maintenance or software manuals).

**(c)** Part number **<partno>** (5.2.1.5.1.3) (Required if available).

**(d)** End Item Code (EIC) **<eic>**, as specified in the FEDLOG (5.2.1.5.1.4) (Required for equipment publications if available, but not for general maintenance or software manuals).

**(e)** Commercial and Government Entity Code (CAGEC) **<cageno>** (F.8.2.1.7) (Required if available).

**e.** Subtitle (when required) **<stitle>** (5.2.1.6).

**f.** Weapon system name (when required) **<weapons_system>** (5.2.1.7). When required, the name of the weapon system to which this publication applies shall be included.

**g.** Cover illustration **<graphic>** (optional; 5.2.1.8).

**h.** (MC) Marine Corps Seal **<graphic>** (only when MC is the proponent; 5.2.1.9).

**i.** Reporting **<reporting>** (5.2.1.10).

**j.** Notices **<notices>** (5.2.1.11).

**(1)** Availability statement **<avail>** (DMWR/NMWR only) (5.2.1.11.1).

**(2)** Supersession notice **<super>** (5.2.1.11.2).

**(3)** Distribution statement **<dist>** (5.2.1.11.3).

**(4)** Export control notice warning (when required) **<export>** (5.2.1.11.4).

**(5)** Destruction notice (when required) **<destr>** (5.2.1.11.5).

**(6)** General purpose notices (when specified) **<general_purpose_notices>** (5.2.1.11.6).

**(a)** Copyright notice (5.2.1.11.6.1).

**(b)** Hazardous materials information (5.2.1.11.6.2).

**(c)** Disclaimer Statement for Manufacturers' Names/Logos (5.2.1.11.6.3).

**k.** CUI Designation Indicator (when required; must stay on the front page) **<cui_set>** (5.2.1.12).

**l.** Service nomenclature **<servnomen>** (5.2.1.13).

**m.** Publication date **<date>** (5.2.1.14).

**n.** (MC) Publication Control Number (PCN) **<pcn>** (if applicable; 5.2.1.15).

The front cover for Marine Corps only manuals (FIGURE 44) shall be used only for Marine Corps only manuals or for multi-service manuals where the Marine Corps is the lead service. For multi-service manuals where Army is the lead service, the Army cover shall be used.

The formats of the front covers for ETMs are shown in the following graphics:

**a.** TMs (FIGURE 41 and FIGURE 42).

**b.** Phased maintenance and preventive maintenance services publications (FIGURE 43).

**c.** Multi-service, including Marine Corps PCN (FIGURE 44).

**d.** DMWR (FIGURE 45).

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**e.** NMWR (FIGURE 46).

**f.** TM with national overhaul standards (FIGURE 47).

**Note**

Figure appearance and tags/ouput may vary slightly based on version of DTD/stylesheets utilized (see applicable DTD or tag definition list for available tags).

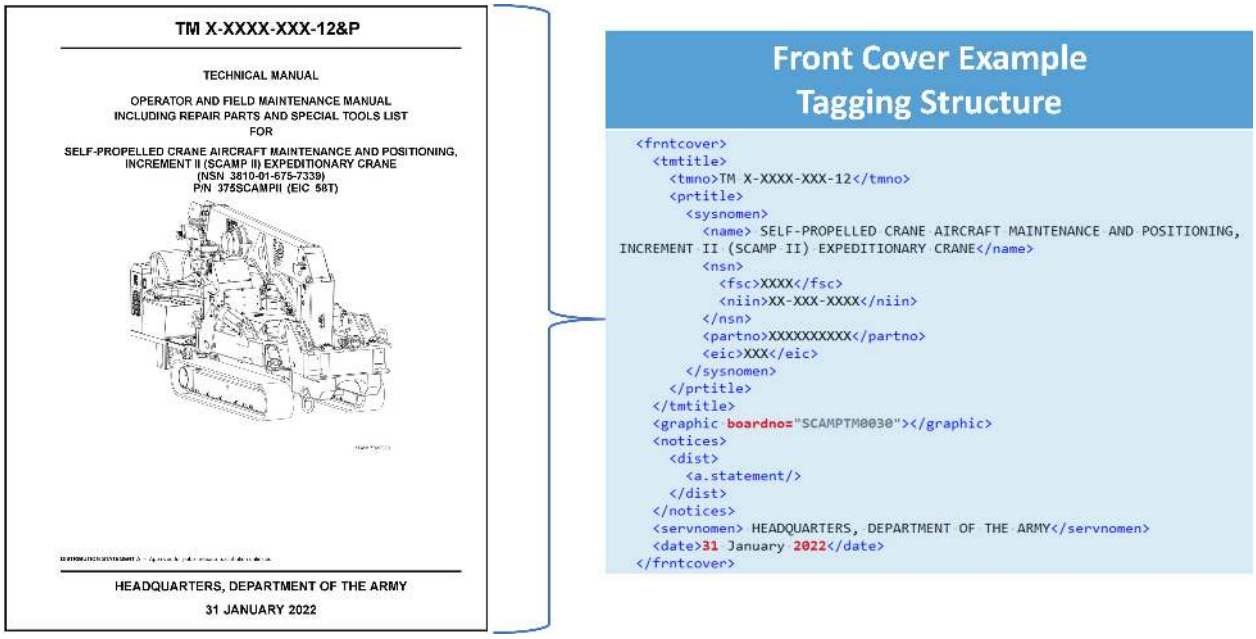


**FIGURE 41. Cover, Army only.**

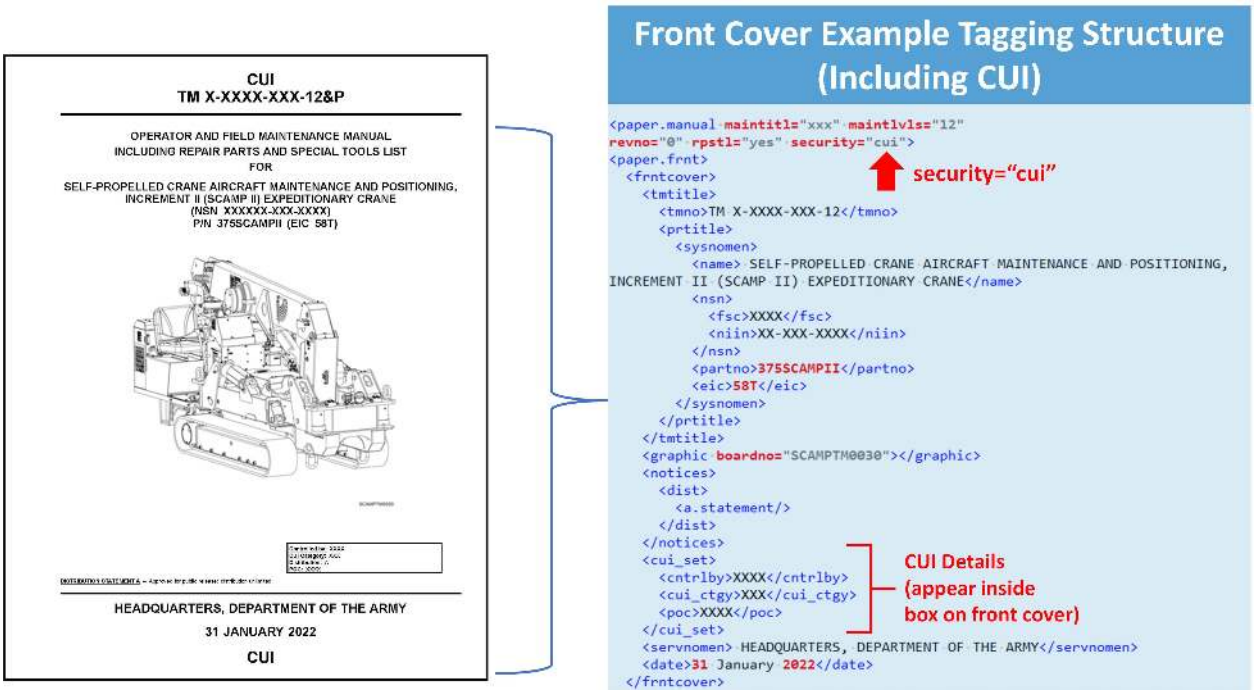


**FIGURE 42. Cover, Army only (including CUI).**

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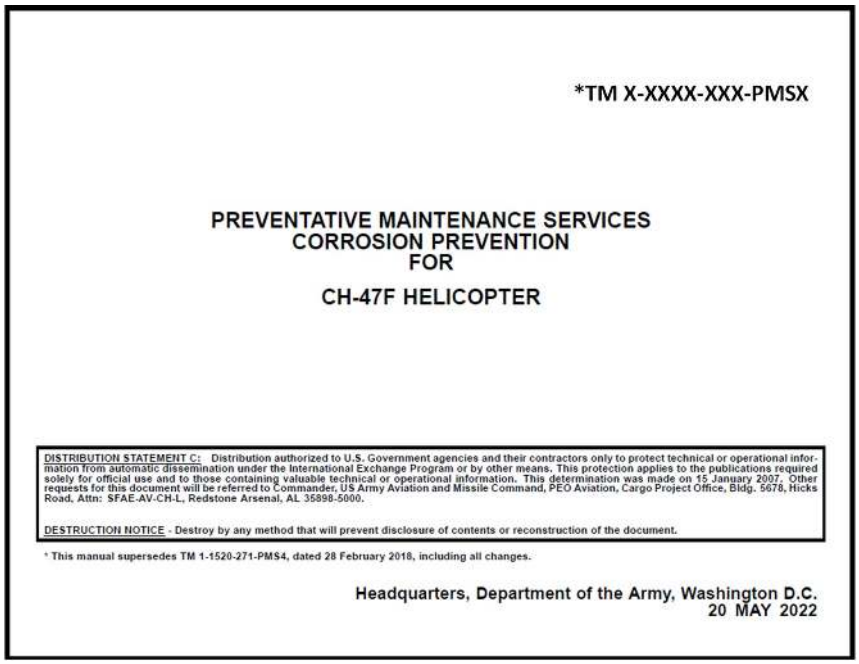


*TM X-XXXX-XXX-PMSX

PREVENTATIVE MAINTENANCE SERVICES
CORROSION PREVENTION
FOR
CH-47F HELICOPTER

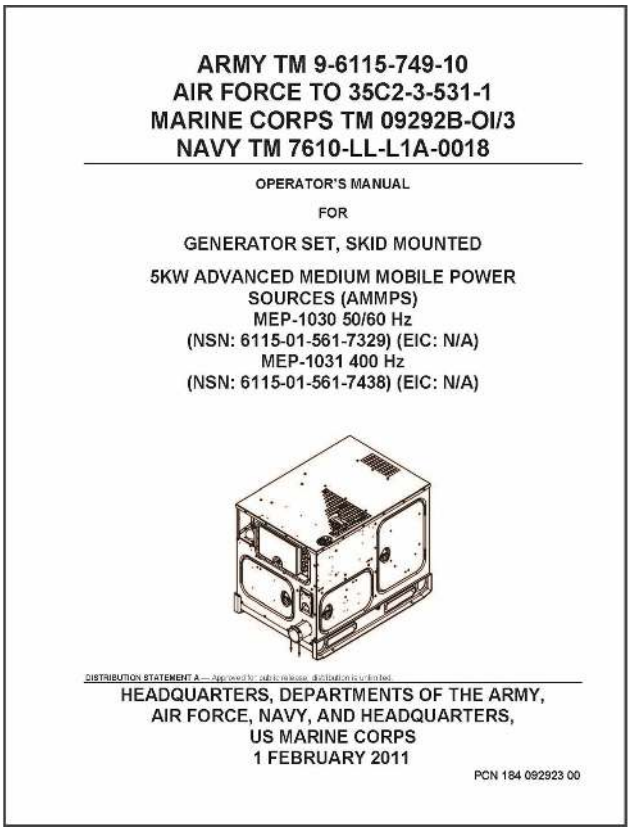
DISTRIBUTION STATEMENT C: Distribution authorized to U.S. Government agencies and their contractors only to protect technical or operational information from automatic dissemination under the International Exchange Program or by other means. This protection applies to the publications required solely for official use and to those containing valuable technical or operational information. This determination was made on 15 January 2007. Other requests for this document will be referred to Commander, US Army Aviation and Missile Command, PEO Aviation, Cargo Project Office, Bldg. 5678, Hicks Road, Attn: SFAE-AV-CH-L, Redstone Arsenal, AL 35898-5000.

DESTRUCTION NOTICE - Destroy by any method that will prevent disclosure of contents or reconstruction of the document.

* This manual supersedes TM 1-1520-271-PMS4, dated 28 February 2018, including all changes.

Headquarters, Department of the Army, Washington D.C.
20 MAY 2022

**FIGURE 43. Cover, phased maintenance and preventive maintenance services TMs.**



ARMY TM 9-6115-749-10
AIR FORCE TO 35C2-3-531-1
MARINE CORPS TM 09292B-OI/3
NAVY TM 7610-LL-L1A-0018

OPERATOR'S MANUAL

FOR

GENERATOR SET, SKID MOUNTED

5KW ADVANCED MEDIUM MOBILE POWER SOURCES (AMMPS)
MEP-1030 50/60 Hz
(NSN: 6115-01-561-7329) (EIC: N/A)
MEP-1031 400 Hz
(NSN: 6115-01-561-7438) (EIC: N/A)

DISTRIBUTION STATEMENT A — Approved for public release; distribution is unlimited.

HEADQUARTERS, DEPARTMENTS OF THE ARMY,
AIR FORCE, NAVY, AND HEADQUARTERS,
US MARINE CORPS
1 FEBRUARY 2011

PCN 184 092923 00

Applicable services / publication numbers

Marine Corps PCN

**FIGURE 44. Cover, multi-service, including Marine Corps PCN.**

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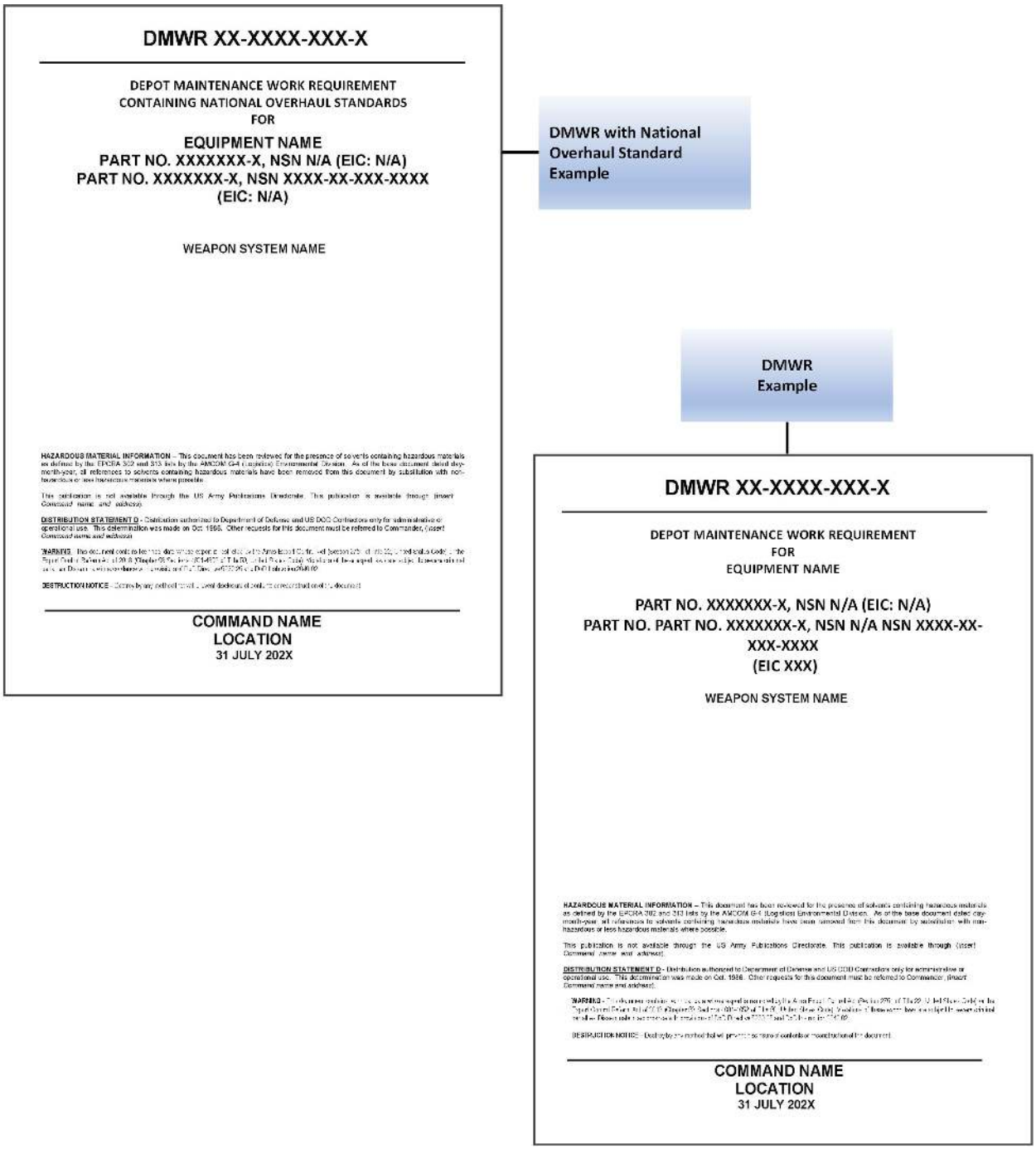


**FIGURE 45. Cover, DMWR with national overhaul standards and DMWR.**

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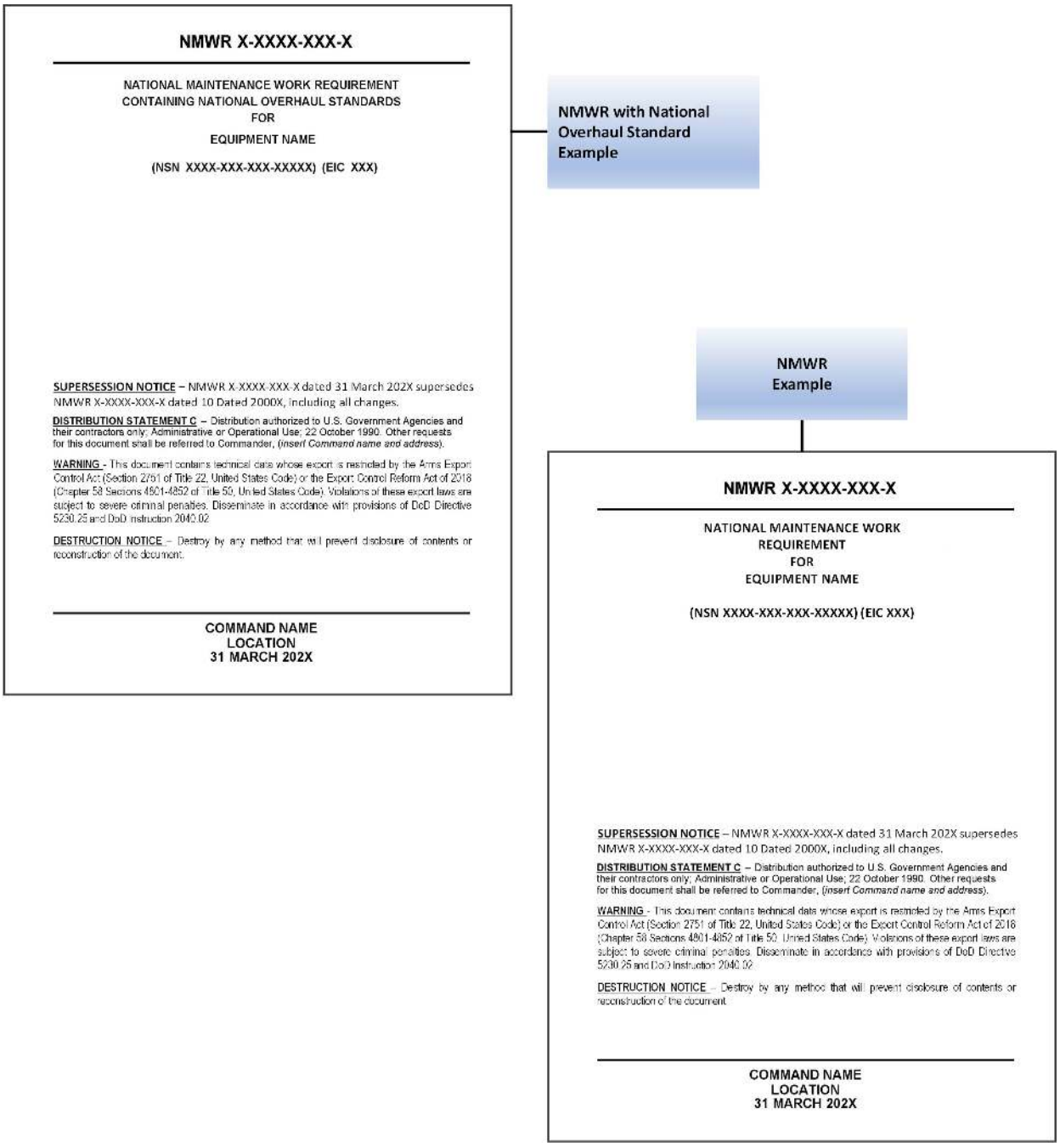


**FIGURE 46. Cover, NMWR with national overhaul standards and NMWR.**

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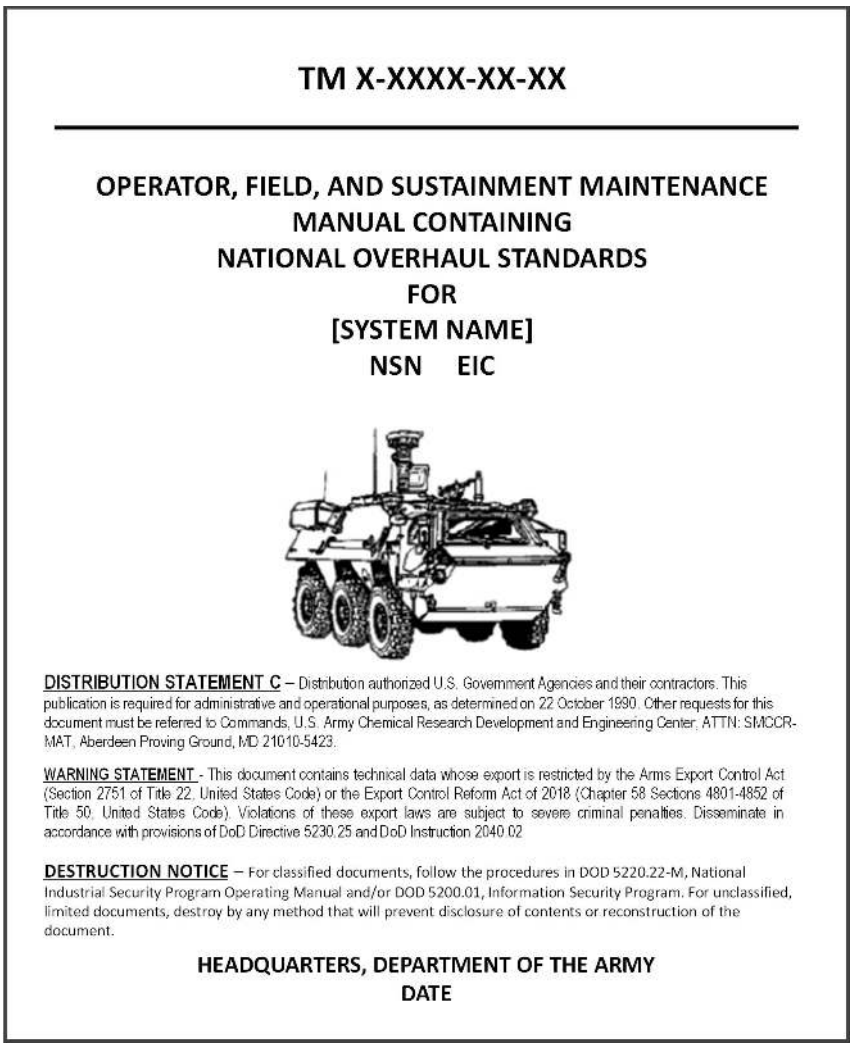

TM X-XXXX-XX-XX

OPERATOR, FIELD, AND SUSTAINMENT MAINTENANCE
MANUAL CONTAINING
NATIONAL OVERHAUL STANDARDS
FOR
[SYSTEM NAME]
NSN EIC

DISTRIBUTION STATEMENT C – Distribution authorized U.S. Government Agencies and their contractors. This publication is required for administrative and operational purposes, as determined on 22 October 1990. Other requests for this document must be referred to Commands, U.S. Army Chemical Research Development and Engineering Center, ATTN: SMCCR-MAT, Aberdeen Proving Ground, MD 21010-5423.

WARNING STATEMENT - This document contains technical data whose export is restricted by the Arms Export Control Act (Section 2751 of Title 22, United States Code) or the Export Control Reform Act of 2018 (Chapter 58 Sections 4801-4852 of Title 50, United States Code). Violations of these export laws are subject to severe criminal penalties. Disseminate in accordance with provisions of DoD Directive 5230.25 and DoD Instruction 2040.02

DESTRUCTION NOTICE – For classified documents, follow the procedures in DOD 5220.22-M, National Industrial Security Program Operating Manual and/or DOD 5200.01, Information Security Program. For unclassified, limited documents, destroy by any method that will prevent disclosure of contents or reconstruction of the document.

HEADQUARTERS, DEPARTMENT OF THE ARMY
DATE

**FIGURE 47. Cover, TM with national overhaul standards.**

### 5.2.1.1 Security classification.

Publications with distributions other than "A" contain CUI information and shall contain the required protective markings. For detailed information on security classification, refer to DoDI 5200.48, Controlled Unclassified Information (CUI).

### 5.2.1.2 (A) Publication number <tmno>.

Electronically delivered TMs shall not include terms such as "IETM," "ETM," or "EM" in the TM number. IETMs shall be numbered in the same manner as ETMs in accordance with DA Pam 25-40. The TM number shall not include words such as "Apache" or "HEMTT." TMs shall be numbered with "&P" (for example, "20&P") for manuals containing a RPSTL.

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### 5.2.1.3 Publication number for multi-service publications<tminfono>.

If the manual is jointly used, each service's number shall be placed on the front cover. Only the proponent service's TM number shall be placed on each page or the top of each frame. The numbers shall be prefixed with the words "Air Force," "Army," "Marine Corps," "Navy" ("NAVSEA" or "NAVAIR"), "Coast Guard," or "Space Force" as applicable. The proponent service's name **<servbranch>** and manual number **<tmno>** shall be placed first. The TM number(s) for the other services shall occur in alphabetical sequence following the proponent service's name and manual number. For example:

| | |
|---|---|
| "ARMY | TM 11-1510-204-24 |
| AIR FORCE | TO 21M-LGM30G-12 |
| MARINE CORPS | TM 12345A-15/1 |
| NAVY (NAVAIR) | AI-F18AA-WRM-070 |
| NAVY (NAVSEA) | SE211-FA-MMA-010/SPS-10A" |

### 5.2.1.4 National overhaul standards statement (TMs/DMWRs/NMWRs with national overhaul standards only).

The following shall be added to the title of TMs/NMWRs/DMWRs that document national overhaul standards for the National Maintenance Program: "Containing National Overhaul Standards for" (FIGURE 45, FIGURE 46, and FIGURE 47).

### 5.2.1.5 Publication title <prtitle>.

The publication title shall appear as specified in TABLE A-I in APPENDIX A. The system name shall consist of the system or item official nomenclature **<sysnomen>** and NSN **<nsn>**. The end item code **<eic>**, model number **<modelno>**, and part number **<partno>** may be included as described below.

#### 5.2.1.5.1 System nomenclature <sysnomen>

The item's system nomenclature shall be included. This name can be found via the H6 Item Directory available at https://www.dla.mil/Working-With-DLA/Federal-and-International-Cataloging/H6/.

#### 5.2.1.5.1.1 Model number <modelno>.

When available, the assigned model number shall be included.

#### 5.2.1.5.1.2 National stock number <nsn> (Required).

The assigned national stock number shall be included. If one is not assigned or is not available, enter "N/A" for the NSN.

#### 5.2.1.5.1.3 Part number <partno>.

When available, the item's part number shall be included.

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#### 5.2.1.5.1.4 End item code <eic>.

When available, the end item code shall be included. Designate with "EIC: N/A" if the item covered by the manual is not an end item or an EIC is not available.

#### 5.2.1.5.1.5 Commercial and Government Entity Code (CAGEC <cageno>.

When available, the item's CAGEC shall be included.

### 5.2.1.6 Subtitle <stitle>.

The subtitle is used for additional clarification ("for XYZ other system," for example).

### 5.2.1.7 Weapon system name <weapons_system>.

A "common name" to better help ID the equipment.

### 5.2.1.8 Cover illustration (optional) <graphic>.

Cover illustrations shall only be used if space permits after all required information is placed on the cover and shall be used for pocket-sized manuals only if the illustration will not cause information to be pushed to the inside front cover.

### 5.2.1.9 (MC) Marine Corps Seal <graphic>.

For Marine Corps only publications, the Department of the Navy, United State Marine Corps seal shall appear below the title on the front cover.

### 5.2.1.10 Reporting <reporting>.

A reporting of errors statement shall be included. There are variations for the reporting of errors statement, including variants for classified and unclassified manuals or various service requirements.

### 5.2.1.11 Notices <notices>.

Front cover notices shall appear in accordance with the following sections.

#### 5.2.1.11.1 Availability statement <avail> (DMWR/NMWR only).

For DMWRs/NMWRs only, the front cover shall contain the following availability statement (italicized text within parentheses shall be replaced with the appropriate information):

"This publication is not available through the St. Louis Media Distribution Division. This publication is available through (*insert the name and address of the proponent activity*)."

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### 5.2.1.11.2 Supersession notice <super>.

When a publication is revised, a supersession notice shall be included on the front cover; it may be placed on the inside front covers of ETMs if necessary. The supersession notice shall have the words "SUPERSESSION NOTICE" underlined preceding the body of the notice.

TABLE XII shows examples of different types of supersession notices that may be used. These are intended as examples only and may not cover all the possible scenarios that may occur. Refer to FIGURE 41 and FIGURE 46 for examples of format and placement of supersession notices For an example of supersession data for IETM disc labels, see 5.6.2.6.

**TABLE XII. Supersession notice text.**

| Type of Pub | Type of Pub Superseding | Supersession Notice Text . Omit "including all changes" if referring to a revision. |
|---|---|---|
| IETM | The previous version of itself (whether the previous version was an ETM or IETM) | "SUPERSESSION NOTICE: This IETM, TM XX-XXXX-XXX-XX&P (EM #), supersedes TM XX-XXXX-XXX-XX&P (EM #), dated (*insert date*)." |
| New IETM | Multiple ETMs | "SUPERSESSION NOTICE: This IETM, TM XX-XXXX-XXX-XX&P (EM#), supersedes TM XX-XXXX-XXX-XX&P volumes 1-3, dated (*insert date*) including all changes, TM XX-XXXX-XXX-12 dated (*insert date*) including all changes, and TM XX-XXXX-XXX-12&P dated (*insert date*) including all changes." |
| ETM | Itself or just one publication | "SUPERSESSION NOTICE: TM XX-XXXX-XXX-XX dated (*insert date*) supersedes TM XX-XXXX-XXX-XX dated (*insert date*), including all changes." |
| ETM | Multiple publications | SUPERSESSION NOTICE: TM XX-XXXX-XXX-XX dated (*insert date*) supersedes TM XX-XXXX-XXX-XX dated (*insert date*) including all changes, TM XX-XXXX-XXX- XX dated (*insert date*), including all changes, and TM XX-XXXX-XXX-XX dated (*insert date*), including all changes. |
| ETM multi-volume | Itself; only some of the volumes are revised and data shift between volumes | "SUPERSESSION NOTICE: TM XX-XXXX-XXX-XX-X supersedes TM XX-XXXX-XXX-XX-X dated (*insert date*) including all changes." Then, on the next volume: "SUPERSESSION NOTICE: TM XX-XXXX-XXX-XX-Y supersedes TM XX-XXXX-XXX-XX-Y dated (*insert date*) including all changes." |
| ETM multi-volume | Multiple volumes | "SUPERSESSION NOTICE: TM XX-XXXX-XXX-XX volumes 1-X supersede TM XXXXXX-XXX-XX volumes 1-Y, dated (*insert date*) including all changes." If superseded volumes have different dates, each volume should be listed separately in the supersession notice. |

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**TABLE XII. Supersession notice text. (continued)**

| Type of Pub | Type of Pub Superseding | Supersession Notice Text . Omit "including all changes" if referring to a revision. |
|---|---|---|
| ETM multi-volume combined into a single, multi-volume publication | Multiple publications | "SUPERSESSION NOTICE: TM XX-XXXX-XXX-XX volumes 1-X supersedes TM XX-XXXX-XXX-XX, dated (*insert date*) including all changes, TM XX-XXXX-XXX-XX dated (*insert date*) including all changes, and TM XX-XXXX-XXX-XX volumes 1-Y, dated (*insert date*) including all changes." |
| ETM with new volume added to existing multi-volume set | Itself or another manual (for example, when a single manual is incorporated into a multi-volume set). *This notice would go on the new volume only. | "SUPERSESSION NOTICE: TM XX-XXXX-XXX-4 supersedes TM XX-XXXX-XXX-XX dated (*insert date*)." Note: If the superseded manual is broken up into the existing volumes, list all the volumes which supersede the manual (for example, "Volumes 1, 3, and 4 supersede TM XX-XXXX-XXX-XX"). |

### 5.2.1.11.3 Distribution statement <dist>.

All TMs, DMWRs, and NMWRs shall have a distribution statement placed on the front cover for each new, revised, or changed publication. The appropriate distribution statement shall be provided by the MATDEV as selected from DoD Instruction (DoDI) 5230.24.

### 5.2.1.11.4 Export control notice warning <export>.

For those publications with export-controlled data, the export control label contained in DoDI 5230.24 shall be included.

### 5.2.1.11.5 Destruction notice <destr>.

All TMs marked with distribution statements "B," "C," "D," "E," or "F" shall be marked with the destruction notice. Destruction notices may be placed on the inside front covers of TMs if necessary. For all publications, the element **<destr>** within **<notices>** shall contain the following text:

"For classified documents, follow the procedures in Part 117, USC 32, National Industrial Security Program Operating Manual (NISPOM) Rule and/or DoDM 5200.01, Information Security Program. For unclassified, limited documents, destroy by any method that will prevent disclosure of contents or reconstruction of the document."

(MC) For Marine Corps only publications, the destruction notice should read:

"DESTRUCTION NOTICE: Destroy by making this publication unreadable, indecipherable, and unrecoverable IAW DoDI 5200.48."

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Check the source to verify that this is the current version before use.

### 5.2.1.11.6 General purpose notice <general_purpose_notices>.

When specified by the MATDEV, additional notices may be included that are not addressed by the notices in 5.2.1.11.6.1 through 5.2.1.11.5. General purpose notices may be placed on the inside front covers of TMs if necessary. General purpose notices shall have a title followed by the notice text.

### 5.2.1.11.6.1 Copyright notice.

When copyrighted material is included in the manual, a copyright notice may be included on the front cover if requested in the copyright release letter.

(MC) For Marine Corps only publications, a copyright notice shall be included in the front matter if copyrighted material is included in the manual. Refer to Section 401(b) of Title 17 for the notice's content.

### 5.2.1.11.6.2 Hazardous materials information.

A general purpose notice may be used to indicate review for hazardous materials in accordance with Executive Order 13423 has been completed. This type of general purpose notice for hazardous materials information shall only be used when the review has been completed. Verbiage similar to the following shall be used:

"HAZARDOUS MATERIALS INFORMATION: This document has been reviewed for the presence of HAZARDOUS CHEMICALS AND TOXIC SUBSTANCES as defined by the EPCRA 302 and 313 lists by (*insert command environmental office*). As of the base document, dated (*insert date*), all references to Hazardous Chemicals and Toxic Substances have been removed from this document by substitution with nonhazardous or less hazardous materials where possible."

### 5.2.1.11.6.3 Disclaimer Statement for Manufacturers' Names/Logos.

If you must use manufacturers' names/logos to prevent rework of graphics or for safety reasons, consult your legal division. A disclaimer such as the following shall be included on the front cover:

"Reference herein to any specific commercial product, process, or service by trade name, trademark, manufacturer, or otherwise, does not constitute or imply endorsement, recommendation, or favoring by the United States Government of one specific commercial product, process, or service by trade name, trademark, manufacturer, or otherwise, over another."

### 5.2.1.12 CUI designation indicator <cui_set>.

The CUI designation indicator must be readily apparent. The first page or cover of any document or material containing CUI, including a document with commingled classified information, will include a CUI designation indicator. This CUI designation indicator is similar to the classification-marking block used for CNSI documents and materials. Documents and materials containing CUI will require a generic "CUI" marking at the top and bottom of each page. Refer to DoDI 5200.48 for additional information.

### 5.2.1.13 Service nomenclature <servnomen>.

All TMs shall include the service or MATDEV's nomenclature.

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### 5.2.1.14 Publication date <date>.

The TM publication date shall be the official publication date assigned by the MATDEV. The day, month, and year shall be given in that sequence (for example, 10 JULY 2022). The publication must be dated within a 60-day window of plus or minus 30 days from the date the publication was submitted to APD for authentication. If a publication is produced in more than one medium, the date must be the same on all media.

(MC) For Marine Corps only publications, the date on the front cover shall be the month and year only.

### 5.2.1.15 (MC Publication control number (<pcn>.

A publication control number shall be placed in the lower right-hand corner of the cover for Marine Corps only manuals and for multi-service manuals involving the Marine Corps.

### 5.2.2 Revision summary <revisionsummary>.

ETMs may have a revision summary. IETMs shall have a revision summary if applicable.

### 5.2.2.1 Revision summary requirements.

A revision summary shall contain a list of WPs by title that have been revised. A revision summary may also be included in a new IETM that is superseding an existing publication or publications.

For each work package (WP) listed, a brief description of the major changes shall be provided.

The revised WPs listed on the change/revision summary shall be linked to the WP containing the change or revised information.

The revision summary shall also indicate those WPs that have been superseded.

If the WP supersedes a WP in the same publication other than itself, the supersession notice shall be as follows (italicized text within parentheses shall be replaced with the appropriate information): "This WP supersedes WP titled (*insert title and hyperlink*), dated (*insert date*)."

If the superseded WP is contained in another publication, the notice shall contain the publication number. If the superseded WP is classified, the supersession information shall be as follows: "This WP supersedes WP titled (insert title and hyperlink), dated (*insert date*), which should be destroyed in accordance with applicable security regulations."

For revisions that are prepared to support, include, or address safety issues (Emergency Changes, Safety of Flight Messages (SOFM)/Safety of Use Messages (SOUM), Maintenance Advisory Messages, etc.), a statement may be included addressing these issues.

### 5.2.3 (MC Signature page <promulgation>.

The signature page shall be included in Marine Corps only publications and any multi-service publication with the Marine Corps as the lead service. The signature page shall follow the Marine Corps cover. For multi-service publications where Army is the lead service, the signature page shall not be included.

The signature page shall be inserted in the publication as a graphic and not as tagged text. The latest signature pages can be found at https://app.mcboss.usmc.mil/suite/sites/publications/page/home.

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### 5.2.4 Warning summary <warnsum>.

When required, a warning summary shall be prepared for all TMs containing warnings. The warning summary should not serve as a "catchall" or contain every warning in the TM; instead, it should focus on the warnings and icons that have major impact throughout the manual. Only one warning summary shall be present within a publication.

As applicable, the warning summary shall consist of the following items used in the TM in the order specified:

1. First aid data **<first_aid>**.
2. Figures that convey crucial safety information and contain context widely applicable within the TM (Aircraft Danger Zone, Keep Out Zone, etc.)**<warninfo>**.
3. Explanation of all general safety warning icons **<safety>**.
4. Explanation of all hazardous materials icons **<haz-icons>**.
5. Descriptions of the hazardous materials warnings **<hazard>**.

In ETMs, the warning summary shall begin on a right-hand page with the title centered at the top.

#### 5.2.4.1 First aid <first_aid>.

Any first aid data required in the TM, and not explicitly included in a warning, shall be included in the warning summary.

The first paragraph of the warning summary shall reference TC 4-02.1. Any additional first aid data not described in TC 4-02.1 shall be described in this section.

(MC) The first paragraph of the warning summary shall reference MCRP 3-40A.9 located at https://app.mcboss.usmc.mil/suite/sites/publications/page/home.

### 5.2.5 Change transmittal for ETMs <chgsheet>.

A change transmittal page, or "change sheet," shall be prepared for each change to a TM and shall be included in the change package (see FIGURE 48 and FIGURE 49). It shall contain the following requirements:

**a.** The change transmittal shall not be numbered and shall be located following the original authentication page.

**b.** The change transmittal shall contain the same title and distribution statement as shown on the front cover of the manual.

**c.** The publication number shall be placed in the top center of all pages of the change transmittal.

**d.** The change transmittal shall list all pages/WPs that have been changed, added, deleted, or superseded.

**e.** When changes are prepared, the change number and date shall be shown on all pages of the change transmittal. The change date shall be dated within a 60-day window of plus or minus 30 days from the date the publication was submitted to APD for authentication.

**f.** Authentication information may appear on the same page as the rest of the change information, or it may appear on a separate page.

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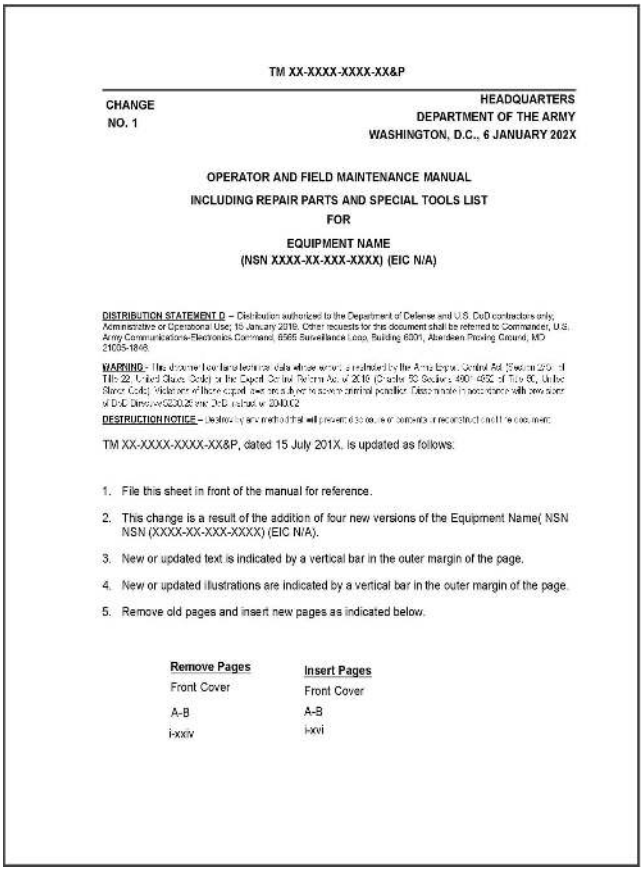
## Note

Figure appearance may vary slightly based on version of DTD/stylesheets utilized.



TM XX-XXXX-XXXX-XX&P

CHANGE
NO. 1

HEADQUARTERS
DEPARTMENT OF THE ARMY
WASHINGTON, D.C., 6 JANUARY 202X

OPERATOR AND FIELD MAINTENANCE MANUAL
INCLUDING REPAIR PARTS AND SPECIAL TOOLS LIST
FOR
EQUIPMENT NAME
(NSN XXXX-XX-XXX-XXXX) (EIC N/A)

DISTRIBUTION STATEMENT D – Distribution authorized to the Department of Defense and U.S. DoD contractors only; Administrative or Operational Use; 15 January 2018. Other requests for this document shall be referred to Commander, U.S. Army Communications-Electronics Command, 6565 Surveillance Loop, Building 6001, Aberdeen Proving Ground, MD 21005-1849.

WARNING – This document contains technical data whose export is restricted by the Arms Export Control Act (Section 2751, et seq. of Title 22, United States Code) or the Export Control Reform Act of 2018 (Chapter 58, Sections 4801-4852, of Title 50, United States Code). Violations of these export laws are subject to severe criminal penalties. Disseminate in accordance with provisions of DoD Directive 5230.25 and DoD Instruction 2040.02.

DESTRUCTION NOTICE – Destroy by any method that will prevent disclosure of contents or reconstruction of the document.

TM XX-XXXX-XXXX-XX&P, dated 15 July 201X, is updated as follows:

1. File this sheet in front of the manual for reference.
2. This change is a result of the addition of four new versions of the Equipment Name( NSN NSN (XXXX-XX-XXX-XXXX) (EIC N/A).
3. New or updated text is indicated by a vertical bar in the outer margin of the page.
4. New or updated illustrations are indicated by a vertical bar in the outer margin of the page.
5. Remove old pages and insert new pages as indicated below.

| Remove Pages | Insert Pages |
|---|---|
| Front Cover | Front Cover |
| A-B | A-B |
| i-xxiv | i-xvi |

**FIGURE 48. Change transmittal (ETMs only), page 1.**



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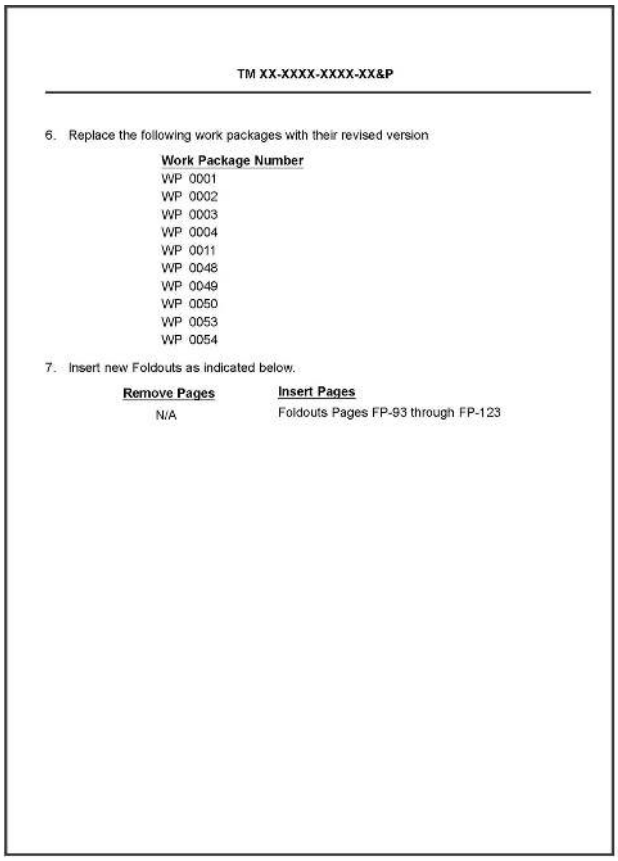

TM XX-XXXX-XXXX-XX&P

6. Replace the following work packages with their revised version

| Work Package Number |
|---|
| WP 0001 |
| WP 0002 |
| WP 0003 |
| WP 0004 |
| WP 0011 |
| WP 0048 |
| WP 0049 |
| WP 0050 |
| WP 0053 |
| WP 0054 |

7. Insert new Foldouts as indicated below.

| Remove Pages | Insert Pages |
|---|---|
| N/A | Foldouts Pages FP-93 through FP-123 |

**FIGURE 49. Change transmittal (ETMs only), page 2.**

## 5.2.6 List of effective pages/WPs <loepwp> (ETMs only).

The list of effective pages/WPs (LOEP) shall be prepared for classified TMs and ammunition data sheets TMs. For all other technical publications, the LOEP is prohibited. When revising an older publication that contains a LOEP, remove it.

When prepared in ETMs, the LOEP shall begin on a right-hand page.

The LOEP provides a quick visual representation of a WP's change history. To assist in reading LOEPs in older publications, refer to FIGURE 50 for an example LOEP for a new publication; refer to FIGURE 51 for an example LOEP for a revised publication; refer to FIGURE 52 for an example LOEP for a changed publication.

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**Note**

Figure appearance may vary slightly based on version of DTD/stylesheets utilized.

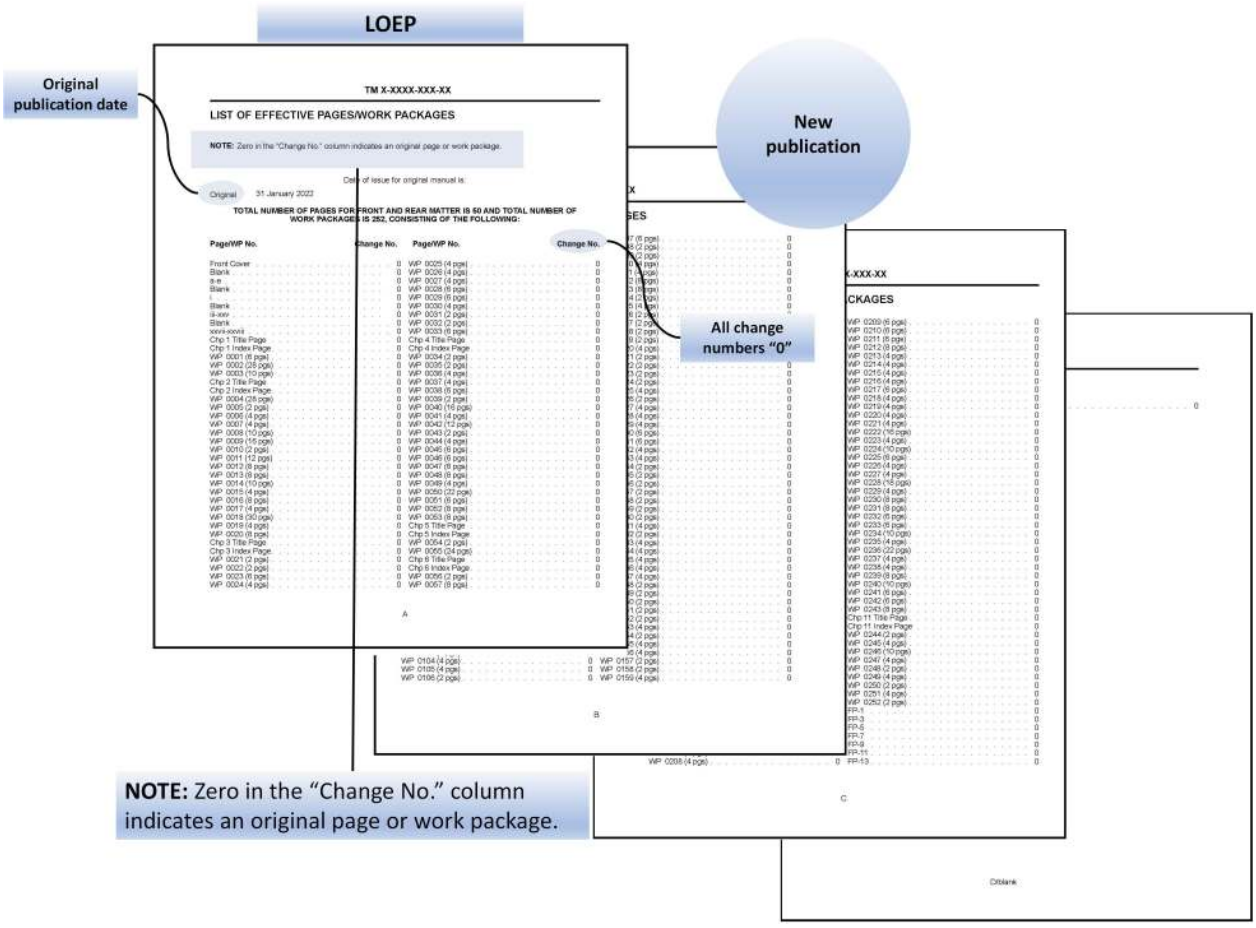


**FIGURE 50. LOEP for a new publication.**

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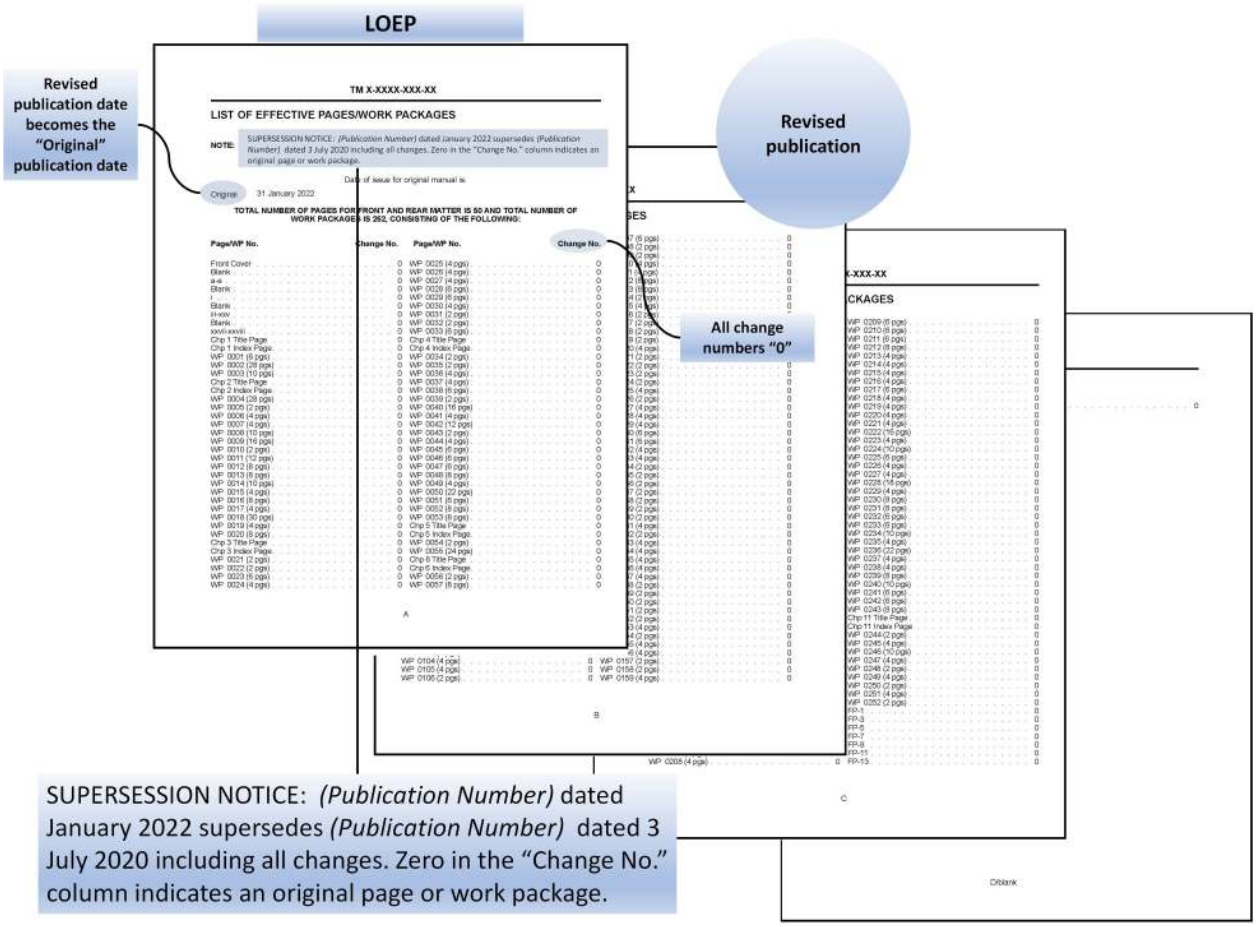


**FIGURE 51. LOEP for a revised publication.**

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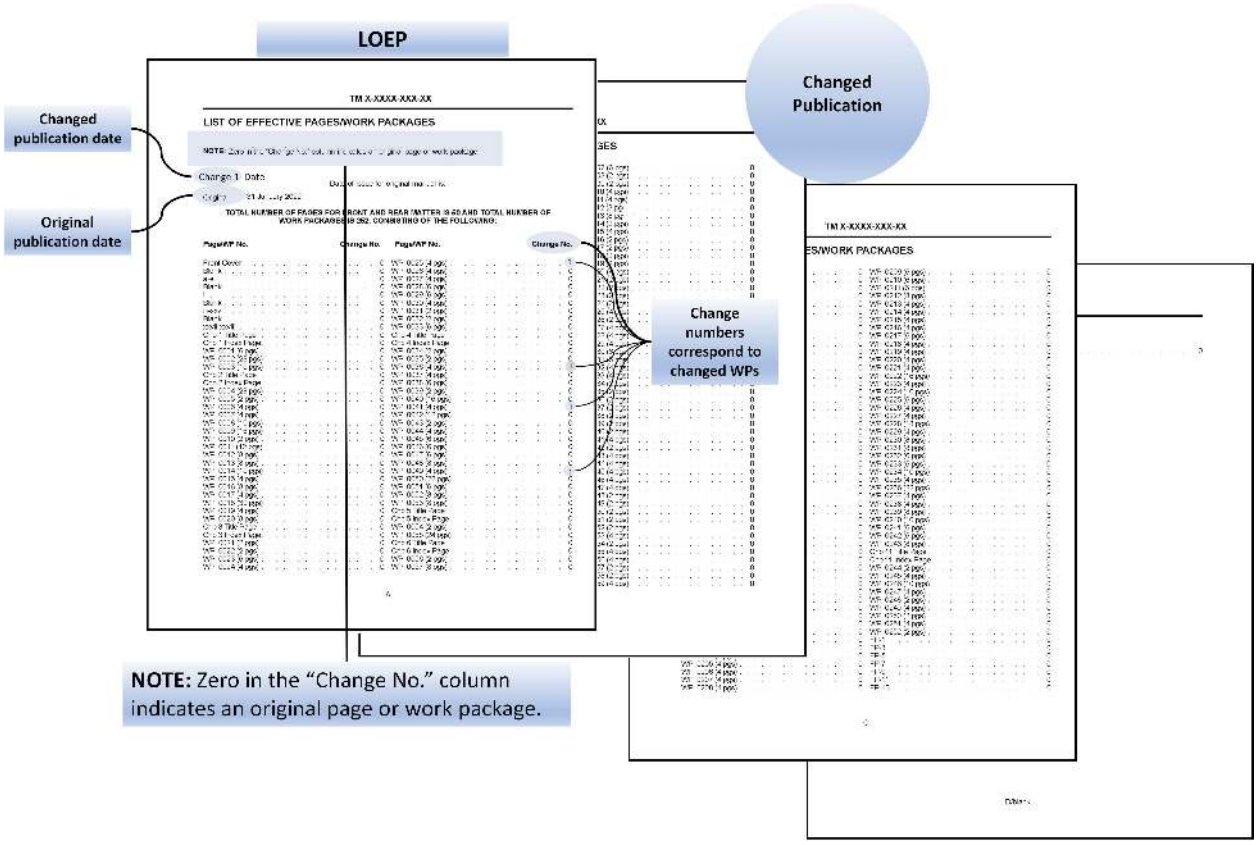


**FIGURE 52. LOEP for a changed publication.**

### 5.2.6.1 Types of pages to be included.

The following types of pages shall be included in a list of effective pages/WPs:

**a.** All front matter pages to include cover, warning summary, table of contents, and "About this manual" info. Page counts and blank backup pages shall be listed.

**b.** Module title pages. Blank backup pages shall be listed.

**c.** All WPs with their page counts (including blank pages). Blank pages within the WPs shall not be listed separately on the list of effective pages.

**d.** Foldout pages.

**e.** Back cover.

### 5.2.6.2 Exempted pages.

The following types of pages shall not be included in a list of effective pages/WPs:

**a.** Change transmittal.

**b.** List of effective pages.

**c.** DA Form 2028.

**d.** Authentication page.

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#### 5.2.6.3 Supersession note (revision only.

For revisions, a supersession note shall be included at the top of the LOEP (see FIGURE 51). This note shall be the same as the supersession notice that appears on the cover and title page of the manual.

#### 5.2.6.4 Dates of issue for original and changes.

At the top of the LOEP, the date of the basic manual and the date of each change that appears in the change number column shall be listed (see FIGURE 52). If no LOEP is included, these dates are included in the General Information WP (B.7.1.1).

#### 5.2.6.5 Total number of pages statement.

After the dates of issue, the LOEP shall contain the following statement:

"THE TOTAL NUMBER OF PAGES FOR FRONT AND REAR MATTER IS (*insert number*) AND THE TOTAL NUMBER OF WORK PACKAGES IS (*insert number*), CONSISTING OF THE FOLLOWING:"

#### 5.2.6.6 Change numbers.

For new publications, the change number is always zero; the LOEP shall have all zeros (see FIGURE 50). When a change is prepared, the appropriate change number shall be placed in the change number column. When a publication is revised, the change numbers shall all be changed back to zero.

#### 5.2.6.7 Listing the pages.

All pages in the book shall be listed except as noted in 5.2.6.2. List each WP by number and put the total number of pages in the WP in parentheses next to the WP number. The words "deleted," "added," or "blank" may be placed next to the page numbers when applicable.

#### 5.2.6.8 Numbering the list of effective pages/WPs.

The LOEP shall have a page "A" for the first page and "B," "C," "D," etc. for additional pages.

### 5.2.7 Table of contents <contents>.

A TOC listing all modules and WPs shall be present for ETMs and IETMs, including DMWRs and NMWRs. They shall have the exact same title and shall be listed in the same order they appear in the TM.

For ETMS, TOCs are auto-generated based on the DTD and stylesheet. In IETMs, the left-hand navigation menu functions as the TOC. This TOC consists of WPs listed in the toc.xml file.

The warning summary shall not be listed on the TOC in ETMs. The "About this manual" information including page number shall be listed on the TOC. TOCs may include figures and tables.

The content and format of the TOC shall be as described in the following:

**a.** The security classification, if any, of modules, WPs, figures, and tables shall be indicated.

**b.** As specified by the MATDEV, figures may be listed in the TOC under the corresponding WP by the figure number and title. A parts information TM shall not include figures in the TOC. When a TM includes the parts

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information module, the listing of RPSTL figures is optional. Foldout figures, if included in the TOC, shall be listed after all the WPs along with the other rear matter.

**c.** As specified by the MATDEV, tables may be listed in the TOC under the corresponding WP by the table number and title.

**d.** The following requirements are applicable to RPSTL entries (FIGURE 53):

**(1)** The Parts Information Introduction WP **<introwp>**.

**(2)** Titles of RPSTL WPs, including the Functional Group Codes (FGCs) as applicable, shall be listed by the same nomenclature and in the same sequence in which they appear in the first tabular listing in the work package. The WP sequence number shall be referenced with each WP title. The figure number may be included in the WP title.

**(3)** When multiple functional groups are under a single RPSTL work package, each functional group tabular list title may be included as a subordinate table of content entry.

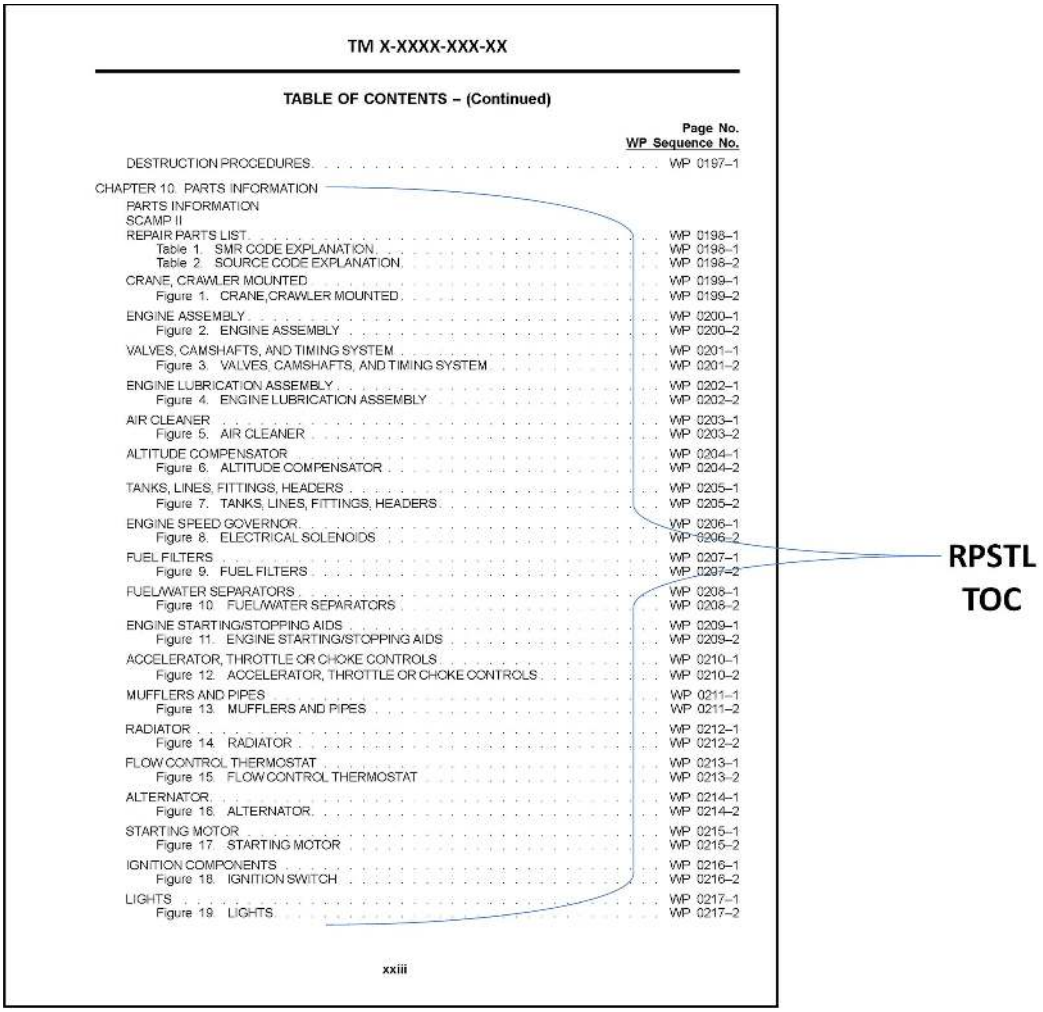
TM X-XXXX-XXX-XX

TABLE OF CONTENTS – (Continued)

Page No.
WP Sequence No.


DESTRUCTION PROCEDURES . . . . . WP 0197–1

CHAPTER 10. PARTS INFORMATION
PARTS INFORMATION
SCAMP II
REPAIR PARTS LIST . . . . . WP 0198–1
Table 1. SMR CODE EXPLANATION . . . . . WP 0198–1
Table 2. SOURCE CODE EXPLANATION . . . . . WP 0198–2
CRANE, CRAWLER MOUNTED . . . . . WP 0199–1
Figure 1. CRANE,CRAWLER MOUNTED . . . . . WP 0199–2
ENGINE ASSEMBLY . . . . . WP 0200–1
Figure 2. ENGINE ASSEMBLY . . . . . WP 0200–2
VALVES, CAMSHAFTS, AND TIMING SYSTEM . . . . . WP 0201–1
Figure 3. VALVES, CAMSHAFTS, AND TIMING SYSTEM . . . . . WP 0201–2
ENGINE LUBRICATION ASSEMBLY . . . . . WP 0202–1
Figure 4. ENGINE LUBRICATION ASSEMBLY . . . . . WP 0202–2
AIR CLEANER . . . . . WP 0203–1
Figure 5. AIR CLEANER . . . . . WP 0203–2
ALTITUDE COMPENSATOR . . . . . WP 0204–1
Figure 6. ALTITUDE COMPENSATOR . . . . . WP 0204–2
TANKS, LINES, FITTINGS, HEADERS . . . . . WP 0205–1
Figure 7. TANKS, LINES, FITTINGS, HEADERS . . . . . WP 0205–2
ENGINE SPEED GOVERNOR . . . . . WP 0206–1
Figure 8. ELECTRICAL SOLENOIDS . . . . . WP 0206–2
FUEL FILTERS . . . . . WP 0207–1
Figure 9. FUEL FILTERS . . . . . WP 0207–2
FUEL/WATER SEPARATORS . . . . . WP 0208–1
Figure 10. FUEL/WATER SEPARATORS . . . . . WP 0208–2
ENGINE STARTING/STOPPING AIDS . . . . . WP 0209–1
Figure 11. ENGINE STARTING/STOPPING AIDS . . . . . WP 0209–2
ACCELERATOR, THROTTLE OR CHOKE CONTROLS . . . . . WP 0210–1
Figure 12. ACCELERATOR, THROTTLE OR CHOKE CONTROLS . . . . . WP 0210–2
MUFFLERS AND PIPES . . . . . WP 0211–1
Figure 13. MUFFLERS AND PIPES . . . . . WP 0211–2
RADIATOR . . . . . WP 0212–1
Figure 14. RADIATOR . . . . . WP 0212–2
FLOW CONTROL THERMOSTAT . . . . . WP 0213–1
Figure 15. FLOW CONTROL THERMOSTAT . . . . . WP 0213–2
ALTERNATOR . . . . . WP 0214–1
Figure 16. ALTERNATOR . . . . . WP 0214–2
STARTING MOTOR . . . . . WP 0215–1
Figure 17. STARTING MOTOR . . . . . WP 0215–2
IGNITION COMPONENTS . . . . . WP 0216–1
Figure 18. IGNITION SWITCH . . . . . WP 0216–2
LIGHTS . . . . . WP 0217–1
Figure 19. LIGHTS . . . . . WP 0217–2


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RPSTL TOC

**FIGURE 53. RPSTL TOC.**

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### 5.2.8 "About this manual" information <about>.

"About this manual" information (which replaces "How to use this manual") shall be included as the last item in the front matter (5.2). It shall be prepared for all TMs except pocket-sized manuals, for which it is optional. Only one "About this manual" section shall be present within a publication.

"About this manual" information shall include, as applicable:

**a.** Information to familiarize the user with special or unusual features of the TM shall be prepared. Coverage shall lead the user through the TM and explain important features of the organization and content.

**b.** Any peculiarities in the basic arrangement of the TM shall be described. "About this manual" information shall not repeat instructions given within the modules and/or WPs.

**c.** For all TMs (excluding operator), "About this manual" shall include a reference to the associated RPSTL and an explanation of how to use the RPSTL in conjunction with the manual.

**d.** For troubleshooting, an explanation on how troubleshooting data are presented in the TM shall be included. If applicable, an explanation on how failure symptom indexes and malfunction codes correspond to maintenance operational checks and troubleshooting procedures for individual systems and components shall be included. If necessary, for multi-volume troubleshooting TMs, examples of the troubleshooting process shall be provided to illustrate how specific troubleshooting volumes and WPs are used together to locate and isolate faults.

**e.** An explanation of warnings, cautions, and notes placement and their meaning.

**f.** An explanation of any symbols (HCP, CSP, etc.) used in the manual.

#### 5.2.8.1 International standardization agreements.

When specified by the MATDEV, the "About this manual" information shall contain the following (italicized text within parentheses shall be replaced with the appropriate information):

**"NOTE**

Certain provisions of this technical manual (*identify by module, work package, paragraph, or similar manner, if appropriate*) are the subject of international standardization agreement. When revision or cancellation of this technical manual is proposed that will modify the international agreement concerned, the Life Cycle Management Command (LCMC) or other publications activity responsible for the publication will take appropriate action through international standardization channels, including departmental standardization offices, to change the agreement or make other appropriate accommodations."

### 5.2.9 Preventive maintenance services and phased maintenance inspection manuals title page with warning data (Aviation only).

For preventive maintenance services and phased maintenance inspection only, the warning data page shall include the reporting errors and recommending improvement statement and the following additional verbatim statement:

**"WARNING**

Certain inspections are Mandatory Safety of Flight requirements, and the inspection intervals cannot be exceeded. In the event these inspections cannot be accomplished at the specified interval, the aircraft condition status symbol will be changed to a red "X." Mandatory Safety of Flight inspection items are printed in bold face type.

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"NOTE

Inspection items contained in this manual are considered the minimum requirements for performing phased maintenance and must be performed. The cumulative effects of inspection deferrals are unknown and could result in catastrophic failure or increased maintenance at a later date. Therefore, the use of special lettering to emphasize Mandatory Safety of Flight Items is not to be construed as authority for deferral of other inspections."

## 5.3 Front matter for multi-volume ETMs.

The first volume in a multi-volume ETM shall contain the following front matter:

1. Front cover **<frntcover>** (5.2.1).
2. When applicable, MC signature page **<promulgation>** (MC only) (5.2.3).
3. Optional comprehensive warning summary **<warnsum>** (5.2.4) for all volumes.
4. When required, change transmittal **<chgsheet>** (5.2.5) that includes only the volume change pages.
5. TOC **<contents>** (5.2.7). The volume contains a complete TOC covering the entire set. Entries shall indicate the volume in which the referenced material appears (for example, "Operator Instructions, Vol. 1").
6. "About this manual" information **<about>** (5.2.8).

The remaining volumes in a multi-volume TM shall contain the following front matter:

1. A front cover **<frntcover>** or **<frontcover_abbreviated>**5.2.1).
2. When applicable, MC signature page **<promulgation>** (MC only) (5.2.3) that includes only the volume change pages.
3. Optional warning summary **<warnsum>** (5.2.4) particular to each volume.
4. When required, change transmittal **<chgsheet>** (5.2.5) that includes only the volume change pages.
5. TOC **<contents>** (5.2.7). Each volume shall contain its own TOC and shall reference companion volumes for the same TM.

Changes to multi-volume ETMs shall be made independently to each volume. The TOC contained in the first volume shall be updated during any change to subsequent volumes. Thus, Volume 1 will have a change package every time any volume is updated if those updates cause TOC changes.

## 5.4 Work package requirements.

### 5.4.1 Work package identification information <wpidinfo>.

All WPs shall include the identification information entries in the following sequential order, as applicable. See FIGURE 54.

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**Note**

Figure appearance and tags/ouput may vary slightly based on version of DTD/stylesheets utilized (see applicable DTD or tag definition list for available tags).

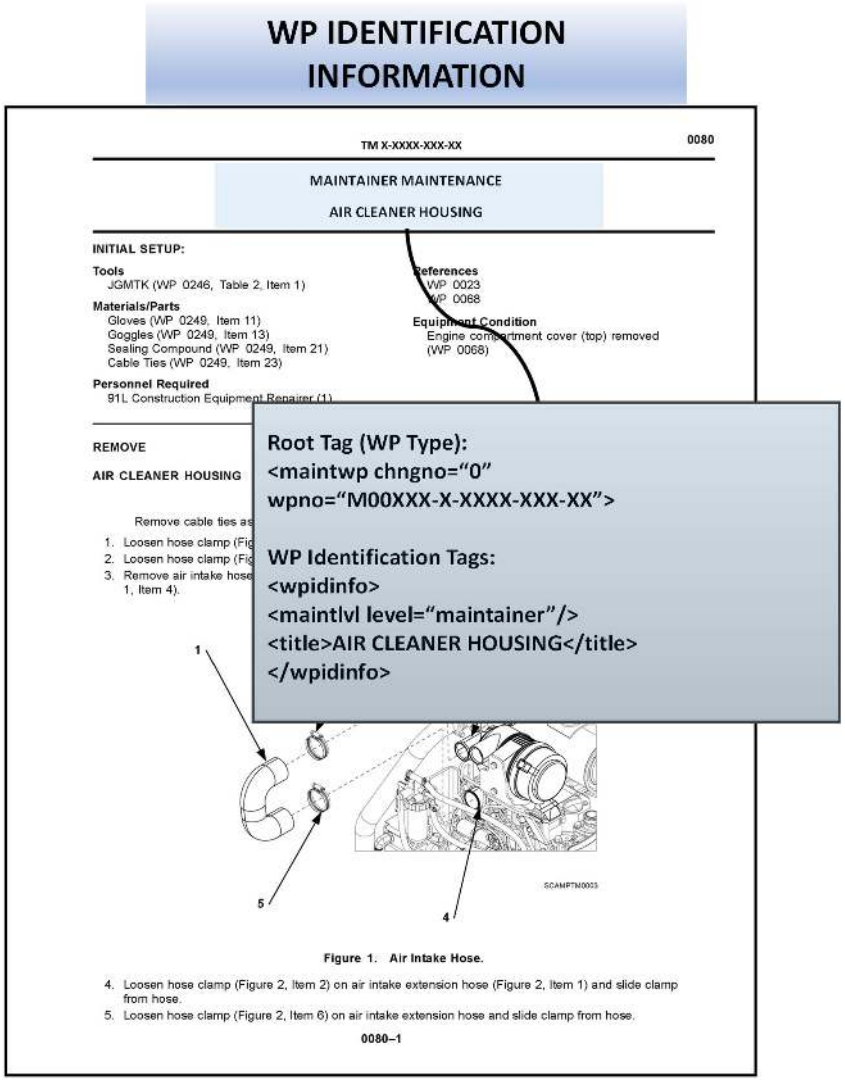


**FIGURE 54. WP identification information**

### 5.4.1.1 Maintenance level <maintlvl>.

The WP shall show the lowest maintenance level authorized to use the work package. TABLE XIII lists requirements that shall be followed when selecting maintenance class for a work package:

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**TABLE XIII. Determining wording for WP maintenance level.**

| Kind of WP | Kind of TM | Wording for maintenance level |
|---|---|---|
| Operator instruction | Operator | "Operator" or "crew" plus the word "instructions" ("operator instructions") |
| Maintenance | Operator or Maintenance TMs, NMWRs, DMWRs, PMCs, or LOs | The type of TM plus "maintenance" ("operator maintenance," "maintainer maintenance," "AMC maintenance," etc.) |
| Troubleshooting | Operator or Maintenance TMs, NMWRs, or DMWRs | The type of TM plus "Troubleshooting" ("maintainer troubleshooting," etc.) |
| MAC | Maintenance TMs | N/A |
| RPSTL | Maintenance TMs only | No lower than "maintainer" with no other words |
| All other types | Operator or maintenance TM, PMC, LO, NMWR, or DMWR | Lowest level authorized for the WP with no additional words (COEI/BII would be "operator," not "operator maintenance" or "operator instructions") |
| Non-procedural | Destruction | "Destruction" |
| Procedural | Destruction | Lowest level authorized to perform the procedure followed by the word "destruction" ("operator destruction,"etc.) |
| Non-procedural | BDAR | "BDAR" |
| Procedural | BDAR | Lowest level authorized to perform the procedure followed by the title ("operator battle damage assessment," "maintainer battle damage repair," etc.) |
| Non-procedural | Software (SUM/SAM) | "user" or "administrator" |
| Software maintenance | Software (SUM/SAM) | "user maintenance" or "administrator maintenance" |
| Software troubleshooting | Software (SUM/SAM) | "user troubleshooting" or "administrator troubleshooting" |

### 5.4.1.2 Work package title <title>.

The title of the individual WP shall be derived from the titles listed in the content selection matrix. APPENDIX A provides content selection matrices while FIGURE 9 and FIGURE 10 depict examples of utilizing the matrices to derive WP titles. Refer to C.5.2.1.1 for instructions when titling operator instructions that shall be integrated into multiple modules.

### 5.4.1.3 Effectivity notice <config>.

If applicable, an effectivity notice shall be included. When the WP does not apply to all configurations of the weapon system/equipment, the applicable configurations **<name>** covered by the WP shall be listed. Omit this requirement if the same task/procedure applies to all configurations. (If certain configurations require different tasks/procedures, separate WPs shall be prepared.)

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#### 5.4.1.4 Multi-service (joint use.

When publications are acquired and specified by the Army for joint use with another or other services (multi-service publications), WPs that do not apply to all services concerned shall be marked to indicate the service(s) to which they apply (for example, LANDING GEAR MAINTENANCE (ARMY ONLY)).

### 5.4.2 Display of WP identification information in IETMs.

Work package identification information shall be displayed as follows:

**a.** Work package title shall always be displayed in the subtitle bar.

**b.** The remaining WP identification information shall be displayed in the inner shell unless context filtering is used.

**c.** If context filtering/login is used in the IETM, the filtered WP identification information shall be in the source data but shall not be displayed.

### 5.4.3 Initial setup <initial_setup>.

The initial setup provides information such as equipment, parts, material, and authorized personnel/time required to perform and complete the task(s) included in the work package. Refer to FIGURE 55 for an example initial setup section as well as the associated tags. The following requirements shall be complied with:

**a.** Unless otherwise specified in this standard, all WPs except those present in front and rear matter shall include initial setup instructions **<initial_setup>**.

**b.** Initial setup shall be at the top of the WP directly following the WP identification information.

**c.** There shall be only one initial setup per work package.

**d.** Multiple tasks shall only be grouped into one WP if all the tasks have the same personnel required (or MOS) and equipment condition initial setup sections(FIGURE 6). The remaining initial setup sections shall be applicable to at least one task in the WP. Tasks shall not be grouped if condition based maintenance applies. Liberal use of references between WPs is encouraged.

**e.** As appropriate, referencing shall be established for all items in the initial setup (5.4.3.1 through 5.4.3.11).

**f.** When no initial setup instructions are required to perform the task(s), the title INITIAL SETUP shall be included with the words "Not Applicable," which is set by selecting the element **<null>**.

**g.** Initial setups shall not contain details such as National Stock Number (NSN), part numbers, Commercial and Government Entity Codes (CAGEC).

**h.** The items in 5.4.3.1 through 5.4.3.11 shall be included in the initial setup in the order given. Items not applicable to the WP may be omitted.

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**Note**

Figure appearance and tags/ouput may vary slightly based on version of DTD/stylesheets utilized (see applicable DTD or tag definition list for available tags).

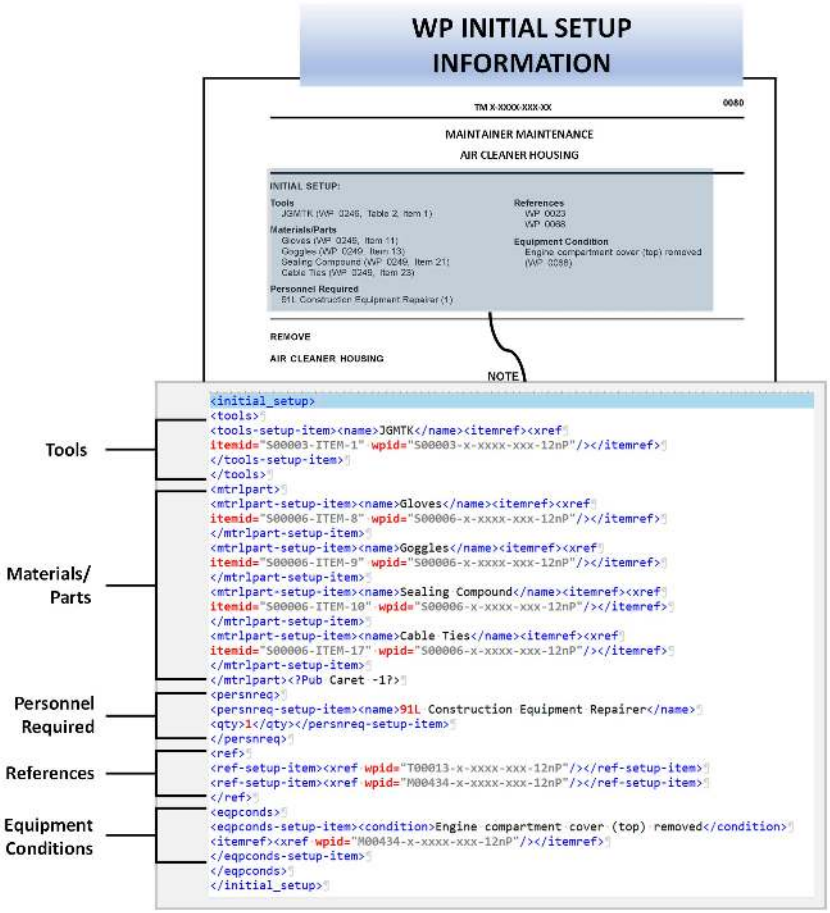


**FIGURE 55. WP initial setup information example.**

### 5.4.3.1 Test equipment <testeqp>.

All test equipment not contained in mechanic-assigned tool kits required to perform the procedure shall be listed by name **<name>**, the WP containing an overall listing of tools and special tools/kits listed by WP number, and item number for publication number **<itemref>**. Referencing will eliminate the need to repeat or update the part and model numbers throughout the publication. Refer to G.7.3.2.4 for more detailed information about test equipment and where it should be listed in supporting information WPs.

### 5.4.3.2 Tools <tools>.

All tools (outlined below) listed in the initial setup sections of troubleshooting and/or maintenance WPs shall be listed in the Tool Identification List WP (G.7.8).

**a. Kits**. Kits may be listed in Table 2 of the MAC. The tool kit (box) assigned (on a one-per-mechanic-by-MOS basis) to the mechanic to be used in the **<name>**. Maintenance of equipment shall be listed by name**<name>**, by the WP sequence number for tool identification list WP (COEI/BII WP sequence number for -10 operator manuals), and by item number **<itemref>**, or publication number **<extref>**. Tools in the kit may be further


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identified by listing them underneath the tool kit in the initial setup with no indentation and linked to the tool identification work package.

**b. Sets/Outfits**. Sets/Outfits may be listed in Table 2 of the MAC. The individual tools from the set or outfit shall be listed in the initial setup by name **<name>**by the WP sequence number for the tool identification list WP (COEI/BII WP sequence number for -10 operator manuals), and by item number **<itemref>**, or publication number **<extref>**(for example, wrench, socket (WP 0123, Item 12).

**c. Common individual tools**. Common individual tools shall not be listed in Table 2 of the MAC. Common individual tools required for performance of the task that are not part of a kit, set, or outfit shall be listed in the initial setup by name **<name>**,by the WP sequence number of the tool identification list WP (COEI/BII WP sequence number for -10 operator manuals), and by item number **<itemref>**or publication number **<extref>**.

### 5.4.3.3 Special tools/kits <spectools>.

Special tools/kits that are used only for the piece of equipment covered in the manual shall be listed in the initial setup by name <**name>** and linked to the tool identification list WP (COEI/BII WP sequence number for - 10 operator manuals). These tools are also in the special tools/kits list but shall not be linked to the RPSTL from initial setup. Special tools/kits may be listed in Table 2 of the MAC. Refer to F.8.6 for information about the special tools/kits list WP.

### 5.4.3.4 Materials <mtrlpart>.

All expendable items, support materials, bulk items, and Critical Safety Items (CSIs) shall be listed by name **<name>**; quantity **<qty>** (if applicable); WP sequence number for WP containing details about the materials/parts (for example, AAL, expendable and durable items list, RPSTL, CSI list, etc.); and item number **<itemref>**, or figure/group number (as applicable). Materials shall be listed in alphabetical order in the initial setup.

Mandatory replacement parts shall be listed under a separate header (5.4.3.5) and shall not be listed under materials header.

Referencing will eliminate the need to repeat or update the part and numbers throughout the TM. For example:

"Material/Parts

Bracket Assembly, Chemical Alarm (WP 0121, Item 4)
Frequency Converter (WP 0122, Item 3)
Grease (WP 0120, Item 5)

Range lock (WP 0120, Item 10)"

### 5.4.3.5 Mandatory replacement parts <mrp>.

Mandatory replacement parts shall be listed in the initial setup separate from other repair parts by name **<name>** and quantity **<qty>** (if applicable). They shall be linked to the mandatory replacement parts list WP in supporting information **<itemref>**.

### 5.4.3.6 Personnel required <persnreq>.

With the exception of DMWR/depot personnel, a list of the type and quantity of personnel required for the task(s) in the WP shall be included. Personnel required for Equipment Conditions and other referenced tasks shall not be included here. If a special skill set is required, the Military Occupational Specialty (MOS) designation shall be included. If a specific skill set is not required, the MOS may be excluded for tasks that do not require a specific skill set (for example, lifting a heavy ramp on a vehicle). When an MOS is included, the MOS name **<name>** and MOS

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designator **<mos>** shall be included. For each MOS, the **<qty>** shall be identified. If MOS is not included, only the number of people required and the quantity shall be listed in the initial setup. If both skilled personnel and additional unskilled helpers are needed for a task, the MOS required and the quantity of that MOS shall be listed first followed by the number of unskilled helpers required:

Example 1 (MOS needed)

"Personnel Required
Artillery Mechanic 91P (2)"

Example 2 (No skill set required)

"Personnel Required
1 person"

Example 3 (1 skilled person and helpers)

"Personnel Required
Artillery Mechanic 91P (1)
Additional personnel (1)"

#### 5.4.3.6.1 Military Occupational Specialties (MOSs) requirements.

All MOSs indicated in a MAC F (maintainer) column within a publication shall align with the Basis of Issue Plan Feeder Data (BOIPFD) DA PAM 700-27. Operator MOSs that do not receive Direct Productive Annual Maintenance Man Hours (DPAMMH) shall not be listed in a MAC F (maintainer) column. Instead, they shall be listed in the MAC C (crew/operator) column in publications. Necessary BII tools and maintenance technician tools/test equipment shall be handled as follows:

**a.** For -10 publications, BII tools and maintenance technician tools/test equipment shall be listed in initial setup in accordance with 5.4.3.1 and 5.4.3.2; these sections shall reference the Additional Authorization List (AAL) WP (G.7.5).

**b.** For -12 and -12&P publications, BII tools and maintenance technician tools/test equipment shall be listed in initial setup in accordance with 5.4.3.1 and 5.4.3.2; these sections shall reference the Tool Identification List (TIL) WP (G.7.8).

**c.** For -20 and -20&P publications, tools and test equipment shall be listed in initial setup in accordance with 5.4.3.1 and 5.4.3.2; these sections shall reference the Tool Identification List (TIL) WP (G.7.8).

### 5.4.3.7 References <ref>.

When necessary, other WPs, TMs, foldouts, and other sources **(<link>/<extref>/<xref>)** that are needed to complete the task shall be listed here. The references shall be linked. Only references not listed in equipment conditions shall be listed.

If the reference is to a WP within the TM, it shall be by WP sequence number and may include the title if needed. For example:

Example 1 (another document):

"References
TM 9-1015-252-20&P"

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Example 2 (WP sequence #):

"References
WP 0056"

Example 3 (Title and WP Sequence #):

"References
Engine Shutdown, WP 0056"

### 5.4.3.8 Equipment conditions <eqpconds>.

Any equipment conditions required before the task can be started shall be listed here and cross- referenced to the appropriate source **(<link>/<extref>/<xref>)** for setting up the condition **<condition>**. Equipment conditions shall be listed in the order that they should be performed. Any task included in equipment conditions shall not be included in the body of the WP as task or procedural steps. For example:

Example 1:

"Equipment Condition
Firing mechanism removed (WP 0010)"

Example 2:

"Equipment Condition Cooling system drained (WP 0090)
Engine compartment cover removed (WP 0112)"
GFCI power cable assembly disconnected from power source"

### 5.4.3.9 Special environmental conditions <specenv>.

Any special environmental conditions (such as ventilation, lighting, or temperature) **<condition>** that are required shall be listed here. The reason **<reason>** that such conditions are needed shall be explained. For example:

"Special Environmental Condition
Darkened area required for testing lights"

### 5.4.3.10 Drawings/diagrams/schematics required <dwgreq>.

When necessary, all drawings, diagrams, and/or schematics required to complete the maintenance task and that are not included in the work package, shall be listed here. Drawings shall be listed by title **<dwgname>** and drawing number **<dwgno>**. For example:

"Drawings Required
Power Supply Schematic (132E470092)"

### 5.4.3.11 Estimated time to complete the task <time.to.comp>.

If required by the MATDEV, the estimated time it will take to complete the task(s) shall be included. Approved Logistics Product Data (LPD), service experience, performance data on similar equipment, and all other Reliability, Availability, and Maintainability (RAM) data available shall be used to estimate the time required to complete the task(s). For example:

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"Time to Complete
8 hours"

### 5.4.4 Task titles.

Each operator, maintenance, and troubleshooting task shall have a title. Maintenance task titles shall include the maintenance task listed in the MAC.

### 5.4.5 End of task/work package statements.

The words "END OF TASK" shall be placed at the end of each task.

The words "END OF WORK PACKAGE" shall be placed immediately following the last data item (for example, text, illustration, etc.) at the end of any work package, except for the following WPs: Repair Parts List, Kits Part List, Bulk Items, Repair Parts for Special Tools/Kits List, and Special Tools/Kits List. For these WPs, the words "END OF FIGURE" shall be placed after the list.

## 5.5 Preparation of rear matter content <rear>.

Requirements for the preparation of the rear matter necessary to supplement the technical content modules and associated WPs in APPENDIX B through APPENDIX N are provided in 5.5.1 through 5.5.4. APPENDIX A provides detailed content and assembly requirements for all TMs covering operation, maintenance, and parts information, at all maintenance levels through sustainment, that shall be tailored to meet system and user requirements.

As applicable, the rear matter shall consist of the following in the order specified:

**a.** Reporting errors and recommending improvements DA Form 2028 **<da2028>** (ETMs only) (5.5.1).

**b.** Authentication **<authent>** (5.5.2).

**c.** Foldout pages (when required) (except RPSTL) **<foldsect>** (5.5.3).

**d.** Back cover **<back>** (5.5.4) that contains the PIN (ETMs only).

### 5.5.1 Reporting errors and recommending improvements Department of the Army (DA) Form 2028 <da2028>.

IETMs with an embedded "Feedback" function may omit the printed copy of DA Form 2028.

For ETMs, one blank DA Form 2028 shall be included and shall precede the back cover of every unclassified TM (except for oversize TMs, pocket-sized TMs, TMs with less than eight pages, and LOs). Providing instructions or sample filled-out copies of this form is at the MATDEV's discretion.

(MC) For Marine Corps only or Marine Corps led TMs, recommendations to publication errors and/or requests for Logistics Product Data (LPD) and suggested technical data changes may be reported by submitting a Change Request in TDM-CATALYST. User must have a registered account to submit a Change Request. For instructions on how to submit a Change Request, go to https://app.mcboss.usmc.mil/suite/sites/support/page/knowledge-base/. Questions or concerns regarding the change request system process should be reported via Support Case within TDM-Publications or by email to catalyst_support@usmc.mil.

### 5.5.2 Authentication.

The authentication page, provided by the MATDEV, shall appear after DA Form 2028 for ETMs. See FIGURE 57.

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The authentication shall appear immediately before any foldouts and the back cover. Authentications shall include the TM number, EM number, and the TM PIN. IETM authentications also shall include the TM number, EM number, and both the TM and EM PINs. See FIGURE 56.

For ETM changes, the authentication for that specific change shall be included on a separate page immediately after the change transmittal in the front matter. The consolidated file also retains the original publication's authentication page in the rear matter.

**Note**

Figure appearance and tags/ouput may vary slightly based on version of DTD/stylesheets utilized (see applicable DTD or tag definition list for available tags).

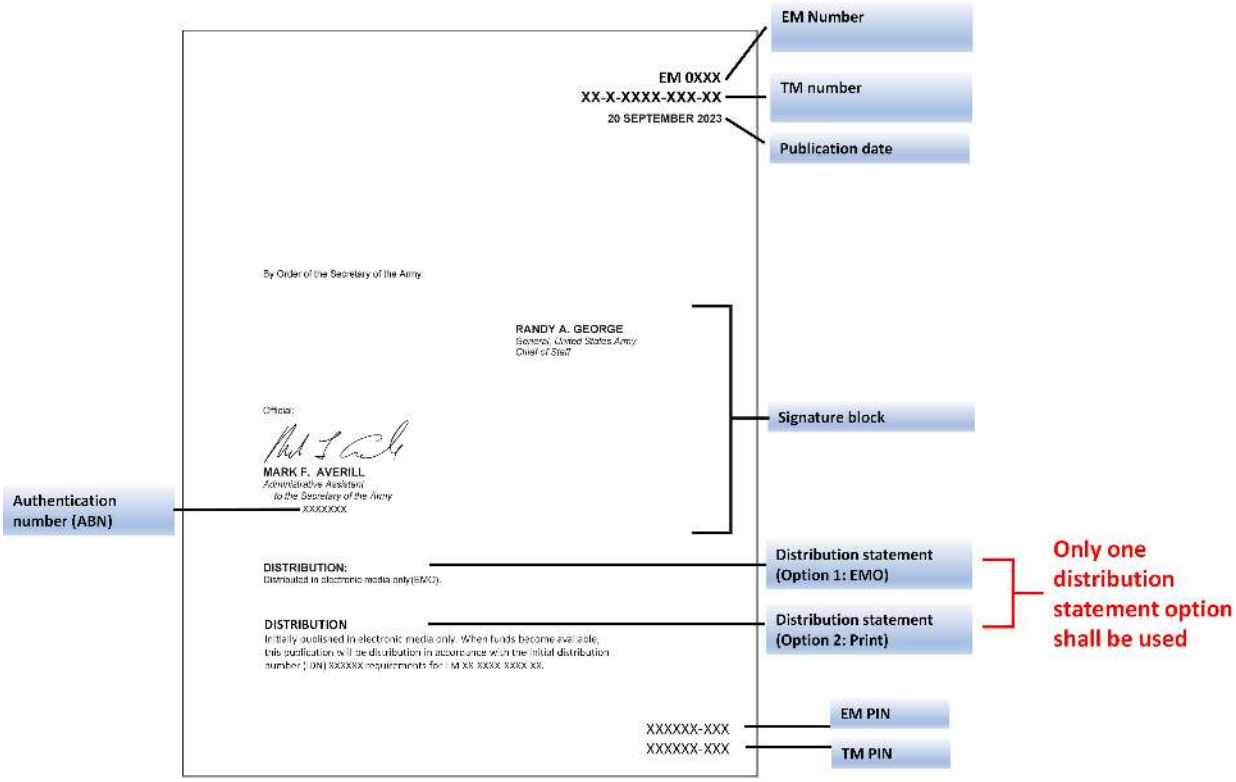


**FIGURE 56. Authentication (IETM).**

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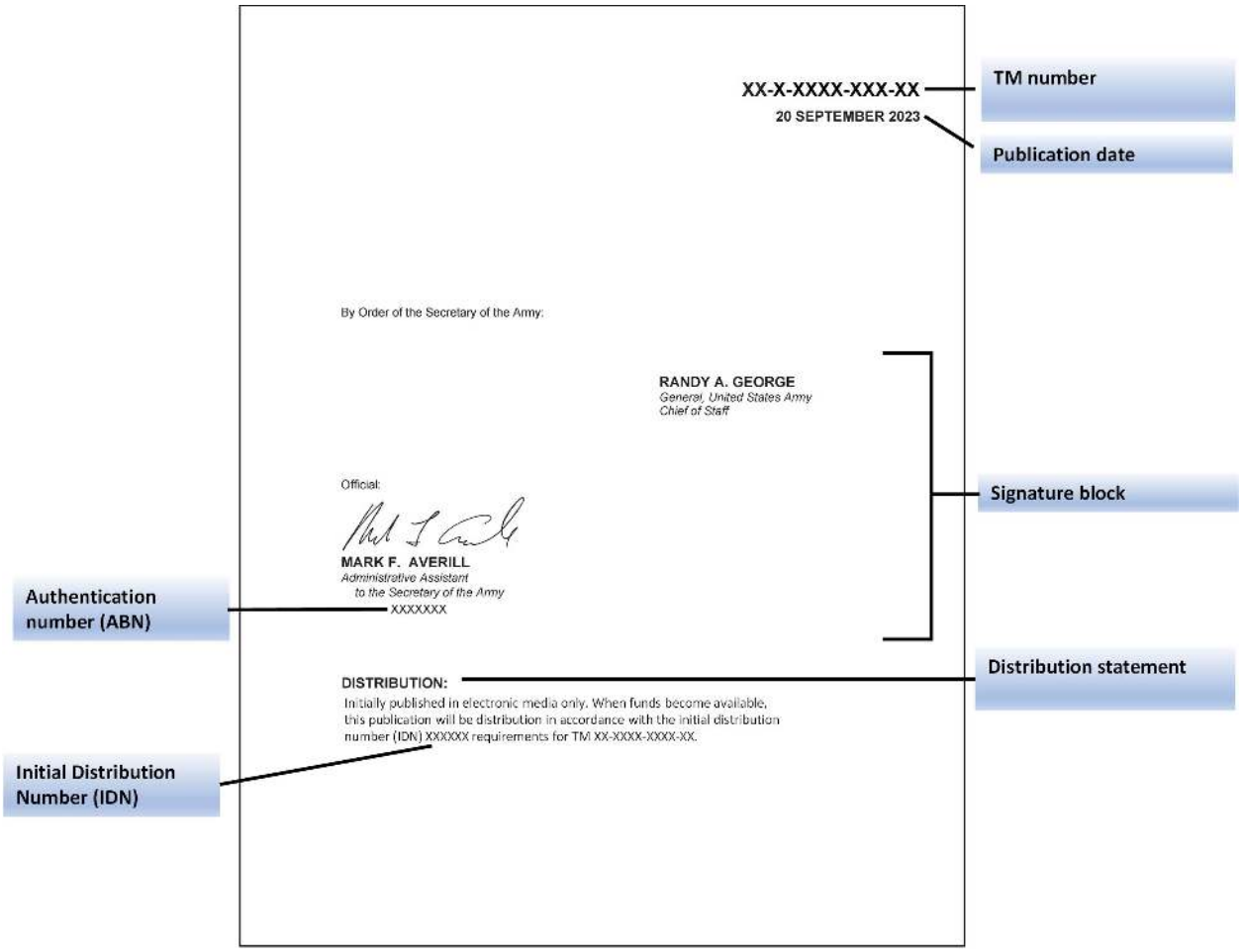


**FIGURE 57. Authentication (ETM).**

### 5.5.3 Foldout pages (Except RPSTL) <foldsect>.

If foldout pages are present, they shall be placed directly before the back cover. Foldout pages shall not be included in RPSTLs or operator-only TMs.

### 5.5.4 Back cover <back> for ETMs.

The outside back cover shall only contain the PIN, except for pocket-sized TMs and classified TMs. For pocket-sized TMs, the outside back cover shall include the TM number. For classified TMs, security classification markings shall be included on the back cover.

The PIN shall be placed in the lower right-hand corner of the back cover (except for Ammunition DMWRs, which do not have PINs).

## 5.6 Preparation of discs.

The disc shall include the IETM file. Viewer software may be provided as a courtesy but is available for download at https://iads.redstone.army.mil and in the Army Software Center. Discs shall be prepared in accordance with the following sections.

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### 5.6.1 Information not contained on the disc face.

The face of the disc shall not contain the following:

**a.** The command seal or emblem of the Department of Defense, Department of the Army, or local command.

**b.** More than two colors. Use of multi-color will be approved only when it makes a decisive contribution to the intended purpose of the product. The criteria and standards for using more than two colors on the face of the disc should follow the guidance in AR 25-30.

**c.** Contractor/company logos or names.

### 5.6.2 Disc label data.

The disc shall have a label that includes at the following information (FIGURE 58):

**a.** Protective markings (5.6.2.1).

**b.** Service nomenclature (5.6.2.2).

**c.** Publication number and Nomenclature (5.6.2.3).

**d.** Short title (5.6.2.4).

**e.** Disc set number, as applicable (for example, "1 of 4") (5.6.2.5).

**f.** Supersession data (5.6.2.6).

**g.** Publication date (5.6.2.7).

**h.** Electronic Media (EM) number (5.6.2.7).

**i.** Initial Distribution Number (IDN) (5.6.2.7).

**j.** Publication Identification Number (PIN) (5.6.2.7).

**k.** (MC) Publication Control Number (PCN) (5.6.2.8).

**l.** Media type (5.6.2.9).

**m.** Operating system requirements (Windows 10 or higher, for example) (5.6.2.10).

**n.** Copyright information, as applicable (5.6.2.11).

**o.** Distribution Statement (5.6.2.11).

**p.** Arms Export Control Notice, as applicable (5.6.2.11).

**q.** Destruction Notice (5.6.2.11.)

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**r.** "See mailer" statement (5.6.2.12).

**s.** Designation box (5.6.2.13).

**FIGURE 58. Disc label.**

### 5.6.2.1 Protective markings.

The classification of the disc shall be placed at the top of disc in all capital letters. The disc shall bear the highest classification of any publications on that disc. If a disc is unclassified, classification may be omitted.

"CUI" shall be placed at the top of discs containing Controlled Unclassified Information.

Refer to DoDM 5200.01, volumes 1–3 and/or DoDI 5200.48 for further guidance on marking the disc classification.

### 5.6.2.2 Service nomenclature.

Service nomenclature shall appear in all capital letters after protective markings.

### 5.6.2.3 Publication number.

The publication number assigned to the IETM shall be listed exactly as assigned without "IETM," "ETM," or "EM" as the publication medium. The publication number does not contain any words.

### 5.6.2.4 Short title.

A short title shall be created for the disc using the equipment's weapon system and official nomenclature of the end item ("SHADOW 200 TACTICAL UNMANNED AIRCRAFT" for example). Do not include NSN, part number, or EIC.

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#### 5.6.2.5 Disc set number.

Discs shall be numbered. If there is one disc, number it "1 of 1." If there are two or more discs in the set, number them "1 of x," "2 of x," "3 of x," etc. All links will be functional across the disc set.

#### 5.6.2.6 Supersession data.

A supersession statement shall appear below the short title.

#### 5.6.2.7 Electronic media (EM) number, publication date, initial distribution number (IDN), and publication identification number (PIN).

EM number, publication date, IDN, and EM PIN shall be placed on the left-center side of the disc face.

#### 5.6.2.8 (MC) Publication Control Number (PCN).

For Marine Corps only IETMs and multi-service IETMs involving the Marine Corps, a PCN shall be put on the disc label. For Marine Corps only IETMs, it shall be put in place of the PIN. For multi-service manuals involving the Marine Corps, it shall appear just below the PIN.

#### 5.6.2.9 Media type.

The media type shall be included beneath the IDN and PIN on the disc label (for example, "This is a DVD").

#### 5.6.2.10 Operating system requirements.

The minimum operating system necessary to use the disc shall appear on the disc label ("Windows 10 or higher," for example).

#### 5.6.2.11 Copyright notice, distribution statement, arms export control notice, and destruction notice.

When one or more of these notices are applicable, the notices shall be identified by name on the disc label. Refer to DoDI 5230.24 for distribution statements and export control notice wording. The notices on the disc face shall be shortened to the notice heading with an asterisk placed in front of each.

#### 5.6.2.12 "See mailer and TM" statement.

The "See mailer and TM" statement points toward the full text for the distribution statement and notices that are noted above it on the disc label.

#### 5.6.2.13 CUI designation indicator.

If applicable, the disc shall include a CUI designation indicator. Refer to DODI 5200.48 for CUI policies, responsibilities, and procedures.

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### 5.6.3 Mailer data.

Discs shall be mailed inside a standard mailer with the following information:

#### 5.6.3.1 Front of mailer.

The front of the mailer (FIGURE 59) shall include the following information:

**a.** Return address.

**b.** Postage paid.

**c.** Official business.

**d.** EM number.

**e.** Publication number.

**f.** Publication date.

**g.** IDN.

**h.** Supersession notice (if applicable).

**i.** Handling instructions.

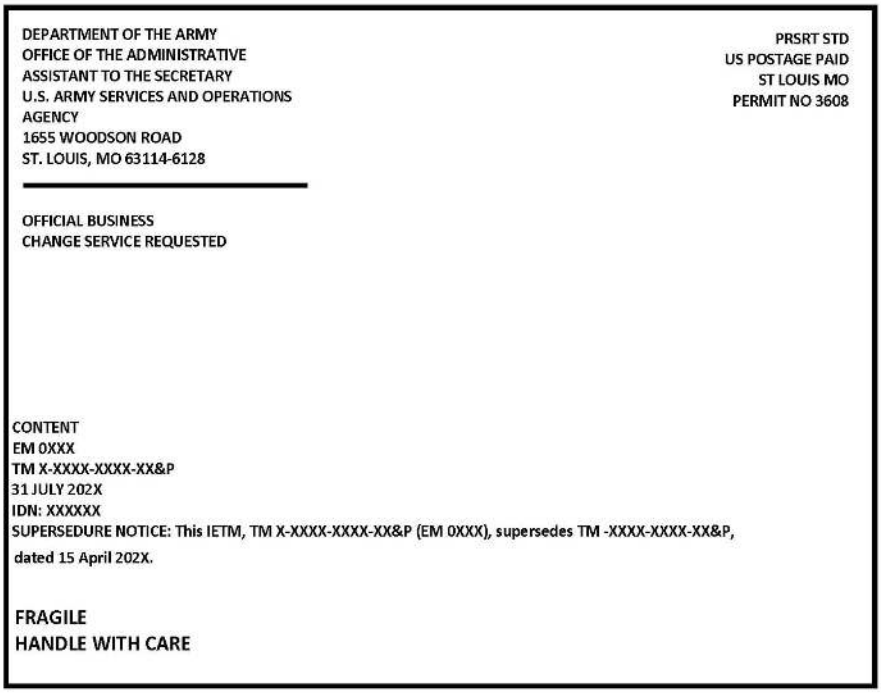


**FIGURE 59. Front of mailer.**

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### 5.6.3.2 Outside spine.

The spine shall contain the EM number, publication number(s) and the date of the publication (FIGURE 60).

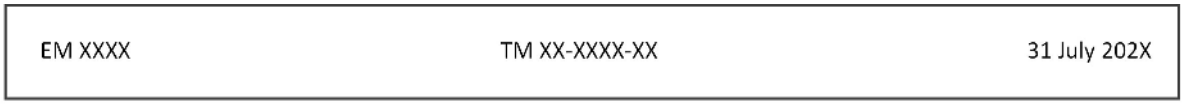



**FIGURE 60. Mailer outside spine.**

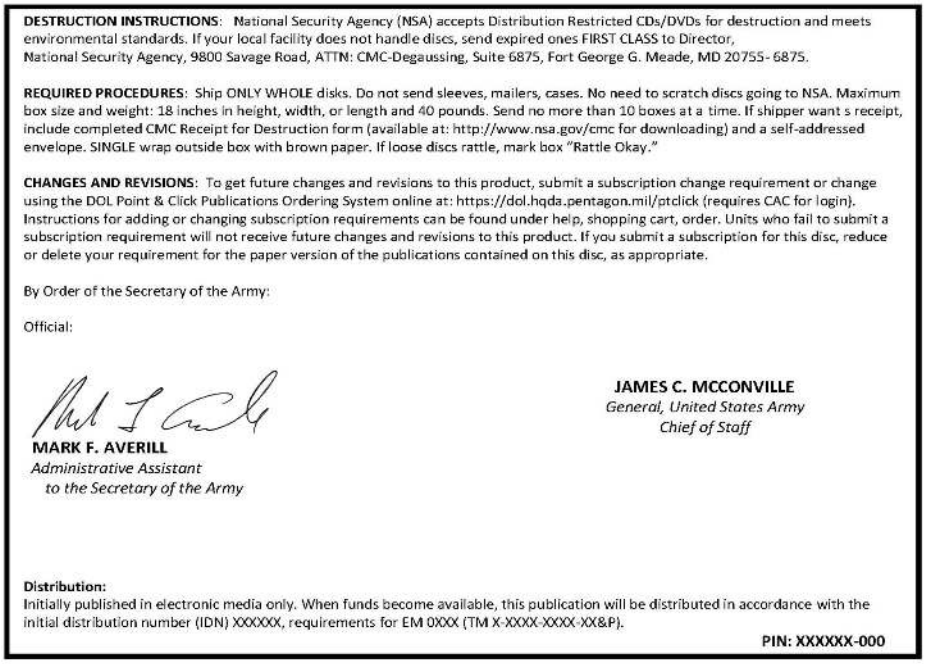
### 5.6.3.3 Back of mailer.

The following information shall be printed verbatim, using boldface and/or capital letters for emphasis as indicated. The authentication of the disc for all services involved shall be printed immediately following the verbatim text.

DESTRUCTION INSTRUCTIONS: National Security Agency (NSA) accepts Distribution Restricted CDs/DVDs for destruction and meets environmental standards. If your local facility does not handle discs, send expired ones FIRST CLASS to Director, National Security Agency, 9800 Savage Road, ATTN: CMC-Degaussing, Suite 6875, Fort George G. Meade, MD 20755–6875.
REQUIRED PROCEDURES: Ship ONLY WHOLE discs. Do not send sleeves, mailers, or cases. No need to scratch discs going to NSA. Maximum box size and weight: 18 inches in height, width, or length, and 40 pounds. Send no more than 10 boxes at a time. If shipper wants a receipt, include completed CMC Receipt for Destruction form (available at https://www. nsa. gov/cmc for downloading) and a self-addressed envelope. SINGLE wrap outside box with brown paper. If loose discs rattle, mark box "Rattle Okay."
CHANGES AND REVISIONS: To get future changes and revisions to this product, submit a subscription change requirement or change using the DOL Point & Click Publications Ordering System online at https://dol. hqda. pentagon. mikl/ptclick). Instructions for adding or changing subscription requirements can be found under help, shopping cart, order. Units who fail to submit a subscription requirement will not receive future changes and revisions to this product. If you submit a subscription for this disc, reduce or delete your requirement for the paper version of the publications contained on this disc (as appropriate).

The Army authentication number shall not be added to the authentication signature block on the mailer. Special distribution information shall be printed on the mailer immediately following the authentication. The EM PIN shall be printed in the lower right-hand corner of the mailer panel (FIGURE 61).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:03Z
Check the source to verify that this is the current version before use.



**DESTRUCTION INSTRUCTIONS**: National Security Agency (NSA) accepts Distribution Restricted CDs/DVDs for destruction and meets environmental standards. If your local facility does not handle discs, send expired ones FIRST CLASS to Director, National Security Agency, 9800 Savage Road, ATTN: CMC-Degaussing, Suite 6875, Fort George G. Meade, MD 20755- 6875.

**REQUIRED PROCEDURES**: Ship ONLY WHOLE disks. Do not send sleeves, mailers, cases. No need to scratch discs going to NSA. Maximum box size and weight: 18 inches in height, width, or length and 40 pounds. Send no more than 10 boxes at a time. If shipper want s receipt, include completed CMC Receipt for Destruction form (available at: http://www.nsa.gov/cmc for downloading) and a self-addressed envelope. SINGLE wrap outside box with brown paper. If loose discs rattle, mark box "Rattle Okay."

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By Order of the Secretary of the Army:

Official:

**MARK F. AVERILL**
*Administrative Assistant*
*to the Secretary of the Army*

**JAMES C. MCCONVILLE**
*General, United States Army*
*Chief of Staff*

**Distribution:**
Initially published in electronic media only. When funds become available, this publication will be distributed in accordance with the initial distribution number (IDN) XXXXXX, requirements for EM 0XXX (TM X-XXXX-XXXX-XX&P).

**PIN: XXXXXX-000**

**FIGURE 61. Back of mailer.**

### 5.6.3.4 Inside left panel.

The security classification or other protective markings (for example, CUI). Publication number(s), full publication title, IDN, and PIN shall be printed on the inside left panel towards the bottom in bold face type, immediately followed by full-text versions of the following: distribution statement, export control warning (if applicable), destruction notice (if applicable), copyright notice (if applicable), and CUI designation box (FIGURE 62).

Destruction Notice. Except for distribution statement A publications, use the following statement verbatim:

"DESTRUCTION NOTICE. For classified documents, follow the procedures in Part 117 of Title 32, Code of Federal Regulations. For unclassified, limited documents, destroy by any method that will prevent disclosure of contents or reconstruction of the document."

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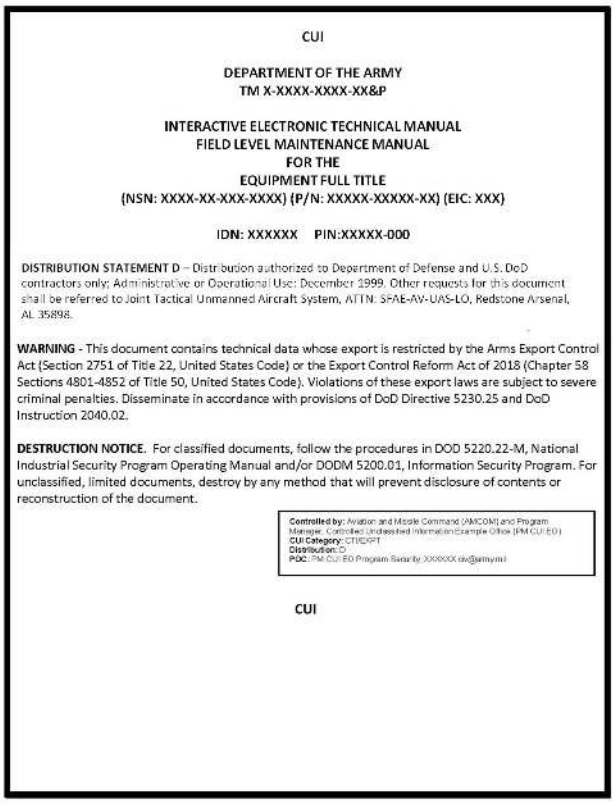

CUI

DEPARTMENT OF THE ARMY
TM X-XXXX-XXXX-XX&P

INTERACTIVE ELECTRONIC TECHNICAL MANUAL
FIELD LEVEL MAINTENANCE MANUAL
FOR THE
EQUIPMENT FULL TITLE
(NSN: XXXX-XX-XXX-XXXX) (P/N: XXXXX-XXXXX-XX) (EIC: XXX)

IDN: XXXXXX PIN:XXXXX-000

**DISTRIBUTION STATEMENT D** – Distribution authorized to Department of Defense and U.S. DoD contractors only; Administrative or Operational Use: December 1999. Other requests for this document shall be referred to Joint Tactical Unmanned Aircraft System, ATTN: SFAE-AV-UAS-LO, Redstone Arsenal, AL 35898.

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**CUI Category:** CTI/EXPT
**Distribution:** D
**POC:** PM CUI EO Program Security XXXXXX.civ@army.mil

CUI

**FIGURE 62. Mailer inside left panel.**

### 5.6.3.5 Inside right panel.

Detailed installation instructions for the IETM, viewer, and any other software required shall be printed on the inside right panel according to the specific software being used. Installation instructions shall include steps for uninstalling previous versions of the viewing software and the IETM. Instructions for uninstalling the disc shall also be included (FIGURE 63).

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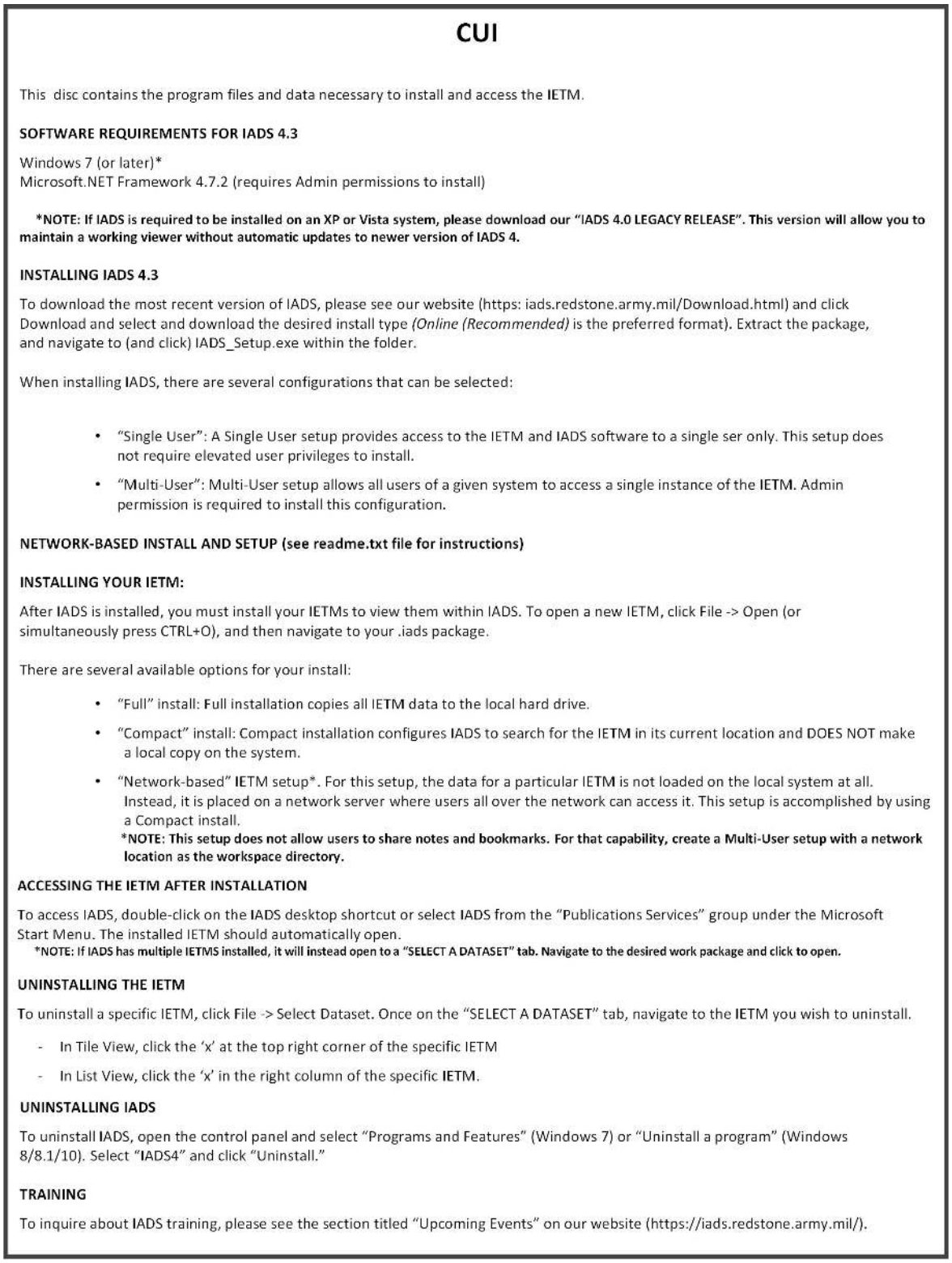

CUI

This disc contains the program files and data necessary to install and access the IETM.

**SOFTWARE REQUIREMENTS FOR IADS 4.3**

Windows 7 (or later)*
Microsoft.NET Framework 4.7.2 (requires Admin permissions to install)

***NOTE: If IADS is required to be installed on an XP or Vista system, please download our "IADS 4.0 LEGACY RELEASE". This version will allow you to maintain a working viewer without automatic updates to newer version of IADS 4.**

**INSTALLING IADS 4.3**

To download the most recent version of IADS, please see our website (https: iads.redstone.army.mil/Download.html) and click Download and select and download the desired install type *(Online (Recommended)* is the preferred format). Extract the package, and navigate to (and click) IADS_Setup.exe within the folder.

When installing IADS, there are several configurations that can be selected:

- "Single User": A Single User setup provides access to the IETM and IADS software to a single ser only. This setup does not require elevated user privileges to install.
- "Multi-User": Multi-User setup allows all users of a given system to access a single instance of the IETM. Admin permission is required to install this configuration.

**NETWORK-BASED INSTALL AND SETUP (see readme.txt file for instructions)**

**INSTALLING YOUR IETM:**

After IADS is installed, you must install your IETMs to view them within IADS. To open a new IETM, click File -> Open (or simultaneously press CTRL+O), and then navigate to your .iads package.

There are several available options for your install:

- "Full" install: Full installation copies all IETM data to the local hard drive.
- "Compact" install: Compact installation configures IADS to search for the IETM in its current location and DOES NOT make a local copy on the system.
- "Network-based" IETM setup*. For this setup, the data for a particular IETM is not loaded on the local system at all. Instead, it is placed on a network server where users all over the network can access it. This setup is accomplished by using a Compact install.
  ***NOTE: This setup does not allow users to share notes and bookmarks. For that capability, create a Multi-User setup with a network location as the workspace directory.**

**ACCESSING THE IETM AFTER INSTALLATION**

To access IADS, double-click on the IADS desktop shortcut or select IADS from the "Publications Services" group under the Microsoft Start Menu. The installed IETM should automatically open.
***NOTE: If IADS has multiple IETMS installed, it will instead open to a "SELECT A DATASET" tab. Navigate to the desired work package and click to open.**

**UNINSTALLING THE IETM**

To uninstall a specific IETM, click File -> Select Dataset. Once on the "SELECT A DATASET" tab, navigate to the IETM you wish to uninstall.

- In Tile View, click the 'x' at the top right corner of the specific IETM
- In List View, click the 'x' in the right column of the specific IETM.

**UNINSTALLING IADS**

To uninstall IADS, open the control panel and select "Programs and Features" (Windows 7) or "Uninstall a program" (Windows 8/8.1/10). Select "IADS4" and click "Uninstall."

**TRAINING**

To inquire about IADS training, please see the section titled "Upcoming Events" on our website (https://iads.redstone.army.mil/).

**FIGURE 63. Mailer inside right panel.**

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# 6 NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

## 6.1 Intended use.

MIL-STD-40051 prescribes requirements applicable to various types of technical publications and the revisions for these publications.

## 6.2 Acquisition requirements.

The acquisition document(s) should cite the following:

**a.** Title, number, and date of this standard.

**b.** Filled-out content selection matrix.

## 6.3 Tailoring guidance.

The acquiring activity in coordination with user representatives should tailor any required options offered herein in accordance with APPENDIX A.

## 6.4 Subject term (key word) listing.

The following terms are to be used to identify the MIL-STD-40051 document during retrieval searches:

**a.** Additional Authorization List (AAL).

**b.** Basic Issue Items (BII).

**c.** Basis of Issue (BOI).

**d.** Components of End Item (COEI).

**e.** Depot Maintenance Work Requirement (DMWR).

**f.** Expendable and Durable Items List.

**g.** eXtensible Markup Language (XML).

**h.** Illustrations.

**i.** Introductory information.

**j.** Maintenance Allocation Chart (MAC).

**k.** Maintenance instructions.

**l.** National Maintenance Work Requirement (NMWR).

**m.** Operator instructions.

**n.** Quality Assurance (QA).

**o.** Repair Parts and Special Tools List (RPSTL).

**p.** Security classification.

**q.** Supporting information.

**r.** Theory of operation.

**s.** Troubleshooting procedures.

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**t.** Work package (WP).

**u.** Work package identification number.

## 6.5 Changes from the previous issue.

Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content without regard to its relationship to previous issues.

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# APPENDIX A

# TECHNICAL MANUAL REQUIREMENTS WITH CONTENT SELECTION MATRICES

## A.1 SCOPE.

This appendix includes Interactive Electronic Technical Manual (IETM) functionality requirements that outline the presentation of information on a computer display. This appendix also includes the requirements for Electronic Technical Manual (ETM) and IETM content selection for all types of equipment, including test and support equipment. This appendix is a mandatory part of this standard. The information contained herein is intended for compliance. These requirements are applicable for all maintenance levels through sustainment, including Depot Maintenance Work Requirements (DMWRs) / National Maintenance Work Requirements (NMWRs).

## A.2 APPLICABLE DOCUMENTS.

The applicable documents in Section 2 apply to this appendix.

## A.3 ACRONYMS AND DEFINITIONS.

The acronyms and terms defined in Section 3 of this standard apply to this appendix.

## A.4 ETM AND IETM REQUIREMENTS.

Sections A.4.1 through A.4.4 discuss requirements for ETM and IETMs.

### A.4.1 Tailoring requirements for technical manuals.

Tailoring of the technical content requirements contained in APPENDIX B through APPENDIX Q is provided in the content matrix tables, TABLE A-II through TABLE A-XX. The tables list all applicable technical content requirements for the development of technical publications.

### A.4.2 Publication titles.

Publication titling shall include the following requirements (as applicable):

**a.** All publications prepared in accordance with this standard shall have a title in accordance with TABLE A-I.

**b.** All publications with a Repair Parts and Special Tools List (RPSTL) shall spell out the RPSTL acronym. Refer to FIGURE A-1.

**c.** All publications with "field" in the title shall include instructions for a maintainer ("field" and "maintainer" are synonymous for publication titling purposes). Refer to FIGURE A-1.

**Note**

Army Training and Doctrine Command (TRADOC) maintainer representative approval is required for any inclusion of sustainment-level maintenance. If approval is granted, "sustainment" may be included in title as described below.

**d.** If a publication contains sustainment information, the publication title shall include "Sustainment" in the title (examples listed below):

**(1)** A -12 TM that includes sustainment information shall be titled "Field and Sustainment Maintenance Manual for *insert system*."

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**(2)** A -12&P TM or -20&P TM with depot parts shall be titled "Field and Sustainment Maintenance Manual with Repair Parts and Special Tools List (RPSTL) Including Depot Repair Parts and Special Tools *insert system*. "

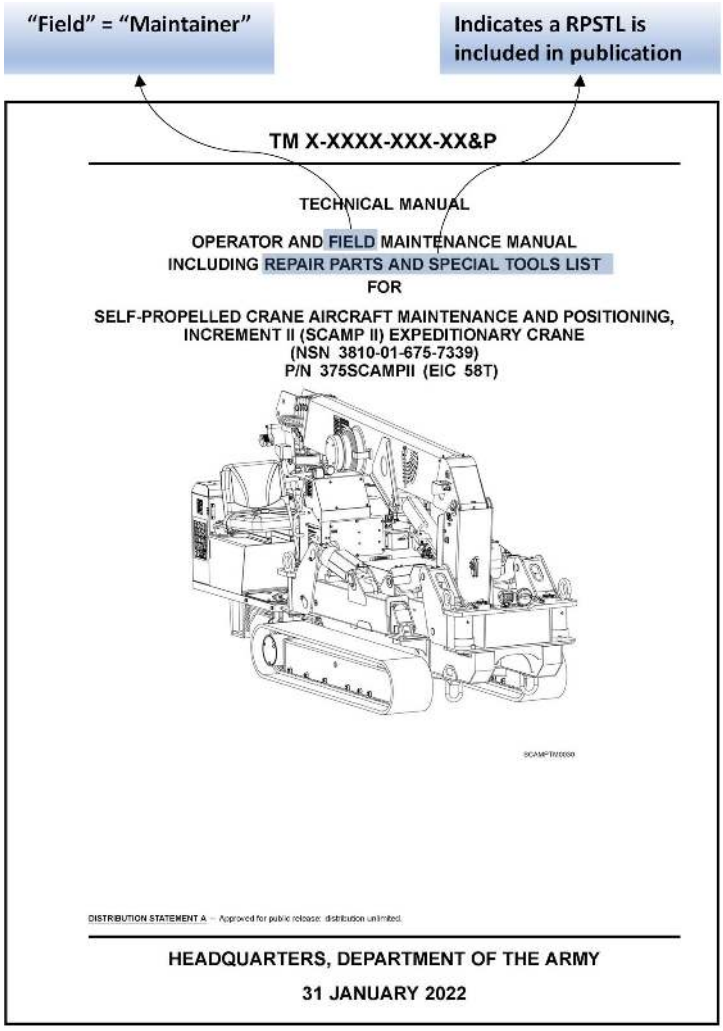


**FIGURE A-1. Example of publication title on TM cover.**

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## A.4.3 Intended use.

This section explains the functions of various elements of technical publication content selection matrices. Information in A.4.3.1 applies to every matrix in this appendix. All publications shall have titles in accordance with TABLE A-I. If work packages (WPs), modules, or components are not included in a matrix for a publication, they shall not be included in that publication. Technical publication content selection matrices become contractually binding when they are included in contract documentation, statements of work, or any other contractual instrument. Refer to A.4.3.2 for acquisition requirements and matrix edits.

### A.4.3.1 Matrix components.

This section outlines matrix components and discusses the meaning of various cell contents.

#### A.4.3.1.1 Matrix components - front and rear matter.

Front and rear matter of a publication lists available components at the beginning of a publication.

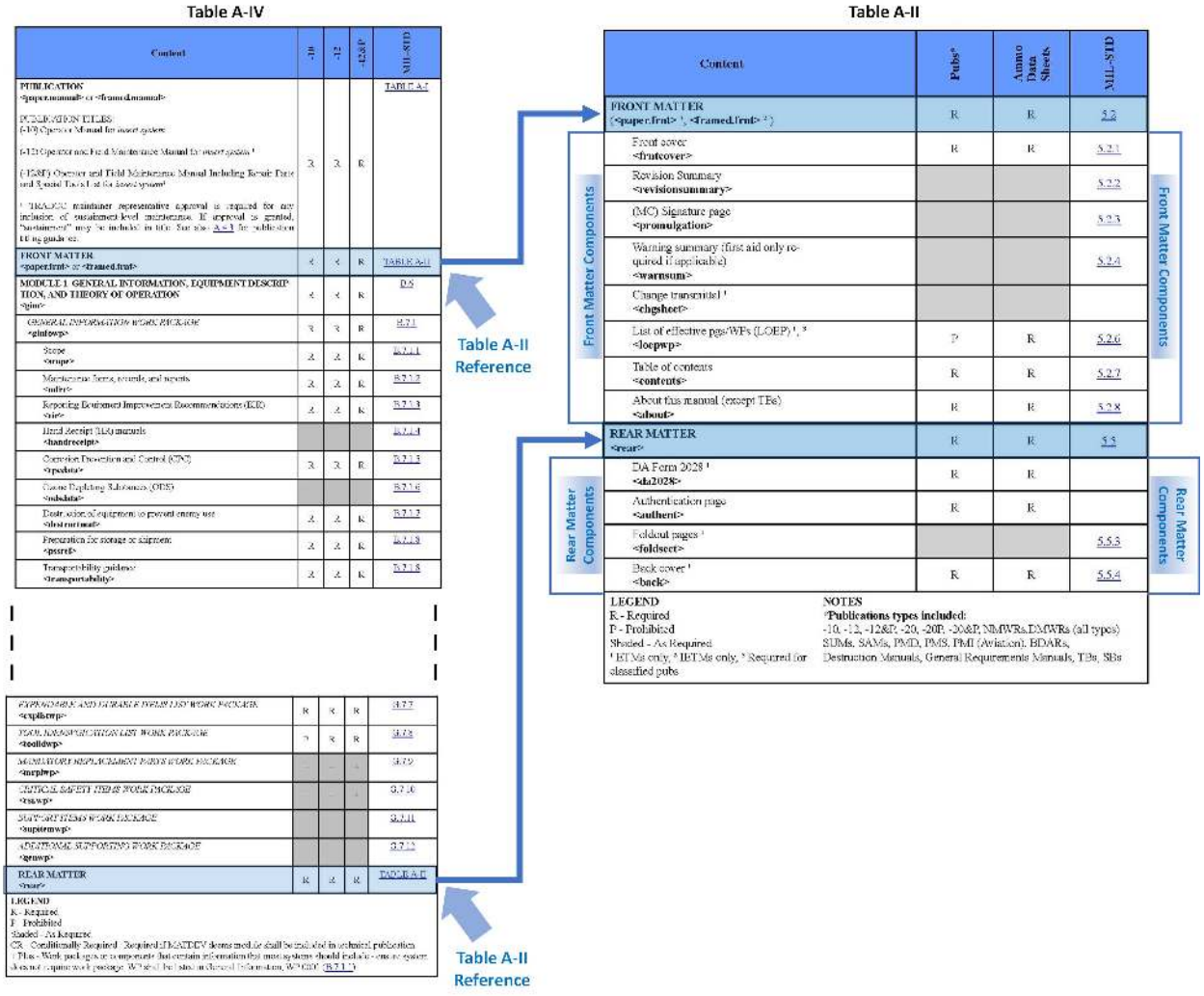


**FIGURE A-2. Front and rear matter.**

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### A.4.3.1.2 Matrix components - individual publication components.

Modules are composed of individual WPs, and publications are composed of individual modules. These components are discussed below and shown in FIGURE A-3.

**a. WP**: Individual components of a publication that make up a module. The content of WPs vary based on module function.

**b. Module**: Individual components that make up a publication. Modules are comprised of individual WPs that provide information relevant to a module type.

**c. Module "X"**: Designates that, if required, at least one of these types of modules shall be in a publication.

The following is also true for module Xs:

- Module Xs that correlate to a shaded cell for a technical publication shall be included based on materiel developer (MATDEV) guidance. See shaded cells discussed in A.4.3.1.4 through A.4.3.1.4.3.
- Publications may contain more than one of module X.
- For publications with multiple levels of maintenance (for example, -12, -12&P, and -20&P TMs), modules may be utilized to divide information according to maintenance level.

  EXAMPLE: A publication with both operator and maintainer information may have a "MAINTENANCE INSTRUCTIONS" module dedicated to an operator and a second module dedicated to a maintainer.
- If more than one of a module X type is needed, then a required ("R") item listed within that module is only required one time. These items may appear in additional modules based on MATDEV guidance; however, it is not required.

  EXAMPLE: If a publication has more than one "MAINTENANCE INSTRUCTIONS" modules for maintenance tasks (one module that covers operator tasks and second module that covers maintainer tasks), only one of these modules is required to have "Service Upon Receipt" WP (for -12 TMs and above). The MATDEV may determine if both modules need this WP type.

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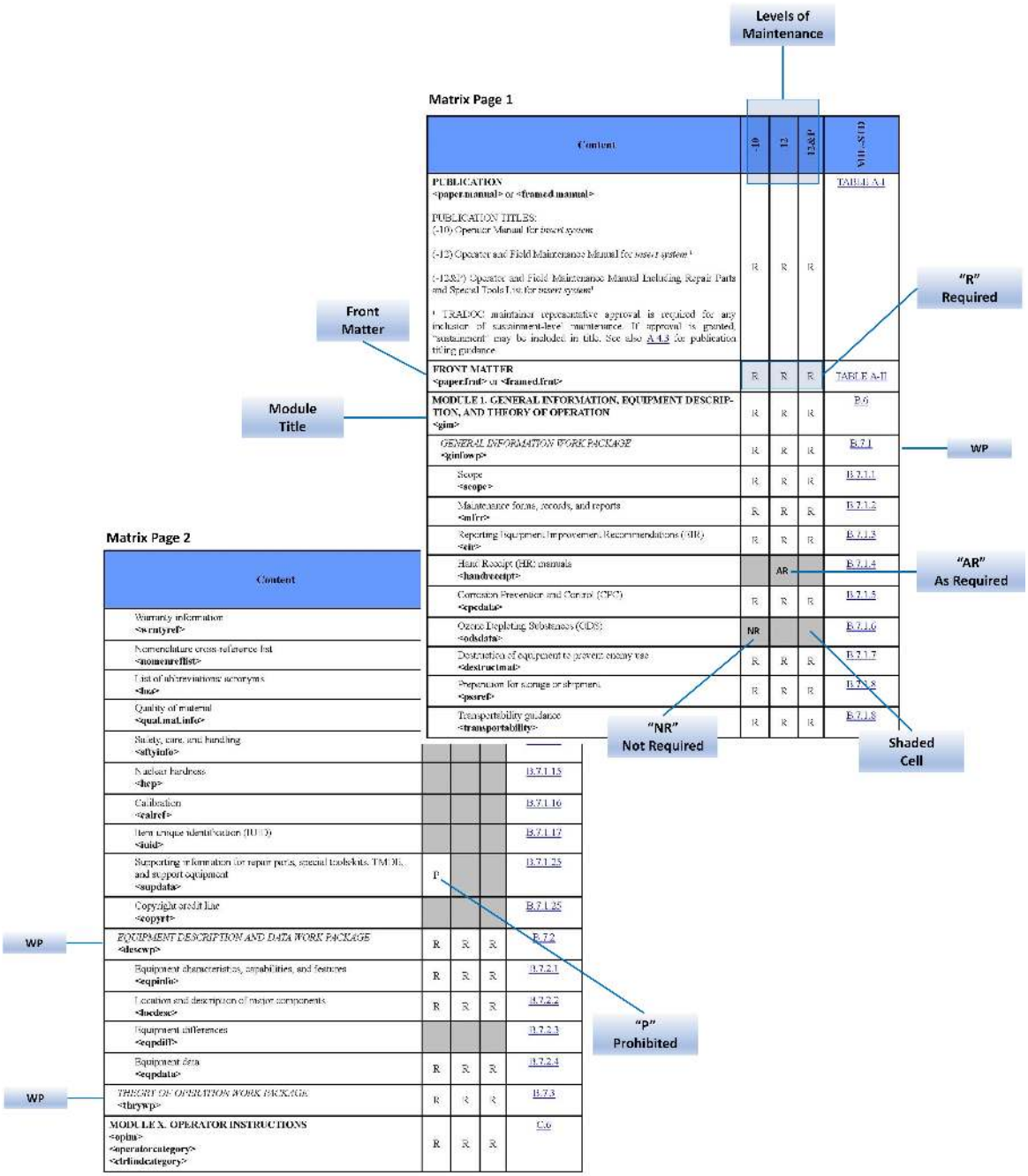


**FIGURE A-3. Content selection matrix components (sheet 1 of 2).**

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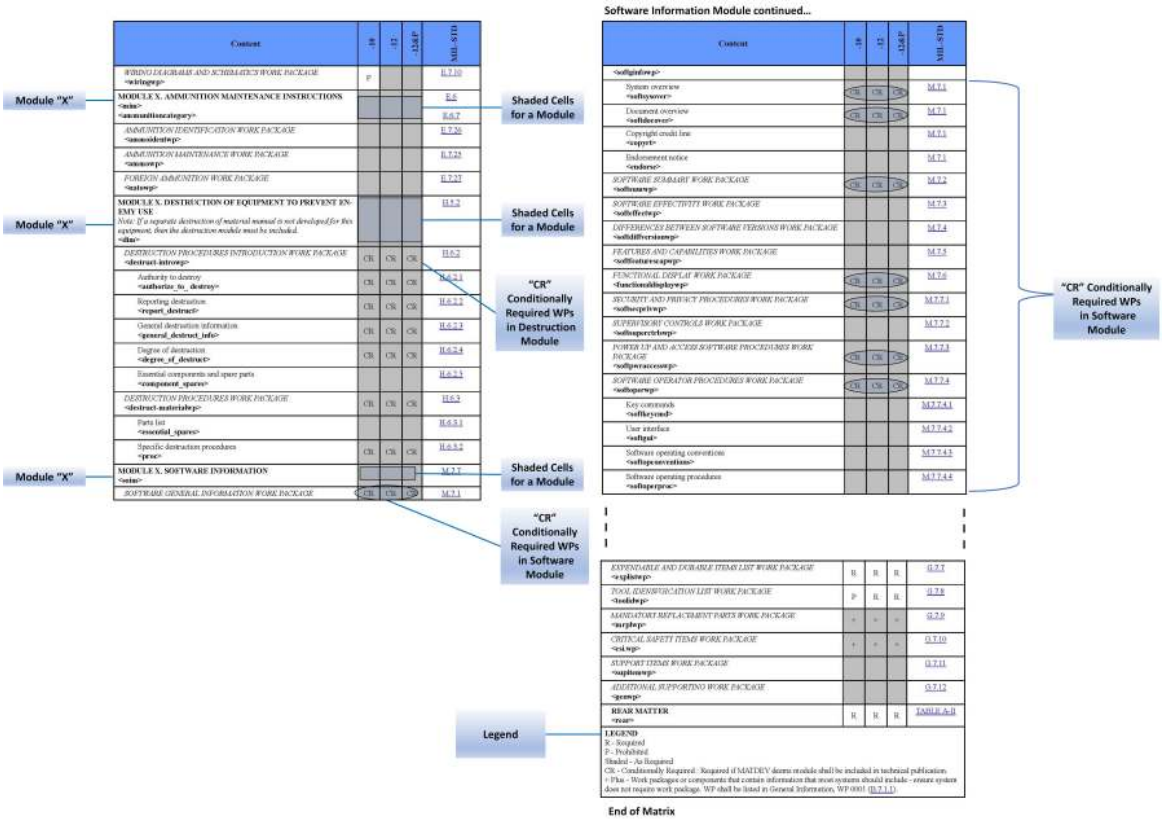


**FIGURE A-4. Content selection matrix components (sheet 2 of 2).**

### A.4.3.1.3 Matrix components - required ("R") and prohibited ("P").

Matrix cells with an "R" designate content that shall be included in a document. Matrix cells with an "P" designate content that is prohibited from inclusion in a document. Both "R" and "P" are listed in publication legends and shown in FIGURE A-3.

### A.4.3.1.4 Matrix components - shaded cells.

Shaded matrix cells designate content that may be included in a document. Shaded cells are listed in publication legends and shown in FIGURE A-3. If a particular equipment system has a need for a WP that is shaded in a matrix, the WP then shall become required ("R") based on MATDEV guidance (consult the MATDEV for any question regarding inclusion of shaded WPs in a publication). Sections A.4.3.1.4.1 through A.4.3.1.4.3 outline specific types of shaded cells.

### A.4.3.1.4.1 Shaded cells with a plus (+) sign.

Shaded cells with a plus (+) sign designate content that contains information that most systems typically include. For these cells, technical writers shall research to ensure a specific equipment system does not require these sections. If content is not included in a technical publication, the "SCOPE" section of WP 0001 General Information shall include a statement that content is not applicable. FIGURE A-5 shows example Mandatory Replacement Parts (MRP) or Critical Safety Items (CSI) entries. If these WPs are not included in a publication, WP 0001 General Information WP, Scope shall list these items as not applicable. Refer to B.7.1.1 for more information.

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| Content | -10 | -12 | -12&P | MIL-STD |
|---|---|---|---|---|
| *POWER DOWN AND EXIT SOFTWARE PROCEDURES WORK PACKAGE* **<softpwrexitwp>** | CR | CR | CR | M.7.7.5 |
| *MESSAGES WORK PACKAGE* **<softmessageswp>** | | | | M.7.8.3 |
| *RECOVERY FROM ERRORS, MALFUNCTIONS, AND EMERGENCIES WORK PACKAGE* **<softerrorswp>** | | | | M.7.8.4 |
| **MODULE X. PARTS INFORMATION** **<pim>** | P | P | R | F.7.2 |
| *INTRODUCTION WORK PACKAGE* **<introwp>** | P | P | R | F.8.1 |
| *REPAIR PARTS LIST WORK PACKAGE* **<plwp>** | P | P | R | F.8.2 |
| *REPAIR PARTS FOR SPECIAL TOOLS WORK PACKAGE* **<stl_partswp>** | P | P | | F.8.3 |
| *KIT PARTS LIST WORK PACKAGE* **<kitswp>** | P | P | | F.8.4 |
| *BULK ITEM WORK PACKAGE* **<bulk_itemswp>** | P | P | | F.8.5 |
| *SPECIAL TOOLS LIST WORK PACKAGE* **<stlwp>** | P | P | | F.8.6 |
| **MODULE X. SUPPORTING INFORMATION** **<sim>** | R | R | R | G.6 |
| *REFERENCES WORK PACKAGE* **<refwp>** | R | R | R | G.7.1 |
| *INTRODUCTION FOR TWO-LEVEL MAINTENANCE ALLOCATION CHART (MAC) WORK PACKAGE* **<macintrowp>** | P | R | R | G.7.2 |
| *MAINTENANCE ALLOCATION CHART (MAC) WORK PACKAGE* **<macwp>** | P | R | R | G.7.3 |
| *COMPONENTS OF END ITEM (COEI) AND BASIC ISSUE ITEMS (BII) LISTS WORK PACKAGE* **<coeibiiwp>** | R | R | R | G.7.4 |
| *ADDITIONAL AUTHORIZATION LIST (AAL) WORK PACKAGE* **<aalwp>** | | | | G.7.5 |
| *EXPENDABLE AND DURABLE ITEMS LIST WORK PACKAGE* **<explistwp>** | R | R | R | G.7.7 |
| *TOOL IDENSVGICATION LIST WORK PACKAGE* **<toolidwp>** | P | R | R | G.7.8 |
| *MANDATORY REPLACEMENT PARTS WORK PACKAGE* **<mrplwp>** | + | + | + | G.7.9 |
| *CRITICAL SAFETY ITEMS WORK PACKAGE* **<csi.wp>** | + | + | + | G.7.10 |
| *SUPPORT ITEMS WORK PACKAGE* **<supitemwp>** | | | | G.7.11 |
| *ADDITIONAL SUPPORTING WORK PACKAGE* **<genwp>** | | | | G.7.12 |
| **REAR MATTER** **<rear>** | R | R | R | TABLE A-II |

**LEGEND**
R - Required
P - Prohibited
Shaded - As Required
CR - Conditionally Required : Required if MAITDEV deems module shall be included in technical publication.
+ Plus - Work packages or components that contain information that most systems should include - ensure system does not require work package. WP shall be listed in General Information, WP 0001 (B.7.1.1).

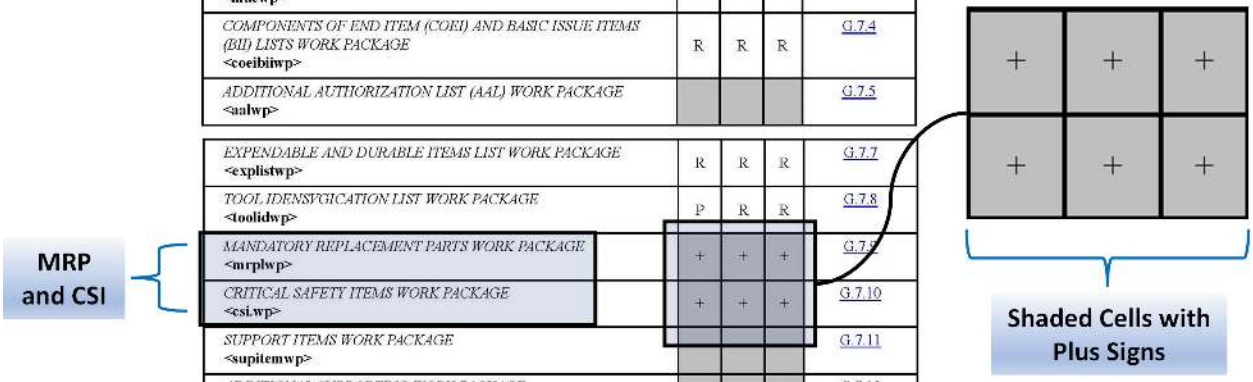


**FIGURE A-5. Mandatory replacement parts shown in matrix.**



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### A.4.3.1.4.2 Shaded cells for illustrated list of manufactured items.

If a system requires manufactured parts, then the shaded cells (shown in FIGURE A-6) do not designate the two WPs (listed below the "Illustrated List of Manufactured Items" entry) as optional. These WPs are required based on system-specific logistics data (regardless of cell shading). Further, both WPs must be represented in the technical publication (introduction and the manufacturing procedures WP) and are designated as Conditionally Required ("CR"). Conditionally required cells are explained further below.

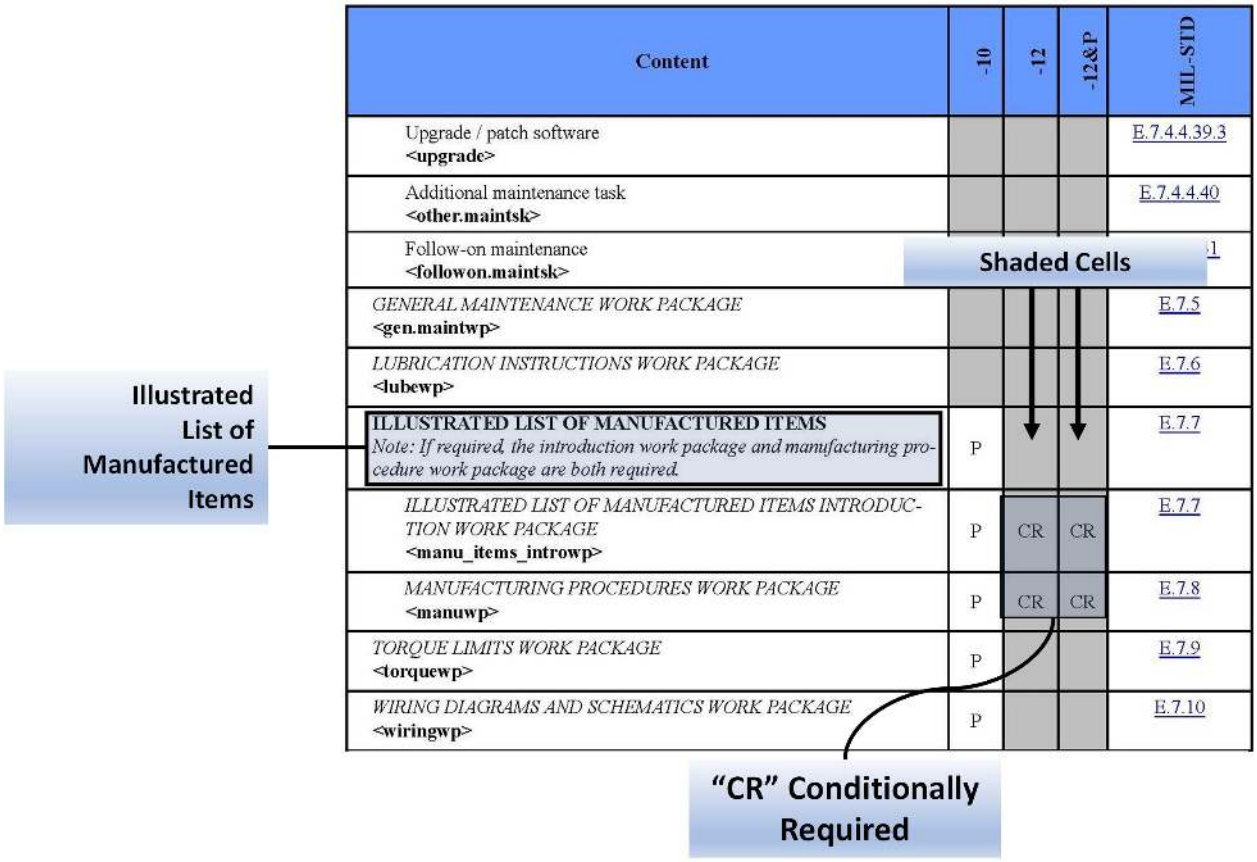


**FIGURE A-6. Illustrated list of manufactured items shown in matrix.**

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#### A.4.3.1.4.3 Shaded cells - conditionally required ("CR" components.

Shaded cells with "CR" designate content that shall be handled differently than items marked with just an "R." These items are conditionally required which means they are only required if the "parent" of that item is included in a publication. For example, see the software information module (Module "X" callout) in FIGURE A-4. As this example demonstrates, all technical publications are not required to include a software information module; thus, this module is shaded in the matrices. However, if a MATDEV decides that this module shall be included in a technical publication, then this module becomes the "parent" and every individual WP (or "children") marked with a "CR" (FIGURE A-4) shall be included in that software module.

### A.4.3.2 Acquisition requirements and matrix edits.

As stated in section A.6.4 above, technical publication content selection matrices become contractually binding when they are included in contract documentation, statements of work, or any other contractual instrument. Refer to https://www.ldac.army.mil for editable matrices that shall be filled out as follows:

**a.** For tables with multiple columns allowing a choice of technical publication type (for example, -10, -12, -12&P), unused columns shall be either removed or blacked out.

**b.** All components with shaded cells shall be identified and labeled as one of the following based on MATDEV directions:

**(1) R**: Designates content that shall be included in a document (shown in FIGURE A-3).

**(2) NR**: Designates content that shall not be included in a document (shown in FIGURE A-3).

**(3) AR**: Designates content that may be included in a document pending later directions by an MATDEV (shown in FIGURE A-3).

**(4) Module X**: Designates modules that may be duplicated in editable matrices (shown in FIGURE A-4).

### A.4.3.3 Example publication components and structure.

FIGURE A-7 is a diagram of an example TM -12&P that shows the following:

**a.** Example modules (**<gim>, <opim>, <tim>**, etc.).

**b.** Example WPs that may be included in modules based on technical publication content selection matrices. The example WPs listed below all belong to the **<gim>** module.

**(1)** General Information WP **<ginfowp>**.

**(2)** Equipment Description and Data WP **<descwp>**.

**(3)** Theory of Operation WP **<thrywp>**.

**Note**

The "Xs" in the example WP identification numbers represent publication numbers.

**c.** Example eXtensible Markup Language (XML) identification numbering (G00001-X-XXXX-XXX-12nP, G00002-X-XXXX-XXX-12nP, etc.) for these WPs is based on guidance in general requirements of this standard. These numbers are the wpno attribute value within each root WP tag (**<xxxwp>**) – located at the beginning of WP tagging structure – and must be unique in each individual WP. The numbers must also be sequential throughout the pub (for example, G00001... through S00100). FIGURE A-8 and FIGURE A-9 provide more tagging details for example **<gim>** and **<opim>** modules and their respective WPs. To aid with configuration management of publications, it is recommended that WP identification numbering include the maintenance level of publications as shown in FIGURE A-7.

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Check the source to verify that this is the current version before use.

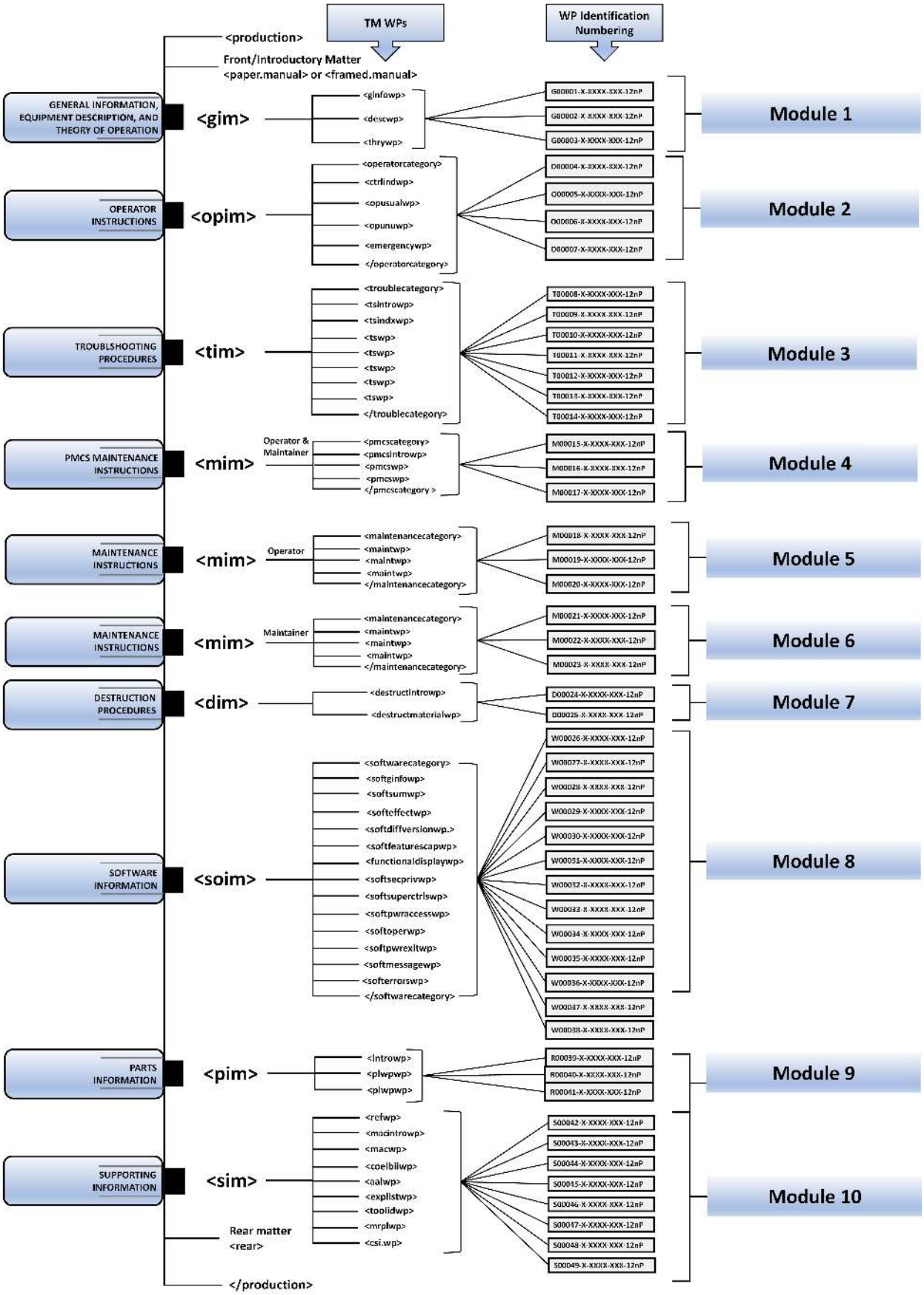


**FIGURE A-7. Example -12&P TM.**

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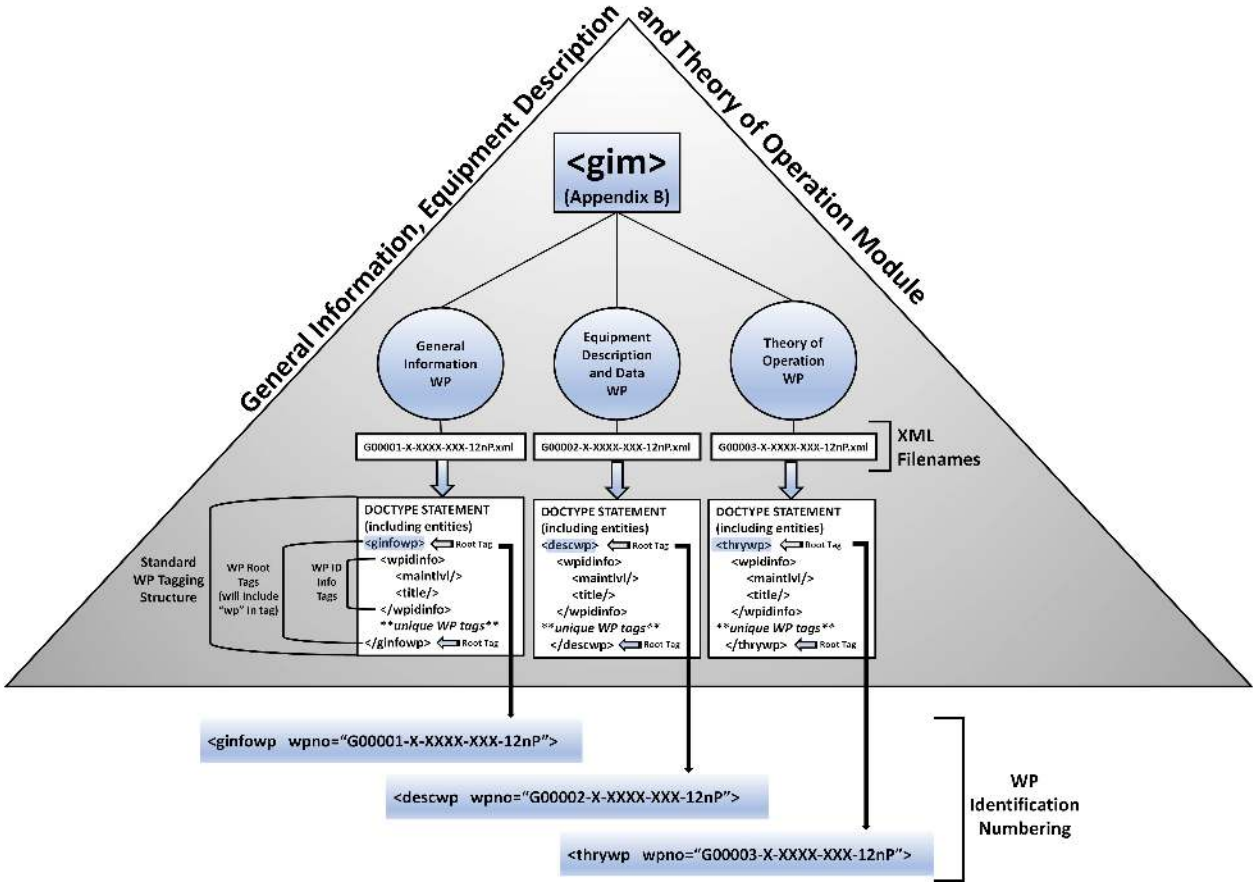


**FIGURE A-8. Example -12&P TM – module 1 (<gim>).**

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Check the source to verify that this is the current version before use.

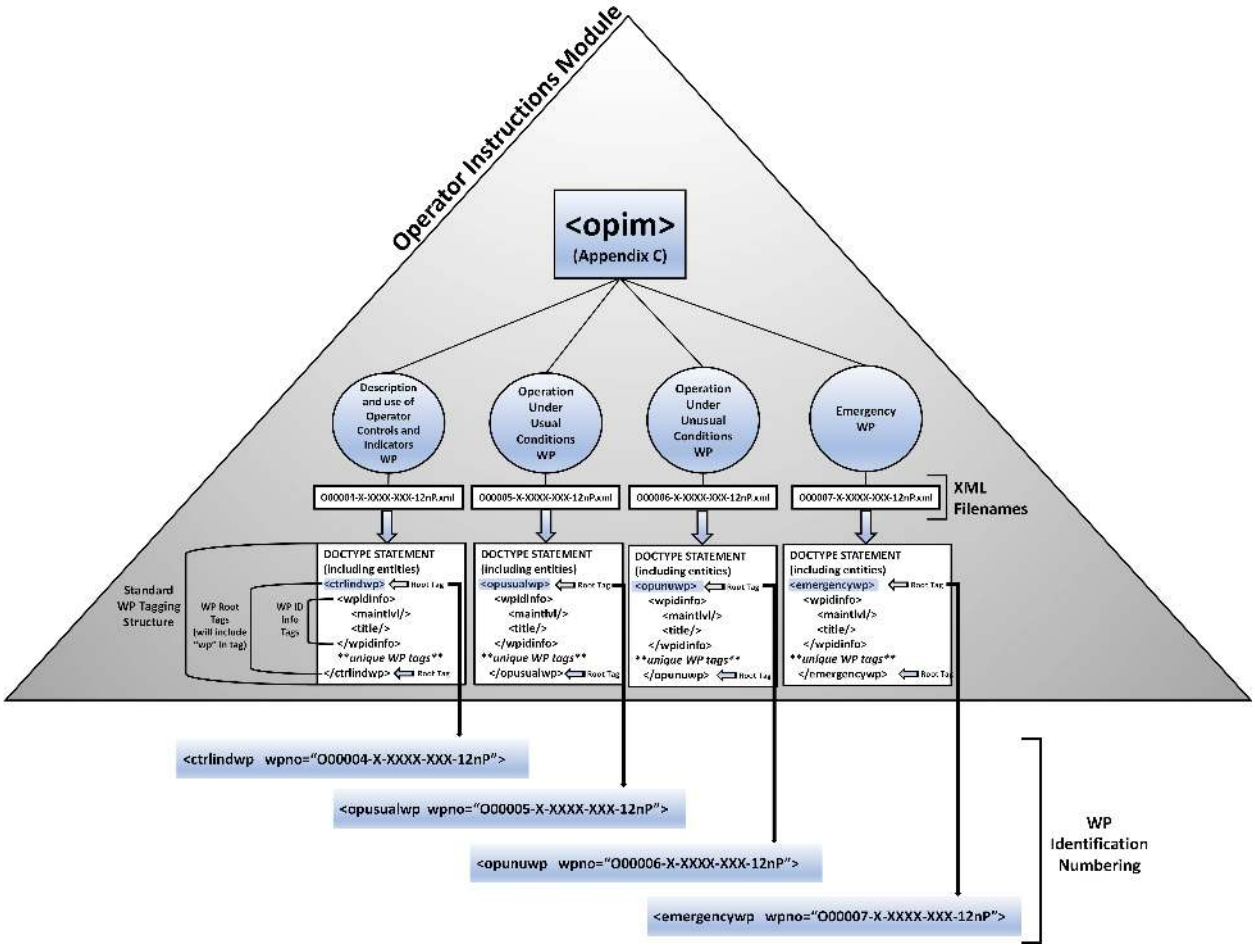


**FIGURE A-9. Example -12&P TM - module 2 (<opim>).**

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### A.4.4 Content selection matrices (tables.

Content selection matrices shall be used to determine content requirements for each type of technical publication. Each table is a matrix that lists components of a technical publication and serves as a checklist to build a publication. Type and complexity of equipment as well as maintenance level determine which components shall be included in the final product. This section describes these matrices and how to use them. All of these matrices are available for download at https://www.ldac.army.mil.

**TABLE A-I. Publication type and associated context matrix tables.**

| PUBLICATION TYPE | REQUIREMENTS MATRIX | PUBLICATION TITLES |
|---|---|---|
| **GENERAL PUBLICATIONS** | | |
| ALL | TABLE A-II | Front Matter and Rear Matter |
| -10 | TABLE A-III | Operator Manual for *insert system* |
| -12 | | Operator and Field Maintenance Manual for *insert system* [1] |
| -12&P | | Operator and Field Maintenance Manual including Repair Parts and Special Tools List (RPSTL) for *insert system* [1] |
| -20 | TABLE A-IV | Field Maintenance Manual for *insert system*[1] |
| -20&P | | Field Maintenance Manual including Repair Parts and Special Tools List (RPSTL) for *insert system* [1] |
| -20P | TABLE A-V | Field Maintenance Repair Parts and Special Tools List (RPSTL) for *insert system* [1] |
| DMWR or NMWR with RPSTL | TABLE A-VI | Depot Maintenance Work Requirements (DMWR) including Repair Parts and Special Tools List (RPSTL) for *insert system* [2] |
| | | National Maintenance Work Requirements (NMWR) including Repair Parts and Special Tools List (RPSTL) for *insert system* [2] |
| DMWR / NMWR | TABLE A-VII | Depot Maintenance Work Requirements (DMWR) for *insert system* [2] |
| | | National Maintenance Work Requirements (NMWR) for *insert system* [2] |
| | | Depot Maintenance Work Requirement Containing National Overhaul Standards *insert system* [2] |
| SUM/SAM | TABLE A-VIII | Software Users Manual (SUM) for *insert system* [2] |
| | | Software Administrators Manual (SAM) for *insert system* [2] |
| | | Software Users and Administrators Manual (SUM/SAM) for *insert system* [2] |
| **ARMAMENT PUBLICATIONS** | | |
| Ammunition DMWR | TABLE A-IX | Depot Maintenance Work Requirements (DMWR) for Demilitarization of *insert system* |
| | | Depot Maintenance Work Requirements (DMWR) for Maintenance of *insert system* |
| Ammunition Data Sheets | TABLE A-X | Army Ammunition Data Sheets for *insert system* |
| | | Army Munitions Equipment Data Sheets for *insert system* |

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**TABLE A-I. Publication type and associated context matrix tables. (continued**

| PUBLICATION TYPE | REQUIREMENTS MATRIX | PUBLICATION TITLES |
|---|---|---|
| **AVIATION PUBLICATIONS** | | |
| -20, -20&P | TABLE A-XI | Field Maintenance Manual for *insert system* [1] |
| | | Field Maintenance Manual including RPSTL for *insert system* [1] |
| Preventive Maintenance | TABLE A-XII | Preventive Maintenance Daily (PMD) Manual for *insert system* |
| | | Preventive Maintenance Services (PMS) Manual for *insert system* |
| Phased Maintenance | TABLE A-XIII | Phased Maintenance Inspection (PMI) Checklist for *insert system* |
| **OTHER MAINTENANCE** | | |
| BDAR | TABLE A-XIV | *Insert maintenance level* Battle Damage Assessment and Repair for *insert system* |
| Destruction | TABLE A-XV | Destruction of Equipment to Prevent Enemy Use for *insert system* |
| Lubrication | TABLE A-XVI | Lubrication Orders for *insert system* |
| Preventive Maintenance | TABLE A-XVII | Preventive Maintenance Checklists (PMC) for *insert system* |
| General Maintenance | TABLE A-XVIII | General Maintenance Manual for *insert system* |
| Hand Receipt | TABLE A-XIX | Hand Receipt Covering Contents of Components of End Item (COEI), Basic Issue Items (BII), and Additional Authorization List (AAL) for *short end item title* |
| Technical Bulletin | TABLE A-XX | Technical Bulletin for *insert system* |

**NOTE**
[1] Army Training and Doctrine Command (TRADOC) maintainer representative approval is required for any inclusion of sustainment-level maintenance. If approval is granted, "sustainment" may be included in title. See also A.6.2 for publication titling guidance.
[2] Includes aviation.

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**TABLE A-II. Front matter and rear matter requirements matrix.**

| Content | Pubs* | Ammo Data Sheets | MIL-STD |
|---|---|---|---|
| **FRONT MATTER**<br>(**<paper.frnt>** [1], **<framed.frnt>** [2] ) | R | R | 5.2 |
| Front cover<br>**<frntcover>** | R | R | 5.2.1 |
| Revision Summary<br>**<revisionsummary>** | | | 5.2.2 |
| (MC) Signature page<br>**<promulgation>** | | | 5.2.3 |
| Warning summary (first aid only required if applicable)<br>**<warnsum>** | | | 5.2.4 |
| Change transmittal [1]<br>**<chgsheet>** | | | 5.2.5 |
| List of effective pgs/WPs (LOEP) [1], [3]<br>**<loepwp>** | P | R | 5.2.6 |
| Table of contents<br>**<contents>** | R | R | 5.2.7 |
| About this manual (except TBs)<br>**<about>** | R | R | 5.2.8 |
| **REAR MATTER**<br>**<rear>** | R | R | 5.5 |
| DA Form 2028 [1]<br>**<da2028>** | R | R | 5.5.1 |
| Authentication page<br>**<authent>** | R | R | 5.5.2 |
| Foldout pages [1]<br>**<foldsect>** | | | 5.5.3 |
| Back cover [1]<br>**<back>** | R | R | 5.5.4 |

**LEGEND**
R - Required
P - Prohibited
Shaded - As Required
[1] ETMs only, [2] IETMs only, [3] Required for classified pubs

**NOTES**
***Publications types included:**
-10, -12, -12&P, -20, -20P, -20&P, NMWRs,DMWRs (all types) SUMs, SAMs, PMD, PMS, PMI (Aviation), BDARs, Destruction Manuals, General Requirements Manuals, TBs, SBs

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**TABLE A-III. Operator and combined operator/maintainer maintenance requirements matrix.**

| Content | -10 | -12 | -12&P | MIL-STD REF |
|---|---|---|---|---|
| **PUBLICATION** (**<production>**)<br>**<paper.manual>** or **<framed.manual>**<br><br>PUBLICATION TITLES:<br>(-10) Operator Manual for *insert system*<br><br>(-12) Operator and Field Maintenance Manual for *insert system* [1]<br><br>(-12&P) Operator and Field Maintenance Manual Including Repair Parts and Special Tools List for *insert system*[1]<br><br>[1] TRADOC maintainer representative approval is required for any inclusion of sustainment-level maintenance. If approval is granted, "sustainment" may be included in title. See also A.4.3 for publication titling guidance. | R | R | R | TABLE A-I |
| **FRONT MATTER**<br>**<paper.frnt>** or **<framed.frnt>** | R | R | R | TABLE A-II |
| **MODULE 1. GENERAL INFORMATION, EQUIPMENT DESCRIPTION, AND THEORY OF OPERATION**<br>**<gim>** | R | R | R | B.6 |
| *GENERAL INFORMATION WORK PACKAGE*<br>**<ginfowp>** | R | R | R | B.7.1 |
| Scope<br>**<scope>** | R | R | R | B.7.1.1 |
| Maintenance forms, records, and reports<br>**<mfrr>** | R | R | R | B.7.1.2 |
| Reporting Equipment Improvement Recommendations (EIR)<br>**<eir>** | R | R | R | B.7.1.3 |
| Hand Receipt (HR) manuals<br>**<handreceipt>** | | | | B.7.1.4 |
| Corrosion Prevention and Control (CPC)<br>**<cpcdata>** | R | R | R | B.7.1.5 |
| Ozone Depleting Substances (ODS)<br>**<odsdata>** | | | | B.7.1.6 |
| Destruction of equipment to prevent enemy use<br>**<destructmat>** | R | R | R | B.7.1.7 |
| Preparation for storage or shipment<br>**<pssref>** | R | R | R | B.7.1.8 |

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**TABLE A-III. Operator and combined operator/maintainer maintenance requirements matrix. (continued**

| Content | -10 | -12 | -12&P | MIL-STD REF |
|---|---|---|---|---|
| Transportability guidance<br>**<transportability>** | R | R | R | B.7.1.8 |
| Warranty information<br>**<wrntyref>** | | | | B.7.1.9 |
| Nomenclature cross-reference list<br>**<nomenreflist>** | | | | B.7.1.10 |
| List of abbreviations/ acronyms<br>**<loa>** | R | R | R | B.7.1.11 |
| Quality of material<br>**<qual.mat.info>** | P | R | R | B.7.1.13 |
| Safety, care, and handling<br>**<sftyinfo>** | | | | B.7.1.14 |
| Nuclear hardness<br>**<hcp>** | | | | B.7.1.15 |
| Calibration<br>**<calref>** | | | | B.7.1.16 |
| Item unique identification (IUID)<br>**<iuid>** | | | | B.7.1.17 |
| Supporting information for repair parts, special tools/kits, TMDE, and support equipment<br>**<supdata>** | P | | | B.7.1.25 |
| Copyright credit line<br>**<copyrt>** | | | | B.7.1.26 |
| *EQUIPMENT DESCRIPTION AND DATA WORK PACKAGE*<br>**<descwp>** | R | R | R | B.7.2 |
| Equipment characteristics, capabilities, and features<br>**<eqpinfo>** | R | R | R | B.7.2.1 |
| Location and description of major components<br>**<locdesc>** | R | R | R | B.7.2.2 |
| Equipment differences<br>**<eqpdiff>** | | | | B.7.2.3 |
| Equipment data<br>**<eqpdata>** | R | R | R | B.7.2.4 |
| *THEORY OF OPERATION WORK PACKAGE*<br>**<thrywp>** | R | R | R | B.7.3 |

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**TABLE A-III. Operator and combined operator/maintainer maintenance requirements matrix. (continued**

| Content | -10 | -12 | -12&P | MIL-STD REF |
|---|---|---|---|---|
| **MODULE X. OPERATOR INSTRUCTIONS**<br>**<opim>**<br>**<operatorcategory>**<br>**<ctrlindcategory>**<br>**<opusualcategory>**<br>**<opunucategory>**<br>**<othercategory>** | R | R | R | C.6 |
| *DESCRIPTION AND USE OF OPERATOR CONTROLS AND INDICATORS WORK PACKAGE*<br>**<ctrlindwp>** | R | R | R | C.7.1 |
| *OPERATION UNDER USUAL CONDITIONS WORK PACKAGE*<br>Operation under usual conditions tasks<br>**<opusualwp>**<br>**<opertsk>** | R | R | R | C.7.2 |
| Security measures for electronic data<br>**<secref>** | | | | C.7.2.1 |
| Siting requirements<br>**<site>** | | | | C.7.2.2 |
| Shelter requirements<br>**<shelter>** | | | | C.7.2.3 |
| Assembly and preparation for use<br>**<prepforuse>** | | | | C.7.2.4 |
| Initial adjustments, before use and self-test<br>**<initial>** | | | | C.7.2.5 |
| Operating procedures<br>**<oper>** | R | R | R | C.7.2.6 |
| Operating auxiliary equipment<br>**<operaux>** | | | | C.7.2.8 |
| Preparation for movement<br>**<prepmove>** | | | | C.7.2.9 |
| *OPERATION UNDER UNUSUAL CONDITIONS WORK PACKAGE*<br>Operation under unusual conditions tasks<br>**<opunuwp>**<br>**<opunutsk>** | + | + | + | C.7.3 |
| Security measures for electronic data<br>**<secref>** | | | | C.7.3.1 |
| Unusual environment/weather | R | R | R | C.7.3.2 |



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**TABLE A-III. Operator and combined operator/maintainer maintenance requirements matrix. (continued**

| Content | -10 | -12 | -12&P | MIL-STD REF |
|---|---|---|---|---|
| **<unusualenv>** | | | | |
| Fording and swimming<br>**<fording>** | | | | C.7.3.3 |
| Interim Chemical, Biological, Radiological, Nuclear and Explosives (CBRNE) decontamination procedures<br>**<decon>** | | | | C.7.3.4 |
| Jamming and Electronic Countermeasures (ECM) procedures<br>**<ecm>** | | | | C.7.3.5 |
| Degraded operation procedures<br>**<degraded>** | | | | C.7.3.6 |
| *EMERGENCY WORK PACKAGE*<br>**<emergencywp>** | | | | C.7.4 |
| *DECAL/DATA PLATE GUIDE WORK PACKAGE*<br>**<stowagewp>** | | | | C.7.5 |
| *ON-VEHICLE EQUIPMENT LOADING PLAN WORK PACKAGE*<br>**<eqploadwp>** | | | | C.7.6 |
| **MODULE X. TROUBLESHOOTING MASTER INDEX**<br>**<tim>**<br>**<masterindexcategory>** | | | | D.8<br>D.8.4 |
| *TROUBLESHOOTING INDEX WORK PACKAGE*<br>**<tsindxwp>** | CR | CR | CR | D.8.3 |
| **MODULE X. TROUBLESHOOTING PROCEDURES**<br>*Note: The notation (*) indicates that at least one of the these content items shall be included.*<br>**<tim>**<br>**<troublecategory>** | | R | R | D.8<br>D.7.2 |
| *TROUBLESHOOTING INTRODUCTION WORK PACKAGE*<br>**<tsintrowp>** | | | | D.8.1 |
| *TROUBLESHOOTING INDEX WORK PACKAGE*<br>**<tsindxwp>** | | | | D.8.3 |
| **OPERATIONAL CHECKOUT WORK PACKAGES*<br>**<opcheckwp>** | | | | D.8.5 |
| **TROUBLESHOOTING WORK PACKAGES*<br>**<tswp>** | | | | D.8.4 |
| **COMBINED OPERATIONAL CHECKOUT AND TROUBLESHOOTING PROCEDURE WORK PACKAGES* | | | | D.8.5.1 |


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**TABLE A-III. Operator and combined operator/maintainer maintenance requirements matrix. (continued**

| Content | -10 | -12 | -12&P | MIL-STD REF |
|---|---|---|---|---|
| **<opcheck-tswp>** | | | | |
| **DIAGNOSTICS WORK PACKAGES*<br>**<diagnosticwp>** | | | | D.8.6 |
| **MODULE X. PMCS MAINTENANCE INSTRUCTIONS**<br>*Note: PMCS may be in its own module or may be combined with other maintenance work packages in a maintenance module but not both places.*<br>**<mim>**<br>**<pmcscategory>** | | | | E.6<br>E.6.2 |
| *PMCS INTRODUCTION WORK PACKAGE*<br>**<pmcsintrowp>** | CR | CR | CR | E.7.1 |
| *PMCS, INCLUDING LUBRICATION INSTRUCTIONS, WORK PACKAGE*<br>**<pmcswp>** | CR | CR | CR | E.7.2 |
| **MODULE X. MAINTENANCE INSTRUCTIONS**<br>*Note: PMCS is required as a minimum in one maintenance module. PMCS may be in its own module or may be combined with other maintenance work packages in a maintenance module but not both places.*<br>**<mim>**<br>**<maintenancepmcscategory>**<br>**<maintenancecategory>** | R | R | R | E.6<br>E.6.2<br>E.6.3 |
| *SERVICE UPON RECEIPT WORK PACKAGE*<br>Service upon receipt tasks<br>**<surwp>**<br>**<surtsk>** | P | R | R | E.7.3<br>E.7.3.1 |
| Siting<br>**<siting>** | | | | E.7.3.1.1 |
| Shelter requirements<br>**<shelter>** | | | | E.7.3.1.2 |
| Service upon receipt of materiel<br>**<surmat>** | | | | E.7.3.1.3 |
| Installation instructions<br>**<install>** | | | | E.7.4.4.31.1 |
| Preliminary serving of equipment<br>**<preserv>** | | | | E.7.3.1.3.3.5 |
| Preliminary checks and adjustment of equipment<br>**<prechkadj>** | | | | E.7.3.1.3.3.6 |
| Preliminary calibration of equipment | | | | E.7.3.1.3.3.7 |

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**TABLE A-III. Operator and combined operator/maintainer maintenance requirements matrix. (continued**

| Content | -10 | -12 | -12&P | MIL-STD REF |
|---|---|---|---|---|
| **<precal>** | | | | |
| Circuit alignment<br>**<calign>** | | | | E.7.3.1.3.3.8 |
| Ammunition service upon receipt<br>**<ammo.sur>** | | | | E.7.3.1.3.3.9 |
| Additional service upon receipt task<br>**<other.surtsk>** | | | | E.7.3.1.3.3.10 |
| Follow-on maintenance<br>**<followon.maintsk>** | | | | E.7.3.1.3.3.11 |
| *EQUIPMENT/USER FITTING INSTRUCTIONS WORK PACKAGE*<br>**<perseqpwp>** | | | | E.7.11 |
| *PMCS INTRODUCTION WORK PACKAGE*<br>**<pmcsintrowp>** | | | | E.7.1 |
| *PMCS WORK PACKAGE*<br>**<pmcswp>** | | | | E.7.2 |
| *MAINTENANCE INSTRUCTION WORK PACKAGES*<br>Maintenance tasks<br>**<maintwp>**<br>**<maintsk>** | R | R | R | E.7.4<br>E.7.4.4 |
| Adjust<br>**<adjust>** | | | | E.7.4.4.1 |
| Align<br>**<align>** | | | | E.7.4.4.2 |
| Arm<br>**<arm>** | | | | E.7.4.4.3 |
| Assemble<br>**<assem>** | | | | E.7.4.4.4.1 |
| Disassemble<br>**<disassem>** | | | | E.7.4.4.4.2 |
| Calibrate<br>**<calibration>** | | | | E.7.4.4.5 |
| Clean<br>**<clean>** | | | | E.7.4.4.6 |
| Configure<br>**<configure>** | | | | E.7.4.4.7 |


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**TABLE A-III. Operator and combined operator/maintainer maintenance requirements matrix. (continued**

| Content | -10 | -12 | -12&P | MIL-STD REF |
|---|---|---|---|---|
| Covering<br>**<cover>** | | | | E.7.4.4.8 |
| External power<br>**<extpwr>** | | | | E.7.4.4.9 |
| Hoisting<br>**<hoist>** | | | | E.7.4.4.10 |
| Inspect<br>**<inspect>** | | | | E.7.4.4.11 |
| Install peripheral device<br>**<installperdev>** | | | | E.7.4.4.12.1 |
| Uninstall peripheral device<br>**<uninstallperdev>** | | | | E.7.4.4.12.2 |
| Jacking<br>**<jack>** | | | | E.7.4.4.13 |
| Load<br>**<load>** | | | | E.7.4.4.14.1 |
| Unload<br>**<unload>** | | | | E.7.4.4.14.2 |
| Lubricate<br>**<lube>** | | | | E.7.4.4.15 |
| Mark<br>**<mark>** | | | | E.7.4.4.16 |
| Mooring<br>**<moor>** | | | | E.7.4.4.17 |
| Nondestructive inspection<br>**<ndi>** | | | | E.7.4.4.18 |
| Overhaul<br>**<overhaul>** | P | | | E.7.4.4.19 |
| Pack<br>**<pack>** | | | | E.7.4.4.20.1 |
| Unpack<br>**<unpack>** | | | | E.7.4.4.20.2 |
| Paint<br>**<paint>** | | | | E.7.4.4.21 |
| Parking | | | | E.7.4.4.22 |

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**TABLE A-III. Operator and combined operator/maintainer maintenance requirements matrix. (continued**

| Content | -10 | -12 | -12&P | MIL-STD REF |
|---|---|---|---|---|
| **<park>** | | | | |
| Place in service **<pis>** | | | | E.7.4.4.23 |
| Preparation for storage **<prepstore>** | + | + | + | E.7.4.4.24 |
| Preparation for shipment **<prepship>** | + | + | + | E.7.4.4.25 |
| Prepare for use **<prepforuse>** | | | | E.7.4.4.26 |
| Preserve **<preservation>** | | | | E.7.4.4.27 |
| Radio interference suppression **<ris>** | | | | E.7.4.4.28 |
| Rebuild **<rebuild>** | P | | | E.7.4.4.29 |
| Remove **<remove>** | | | | E.7.4.4.31.1 |
| Install **<install>** | | | | E.7.4.4.31.2 |
| Repair **<repair>** | | | | E.7.4.4.32 |
| Replace **<replace>** | | | | E.7.4.4.33 |
| Service **<service>** | | | | E.7.4.4.34 |
| Sling loading **<sling>** | | | | E.7.4.4.35 |
| Test **<test>** | | | | E.7.4.4.36 |
| Towing **<tow>** | | | | E.7.4.4.37 |
| Transport **<transport>** | + | + | + | E.7.4.4.38 |
| Additional maintenance task **<other.maintsk>** | | | | E.7.4.4.40 |

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**TABLE A-III. Operator and combined operator/maintainer maintenance requirements matrix. (continued**

| Content | -10 | -12 | -12&P | MIL-STD REF |
|---|---|---|---|---|
| Follow-on maintenance<br>**<followon.maintsk>** | | | | E.7.4.4.41 |
| *SOFTWARE MAINTENANCE WORK PACKAGES*<br>Software maintenance tasks<br>**<maintwp>**<br>**<softmaintsk>** | | | | E.7.4<br>E.7.4.4.39 |
| Configure software<br>**<softconfig>** | | | | E.7.4.4.39.1 |
| Debug software<br>**<debug>** | | | | E.7.4.4.39.2 |
| Install software<br>**<install>** | | | | E.7.4.4.31.1 |
| Remove software<br>**<remove>** | | | | E.7.4.4.31 |
| Upgrade / patch software<br>**<upgrade>** | | | | E.7.4.4.39.3 |
| Additional maintenance task<br>**<other.maintsk>** | | | | E.7.4.4.40 |
| Follow-on maintenance<br>**<followon.maintsk>** | | | | E.7.4.4.41 |
| *GENERAL MAINTENANCE WORK PACKAGE*<br>**<gen.maintwp>** | | | | E.7.5 |
| *LUBRICATION INSTRUCTIONS WORK PACKAGE*<br>**<lubewp>** | | | | E.7.6 |
| **ILLUSTRATED LIST OF MANUFACTURED ITEMS**<br>*Note: If required, the introduction work package and manufacturing procedure work package are both required.* | P | | | E.7.7 |
| *ILLUSTRATED LIST OF MANUFACTURED ITEMS INTRODUCTION WORK PACKAGE*<br>**<manu_items_introwp>** | P | CR | CR | E.7.7 |
| *MANUFACTURING PROCEDURES WORK PACKAGE*<br>**<manuwp>** | P | CR | CR | E.7.8 |
| *TORQUE LIMITS WORK PACKAGE*<br>**<torquewp>** | P | | | E.7.9 |
| *WIRING DIAGRAMS AND SCHEMATICS WORK PACKAGE*<br>**<wiringwp>** | P | | | E.7.10 |

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**TABLE A-III. Operator and combined operator/maintainer maintenance requirements matrix. (continued**

| Content | -10 | -12 | -12&P | MIL-STD REF |
|---|---|---|---|---|
| **MODULE X. AUXILIARY EQUIPMENT MAINTENANCE INSTRUCTIONS**<br>**<mim>**<br>**<auxililarycategory>** | | | | E.6<br>E.6.6 |
| *AUXILIARY EQUIPMENT MAINTENANCE WORK PACKAGE*<br>**<auxeqpwp>** | CR | CR | CR | E.7.24 |
| **ILLUSTRATED LIST OF MANUFACTURED ITEMS**<br>*Note: If required, the introduction work package and manufacturing procedure work package are both required.* | P | | | E.7.7 |
| *ILLUSTRATED LIST OF MANUFACTURED ITEMS INTRODUCTION WORK PACKAGE*<br>**<manu_items_introwp>** | P | CR | CR | E.7.7 |
| *MANUFACTURING PROCEDURES WORK PACKAGE*<br>**<manuwp>** | P | CR | CR | E.7.8 |
| *TORQUE LIMITS WORK PACKAGE*<br>**<torquewp>** | P | | | E.7.9 |
| *WIRING DIAGRAMS AND SCHEMATICS WORK PACKAGE*<br>**<wiringwp>** | P | | | E.7.10 |
| **MODULE X. AMMUNITION MAINTENANCE INSTRUCTIONS**<br>**<mim>**<br>**<ammunitioncategory>** | | | | E.6<br>E.6.7 |
| *AMMUNITION IDENTIFICATION WORK PACKAGE*<br>**<ammoidentwp>** | | | | E.7.25 |
| *AMMUNITION MAINTENANCE WORK PACKAGE*<br>**<ammowp>** | | | | E.7.26 |
| *FOREIGN AMMUNITION WORK PACKAGE*<br>**<natowp>** | | | | E.7.27 |
| **MODULE X. DESTRUCTION OF EQUIPMENT TO PREVENT ENEMY USE**<br>*Note: If a separate destruction of material manual is not developed for this equipment, then the destruction module must be included.*<br>**<dim>** | | | | H.5.2 |
| *DESTRUCTION PROCEDURES INTRODUCTION WORK PACKAGE*<br>**<destruct-introwp>** | CR | CR | CR | H.6.2 |
| Authority to destroy<br>**<authorize_to_ destroy>** | CR | CR | CR | H.6.2.1 |

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**TABLE A-III. Operator and combined operator/maintainer maintenance requirements matrix. (continued**

| Content | -10 | -12 | -12&P | MIL-STD REF |
|---|---|---|---|---|
| Reporting destruction<br>**<report_destruct>** | CR | CR | CR | H.6.2.2 |
| General destruction information<br>**<general_destruct_info>** | CR | CR | CR | H.6.2.3 |
| Degree of destruction<br>**<degree_of_destruct>** | CR | CR | CR | H.6.2.4 |
| Essential components and spare parts<br>**<component_spares>** | | | | H.6.2.5 |
| *DESTRUCTION PROCEDURES WORK PACKAGE*<br>**<destruct-materialwp>** | CR | CR | CR | H.6.3 |
| Parts list<br>**<essential_spares>** | | | | H.6.3.1 |
| Specific destruction procedures<br>**<proc>** | CR | CR | CR | H.6.3.2 |
| **MODULE X. BATTLE DAMAGE ASSESSMENT AND REPAIR INFORMATION**<br>**<bim>** | P | | | I.6.1 |
| *BDAR UNIQUE GENERAL INFORMATION WORK PACKAGE*<br>**<bdar-geninfowp>** | P | CR | CR | I.7.2 |
| Standards and practices<br>**<bdar-std-practices>** | P | CR | CR | I.7.2.1 |
| Tasks and responsibilities<br>**<bdar-task-resp>** | P | | | I.7.2.2 |
| Combat threats (aviation only)<br>**<bdar-combat-threat>** | P | | | I.7.2.3 |
| *BATTLE DAMAGE ASSESSMENT WORK PACKAGE*<br>**<damage-assesswp>** | P | CR | CR | I.7.3 |
| *BATTLE DAMAGE REPAIR WORK PACKAGE*<br>**<genrepairwp>** | P | | | I.7.4 |
| Introduction<br>**<geninfo>** | P | CR | CR | I.7.4 |
| Repair procedure<br>**<bdar-repair-proc>** | P | CR | CR | I.7.4.1 |
| *SPECIAL OR FABRICATED TOOLS WORK PACKAGE*<br>**<bdartoolswp>** | | | | I.7.6 |


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**TABLE A-III. Operator and combined operator/maintainer maintenance requirements matrix. (continued**

| Content | -10 | -12 | -12&P | MIL-STD REF |
|---|---|---|---|---|
| *SUBSTITUTE MATERIALS/PARTS WORK PACKAGE* **<substitute-matwp>** | | | | I.7.8 |
| **MODULE X. SOFTWARE INFORMATION** **<soim>** | | | | M.7.7 |
| *SOFTWARE GENERAL INFORMATION WORK PACKAGE* **<softginfowp>** | CR | CR | CR | M.7.1 |
| System overview **<softsysover>** | CR | CR | CR | M.7.1 |
| Document overview **<softdocover>** | CR | CR | CR | M.7.1 |
| Copyright credit line **<copyrt>** | | | | M.7.1 |
| Endorsement notice **<endorse>** | | | | M.7.1 |
| *SOFTWARE SUMMARY WORK PACKAGE* **<softsumwp>** | CR | CR | CR | M.7.2 |
| *SOFTWARE EFFECTIVITY WORK PACKAGE* **<softeffectwp>** | | | | M.7.3 |
| *DIFFERENCES BETWEEN SOFTWARE VERSIONS WORK PACKAGE* **<softdiffversionwp>** | | | | M.7.4 |
| *FEATURES AND CAPABILITIES WORK PACKAGE* **<softfeaturescapwp>** | | | | M.7.5 |
| *FUNCTIONAL DISPLAY WORK PACKAGE* **<functionaldisplaywp>** | CR | CR | CR | M.7.6 |
| *SECURITY AND PRIVACY PROCEDURES WORK PACKAGE* **<softsecprivwp>** | CR | CR | CR | M.7.7.1 |
| *SUPERVISORY CONTROLS WORK PACKAGE* **<softsuperctrlswp>** | | | | M.7.7.2 |
| *POWER UP AND ACCESS SOFTWARE PROCEDURES WORK PACKAGE* **<softpwraccesswp>** | CR | CR | CR | M.7.7.3 |
| *SOFTWARE OPERATOR PROCEDURES WORK PACKAGE* **<softoperwp>** | CR | CR | CR | M.7.7.4 |
| Key commands **<softkeycmd>** | | | | M.7.7.4.1 |


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**TABLE A-III. Operator and combined operator/maintainer maintenance requirements matrix. (continued**

| Content | -10 | -12 | -12&P | MIL-STD REF |
|---|---|---|---|---|
| User interface<br>**<softgui>** | | | | M.7.7.4.2 |
| Software operating conventions<br>**<softopconventions>** | | | | M.7.7.4.3 |
| Software operating procedures<br>**<softoperproc>** | | | | M.7.7.4.4 |
| *POWER DOWN AND EXIT SOFTWARE PROCEDURES WORK PACKAGE*<br>**<softpwrexitwp>** | CR | CR | CR | M.7.7.5 |
| *MESSAGES WORK PACKAGE*<br>**<softmessageswp>** | | | | M.7.8.3 |
| *RECOVERY FROM ERRORS, MALFUNCTIONS, AND EMERGENCIES WORK PACKAGE*<br>**<softerrorswp>** | | | | M.7.8.4 |
| **MODULE X. PARTS INFORMATION**<br>**<pim>** | P | P | R | F.7.2 |
| *INTRODUCTION WORK PACKAGE*<br>**<introwp>** | P | P | R | F.8.1 |
| *REPAIR PARTS LIST WORK PACKAGE*<br>**<plwp>** | P | P | R | F.8.2 |
| *REPAIR PARTS FOR SPECIAL TOOLS WORK PACKAGE*<br>**<stl_partswp>** | P | P | | F.8.3 |
| *KIT PARTS LIST WORK PACKAGE*<br>**<kitswp>** | P | P | | F.8.4 |
| *BULK ITEM WORK PACKAGE*<br>**<bulk_itemswp>** | P | P | | F.8.5 |
| *SPECIAL TOOLS LIST WORK PACKAGE*<br>**<stlwp>** | P | P | | F.8.6 |
| **MODULE X. SUPPORTING INFORMATION**<br>**<sim>** | R | R | R | G.6 |
| *REFERENCES WORK PACKAGE*<br>**<refwp>** | R | R | R | G.7.1 |
| *INTRODUCTION FOR TWO-LEVEL MAINTENANCE ALLOCATION CHART (MAC) WORK PACKAGE*<br>**<macintrowp>** | P | R | R | G.7.2 |

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**TABLE A-III. Operator and combined operator/maintainer maintenance requirements matrix. (continued**

| Content | -10 | -12 | -12&P | MIL-STD REF |
|---|---|---|---|---|
| *MAINTENANCE ALLOCATION CHART (MAC) WORK PACKAGE* **<macwp>** | P | R | R | G.7.3 |
| *COMPONENTS OF END ITEM (COEI) AND BASIC ISSUE ITEMS (BII) LISTS WORK PACKAGE* **<coeibiiwp>** | R | R | R | G.7.4 |
| *ADDITIONAL AUTHORIZATION LIST (AAL) WORK PACKAGE* **<aalwp>** | | | | G.7.5 |
| *EXPENDABLE AND DURABLE ITEMS LIST WORK PACKAGE* **<explistwp>** | R | R | R | G.7.7 |
| *TOOL IDENTIFICATION LIST WORK PACKAGE* **<toolidwp>** | P | R | R | G.7.8 |
| *MANDATORY REPLACEMENT PARTS WORK PACKAGE* **<mrplwp>** | + | + | + | G.7.9 |
| *CRITICAL SAFETY ITEMS WORK PACKAGE* **<csi.wp>** | + | + | + | G.7.10 |
| *SUPPORT ITEMS WORK PACKAGE* **<supitemwp>** | | | | G.7.11 |
| *ADDITIONAL SUPPORTING WORK PACKAGE* **<genwp>** | | | | G.7.12 |
| **REAR MATTER** **<rear>** | R | R | R | TABLE A-II |

**LEGEND**
R - Required (see A.4.3.1.3).
P - Prohibited (see A.4.3.1.3).
Shaded - May be Required (see A.4.3.1.4).
CR - Conditionally Required (see A.4.3.1.4.3).
+ Plus - Typically Required (see A.4.3.1.4.1). WP shall be listed in General Information, WP 0001 (see B.7.1.1).

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**TABLE A-IV. Maintainer maintenance requirements matrix.**

| Content | -20 | -20&P | MIL-STD REF |
|---|---|---|---|
| **PUBLICATION** (**<production>**)<br>**<paper.manual>** or **<framed.manual>**<br><br>PUBLICATION TITLES:<br>(-20) Field Maintenance Manual for *insert system* [1]<br><br>(-20&P) Field Maintenance Manual Including Repair Parts and Special Tools List for *insert system* [1]<br><br>[1] TRADOC maintainer representative approval is required for any inclusion of sustainment-level maintenance. If approval is granted, "sustainment" may be included in title. See also A.4.3 for publication titling guidance. | R | R | TABLE A-I |
| **FRONT MATTER**<br>**<paper.frnt>** or **<framed.frnt>** | R | R | TABLE A-II |
| **MODULE 1. GENERAL INFORMATION, EQUIPMENT DESCRIPTION, AND THEORY OF OPERATION**<br>**<gim>** | R | R | B.6 |
| *GENERAL INFORMATION WORK PACKAGE*<br>**<ginfowp>** | R | R | B.7.1 |
| Scope<br>**<scope>** | R | R | B.7.1.1 |
| Maintenance forms, records, and reports<br>**<mfrr>** | R | R | B.7.1.2 |
| Reporting Equipment Improvement Recommendations (EIR)<br>**<eir>** | R | R | B.7.1.3 |
| Hand Receipt (HR) manuals<br>**<handreceipt>** | | | B.7.1.4 |
| Corrosion Prevention and Control (CPC)<br>**<cpcdata>** | R | R | B.7.1.5 |
| Ozone Depleting Substances (ODS)<br>**<odsdata>** | | | B.7.1.6 |
| Destruction of equipment to prevent enemy use<br>**<destructmat>** | R | R | B.7.1.7 |
| Preparation for storage or shipment<br>**<pssref>** | R | R | B.7.1.8 |
| Transportability guidance<br>**<transportability>** | R | R | B.7.1.8 |
| Warranty information | | | B.7.1.9 |

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**TABLE A-IV. Maintainer maintenance requirements matrix. (continued**

| Content | -20 | -20&P | MIL-STD REF |
|---|---|---|---|
| **<wrntyref>** | | | |
| Nomenclature cross-reference list<br>**<nomenreflist>** | | | B.7.1.10 |
| List of abbreviations/ acronyms<br>**<loa>** | R | R | B.7.1.11 |
| Quality of material<br>**<qual.mat.info>** | R | R | B.7.1.13 |
| Safety, care, and handling<br>**<sftyinfo>** | | | B.7.1.14 |
| Nuclear hardness<br>**<hcp>** | | | B.7.1.15 |
| Calibration<br>**<calref>** | | | B.7.1.16 |
| Item unique identification (IUID)<br>**<iuid>** | | | B.7.1.17 |
| Supporting information for repair parts, special tools/kits, TMDE, and support equipment<br>**<supdata>** | | | B.7.1.25 |
| Copyright credit line<br>**<copyrt>** | | | B.7.1.26 |
| *EQUIPMENT DESCRIPTION AND DATA WORK PACKAGE*<br>**<descwp>** | R | R | B.7.2 |
| Equipment characteristics, capabilities, and features<br>**<eqpinfo>** | R | R | B.7.2.1 |
| Location and description of major components<br>**<locdesc>** | R | R | B.7.2.2 |
| Equipment differences<br>**<eqpdiff>** | | | B.7.2.3 |
| Equipment data<br>**<eqpdata>** | R | R | B.7.2.4 |
| *THEORY OF OPERATION WORK PACKAGE*<br>**<thrywp>** | R | R | B.7.3 |
| **MODULE X. TROUBLESHOOTING MASTER INDEX**<br>**<tim>**<br>**<masterindexcategory>** | | | D.8<br><br>D.8.4 |



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**TABLE A-IV. Maintainer maintenance requirements matrix. (continued**

| Content | -20 | -20&P | MIL-STD REF |
|---|---|---|---|
| *TROUBLESHOOTING INDEX WORK PACKAGE*<br>**<tsindxwp>** | CR | CR | D.8.3 |
| **MODULE X. TROUBLESHOOTING PROCEDURES**<br>*Note: The notation (*) indicates that at least one of the these content items shall be included.*<br>**<tim>**<br>**<troublecategory>** | R | R | D.8<br>D.7.2 |
| *TROUBLESHOOTING INTRODUCTION WORK PACKAGE*<br>**<tsintrowp>** | | | D.8.1 |
| *TROUBLESHOOTING INDEX WORK PACKAGE*<br>**<tsindxwp>** | | | D.8.3 |
| **OPERATIONAL CHECKOUT WORK PACKAGES*<br>**<opcheckwp>** | | | D.8.5 |
| **TROUBLESHOOTING WORK PACKAGES*<br>**<tswp>** | | | D.8.4 |
| **COMBINED OPERATIONAL CHECKOUT AND TROUBLESHOOTING PROCEDURE WORK PACKAGES*<br>**<opcheck-tswp>** | | | D.8.5.1 |
| **DIAGNOSTICS WORK PACKAGES*<br>**<diagnosticwp>** | | | D.8.6 |
| **MODULE X. PMCS MAINTENANCE INSTRUCTIONS**<br>*Note: PMCS may be in its own module or may be combined with other maintenance work packages in a maintenance module but not both places.*<br>**<mim>**<br>**<pmcscategory>** | | | E.6<br>E.6.2 |
| *PMCS INTRODUCTION WORK PACKAGE*<br>**<pmcsintrowp>** | CR | CR | E.7.1 |
| *PMCS, INCLUDING LUBRICATION INSTRUCTIONS, WORK PACKAGE*<br>**<pmcswp>** | CR | CR | E.7.2 |
| **MODULE X. MAINTENANCE INSTRUCTIONS**<br>*Note: PMCS is required as a minimum in one maintenance module. PMCS may be in its own module or may be combined with other maintenance work packages in a maintenance module but not both places.*<br>**<mim>**<br>**<maintenancepmcscategory>**<br>**<maintenancecategory>** | R | R | E.6<br>E.6.2<br>E.6.3 |
| *SERVICE UPON RECEIPT WORK PACKAGE* | R | R | E.7.3 |



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**TABLE A-IV. Maintainer maintenance requirements matrix. (continued**

| Content | -20 | -20&P | MIL-STD REF |
|---|---|---|---|
| Service upon receipt tasks<br>**<surwp>**<br>**<surtsk>** | | | E.7.3.1 |
| Siting<br>**<siting>** | | | E.7.3.1.1 |
| Shelter requirements<br>**<shelter>** | | | E.7.3.1.2 |
| Service upon receipt of materiel<br>**<surmat>** | | | E.7.3.1.3 |
| Installation instructions<br>**<install>** | | | E.7.4.4.31.1 |
| Preliminary serving of equipment<br>**<preserv>** | | | E.7.3.1.3.3.5 |
| Preliminary checks and adjustment of equipment<br>**<prechkadj>** | | | E.7.3.1.3.3.6 |
| Preliminary calibration of equipment<br>**<precal>** | | | E.7.3.1.3.3.7 |
| Circuit alignment<br>**<calign>** | | | E.7.3.1.3.3.8 |
| Ammunition service upon receipt<br>**<ammo.sur>** | | | E.7.3.1.3.3.9 |
| Additional service upon receipt task<br>**<other.surtsk>** | | | E.7.3.1.3.3.10 |
| Follow-on maintenance<br>**<followon.maintsk>** | | | E.7.3.1.3.3.11 |
| *EQUIPMENT/USER FITTING INSTRUCTIONS WORK PACKAGE*<br>**<perseqpwp>** | | | E.7.11 |
| *PMCS INTRODUCTION WORK PACKAGE*<br>**<pmcsintrowp>** | | | E.7.1 |
| *PMCS WORK PACKAGE*<br>**<pmcswp>** | | | E.7.2 |
| *MAINTENANCE INSTRUCTION WORK PACKAGES*<br>Maintenance tasks<br>**<maintwp>**<br>**<maintsk>** | R | R | E.7.4<br>E.7.4.4 |
| Adjust<br>**<adjust>** | | | E.7.4.4.1 |

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**TABLE A-IV. Maintainer maintenance requirements matrix. (continued**

| Content | -20 | -20&P | MIL-STD REF |
|---|---|---|---|
| Align<br>**<align>** | | | E.7.4.4.2 |
| Arm<br>**<arm>** | | | E.7.4.4.3 |
| Assemble<br>**<assem>** | | | E.7.4.4.4.1 |
| Disassemble<br>**<disassem>** | | | E.7.4.4.4.2 |
| Calibrate<br>**<calibration>** | | | E.7.4.4.5 |
| Clean<br>**<clean>** | | | E.7.4.4.6 |
| Configure<br>**<configure>** | | | E.7.4.4.7 |
| Covering<br>**<cover>** | | | E.7.4.4.8 |
| External power<br>**<extpwr>** | | | E.7.4.4.9 |
| Hoisting<br>**<hoist>** | | | E.7.4.4.10 |
| Inspect<br>**<inspect>** | | | E.7.4.4.11 |
| Install peripheral device<br>**<installperdev>** | | | E.7.4.4.12.1 |
| Uninstall peripheral device<br>**<uninstallperdev>** | | | E.7.4.4.12.2 |
| Jacking<br>**<jack>** | | | E.7.4.4.13 |
| Load<br>**<load>** | | | E.7.4.4.14.1 |
| Unload<br>**<unload>** | | | E.7.4.4.14.2 |
| Lubricate<br>**<lube>** | | | E.7.4.4.15 |
| Mark<br>**<mark>** | | | E.7.4.4.16 |



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**TABLE A-IV. Maintainer maintenance requirements matrix. (continued**

| Content | -20 | -20&P | MIL-STD REF |
|---|---|---|---|
| Mooring<br>**<moor>** | | | E.7.4.4.17 |
| Nondestructive inspection<br>**<ndi>** | | | E.7.4.4.18 |
| Overhaul<br>**<overhaul>** | | | E.7.4.4.19 |
| Pack<br>**<pack>** | | | E.7.4.4.20.1 |
| Unpack<br>**<unpack>** | | | E.7.4.4.20.2 |
| Paint<br>**<paint>** | | | E.7.4.4.21 |
| Parking<br>**<park>** | | | E.7.4.4.22 |
| Place in service<br>**<pis>** | | | E.7.4.4.23 |
| Preparation for storage<br>**<prepstore>** | + | + | E.7.4.4.24 |
| Preparation for shipment<br>**<prepship>** | + | + | E.7.4.4.25 |
| Prepare for use<br>**<prepforuse>** | | | E.7.4.4.26 |
| Preserve<br>**<preservation>** | | | E.7.4.4.27 |
| Radio interference suppression<br>**<ris>** | | | E.7.4.4.28 |
| Rebuild<br>**<rebuild>** | | | E.7.4.4.29 |
| Remove<br>**<remove>** | | | E.7.4.4.31.1 |
| Install<br>**<install>** | | | E.7.4.4.31.2 |
| Repair<br>**<repair>** | | | E.7.4.4.32 |
| Replace<br>**<replace>** | | | E.7.4.4.33 |



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**TABLE A-IV. Maintainer maintenance requirements matrix. (continued**

| Content | -20 | -20&P | MIL-STD REF |
|---|---|---|---|
| Service<br>**<service>** | | | E.7.4.4.34 |
| Sling loading<br>**<sling>** | | | E.7.4.4.35 |
| Test<br>**<test>** | | | E.7.4.4.36 |
| Towing<br>**<tow>** | | | E.7.4.4.37 |
| Transport<br>**<transport>** | + | + | E.7.4.4.38 |
| Additional maintenance task<br>**<other.maintsk>** | | | E.7.4.4.40 |
| Follow-on maintenance<br>**<followon.maintsk>** | | | E.7.4.4.41 |
| *SOFTWARE MAINTENANCE WORK PACKAGES*<br>Software maintenance tasks<br>**<maintwp>**<br>**<softmaintsk>** | | | E.7.4<br>E.7.4.4.39 |
| Configure software<br>**<softconfig>** | | | E.7.4.4.39.1 |
| Debug software<br>**<debug>** | | | E.7.4.4.39.2 |
| Install software<br>**<install>** | | | E.7.4.4.31.1 |
| Remove software<br>**<remove>** | | | E.7.4.4.31 |
| Upgrade / patch software<br>**<upgrade>** | | | E.7.4.4.39.3 |
| Additional maintenance task<br>**<other.maintsk>** | | | E.7.4.4.40 |
| Follow-on maintenance<br>**<followon.maintsk>** | | | E.7.4.4.41 |
| *GENERAL MAINTENANCE WORK PACKAGE*<br>**<gen.maintwp>** | | | E.7.5 |
| *LUBRICATION INSTRUCTIONS WORK PACKAGE*<br>**<lubewp>** | | | E.7.6 |

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**TABLE A-IV. Maintainer maintenance requirements matrix. (continued**

| Content | -20 | -20&P | MIL-STD REF |
|---|---|---|---|
| **ILLUSTRATED LIST OF MANUFACTURED ITEMS**<br>*Note: If required, the introduction work package and manufacturing procedure work package are both required.* | | | E.7.7 |
| *ILLUSTRATED LIST OF MANUFACTURED ITEMS INTRODUCTION WORK PACKAGE*<br>**<manu_items_introwp>** | CR | CR | E.7.7 |
| *MANUFACTURING PROCEDURES WORK PACKAGE*<br>**<manuwp>** | CR | CR | E.7.8 |
| *TORQUE LIMITS WORK PACKAGE*<br>**<torquewp>** | | | E.7.9 |
| *WIRING DIAGRAMS AND SCHEMATICS WORK PACKAGE*<br>**<wiringwp>** | | | E.7.10 |
| **MODULE X. AUXILIARY EQUIPMENT MAINTENANCE INSTRUCTIONS**<br>**<mim>**<br>**<auxiliarycategory>** | | | E.6<br>E.6.6 |
| *AUXILIARY EQUIPMENT MAINTENANCE WORK PACKAGE*<br>**<auxeqpwp>** | | | E.7.24 |
| **ILLUSTRATED LIST OF MANUFACTURED ITEMS**<br>*Note: If required, the introduction work package and manufacturing procedure work package are both required.* | | | E.7.7 |
| *ILLUSTRATED LIST OF MANUFACTURED ITEMS INTRODUCTION WORK PACKAGE*<br>**<manu_items_introwp>** | CR | CR | E.7.7 |
| *MANUFACTURING PROCEDURES WORK PACKAGE*<br>**<manuwp>** | CR | CR | E.7.8 |
| *TORQUE LIMITS WORK PACKAGE*<br>**<torquewp>** | | | E.7.9 |
| *WIRING DIAGRAMS AND SCHEMATICS WORK PACKAGE*<br>**<wiringwp>** | | | E.7.10 |
| **MODULE X. AMMUNITION MAINTENANCE INSTRUCTIONS**<br>**<mim>**<br>**<ammunitioncategory>** | | | E.6<br>E.6.7 |
| *AMMUNITION IDENTIFICATION WORK PACKAGE*<br>**<ammoidentwp>** | | | E.7.25 |
| *AMMUNITION MAINTENANCE WORK PACKAGE*<br>**<ammowp>** | | | E.7.26 |


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**TABLE A-IV. Maintainer maintenance requirements matrix. (continued**

| Content | -20 | -20&P | MIL-STD REF |
|---|---|---|---|
| *FOREIGN AMMUNITION WORK PACKAGE*<br>**<natowp>** | | | E.7.27 |
| **MODULE X. DESTRUCTION OF EQUIPMENT TO PREVENT ENEMY USE**<br>*Note: If a separate destruction of material manual is not developed for this equipment, then the destruction module must be included.*<br>**<dim>** | | | H.5.2 |
| *DESTRUCTION PROCEDURES INTRODUCTION WORK PACKAGE*<br>**<destruct-introwp>** | CR | CR | H.6.2 |
| Authority to destroy<br>**<authorize_to_ destroy>** | CR | CR | H.6.2.1 |
| Reporting destruction<br>**<report_destruct>** | CR | CR | H.6.2.2 |
| General destruction information<br>**<general_destruct_info>** | CR | CR | H.6.2.3 |
| Degree of destruction<br>**<degree_of_destruct>** | CR | CR | H.6.2.4 |
| Essential components and spare parts<br>**<component_spares>** | | | H.6.2.5 |
| *DESTRUCTION PROCEDURES WORK PACKAGE*<br>**<destruct-materialwp>** | CR | CR | H.6.3 |
| Parts list<br>**<essential_spares>** | | | H.6.3.1 |
| Specific destruction procedures<br>**<proc>** | CR | CR | H.6.3.2 |
| **MODULE X. BATTLE DAMAGE ASSESSMENT AND REPAIR INFORMATION**<br>**<bim>** | | | I.6.1 |
| *BDAR UNIQUE GENERAL INFORMATION WORK PACKAGE*<br>**<bdar-geninfowp>** | CR | CR | I.7.2 |
| Standards and practices<br>**<bdar-std-practices>** | CR | CR | I.7.2.1 |
| Tasks and responsibilities<br>**<bdar-task-resp>** | | | I.7.2.2 |
| Combat threats (aviation only)<br>**<bdar-combat-threat>** | | | I.7.2.3 |



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**TABLE A-IV. Maintainer maintenance requirements matrix. (continued**

| Content | -20 | -20&P | MIL-STD REF |
|---|---|---|---|
| *BATTLE DAMAGE ASSESSMENT WORK PACKAGE* **<damage-assesswp>** | CR | CR | I.7.3 |
| *BATTLE DAMAGE REPAIR WORK PACKAGE* **<genrepairwp>** | | | I.7.4 |
| Introduction **<geninfo>** | CR | CR | I.7.4 |
| Repair procedure **<bdar-repair-proc>** | CR | CR | I.7.4.1 |
| *SPECIAL OR FABRICATED TOOLS WORK PACKAGE* **<bdartoolswp>** | | | I.7.6 |
| *SUBSTITUTE MATERIALS/PARTS WORK PACKAGE* **<substitute-matwp>** | | | I.7.8 |
| **MODULE X. SOFTWARE INFORMATION** **<soim>** **<softwarecategory>** | | | M.7.7 |
| *SOFTWARE GENERAL INFORMATION WORK PACKAGE* **<softginfowp>** | CR | CR | M.7.1 |
| System overview **<softsysover>** | CR | CR | M.7.1 |
| Document overview **<softdocover>** | CR | CR | M.7.1 |
| Copyright credit line **<copyrt>** | | | M.7.1 |
| Endorsement notice **<endorse>** | | | M.7.1 |
| *SOFTWARE SUMMARY WORK PACKAGE* **<softsumwp>** | CR | CR | M.7.2 |
| *SOFTWARE EFFECTIVITY WORK PACKAGE* **<softeffectwp>** | | | M.7.3 |
| *DIFFERENCES BETWEEN SOFTWARE VERSIONS WORK PACKAGE* **<softdiffversionwp>** | | | M.7.4 |
| *FEATURES AND CAPABILITIES WORK PACKAGE* **<softfeaturescapwp>** | | | M.7.5 |
| *FUNCTIONAL DISPLAY WORK PACKAGE* **<functionaldisplaywp>** | CR | CR | M.7.6 |

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**TABLE A-IV. Maintainer maintenance requirements matrix. (continued**

| Content | -20 | -20&P | MIL-STD REF |
|---|---|---|---|
| *SECURITY AND PRIVACY PROCEDURES WORK PACKAGE* **<softsecprivwp>** | CR | CR | M.7.7.1 |
| *SUPERVISORY CONTROLS WORK PACKAGE* **<softsuperctrlswp>** | | | M.7.7.2 |
| *POWER UP AND ACCESS SOFTWARE PROCEDURES WORK PACKAGE* **<softpwraccesswp>** | CR | CR | M.7.7.3 |
| *SOFTWARE OPERATOR PROCEDURES WORK PACKAGE* **<softoperwp>** | CR | CR | M.7.7.4 |
| Key commands **<softkeycmd>** | | | M.7.7.4.1 |
| User interface **<softgui>** | | | M.7.7.4.2 |
| Software operating conventions **<softopconventions>** | | | M.7.7.4.3 |
| Software operating procedures **<softoperproc>** | | | M.7.7.4.4 |
| *POWER DOWN AND EXIT SOFTWARE PROCEDURES WORK PACKAGE* **<softpwrexitwp>** | CR | CR | M.7.7.5 |
| *MESSAGES WORK PACKAGE* **<softmessageswp>** | | | M.7.8.3 |
| *RECOVERY FROM ERRORS, MALFUNCTIONS, AND EMERGENCIES WORK PACKAGE* **<softerrorswp>** | | | M.7.8.4 |
| **MODULE X. PARTS INFORMATION** **<pim>** | P | R | F.7.2 |
| *INTRODUCTION WORK PACKAGE* **<introwp>** | P | R | F.8.1 |
| *REPAIR PARTS LIST WORK PACKAGE* **<plwp>** | P | R | F.8.2 |
| *REPAIR PARTS FOR SPECIAL TOOLS WORK PACKAGE* **<stl_partswp>** | P | | F.8.3 |
| *KIT PARTS LIST WORK PACKAGE* **<kitswp>** | P | | F.8.4 |



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**TABLE A-IV. Maintainer maintenance requirements matrix. (continued**

| Content | -20 | -20&P | MIL-STD REF |
|---|---|---|---|
| *BULK ITEM WORK PACKAGE*<br>**<bulk_itemswp>** | P | | F.8.5 |
| *SPECIAL TOOLS LIST WORK PACKAGE*<br>**<stlwp>** | P | | F.8.6 |
| **MODULE X. SUPPORTING INFORMATION**<br>**<sim>** | R | R | G.6 |
| *REFERENCES WORK PACKAGE*<br>**<refwp>** | R | R | G.7.1 |
| *INTRODUCTION FOR TWO-LEVEL MAINTENANCE ALLOCATION CHART (MAC) WORK PACKAGE*<br>**<macintrowp>** | R | R | G.7.2 |
| *MAINTENANCE ALLOCATION CHART (MAC) WORK PACKAGE*<br>**<macwp>** | R | R | G.7.3 |
| *EXPENDABLE AND DURABLE ITEMS LIST WORK PACKAGE*<br>**<explistwp>** | R | R | G.7.7 |
| *TOOL IDENTIFICATION LIST WORK PACKAGE*<br>**<toolidwp>** | R | R | G.7.8 |
| *MANDATORY REPLACEMENT PARTS WORK PACKAGE*<br>**<mrplwp>** | + | + | G.7.9 |
| *CRITICAL SAFETY ITEMS WORK PACKAGE*<br>**<csi.wp>** | + | + | G.7.10 |
| *SUPPORT ITEMS WORK PACKAGE*<br>**<supitemwp>** | | | G.7.11 |
| *ADDITIONAL SUPPORTING WORK PACKAGE*<br>**<genwp>** | | | G.7.12 |
| **REAR MATTER**<br>**<rear>** | R | R | TABLE A-II |

**LEGEND**
R - Required (see A.4.3.1.3).
P - Prohibited (see A.4.3.1.3).
Shaded - May be Required (see A.4.3.1.4).
CR - Conditionally Required (see A.4.3.1.4.3).
+ Plus - Typically Required (see A.4.3.1.4.1). WP shall be listed in General Information, WP 0001 (see B.7.1.1).



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**TABLE A-V. Stand-alone RPSTL requirements matrix.**

| Content | -20P | MIL-STD REF |
|---|---|---|
| **PUBLICATION** (**<production>**)<br>**<paper.manual>** or **<framed.manual>**<br><br>Publication title:<br><br>(-20P) Field Maintenance Repair Parts and Special Tools List for *insert system* [1]<br><br>[1] TRADOC maintainer representative approval is required for any inclusion of sustainment-level maintenance. If approval is granted, "sustainment" may be included in title. See also A.4.3 for publication titling guidance. | R | TABLE A-I |
| **FRONT MATTER**<br>**<paper.frnt>** or **<framed.frnt>** | R | TABLE A-II |
| **MODULE X. PARTS INFORMATION**<br>**<pim>** | R | F.7.2 |
| *INTRODUCTION WORK PACKAGE*<br>**<introwp>** | R | F.8.1 |
| *REPAIR PARTS LIST WORK PACKAGE*<br>**<plwp>** | R | F.8.2 |
| *REPAIR PARTS FOR SPECIAL TOOLS WORK PACKAGE*<br>**<stl_partswp>** | | F.8.3 |
| *KIT PARTS LIST WORK PACKAGE*<br>**<kitswp>** | | F.8.4 |
| *BULK ITEM WORK PACKAGE*<br>**<bulk_itemswp>** | | F.8.5 |
| *SPECIAL TOOLS LIST WORK PACKAGE*<br>**<stlwp>** | | F.8.6 |
| **REAR MATTER**<br>**<rear>** | R | TABLE A-II |
| **LEGEND**<br>R - Required (see A.4.3.1.3).<br>Shaded - May be Required (see A.4.3.1.4). | | |



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**TABLE A-VI. DMWR / NMWR including RPSTL requirements matrix.**

| Content | DMWR with RPSTL | NMWR with RPSTL | MIL-STD REF |
|---|---|---|---|
| **PUBLICATION (<production>)**<br>**<paper.manual>** or **<framed.manual>**<br><br>PUBLICATION TITLES:<br>DMWR including RPSTL for *insert system*<br><br>NMWR including RPSTL for *insert system* | R | R | TABLE A-I |
| **FRONT MATTER**<br>**<paper.frnt>** or **<framed.frnt>** | R | R | TABLE A-II |
| **MODULE 1. GENERAL INFORMATION, EQUIPMENT DESCRIPTION, AND THEORY OF OPERATION**<br>**<gim>** | R | R | B.6 |
| *GENERAL INFORMATION WORK PACKAGE*<br>**<ginfowp>** | R | R | B.7.1 |
| Scope<br>**<scope>** | R | R | B.7.1.1 |
| Maintenance forms, records, and reports<br>**<mfrr>** | R | R | B.7.1.2 |
| Reporting Equipment Improvement Recommendations (EIR)<br>**<eir>** | R | R | B.7.1.3 |
| Hand Receipt (HR) manuals<br>**<handreceipt>** | | | B.7.1.4 |
| Corrosion Prevention and Control (CPC)<br>**<cpcdata>** | R | R | B.7.1.5 |
| Ozone Depleting Substances (ODS)<br>**<odsdata>** | | | B.7.1.6 |
| Destruction of equipment to prevent enemy use<br>**<destructmat>** | R | R | B.7.1.7 |
| Preparation for storage or shipment<br>**<pssref>** | R | R | B.7.1.8 |
| Transportability guidance<br>**<transportability>** | R | R | B.7.1.8 |
| Warranty information<br>**<wrntyref>** | | | B.7.1.9 |


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**TABLE A-VI. DMWR / NMWR including RPSTL requirements matrix. (continued**

| Content | DMWR with RPSTL | NMWR with RPSTL | MIL-STD REF |
|---|---|---|---|
| Nomenclature cross-reference list **<nomenreflist>** | | | B.7.1.10 |
| List of abbreviations/ acronyms **<loa>** | R | R | B.7.1.11 |
| Quality Assurance (QA) **<qainfo>** | | | B.7.1.12 |
| Quality of material **<qual.mat.info>** | R | R | B.7.1.13 |
| Safety, care, and handling **<sftyinfo>** | | | B.7.1.14 |
| Nuclear hardness **<hcp>** | | | B.7.1.15 |
| Calibration **<calref>** | | | B.7.1.16 |
| Item unique identification (IUID) **<iuid>** | | | B.7.1.17 |
| Engineering Change Proposals (ECP) **<ecp>** | R | R | B.7.1.18 |
| Modifications **<modification>** | | | B.7.1.19 |
| Deviations and exceptions **<deviation>** | R | R | B.7.1.20 |
| Mobilization requirements **<mobreq>** | R | R | B.7.1.21 |
| Mandatory replacement parts (MRP) **<mrpref>** | | | B.7.1.22 |
| Critical safety items (CSI) **<csireq>** | | | B.7.1.23 |
| Cost considerations **<cost>** | R | R | B.7.1.24 |
| Supporting information for repair parts, special tools/ kits, TMDE, and support equipment **<supdata>** | | | B.7.1.25 |
| Copyright credit line | | | B.7.1.26 |

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**TABLE A-VI. DMWR / NMWR including RPSTL requirements matrix. (continued**

| Content | DMWR with RPSTL | NMWR with RPSTL | MIL-STD REF |
|---|---|---|---|
| **<copyrt>** | | | |
| *EQUIPMENT DESCRIPTION AND DATA WORK PACKAGE*<br>**<descwp>** | R | R | B.7.2 |
| Equipment characteristics, capabilities, and features<br>**<eqpinfo>** | R | R | B.7.2.1 |
| Location and description of major components<br>**<locdesc>**<br>(optional for component DMWRs) | R | R | B.7.2.2 |
| Equipment differences<br>**<eqpdiff>**<br>(required if there are differences) | | | B.7.2.3 |
| Equipment data<br>**<eqpdata>** | R | R | B.7.2.4 |
| *THEORY OF OPERATION WORK PACKAGE*<br>**<thrywp>** | | | B.7.3 |
| **MODULE X. DMWR/NMWR TROUBLESHOOTING PROCEDURES**<br>*Note: The notation (*) indicates that at least one of the these content items shall be included.*<br>**<tim><troubledmwrnmwrcategory>** | R | R | D.8<br>D.7.2 |
| *TROUBLESHOOTING INTRODUCTION WORK PACKAGE*<br>**<tsintrowp>** | | | D.8.1 |
| *TROUBLESHOOTING INDEX WORK PACKAGE*<br>**<tsindxwp>** | | | D.8.3 |
| *PRESHOP ANALYSIS WORK PACKAGE*<br>**<pshopanalwp>** | R | R | D.8.7 |
| *COMPONENT CHECKLIST WORK PACKAGE*<br>**<compchklistwp>** | | | D.8.8 |
| **OPERATIONAL CHECKOUT WORK PACKAGES*<br>**<opcheckwp>** | | | D.8.5 |
| **TROUBLESHOOTING WORK PACKAGES*<br>**<tswp>** | | | D.8.4 |
| **COMBINED OPERATIONAL CHECKOUT AND TROUBLESHOOTING PROCEDURE WORK PACKAGES* | | | D.8.5.1 |



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**TABLE A-VI. DMWR / NMWR including RPSTL requirements matrix. (continued**

| Content | DMWR with RPSTL | NMWR with RPSTL | MIL-STD REF |
|---|---|---|---|
| **<opcheck-tswp>** | | | |
| **DIAGNOSTICS WORK PACKAGES*<br>**<diagnosticwp>** | | | D.8.6 |
| **MODULE X. DEPOT MAINTENANCE INSTRUCTIONS**<br>**<mim>**<br>**<depotcategory>** | R | R | E.6<br>E.6.4 |
| *PRESERVATION, PACKAGING AND MARKING GENERAL INFORMATION WORK PACKAGE*<br>**<ppmgeninfowp>** | R | R | E.7.19 |
| *EQUIPMENT / USER FITTING INSTRUCTIONS WORK PACKAGE*<br>**<perseqpwp>** | | | E.7.11 |
| *MAINTENANCE INSTRUCTION WORK PACKAGES*<br>Maintenance tasks<br>**<maintwp>**<br>**<maintsk>** | R | R | E.7.4<br>E.7.4.4 |
| Adjust<br>**<adjust>** | | | E.7.4.4.1 |
| Align<br>**<align>** | | | E.7.4.4.2 |
| Arm<br>**<arm>** | | | E.7.4.4.3 |
| Assemble<br>**<assem>** | | | E.7.4.4.4.1 |
| Disassemble<br>**<disassem>** | | | E.7.4.4.4.2 |
| Calibrate<br>**<calibration>** | | | E.7.4.4.5 |
| Clean<br>**<clean>** | | | E.7.4.4.6 |
| Configure<br>**<configure>** | | | E.7.4.4.7 |
| Covering<br>**<cover>** | | | E.7.4.4.8 |
| External power | | | E.7.4.4.9 |



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**TABLE A-VI. DMWR / NMWR including RPSTL requirements matrix. (continued**

| Content | DMWR with RPSTL | NMWR with RPSTL | MIL-STD REF |
|---|---|---|---|
| **<extpwr>** | | | |
| Hoisting<br>**<hoist>** | | | E.7.4.4.10 |
| Inspect<br>**<inspect>** | | | E.7.4.4.11 |
| Install peripheral device<br>**<installperdev>** | | | E.7.4.4.12.1 |
| Uninstall peripheral device<br>**<uninstallperdev>** | | | E.7.4.4.12.2 |
| Jacking<br>**<jack>** | | | E.7.4.4.13 |
| Load<br>**<load>** | | | E.7.4.4.14.1 |
| Unload<br>**<unload>** | | | E.7.4.4.14.2 |
| Lubricate<br>**<lube>** | | | E.7.4.4.15 |
| Mark (for example, "data plates")<br>**<mark>** | | | E.7.4.4.16 |
| Mooring<br>**<moor>** | | | E.7.4.4.17 |
| Nondestructive inspection<br>**<ndi>** | | | E.7.4.4.18 |
| Overhaul<br>**<overhaul>** | | | E.7.4.4.19 |
| Pack<br>**<pack>** | | | E.7.4.4.20.1 |
| Unpack<br>**<unpack>** | | | E.7.4.4.20.2 |
| Paint<br>**<paint>** | | | E.7.4.4.21 |
| Parking<br>**<park>** | | | E.7.4.4.22 |
| Place in service | | | E.7.4.4.23 |



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**TABLE A-VI. DMWR / NMWR including RPSTL requirements matrix. (continued**

| Content | DMWR with RPSTL | NMWR with RPSTL | MIL-STD REF |
|---|---|---|---|
| **<pis>** | | | |
| Preparation for storage **<prepstore>** | + | + | E.7.4.4.24 |
| Preparation for shipment **<prepship>** | + | + | E.7.4.4.25 |
| Prepare for use **<prepforuse>** | | | E.7.4.4.26 |
| Preserve **<preservation>** | | | E.7.4.4.27 |
| Radio interference suppression **<ris>** | | | E.7.4.4.28 |
| Rebuild **<rebuild>** | | | E.7.4.4.29 |
| Remove **<remove>** | | | E.7.4.4.31.1 |
| Install **<install>** | | | E.7.4.4.31.2 |
| Repair **<repair>** | | | E.7.4.4.32 |
| Replace **<replace>** | | | E.7.4.4.33 |
| Service **<service>** | | | E.7.4.4.34 |
| Sling loading **<sling>** | | | E.7.4.4.35 |
| Test **<test>** | | | E.7.4.4.36 |
| Towing **<tow>** | | | E.7.4.4.37 |
| Transport **<transport>** | + | + | E.7.4.4.38 |
| Additional maintenance task **<other.maintsk>** | | | E.7.4.4.40 |
| Follow-on maintenance | | | E.7.4.4.41 |



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**TABLE A-VI. DMWR / NMWR including RPSTL requirements matrix. (continued**

| Content | DMWR with RPSTL | NMWR with RPSTL | MIL-STD REF |
|---|---|---|---|
| **<followon.maintsk>** | | | |
| *SOFTWARE MAINTENANCE WORK PACKAGES*<br>Software maintenance tasks<br>**<maintwp>**<br>**<softmaintsk>** | | | E.7.4<br>E.7.4.4.39 |
| Configure software<br>**<softconfig>** | | | E.7.4.4.39.1 |
| Debug software<br>**<debug>** | | | E.7.4.4.39.2 |
| Install software<br>**<install>** | | | E.7.4.4.31.1 |
| Remove software<br>**<remove>** | | | E.7.4.4.31 |
| Upgrade / patch software<br>**<upgrade>** | | | E.7.4.4.39.3 |
| Additional maintenance task<br>**<other.maintsk>** | | | E.7.4.4.40 |
| Follow-on maintenance<br>**<followon.maintsk>** | | | E.7.4.4.41 |
| *GENERAL MAINTENANCE WORK PACKAGE*<br>**<gen.maintwp>** | | | E.7.5 |
| *LUBRICATION INSTRUCTIONS WORK PACKAGE*<br>**<lubewp>** | | | E.7.6 |
| *FACILITIES WORK PACKAGE*<br>**<facilwp>** | | | E.7.20 |
| *OVERHAUL INSPECTION PROCEDURES WORK PACKAGE*<br>**<oipwp>** | | | E.7.21 |
| *DEPOT MOBILIZATION REQUIREMENTS WORK PACKAGE*<br>**<mobilwp>** | | | E.7.22 |
| *QUALITY ASSURANCE REQUIREMENTS WORK PACKAGE*<br>**<qawp>** | R | R | E.7.23 |


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**TABLE A-VI. DMWR / NMWR including RPSTL requirements matrix. (continued**

| Content | DMWR with RPSTL | NMWR with RPSTL | MIL-STD REF |
|---|---|---|---|
| **ILLUSTRATED LIST OF MANUFACTURED ITEMS**<br>*Note: If required, the introduction work package and manufacturing procedure work package are both required.* | | | E.7.7 |
| *ILLUSTRATED LIST OF MANUFACTURED ITEMS INTRODUCTION WORK PACKAGE*<br>**<manu_items_introwp>** | CR | CR | E.7.7 |
| *MANUFACTURING PROCEDURES WORK PACKAGE*<br>**<manuwp>** | CR | CR | E.7.8 |
| *TORQUE LIMITS WORK PACKAGE*<br>**<torquewp>** | | | E.7.9 |
| *WIRING DIAGRAMS AND SCHEMATICS WORK PACKAGE*<br>**<wiringwp>** | | | E.7.10 |
| **MODULE X. AUXILIARY EQUIPMENT MAINTENANCE INSTRUCTIONS**<br>**<mim>**<br>**<auxiliarycategory>** | | | E.6<br>E.6.6 |
| *AUXILIARY EQUIPMENT MAINTENANCE WORK PACKAGE*<br>**<auxeqpwp>** | | | E.7.24 |
| **ILLUSTRATED LIST OF MANUFACTURED ITEMS**<br>*Note: If required, the introduction work package and manufacturing procedure work package are both required.* | | | E.7.7 |
| *ILLUSTRATED LIST OF MANUFACTURED ITEMS INTRODUCTION WORK PACKAGE*<br>**<manu_items_introwp>** | CR | CR | E.7.7 |
| *MANUFACTURING PROCEDURES WORK PACKAGE*<br>**<manuwp>** | CR | CR | E.7.8 |
| *TORQUE LIMITS WORK PACKAGE*<br>**<torquewp>** | | | E.7.9 |
| *WIRING DIAGRAMS AND SCHEMATICS WORK PACKAGE*<br>**<wiringwp>** | | | E.7.10 |


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**TABLE A-VI. DMWR / NMWR including RPSTL requirements matrix. (continued**

| Content | DMWR with RPSTL | NMWR with RPSTL | MIL-STD REF |
|---|---|---|---|
| **MODULE X. AMMUNITION MAINTENANCE INSTRUCTIONS**<br>**<mim>**<br>**<ammunitioncategory>** | | | E.6<br>E.6.7 |
| *AMMUNITION IDENTIFICATION WORK PACKAGE*<br>**<ammoidentwp>** | | | E.7.25 |
| *AMMUNITION MAINTENANCE WORK PACKAGE*<br>**<ammowp>** | | | E.7.26 |
| *FOREIGN AMMUNITION WORK PACKAGE*<br>**<natowp>** | | | E.7.27 |
| **MODULE X. SOFTWARE INFORMATION**<br>**<soim>** | | | M.7.7 |
| *SOFTWARE GENERAL INFORMATION WORK PACKAGE*<br>**<softginfowp>** | CR | CR | M.7.1 |
| System overview<br>**<softsysover>** | CR | CR | M.7.1 |
| Document overview<br>**<softdocover>** | CR | CR | M.7.1 |
| Copyright credit line<br>**<copyrt>** | | | M.7.1 |
| Endorsement notice<br>**<endorse>** | | | M.7.1 |
| *SOFTWARE SUMMARY WORK PACKAGE*<br>**<softsumwp>** | CR | CR | M.7.2 |
| *SOFTWARE EFFECTIVITY WORK PACKAGE*<br>**<softeffectwp>** | | | M.7.3 |
| *DIFFERENCES BETWEEN SOFTWARE VERSIONS WORK PACKAGE*<br>**<softdiffversionwp>** | | | M.7.4 |
| *FEATURES AND CAPABILITIES WORK PACKAGE*<br>**<softfeaturescapwp>** | CR | CR | M.7.5 |
| *FUNCTIONAL DISPLAY WORK PACKAGE*<br>**<functionaldisplaywp>** | CR | CR | M.7.6 |
| *SECURITY AND PRIVACY PROCEDURES WORK PACKAGE* | CR | CR | M.7.7.1 |



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**TABLE A-VI. DMWR / NMWR including RPSTL requirements matrix. (continued**

| Content | DMWR with RPSTL | NMWR with RPSTL | MIL-STD REF |
|---|---|---|---|
| **<softsecprivwp>** | | | |
| *SUPERVISORY CONTROLS WORK PACKAGE* **<softsuperctrlswp>** | | | M.7.7.2 |
| *POWER UP AND ACCESS SOFTWARE PROCEDURES WORK PACKAGE* **<softpwraccesswp>** | CR | CR | M.7.7.3 |
| *SOFTWARE OPERATOR PROCEDURES WORK PACKAGE* **<softoperwp>** | CR | CR | M.7.7.4 |
| Key commands **<softkeycmd>** | | | M.7.7.4.1 |
| User interface **<softgui>** | | | M.7.7.4.2 |
| Software operating conventions **<softopconventions>** | | | M.7.7.4.3 |
| Software operating procedures **<softoperproc>** | | | M.7.7.4.4 |
| *POWER DOWN AND EXIT SOFTWARE PROCEDURES WORK PACKAGE* **<softpwrexitwp>** | CR | CR | M.7.7.5 |
| *MESSAGES WORK PACKAGE* **<softmessageswp>** | | | M.7.8.3 |
| *RECOVERY FROM ERRORS, MALFUNCTIONS, AND EMERGENCIES WORK PACKAGE* **<softerrorswp>** | | | M.7.8.4 |
| **MODULE X. PARTS INFORMATION** **<pim>** | R | R | F.7.2 |
| *INTRODUCTION WORK PACKAGE* **<introwp>** | R | R | F.8.1 |
| *REPAIR PARTS LIST WORK PACKAGE* **<plwp>** | R | R | F.8.2 |
| *REPAIR PARTS FOR SPECIAL TOOLS WORK PACKAGE* **<stl_partswp>** | | | F.8.3 |
| *KIT PARTS LIST WORK PACKAGE* **<kitswp>** | | | F.8.4 |


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**TABLE A-VI. DMWR / NMWR including RPSTL requirements matrix. (continued**

| Content | DMWR with RPSTL | NMWR with RPSTL | MIL-STD REF |
|---|---|---|---|
| *BULK ITEMS WORK PACKAGE*<br>**<bulk_itemswp>** | | | F.8.5 |
| *SPECIAL TOOLS LIST WORK PACKAGE*<br>**<stlwp>** | | | F.8.6 |
| **MODULE X. SUPPORTING INFORMATION**<br>Note: Applicable supporting information work packages shall be arranged in the order in which they are presented here and numbered accordingly.<br>**<sim>** | R | R | G.6 |
| *REFERENCES WORK PACKAGE*<br>**<refwp>** | R | R | G.7.1 |
| *EXPENDABLE AND DURABLE ITEMS LIST WORK PACKAGE*<br>**<explistwp>** | R | R | G.7.7 |
| *TOOL IDENTIFICATION LIST WORK PACKAGE*<br>**<toolidwp>** | R | R | G.7.8 |
| *MANDATORY REPLACEMENT PARTS WORK PACKAGE*<br>**<mrplwp>** | + | + | G.7.9 |
| *CRITICAL SAFETY ITEMS WORK PACKAGE*<br>**<csi.wp>** | + | + | G.7.10 |
| *SUPPORT ITEMS WORK PACKAGE*<br>**<supitemwp>** | | | G.7.11 |
| *ADDITIONAL SUPPORTING WORK PACKAGE*<br>**<genwp>** | | | G.7.12 |
| **REAR MATTER**<br>**<rear>** | R | R | TABLE A-II |

**LEGEND**
R - Required (see A.4.3.1.3).
P - Prohibited (see A.4.3.1.3).
Shaded - May be Required (see A.4.3.1.4).
CR - Conditionally Required (see A.4.3.1.4.3).
+ Plus - Typically Required (see A.4.3.1.4.1). WP shall be listed in General Information, WP 0001 (see B.7.1.1).



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**TABLE A-VII. DMWR / NMWR requirements matrix.**

| Content | DMWR | NMWR | MIL-STD REF |
|---|---|---|---|
| **PUBLICATION** (**<production>**)<br>**<paper.manual>** or **<framed.manual>**<br><br>PUBLICATION TITLES:<br>DMWR for *insert system*<br><br>NMWR for *insert system* | R | R | TABLE A-I |
| **FRONT MATTER**<br>**<paper.frnt>** or **<framed.frnt>** | R | R | TABLE A-II |
| **MODULE 1. GENERAL INFORMATION, EQUIPMENT DESCRIPTION, AND THEORY OF OPERATION**<br>**<gim>** | R | R | B.6 |
| *GENERAL INFORMATION WORK PACKAGE*<br>**<ginfowp>** | R | R | B.7.1 |
| Scope<br>**<scope>** | R | R | B.7.1.1 |
| Maintenance forms, records, and reports<br>**<mfrr>** | R | R | B.7.1.2 |
| Reporting Equipment Improvement Recommendations (EIR)<br>**<eir>** | R | R | B.7.1.3 |
| Hand Receipt (HR) manuals<br>**<handreceipt>** | | | B.7.1.4 |
| Corrosion Prevention and Control (CPC)<br>**<cpcdata>** | R | R | B.7.1.5 |
| Ozone Depleting Substances (ODS)<br>**<odsdata>** | | | B.7.1.6 |
| Destruction of equipment to prevent enemy use<br>**<destructmat>** | R | R | B.7.1.7 |
| Preparation for storage or shipment<br>**<pssref>** | R | R | B.7.1.8 |
| Transportability guidance<br>**<transportability>** | R | R | B.7.1.8 |
| Warranty information<br>**<wrntyref>** | | | B.7.1.9 |
| Nomenclature cross-reference list<br>**<nomenreflist>** | | | B.7.1.10 |

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**TABLE A-VII. DMWR / NMWR requirements matrix. (continued**

| Content | DMWR | NMWR | MIL-STD REF |
|---|---|---|---|
| List of abbreviations/ acronyms **<loa>** | R | R | B.7.1.11 |
| Quality Assurance (QA) **<qainfo>** | | | B.7.1.12 |
| Quality of material **<qual.mat.info>** | R | R | B.7.1.13 |
| Safety, care, and handling **<sftyinfo>** | | | B.7.1.14 |
| Nuclear hardness **<hcp>** | | | B.7.1.15 |
| Calibration **<calref>** | | | B.7.1.16 |
| Item unique identification (IUID) **<iuid>** | | | B.7.1.17 |
| Engineering Change Proposals (ECP) **<ecp>** | R | R | B.7.1.18 |
| Modifications **<modification>** | | | B.7.1.19 |
| Deviations and exceptions **<deviation>** | R | R | B.7.1.20 |
| Mobilization requirements **<mobreq>** | R | R | B.7.1.21 |
| Mandatory replacement parts (MRP) **<mrpref>** | | | B.7.1.22 |
| Critical safety items (CSI) **<csireq>** | | | B.7.1.23 |
| Cost considerations **<cost>** | R | R | B.7.1.24 |
| Supporting information for repair parts, special tools/ kits, TMDE, and support equipment **<supdata>** | | | B.7.1.25 |
| Copyright credit line **<copyrt>** | | | B.7.1.26 |
| *EQUIPMENT DESCRIPTION AND DATA WORK PACKAGE* **<descwp>** | R | R | B.7.2 |

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**TABLE A-VII. DMWR / NMWR requirements matrix. (continued**

| Content | DMWR | NMWR | MIL-STD REF |
|---|---|---|---|
| Equipment characteristics, capabilities, and features **<eqpinfo>** | R | R | B.7.2.1 |
| Location and description of major components **<locdesc>** | R | R | B.7.2.2 |
| Equipment differences **<eqpdiff>** | R | R | B.7.2.3 |
| Equipment data **<eqpdata>** | R | R | B.7.2.4 |
| *THEORY OF OPERATION WORK PACKAGE* **<thrywp>** | | | B.7.3 |
| **MODULE X. DMWR/NMWR TROUBLESHOOTING PROCEDURES** *Note: The notation (*) indicates that at least one of the these content items shall be included.* **<tim><troubledmwrnmwrcategory>** | R | R | D.8<br>D.7.2 |
| *TROUBLESHOOTING INTRODUCTION WORK PACKAGE* **<tsintrowp>** | | | D.8.1 |
| *TROUBLESHOOTING INDEX WORK PACKAGE* **<tsindxwp>** | | | D.8.3 |
| *PRESHOP ANALYSIS WORK PACKAGE* **<pshopanalwp>** | R | R | D.8.7 |
| *COMPONENT CHECKLIST WORK PACKAGE* **<compchklistwp>** | | | D.8.8 |
| **OPERATIONAL CHECKOUT WORK PACKAGES* **<opcheckwp>** | | | D.8.5 |
| **TROUBLESHOOTING WORK PACKAGES* **<tswp>** | | | D.8.4 |
| **COMBINED OPERATIONAL CHECKOUT AND TROUBLESHOOTING PROCEDURE WORK PACKAGES* **<opcheck-tswp>** | | | D.8.5.1 |
| **DIAGNOSTICS WORK PACKAGES* **<diagnosticwp>** | | | D.8.6 |
| **MODULE X. DEPOT MAINTENANCE INSTRUCTIONS** **<mim>** **<depotcategory>** | R | R | E.6<br>E.6.4 |
| *PRESERVATION, PACKAGING AND MARKING GENERAL INFORMATION WORK PACKAGE* | R | R | E.7.19 |


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**TABLE A-VII. DMWR / NMWR requirements matrix. (continued**

| Content | DMWR | NMWR | MIL-STD REF |
|---|---|---|---|
| **<ppmgeninfowp>** | | | |
| *EQUIPMENT / USER FITTING INSTRUCTIONS WORK PACKAGE*<br>**<perseqpwp>** | | | E.7.11 |
| *MAINTENANCE INSTRUCTION WORK PACKAGES*<br>Maintenance tasks<br>**<maintwp>**<br>**<maintsk>** | R | R | E.7.4<br>E.7.4.4 |
| Adjust<br>**<adjust>** | | | E.7.4.4.1 |
| Align<br>**<align>** | | | E.7.4.4.2 |
| Arm<br>**<arm>** | | | E.7.4.4.3 |
| Assemble<br>**<assem>** | | | E.7.4.4.4.1 |
| Disassemble<br>**<disassem>** | | | E.7.4.4.4.2 |
| Calibrate<br>**<calibration>** | | | E.7.4.4.5 |
| Clean<br>**<clean>** | | | E.7.4.4.6 |
| Configure<br>**<configure>** | | | E.7.4.4.7 |
| Covering<br>**<cover>** | | | E.7.4.4.8 |
| External power<br>**<extpwr>** | | | E.7.4.4.9 |
| Hoisting<br>**<hoist>** | | | E.7.4.4.10 |
| Inspect<br>**<inspect>** | | | E.7.4.4.11 |
| Install peripheral device<br>**<installperdev>** | | | E.7.4.4.12.1 |
| Uninstall peripheral device<br>**<uninstallperdev>** | | | E.7.4.4.12.2 |



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**TABLE A-VII. DMWR / NMWR requirements matrix. (continued**

| Content | DMWR | NMWR | MIL-STD REF |
|---|---|---|---|
| Jacking<br>**<jack>** | | | E.7.4.4.13 |
| Load<br>**<load>** | | | E.7.4.4.14.1 |
| Unload<br>**<unload>** | | | E.7.4.4.14.2 |
| Lubricate<br>**<lube>** | | | E.7.4.4.15 |
| Mark (for example, "data plates")<br>**<mark>** | | | E.7.4.4.16 |
| Mooring<br>**<moor>** | | | E.7.4.4.17 |
| Nondestructive inspection<br>**<ndi>** | | | E.7.4.4.18 |
| Overhaul<br>**<overhaul>** | | | E.7.4.4.19 |
| Pack<br>**<pack>** | | | E.7.4.4.20.1 |
| Unpack<br>**<unpack>** | | | E.7.4.4.20.2 |
| Paint<br>**<paint>** | | | E.7.4.4.21 |
| Parking<br>**<park>** | | | E.7.4.4.22 |
| Place in service<br>**<pis>** | | | E.7.4.4.23 |
| Preparation for storage<br>**<prepstore>** | + | + | E.7.4.4.24 |
| Preparation for shipment<br>**<prepship>** | + | + | E.7.4.4.25 |
| Prepare for use<br>**<prepforuse>** | | | E.7.4.4.26 |
| Preserve<br>**<preservation>** | | | E.7.4.4.27 |
| Radio interference suppression<br>**<ris>** | | | E.7.4.4.28 |

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**TABLE A-VII. DMWR / NMWR requirements matrix. (continued**

| Content | DMWR | NMWR | MIL-STD REF |
|---|---|---|---|
| Rebuild<br>**<rebuild>** | | | E.7.4.4.29 |
| Remove<br>**<remove>** | | | E.7.4.4.31.1 |
| Remove<br>**<remove>** | | | E.7.4.4.31.2 |
| Repair<br>**<repair>** | | | E.7.4.4.32 |
| Replace<br>**<replace>** | | | E.7.4.4.33 |
| Service<br>**<service>** | | | E.7.4.4.34 |
| Sling loading<br>**<sling>** | | | E.7.4.4.35 |
| Test<br>**<test>** | | | E.7.4.4.36 |
| Towing<br>**<tow>** | | | E.7.4.4.37 |
| Transport<br>**<transport>** | + | + | E.7.4.4.38 |
| Additional maintenance task<br>**<other.maintsk>** | | | E.7.4.4.40 |
| Follow-on maintenance<br>**<followon.maintsk>** | | | E.7.4.4.41 |
| *SOFTWARE MAINTENANCE WORK PACKAGES*<br>Software maintenance tasks<br>**<maintwp>**<br>**<softmaintsk>** | | | E.7.4<br>E.7.4.4.39 |
| Configure software<br>**<softconfig>** | | | E.7.4.4.39.1 |
| Debug software<br>**<debug>** | | | E.7.4.4.39.2 |
| Install software<br>**<install>** | | | E.7.4.4.31.1 |
| Remove software<br>**<remove>** | | | E.7.4.4.31 |


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**TABLE A-VII. DMWR / NMWR requirements matrix. (continued**

| Content | DMWR | NMWR | MIL-STD REF |
|---|---|---|---|
| Upgrade / patch software **<upgrade>** | | | E.7.4.4.39.3 |
| Additional maintenance task **<other.maintsk>** | | | E.7.4.4.40 |
| Follow-on maintenance **<followon.maintsk>** | | | E.7.4.4.41 |
| *GENERAL MAINTENANCE WORK PACKAGE* **<gen.maintwp>** | | | E.7.5 |
| *LUBRICATION INSTRUCTIONS WORK PACKAGE* **<lubewp>** | | | E.7.6 |
| *FACILITIES WORK PACKAGE* **<facilwp>** | | | E.7.20 |
| *OVERHAUL INSPECTION PROCEDURES WORK PACKAGE* **<oipwp>** | | | E.7.21 |
| *DEPOT MOBILIZATION REQUIREMENTS WORK PACKAGE* **<mobilwp>** | | | E.7.22 |
| *QUALITY ASSURANCE REQUIREMENTS WORK PACKAGE* **<qawp>** | R | R | E.7.23 |
| **ILLUSTRATED LIST OF MANUFACTURED ITEMS** *Note: If required, the introduction work package and manufacturing procedure work package are both required.* | | | E.7.7 |
| *ILLUSTRATED LIST OF MANUFACTURED ITEMS INTRODUCTION WORK PACKAGE* **<manu_items_introwp>** | CR | CR | E.7.7 |
| *MANUFACTURING PROCEDURES WORK PACKAGE* **<manuwp>** | CR | CR | E.7.8 |
| *TORQUE LIMITS WORK PACKAGE* **<torquewp>** | | | E.7.9 |
| *WIRING DIAGRAMS AND SCHEMATICS WORK PACKAGE* **<wiringwp>** | | | E.7.10 |
| **MODULE X. AUXILIARY EQUIPMENT MAINTENANCE INSTRUCTIONS** | | | E.6 |

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**TABLE A-VII. DMWR / NMWR requirements matrix. (continued**

| Content | DMWR | NMWR | MIL-STD REF |
|---|---|---|---|
| **<mim>**<br>**<auxiliarycategory>** | | | E.6.6 |
| *AUXILIARY EQUIPMENT MAINTENANCE WORK PACKAGE*<br>**<auxeqpwp>** | | | E.7.24 |
| **ILLUSTRATED LIST OF MANUFACTURED ITEMS**<br>*Note: If required, the introduction work package and manufacturing procedure work package are both required.* | | | E.7.7 |
| *ILLUSTRATED LIST OF MANUFACTURED ITEMS INTRODUCTION WORK PACKAGE*<br>**<manu_items_introwp>** | CR | CR | E.7.7 |
| *MANUFACTURING PROCEDURES WORK PACKAGE*<br>**<manuwp>** | CR | CR | E.7.8 |
| *TORQUE LIMITS WORK PACKAGE*<br>**<torquewp>** | | | E.7.9 |
| *WIRING DIAGRAMS AND SCHEMATICS WORK PACKAGE*<br>**<wiringwp>** | | | E.7.10 |
| **MODULE X. AMMUNITION MAINTENANCE INSTRUCTIONS**<br>**<mim>**<br>**<ammunitioncategory>** | | | E.6<br>E.6.7 |
| *AMMUNITION IDENTIFICATION WORK PACKAGE*<br>**<ammoidentwp>** | | | E.7.25 |
| *AMMUNITION MAINTENANCE WORK PACKAGE*<br>**<ammowp>** | | | E.7.26 |
| *FOREIGN AMMUNITION WORK PACKAGE*<br>**<natowp>** | | | E.7.27 |
| **MODULE X. SOFTWARE INFORMATION**<br>**<soim>** | | | M.7.7 |
| *SOFTWARE GENERAL INFORMATION WORK PACKAGE*<br>**<softginfowp>** | CR | CR | M.7.1 |
| System overview<br>**<softsysover>** | CR | CR | M.7.1 |
| Document overview<br>**<softdocover>** | CR | CR | M.7.1 |


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**TABLE A-VII. DMWR / NMWR requirements matrix. (continued**

| Content | DMWR | NMWR | MIL-STD REF |
|---|---|---|---|
| Copyright credit line<br>**<copyrt>** | | | M.7.1 |
| Endorsement notice<br>**<endorse>** | | | M.7.1 |
| *SOFTWARE SUMMARY WORK PACKAGE*<br>**<softsumwp>** | CR | CR | M.7.2 |
| *SOFTWARE EFFECTIVITY WORK PACKAGE*<br>**<softeffectwp>** | | | M.7.3 |
| *DIFFERENCES BETWEEN SOFTWARE VERSIONS WORK PACKAGE*<br>**<softdiffversionwp>** | | | M.7.4 |
| *FEATURES AND CAPABILITIES WORK PACKAGE*<br>**<softfeaturescapwp>** | CR | CR | M.7.5 |
| *FUNCTIONAL DISPLAY WORK PACKAGE*<br>**<functionaldisplaywp>** | CR | CR | M.7.6 |
| *SECURITY AND PRIVACY PROCEDURES WORK PACKAGE*<br>**<softsecprivwp>** | CR | CR | M.7.7.1 |
| *SUPERVISORY CONTROLS WORK PACKAGE*<br>**<softsuperctrlswp>** | | | M.7.7.2 |
| *POWER UP AND ACCESS SOFTWARE PROCEDURES WORK PACKAGE*<br>**<softpwraccesswp>** | CR | CR | M.7.7.3 |
| *SOFTWARE OPERATOR PROCEDURES WORK PACKAGE*<br>**<softoperwp>** | CR | CR | M.7.7.4 |
| Key commands<br>**<softkeycmd>** | | | M.7.7.4.1 |
| User interface<br>**<softgui>** | | | M.7.7.4.2 |
| Software operating conventions<br>**<softopconventions>** | | | M.7.7.4.3 |
| Software operating procedures<br>**<softoperproc>** | | | M.7.7.4.3 |
| *POWER DOWN AND EXIT SOFTWARE PROCEDURES WORK PACKAGE*<br>**<softpwrexitwp>** | CR | CR | M.7.7.5 |



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**TABLE A-VII. DMWR / NMWR requirements matrix. (continued**

| Content | DMWR | NMWR | MIL-STD REF |
|---|---|---|---|
| *MESSAGES WORK PACKAGE*<br>**<softmessageswp>** | | | M.7.8.3 |
| *RECOVERY FROM ERRORS, MALFUNCTIONS, AND EMERGENCIES WORK PACKAGE*<br>**<softerrorswp>** | | | M.7.8.4 |
| **MODULE X. PARTS INFORMATION**<br>**<pim>** | P | P | F.7.2 |
| *INTRODUCTION WORK PACKAGE*<br>**<introwp>** | P | P | F.8.1 |
| *REPAIR PARTS LIST WORK PACKAGE*<br>**<plwp>** | P | P | F.8.2 |
| *REPAIR PARTS FOR SPECIAL TOOLS WORK PACKAGE*<br>**<stl_partswp>** | P | P | F.8.3 |
| *KIT PARTS LIST WORK PACKAGE*<br>**<kitswp>** | P | P | F.8.4 |
| *BULK ITEMS WORK PACKAGE*<br>**<bulk_itemswp>** | P | P | F.8.5 |
| *SPECIAL TOOLS LIST WORK PACKAGE*<br>**<stlwp>** | P | P | F.8.6 |
| **MODULE X. SUPPORTING INFORMATION**<br>Note: Applicable supporting information work packages shall be arranged in the order in which they are presented here and numbered accordingly.<br>**<sim>** | R | R | G.6 |
| *REFERENCES WORK PACKAGE*<br>**<refwp>** | R | R | G.7.1 |
| *EXPENDABLE AND DURABLE ITEMS LIST WORK PACKAGE*<br>**<explistwp>** | R | R | G.7.7 |
| *TOOL IDENTIFICATION LIST WORK PACKAGE*<br>**<toolidwp>** | R | R | G.7.8 |
| *MANDATORY REPLACEMENT PARTS WORK PACKAGE*<br>**<mrplwp>** | + | + | G.7.9 |
| *CRITICAL SAFETY ITEMS WORK PACKAGE*<br>**<csi.wp>** | + | + | G.7.10 |
| *SUPPORT ITEMS WORK PACKAGE*<br>**<supitemwp>** | | | G.7.11 |

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**TABLE A-VII. DMWR / NMWR requirements matrix. (continued**

| Content | DMWR | NMWR | MIL-STD REF |
|---|---|---|---|
| *ADDITIONAL SUPPORTING WORK PACKAGE*<br>**<genwp>** | | | G.7.12 |
| **REAR MATTER**<br>**<rear>** | R | R | TABLE A-II |

**LEGEND**
R - Required (see A.4.3.1.3).
P - Prohibited (see A.4.3.1.3).
Shaded - May be Required (see A.4.3.1.4).
CR - Conditionally Required (see A.4.3.1.4.3).
+ Plus - Typically Required (see A.4.3.1.4.1). WP shall be listed in General Information, WP 0001 (see B.7.1.1).


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**TABLE A-VIII. Software requirements matrix.**

| Content | SUM | SAM | SUM/SAM | MIL-STD REF |
|---|---|---|---|---|
| **PUBLICATION** (**<production>**)<br>**<sum>** or **<sam>**<br><br>PUBLICATION TITLES:<br>(SUM) Software Users Manual for *insert system* .<br><br>(SAM) Software Administrators Manual for *insert system* .<br><br>(SUM/SAM) Software Users and Administrators Manual for *insert system* . | R | R | R | TABLE A-I |
| **FRONT MATTER**<br>**<paper.frnt>** or **<framed.frnt>** | R | R | R | TABLE A-II |
| **MODULE 1. SOFTWARE GENERAL INFORMATION**<br>**<gim>** | R | R | R | M.6.2.1 |
| *GENERAL INFORMATION WORK PACKAGE*<br>**<softginfowp>** | R | R | R | M.7.1 |
| Scope<br>**<scope>** | R | R | R | M.7.1 |
| Maintenance forms, records, and reports<br>**<mfrr>** | R | R | R | M.7.1 |
| Reporting Equipment Improvement Recommendations (EIR)<br>**<eir>** | R | R | R | M.7.1 |
| System overview<br>**<softsysover>** | R | R | R | M.7.1 |
| Document overview<br>**<softdocover>** | R | R | R | M.7.1 |
| Warranty information<br>**<wrntyref>** | | | | M.7.1 |
| Destruction of software to prevent enemy use<br>**<destructmat>** | R | R | R | M.7.1 |
| Nomenclature crossreference list<br>**<nomenreflist>** | | | | M.7.1 |
| List of abbreviations / acronyms<br>**<loa>** | R | R | R | M.7.1 |
| Copyright credit line<br>**<copyrt>** | | | | M.7.1 |

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**TABLE A-VIII. Software requirements matrix. (continued**

| Content | SUM | SAM | SUM/SAM | MIL-STD REF |
|---|---|---|---|---|
| Endorsement notice<br>**<endorse>** | | | | M.7.1 |
| *SOFTWARE SUMMARY WORK PACKAGE*<br>**<softsumwp>** | R | R | R | M.7.2 |
| Software application<br>**<soft_app>** | R | R | R | M.7.2 |
| Software inventory<br>**<soft_inventory>** | R | R | R | M.7.2 |
| Software environment<br>**<soft_environment>** | | R | R | M.7.2 |
| Security and privacy<br>**<soft_secpriv>** | R | R | R | M.7.2 |
| Supervisory controls<br>**<soft_superctrls>** | | R | R | M.7.2 |
| Assistance and problem reporting<br>**<soft_assistreport>** | R | R | R | M.7.2 |
| *SOFTWARE EFFECTIVITY WORK PACKAGE*<br>**<softeffectwp>** | | | | M.7.3 |
| *DIFFERENCES BETWEEN SOFTWARE VERSIONS WORK PACKAGE*<br>**<softdiffversionwp>** | | | | M.7.4 |
| **MODULE 2. SOFTWARE DESCRIPTION / DATA**<br>**<softdescdata>** | R | R | R | M.6.2.2 |
| *FEATURES AND CAPABILITIES WORK PACKAGE*<br>**<softfeaturescapwp>** | | | | M.7.5 |
| *FUNCTIONAL DISPLAY WORK PACKAGE*<br>**<functionaldisplaywp>** | R | R | R | M.7.6 |
| **MODULE X. SOFTWARE OPERATOR INSTRUCTIONS**<br>**<sopim>** | R | R | R | M.6.2.3 |
| *SECURITY AND PRIVACY PROCEDURES WORK PACKAGE*<br>**<softsecprivwp>** | R | R | R | M.7.7.1 |
| *SUPERVISORY CONTROLS WORK PACKAGE*<br>**<softsuperctrlswp>** | | R | R | M.7.7.2 |
| *POWERUP AND ACCESS SOFTWARE PROCEDURES WORK PACKAGE*<br>**<softpwraccesswp>** | | | | M.7.7.3 |


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**TABLE A-VIII. Software requirements matrix. (continued**

| Content | SUM | SAM | SUM/SAM | MIL-STD REF |
|---|---|---|---|---|
| *SOFTWARE OPERATOR PROCEDURES WORK PACKAGE* **<softoperwp>** | R | R | R | M.7.7.4 |
| Key Commands **<softkeycmd>** | | | | M.7.7.4.1 |
| User Interface **<softgui>** | | | | M.7.7.4.2 |
| Software operating conventions **<softopconventions>** | | | | M.7.7.4.3 |
| Software operating procedures **<softoperproc>** | | | | M.7.7.4.4 |
| *POWER DOWN AND EXIT SOFTWARE PROCEDURES WORK PACKAGE* **<softpwrexitwp>** | R | R | R | M.7.7.5 |
| **MODULE X. SOFTWARE TROUBLESHOOTING PROCEDURES** *Note: The notation (*) indicates that at least one of the these content items shall be included.* **<tim>** **<troublesoftcategory>** | | R | R | M.7.8 |
| *TROUBLESHOOTING INTRODUCTION WORK PACKAGE* **<tsintrowp>** | | R | R | M.7.8.1 |
| *TROUBLESHOOTING INDEX WORK PACKAGE* **<tsindexwp>** | | R | R | M.7.8.2 |
| **MESSAGES WORK PACKAGE* **<softmessageswp>** | | R | R | M.7.8.3 |
| **RECOVERY FROM ERRORS, MALFUNCTIONS, AND EMERGENCIES WORK PACKAGE* **<softerrorswp>** | | | | M.7.8.4 |
| **TROUBLESHOOTING WORK PACKAGE* **<tswp>** | | | | M.7.8.5 |
| **MODULE X. SOFTWARE PMCS** **<mim>** **<maintenancepmcscategory>** | | | | M.7.9.3 M.7.9.3.2.1 |
| *PMCS INTRODUCTION WORK PACKAGE* **<pmcsintrowp>** | CR | CR | CR | M.7.9.3.1 |
| *PMCS WORK PACKAGE* **<pmcswp>** | CR | CR | CR | M.7.9.3.2 |


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**TABLE A-VIII. Software requirements matrix. (continued**

| Content | SUM | SAM | SUM/SAM | MIL-STD REF |
|---|---|---|---|---|
| **MODULE X. SOFTWARE MAINTENANCE INSTRUCTIONS**<br>**<mim>**<br>**<softmaintcategory>** | | R | R | M.7.9<br>M.7.9.1 |
| *MAINTENANCE INSTRUCTION WORK PACKAGES*<br>Maintenance tasks<br>**<maintwp>**<br>**<softmaintsk>** | | | | E.7.4<br>E.7.4.4.39 |
| Configure software<br>**<softconfig>** | | | | E.7.4.4.39.1 |
| Debug software<br>**<debug>** | | | | E.7.4.4.39.2 |
| Install peripheral device<br>**<installperdev>** | | | | E.7.4.4.12.1 |
| Install software<br>**<install>** | | | | E.7.4.4.31.1 |
| Remove software<br>**<remove>** | | | | E.7.4.4.31 |
| Repair software<br>**<repair>** | | | | E.7.4.4.32 |
| Test<br>**<test>** | | | | E.7.4.4.36 |
| Upgrade/patch software<br>**<upgrade>** | | | | E.7.4.4.39.3 |
| Uninstall peripheral device<br>**<uninstallperdev>** | | | | E.7.4.4.12.2 |
| Additional maintenance task<br>**<other.maintsk>** | | | | E.7.4.4.40 |
| **MODULE X. DESTRUCTION TO PREVENT ENEMY USE**<br>**<dim>** | | | | M.6.2.6 |
| **MODULE X. SUPPORTING INFORMATION**<br>**<sim>** | R | R | R | M.6.2.7 |
| *REFERENCES WORK PACKAGE*<br>**<refwp>** | R | R | R | M.7.11.1 |
| *INTRODUCTION FOR TWO-LEVEL MAINTENANCE ALLOCATION CHART (MAC) WORK PACKAGE*<br>**<macintrowp>** | | | | M.7.11.2 |

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**TABLE A-VIII. Software requirements matrix. (continued**

| Content | SUM | SAM | SUM/SAM | MIL-STD REF |
|---|---|---|---|---|
| *MAINTENANCE ALLOCATION CHART (MAC) WORK PACKAGE* **<macwp>** | | | | M.7.11.3 |
| *SOFTWARE BASIC ISSUE ITEMS (BII) LISTS WORK PACKAGE* **<softbiiwp>** | R | R | R | M.7.11.4 |
| *ADDITIONAL AUTHORIZATION LIST (AAL) WORK PACKAGE* **<aalwp>** | | | | M.7.11.5 |
| *EXPENDABLE AND DURABLE ITEMS LIST WORK PACKAGE* **<explistwp>** | | | | M.7.11.6 |
| *ADDITIONAL SUPPORTING WORK PACKAGE* **<genwp>** | | | | M.7.11.7 |
| **REAR MATTER** **<rear>** | R | R | R | TABLE A-II |

**LEGEND**
R - Required (see A.4.3.1.3).
P - Prohibited (see A.4.3.1.3).
Shaded - May be Required (see A.4.3.1.4).
CR - Conditionally Required (see A.4.3.1.4.3).



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**TABLE A-IX. Ammunition DMWR for demilitarization / maintenance requirements matrix.**

| Content | Demil / Maint | MIL-STD REF |
|---|---|---|
| **PUBLICATION** (**<production>**)<br>**<dmwr_ammo>**<br><br>PUBLICATION TITLES:<br>DMWR for Demilitarization of *insert system*<br><br>DMWR for Maintenance of *insert system* | R | TABLE A-I |
| **FRONT MATTER**<br>**<paper.frnt>** or **<framed.frnt>** | R | TABLE A-II |
| **MODULE 1. GENERAL INFORMATION AND DMWR INTRODUCTION**<br>**<gim>** | R | L.5.1 |
| *GENERAL INFORMATION WORK PACKAGE*<br>**<ginfowp>** | R | L.6.1 |
| Scope<br>**<scope>** | R | L.6.1 |
| Maintenance forms, records, and reports<br>**<mfrr>** | R | L.6.1 |
| Corrosion Prevention and Control (CPC)<br>**<cpcdata>** | | L.6.1 |
| List of abbreviations<br>**<loa>** | R | L.6.1 |
| *DMWR INTRODUCTION WORK PACKAGE*<br>**<dmwr_introwp>** | R | L.6.2 |
| Work planning<br>**<work_planning>** | R | L.6.2 |
| Disposition<br>**<disposition>** | R | L.6.2 |
| Equipment<br>**<equipment>** | R | L.6.2 |
| Safety requirements<br>**<sfty_req>** | R | L.6.2 |
| Protection against general hazards<br>**<gen_hazards>** | R | L.6.2 |
| Protection against specific hazards<br>**<spec_hazards>** | R | L.6.2 |



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**TABLE A-IX. Ammunition DMWR for demilitarization / maintenance requirements matrix. (continued**

| Content | Demil / Maint | MIL-STD REF |
|---|---|---|
| Hazard analysis **<haz_analysis>** | R | L.6.2 |
| Environmental regulation compliance **<erc>** | R | L.6.2 |
| Resource conversation and recover regulations **<rcrr>** | R | L.6.2 |
| Resource recovery **<resource_recovery>** | R | L.6.2 |
| Reporting requirements **<reporting_req>** | R | L.6.2 |
| Tabulated data **<tabdata>** | R | L.6.2 |
| Flowchart **<flowchart>** | | L.6.2 |
| **MODULE X. MAINTENANCE REQUIREMENTS** **<mim>** | R | L.5.2 |
| *MAINTENANCE REQUIREMENTS WORK PACKAGE* **<dmwr_operationalreqwp>** | R | L.6.3 |
| Safety requirements **<special_sfty>** | R | L.6.3 |
| Procedural steps **<op_steps>** | R | L.6.3 |
| Flowchart **<flowchart>** | | L.6.3 |
| **MODULE X. QUALITY ACCEPTANCE REQUIREMENTS** **<mim>** | | L.5.3 |
| *QUALITY ACCEPTANCE REQUIREMENTS WORK PACKAGE* **<dmwr_qarwp>** | CR | L.6.4 |
| Ammunition maintenance **<maintenance_ qar>** | CR | L.6.4 |
| **MODULE X. SUPPORTING INFORMATION** **<sim>** | R | L.5.4 |
| *REFERENCES WORK PACKAGE* **<refwp>** | R | L.6.5 |
| *EXPENDABLE AND DURABLE ITEMS LIST WORK PACKAGE* | R | L.6.6 |

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**TABLE A-IX. Ammunition DMWR for demilitarization / maintenance requirements matrix. (continued**

| Content | Demil / Maint | MIL-STD REF |
|---|---|---|
| **<explistwp>** | | |
| *TOOL IDENTIFICATION LIST WORK PACKAGE*<br>**<toolidwp>** | | L.6.7 |
| *AMMUNITION PECULIAR EQUIPMENT (APE) WORK PACKAGE*<br>**<genwp>** | | L.6.8 |
| *SPECIAL EQUIPMENT, LOCALLY AVAILABLE MATERIALS, AND SPECIAL FACILITIES WORK PACKAGE*<br>**<genwp>** | R | L.6.9 |
| *APPROVED INTRAPLANT TRANSFER EQUIPMENT WORK PACKAGE*<br>**<genwp>** | R | L.6.10 |
| *CHEMICALLY TREATED PACKING MATERIALS WORK PACKAGE*<br>**<genwp>** | R | L.6.11 |
| *ENVIRONMENTAL REQUIREMENTS WORK PACKAGE*<br>**<genwp>** | R | L.6.12 |
| *HAZARD ANALYSIS WORK PACKAGE*<br>**<genwp>** | R | L.6.13 |
| *HAZARDOUS COMPONENT SAFETY DATA STATEMENT (HCSDS) WORK PACKAGE*<br>**<genwp>** | R | L.6.14 |
| *OTHER SUPPORTING INFORMATION WORK PACKAGE*<br>**<genwp>** | | L.6.15 |
| **REAR MATTER**<br>**<rear>** | R | TABLE A-II |

**LEGEND**
R - Required (see A.4.3.1.3).
Shaded - May be Required (see A.4.3.1.4).
CR - Conditionally Required (see A.4.3.1.4.3).



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**TABLE A-X. Ammunition / munitions equipment data sheets requirements matrix.**

| Content | Data Sheets | MIL-STD REF |
|---|---|---|
| **PUBLICATION** (**<production>**)<br>**<ammo_ds>**<br><br>PUBLICATION TITLES:<br>Ammunition Data Sheets for *insert system*<br><br>Munitions Equipment Data Sheets for *insert system* | R | TABLE A-I |
| **FRONT MATTER**<br>**<paper.frnt>** or **<framed.frnt>** | R | TABLE A-II |
| **MODULE 1. GENERAL INFORMATION AND DATA SHEETS INTRODUCTION**<br>**<ammods_gim>** | R | P.5.1.1<br><br>P.5.2.1 |
| *GENERAL INFORMATION WORK PACKAGE*<br>**<ginfowp>** | R | P.6.1 |
| Scope<br>**<scope>** | R | P.6.1 |
| Maintenance forms, records, and reports<br>**<mfrr>** | R | P.6.1 |
| List of abbreviations<br>**<loa>** | R | P.6.1 |
| *DATA SHEETS INTRODUCTION*<br>**<ammods_introwp>** | R | P.6.2 |
| **MODULE X. ARMY AMMUNITION DATA SHEETS**<br>**<ammods_mim>** | R | P.5.1.2<br><br>P.5.2.2 |
| *ARMY AMMUNITION DATA SHEETS WORK PACKAGE*<br>**<ammo_dswp>** | R | P.6.3 |
| Figure of ammunition item or munitions equipment<br>**<ammods.fig>** | | P.6.3 |
| Use<br>**<ammods.use>** | | P.6.3 |
| Description<br>**<ammods.desc>** | | P.6.3 |
| Functioning<br>**<ammods.func>** | | P.6.3 |
| Differences between models<br>**<ammods.diff>** | | P.6.3 |

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**TABLE A-X. Ammunition / munitions equipment data sheets requirements matrix. (continued**

| Content | Data Sheets | MIL-STD REF |
|---|---|---|
| Tabulated data<br>**<ammods.tabdata>** | | P.6.3 |
| Performance<br>**<ammods.perf>** | | P.6.3 |
| Temperature limits<br>**<ammods.temp>** | | P.6.3 |
| Drawings/specifications<br>**<ammods.draw>** | | P.6.3 |
| Packing configuration<br>**<ammods.packcfg>** | | P.6.3 |
| Packing data<br>**<ammods.pack>** | | P.6.3 |
| Shipping and storage data<br>**<ammods.shipstor>** | | P.6.3 |
| Logistics data<br>**<ammods.logdata>** | | P.6.3 |
| Limitations<br>**<ammods.limit>** | | P.6.3 |
| Remarks<br>**<ammods.rem>** | | P.6.3 |
| References<br>**<ammods.refs>** | | P.6.3 |
| Associated equipment<br>**<ammods.eqp>** | | P.6.3 |
| Kits<br>**<ammods.kits>** | | P.6.3 |
| **MODULE X. SUPPORTING INFORMATION**<br>**<ammods.sim>** | R | P.5.1.3<br>P.5.2.3 |
| *REFERENCES WORK PACKAGE*<br>**<refwp>** | R | P.6.4 |
| *DELETED ITEMS WORK PACKAGE*<br>**<genwp>** | | P.6.5 |
| *OPERATIONAL INDEX WORK PACKAGE*<br>**<genwp>** | | P.6.6 |


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**TABLE A-X. Ammunition / munitions equipment data sheets requirements matrix. (continued**

| Content | Data Sheets | MIL-STD REF |
|---|---|---|
| *PREPARATION AND HANDLING OF AMMUNITION PECULIAR EQUIPMENT (APE) FOR SHIPMENT AND STORAGE WORK PACKAGE* **<genwp>** | | P.6.7 |
| *OTHER SUPPORTING INFORMATION WORK PACKAGES* **<genwp>** | | P.6.8 |
| **REAR METTER** | R | TABLE A-II |
| **LEGEND**<br>R - Required (see A.4.3.1.3).<br>Shaded - May be Required (see A.4.3.1.4).<br>CR - Conditionally Required (see A.4.3.1.4.3). | | |



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**TABLE A-XI. Aviation maintainer maintenance requirements matrix (aircraft only.**

| Content | -20 | -20&P | MIL-STD REF |
|---|---|---|---|
| **PUBLICATION** (<production>)<br>**<framed.manual>**<br><br>PUBLICATION TITLES:<br>(-20) Field Maintenance Manual for *insert system* [1]<br><br>(-20&P) Field Maintenance Manual Including Repair Parts and Special Tools List for *insert system* [1]<br><br>[1] TRADOC maintainer representative approval is required for any inclusion of sustainment-level maintenance. If approval is granted, "sustainment" may be included in title. See also A.4.3 for publication titling guidance. | R | R | TABLE A-I |
| **FRONT MATTER**<br>**<framed.frnt>** | R | R | TABLE A-II |
| **MODULE 1. GENERAL INFORMATION, EQUIPMENT DESCRIPTION, AND THEORY OF OPERATION**<br>**<gim>** | R | R | B.6 |
| *GENERAL INFORMATION WORK PACKAGE*<br>**<ginfowp>** | R | R | B.7.1 |
| Scope<br>**<scope>** | R | R | B.7.1.1 |
| Maintenance forms, records, and reports<br>**<mfrr>** | R | R | B.7.1.2 |
| Reporting Equipment Improvement Recommendations (EIR)<br>**<eir>** | R | R | B.7.1.3 |
| Hand Receipt (HR) manuals<br>**<handreceipt>** | | | B.7.1.4 |
| Corrosion Prevention and Control (CPC)<br>**<cpcdata>** | R | R | B.7.1.5 |
| Ozone Depleting Substances (ODS)<br>**<odsdata>** | | | B.7.1.6 |
| Destruction of equipment to prevent enemy use<br>**<destructmat>** | R | R | B.7.1.7 |
| Preparation for storage or shipment<br>**<pssref>** | R | R | B.7.1.8 |
| Transportability guidance<br>**<transportability>** | R | R | B.7.1.8 |



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**TABLE A-XI. Aviation maintainer maintenance requirements matrix (aircraft only. (continued**

| Content | -20 | -20&P | MIL-STD REF |
|---|---|---|---|
| Warranty information<br>**<wrntyref>** | | | B.7.1.9 |
| Nomenclature cross-reference list<br>**<nomenreflist>** | | | B.7.1.10 |
| List of abbreviations/acronyms<br>**<loa>** | R | R | B.7.1.11 |
| Quality Assurance (QA)<br>**<qainfo>** | R | R | B.7.1.12 |
| Quality of material<br>**<qual.mat.info>** | R | R | B.7.1.13 |
| Safety, care, and handling<br>**<sftyinfo>** | R | R | B.7.1.14 |
| Nuclear hardness<br>**<hcp>** | | | B.7.1.15 |
| Calibration<br>**<calref>** | | | B.7.1.16 |
| Item unique identification (IUID)<br>**<iuid>** | | | B.7.1.17 |
| Mandatory replacement parts (MRP)<br>**<mrpreq>** | | | B.7.1.22 |
| Critical safety items<br>**<csireq>** | | | B.7.1.23 |
| Supporting information for repair parts, special tools/kits, TMDE, and support equipment<br>**<supdata>** | | | B.7.1.25 |
| Copyright credit line<br>**<copyrt>** | | | B.7.1.26 |
| *EQUIPMENT DESCRIPTION AND DATA WORK PACKAGE*<br>**<descwp>** | R | R | B.7.2 |
| Equipment characteristics, capabilities, and features<br>**<eqpinfo>** | R | R | B.7.2.1 |
| Location and description of major components<br>**<locdesc>** | R | R | B.7.2.2 |
| Equipment differences<br>**<eqpdiff>** | | | B.7.2.3 |


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**TABLE A-XI. Aviation maintainer maintenance requirements matrix (aircraft only. (continued**

| Content | -20 | -20&P | MIL-STD REF |
|---|---|---|---|
| Equipment data<br>**<eqpdata>** | R | R | B.7.2.4 |
| *THEORY OF OPERATION WORK PACKAGE*<br>**<thrywp>** | R | R | B.7.3 |
| **MODULE X. TROUBLESHOOTING MASTER INDEX**<br>**<tim>**<br>**<masterindexcategory>** | | | D.8<br>D.8.4 |
| *TROUBLESHOOTING INDEX WORK PACKAGE*<br>**<tsindxwp>** | CR | CR | |
| **MODULE X. TROUBLESHOOTING PROCEDURES**<br>*Note: The notation (*) indicates that at least one of the these content items shall be included.*<br>**<tim>**<br>**<troublecategory>** | R | R | D.8<br>D.7.2 |
| *TROUBLESHOOTING INTRODUCTION WORK PACKAGE*<br>**<tsintrowp>** | R | R | D.8.1 |
| *TECHNICAL DESCRIPTION WORK PACKAGE*<br>**<techdescwp>** | R | R | D.8.2 |
| *TROUBLESHOOTING INDEX WORK PACKAGE*<br>**<tsindxwp>** | | | D.8.3 |
| **OPERATIONAL CHECKOUT WORK PACKAGES*<br>**<opcheckwp>** | | | D.8.5 |
| **TROUBLESHOOTING WORK PACKAGES*<br>**<tswp>** | | | D.8.4 |
| **COMBINED OPERATIONAL CHECKOUT AND TROUBLE-SHOOTING PROCEDURE WORK PACKAGES*<br>**<opcheck-tswp>** | | | D.8.5.1 |
| **DIAGNOSTICS WORK PACKAGES*<br>**<diagnosticwp>** | | | D.8.6 |
| **MODULE X. PMCS MAINTENANCE INSTRUCTIONS**<br>*Note: PMCS is required as a minimum in one maintenance module. PMCS may be in its own module or may be combined with other maintenance work packages in a maintenance module but not both places.*<br>**<mim>**<br>**<pmcscategory>** | | | E.6<br>E.6.2 |
| *PMCS INTRODUCTION WORK PACKAGE*<br>**<pmcsintrowp>** | CR | CR | E.7.1 |


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**TABLE A-XI. Aviation maintainer maintenance requirements matrix (aircraft only. (continued**

| Content | -20 | -20&P | MIL-STD REF |
|---|---|---|---|
| *PMCS WORK PACKAGE*<br>**<pmcswp>** | CR | CR | E.7.2 |
| **MODULE X. AIRCRAFT MAINTENANCE INSTRUCTIONS**<br>*Note: PMCS is required as a minimum in one maintenance module. PMCS may be in its own module or may be combined with other maintenance work packages in a maintenance module but not both places.*<br>**<mim>**<br>**<aviationcategory>** | R | R | E.6.5 |
| *PRESERVATION, PACKAGING, AND MARKING GENERAL INFORMATION WORK PACKAGE*<br>**<ppmgeninfowp>** | | | E.7.19 |
| *SERVICE UPON RECEIPT WORK PACKAGE*<br># Service upon receipt work package required for Uncrewed Aerial System (UAS) with ground equipment. Optional for other aviation systems.<br>Service upon receipt tasks<br>**<surwp>**<br>**<surtsk>** | # | # | E.7.3<br>E.7.3.1 |
| Siting<br>**<siting>** | | | E.7.3.1.1 |
| Shelter requirements<br>**<shelter>** | | | E.7.3.1.2 |
| Service upon receipt of materiel<br>**<surmat>** | | | E.7.3.1.3 |
| Installation instructions<br>**<install>** | | | E.7.4.4.31.1 |
| Preliminary serving of equipment<br>**<preserv>** | | | E.7.3.1.3.3.5 |
| Preliminary checks and adjustment of equipment<br>**<prechkadj>** | | | E.7.3.1.3.3.6 |
| Preliminary calibration of equipment<br>**<precal>** | | | E.7.3.1.3.3.7 |
| Circuit alignment<br>**<calign>** | | | E.7.3.1.3.3.8 |
| Ammunition service upon receipt<br>**<ammo.sur>** | | | E.7.3.1.3.3.9 |
| Additional service upon receipt task<br>**<other.surtsk>** | | | E.7.3.1.3.3.10 |

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**TABLE A-XI. Aviation maintainer maintenance requirements matrix (aircraft only. (continued**

| Content | -20 | -20&P | MIL-STD REF |
|---|---|---|---|
| Follow-on maintenance<br>**<followon.maintsk>** | | | E.7.3.1.3.3.11 |
| *EQUIPMENT/USER FITTING INSTRUCTIONS WORK PACKAGE*<br>**<perseqpwp>** | | | E.7.11 |
| *PMI WORK PACKAGE*<br>**<pmiwp>** | R | R | E.7.12 |
| *MAINTENANCE INSTRUCTION WORK PACKAGES*<br>Maintenance tasks<br>**<maintwp>**<br>**<maintsk>** | R | R | E.7.4<br>E.7.4.4 |
| Adjust<br>**<adjust>** | | | E.7.4.4.1 |
| Align<br>**<align>** | | | E.7.4.4.2 |
| Arm<br>**<arm>** | | | E.7.4.4.3 |
| Assemble<br>**<assem>** | | | E.7.4.4.4.1 |
| Disassemble<br>**<disassem>** | | | E.7.4.4.4.2 |
| Calibrate<br>**<calibration>** | | | E.7.4.4.5 |
| Clean<br>**<clean>** | | | E.7.4.4.6 |
| Configure<br>**<configure>** | | | E.7.4.4.7 |
| Covering<br>**<cover>** | | | E.7.4.4.8 |
| External power<br>**<extpwr>** | | | E.7.4.4.9 |
| Hoisting<br>**<hoist>** | | | E.7.4.4.10 |
| Inspect<br>**<inspect>** | | | E.7.4.4.11 |
| Install peripheral device<br>**<installperdev>** | | | E.7.4.4.12.1 |



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**TABLE A-XI. Aviation maintainer maintenance requirements matrix (aircraft only. (continued**

| Content | -20 | -20&P | MIL-STD REF |
|---|---|---|---|
| Uninstall peripheral device<br>**<uninstallperdev>** | | | E.7.4.4.12.2 |
| Jacking<br>**<jack>** | | | E.7.4.4.13 |
| Load<br>**<load>** | | | E.7.4.4.14.1 |
| Unload<br>**<unload>** | | | E.7.4.4.14.2 |
| Lubricate<br>**<lube>** | | | E.7.4.4.15 |
| Mark<br>**<mark>** | | | E.7.4.4.16 |
| Mooring<br>**<moor>** | | | E.7.4.4.17 |
| Nondestructive inspection<br>**<ndi>** | | | E.7.4.4.18 |
| Overhaul<br>**<overhaul>** | | | E.7.4.4.19 |
| Pack<br>**<pack>** | | | E.7.4.4.20.1 |
| Pack<br>**<pack>**<br>Unpack<br>**<unpack>** | | | E.7.4.4.20.2 |
| Paint<br>**<paint>** | | | E.7.4.4.21 |
| Parking<br>**<park>** | | | E.7.4.4.22 |
| Place in service<br>**<pis>** | | | E.7.4.4.23 |
| Preparation for storage<br>**<prepstore>** | + | + | E.7.4.4.24 |
| Preparation for shipment<br>**<prepship>** | + | + | E.7.4.4.25 |
| Prepare for use<br>**<prepforuse>** | | | E.7.4.4.26 |


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**TABLE A-XI. Aviation maintainer maintenance requirements matrix (aircraft only. (continued**

| Content | -20 | -20&P | MIL-STD REF |
|---|---|---|---|
| Preserve<br>**<preservation>** | | | E.7.4.4.27 |
| Radio interference suppression<br>**<ris>** | | | E.7.4.4.28 |
| Rebuild<br>**<rebuild>** | | | E.7.4.4.29 |
| Remove<br>**<remove>** | | | E.7.4.4.31.1 |
| Install<br>**<install>** | | | E.7.4.4.31.2 |
| Repair<br>**<repair>** | | | E.7.4.4.32 |
| Replace<br>**<replace>** | | | E.7.4.4.33 |
| Service<br>**<service>** | | | E.7.4.4.34 |
| Sling loading<br>**<sling>** | | | E.7.4.4.35 |
| Test<br>**<test>** | | | E.7.4.4.36 |
| Towing<br>**<tow>** | | | E.7.4.4.37 |
| Transport<br>**<transport>** | + | + | E.7.4.4.38 |
| Additional maintenance task<br>**<other.maintsk>** | | | E.7.4.4.40 |
| Follow-on maintenance<br>**<followon.maintsk>** | | | E.7.4.4.41 |
| *SOFTWARE MAINTENANCE WORK PACKAGES*<br>Software maintenance tasks<br>**<maintwp>**<br>**<softmaintsk>** | | | E.7.4<br>E.7.4.4.39 |
| Configure software<br>**<softconfig>** | | | E.7.4.4.39.1 |
| Debug software<br>**<debug>** | | | E.7.4.4.39.2 |



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**TABLE A-XI. Aviation maintainer maintenance requirements matrix (aircraft only. (continued**

| Content | -20 | -20&P | MIL-STD REF |
|---|---|---|---|
| Install software<br>**<install>** | | | E.7.4.4.31.1 |
| Remove software<br>**<remove>** | | | E.7.4.4.31 |
| Upgrade / patch software<br>**<upgrade>** | | | E.7.4.4.39.3 |
| Additional maintenance task<br>**<other.maintsk>** | | | E.7.4.4.40 |
| Follow-on maintenance<br>**<followon.maintsk>** | | | E.7.4.4.41 |
| *GENERAL MAINTENANCE WORK PACKAGE*<br>**<gen.maintwp>** | | | E.7.5 |
| *OVERHAUL AND RETIREMENT SCHEDULE WORK PACKAGE*<br>**<orschwp>** | R | R | E.7.16 |
| *LUBRICATION INSTRUCTIONS WORK PACKAGE*<br>**<lubewp>** | | | E.7.6 |
| **ILLUSTRATED LIST OF MANUFACTURED ITEMS**<br>*Note: If required, the introduction work package and manufacturing procedure work package are both required.* | | | E.7.7 |
| *ILLUSTRATED LIST OF MANUFACTURED ITEMS INTRO-DUCTION WORK PACKAGE*<br>**<manu_items_introwp>** | CR | CR | E.7.7 |
| *MANUFACTURING PROCEDURES WORK PACKAGE*<br>**<manuwp>** | CR | CR | E.7.8 |
| *TORQUE LIMITS WORK PACKAGE*<br>**<torquewp>** | | | E.7.9 |
| *AIRCRAFT INVENTORY MASTER GUIDE WORK PACKAGE*<br>**<inventorywp>** | R | R | E.7.13 |
| *STORAGE OF AIRCRAFT WORK PACKAGE*<br>**<storagewp>** | R | R | E.7.14 |
| *WEIGHING AND LOADING WORK PACKAGE*<br>**<wtloadwp>** | R | R | E.7.15 |
| *WIRING DIAGRAMS AND SCHEMATICS WORK PACKAGE*<br>**<wiringwp>** | | | E.7.10 |
| **MODULE X. AUXILIARY EQUIPMENT MAINTENANCE INSTRUCTIONS**<br>**<mim>** | | | E.6<br>E.6.6 |

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**TABLE A-XI. Aviation maintainer maintenance requirements matrix (aircraft only. (continued**

| Content | -20 | -20&P | MIL-STD REF |
|---|---|---|---|
| **<auxiliarycategory>** | | | |
| *AUXILIARY EQUIPMENT MAINTENANCE WORK PACKAGE* **<auxeqpwp>** | | | E.7.24 |
| **ILLUSTRATED LIST OF MANUFACTURED ITEMS** *Note: If required, the introduction work package and manufacturing procedure work package are both required.* | | | E.7.7 |
| *ILLUSTRATED LIST OF MANUFACTURED ITEMS INTRODUCTION WORK PACKAGE* **<manu_items_introwp>** | CR | CR | E.7.7 |
| *MANUFACTURING PROCEDURES WORK PACKAGE* **<manuwp>** | CR | CR | E.7.8 |
| *TORQUE LIMITS WORK PACKAGE* **<torquewp>** | | | E.7.9 |
| *WIRING DIAGRAMS AND SCHEMATICS WORK PACKAGE* **<wiringwp>** | | | E.7.10 |
| **MODULE X. AMMUNITION MAINTENANCE INSTRUCTIONS <mim> <ammunitioncategory>** | | | E.6<br>E.6.7 |
| *AMMUNITION IDENTIFICATION WORK PACKAGE* **<ammoidentwp>** | | | E.7.25 |
| *AMMUNITION MAINTENANCE WORK PACKAGE* **<ammowp>** | | | E.7.26 |
| *FOREIGN AMMUNITION WORK PACKAGE* **<natowp>** | | | E.7.27 |
| **MODULE X. AIRCRAFT PMS/PMD GENERAL INFORMATION <gim>** | | | E.6.12.1.1 |
| *GENERAL INFORMATION WORK PACKAGE* **<pms-ginfowp>** | | | E.6.12.1.1 |
| **MODULE X. AIRCRAFT PMS/PMD GENERAL INFORMATION <mim> <pmscategory>** | | | E.6.12.1.2 |
| *PMS/PMD INSPECTION WORK PACKAGE* **<pms-inspecwp>** | | | E.6.12.1.2 |
| **MODULE X. AIRCRAFT PHASED MAINTENANCE GENERAL INFORMATION <gim>** | | | E.6.12.2.1 |

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**TABLE A-XI. Aviation maintainer maintenance requirements matrix (aircraft only. (continued**

| Content | -20 | -20&P | MIL-STD REF |
|---|---|---|---|
| *GENERAL INFORMATION WOK PACKAGE*<br>**<pm-ginfowp>** | | | E.6.12.2.1 |
| **MODULE X. AIRCRAFT PHASED MAINTENANCE**<br>**<mim>**<br>**<checklistcategory>** | | | E.6.12.2.2 |
| *PM INSPECTION WORK PACKAGE*<br>**<pmi-cklistwp>** | | | E.6.12.2.2 |
| **MODULE X. TEST AND INSPECTION MAINTENANCE INSTRUCTIONS**<br>**<mim>**<br>**<testinspectioncategory>** | | | E.6.8 |
| *MAINTENANCE WORK PACKAGES*<br>**<maintwp>** | | | E.7.4 |
| *Inspection*<br>**<inspect>** | | | E.7.4.4.11 |
| *Test*<br>**<test>** | | | E.7.4.4.36 |
| **MODULE X. DESTRUCTION OF EQUIPMENT TO PREVENT ENEMY USE**<br>*Note: If a separate destruction of material manual is not developed for this equipment, then the destruction module must be included.*<br>**<dim>** | | | H.5.2 |
| *DESTRUCTION PROCEDURES INTRODUCTION WORK PACKAGE*<br>**<destruct-introwp>** | CR | CR | H.6.2 |
| Authority to destroy<br>**<authorize_to_ destroy>** | CR | CR | H.6.2.1 |
| Reporting destruction<br>**<report_destruct>** | CR | CR | H.6.2.2 |
| General destruction information<br>**<general_destruct_info>** | CR | CR | H.6.2.3 |
| Degree of destruction<br>**<degree_of_destruct>** | CR | CR | H.6.2.4 |
| Essential components and spare parts<br>**<component_spares>** | | | H.6.2.5 |
| *DESTRUCTION PROCEDURES WORK PACKAGE*<br>**<destruct-materialwp>** | CR | CR | H.6.3 |


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**TABLE A-XI. Aviation maintainer maintenance requirements matrix (aircraft only. (continued**

| Content | -20 | -20&P | MIL-STD REF |
|---|---|---|---|
| Parts list<br>**<essential_spares>** | | | H.6.3.1 |
| Specific destruction procedures<br>**<proc>** | CR | CR | H.6.3.2 |
| **MODULE X. BATTLE DAMAGE ASSESSMENT AND REPAIR INFORMATION**<br>**<bim>** | | | I.6.1 |
| *BDAR UNIQUE GENERAL INFORMATION WORK PACKAGE*<br>**<bdar-geninfowp>** | CR | CR | I.7.2 |
| Standards and practices<br>**<bdar-std-practices>** | CR | CR | I.7.2.1 |
| Tasks and responsibilities<br>**<bdar-task-resp>** | | | I.7.2.2 |
| Combat threats (aviation only)<br>**<bdar-combat-threat>** | | | I.7.2.3 |
| *BATTLE DAMAGE ASSESSMENT WORK PACKAGE*<br>**<damage-assesswp>** | CR | CR | I.7.3 |
| *BATTLE DAMAGE REPAIR WORK PACKAGE*<br>**<genrepairwp>** | | | I.7.4 |
| Introduction<br>**<geninfo>** | CR | CR | I.7.4 |
| Repair procedure<br>**<bdar-repair-proc>** | CR | CR | I.7.4.1 |
| *SPECIAL OR FABRICATED TOOLS WORK PACKAGE*<br>**<bdartoolswp>** | | | I.7.6 |
| *SUBSTITUTE MATERIALS/PARTS WORK PACKAGE*<br>**<substitute-matwp>** | | | I.7.8 |
| **MODULE X. SOFTWARE INFORMATION**<br>**<soim>** | | | M.7.7 |
| *SOFTWARE GENERAL INFORMATION WORK PACKAGE*<br>**<softginfowp>** | CR | CR | M.7.1 |
| System overview<br>**<softsysover>** | CR | CR | M.7.1 |
| Document overview<br>**<softdocover>** | CR | CR | M.7.1 |


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**TABLE A-XI. Aviation maintainer maintenance requirements matrix (aircraft only. (continued**

| Content | -20 | -20&P | MIL-STD REF |
|---|---|---|---|
| Copyright credit line<br>**<copyrt>** | CR | CR | M.7.1 |
| Endorsement notice<br>**<endorse>** | CR | CR | M.7.1 |
| *SOFTWARE SUMMARY WORK PACKAGE*<br>**<softsumwp>** | CR | CR | M.7.2 |
| *SOFTWARE EFFECTIVITY WORK PACKAGE*<br>**<softeffectwp>** | | | M.7.3 |
| *DIFFERENCES BETWEEN SOFTWARE VERSIONS WORK PACKAGE*<br>**<softdiffversionwp>** | | | M.7.4 |
| *FEATURES AND CAPABILITIES WORK PACKAGE*<br>**<softfeaturescapwp>** | | | M.7.5 |
| *FUNCTIONAL DISPLAY WORK PACKAGE*<br>**<functionaldisplaywp>** | CR | CR | M.7.6 |
| *SECURITY AND PRIVACY PROCEDURES WORK PACKAGE*<br>**<softsecprivwp>** | CR | CR | M.7.7.1 |
| *SUPERVISORY CONTROLS WORK PACKAGE*<br>**<softsuperctrlswp>** | | | M.7.7.2 |
| *POWER UP AND ACCESS SOFTWARE PROCEDURES WORK PACKAGE*<br>**<softpwraccesswp>** | CR | CR | M.7.7.3 |
| *SOFTWARE OPERATOR PROCEDURES WORK PACKAGE*<br>**<softoperwp>** | CR | CR | M.7.7.4 |
| Key commands<br>**<softkeycmd>** | | | M.7.7.4.1 |
| User interface<br>**<softgui>** | | | M.7.7.4.2 |
| Software operating conventions<br>**<softopconventions>** | | | M.7.7.4.3 |
| Software operating procedures<br>**<softoperproc>** | | | M.7.7.4.4 |
| *POWER DOWN AND EXIT SOFTWARE PROCEDURES WORK PACKAGE*<br>**<softpwrexitwp>** | CR | CR | M.7.7.5 |

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**TABLE A-XI. Aviation maintainer maintenance requirements matrix (aircraft only. (continued**

| Content | -20 | -20&P | MIL-STD REF |
|---|---|---|---|
| *MESSAGES WORK PACKAGE* **<softmessageswp>** | | | M.7.8.3 |
| *RECOVERY FROM ERRORS, MALFUNCTIONS, AND EMERGENCIES WORK PACKAGE* **<softerrorswp>** | | | M.7.8.4 |
| **MODULE X. PARTS INFORMATION** **<pim>** | P | R | F.7.2 |
| *INTRODUCTION WORK PACKAGE* **<introwp>** | P | R | F.8.1 |
| *REPAIR PARTS LIST WORK PACKAGE* **<plwp>** | P | R | F.8.2 |
| *REPAIR PARTS FOR SPECIAL TOOLS WORK PACKAGE* **<stl_partswp>** | P | | F.8.3 |
| *KIT PARTS LIST WORK PACKAGE* **<kitswp>** | P | | F.8.4 |
| *BULK ITEM WORK PACKAGE* **<bulk_itemswp>** | P | | F.8.5 |
| *SPECIAL TOOLS LIST WORK PACKAGE* **<stlwp>** | P | | F.8.6 |
| **MODULE X. SUPPORTING INFORMATION** **<sim>** | R | R | G.6 |
| *REFERENCES WORK PACKAGE* **<refwp>** | R | R | G.7.1 |
| *AVIATION MAINTENANCE ALLOCATION CHART (MAC) INTRODUCTION WORK PACKAGE* **<macintrowp>** | R | R | G.7.2 |
| *AVIATION MAINTENANCE ALLOCATION CHART (MAC) WORK PACKAGE* **<macwp>** | R | R | G.7.3 |
| *EXPENDABLE AND DURABLE ITEMS LIST WORK PACKAGE* **<explistwp>** | R | R | G.7.7 |
| *TOOL IDENTIFICATION LIST WORK PACKAGE* **<toolidwp>** | R | R | G.7.8 |
| *MANDATORY REPLACEMENT PARTS WORK PACKAGE* **<mrplwp>** | + | + | G.7.9 |



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**TABLE A-XI. Aviation maintainer maintenance requirements matrix (aircraft only. (continued**

| Content | -20 | -20&P | MIL-STD REF |
|---|---|---|---|
| *CRITICAL SAFETY ITEMS WORK PACKAGE*<br>**<csi.wp>** | + | + | G.7.10 |
| *SUPPORT ITEMS WORK PACKAGE*<br>**<supitemwp>** | | | G.7.11 |
| *ADDITIONAL SUPPORTING WORK PACKAGE*<br>**<genwp>** | | | G.7.12 |
| **REAR MATTER**<br>**<rear>** | R | R | TABLE A-II |

**LEGEND**
R - Required (see A.4.3.1.3).
P - Prohibited (see A.4.3.1.3).
Shaded - May be Required (see A.4.3.1.4).
CR - Conditionally Required (see A.4.3.1.4.3).
+ Plus - Typically Required (see A.4.3.1.4.1). WP shall be listed in General Information, WP 0001 (see B.7.1.1).
# Service upon receipt work package required for Uncrewed Aerial System (UAS) with ground equipment. Optional for other aviation systems.



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**TABLE A-XII. Aircraft PMS or PMD requirements matrix.**

| Content | PMS or PMD | MIL-STD REF |
|---|---|---|
| **PUBLICATION** (**<production>**)<br>**<pms>** or **<pmd>**<br><br>PUBLICATION TITLES:<br>Preventive Maintenance Services (PMS) Manual for *insert system*<br><br>Preventive Maintenance Daily (PMD) Manual for *insert system* | R | TABLE A-I |
| **FRONT MATTER**<br>**<paper.frnt>** or **<framed.frnt>** | R | TABLE A-II |
| **MODULE I. GENERAL INFORMATION**<br>**<gim>** | R | E.6.12.1.1 |
| *GENERAL INFORMATION WORK PACKAGE*<br>**<pms-ginfowp>** | R | E.6.12.1.1 |
| Maintenance activities<br>**<scope>** | R | E.6.12.1.1 |
| General information<br>**<pms-geninfo>** | R | E.6.12.1.1 |
| **MODULE X. PREVENTIVE MAINTENANCE SERVICES MAINTENANCE INFORMATION MODULE**<br>**<mim>**<br>**<pmscategory>** | R | E.6.12.1.2 |
| *PMS/PMD INSPECTION WORK PACKAGE*<br>**<pms-inspecwp>** | R | E.6.12.1.2 |
| **REAR MATTER**<br>**<rear>** | R | TABLE A-II |
| **LEGEND**<br>R - Required (see A.4.3.1.3). | | |



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**TABLE A-XIII. Aircraft PMI requirements matrix.**

| Content | PMI | MIL-STD REF |
|---|---|---|
| **PUBLICATION** (**<production>**)<br>**<pmi>**<br><br>PUBLICATION TITLE:<br>Phased Maintenance Inspection (PMI) Manual for *insert system* | R | TABLE A-I |
| **FRONT MATTER**<br>**<paper.frnt>** or **<framed.frnt>** | R | TABLE A-II |
| **MODULE I. PHASED MAINTENANCE INFORMATION MODULE**<br>**<gim>** | R | E.6.12.2.1 |
| *GENERAL INFORMATION WORK PACKAGE*<br>**<pm-ginfowp>** | R | E.6.12.2.1 |
| **MODULE X. PHASED MAINTENANCE INSTRUCTIONS MODULE**<br>**<mim>**<br>**<checklistcategory>** | R | E.6.12.2.2 |
| *PHASED MAINTENANCE CHECKLIST WORK PACKAGE*<br>**<pmi-cklistwp>** | R | E.6.12.2.2 |
| **REAR MATTER**<br>**<rear>** | R | TABLE A-II |
| **LEGEND**<br>R - Required (see A.4.3.1.3).<br>Shaded - May be Required (see A.4.3.1.4). | | |



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**TABLE A-XIV. Battle damage assessment and repair (BDAR requirements matrix.**

| Content | BDAR | MIL-STD REF |
|---|---|---|
| **PUBLICATION** (**<production>**)<br>**<bdar>**<br><br>PUBLICATION TITLE:<br>Battle Damage Assessment and Repair for *insert system* | R | TABLE A-I |
| **FRONT MATTER**<br>**<paper.frnt>** or **<framed.frnt>** | R | TABLE A-II |
| **MODULE 1. GENERAL INFORMATION**<br>**<gim>** | R | I.6.2 |
| *BDAR GENERAL INFORMATION WORK PACKAGE*<br>**<ginfowp>** | R | I.7.1 |
| *BDAR UNIQUE GENERAL INFORMATION WORK PACKAGE*<br>**<bdar-geninfowp>** | R | I.7.2 |
| Standards and practices<br>**<bdar-std-practices>** | R | I.7.2.1 |
| Tasks and responsibilities<br>**<bdar-task-resp>** | R | I.7.2.2 |
| Combat threats (aviation only)<br>**<bdar-combat-threat>** | | I.7.2.3 |
| **MODULE 2. BATTLE DAMAGE ASSESSMENT INSTRUCTIONS MODULE**<br>**<baim>** | R | I.6.3 |
| *BATTLE DAMAGE ASSESSMENT WORK PACKAGE*<br>**<damage-assesswp>** | R | I.7.3 |
| Introduction<br>**<intro>** | R | I.7.3 |
| General fault assessment tables<br>**<table>** | R | I.7.3.1 |
| **MODULE X. BATTLE DAMAGE REPAIR INSTRUCTIONS MODULE**<br>**<brim>** | R | I.6.4 |
| *BATTLE DAMAGE REPAIR WORK PACKAGE*<br>**<genrepairwp>** | R | I.7.4.1 |
| Introduction<br>**<geninfo>** | R | I.7.4.1 |
| Repair procedure<br>**<bdar-repair-proc>** | R | I.7.4.1 |



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**TABLE A-XIV. Battle damage assessment and repair (BDAR requirements matrix. (continued**

| Content | BDAR | MIL-STD REF |
|---|---|---|
| **MODULE X. SUPPORTING INFORMATION**<br>**<sim>** | R | I.6.5 |
| *REFERENCES WORK PACKAGE*<br>**<refwp>** | R | I.7.5 |
| *SPECIAL OR FABRICATED TOOLS WORK PACKAGE*<br>**<bdartoolswp>** | | I.7.6 |
| *EXPENDABLE AND DURABLE ITEMS WORK PACKAGE*<br>**<explistwp>** | R | I.7.7 |
| *SUBSTITUTE MATERIALS/PARTS WORK PACKAGE*<br>**<substitute-matwp>** | R | I.7.8 |
| **REAR MATTER**<br>**<rear>** | R | TABLE A-II |
| **LEGEND**<br>R - Required (see A.4.3.1.3).<br>Shaded - May be Required (see A.4.3.1.4). | | |

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**TABLE A-XV. Destruction of equipment to prevent enemy use requirements matrix.**

| Content | Destruction | MIL-STD REF |
|---|---|---|
| **PUBLICATION** (**<production>**)<br>**<destruction_manual>**<br><br>PUBLICATION TITLE:<br>Publication title: Destruction of Equipment to Prevent Enemy Use for *insert system* | R | TABLE A-I |
| **FRONT MATTER**<br>**<paper.frnt>** or **<framed.frnt>** | R | TABLE A-II |
| **MODULE 1. GENERAL INFORMATION**<br>**<gim>** | R | H.5.1 |
| *GENERAL INFORMATION WORK PACKAGE*<br>**<ginfowp>** | R | H.6.1 |
| *DESTRUCTION PROCEDURES INTRODUCTION WORK PACKAGE*<br>**<destruct-introwp>** | R | H.6.2 |
| Authority to destroy<br>**<authorize_to_ destroy>** | R | H.6.2.1 |
| Reporting destruction<br>**<report_destruct>** | R | H.6.2.2 |
| General destruction information<br>**<general_destruct_info>** | R | H.6.2.3 |
| Degree of destruction<br>**<degree_of_ destruct>** | R | H.6.2.4 |
| Essential component and spare parts<br>**<component_ spares>** | | H.6.2.5 |
| **MODULE X. DESTRUCTION OF EQUIPMENT TO PREVENT ENEMY USE**<br>**<dim>** | R | H.5.1 |
| *DESTRUCTION PROCEDURES WORK PACKAGE*<br>**<destruct-materialwp>** | R | H.6.3 |
| Parts list<br>**<essential_ spares>** | | H.6.3.1 |
| Specific destruction procedures<br>**<proc>** | R | H.6.3.2 |
| **REAR MATTER**<br>**<rear>** | R | TABLE A-II |
| **LEGEND**<br>R - Required (see A.4.3.1.3).<br>Shaded - May be Required (see A.4.3.1.4). | | |

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**TABLE A-XVI. Lubrication orders (LO requirements matrix.**

| Content | LO | MIL-STD REF |
|---|---|---|
| **PUBLICATION** (**<production>**)<br>**<lubeorder>**<br><br>PUBLICATION TITLE:<br>Lubrication Orders for *insert system* | R | TABLE A-I |
| **FRONT MATTER**<br>**<paper.frnt>** or **<framed.frnt>** | R | TABLE A-II |
| **MODULE 1. GENERAL INFORMATION**<br>**<lo_gim>** | R | K.6.1 |
| *INTRODUCTION WORK PACKAGE*<br>**<lointrowp>** | R | K.7.1 |
| **MODULE 2. MAINTENANCE INSTRUCTIONS**<br>**<lo_mim>** | R | K.6.2 |
| *LUBRICATION PROCEDURES WORK PACKAGE*<br>**<lubewp>** | R | |
| **MODULE 3. SUPPORTING INFORMATION**<br>**<sim>** | R | K.6.3 |
| *REFERENCE WORK PACKAGE*<br>**<refwp>** | | K.7.3 |
| *TOOL IDENTIFICATION WORK PACKAGE*<br>**<toolidwp>** | | K.7.4 |
| *LUBRICANTS AND MILITARY SYMBOLS WORK PACKAGE*<br>**<lubricantsymbolswp>** | R | K.7.5 |
| *LUBRICANT TYPES WORK PACKAGE*<br>**<lubetypeswp>** | R | K.7.6 |
| *SPECIAL NOTES WORK PACKAGE*<br>**<lospecnoteswp>** | R | K.7.7 |
| **REAR MATTER**<br>**<rear>** | R | TABLE A-II |
| **LEGEND**<br>R - Required (see A.4.3.1.3).<br>Shaded - May be Required (see A.4.3.1.4). | | |



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**TABLE A-XVII. PMC requirements matrix.**

| Content | PMC -10 | PMC -12 | PMC -20 | MIL-STD REF |
|---|---|---|---|---|
| **PUBLICATION** (**<production>**)<br>**<pmc>**<br><br>PUBLICATION TITLE:<br>Preventive Maintenance Checklists (PMC) for *insert system* | R | R | R | TABLE A-I |
| **FRONT MATTER**<br>**<frntcover>** | R | R | R | TABLE A-II |
| *PMCS INTRODUCTION WORK PACKAGE*<br>**<pmcsintrowp>** | R | R | R | J.3.1 |
| *PMCS WORK PACKAGE*<br>**<pmcswp>** | R | R | R | J.3.2 |
| **REAR MATTER**<br>**<rear>** | R | R | R | TABLE A-II |
| **LEGEND**<br>R - Required (see A.4.3.1.3).<br>Shaded - May be Required (see A.4.3.1.4). | | | | |



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**TABLE A-XVIII. General maintenance requirements matrix.**

| Content | Gen Maint | MIL-STD REF |
|---|---|---|
| **PUBLICATION** (**<production>**)<br>**<genmaintman>**<br><br>PUBLICATION TITLE:<br>General Maintenance for *insert system* | R | TABLE A-I |
| **FRONT MATTER**<br>**<paper.frnt>** or **<framed.frnt>** | R | TABLE A-II |
| **MODULE 1. GENERAL INFORMATION**<br>**<gim>** | R | N.4.1 |
| *GENERAL INFORMATION WORK PACKAGE*<br>**<genmaint_ginfowp>** | R | N.5.1 |
| Scope<br>**<scope>** | R | N.5.1 |
| Maintenance forms, records, and reports<br>**<mfrr>** | R | N.5.1 |
| Equipment Improvement Recommendations (EIR)<br>**<eir>** | R | N.5.1 |
| Policy<br>**<policy>** | | N.5.1 |
| Safety<br>**<safety>** | | N.5.1 |
| Warranty information<br>**<wrntyref>** | | N.5.1 |
| Nomenclature crossreference list<br>**<nomenreflist>** | | N.5.1 |
| List of abbreviation/acronyms<br>**<loa>** | R | N.5.1 |
| *EQUIPMENT DESCRIPTION WORK PACKAGE*<br>**<descwp>** | | N.5.2 |
| **MODULE 2. MAINTENANCE INSTRUCTIONS**<br>**<mim>**<br>**<maintenancecategory>** | R | E.6.3 |
| *MAINTENANCE INSTRUCTION WORK PACKAGES*<br>Maintenance tasks<br>**<maintwp>**<br>**<maintsk>** | R | N.5.3 |
| Adjust | | E.7.4.4.1 |

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**TABLE A-XVIII. General maintenance requirements matrix. (continued**

| Content | Gen Maint | MIL-STD REF |
|---|---|---|
| **<adjust>** | | |
| Align<br>**<align>** | | E.7.4.4.2 |
| Arm<br>**<arm>** | | E.7.4.4.3 |
| Assemble<br>**<assem>** | | E.7.4.4.4.1 |
| Disassembly<br>**<disassem>** | | E.7.4.4.4.2 |
| Calibrate<br>**<calibration>** | | E.7.4.4.5 |
| Clean<br>**<clean>** | | E.7.4.4.6 |
| Configure<br>**<configure>** | | E.7.4.4.7 |
| Covering<br>**<cover>** | | E.7.4.4.8 |
| External power<br>**<extpwr>** | | E.7.4.4.9 |
| Hoisting<br>**<hoist>** | | E.7.4.4.10 |
| Inspect<br>**<inspect>** | | E.7.4.4.11 |
| Install peripheral device<br>**<installperdev>** | | E.7.4.4.12.1 |
| Uninstall peripheral device<br>**<uninstallperdev>** | | E.7.4.4.12.2 |
| Jacking<br>**<jack>** | | E.7.4.4.13 |
| Load<br>**<load>** | | E.7.4.4.14.1 |
| Unload<br>**<unload>** | | E.7.4.4.14.2 |
| Lubricate<br>**<lube>** | | E.7.4.4.15 |
| Mark | | E.7.4.4.16 |

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**TABLE A-XVIII. General maintenance requirements matrix. (continued**

| Content | Gen Maint | MIL-STD REF |
|---|---|---|
| **<mark>** | | |
| Mooring<br>**<moor>** | | E.7.4.4.17 |
| Nondestructive inspection<br>**<ndi>** | | E.7.4.4.18 |
| Overhaul<br>**<overhaul>** | | E.7.4.4.19 |
| Pack<br>**<pack>** | | E.7.4.4.20.1 |
| Unpack<br>**<unpack>** | | E.7.4.4.20.2 |
| Paint<br>**<paint>** | | E.7.4.4.21 |
| Parking<br>**<park>** | | E.7.4.4.22 |
| Place in service<br>**<pis>** | | E.7.4.4.23 |
| Preparation for storage<br>**<prepstore>** | | E.7.4.4.24 |
| Preparation for shipment<br>**<prepship>** | | E.7.4.4.25 |
| Prepare for use<br>**<prepforuse>** | | E.7.4.4.26 |
| Preserve<br>**<preservation>** | | E.7.3.1.3.3.5 |
| Rebuild<br>**<rebuild>** | | E.7.4.4.29 |
| Remove<br>**<remove>** | | E.7.4.4.31.1 |
| Install<br>**<install>** | | E.7.4.4.31.2 |
| Repair<br>**<repair>** | | E.7.4.4.32 |
| Replace<br>**<replace>** | | E.7.4.4.33 |

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**TABLE A-XVIII. General maintenance requirements matrix. (continued**

| Content | Gen Maint | MIL-STD REF |
|---|---|---|
| Radio interferences suppression<br>**<ris>** | | E.7.4.4.28 |
| Service<br>**<service>** | | E.7.4.4.34 |
| Sling loading<br>**<sling>** | | E.7.4.4.35 |
| Test<br>**<test>** | | E.7.4.4.36 |
| Towing<br>**<tow>** | | E.7.4.4.37 |
| Transport<br>**<transport>** | | E.7.4.4.38 |
| Additional maintenance task<br>**<other.maintsk>** | | E.7.4.4.40 |
| Follow-on maintenance<br>**<followon.maintsk>** | | E.7.4.4.41 |
| *GENERAL MAINTENANCE WORK PACKAGE*<br>**<gen.maintwp>** | | N.5.3<br>E.7.5 |
| **MODULE X. SUPPORTING INFORMATION**<br>*Note: Applicable supporting information work packages shall be arranged in the order in which they are presented here and numbered accordingly.*<br>**<sim>** | R | N.4.3 |
| *REFERENCES WORK PACKAGE*<br>**<refwp>** | R | N.5.4 |
| *EXPENDABLE AND DURABLE ITEMS LIST WORK PACKAGE*<br>**<explistwp>** | | N.5.5 |
| *ADDITIONAL SUPPORTING WORK PACKAGE*<br>**<genwp>** | | N.5.6 |
| **REAR MATTER**<br>**<rear>** | R | TABLE A-II |
| **LEGEND**<br>R - Required (see A.4.3.1.3).<br>Shaded - May be Required (see A.4.3.1.4). | | |



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**TABLE A-XIX. Hand receipt requirements matrix.**

| Content | HR | MIL-STD REF |
|---|---|---|
| **PUBLICATION** (**<production>**)<br>**<hand_receipt>**<br><br>PUBLICATION TITLE:<br>Hand Receipt Covering Contents of Components of End Item (COEI), Basic Issue Items (BII), and Additional Authorization List (AAL) for *short end item title* | R | TABLE A-I |
| **FRONT MATTER**<br>**<paper.frnt>** or **<framed.frnt>** | R | TABLE A-II |
| **MODULE 1. INTRODUCTION**<br>**<hr_gim>** | R | O.4.1 |
| *HR INTRODUCTION WORK PACKAGE*<br>**<hrintrowp>** | R | O.5.1 |
| Scope<br>**<scope>** | R | O.5.1 |
| General – explains DA Form 2062 and purpose<br>**<para0>** | R | O.5.1 |
| Explanation of Blocks and Columns (DA For 2062)<br>**<para0>** | R | O.5.1 |
| Authorization Documents for COEI, BII, and AAL<br>**<para0>** | R | O.5.1 |
| Additional information<br>**<para0>** | | O.5.1 |
| **MODULE II. HAND RECEIPT**<br>**<hr_sim>** | R | O.4.2 |
| *HR WORK PACKAGE*<br>**<hrwp>** | R | O.5.2 |
| Paragraph<br>**<para0>** | R | O.5.2 |
| Technical content<br>**<para0>** | R | O.5.2 |
| Overprinted DA Form 2062<br>**<para0>** | R | O.5.2.1 |

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**TABLE A-XIX. Hand receipt requirements matrix. (continued**

| Content | HR | MIL-STD REF |
|---|---|---|
| **REAR MATTER**<br>**<rear>** | R | TABLE A-II |
| **LEGEND**<br>R - Required (see A.4.3.1.3).<br>Shaded - May be Required (see A.4.3.1.4). | | |

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**TABLE A-XX. Technical bulletin requirements matrix.**

| Content | TB | MIL-STD REF |
|---|---|---|
| **PUBLICATION** (**<production>**)<br>**<tech_bltn>**<br><br>PUBLICATION TITLE:<br>Technical Bulletin for *insert system* | R | TABLE A-I |
| **FRONT MATTER**<br>**<paper.frnt>** or **<framed.frnt>** | R | TABLE A-II |
| **MODULE 1. GENERAL INFORMATION AND EQUIPMENT DESCRIPTION**<br>**<tb_gim>** | R | Q.4.1 |
| *GENERAL INFORMATION WORK PACKAGE*<br>**<ginfowp>** | R | Q.5.1 |
| Scope<br>**<scope>** | R | B.7.1.1 |
| Maintenance forms, records, and reports<br>**<mfrr>** | | B.7.1.2 |
| Reporting Equipment Improvement Recommendations (EIR)<br>**<eir>** | | B.7.1.3 |
| Hand Receipt (HR) manuals<br>**<handreceipt>** | | B.7.1.4 |
| Corrosion Prevention and Control (CPC)<br>**<cpcdata>** | | B.7.1.5 |
| Ozone Depleting Substances (ODS)<br>**<odsdata>** | | B.7.1.6 |
| Destruction of equipment to prevent enemy use<br>**<destructmat>** | | B.7.1.7 |
| Preparation for storage or shipment<br>**<pssref>** | | B.7.1.8 |
| Transportability guidance<br>**<transportability>** | | B.7.1.8 |
| Warranty information<br>**<wrntyref>** | | B.7.1.9 |
| Nomenclature cross-reference list<br>**<nomenreflist>** | | B.7.1.10 |
| List of abbreviations/ acronyms<br>**<loa>** | | B.7.1.11 |



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**TABLE A-XX. Technical bulletin requirements matrix. (continued**

| Content | TB | MIL-STD REF |
|---|---|---|
| Quality of material<br>**<qual.mat.info>** | | B.7.1.13 |
| Safety, care, and handling<br>**<sftyinfo>** | | B.7.1.14 |
| Nuclear hardness<br>**<hcp>** | | B.7.1.15 |
| Calibration<br>**<calref>** | | B.7.1.16 |
| Item unique identification (IUID)<br>**<iuid>** | | B.7.1.17 |
| Supporting information for repair parts, special tools/kits, TMDE, and support equipment<br>**<supdata>** | | B.7.1.25 |
| Copyright credit line<br>**<copyrt>** | | B.7.1.26 |
| *EQUIPMENT DESCRIPTION AND DATA WORK PACKAGE*<br>**<descwp>** | | Q.5.2 |
| Equipment characteristics, capabilities, and features<br>**<eqpinfo>** | | B.7.2.1 |
| Location and description of major components<br>**<locdesc>** | | B.7.2.2 |
| Equipment differences<br>**<eqpdiff>** | | B.7.2.3 |
| Equipment data<br>**<eqpdata>** | | B.7.2.3 |
| **MODULE X. MAINTENANCE INSTRUCTIONS**<br>**<tb_mim>** | | Q.4.2 |
| *MAINTENANCE INSTRUCTION WORK PACKAGES*<br>Maintenance tasks<br>**<maintwp>**<br>**<maintsk>** | | E.7.4<br>E.7.4.4 |
| Inspect<br>**<inspect>** | | E.7.4.4.11 |
| Remove<br>**<remove>** | | E.7.4.4.31.1 |
| Install<br>**<install>** | | E.7.4.4.31.2 |

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**TABLE A-XX. Technical bulletin requirements matrix. (continued**

| Content | TB | MIL-STD REF |
|---|---|---|
| Pack<br>**<pack>** | | E.7.4.4.20.1 |
| Unpack<br>**<unpack>** | | E.7.4.4.20.2 |
| Preparation for shipment<br>**<prepship>** | | E.7.4.4.25 |
| Transport<br>**<transport>** | | E.7.4.4.38 |
| Load<br>**<load>** | | E.7.4.4.14.1 |
| Unload<br>**<unload>** | | E.7.4.4.14.2 |
| *WIRING DIAGRAMS AND SCHEMATICS WORK PACKAGE*<br>**<wiringwp>** | | Q.5.4 |
| *ADDITIONAL MAINTENANCE WORK PACKAGE*<br>**<gen.maintwp>** | | Q.5.5 |
| **MODULE X. SUPPORTING INFORMATION**<br>**<tb_sim>** | | Q.4.2 |
| *REFERENCES WORK PACKAGE*<br>**<refwp>** | CR | Q.5.6 |
| *EXPENDABLE AND DURABLE ITEMS LIST WORK PACKAGE*<br>**<explistwp>** | | Q.5.7 |
| *ADDITIONAL SUPPORTING WORK PACKAGE*<br>**<genwp>** | | |
| **REAR MATTER**<br>**<rear>** | R | TABLE A-II |
| **LEGEND**<br>R - Required (see A.4.3.1.3).<br>Shaded - May be Required (see A.4.3.1.4).<br>CR - Conditionally Required (see A.4.3.1.4.3). | | |

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## A.5 IETM FUNCTIONALITY GENERAL REQUIREMENTS.

### A.5.1 Physical IETM layout and styles.

Sections A.5.1.1 through A.5.1.10 discuss IETM layout and style parameters.

#### A.5.1.1 Screen layout.

The screen can have an outer shell, title bar, and an inner shell as described in this section and shown in FIGURE A-10.

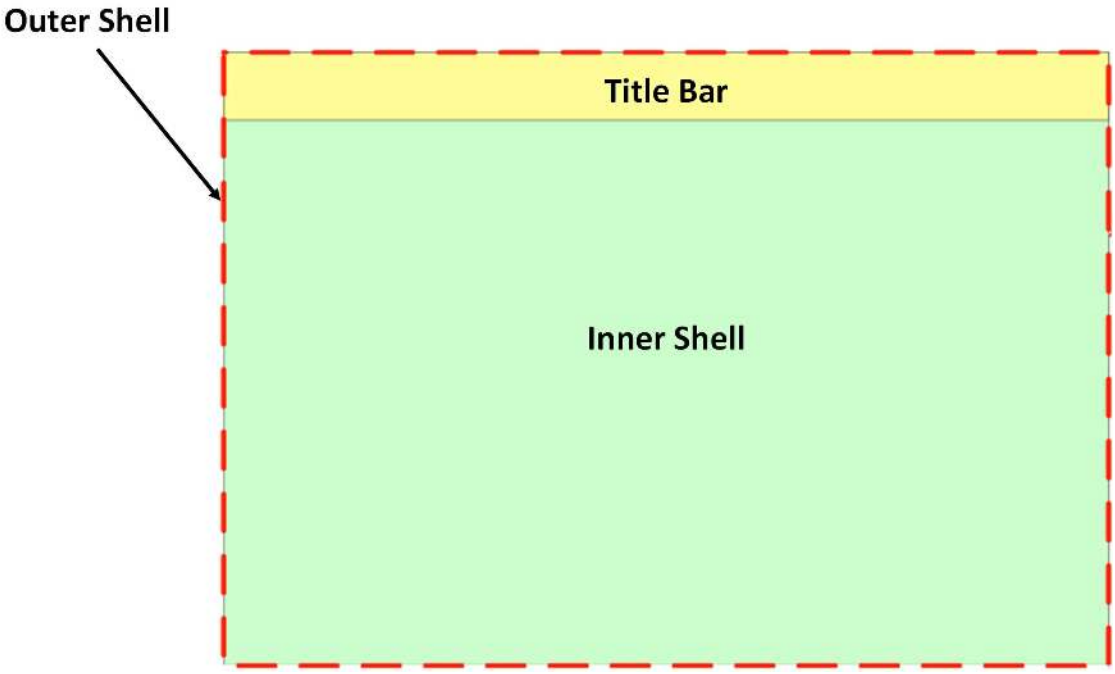


**FIGURE A-10. Screen layout.**

**a. Outer shell**.

The outer shell is the application/browser window and is the portion of the screen that surrounds the title bar and inner shell.

**b. Title bar**.

A horizontal title bar shall appear at the top of the outer shell. The title bar can include the name of the application and can also include security classification markings, the WP sequence number and the WP title of the WP currently being viewed.

**c. Inner shell**.

The inner shell (shown in FIGURE A-11) is the portion of the IETM that provides the primary display area. The inner shell shall contain, as a minimum, the table of contents panel, navigation panel and the main content area. Optionally it can include a Reset area.



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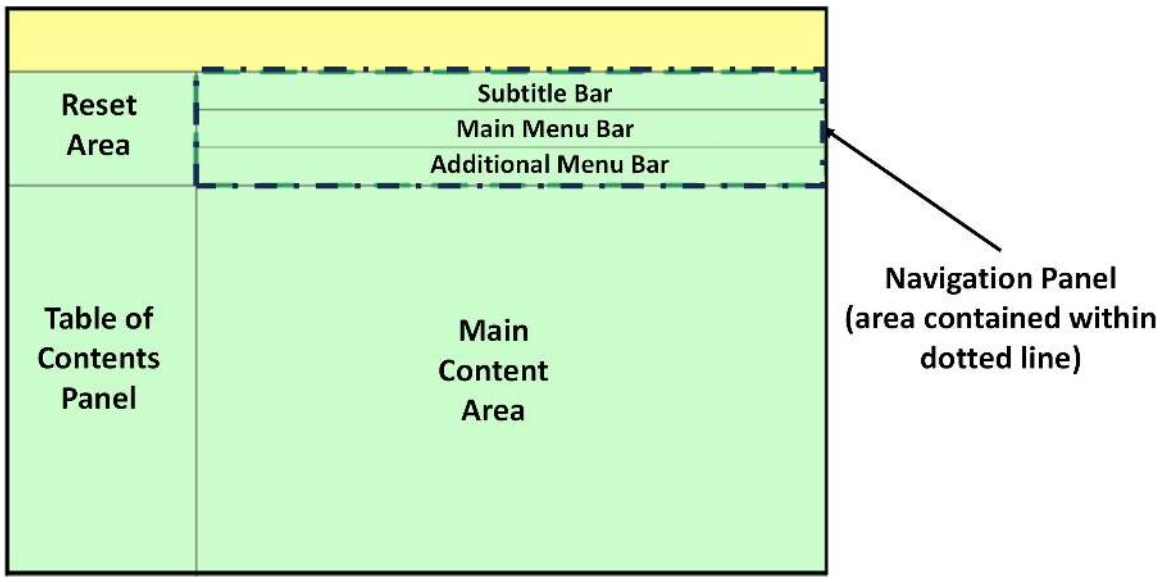


**FIGURE A-11. Inner shell components.**

**d. Table of contents panel**.

The table of contents panel, located to the left of the main content area, is where the navigation interactions shall appear. A table or list of all key entry points (for example, modules and WPs) shall be made available for user access. This panel shall be resizable and can be toggled on and off.

**e. Reset area**.

The reset area allows a user to return the IETM view back to its default settings. The reset area also provides a special mechanism for navigation and preferences. Items in this area are provided using a function menu. For example, the reset area can provide a menu allowing the user to navigate the IETM in different ways, such as, by part number, list of graphics, revision summary, etc. The reset area can be toggled on and off but shall always be accessible. If toggled off, the reset area shall be easy to find by the user. This could be achieved via an icon such as a compass rose or by a right click menu. Clicking on this area can provide a function menu.

The reset area also provides a means for selecting preferences. The reset area menu may allow the user to toggle screen areas on and off. It is recommended that a menu item be grayed out if the user is not permitted to toggle that particular screen area off.

### A.5.1.2 Navigation panel.

The navigation panel can appear either above or below the main content area. The position of the navigation panel shall, as far as possible, remain consistent throughout the application. The navigation panel is comprised of multiple bars in the following sequence:

**a.** Sub title bar (optional)

The subtitle bar can be used for additional information, from the identification and status sections such as modification state or classification, when it has not been presented in the title bar.

**b.** Main menu bar (mandatory)

The main menu bar consists of a basic set of functions. If required by the project, these functions (or any combination of them) can be mandated. There are three ways to present the standard functions:

- Text based - the function is given a text label
- Graphic based - the function is represented by an icon

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- Text and graphic - a combination of an icon and text is used

The main menu bar can contain the functions as illustrated in FIGURE A-12. When used, icons shall be developed to be representative of the function that they perform. Their meaning shall be apparent with little confusion in the mind of the user. Icons are not mandated because the end user and operating environment have to be considered before deciding upon the final set. These relate to the basic functions previously described in this chapter with some additional icons for further expansion of the application.

Cascading menus can appear as a child of a function when selected. In a drop-down menu, this appears next to the function selected. There can be several levels of cascading menus.

Functions that are not active during any rendering shall be presented as disabled. The bar shall remain accessible.

**FIGURE A-12. Navigation panel.**

**(1) Previous**.

This navigates the user through the data in a sequential manner backward.

**(2) Next**.

This navigates the user through the data in a sequential manner forward.

**(3) Table of contents**.

A function that toggles the Table of contents panel on and off.

**(4) History**.

A navigational function that displays previously viewed data in a sequential manner (for example, data module visited).

**(5) Search**.

A function that displays a search dialog box.

**(6) Print**.

A function that allows a WP to be printed (for example, by use of a print dialog box). By decision, if information is classified the print function can be disabled and the print button grayed out.

**(7) Feedback**.

A function that allows feedback to be gained from the user either on an individual module, publication or a complete WP.

**(8) Exit**.

A function that allows exit from the IETM, where appropriate, the following modes can be available: Complete (exists the application) and Suspend (suspends the session for the user).

**(9) Help**.

A function that displays a help file. The help can contain viewer usage information.

**(10) IDSTATUS**.

A function that shows part or the whole of the identification and status section of the WP or publication.

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**(11) Additional Information Bar**.

The additional information bar can be used if additional functions are required (for example, ordering of spares). It is presented below the main menu bar and shall include the functionality to be toggled on and off.

### A.5.1.3 Style and format.

Style and format shall conform to the general principles defined in this section.

**a. Use of colors**.

It is recommended to design use of colors for the least capable display device (8-bit, monochrome devices, etc). The use of color shall be controlled by human factors studies or usability testing. The operational environment and equipment in use shall dictate some rules regarding the colors used. There can be operational considerations such as night ops, red-light conditions and where color has special meaning. The recommended color scheme for standard text display shall be black on a white background.

**b. Typefaces**.

The use of font size shall be adequate for the user to be able to read the technical content. Attention shall be paid to the display device and working environment.

### A.5.1.4 Dialog box.

A dialog box is used to obtain information from the user and provide feedback to the user. When possible, it shall appear in the screen center. The dialog box appearance shall be consistent throughout the IETM.

A dialog box can contain controls. A user interacts with the dialog box through named controls. Typical controls are as follows:

- Text input box
- Push button
- Checkbox
- Radio button
- Selection box

Controls can have tool tips. Tool tips display further information about the purpose of the control. A tool tip appears when the user hovers over the control with the mouse pointer. Refer to FIGURE A-13 for an example dialog box which contains various named controls types and a tool tip.



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**FIGURE A-13. Dialog box example.**

### A.5.1.4.1 Text input box

A text input box allows the end user the ability to provide information to the application in the form of free text and/or numerical values.

Wherever possible, the data input field prompt shall be placed on the same line as the data entry area. Formatting for prompts shall be distinctive. Prompts shall be placed in close proximity to their respective data entry area.

When required, an acceptable value criteria shall be provided through the element **<validate>** expression on the element **<userEntry>** expression. The acceptable values specified by the **<validate>** expression shall be displayed in close proximity to the text input box, such as appended to the prompt or directly below the text input box. As a minimum, if the user enters a value outside of the acceptable values criteria **<validate>** expression, a message shall be displayed to the user identifying the acceptable values. Upon acknowledging the message, the text input box shall be displayed to give the user another chance to enter a valid value.

### A.5.1.4.2 Push button.

The push button executes a function. Examples of push buttons are "OK", "Calculator" (executes a calculator program), "Figure" (displays associated graphic), and "IDSTATUS" shows identification and status information. A dialog box has five pre-defined push button functions and shall be displayed in the following order centered along the dialog box bottom:

- Submit (mandatory)
- Cancel/Abort (mandatory)
- Reset (optional)
- Help (optional)
- Generic (optional)

#### A.5.1.4.2.1 Submit.

The submit push button is used to submit the data in the dialog box to the IETM, closes the dialog box, and performs the next action. Submit push buttons can have several different text labels. Examples of submit push buttons are:

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- "OK" - the user enters the prompted data before submitting to the logic engine
- "Search" - the user enters search criteria before submitting to the search engine
- "Comment" - the user enters comments on the dialog box before submitting to the comments service

#### A.5.1.4.2.2 Cancel.

The cancel push button is used to void information entered, to void executing the function, or to close the dialog box, and return control to the prescribed action as identified for the function.

#### A.5.1.4.2.3 Reset.

The reset push button, when activated, is used to reset controls on the dialog box to their default values. An example of a reset push button is to "Clear" the contents of a search dialog box ready for a new search.

#### A.5.1.4.2.4 Help.

The help push button, when activated, is used to provide further information about the dialog box.

#### A.5.1.4.2.5 Generic

A generic push button, when activated, executes a defined function. A generic push button shall not close the dialog box or display along the dialog box bottom. An example of a generic push button is to execute a calculator to help compute an answer for the dialog box.

### A.5.1.4.3 Checkboxes.

Checkboxes provide "On/Off" toggle switches. A switch is "On" when the box is checked. It is possible to have none or several checkboxes with the same control name thus allowing users to select several values for the same property. An example of a checkbox function in an IETM is selecting applicability values for an applicability filter.

### A.5.1.4.4 Radio buttons.

Radio buttons are similar to checkboxes except that where a group of radio buttons exist, only one of the options can be selected. At all times, exactly one radio button in a group is set to "Checked."

Radio buttons can be used to implement a Yes-No dialog box. Two radio buttons are labeled with "Yes" and "No" respectively, and a submit button (for example, "Submit" or "OK") is used to submit the response to the IETM.

### A.5.1.4.5 Selection boxes.

Selection boxes Selection boxes are used when the user is required to select a value from a scrollable list box. Selection boxes can be set so that a user can pick only various options.

## A.5.1.5 Lists.

Lists shall conform to the general requirements as identified below with the exception when those specific measurements for indentation and spacing cannot apply.

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### A.5.1.5.1 General list information.

Lists consist of two main parts, a title and the list items. The title is optional.

The list title line shall be aligned to the left type limit and presented in 10/11 pt. Bold shall be used for titles for random lists and sequential lists. Italic shall be used for definition lists and legends.

The normal leading from a preceding headline, paragraph, list item, figure title line, table footer (closing line), last line in a warning, caution or note, etc, to the list title line shall be used. The leading between the title line and the first list item shall be 14 pt (equal to an extra 4 pt leading added after the title).

The basic value for list item leading is 10/11 pt. The values of the extra leadings between and after list items are given in the following paragraphs.

**Note**

The prefixes are not entered in the data modules when data modules are written in an XML editor or a WYSIWYG-editor with an automatic function for presentation.

### A.5.1.5.2 Random lists - unordered lists.

Unordered list items at level 1, start with an en-dash [–] as the prefix aligning the left type limit. The text shall be indented 7 mm.

The list items within unordered lists on all three levels shall be separated by normal leading, 11 pt, giving no extra leading between the same levels of unordered list items.

An extra leading of 8 pt shall be added after the last list item on each of the three levels.

- Subunordered (level 2) list items start with a bullet [●] as the prefix indented 7 mm from the left type limit.
- The text shall be indented 7 mm further (14 mm from left type limit).
  - Sub-subunordered (level 3) list items starts with en-dash [–] as the prefix indented 14 mm from the left type limit. The text shall be indented 7 mm further (21 mm from left type limit).
  - Each list item, on any level, can consist of one or more paragraphs.

    The paragraphs within a list item shall be separated by an extra leading of 8 pt. This applies on all levels.

    An extra leading of 8 pt shall be added after the last paragraph in a list item.

### A.5.1.5.3 Random lists - simple lists.

Simple lists are presented just like an unordered list (A.5.1.5.2) at level 1, but without any prefix.

### A.5.1.5.4 Sequential lists - ordered lists.

General there are two levels of ordered lists, both indented with a multiple of 7 mm.

Only one ordered list shall be placed under a sidehead. One subordered list can follow each ordered list item.

Ordered lists shall not be used to provide procedural step information.

1. Ordered lists shall be numbered with Arabic numerals. The number, starting with "1", shall align the left type limit.
2. The text shall be indented 7 mm.
3. The additional leading after each list item on both levels shall be the same as for text paragraphs, 8 pt.

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4. There shall be no period [.] after the last digit of the number.

   4.1 Subordered lists shall be numbered. The number shall start with the number of the preceding item followed by a number starting with "1". The numbers shall be separated by a period [.].

   4.2 The number in subordered lists shall be indented 7 mm from the left type limit. The text shall be indented 7 mm further (14 mm from left type limit). The text indent can be extended in steps of 7 mm as necessary, for example if one of the numbers is wider than the character 9 (for example, 6.12, 12.6, 10.10). The indentation shall be consistent within each ordered or subordered list.
   Each list item, on any level, can consist of one or more paragraphs and shall be separated by an extra leading of 8 pt.

5. Ordered lists can include subunordered (level 2) and sub-subunordered lists.

   - Subunordered (level 2) list items can be included in ordered lists.
   - Each subunordered and sub-subunordered list can consist of one or more paragraphs.
     – Sub-subunordered (level 3) list items can also be included in ordered and subordered lists.
     – Each subunordered and sub-subunordered list can consist of one or more paragraphs.

## A.5.1.6 Steps.

Steps shall conform to the general principles defined in this section.

**a. Procedural steps**.

It is preferred that procedural steps and their corresponding illustrations be presented together. When this is not feasible or it affects the user's perception of the screen layout (such as a scrolling screen or "cluttering" when using a small screen) the illustration or inline an icon can be placed inline to allow the display of the illustration. Steps shall be presented such that the order cannot be misunderstood. Screen stacking (for example, several open windows) to present multiple steps is not permitted.

**b. Keep with next.**

Procedural, fault isolation, and crew steps with the attribute keepWithNext set to "1" are presented with the next sibling step (if one exists) and therefore, all children of the step for which the attribute is set are kept together as well (if possible).

## A.5.1.7 Tables.

Tables shall conform to the general principles defined in this section.

**a. Location**.

Tables can appear within the body (inline) of the IETM or in their own separate window (pop-up). When pop-up tables are used an icon shall identify the table. The icon serves as a push button to open up a table pane.

**b. Typeface**.

Typefaces shall be the same as the rest of a WP.

**c. Headers and footers**.

The header shall always be visible so that it does not scroll away while rows are scrolled. Where appropriate the footer can remain visible so that it does not scroll away while rows are scrolled.

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**d. Footnotes**.

Table footnotes marker, presented either in superscripted format (superscripted numbers are recommended) in parentheses following the item, or by any other means, shall have a link to the related footnote presented in the table footer or display a pop-up when the mouse "hovers" over the footnote marker in the table cell.

**e. Background**.

It is preferred that the background be white. Where the table is long, it can be acceptable to change the background colors of alternate rows to aid readability.

**f. Links to table cells**.

Only the referencing text within a table cell shall be hyperlinked. Do not link the entire cell. Reference to a table cell or row shall scroll directly to the referenced cell or row. This type of reference is not presented in a printed output.

### A.5.1.8 Hyperlinks.

Hyperlinks shall conform to the general principles defined in this section.

**a. Hyperlink presentation**.

Standard web practices shall be followed to indicate the presence of hyperlinks. Blue underlined text shall be used to indicate unvisited links and purple underlined text shall t be used to indicate visited links.

**b. Graphic links (hotspots)**.

The following are the four acceptable modes of visual indication of hotspots in graphics:

- persistent visual indication that an area is hot
- cursor changes shape or color when cursor is over the hotspot area
- object changes shape or color when cursor is over the hotspot area
- pop-up appears while cursor is over the hotspot area (for example, RPSTL callout expands)

### A.5.1.9 Warnings and cautions.

Warnings and cautions shall conform to the general principles defined in this section.

**a. Presentation and location**.

Warnings and cautions shall be presented inline either by the complete warning or caution or by an icon. When icons are used the warning or caution shall pop-up when the icon is pushed. Warnings and cautions and any associated icons shall be displayed prominently in the main content area. They shall be treated with particular formatting requirements to ensure the user’s attention.

The warning and caution, and associated symbols, or icons shall be visible to the user whenever the warning or caution is applicable to the information (procedure, etc) is viewed. Any warnings or cautions that are applicable to a complete WP (for example, included and presented in safety conditions within preliminary requirements) shall be displayed when a WP is opened.

**b. Pop-up warnings and cautions**.

Pop-up warnings and cautions shall appear in the center of the user viewing area, to alert the user of a specific condition. A system can display all applicable pop-ups as stacked window frames. The user shall be able to again see the pop-up after they are acknowledged by clicking on an applicable icon. See FIGURE A-14 and FIGURE A-15

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**FIGURE A-14. Warning and a caution with acknowledge buttons.**

**FIGURE A-15. Warning and caution icons.**

**c. Safety icons and symbols**.

Hazardous material icons are optional. The meaning of the hazardous material icon can pop-up by "hovering" the mouse over the icon.

**d. Illustrations - figures**.

Illustrations can appear within the body (inline) of the IETM or in their own separate window (pop-up). When pop-up illustrations are used an icon shall identify the "hidden" illustration. The icon serves as a push button to open up an illustration pane.

Illustrations can be displayed using as much screen space as is available. The IETM presentation system shall provide the user with the ability to view graphic objects with pan, zoom, expand, and magnify.

If required, hovering over an area of a graphic can provide some means of descriptive data. For example, if used in a wiring data graphic it can be some information related to the cable.

Right-mouse clicking can optionally display a menu of options to include related remove/replace/repair procedures, part ordering, training, etc.


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### A.5.1.10 Printed output from IETM.

The IETM application is designed for the purpose of displaying data electronically. It shall be discouraged for use as a publishing engine. However, if there is the requirement to print any data the following guidelines shall be adhered to.

Consideration could be given to the printed output of work packages. This would require that any applicable caveats be printed. The function to provide a complete publication output is strongly discouraged. When printing a data module from the IETM application, a header and footer shall be printed on all pages that are output. The following information can be displayed on them:

- WP number
- the date and time of printing
- the security classification
- a caveat to state the information is only applicable at the time of printing and is not subjected to amendment action and that it shall be "destroyed after use"
- the page number, starting with "Page 1"
- the words "End of work package" on the last page

## A.6 IETM FUNCTIONALITY DETAILED REQUIREMENTS.

The specific level of functionality and user interaction to be provided in IETMs shall be in accordance with the functionality matrix requirements provided in A.6.1. The functionality matrix (TABLE A-XXI) shall be tailored to user requirements for each system.

### A.6.1 Background use of the functionality matrix.

The functionality matrix provides a standard format for documenting the functional needs of a technical data project. It provides also standard definitions for each functionality, so that companies and customers can clearly state their needs and capabilities.

### A.6.1.1 Functionality categories.

The matrix is divided into the following categories based on functionality type.

- Functionality

  The functionality column lists possible technical publications functionalities. The following summarizes the functional categories at a high level.

- Access

  These are the functionalities that allow or restrict users to view specific data.

- Annotation

  These are the functionalities that have to do with the user’s ability to add notations or other marks associated with the data. Bookmarks are a type of annotation.

- Delivery and distribution

  Delivery is the method of moving tech data from a contracted company to the contracted client and distribution is the method of moving tech data from an initial point to all the end users of the tech data. Selection of the desired media and method will drive costs; however, the most significant consideration is the readiness of the infrastructure at the user level for whatever method is chosen.


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- Diagnostics and prognostics

  Diagnostics include procedures for identification of a fault that lead to a corrective action or maintenance procedure. Diagnostics span from basic stand-alone troubleshooting procedures to integration with the product and other maintenance systems. Prognostics include the capability to predict system degradation or failure based on maintenance and operational data input and real time monitoring. Diagnostics/prognostics can be a significant benefit in reducing maintenance times and total ownership costs; however, they can also be a sizable cost driver in publications development.

- External processes

  The digital environment has the potential to provide greater functionality by interacting with external processes to retrieve and transmit information.

- Graphics

  Various levels of graphics display, interactivity and navigation can be implemented. Examples of these functionalities include.

  - Point and click on a locator graphic for the purpose of navigating to graphic details

  - Links to textual or tabular information

  - Highlighting specific connections in a circuit

  - System simulation expressed graphically

- Linking

  Basic linking functionality is defined as essentially link access or connections to the data within the publication such as from the table of contents to the applicable section. Additional linkage such as cross-references can require some additional effort to maintain. Linking to data items externally can also be accomplished. These links can be to resources such as material handling information or for integration with other related information.

- Navigation and tracking

  Different navigation methods enable linear and non-linear access thru the data. Features such as "next" and "previous", search and the use of bookmarks are considered to be relatively fundamental and consistent with most Web-based data presentation techniques. Higher complexity navigation techniques include dialog driven interaction, voice activated commands and various filtering techniques. Examples of filtering characteristics are model number, identification number (for example, tail number), modification performed and user qualifications.

- Printing

  Some publications by their nature are intended for use in an electronic environment, with print functionality limited primarily to task oriented and screen print output. By defining a hard copy output that more closely resembles a paper technical publication, the resulting costs and complexity rise.

- Special content

  The inclusion of additional data types such as audio, motion video, and animations are accommodated relatively easily by most presentation systems; however, content generation often cost more. Performance issues can also occur, which can have further cost implications.

- Updates

  There are a number of different update methodologies that affect life cycle costs and include revisions, changes, and urgent changes (Rapid Action Changes (RACs), etc). These can include change markings or other change indications. Updates include any data delivery after the initial delivery.

- User operation mode

  These functionalities have to do with infrastructure and a user’s ability to connect with the source of the data.

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### A.6.2 References.

Wherever possible references are made in the functionality definitions to indicate related material. When a reference is not possible, a notation is made to indicate that the functionality is viewer, infrastructure, or external software functionality that is not enabled.

### A.6.3 Matrix columns.

The horizontal axis contains the column headings: Functionality, Complexity , Requirement, All data sets, and a list of the information sets.

#### A.6.3.1 Complexity.

There are two columns that indicate a relative measure of difficulty and probable higher cost associated with a particular functionality (the higher the complexity factor, the more difficult implementation and a more probable higher cost). In many cases the complexity level is different depending upon a page-oriented or IETM implementation. Separate columns are provided that indicate the complexities in these two implementations. The absence of a complexity factor indicates that the functionality is generally not available or desirable.

PDF publications are presented in a linear or document oriented manner. The data author largely predefines the sequence of the data presentation. PDF publications can be designed for traditional printing or for electronic display. Electronically displayed page oriented publications are sometimes called Electronic Technical Manuals (ETMs).

IETM data is displayed in a non-linear fashion. There are high levels of interactivity between the data and the user. The sequence of presentation is dictated by inputs by the user or external sources and events. IETM implementations typically require the use of the process data module and a logic engine.

The complexity factors provided are intended to provide relative complexities (and relative costs) for implementing various functionalities. The complexity factors range between one and five, with one being the least complex, to implement and five being the most complex.

#### A.6.3.2 Requirement.

The requirement column is to be used by the project team to indicate the need for a specific functionality. If desired, the acquisition agent can also use the requirement column (in the spreadsheet version of the matrix) to capture and coordinate notes reflecting the importance of the functionality to the project (shall have, nice to have, do not want, etc) prior to making final requirements decisions. The spreadsheet version of the matrix provides a usable format for this purpose.

#### A.6.3.3 All information sets.

The All information sets column is used to identify those functionalities that typically apply universally across all information sets. These functionalities are identified with an "A." These are in contrast to functionalities that can be applied to specific identified information sets.

#### A.6.3.4 Information sets.

The Information sets columns are to be used by the project team to indicate, which data sets should include the specified functionality. A project or organization can modify the matrix to match the information sets used in their actual project or organization.

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### A.6.3.5 Front matter.

Front matter is the information traditionally displayed in the front of paper publications. It can include the title information, "A" page or change information, verification status, export control notice, warning, safety summary, title page, list of pages or data modules, change records or highlights, access illustration, list of abbreviations, list of terms, list of symbols, technical standards records, table of contents, general data, function (of an IETM/Publication), general warnings cautions and related safety data, numeric indexes, alphabetic and alphanumeric indexes, list of special materials, list of dangerous materials, list of related data, introduction, supplier lists, configuration, copyright, etc. The complexity, practicality and desirability of front matter content vary greatly depending on the data acquired.

## A.6.4 Functionality definitions.

The following tables provide descriptions of each of the functionalities identified in the matrix as well as some considerations to be made and the range of capabilities that can be addressed therein. These tables provide identification of the category, title of the functionality, a definition of the functionality and in many cases, an example.

### A.6.4.1 Functionality definitions - access.

#### A.6.4.1.1 Login.

This is the capability to identify a user for tracking, authorization or other purposes. This is an IETM viewer functionality. This functionality requires the inclusion of the Exit functionality. Consideration shall be given to determine if specific profiles, user IDs, roles and permissions will be required to permit usage. A password for log on is not necessary. It should be kept in mind, that if there is a higher-level infrastructure controlling access, it should be the controlling authority. Data access based on the training and skill of the maintainer can require that the system link into the maintainer’s training records. In a stand-alone mode, the amount of data required on the stand-alone device is considerable and complex. Consideration shall be given to the maintenance of this data in multiple locations.

#### A.6.4.1.2 Suspend and restart.

This is the capability to suspend a session and provide the user with the ability to restart at the point of suspense. This functionality requires the inclusion of the Login functionality. This capability extends beyond a user session and retains session data. This is an IETM viewer functionality. During the performance of a maintenance session, a part is removed. The replacement part is not available in supply. The session is suspended and subsequently restarted after the part becomes available.

#### A.6.4.1.3 Exit.

Function to close the session. This is an IETM viewer functionality. If suspend is used, the current state tables and position are maintained, otherwise all information concerning state tables and position is cleared.

### A.6.4.2 Functionality definitions - annotation.

#### A.6.4.2.1 Action Complete Indicator (checkbox.

The functionality used to indicate the completion of an action. This functionality is normally associated with, but not limited to use with, checklists. Other methods can be used (for example, "sign off"). The data entered via the action

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complete indicator can be retained for state table or audit trail purposes depending on project and organization requirements.

### A.6.4.2.2 Global data annotation.

This functionality allows the entry, storage and display of globally applicable supplemental data. This is an IETM viewer functionality.

Globally accessible annotations should be limited in scope and required to be approved by the end user's QA process and be made available to the development contractor. Globally accessible annotation is a quality maintenance supplement indicating the new type of lubricant to use. Some users of data on CD-ROM might use these global data annotations to provide updates to data by quickly revising the CD-ROM to contain the "sticky" notes.

### A.6.4.2.3 Local data annotation.

This functionality allows the entry, storage and display of locally applicable supplemental data. This is an IETM viewer functionality.

These annotations should be limited in scope and follow local approval process.

Example - The maintainer can be required to use additional filtering in a desert environment.

### A.6.4.2.4 Personal annotation.

This capability allows an individual user to add supplemental information to the data. This is an IETM viewer functionality. This functionality requires the inclusion of the Login functionality.

Personal annotation can be added or deleted at the end user's discretion and it is suggested that it is not retained at the end of the user session. It is not the intent that the personal annotation persists. See FIGURE A-16.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:03Z
Check the source to verify that this is the current version before use.

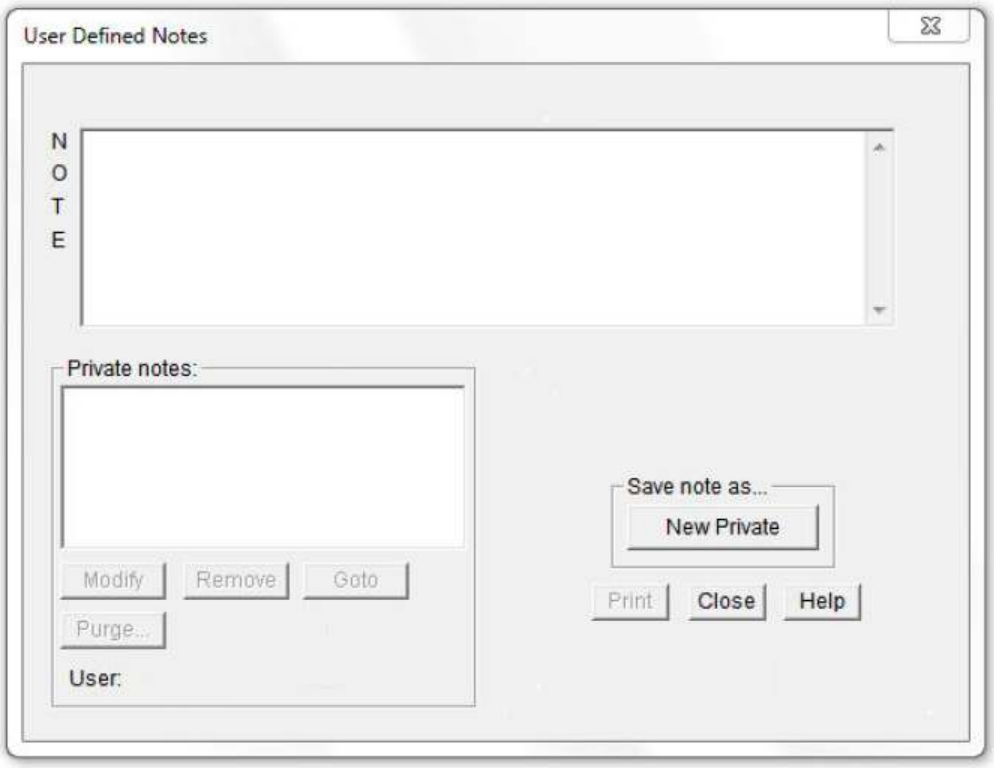


**FIGURE A-16. Personal annotations**

### A.6.4.2.5 Redlining text.

Capability provided during the publication development and verification process to annotate text using markings for deletions and insertions. The capability exists to save and attribute redline markups. These are saved for feedback only for the improvement of the publication. This is an IETM viewer functionality.

### A.6.4.2.6 Redlining graphics.

This is a capability to annotate graphics using an overlay freehand-type drawing facility. It can be used to report errors during the publication development and verification process. The capability exists to save and attribute redline markups. These are saved for feedback only for the improvement of the publication. This is an IETM viewer functionality. See FIGURE A-17.



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Check the source to verify that this is the current version before use.

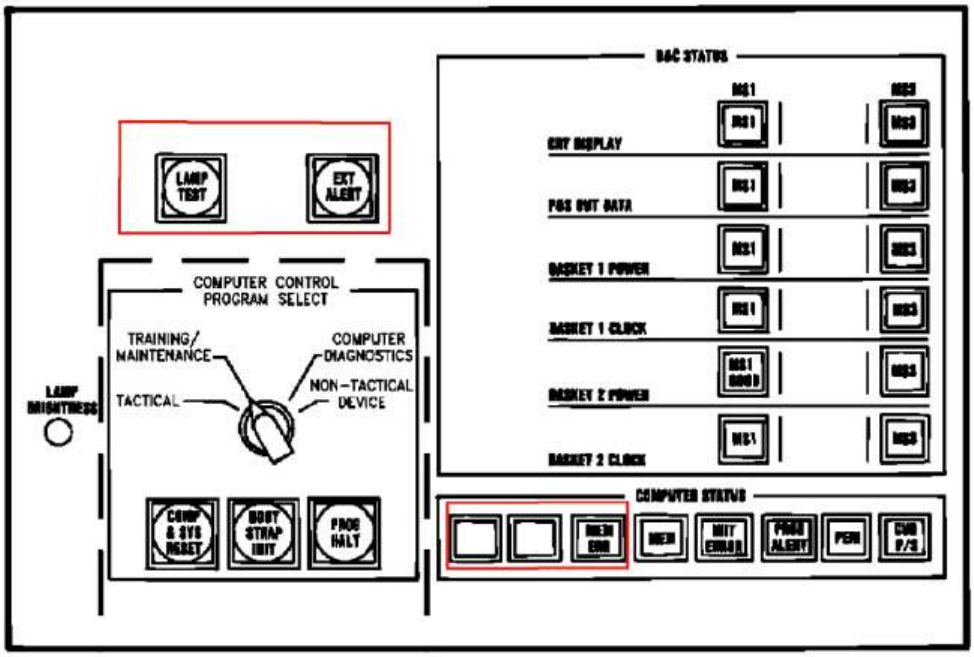


**FIGURE A-17. Redlining graphics.**

### A.6.4.3 Functionality definitions - delivery and distribution.

#### A.6.4.3.1 Printed publication.

A paper publication compiled from data modules.

#### A.6.4.3.2 Physical media.

The capability to deliver and/or distribute data on hard media such as CD-ROM, DVD, hard drive. This is an infrastructure functionality.

Such physical distribution methods typically entail issuance of a complete database thus replacing the data in use. This replacement constitutes a block change update and is generally performed on a periodic basis.

#### A.6.4.3.3 Network distribution.

Distribution via the Internet (connections to the World Wide Web (www))/Intranet (internal to one network) and consists of direct transfer from one computing system to another. This is an infrastructure functionality.

Delivery could be via secure FTP, (HTTP) or other transfer protocols. Bandwidth, security and operational deployment considerations should be addressed. This could enable near real-time updates.



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### A.6.4.4 Functionality definitions - diagnostics and prognostics.

#### A.6.4.4.1 Diagnostics - user determined entry to data.

Tasking for troubleshooting procedures is provided through primarily textual references. "If statements" provide alternatives in a narrative form. This makes use of a predefined fault tree. The user determines starting point for maintenance action.

#### A.6.4.4.2 Diagnostics - software driven entry to data.

The appropriate maintenance action starting point is software determined thru use of an inference or logic engine. The user is provided with the appropriate starting point for fault isolation. Various inputs from personnel and system and multiple fault codes are analyzed. The analysis will determine if a relationship exists between these fault codes and information with the appropriate action as a result. This is an external software functionality.

#### A.6.4.4.3 Dynamic diagnostics.

Dynamic diagnostics This is diagnostic capabilities that use onboard monitoring devices (for example, Built-In Test (BIT)) and/or support/test equipment to provide enhanced capability for fault detection and isolation. Dynamic diagnostics direct fault isolation and troubleshooting based on results returned from the product rather than inputs received from the maintainer. With this type of diagnostics, there are no predefined paths in the troubleshooting data and the paths are generally model-based. This is an external software functionality.

#### A.6.4.4.4 Wire/Fluid system tracing.

This is the capability to select wires, fluid, pneumatic or (HVAC) line in a diagram or schematic and have continuity highlighted thru the circuit or schematic. See FIGURE A-18 and FIGURE A-19.

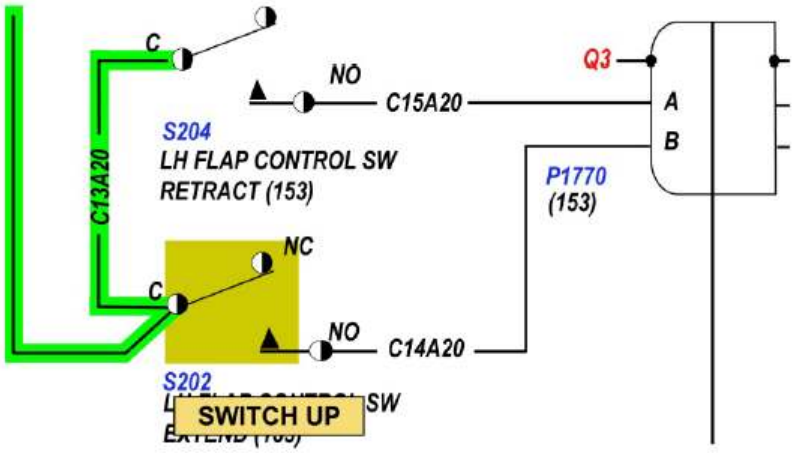


**FIGURE A-18. Wire tracing, switch up.**

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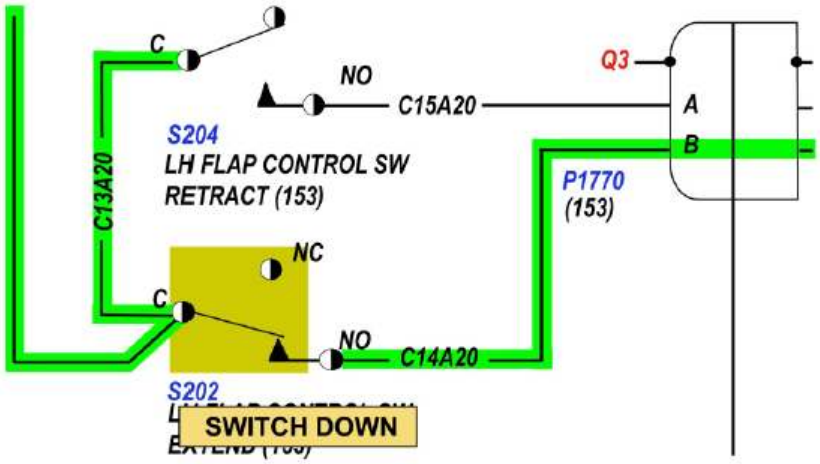


**FIGURE A-19. Wire tracing, switch down.**

### A.6.4.4.5 System simulation.

This functionality includes the capability to represent the behavior or characteristics of the system function/ malfunction to determine or reenact the problem. System simulation allows the user to introduce stimulus (such as pressure, valve positions, temperatures, voltages, sensor inputs, switch positions) and present the results in a manner that models the system behaviors. These capabilities can be used to model hydraulic, fuel, pneumatic and other systems. This is an external software functionality. See FIGURE A-20.

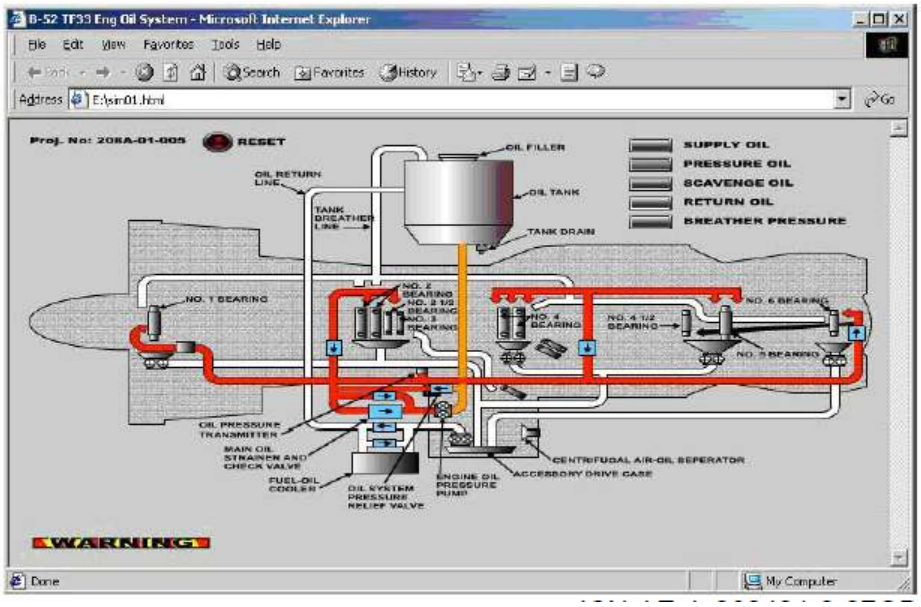


**FIGURE A-20. System simulation.**

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### A.6.4.4.6 Prognostics.

Prognostics are the prediction of component degradation or impending failure, which will allow maintenance personnel to replace components based on their actual condition. The goal is autonomic logistics, which uses electronic information collected from the Product system to determine, plan and perform preventive maintenance. This is an external software functionality. This functionality requires the inclusion of the retrieval, transmittal and maintenance data collection functionalities.

## A.6.4.5 Functionality definitions - external processes.

### A.6.4.5.1 Parts ordering.

Parts ordering This functionality enables parts ordering from within the IETM presentation and is integrated with the supply system. This is an external software functionality. This functionality requires inclusion of the transmittal functionality. This functionality would not circumvent the supply system.

### A.6.4.5.2 Deficiency / improvement report transmittal.

It provides a method for users to transmit errors and recommended changes to the data and/or equipment. This is an external software functionality. This functionality requires inclusion of the transmittal functionality.

The complexity of this will be determined by the level of integration with the deficiency reporting system and the type of reporting structure:

- Paper trail reporting
- Electronic reporting

### A.6.4.5.3 Maintenance data collection.

This functionality provides for the capability to capture and transmit configuration change data (ie, removed and installed part number information), tasks authorized, tasks performed, results of that work, other observed problem, etc. This is an external software functionality. This functionality requires inclusion of the Transmittal functionality.

This update can feed an external data repository or maintenance application. The state table or audit trail can be used to accumulate information for maintenance data collection.

### A.6.4.5.4 Operator debriefing.

An interface with a system or the product itself, to identify and report observed behaviors of the product. This is an external software functionality. This functionality requires inclusion of the Transmittal functionality.

The actual debriefing activity can be carried out by a human operator or the product itself.

### A.6.4.5.5 Resource scheduling.

Interface with a system designed to assign tasks, personnel and items related to maintenance of the Product. This functionality requires inclusion of the transmittal and retrieval functionalities.

Resource scheduling can be coordinated among multiple human or software systems. These systems act against a set of constraints to minimize conflicts. Automatic resource scheduling is characteristic of a complex autonomic logistics system.

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Example - An engine maintainer’s planned maintenance activity can be modified from "test and repair" to "remove and replace" based on reliability data, mission availability requirements and spares availability data from a ship within range.

### A.6.4.5.6 Knowledge management.

This is an interface with a system that models an organization's knowledge assets and environment to enhance its ability to deliver on its mission. This is an external software functionality. This functionality requires inclusion of the transmittal and retrieval functionalities.

Knowledge management provides for information tracking, access and synthesis in coordination with organizational culture, values and guidance. Additionally, a knowledge management system can indicate changes to maintenance actions based on given conditions and lessons learned. For example: A particular system failure can be cause by one of three things, usually it is caused by an electrical failure. However, industry conditions, a clogged air filter is the most likely cause of failure. A knowledge management system will adjust the order of procedures accordingly, so that the air filter is checked first in a dusty environment.

Example - The technical maintenance data gathered thru audit logs can suggest that a restructuring of the logistics footprint is needed to maximize efficiency.

### A.6.4.5.7 Other external applications.

The other external applications functionality is provided so that customers can indicate specific external applications that are required as part of the IETM or data functionality. This is an external software functionality.

## A.6.4.6 Functionality definitions - graphics functionality.

### A.6.4.6.1 Pan, zoom, expand, rotate, magnify.

Controls are provided to perform pan, zoom, expand, rotate and magnify on graphic components. This is an IETM viewer function. Additional functionality can include spyglass view, text search, graphics and window re-sizing.

Consideration should be given to the quality or limitations of the source data. See FIGURE A-21.

**FIGURE A-21. Pan, zoom, expand, rotate, magnify.**

Example - This depicts icons that might be used for the identified function.

### A.6.4.6.2 Assembly/disassembly.

Graphical figures that can be used to allow virtual assembly, disassembly, removal and installation of parts of the system. These can be implemented thru linked drawings or thru manipulation of modeled vector graphics. See FIGURE A-22.



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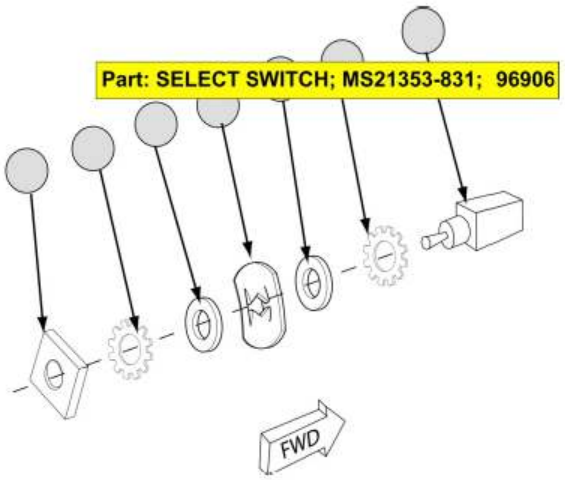


**FIGURE A-22. Assembly/disassembly.**

### A.6.4.6.3 Locator graphics.

Locator graphics show where a component is located relative to other components. See FIGURE A-23.

**FIGURE A-23. Locator graphic.**

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#### A.6.4.6.4 3D modeling.

This is a modeling of the system using 3D, solid object graphical figures.

3D models can allow virtual assembly, disassembly, removal and installation of parts of the system using animation, simulation and/or virtual reality concepts. Levels of capability can include fly thru (navigation thru a 3D model) type viewing. The degree of simulation, animation and virtual reality concepts can dramatically affect cost.

### A.6.4.7 Functionality definitions - linking.

#### A.6.4.7.1 External references.

External references are links to data outside the current data module or view.
This includes links to supporting technical data that is helpful to the user, but not part of the traditional technical publication. Examples include general publications, part and process publications, commodity books, commercial publications, reference material, training data, engineering drawings.

Consideration shall be given for bidirectional links. A bi-directional link returns the user to the original referenced link after completing the action or procedure.

For example, a link from a data module to a publication module containing a related procedure is an external link. It should be noted that references to some data can require network connectivity.

#### A.6.4.7.2 Internal references.

Internal references are links to another place in the same data module or view.

Examples - Links to figures and tables.

#### A.6.4.7.3 Hot reference.

This is the capability to display additional content such as acronyms, tool tips, etc.

Example - A "mouse over" of the acronym IETM would generate popup text box with "IETM."

#### A.6.4.7.4 Link to separate parts data.

Linkage from a maintenance task or narrative can be provided to a separate parts display (illustrated parts breakdown/catalog) in the current or separate window.

This external link type links the user to a specific point within the parts data.

#### A.6.4.7.5 Table of contents, lists of figures, tables and photos.

A categorized items list with links to the corresponding data within the technical publication.

#### A.6.4.7.6 Hot spotting.

Graphics with links to other relevant information.
Links can be to related text such as procedural information, IPD data; or to related graphics such as breakdown illustrations showing greater detail, next higher assembly, etc.

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Check the source to verify that this is the current version before use.

Example - The figure below shows a hot reference from a graphic to related text. See FIGURE A-24.

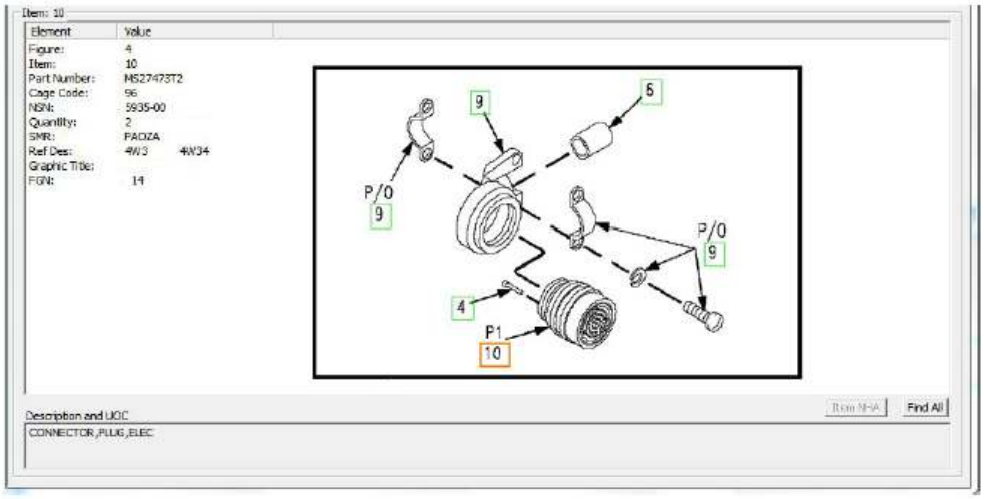


**FIGURE A-24. Hot spotting.**

## A.6.4.8 Functionality definitions - navigation and tracking.

### A.6.4.8.1 Next and previous.

"Next" is a navigational function that, when invoked, displays the subsequent data. "Previous" is a navigational function that, when invoked, displays the preceding data. This is an IETM viewer functionality.

"Next" can be a re-navigation of previously viewed data with no impact on state information (browse); or a progression to the next data in sequence. "Previous" can be a re-navigation of previously viewed data with no impact on state information (browse); or a reversal of steps where state table information is modified accordingly. "Next" and "previous" will normally not be implemented independent of each other. These functionalities, particularly when implemented without the process data module using a state table, are sometimes referred to as "forward" or "back".

### A.6.4.8.2 Return (chronological).

This is a navigational function that allows a user to get back to the link source from the link destination. This is an IETM viewer functionality.

This functionality could be implemented in such a way to allow a user to navigate back to the start of one of several branches or sub-branches.

### A.6.4.8.3 History of traversed links

A navigational feature that tracks and lists each location (link) a user sees along the navigational path thru an IETM. This is an IETM viewer functionality.

In many cases, the user has the ability to bring the list up and use each location (link) in the history list, as a link back to a point in the path. It provides the user a way to move forward and backward along previously traversed links and a quick way to return to a home document. This feature is useful when flipping back and forth between several data types or components of the unit under maintenance.

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Check the source to verify that this is the current version before use.

Example - The user opens the IETM to the front matter:

**a.** Follows the link to the section on the landing gear

**b.** Follows a reference link to an adjustment procedure.

**c.** The history of traversed links will be discretely listed as C, B, A.

### A.6.4.8.4 User creation of bookmarks.

This is a navigational feature that allows the user to flag certain locations for later access. It allows the user to build their own index of links to specific locations in the data. This is an IETM viewer functionality.

Associated advanced features include login-specific bookmarks, so different people using the same presentation device have their own unique set of bookmarks. This is a digital means of implementing the dog-eared page or the paperclip used in paper publications.

Example - The maintenance crew member that generally does the preventive maintenance checks and services might have the bookmarks to those tasks. Since that crewmember might also do the rotor tension adjustment, he would also have a bookmark to that task.

### A.6.4.8.5 System/subsystem navigation.

This is a navigational feature allowing the user to follow a top-down path thru the system breakdown structure. The user follows a physical or functional breakdown to the next lower assembly and then to the next lower assembly from that. This is an IETM viewer functionality.

Example - A helicopter mechanic might begin the navigation of an IETM at the helicopter level. Next step would be to go down to the airframe. From the airframe, the mechanic might pick the cockpit. Next subsystem might be the Pilots seat and the final topic might be the forward/rearward adjustment.

### A.6.4.8.6 Restore initial navigation view.

This is a function that returns all navigational and other frames to their initial view. This is an IETM viewer functionality.

In some IETM implementations, a user will have the capability to hide navigation or other frames of the display. The restore initial navigation view functionality returns all hidden frames to their default positions. The ability to initiate the restore initial navigation view function should always be available to the user.

### A.6.4.8.7 Audit trail.

This functionality provides the capability to capture all user and IETM interaction including dialog interaction entries. This is an IETM viewer functionality. See FIGURE A-25.

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Check the source to verify that this is the current version before use.

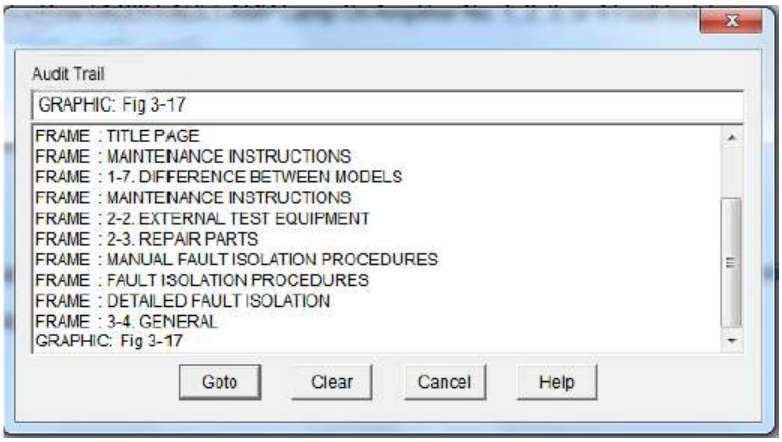


**FIGURE A-25. Audit trail #1.**

### A.6.4.8.8 Graphical navigation.

Graphical navigation This is a navigation of the data thru graphical representation of the product and its components. (The use of hot spotting is a prerequisite).

This can be a type of system/subsystem navigation.

Example - From a graphical overview of the aircraft, the user selects a wing. A graphical overview of the wing is presented. The user then selects the flaps. A graphical overview of the flaps is presented. The user selects the actuator. Information on the actuator is presented.

### A.6.4.8.9 Dialog-driven interaction.

A navigational feature used with the process data module that allows the user to directly feed information to the IETM environment. The IETM would cue the user to input specific data. The feature implies that at some point the IETM would then respond to the information entered.

Example - During a troubleshooting procedure, the IETM would open a dialog box that states "Enter the voltage reading at TP 5". If the user types in 5 (a nominal value), the IETM jumps to the next step in the procedure. Alternately, if the user enters 0 (a fault level reading for a short) the IETM might jump to a remove and replace procedure.

### A.6.4.8.10 Voice-activated commands.

A feature that enables the user to navigate thru the IETM by predetermined voice commands. This is an IETM viewer functionality.

The complexity of this feature depends upon the extent of voice tagging in the IETM and the quality of the voice recognition software.

### A.6.4.8.11 Search - full text.

This is a navigational feature that allows the user to search an IETM for any words or phrases. This feature does not depend upon the predefinition of key words. This is an IETM viewer functionality.

Example - In searching for "IFF" the user can find "IFF", "difference", "TIFF", etc, depending on the search criteria.

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### A.6.4.8.12 Search - sser defined boolean

A search feature that permits the logical association of terms by use of the Boolean operators (such as AND, OR, etc) to narrow the results of the search. This is an IETM viewer functionality.

User defined Boolean search functionality is a possible extension of a full text search.

### A.6.4.8.13 Search - across multiple databases/files.

A search feature that allows the user to look for words or phrases in more than one database. This is an IETM viewer functionality.

### A.6.4.8.14 Search - context.

A feature that allows the user to search for words or phrases within predefined subsets of an IETM or data sources. This is an IETM viewer functionality.

It can be necessary to identify the predefined subsets by use of markup. See FIGURE A-26.

**FIGURE A-26. Search.**

### A.6.4.8.15 Search - keyword.

A navigational feature that allows the user to search an IETM for occurrences of a specific word. System specific "key words" are predefined with links to their location in the data. This is an IETM viewer functionality.

A key word search provides benefits similar to an index in a paper document. It can also use other search functionalities such as Boolean, search across multiple data bases, etc.

Example - A user might search for the term "IFF". The Key word search would locate each pre-identified occurrence of the term in the data. This will find all occurrences of IFF that have been predefined as a key word, but will not find all occurrences of the letters IFF.


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Check the source to verify that this is the current version before use.

### A.6.4.8.16 Filter content per applicability.

Remove or de-emphasize content in the display based on information about the product of interest to the user."

The user has provided input that they are working on a particular aircraft ("serial number 001") with certain conditions present ("POST Service Bulletin 23- 1100"). Only content relevant to aircraft "001" with SB 23-100 implemented is displayed to the user. Alternatively content not relevant is de-emphasized in some manner (for example, is shown in a lighter colored text)."

### A.6.4.8.17 Simultaneous display of multiple content objects.

A functionality that establishes a relationship between content objects (text, tables, graphics, etc) requiring simultaneous display.

The text and its associated supporting graphic shall be displayed concurrently to provide complete information to the user.

### A.6.4.8.18 Tear off window.

This is the capability to capture an existing pane/screen image and then allow the user to navigate forward, while retaining the captured image for later reference. This is an IETM viewer functionality. See FIGURE A-27.

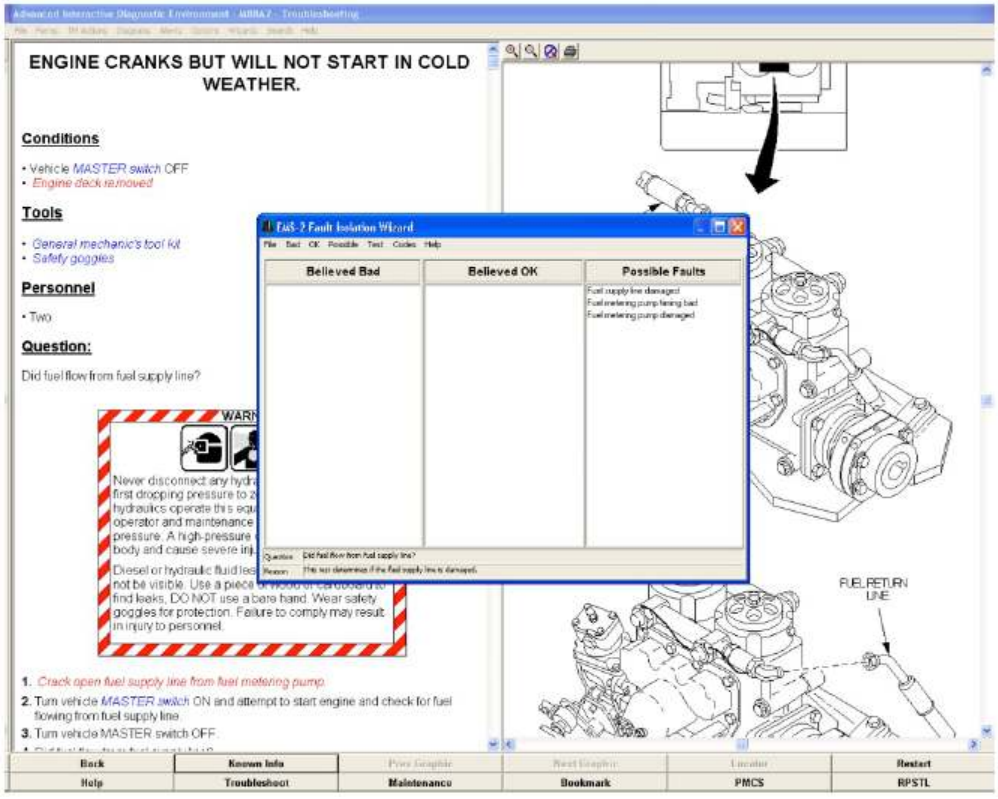


**FIGURE A-27. Tear off window.**



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Check the source to verify that this is the current version before use.

### A.6.4.9 Functionality definitions - printing.

#### A.6.4.9.1 Print screen.

This is the capability to print only the screen currently being viewed. This is an IETM viewer functionality. Beyond the printed technical data, considerations should be given to time/date stamps, destruction notices, destruction dates, destruction requirements, security and derivative classification markings, associated warnings and cautions, etc.

#### A.6.4.9.2 Data module specific printing.

This is the capability to print a single data module in its entirety. This is an IETM viewer functionality. Beyond the printed technical data, considerations should be given to time/date stamps, classified security and derivative classification marks, destruction notices, destruction dates and destruction requirements. Print linked data.

#### A.6.4.9.3 Print linked data.

This capability to print any linked data on a given task/location. This should be limited to one level of linking. Traversing lower than one layer greatly increases the complexity. Beyond the printed technical data, considerations should be given to time/date stamps, persistent warnings and cautions, classified security and derivative classification marks, destruction notices, destruction dates and destruction requirements.

#### A.6.4.9.4 Fully formatted/ book version.

This is the capability to print a complete publication (Publication) from individual data modules. This implies the capability to produce the output in compliance with a predefined format for printed publications.

Beyond the printed technical data, considerations should be given to time/date stamps, classified security and derivative classification marks, destruction notices, destruction dates and destruction requirements. When the document exists as an IETM, in addition to the cost incurred, this will sub-optimize both the IETM and the printed technical publication.

### A.6.4.10 Functionality definitions - special content.

#### A.6.4.10.1 Acknowledgement of warnings and cautions.

This is the capability of the user to acknowledge warnings, cautions and notes in the technical data.

Warnings and cautions shall be readily identified and can require specific operator acknowledgment prior to proceeding with the presented data.

#### A.6.4.10.2 Emergency procedures.

A functionality that will call up and display all emergency procedures associated with a system, subsystem or sub-subsystem.

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Check the source to verify that this is the current version before use.

### A.6.4.10.3 Photos.

This is the capability to include photographic images in the technical data.

Photos can be included to show a specific visual representation of actual systems. Presentation standards (such as resolution, the use of color, file size and file format) shall be addressed as well as content and context of the photo.

### A.6.4.10.4 Audio.

This is the capability to provide sound within the technical data.
Examples - operator warnings and cautions, presentation clips or complete audio representation of the data.
Presentation standards shall be addressed as well as content, duration and context of the audio. Consideration shall be given to the anticipated environment to determine the usability of audio.

### A.6.4.10.5 Motion video.

This is the capability to provide video within the technical data.
Motion video should be used as a supplement to the procedural text. Presentation standards shall be addressed as well as content, duration and context of the video.

### A.6.4.10.6 Animation.

This is the capability to provide animated graphics within the technical data.
Animation should be used as a supplement to the procedural text. Animation can be included to show a variety of system functions from theory of operation (hydraulic flow) to maintenance procedures (how to access a specific part). Presentation standards shall be addressed as well as content, duration and context of the animation.

### A.6.4.10.7 Content sensitive help (Technical data help

Help information based on the data being presented or the tasks being performed.
Examples - Lists of acronyms, lists of abbreviations and definitions of acronyms and abbreviations. The type of help pertains to the particular subject matter of the data, such as the specific product. Content sensitive help can be delivered thru pop-ups, dedicated help screens or pull down menus.

### A.6.4.10.8 Context sensitive help (viewer help.

Help information for the IETM operation including the features and functions of the IETM viewer. This is an IETM viewer functionality.

Example - In many systems, right-clicking via the mouse or hovering the cursor over a particular graphic or menu item will cause a "tool tip" to pop up, providing help or a description for the specified feature. See FIGURE A-28.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:03Z
Check the source to verify that this is the current version before use.

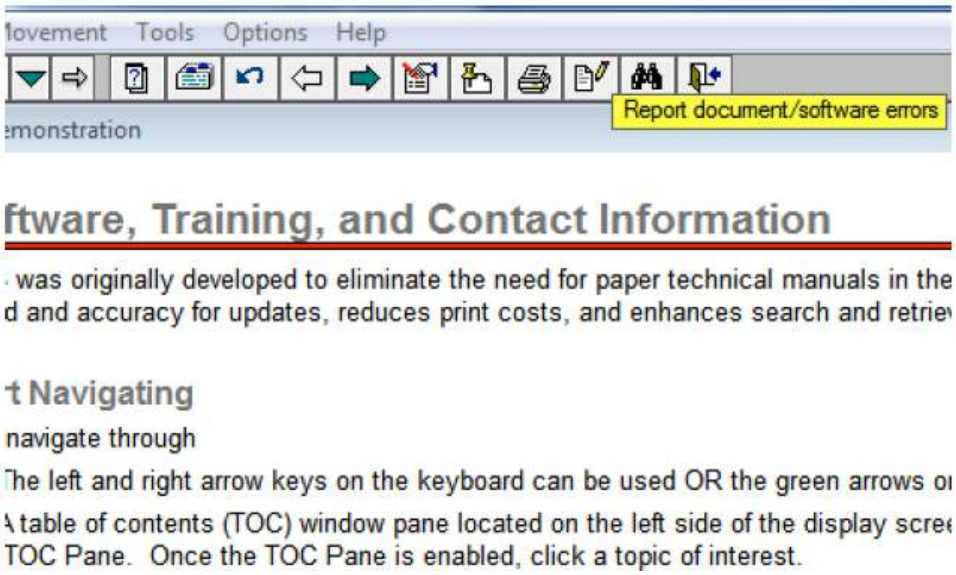


**FIGURE A-28. Context sensitive help.**

### A.6.4.10.9 User training.

This is the capability to integrate or link maintenance and/or operational training data with the IETM. This is both a data and IETM viewer functionality.

User training complexity can range from online access during IETM run time by linking to a CBT database to run complete training lessons.

Access of the data can be thru a link to an external module or integrated with the IETM. Advanced capabilities can be to monitor usage and training access for the purpose of tracking user competency. Specific training courseware and or actions are not necessarily part of the data.

### A.6.4.11 Functionality definitions - updates.

### A.6.4.11.1 Passive change indications and markings.

This is the capability to identify changes in the technical data thru a list of changes only and not thru visible marks within the actual technical data.

Examples - List of Changes, Highlights Page or summary of changed data. Display and tracking of change information can drive costs.

### A.6.4.11.2 Active change indications and markings.

This is the capability to discretely identify and visibly mark each change in the data.

Complexity can be affected by the methods of display, change identification and when they are removed or suppressed. The IETM can also include passive change markings. Display and tracking of change information can drive costs.

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### A.6.4.11.3 Full change.

The capability to provide a complete replacement of the data previously distributed.

A full change can contain detailed change markings (or a change summary).

### A.6.4.11.4 Block cycle and urgent changes.

The capability to provide regularly scheduled updates to the technical data. Changes from all sources are consolidated and issued at regularly scheduled intervals. Urgent changes are interim updates between scheduled block cycle updates.

If block cycle updates are contracted, a company is responsible for providing both regularly scheduled updates to the data as well as urgent off-cycle updates as needed.

### A.6.4.11.5 Near real time updates.

The capability to quickly provide data updates to users rather than depending on a predetermined schedule.

This functionality requires the inclusion of the Network distribution and Network connectivity functionalities.

## A.6.4.12 Functionality definitions - user operation model.

### A.6.4.12.1 Web browser viewable.

This is technical data that can primarily be viewed thru a COTS Web browser. This is an infrastructure functionality.

The functionality selected in this matrix will determine the level of complexity and cost of implementing a Web browser viewable application. This can be accomplished directly thru the Web browser (for example, Internet Explorer™) or thru helper applications (for example, Adobe Acrobat ™) or plug-ins (for example, IsoView ™). Consideration will be required for a specific Web browser. Different implementations include remote access to a Web server, opening static HTML pages locally and Web server emulation on the client viewer.

### A.6.4.12.2 Stand-alone mode

This is the capability to access the IETM via the hard drive on the stand alone device after uploading data from a network or thru a transfer media such as CD-ROM or DVD.

Consideration should be given for the update capabilities of this operating method. When in stand-alone mode, full update capability entails the update of the complete data after uploading from a network or thru a transfer media such as CD-ROM or DVD. This method provides the lowest cost impact. When in stand-alone mode, a partial update capability entails the update of only the changed data from the previous release after uploading from a network or thru a transfer media such as CD-ROM or DVD.

### A.6.4.12.3 Network connectivity.

This is the capability to access to the IETM via a network infrastructure. The data can be downloaded to or viewed on the client device. The data changes are installed on the host server and updates are transmitted via the network. This is infrastructure functionality.

Data changes are/can be incorporated as complete or partial updates.

Selection of functionality using the functionality matrix — Decide on which functionality is required in the technical publications.

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## A.6.5 IETM functionality matrix.

The IETM functionality matrix is a tool to define the intended use and capabilities of a equipment system data. This matrix provides a standard format for documenting the functional needs of a project. Section A.6.4 provides standard definitions for each functionality. These requirements are applicable for all maintenance levels through overhaul (depot), including DMWRs/NMWRs. The definitions found in section A.6.4 are included to aid decision making processes, and a completed matrix shall reflect final decisions made after coordination with the MATDEV.

### A.6.5.1 Use of the matrix.

The vertical axis of the matrix lists all the possible functionalities of IETM technical publications. The rows containing functionalities required by the Army are pre-populated with an "R" (indicating Required) in the Requirement column. The horizontal axis includes the information sets that can contain the desired functionalities. The matrix provided in this appendix has been tailored so the listed information sets match typical Army information sets. The functionality matrix is available at https://www.ldac.army.mil/mil40051/S1000D.cfm. The matrix can be tailored to include additional required information as required.

### A.6.5.2 Matrix input.

There are a number of possible ways to document input to the matrix:

**a.** Binary. A simple yes or no (or check mark) relative to the requirement for each functionality may be inserted.

**b.** Degree of need. The following input may be inserted: R = Required functionality, N = Nice to have functionality, or P = Prohibited for each functionality.

**c.** Color coding. A project may develop a color coding scheme (or other code) for input to express additional functionality requirements (for example, whether or not a specific functionality is needed, but only in certain unique uses or implementations).

**d.** Other. Inputs may be creative; other guidance for completing the matrix may be inserted, as needed.


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**TABLE A-XXI. IETM Functionality Matrix.**

| Functionality | Complexity | Requirement | All Data Sets | Operator / Operator & Field (-10/-12/-12&P) | Field (-20/-20&P) | Field (-20P) | NMWR/DMWR | SUM/SAM | DMWRs (Demilitarization) | Data Sheets | Aviation Field (-20/-20&P) | Aircraft PMD/PMS | Aircraft PMI Checklist | BDAR | Destruction of Equipment to Prevent Enemy Use | Lubrication Order | Preventive Maintenance Checklist | General Maintenance | Hand Receipt | Technical Bulletin |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Functionality Category: Access | | | | | | | | | | | | | | | | | | | | |
| Login | 2 | | A | | | | | | | | | | | | | | | | | |
| Suspend and Restart | 1 | | A | | | | | | | | | | | | | | | | | |
| Exit | 1 | R | A | | | | | | | | | | | | | | | | | |
| Functionality Category: Annotation | | | | | | | | | | | | | | | | | | | | |
| Action complete indicator (check box) | 1 | | | | | | | | | | | | | | | | | | | |
| Local data annotation | 2 | | A | | | | | | | | | | | | | | | | | |
| Personal annotation | 1 | | A | | | | | | | | | | | | | | | | | |
| Redlining text | 3 | | A | | | | | | | | | | | | | | | | | |
| Redlining graphics | 3 | | A | | | | | | | | | | | | | | | | | |



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**TABLE A-XXI. IETM Functionality Matrix. (continued)**

| Functionality | Complexity | Requirement | All Data Sets | Operator / Operator & Field (-10/-12/-12&P) | Field (-20/-20&P) | Field (-20P) | NMWR/DMWR | SUM/SAM | DMWRs (Demilitarization) | Data Sheets | Aviation Field (-20/-20&P) | Aircraft PMD/PMS | Aircraft PMI Checklist | BDAR | Destruction of Equipment to Prevent Enemy Use | Lubrication Order | Preventive Maintenance Checklist | General Maintenance | Hand Receipt | Technical Bulletin |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Functionality Category: Deliver and Distribution | | | | | | | | | | | | | | | | | | | | |
| Printed publication | 5 | | A | | | | | | | | | | | | | | | | | |
| Physical media | 1 | | A | | | | | | | | | | | | | | | | | |
| Network distribution | 2 | | A | | | | | | | | | | | | | | | | | |
| Functionality Category: Diagnostics and Prognostics | | | | | | | | | | | | | | | | | | | | |
| Diagnostics - User determined entry to data | 1 | | | | | | | | | | | | | | | | | | | |
| Diagnostics - Software driven entry to data | 2 | | | | | | | | | | | | | | | | | | | |
| Dynamic diagnostics | 5 | | | | | | | | | | | | | | | | | | | |



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**TABLE A-XXI. IETM Functionality Matrix. (continued)**

| Functionality | Complexity | Requirement | All Data Sets | Operator / Operator & Field (-10/-12/-12&P) | Field (-20/-20&P) | Field (-20P) | NMWR/DMWR | SUM/SAM | DMWRs (Demilitarization) | Data Sheets | Aviation Field (-20/-20&P) | Aircraft PMD/PMS | Aircraft PMI Checklist | BDAR | Destruction of Equipment to Prevent Enemy Use | Lubrication Order | Preventive Maintenance Checklist | General Maintenance | Hand Receipt | Technical Bulletin |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Wire/Fluid system tracing | 4 | | | | | | | | | | | | | | | | | | | |
| System simulation | 4 | | | | | | | | | | | | | | | | | | | |
| Prognostics | 5 | | | | | | | | | | | | | | | | | | | |
| Functionality Category: External Processes | | | | | | | | | | | | | | | | | | | | |
| Transmittal | 3 | | | | | | | | | | | | | | | | | | | |
| Retrieval | 2 | | | | | | | | | | | | | | | | | | | |
| Parts ordering | 3 | R | | | | | | | | | | | | | | | | | | |
| Deficiency/ Improvement report transmittal | 3 | R | A | | | | | | | | | | | | | | | | | |
| Maintenance data collection | 3 | | | | | | | | | | | | | | | | | | | |
| Operator debriefing | 3 | | | | | | | | | | | | | | | | | | | |



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**TABLE A-XXI. IETM Functionality Matrix. (continued)**

| Functionality | Complexity | Requirement | All Data Sets | Operator / Operator & Field (-10/-12/-12&P) | Field (-20/-20&P) | Field (-20P) | NMWR/DMWR | SUM/SAM | DMWRs (Demilitarization) | Data Sheets | Aviation Field (-20/-20&P) | Aircraft PMD/PMS | Aircraft PMI Checklist | BDAR | Destruction of Equipment to Prevent Enemy Use | Lubrication Order | Preventive Maintenance Checklist | General Maintenance | Hand Receipt | Technical Bulletin |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Resource scheduling | 3 | | | | | | | | | | | | | | | | | | | |
| Knowledge management | 5 | | | | | | | | | | | | | | | | | | | |
| Functionality Category: Graphics Functionality | | | | | | | | | | | | | | | | | | | | |
| Pan, zoom, ex-pand, magnify | 1 | | A | | | | | | | | | | | | | | | | | |
| Assembly/ Disassembly | 2 | | | | | | | | | | | | | | | | | | | |
| Locator graphics | 1 | | | | | | | | | | | | | | | | | | | |
| 3D modeling | 4 | | | | | | | | | | | | | | | | | | | |
| Functionality Category: Linking | | | | | | | | | | | | | | | | | | | | |
| External reference | 2 | | A | | | | | | | | | | | | | | | | | |



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**TABLE A-XXI. IETM Functionality Matrix. (continued)**

| Functionality | Complexity | Requirement | All Data Sets | Operator / Operator & Field (-10/-12/-12&P) | Field (-20/-20&P) | Field (-20P) | NMWR/DMWR | SUM/SAM | DMWRs (Demilitarization) | Data Sheets | Aviation Field (-20/-20&P) | Aircraft PMD/PMS | Aircraft PMI Checklist | BDAR | Destruction of Equipment to Prevent Enemy Use | Lubrication Order | Preventive Maintenance Checklist | General Maintenance | Hand Receipt | Technical Bulletin |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Internal references | 1 | | A | | | | | | | | | | | | | | | | | |
| Hot reference | 2 | | A | | | | | | | | | | | | | | | | | |
| Link to separate parts data | 2 | | A | | | | | | | | | | | | | | | | | |
| TOC, list of figures, tables and photos | 1 | R | A | | | | | | | | | | | | | | | | | |
| Hot spotting | 3 | | A | | | | | | | | | | | | | | | | | |
| Functionality Category: Navigation and Tracking | | | | | | | | | | | | | | | | | | | | |
| Next and previous | 1 | R | A | | | | | | | | | | | | | | | | | |
| Return (Chronological) | 1 | | A | | | | | | | | | | | | | | | | | |
| History of traversed link | 1 | R | A | | | | | | | | | | | | | | | | | |



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**TABLE A-XXI. IETM Functionality Matrix. (continued)**

| Functionality | Complexity | Requirement | All Data Sets | Operator / Operator & Field (-10/-12/-12&P) | Field (-20/-20&P) | Field (-20P) | NMWR/DMWR | SUM/SAM | DMWRs (Demilitarization) | Data Sheets | Aviation Field (-20/-20&P) | Aircraft PMD/PMS | Aircraft PMI Checklist | BDAR | Destruction of Equipment to Prevent Enemy Use | Lubrication Order | Preventive Maintenance Checklist | General Maintenance | Hand Receipt | Technical Bulletin |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| User creation of bookmarks | 1 | | A | | | | | | | | | | | | | | | | | |
| System/Subsys-tem navigation | 1 | | A | | | | | | | | | | | | | | | | | |
| Restore initial navigation view | 1 | R | A | | | | | | | | | | | | | | | | | |
| Audit trail | 2 | | A | | | | | | | | | | | | | | | | | |
| Graphical navigation | 2 | | | | | | | | | | | | | | | | | | | |
| Dialog-driven interaction | 3 | | | | | | | | | | | | | | | | | | | |
| Voice-Activated commands | 3 | | A | | | | | | | | | | | | | | | | | |
| Search - Full text | 1 | R | A | | | | | | | | | | | | | | | | | |
| Search - User de-fined Boolean | 1 | | A | | | | | | | | | | | | | | | | | |



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**TABLE A-XXI. IETM Functionality Matrix. (continued)**

| Functionality | Complexity | Requirement | All Data Sets | Operator / Operator & Field (-10/-12/-12&P) | Field (-20/-20&P) | Field (-20P) | NMWR/DMWR | SUM/SAM | DMWRs (Demilitarization) | Data Sheets | Aviation Field (-20/-20&P) | Aircraft PMD/PMS | Aircraft PMI Checklist | BDAR | Destruction of Equipment to Prevent Enemy Use | Lubrication Order | Preventive Maintenance Checklist | General Maintenance | Hand Receipt | Technical Bulletin |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Search - Across multiple data-bases/files | 3 | | A | | | | | | | | | | | | | | | | | |
| Search - Context | 2 | | A | | | | | | | | | | | | | | | | | |
| Search - Keyword | 2 | | A | | | | | | | | | | | | | | | | | |
| Filter content per applicability | 1 | | A | | | | | | | | | | | | | | | | | |
| Simultaneous dis-play of multiple content objects | 2 | | A | | | | | | | | | | | | | | | | | |
| Tear off window | 2 | | A | | | | | | | | | | | | | | | | | |
| Functionality Category: Printing | | | | | | | | | | | | | | | | | | | | |
| Print Screen | 1 | R | A | | | | | | | | | | | | | | | | | |
| Data module spe-cific printing | 2 | R | A | | | | | | | | | | | | | | | | | |



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1. Printed operator's manuals are required per AR 25-30.

**TABLE A-XXI. IETM Functionality Matrix. (continued)**

| Functionality | Complexity | Requirement | All Data Sets | Operator / Operator & Field (-10/-12/-12&P) | Field (-20/-20&P) | Field (-20P) | NMWR/DMWR | SUM/SAM | DMWRs (Demilitarization) | Data Sheets | Aviation Field (-20/-20&P) | Aircraft PMD/PMS | Aircraft PMI Checklist | BDAR | Destruction of Equipment to Prevent Enemy Use | Lubrication Order | Preventive Maintenance Checklist | General Maintenance | Hand Receipt | Technical Bulletin |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Print linked data | 2 | | A | | | | | | | | | | | | | | | | | |
| Fully formatted/ book version | 5 | | A | | | R[1] | R[1] | R[1] | | | | | | | | | | | | |
| Functionality Category: Special Content | | | | | | | | | | | | | | | | | | | | |
| Front matter | 1 | | A | | | | | | | | | | | | | | | | | |
| Supporting technical data | 2 | | A | | | | | | | | | | | | | | | | | |
| Alerts | 1 | | A | | | | | | | | | | | | | | | | | |
| Emergency procedures | 2 | | A | | | | | | | | | | | | | | | | | |
| Photos | 1 | | | | | | | | | | | | | | | | | | | |
| Audio | 2 | | | | | | | | | | | | | | | | | | | |
| Motion video | 3 | | | | | | | | | | | | | | | | | | | |
| Animation | 4 | | | | | | | | | | | | | | | | | | | |



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**TABLE A-XXI. IETM Functionality Matrix. (continued)**

| Functionality | Complexity | Requirement | All Data Sets | Operator / Operator & Field (-10/-12/-12&P) | Field (-20/-20&P) | Field (-20P) | NMWR/DMWR | SUM/SAM | DMWRs (Demilitarization) | Data Sheets | Aviation Field (-20/-20&P) | Aircraft PMD/PMS | Aircraft PMI Checklist | BDAR | Destruction of Equipment to Prevent Enemy Use | Lubrication Order | Preventive Maintenance Checklist | General Maintenance | Hand Receipt | Technical Bulletin |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Content sensitive help (Tech data help) | 1 | R | A | | | | | | | | | | | | | | | | | |
| Context sensitive help (Viewer) | 2 | R | A | | | | | | | | | | | | | | | | | |
| User training | 3 | | A | | | | | | | | | | | | | | | | | |
| Functionality Category: Updates | | | | | | | | | | | | | | | | | | | | |
| Passive change indications and markings | 1 | R | A | | | | | | | | | | | | | | | | | |
| Active change indications and markings | 2 | R | A | | | | | | | | | | | | | | | | | |
| Full change | 1 | | A | | | | | | | | | | | | | | | | | |
| Block cycle and urgent changes | 2 | | A | | | | | | | | | | | | | | | | | |



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**TABLE A-XXI. IETM Functionality Matrix. (continued)**

| Functionality | Complexity | Requirement | All Data Sets | Operator / Operator & Field (-10/-12/-12&P) | Field (-20/-20&P) | Field (-20P) | NMWR/DMWR | SUM/SAM | DMWRs (Demilitarization) | Data Sheets | Aviation Field (-20/-20&P) | Aircraft PMD/PMS | Aircraft PMI Checklist | BDAR | Destruction of Equipment to Prevent Enemy Use | Lubrication Order | Preventive Maintenance Checklist | General Maintenance | Hand Receipt | Technical Bulletin |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Near real time updates | 2 | | A | | | | | | | | | | | | | | | | | |
| Functionality Category: User Operation Mode | | | | | | | | | | | | | | | | | | | | |
| Web browser viewable | 3 | | A | | | | | | | | | | | | | | | | | |
| Stand alone mode | 1 | R | A | | | | | | | | | | | | | | | | | |
| Network connectivity | 2 | | A | | | | | | | | | | | | | | | | | |



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# APPENDIX B

# GENERAL INFORMATION, EQUIPMENT DESCRIPTION, AND THEORY OF OPERATION

## B.1 SCOPE.

This appendix establishes the technical content requirements for the preparation of general information, equipment description, and theory of operation data for major equipment systems and their related systems, subsystems, equipment, assemblies, components, Shop Replaceable Units (SRUs), and Line Replaceable Units (LRUs). This appendix is a mandatory part of this standard. The information contained herein is intended for compliance.

**Note**

To be compliant with this standard, publications shall be developed within **<production>** (opening) and **</production>** (closing) root tags.

When preparing information in accordance with this appendix, all references shall be linked using the appropriate tags (**<xref>**, **<extref>**, etc.). Links shall not be typed as text without tagging.

### B.1.1 Content tailoring using Appendix A matrices.

Applicability of some requirements contained in this standard are the responsibility of the materiel developer (MATDEV) (refer to AR 750–1, AR 700–127) and shall be accomplished using matrices referenced in TABLE A-I. The applicability of requirements are also designated by statements including, but not limited to, the following:

**a.** "...as specified by the MATDEV."

**b.** "...as required by the MATDEV."

**c.** "Refer to the MATDEV for..."

**d.** "Contact MATDEV if..."

**e.** "The MATDEV shall determine..."

**f.** "...as needed."

#### B.1.1.1 Front and rear matter.

Front and rear matter requirements for documents discussed in this appendix shall be determined in accordance with TABLE A-II.

## B.2 APPLICABLE DOCUMENTS.

The applicable documents in Section 2 apply to this appendix.

## B.3 ACRONYMS AND DEFINITIONS.

The acronyms and terms defined in Section 3 of this standard apply to this appendix.

## B.4 GENERAL INFORMATION, EQUIPMENT DESCRIPTION, AND THEORY OF OPERATION APPLICABILITY.

Requirements contained in this appendix are applicable to all maintenance levels unless specifically labeled in bold and in parentheses. An explanation of applicable DA maintenance levels is provided in Section 3.

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## B.5 GENERAL INFORMATION, EQUIPMENT DESCRIPTION, AND THEORY OF OPERATION FUNCTIONALITY.

The general information, equipment description, and theory of operation module **<gim>** shall appear as the first module in a technical publication and shall be prepared for major equipment systems, applicable support equipment, as well as any applicable interface equipment. This module shall provide the user with a physical description of equipment and functionally explain how the equipment operates as well as the theory behind its operational design. The general information work package (WP) requirements are similar across most platforms. However, the equipment description and theory of operation WPs shall be developed in narrative or tabular form, or by whatever method is most simple or effective for conveying the specific application. For complex systems, these two WP types may be duplicated to better present information. Descriptive information shall not contain any procedural data or warnings or cautions. When necessary for clarity or improved understanding, illustrations shall be used to support the narrative or tabular information. See TABLE B-I for a list of module WP components in many general equipment publications, and see TABLE B-II for a list of module WP components for relatively specialized publications discussed in this appendix as well as other appendices.

**TABLE B-I. General information, equipment description, and theory of operation WPs.**

| WP | WP Tag | Appendix Section |
|---|---|---|
| General information WP [1] | **<ginfowp>** | B.7.1 |
| Equipment description and data WP [2] | **<descwp>** | B.7.2 |
| Theory of operation WP [2] | **<thrywp>** | B.7.3 |

[1] WP type duplication is prohibited.

[2] WP type frequently duplicated in a module for effective information display.

**TABLE B-II. Other general information and equipment description WPs.**

| WP | WP Tag | Appendix Section |
|---|---|---|
| General Information WPs | | |
| Aircraft preventive maintenance services manual or preventive maintenance daily manual only [2] | **<pms-ginfowp>** | B.7.4 |
| Aircraft phased maintenance inspection manual only [2] | **<pm-ginfowp>** | B.7.5 |
| General information WP [1] (Software Information Modules / Publications) | **<softginfowp>** | M.7.1 |
| General information WP [1] (General Maintenance Manuals) | **<genmaint_ginfowp>** | N.4.1 |
| Unique general information WP [1] (Battle Damage Assessment Reports (BDARs) | **<bdar-geninfowp>** | I.6.2 |
| General information WP [1] (Technical Bulletins) | **<ginfowp>** | Q.5.1 |
| Equipment Description WPs | | |
| Software summary WP (Software Information Modules / Publications) | **<softsumwp>** | M.7.2 |


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**TABLE B-II. Other general information and equipment description WPs. (continued**

| WP | WP Tag | Appendix Section |
|---|---|---|
| Software effectivity WP<br>(Software Information Modules / Publications) | **<softeffectwp>** | M.7.3 |
| Differences between versions WP<br>(Software Information Modules / Publications) | **<softdiffversionwp>** | M.7.4 |
| Equipment description and data WP(s)<br>(Technical Bulletins) | **<descwp>** | Q.5.2 |

[1] WP type duplication is prohibited.

[2] WP type for aircraft publications only.

## B.6 DEVELOPMENT OF GENERAL INFORMATION MODULES.

A general information module **<gim>** shall be prepared and shall contain the following WPs as required by publication type (see A.4.4). See FIGURE B-1 for a diagram of components of a technical publication containing this module along with WPs available within the **<gim>** module.

**a.** General information WP **<ginfowp>** (B.7.1).

**b.** Equipment description and data WP **<descwp>** (B.7.2).

**c.** Theory of operation WP **<thrywp>** (B.7.3).

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**Note**

Tags/output shown in figure(s) below may vary slightly based on version of DTD/stylesheets utilized (see applicable DTD or tag definition list for available tags).

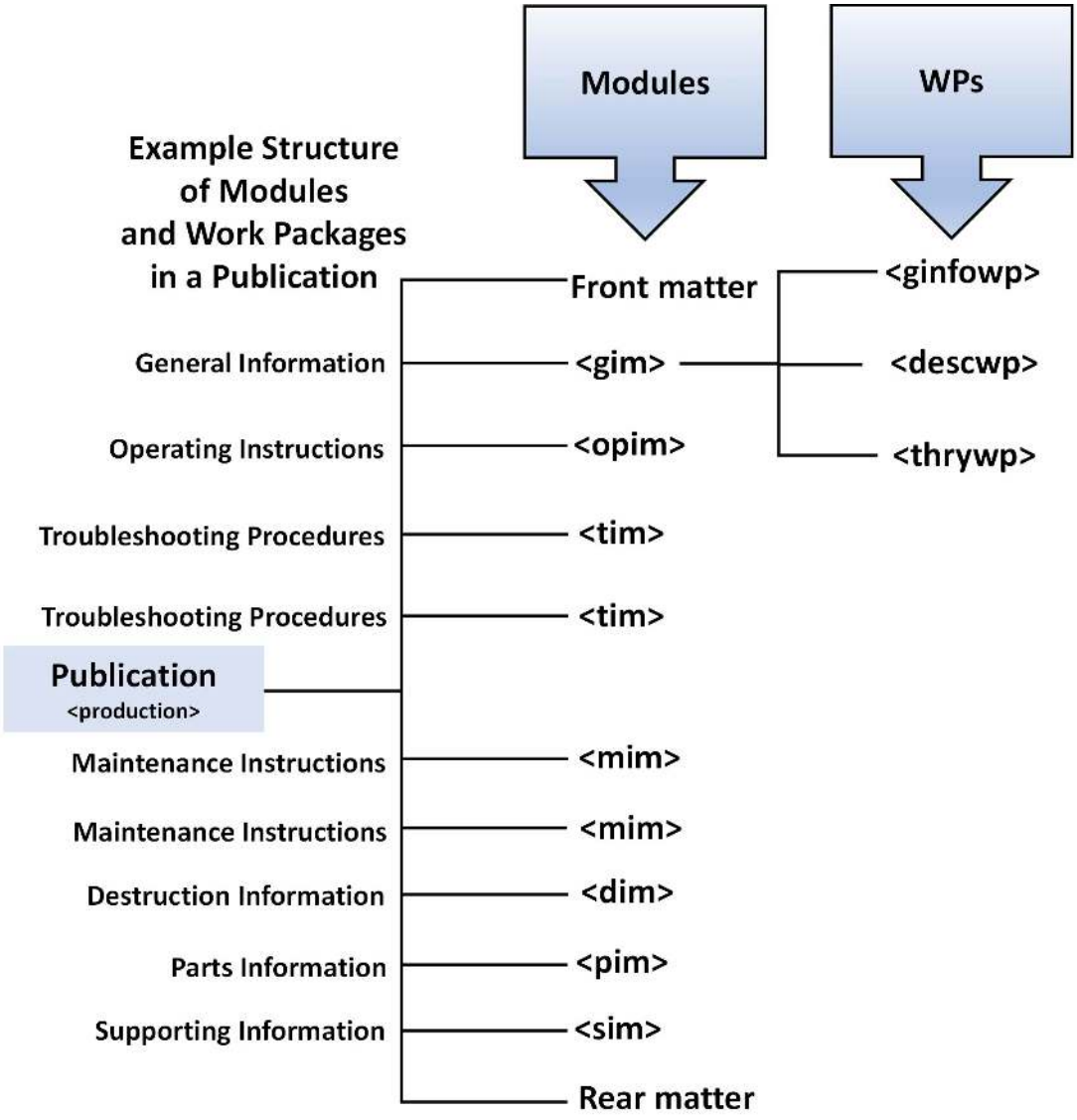


**FIGURE B-1. General information, equipment description and data, and theory of operation WPs modules and WPs.**



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## B.7 DEVELOPMENT OF GENERAL INFORMATION, EQUIPMENT DESCRIPTION, AND THEORY WORK PACKAGES.

General information, equipment description, and theory of operation data shall be prepared as required by publication type (see A.4.4) and divided into the following types of WPs included in a general information **<gim>** module. Nomenclature used to identify the equipment system, major equipment, components, and applicable support and interface equipment shall remain consistent throughout the publication and among all WPs.

### B.7.1 General information work package <ginfowp>.

The general information WP shall be prepared and contain general introductory information about an equipment system. FIGURE B-1 through FIGURE B-5 depict some sections of a general information WP and their associated tags. General information WP components are listed below:

**a.** Work package identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** Initial setup **<initial_setup>** is not required for this WP.

**c.** General information sections discussed in B.7.1.1 through B.7.1.25 as applicable for an equipment system.

#### B.7.1.1 Scope <scope>.

A brief statement shall be prepared to tell what is covered in the technical publication (see FIGURE B-2). As applicable, the following information shall also be included:

**a.** Type of manual.

**b.** Model number(s) and equipment name(s).

**c.** Purpose of equipment.

**d.** Special inclusions in the manual, such as drill procedures or on-vehicle loading plans.

**e.** List of WPs/components not included in technical publication (items marked with a "+" in the content selection matrices in APPENDIX A). Specify whether the item is or is not included in an equipment system.

**f.** Dates of changes for changed WPs.

For example:

"Change 2.....date"

"Change 1.....date"

"Original.....date"

#### B.7.1.2 Maintenance forms, records, and reports <mfrr>

A brief statement (listed below) shall be included based on publication type (see FIGURE A-5).

**Note**

All listed forms may not be required in a technical publication. Review options listed to determine forms needed for a particular publication.

**a.** (A) Army only technical publication. The following statement shall be included:

**"MAINTENANCE FORMS, RECORDS, AND REPORTS**

Department of the Army forms and procedures used for equipment maintenance will be those prescribed by (list as applicable) DA Pam 750-8, The Army Maintenance Management System (TAMMS) Users Manual;


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DA Pam 738-751, Functional Users Manual for the Army Maintenance Management Systems – Aviation (TAMMS-A); or AR 700-138, Army Logistics Readiness and Sustainability."

**b.** (MC) Marines only technical publication. The following statement shall be included:

**"MAINTENANCE FORMS, RECORDS, AND REPORTS**

Maintenance forms and records used by Marine Corps personnel are prescribed by TM 4700-15/1."

**c.** Multi-Service technical publication.

The following statements shall be included only for multi-service technical publications and shall use only applicable services (for example, if the Navy does not use the publication, do not include a statement for that Service):

**"MAINTENANCE FORMS, RECORDS, AND REPORTS**

(A) Department of the Army forms and procedures used for equipment maintenance will be those prescribed by (list as applicable) DA Pam 750-8, The Army Maintenance Management System (TAMMS) Users Manual; DA Pam 738-751, Functional Users Manual for the Army Maintenance Management Systems - Aviation; or AR 700-138, Army Logistics Readiness and Sustainability.

(MC) Maintenance forms and records used by Marine Corps personnel are prescribed by TM 4700-15/1.

(F) Maintenance forms and records used by Air Force personnel are prescribed in AFI 21-101 and the applicable TO 00-20 Series Technical Orders.

(N) Navy users should refer to their service directives to determine applicable maintenance forms and records to be used."

**d.** (A) Army ammunition. The following statement shall be added:

“Accidents involving injury to personnel or damage to materiel will be reported in accordance with AR 385-10, The Army Safety Program. Explosive and ammunition malfunctions will be reported in accordance with AR 75-1, Malfunctions Involving Ammunition and Explosives.”

**e.** When applicable, add references to DA Pam 742-1, Ammunition Surveillance Procedures.

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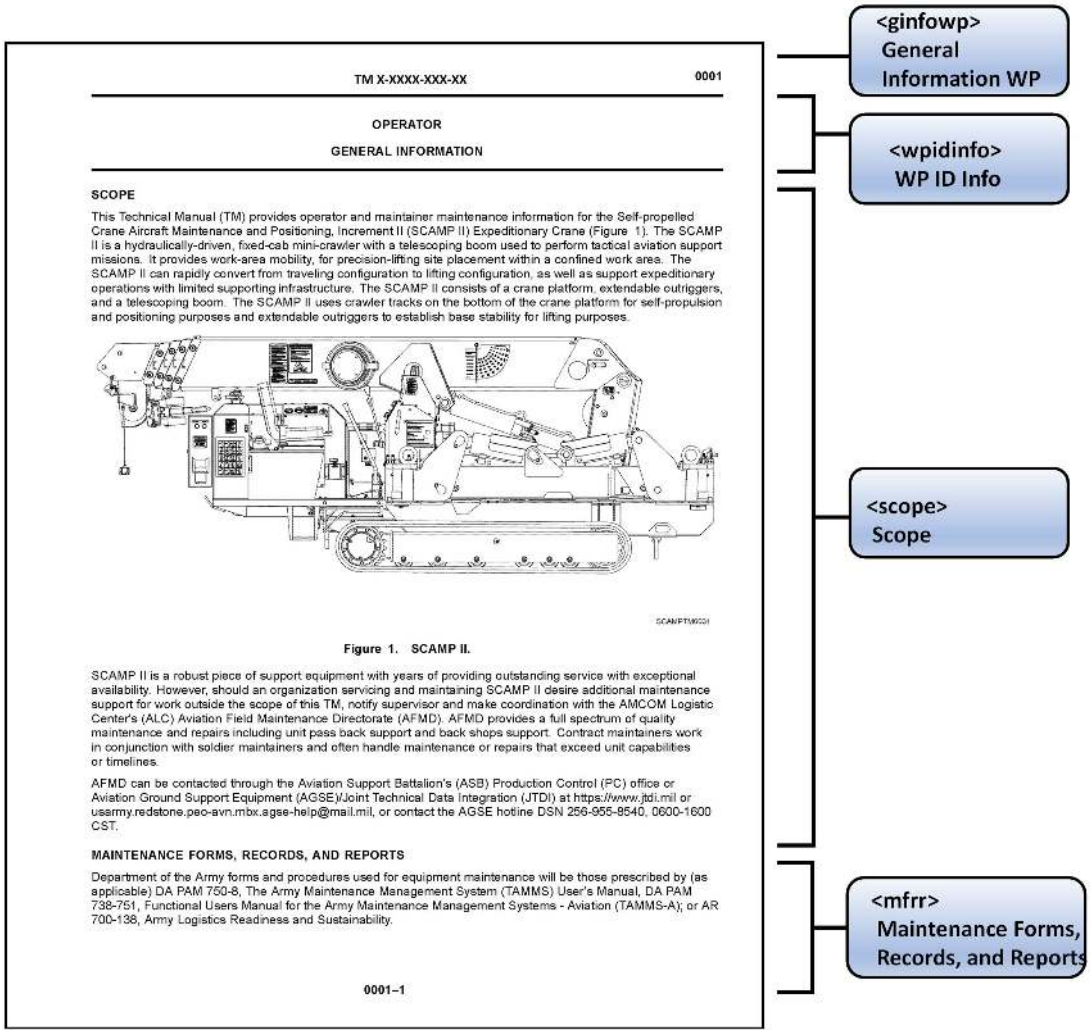


**FIGURE B-2. General information WP (sheet 1 of 4).**

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### B.7.1.3 Reporting equipment improvement recommendations (EIR <eir>.

The following statements in the sections below shall be included (italicized text within parentheses shall be replaced with the appropriate information).

#### B.7.1.3.1 Aviation and Missiles Systems EIR.

**"REPORTING EQUIPMENT IMPROVEMENT RECOMMENDATIONS (EIR**

If your equipment needs improvement, let us know. Send us an EIR. You, the user, are the only one who can tell us what you do not like about your equipment. Let us know why you do not like the design or performance.

All AMCOM (Aviation and Missile Command) Deficiency Reports (DRs), (Warranty, EIR, and Product Quality Deficiency Reports (PQDRs)) must be submitted through the Joint Deficiency Reporting System (JDRS) at https://jdrs.mil/.

If you do not have Internet access, you may submit your information using an SF 368 (PQDR). You can send your SF 368 using email, regular mail, or fax using the addresses/fax numbers specified in DA Pam 738-751, Functional Users Manual for the Army Maintenance Management Systems - Aviation for aviation systems. We will send you a reply."

#### B.7.1.3.2 Non-Aviation and Missiles Systems EIR.

**"REPORTING EQUIPMENT IMPROVEMENT RECOMMENDATIONS (EIR**

If your equipment needs improvement, let us know. Send us an EIR. You, the user, are the only one who can tell us what you do not like about your equipment. Let us know why you do not like the design or performance.

All non-Aviation/Missile EIRs and PQDRs must be submitted through the Product Data Reporting and Evaluation Program (PDREP) website (https://www.pdrep.csd.disa.mil).

If you do not have Internet access, you may submit your information using an SF 368 (Product Quality Deficiency Report). You can send your SF 368 using email, regular mail, or fax using the addresses/fax numbers specified in DA Pam 750-8, The Army Maintenance Management System (TAMMS) Users Manual. We will send you a reply."

#### B.7.1.3.3 Marine Corps EIR.

**"REPORTING EQUIPMENT IMPROVEMENT RECOMMENDATIONS (EIR**

If your equipment needs improvement, let us know. Send us an EIR. You, the user, are the only one who can tell us what you do not like about your equipment. Let us know why you do not like the design or performance. SF Form 368, Product Quality Deficiency Report must be submitted through the Product Data Reporting and Evaluation Program (PDREP) website (https://www.pdrep.csd.disa.mil). Information on the PQDR can be found at https://www.logcom.marines.mil/logcom/centers/QMC/Quality-Management. Question or comments can be sent to smblogcompqdrstracking@usmc.mil ."

### B.7.1.4 Hand Receipt (HR manuals <handreceipt>.

If the equipment supported by the technical publication uses resources that require a hand receipt and a hand receipt manual has been prepared, the following statement shall be included in the maintainer maintenance or AMC manuals and below general information WP. For IETMs, a separate PDF file shall be used for hand receipt information contained on the IETM disk. Hand receipt information shall not be used in place of COEI, BII, and AAL information. A link to the information shall be provided.

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**"HAND RECEIPT (HR MANUALS**

This manual has a companion document with a publication number followed by "-HR" (which stands for "Hand Receipt"). That technical publication consists of preprinted hand receipts that list end item related equipment (for example, Components of End Item (COEI), Basic Issue Items (BIIs), and Additional Authorization List (AAL)) that must be accounted for. As an aid to property accountability, additional HR manuals may be requisitioned through normal publication channels."

### B.7.1.5 Corrosion Prevention and Control (CPC <cpcdata>.

CPC information shall be included in the general information WP. Refer to AR 750-59 and AR 750-1 for further information on CPC. CPC information shall consist of the following:

**a.** CPC boilerplate statement (B.7.1.5.1).

**b.** SF 368 boilerplate statement (B.7.1.5.2).

**c.** References to relevant maintenance tasks (E.7.4.4), WPs (E.7), or DA publications.

### B.7.1.5.1 CPC boilerplate statement.

"CORROSION PREVENTION AND CONTROL (CPC)

Army regulation AR 750-59 defines corrosion as "the impairment, degradation, or damage of materials (metallic and nonmetallic) as a result of exposure to a natural or induced environment owing to the individual or combined effects of chemical, electrochemical, biological, or physical attacks on the material." Corrosion is commonly thought of as occurring on metals, but degradation of other materials like composites, plastics, rubber, wood, and fabrics is also considered corrosion as defined per the AR. Regular washing, inspection, corrosion inhibiting compound application, and spot painting will play a significant role in the longevity, safety, and operability of Army assets.

See [*insert specific publication* ] TM-1-1500-344-23-1 thru -4 (AMCOM systems); TB 43-0213 (TACOM systems); and TB 43-0118 (CECOM systems) for further information on corrosion inspection, visually identifying the forms and severity of corrosion, and reporting corrosion issues. When corrosion is identified as a contributing factor to component failures, or materiel requires corrosion repair, spot painting, or Custom Industrial Coatings (CIC) application, it shall be recorded as corrosion in the inspection record and the Global Combat Support System – Army (GCSS-A) Cause Code of "170 – Rusted/Corroded" shall be used when requesting/performing maintenance." The applicable GCSS-A Object Part Code (for example, 12 = Brakes) must also be identified to denote and track the parts being repaired or replaced due to corrosion.

CARC Spot Painting; see TM 43-0242 "Chemical Agent Resistant Coating Spot Painting for Army Ground Equipment"

CARC Paint Booth Operations; see TM 43-0139 "Chemical Agent Resistant Coating (CARC) Application and Quality Assurance on Army Ground Equipment."

### B.7.1.5.2 SF 368 boilerplate statement.

One of the following statements shall be included verbatim in the CPC information after the CPC boilerplate information:

**a.** For non-aviation systems:

"If a corrosion problem is identified, it can be reported as an EIR or PQDR. Use of key words such as "corrosion," "rust," "deterioration," or "cracking" will ensure that the information is identified as a CPC problem. SF 368, Product Quality Deficiency Report should be submitted to the address specified in DA Pam 750-8, The Army Maintenance Management System (TAMMS) Users Manual."

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**b.** For aviation systems:

"If a corrosion problem is identified, it can be reported as an EIR or PQDR. Use of key words such as "corrosion," "rust," "deterioration," or "cracking" will ensure that the information is identified as a CPC problem. SF 368, Product Quality Deficiency Report should be submitted to the address specified in DA Pam 738-751, Functional Users Manual for The Army Maintenance Management System-Aviation (TAMMS-A)."

### B.7.1.5.3 References to relevant tasks, WPs, and DA publications.

References may be included to relevant tasks, WPs, and DA publications. For aviation systems reference shall be included to TM 1-1500-344-23, volumes 1-4 (Cleaning and Corrosion Control). For Army Ground Equipment TMs, reference shall be included to TB 43-0213 (Corrosion Prevention and Control (CPC) for Army Ground Equipment). If applicable, reference to TM 43-0139 (Painting Instructions for Army Materiel) and TM 43-0242 (Chemical Agent Resistant Coating Spot Painting for Army Ground Equipment) shall be included. For all Army ground equipment, these references are applicable and the following language shall be included:

"AR 750-59, Corrosion Prevention and Control of Army Materiel, establishes requirements for CPC to include painting. In general terms, it requires equipment and materiel publications contain current and appropriate CPC information to maintain the systems and references to existing technical publications for CPC such as TM 43-0139, TM 43-0242, TB 43-0213, and others. See AR 750-59 for further information on CPC requirements.

AR 750-1, Army Materiel Maintenance Policy, establishes the Army Chemical Agent Resistant Coating (CARC), Camouflage, and Marking Program and related painting policies. It implements CARC as the approved coating for all combat and combat support equipment; tactical vehicles; aircraft, including unmanned; essential Ground Support Equipment; and repairable containers.

AR 750-1 prohibits full system re-painting for aesthetic or cosmetic purposes and states at least 25 percent of the system coating should be missing/unserviceable before full repainting, and mandates touch-up painting for ground equipment be with CARC in accordance with TM 43-0242. Full system painting and repainting may only be performed with equipment and paint booths meeting Occupational Safety & Health Administration (OSHA) standards in accordance with TM 43-0139. See AR 750-1 for further information on maintenance policies and requirements.

Refer to TB 43-0213 for additional corrosion prevention and control information, materials, and instructions. Refer to TM 43-0139 for full system CARC painting/repainting information, materials, and instructions. Refer to TM 43-0242 for CARC touch-up information, materials, and instructions. Search "corrosion" on the Unit Training Assistance Program (UTAP) website at https://utap.army.mil/ for CPC related instructional videos."

### B.7.1.6 Ozone Depleting Substances (ODSs) <odsdata>.

The use of Class 1 ODS for new acquisitions has been curtailed by Executive Order, Public Law, and related Army policy. ODSs are listed in Title VI of the Clean Air Act. For systems procured and fielded prior to the date these became effective (June 1993) that use a Class 1 ODS, a listing of those substances required to operate and maintain the system shall be included in the manual. After June 1993, this requirement applies to any system procured or fielded that requires the use of a Class 1 ODS, where the use of the ODS has been properly documented and waived. The procuring activity will provide a list of Class 1 ODS upon request.

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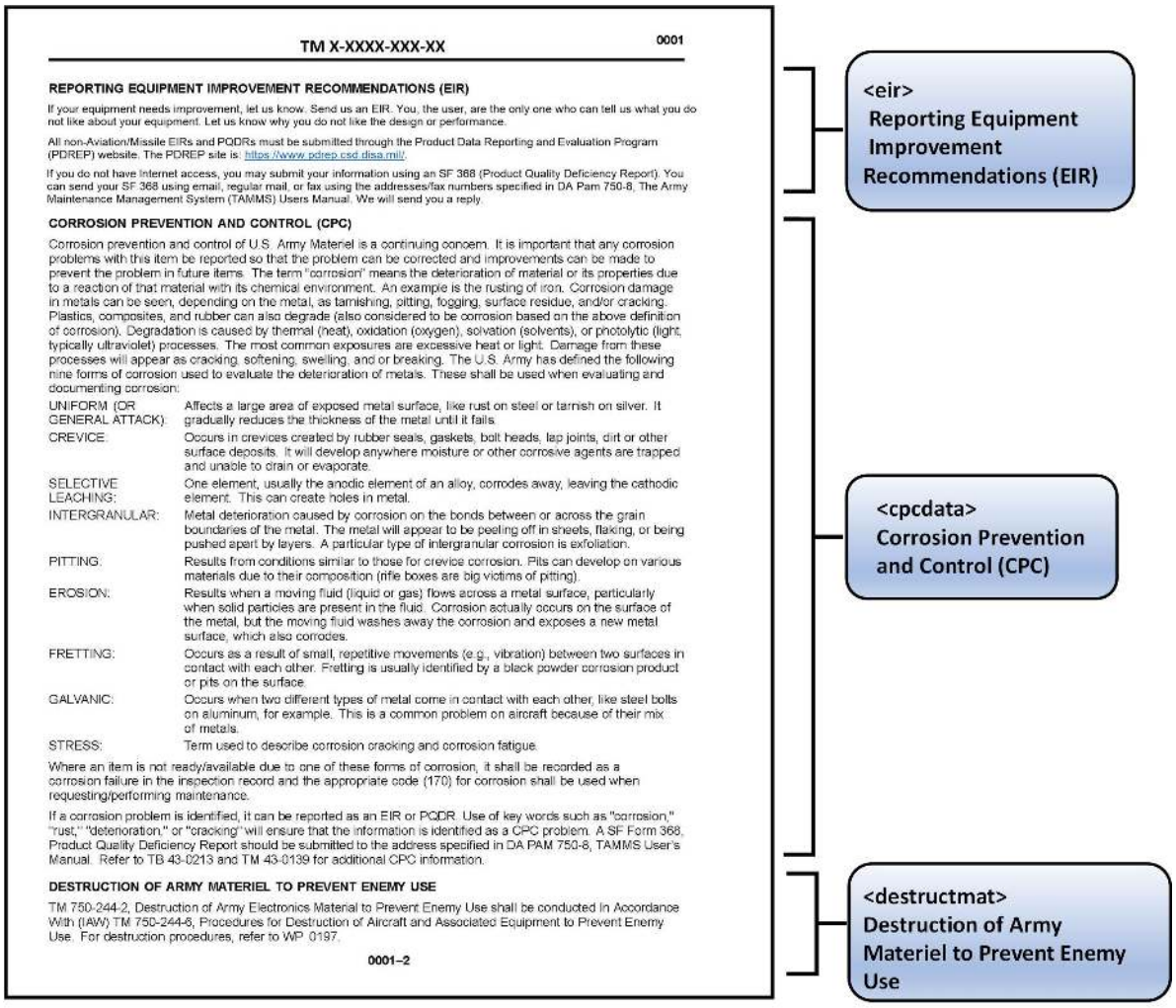
### B.7.1.7 Destruction of equipment to prevent enemy use <destructmat>.

Reference shall be made to the appropriate technical publications or WP(s) covering the destruction of equipment to prevent enemy use. Refer to the MATDEV for destruction information guidance as needed.

**Note**

Figure appearance and tags/output may vary slightly based on version of DTD/stylesheets utilized (see applicable DTD or tag definition list for available tags).



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**REPORTING EQUIPMENT IMPROVEMENT RECOMMENDATIONS (EIR)**

If your equipment needs improvement, let us know. Send us an EIR. You, the user, are the only one who can tell us what you do not like about your equipment. Let us know why you do not like the design or performance.

All non-Aviation/Missile EIRs and PQDRs must be submitted through the Product Data Reporting and Evaluation Program (PDREP) website. The PDREP site is: https://www.pdrep.csd.disa.mil/.

If you do not have Internet access, you may submit your information using an SF 368 (Product Quality Deficiency Report). You can send your SF 368 using email, regular mail, or fax using the addresses/fax numbers specified in DA Pam 750-8, The Army Maintenance Management System (TAMMS) Users Manual. We will send you a reply.

**CORROSION PREVENTION AND CONTROL (CPC)**

Corrosion prevention and control of U.S. Army Materiel is a continuing concern. It is important that any corrosion problems with this item be reported so that the problem can be corrected and improvements can be made to prevent the problem in future items. The term "corrosion" means the deterioration of material or its properties due to a reaction of that material with its chemical environment. An example is the rusting of iron. Corrosion damage in metals can be seen, depending on the metal, as tarnishing, pitting, fogging, surface residue, and/or cracking. Plastics, composites, and rubber can also degrade (also considered to be corrosion based on the above definition of corrosion). Degradation is caused by thermal (heat), oxidation (oxygen), solvation (solvents), or photolytic (light, typically ultraviolet) processes. The most common exposures are excessive heat or light. Damage from these processes will appear as cracking, softening, swelling, and or breaking. The U.S. Army has defined the following nine forms of corrosion used to evaluate the deterioration of metals. These shall be used when evaluating and documenting corrosion:

| | |
|---|---|
| UNIFORM (OR GENERAL ATTACK): | Affects a large area of exposed metal surface, like rust on steel or tarnish on silver. It gradually reduces the thickness of the metal until it fails. |
| CREVICE: | Occurs in crevices created by rubber seals, gaskets, bolt heads, lap joints, dirt or other surface deposits. It will develop anywhere moisture or other corrosive agents are trapped and unable to drain or evaporate. |
| SELECTIVE LEACHING: | One element, usually the anodic element of an alloy, corrodes away, leaving the cathodic element. This can create holes in metal. |
| INTERGRANULAR: | Metal deterioration caused by corrosion on the bonds between or across the grain boundaries of the metal. The metal will appear to be peeling off in sheets, flaking, or being pushed apart by layers. A particular type of intergranular corrosion is exfoliation. |
| PITTING: | Results from conditions similar to those for crevice corrosion. Pits can develop on various materials due to their composition (rifle boxes are big victims of pitting). |
| EROSION: | Results when a moving fluid (liquid or gas) flows across a metal surface, particularly when solid particles are present in the fluid. Corrosion actually occurs on the surface of the metal, but the moving fluid washes away the corrosion and exposes a new metal surface, which also corrodes. |
| FRETTING: | Occurs as a result of small, repetitive movements (e.g., vibration) between two surfaces in contact with each other. Fretting is usually identified by a black powder corrosion product or pits on the surface. |
| GALVANIC: | Occurs when two different types of metal come in contact with each other, like steel bolts on aluminum, for example. This is a common problem on aircraft because of their mix of metals. |
| STRESS: | Term used to describe corrosion cracking and corrosion fatigue. |

Where an item is not ready/available due to one of these forms of corrosion, it shall be recorded as a corrosion failure in the inspection record and the appropriate code (170) for corrosion shall be used when requesting/performing maintenance.

If a corrosion problem is identified, it can be reported as an EIR or PQDR. Use of key words such as "corrosion," "rust," "deterioration," or "cracking" will ensure that the information is identified as a CPC problem. A SF Form 368, Product Quality Deficiency Report should be submitted to the address specified in DA PAM 750-8, TAMMS User's Manual. Refer to TB 43-0213 and TM 43-0139 for additional CPC information.

**DESTRUCTION OF ARMY MATERIEL TO PREVENT ENEMY USE**

TM 750-244-2, Destruction of Army Electronics Material to Prevent Enemy Use shall be conducted in Accordance With (IAW) TM 750-244-6, Procedures for Destruction of Aircraft and Associated Equipment to Prevent Enemy Use. For destruction procedures, refer to WP 0197.

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**FIGURE B-3. General information WP (sheet 2 of 4).**

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### B.7.1.8 Preparation for storage or shipment <pssref> and transportability guidance <transportability>.

Reference shall be made to the preparation for storage, preparation for shipment, and transport WPs found in the technical publication, if they exist, and/or to applicable U.S. Army authenticated publications containing this guidance. If the relevant WPs are in another DA-authenticated publication, reference shall be made to that publication. Reference shall not be made to any Surface Deployment and Distribution Center/ Transportation Engineering Agency (SDDC/TEA) (formerly Military Traffic Management Command Transportation Engineering Agency (MTMC/TEA)) publications. If there are no requirements related to storage, shipment, or transportability, the following statement shall be inserted in these paragraphs:

"There are no requirements related to storage or shipment for this equipment."

"There are no requirements related to transportability for this equipment."

### B.7.1.9 Warranty information <wrntyref>.

When the technical publication covers equipment that is under warranty and a Warranty Technical Bulletin (WTB) is published, the applicable WTB shall be referenced. When a WTB is not published, the following statement shall be included (italicized text within parentheses shall be replaced with the appropriate information):

**"WARRANTY INFORMATION**

This equipment is warranted for (*insert miles or other timeframe as appropriate*). The warranty starts on the date found in block 23 of DA Form 2408-9, Equipment Control Record. Report all defects to your supervisor, who will take appropriate action."

### B.7.1.10 Nomenclature cross-reference list <nomenreflist>.

A cross-reference list shall be prepared when unofficial nomenclature (common name) is approved by the MATDEV activity.

### B.7.1.11 List of abbreviations/acronyms <loa>.

A list of all abbreviations, acronyms, signs, or symbols used in the manual shall be prepared. Warning icons are defined in the Warning Summary (if included). For **aircraft only**, a statement shall be prepared that abbreviations are in accordance with abbreviations contained in the Abbreviations, Brevity Codes, and Acronyms (ABCA) tool at https://armypubs.army.mil/, except when the abbreviation stands for a marking actually found in the aircraft.

**Note**

To improve usability, in-text abbreviations/acronyms throughout a technical publication may be linked to the list of abbreviations/acronyms in this WP allowing for quick access to definitions.

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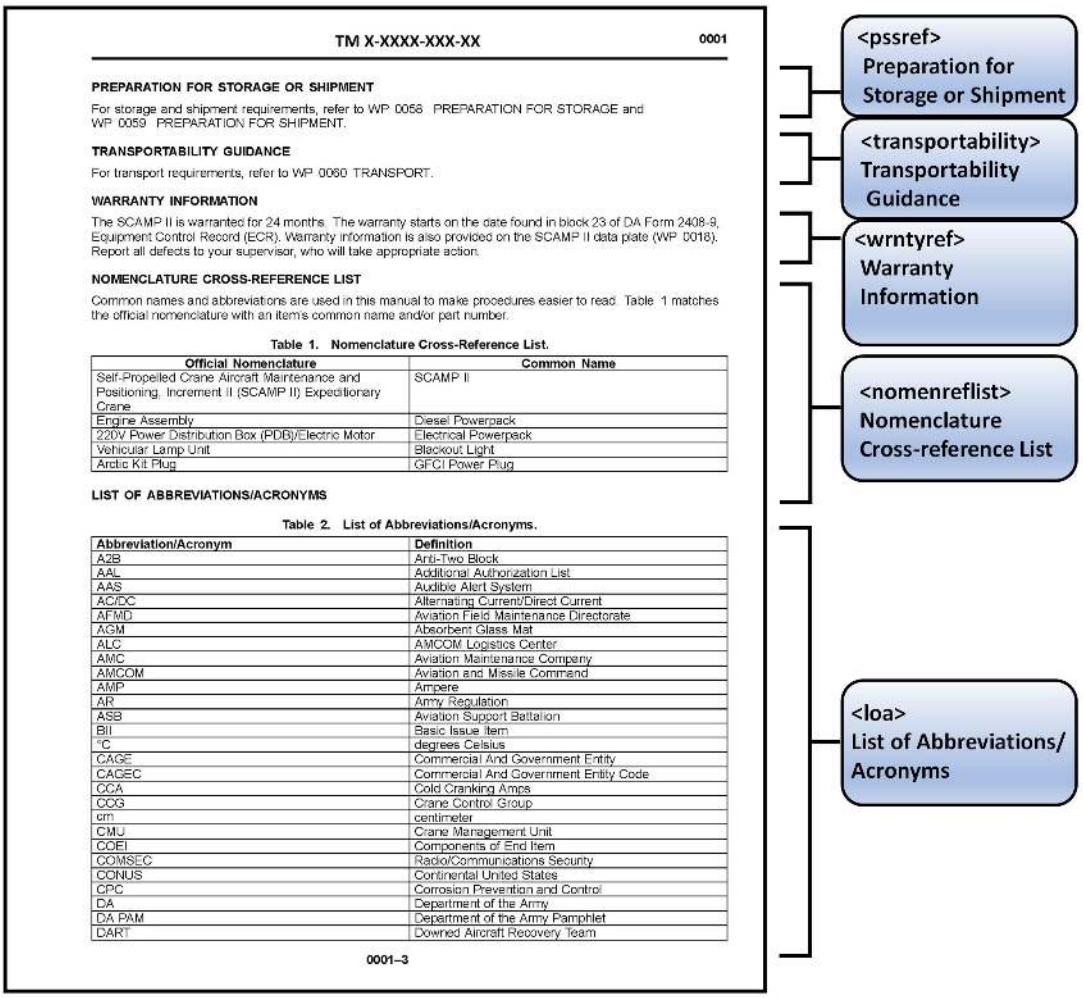
## B.7.1.11.1 Common abbreviations/acronyms.

Common (or standard) abbreviations/acronyms that are widely known and accepted (inch, pound, am/pm, laser, sonar, etc.) may not be defined in the list of abbreviations/acronyms. Contact Equipment Publication Control Office (EPCO) representatives about whether a potential entry is considered common.

**Note**

Figure appearance and tags/output may vary slightly based on version of DTD/stylesheets utilized (see applicable DTD or tag definition list for available tags).



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**PREPARATION FOR STORAGE OR SHIPMENT**

For storage and shipment requirements, refer to WP 0058 PREPARATION FOR STORAGE and WP 0059 PREPARATION FOR SHIPMENT.

**TRANSPORTABILITY GUIDANCE**

For transport requirements, refer to WP 0060 TRANSPORT.

**WARRANTY INFORMATION**

The SCAMP II is warranted for 24 months. The warranty starts on the date found in block 23 of DA Form 2408-9, Equipment Control Record (ECR). Warranty information is also provided on the SCAMP II data plate (WP 0018). Report all defects to your supervisor, who will take appropriate action.

**NOMENCLATURE CROSS-REFERENCE LIST**

Common names and abbreviations are used in this manual to make procedures easier to read. Table 1 matches the official nomenclature with an item's common name and/or part number.

Table 1. Nomenclature Cross-Reference List.

| Official Nomenclature | Common Name |
|---|---|
| Self-Propelled Crane Aircraft Maintenance and Positioning, Increment II (SCAMP II) Expeditionary Crane | SCAMP II |
| Engine Assembly | Diesel Powerpack |
| 220V Power Distribution Box (PDB)/Electric Motor | Electrical Powerpack |
| Vehicular Lamp Unit | Blackout Light |
| Arctic Kit Plug | GFCI Power Plug |

**LIST OF ABBREVIATIONS/ACRONYMS**

Table 2. List of Abbreviations/Acronyms.

| Abbreviation/Acronym | Definition |
|---|---|
| A2B | Anti-Two Block |
| AAL | Additional Authorization List |
| AAS | Audible Alert System |
| AC/DC | Alternating Current/Direct Current |
| AFMD | Aviation Field Maintenance Directorate |
| AGM | Absorbent Glass Mat |
| ALC | AMCOM Logistics Center |
| AMC | Aviation Maintenance Company |
| AMCOM | Aviation and Missile Command |
| AMP | Ampere |
| AR | Army Regulation |
| ASB | Aviation Support Battalion |
| BII | Basic Issue Item |
| °C | degrees Celsius |
| CAGE | Commercial And Government Entity |
| CAGEC | Commercial And Government Entity Code |
| CCA | Cold Cranking Amps |
| CCG | Crane Control Group |
| cm | centimeter |
| CMU | Crane Management Unit |
| COEI | Components of End Item |
| COMSEC | Radio/Communications Security |
| CONUS | Continental United States |
| CPC | Corrosion Prevention and Control |
| DA | Department of the Army |
| DA PAM | Department of the Army Pamphlet |
| DART | Downed Aircraft Recovery Team |

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<pssref> Preparation for Storage or Shipment

<transportability> Transportability Guidance

<wrntyref> Warranty Information

<nomenreflist> Nomenclature Cross-reference List

<loa> List of Abbreviations/ Acronyms

**FIGURE B-4. General information WP (sheet 3 of 4).**

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### B.7.1.12 Quality Assurance (QA (Depot Maintenance Work Requirements (DMWRs / National Maintenance Work Requirements (NMWRs and aviation only <qainfo>.

When specified by the MATDEV, reference shall be made to pertinent QA information or include the appropriate general QA information. If QA information is not referenced but is included in the manual, it shall be stated that the text of each quality assurance procedure or step in the manual is preceded and highlighted by the addition of "QA check." For aircraft maintenance publications, include a reference to ATP 3-04.7 (Army Aviation Maintenance).

### B.7.1.13 Quality of material <qual.mat.info>.

A statement or statements similar to the following shall be included (italicized text within parentheses shall be replaced with the appropriate information). Manuals requiring this information contain an "R" in the matrix:

"Material used for replacement, repair, or modification must meet the requirements of this (*insert manual*). If quality of material requirements are not stated in this (*insert manual*), the material must meet the requirements of the drawings, standards, specifications, or approved engineering change proposals applicable to the subject equipment."

### B.7.1.14 Safety, care, and handling <sftyinfo>.

The following general precautions and safety regulations may be prepared:

**a. (Ammunition Technical Publications)** Information shall be prepared to comply with DA Pam 385-63. References to applicable ARs for range safety danger zones during training and combat shall be included. A reference to AR 385-10, AR 385-63, and DA Pam 385-64 shall be made for general ammunition care, handling, and safety.

**b.** For publications covering equipment with radioactive parts or components, information shall be prepared to comply with Nuclear Regulatory Commission provisions, and references to applicable ARs and safety publications on radioactive materials shall be included. If additional coverage on radioactive materials is needed, but is not included in applicable publications, instructions shall be prepared as required. In addition, the following information shall be prepared for inclusion throughout the technical publication.

**(1)** Nuclear warning notices. These shall be placed at the beginning of any instruction covering procedures that will expose personnel to a nuclear radiation hazard.

**(2)** Procedures to be followed before maintenance actions or in the event of breakage of radioactive parts or components. These include safety, care, and handling instructions.

**(3)** Radioactive parts or components. These shall be shown and identified on a parts location diagram or illustration. Warning notices shall be included.

**(4)** A list of radioactive parts or components and the type and quantity of radioactive material involved. These shall be included as part of equipment data (refer to B.7.2.4).

**(5)** Instructions for the disposal of radioactive material, such as the requirement to double bag all broken tritium sources in plastic.

**c.** ESD control standards for the protection of electrical and electronic parts, assemblies, and equipment shall be prepared. The ESD classes shall be identified. For classifications of ESD marking procedures, refer to ANSI ESD TR20.20 and ANSI ESD S20.20.

**d. (DMWRs/NMWRs only)** When applicable, reference shall be made to the electromagnetic compatibility standards (for example, MIL-STD-461) that apply to the equipment covered in the DMWR/NMWR.

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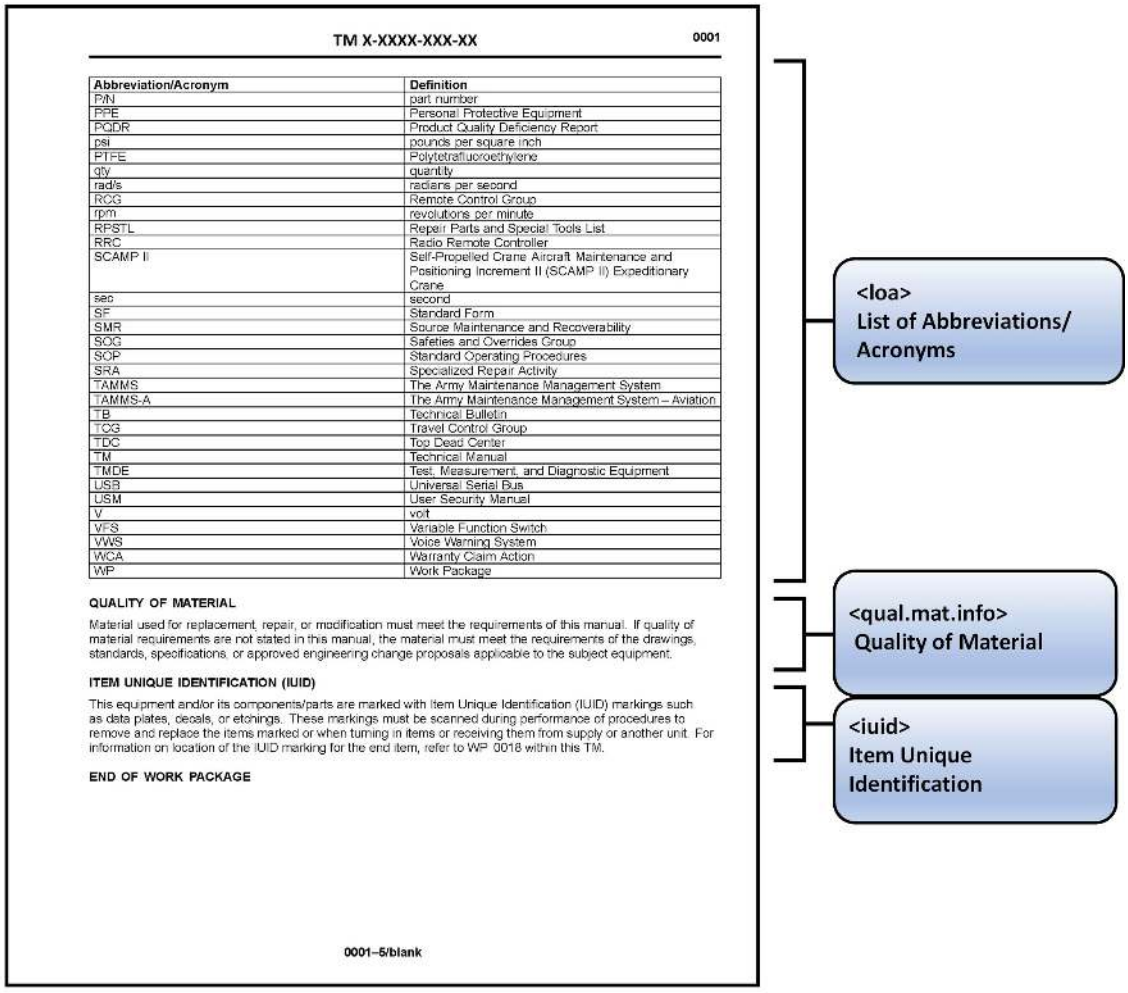


**FIGURE B-5. General information WP (sheet 4 of 4).**

## B.7.1.15 Nuclear hardness <hcp>.

If equipment covered in the technical publication has nuclear survivability requirements (for example, overpressure and burst, thermal radiation, electromagnetic pulse, or transient radiation effects on electronics), it shall be so stated (refer to 4.15.10.2 for marking HCP procedures). The following statement shall be included.

**"NUCLEAR HARDNESS**

All hardness critical process (HCP) procedures in this manual are marked with the acronym "HCP" as follows:

1. When an entire task, including all paragraphs and procedures, is considered hardness critical, only the task title will be marked by the acronym "HCP." This will be placed before the title.
2. When only certain processes and steps within the WP are hardness critical, only the applicable processes and steps will be marked by placement of the acronym "HCP" between each applicable step number and the text."

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### B.7.1.16 Calibration <calref>.

Equipment requiring calibration shall be identified, and reference shall be made to the publication containing the applicable calibration procedure.

### B.7.1.17 Item Unique Identification (IUID <iuid>.

If the equipment covered by the manual or any of its components/parts have IUID markings, a statement similar to the following shall be included:

**"ITEM UNIQUE IDENTIFICATION**

This equipment and/or its components/parts are marked with Item Unique Identification (IUID) markings such as data plates, decals, or etchings. These markings must be scanned during performance of procedures to remove and replace the items marked or when turning in items or receiving them from supply or another unit. For information on location of the IUID marking for the end item, refer to the decal/data plate guide contained in the operator manual for the equipment."

### B.7.1.18 Engineering Change Proposal (ECPs (DMWR/NMWR only <ecp>.

The following statement shall be included (italicized text within parentheses shall be replaced with the appropriate information):

**"ENGINEERING CHANGE PROPOSALS**

Engineering Change Proposals (ECPs) will be submitted in accordance with AR 70-1 directly to (*enter the name and address of the responsible command or activity*). A reply will be furnished to you."

### B.7.1.19 Modification list (DMWR/NMWR only <modification>.

Modification Work Orders (MWOs) and ECPs shall be identified for all modifications that have been incorporated into the work required by the DMWR/NMWR. MWOs shall be reported as outlined in AR 750-10. The applicable MWOs and the ECPs shall be listed by title and number. These listings shall be supplied by the major subordinate command. Alternatively, a statement shall be made stating that the modifications must be applied during the overhaul of the item. For example (italicized text within parentheses shall be replaced with the appropriate information):

**"MODIFICATIONS**

All Modification Work Orders (MWOs), all minor alteration procedures (MAPs) specified in the contract/work directive, and all Engineering Change Proposals (ECPs) listed in the (*insert DMWR or NMWR*) must be applied during the overhaul of the item."

### B.7.1.20 Deviations and exceptions (DMWR/NMWR only <deviation>.

The following statement shall be included (italicized text within parentheses shall be replaced with the appropriate information):

**"DEVIATIONS AND EXCEPTIONS**

Requests for deviations or exceptions to this (*insert Depot Maintenance Work Requirement (DMWR) or National Maintenance Work Requirement (NMWR)*) will be processed in accordance with International Standards Organization (ISO) 9000 Series standards, or equivalent."

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### B.7.1.21 Mobilization requirements (DMWR/NMWR only <mobreq>.

The following statement shall be included (italicized text within parentheses shall be replaced with the appropriate information):

**"MOBILIZATION REQUIREMENTS**

All requirements of this (*insert DMWR or NMWR*) will be exempted or revised in the event of mobilization. Only those procedures necessary to return the (*insert equipment name*) to a serviceable condition will be performed. The exemptions and revisions are explained in supporting information WP (*insert appropriate WP sequence number*)."

### B.7.1.22 Mandatory Replacement Parts (MRPs <mrpref>

This paragraph shall reference the mandatory replacement parts list work package if it exists. If there are no MRPs for equipment covered by the manual, insert the following statement in this paragraph:

"There are no mandatory replacement parts for (insert equipment name)."

Additionally, specify whether MRP is or is not included in an equipment system in accordance with B.7.1.1.

### B.7.1.23 Critical Safety Items (CSI requirements <csireq>.

As specified by the MATDEV, the following statement shall be included, if applicable:

**"CRITICAL SAFETY ITEMS (CSI PROGRAM**

Parts, assemblies, or installations identified under the CSI program require special handling during maintenance or overhaul (M&O). Throughout the M&O procedures, warnings are included emphasizing critical instructions to be followed. These warnings are identified as CSI warnings.

A critical safety item is defined as:

A part, assembly, installation, or production system with one or more critical or critical safety characteristics that, if missing or not conforming to the design data, quality requirements, or overhaul and maintenance documentation, would result in an unsafe condition that could cause loss or serious damage to the end item or major components, loss of control, uncommanded engine shutdown, or serious injury or death to personnel. Unsafe conditions relate to hazard severity categories I and II of MIL-STD-882 and include items determined to be "life-limited," "fracture critical," "fatigue-sensitive," etc. The determining factor in Aviation CSI is the consequence of failure, not the probability that the failure or consequence would occur. All CSIs shall be handled and managed as prescribed in DoDI 4140.01 and DA Pam 95-9. Throughout the maintenance tasks, "CRITICAL SAFETY ITEM" alerts will precede the procedural step that includes a CSI, emphasizing that this part or parts requires special handling during maintenance."

If there are no CSIs for equipment covered by the manual, insert the following statement in this paragraph:

"There are no critical safety items for this equipment."

Additionally, specify whether CSI is or is not included in an equipment system in accordance with B.7.1.1.

### B.7.1.24 Cost considerations (DMWR/NMWR only <cost>.

The following statement shall be included (italicized text within parentheses shall be replaced with the appropriate information):

**"COST CONSIDERATIONS**

This work requirement shall be the basis for establishing the extent of overhaul while taking into consideration cost factors. A determination shall be made on all subassemblies/assemblies to replace worn or damaged components that

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are available in supply if acquisition cost is less than the cost to repair and restore to the (*insert DMWR or NMWR*) standard. The cost to repair/restore any individual item with an established Maintenance Expenditure Limit (MEL) to the (*insert DMWR or NMWR*) standard shall not exceed the MEL, unless a waiver has been approved in accordance with AR 750-1. This requirement does not apply to items exempted from MEL in accordance with AR 750-1."

### B.7.1.25 Supporting information for repair parts, special tools/kits, Test Measurement, and Diagnostic Equipment (TMDE and support equipment (Maintainer/AMC and above <supdata>.

When applicable, the following information shall include a reference to the common tools and equipment; special tools/kits, TMDE, and support equipment; and the repair parts as shown in the following paragraphs. The information in B.7.1.25.1 through shall be included.

### B.7.1.25.1 Common tools and equipment.

The following statement shall be included:

**"COMMON TOOLS AND EQUIPMENT**

For authorized common tools and equipment, refer to the Modified Table of Organization and Equipment (MTOE), Common Table of Allowances (CTA) 50-970, Expendable/Durable Items (Except: Medical, Class V, Repair Parts, and Heraldic Items); CTA 50-909, Field and Garrison Furnishings and Equipment; or CTA 8-100, Army Medical Department Expendable/Durable Items; as applicable to your unit."

### B.7.1.25.2 Special tools/kits, Test, Measurement, and Diagnostic Equipment (TMDE, and support equipment.

A reference to the Repair Parts and Special Tools List (RPSTL) and Maintenance Allocation Chart (MAC) shall be included. When no special tools/kits or equipment are required, it shall be so stated. If tools are to be fabricated, reference shall be made to the Illustrated List of Manufactured Items WP (refer to E.7.8).

### B.7.1.25.3 Repair parts.

One of the following statements shall be included (italicized text within parentheses shall be replaced with the appropriate information):

"Repair parts are listed and illustrated in (*insert publication number of RPSTL*)."

Or

"Repair parts are listed and illustrated in the parts information WP(s) (*insert WP or range of WPs*)."

### B.7.1.26 Copyright credit line <copyrt>

Technical publications should not contain copyrighted material except as specified in the Federal Acquisition Regulations and Defense Federal Acquisition Regulation Supplement. When copyrighted material is included in a publication, the publication author shall obtain prior written permission from the copyright owner or authorized agent for its use. The written permission shall contain a statement declaring whether or not a copyright credit line is required. When a copyright credit line is required, the information shall appear as the last paragraph of the general information WP.

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### B.7.1.26.1 Proprietary names.

Trade names, copyrighted names, or other proprietary names applying exclusively to the product of one company shall not be used unless the items cannot be adequately described without using the proprietary names because of the technical involvement, construction, or composition. In such instances, one commercial product shall be listed, followed by the words "or equal" (except for aviation manuals and some software manuals). The same shall apply to manufacturers' part numbers or drawing numbers for minor parts where it is impractical to specify the exact requirements. If possible, the particular characteristics required for the "or equal" products shall be defined. If the names/logos appear in graphics or screen shots used in the technical publicaiton and it is cost prohibitive to modify all the graphics, the graphics may be used as-is and a disclaimer statement shall be placed on the front cover. Contact MATDEV if unsure if different but "equal" products may be used.

### B.7.1.26.2 Advertising.

Publication material shall not contain advertising matter.

## B.7.2 Equipment description and data work package <descwp>.

The equipment description and data WP shall be prepared and shall contain pertinent equipment description and data about an equipment system. This WP type may be duplicated in a module for effective information display. If the descriptive data are provided in a separate operator manual, a paragraph referencing the equipment description and data in the operator manual shall suffice. Additional equipment description and data required for a higher maintenance level, but not included in the operator manual, shall be included. This WP shall not contain any operator or maintenance procedures. FIGURE B-6 shows an example equipment description and data WP and the associated tags. Equipment description and data WP components are listed below:

**a.** WP identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** Initial setup **<initial_setup>** is not required for this WP.

**c.** Equipment description and data information sections discussed in B.7.2.1 through B.7.2.4 as applicable for an equipment system.

### B.7.2.1 Equipment characteristics, capabilities, and features <eqpinfo>.

An overall description of the equipment **<eqpdesc>** shall be prepared, including general capabilities, special features, and other like information (for example, applications, limitations) helpful in the operation and maintenance of the equipment. Unless otherwise directed, the information may be in narrative or tabular format. Additional description requirements are outlined by the following:

**a.** The equipment type shall be stated, as shall the following equipment features: portability or mobility, operational and special environment, and remote control.

**b.** Components and their functions shall not be described unless essential to continuity. For functional data, reference shall be made to the theory of operation.

**c.** When the equipment covered varies in scope and application or has several applications within an end item, a brief explanation of the multiple uses and a simple diagram showing all aspects of a typical application shall be prepared.


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### B.7.2.2 Location and description of major components <locdesc>.

Equipment location information shall be prepared. It shall include external and internal views of the equipment used to show general features and all major components. This information shall not duplicate information contained in the equipment data requirements and the equipment characteristics, capabilities, and features.

**a.** The equipment and weapons systems configuration shall be described as follows:

**(1)** A description of system areas and compartments shall be prepared. The system equipment and components contained in the areas shall be identified. To identify and locate the listed system equipment, the configuration description shall be supported by separate illustrations of each compartment and area. For aircraft only, a station diagram showing fuselage station, water line, and butt line, etc., shall be included.

**(2)** The subsystems or equipment comprising the system shall be identified and described. Other equipment that is installed in the subject system compartments and areas does not need to be listed in the text or called out in the illustrations if it does not directly affect the operation or maintenance of the subject system. Descriptions of operator-attended equipment shall include general statements about the nature and purpose of the controls and indicators. The text shall be supported by illustrations.

**(3)** Descriptions and illustrations of associated systems' equipment shall be limited to the major units of that equipment. The descriptions shall be more concise than those of the subject system's equipment; otherwise, the same requirements shall apply. In the descriptions, emphasis shall be placed on the associated system equipment that constitutes operational or functional interfaces with the subject system. Such units shall be included in the system illustrations.

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**Note**

Figure appearance and tags/output may vary slightly based on version of DTD/stylesheets utilized (see applicable DTD or tag definition list for available tags).

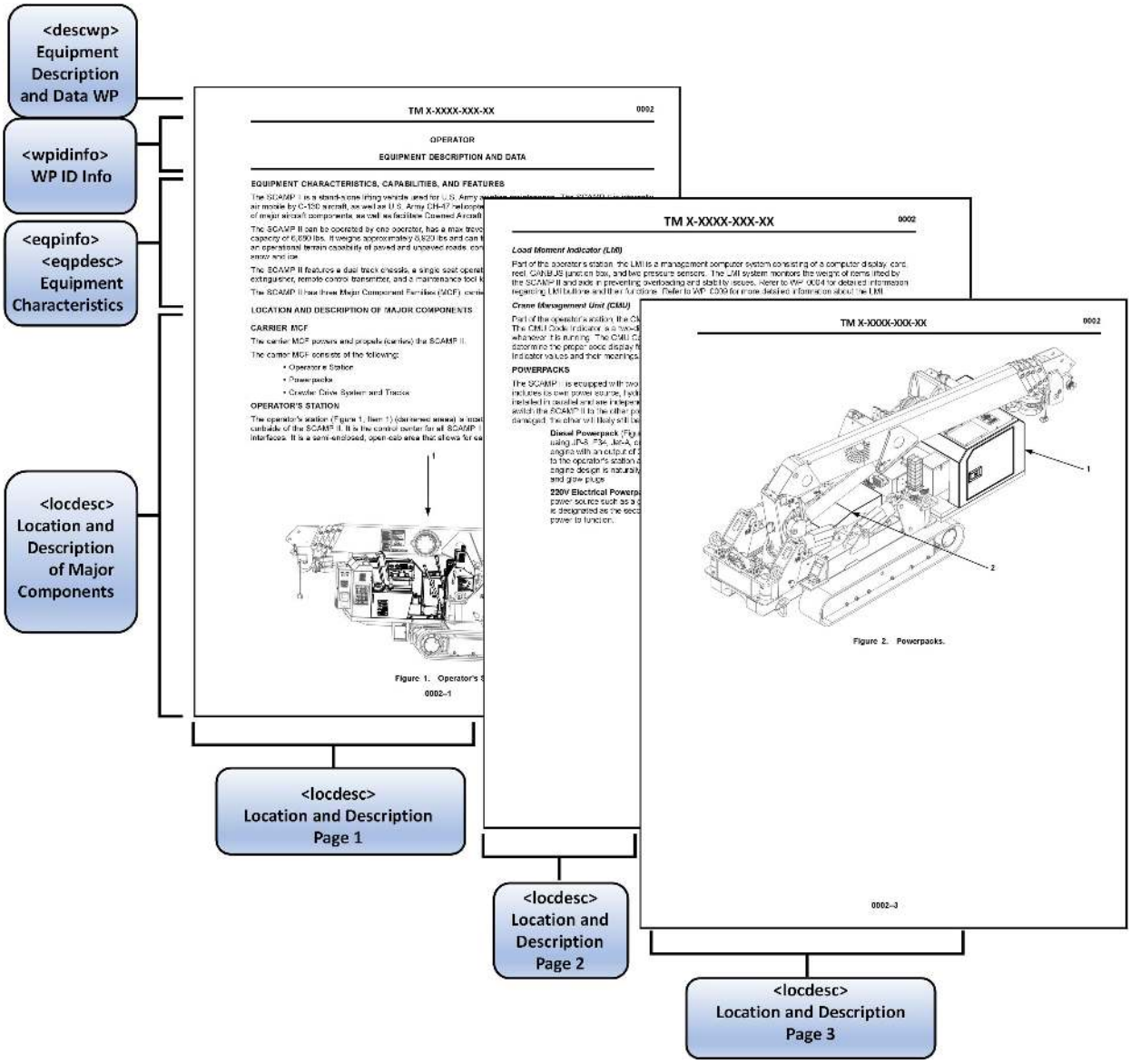


**FIGURE B-6. Equipment description and data WP.**



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### B.7.2.3 Equipment differences <eqpdiff>.

Equipment differences shall be prepared as needed and shall include the following:

**a.** Differences between models that affect operation, maintenance, or interchangeability shall be described to allow for easy identification by the user.

**b.** Differences within the same model (options, upgrades, etc.) shall be related explicitly to equipment part number or serial number ranges to allow for easy identification of the specific equipment configuration involved.

Non-specific terms such as "on later equipment," "on later models," and "on early serial numbers" shall not be used. References to other WPs such as "Decals and instruction plates" and content filtering through applicability may be used to supplement the above requirements.

### B.7.2.4 Equipment data <eqpdata>.

Equipment data shall be prepared and include the following:

**a.** Performance data shall be prepared, including numerical and other standard-related data applying to operational and maintenance tasks. The equipment data shall summarize the specific capabilities and limitations of the equipment and other critical data needed by the technical publication user for maintenance of the equipment. Vehicle and cargo space dimensions and metric and other equivalents shall be included.

**b.** For systems, a list of the environmental control requirements, such as limited temperature, humidity, or other limited conditions shall be prepared. Reference shall be made to the WP(s) containing information on any damage to be expected from exceeding these limits and procedures for minimizing the damage.

**c.** A summary shall be prepared that lists the effects of weather conditions on equipment that could affect system capability or cause equipment damage. This summary shall include references to any special servicing procedures that must be accomplished because of climatic changes, such as adding antifreeze to coolants.

**d.** Instructions for the use, transportation, handling, storage, or disposal of such substances as fuels, toxic and hazardous substances, chemicals, and ammunition shall be prepared. These instructions shall meet the applicable requirements of the Federal Environmental Protection Standards (standards to be provided by the MATDEV).

**e.** The energy efficiency rating shall be included for products that directly consume energy in normal operations and that commonly have a method of expressing energy efficiency.

### B.7.3 Theory of operation work package <thrywp>.

A theory of operation WP shall be prepared as required by publication type (see A.4.4) to provide a maintenance technician with adequate background information to support and perform maintenance tasks and troubleshooting on an equipment system and/or its components. This WP type may be duplicated in a module for effective information display. DMWRs/NMWRs shall include this WP as required by the MATDEV. The amount of detail and complexity of the theory of operation presentation shall be in accordance with the Logistics Product Data (LPD) maintenance concept, the MAC, or an approved maintenance plan. Theory of operation shall be provided as described in B.7.3.1 through B.7.3.1.3. This WP shall not contain any operator or maintenance procedures. FIGURE B-7 shows an example theory of operation WP and the associated tags. The equipment theory of operation WP components are listed below:

**a.** WP identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** Initial setup **<initial_setup>** is not required for this WP.

**c.** Introduction **<intro>** may precede theory of operation narrative (as necessary).


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**d.** Equipment system theory information discussed in B.7.3.1 through B.7.3.1.3 as applicable for an equipment system.

**Note**

Figure appearance and tags/output may vary slightly based on version of DTD/stylesheets utilized (see applicable DTD or tag definition list for available tags).

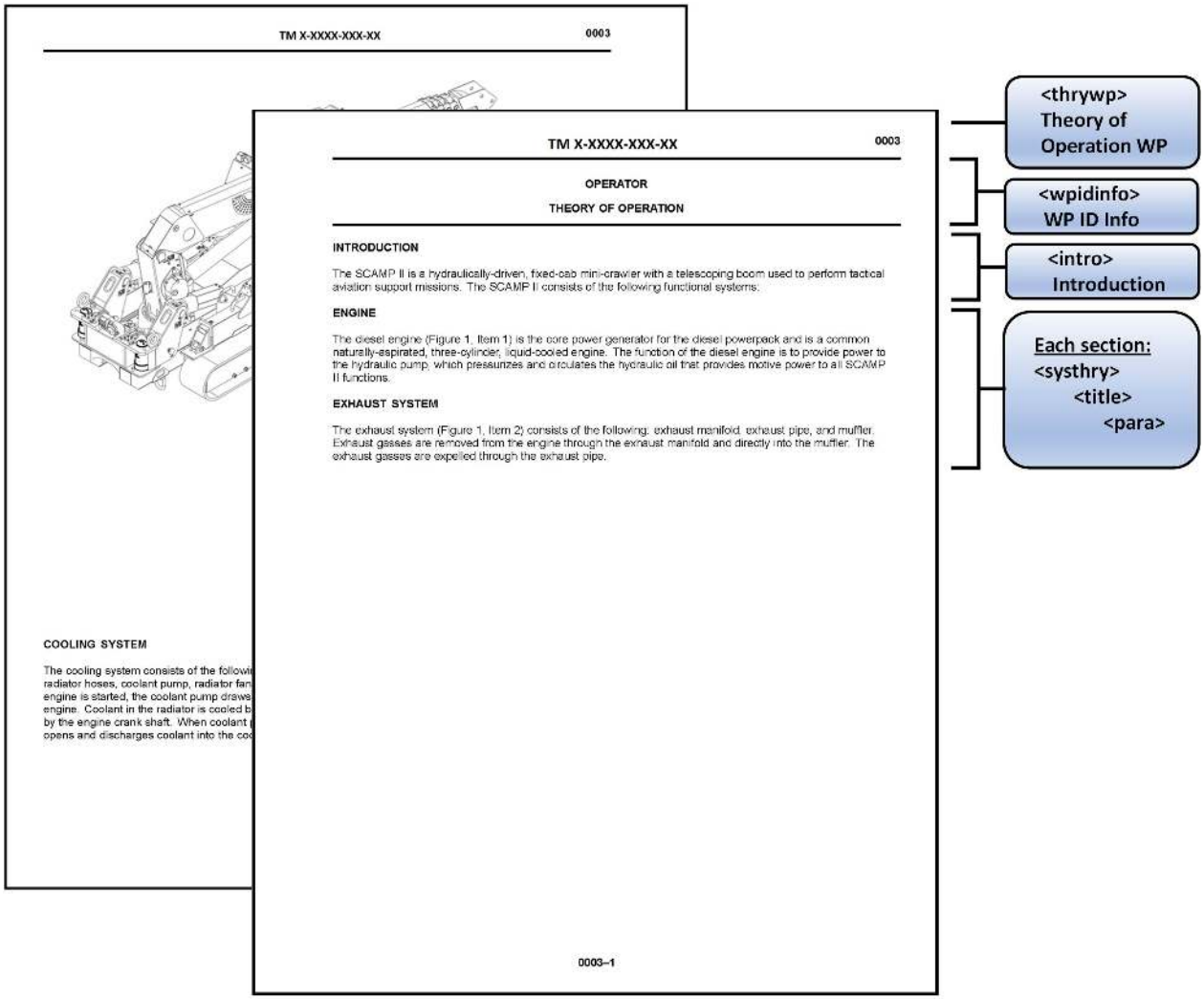


**FIGURE B-7. Theory of operation WP.**

## B.7.3.1 Theory presentation.

Theory of operation shall consist of a narrative to explain the functional operation of equipment (electrical/ electronic, hydraulic, pneumatic, and mechanical). All theory of operation WPs (both simple and complex) shall explain operation to a depth necessary to support a technician in fault isolation to the level directed by LPD. Theory shall not be prepared for non-repairable (throw-away) system components. Block diagrams, functional flow diagrams, schematics, and other illustrations may be included to support the text. If the technical publication covers more than one model of equipment, or more than one configuration of an equipment system, the differences shall be explained or separate WPs may be used. Depending on relative system intricacy, theory WPs shall be presented as simple or complex.

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### B.7.3.1.1 Simple theory presentation.

For systems with LPD requiring simple maintenance of easily repairable items, simple theory presentation shall be utilized. All simple theory **<systhry>** may be included in one single WP or multiple WPs (depending on number of models/configurations). Operator-only technical publications (-10 TMs) typically contain simple theory WPs and include one or more straightforward system-centered WP(s) outlining how major system components functionally work together to perform a specific intended function. Operator and maintainer technical publications (-12 TMs) typically contain one or more straightforward system-centered WP(s) for operators and are accompanied by one or more complex theory WPs detailing maintainer-level theory of operation outlined in B.7.3.1.2. For simple theory WPs, various equipment graphics, block diagrams, flow diagrams, and charts/tables etc. may be utilized to improve usability.

### B.7.3.1.2 Complex theory presentation.

For systems with LPD requiring maintenance of internal components and/or components with maintainer-level repairability, complex theory presentation shall be utilized. This presentation requires more detailed information outlining not only how major system components functionally work together but also significant narrative covering the interworking of components and the logical flow of information, data, signal, power, etc. through a system. Significant input, output, and control signals, supply voltages, and power supply output voltages shall be identified. When available, wiring diagrams (FIGURE B-8) shall be integrated to show flow of power, data, etc. through a system or subsystem.

### B.7.3.1.3 Simple and complex theory presentation.

When applicable, both simple and complex theory WPs shall present information in the following ways:

**a.** First present theory for an equipment end item and then present theory for its major subsystem(s) and (**<ssysthry>**) subcomponents (Line Replaceable Units (LRUs) **<lruthry>** and Shop Replaceable Units (SRUs) **<sruthry>**. Subcomponent theory narrative may be contained in separate WPs.

**b.** For aircraft systems, include separate theory of operation WPs for each aircraft.

**c.** Incorporate all details behind the operation of a system, which may include concepts such as:

**(1)** If a system operates in more than one mode, each mode shall be explained and supported with relevant block diagrams, etc.

**(2)** If a system contains detailed circuitry and/or signal, data, or power flow, etc. involving repairable components covered in a publication, these components, and their respective functionality (include circuitry, signal, data, etc.) shall be discussed.

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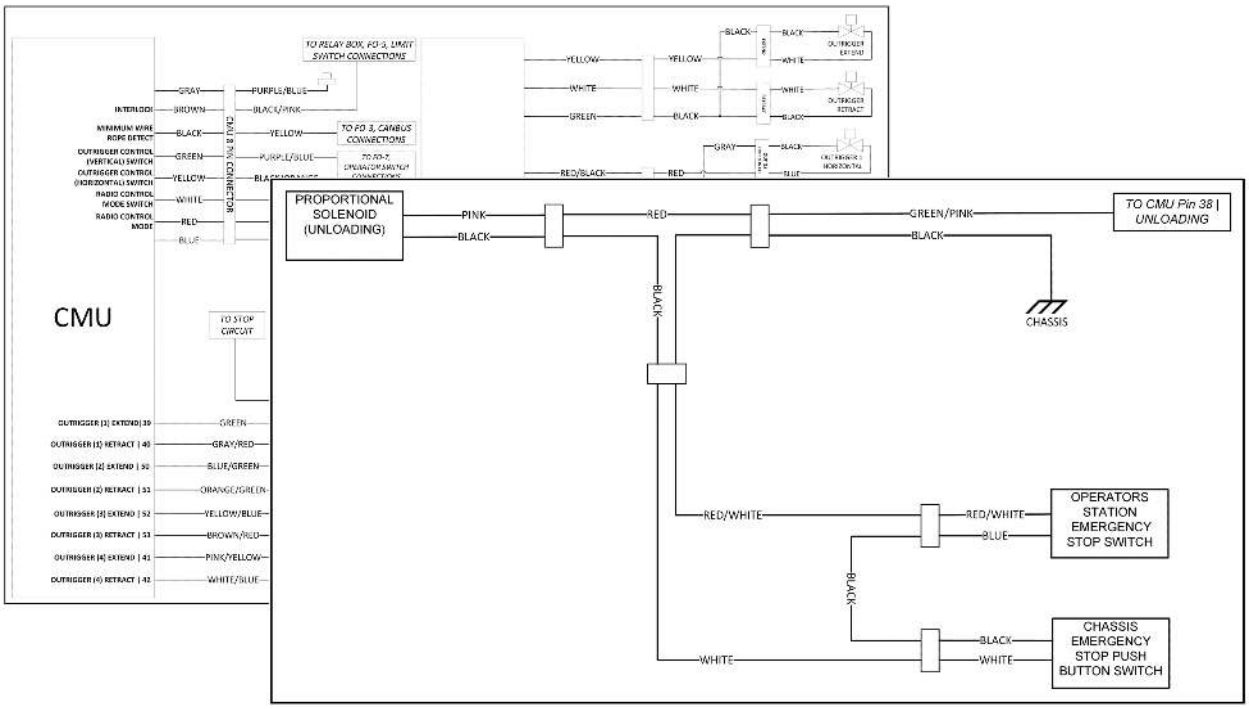


**FIGURE B-8. Theory of operation wiring diagrams.**

## B.7.4 General information work package (aircraft preventive maintenance services manual or preventive maintenance daily manual only) <pms-ginfowp>.

This WP shall be prepared for aircraft preventive maintenance services manual or preventive maintenance daily manual only and shall contain the content requirements provided in B.7.4.1 and B.7.4.2. This general information WP components are listed below:

**a.** Work package identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** Initial setup **<initial_setup>** is not required for this WP.

**c.** Maintenance activities **<scope>** shall be included.

**d.** General information **<pms-geninfo>** shall be included.

### B.7.4.1 Maintenance activities <scope>.

The following text within quotes shall be included verbatim (italicized text within parentheses shall be replaced with the appropriate information).

**"SCOPE**

The Preventive Maintenance Services Inspection Checklist WP contains complete requirements for a (*insert specific inspection interval(s) here*) for the (*insert specific equipment here*). It does not contain instructions for repair, adjustment, or other means of rectifying conditions, nor does it contain instruction for troubleshooting to find causes for malfunctioning. Specific tolerances, limits, etc., can be found in the applicable maintenance manuals."


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## B.7.4.2 General information <pms-geninfo>.

The following text within quotes shall be included verbatim (italicized text within parentheses shall be replaced with the appropriate information).

**"INSPECTION REQUIREMENTS**

The inspection requirements contained in this WP are stated in such a manner as to establish when certain equipment is to be inspected and what conditions are desired/undesired. Compliance with the provisions outlined herein is required in order to ensure that latent defects are discovered and corrected before malfunctioning or serious trouble results. Inspection requirements are arranged, as nearly as possible, according to the manner in which they will be performed. The requirements are divided into groups and listed under the area heading in the "About this manual" portion of this manual and Figure (*insert figure number here*).

**INSPECTION INTERVALS**

The (*insert inspection interval here*) inspection will be performed every (*insert the specific aircraft hours here*) flight hours or (*insert specific calendar days here*) days, whichever comes first. The (*insert the specific aircraft hours here*) will not be extended except in actual operational emergencies. In no case shall the aircraft intentionally be scheduled for a flight that will cause it to exceed the (*insert the specific aircraft hours here*) inspection due time. The (*insert specific calendar days here*) interval is a full (*insert the number of weeks here if applicable*) weeks. That is, if a (*insert specific calendar days here*) is done on Tuesday, the next (*insert specific calendar days here*) days inspection will not be due until (*insert the specific day here*) (*insert the specific number of weeks here*) later.

**SPECIFIC NON-INSTALLED EQUIPMENT ON AIRCRAFT**

This WP may contain inspection requirements applicable to specific equipment not installed on your aircraft. Those requirements should be disregarded.

**DA FORMS**

DA Form 2408-13-1 will be used to record all deficiencies or shortcomings discovered during the (*insert specific inspection interval here*). Use DA Pam 738-751 to properly complete this form.

**SPECIAL INSTRUCTIONS**

The (*insert inspection interval here*) will not be exceeded except in actual operational emergencies. When operational emergencies require aircraft operation beyond the normal inspection due-time, a circled red "X" status symbol and an appropriate statement (to include authority) must be entered in Part I, Fault Information block of DA Form 2408-13-1 (Aircraft Inspection and Maintenance Record) until such time as the inspection is complete.

When inspections are delayed to meet emergency requirements, commanders will ensure that the aircraft status symbol reverts to a red "X" and that delayed inspections are accomplished immediately upon termination of the actual emergency. When unusual local conditions of environment, use, mission, experience of flight crew and maintenance personnel, periods of inactivity, etc., are encountered; the maintenance officer will, at his discretion, increase the scope and/or frequency of maintenance of inspections as necessary to ensure safe flight.

Aircraft that are down, Not Mission Capable Supply (NMCS), or Not Mission Capable Maintenance (NMCM), are deferred from the (*insert inspection interval here*) inspection until the aircraft is returned to flyable status. When the NMCS and/or NMCM condition is cleared from the aircraft that has been deferred, the (*insert inspection interval here*) must be done before the first flight. It is the maintenance office's responsibility to determine those inspections necessary during NMCS and/or NMCM to preserve the aircraft. Maintenance situations and climates vary too much to permit a definition of an adequate inspection of the aircraft in NMCS and/or NMCM status.

Accessing procedures and detailed inspection criteria can be found in the applicable maintenance manuals. Unless otherwise directed, removed panels and opened doors will be reinstalled and closed upon completion of each area inspection.

The total man-hour (M/H) requirements for a complete (*insert inspection interval here*) inspection is (*insert total number of man-hours here*) M/H.

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**REPORTING ERRORS AND RECOMMENDING IMPROVEMENTS**

You can help improve this TM. If you find any mistakes or if you know of a way to improve the procedures, please let us know. Mail the DA Form 2028 directly to: ( *insert mailing address*). You may also send in your recommended changes using electronic mail, by fax, or by the World Wide Web. Our fax number is (*insert DSN and commercial number of proponent*). Our email address is (*insert email address of proponent*) . Instructions for sending an electronic DA Form 2028 may be found at the back of the applicable technical manual. For World Wide Web, use https://amcom2028.redstone.army.mil. A reply will be furnished to you.

**OZONE DEPLETING CHEMICALS**

See necessary Materials Safety Data Sheet (MSDS) for chemical types. Insert appropriate ODC statement as necessary. *Insert appropriate ODC statement as necessary.*

**HAZARDOUS MATERIALS (HAZMAT**

See necessary MSDS for hazardous materials. Insert appropriate ODC statement as necessary. *Insert appropriate HAZMAT statement here as necessary.*

**INSPECTION AREAS**

Inspection areas are shown in (*enter WP's title and figure number*) . "

## B.7.5 General information work package (aircraft phased maintenance inspection manual only <pm-ginfowp>.

This WP shall be prepared for preventive maintenance inspection manuals and shall contain the content requirements provided in B.7.5.1 and B.7.5.1.1. This general information WP components are listed below:

**a.** WP identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** Initial setup **<initial_setup>** is not required for this WP.

**c.** General information **<geninfo>** (B.7.5.1) shall be prepared.

## B.7.5.1 Phased schedule.

One of the following shall be included verbatim as applicable (italicized text within parentheses shall be replaced with the appropriate information):

**"PHASED SCHEDULE**

The phased maintenance inspection checklist contains requirements for inspection of the *insert aircraft model* aircraft on a phased schedule having a (*insert flight hour cycle*) hour (*flight hours*) cycle with (*insert phase hours*) hour phases. Each requirement included herein is designated for accomplishment at least once, but not more than (*insert number of phases*) times during the (*insert flight hour cycle*) hour cycle."

or

**"PROGRESSIVE PHASED MAINTENANCE SCHEDULE**

The progressive phased maintenance inspection checklist contains requirements for inspection of the (*insert aircraft model*) aircraft on a phased schedule of (*insert inspection interval hour intervals.*)".

## B.7.5.1.1 Additional general information.

The following additional text shall be included verbatim (italicized text within parentheses shall be replaced with the appropriate information):


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**"EXCEEDING THE PHASED SCHEDULE**

The phased maintenance inspection intervals designated are the maximum and shall not be exceeded except in actual operational emergencies as explained herein. It is the Commander's responsibility to determine (on an individual aircraft basis) when inspection intervals may be exceeded. For this purpose, operational emergencies are conditions of combat, or conditions of disaster that necessitate flight to evacuate aircraft or personnel. When air- craft are operated beyond the normal inspection due time because of such emergency situations, a circled red "X" status symbol and an appropriate statement (to include authority) must be entered on the appropriate aircraft form as specified in DA Pam 738-751 until such time as the inspection is complete. When inspections are delayed to meet emergency requirements, Commanders will ensure that the aircraft status symbol reverts to a red "X" and that delayed inspections are accomplished immediately upon termination of the actual emergency. When unusual local conditions (use, type of mission, personnel, periods of inactivity, environmental conditions, etc.) dictate, it is the prerogative and responsibility of the Maintenance Officer to increase the scope and/or frequency of maintenance or inspection as necessary to ensure safe operation (TM 1-1500-328-23).

**MAINTENANCE ACTIVITIES**

The inspections prescribed by this checklist will be accomplished at specified phases by Aviation Maintenance Company (AMC) activities with assistance of Aviation Support Battalion (ASB) and Depot Maintenance activities when required. The inspection of the part/component is visual unless stated otherwise.

**LIMITATIONS**

The checklist does not contain instructions for repair, adjustment, or other means of rectifying conditions. Neither does it contain special tolerances, limits, or instructions for special troubleshooting to find causes for malfunctions. Such data will be obtained from the latest issue of the aircraft (*insert applicable aircraft technical manuals*) series Maintenance Manuals.

**CHANGEOVER TO THE PHASED MAINTENANCE SYSTEM**

Changeover shall be accomplished in accordance with instructions provided in (*insert appropriate TM/TB*) entitled, (*insert title*). The requirements of this TM/TB must be accomplished before implementation of Phase 1 inspection requirements specified in this checklist.

**PRE-INSPECTION MAINTENANCE TEST FLIGHT (MTF**

A pre-inspection MTF to duplicate non-hazardous equipment problems, determine unsatisfactory conditions, determine equipment operation problems, etc., is recommended before start of aircraft disassembly for phased maintenance inspection. However, the decision to perform the pre-inspection MTF shall be the responsibility of the unit Maintenance Officer.

**SPECIAL INSPECTIONS, CALENDAR INSPECTIONS AND LUBRICATION REQUIREMENTS**

Special inspections, calendar inspections, and lubrication requirements contained in (*insert applicable aircraft technical manual*) and those listed on the aircraft's DA Form 2408-18 shall be reviewed and accomplished in accordance with the "inspection due" requirements specified in those documents.

**TIME BETWEEN OVERHAUL (TBO AND RETIREMENT LIFE ITEMS CHECK**

Before the start of the applicable phased maintenance inspection, a check will be made of components and their remaining operating hours before removal. The latest issue of the aircraft's (*insert applicable aircraft technical manual*) and DA Form 2408-16 shall be referred to for a complete listing of components and their TBO and retirement life.

**USING THE PHASED INSPECTION CHECKLIST**

A new checklist shall be used each time phased maintenance is due on the aircraft. This checklist is arranged such that it can be separated by area and distributed to the maintenance crew. For use of the checklist, refer to DA Pam 738-751.

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**FINAL RECORDS CHECK**

After all corrective actions have been completed and following completion of the phased inspection, the Technical Inspector or designated supervisor shall verify that all applicable forms and records have been properly updated. All uncorrected faults shall be entered on applicable aircraft forms in accordance with DA Pam 738-751. A Final Records Checklist shall be used to ensure forms and records have been inspected for completeness and accuracy before release of the aircraft from the phased maintenance inspection. The Personal Identification (PID) of the inspector verifying the final records check shall be entered adjacent to the indicated form or record on the Final Records Checklist. The PID entered shall be registered on the Signature Sheet adjacent to that person's signature.

**MAINTENANCE OPERATIONAL CHECKS**

After the completion of any required corrective actions to any of the components of a functional system of the aircraft, maintenance operational checks (MOCs) shall be performed on that system to determine the effectiveness of the maintenance actions performed and to verify the proper operation of that system. These MOCs shall be performed in accordance with TM 1-1500-328-23. DA Form 2408-13-1 will be used to record and sign off on the MOC performed.

**MAINTENANCE TEST FLIGHT**

When all required inspections have been accomplished and initialed in accordance with the previously mentioned procedure, the Maintenance Test Flight (MTF) shall be performed in accordance with the requirements of (*insert applicable aircraft technical manuals*) and TM 1-1500-328-23 using the MTF form in the MTF TM.

**CHECKLIST DISTRIBUTION**

The completion of each phased maintenance inspection shall be recorded on applicable forms as prescribed by DA Pam 738-751. The signed checklist, together with all forms prescribed by DA Pam 738-751, shall be filed. Disposition shall be in accordance with DA Pam 738-751 or specific instructions in the applicable aircraft TM.

**INSPECTION AREAS**

(*Insert WP title and figure number*) reflects the inspection areas of the (*insert applicable aircraft mode*) aircraft. Those areas are titled as shown. Figure (*insert number*) shows the location of access doors and panels that require removal at various phased maintenance inspections.

**REPORTING ERRORS AND RECOMMENDING IMPROVEMENTS**

(*Insert appropriate reporting errors statement here.* )"

## B.8 NOTES.

The notes in Section 6 apply to this appendix.

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# APPENDIX C

# OPERATOR INSTRUCTIONS

## C.1 SCOPE.

This appendix establishes the technical content requirements for the preparation of operating instructions for major weapon systems, and their related systems, subsystems, equipment, assemblies, components, Shop Replaceable Units (SRUs), and Line Replaceable Units (LRUs). This appendix is a mandatory part of this standard. The information contained herein is intended for compliance.

**Note**

To be compliant with this standard, publications shall be developed within **<production>** (opening) and **</production>** (closing) root tags.

When preparing information in accordance with this appendix, all references shall be linked using the appropriate tags (**<xref>**, **<extref>**, etc.). Links shall not be typed as text without tagging.

### C.1.1 Content tailoring using Appendix A matrices.

Applicability of some requirements contained in this standard are the responsibility of the materiel developer (MATDEV) (refer to AR 750–1, AR 700–127) and shall be accomplished using matrices referenced in TABLE A-I. The applicability of requirements are also designated by statements including, but not limited to, the following:

**a.** "...as specified by the MATDEV."

**b.** "...as required by the MATDEV."

**c.** "Refer to the MATDEV for..."

**d.** "Contact MATDEV if..."

**e.** "The MATDEV shall determine..."

**f.** "...as needed."

#### C.1.1.1 Front and rear matter.

Front and rear matter requirements for documents discussed in this appendix shall be determined in accordance with TABLE A-II.

## C.2 APPLICABLE DOCUMENTS.

The applicable documents in Section 2 apply to this appendix.

## C.3 ACRONYMS AND DEFINITIONS.

The acronyms and terms defined in Section 3 of this standard apply to this appendix.

## C.4 OPERATING INSTRUCTIONS APPLICABILITY.

Requirements contained in this appendix are applicable to operators only. An explanation of applicable DA maintenance levels is provided in Section 3.

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## C.5 OPERATING INSTRUCTIONS FUNCTIONALITY.

The operating instructions module **<opim>** consists of a sequence of work packages (WPs) that define the functionality of equipment controls and indicators and provide instructions for authorized procedures in both normal and emergency conditions. Technical publications may have one singular operating instructions module (C.5.1) or multiple operating instructions modules (C.5.2) depending on system operating complexity. See TABLE C-I for a list of module WP components, and see TABLE C-II for a list of software operator instructions WPs discussed in APPENDIX G.

**TABLE C-I. Operator instructions WPs.**

| WP | WP Tag | Appendix Section |
|---|---|---|
| Description and use of controls and indicators WP [1] | **<ctrlindwp>** | C.7.1 |
| Operation under usual conditions WP [1] | **<opusualwp>** | C.7.2 |
| Operation under unusual conditions WP | **<opunuwp>** | C.7.3 |
| Emergency WP | **<emergencywp>** | C.7.4 |
| Decal/data plate guide WP | **<stowagewp>** | C.7.5 |
| On-vehicle equipment loading plan WP | **<eqploadwp>** | C.7.6 |

[1] WP type frequently duplicated in a module for effective information display.

**TABLE C-II. Software operator instructions WPs.**

| WP | WP Tag | Appendix Section |
|---|---|---|
| Functional display WP | **<functionaldisplaywp>** | M.7.6 |
| Security and privacy procedures WP | **<softsecprivwp>** | M.7.7.1 |
| Supervisory controls WP | **<softsuperctrlswp>** | M.7.7.2 |
| Power up and access software procedures WP | **<softpwraccesswp>** | M.7.7.3 |
| Software operation WP [1] | **<softoperwp >** | M.7.7.4 |
| Power down and exit software procedures WP | **<softpwrexitwp>** | M.7.7.5 |

[1] WP type frequently duplicated in a module for effective information display.


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### C.5.1 Singular operating instructions module in a technical publication.

The operating instructions module shall appear as the second module in an operator publication or a combined operator/maintainer publication and is broken into the WP types listed in TABLE C-I. Some WPs within this module may be duplicated to better present operator information and instructions. When there is a large number of operator WPs where separation into multiple modules would be beneficial to improve usability, the publication shall include multiple modules (C.5.2).

### C.5.2 Multiple operating instructions modules in a technical publication.

The operating instructions modules shall appear as the second, third, fourth, etc. modules in an operator publication or a combined operator/maintainer publication and include the WP types listed in TABLE C-I. Multiple operating instructions modules **<opim>** shall be used when a system has a significant number of controls and indicator WPs, operation under usual conditions WPs, and/or, possibly operation under unusual conditions WPs (FIGURE C-2). Organizing WPs into separate modules with separate category tags **<ctrlindcategory>**, **<opusualcategory>**, **<opunusualcategory>**, and **<othercategory>** improves usability. More specifically, with this organization, the user is easily able to identify which WPs belong to usual conditions, for example, and which WPs belong to unusual conditions without repetition of the "Operation Under Usual (or Unusual) Conditions – Instruction Title" titling format for every WP (see section C.5.2.1).

#### C.5.2.1 Operating instruction titling.

**Note**

Concepts discussed in this section are suggested methods of titling operating instruction sections. Contact the EPCO for any titling-related inquiries.

In technical publications, some operating instruction WP types are frequently utilized numerous times – especially Control and Indicator WPs **<ctrlindwp>** and Operation Under Usual Conditions WPs **<opusualwp>**. The guidance in this section serves to improve the usability of operating instructions modules ETMs and IETMs.

##### C.5.2.1.1 WP titling format for large publications.

The important portion of a title ("– Instruction Title") shall be listed first (see FIGURE C-1). For small publications, the structure shown in FIGURE C-3 may be used. However, for large publications with numerous operator instructions WPs, usability is significantly impacted by this title formatting. For these large publications (ETMs and IETMs), WPs may be organized in separate modules (structure shown in FIGURE C-4) with qualifying text removed. For example, "OPERATION UNDER USUAL CONDITIONS - STAND ALONE LARGE (SAL) ORDER OF ASSEMBLY" and "OPERATION UNDER USUAL CONDITIONS - STAND ALONE LARGE (SAL) ORDER OF DISASSEMBLY" shall be titled "STAND ALONE LARGE (SAL) ORDER OF ASSEMBLY" and "SAL ORDER OF DISASSEMBLY" (FIGURE C-2, right side).

##### C.5.2.1.2 Operating instructions modules title pages.

Module title pages often aid usability for technical publications with multiple operating instructions modules. This is only beneficial for Electronic Technical Manuals (ETMs) that are PDFs (and may be printed). Interactive Electronic Technical Manuals (IETMs) may not include title pages for publications with multiple operating instructions modules.

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**Note**

One category tag **<operatorcategory>** may be used for technical publications with a singular operating instructions module. Also, figure appearance may vary slightly based on version of DTD/ stylesheets utilized (see applicable DTD or tag definition list for available tags).

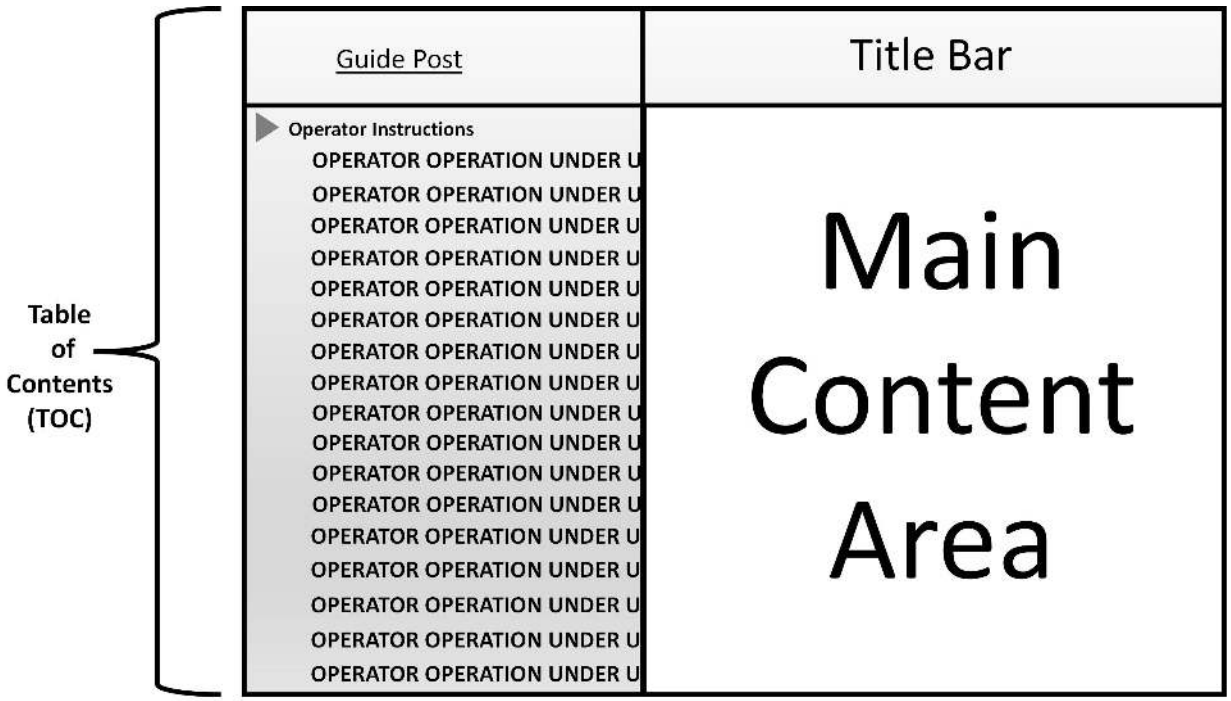


**FIGURE C-1. Screen display with one operator instructions module.**

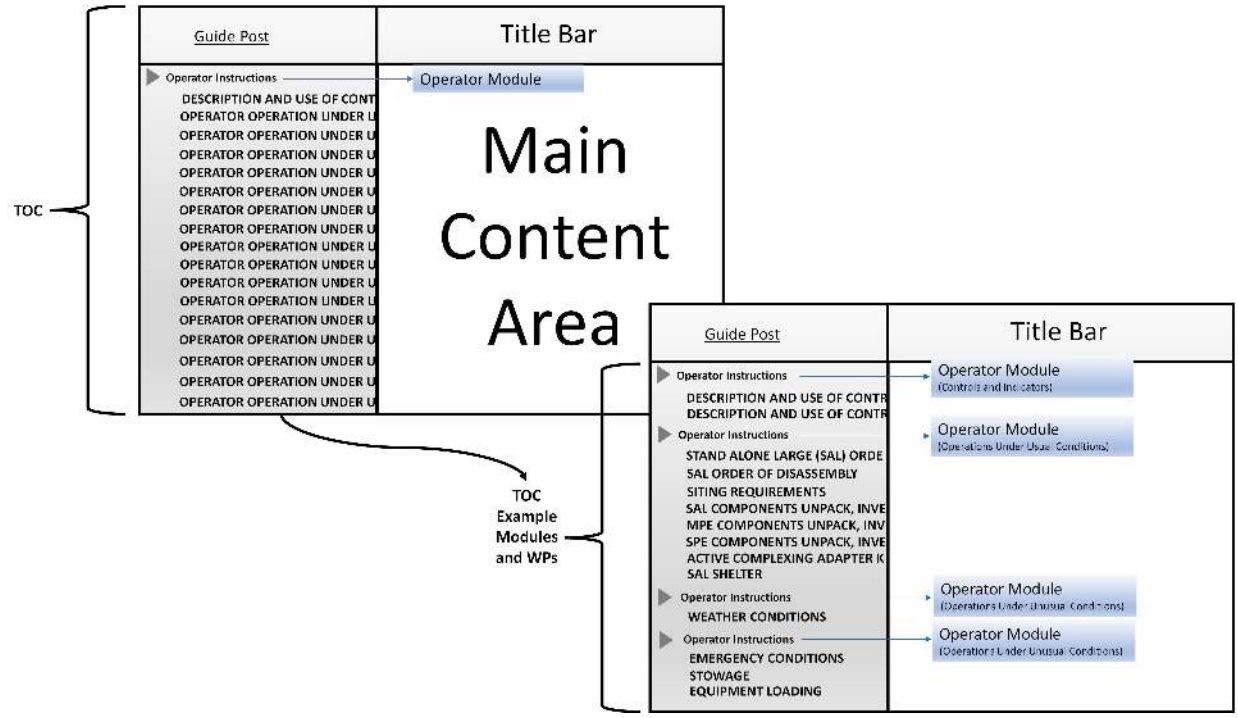


**FIGURE C-2. Screen display with multiple operator instructions modules.**



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## C.6 DEVELOPMENT OF OPERATING INSTRUCTIONS MODULES.

One or more operating instructions module(s) **<opim>** shall be prepared as required by publication type (see A.4.4). These modules shall consist of WPs identified in C.6.1 and C.6.2.

### C.6.1 Development of a singular operating instructions module in a publication.

The following WPs shall be prepared as required by publication type (see A.4.4) for a singular operating instructions module within **<operatorcategory>** category tags. Refer to FIGURE C-3 for a graphical representation of WP organization and tags discussed in this section.

**a.** Description and use of controls and indicators WP **<cntrlindwp>** (C.7.1).

**b.** Operation under usual conditions WP **<opusualwp>** (C.7.2).

**c.** Operation under unusual conditions WP **<opunuwp>** (C.7.3).

**d.** Emergency WP **<emergencywp>** (C.7.4).

**e.** Decal / data plate guide WP **<stowagewp>** (C.7.5).

**f.** On-vehicle equipment loading plan WP **<eqploadwp>** (C.7.6).

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**Note**

Tags/output shown in figure(s) below may vary slightly based on version of DTD/stylesheets utilized (see applicable DTD or tag definition list for available tags).

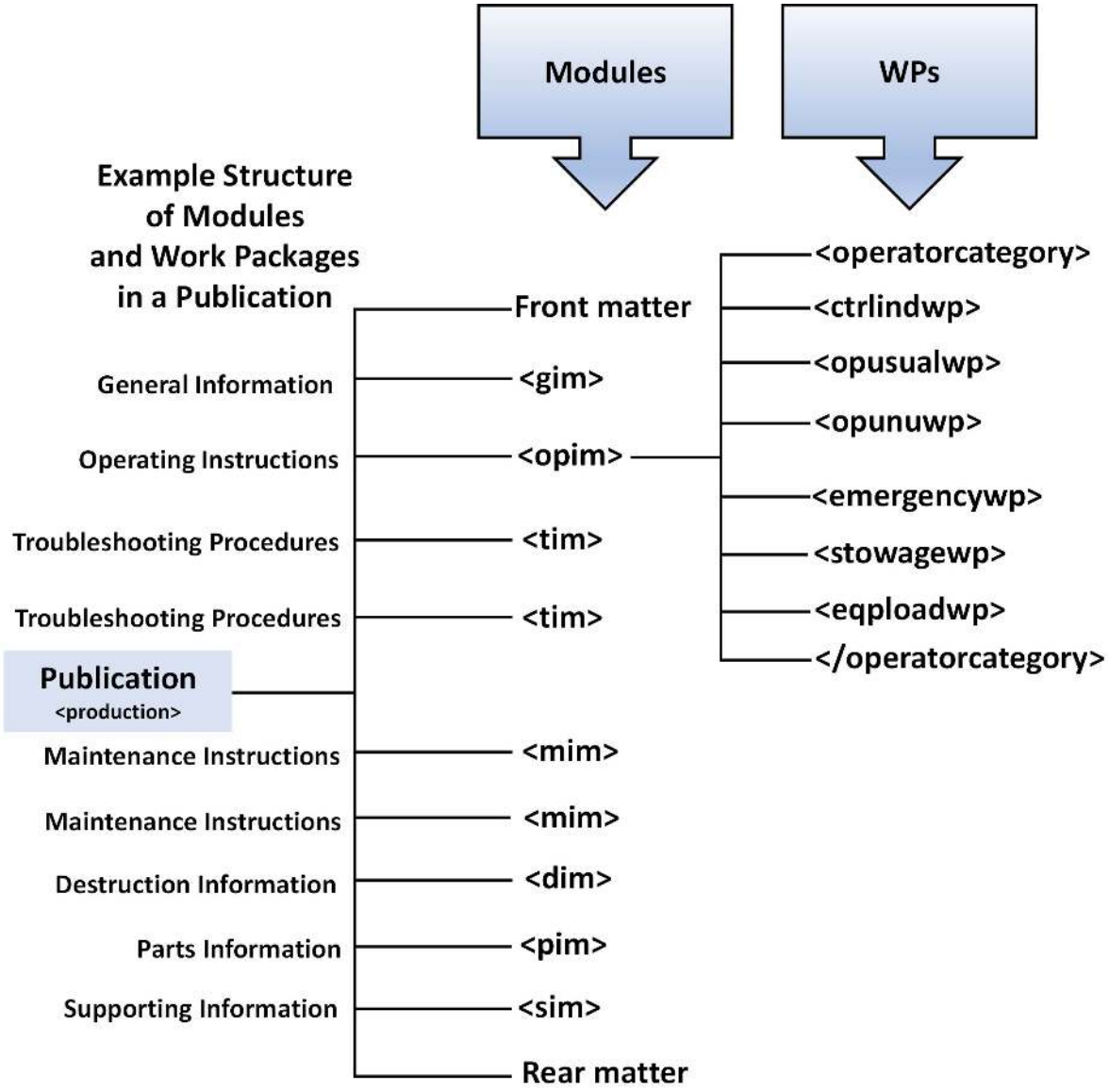


**FIGURE C-3. Operating instructions module and WPs (single <opim> module).**



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## C.6.2 Development of multiple operating instructions modules in a publication.

The modules and WPs listed in sections C.6.2.1 through C.6.2.4 shall be prepared as required by publication type (see A.4.4) for a technical publication with multiple operating instructions modules. Refer to FIGURE C-4 for a graphical representation of WP organization and tags discussed in this section.

### C.6.2.1 Description and use of controls and indicators module.

The following WPs shall be prepared for a description and use of controls and indicators module within **<cntrlindcategory>** category tags.

**a.** Description and use of controls and indicators WP **<cntrlindwp>** (C.7.1).

**b.** Additional description and use of controls and indicator WP(s) as needed.

### C.6.2.2 Operation under usual conditions module.

The following WPs shall be prepared for an operation under usual conditions module within **<opusualcategory>** category tags.

**a.** Operation under usual conditions WP **<opusualwp>** (C.7.2).

**b.** Additional operation under usual conditions WP(s) as needed.

### C.6.2.3 Operation under unusual conditions module.

The following WPs shall be prepared for an operation under unusual conditions module within **<opunucategory>** category tags.

**a.** Operation under unusual conditions WP **<opunuwp>** (C.7.3).

**b.** Additional operation under unusual conditions WP(s) as needed.

### C.6.2.4 Other operating instructions module.

The following WPs within **<othercategory>** category tags shall be prepared as required by publication type (see A.4.4) for a module containing other operating instructions.

**a.** Emergency WP **<emergencywp>** (C.7.4).

**b.** Decal / data plate guide WP **<stowagewp>** (C.7.5).

**c.** On-vehicle equipment loading plan WP **<eqploadwp>** (C.7.6).



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**Note**

Multiple category tags shall be used for technical publications with a multiple operating instructions modules. Also, tags/output may vary slightly based on version of DTD/stylesheets utilized (see applicable DTD or tag definition list for available tags).

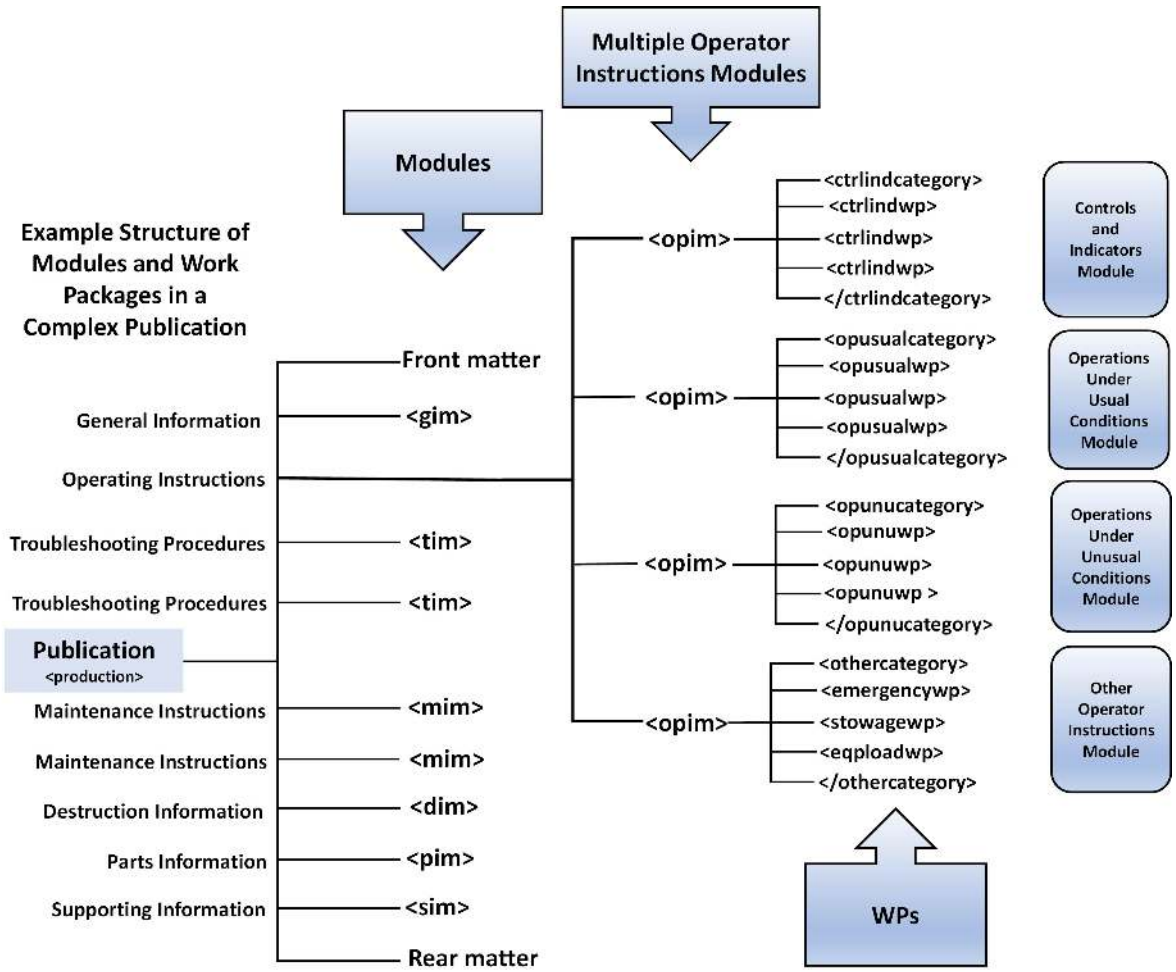


**FIGURE C-4. Operating instructions modules and WPs (multiple <opim> modules).**

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## C.7 DEVELOPMENT OF OPERATING INSTRUCTIONS WORK PACKAGES.

The following types of operator WPs shall be prepared as required by publication type (see A.4.4) and as applicable for an equipment system.

### C.7.1 Description and use of controls and indicators work package <ctrlindwp>.

Description and use of controls and indicators WPs shall be prepared for each equipment, assembly, or control panel having controls and indicators. Controls allow a user to input or "control" system data while indicators serve as a way to output or "indicate" system data, graphics, and information. This WP may also include information about ports and connectors as controls and indicators. Controls and indicators WPs shall contain graphic depictions of equipment and shall describe the equipment using a tabular option or a narrative option. The same format shall be used throughout the WP(s). FIGURE C-5 shows these two WP format options and the associated tags. Controls and indicators WP components are listed below:

**a.** WP identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** Initial setup **<initial_setup>** is not required for this WP.

**c.** Controls and indicators information **<ctrlindtab>** (C.7.1.1) or **<ctrlinddesc>** (C.7.1.2).

**Note**

Figure appearance and tags/output may vary slightly based on version of DTD/stylesheets utilized. Also, for the tabular option, figures and tables do not have to contain in-text references as is required for the rest of a technical publication.

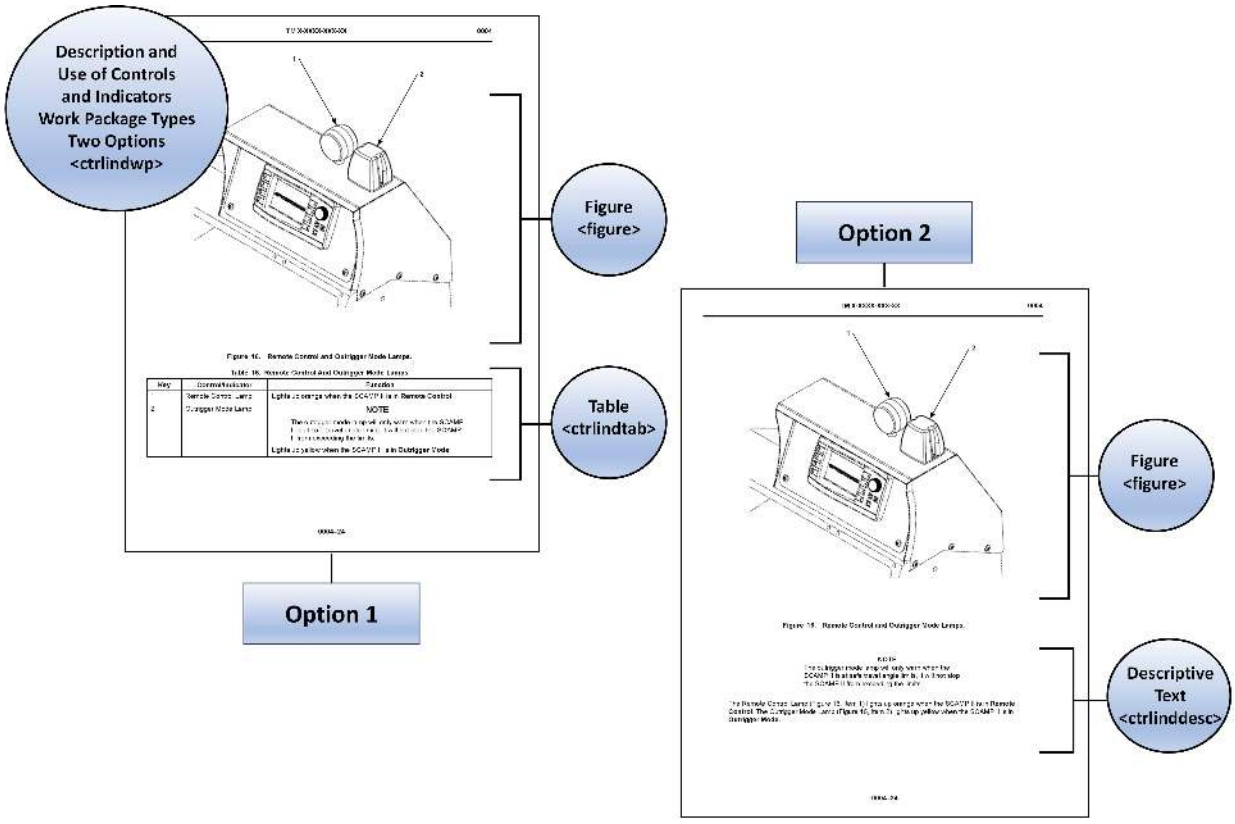


**FIGURE C-5. Controls and indicators WP options.**

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### C.7.1.1 Tabular option <ctrlindtab>.

This WP shall start with a short introduction **<intro>** that defines the basic system, area, or other breakdown. The introduction shall be followed by one or more controls and indicators tables **<ctrlindtab>** with the associated figure **<figure>** for each control and indicator.

**Note**

A **<ctrlind>**tag shall not be nested inside another **<ctrlind>** tag.

The number of controls and indicators tables depends on the following factors: system complexity, different users (crew members/stations) or configuration variations. Each control and indicator listed as a table entry must include the information listed below. More than one figure may be used to improve user understanding. FIGURE C-6 through FIGURE C-10 show an example tabular option WP and the associated tags.

**a.** Index Number **<key>** : Correlates to callouts depicted on a figure.

**b.** Name **<ctrl>** : Any controls and indicators that are not labeled, such as an accelerator or brake pedals, shall be identified. Each control and indicator shall be named as it appears on the equipment (for example, **POWER** button or **PWR** button).

**c.** Function **<function>** : Describes the function of the control and indicator.

### C.7.1.2 Description narrative option <ctrlinddesc>.

This option provides a narrative approach to describe each control and indicator. This textual approach shall begin with a figure **<figure>** illustrating the control and indicator that is being described. The figure shall be followed by one or more paragraphs **<ctrlinddesc>** describing each control or indicator show in the figure. The narrative option for controls and indicators shall contain the same items listed in C.7.1.1. More than one figure may be used to improve user understanding.


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**Note**

Figure appearance and tags/output may vary slightly based on version of DTD/stylesheets utilized (see applicable DTD or tag definition list for available tags).

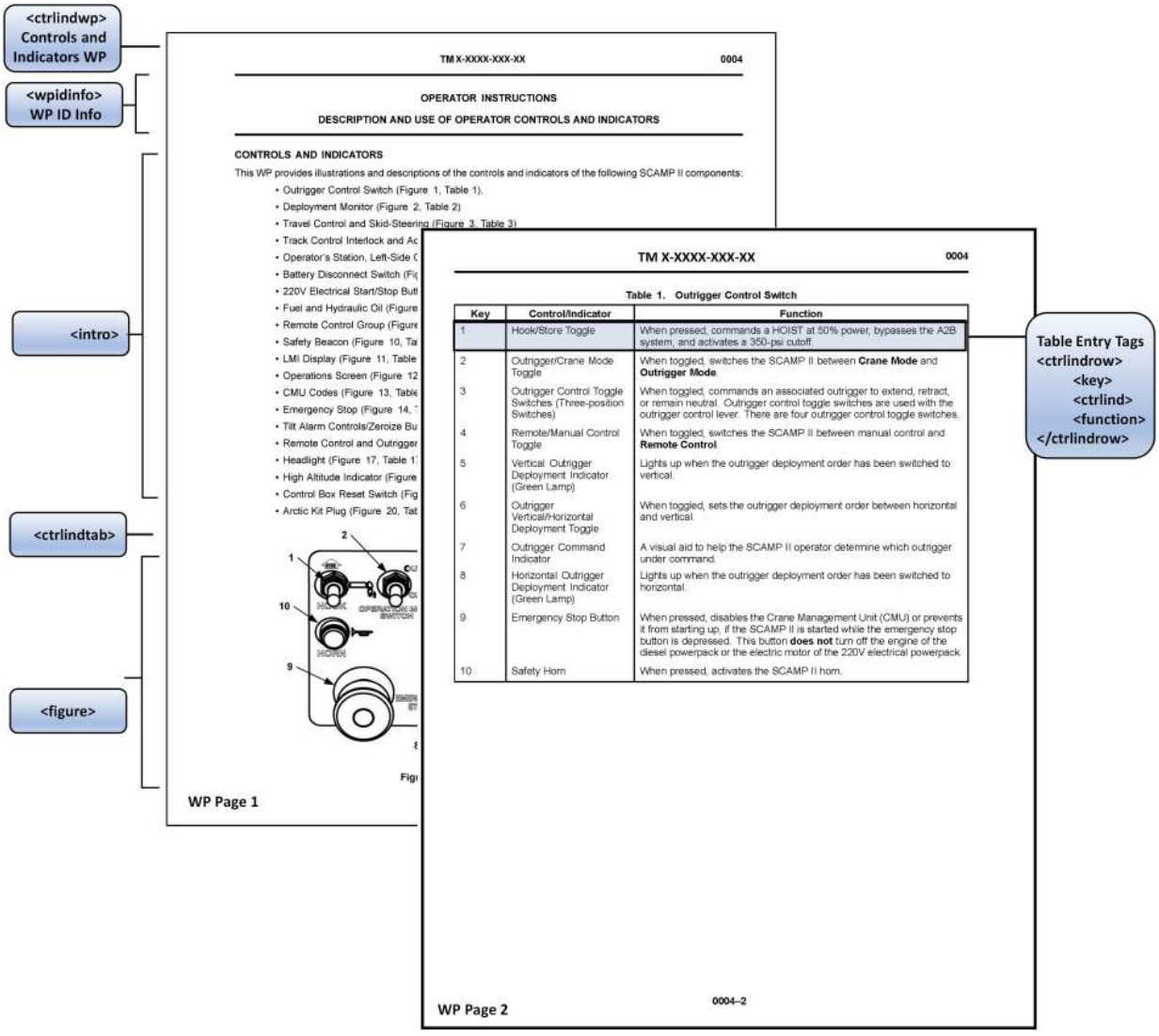


**FIGURE C-6. Controls and indicators WP (tabular option) (sheet 1 of 5).**

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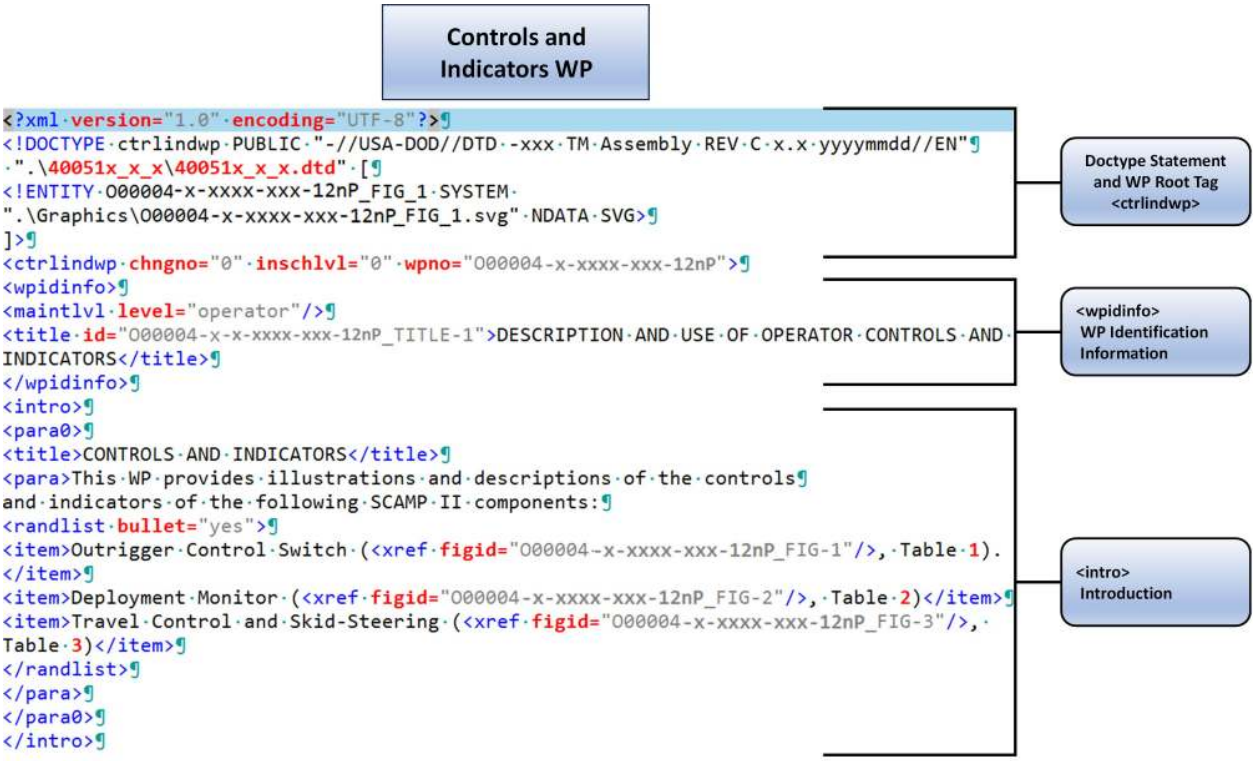


**FIGURE C-7. Controls and indicators WP (tabular option) (sheet 2 of 5).**

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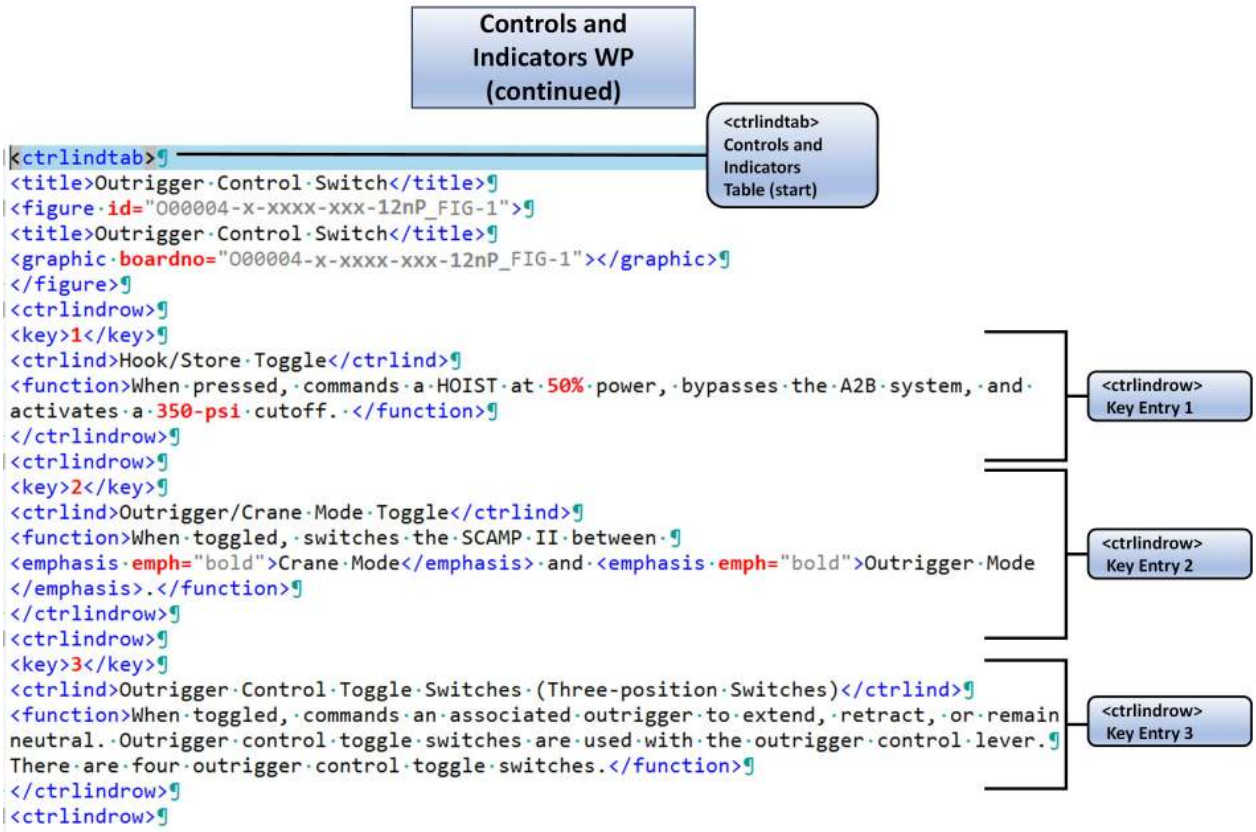


```
<ctrlindtab>¶
<title>Outrigger Control Switch</title>¶
<figure id="O00004-x-xxxx-xxx-12nP_FIG-1">¶
<title>Outrigger Control Switch</title>¶
<graphic boardno="O00004-x-xxxx-xxx-12nP_FIG-1"></graphic>¶
</figure>¶
<ctrlindrow>¶
<key>1</key>¶
<ctrlind>Hook/Store Toggle</ctrlind>¶
<function>When pressed, commands a HOIST at 50% power, bypasses the A2B system, and
activates a 350-psi cutoff. </function>¶
</ctrlindrow>¶
<ctrlindrow>¶
<key>2</key>¶
<ctrlind>Outrigger/Crane Mode Toggle</ctrlind>¶
<function>When toggled, switches the SCAMP II between ¶
<emphasis emph="bold">Crane Mode</emphasis> and <emphasis emph="bold">Outrigger Mode
</emphasis>.</function>¶
</ctrlindrow>¶
<ctrlindrow>¶
<key>3</key>¶
<ctrlind>Outrigger Control Toggle Switches (Three-position Switches)</ctrlind>¶
<function>When toggled, commands an associated outrigger to extend, retract, or remain
neutral. Outrigger control toggle switches are used with the outrigger control lever.¶
There are four outrigger control toggle switches.</function>¶
</ctrlindrow>¶
<ctrlindrow>¶
```

**FIGURE C-8. Controls and indicators WP (tabular option) (sheet 3 of 5).**

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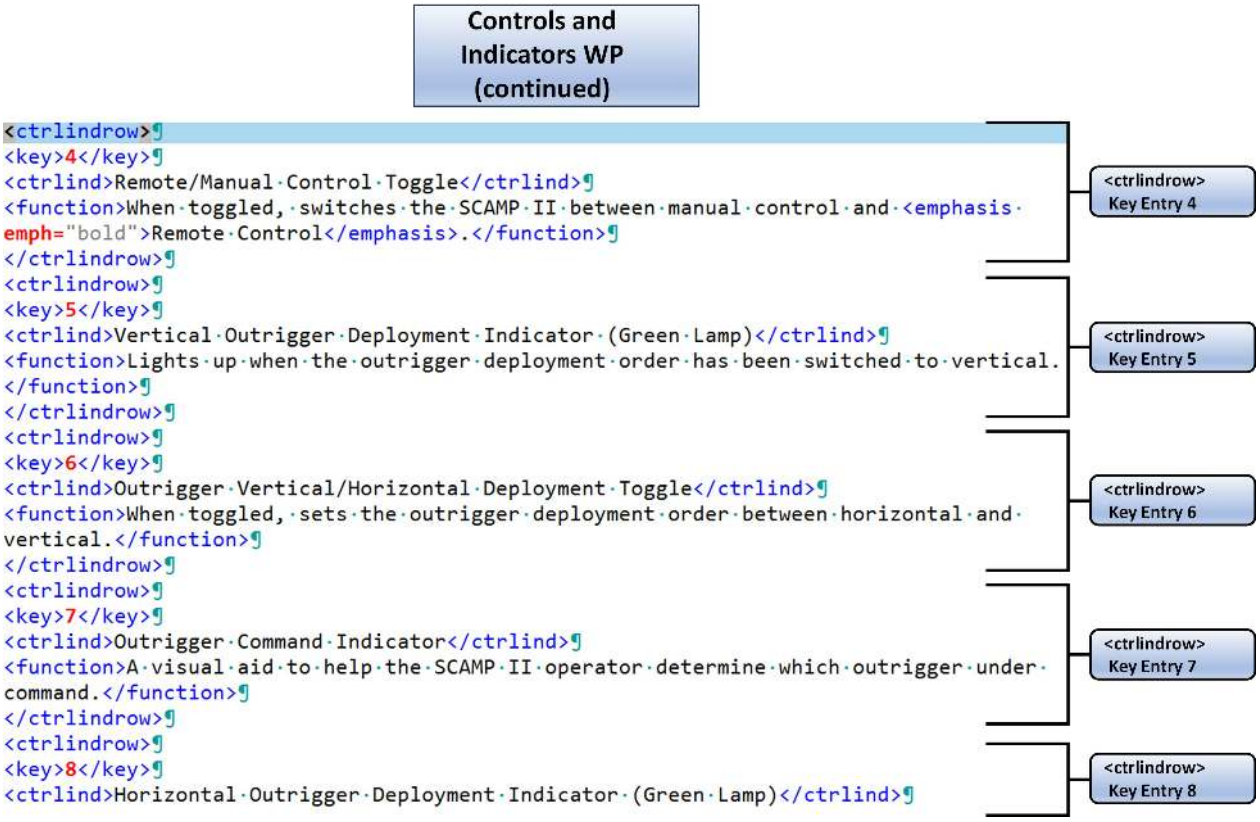


**FIGURE C-9. Controls and indicators WP (tabular option) (sheet 4 of 5).**

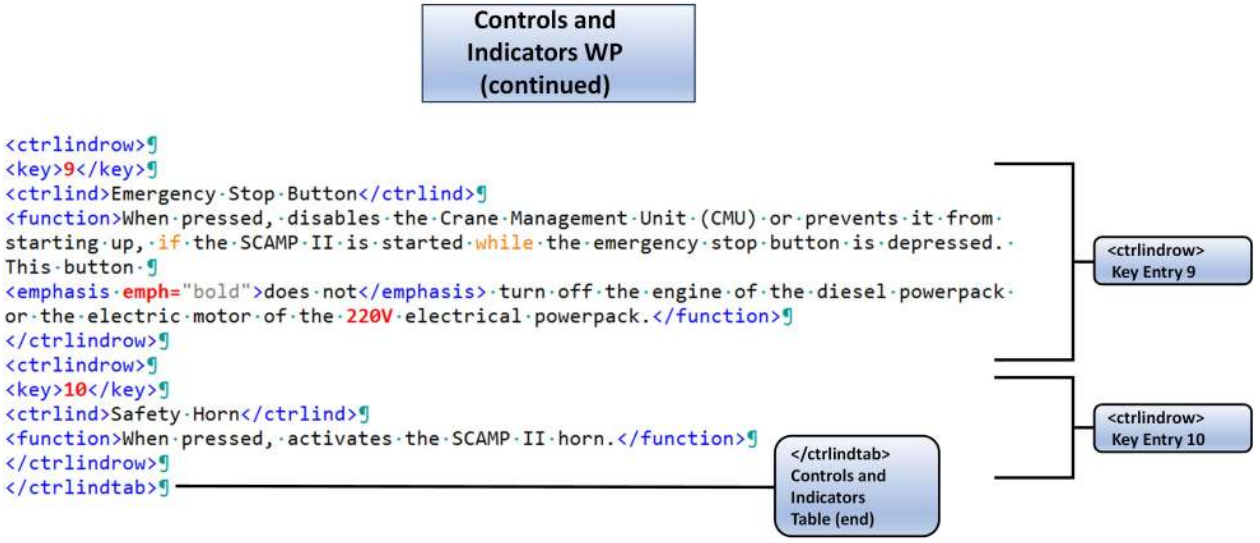


**FIGURE C-10. Controls and indicators WP (tabular option) (sheet 5 of 5).**

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## C.7.2 Operation under usual conditions work package <opusualwp>.

Instructions to operate the equipment system and auxiliary equipment in all modes of operation shall be prepared. Any combination of control settings that will create a hazard to personnel or cause damage to equipment shall be preceded by a warning or caution. Instructions to ensure proper grounding of equipment shall be prepared. If a technical publication has numerous operation under usual conditions WPs, the publication shall be constructed so that all **<opusualwp>** WP types are grouped together in one module (between **<opusualcategory>** tags) and separate from other operator WPs (see C.5.2). Note the organization of the remaining operation WP types into other modules. Operation under usual conditions WP components are listed below. Refer to FIGURE C-11 for an example of operation under usual conditions WP tagging structure.

**a.** WP identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** Initial setup **<initial_setup>** is required for this WP (5.4.3).

**c.** Operation task **<opertsk>** is required for each task included in this WP.

**d.** Task options listed in C.7.2.1 thru C.7.2.9 as applicable for an equipment system.

### C.7.2.1 Security measures for electronic data <secref>.

Instructions for handling, loading, purging, overwriting, or unloading classified electronic data under usual conditions shall be developed when the systems are classified, have non-volatile on-board memory that is required to be cleared prior to transportation, or for any other action that might compromise the data as the result of being accessed by unauthorized personnel. Instructions shall meet the requirements of current regulations as they pertain to automation security.

### C.7.2.2 Siting requirements <site>.

When siting instructions specific to the equipment exist, these instructions shall be prepared. Operational features shall be considered, such as the following:

**a.** Location.

**b.** Proximity to power sources.

**c.** Effective ranges.

**d.** Terrain requirements to avoid screening reflections, ground clutter, and other poor operational conditions due to terrain.

**e.** Technical requirements.

**f.** Shelter locations.

**g.** Compensating for adverse siting conditions.

**h.** Orientation to a baseline during siting when the equipment contains large components, such as towers and antennas.

**i.** Mobile equipment oriented during installation.

### C.7.2.3 Shelter requirements <shelter>.

When equipment is normally housed in a permanent or semi-permanent shelter (other than a military truck, van, or transportable shelter) during use, the following information shall be prepared:

**a.** Amount of floor, wall, and height space required to house the equipment.

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**b.** A plan for a typical layout.

**c.** Required weight capacity of the building floor.

**d.** Dimensions required for installed equipment.

**e.** Total weight that the floor must support and the area in square feet over which the total weight will be distributed.

**f.** Environmental conditions (for example, venting).

**g.** Power requirements.

**h.** Unusual requirements specific to the equipment, such as air-conditioning.

**i.** Architectural and engineering data on beam sizes, lengths, bending moments, and required supports shall not be included.

### C.7.2.4 Assembly and preparation for use <prepforuse>.

Procedures shall be prepared when unpacking, assembly, and installation is required. When the equipment is shipped or delivered in specially designed containers, unpacking instructions shall be prepared. If the containers are to be used again, kept for future use, turned in to supply, or if any special disposition is required, the necessary procedures shall be prepared. Assembly and installation procedures shall be prepared when needed. These instructions shall be supported by illustrations.

### C.7.2.5 Initial adjustments before use and self-test <initial>.

Procedures shall be prepared for any routine checks, self-test, or adjustments that the operator must make before putting the equipment in operation is required.

### C.7.2.6 Operating procedures <oper>.

The operating instructions listed below shall be prepared, as applicable. FIGURE C-11 shows an example operation under usual conditions WP, and the associated tags.

**a.** All steps necessary to bring the equipment from OFF through STANDBY condition to full operation, including all necessary warnings and cautions.

**b.** Procedures for each mode of operation, for example, manual, automatic, local, remote, etc. The use and relative advantage of each mode shall also be described.

**c.** Description of the equipment's anti-jamming and interference reduction features, the advantage of each feature, and the operating procedures to be followed. Supporting illustrations (such as indicator displays, waveforms, etc.) that provide typical observations of jamming and interference for evaluation by the operator shall be included.

**d.** Operator turn-off procedures, including all steps necessary to bring the equipment from full operation through STANDBY to OFF condition.

**e.** Operating procedures for malfunctions (for example, misfire, hangfire, etc.) applicable to ammunition.

**f.** Operating procedures explaining how the equipment is operated in conjunction with auxiliary equipment or how it operates when integrated with other equipment.

**g.** When specified by the MATDEV, operating procedures containing the identification, loading, initializing, and downloading of applicable operational and diagnostic software shall be included. Identification of the software shall include the purpose, configuration applicability, and version information. Procedures that verify that the

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proper software has been loaded and is operating properly shall also be included. Examples of specific types of data that may be applicable to these WPs are:

**(1)** Descriptions of screen data and interpretation of message formats.

**(2)** Operator actions based on screen display.

**(3)** Data entry by the operator.

**(4)** Saving or purging data.

**(5)** Processing of messages.

**(6)** Software transfer procedures.

**(7)** Reviewing message and entry formats.

**Note**

Figure appearance and tags/output may vary slightly based on version of DTD/stylesheets utilized (see applicable DTD or tag definition list for available tags).

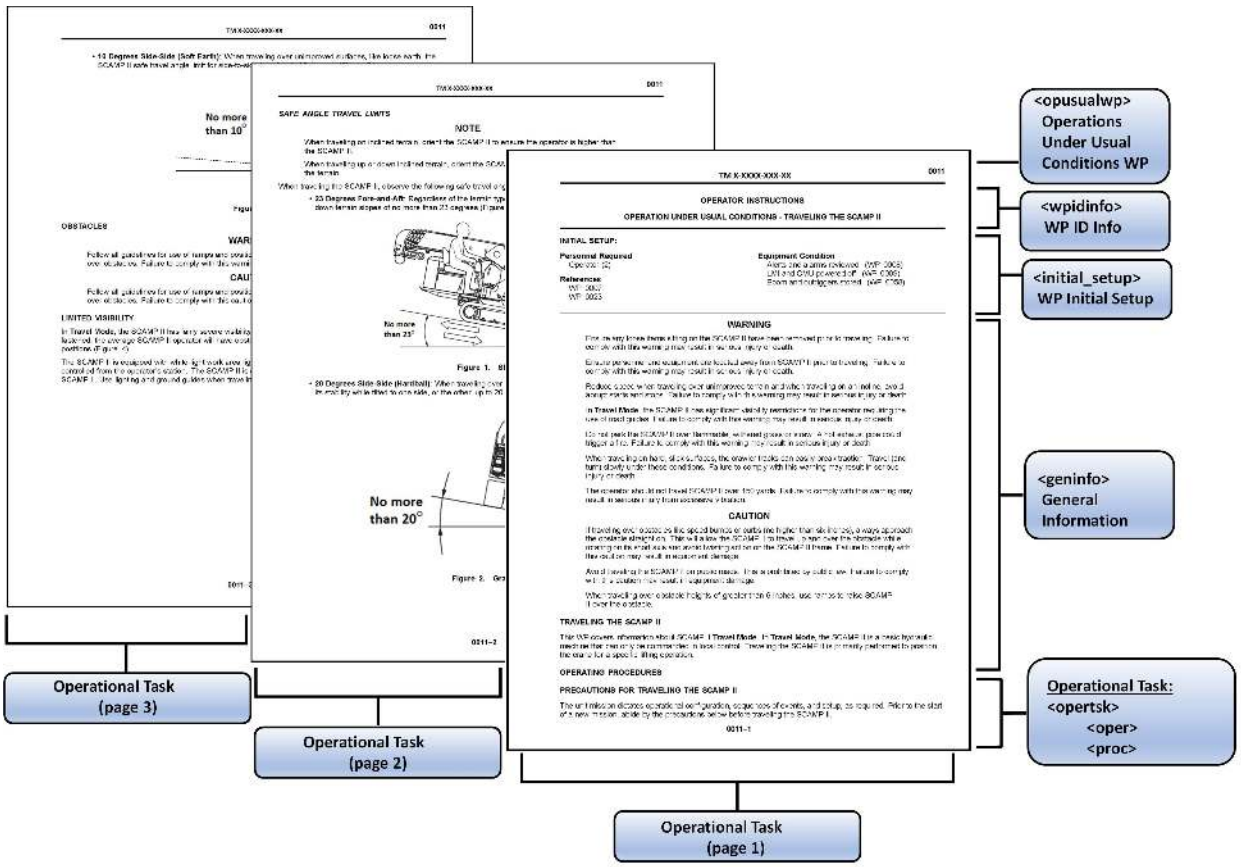


**FIGURE C-11. Operation under usual conditions WP.**

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### C.7.2.7 Operating procedure considerations.

The following considerations should be considered when preparing operating procedures:

**a.** Inclusion of initial safety requirements (actions, inspections, and emergency power off, etc).

**b.** Installation of required accessory equipment.

**c.** Level of repair assignment for a procedure.

**d.** Confirmation of inputs such as power, coolant, air, signal, etc. (including any required values).

**e.** Development of adjustment procedures prior to equipment power-on procedures.

**f.** Development of procedures to determine operational readiness and associated indicates from Built-In-Test (BIT) indicators such as lamps, gauges, displays, recorder readouts, etc.

**g.** Incorporation of status milestones in operational status in a procedure (i.e., "The generator is now in "STANDBY.")

**h.** Inclusion of visual or audible observations that occur as a result of operator action (i.e., boom lowering, blower motor running, etc.).

### C.7.2.8 Operating auxiliary equipment <operaux>.

If applicable, procedures shall be prepared for equipment that is not included in a system's equipment listing (or "end item"). This equipment is referred to as "auxiliary equipment." As necessary, procedures shall be developed for performing any initial adjustments and power on for auxiliary equipment, operating auxiliary equipment, and switching auxiliary equipment to standby or shutdown. If any of these procedures are published in another technical publication, references to that publication shall be included in accordance with general requirements section "other documents" **<extref>**.

### C.7.2.9 Preparation for movement <prepmove>.

Preparation for movement procedures shall be prepared if the equipment is designed for movement and it can be readied for movement by the operator. Procedures shall be prepared for actions such as disassembly, folding, and telescoping. Illustrations shall be prepared, as required, to support the text. This information shall not duplicate the "assembly and preparation for use" requirements contained in C.7.2.4.

## C.7.3 Operation under unusual conditions work package <opunuwp>.

Instructions shall be prepared for operation under unusual conditions. Preventive or protective measures to be taken beyond the operator capabilities shall be identified. Instructions to ensure proper grounding of equipment shall be prepared, as applicable. If there are no requirements related to operation under unusual conditions, this WP shall be listed in WP 0001, General Information as a WP not included in the publication (see B.7.1.1). Operation under unusual conditions WP components are listed below. Refer to FIGURE C-12 and FIGURE C-13 for an examples of operation under unusual conditions WP tagging structure. FIGURE C-12 depicts this WP in IETM format.

**a.** WP identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** Initial setup **<initial_setup>** is required for this WP (5.4.3).

**c.** Task options listed in C.7.3.1 thru C.7.3.6 as applicable for an equipment system.



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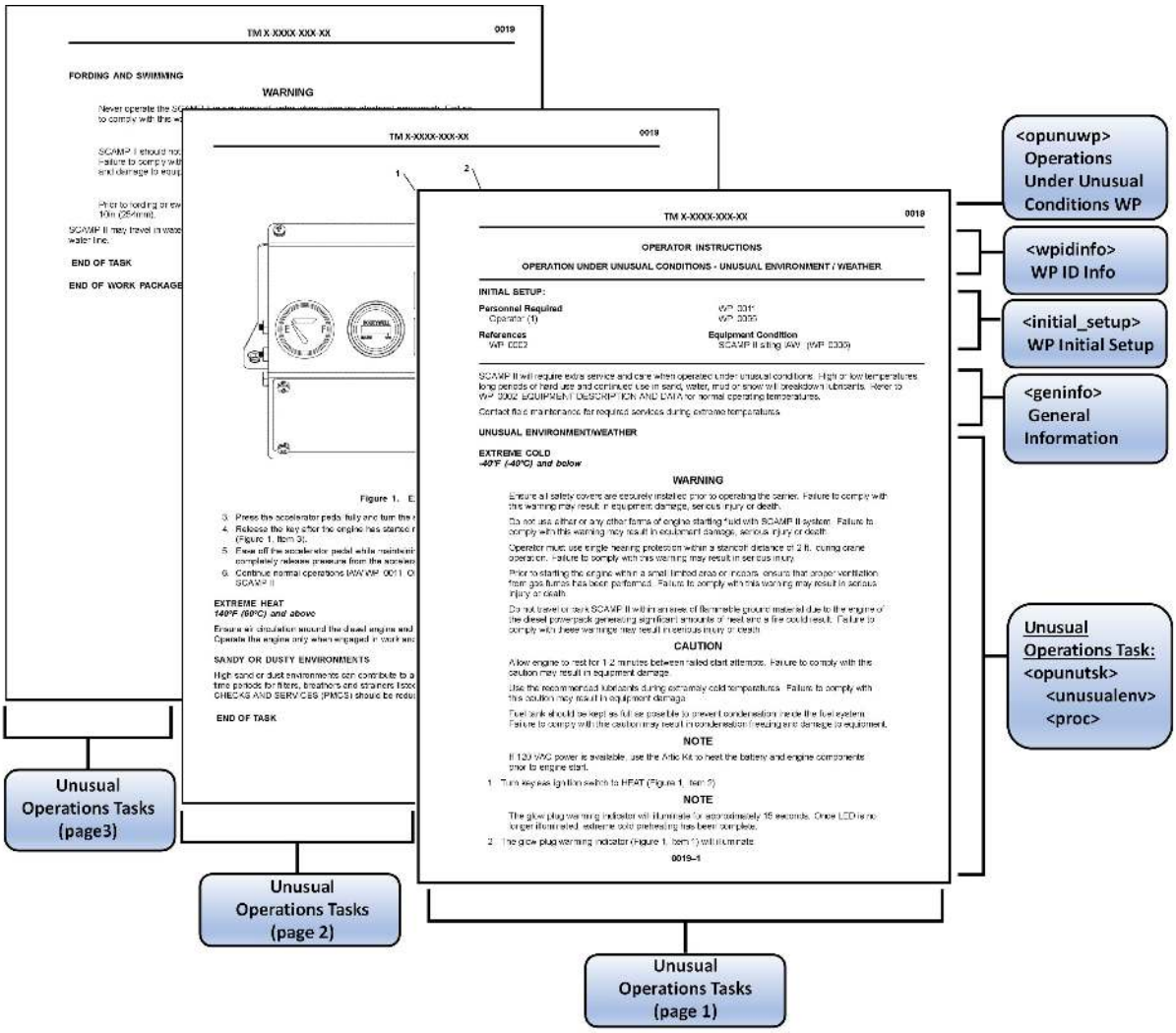


**FIGURE C-12. Operation under unusual conditions WP (sheet 1 of 2).**

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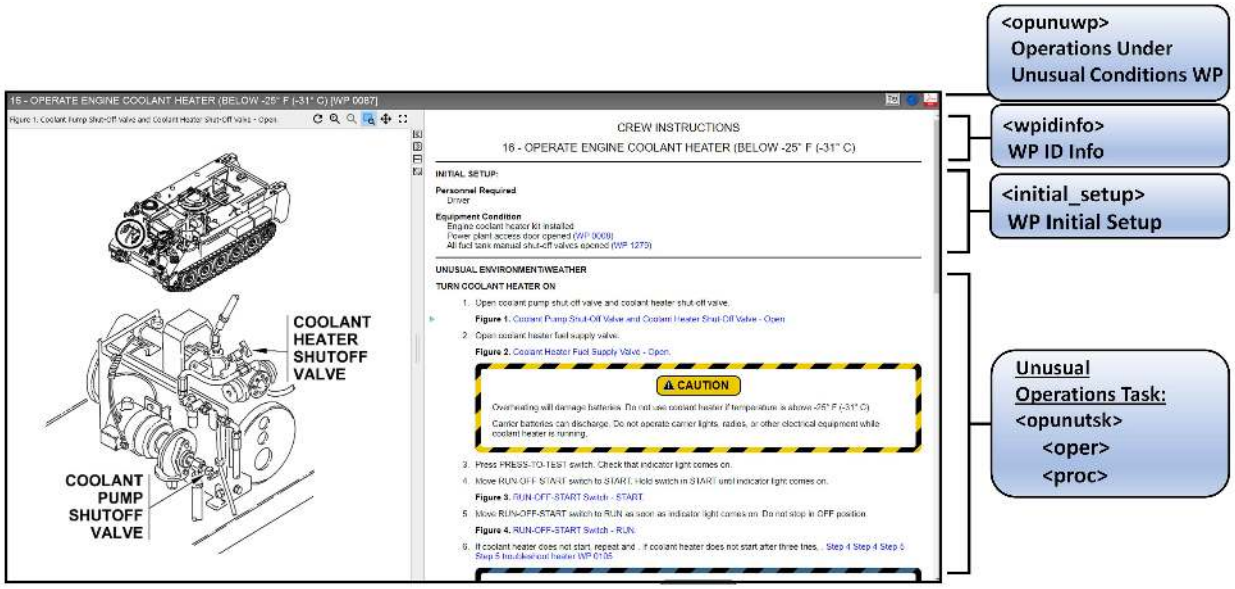


**FIGURE C-13. Operation under unusual conditions WP (sheet 2 of 2).**

## C.7.3.1 Security measures for electronic data <secref>.

Instructions for handling, loading, purging, overwriting, or unloading classified electronic data under unusual conditions shall be provided. These instructions shall be developed when the systems are classified. Instructions shall meet the requirements of current regulations as they pertain to automation security. Procedures shall include but are not limited to:

**a.** Clearing non-volatile on-board memory that is required to be cleared before transport.

**b.** Any other action that allows the data to be accessed by unauthorized personnel.

## C.7.3.2 Unusual environment/weather <unusualenv>.

Procedures shall be prepared for operation under conditions of extreme moist heat, extreme dry heat, extreme cold, salt air, sea spray, dust storms, sandstorms, high altitudes, snow, mud, and other similar adverse environmental/ weather conditions. Ranges of environmental/weather operating conditions considered for the system addressed shall be defined.

## C.7.3.3 Fording and swimming <fording>.

If applicable, procedures for fording and swimming the equipment shall be provided.

## C.7.3.4 Interim Chemical, Biological, Radiological, Nuclear and Explosives (CBRNE) decontamination procedures <decon>.

As applicable and specified by the MATDEV, general CBRNE decontamination procedures to be performed until CBRNE decontamination facilities are available shall be prepared. Decontamination procedures shall be prepared so that mission completion is emphasized; therefore, interim decontamination procedures shall be developed for period of time between initial contamination and availability to CBRNE decontamination facilities. Other decontamination publications shall be referenced only when necessary.

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### C.7.3.5 Jamming and Electronic Countermeasures (ECM procedures <ecm>.

As applicable, procedures shall be prepared for operation of the equipment in an ECM environment through transmitted and reflected deception signals and through transmitted and reflected jamming.

### C.7.3.6 Degraded operation procedures <degraded>.

When operation of the equipment in a degraded condition is required, procedures shall be prepared for temporarily adapting the equipment and the operating procedures to meet the reduction of power, partial failure, failure of a portion of the equipment, or similar conditions.

## C.7.4 Emergency work package <emergencywp>.

Emergency procedures for, but not limited to, operating, and shutting down equipment during emergency conditions shall be prepared as applicable for an equipment system. Emergency WPs shall be marked as specified in 4.15.10.6.3. See FIGURE C-14 for an example emergency WP and the associated tagging structure. Emergency WP components are listed below.

**a.** WP identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** Initial setup **<initial_setup>** is required for this WP (5.4.3).

**c.** Emergency task options listed in C.7.4.1 thru C.7.4.3 as applicable for an equipment system.

### C.7.4.1 Emergency operation <emergency>.

Procedures covering operation of the equipment during emergency conditions (control failure, air failure, lube oil failure, loss of cooling water, etc.) shall be provided. Emergency operating instructions shall be included. A warning or a caution to return the equipment to proper operation when the emergency is over shall also be included.

### C.7.4.2 Emergency shutdown <emergency>.

Procedures to turn the equipment off during an emergency (fire, water, smoke, hazard to personnel, loss of coolant, normal power, etc.) shall be provided.

### C.7.4.3 Vehicle recovery.

For vehicle manuals, information related to vehicle recovery and towing shall be included in the emergency WP.

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**Note**

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Contact the Equipment Publications Control Office (EPCO) with any border display issues.

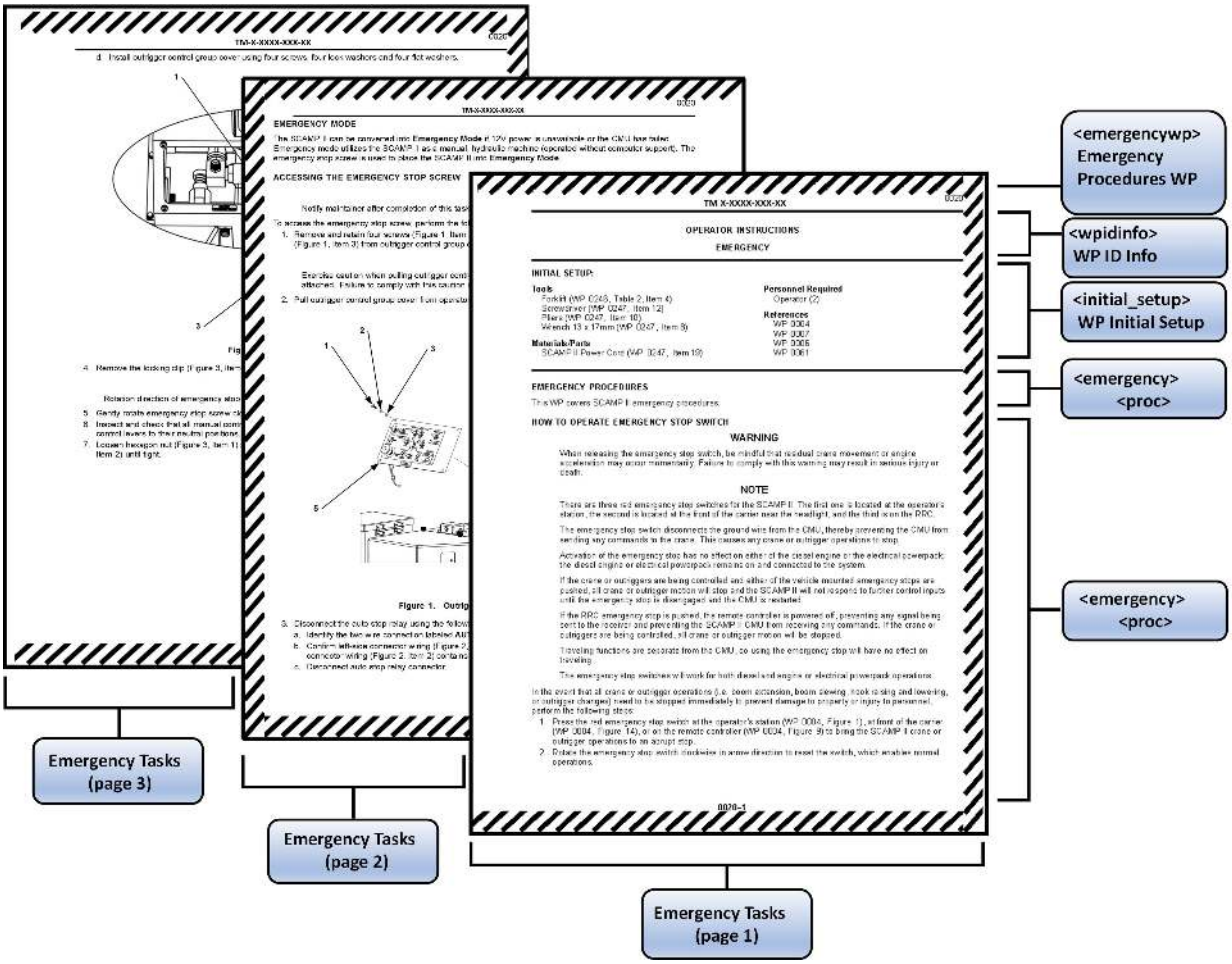


**FIGURE C-14. Emergency operation WP.**

## C.7.5 Decal / data plate guide work package <stowagewp>.

This WP shall be prepared as directed by the MATDEV. The guide plan shall include information provided by the MATDEV. Refer to FIGURE C-15 through FIGURE C-17 for an example of a decal / data plate guide WP and the associated tagging structure. Decal/data plate guide WP components are listed below.

**a.** WP identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** Initial setup **<initial_setup>** is not required for this WP (5.4.3).

**c.** A brief scope statement **<intro>** shall be prepared explaining the purpose of the WP.

**d.** Decal / data plate guide **<decalinfo>** shall be prepared identifying all major decals and data plates (C.7.5.1).

**e.** Stowage information **<stowinfo>** may be prepared as needed explaining how items shall be secured during transit.

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## C.7.5.1 Decal / data plate guide <decalinfo>.

Data on the location of all decals and data plates including Item Unique Identification (IUID) markings, if applicable, shall be prepared. As applicable, illustrations detailing the locations of the decals and data plates shall be included. Information and illustrations for labels/decals shall be an exact representation as what appears on the equipment. FIGURE C-15 through FIGURE C-17 provide example orientation graphics for important decals as well as larger depictions of these same graphics to emphasize details.

**Note**

Figure appearance and tags/output may vary slightly based on version of DTD/stylesheets utilized (see applicable DTD or tag definition list for available tags).

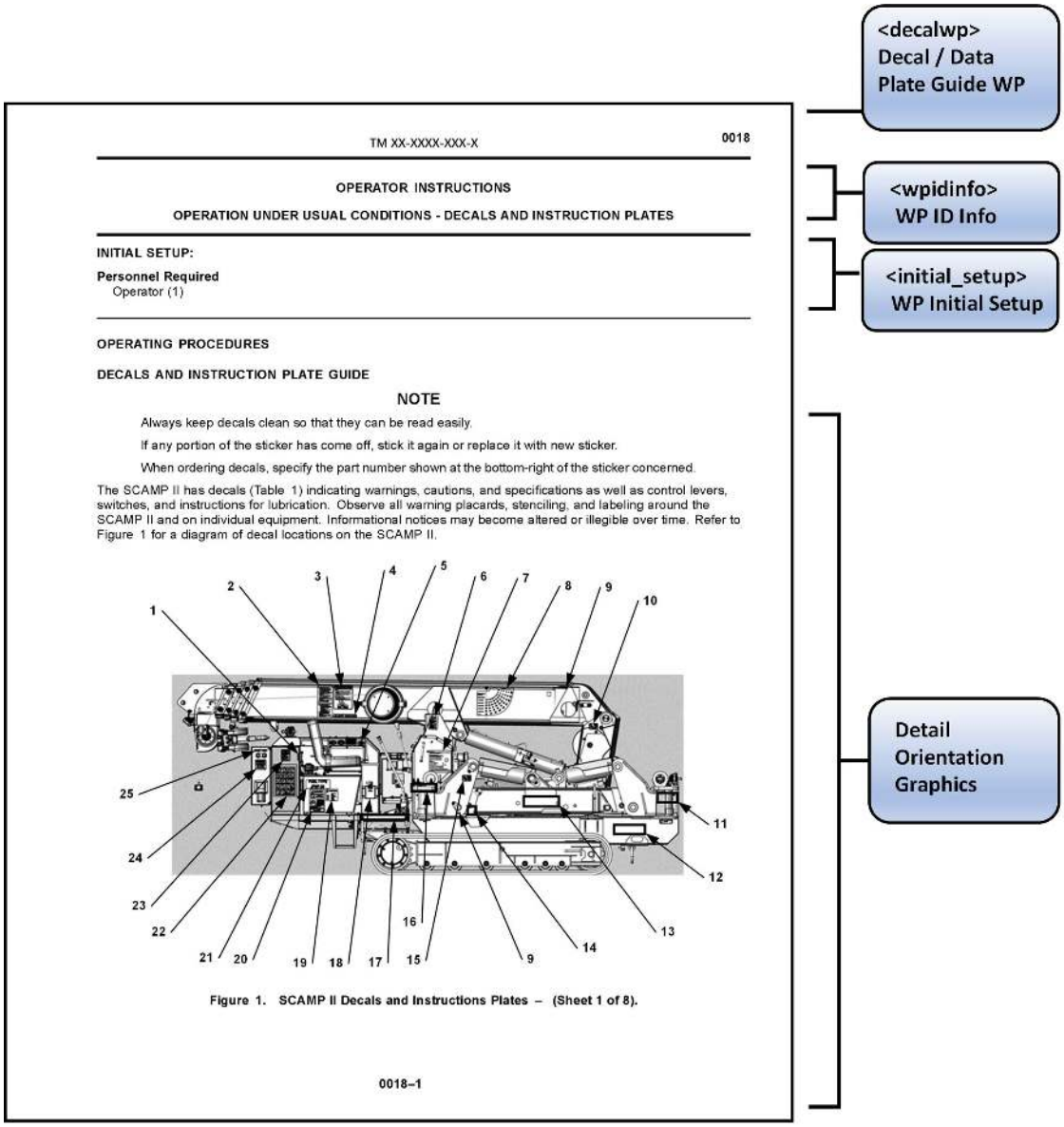


**FIGURE C-15. Decal and instructions WP (sheet 1 of 3).**

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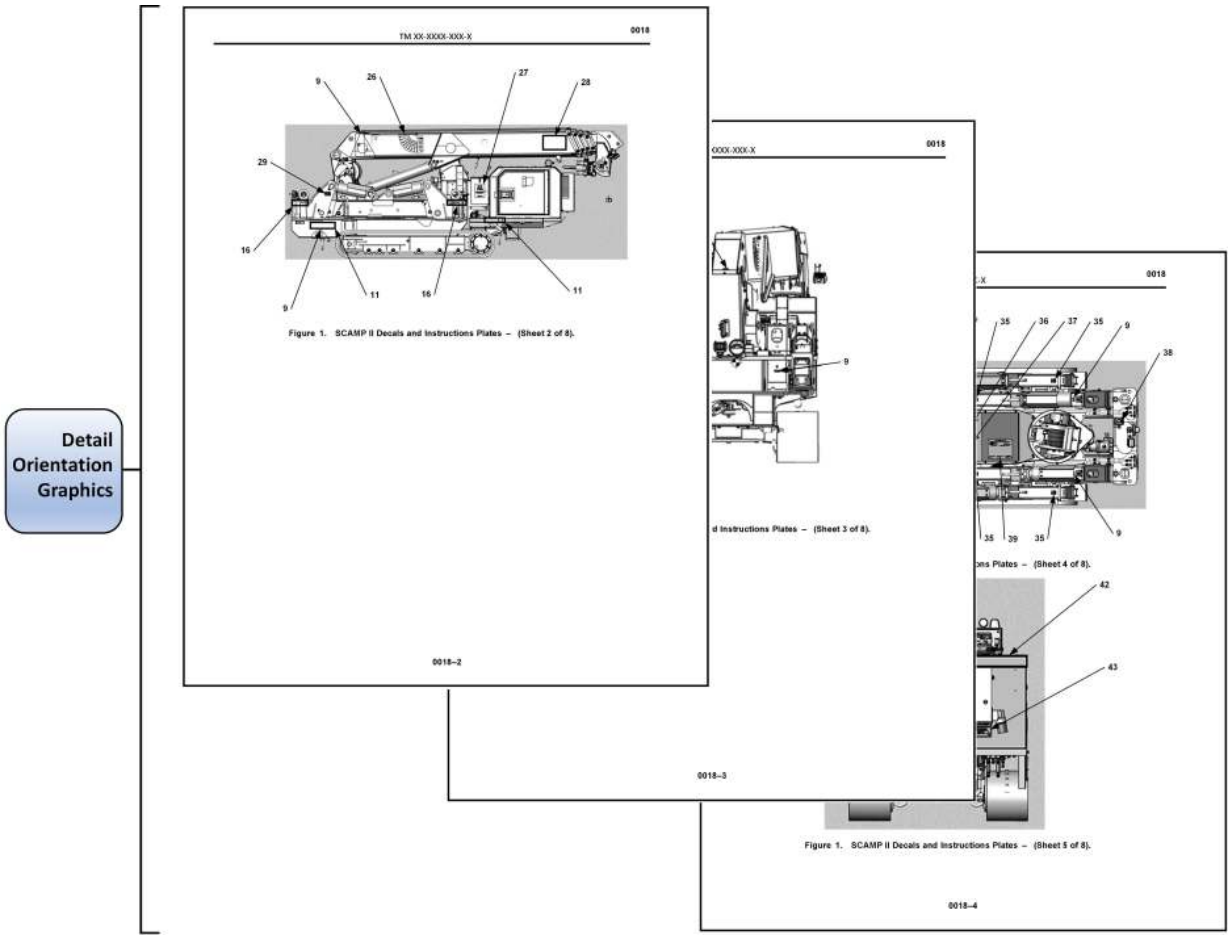


**FIGURE C-16. Decal and instructions WP (sheet 2 of 3).**

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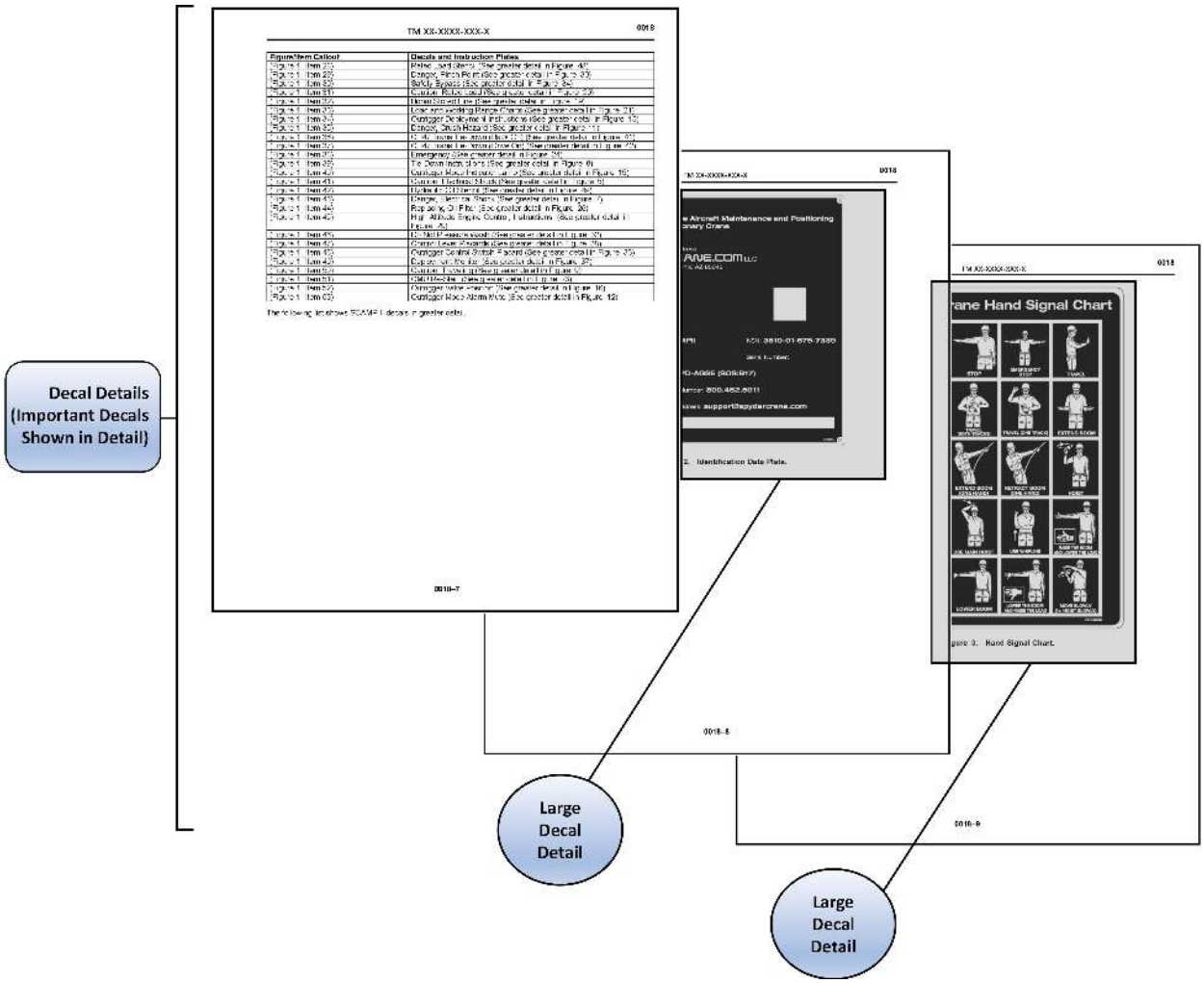


**FIGURE C-17. Decal and instructions WP (sheet 3 of 3).**

## C.7.6 On-vehicle equipment loading plan work package <eqploadwp>.

This WP shall be prepared as applicable for an equipment system and shall include information provided by the MATDEV. On-vehicle equipment loading plan WP components are listed below.

**a.** WP identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** Initial setup **<initial_setup>** is not required for this WP (5.4.3).

**c.** A brief scope statement **<intro>** shall be prepared explaining the purpose of the WP.

**d.** An illustrated loading plan list(s) **<loaddesc>** shall be prepared. This plan shall include an illustration or illustrations identifying and locating the on-vehicle equipment. External and internal views shall be used if necessary. As applicable, both tactical and non-tactical situation loading configurations shall be shown.

## C.8 NOTES.

The notes in Section 6 apply to this appendix.

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# APPENDIX D

# TROUBLESHOOTING PROCEDURES

## D.1 SCOPE.

This appendix explains the types of possible troubleshooting methods available within MIL-STD-40051 and how to choose which of these methods is best to use depending on the equipment system and its subcomponents. Choosing the right type of troubleshooting methodology allows for efficient and accurate fault isolation tailored to specific Army equipment and available testing approaches for an equipment system, its assemblies, subassemblies, wiring, junction boxes, accessories, etc. Troubleshooting is a required component of technical publications, and this appendix establishes the required content applicable for all maintenance levels through depot including DMWRs and NMWRs. This appendix is a mandatory part of this standard. The information contained herein is intended for compliance.

**Note**

To be compliant with this standard, publications shall be developed within **<production>** (opening) and **</production>** (closing) root tags.

When preparing information in accordance with this appendix, all references shall be linked using the appropriate tags (**<xref>**, **<extref>**, etc.). Links shall not be typed as text without tagging.

### D.1.1 Content tailoring using Appendix A matrices.

Applicability of some requirements contained in this standard are the responsibility of the materiel developer (MATDEV) (refer to AR 750–1, AR 700–127) and shall be accomplished using matrices referenced in TABLE A-I. The applicability of requirements are also designated by statements including, but not limited to, the following:

**a.** "...as specified by the MATDEV."

**b.** "...as required by the MATDEV."

**c.** "Refer to the MATDEV for..."

**d.** "Contact MATDEV if..."

**e.** "The MATDEV shall determine..."

**f.** "...as needed."

#### D.1.1.1 Front and rear matter.

Front and rear matter requirements for documents discussed in this appendix shall be determined in accordance with TABLE A-II.

## D.2 APPLICABLE DOCUMENTS.

The applicable documents in Section 2 apply to this appendix.

## D.3 ACRONYMS AND DEFINITIONS.

The acronyms and terms defined in Section 3 of this standard apply to this appendix.

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## D.4 TROUBLESHOOTING PROCESS.

All troubleshooting procedures begin with an observed problem through either visual inspection or through other means of fault/malfunction determination. Testing and troubleshooting procedures shall guide a user, in as practical a manner as possible, to the system, subsystem, assembly, component, Shop Replaceable Unit (SRU), or Line Replaceable Unit (LRU), or further to the replaceable part, interconnecting wire, or mechanical linkage that caused the fault/malfunction. Troubleshooting procedures shall be developed to the extent required to maintain systems subsystems, equipment, components, and support equipment at the authorized maintenance level in accordance with the Logistics Product Data (LPD) and Source, Maintenance, and Recoverability (SMR) codes developed for the equipment and listed in the RPSTL. The method used for identifying system equipment testing shall be explained.

All corrective actions in troubleshooting work packages (WPs) shall be authorized by the Maintenance Allocation Chart (MAC) up to the maintenance level(s) included in the publication.

**Note**

In addition to external causes for malfunctions, troubleshooting should also identify symptoms resulting from failure of every spare and repair part authorized for replacement at the user level. Troubleshooting procedures shall be prepared assuming one malfunction at a time is being corrected.

Other factors to be considered in the development of troubleshooting procedures include, but are not limited to, the following:

**TABLE D-I. Troubleshooting Factors**

| | |
|---|---|
| Complexity of equipment to troubleshoot | Replaceable component and part reliability |
| Technical experience of target audience | Ease of testing |
| User environment | Test access time |
| System quick-turnaround requirements | Test time |
| Test equipment requirements and availability | Equipment pre-condition requirements |
| User-driven versus diagnostic testing | |

### D.4.1 Methodology to determine what type of troubleshooting WP to develop.

Many factors determine what type of troubleshooting WP is appropriate for an LRU. Two major factors are technical assessments and a report called the Failure Mode, Effects, and Criticality Analysis (FMECA) Requirements. These two factors are discussed in sections D.4.1.1 and D.4.1.2. Troubleshooting module(s) shall be divided into WPs that shall stand alone and are broken into the WP types listed in TABLE D-II. Additionally, TABLE D-III lists software troubleshooting WPs discussed in APPENDIX G. Prior to starting development of troubleshooting procedures, refer to the sections listed in TABLE D-II for required components to build various WP types.

**TABLE D-II. Troubleshooting WPs.**

| WP | WP Tag | Appendix Section |
|---|---|---|
| Troubleshooting introduction WP | **<tsintrowp>** | D.8.1 |
| Technical description WP [2] | **<techdescwp>** | D.8.2 |
| Troubleshooting index WP | **<tsindxwp>** | D.8.3 |
| Manual (non-integrated) troubleshooting WP [1] | **<tswp>** | D.8.4 |

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**TABLE D-II. Troubleshooting WPs. (continued**

| WP | WP Tag | Appendix Section |
|---|---|---|
| Operational checkout WP [1] | **<opcheckwp>** | D.8.5 |
| Combined operational checkout and troubleshooting WP [1] | **<opcheck-tswp>** | D.8.5.1 |
| Diagnostic (integrated) troubleshooting WP [1] | **<diagnosticwp>** | D.8.6 |
| Preshop analysis WP [2] | **<pshopanalwp>** | D.8.7 |
| Component checklist WP [2] | **<compchklistwp>** | D.8.8 |

[1] WP type frequently duplicated in a module for effective information display.

[2] DMWRs/NMWRs Only

**TABLE D-III. Software troubleshooting WPs.**

| WP | WP Tag | Appendix Section |
|---|---|---|
| Messages WP(s) (Required for SAM only) | **<softmessageswp>** | M.7.8.3 |
| Recovery from errors, malfunctions, and emergencies WP | **<softerrorswp>** | M.7.8.4 |

### D.4.1.1 Technical assessments.

The question of whether manual (non-integrated) troubleshooting or diagnostic (integrated) troubleshooting shall be used can be answered with technical assessments of the complexity of a system/subsystem that has a fault isolation requirement. Technical assessments include the following considerations:

1. Failure Detection Cues: Failure detection cues may include visual or audible warning devices, automatic sensing devices, sensing instrumentation, or other unique indications. Failure detection methods can also be used to measure technical complexity.
2. Technical Complexity: Variations in technical design complexity dictate which authoring strategy should be used for troubleshooting development. Simple systems/subsystems do not require state table manipulation or a logic model to perform fault isolation. For simple systems, **<faultproc>** is likely the preferred troubleshooting method.
3. Fault Isolation Methodology: Fault isolation methodology favors the most direct procedure that allows an operator/maintainer to isolate a malfunction or failure. Fault isolation starts with the initial symptom that presents itself and proceeds to further specific actions taken, such as performing a Built-In-Test (BIT) procedure. Any fault isolation that requires class II network communications with LRUs will require a state table and logic model. The send and receive parameters that perform BIT functions will be needed to perform integrated diagnostics.

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### D.4.1.2 Failure Mode, Effects and Criticality Analysis (FMECA.

The FMECA has its own Logistics Support Analysis (LSA) report that identifies potential equipment problems and malfunctions before they occur. The FMECA provides two primary revelations: methods of detection and probability of occurrence. Detection method is important because it could possibly serve as an indicator for task complexity, and, thus, guide which WP type shall be used. Probability of occurrence is important because equipment that breaks frequently should be tested first in a WP prior to testing other system components. The FMECA allows these specialized troubleshooting development strategies which reduce the Mean Time to Repair (MTTR) and ensure the appropriate troubleshooting type is utilized based on system complexity.

## D.4.2 Troubleshooting flowchart use.

Functional flowcharts, as directed by the MATDEV, may be used as a stand-alone troubleshooting method depending on system complexity only if all text in the flowchart is searchable text (static flowchart graphics are prohibited).

**Note**

Since references within flowcharts are linked, ensure all references (figures, item numbers, WPs, etc.) are correct for every version of a publication.

Use of other symbols not shown in this section shall be approved by the Logistics Data Analysis Center (LDAC) prior to use.

Functional flowcharts shall only be used as a stand-alone solution if the troubleshooting procedure can clearly and comprehensively be depicted in graphical flowchart form and may be inserted as one or more graphics into the WP. Flowcharts shall include all warnings, cautions, or notes displayed in or above shapes containing the relevant step. FIGURE D-1 (including "Using Troubleshooting Charts" and text) and FIGURE D-2 shall be included in technical publications with flowcharts and shall serve as a legend identifying flowchart shapes and the types of data these shapes contain.

FIGURE D-2 shows an example functional flowchart depicting the same troubleshooting procedure **<logicproc>** discussed in D.5.2. Step 1 is represented in a trapezoid and directs the user to "Park SCAMP II on flat level surface…" and serves as the **<test>** element of this procedure. Once the test actions are complete, the question "Does the tilt sensor blink…" serves as the condition/indication **<indication>** of this procedure and guides the user to decide on a "YES" or "NO" answer **<answer>** to the condition/indication. The condition/indication is represented in a diamond. The answer could then lead the user to another test (or series of tests) **<test>** or, as shown in FIGURE D-2, a malfunction **<malfunc>** and corrective action **<action>** (rectangles).

**Using Troubleshooting Charts**

Troubleshooting flowcharts guide the user through a sequence of steps to isolate faults. The table describes the symbols used in the flow charts and their meaning. Troubleshooting begins at the START oval on each chart and continues through the flow chart until the fault is corrected, or until directed to perform a specific action by a defined process box.

**Note**

Use of other symbols not shown in next figure ("Functional flowchart") shall be approved by the Logistics Data Analysis Center (LDAC) prior to use.

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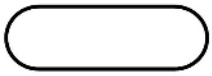
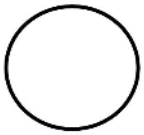
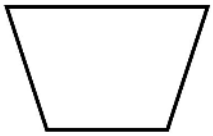
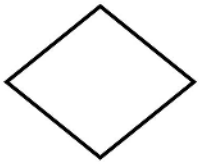
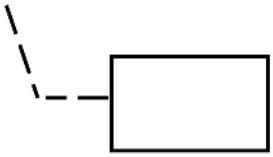
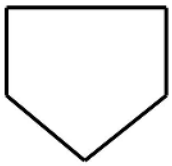
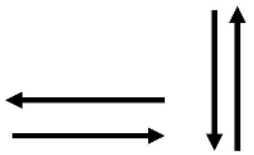
| SYMBOL | NAME | DEFINITION |
|---|---|---|
|  | TERMINAL | ENTRY (START OR EXIT (STOP)) POINT OF THE FLOWCHART |
|  | CONNECTOR | WITHOUT A NUMBER AND LETTER IN THE SYMBOL, IT CONNECTS TWO OR MORE FLOW LINES. A PAIR OF THESE SYMBOLS WITH THE SAME NUMBER AND/OR LETTER INSIDE, CONNECTS TWO POINTS OF THE CHART TO GO TO. DECIMAL NUMBERS ARE USED IN THE SYMBOLS TO DISTINGUISH CONNECTIONS WHEN THERE ARE MORE THAN ONE ON A PAGE |
|  | OPERATION | SHOWS AN OPERATION THAT THE USER OF THE CHART MUST PERFORM BEFORE CONTINUING WITH THE CHART |
|  | DECISION | POINT AT WHICH THE USER DECIDES WHICH OF TWO PATHS TO FOLLOW DEPENDING UPON WHAT INDICATION, DISPLAY OR CONDITION EXISTS |
|  | NOTE | PROVIDES EXTRA INFORMATION FOR AN OPERATION OR DECISION TO WHICH IT IS CONNECTED BY A BROKEN LINE |
| | DEFINED PROCESS | DIRECTS THE USER OF THE FLOW CHART TO PERFORM A A SPECIFIC OPERATION |
|  | OFF-PAGE REFERENCE | THE NUMBER REFERENCED IN THIS SYMBOL IS THE SHEET NUMBER OF A FLOWCHART TO GO TO |
|  | PATH FLOW | ARROWHEADS INDICATE THE PATH FLOW BETWEEN SYMBOLS |

**FIGURE D-1. Functional flowchart (sheet 1 of 2).**

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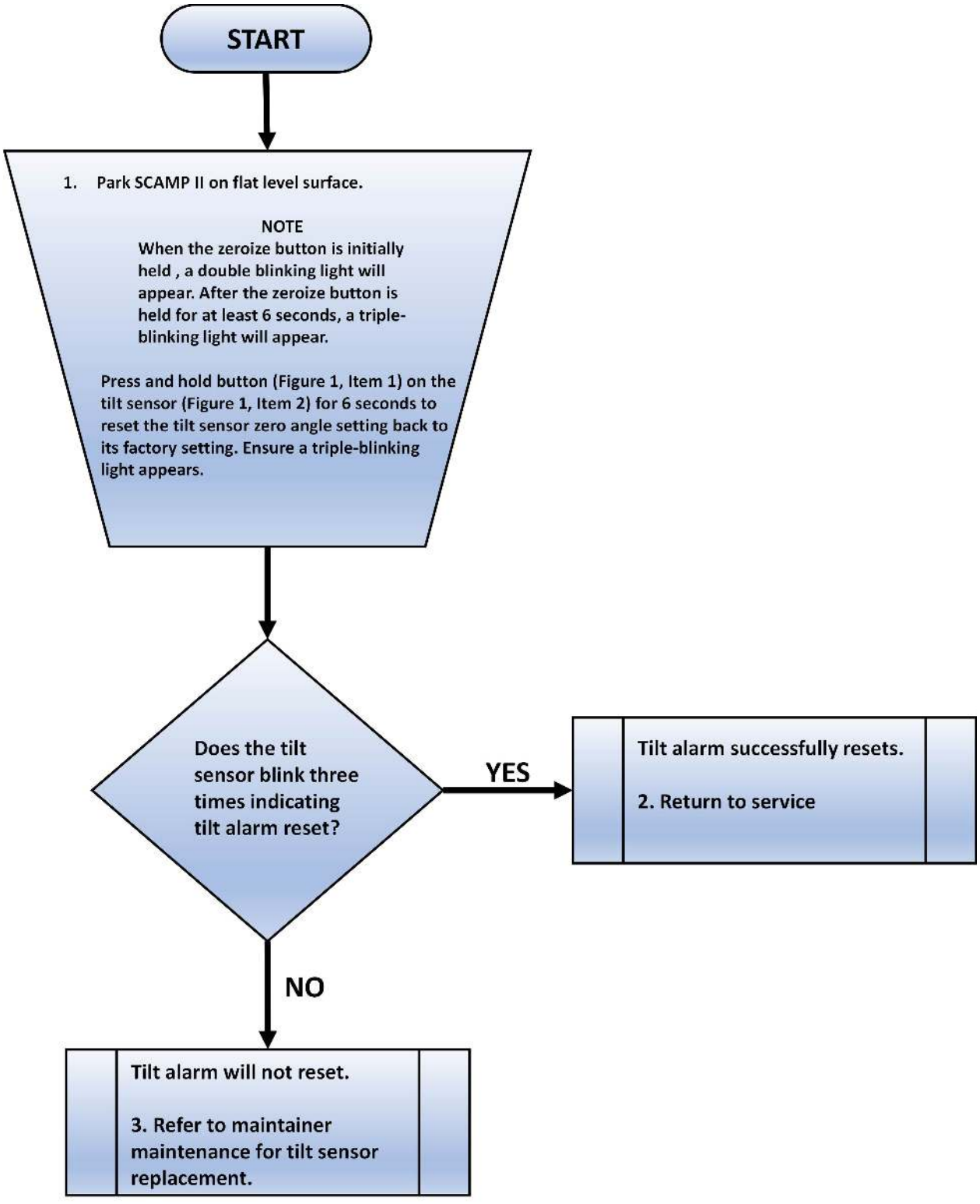


**FIGURE D-2. Functional flowchart (sheet 2 of 2).**

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## D.5 TROUBLESHOOTING METHODOLOGY.

This standard includes two types of troubleshooting methodologies: manual troubleshooting and diagnostic troubleshooting. Manual troubleshooting guides a user to a fault through either iterative system testing exercises and/or predetermined fault identification listings. Manual troubleshooting typically involves technical data that is able to be displayed as a sequential "path" through a fault utilizing a specific order of steps that must be followed. Diagnostic troubleshooting directs a user to utilize diagnostic testing capabilities inherent to a system. The specific nature and location of a fault as well as system complexity determines the method of troubleshooting as well as the types of procedures that shall be utilized.

**Wiring diagrams and schematics**: Often use of wiring diagrams and schematics may aid users in understanding troubleshooting procedures. Refer to E.7.10 for more information about the development of wiring diagram and schematics WPs **<wiringwp>** (which may be referenced from troubleshooting WPs as applicable). Refer to E.7.10.9 for more information about system and subsystem schematics.

### D.5.1 Integrated vs. non-integrated troubleshooting.

Manual troubleshooting utilizes predominantly non-automatic testing, while diagnostic troubleshooting utilizes semi-automatic or fully automatic testing capabilities of a system. More specifically, the primary difference between manual and diagnostic troubleshooting methodologies is the level of hands-on testing required to troubleshoot an LRU, which is directly based on whether an LRU has "non-integrated" or "integrated" testing capabilities. An LRU with "non-integrated" testing capability requires a user to utilize the ETM/IETM for test guidance and must use testing hardware to manually get a test result. This test can result from diagnostic software or a manual test procedure that does not require diagnostic software.

An LRU with "integrated" testing capabilities has three primary components: IETM software, diagnostic software, and diagnostic test equipment that connects to a vehicle or some other equipment. The IETM software and diagnostic software are able to transmit data and receive requests through a communications link (or "comms link"), also referred to as an Application Programming Interface (API). WPs developed for manual system testing **<tswp>** are appropriate for non-integrated testing, while integrated testing is more complex and, thus, requires diagnostic troubleshooting WPs **<diagnosticwp>**. Diagnostic troubleshooting WPs **<diagnosticwp>** shall only be utilized for IETMs (not ETMs). See FIGURE D-3 for a diagram of the relationships between troubleshooting methods and associated procedure types. Both types of testing—integrated and non-integrated—are useful depending on system complexity and testing capability, and both may be used to prepare troubleshooting procedures for a system.

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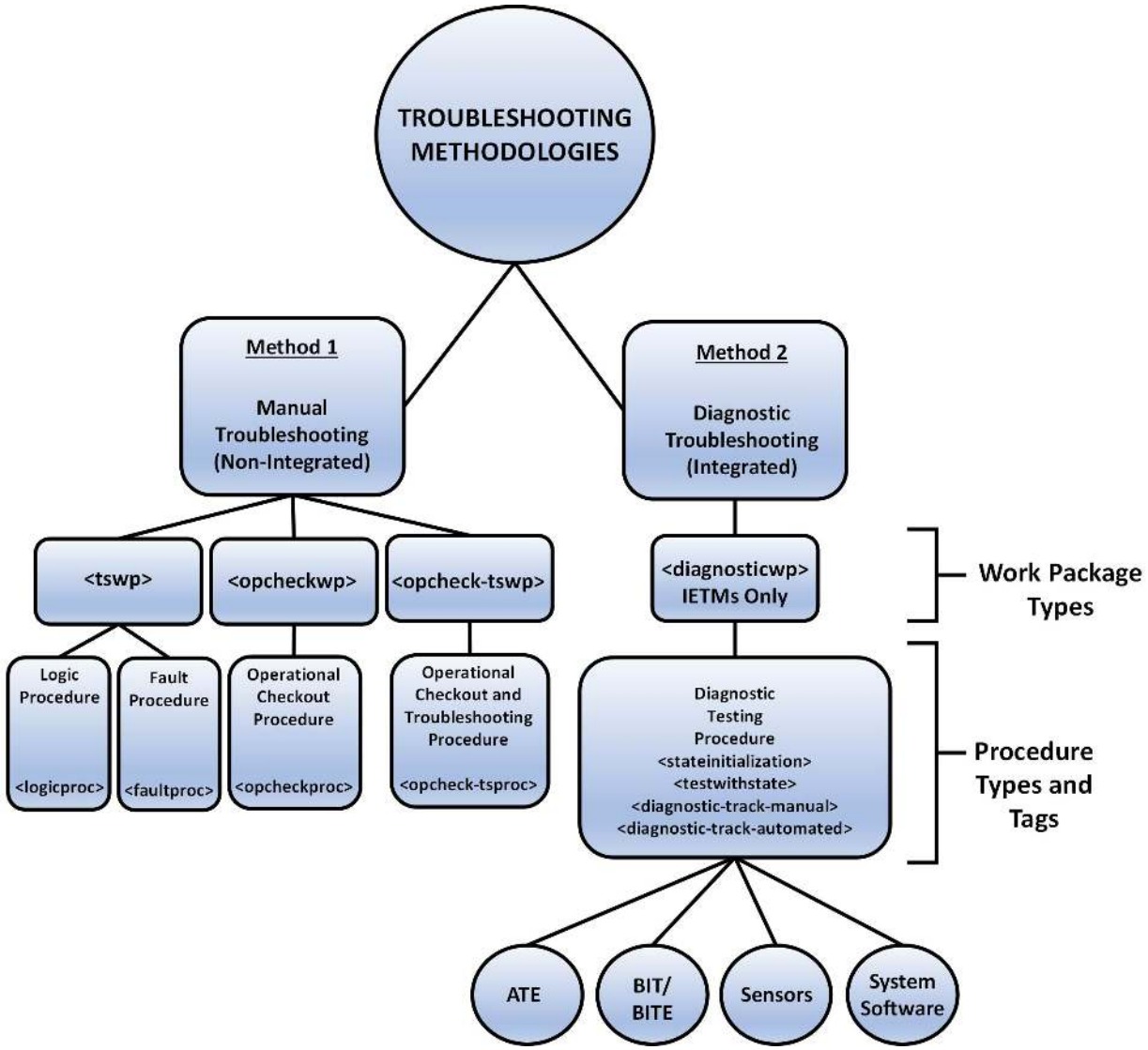


**FIGURE D-3. Troubleshooting methods and procedure types.**

## D.5.2 Manual (non-integrated) troubleshooting methodology.

Manual troubleshooting begins with an observed problem, a fault symptom, or malfunction and then determines the next action depending on the following: availability of information such as equipment state; test results; user interaction; and/or historical data/probability and then concludes with a diagnosis to a single fault symptom or malfunction. Manual troubleshooting utilizes non-integrated, predominately hands-on testing approaches to arrive at a single fault and shall be prepared using one of two available procedure options: logic procedures **<logicproc>** and fault procedures **<faultproc>**. These testing options are discussed in further detail in D.8.4. Operational checkout **<opcheck>** procedures consist of a series of test procedures to determine whether corrective actions rectified a fault, and the system can return to normal operation. Operational checkout procedures are also included in manual troubleshooting and discussed in further detail in D.8.5. FIGURE D-3 shows logic, fault, and operational checkout procedures all under the "Method 1, Manual Troubleshooting" block. All these procedure types heavily rely upon user action.

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### D.5.3 Diagnostic (integrated troubleshooting methodology.

Like manual troubleshooting, diagnostic troubleshooting also begins with an observed problem; however, the mechanism of observation is significantly different from manual troubleshooting. Diagnostic troubleshooting leverages integrated system monitoring technology that constantly measures operating parameters in real time. When a fault is identified in a system, the system monitoring software creates a Diagnostic Trouble Code (DTC), and troubleshooting utilizing the DTC data shall be accomplished with a diagnostic troubleshooting WP **<diagnosticwp>**. Diagnostic troubleshooting utilizes primarily semi-automatic and/or automatic testing capabilities and is only available on select weapon systems.

Thus, diagnostic troubleshooting shall only be used for systems that have sensors, diagnostic software, and other types of Automatic Test Equipment (ATE). Diagnostic testing is discussed further in D.8.6. When necessary, both manual and diagnostic troubleshooting methods may be used on the same weapon system/equipment. Based on the specific nature and location of a fault as well as system complexity, both manual and diagnostic troubleshooting may be used within the same troubleshooting procedure. Equipment sophistication, available sensors, and Built-In-Test (BIT) / Built-In Test Equipment (BITE) /ATE shall determine the chosen troubleshooting method(s). FIGURE D-3 shows the "Method 2, Diagnostic Troubleshooting" block and the associated semi-automatic and fully automatic testing types that may be available on some systems.

### D.5.4 Linear vs. non-linear troubleshooting.

Linear troubleshooting predominantly employs a system test concept whereby operators/maintainers functionally test equipment in a specific order using end-to-end checks of the system or equipment to ensure it performs the function it was intended to do. With linear troubleshooting, elements of a system are tested by answering questions with either "YES" or "NO" responses that guide the user to one of the following:

1. A next step (or series of steps) or a malfunction (or series of malfunctions) and corrective action **<logicproc>**.
2. A symptom (or series of symptoms) and the associated malfunction (or series of malfunctions) along with a corrective action **<faultproc>**.

Non-linear troubleshooting is relatively more complex than linear troubleshooting and heavily involves a system test concept whereby diagnostic software detects hardware faults and interacts with IETM software to diagnose malfunctions.TABLE D-IV identifies troubleshooting WP types and the corresponding method of troubleshooting associated with each WP type.

**TABLE D-IV. Troubleshooting WPs and testing methodology.**

| TS WP Type | WP Tag (s) | Procedure Tags | Linear Logic | Non-Linear Logic | System Monitoring | LRU Integration |
|---|---|---|---|---|---|---|
| Manual | <tswp> | <faultproc> | X | | | |
| | | <logicproc> | X | X | X | |
| | | <opcheckproc> | X | | | |
| Diagnostic | <diagnosticwp> | <testwithstate> | | X | X | X |
| | | <diagnostic-track-manual> | | | | |
| | | <diagnostic-track-automated> | | | | |


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## D.6 TYPES OF TROUBLESHOOTING PROCEDURES.

### D.6.1 Troubleshooting procedures content parameters and requirements.

Troubleshooting procedures shall contain all essential and pertinent information that would be included in any other form of maintenance procedure. This includes warnings, cautions, notes, power turn-on procedures, pre-checkout procedures, reference diagrams, and initial switch settings. Troubleshooting procedures shall be prepared assuming one malfunction at a time is being corrected. FIGURE D-4 shows an example troubleshooting flowchart that identifies a system malfunction followed by a series of steps organized into blocks. Each block includes steps to test one possible issue at a time followed by a condition/indication question posing if the malfunction has been corrected. Answers (in this example, "YES" and "NO") to this condition/indication dictate any further steps to perform toward the ultimate goal of malfunction correction. Manual troubleshooting as well as some non-automatic troubleshooting procedures employ this type of fault isolation methodology.

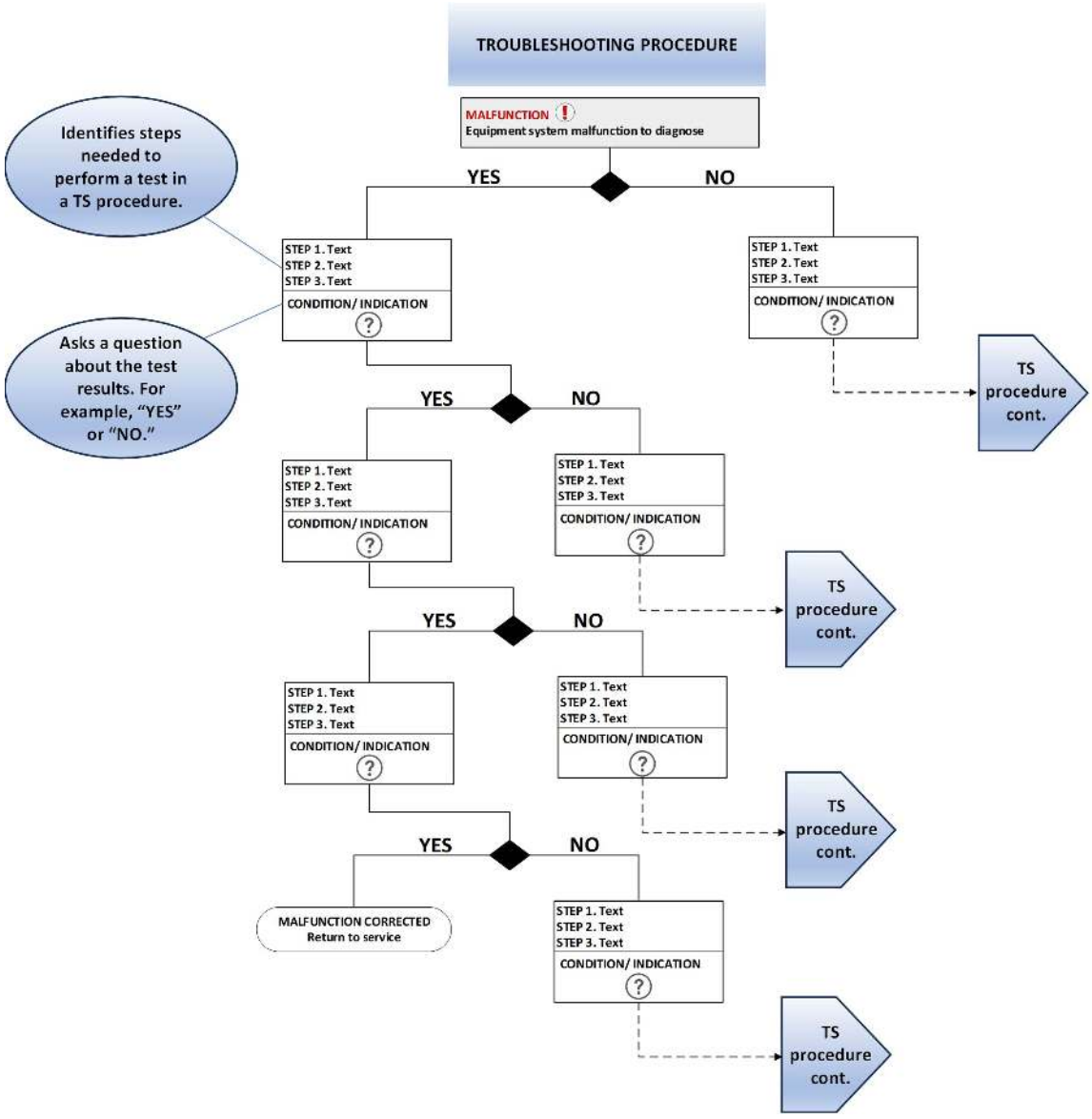


**FIGURE D-4. Troubleshooting template.**

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Before beginning troubleshooting procedures, the operator/maintainer shall be instructed to perform any applicable self-tests, alignments, and inspections. As applicable, an operational check shall be specified to be performed after the fault is corrected to ensure correct operation of the system. Troubleshooting manual instructions shall be prepared following these general requirements:

**a.** A concise explanation of the testing and troubleshooting format and an explanation of how to use the testing and troubleshooting procedures with the malfunction/symptom index, when applicable, shall be included.

**b.** The location for each component, accessory, connector, or junction box in the system under test shall be provided or a reference to the equipment description and data WP shall be included. The text and illustrations, as necessary, shall identify every test connector or other test point to be used in the test.

**c.** A complete list of test options shall be stipulated by the troubleshooting procedure. Any self-tests that are associated with the system shall be listed. Self-test schemes shall be described as the primary troubleshooting tool, with manual or automatic troubleshooting prepared to supplement the instructions where the self-test leaves off or fails to locate the malfunction. The procedure shall be built using system self-tests before using external test equipment.

**d.** Test setup procedures and post-test tear down procedures shall be included.

**e.** Pre-test procedures (for example, system turn on, initial adjustments, etc.) shall be referenced either in the troubleshooting WP initial setup or in initial steps prior to test procedures.

**f.** Complete step-by-step troubleshooting procedures, including instructions required for use and application of installed online testing equipment, shall be included. Procedures shall consider controls, test point accessibility, indicator displays, and the feasibility of using BITE or automated test equipment where available.

**g.** Test procedures (for example, identification of time required to run and complete the system test, and an indication of any possible mid-test interruptions or stoppages and how to respond to them) shall be included.

**h.** Backup diagrams showing all test points, input and output signals, logic charts, schematics, signal flow diagrams, tables, and other illustrations as required for comprehensible understanding of the procedures shall be included. (ETMs only) When possible, depending on diagram scale, include graphics within troubleshooting WPs. These diagrams may also be referenced as part of other troubleshooting WPs, a maintenance WP, or integrated into the publication as foldouts. Refer to APPENDIX E for more information about wiring diagrams and schematics **<wiringwp>**.

**i.** Any information that will aid the operator/maintainer, such as waveforms; resistance data; fluid pressures; voltage levels; references to test diagrams, functional diagrams, text, etc.; and alignment procedures, checkout procedures, or other scheduled maintenance procedures shall be included. Connector numbers, pin designations, etc., shall be identified.

**j.** Special attention shall be given to interface wiring fault isolation procedures. Wiring fault isolation procedures shall include the following types of data, as applicable:

**(1)** Specific wire reading access points and resistances for wiring components (where practical).

**(2)** Wire-to-wire and wire-to-ground criteria for circuit integrity (including splice location information).

**(3)** Special wire definition where required (including interconnecting criteria for proper sealing or terminal application), and special notations where wire harnesses should be completely replaced and not repaired.

**(4)** It is also essential when developing fault isolation procedures to provide or refer to ground zone tables, which include type, location, and wires connected; charts for both connectors and terminal boards; and a wire number log to identify any wire with its prime wiring diagram.

## D.6.2 Troubleshooting procedures development.

This section discusses various types of troubleshooting procedures. Troubleshooting WP types determine which procedures are available for use.

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### D.6.2.1 Troubleshooting organization.

Troubleshooting procedures shall first direct the user to perform any applicable self-tests, alignments, and inspections prior to beginning any other troubleshooting procedure to test equipment failure. Troubleshooting procedures shall then direct users on detection, isolation, and correction of equipment failures for weapon systems and their subsystems. See D.6.1 for troubleshooting procedures content parameters and requirements. WPs will relate either to a specific symptom or to a system, assembly, or component. WPs related to a system of some complexity may contain more than one set of troubleshooting procedures directed to specific subsystems. Each procedure shall end with one of the following:

**a.** When the corrective action can be performed at the same level of maintenance authorized to troubleshoot, the troubleshooting procedure shall end with a corrective action that includes a reference to the appropriate maintenance WP and to a WP (operational checkout, test or operating instructions) to confirm the malfunction was corrected.

**b.** When the corrective action cannot be performed at the same level of maintenance authorized to troubleshoot, then the troubleshooting procedure shall end with the following statement, "Send to next level of maintenance."

Troubleshooting procedures, when necessary, shall be accompanied by schematics, signal flow diagrams, waveforms, tables, and other illustrations that provide a comprehensive understanding of the procedures. When schematics are required as backup data, they shall be referenced, or they may be contained in the same WP. The schematics shall integrate fluid, mechanical, electrical, and electronic components. Illustrations may also be included that locate and identify the controls and displays used to perform the testing and troubleshooting procedures. See section for discussion of system-level schematics and connector end view libraries (E.7.10).

### D.6.2.2 Manual troubleshooting procedures <tswp>.

Manual troubleshooting shall include detailed inspection and troubleshooting information and shall include or reference functional descriptions of subsystems being diagnosed to aid the user. For manual troubleshooting, initial troubleshooting for a fault shall utilize one or both types of procedure types: Logic **<logicproc>** and Fault **<faultproc>**. Selection of procedure type shall depend on the type of system, equipment, or assembly / subassembly being addressed; the target audience description; and the maintenance level of the user.

Manual troubleshooting procedures can be structured differently and therefore contain different content elements based on the complexity of the troubleshooting to be performed. Once selected, the troubleshooting procedure shall be prepared in accordance with the requirements specified in this standard.

#### D.6.2.2.1 Logic troubleshooting procedures <logicproc>.

Logic troubleshooting shall be utilized when the user can effectively test elements of a system by answering questions such as "yes/no, pass/fail, true/not true," etc. Users shall employ a manual testing approach to determine the root cause of a fault and shall identify a list of malfunction symptoms that serve as specific points of entry into the testing/troubleshooting cycle. This list of malfunctions shall aim to reduce maintenance downtime and decrease fault detection time. Logic troubleshooting shall utilize a series of tests **<test>** (steps and substeps) that lead to an indication or condition **<indication>** (usually stated in the form of a question). Based on these indications or conditions, a response **<answer>** is provided that will guide the user to either the next step or a series of steps **<test>**, or to a malfunction **<malfunc>** and corrective action **<action>**. This process is continued until the entire troubleshooting procedure is completed. The corrective action shall include a reference to the WP or paragraph **<xref>** that contains the data to perform the corrective action. See D.8.4 for creating manual troubleshooting WPs.

Refer to FIGURE D-5 as an example of logic troubleshooting as it appears as a series of text-based branches (FIGURE D-2 shows this same procedure as a flowchart). FIGURE D-5, step 1 directs the user to "Park SCAMP II [small crane with tracks for movement and outriggers for stabilization] on flat level surface" and "Park and hold the zeroize button…" This first step serves as the **<test>** element of this simple example procedure. Once the test actions

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are complete, the question "Does the tilt sensor blink…" serves as the condition/indication **<indication>** of this procedure and guides the user to decide on a "YES" or "NO" answer **<answer>** to the condition/indication. The answer could then lead the user to another test (or series of tests) **<test>** or, as shown in FIGURE D-2 , a malfunction **<malfunc>** and corrective action **<action>**. Refer to FIGURE D-12 through FIGURE D-14 for tagging details associated with FIGURE D-5.

**Note**

Troubleshooting steps/procedures may be repeated, and the words "TESTING BRANCH" are used to break up troubleshooting into branches to orient the user while completing a procedure. This content organization also makes commenting and linking in and out of troubleshooting procedures easier.

Logic procedures may be presented in as functional flowcharts as discussed in D.4.2.

Figure appearance may vary slightly based on version of DTD/stylesheets utilized.

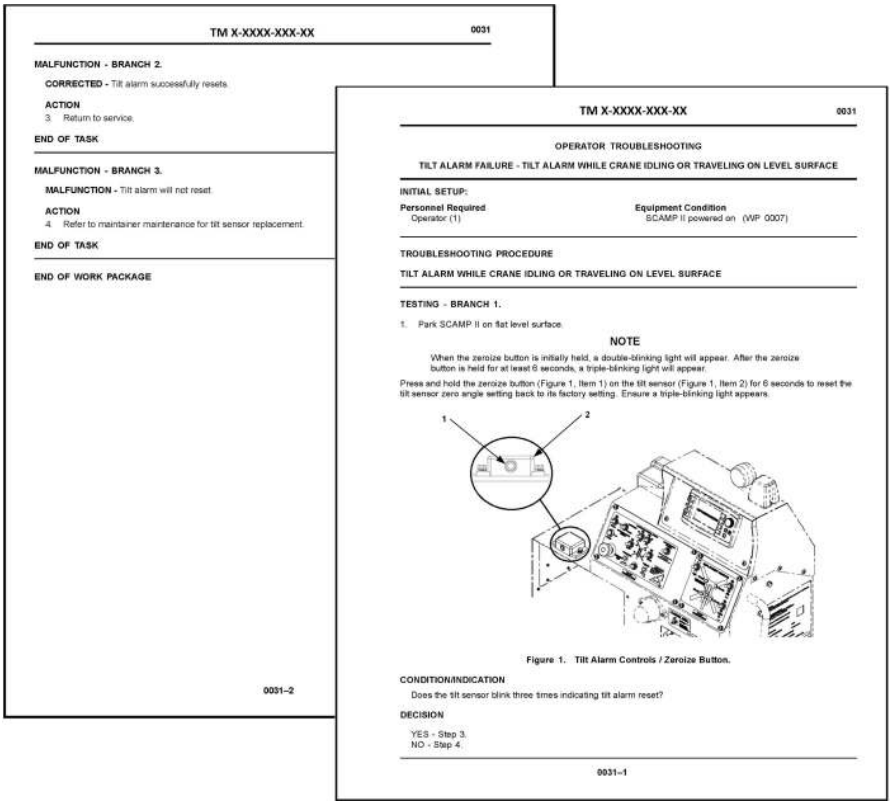


**FIGURE D-5. Logic troubleshooting procedure.**

## D.6.2.2.2 Fault troubleshooting procedures <faultproc>.

While logic troubleshooting shall be utilized when the user can effectively test elements of a system by answering questions with either "YES" or "NO" responses, fault troubleshooting procedures shall be utilized when a specific known SYMPTOM or series symptoms **<symptom>** and the associated MALFUNCTION or series of malfunctions **<malfunc>** that may have caused the fault(s) can effectively serve to restore a system to operational status. For each probable malfunction identified, a corrective action or actions **<action>** shall be stated that consists of a single test step or list of testing steps within the troubleshooting WP. These steps may include references to WPs **<xref>/<link>** that contain the data to perform a corrective action. See example **<faultproc>** troubleshooting tagging structure in FIGURE D-15. A malfunction-specific troubleshooting index WP may be created that either lists known

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malfunctions in alphabetical order or lists symptoms by subsystem categories. This index is discussed further in D.6.2.3.4.

### D.6.2.3 Operational checkout procedures <opcheckproc>.

Operational checkout procedures shall be developed to subject an aircraft, or other type of major equipment system, subsystems, components, accessories, and items of equipment to prescribed conditions to determine if they will function in accordance with predetermined test parameters. Operational checkout procedures shall follow fault repairs as needed. An operational checkout WP may include the following:

**a.** Test set hookup and disconnect procedures.

**b.** Index of test set message words.

**c.** Reference index of test set fault codes and related actions.

**d.** Further testing procedures related to the message words and fault codes.

**e.** BIT/BITE fault codes (may be referenced from diagnostic troubleshooting).

#### D.6.2.3.1 Stand-alone vs. combined procedures.

Operational checkout and troubleshooting procedures may be combined and contained in the same WP or may be developed in separate operational checkout and troubleshooting WPs based on the following factors: equipment complexity; type of required test equipment; equipment self-test or BIT capability; testing complexity; clarity and usability; and whether an operational checkout WP needs to be repeatedly referenced (from troubleshooting or maintenance WPs). Operational checkout WPs shall stand alone when global system checks (testing of system as a whole) are performed. These checks may include tests to determine if the entire system powers on or tests to determine if signals are received, etc. In all other instances, operational checkout WPs may be combined with troubleshooting procedures where necessary.

#### D.6.2.3.2 Operational checkout test procedures <opcheck>.

Operational checkout procedures **<testproc>** shall consist of a series of numbered steps **<step1>** and substeps **<step2> - <step6>** that lead to an indication or condition **<indication>**. Based on the indications or conditions, a corrective action **<action>** shall be provided. This corrective action can either be stated as a specific remedy or can be a reference **<xref>/<link>** to a detailed troubleshooting procedure WP. This process is continued until the complete operational checkout procedure is completed. See D.8.5 for creating operational checkout WPs.

When it is determined that the operational checkout procedures shall be separate from the troubleshooting procedures, the operational checkout procedures shall be included under the heading "OPERATIONAL CHECKOUT."

FIGURE D-6, step 1 directs the user to "Test outrigger pre-deployment IAW WP 0012…." This first step serves as the **<step1>** element of this procedure. Once the test action is complete, the statements "Outrigger does not easily rotate…" and "Pivot pin does not rotate…" as well as "Deployment pivot pin does not insert…" serve as the possible conditions/indications **<indication>** of this procedure. The user is then guided to a corrective action **<action>** based on the observed condition/indication.

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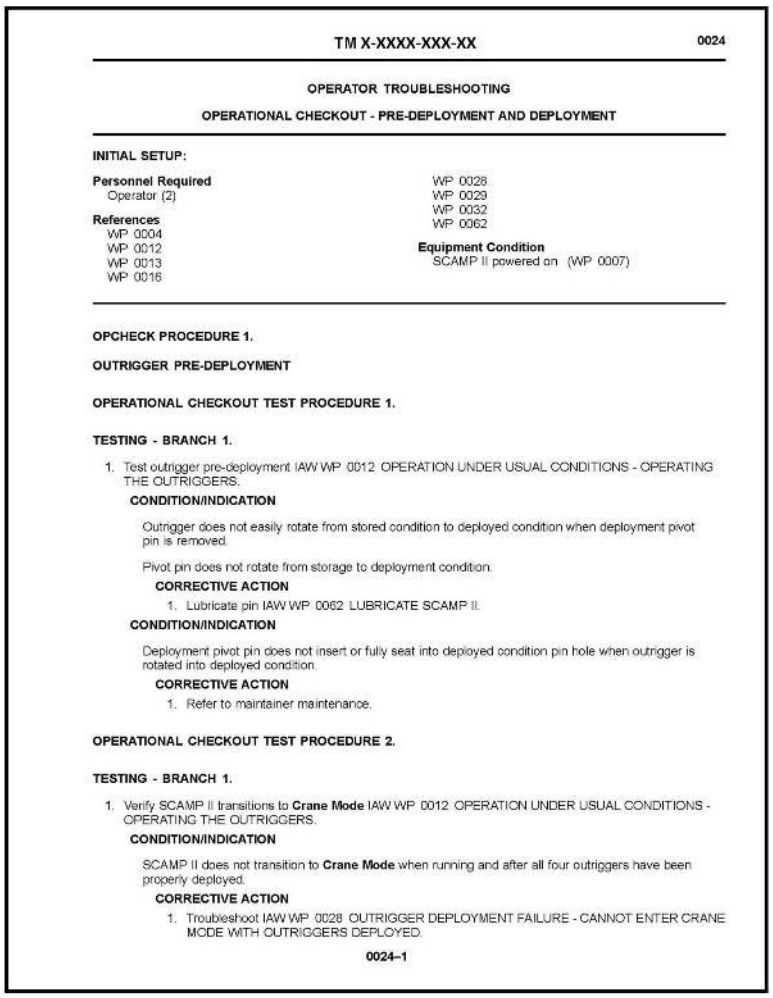
**Note**

Like logic troubleshooting **<logicproc>**, fault troubleshooting procedures display the words "TESTING BRANCH" to break up troubleshooting into branches to orient the user while completing a procedure. This content organization also makes commenting and linking in and out of troubleshooting procedures easier.

If an operational checkout WP utilizes ATE, and a Test Program Set has been developed, the operational checkout and troubleshooting procedures contained in the Test Program Set shall not be duplicated. A reference to the Test Program Set shall be provided. Operational checkout for DMWRs/ NMWRs shall be developed as specified by the MATDEV.

Figure appearance may vary slightly based on version of DTD/stylesheets utilized (see applicable DTD or tag definition list for available tags).



TM X-XXXX-XXX-XX 0024

OPERATOR TROUBLESHOOTING

OPERATIONAL CHECKOUT - PRE-DEPLOYMENT AND DEPLOYMENT

INITIAL SETUP:

Personnel Required
Operator (2)

References
WP 0004
WP 0012
WP 0013
WP 0016
WP 0028
WP 0029
WP 0032
WP 0062

Equipment Condition
SCAMP II powered on (WP 0007)

OPCHECK PROCEDURE 1.

OUTRIGGER PRE-DEPLOYMENT

OPERATIONAL CHECKOUT TEST PROCEDURE 1.

TESTING - BRANCH 1.

1. Test outrigger pre-deployment IAW WP 0012 OPERATION UNDER USUAL CONDITIONS - OPERATING THE OUTRIGGERS.
   CONDITION/INDICATION
   Outrigger does not easily rotate from stored condition to deployed condition when deployment pivot pin is removed.
   Pivot pin does not rotate from storage to deployment condition.
   CORRECTIVE ACTION
   1. Lubricate pin IAW WP 0062 LUBRICATE SCAMP II.
   CONDITION/INDICATION
   Deployment pivot pin does not insert or fully seat into deployed condition pin hole when outrigger is rotated into deployed condition.
   CORRECTIVE ACTION
   1. Refer to maintainer maintenance.

OPERATIONAL CHECKOUT TEST PROCEDURE 2.

TESTING - BRANCH 1.

1. Verify SCAMP II transitions to **Crane Mode** IAW WP 0012 OPERATION UNDER USUAL CONDITIONS - OPERATING THE OUTRIGGERS.
   CONDITION/INDICATION
   SCAMP II does not transition to **Crane Mode** when running and after all four outriggers have been properly deployed.
   CORRECTIVE ACTION
   1. Troubleshoot IAW WP 0028 OUTRIGGER DEPLOYMENT FAILURE - CANNOT ENTER CRANE MODE WITH OUTRIGGERS DEPLOYED.

0024–1

**FIGURE D-6. Operational checkout procedure.**

## D.6.2.3.3 Test set message word index <messageindx>.

The message word index shall consist of a series of test set messages or bit-code words with message word description. Based on the message or bit-code word, a corrective action shall be stated. This corrective action can either be stated as a specific remedy or can be a reference **<xref>** to a detailed troubleshooting procedure WP.

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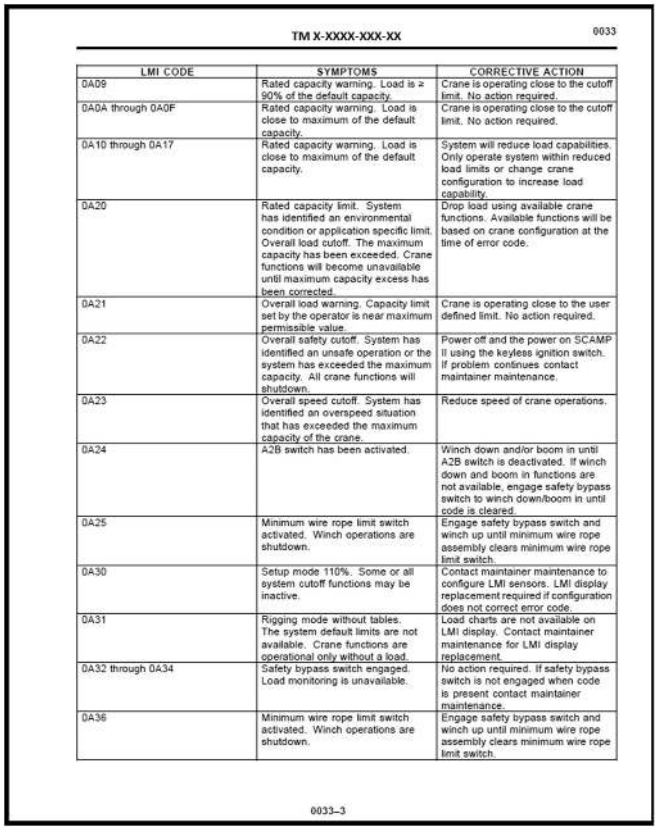
### D.6.2.3.4 Fault code reference index <faultreports>.

The fault code reference index shall consist of a fault code or codes that lead to a corrective action. This corrective action can either be stated as a specific remedy or can be a reference **<xref>/<link>** to a maintenance WP. If applicable, additional follow-on operational testing procedures **<follow-on>** shall be included based on the corrective action. See example fault reference code index in FIGURE D-7.

**Note**

Figure appearance may vary slightly based on version of DTD/stylesheets utilized (see applicable DTD or tag definition list for available tags).



TM X-XXXX-XXX-XX 0033

| LMI CODE | SYMPTOMS | CORRECTIVE ACTION |
|---|---|---|
| 0A09 | Rated capacity warning. Load is ≥ 90% of the default capacity. | Crane is operating close to the cutoff limit. No action required. |
| 0A0A through 0A0F | Rated capacity warning. Load is close to maximum of the default capacity. | Crane is operating close to the cutoff limit. No action required. |
| 0A10 through 0A17 | Rated capacity warning. Load is close to maximum of the default capacity. | System will reduce load capabilities. Only operate system within reduced load limits or change crane configuration to increase load capability. |
| 0A20 | Rated capacity limit. System has identified an environmental condition or application specific limit. Overall load cutoff. The maximum capacity has been exceeded. Crane functions will become unavailable until maximum capacity excess has been corrected. | Drop load using available crane functions. Available functions will be based on crane configuration at the time of error code. |
| 0A21 | Overall load warning. Capacity limit set by the operator is near maximum permissible value. | Crane is operating close to the user defined limit. No action required. |
| 0A22 | Overall safety cutoff. System has identified an unsafe operation or the system has exceeded the maximum capacity. All crane functions will shutdown. | Power off and the power on SCAMP II using the keyless ignition switch. If problem continues contact maintainer maintenance. |
| 0A23 | Overall speed cutoff. System has identified an overspeed situation that has exceeded the maximum capacity of the crane. | Reduce speed of crane operations. |
| 0A24 | A2B switch has been activated. | Winch down and/or boom in until A2B switch is deactivated. If winch down and boom in functions are not available, engage safety bypass switch to winch down/boom in until code is cleared. |
| 0A25 | Minimum wire rope limit switch activated. Winch operations are shutdown. | Engage safety bypass switch and winch up until minimum wire rope assembly clears minimum wire rope limit switch. |
| 0A30 | Setup mode 110%. Some or all system cutoff functions may be inactive. | Contact maintainer maintenance to configure LMI sensors. LMI display replacement required if configuration does not correct error code. |
| 0A31 | Rigging mode without tables. The system default limits are not available. Crane functions are operational only without a load. | Load charts are not available on LMI display. Contact maintainer maintenance for LMI display replacement. |
| 0A32 through 0A34 | Safety bypass switch engaged. Load monitoring is unavailable. | No action required. If safety bypass switch is not engaged when code is present contact maintainer maintenance. |
| 0A36 | Minimum wire rope limit switch activated. Winch operations are shutdown. | Engage safety bypass switch and winch up until minimum wire rope assembly clears minimum wire rope limit switch. |

0033–3

**FIGURE D-7. Fault code reference index.**

### D.6.2.3.5 Operational checkout and troubleshooting procedures <opcheck-tsproc>.

When advantageous, operational checkout and troubleshooting procedures may be combined into one WP depending on system complexity, equipment specifications, available testing options, and publication composition. When combined, operational checkout and troubleshooting procedures together shall do the following: (1) verify proper operation to prescribed standards, and (2) detect, isolate, and correct system and equipment failures and malfunctions.

When it is determined that the troubleshooting procedures shall be separate from the operational checkout procedures, the troubleshooting procedures shall be included under the heading "TROUBLESHOOTING." Combined operational checkout and troubleshooting for DMWRs/NMWRs shall be developed as specified by the MATDEV.

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### D.6.3 Diagnostic (integrated troubleshooting procedures.

Many systems have been designed to use semi-automatic/automatic self-test equipment such as sensors and fault-detecting software. If equipment/system has self-testing capabilities, they shall be used to provide critical information on system operation or discrepancies. These systems are designed and programmed for rapid electronic testing in the interest of reducing maintenance downtime to fault isolate and repair. Many systems/pieces of equipment have been designed with BITE capabilities, which identify faults to the operator or maintainer. BITE faults may be further isolated using diagnostic software or other troubleshooting procedures. When diagnostic software is used to isolate BIT faults, the required software shall be identified in the technical publication. APPENDIX M guides development of a software information module, which includes potentially relevant error recovery, malfunction, and emergency screens in various required WPs. Troubleshooting WPs shall reference software information WPs as applicable to prevent duplication of information.

**Note**

Software troubleshooting WPs shall have the word "user" or "administrator" followed by the words "software troubleshooting" and then the WP title. Any that apply to both user and administrator will use the word "user".

See D.8.6 for creating diagnostic troubleshooting WPs. See FIGURE D-20 for an example IETM troubleshooting screen.

#### D.6.3.1 Modes of diagnostic troubleshooting.

Technical publications shall support one of two modes of diagnostics: simple and complex. Linear data shall use the simple diagnostic mode. Nonlinear data shall use either mode as specified by the MATDEV.

**a.** Simple diagnostic mode: Follows the same binary, linear ("yes" / "no"; "true" / "false") logic as discussed in manual troubleshooting (D.8.4) and does not support state table functionality (D.6.3.2 and D.6.3.3.).

**b.** Complex diagnostic mode: Provides direct interface between an Interactive Electronic Technical Manual (IETM), diagnostic software that supports testing, and test equipment/hardware. Complex diagnostics also provides dynamic support to system changes through manipulation of a state table (D.6.3.2 and D.6.3.3). This mode also allows for evaluation and action based on multiple inputs and allows for multiple branching in one troubleshooting procedure. The MATDEV shall specify desired features in their diagnostics through the IETM Functionality Matrix (APPENDIX A).

#### D.6.3.2 State tables.

A state table provides the IETM and/or the user with information on a condition of a task being performed or changes in system or user-defined variables. Input to the state table may come from various sources such as the following:

**a.** Initial settings from the technical publication data, entered by the author during publication development.

**b.** Input from installed systems/subsystems through direct interconnection with the system/subsystem or data bus connectivity. This input may be either values obtained during initialization of the system or as a result of user or other input obtained during a diagnostic or operational test.

**c.** Input provided by the technical publication (end) user through menu selection choices or direct input through dialogs provided by the IETM.

#### D.6.3.3 State table options.

As a minimum, a state table shall allow storing system and user input. State table information may be used by the system to perform tests, evaluate, provide feedback, make comparisons with other stored information, or provide

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alternative courses of action. In more sophisticated systems, the state table may be tied into specific item status file (s). This more robust state table would store information from all ongoing maintenance and provide all users with the condition of equipment at any specific point in time (for example, in service, awaiting parts, outstanding maintenance actions [open panels or fuel cells or other conditions that might preclude some maintenance from being performed or allow skipping some preconditions as they have been previously completed]).

### D.6.3.4 Diagnostic testing.

Diagnostic testing may be an entire automatic system test **<diagnostic-track-automated>**), a series of manual steps required to obtain some level of fault identification **<diagnostic-track-manual>**, or both of these scenarios. Additionally, both of these diagnostic testing types – automated and manual – may be utilized in the same WP. The following types of information shall be included in diagnostic testing (as applicable):

**a.** Warnings, cautions, or notes that would apply to the entire diagnostic procedure.

**b.** Configuration unique hookup **<hookup>** or conditional hookup (depending on state table information) **<hookup-alt>** requirements not identified in the initial setup.

**c.** Reason(s) for performing the test **<reasonfortest>**.

**d.** Test methods consisting of a single test selected from the following:

**(1)** Complex test **<testwithstate>**.

**(2)** Conditional complex test **<testwithstate-alt>**.

**e.** Testing conclusion consisting of the following:

**(1)** Testing conclusion (Reset): Follow-on actions that need to be taken to "reset" the system so the next tests can be executed. These actions may require installing disconnected harnesses, removing jumper wires, flipping switches, master power off, etc.

**(2)** Testing conclusion (Disconnect): Upon a testing conclusion, any test equipment not required for the next diagnostic test shall be removed through a disconnection procedure **<disconnect>** or conditional disconnection **<disconnect-alt>** procedure.

### D.6.3.5 Troubleshooting integrated systems.

When several systems are dependent upon each other for proper operation, the interdependent systems (as a unit) are identified as an integrated system. Development of testing and troubleshooting for integrated systems shall be determined by whether the systems have self-test or BIT capabilities or require the use of a system peculiar test set or common test equipment. Troubleshooting of integrated systems require more specifics of which components are tested and by which method (manual or diagnostic).

### D.6.3.6 Integrated systems with self-test or BIT capability.

Testing and troubleshooting procedures shall identify components or functions that are tested as well as any additionally input required for proper testing (power parameters, signals, motion, air, hydraulic, etc.). If wiring test are included, they shall have defined testing parameters (resistance tolerances, open definitions, wire-to-wire, and wire-to-ground resistances, and any unique wire criteria). Fault criteria for a failure indication shall be specified for integrated system troubleshooting. Any integrated system that requires the use of peculiar test sets shall specify any requirements for special cables or support equipment. Any integrated system for only requires common test equipment shall focus on actual readings or signal requirements to ensure sources of common test equipment are not restricted.

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### D.6.4 Fault reporting and fault isolation testing.

Fault reporting provides operating personnel with a standardized method to report malfunctions and fault symptoms. Fault isolation information is designed for use in rapid isolation of faults revealed during an operational mission or when an aircraft/equipment system is in an operational configuration. These data shall instruct maintenance personnel about maintenance actions to perform to correct a fault. Fault reporting information and fault isolation testing data are designed to be used together. Fault isolation information coverage shall be limited to faults identified in the fault reporting data that require specific procedures to isolate the cause. Fault reporting data shall reference the fault isolation data to the maximum extent practical for isolation of indicated malfunctions.

#### D.6.4.1 Fault reporting and fault isolation data.

Troubleshooting fault reporting and fault isolation data includes detailed testing and troubleshooting results for each weapon system's equipment, systems, components, and support equipment. When applicable, integrated system testing and troubleshooting data for aircraft and major weapon systems shall also be included.

**a.** Manually tested systems. Lookup tables shall be prepared so that a maintainer may properly isolate and interpret and fault and correct a malfunction. Tables shall provide indication, observed equipment symptom, and corrective action.

**b.** Diagnostically tested systems. Software status indication coding tables shall be prepared so that a maintainer may properly interpret software data and isolate and correct malfunctions. Tables shall provide indication, observed equipment symptom, and corrective action.

## D.7 DEVELOPMENT OF TROUBLESHOOTING MODULES.

Some WPs in a technical publication are grouped together to form a category, and these categories are based on the type and complexity of equipment in a system. A technical publication troubleshooting module **<tim>** shall be prepared as required by publication type (see A.4.4) and shall include one or more of the categories shown in FIGURE D-8 and listed in D.7.1 thru D.7.3. FIGURE D-9 shows an example structure of **<troublecategory>** module and the available WPs in a publication.

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**Note**

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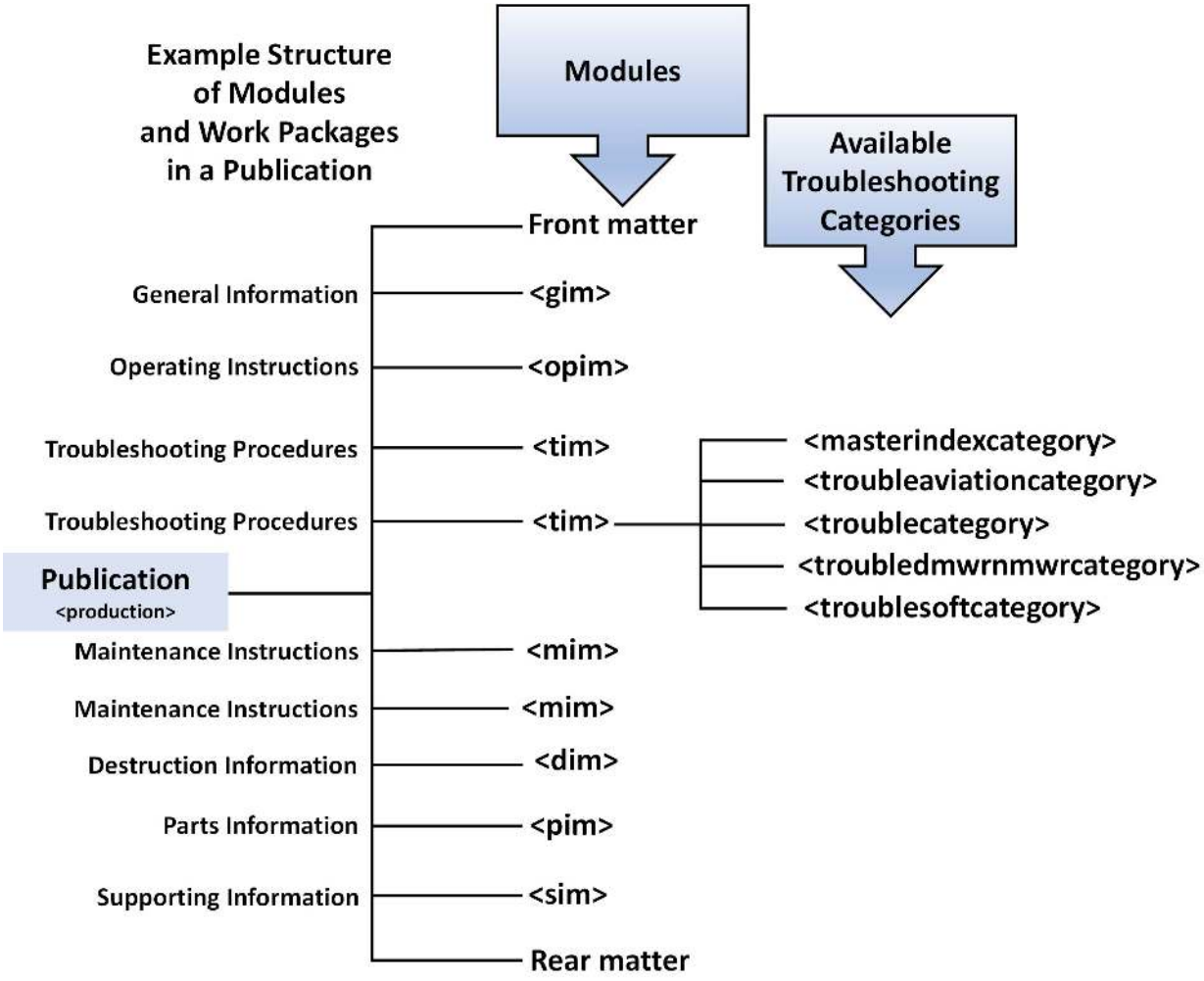


**FIGURE D-8. Troubleshooting modules and categories.**



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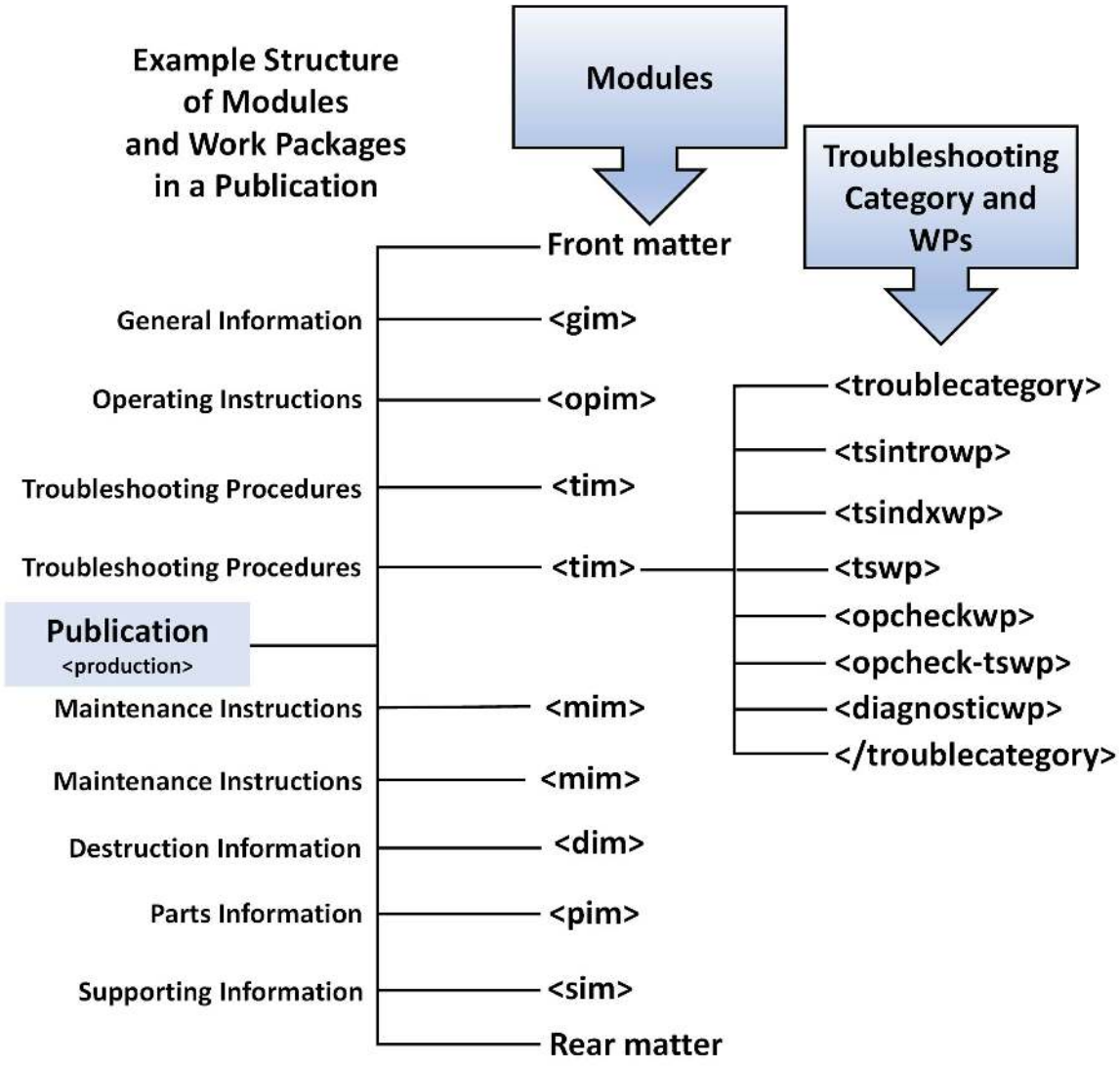


**FIGURE D-9. Troubleshooting modules, categories, and WPs.**

### D.7.1 Aviation testing and troubleshooting category (aircraft TMs only) <troubleaviationcategory>.

This troubleshooting category shall contain the WPs listed below (in the order specified) as required by publication type (see A.4.4) and as applicable for an equipment system.

**a.** Introduction WP **<tsintrowp>** (D.8.1).

**b.** Technical description WP **<techdescwp>** (D.8.2).

**c.** Troubleshooting index WP **<tsindxwp>** (D.8.3).

**d.** Operational checkout WPs **<opcheckwp>** (D.8.5).

**e.** Combined operational checkout and troubleshooting WP **<opcheck-tswp>** (D.8.5.1).

**f.** Manual troubleshooting WP **<tswp>** (D.8.4).

**g.** Diagnostic troubleshooting WP **<diagnosticwp>** (D.8.6).



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### D.7.2 Standard testing and troubleshooting category <troublecategory>.

This troubleshooting category shall contain the WPs listed below (in the order specified) as required by publication type (see A.4.4) and as applicable for an equipment system.

**a.** Introduction WP **<tsintrowp>** (D.8.1).

**b.** Troubleshooting index WP **<tsindxwp>** (D.8.3).

**c.** Operational checkout WPs **<opcheckwp>** (D.8.5).

**d.** Combined operational checkout and troubleshooting WP **<opcheck-tswp>** (D.8.5.1).

**e.** Manual troubleshooting WP **<tswp>** (D.8.4).

### D.7.3 Depot Maintenance Work Requirements (DMWRs) / National Maintenance Work Requirements (NMWRs) testing and troubleshooting category (depot only) <troubledmwrnmwrcategory>.

This troubleshooting category shall contain the WPs listed below (in the order specified) as required by publication type (see A.4.4) and as applicable for an equipment system.

**a.** Introduction WP **<tsintrowp>** (D.8.1).

**b.** Troubleshooting index WP **<tsindxwp>** (D.8.3).

**c.** Operational checkout WPs **<opcheckwp>** (D.8.5).

**d.** Combined operational checkout and troubleshooting WP **<opcheck-tswp>** (D.8.5.1).

**e.** Manual troubleshooting WP **<tswp>** (D.8.4).

**f.** Diagnostic troubleshooting WP **<diagnosticwp>** (D.8.6).

**g.** Preshop analysis WP **<pshopanalwp>** (D.8.7).

**h.** Component checklist WP **<compchklistwp>** (D.8.8).

### D.7.4 Master index testing and troubleshooting category <masterindexcategory>.

The troubleshooting category shall only be utilized when developing a technical publication with a master troubleshooting index (D.8.3.2.1) within an troubleshooting index WP (D.8.3.1).

## D.8 DEVELOPMENT OF TROUBLESHOOTING WORK PACKAGES.

This section covers the general components of the following example troubleshooting WPs: troubleshooting intro WP **<tsintrowp >**, technical description WP **<techdescwp>** (aircraft only), troubleshooting index WP **<tsindxwp>**, manual troubleshooting WP **<tswp>**, operational checkout WP **<opcheckwp>**, **<diagnosticwp>**, preshop analysis WP **<pshopanalwp>**, and component checklist WP **<compchklistwp>**. Some required tags of a few these WPs are also shown to provide an overview of the general structure of these WPs.

### D.8.1 Troubleshooting introduction work package <tsintrowp>.

This WP shall be developed for aviation systems and may be developed for non-aviation systems. This WP shall describe the testing and troubleshooting process used to perform troubleshooting and shall include information on the methods used to perform troubleshooting. The general flow of the troubleshooting process shall be described, and the general methods used to perform testing and troubleshooting shall be included. Any information unique to troubleshooting electrical subsystems and electronic equipment shall also be described. If a troubleshooting index


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**<tsindxwp>** is used, an explanation of the index shall be provided. FIGURE D-10 shows an example troubleshooting introduction WP and associated tags. The following components shall be included in troubleshooting introduction WPs.

**a.** WP identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** WP initial setup **<initial_setup>** is not required for this WP.

**c.** Troubleshooting introduction information.

**Note**

Figure appearance and tags/output may vary slightly based on version of DTD/stylesheets utilized (see applicable DTD or tag definition list for available tags).

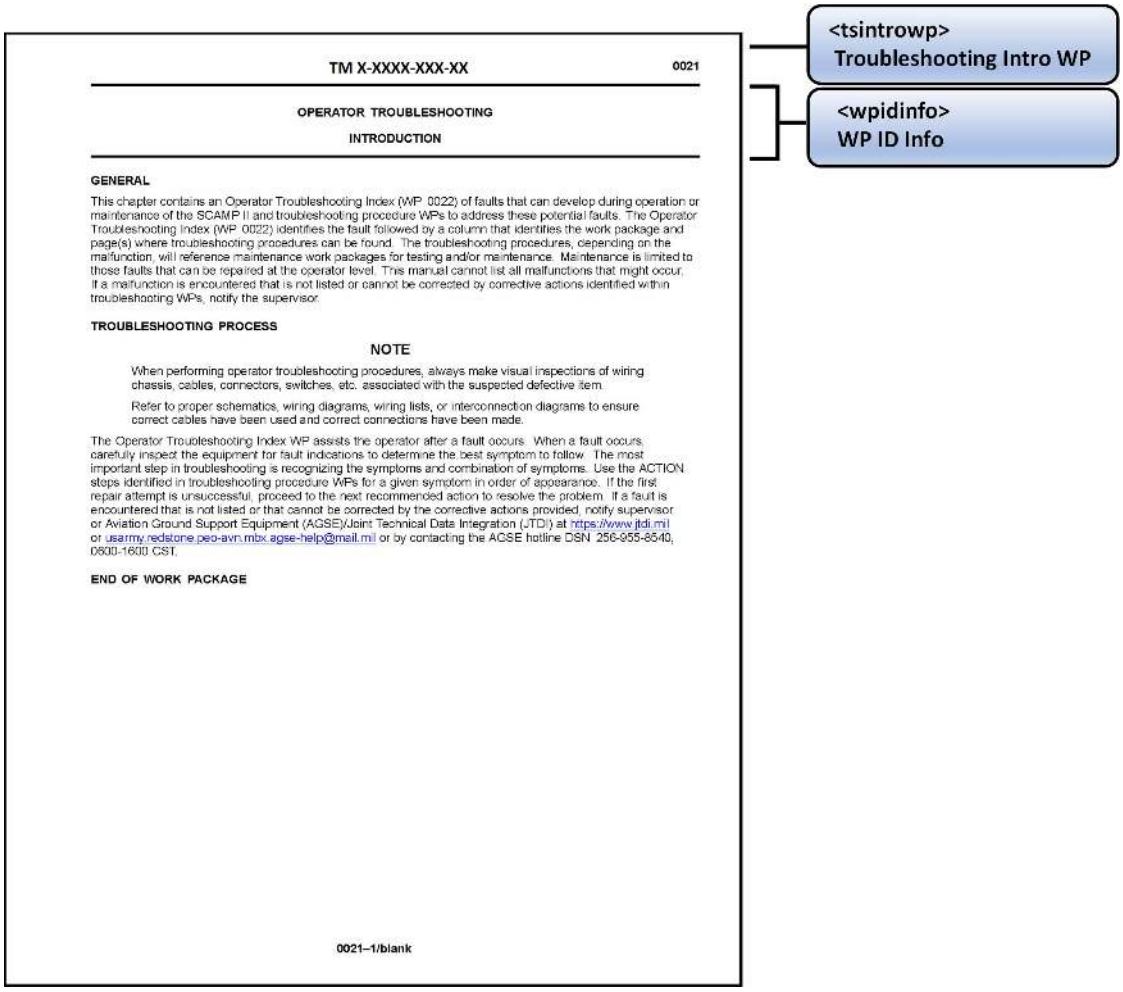


**FIGURE D-10. Troubleshooting introduction WP example.**

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## D.8.2 Technical description work package <techdescwp> (Aircraft Maintenance Manuals).

A technical description WP may be developed as applicable for an equipment system. The following components shall be included in technical description WPs.

**a.** WP identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** WP initial setup **<initial_setup>** is required for this WP (5.4.3).

**c.** The following WP components may be included in technical description WPs, as applicable:

**(1)** Equipment description and data **<descproc>**. When equipment description and data information is required to support the testing and troubleshooting procedures, it shall be prepared in accordance with the requirements provided in B.7.2, as applicable. If this information is provided in another technical publication, a reference to the publication may be included in lieu of including the descriptive data.

**(2)** Theory of operation **<thryproc>**. When theory of operation is required to support the troubleshooting procedures, it shall be prepared in accordance with the requirements provided in B.7.3, as applicable. If this information is provided in another technical publication, a reference to the publication may be included in lieu of including the theory data.

## D.8.3 Troubleshooting index types.

There are two troubleshooting index types: general troubleshooting indexes (D.8.3.1) and malfunction-specific indexes (D.8.3.2). General troubleshooting indexes list all troubleshooting topics covered in a module and do not group topics together based on malfunction type. Malfunction-specific indexes are used when all probable faults have been determined and described. Malfunction-specific indexes shall use the exact description of the fault or symptom as was used in the troubleshooting procedures.

### D.8.3.1 Troubleshooting index WP <tsindxwp>.

When applicable, one troubleshooting index WP shall be prepared for all troubleshooting for the system/equipment and shall appear after the troubleshooting introduction WP. FIGURE D-11 shows an example troubleshooting index WP and associated tags. This WP shall be prepared as directed by the MATDEV and consist of either a malfunction/symptom index **<tsindx.symptom>/<tsindx.messageword>** or a system/subsystem index **<tsindx. system>**. The following components shall be included in troubleshooting index WPs.

**a.** WP identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** WP initial setup **<initial_setup>** is not required for this WP.

**c.** Malfunction/symptom index **<tsindx.symptom>/<tsindx.messageword>** or system/subsystem index **<tsindx. system>**.

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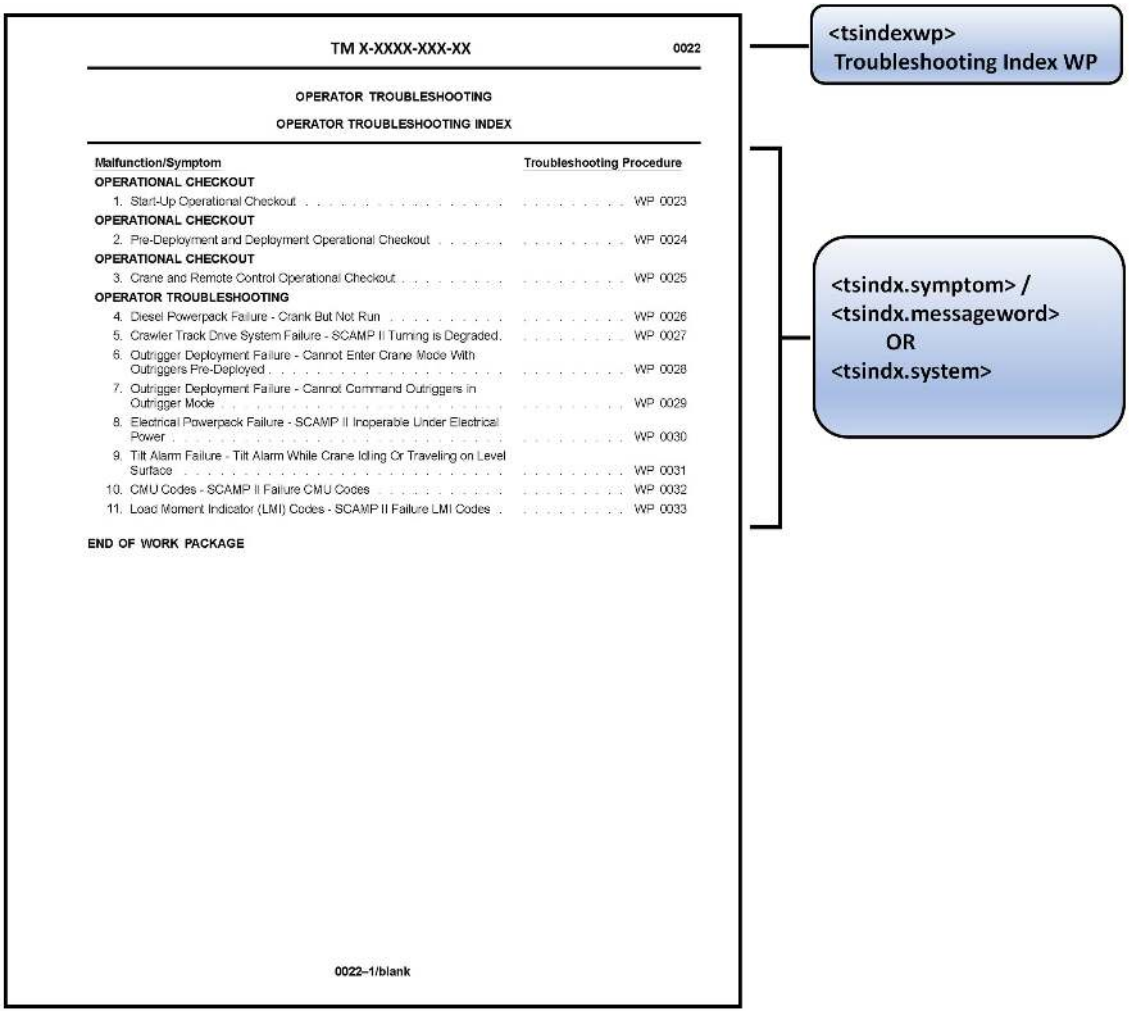


**FIGURE D-11. Troubleshooting index WP example.**

## D.8.3.2 Malfunction/symptom index <tsindx.symptom>/<tsindx.messageword>.

Malfunction/symptom indexes help users pinpoint possible malfunctions based on symptoms, BIT test codes, systems, subsystems, etc. The malfunction/symptom index shall include the following data:

**a.** For simple systems, list all fault symptoms or known malfunctions in alphabetical order by malfunction/symptom **<malfunc>** or by built-in test code/fault message word **<messageword>**. Reference this information to the applicable testing and troubleshooting WP sequence number **<xref>/<link>/<extref>** or the required corrective action **<action>**.

**b.** For complex systems, list symptoms by subsystem categories **<tsindx.symptom-category>/<tsindx.messageword-category>**, if necessary, and use codes such as FGC that help identify specific items. Group symptoms to common subsystem areas both in the malfunction/symptom index and in the troubleshooting procedures. For example, if a system has a data link, communications, radar, display, and tracking subsystems, the symptoms would be grouped into each related subsystem. All fault symptoms of a communications nature

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would fall under the communications subsystem. The symptoms may be further divided into functions within the communications subsystem that would be common. The same would be done for radar, data link, display, and tracking subsystems. Subsystem categories shall be listed in alphabetical order or by code.

**c.** Catalog malfunctions/symptoms by method of detection if this aids usability.

**d.** Fault symptom descriptions (titles) shall be standardized between malfunction/symptom index WPs and troubleshooting procedures WPs.

#### D.8.3.2.1 Master malfunction/symptom index <tsindx.system>.

When applicable, one troubleshooting malfunction/symptom index WP shall be prepared for all troubleshooting for the system/equipment. Refer to D.8.4.

#### D.8.3.2.2 System/subsystem index <tsindx.system>.

This index shall consist of a list of specific systems, subsystems, assemblies, and components requiring troubleshooting, referenced to the applicable testing and troubleshooting WP sequence number **<xref>/<link>/<extref>** or required corrective action **<action>**.

### D.8.4 Manual (non-integrated) troubleshooting work package <tswp>.

Manual troubleshooting WPs shall be developed so that operator/crew and maintainer or field maintenance personnel can perform all required troubleshooting to isolate faults to a single fault/failure and part. FIGURE D-12 through FIGURE D-14 show a manual troubleshooting **<logicproc>** example. FIGURE D-15 shows a manual troubleshooting **<faultproc>** example. The following components shall be included in troubleshooting WPs.

**a.** WP identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** WP initial setup **<initial_setup>** is required for this WP (5.4.3).

**c.** Troubleshooting procedures **<tsproc>**. The following WP components may be included in troubleshooting WP when required:

**(1)** Introduction **<intro>**. When required, an introduction shall be included explaining how the troubleshooting procedures are to be used to perform checkout and troubleshooting and how they relate to the associated maintenance WPs that include the corrective actions that will return the equipment to proper operation.

**(2)** General procedures and precautions **<proc>**. When required, any general procedures that must be performed prior to checkout and precautions that must be taken during the performance of the checkout procedure shall be included.

**(3)** Component Checklist: When required, the introduction **<intro>** shall contain a component checklist preceded by a brief explanation of its use.

**(4)** Pretest setup procedures **<hookup>**. When required, procedures for connecting any test and accessory equipment, including cable connections, shall be included. Procedures for the initial setting of controls shall also be provided.

**(5)** Post-operational shutdown procedures **<disconnect>**. When required, procedures to return the aircraft, aircraft system, or equipment to its normal configuration prior to operational checkout or troubleshooting setup shall be included.

**(6)** Follow-on maintenance **<followon.maintsk>** (E.7.3.1.3.3.11). When required, instructions or reference to appropriate WPs related to any follow-on maintenance shall be included.

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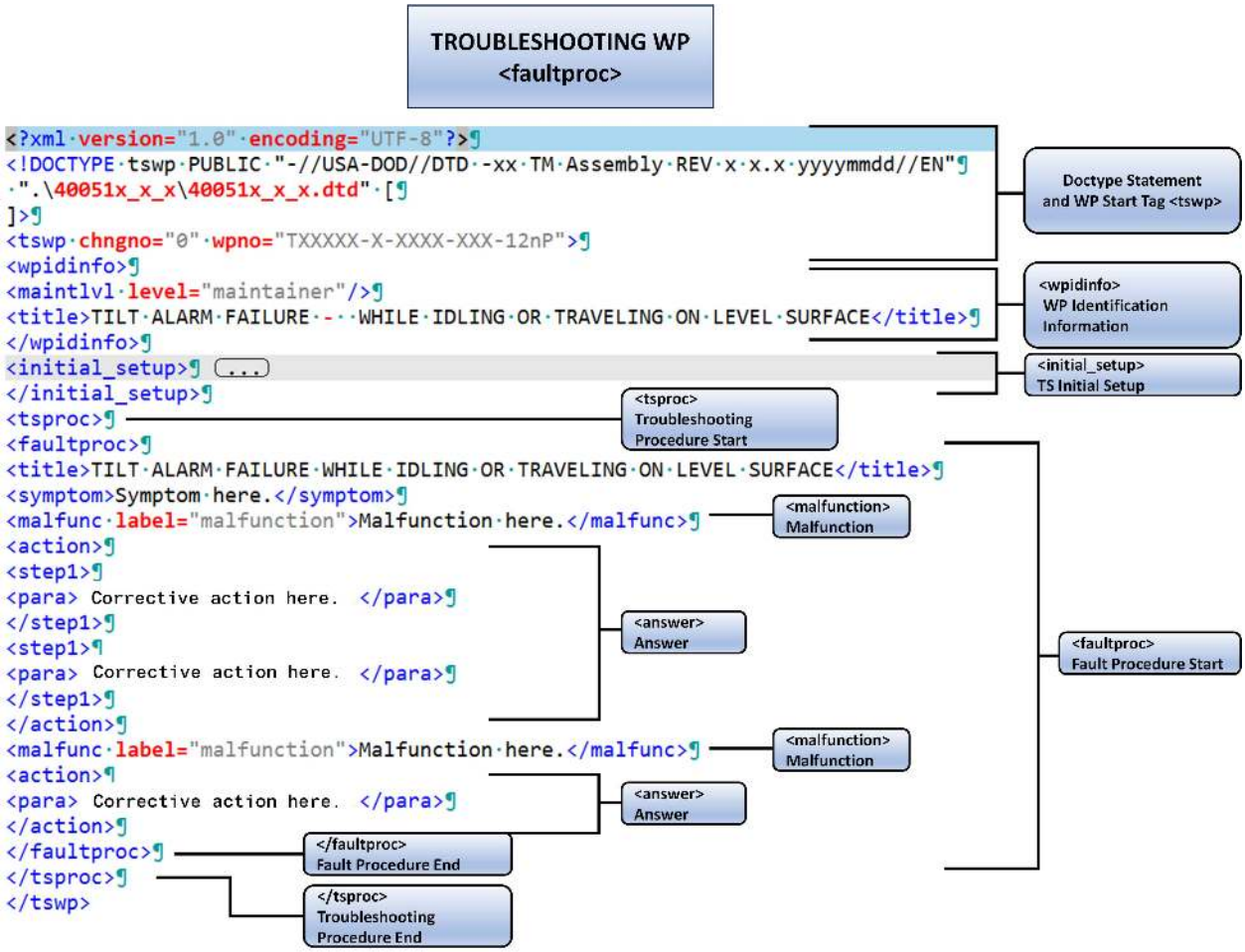


**FIGURE D-12. Manual troubleshooting WP example.**

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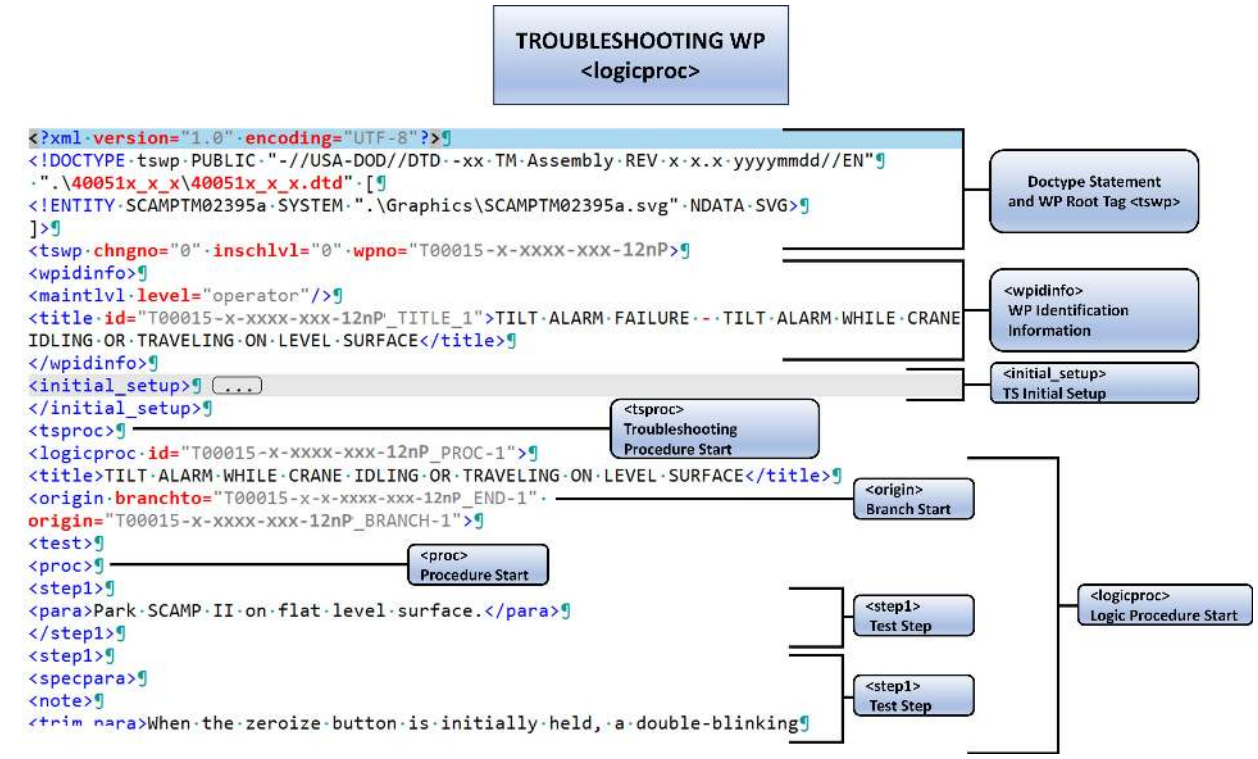


**FIGURE D-13. Manual troubleshooting WP example <logicproc> (sheet 1 of 2).**

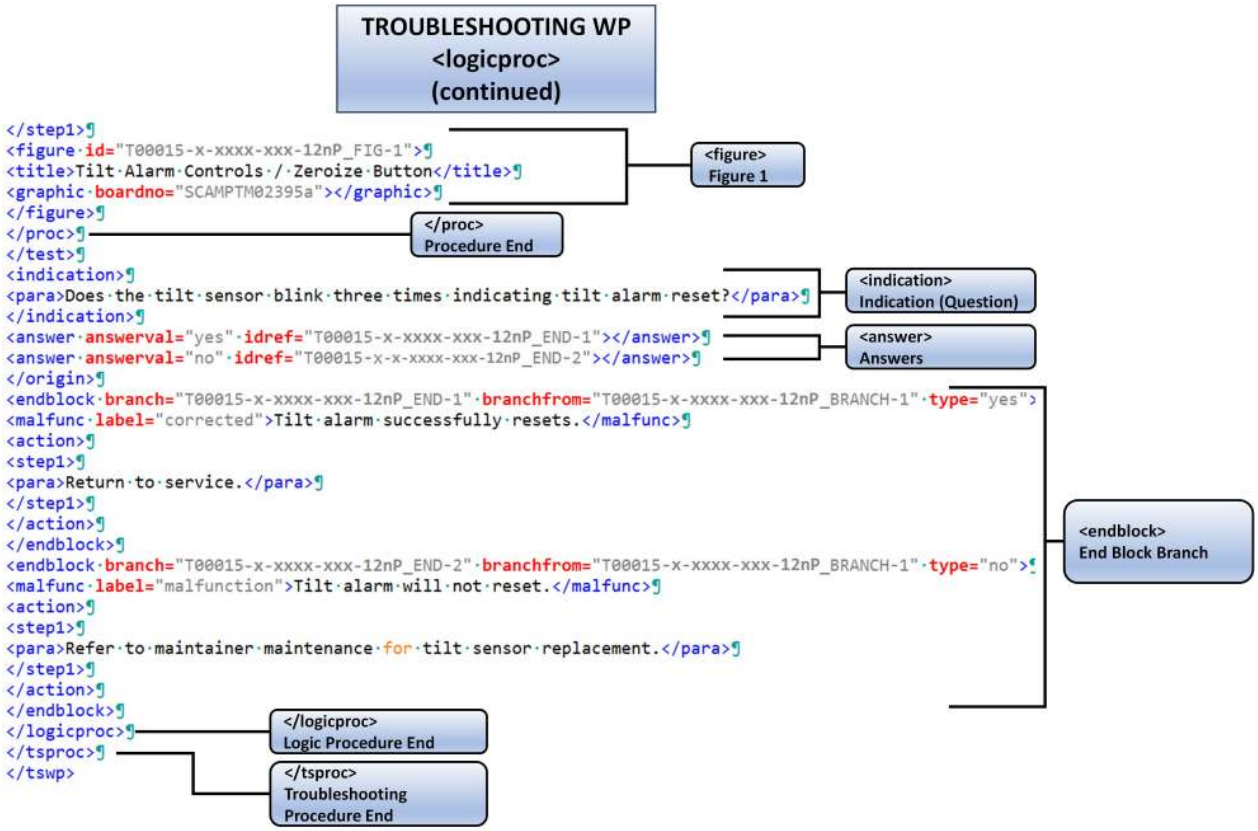


**FIGURE D-14. Manual troubleshooting WP example <logicproc> (sheet 2 of 2).**

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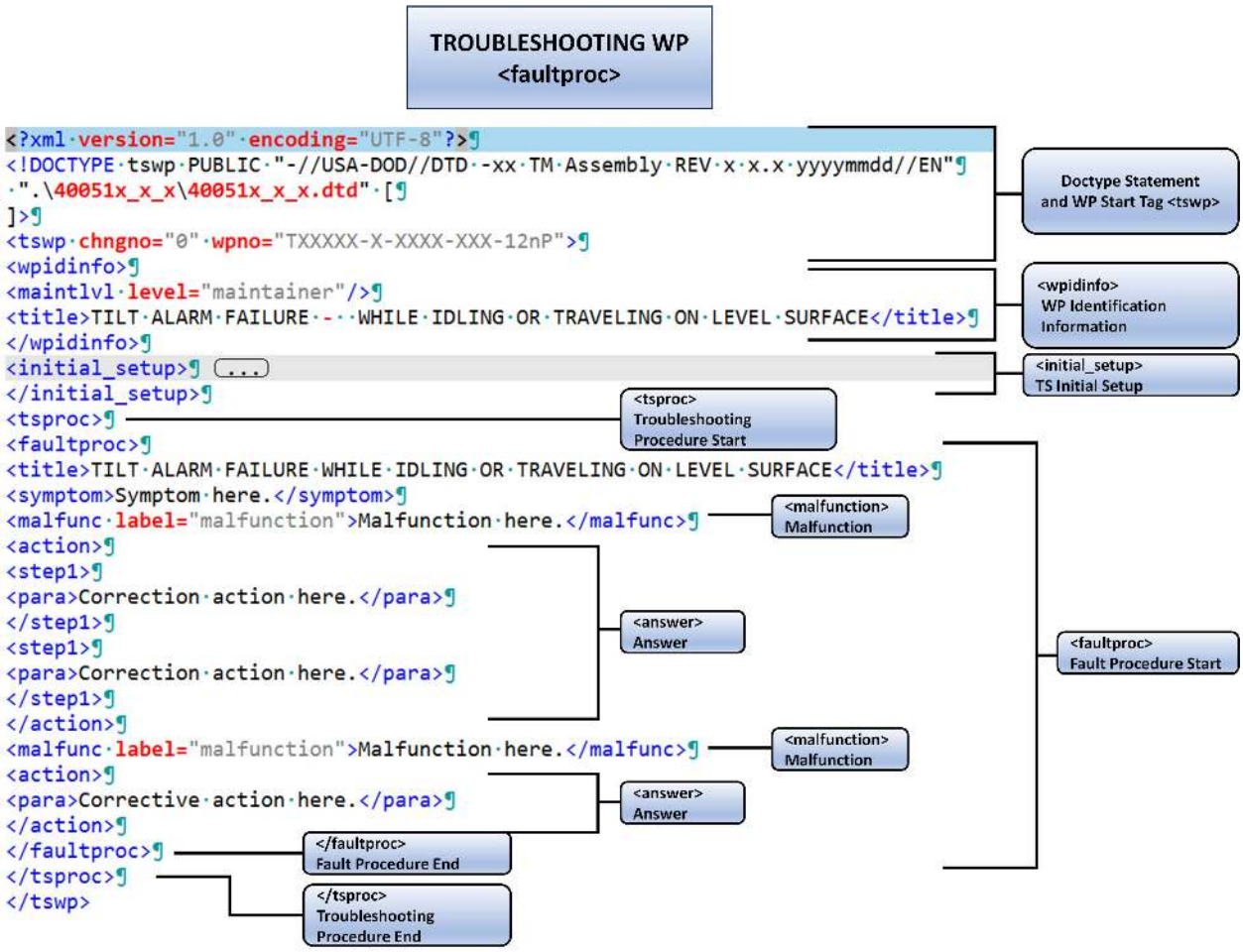


**FIGURE D-15. Manual troubleshooting WP example <faultproc>.**

## D.8.5 Operational checkout work package <opcheckwp>.

Operational checkout WPs shall be developed to subject equipment systems and/or their components to prescribed conditions to determine if they will function in accordance with predetermined test parameters. When applicable, these WPs shall be referenced to test if troubleshooting and/or maintenance procedures in a publication rectified a problem. FIGURE D-16 through FIGURE D-19 show an example operational checkout WP and associated tags. The following components shall be included in operational checkout WPs.

**a.** WP identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** WP initial setup **<initial_setup>** is required for this WP (5.4.3).

**c.** Introduction **<intro>**. When required, an introduction shall be included explaining how the operational checkout and/or troubleshooting procedures are to be used to perform checkout and/or troubleshooting and how they relate to the associated maintenance WPs that include the corrective actions that will return the equipment to proper operation.

**d.** General procedures and precautions **<proc>**. When required, any general procedures that must be performed prior to checkout and precautions that must be taken during the performance of the checkout procedure shall be included.

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**e.** Pretest setup procedures **<hookup>**. When required, procedures for connecting any test and accessory equipment, including cable connections, shall be included. Procedures for the initial setting of controls shall also be provided.

**f.** Operational checkout procedures **<opcheckproc>**. The selection of an operational checkout type shall be based on the type of system, equipment, or assembly/subassembly being addressed, the target audience, and the maintenance level of the operator/technician. Based on the complexity of the operational checkout to be performed, operational checkout procedures can be structured in one of three ways and therefore contain different content elements:

**(1)** Operational checkout test procedures **<testproc>** (D.6.2.3.2).

**(2)** Test set message word index **<messageindx>** (D.6.2.3.3.

**(3)** Fault code reference index **<faultreports>** (D.6.2.3.4).

**g.** Post-operational shutdown procedures **<disconnect>**. When required, procedures to return the aircraft, aircraft system, or equipment to its normal configuration prior to operational checkout or troubleshooting setup shall be included.

**h.** Follow-on maintenance **<follow-on>** (E.7.3.1.3.3.11). When required, instructions or reference to appropriate WPs related to any follow-on maintenance shall be included. Follow-on maintenance shall be included after maintenance has been performed on equipment and shall typically restore equipment to a state prior to what is listed in equipment conditions.

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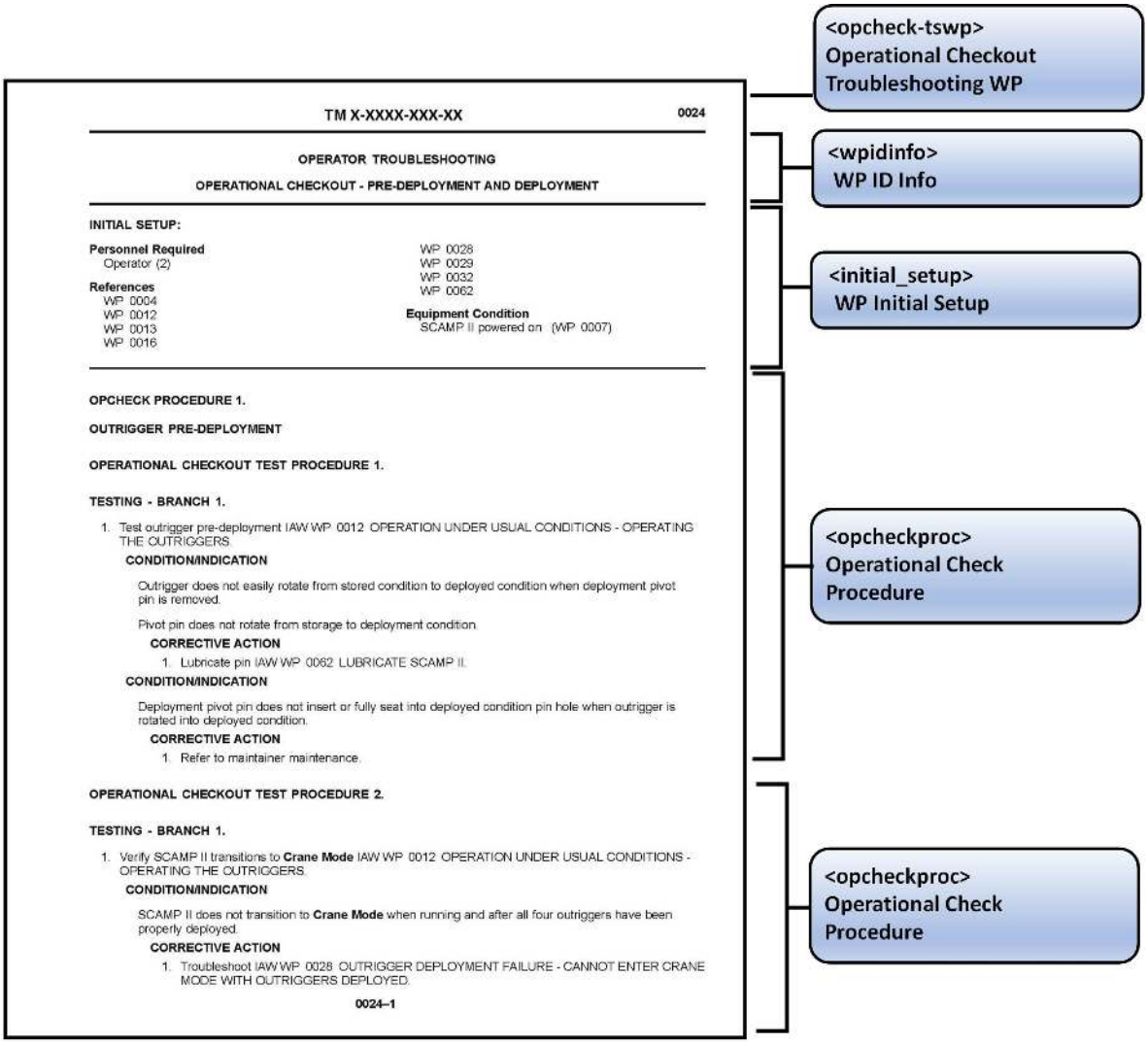


**FIGURE D-16. Operational checkout WP example.**

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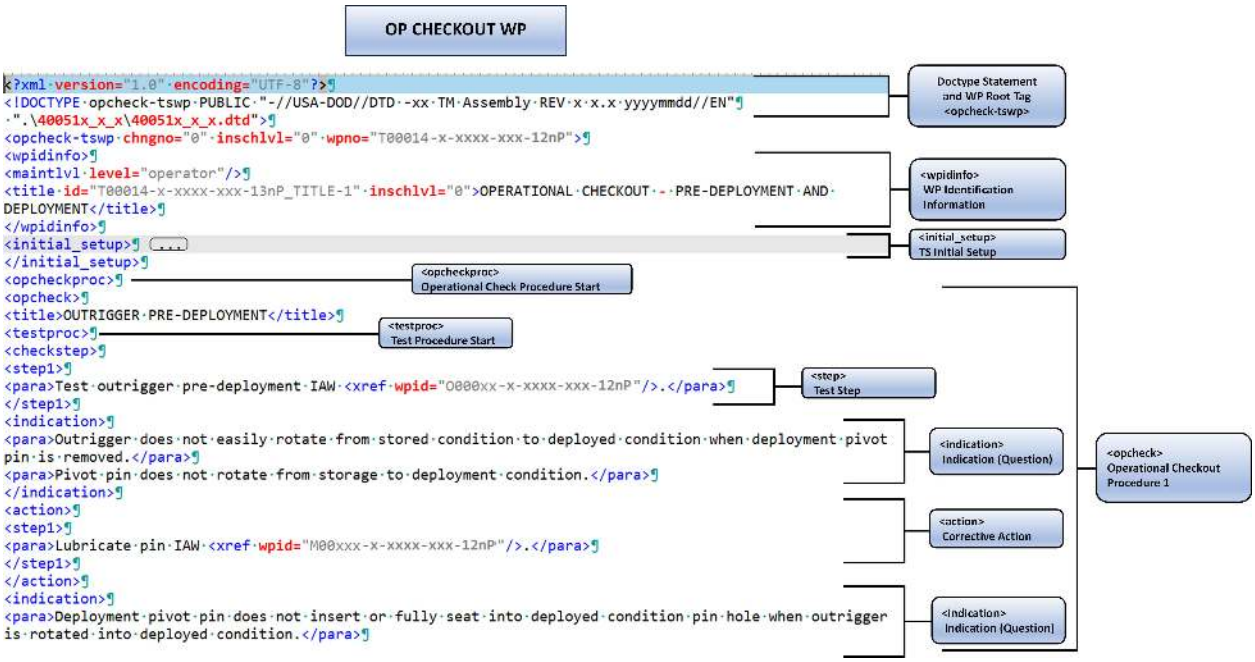


**FIGURE D-17. Operational checkout WP tags (1 of 3).**

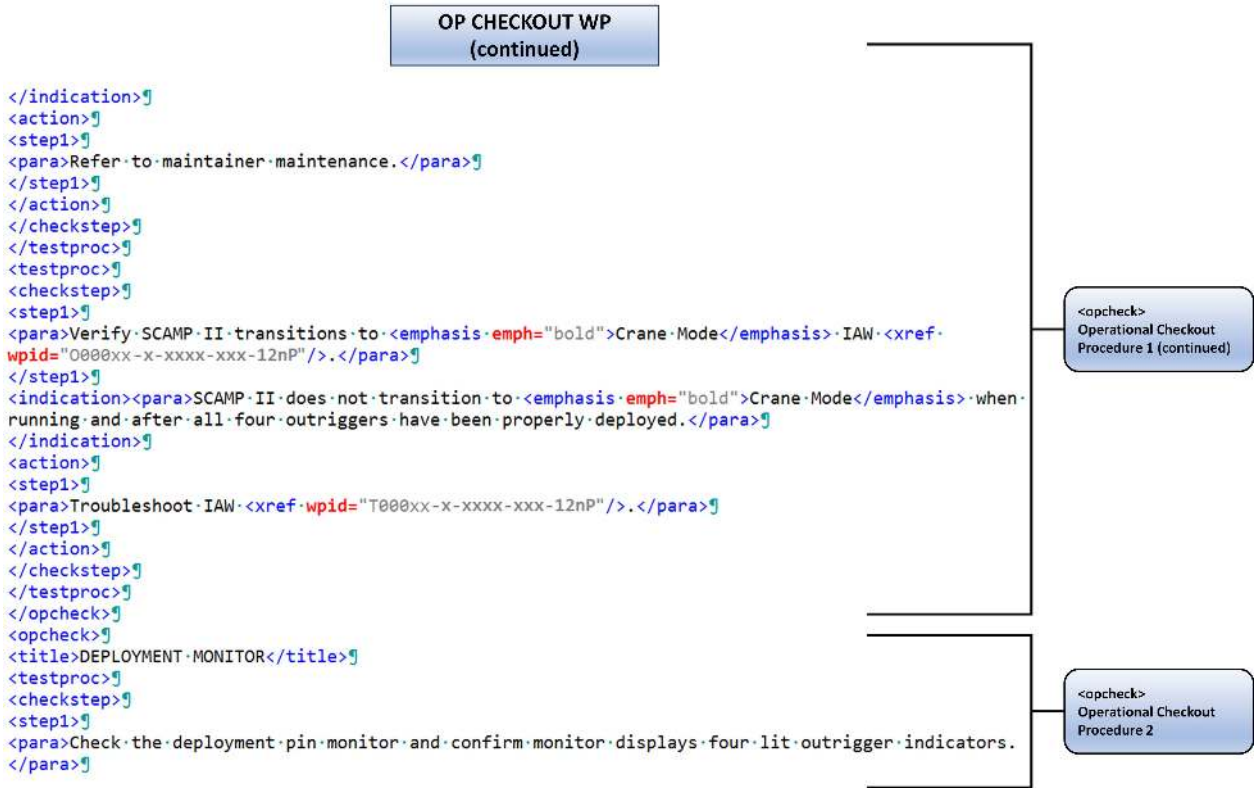


**FIGURE D-18. Operational checkout WP tags (2 of 3).**

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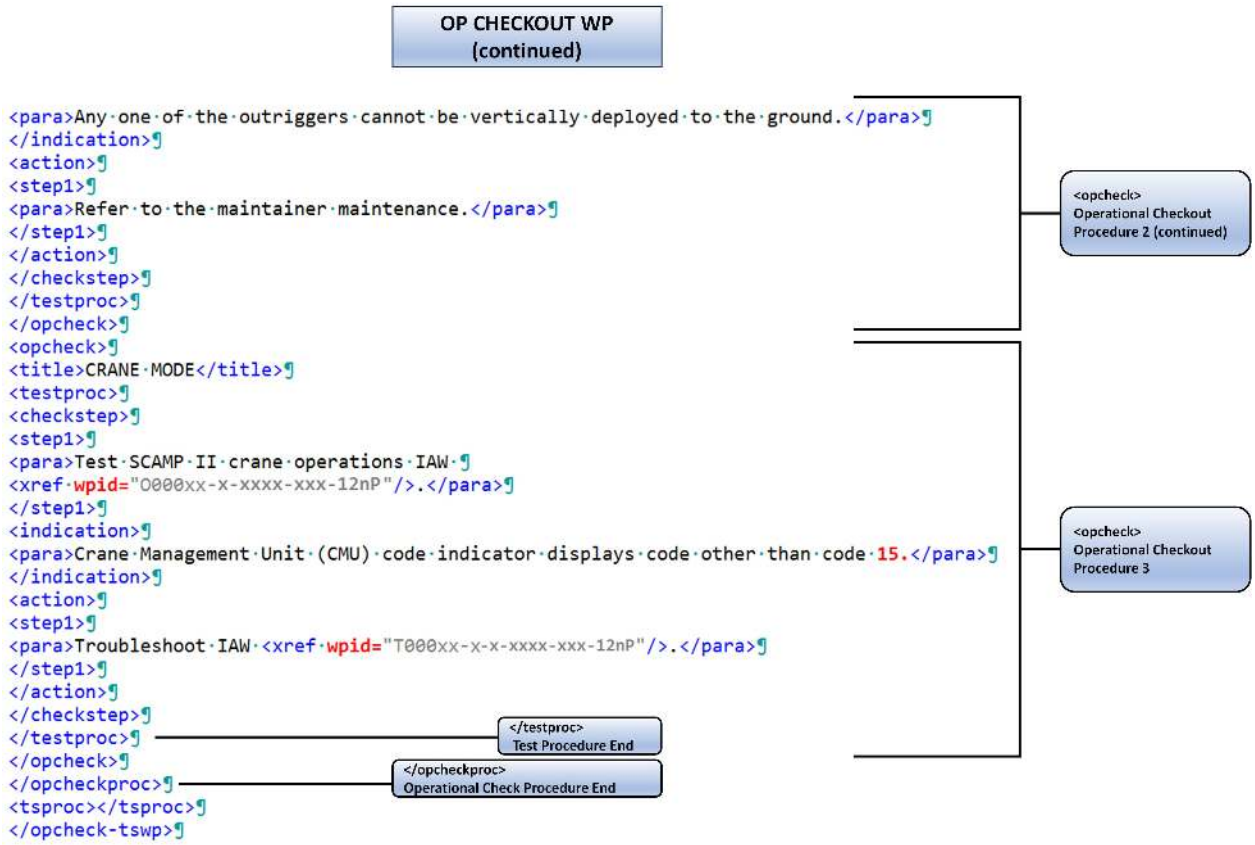


**FIGURE D-19. Operational checkout WP tags (3 of 3).**

### D.8.5.1 Combined operational checkout and troubleshooting WP <opcheck-tswp>.

Operational checkout procedures and troubleshooting procedures to verify proper operation to prescribed standards and for detecting, isolating, and correcting system and equipment failures and malfunctions may be developed as applicable for an equipment system. Combined operational checkout and troubleshooting for DMWRs/NMWRs shall be developed as specified by the MATDEV. The following components shall be included in combined WPs.

**a.** WP identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** WP initial setup **<initial_setup>** is required for this WP (5.4.3).

**c.** Introduction **<intro>**. When required, an introduction shall be included explaining how the operational checkout and/or troubleshooting procedures are to be used to perform checkout and/or troubleshooting and how they relate to the associated maintenance WPs that include the corrective actions that will return the equipment to proper operation.

**d.** General procedures and precautions **<proc>**. When required, any general procedures that must be performed prior to checkout and precautions that must be taken during the performance of the checkout procedure shall be included.

**e.** Pretest setup procedures **<hookup>**. When required, procedures for connecting any test and accessory equipment, including cable connections, shall be included. Procedures for the initial setting of controls shall also be provided.

**f.** Operational checkout and troubleshooting procedures **<opcheckproc>**. Operational checkout and troubleshooting procedures may be combined in a single procedure or may be prepared as a separate operational checkout procedure and a separate troubleshooting procedure:

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**(1)** Combined operational checkout and troubleshooting procedures. Combined operational checkout and troubleshooting procedures shall consist of a series of test procedures **<testproc>** (D.6.2.3.2) that lead to either a normal or abnormal condition or indication.

**(a)** Normal condition or indication – the operational checkout continues until the complete checkout is completed or until an abnormal condition or indication is observed.

**(b)** Abnormal condition or indication – a malfunction **<malfunc>** or a series of malfunctions is provided. For each malfunction, the possible correction actions **<action>** shall be provided with a link to a WP to perform the corrective action (as necessary). This process is continued until the entire troubleshooting procedure is completed.

**(2)** Separate operational checkout and troubleshooting procedures. When it is determined that the operational checkout procedures **<opcheckproc>** shall be separate from the troubleshooting procedures **<tsproc>**, the operational checkout procedures shall be developed in accordance with D.8.5 and the troubleshooting procedures shall be developed in accordance with D.8.4.

**g.** Post-operational shutdown procedures **<disconnect>**. When required, procedures to return the aircraft, aircraft system, or equipment to its normal configuration prior to operational checkout or troubleshooting setup shall be included.

**h.** Follow-on maintenance **<follow-on>** (E.7.3.1.3.3.11). When required, instructions or reference to appropriate WPs related to any follow-on maintenance shall be included. Follow-on maintenance shall be included after maintenance has been performed on equipment and shall typically restore equipment to a state prior to what is listed in equipment conditions.

## D.8.6 Diagnostic troubleshooting work package <diagnosticwp> (IETMs only).

Diagnostic troubleshooting WPs (shown in FIGURE D-20 through FIGURE D-23) shall be developed for all systems that require complex evaluation based on multiple inputs and/or system integrated self-testing. The procedures in these electronic WPs interactively diagnose a fault or symptom and direct the user to the corrective action(s). Troubleshooting WPs created for ETMs require the same tagging structure used to create IETM troubleshooting procedures.

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**Note**

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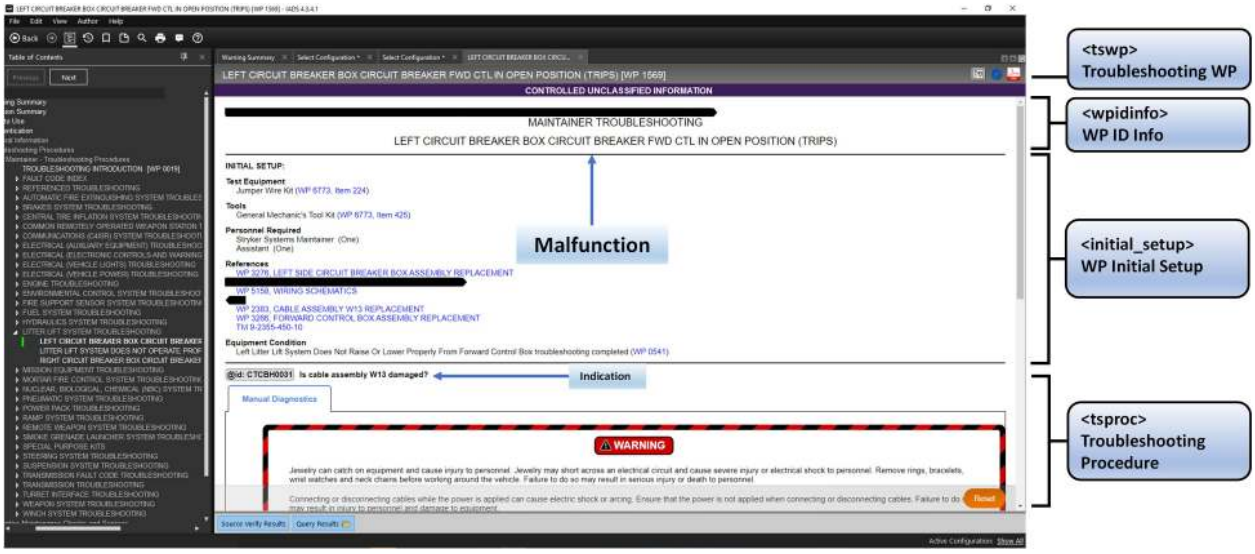


**FIGURE D-20. IETM screen example (sheet 1 of 4).**

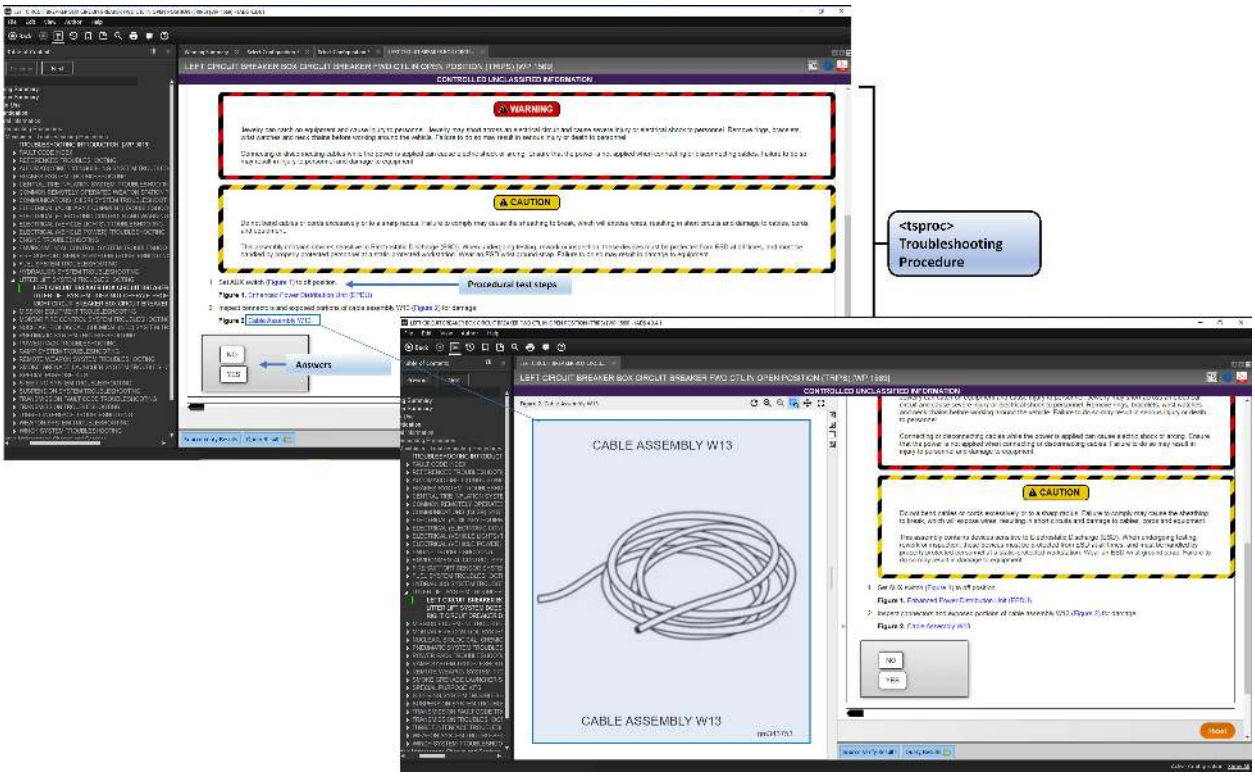


**FIGURE D-21. IETM screen example (sheet 2 of 4).**

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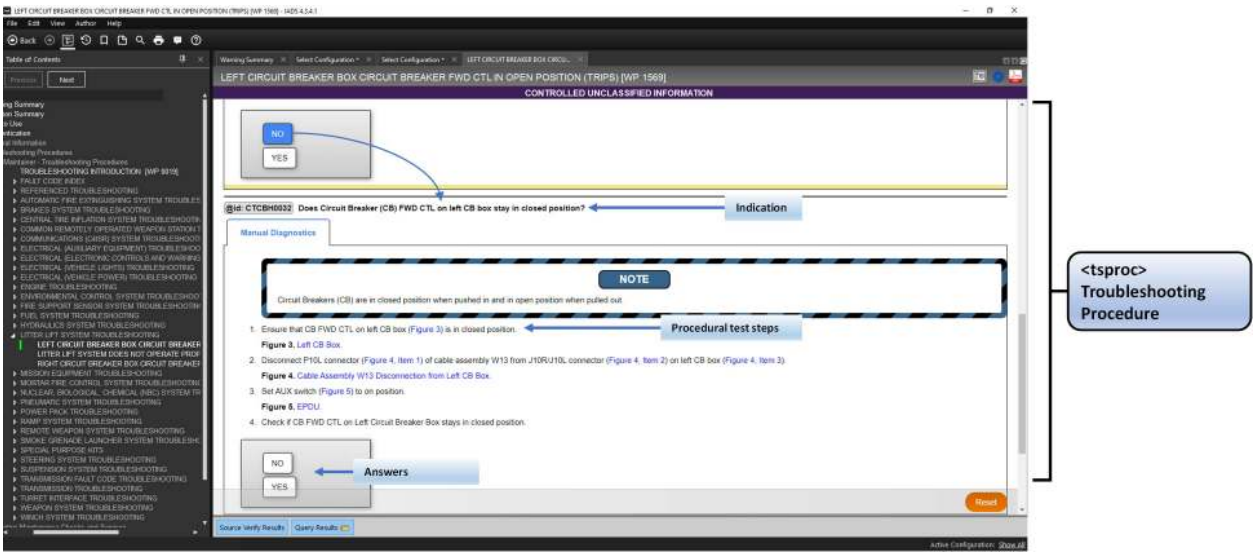


**FIGURE D-22. IETM screen example (sheet 3 of 4).**

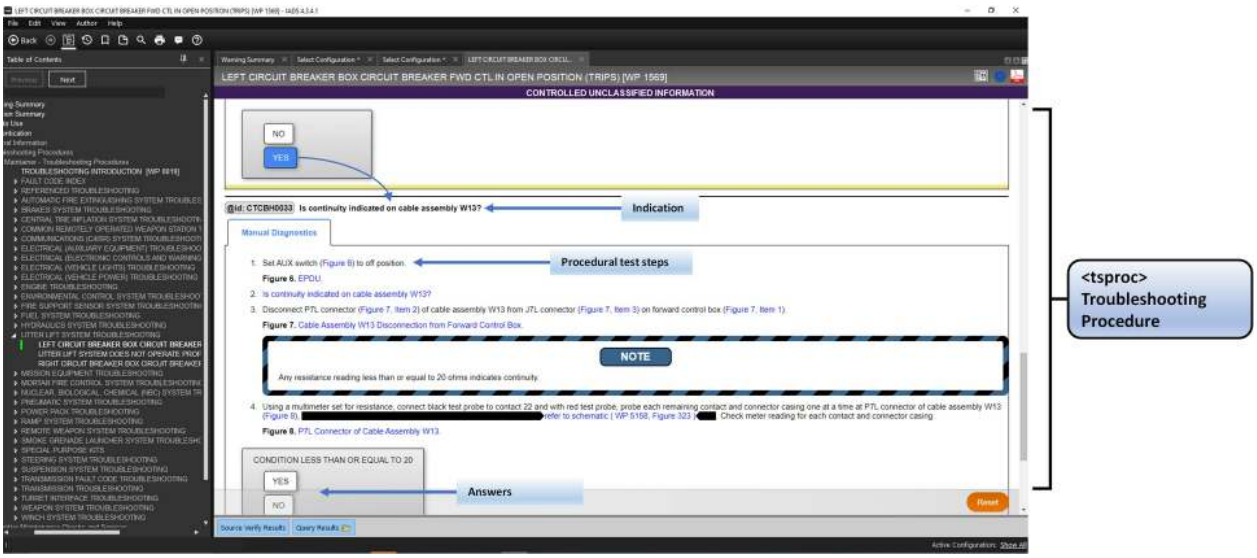


**FIGURE D-23. IETM screen example (sheet 4 of 4).**

In this standard, some specific diagnostic WP tags and their functions are discussed below. The following components shall be included in troubleshooting WPs.

**a.** WP identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** WP initial setup **<initial_setup>** is required for this WP (5.4.3).

**c.** Diagnostic testing procedure(s). The following WP components may be included in troubleshooting WP when required:

**(1)** Definition tag **<stateinitialization>**. First tag in diagnostic testing procedure that defines all variables that will be used for evaluation in the WP.

**(2)** Initial test tag **<testwithstate>**. Tag in diagnostic testing procedure that allows for specification of testing type choice: manual, automatic, or both.

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**(3)** Manual test type **<diagnostic-track-manual>**. Tag identifies manual diagnostic testing, which is used when testing is less complex and/or necessary software is not available for automatic testing. Manual testing does not include interaction with IETM software.

**(4)** Automatic test type **<diagnostic-track-automated>**. Tag identifies automatic diagnostic testing, which is used testing is complex and an external diagnostic software suite to communicate with API from IETM software is available.

### D.8.6.1 Integrated system troubleshooting procedures WP.

When specified by the MATDEV, integrated system troubleshooting procedures shall be developed. Troubleshooting procedures that involve more than one system or more than one major subsystem and cannot be logically placed in one of the individual system/ subsystem troubleshooting information WPs shall be covered in this type of WP.

## D.8.7 Preshop analysis work package <pshopanalwp> (DMWR/NMWR only).

Preshop analysis shall apply when data indicate that an inspection or test is more effective in determining the useful life of a system, subsystem, or component than a mandatory disassembly. FIGURE D-24 shows an example preshop analysis WP. The following components shall be included in preshop analysis WPs.

**a.** WP identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** WP initial setup **<initial_setup>** is required for this WP (5.4.3).

**c.** Scope **<scope>**. The purpose and coverage of the preshop analysis shall be stated.

**d.** Preshop preparation procedures **<proc>**:

**(1)** Unpacking and special handling. Procedures shall be prepared for removing the item, assemblies, subassemblies, or components from the shipping containers and packaging material. Instructions shall be prepared on any needed handling requirements for hazardous material, electrostatic sensitive devices, precious metal content, classified material, or critical material. Instructions shall also be prepared for any special condemnation procedures for the item and its assemblies and subassemblies.

**(2)** Checking attached documents. Instructions shall be prepared for checking all tags, forms, and documents attached to the item to determine the reason for its return and to identify any other obvious faults or damage.

**(3)** External inspection. Procedures shall be prepared for external inspection of the item to determine if it is complete and if there is any obvious external damage.

**(4)** Cleaning and preservation. Instructions shall be prepared for cleaning the item to prepare it for preshop analysis testing. The instructions shall include the procedures for any temporary preservation or corrosion protection measures needed to protect the item until the work required is started.

**e.** Preshop analysis procedures **<pshopanal>**. The MATDEV shall determine if the preshop analysis procedures shall be a narrative or be structured as a checklist. The checklist shall permit the inclusion of the name and signature of the person performing the analysis and any remarks that are required based on the results of the analysis. If a narrative preshop analysis is not provided, a printable checklist shall be provided. When specified by the MATDEV, an electronic checklist shall be provided in lieu of the narrative or printable checklist.

**(1)** Narrative **<proc>**. Preshop analysis text shall be presented in procedural format. Test and analysis procedures shall be presented in a logical sequence not to cause any unnecessary disassembly and in the order in which they should be performed. Each procedure shall be identified by a step number. Procedures shall be arranged in groups by major components, assemblies, and subassemblies. Each group shall be headed with an applicable title.

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**(2)** Checklist **<chklist>**. The checklist shall include the following data:

**(a)** Cover sheet / frame. The cover sheet/frame shall contain an area to record the following item information: part number **<partno>** ; serial number **<serialno>**; NSN **<nsn>** ; modifications required **<modreq>** ; reason for repair **<reason>** ; unpacking of secondary items required **<secitem>** ; review of tags **<revtag>** or forms **<revform>** with the item, name **<name>** , and signature **<sig>** of the person doing the analysis, and date **<date>**. See example in FIGURE D-24.

**(b)** Introduction **<intro>**. When necessary, the table of tests and inspections shall be preceded by a brief explanation of its use.

**(c)** Table of test and inspections **<pshopck.tab>**. This table shall have an entry for each test and inspection procedure. Each entry shall have, as a minimum, the following information: inspection point (the item or area to be inspected), condition, action, remarks, and identification of the personnel performing the inspection. If the procedure is too complex or lengthy to be included in the checklist, a reference to the WP where the procedures or actions are provided shall be included in the checklist. See example in FIGURE D-25.

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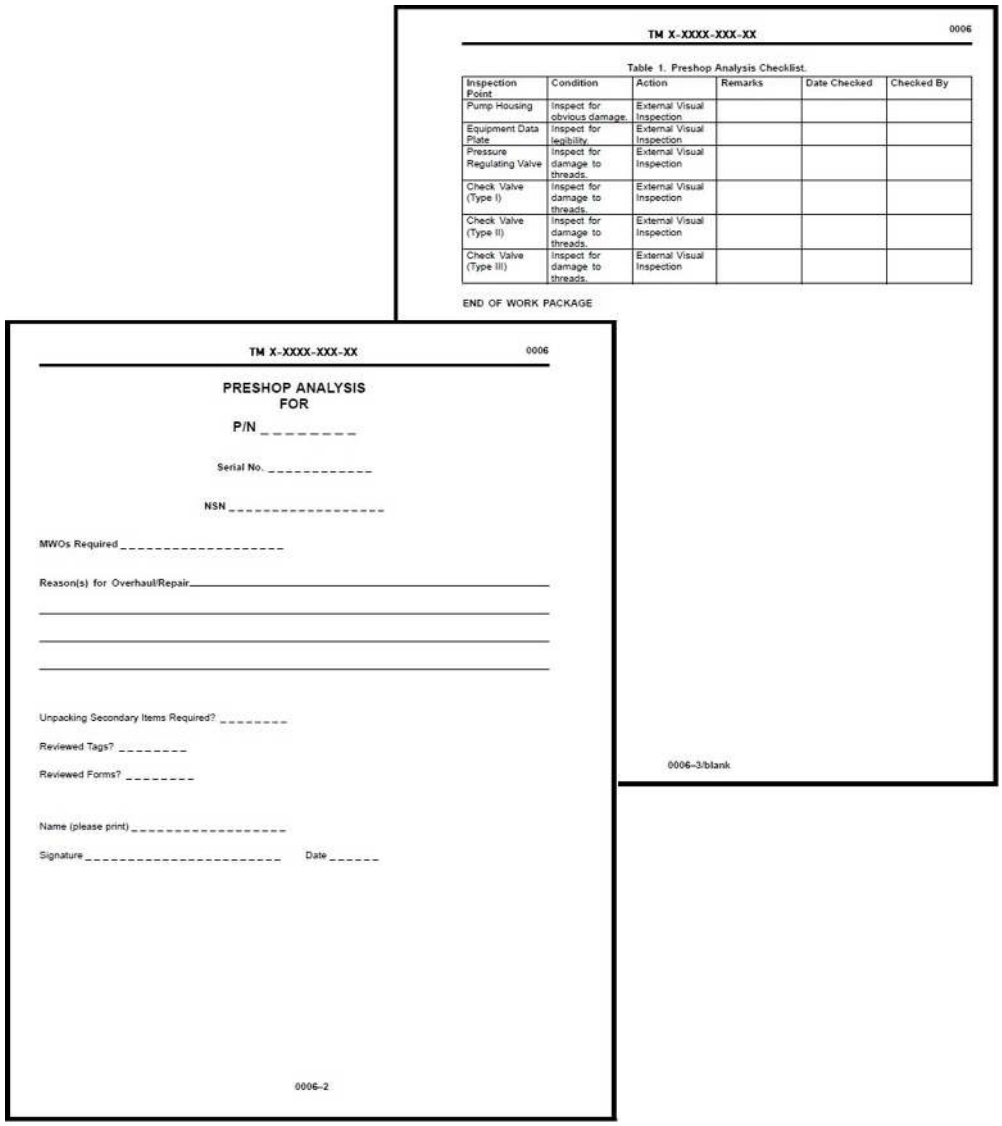
**Note**

Figure appearance may vary slightly based on version of DTD/stylesheets utilized.



TM X-XXXX-XXX-XX 0006

Table 1. Preshop Analysis Checklist.

| Inspection Point | Condition | Action | Remarks | Date Checked | Checked By |
|---|---|---|---|---|---|
| Pump Housing | Inspect for obvious damage. | External Visual Inspection | | | |
| Equipment Data Plate | Inspect for legibility. | External Visual Inspection | | | |
| Pressure Regulating Valve | Inspect for damage to threads. | External Visual Inspection | | | |
| Check Valve (Type I) | Inspect for damage to threads. | External Visual Inspection | | | |
| Check Valve (Type II) | Inspect for damage to threads. | External Visual Inspection | | | |
| Check Valve (Type III) | Inspect for damage to threads. | External Visual Inspection | | | |

END OF WORK PACKAGE

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TM X-XXXX-XXX-XX 0006

PRESHOP ANALYSIS
FOR

P/N _____

Serial No. _____

NSN _____

MWOs Required _____

Reason(s) for Overhaul/Repair _____

Unpacking Secondary Items Required? _____

Reviewed Tags? _____

Reviewed Forms? _____

Name (please print) _____

Signature _____ Date _____

0006-2

**FIGURE D-24. Preshop analysis cover sheet and table of test and inspections example.**

## D.8.8 Component checklist work package <compchklistwp> (DMWR/NMWR only).

A component checklist WP shall be prepared when required to support the preshop analysis procedures. In addition to the main components, subcomponents may be listed. FIGURE D-25 shows an example component checklist WP. The following components shall be included in component checklist WPs.

**a.** WP identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** WP initial setup **<initial_setup>** is required for this WP (5.4.3).

**c.** Introduction **<intro>** may be included as necessary.

The introduction shall provide a brief explanation of the use of the component checklist.



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**d.** Component checklist **<compchklist>** :

**(1)** Name/nomenclature of the equipment/item **<name>**. [Required]

**(2)** Serial number **<serialno>**. [As Applicable]

**(3)** Date received **<daterec>**. [As Applicable]

**(4)** Received from (identifying unit) **<recfrom>**. [As Applicable]

**(5)** Component name **<compname>**. [As Applicable]

**(6)** NSN **<nsn>**. [As Applicable]

**(7)** Part number/ CAGEC **<partno>/<cageno>**. [As Applicable]

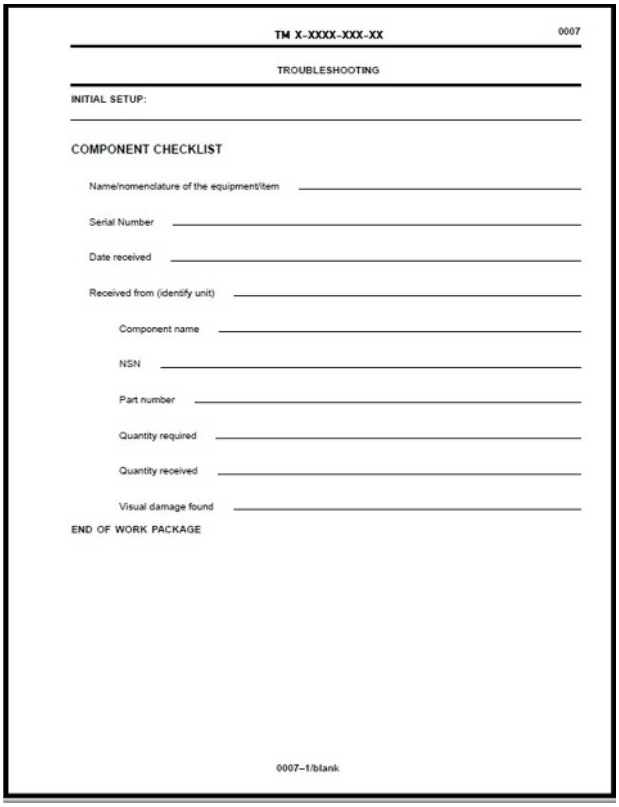
**(8)** Quantity required **<qty>**. [As Applicable]

**(9)** Quantity received **<qtyrec>**. [As Applicable]

**(10)** Visual damage found **<damage>**. [As Applicable]

**Note**

Figure appearance may vary slightly based on version of DTD/stylesheets utilized.


TM X-XXXX-XXX-XX 0007

TROUBLESHOOTING

INITIAL SETUP:

COMPONENT CHECKLIST

Name/nomenclature of the equipment/item

Serial Number

Date received

Received from (identify unit)

Component name

NSN

Part number

Quantity required

Quantity received

Visual damage found

END OF WORK PACKAGE

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**FIGURE D-25. Component checklist WP example.**

## D.9 NOTES.

The notes in Section 6 apply to this appendix.



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# APPENDIX E

# MAINTENANCE INSTRUCTIONS

## E.1 SCOPE.

This appendix explains the types of maintenance instructions available within MIL-STD-40051 and how to prepare maintenance instructions tailored to specific levels of repair. Maintenance tasks shall be prepared for equipment systems and their related subsystems, assemblies, subassemblies, components, wiring, junction boxes, accessories, Shop Replaceable Units (SRUs), and Line Replaceable Units (LRUs). Maintenance tasks and supporting illustrations shall be prepared so that personnel can perform all required maintenance authorized by the Maintenance Allocation Chart (MAC) up to the maintenance level(s) included in the publication. Maintenance is a required component of technical publications, and this appendix establishes the required content applicable for all maintenance levels including DMWRs and NMWRs. This appendix is a mandatory part of this standard. The information contained herein is intended for compliance.

**Note**

To be compliant with this standard, publications shall be developed within **<production>** (opening) and **</production>** (closing) root tags.

When preparing information in accordance with this appendix, all references shall be linked using the appropriate tags (**<xref>**, **<extref>**, etc.). Links shall not be typed as text without tagging.

### E.1.1 Content tailoring using Appendix A matrices.

Applicability of some requirements contained in this standard are the responsibility of the materiel developer (MATDEV) (refer to AR 750–1, AR 700–127) and shall be accomplished using matrices referenced in TABLE A-I. The applicability of requirements are also designated by statements including, but not limited to, the following:

**a.** "...as specified by the MATDEV."

**b.** "...as required by the MATDEV."

**c.** "Refer to the MATDEV for..."

**d.** "Contact MATDEV if..."

**e.** "The MATDEV shall determine..."

**f.** "...as needed."

#### E.1.1.1 Front and rear matter.

Front and rear matter requirements for documents discussed in this appendix shall be determined in accordance with TABLE A-II.

## E.2 APPLICABLE DOCUMENTS.

The applicable documents in Section 2 apply to this appendix.

## E.3 ACRONYMS AND DEFINITIONS.

The acronyms and terms defined in Section 3 of this standard apply to this appendix.

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## E.4 MAINTENANCE APPLICABILITY.

Requirements contained in this appendix are applicable to all maintenance levels unless specifically labeled in bold and in parentheses. These labels designate the level of personnel authorized to perform maintenance tasks. There are two primary maintenance levels covered in this standard: Field and Sustainment. Each level makes a different contribution to the overall system. Field-level maintenance is generally characterized by on (near) system maintenance utilizing LRUs and component replacement in the owning unit using tools and test equipment found in the unit. Sustainment-level maintenance is generally characterized by "off system" component repair or end item repair and return to the supply system, or by exception, back to the owning unit. Each of these two primary levels includes two sublevels. Field maintenance includes crew/operator and maintainer maintenance personnel while sustainment includes below depot and depot level personnel. The requirements are applicable for all maintenance levels through depot, including DMWRs and NMWRs. FIGURE E-2 shows these levels of maintenance in column 4.

## E.5 MAINTENANCE METHODOLOGY.

Maintenance plays an integral role in restoring equipment to operational status. Preventive Maintenance Checks and Services (PMCS) is the foundation of field-level maintenance and includes all checks and services that must be performed by both crew/operator and maintainer personnel before, during, and after operation. Crew/operator and maintainer maintenance instructions enable a technician to perform maintenance on equipment systems and associated assemblies to bring assets to mission-capable status. All maintenance shall be developed in accordance with Logistics Product Data (LPD) as well as Source Maintenance and Recoverability (SMR) codes listed in the Repair Parts and Special Tools List (RPSTL). More specifically, maintenance shall be developed in accordance with the third and fourth SMR code positions, which define levels of maintenance authorized to maintain and use an item and denote the lowest maintenance level with the capability and resources to perform a complete repair, respectively. E.5.1 provides more detail about maintenance within technical publication development.

### E.5.1 Technical Publication Development Overview.

FIGURE E-1 provides a holistic diagram of technical publication development and the centralized role maintenance instructions play within this process. Logistics product data (LPD) is developed by provisioners and other logistics data specialists, and this data is output in the form of various reports (LSAR 030, LSAR 004, etc.). This LPD informs the development of the MAC that appears in the supporting information (APPENDIX G) of a publication and the RPSTL. MAC entries dictate which maintenance tasks shall appear in maintenance instruction modules discussed in this appendix; and, furthermore, maintenance tasks, MAC entries, and RPSTL components shall have traceability through a publication based on Functional Group Code (FGC) designations. Technical publication alignment among maintenance instructions, the MAC, and the RPSTL is discussed further in APPENDIX F. During normal operation of an equipment system, PMCS is performed and components malfunction. PMCS instructions and troubleshooting procedures to correct malfunctions (APPENDIX D) shall have references directly leading to maintenance instructions. Maintenance instructions then serve to restore equipment functionality to accomplish a mission.

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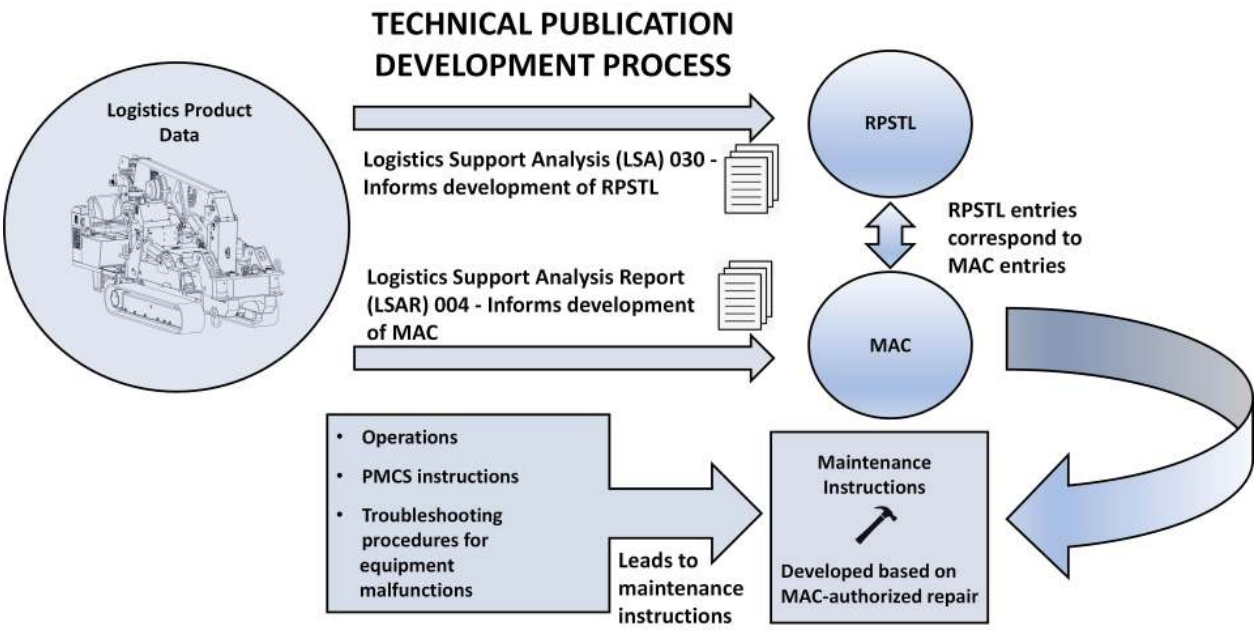


**FIGURE E-1. Technical publication development.**

## E.5.2 Maintenance based on publication type.

The designated technical publication type (-10, -12, or -20) indicates the types and levels of maintenance covered in a publication (see TABLE A-I, APPENDIX A). TABLE E-II outlines the basic levels of instruction provided in each publication type.

### E.5.2.1 Technical publication type options.

There are a number of factors (TABLE E-I) that can influence the type of technical publication (TABLE E-II) developed for an equipment system. Levels of maintenance required to service the equipment is the primary factor; however other factors may include those listed in TABLE E-I (based on equipment). The MATDEV ultimately determines the types of publications developed.

**TABLE E-I. Technical publication factor considerations.**

| Factor | Factor Considerations |
|---|---|
| Required Military Occupational Specialty (MOS) | For equipment with one MOS for operator and maintainer, one -12 TM (instead of two TMs) may be developed since one type of personnel can perform all instructions. |
| Complexity | Large complex systems may benefit from the division of TMs into a -10 TM (operator) and a -20 TM (maintainer and sustainment) given the likelihood of numerous and lengthy work packages (WP)s. |
| Fielding schedule | The fielding schedule for equipment may benefit from the development of a -10 TM and -20 TM since they can be divided and released at separate times. A -10 TM can be produced and released prior to the -20 TM depending on fielding requirements. |


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**TABLE E-II. Technical publication type options.**

| TM Type | Level of Instruction |
|---|---|
| **-10** | Operator instructions and operator maintenance (if applicable) |
| **-12** | Operator instructions as well as operator, maintainer, and sustainment maintenance (including, if applicable, below-depot (sustainment) maintenance) |
| **-12&P** | Operator instructions as well as operator, maintainer, and sustainment maintenance (including, if applicable, below-depot (sustainment) maintenance) and a Repair Parts and Special Tools List (RPSTL) |
| **-20** | Field maintenance |
| | Field and sustainment maintenance (including, if applicable, below-depot (sustainment) maintenance) |
| **-20&P** | Field maintenance as well as a RPSTL |
| | Field and sustainment maintenance (including, if applicable, below-depot (sustainment) maintenance) and a RPSTL |
| **-20P** | RPSTL only for field repairable parts |
| | RPSTL only for field and sustainment repairable parts (including, if applicable, below-depot (sustainment) parts) |

### E.5.2.2 Maintenance alignment with MAC.

Maintenance instructions included in the main body of a technical publication must align with the MAC. This alignment, also known as a "crosswalk," means every entry in the MAC shall have a corresponding work package (WP) containing the relevant maintenance instruction (unless stipulated otherwise in MAC remarks codes). These maintenance WPs shall be arranged to coincide with the FGC, or top-down breakdown sequence, followed in the MAC. The MAC consists of an end item's maintenance-significant assemblies and subassemblies organized by Functional Group Code (FGC) (FIGURE E-2, column 1, FGC "00"). The MAC shall contain all removal, repair, replacement, and any other related maintenance through sustainment for both hardware and software maintenance tasks (FIGURE E-2, columns 3). For each repairable assembly or subassembly (FIGURE E-2, column 2), the MAC shall identify the maintenance task to be performed, the level of responsibility for the function, and the needed tools.

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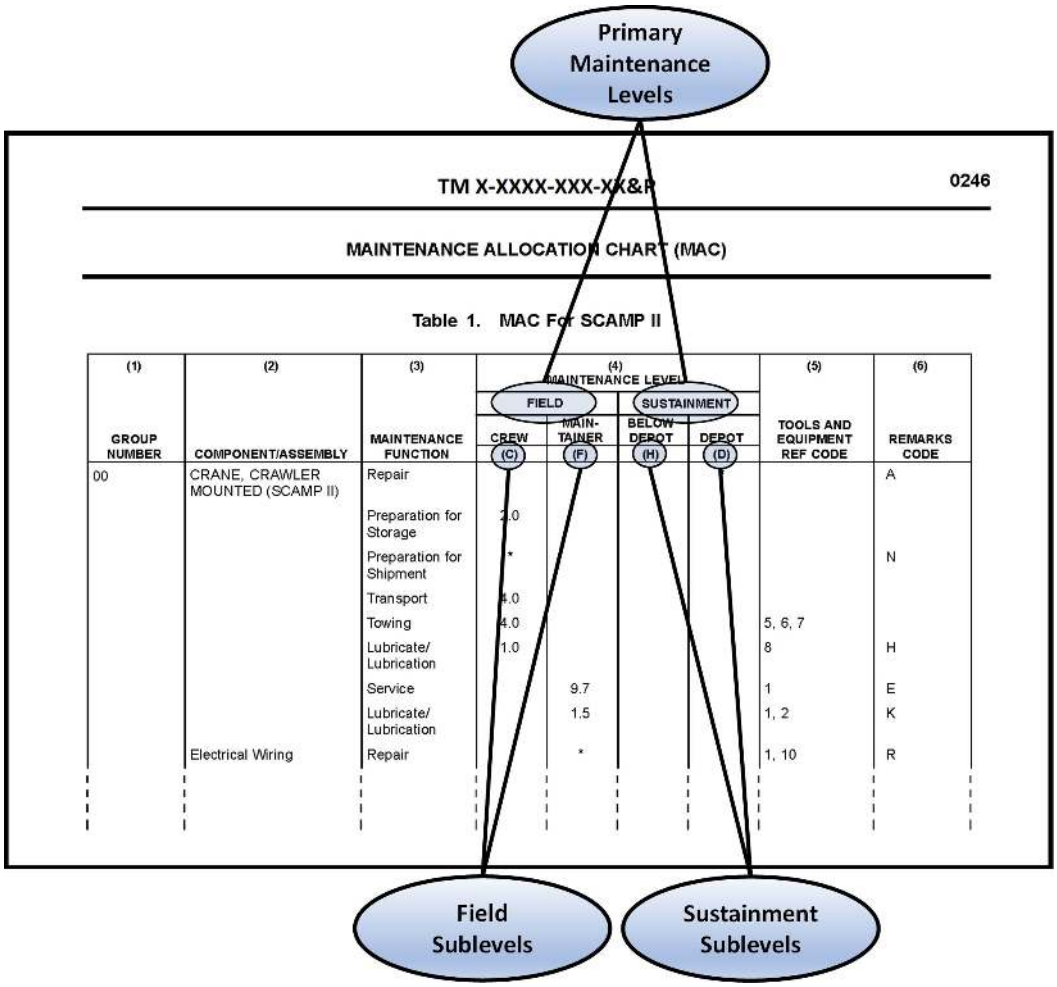


**FIGURE E-2. Maintenance levels.**

## E.5.3 MAINTENANCE REFERENCING.

Referencing shall be used to ensure maintenance tasks reference not only other maintenance tasks but also directly correspond to task entries in the MAC (as discussed in E.5.3.2). Sections E.5.3.1 and E.5.3.2 discuss referencing in more detail.

### E.5.3.1 Referencing maintenance tasks throughout a technical publication.

Maintenance tasks shall be referenced throughout the technical publication as needed. PMCS and/or troubleshooting WPs (shown in FIGURE E-3, top section) shall be developed to ensure that, when necessary, instructions lead to and reference repair and replace maintenance tasks in maintenance WPs (shown in FIGURE E-3). Referencing within maintenance WPs and modules shall be utilized as much as possible to prevent procedural redundancy. For example, FIGURE E-3 (bottom section) shows a maintenance WP referencing other tasks within the same maintenance WP. The replace task contains two steps; one step references the removal task for the battery in the same WP, and the second step references the installation task for the battery in the same WP. FIGURE E-4 shows a maintenance WP (WP 0105, BATTERY) referencing another maintenance WP (WP 0129, BATTERY HEATING PAD) to complete maintenance tasks.

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**Note**

Some maintenance tasks, such as remove, install, etc. exist to simply to facilitate these repair tasks.

All maintenance tasks are included in the technical publication based on logistics data including SMR codes and shall be referenced as needed. This appendix and APPENDIX G discuss these maintenance tasks in detail.

**Note**

Figure appearance may vary slightly based on version of DTD/stylesheets utilized.

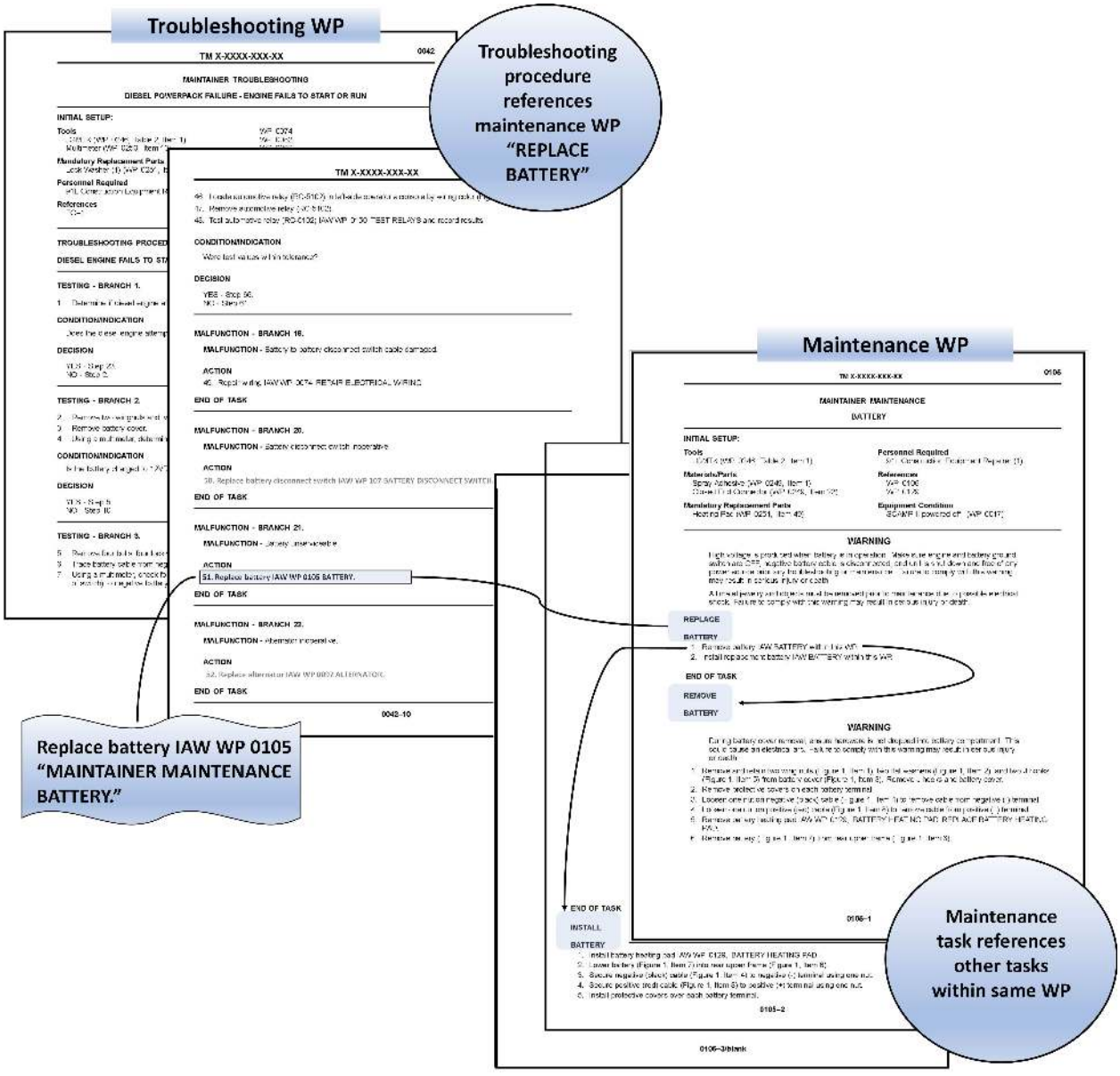


**FIGURE E-3. Referencing example (sheet 1 of 3).**

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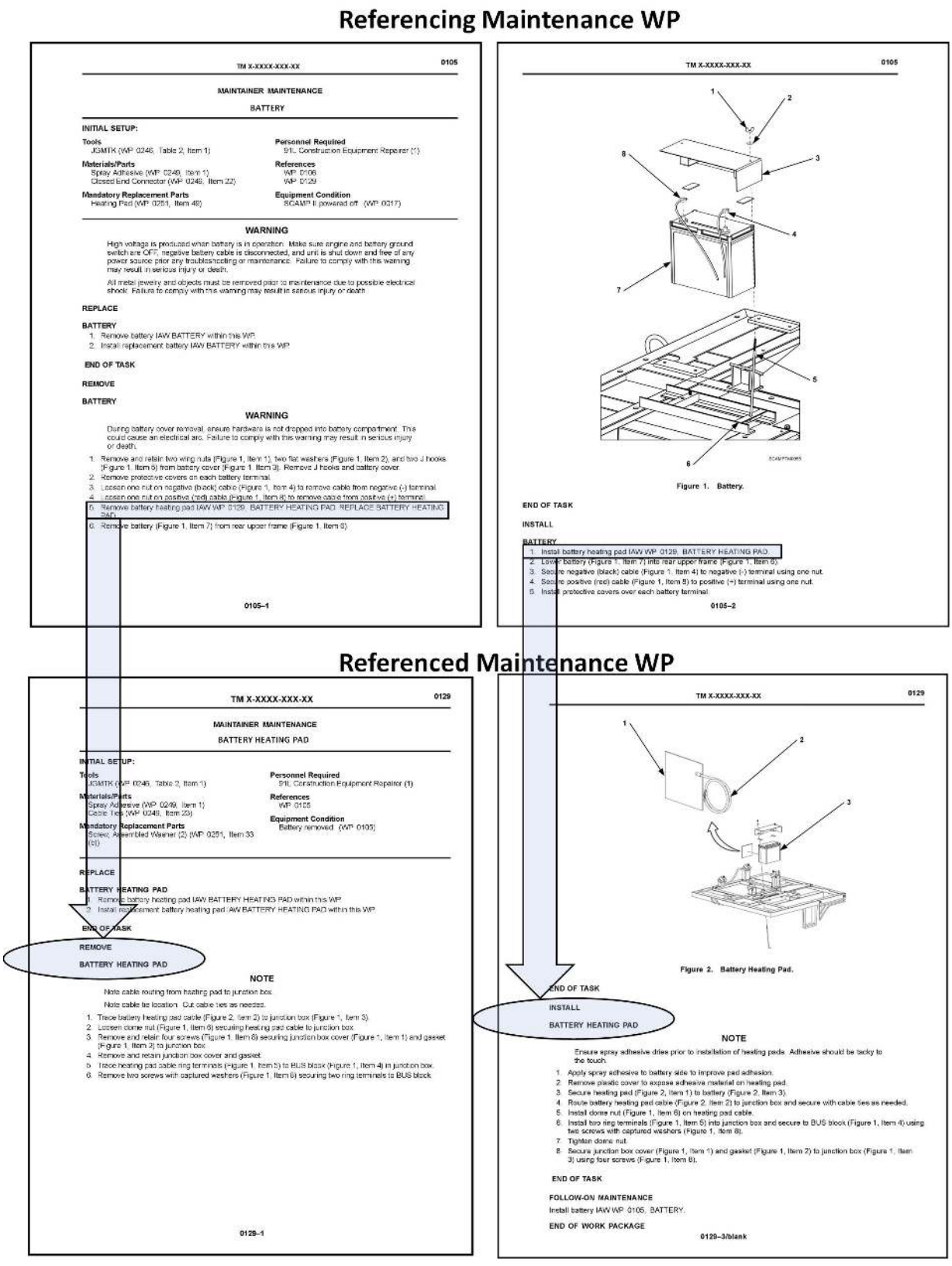


**FIGURE E-4. Referencing example (sheet 2 of 3).**

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## E.5.3.2 Referencing MAC entries to maintenance tasks.

In maintenance instruction WPs, all maintenance tasks with corresponding MAC entries shall be linked (referenced) directly to their respective task entries in the MAC as frequently as possible using the appropriate tags (also discussed in G.7.3.2.3). This practice serves to ensure task/MAC alignment with tasks represented in a publication (shown in FIGURE E-5).

**Note**

Not all tasks (for example minor tasks such as cleaning, adjusting, etc.) will necessarily be represented as a line item in the MAC. MAC entries are determined by logistics product data (LPD).

Figure appearance may vary slightly based on version of DTD/stylesheets utilized.

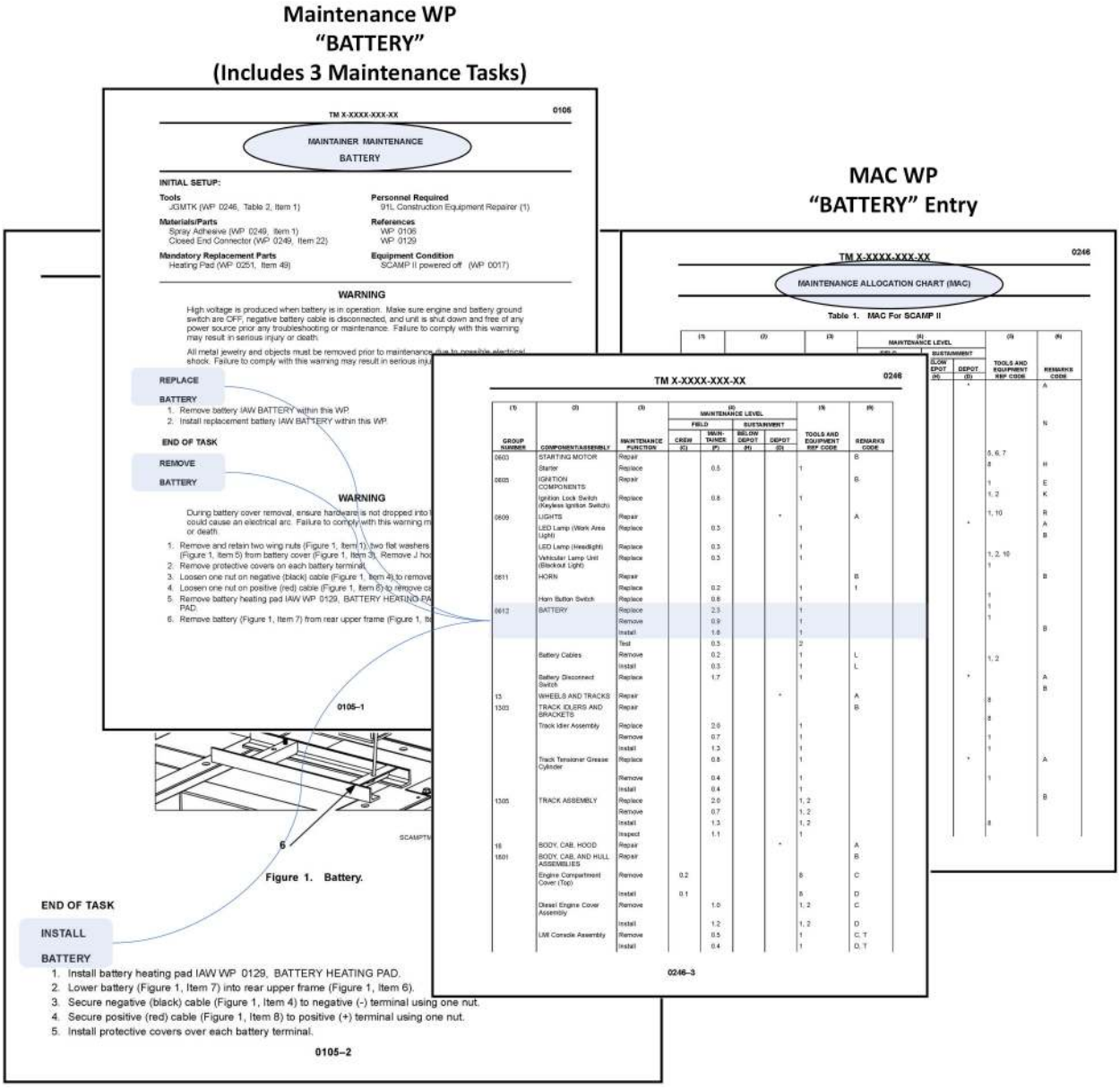


**FIGURE E-5. Referencing example (sheet 3 of 3).**

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## E.5.4 MAINTENANCE MODULE COMPONENTS.

Maintenance instructions developed in accordance with this appendix shall comprise the maintenance module(s) of a technical publication. Refer to APPENDIX A for content selection matrices that designate required, prohibited, and as needed sections of a maintenance module. Maintenance module(s) shall be divided into WPs that shall stand alone and are broken into the WP types listed in TABLE E-III. Prior to starting development of maintenance instructions, refer to the sections listed in TABLE E-III for required components to build various WP types.

**TABLE E-III. Maintenance WPs.**

| WP | WP Tag | Appendix Section | Applicability |
|---|---|---|---|
| PMCS introduction WP | **<pmcsintrowp>** | E.7.1 | **All Equipment Systems** |
| PMCS WP [1] | **<pmcswp>** | E.7.2 | |
| Service upon receipt WP | **<surwp>** | E.7.3 | |
| Maintenance instruction WP [1] | **<maintwp>** | E.7.4 | |
| General maintenance WP | **<gen.maintwp>** | E.7.5 | |
| Lubrication instructions WP | **<lubewp>** | E.7.6 | |
| Illustrated list of manufactured items introduction WP | **<manu_items_introwp>** | E.7.7 | |
| Manufacturing instruction WP | **<manuwp>** | E.7.8 | |
| Torque limits WP | **<torquewp>** | E.7.9 | |
| Wiring diagrams and schematics WP | **<wiringwp>** | E.7.10 | |
| Equipment/user fitting instructions WP | **<perseqpwp>** | E.7.11 | |
| Preventive maintenance inspections WP | **<pmiwp>** | E.7.12 | **Aircraft Only** |
| Aircraft inventory master guide WP | **<inventorywp>** | E.7.13 | |
| Storage of aircraft WP | **<storagewp>** | E.7.14 | |
| Weighing and loading WP | **<wtloadwp>** | E.7.15 | |
| Overhaul and retirement schedule WP | **<orschwp>** | E.7.16 | |
| Preventive maintenance services / Preventive maintenance daily inspection WP | **<pms-inspecwp>** | E.7.17 | |
| Phased maintenance checklist WP | **<pmi-cklistwp>** | E.7.18 | |
| Preservation, packaging, and marking general information WP | **<ppmgeninfowp>** | E.7.19 | **DMWR/NMWR Only** |
| Facilities WP | **<facilwp>** | E.7.20 | |
| Overhaul inspection procedures (OIP) WP | **<oipwp>** | E.7.21 | |
| Depot mobilization requirements WP | **<mobilwp>** | E.7.22 | |
| Quality assurance (QA) requirements WP | **<qawp>** | E.7.23 | |

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**TABLE E-III. Maintenance WPs. (continued**

| WP | WP Tag | Appendix Section | Applicability |
|---|---|---|---|
| Auxiliary equipment maintenance WP | **<auxeqwp>** | E.7.24 | **AUX Only** |
| Ammunition identification information WP | **<ammoidentwp>** | E.7.25 | |
| Ammunition maintenance WP | **<ammowp>** | E.7.26 | **Ammo Only** |
| Foreign ammunition WP | **<natowp>** | E.7.27 | |

[1] WP type frequently duplicated in a module for effective information display.

## E.6 DEVELOPMENT OF MAINTENANCE MODULES.

Some WPs in a technical publication are grouped together to form a category, and these categories are based on the type and complexity of equipment in a system. A technical publication maintenance module **<mim>** shall include one or more of the categories shown in FIGURE E-6 and listed in E.6.1 thru E.6.12.2. FIGURE E-7 through FIGURE E-9 shows example structures of commonly-used maintenance module categories, such as **<maintenancecategory>**, and the available WPs based on each category type in a publication. These figures also depict publications that may include both operator and maintainer modules (thus two **<mim>** tags). FIGURE E-10 shows that troubleshooting modules and maintenance modules may alternate in order in a publication or be grouped together. FIGURE E-11 shows an example structure of module categories in a publication that may include not only operator and maintainer modules, but also a sustainment maintenance module for any below depot maintenance performed in the field by a solider (thus three **<mim>** tags). The MATDEV shall determine any applicable below depot maintenance that shall be included in a publication.

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**Note**

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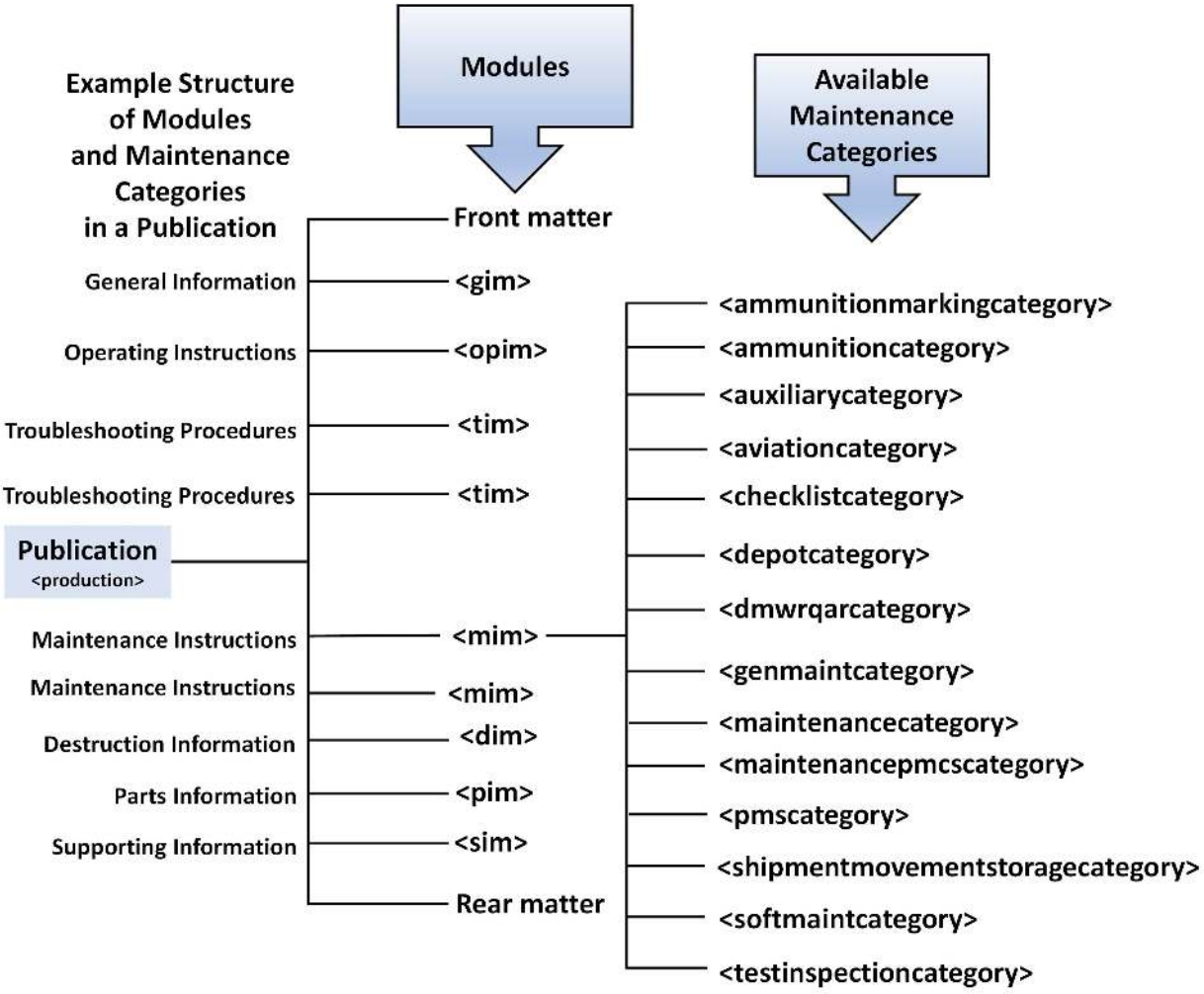


**FIGURE E-6. Maintenance modules and categories.**

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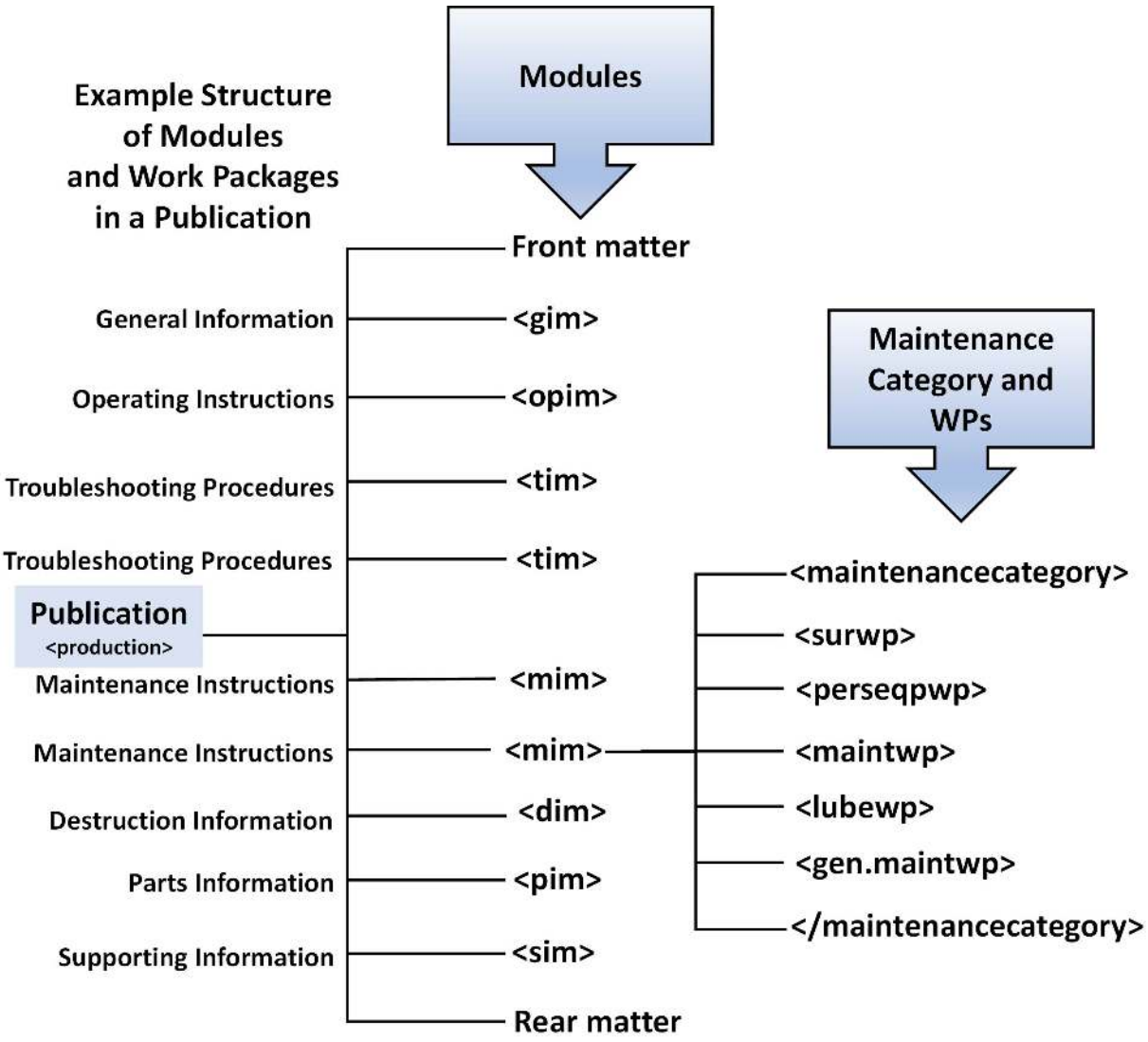


**FIGURE E-7. Maintenance modules, categories, and WPs (<maintenancecategory>).**

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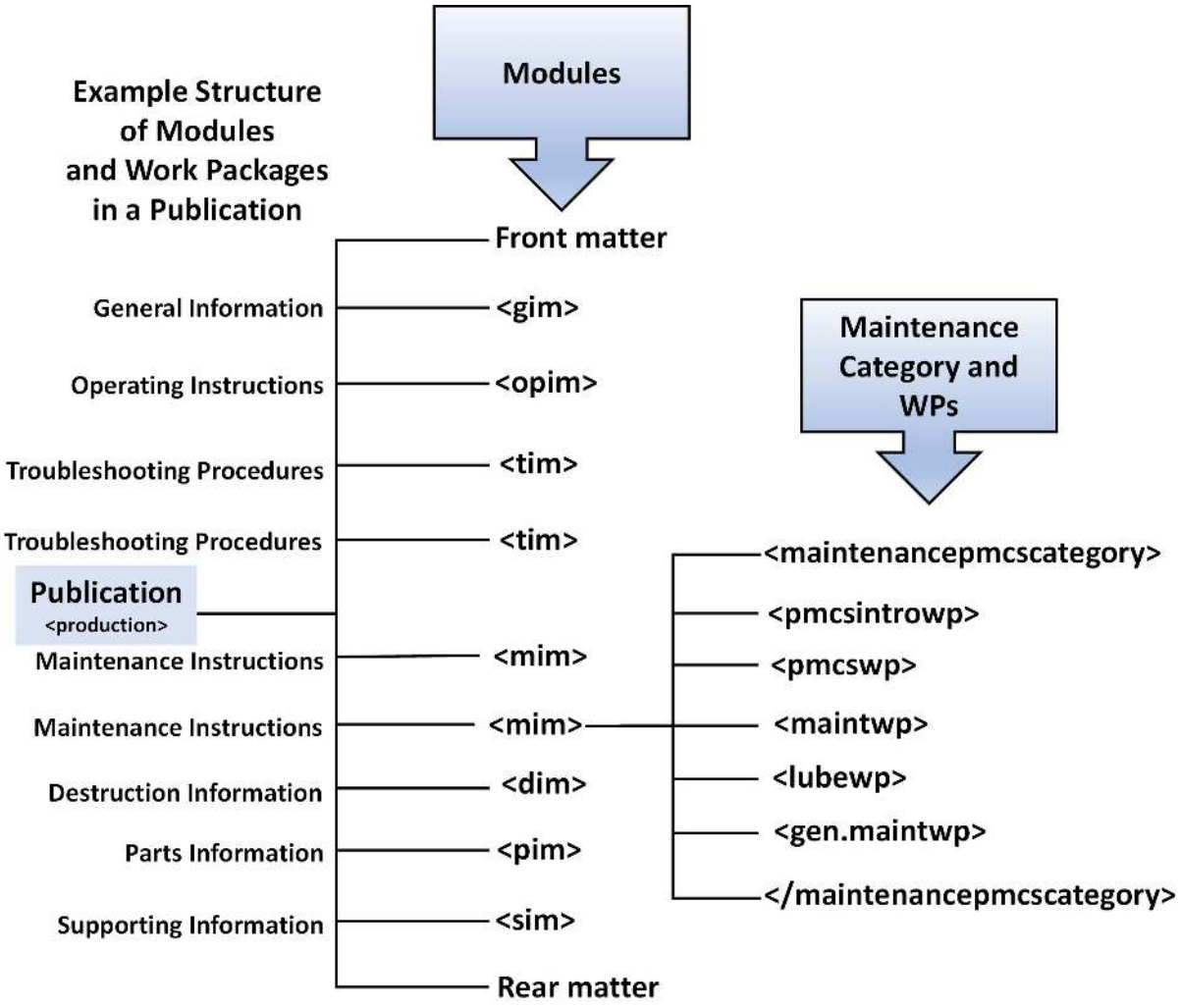


**FIGURE E-8. Maintenance modules, categories, and WPs (<maintenancepmcscategory>).**



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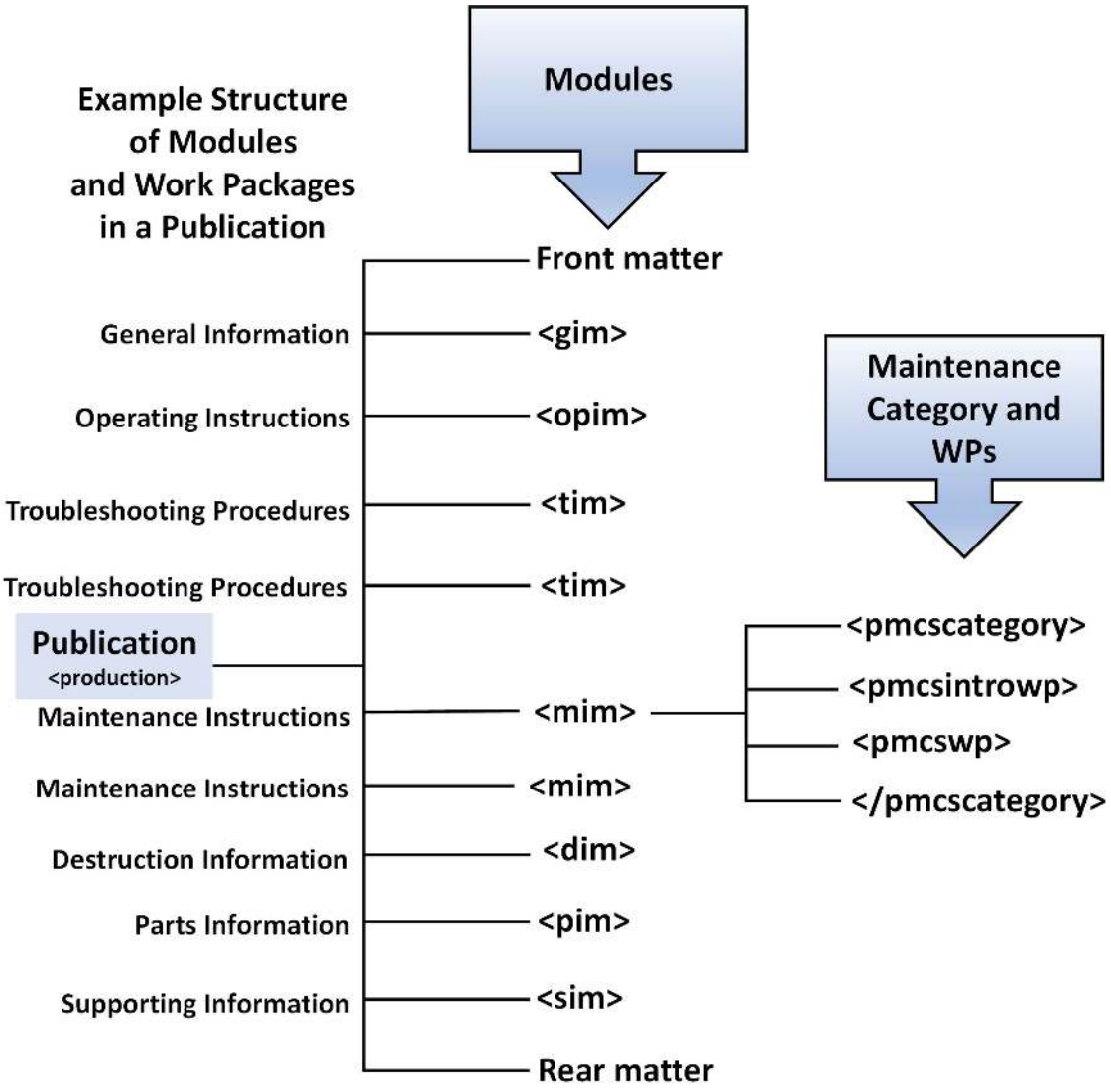


**FIGURE E-9. Maintenance modules, categories, and WPs (<pmcscategory>).**

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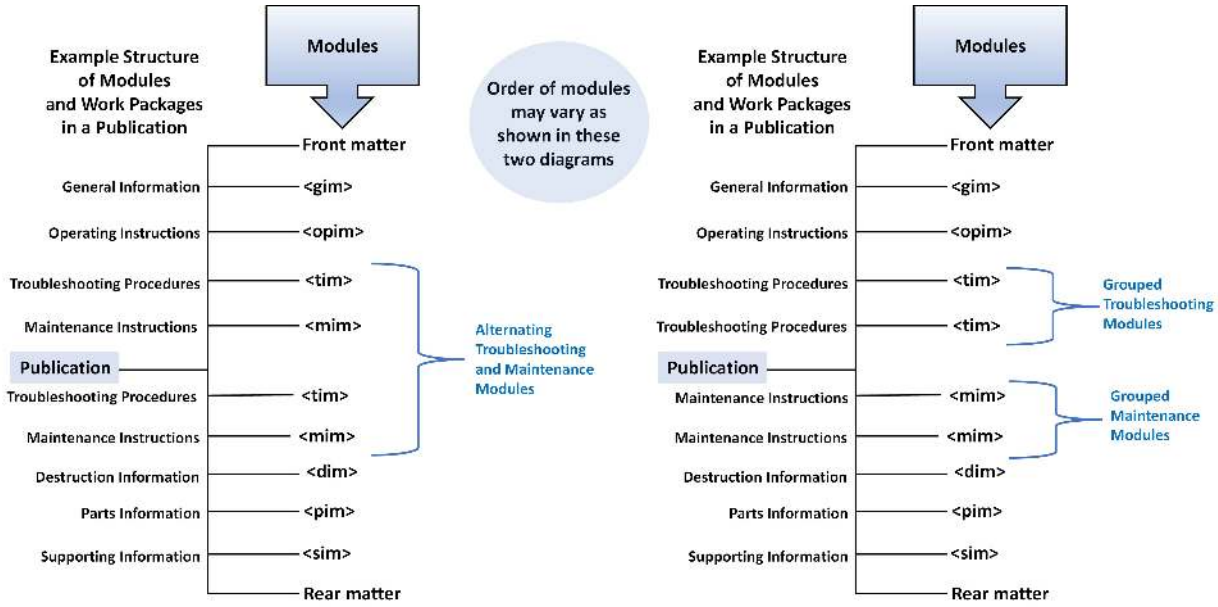


**FIGURE E-10. Module order options in a publication.**

## Note

The graphic below depicting multiple **<mim>** modules serves to emphasize that sustainment may be contained in its own module separate from other maintenance as needed. Note; however, that other categories (such as**<maintenancepmcscategory>**, etc. ) are permitted with use of additional **<mim>** modules.

Tags/output shown in figure(s) below may vary slightly based on version of DTD/stylesheets utilized (see applicable DTD or tag definition list for available tags).



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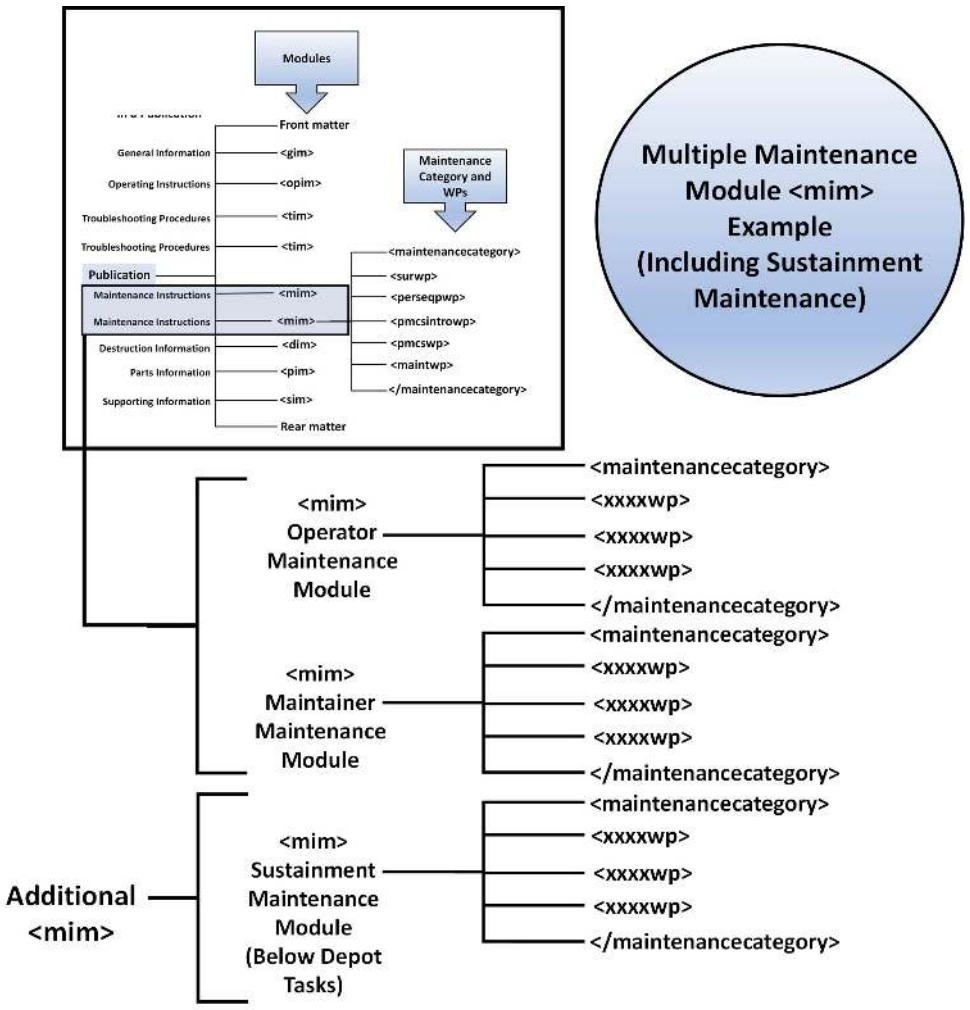


**FIGURE E-11. Maintenance modules, categories, and WPs (including sustainment maintenance module).**

## E.6.1 PMCS module (except for DMWRs and NMWRs) <mim>.

This maintenance module contains only the PMCS requirements and shall be used only when PMCS will be in a separate module **<mim>** by itself, and the remaining maintenance WPs will be in a separate module or modules. The PMCS module shall contain WPs listed below (in the order specified) between **<pmcscategory>** tags.

**a.** PMCS Introduction WP **<pmcsintrowp>** (E.7.1).

**b.** PMCS WP **<pmcswp>** (E.7.2).

## E.6.2 Equipment system /equipment maintenance with required PMCS module (except for DMWRs and NMWRs) <mim>.

This maintenance module shall be used when PMCS is combined with other maintenance WPs in a single module **<mim>**. This maintenance module shall contain WPs listed below (in the order specified) as required by publication type (see A.4.4) and as applicable for an equipment system between **<maintenancepmcscategory>** category tags (example shown in FIGURE E-12).



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**a.** Service upon receipt WP **<surwp>** (maintainer level and above) (E.7.3).

**b.** Equipment/User fitting Instruction WP **<perseqpwp>** (E.7.11).

**c.** PMCS introduction WP **<pmcsintrowp>** (E.7.1).

**d.** PMCS WP **<pmcswp>** (E.7.2).

**e.** The following WPs types occur in no specific order:

**(1)** Maintenance WP **<maintwp>** (E.7.4).

**(2)** General maintenance WP **<gen.maintwp>** (E.7.5).

**(3)** Lubrication instructions WP **<lubewp>** (E.7.6).

**f.** Illustrated list of manufactured items WP **<manuwp>** (maintainer level and above) (E.7.8).

**g.** Torque limits WP **<torquewp>** (maintainer level and above) (E.7.9).

**h.** Wiring diagrams and schematics WP **<wiringwp>** (maintainer level and above) (E.7.10).

### E.6.3 Equipment system / equipment maintenance without PMCS module (except for aircraft TMs, DMWRs, and NMWRs) <mim>.

This maintenance module shall be used for maintenance modules **<mim>** that do not contain PMCS. In addition to this maintenance module, either the PMCS or maintenance with PMCS category shall also be developed. This maintenance module shall contain the following WPs (in the order specified) as required by publication type (see A.4.4) and as applicable for an equipment system between **<maintenancecategory>** (example shown in FIGURE E-12).

**a.** Service upon receipt WP **<surwp>** (maintainer level and above) (E.7.3).

**b.** Equipment/User fitting instruction WP **<perseqpwp>** (E.7.11).

**c.** The following WPs types occur in no specific order:

**(1)** Maintenance WP **<maintwp>** (E.7.4).

**(2)** General maintenance WP **<gen.maintwp>** (E.7.5).

**(3)** Lubrication instructions WP **<lubewp>** (E.7.6).

**d.** Illustrated list of manufactured items WP **<manuwp>** (maintainer level and above) E.7.8).

**e.** Torque limits WP **<torquewp>** (maintainer level and above) (E.7.9).

**f.** Wiring diagrams and schematics WP **<wiringwp>** (maintainer level and above) (E.7.10).

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Check the source to verify that this is the current version before use.

**Note**

Tags/output shown in figure(s) below may vary slightly based on version of DTD/stylesheets utilized (see applicable DTD or tag definition list for available tags).

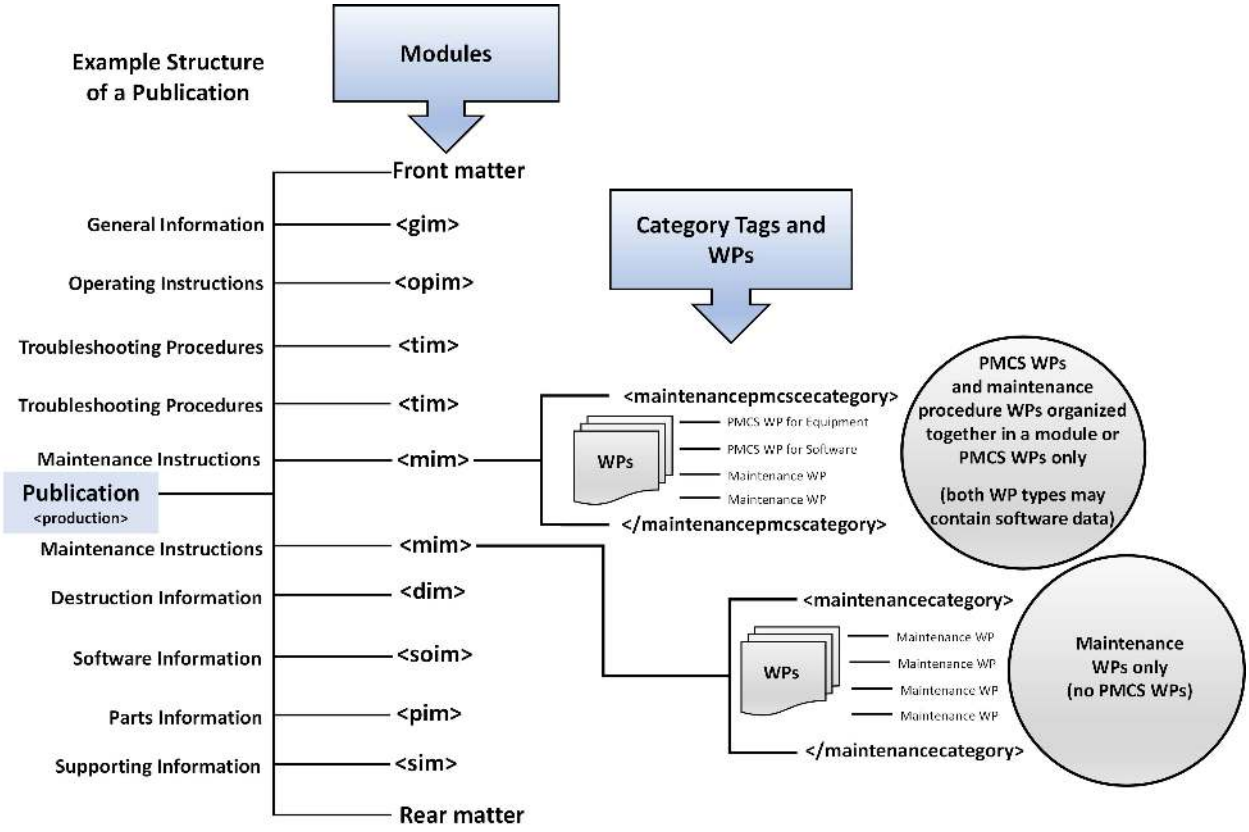


**FIGURE E-12. Example Publication including PMCS maintenance module.**

## E.6.4 Depot weapon system / equipment maintenance module <mim>.

Unless otherwise specified by the MATDEV, the depot maintenance module shall contain the WPs below (in the order specified) as required by publication type (see A.4.4) and as applicable for an equipment system between **<depotcategory>** tags.

**a.** Preservation, packaging, and marking general information WP **<ppmgeninfowp>** (E.7.19).

**b.** Equipment/User Fitting Instruction WP **<perseqpwp>** (E.7.11).

**c.** The following WPs types occur in no specific order:

**(1)** Maintenance WP **<maintwp>** (E.7.4).

**(2)** General maintenance WP **<gen.maintwp>** (E.7.5).

**(3)** Lubrication instructions WP **<lubewp>** (E.7.6).

**d.** Facilities WP **<facilwp>** (E.7.20).

**e.** Overhaul inspection procedures (OIPs) WP **<oipwp>** (E.7.21).

**f.** Depot mobilization requirements WP **<mobilwp>** (E.7.22).

**g.** QA requirements WP **<qawp>** (E.7.23).

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**h.** Illustrated list of manufactured items **<manuwp>** (E.7.8).

**i.** Torque limits WP **<torquewp>** (E.7.9).

**j.** Aircraft inventory master guide WP **<inventorywp>** (E.7.13) (**Aircraft only**).

**k.** Storage of aircraft WP **<storagewp>** (E.7.14) (**Aircraft only**).

**l.** Wiring diagrams and schematics WP **<wiringwp>** (E.7.10).

### E.6.5 Aircraft maintenance module (aircraft TMs, DMWRs, and NMWRs) <mim>.

This maintenance module shall contain the WPs listed below (in the order specified) as required by publication type (see A.4.4) and as applicable for an equipment system between **<aviationcategory>** tags.

**a.** Preservation, packaging, and marking general information WP **<ppmgeninfowp>** (E.7.19).

**b.** Service upon receipt WP **<surwp>** (maintainer level and above) (E.7.3).

**c.** Equipment/User Fitting Instruction WP **<perseqpwp>** (E.7.11).

**d.** The following WPs types occur in no specific order:

**(1)** Maintenance WP **<maintwp>** (E.7.4).

**(2)** General maintenance WP **<gen.maintwp>** (E.7.5).

**(3)** Lubrication instructions WP **<lubewp>** (E.7.6).

**(4)** Preventive maintenance inspections WP **<pmiwp>** (E.7.12)_.

**e.** Overhaul and retirement schedule WP **<orschwp>** (E.7.16).

**f.** Illustrated list of manufactured items **<manu_items_introwp>** (E.7.7).

**g.** Torque limits WP **<torquewp>** (maintainer level and above) (E.7.9).

**h.** Aircraft inventory master guide WP **<inventorywp>** (E.7.13).

**i.** Storage of aircraft WP **<storagewp>** (E.7.14).

**j.** Weighing and loading WP (ASB only) **<wtloadwp>** (E.7.15).

**k.** Wiring diagrams and schematics WP **<wiringwp>** (maintainer level and above) (E.7.10).

### E.6.6 Auxiliary equipment maintenance module <mim>.

This maintenance module shall contain the WPs listed below (in the order specified) as required by publication type (see A.4.4) and as applicable for an equipment system between **<auxiliarycategory>** tags.

**a.** Auxiliary equipment maintenance WP **<auxeqpwp>** (E.7.24).

**b.** Illustrated list of manufactured items **<manuwp>** (maintainer level and above) (E.7.8).

**c.** Torque limits WP **<torquewp>** (maintainer level and above) (E.7.9).

**d.** Wiring diagrams and schematics WP **<wiringwp>** (maintainer level and above) (E.7.10).

### E.6.7 Ammunition maintenance module <mim>.

This maintenance module shall be used for equipment system publications that contain ammunition maintenance WPs and shall not be used for separate ammunition maintenance manuals. This maintenance module shall contain

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the following WPs listed below (in the order specified) as required by publication type (see A.4.4) and as applicable for an equipment system between **<ammunitioncategory>** tags.

**a.** Service upon receipt WP **<surwp>** (maintainer level and above) (E.7.3).

**b.** The following WPs types occur in no specific order:

**(1)** Ammunition identification WP **<ammoidentwp>** (E.7.25).

**(2)** Ammunition maintenance WP **<ammowp>** (E.7.26).

**(3)** Foreign ammunition WP **<natowp>** (E.7.27).

### E.6.8 Test and inspection maintenance module <mim>.

This maintenance category shall contain test and inspection maintenance WP(s) **<maintwp>** (E.7.4) between **<testinspectioncategory>** tags.

### E.6.9 Shipment / movement and storage maintenance module <mim>.

This maintenance category shall contain shipment/movement and storage maintenance WP(s) **<maintwp>** (E.7.4) between **<shipmentmovementstoragecategory>** tags.

### E.6.10 Software maintenance instructions module <mim>.

This maintenance module shall contain software maintenance WPs **<maintwp>** (E.7.4) between **<softmaintcategory>** tags.

### E.6.11 General maintenance module <mim>.

This maintenance module shall contain the maintenance WP **<maintwp>** (E.7.4) and the general maintenance WP **<gen.maintwp>** between **<genmaintcategory>** tags.

### E.6.12 Preventive and phased maintenance modules (aircraft maintenance services only).

Modules for the following shall be prepared and shall contain the following WPs as required by publication type (see A.4.4).

**a.** Preventive maintenance services manual **<pms>** (E.6.12.1).

**b.** Preventive maintenance daily manual **<pmd>** (E.6.12.1).

**c.** Phased maintenance inspection **<pmi>** (E.6.12.2).

#### E.6.12.1 Preventive maintenance services <pms> and daily <pmd> modules (aircraft maintenance services only).

An aircraft preventive maintenance services manual **<pms>** or an aircraft preventive maintenance daily manual **<pmd>** shall be prepared and shall contain the following WPs. See FIGURE E-13 for a diagram of modules and available WPs in these publications.

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### E.6.12.1.1 Preventive maintenance services <pms> or daily <pmd> general information module <gim>.

A preventive maintenance services or preventive maintenance daily general information module shall be prepared and shall contain the following WP.

**a.** General information WP **<pms-ginfowp>** (B.7.4).

### E.6.12.1.2 Preventive maintenance services <pms> or daily <pmd> inspection maintenance module <mim>.

A preventive maintenance services or preventive maintenance daily maintenance module shall be prepared and shall contain the following WP(s) between **<pmscategory>** tags.

**a.** PMS/PMD inspection WP **<pms-inspecwp>** (E.7.17).

**b.** Additional PMS/PMD inspection WP(s) as needed.

**Note**

Tags/output shown in figure(s) below may vary slightly based on version of DTD/stylesheets utilized (see applicable DTD or tag definition list for available tags).

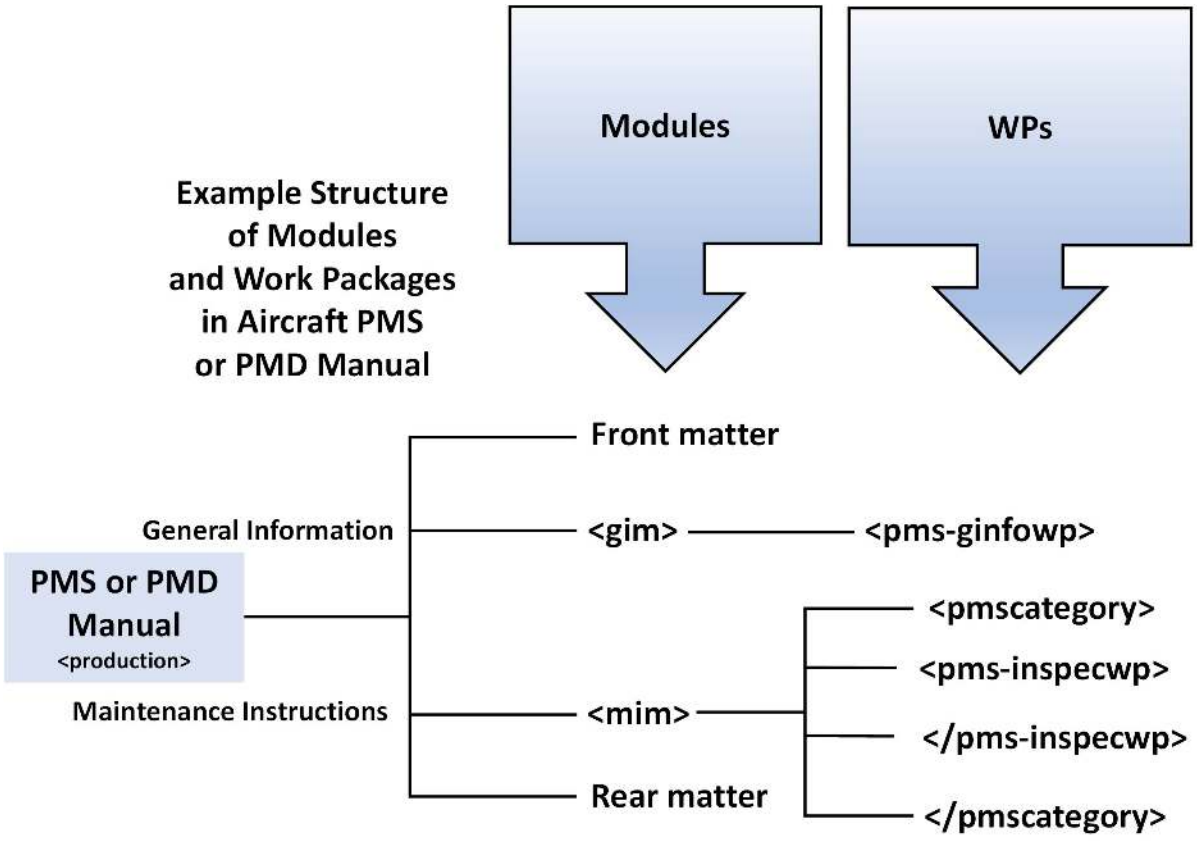


**FIGURE E-13. PMS/PMD manual modules and WPs.**

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### E.6.12.2 Phased maintenance inspection modules (aircraft maintenance services only.

A phased maintenance inspection manual **<pmi>** shall be prepared and shall contain the following modules. See FIGURE E-14 for a diagram of modules and available WPs in these publications.

#### E.6.12.2.1 Phased maintenance inspection general information module <gim>.

A phased maintenance inspection general information module shall be prepared and shall contain the following WP.

**a.** General information WP **<pm-ginfowp>** (B.7.5).

#### E.6.12.2.2 Phased maintenance inspection maintenance module <mim>.

An aircraft phased maintenance instructions module shall be prepared and shall contain the following WP(s) between **<checklistcategory>** tags.

**a.** Phased maintenance checklist WP **<pmi-cklistwp>** (E.7.18).

**b.** Additional phased maintenance checklist WP(s) as needed.

**Note**

Tags/output shown in figure(s) below may vary slightly based on version of DTD/stylesheets utilized (see applicable DTD or tag definition list for available tags).

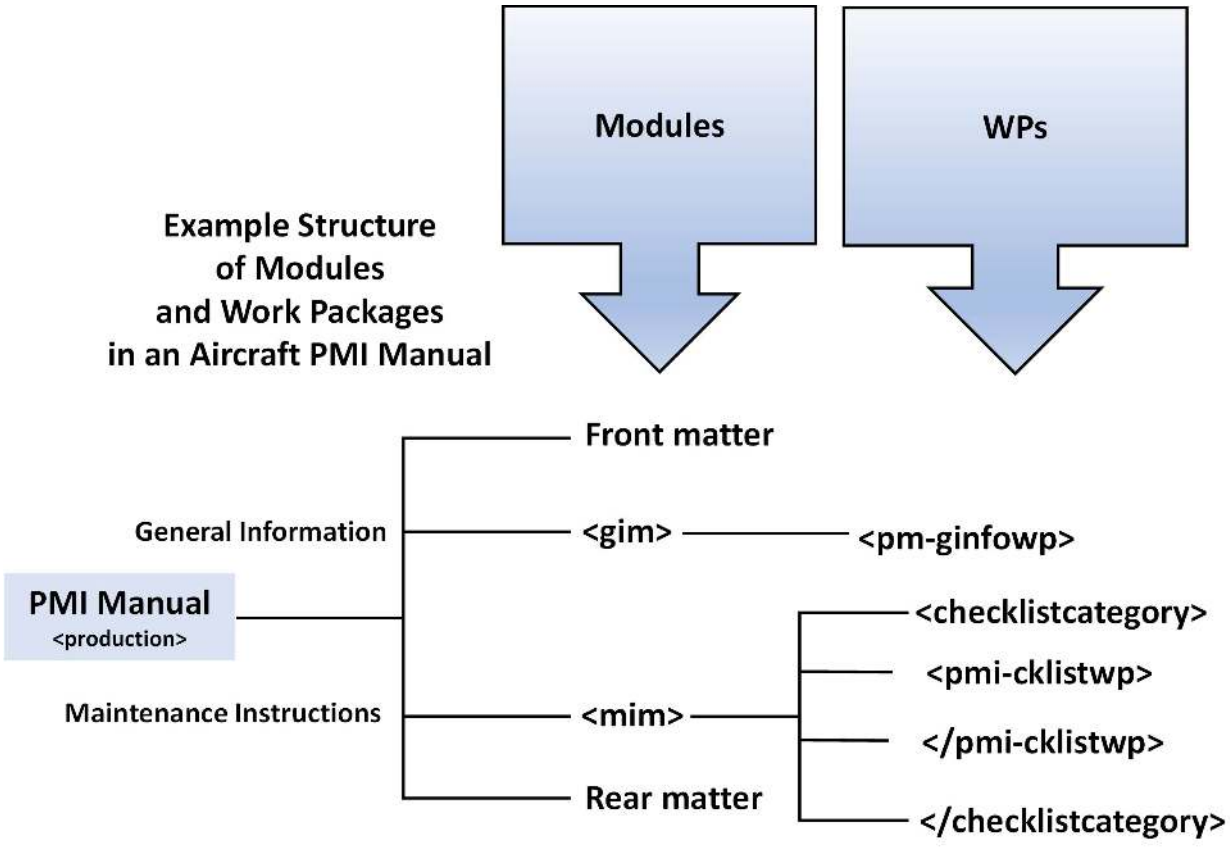


**FIGURE E-14. PMI manual.**

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Check the source to verify that this is the current version before use.

## E.7 DEVELOPMENT OF MAINTENANCE WORK PACKAGES.

Maintenance instructions shall be contained in individual WPs and shall contain the necessary instructions to perform tasks stipulated in the MAC. This section of the standard describes required content development parameters for each kind of maintenance instruction and WP.

**a. Titling.** All maintenance WPs should have the maintenance class followed by the word "maintenance" and then the WP title. See for more information E.7.4.2 for more WP titling guidance.

**b. Grouping tasks.** Multiple tasks shall only be grouped into one WP if all the tasks have the same following initial setup sections: personnel required (or MOS) and equipment condition (see FIGURE E-15). The remaining initial setup sections shall be applicable to at least one task in the WP. Tasks shall not be grouped if condition-based maintenance applies. Liberal use of references between WPs is encouraged.

**Note**

Figure appearance may vary slightly based on version of DTD/stylesheets utilized.

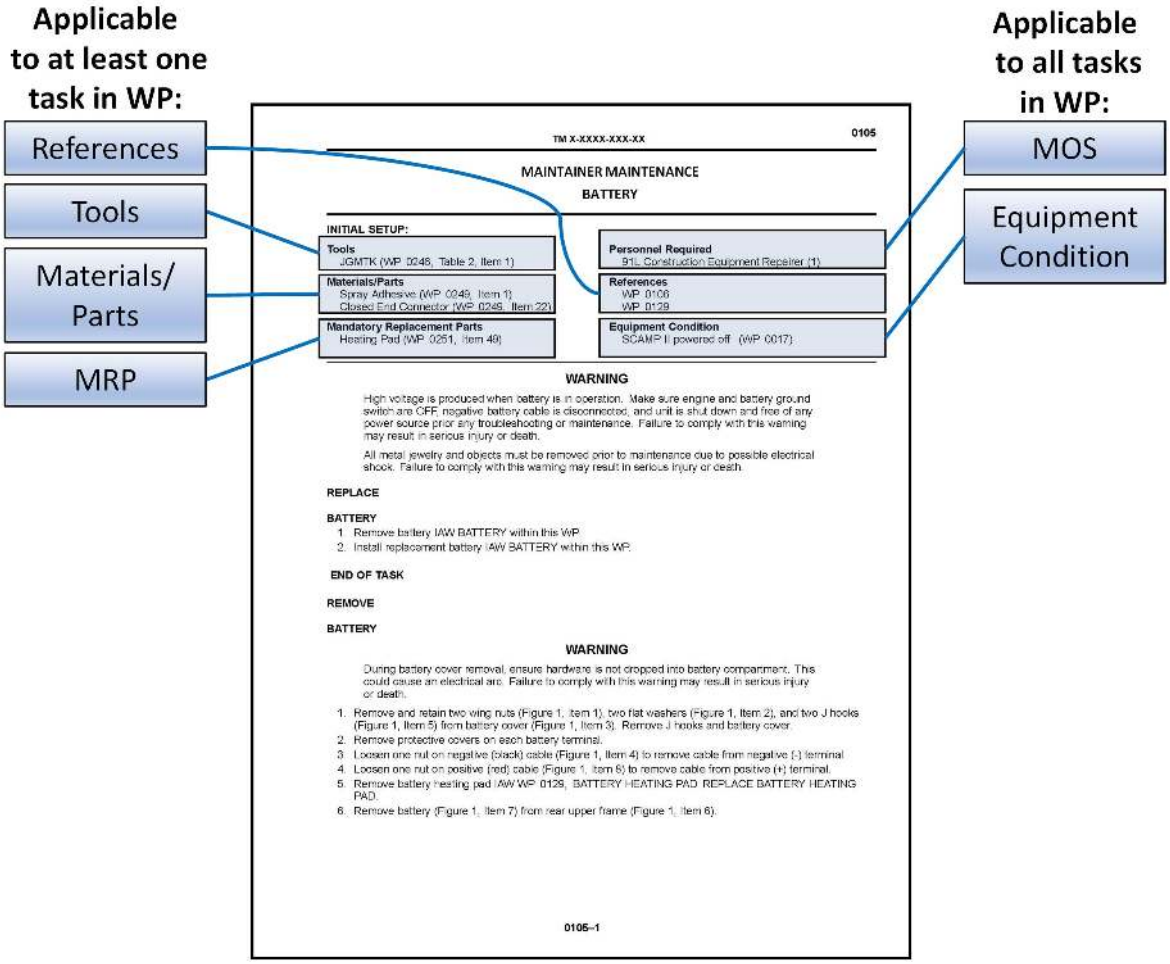


**FIGURE E-15. Initial setup applicability in maintenance WPs.**

**c. Follow on maintenance tasks.** Any follow-on maintenance that must be performed after maintenance instructions are completed shall be included or referenced (for example, disconnect external power, perform operational checks, etc.). When the follow-on maintenance is extensive, it shall be contained in a separate WP and a reference shall be made to the applicable WP.

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**d. End of WP statement.** The words "END OF WORK PACKAGE" shall be placed below the last data item (for example, text, illustration, etc.) of the work package containing the maintenance procedure(s).

## E.7.1 PMCS introduction work package <pmcsintrowp>.

A PMCS introduction shall be prepared as required by publication type (see A.4.4) and as applicable for an equipment system. This WP shall explain the purpose and use of the PMCS data, but shall not contain any maintenance tasks. Additionally, this WP may contain reference/links to the PMCS data by interval but shall not contain reference/links to any other maintenance tasks.FIGURE E-16 shows an example PMCS introduction WP and associated tags. PMCS introduction data may include the following information:

**a.** WP identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** WP initial setup **<initial_setup>** is not required for this WP.

**c.** PMCS introduction data may include the following information:

**(1) DA Form 2404.** An explanation shall be prepared for each PMCS entry. The explanation for the item numbers shall detail how the item numbers are used when recording results of PMCS on DA Form 2404, equipment Inspection and Maintenance Worksheet.

**(2) Lubrication Instructions.** If lubrication instructions are included in the PMCS data, the requirements contained in APPENDIX K shall be used. Only lubrication information such as intervals, lubricant types, etc. shall be included in the introduction. Lubrication instructions shall be included in the PMCS WP. No lubrication instructions shall be contained in the PMCS introduction WP.

**(3) Corrosion Prevention Control.** Information concerning CPC shall be prepared. This information shall contain a reference to the CPC information in the general information WP. In addition, this information shall contain inspection requirements for corrosion. When items are determined to be not ready or available because of the presence of one (or more) forms of corrosion, these items shall be recorded as corrosion failures in the inspection record and the appropriate code as given in DA Pam 750-8 or DA Pam 738-751 will be used when requesting/ performing maintenance activities. In addition, if the inclusion of such instructions is applicable, a statement shall be prepared that states that the instructions are mandatory.

**(4) Fluids.** When the equipment contains fluids, such as lubrication oil or hydraulic fluid, leakage criteria shall be prepared for the PMCS introduction as follows and referred to in the NOT MISSION CAPABLE IF: column (italicized text within parentheses shall be replaced with the appropriate information).

**"FLUID LEAKAGE**

It is necessary for you to know how fluid leakage affects the status of the (*enter component/equipment name*). Following are types/classes of leakage you need to know to be able to determine the status of the (*enter component/equipment name*). Learn these leakage definitions and remember—when in doubt, notify your supervisor. Equipment operation is allowed with minor leakage (*Class I or II*). Consideration must be given to fluid capacity in the item/system being checked/inspected. When in doubt, notify your supervisor.

When operating with Class I or II leaks, continue to check fluid levels as required in the PMCS.

Class III leaks should be reported immediately to your supervisor.

1. Class I. Seepage of fluid (as indicated by wetness or discoloration) but not great enough to form drops.
2. Class II. Leakage of fluid great enough to form drops, but not enough to cause drops to drip from the item being checked/inspected.
3. Class III. Leakage of fluid great enough to form drops that fall from item being checked/inspected."

**(5) Family of Vehicles (FOV) Service Intervals.** The following service intervals for the vehicles listed below shall be utilized as necessary.

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| High Mobility Multipurpose Wheeled Vehicle (HMMWV) FOV Service Intervals | |
|---|---|
| **Current TM Service Intervals** | **Revised Service Intervals** |
| 3K miles or every 6 months, whichever comes first (semi-annual) | 6K miles |
| 6K miles or every 12 months, whichever comes first (annual) | 6K miles |
| 12K miles or every 24 months, whichever comes first (biennial) | 12K miles or every 24 months, whichever comes first (biennial) |

| Family of Medium Tactical Vehicles (FMTV) FOV Service Intervals | |
|---|---|
| **Current TM Service Intervals** | **Revised Service Intervals** |
| 3K miles or every 6 months, whichever comes first (semi-annual) | 6K miles |
| 6K miles or every 12 months, whichever comes first (annual) | 6K miles |
| 12K miles or every 24 months, whichever comes first (biennial) | 12K miles or every 24 months, whichever comes first (biennial) |

| Heavy Expanded Mobility Tactical Truck (HEMTT) FOV Service Intervals | |
|---|---|
| **Current TM Service Intervals** | **Revised Service Intervals** |
| 3K miles or every 6 months, whichever comes first (semi-annual) | 6K miles |
| 6K miles or every 12 months, whichever comes first (annual) | 6K miles |
| 12K miles or every 24 months, whichever comes first (biennial) | 12K miles or every 24 months, whichever comes first (biennial) |

| Palletized Load System (PLS) FOV Service Intervals | |
|---|---|
| **Current TM Service Intervals** | **Revised Service Intervals** |
| 3K miles or every 6 months, whichever comes first (semi-annual) | 6K miles |
| 6K miles or every 12 months, whichever comes first (annual) | 6K miles |
| 12K miles or every 24 months, whichever comes first (biennial) | 12K miles or every 24 months, whichever comes first (biennial) |

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**Note**

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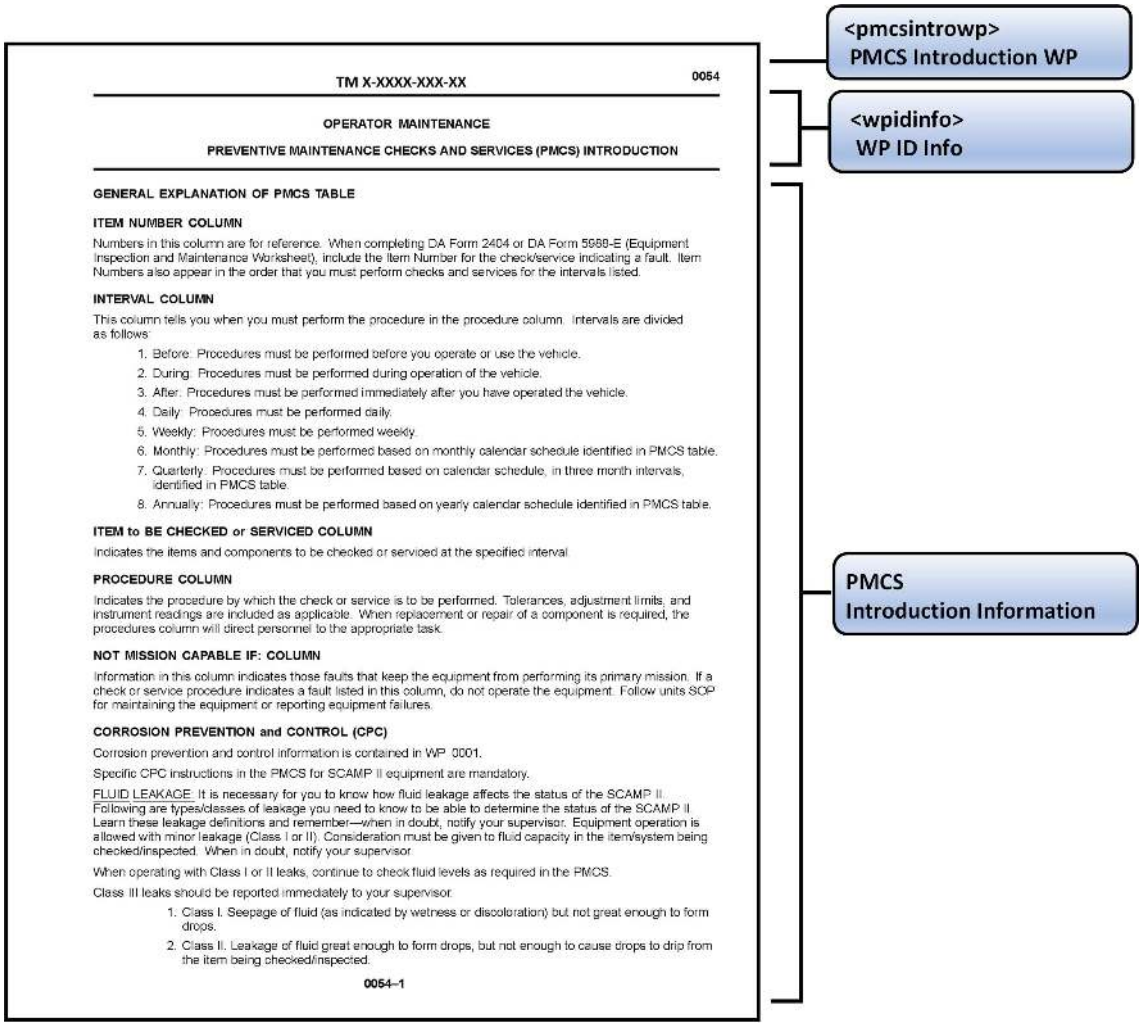


**FIGURE E-16. PMCS introduction WP example.**

## E.7.2 PMCS work package <pmcswp> (except aircraft TMs, DMWRs, and NMWRs).

A PMCS WP shall be prepared as required by publication type (see A.4.4) and as applicable for an equipment system. The PMCS shall be prepared for operator manuals and as required for other maintenance levels. PMCS shall be based upon the principles of Reliability Centered Maintenance (RCM) logic. It shall include PMCS information and applicable scheduled corrosion inspections. Lubrication instructions may be included in the PMCS information, or a separate lubrication order may be prepared (refer to APPENDIX K).

PMCS instructions shall include step-by-step instructions, supporting illustrations and references to any other information required to perform each check or service. This may include lubrication, appropriate tolerances adjustment limits, and instrument gauge readings. When specified by the MATDEV, illustrations shall be prepared to identify the location or the process of the task being performed and shall be integrated with the instructions. Whenever replacement or repair is recommended, the maintenance task shall be referenced. PMCS procedural steps

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shall be numbered in accordance with 4.12.8. An example PMCS table is shown in FIGURE E-17.FIGURE E-18 through FIGURE E-20 shows an example PMCS WP and the associated tags.

The following components shall be included in PMCS WPs.

**a.** WP identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** WP initial setup **<initial_setup>** is required for this WP (5.4.3).

**c.** PMCS data discussed in E.7.2.1 through E.7.2.6.

### E.7.2.1 Item Number <itemno>.

Item numbers (ITEM NO.) shall be assigned to the PMCS instructions. The PMCS instructions shall be arranged in a logical sequence requiring minimum time and motion on the part of the person(s) performing them and shall be so arranged that minimum interference will occur between persons performing the checks simultaneously on the same end item. Item numbers shall be Arabic numbers starting with 1 and continuing sequentially and shall not use decimals or letters. See FIGURE E-17, Item 1 for an example.

### E.7.2.2 PMCS intervals.

The designated interval (INTERVAL) when each check is to be performed shall be included (see FIGURE E-17, Item 2). Some intervals are time-based (E.7.2.2.1) while other intervals are based on measurements other than time (E.7.2.2.2).

Instructions done first or most frequently shall appear first in the PMCS list of checks and services. Each PMCS item (row) shall have one interval and like intervals shall be grouped together. For example, time-based PMCS intervals shall group "before" entries together and "during" entries together, etc.

#### E.7.2.2.1 Intervals <interval> (time-based).

For some equipment systems, PMCS shall be performed at time-based intervals. The time-based PMCS intervals that can be used are listed in TABLE E-IV. Contact the MATDEV with any questions regarding intervals for a specific equipment system.

**TABLE E-IV. PMCS intervals.**

| Interval | Maintenance Type | Maintenance Action |
|---|---|---|
| Before | General | Maintenance checks performed on equipment systems prior to operation |
| During | General | Maintenance checks performed on equipment systems during operation |
| After | General | Maintenance checks performed on equipment systems after operation |
| Daily | General | Maintenance checks on equipment systems performed every day |
| Weekly | General | Maintenance checks on aircraft systems performed every week |
| Monthly | General | Maintenance checks on aircraft systems performed every month |
| Quarterly | General | Maintenance checks performed on equipment systems four times a year |

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**TABLE E-IV. PMCS intervals. (continued**

| Interval | Maintenance Type | Maintenance Action |
|---|---|---|
| Semiannually | General | Maintenance checks performed on equipment systems twice a year |
| Annually | General | Maintenance checks performed on equipment systems every year |
| Biennially | General | Maintenance checks performed on equipment systems every two years |
| Periodic | General | Maintenance checks performed on equipment systems repeatedly from time to time |
| Intermediate | Aviation only | Maintenance checks on aircraft systems performed approximately every 25 hours of flying time (light checks) * |
| Man-hour/day | Aviation only | Maintenance checks on aircraft systems performed approximately every 25 – 50 hours of flight time (routine checks) * |
| Phased | Aviation only | Maintenance checks on aircraft systems performed approximately every 100 hours of flight time; performed to break up inspections into sections to reduce time an aircraft spends grounded (staggered checks) * |
| Other | General | Maintenance checks performed on equipment systems as needed based on system-dependent requirements |
| * Contact MATDEV for specific guidance on inspection and service intervals and other requirements. | | |

### E.7.2.2.2 Other intervals <intervals> (non-time based).

For some equipment systems, PMCS shall be performed at intervals other than time-based intervals. For example, some ground vehicles may have been previously required to undergo PMCS semi-annually or every 3,000 miles (whichever came first). Current PMCS requirements may stipulate these same ground vehicles undergo PMCS every 3,000 miles regardless of time. Contact the MATDEV with any questions regarding intervals for a specific equipment system.

### E.7.2.3 Man-hours <manhours>.

When specified by the MATDEV, man-hours (MAN-HOUR) required to complete all prescribed lubrication services shall be included. Man-hours shall be stated to the nearest 10th of an hour.

### E.7.2.4 Item to be checked or serviced <checked>.

The items listed (ITEM TO BE CHECKED OR SERVICED) shall be identified in as few words as possible to clearly identify the item. Usually the common name (for example, bumper, gas can and mounting bracket, front axle, etc.) will be enough. See FIGURE E-17, Item 3 for an example.

### E.7.2.5 Procedure <pmcsproc>.

The instructions (procedure) include specific step-by-step instructions to check or service item listed in E.7.2.4. See FIGURE E-17, Item 4 for an example.

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### E.7.2.6 Not mission capable if <eqpnotavail>.

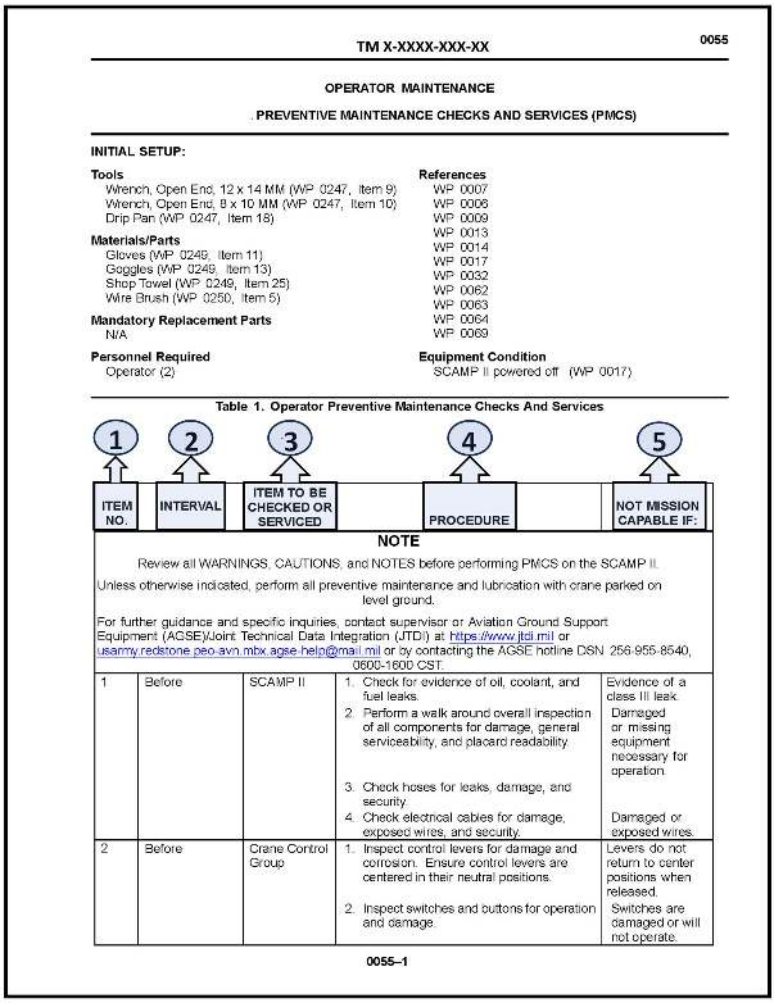
If a PMCS item has no mission capable criteria, a brief statement shall be provided to detail the condition (NOT MISSION CAPABLE IF:) (for example, malfunction, shortage) that would cause the equipment to be less than fully ready to perform its assigned mission. If the instruction contains detailed steps, the statement shall be placed opposite the applicable step. See FIGURE E-17, Item 5 for an example.

### E.7.2.7 Preventive Maintenance Checklist (PMC.

When specified by the MATDEV, a PMC shall be prepared as specified in APPENDIX J.

**Note**

Figure appearance may vary slightly based on version of DTD/stylesheets utilized.



TM X-XXXX-XXX-XX 0055

OPERATOR MAINTENANCE

PREVENTIVE MAINTENANCE CHECKS AND SERVICES (PMCS)

INITIAL SETUP:

Tools
Wrench, Open End, 12 x 14 MM (WP 0247, Item 9)
Wrench, Open End, 8 x 10 MM (WP 0247, Item 10)
Drip Pan (WP 0247, Item 18)

Materials/Parts
Gloves (WP 0249, Item 11)
Goggles (WP 0249, Item 13)
Shop Towel (WP 0249, Item 25)
Wire Brush (WP 0250, Item 5)

Mandatory Replacement Parts
N/A

Personnel Required
Operator (2)

References
WP 0007
WP 0006
WP 0009
WP 0013
WP 0014
WP 0017
WP 0032
WP 0062
WP 0063
WP 0064
WP 0069

Equipment Condition
SCAMP II powered off (WP 0017)

Table 1. Operator Preventive Maintenance Checks And Services

| 1 ITEM NO. | 2 INTERVAL | 3 ITEM TO BE CHECKED OR SERVICED | 4 PROCEDURE | 5 NOT MISSION CAPABLE IF: |
|---|---|---|---|---|
| | | NOTE<br>Review all WARNINGS, CAUTIONS, and NOTES before performing PMCS on the SCAMP II.<br>Unless otherwise indicated, perform all preventive maintenance and lubrication with crane parked on level ground.<br>For further guidance and specific inquiries, contact supervisor or Aviation Ground Support Equipment (AGSE)/Joint Technical Data Integration (JTDI) at https://www.jtdi.mil or usarmy.redstone.peo.avn.mbx.agse-help@mail.mil or by contacting the AGSE hotline DSN 256-955-8540, 0600-1600 CST. | | |
| 1 | Before | SCAMP II | 1. Check for evidence of oil, coolant, and fuel leaks.<br>2. Perform a walk around overall inspection of all components for damage, general serviceability, and placard readability.<br>3. Check hoses for leaks, damage, and security.<br>4. Check electrical cables for damage, exposed wires, and security. | Evidence of a class III leak.<br>Damaged or missing equipment necessary for operation.<br>Damaged or exposed wires. |
| 2 | Before | Crane Control Group | 1. Inspect control levers for damage and corrosion. Ensure control levers are centered in their neutral positions.<br>2. Inspect switches and buttons for operation and damage. | Levers do not return to center positions when released.<br>Switches are damaged or will not operate. |

0055–1

**FIGURE E-17. PMCS table in WP.**

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**Note**

Figure appearance and tags/output may vary slightly based on version of DTD/stylesheets utilized (see applicable DTD or tag definition list for available tags).

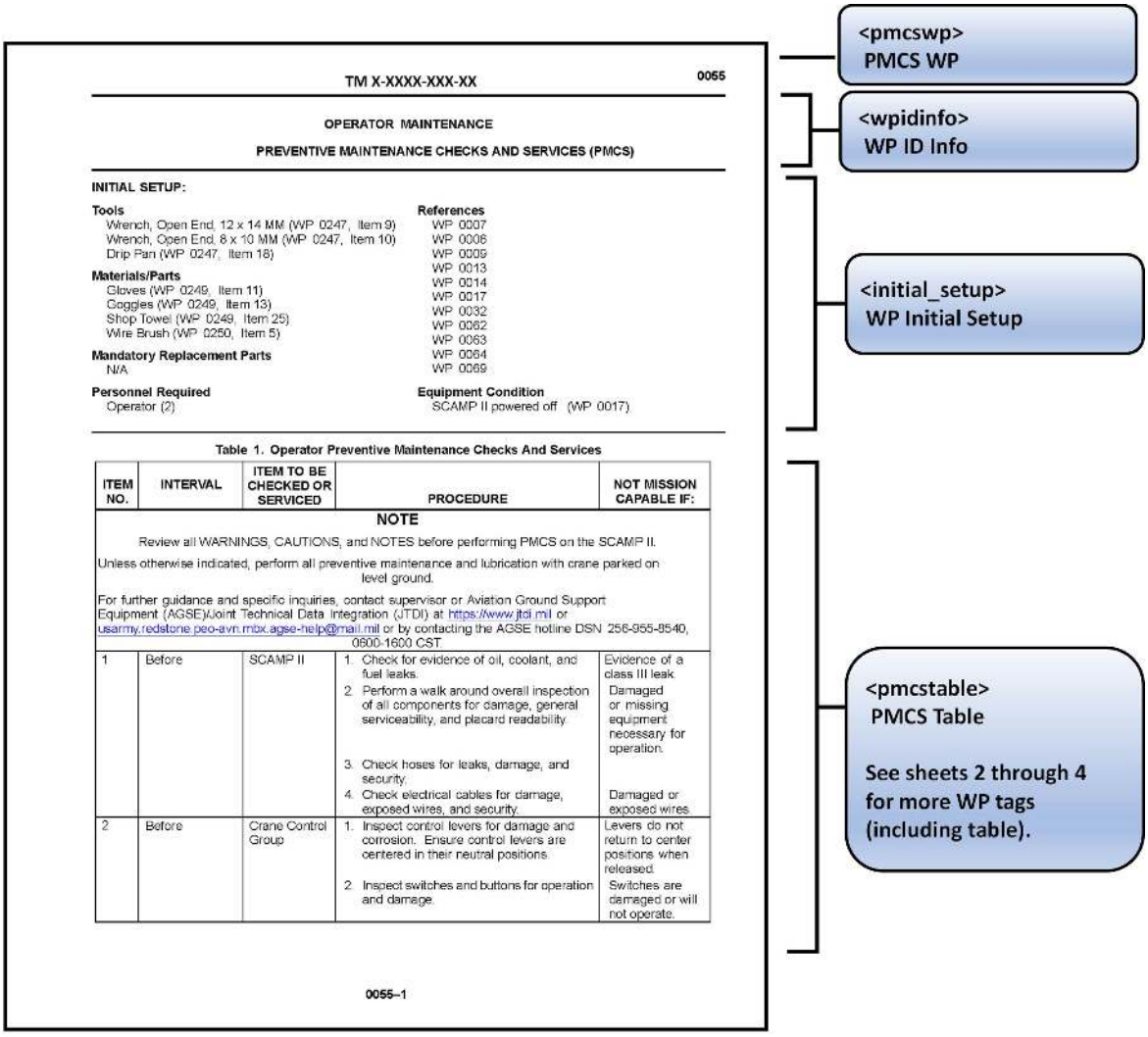


**FIGURE E-18. PMCS WP example (sheet 1 of 3).**

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PMCS WP

```
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE pmcswp PUBLIC "-//USA-DOD//DTD xx TM Assembly REV x x.x yyyymmdd//EN"
".\40051x_x_x\40051x_x_x.dtd" [
<!ENTITY SCAMPTM02394d SYSTEM ".\Graphics\SCAMPTM02394d.svg" NDATA SVG>
]>
<pmcswp chngno="0" wpno="M00023-x-xxxx-xxx-12nP">
<wpidinfo>
<maintlvl level="operator"/>
<title id="M00023-x-xxxx-xxx-12nP_TITLE_1">PREVENTIVE MAINTENANCE CHECKS AND SERVICES
(PMCS)</title>
</wpidinfo>
<initial_setup> ...
</initial_setup>
<pmcstable>
<title>Preventive Maintenance Checks and Services</title>
<pmcs-entry>
<itemno>1</itemno>
<interval>Before</interval>
<checked>SCAMP II</checked>
<pmcsproc>
<pmcsstep1>
<para>Check for evidence of oil, coolant, and fuel leaks.</para>
<eqpnotavail>
<trim.para>Evidence of a class III leak.</trim.para>
</eqpnotavail>
</pmcsstep1>
<pmcsstep1>
```

Doctype Statement and WP RootTag <pmcswp>

<wpidinfo> WP Identification Information

<initial_setup> PMCS Initial Setup

<pmcstable> PMCS Table

<pmcsproc> PMCS Procedure Start

<eqpnotavail> Equipment Not Available If

<pmcs-entry> PMCS Entry 1

FIGURE E-19. PMCS WP example (sheet 2 of 3).

PMCS WP (continued)

```
<pmcsstep1>
<para>Check for evidence of oil, coolant, and fuel leaks.</para>
<eqpnotavail>
<trim.para>Evidence of a class III leak.</trim.para>
</eqpnotavail>
</pmcsstep1>
<pmcsstep1>
<para>Perform a walk around overall inspection of all components for damage, general
serviceability, and placard readability.</para>
<eqpnotavail>
<trim.para>Damaged or missing equipment necessary for operation.</trim.para>
</eqpnotavail>
</pmcsstep1>
<pmcsstep1>
<para>Check hoses for leaks, damage, and security.</para>
</pmcsstep1>
<pmcsstep1>
<para>Check electrical cables for damage, exposed wires, and security.</para>
<eqpnotavail>
<trim.para>Damaged or exposed wires.</trim.para>
</eqpnotavail>
</pmcsstep1>
</pmcsproc>
</pmcs-entry>
<pmcs-entry>
<itemno>2</itemno>
<interval>Before</interval>
```

<pmcs-entry> PMCS Entry 1 (continued)

<pmcs-entry> PMCS Entry 2

FIGURE E-20. PMCS WP example (sheet 3 of 3).

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## E.7.3 Service upon receipt work package <surwp> (maintainer level and above.

One or more service upon receipt WPs shall be prepared as required by publication type (see A.4.4) and shall contain information required for the user to ensure that equipment will be adequately inspected, serviced, and operationally tested before it is subjected to use. Each WP shall contain a single service upon receipt task(s) **<surtsk>**. FIGURE E-21 shows an example service upon receipt WP and the associated tags. The following components shall be included in service upon receipt WPs.

**a.** WP identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** WP initial setup **<initial_setup>** is required for this WP (5.4.3).

**c.** Service upon receipt task **<surtsk>** as described in E.7.3.1 through E.7.3.1.3.3.11. Each WP shall contain a single service upon receipt task.

**Note**

Figure appearance and tags/output may vary slightly based on version of DTD/stylesheets utilized (see applicable DTD or tag definition list for available tags).

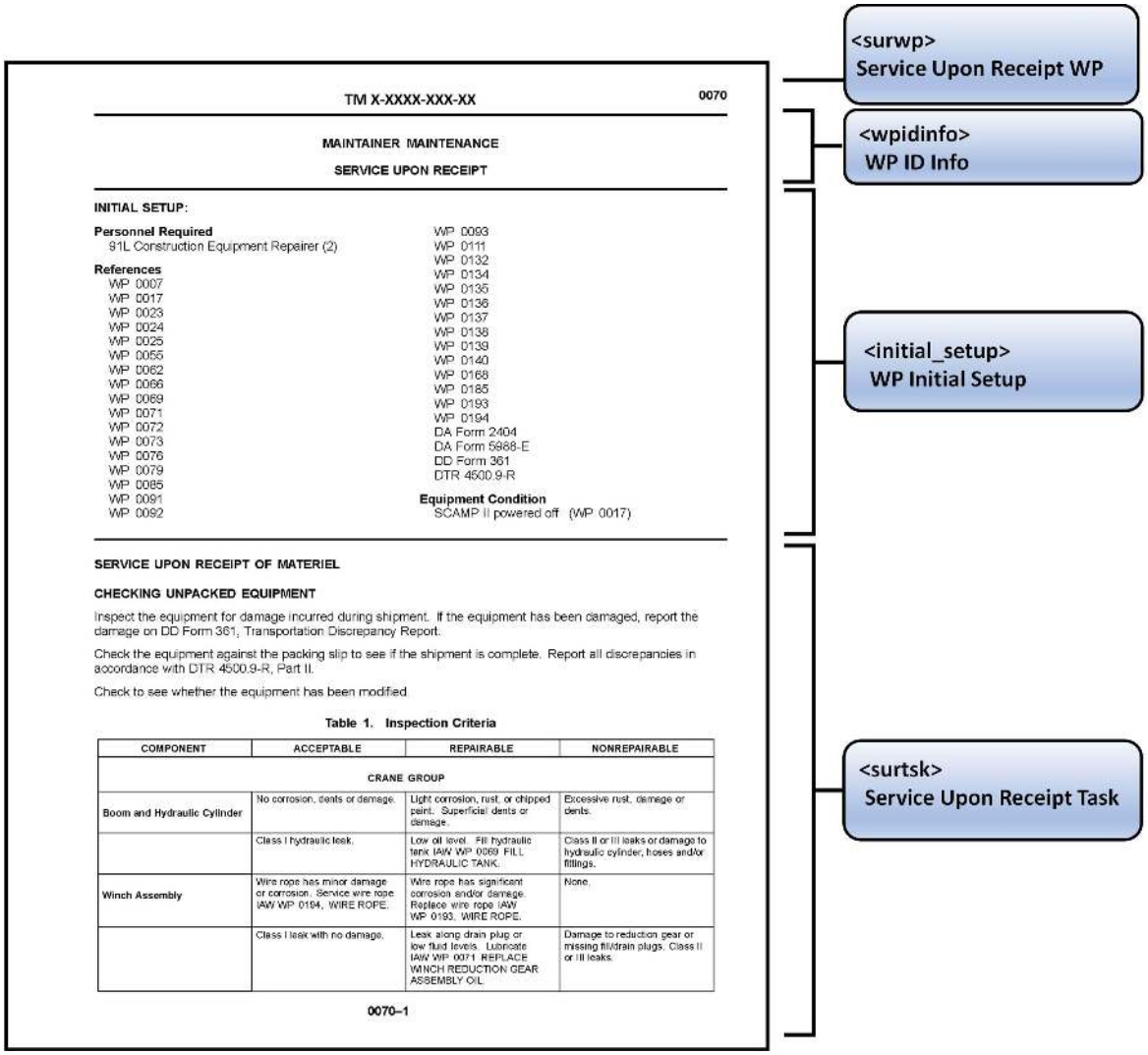


**FIGURE E-21. Service upon receipt WP example.**

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### E.7.3.1 Service upon receipt tasks <surtsk>.

For equipment that requires extensive service upon receipt, the following tasks described in E.7.3.1.1 through E.7.3.1.3.3.11 shall be prepared and shall be placed in individual WPs. Instructions for ammunition service upon receipt are contained in E.7.3.1.3.3.9. If these tasks reside in an existing WP/manual, reference may made to them.

#### E.7.3.1.1 Siting <siting>.

Siting instructions peculiar to the equipment shall be prepared, as applicable. In preparing the instructions, operational and maintenance features shall be considered, such as the following:

**a.** Location.

**b.** Proximity to power sources.

**c.** Effective ranges.

**d.** Terrain requirements to avoid screening, reflections, ground clutter, and other poor operational conditions due to terrain.

**e.** Technical requirements.

**f.** Shelter locations.

**g.** Compensation for adverse siting conditions.

**h.** When the equipment contains large components such as towers and antennas that require orientation to a baseline during siting.

**i.** Orientation of mobile equipment during installation.

#### E.7.3.1.2 Shelter requirements <shelter>.

When equipment is normally housed in a permanent or semi-permanent shelter (other than a military truck, van, or transportable shelter) during use, the following information shall be prepared:

**a.** Amount of floor, wall, and height space required.

**b.** A plan for a typical layout.

**c.** Required weight capacity of the building floor.

**d.** Dimensions required for installed equipment.

**e.** Total weights that the floor must support and the area in square feet over which the total weight will be distributed.

**f.** Environmental conditions (for example, venting).

**g.** Power requirements.

**h.** Unusual requirements specific to equipment, such as air conditioning.

**i.** Architectural and engineering data on beam sizes, lengths, bending moments, and required supports shall not be included.

#### E.7.3.1.3 Service upon receipt of materiel <surmat>.

The following instructions shall be prepared to service upon receipt of materiel.

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### E.7.3.1.3.1 Unpacking <unpack>.

Instructions for unpacking materiel or equipment shall be prepared.

### E.7.3.1.3.2 Checking unpacked equipment <chkeqp>.

Instructions shall be prepared for a condition check of the shipment (including that of pallets, containers, boxes, and legibility of markings). These instructions may be contained in a table (standard format information per 4.15.8). The following data shall be included:

**a.** Packaging material **<crit.insp.tab>**. For each item **<eqpitem>** of a component requiring inspection, the following conditions shall be provided: acceptable <accept>, repairable <repairable>, and non-repairable **<nonrepairable>**.

**b.** Equipment components **<pecul.insp.tab>**. A table shall be provided listing, by location **<location>**, each item **<eqpitem>** of a component **<compntassem>**requiring inspection. For each of these items, an inspection action **<step1>** shall be provided and, if applicable, a reference **<remarks>** shall be made to another WP. In addition, the following shall be inserted:

"Inspect the equipment for damage incurred during shipment. If the equipment has been damaged, report the damage on DD Form 361, Transportation Discrepancy Report. Check the equipment against the packing slip to see if the shipment is complete. Report all discrepancies in accordance with DTR 4500.9-R, Part II. Check to see whether the equipment has been modified."

**c.** Processing unpacked equipment **<processeqp>**. Instructions shall be prepared for processing the unpacked equipment (for example, removing excess lubricant from a new rifle), provided they do not conflict with any warranty provisions. The following information shall be prepared, as applicable:

**(1)** Any special skills required by processing personnel.

**(2)** All caustic, corrosive, and/or toxic material used during processing shall be identified and applicable warnings and cautions given.

**(3)** Instructions on safe disposal of waste products generated during processing actions.

**(4)** Man-hour requirements and total man-hours required for processing the equipment.

### E.7.3.1.3.3 Installation instructions.

Instructions shall be prepared to install the equipment properly. These instructions shall include which tools are to be used to make the necessary interconnections, to lubricate, calibrate, and adjust the equipment. Instructions for cabling and wiring shall include the following:

**a.** Cable diagrams shall be included or referenced as necessary. When cable assemblies are not supplied but are required for bench test setup, instructions shall be prepared in the manufactured items WP (refer to E.7.8) for fabricating interconnecting cable assemblies.

**(1)** Instructions shall be prepared for any mating connectors that call for a special instruction either to make the proper connection or to prevent damage to the connector. Warnings and cautions shall be included where necessary.

**(2)** A wiring diagram shall be prepared that fully identifies, by either color code or wire number (if applicable), each wire to be connected. This diagram shall show the location of each pertinent terminal. The terminal(s) shall be identified by number or other marking, if available, or by position if neither is available. Where appropriate, voltage readings shall be annotated.

**(3)** All alternate connection patterns required for various modes of operation shall be shown and explained.

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**(4)** Only one diagram shall be used to illustrate interconnection patterns that appear more than once within the same equipment.

**b.** For installation of plug-in items, diagrams shall be prepared or referenced showing the location of items that are not installed in the equipment when received. Instructions shall be prepared whenever special techniques or connections are required.

### E.7.3.1.3.3.1 Installation of equipment.

Installation instructions (including placing, mounting, and attaching) shall be prepared for cable and wiring interconnections as well as proper use of special tools/kits. Installation instructions shall identify all dimensions that must be maintained in placing, mounting, or attaching items. When initial adjustments can be made efficiently during installation, such adjustments shall be included. For equipment designed and intended for use in more than one type of installation (for example, field, fixed station, and mobile), instructions shall be prepared for each type of installation involved. Performance of any step in the installation instructions that requires the assistance of personnel from a higher level of maintenance shall be detailed. This shall be stated in a note like the following (italicized text within parentheses shall be replaced with the appropriate information):

"Note

The following installation instruction must be made with the assistance of (*insert level*) maintenance personnel (*include Military Occupational Specialty, if applicable*)."

Installation instructions shall include instructions for (*as applicable*):

**a.** All required installation options (*for example, ESD control requirements*).

**b.** Accessory items.

**c.** Auxiliary items (*those that extend or increase equipment capability*).

**d.** Grounding of the equipment for both safety and proper operation.

**e.** Torque requirements.

### E.7.3.1.3.3.2 Special applications.

Installation instructions that are common to all special applications of a system shall be prepared. Details resulting from the installation shall be omitted if they are specific only to the equipment into which the system is being installed (for example, special treatment required when the system is installed in a vehicle or aircraft).

### E.7.3.1.3.3.3 Van and shelter installations.

When the equipment is permanently installed in vans or shelters, installations instructions will not need to be prepared. The following information shall be prepared only to the extent required for the applicable level of maintenance:

**a.** Instructions shall be prepared for the removal, installation and/or replacement of each nonpermanent unit.

**b.** Diagrams and instructions shall be prepared that pertain to electrical and interconnection wiring exclusive of wiring specific to the equipment on which the installation is being made (for example, headlight, ignition wiring).

**c.** Instructions shall be prepared for cable run locations, equipment locations, circuit breaker panels, and other similar details.

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### E.7.3.1.3.3.4 Assembly of equipment <assem>.

Instructions shall be prepared for assembling equipment that has been shipped unassembled.

**a. When the equipment is to be shelf or rack mounted.** Instructions shall also be prepared for assembly of the rack, if necessary, and for installation of the equipment in the rack. As applicable, power requirements, connections, and initial control settings needed for installation purposes shall be included.

**b. When the equipment is shipped or delivered in specially designed containers.** Unpacking instructions shall be prepared as detailed in E.7.3.1.3.1.

**c. For security measures for electronic data.** Instructions shall be prepared for handling, loading, purging, overwriting, or unloading classified electronic data under usual conditions. Instructions shall meet current security regulations as they pertain to automation security

### E.7.3.1.3.3.5 Preliminary servicing of equipment <preserv>.

Instructions shall be prepared for all preliminary services required on newly installed equipment. This should include but not be limited to the following: lubrication, wiring, and fueling, etc.

### E.7.3.1.3.3.6 Preliminary checks and adjustment of equipment <prechkadj>.

Instructions shall be prepared for all checks and adjustments to be made on newly installed equipment. Information on the location of items such as controls and check points shall be prepared or referenced. Instructions shall be prepared for checks and adjustments that must be made before the equipment is put into operation and for all other checks required to ensure proper operation of the equipment. These instructions shall include but not be limited to the following (as applicable):

**a.** Checks for interconnections.

**b.** Checks for grounding, including earth ground connections, earth conditioning for conduction, as well as a check of the grounding circuit for negligible resistance.

**c.** Checks for adequate clearance for rotating or moving devices.

**d.** Checks of initial settings of all controls that must be preset before power is to be applied.

**e.** All other checks needed to determine that power can be applied without injuring personnel or damaging the equipment.

**f.** Firm seating and connection of all plug-in parts, mating connectors, jacks, and plugs.

**g.** Cable and wire harness routing, dressing, and fastening.

**h.** ESD control standards and cautions against damaging transistors, diodes, and other electrically sensitive items.

**i.** Replacement of all covers, inspection and access doors, and plates.

**j.** Operation of safety interlocks and switches.

**k.** Operation of ventilating louvers and intake and exhaust ports.

**l.** Operation and content of liquid cooling systems.

**m.** Lubricants and Corrosion Prevention Control (CPC) instructions.

**n.** Switch and control settings that are preset at installation (installer's adjustments).

**o.** Presetting and adjustment of automatic controls.

**p.** Terminal connections.

**q.** Required terminal or capacitor strapping.

**r.** Preliminary test measurements.

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**s.** Presetting operator controls.

**t.** Normal operating checks.

**u.** After-installation orientation.

**v.** Burn-in of parts.

**w.** After-operations shutdown, checks, and inspections.

### E.7.3.1.3.3.7 Preliminary calibration of equipment <precal>.

Instructions shall be prepared for all calibration to be made on newly installed equipment.

### E.7.3.1.3.3.8 Circuit alignment <calign>.

Instructions shall be prepared for circuit alignment. Applicable instructions shall be prepared in the following order:

**a.** External connections **<extconn>**. Connections to external lines that are required for each installation option shall be included. Connection instructions shall conform to the requirements for installing wiring and cabling interconnections.

**b.** Switch settings, patch panel connections, and internal control settings **<setconn>**. Instructions shall be prepared for all switch settings, patch panel connections, and internal control settings required for each installation option and mode of operation.

**c.** Alignment instructions **<alignpcroc>**. Instructions shall be prepared for all alignment instructions, including any variations required for different installation options and modes of operation.

### E.7.3.1.3.3.9 Ammunition service upon receipt <ammo.sur>.

Instructions shall be prepared for performing the tasks needed to perform upon receipt of ammunition. Instructions and information shall be provided that can include receipt of correct items, deficiencies, safety precautions, identification, and packaging and ammunition defects.

### E.7.3.1.3.3.10 Other service upon receipt tasks <other.surtsk>.

Additional service upon receipt tasks may be developed when the specific type of service upon receipt tasks are not covered as described in E.7.3.1.3.3 through E.7.3.1.3.3.9. If additional service upon receipt tasks are used, the MATDEV shall submit to LDAC the requirements for this service upon receipt task type for possible incorporation within future revisions to this standard.

### E.7.3.1.3.3.11 Follow-on maintenance <followon.maintsk>.

As applicable, instructions shall be prepared or references to the applicable WP(s) for any follow-on maintenance required and shall be the last information in the WP. See E.7.4.4.41.

## E.7.4 Maintenance instruction work package <maintwp>.

Maintenance instruction WP(s) shall be prepared and shall be functionally divided into individual maintenance WP (s) **<maintwp>** containing one or more complete start-to-finish maintenance tasks **<maintsk>**. These maintenance WPs shall be in the order listed in the MAC and shall use the same task titles **<title>** as shown in the MAC.

Every entry in the MAC shall be contained in a maintenance task within the maintenance publications (TMs, NMWR, DMWR, SUM, SAM, etc.) for the system. Maintenance WPs associated with a MAC entry may be in a separate publication. For example, for a system with a -10 publication and a -20 publication, operator maintenance tasks listed in the -20 MAC shall appear in the -10.

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Technical content structure for these WP shall be consistent across WPs. Illustrations shall be prepared to identify the location or the process of the task being performed and shall be integrated with the procedures. FIGURE E-22 through FIGURE E-24 show an example maintenance instruction WP and the associated tags. The following components shall be included in maintenance instruction WPs.

**a.** WP identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** WP initial setup **<initial_setup>** is required for this WP (5.4.3).

**c.** Maintenance instruction task(s) **<maintsk>** shall be prepared. FIGURE E-22 through FIGURE E-24 show the general structure of one example WP containing multiple tasks that are all relevant to one piece of equipment.

**Note**

Figure appearance and tags/output may vary slightly based on version of DTD/stylesheets utilized (see applicable DTD or tag definition list for available tags).

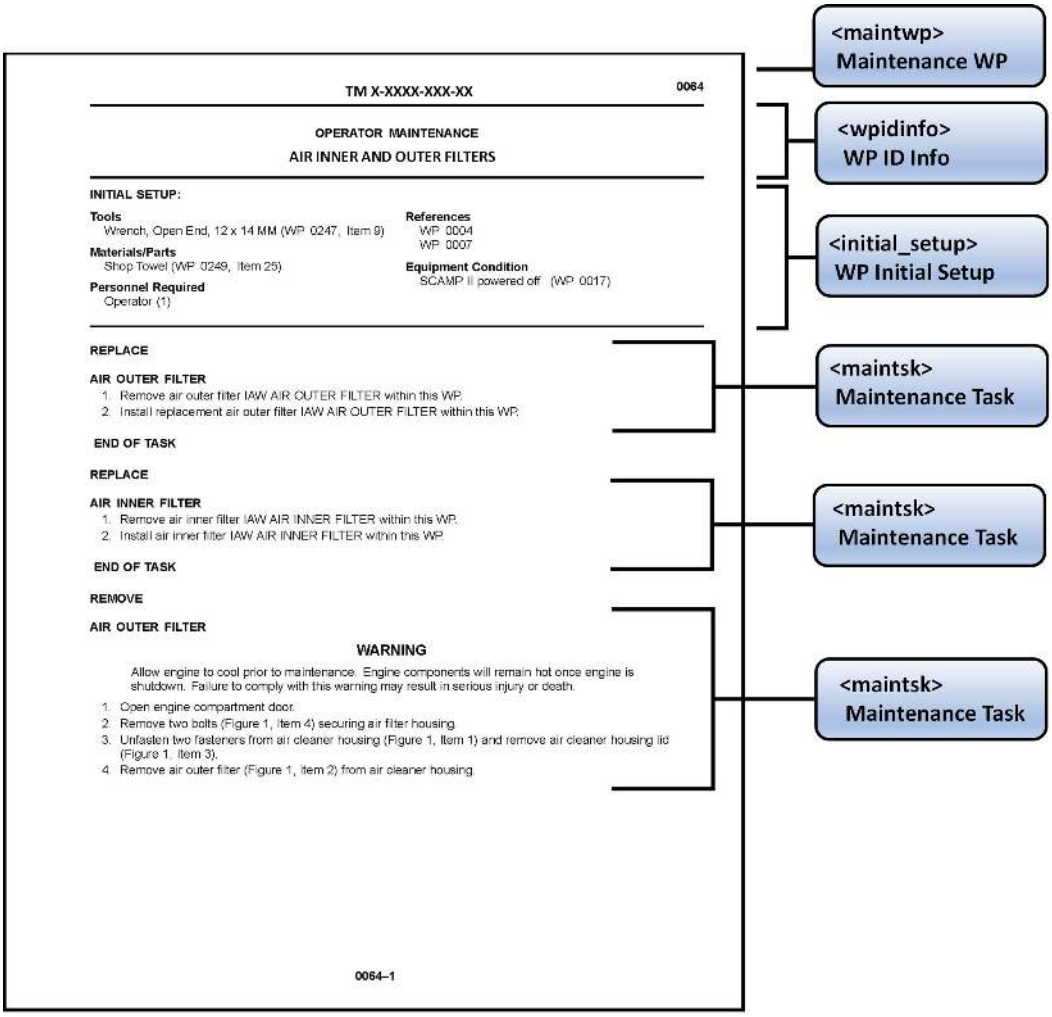


**FIGURE E-22. Maintenance WP example (sheet 1 of 3).**

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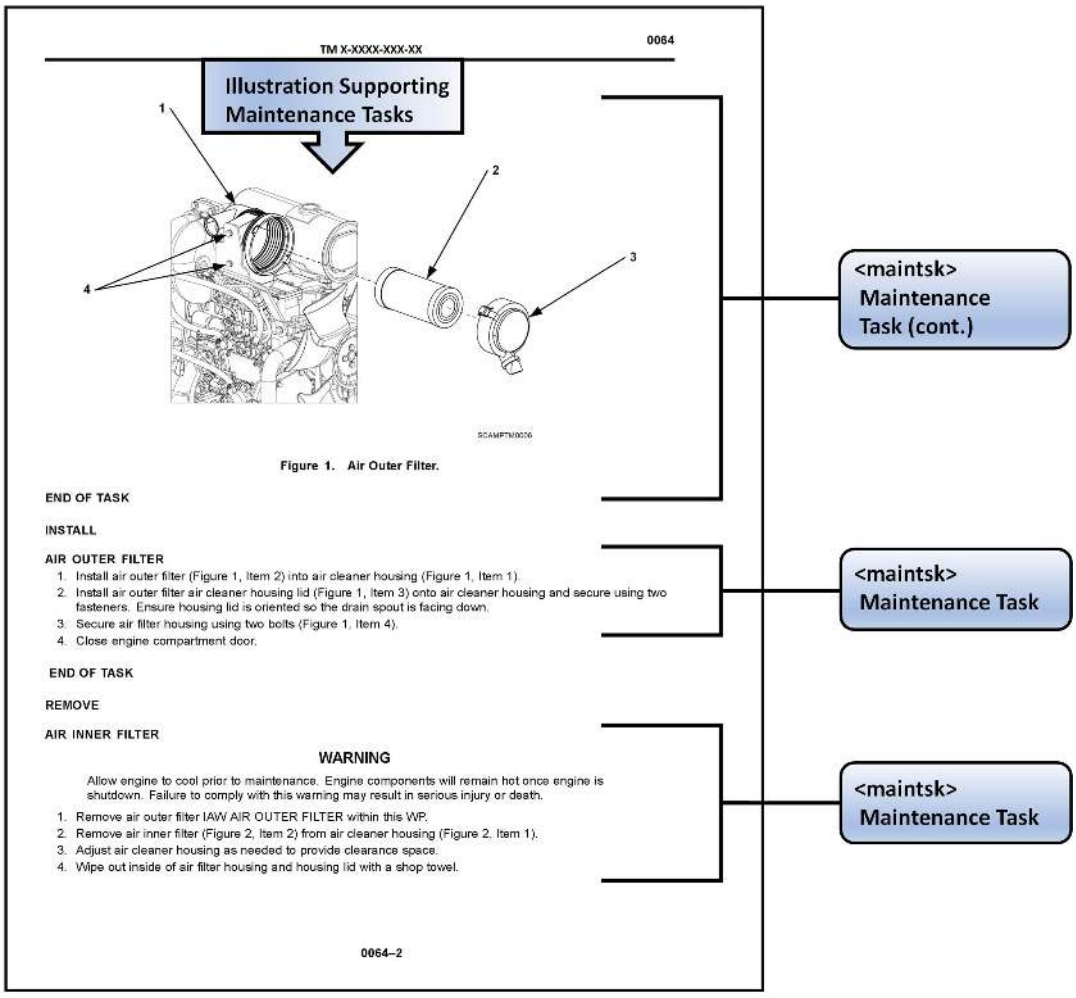


**FIGURE E-23. Maintenance WP example (sheet 2 of 3).**

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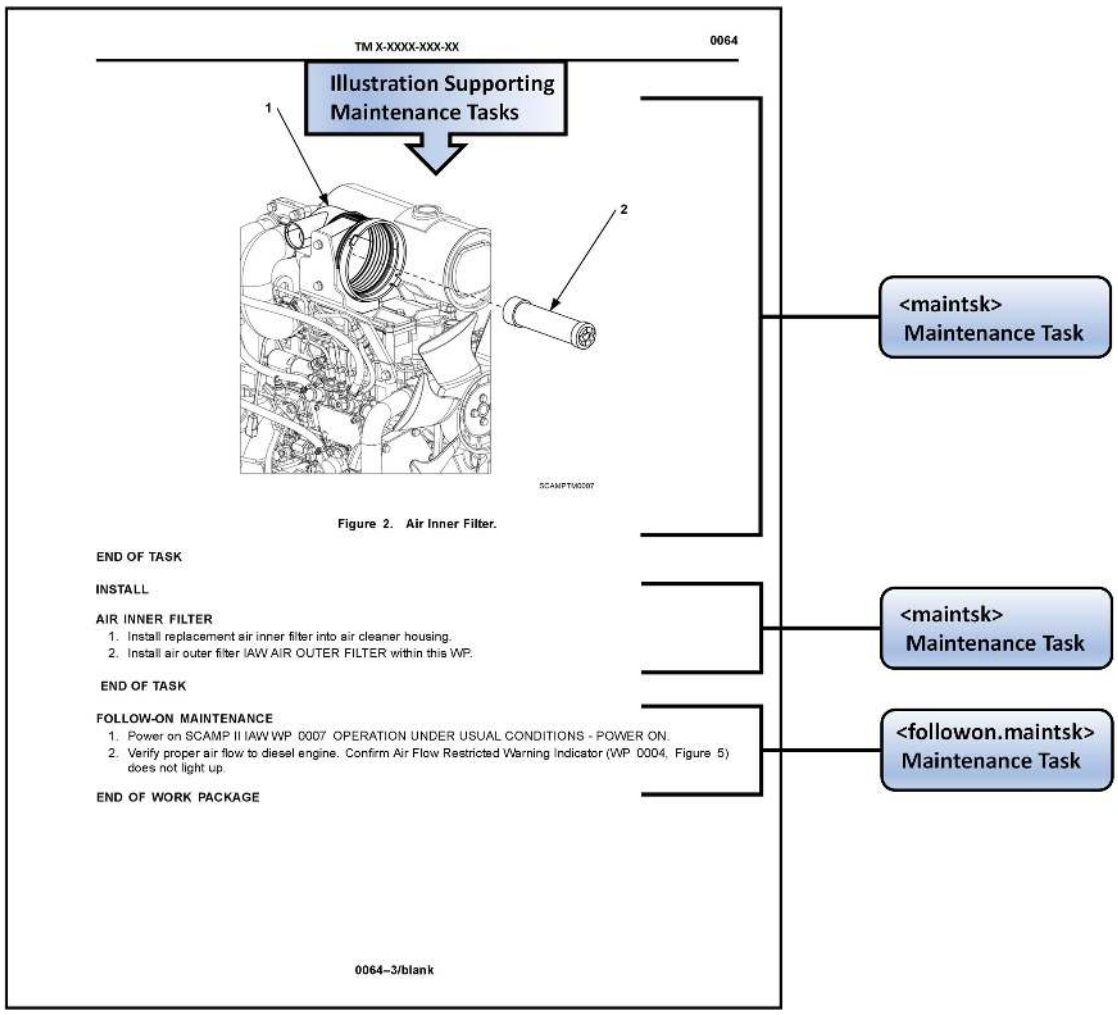


**FIGURE E-24. Maintenance WP example (sheet 3 of 3).**

## E.7.4.1 Maintenance illustrations.

Maintenance illustrations may be used to support or clarity text. These illustrations may be line drawings (shown in FIGURE E-23 and FIGURE E-24), photographs, engineering drawings, diagrams, charts and graphs, and tools and test equipment illustrations. In some cases, wiring diagram and schematics WPs **<wiringwp>** (E.7.10) may be created to outline specific wire routing and details. These WPs shall typically have greater wiring details than provided in maintenance WPs **<maintwp>** and may be referenced by maintenance WPs. Illustrations developed to support operator and maintainer instructions shall conform to the following stipulations outlined in 4.17.

### E.7.4.1.1 Tools and test equipment illustrations.

Only uncommon or unusual uses and connections for test purposes shall be illustrated if essential to avoid misunderstanding. Unusual operations shall also be illustrated. Special tools/kits and test equipment shall be illustrated. Standard tools and test equipment as well as their standard uses shall not be illustrated.



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## E.7.4.2 Maintenance WP titling.

Maintenance task titles shall match MAC component names to ensure traceability between MAC entries and corresponding maintenance (discussed further in F.6.4). Grouping of tasks in a WP shall depend considerations such as relevancy of the tasks to each other (FGCs tasks grouped together); for ETMs, the length of a WP when multiple tasks are included; and initial setup stipulations discussed in E.7. The following guideline shall serve as the suggested method of titling WPs that contain maintenance tasks: a maintenance WP shall simply be titled "OPERATOR MAINTENANCE" or "MAINTAINER MAINTENANCE," depending on level of maintenance, followed by the component name.

**Note**

The titles "OPERATOR MAINTENANCE" AND "MAINTAINER MAINTENANCE" titles are automatically generated by the stylesheet once the level of maintenance in a WP is assigned.

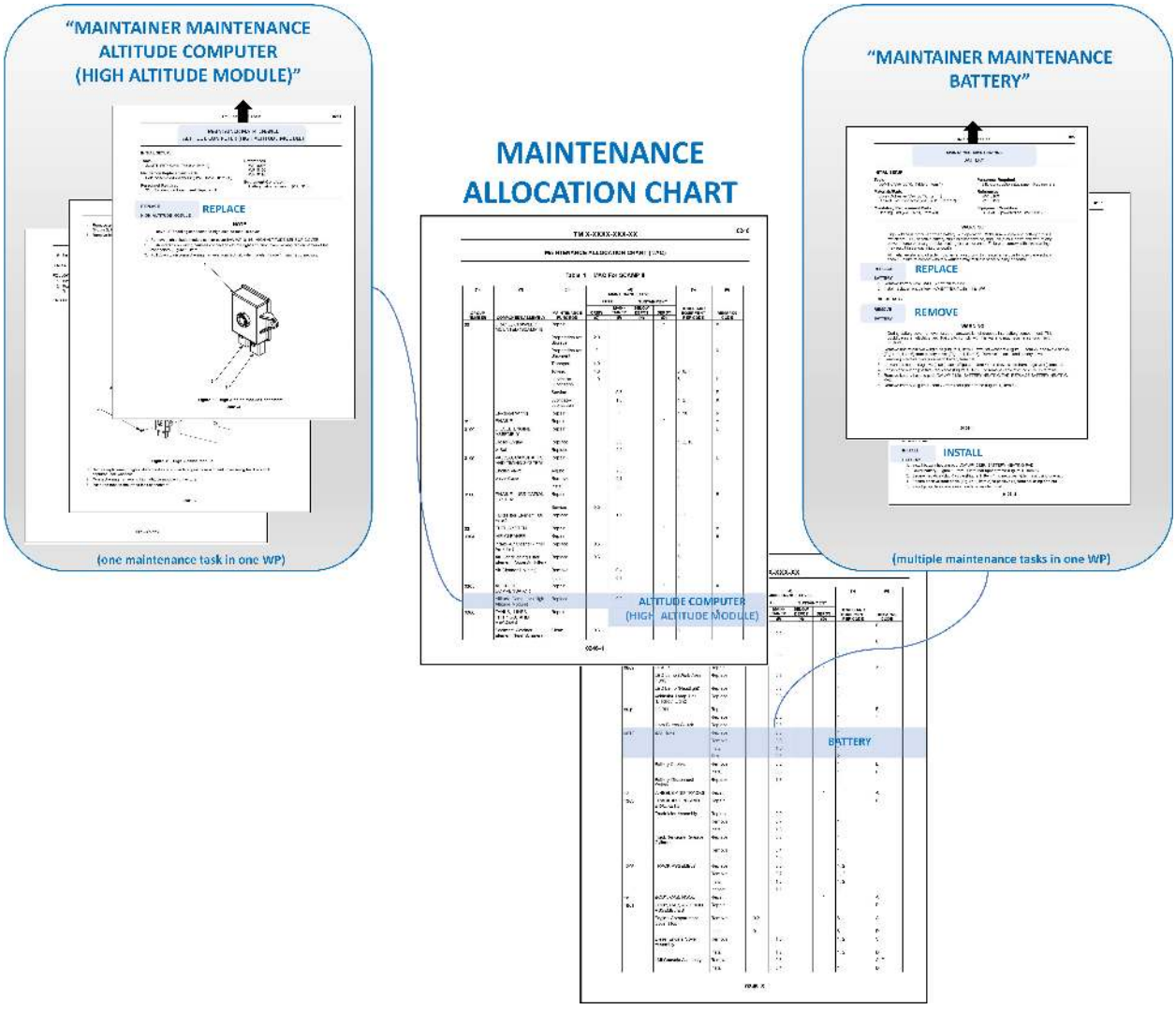


**FIGURE E-25. Maintenance WP titling.**



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### E.7.4.3 Maintenance instruction guidelines.

General maintenance instruction guidelines are listed below. Additional mandatory or unique technical information or additional explanations may be required to be included in the maintenance tasks listed in E.7.4.4.

**a.** Maintenance instructions shall reference all WPs required for any unusual or critical steps such as specifying QA checks (depot and aviation only), care and handling of ESD sensitive items and all hazardous material (for example, ammunition, radioactive components or materials, including prevention of deterioration due to rough handling, exposure to adverse weather conditions, or other hazards). Visual inspection and safety criteria shall be prepared to determine item serviceability. When applicable, instructions shall contain references to the WPs for disposition of defective ammunition.

**b.** WPs shall be prepared for use of cleaning materials and paint authorized for use in the specified maintenance operations. When a tool is unusual or abnormal, it shall be described. Other tools, except for tools in a kit, may be described.

**c.** When specific to the equipment, applicable CPC procedural steps shall be included, or the WP shall reference applicable CPC publications.

**d.** NSNs shall not be used in procedural steps, illustrations, or legends of maintenance WPs.

**e.** P/Ns shall not be used in procedural steps, illustrations, or legends, except when essential for identification.

**f.** The general maintenance WP shall contain common, general, or standard maintenance procedures (for example, painting, cleaning, marking) applicable to other maintenance WPs contained within a technical publication that require this general maintenance procedure to complete the task. Maintenance tasks shall not be included. This WP may be referenced in other maintenance WPs.

**g.** Aviation maintenance TMs shall reference TM 1-1500-204-23, Volumes 1 through 11, as applicable.

**h.** The maintenance instructions shall be prepared to include required environmental control data and information. Instructions shall be prepared for information on any special maintenance required under extreme temperature, altitude, and humidity conditions within the limits established by the design specification for the equipment.

**i.** When maintenance tasks are updated during a change/revision cycle, the MAC shall be updated to reflect any changes made in the maintenance instructions (for example, new tasks, deleted tasks, changes in times, etc.).

**j.** Specific instructions shall be prepared for lockwiring, installing cotter pins, use of sealing compounds, lubricants or CPCs, and similar operations with applicable references to the expendable and durable items list.

**k.** Instructions shall not be prepared for separation of bonded, press-fitted, soldered, welded, or riveted parts, or the removal of electronic circuitry parts, unless such removal is necessary to clean, inspect, or test separately.

**l.** If servicing (for example, pressurizing and charging with gas, lubrication, etc.) is required upon completion of a maintenance task, include this information as part of the task.

**m.** Warnings and cautions shall be included whenever chemicals or cleaning compounds are used or combined that may result in a dangerous or hazardous mixture. Whether the danger is to personnel or equipment, it shall be identified and the effect (for example, gases, fumes, caustic, and fire) shall be stated.

**n.** Any mandatory replacement parts required shall be indicated in the maintenance WP procedural step(s) and reference made in the initial setup to the mandatory replacement parts WP in supporting information for those parts.

**o.** Torque requirements, values, and sequences shall be indicated. Only critical torques **<torque>** shall be indicated in task steps. All noncritical torques will be covered by the Torque Limits WP and a reference to the WP shall be provided. Torque values shall be given for all structural attaching hardware, fluid couplings (fuel, oil, hydraulic, pneumatic, etc.), and connections. Torque values shall include torque correction factors when crowfoot extensions, thread lubricants, and cadmium-plated screws or nuts are used. Torque values identified in the tasks must reflect torque wrenches authorized to personnel targeted to perform tasks. Upon completion of torque action, instructions shall be prepared on use of an orientation mark (striping).

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**p.** Such terms as "reverse the disassembly instructions" or "installation is the reverse of removal" shall not be used in any maintenance task.

**q. (Depot and aviation maintenance only).** Maintenance instructions or steps that have a major QA effect shall be preceded by a statement (such as "QA check") to identify them.

**r. (DMWRs/NMWRs only).** For items that have parts with specific characteristics, wear limits, specified performance requirements, or fatigue characteristics or tolerances, OIPs shall be included in any applicable maintenance task.

### E.7.4.4 Maintenance tasks <maintsk>.

Maintenance tasks include step-by-step instructions for performing maintenance tasks such as replacing a part, repairing an assembly, cleaning equipment, or removing an item of equipment, etc. Procedural maintenance steps shall be presented in logical sequence and reflect the sequence in which the tasks are actually performed. Procedural steps shall be placed immediately after task or instruction title, or, when applicable, after a small paragraph that introduces the procedural steps. Individual maintenance task types and their definitions are discussed in sections E.7.4.4.1 through E.7.4.4.41.

#### E.7.4.4.1 Adjust <adjust>.

Adjustment instructions shall be prepared for the item to maintain or regulate, within prescribed limits, by bringing into proper position, or by setting the operating characteristics to specified parameters before operating the part, system, or end item.

#### E.7.4.4.2 Align <align>.

Detailed alignment instructions shall be prepared to adjust specified variable elements of an item to bring about optimum or desired performance.

#### E.7.4.4.3 Arm <arm>.

Instructions shall be prepared for arming/activation of ammunition (for example, ammunition, mines, etc.) prior to use.

#### E.7.4.4.4 Assemble <assem> and dissassemble <disassem>.

##### E.7.4.4.4.1 Assemble <assem>.

Step-by-step instructions shall be prepared for assembling items disassembled or removed that make up the components, assemblies, or subassemblies.

**a.** Instructions shall be prepared for assembling precision-matched or mated parts marked during disassembly.

**b.** Instructions shall be prepared for checking and recording gear wear patterns, backlash, shimming requirements, and the indexing of parts to ensure proper alignment during assembly. The purpose of shims shall be given (for example, adjust backlash, prevent metallurgical reaction, etc.).

**c.** Torque requirements, values, and sequences shall be indicated. Only critical torques **<torque>** shall be indicated in task steps. All non-critical torques will be covered by the Torque Limits WP (E.7.9) Torque values shall be given for all structural attaching hardware, fluid couplings (fuel, oil, hydraulic, pneumatic, etc.), and connections. Torque values shall include torque correction factors when crowfoot extensions, thread lubricants, and cadmium-plated screws or nuts are used. Torque values identified in the tasks must reflect torque wrenches

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authorized to personnel targeted to perform tasks. Upon completion of torque action, instructions shall be prepared on use of an orientation mark (striping).

**d.** Instructions such as "reverse the disassembly instruction" shall not be used.

**e.** ESD standards, ESD sensitive items along with the protective and control measures to be taken, and CPC instructions shall be identified.

**f.** For ammunition, direction shall be provided if the assembly instruction results in a logistics change (NSN, DoDIC, etc.) to the end item.

### E.7.4.4.4.2 Disassemble <disassem>.

Instructions shall be prepared to take apart components, assemblies, or subassemblies to the extent specified by the MAC and SMR coded items. Instructions shall be prepared for precision-matched or mated components assemblies, subassemblies, or parts (other than common hardware), including ESD sensitive items, to ensure they will be marked, handled, and stored to preclude damage and to ensure assembly and installation in their matched positions. For ammunition, direction shall be provided if the disassembly instruction results in a logistics change (NSN, DoDIC, etc.) to the end item or component(s).

### E.7.4.4.5 Calibrate <calibration>.

Instructions to determine and cause corrections to be made or to be adjusted on instruments of test, measuring, and diagnostic equipment used in precision measurement. It consists of comparisons of two instruments, one of which is a certified standard of known accuracy, to detect and adjust any discrepancy in the accuracy of the instrument being compared. Equipment that requires calibration after assembly or installation shall be indicated. Reference shall be made to the publication containing the applicable calibration instruction.

### E.7.4.4.6 Clean <clean>.

Step-by-step instructions on how to remove dirt, corrosion, or other contaminants from equipment shall be prepared. All cleaning instructions, methods, special equipment, and materials shall be specified. Instructions shall be prepared for corrosion prevention treatment of metal parts after cleaning.

**a.** All materials used in the cleaning and corrosion prevention of equipment, components, or parts shall be referenced and contained in the expendable and durable items list (refer to APPENDIX G).

**b.** Cleaning materials used for the cleaning of systems, subsystems, and components to prepare them for painting, bonding, applying sealants or adhesives, and the removal thereof shall be Hazardous Air Pollutant (HAP) Free. The use of HAP-containing cleaner(s) is considered a serious risk to human health and the environment due to potential impacts on installations that are required to perform the specific cleaning tasks. If a HAP-containing cleaner must be used due to performance/technical requirements, then it shall be formally approved by the risk acceptance authority for serious-level risks, as identified in the System Safety program and MIL-STD-882.

**c.** Instructions shall include cautions to avoid damage of components and to prevent the entrance of water or other solvents into electrical components, ducts, or similar openings.

**d.** Warnings and cautions shall be prepared whenever chemicals or cleaning compounds are used or combined that may result in a dangerous or hazardous mixture. Any danger to personnel or equipment shall be identified and the effect (for example, gases, fumes, caustic, and fire) shall be stated.

**e.** For aircraft, detailed instructions shall be prepared for cleaning and washing the entire aircraft. Instructions shall be prepared for the removal of the battery, the relief tube, and power plant. Removal instructions for armament exhaust deposits or other items or material as necessary shall be provided. Instructions shall also be prepared regarding components that require relubrication after the aircraft has been washed or steam cleaned.

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**f.** Cleaning methods or materials shall not cause corrosion, create conditions that promote corrosion, or remove/negate any in-place corrosion prevention methods. If normal cleaning (for example, pressure washing) has the potential to remove coatings, CPCs or other corrosion prevention materials instructions (or reference to the appropriate methods) shall be provided on how to properly restore the corrosion prevention to the affected system(s).

**g.** The standard cleaning practice shall include instructions for proper cleanout and draining of enclosed areas. This shall include but is not limited to cleaning drain holes; removing drain plugs; and opening covers, hatches, etc. The instruction should clearly state when to open access points; how to clean enclosed areas; how to verify that such spaces are clean and dry; and how to reinstall drain plugs, covers, hatches, etc.

### E.7.4.4.7 Configure <configure>.

Instructions shall be prepared to enable hardware or an end item to perform a different type of function or mission.

### E.7.4.4.8 Covering <cover>.

Instructions shall be prepared to place a protective wrapping over a vehicle to protect it from environmental conditions or to hide (for example, camouflage) it.

### E.7.4.4.9 External power <extpwr>.

Instructions shall be prepared on how to apply electrical power from any authorized power source (for example, external generator or facility power).

### E.7.4.4.10 Hoisting <hoist>.

Instructions shall be prepared to allow a vehicle equipment or ammunition to be raised by cables or ropes through attaching points.

### E.7.4.4.11 Inspect <inspect>.

Instructions detailing all required inspections to determine the serviceability of an item by comparing its physical, mechanical, and/or electrical characteristics with established standards through examination (for example, by sight, sound, or feel) shall be prepared. Special inspection requirements cited below shall be included as necessary.

### E.7.4.4.11.1 Inspect during assembly.

Instructions shall be prepared for testing and inspection during or after assembly to ensure proper assembly of the item. Correct methods of testing, instructions for making tolerance checks, and instructions for inspection of distance measurements (for example, clearance, end play, backlash) shall be prepared. Measurement criteria and tolerances shall reflect the Test Measurement and Diagnostic Equipment (TMDE) available to the user.

### E.7.4.4.11.2 Inspect of ammunition, chemical ammunition, or components including those that contain radioactive materials (Maintainer, below depot sustainment, or ASB only).

The following information shall be prepared for ammunition, chemical ammunition, or components including those that contain radioactive materials:

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**a.** A statement shall be included that inspection criteria are provided to ensure that performed maintenance will restore items to an acceptable level. At a minimum, the types of inspection instructions shall include a pre-maintenance inspection to be conducted during unpacking, in-process inspections, and final acceptance inspection. Regulations and technical publications relating to policy responsibility and instructions applicable to ammunition stockpile reliability, ammunition surveillance, and quality evaluation programs shall be referenced. When approved by the MATDEV, these instructions contained in other publications shall be included in the task.

**b.** Visual inspection criteria shall be prepared for the packing of the items in conformance with the inspection criteria noted in "a" above.

**c.** Detailed instructions and criteria shall be prepared for function testing. When test fixtures must be fabricated, diagrams and instructions for the fabrication shall be prepared. Where ammunition is required for function testing weapons, it shall be identified by Department of Defense Ammunition Code (DoDAC), NSN, and nomenclature. This shall also include dummy rounds.

### E.7.4.4.11.3 Specific instructions for inspection of radioactive ammunition, chemical ammunition, or components.

Regulations and technical publications relating to policy responsibility and instructions applicable to radioactive materials instructions shall be referenced.

**a.** Instructions shall be prepared for inspection methods or techniques used to detect defective components or end items being processed. Classification of Materiel Defects tables **<defect.tab>** shall be prepared for ammunition components and packaging material. A classification of defects (for example, minor, major, or critical) for both functioning and non-functioning categories shall be included. The tabulated data shall include the following entries:

**(1)** A list of categories of defect **<defectype>** critical, major, minor) by the defects attributable to each component **<condition>**.

**(2)** The corrective action to be taken **<actionreq>** or a reference **<xref>/<link>** to the corrective action.

**(3)** The inspection methods **<insp-method>** used to determine if corrective action was accomplished.

**(4)** The acceptable quality level **<acceptqual>** established for each defect.

**b.** Instructions shall be prepared to establish a uniform system of examination for deterioration or damage. Definitions shall be prepared to explain minor, major, and critical defects. When appropriate, lower maintenance levels shall be included.

**c.** Instructions for disposition of lots shall be prepared and shall be as specified by the MATDEV. The following statements shall be included in the technical publication verbatim (italicized text in parentheses shall be replaced with the appropriate information):

**(1)** Each lot of material shall be inspected and screened 100 percent if one critical non-functioning defect is observed. If a critical functioning defect occurs, save the remaining pieces and components; suspend the lot from local issue and use. Submit malfunction reports as prescribed in AR 75-1. Disposition instructions will be furnished by the U.S. Army Materiel Command.

**(2)** A lot of materiel is acceptable for issue if the acceptable criteria as indicated in (*insert applicable table number*) are met.

**(3)** Report all lots of materiel rejected under applicable serviceability table for disposition instructions to: Commander, U.S. Army TACOM Life Cycle Management Command, Chemical/Biological Defense, ATTN: AMTA-LCW-C, 6501 E. 11 Mile Road, Warren, MI 48397-5000. Include a statement describing the capability and workload situation of your organization as to whether you are capable of reworking/ demilitarizing the item."

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### E.7.4.4.11.4 Specific instructions for inspection for non-radioactive ammunition, chemical ammunition, or components.

Criteria shall be prepared for inspection methods used to detect defective components or end items that have corrective action (every corrective action will have a corresponding WP). An Acceptable, Repairable, Irreparable Criteria table shall be prepared for ammunition components and packaging material.

**a.** The default method for inspection is visual. If a method other than visual is required, it will be annotated in the defects column with a superscript. The tabulated data shall include the following entries:

**(1)** A list of components that have a correction action WP associated with it.

**(2)** The condition that is acceptable.

**(3)** The condition that requires repair.

**(4)** Reference **<xref>/<link>** to the corrective action.

**(5)** The condition that is irreparable.

**b.** The following statement shall be included in the technical publication verbatim:

"The Munitions History Program (MHP) or DA Pam 742-1, Ammunition Surveillance Procedures, criteria shall be used with this WP in order to properly inspect all ammunition and components."

**c.** Instructions for disposition of ammunition or components shall be prepared and shall be as specified by the MATDEV. The following statement shall be included in the technical publication verbatim:

"Disposition of each ammunition or component, including reporting, shall be in accordance with DA Pam 742-1."

### E.7.4.4.11.5 Pre-embarkation inspection of materiel in units alerted for overseas movement.

If applicable, pre-embarkation inspection instructions shall be prepared. They shall be as specified by the MATDEV.

### E.7.4.4.11.6 Inspection of installed items.

Instructions shall be prepared for inspection of components, assemblies, or parts installed on the equipment. Instructions shall indicate that inspection will be performed with the item in its normally installed position or condition. The instructionsshall consider accessibility and visibility of the item being inspected. The purpose of the inspection shall be stated (for example, to determine if the item is damaged, deteriorated, or incomplete to the extent that it should be replaced or repaired). Instructions shall be prepared for inspecting solder joints on an electronic item, welds on an armored vehicle, fluid leakage on vehicles, connectors on electronic devices, and other items to identify defects that must be corrected.

### E.7.4.4.11.7 Inspection-acceptance and rejection criteria.

Inspection requirements shall be prepared to include acceptance and rejection information sufficient to determine that new, repaired, and used components, assemblies, and subassemblies conform to wear limits, fits, and tolerances established.

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### E.7.4.4.12 Install peripheral device <installperdev> and uninstall peripheral device <uninstallperdev>.

#### E.7.4.4.12.1 Install peripheral device <installperdev>.

Instructions shall be prepared for installing peripheral devices such as printers, scanner, modems, jamming devices, antennas, etc.

#### E.7.4.4.12.2 Uninstall peripheral device <uninstallperdev>.

Instructions shall be prepared for uninstalling peripheral devices such as printers, scanner, modems, jamming devices, antennas, etc.

### E.7.4.4.13 Jacking <jack>.

Instructions shall be prepared for placement of jack stands or supporting devices and for raising or lifting a vehicle to facilitate maintenance.

### E.7.4.4.14 Load <load> and unload <unload>.

#### E.7.4.4.14.1 Load <load>.

Instructions for placing assets onto a transportation medium (for example, vehicle, aircraft, pallet, container) or ammunition into a weapon/weapon system shall be prepared as required to support the specific equipment.

**a.** For transportation, the act of placing assets onto a transportation medium (for example, vehicle, aircraft, pallet, container).

**b.** For ammunition, the act of placing ammunition into a weapon/weapon system.

#### E.7.4.4.14.2 Unload <unload>.

Instructions for removing assets from a transportation medium (for example, vehicle, aircraft, pallet, container) or ammunition from a weapon/weapon system shall be prepared as required to support the specific equipment.

**a.** For transportation, the act of removing assets from a transportation medium (for example, vehicle, aircraft pallet, container).

**b.** For ammunition, the act of removing ammunition from a weapon/weapon system.

### E.7.4.4.15 Lubricate <lube>.

Pertinent mandatory lubrication instructions, CPC instructions and general lubrication instructions not contained elsewhere shall be prepared and appear in this section (refer to APPENDIX K).

### E.7.4.4.16 Mark <mark>.

For non-ammunition, instructions shall be prepared placing identifying information on the equipment or item. This may be done after repair or when required due to normal wear. For ammunition, the following information shall be prepared as a minimum:

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**a.** Any special sequence of actions necessary to protect the ammunition, packaging, or components being marked.

**b.** Detailed step-by-step instructions shall be provided on the proper method to mark the ammunition and/or packaging.

**c.** The appropriate references, for example, drawings, specifications, etc., shall be provided to ensure correct color, location, and size of markings.

**d.** For DMWRs/NMWRs, a reference to the preservation, packaging, and marking general information WP shall be included. For DMWRs/NMWRs, this WP shall be used if instructions for marking IUID are required.

### E.7.4.4.17 Mooring <moor>.

Instructions shall be prepared to secure a vehicle by chains, ropes, or other means to protect the vehicle from environmental conditions or secure for transportation.

### E.7.4.4.18 Nondestructive Inspection (NDI) <ndi>.

Step-by-step instructions on preparation and accomplishment of inspections or tests that do not destroy or damage the item or equipment.

**a.** The reject criteria shall be specified in all cases. This shall be done by means of a blanket statement, individual criteria for a part, or a combination of both.

**b.** When several NDI methods are permitted, the relative order of preference shall be specified.

**c.** Instructions shall be prepared for removing primer and/or paint for technical publications that require the removal process as part of NDI instructions. If a part requires a special process, this instruction must be contained within the NDI instructions for that part.

**d.** Cleaning requirements before, during, and after NDI shall be specified. If a part has a built-in bearing, then an instruction shall be prepared to ensure protection of the bearing for the NDI instruction.

**e.** The following requirements apply to **aircraft NDI TMs only.**

**(1)** Instructions for use of visible dye penetrants shall not be included as part of NDI instructions unless specified otherwise by the MATDEV activity. When required, refer to TM 1-1500-335-23 for preparation of those instructions.

**(2)** When specified by the MATDEV, TM 1-1500-335-23 shall be the only NDI document referenced in the NDI instructions. The technical provisions of this TM shall be followed. Individual NDI instructions shall be specified for each part requiring NDI. In order to satisfy this requirement, the following shall be prepared:

**(a)** If penetrant is required, the applicable process in TM 1-1500-335-23 shall be identified.

**(b)** If magnetic particle inspection is required, the specific TM 1-1500-335-23 method, the type of magnetization, and amount of current or ampere turns shall be provided.

### E.7.4.4.19 Overhaul <overhaul>.

Instructions shall be prepared to restore an item to a completely serviceable/operational condition with a measurable (expected) life as required by maintenance standards in the appropriate technical publications. This process involves inspection and diagnosis according to the DMWR or a similar technical direction that identifies all components exhibiting wear and directs the replacement or adjustment of those items to applicable equipment specifications. "Overhaul" is normally the highest degree of maintenance performed by the Army. "Overhaul" does not normally return an item to a like new condition.

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### E.7.4.4.20 Pack <pack> and unpack <unpack>.

#### E.7.4.4.20.1 Pack <pack>.

Instructions shall be prepared detailing how to place an item into a container for either storage or shipment after service and other maintenance operations have been completed. For ammunition, the following information shall be prepared as a minimum:

**a.** Any special sequence of action necessary to protect the ammunition.

**b.** If a specially designed reusable container is involved for either the end item or components that are authorized for replacement, instructions shall be prepared to report or reenter the empty container through supply channels.

**c.** When providing packaging instructions, the following information shall be included as necessary: part number/ drawing number, CAGEC, and drawing title.

**d.** For DMWR/NMWRs, a reference to the preservation, packaging, and marking general information WP shall be included.

#### E.7.4.4.20.2 Unpack <unpack>.

Instructions shall be prepared detailing how to remove an item from a storage or shipping container or other shipping device prior to service or other maintenance operations. If the containers are to be used again, kept for future use, turned into supply, or require a special disposition method, the necessary instructions for reassembly of the container shall be prepared. These instructions shall be supported by illustrations. For ammunition, the following information shall be prepared as a minimum:

**a.** Any special sequence of action necessary to protect the ammunition.

**b.** If a specially designed reusable container is involved for either the end item or components that are authorized for replacement, instructions shall be prepared to report or re-enter the empty container through supply channels.

### E.7.4.4.21 Paint <paint>.

Instructions shall be prepared for required painting, refinishing, and marking of assembled components, assemblies, subassemblies, or end item. Reference may be made to TM 1-1500-345-23, TB 43-0118, TM 43-0139, or other documents. Instructions shall also be prepared for any final painting, refinishing, and marking that could not be done during the overhaul instructions.

### E.7.4.4.22 Parking <park>.

Instructions shall be prepared to safely place a vehicle in a lot, ramp area, or other designated location.

### E.7.4.4.23 Place in service <pis>.

Instructions shall be prepared for actions not previously provided in a service upon receipt WP (E.7.3) that may be required for an assembly, component, or end item. Instructions shall be prepared such as removal of an item from storage and preparation for installation on an end item. Final servicing checks, calibration, leak checks, charging, pressurizing, and operational checks shall be prepared.

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### E.7.4.4.24 Preparation for storage <prepstore>.

This task shall be prepared and as applicable, shall include the following for both short-term and long-term storage:

**a.** Instructions for security instructions and special storage requirements for sensitive items (security, terrorism, etc.) related to storing the equipment.

**b.** Instructions for special preservation, packaging, packing, marking, ESD protective and control measures required for storage. These shall include the use of specially designed reusable containers.

**c.** Instructions on special use of corrosion-preventive compounds, moisture barriers, and desiccant materials required for storage.

**d.** Instructions for applying special identifying and cautionary markings to storage containers. These shall include security classification, special temperature requirements, and shelf life.

**e.** Instructions will be provided by the MATDEV activity for placing equipment in and for removing it from administrative storage.

**f.** (Ammunition only) Instructions for basic load storage, quantity-distance class, storage compatibility groupings, storage temperatures, stacking limits, and other pertinent storage requirements per DA Pam 385-64.

**g.** Instructions for aviation ground support equipment requirements to include a reference to TM 1-1500-204-23 for general technical information for preparation for storage.

**h.** For wheeled and tracked vehicles, refer to MIL-STD-3003 for further guidance related to storage.

Reference shall not be made to any Surface Deployment and Distribution Command (formerly Military Transportation Management Command) Transportation Engineering Agency (SDDC/TEA) (formerly MTMC/TEA) publications. If there are no requirements related to the preparation for storage, a statement shall be inserted in the general information WP (refer to B.7.1).

### E.7.4.4.25 Preparation for shipment <prepship>.

This task shall be prepared and as applicable, shall include the following:

**a.** Instructions for security instructions and special transportation requirements for sensitive items (security, terrorism, etc.) related to shipping the equipment.

**b.** Instructions for special preservation, packaging, packing, marking, ESD protective and control measures required for shipping. These shall include the use of specially designed reusable containers.

**c.** Instructions on special use of corrosion-preventive compounds, moisture barriers, and desiccant materials required for shipping.

**d.** Instructions for applying special identifying, shipping, and cautionary markings to shipping containers. These shall include security classification, special temperature requirements, and shelf life.

**e.** Instructions should cover any component removal, fluid removal, etc. required to ship/transport the item.

**f.** (Ammunition only) Instructions for basic load shipping, quantity-distance class, shipping compatibility groupings, storage temperatures, stacking limits, and other pertinent shipping requirements per DA Pam 385-64.

**g.** Instructions for aviation ground support equipment requirements to include a reference to TM 1-1500-204-23 for general technical information for preparation for shipment.

**h.** For wheeled and tracked vehicles, refer to MIL-STD-3003 for further guidance related to shipments (refer to B.7.1.8).

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If there are no requirements related to the preparation for shipment, a statement shall be inserted in the general information WP (refer to B.7.1).

### E.7.4.4.26 Prepare for use <prepforuse>.

As applicable, instructions shall be prepared for assembly or other tasks required to prepare the equipment for use after it has been unpacked such as power requirements, connections, and initial control settings needed for installation purposes.

For security measures for electronic data, instructions shall be prepared for handling, loading, purging, overwriting, or unloading classified electronic data under usual conditions. Instructions shall meet current security regulations as they pertain to automation security.

### E.7.4.4.27 Preserve <preservation>.

Instructions shall be prepared for all authorized methods to treat systems and equipment, whether installed or stored, to keep them in a satisfactory condition. For DMWR/NMWRs, a reference to the preservation, packaging, and marking general information WP shall be included.

### E.7.4.4.28 Radio interference suppression <ris>.

Instructions shall be prepared for primary components in the suppression system. The instructions shall also include the replacement of these primary components. Secondary components shall be referenced to pertinent maintenance instructions that contain the removal and installation instructions. Instructions shall be prepared for testing radio interference suppression components.

### E.7.4.4.29 Rebuild <rebuild>.

Instructions shall be prepared for the restoration of unserviceable equipment to a like new condition in accordance with original manufacturing standards. "Rebuild" is the highest degree of materiel maintenance applied to Army equipment. The "rebuild" operation includes the act of returning to zero those age measurements (for example, hours/miles) considered in classifying Army equipment/components.

### E.7.4.4.30 Replace task vs. remove task vs. install task.

Use of replace, remove, and install tasks are easily misunderstood. FIGURE E-26 shows a cup, a plate, and a table to illustrate a simple way to think about these three tasks to ensure they are used correctly. The original configuration of components when all parts are serviceable is represented by FIGURE E-26, Item 1. Refer to FIGURE E-28 for a simple MAC example with accompanying WPs that illustrate concepts discussed below.

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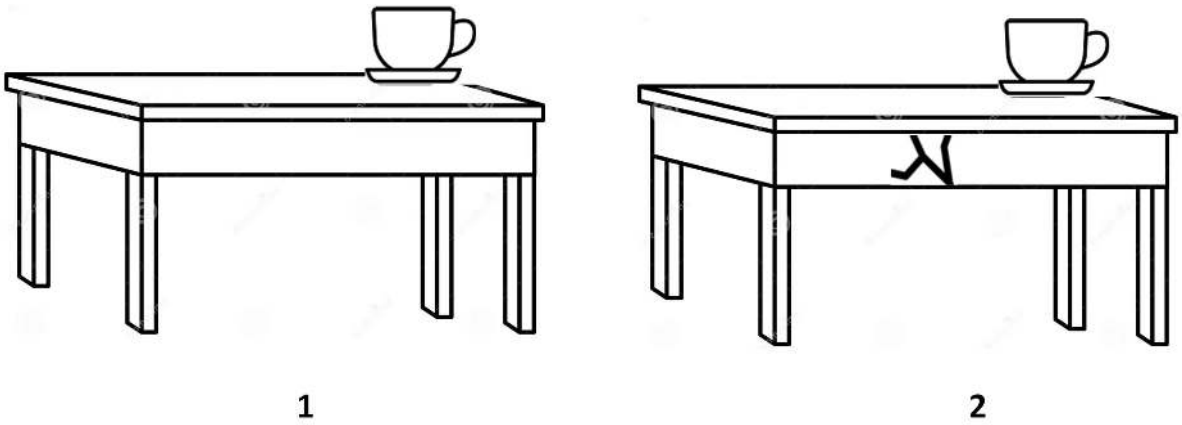


**FIGURE E-26. Illustration of remove, install and replace tasks (sheet 1 of 2).**

**a.** Replace **<replace>** : Instructions to replace an unserviceable (or broken component). If the cup breaks (FIGURE E-27, Item 3), then a simple replacement task for the cup is all that is needed because no component prevents or is "in the way of" performing this task. If the plate breaks (FIGURE E-27, Item 4), the cup sitting on the plate is "in the way of" performing the plate replacement task. If the table breaks (FIGURE E-26, Item 2), the plate and the cup sitting on the table are "in the way" of performing the table replacement task.

**b.** Remove **<remove>** : Instructions to remove a component preventing other maintenance. The item being removed is still serviceable. In the case of the broken plate, a cup removal task must be prepared to access the broken plate. Then a replacement task for the plate itself must be prepared.

**Note**

If the cup is only prohibiting maintenance on one component in a technical publication (the plate in FIGURE E-27, Item 4), then the cup removal task shall be incorporated into plate replacement task steps as part of the plate replacement instruction. However, if the cup is prohibiting maintenance on more than one component in a publication, for example, the broken table (FIGURE E-26, Item 2), then the cup removal instruction shall stand alone as its own task and replacement tasks for both the plate and the table shall reference the cup removal task as necessary to prevent duplication of steps in a publication.

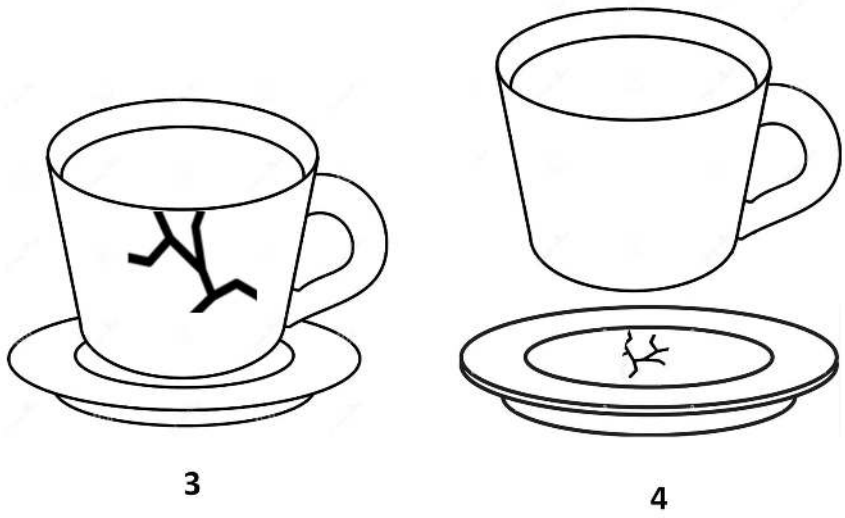


**FIGURE E-27. Illustration of remove, install and replace tasks (sheet 2 of 2).**

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**c.** Install **<install>**: Instructions to install a component that prevented other maintenance. The item being installed retained serviceability. In the case of the broken plate, a cup installation task must be prepared to return equipment to its original configuration after the plate replacement task. In the case of the broken table, a plate installation task and a cup installation task must be prepared to return equipment to its original configuration after the table replacement task.

**Note**

Just like is stated with the remove task above, if the cup only prohibited maintenance on one component in a technical publication (the plate in FIGURE E-22, Item 12), then the cup install task shall be incorporated into plate replacement task steps as part of the plate replacement instruction. However, if the cup is prohibiting maintenance on more than one component in a publication, for example, the broken table (FIGURE E-26, Item 2), then the cup install instruction shall stand alone as its own task and replacement tasks for both the plate and the table shall reference as the cup installation task as necessary to prevent duplication of steps in the publication.

When appropriate, equipment condition (in WP initial setup) and follow-on maintenance may also direct the completion of initial and/or final maintenance tasks instead of the WP organization shown in the figure below.

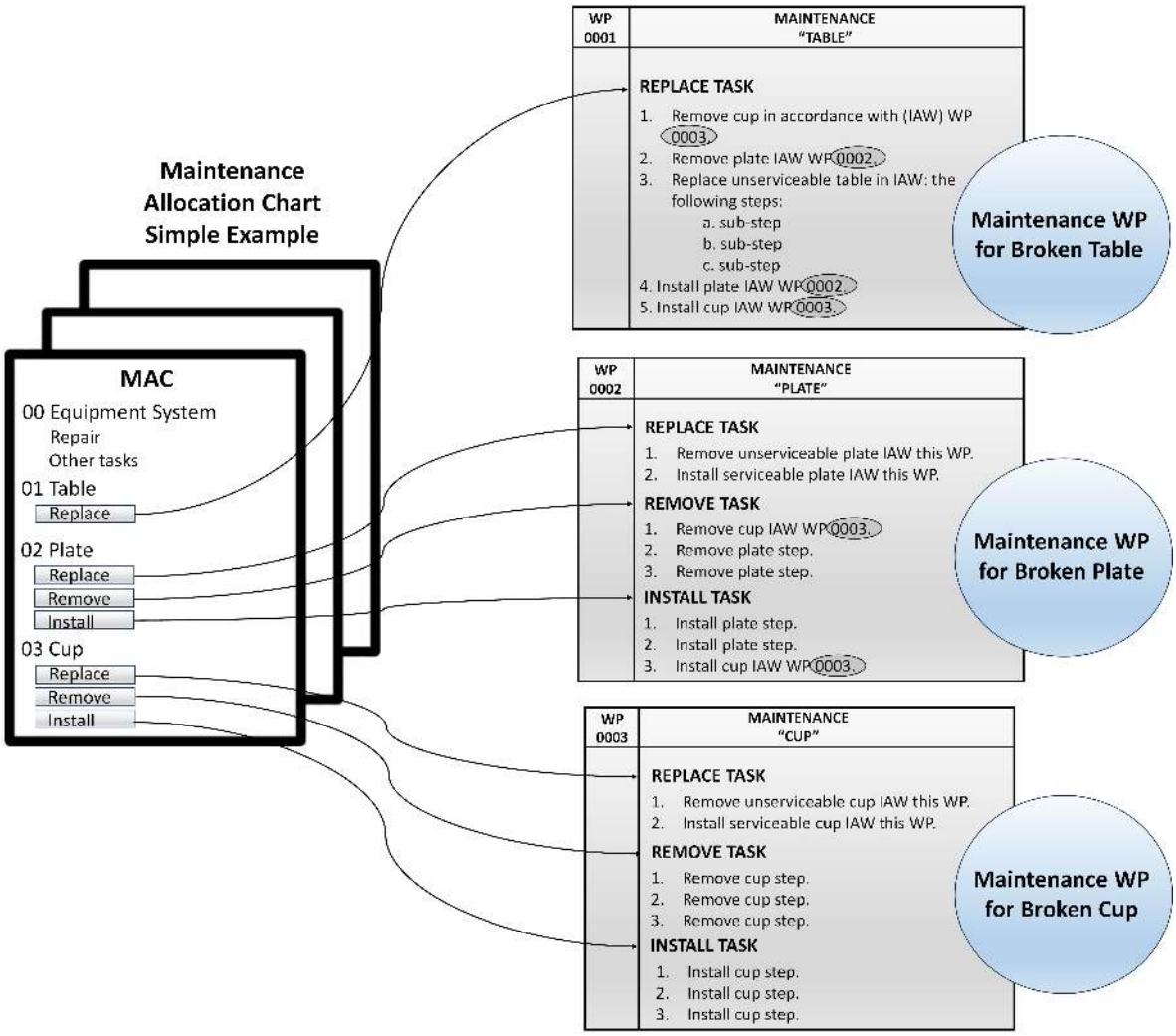


**FIGURE E-28. Simple example MAC and maintenance WPs.**

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### E.7.4.4.31 Remove <remove> and install <install>.

#### E.7.4.4.31.1 Remove <remove>

Instructions shall be prepared to take a component off an asset to facilitate other maintenance on a different component or on the same component (except for replace and repair.) If a component is removed only to repair or replace it, the removal instruction shall be incorporated into the "repair" or "replace" task, rather than using a separate "remove" task. A "remove" task typically requires an "install" task. The "remove" task is authorized by the LPD/ MAC and the assigned maintenance level is shown as the third position code of the SMR code. See E.7.4.4.30 for a simplified explanation of this task.

**a.** Instructions shall be prepared in the logical removal sequence. Instructions shall be prepared for checking and recording gear wear patterns, backlash, ESD protective control measures, measurements and tolerances for determining thickness of shims and purpose for shims, and separating and indexing parts for the assembly. Instructions shall identify items that must be matched or precision-mated when installed at a later time.

**b. (DMWR/NMWR only).** Instructions shall be prepared for recording the condition of the item/assembly, marking, handling, and storing the item.

**c. (Software only).** The "remove" task shall be used for removing software from workstation/viewing hardware.

#### E.7.4.4.31.2 Install <install>.

Instructions shall be prepared for the placing, positioning, or otherwise locating a component to make it part of a higher-level end item. If a component is only installed after repair or to replace it, a separate "install" task shall not be prepared. The install instruction shall be incorporated into the "repair" or "replace" task. The "install" task is authorized by the LPD/MAC and the assigned maintenance level is shown as the third position code of the SMR code. Illustrations shall be used to support and clarify the text. See E.7.4.4.30 for a simplified explanation of this task.

**a.** Instructions shall be referenced for painting, refinishing, and marking the item before its installation in the next higher assembly of the equipment.

**b.** Inspection instructions shall be prepared for checking alignment and adjustment of the item during the installation sequence. These instructions shall include a statement that adjustment, servicing, testing, and/or an operational check is required.

**c.** Instructions such as "reverse the removal instruction" shall not be used.

**d.** Specific instructions shall be prepared for lockwiring, installing cotter pins, use of sealing compounds, lubricants, or CPCs and similar operations with applicable references to the expendable and durable items list.

**e.** Instructions shall identify any mandatory replacement parts or items required during installation. Reference shall be made in the maintenance WP initial setup to the Mandatory Replacement Parts List (refer to G.7.9.1).

**f. (Software only).** The "install" task shall be used for installing software to workstation/viewing hardware.

### E.7.4.4.32 Repair <repair>.

Instructions for repair actions required to restore an item to a completely serviceable or fully mission capable status. "Repair" may contain references to separate "remove" and "install" tasks if the component is removed/installed for purposes other than to repair it. "Repair" shall not be used for replacement action. The "repair" task is authorized by the LPD/MAC and the assigned maintenance level is shown as the fourth position code of the SMR code. See APPENDIX G for further discussion about how repair and other tasks shall be listed in the MAC.

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### E.7.4.4.33 Replace <replace>.

Instructions to take off an unserviceable component and put a serviceable component in its place. "Replace" may contain references to separate "remove" and "install" tasks if the component is removed/installed for purposes other than to replace it with a new component. The "replace" task is authorized by the LPD/MAC and the assigned maintenance level is shown as the third position code of the SMR code. "Replace" shall not be used for software to remove an old version and replace with new version. "Remove"/ "install" and/or "upgrade" shall be used for this purpose. See E.7.4.4.30 for a simplified explanation of this task.

### E.7.4.4.34 Service <service>.

Instructions shall be prepared for replenishment of fuel; oil; hydraulic or other fluids; oxygen, nitrogen, or other gases; and tire pressure. They shall also include any other such items and materials (except for lubricants) required for complete servicing of the equipment.

Servicing instructions shall be supplemented with a diagram showing locations of regular and emergency servicing points. Items located on each side of the equipment that require servicing shall be illustrated and identified as right and left side. NO STEP areas on walkways leading to any tank (in an aircraft) shall be indicated and necessary cautions shall be included.

All expendable and durable items used in the servicing instructions shall be referenced and contained in the expendable and durable items list (refer to APPENDIX G) by standard nomenclature, P/N, and CAGEC. A servicing diagram shall be referenced or included to support the instructions when required.

The warnings and cautions to observe in servicing a particular system tank or reservoir (for example, grounding and prevention of fire hazards) shall be stated clearly. Instructions shall be prepared regarding access to any out-of-the-way or unusual places requiring service.

### E.7.4.4.35 Sling loading <sling>.

Instructions shall be prepared to place a sling around a vehicle, equipment, or ammunition to allow it to be raised.

### E.7.4.4.36 Test <test>.

Instructions shall be prepared, as applicable, to verify serviceability by measuring the mechanical, pneumatic, hydraulic, chemical, electrical, or electronic characteristics of components, assemblies, and subassemblies and comparing those characteristics with prescribed standards. For software, instructions shall be prepared as applicable to verify usability/operability/functionality of the software. (DMWR/NMWR only) Information shall be prepared for final testing of the highest assembly or equipment/ end item involved to ensure the parameters of RAM and durability are met. The following instructions shall be prepared:

a. **Inspection.** Inspection instructions (refer to E.7.4.4.11) shall be prepared that are required before final testing to ensure the item is complete and ready for final testing. Instructions shall be prepared for any minor preparation tasks needed before final testing.

b. **Lubrication.** Any final lubrication instructions (refer to E.7.4.4.15) that need to be done before final testing shall be prepared.

c. **Final test instructions.** Test instructions, performance standards, and tolerances shall be prepared to establish that the equipment is adequately overhauled and ready for issue without qualifications. The instructions shall list all tools, TMDE, jigs, fixtures, and other support items required for the test in the initial setup information. Operating instructions shall be prepared for special test equipment where necessary. Instructions shall be prepared for minor adjustments that can be done without disassembling equipment. Complete instructions shall be prepared for burn-in or run-in tests.

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**d. Final painting, refinishing, and marking.** Instructions shall be prepared for any final painting (refer to E.7.4.4.21), refinishing, and marking (refer to E.7.4.4.16) that could not be done during the overhaul instructions. The materials and tools required to do the job shall be identified. Depot-level maintenance shall include data plate replacement data. For data plates that require replacement, the type of material shall be indicated. Detailed preparation and attachment instructions shall be prepared. The instructions for stamping data plates shall include the initials of the facility performing the overhaul or modification, the contact number (if applicable), the date of overhaul or modification, the part number, and the total operating time since new (if applicable). The instructions shall specify the letter and figure sizes and indicate their placement (adjustment to manufacturer's data). The following statement shall be inserted:

"When sufficient space is not available on the existing data plate to add information, the plate shall be replaced, and all pertinent data transferred to the new plate. Data shall not be stamped directly on any part, assembly, or item of equipment except when approved by the Government."

## E.7.4.4.37 Towing <tow>.

Instructions shall be prepared to connect one vehicle to another for the purpose of having one vehicle moved through the motive power of the other vehicle.

## E.7.4.4.38 Transport <transport>.

This task shall be prepared and as applicable, shall include the following:

**a.** Requirements for dimensions, weights, and types of transport that can/can't be used.

**b.** Instructions for transporting the equipment via air, sea, land, and rail. For vehicles, this includes instructions for self-transport (i.e., it can be driven, flown, or sailed to its destination). Instructions should cover any chocking, bracing, tiedown, etc. required to ship/transport the item.

**c.** Instructions for loading and unloading the equipment.

**d.** Instructions for instructions on the proper handling, blocking, and bracing of basic load ammunition when being transported in trucks and other tactical vehicles.

**e. (Ammunition only).** Instructions for basic load shipping, quantity-distance class, shipping compatibility groupings, storage temperatures, stacking limits, and other pertinent shipping requirements per DA Pam 385-64.

**f.** Reference shall not be made to any Surface Deployment and Distribution Command (formerly Military Transportation Management Command) Transportation Engineering Agency (SDDC/TEA) (formerly MTMC/TEA) publications. If there are no requirements related to transport of the equipment, a statement shall be inserted in the general information WP (refer to B.7.1.1).

## E.7.4.4.39 Software tasks <softmaintsk>.

Software tasks include step-by-step instructions for performing maintenance tasks related to software. Individual maintenance task types and their definitions are discussed in sections E.7.4.4.39.1 through E.7.4.4.41.

## E.7.4.4.39.1 Configure software <softconfig>.

Instructions for configuring the software for different uses/purposes and/or different users shall be prepared. This includes creating accounts in the software.

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### E.7.4.4.39.2 Debug <debug>.

Instructions for locating software bugs and removing those bugs/correcting errors shall be prepared.

### E.7.4.4.39.3 Upgrade/patch <upgrade>.

Instructions for performing software upgrades and/or installing software patches shall be prepared.

### E.7.4.4.40 Additional maintenance tasks <other.maintsk>.

Additional maintenance tasks may be developed when desired maintenance tasks are not covered in E.7.4.4.1 through E.7.4.4.39.3. If additional maintenance tasks are used, the MATDEV shall submit to LDAC the requirements for this maintenance task type for possible incorporation within future revisions to this standard.

### E.7.4.4.41 Follow-on maintenance <followon.maintsk>.

As applicable, instructions shall be prepared or references to the applicable WP (s) for any follow-on maintenance required and shall be the last information in the WP. Follow-on is a maintenance condition that must be accomplished following the completion of a task to clean up or undo actions performed during the task.

For example, to fix a component a task might require that an access panel be removed. The panel would then need to be installed as a follow-on action. This task might be performed sometime after the repair task is completed but not immediately after the repair task. Other maintenance tasks might be performed in the same area before the follow-on task is accomplished.

## E.7.5 General maintenance work package <gen.maintwp>.

This WP shall be prepared as directed by the MATDEV. It shall contain common, general, or standard maintenance instructions (procedures) **<proc>** (for example, painting, cleaning, marking) applicable to other maintenance WPs contained within the technical publication that require this general maintenance instruction to complete the task. Maintenance tasks listed in E.7.4 shall not be included. This WP may be referenced in other maintenance WPs. The following components shall be included in general maintenance WPs.

**a.** WP identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** WP initial setup **<initial_setup>** is required for this WP (5.4.3).

**c.** Instructions to perform a specific common, general, or standard maintenance procedure **<proc>** shall be prepared or referenced.

## E.7.6 Lubrication instructions work package <lubewp>.

This WP shall be prepared as directed by the MATDEV. The following components shall be included in lubrication instructions WPs.

**a.** WP identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** WP initial setup **<initial_setup>** is required for this WP (5.4.3).

**c.** Lubrication schedules (charts) (E.7.6.1) shall be prepared to present all applications, instructions, lubricants, and lubrication points to completely lubricate equipment.

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### E.7.6.1 Lubrication charts.

**a.** Lubrication charts shall consist of a main drawing prepared as a three-dimensional (3D) diagram. They shall consist of enlarged or detailed views as are considered necessary to identify items that otherwise would be obscured. They shall show all lubrication requirements for all parts of the equipment requiring periodic lubrication other than those lubricated by the main engine oil system. The charts shall also indicate type of lubricant, method of application, and frequency.

**b.** Use of black silhouette figures representing a likeness of the tool used in the application (oil can, grease gun, brush, or hand) shall be the accepted means of presenting application methods on the lubrication chart.

**c.** Abbreviations, as specified in MIL-HDBK-275 (aviation) or MIL-HDBK-113 (non-aviation), shall be used to present lubricant types. In the event a lubricant does not have an abbreviation listed in MIL-HDBK-275 or MIL-HDBK-113, the abbreviation shall be provided by the MATDEV. Assigned application symbols, type abbreviations, and frequency shall be placed within the standard lubrication symbols.

**d.** Each application symbol and lubricant abbreviation used shall be defined. Notes may be used to specify any other than normal requirements.

## E.7.7 Illustrated list of manufactured items introduction work package <manu_items_introwp> (Maintainer/AMC or above).

The illustrated list of manufactured items information shall be prepared when there are any items required to support maintenance or operation coded with an "M" in the source code of the SMR contained in the RPSTL. The following components shall be included in these WPs.

**a.** WP identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** WP initial setup **<initial_setup>** is not required for this WP.

**c.** Information described in E.7.7.1 and E.7.7.2.

### E.7.7.1 Introduction <intro>.

The introduction shall contain the following information verbatim (italicized text within parentheses shall be replaced with the appropriate information):

**"ILLUSTRATED LIST OF MANUFACTURED ITEMS INTRODUCTION**

**Scope**

This WP includes complete instructions for making items authorized to be manufactured or fabricated at the *enter applicable maintenance level* .

**How to Use the Index of Manufactured Items**

A part number index in alphanumeric order is provided for cross-referencing the part number of the item to be manufactured to the information that covers fabrication criteria.

**Explanation of the Illustrations of Manufactured Items**

All instructions needed by maintenance personnel to manufacture the item shall be provided and shall include illustrations as required. (*When applicable, a reference to the associated parts information TM or parts information WP shall be entered here*.) All bulk materials needed for manufacture of an item are listed by part number or specification number in a tabular list on the illustration."

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### E.7.7.2 Index of manufactured items <manuindx>.

An index of P/Ns or drawing numbers shall be prepared. This index shall list P/Ns and/or drawing numbers, in alphanumeric order, along with the name of the part for all items illustrated in this WP. The WP number to the manufactured items WP containing the manufacturing instructions shall be included.

## E.7.8 Manufacturing instruction work package <manuwp>.

A WP shall be prepared for each manufactured item. The following components shall be included in these WP.

**a.** WP identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** WP initial setup **<initial_setup>** is required for this WP (5.4.3).

**c.** Information described in E.7.8.1.

### E.7.8.1 Instructions for manufactured items <manuitem>.

The following instructions for manufactured items shall be shall be prepared.

**a.** Illustrations that contain sufficient views to portray all features of the item (as required). Refer to FIGURE E-29 for detailed illustration guidelines.

**b.** All instructions (explanatory text and list of bulk materials) needed by maintenance personnel to manufacture the item shall supplement the illustrations and shall contain the following data:

**(1)** All dimensional, location, and processing instructions needed to manufacture the item shall be included (e. g., 30 in. long, top surface, primer coating).

**(2)** A description of the item to be manufactured, including the P/N and name.

**(3)** A list of bulk materials needed to manufacture the item shall be prepared. The list of bulk materials shall consist of the P/N, CAGEC and NSN, or specification number of the raw bulk material to be used in manufacture of the item. The list shall cite the technical characteristics (for example, standards, specifications, conditions, dimensions, and any other pertinent data).

**(4)** When applicable, a reference shall be made to the associated RPSTL, RPSTL TM, or Repair Parts List WP (for combined TMs).

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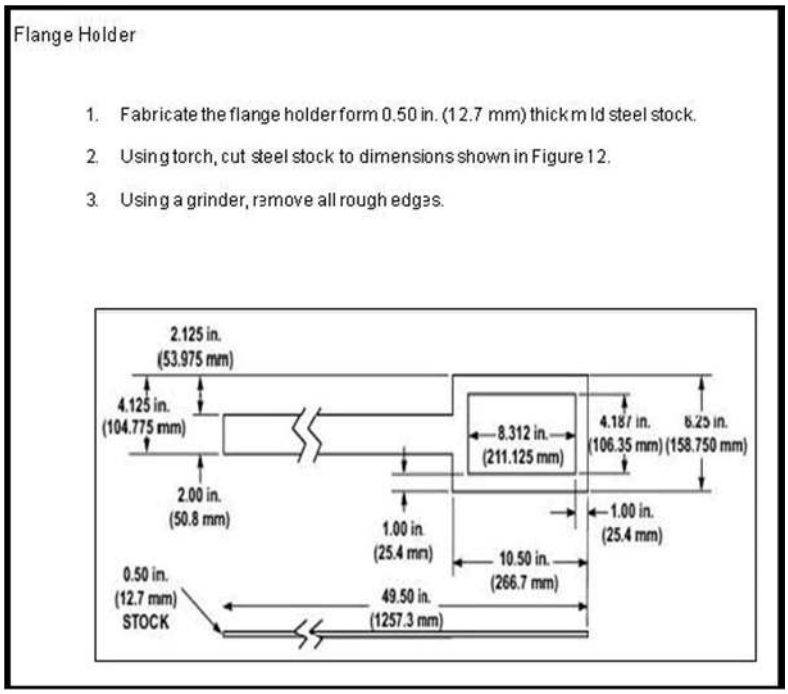


**FIGURE E-29. Example of an illustrated list of manufactured items (dimensions for flange holder).**

## E.7.9 Torque limits work package <torquewp> (Maintainer/ AMC and above).

This WP shall be prepared as directed by the MATDEV. Information shall be prepared to provide applicable torque values **<torque>** (expressed in foot or inch pound terms and/or metric terms), data as to bolt grade markings such as SAE markings and their proper identification, and specific torque sequencing requirements. The following components shall be included in these WPs.

**a.** WP identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** WP initial setup **<initial_setup>** is required for this WP (5.4.3).

**c.** Information described in E.7.9.1 through E.7.9.2.

### E.7.9.1 Torque limits introduction <intro>.

Information shall be prepared to include the scope or how to use the WP **<intro>**.

### E.7.9.2 Torque instructions <torqueval>.

Specific instructions such as torque limits **<torqueval>** for dry and wet fasteners, fastener sizes and thread patterns, etc., shall be prepared.

## E.7.10 Wiring diagrams and schematics work package <wiringwp> (Maintainer/AMC and above).

This WP shall be prepared as directed by the MATDEV. It shall include wiring and cable provisions contained in the equipment/end item, including all systems or equipment that can be installed or removed later (for example, mission-

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related systems/equipment). Applicability of diagrams shall be explained in relation to equipment configuration. The following components shall be included these WPs.

**a.** WP identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** WP initial setup **<initial_setup>** is not required for this WP.

**c.** Information described in E.7.10.1 through E.7.10.9.1.

### E.7.10.1 Wiring introduction <intro>.

Information shall be prepared to include the scope **<intro>** of the WP. A statement shall be included explaining that wiring diagrams and essential wiring information are provided for all electrical and electronic systems and circuits.

### E.7.10.2 Wiring diagrams index.

A table that lists the wiring diagrams by foldout number and title may be included in the introduction. The wiring diagrams index list shall include the sheet numbers as applicable and the page number for the foldout.

### E.7.10.3 Abbreviations <abbrev>.

A statement shall be prepared that abbreviations are in accordance with ASME Y14.38, except when the abbreviation stands for a marking actually found in the equipment. A table listing the abbreviations used on the wiring diagram may be included on this work package. The table shall include the abbreviation and its definition.

### E.7.10.4 Component descriptions with related schematic locations table <component_desc>.

A table may be prepared to assist users in finding components. It shall contain component descriptions in alphabetical order, foldout sheet number, grid number, and the official name that appears on the wiring diagram.

### E.7.10.5 Wire identification <wireid>.

Identification of wires by number shall be explained. A list of circuit designators and a wire identification diagram shall be prepared. Additional wiring tables and diagrams may be included as discussed in E.7.10.4 thru E.7.10.6.

### E.7.10.6 SAE wire designations.

A table (FIGURE E-30) explaining the SAE wire designations used in the wiring diagrams may be included in the wiring diagrams WP.

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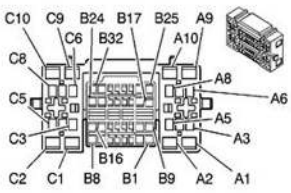
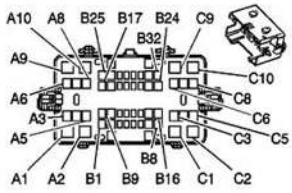
Connector End Views

Circuit Functional Name Table

Wiring Harnes W102 P1 to Wiring Harness W103 P4




| Connector Information | • 52-Way F (BLK) | | | Connector Information | • 52-Way M (BLK) | | |
|---|---|---|---|---|---|---|---|
| Pin | Wire Color | Circuit No. | Function | Pin | Wire Color | Circuit No. | Function |
| A1 | — | — | Not Used | A1 | — | — | Not Used |
| A2 | PNK | 3 | Ignition 1 Voltage | A2 | PNK | 3 | Ignition 1 Voltage |
| A3 | LT GRN | 1763 | Position Signal A | A3 | LT GRN | 1763 | Position Signal A |
| | | | | | LT GRN | 1763 | Position Signal A |
| A4 | PPL | 1358 | Switch Signal | A4 | DK GRN/WHT | 1358 | Switch Signal |
| A5 | BLK | 1851 | Ground | A5 | BLK/WHT | 1851 | Ground |
| A6 | LT BLU | 1764 | Position Signal B | A6 | LT BLU | 1764 | Position Signal B |
| | | | | | LT BLU | 1764 | Position Signal B |
| A7 | WHT | 1765 | Position Marker Pulse Signal | A7 | WHT | 1765 | Position Marker Pulse Signal |
| | | | | | WHT | 1765 | Position Marker Pulse Signal |
| A8 | PNK | 639 | Ignition 1 Voltage | A8 | PNK | 639 | Ignition 1 Voltage |
| A9 | — | — | Not Used | A9 | — | — | Not Used |
| A10 | RED | 142 | Battery Positive Voltage | A10 | RED | 142 | Battery Positive Voltage |
| B1 | GRY/BLK | 87 | Switch Signal (w/K34) | B1 | GRY/BLK | 87 | Switch Signal (w/K34) |
| B2 | YEL | 1327 | Fuel Signal | B2 | YEL | 1327 | Fuel Signal |
| B3 | BLK/YEL | 28 | Horn Relay Control | B3 | BLK | 28 | Horn Relay Control |
| B4 | LT BLU | 1059 | Position Sensor Signal 1 | B4 | LT BLU | 1059 | Position Sensor Signal 1 |
| B5 | LT BLU | 553 | Switch Signal | B5 | LT BLU | 553 | Switch Signal |
| B6 | BRN/WHT | 230 | Instrument Panel Lamps Control | B6 | BRN/WHT | 230 | Instrument Panel Lamps Control |
| B7 | GRY | 478 | Windshield Wiper Switch Supply Voltage | B7 | GRY | 478 | Windshield Wiper Switch Supply Voltage |
| B8 | BLK | 1050 | Ground | B8 | BLK | 1050 | Ground |
| B9 | TAN | 1851 | Ground | B9 | BLK/WHT | 1851 | Ground |
| B10 | DK BLU | 84 | Control Switch Signal (w/K34) | B10 | DK BLU | 84 | Control Switch Signal (w/K34) |
| B11 | — | — | Not Used | B11 | — | — | Not Used |

**FIGURE E-30. Wire designations table.**

## E.7.10.7 Wire color designations <wire_color>.

As applicable, a table may be prepared that explains the color codes used on the wiring diagrams.

## E.7.10.8 Wiring diagrams <wiringdiag>.

As specified by the MATDEV, wiring diagrams shall be prepared for all electrical and electronic systems and circuits. FIGURE E-31 shows an example wiring diagram where wire colors and end-point connections are identified. The MATDEV shall specify wiring diagram details that best aid the user.

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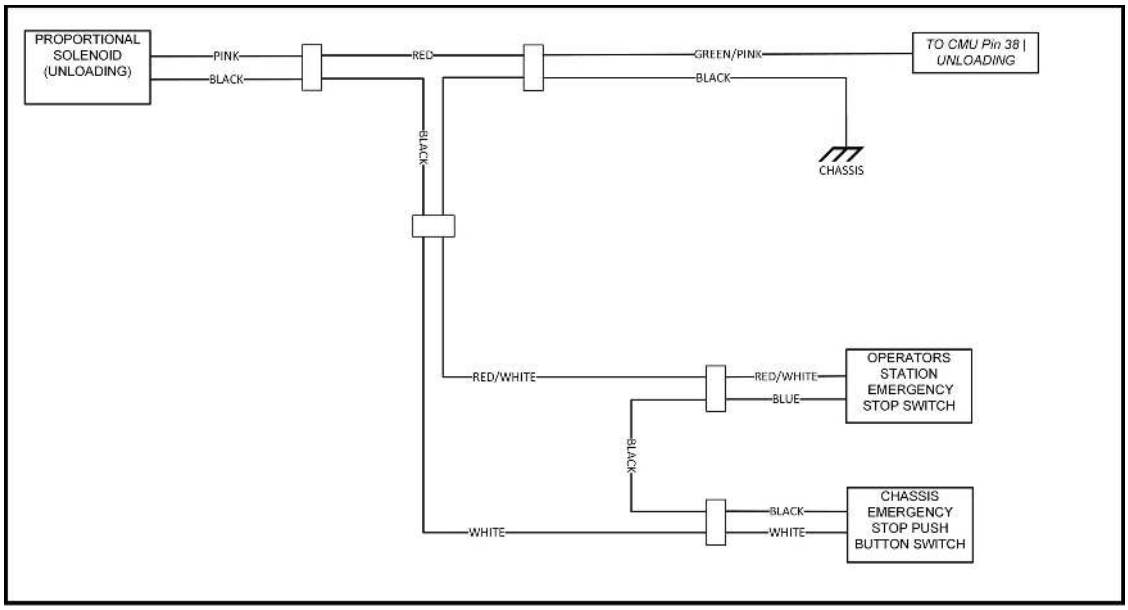


**FIGURE E-31. Wiring WP illustration example.**

## E.7.10.9 System and subsystem schematic development.

Schematics such as wiring schematics and connector libraries shall be developed as needed when a user would benefit from their inclusion for complex systems. Schematic development offering wiring diagrams showing port connections as well as detailed connector pins, etc. shall focus only on what is needed for an Army maintainer. Use of commercial engineering architecture schematics depicting all end item wiring with much greater detail than necessary for Army authorized system maintenance shall be minimized. The goal is maximizing schematic usability and minimizing confusion for the user. Schematics may be included within WPs or as integrated into a technical publication as foldouts depending on graphic size requirements. Some global schematic development suggestions are included in this standard below.

### E.7.10.9.1 System schematic.

Many complex systems that require schematics may benefit from one centralized schematic that orients a user and shows comprehensive wiring relevant to an Army field-level maintainer. These system schematics may be developed to show an entire system or rather a subcomponent of a system depending on troubleshooting requirements. System schematics could then link to other smaller schematics showing the wiring of components such as fuse boxes and power distribution units. FIGURE E-20 shows an example system schematic graphic. Each callout identifies subcomponents of the system that would benefit from detailed schematics linked from this system-level graphic.


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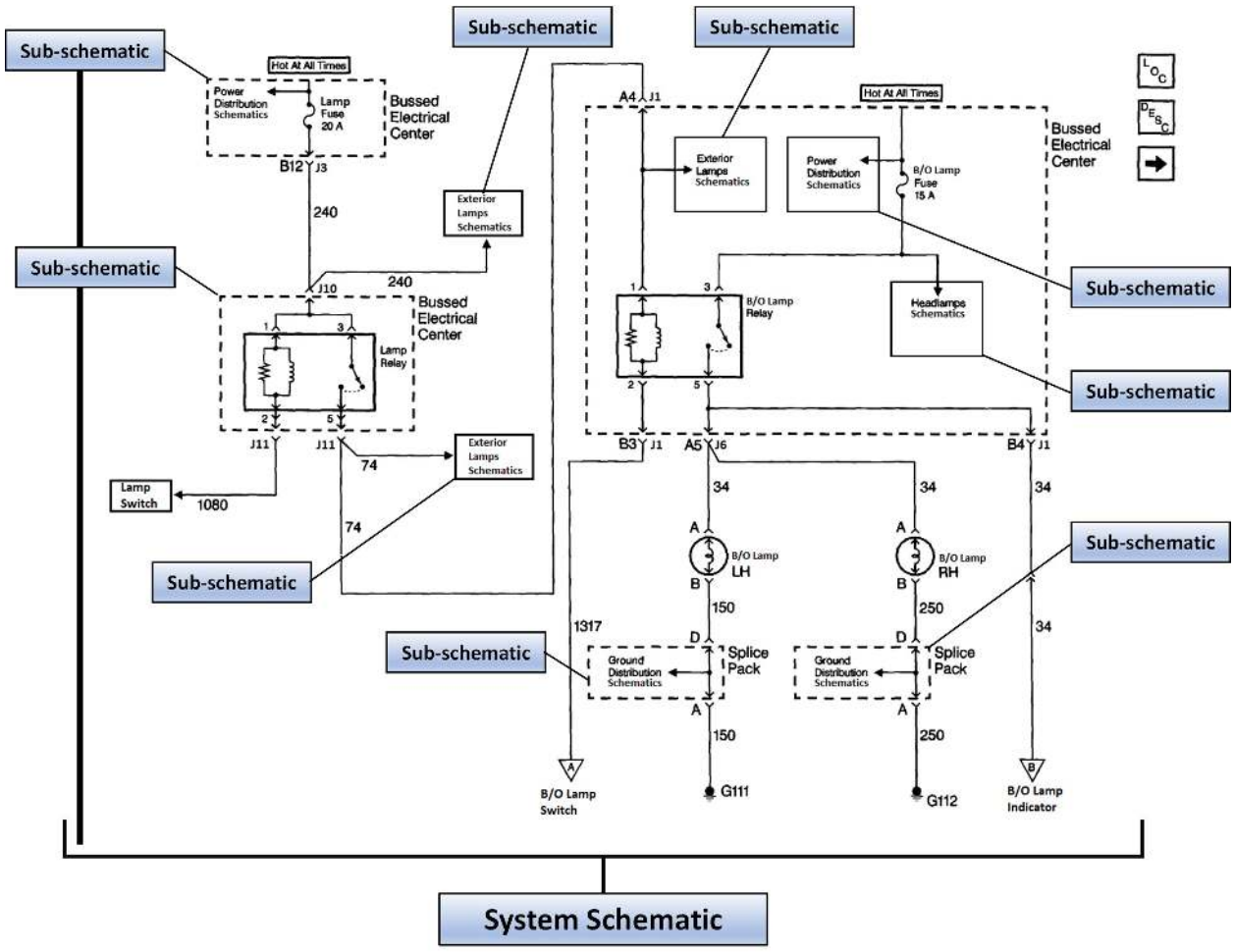


**FIGURE E-32. System schematic example.**

## E.7.11 Equipment/user fitting instructions work package <perseqpwp>.

As applicable, equipment/user fitting instructions for personal use equipment shall be prepared. The following components shall be included these WPs.

**a.** WP identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** WP initial setup **<initial_setup>** is required for this WP (5.4.3).

**c.** Equipment/user fitting instructions for personal use equipment.

## E.7.12 Preventive maintenance inspections work package <pmiwp> (Aircraft Only).

This WP contains complete requirements for special inspections, overhaul and retirement schedule, and standards of serviceability applicable to the aircraft. The following components shall be included these WPs.

**a.** WP identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** WP initial setup **<initial_setup>** is required for this WP (5.4.3).

**c.** Information described in E.7.12.1 through E.7.12.2.

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### E.7.12.1 PMI general information.

The following paragraph shall be inserted (italicized text within parentheses shall be replaced with the appropriate information):

**"GENERAL INFORMATION**

This WP contains complete requirements for special inspections, overhaul and retirement schedule, and standards of serviceability applicable to the aircraft. The inspections prescribed in this WP shall be accomplished at specified periods by aviation maintenance companies, with the assistance of aviation support battalions when required. Complete Daily, Intermediate, Periodic, or Phased inspections are contained in the (*insert applicable aircraft inspection checklist TM*)."

### E.7.12.2 Standards of serviceability.

The following paragraph shall be inserted:

"Standards of serviceability to be used in the day-to-day inspection and maintenance of the aircraft can be found as fits, tolerances, wear limits, and specifications in the aircraft maintenance manuals. Standards of serviceability for transfer to aircraft are contained in TM 1-1500-328-23."

### E.7.12.3 Special instructions.

**a. Definition and general information.** The following paragraph shall be inserted:

"This information supplements scheduled inspections as outlined in the applicable aircraft inspection checklists. Inspection of items that are required to be inspected at intervals not compatible with airframe operating time or airframe inspection intervals is also included. Refer to DA Pam 738-751 (Functional Users Manual for the Army Maintenance

Management System-Aviation (TAMMS-A)) for applicable forms, records, and worksheets required for these inspection intervals. Typical examples of this type of inspection are as follows:

**(1)** Inspections that are solely contingent upon specific conditions or incidents that occur (for example, hard landings, over speed, or sudden stoppage), wherein immediate inspection is required to ensure safe flight.

**(2)** Inspection of components or airframe on a calendar basis, for example, first aid kits, weight and balance check, aircraft inventory."

**b. Requirements.** Components and other items that qualify under the criteria for special inspections, as detailed previously, or over speed shall be included. These inspections shall be grouped under specific aircraft areas. A line drawing of the aircraft or accessory showing sequence for inspection by area shall be included. The area identified shall include all surfaces, materials, components, and equipment pertaining to that specific location. The following inspection data entries shall be included, as applicable. The information entries shall be placed in a table **<pmi.pecul.tab>**.

**(1)** Aircraft serial or tail number **<serialno>**.

**(2)** Date of inspection **<date>**.

**(3)** Area number **<areano>**.

**(4)** Inspection number **<itemno>**.

**(5)** Inspection interval **<interval>**.

**(6)** Name of component being inspected **<compname>**.

**(7)** Inspection instruction **<proc>**.

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## E.7.13 Aircraft inventory master guide work package <inventorywp> (Aircraft Only.

This WP shall be prepared as directed by the MATDEV. Information shall be prepared on standard inventory instructions to allow determination of inventoriable items of installed and loose equipment authorized and required by the specific aircraft in performance of its mission. The following components shall be included these WPs.

**a.** WP identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** WP initial setup **<initial_setup>** is required for this WP (5.4.3).

**c.** Information described in E.7.13.1 and E.7.13.2.

### E.7.13.1 Aircraft inventory master guide

**a.** Introduction **<intro>**:

A short explanation of the scope and purpose of the WP shall be prepared. Information pertaining to the necessary steps to ensure the list is accurate, exact, and complete (for example, research of authorized changes, Modification Work Orders (MWOs), additions/deletions for special mission requirements) shall be included. The introduction shall include a reference to DA Pam 738-751 for applicable forms and records.

**b.** Security **<security>**:

It shall be stated here that aircraft inventory records should be unclassified, but, if necessary, any classification of the contents shall be in accordance with the existing security regulations.

**c.** Inventoriable items **<inventoriable>**:

- Listed items — The selection of inventoriable items to be listed is to be without regard to the agency (governmental or contractual) furnishing the items. Items to be listed are outlined below.

  **(1)** Items essential to the execution of the designated mission of the aircraft, such as electronic, photographic, armament, special mission instruments, and safety and comfort equipment.

  **(2)** Loose equipment delivered with the aircraft and items subject to pilferage or readily converted to personal use.

  **(3)** Modification kits that are reissued or distributed to using organizations for installation and that are not immediately placed in use. These shall be recorded on the affected aircraft's DA Form 2408-17, Aircraft Inventory Record, and identified as loose equipment until modification is completed.

  **(4)** Equipment required for operation in a specific environment.

- Excluded items— Items to be excluded are outlined below.

  **(1)** Non-accountable items coded as expendable in the applicable stock lists.

  **(2)** Personal issue or items furnished on unit allowance or other authority.

  **(3)** Items or components considered as basic or integral parts of the airframe or basic aircraft, such as engines, propellers, wheels, and standard instruments.

  **(4)** Equipment publications, checklists, and aircraft forms.

### E.7.13.2 Periods of inventory <prdinv>.

The following text shall be included verbatim:

**"PERIODS OF INVENTORY**

Inventoriable items shall be checked against the Aircraft Inventory Record, DA Form 2408-17, at the following periods:

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1. Upon receipt.
2. Before transfer of the aircraft to another organization.
3. Upon placing aircraft in storage and upon removal from storage. Aircraft need not be inventoried while in storage.
4. Twelve months after last inventory.

## E.7.14 Aircraft storage work package <storagewp> (Aircraft Only).

The storage of aircraft WP(s) shall be prepared as directed by the MATDEV. The following components shall be included these WPs.

**a.** WP identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** WP initial setup **<initial_setup>** is required for this WP (5.4.3).

**c.** Information described in E.7.14.1 through E.7.14.2.

### E.7.14.1 Storage of aircraft general information.

The following text shall be included verbatim:

"**STORAGE OF AIRCRAFT GENERAL INFORMATION**

**Components Involved in an Accident**

Any component removed for reason of accident shall not be preserved, but shall be shipped in the same condition it was in after the accident.

**Categories of Storage**

1. Flyable storage – no time limit.
2. Short term (administrative storage) – 1 to 45 days.
3. Intermediate storage – 46 to 180 days."

### E.7.14.2 Flyable storage <flyable>, storage term storage <short>, and intermediate storage <intermediate>.

A general discussion shall be prepared for each category of aircraft storage, to include considerations for selection of the appropriate category (for example, ground operation, motoring of engines, and other required maintenance for which personnel and materials are needed) and steps to be taken for care of the aircraft during exceptionally wet weather.

For each category of aircraft storage, all essential information shall be prepared to include all instructions for preparing the complete aircraft for storage and removal from storage. It shall exclude any information on when or why the aircraft is stored. Each category of storage shall reference to inspection documents and inspection instructions to be conducted before, during, and after storage.

## E.7.15 Weighing and loading work package (ASB only) <wtloadwp>.

The weighing and loading WP(s) shall be prepared. It shall provide description, information, and instructions for aircraft weighing, balancing, and loading. The following components shall be included these WPs.

**a.** WP identification information **<wpidinfo>** is required for this WP (5.4.1).

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**b.** WP initial setup **<initial_setup>** is required for this WP (5.4.3).

**c.** Information described in E.7.15.1 through E.7.15.3.

### E.7.15.1 Weighing and loading general information.

The following text shall be included verbatim:

"**Scope**

This WP contains description, information, and instructions for aircraft weighing and loading."

### E.7.15.2 Weighing information <formchart>.

Instructions shall be included for preparing the aircraft, weighing the aircraft in the basic weight condition, performing calculations, and using and recording data on DD Form 365-1 (Basic Weight Checklist) and DD Form 365-2 (Aircraft Weighing Record). Instructions shall include setup requirements, instructions for positioning the aircraft in the weighing area, and assembly of the aircraft weighing equipment. Illustrations shall be prepared to support the text, including a two-view chart diagram. A reference may be made to TM 55-1500-342-23 for additional information governing weight and balance of aircraft, forms, and records.

### E.7.15.3 Loading information <weightinst>.

Descriptions and instructions shall be prepared for aircraft loading and for computing weight and balance information. Sufficient information and data shall be provided so that an aviator, knowing the basic weight and moment of the aircraft, can compute any combination of weight and balance using the prescribed charts and forms. Reference shall be made to AR 95-1, DA Pam 738-751, and TM 55-1500-342-23 for additional information governing weight and balance of aircraft, forms, and records. Data shall include fundamental principles of loading. An illustration of aircraft compartments and stations shall be included. Reference shall be made to DD Form 365-1 for a more complete listing of compartments and equipment that comprise the basic weight of the aircraft. Loading information shall include weight and balance characteristics, center of gravity limits, weight/balance and loading, and weight and moment tables for load items such as crew, fuel, cargo, and armament.

## E.7.16 Overhaul and retirement schedule work package <orschwp> (Aircraft Only).

A WP identifying the criteria to overhaul or retire an aircraft or aircraft components shall be prepared. The following components shall be included these WPs.

**a.** WP identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** WP initial setup **<initial_setup>** is not required for this WP.

**c.** Information described in E.7.16.1.

### E.7.16.1 Overhaul and retirement <orsch>.

The overhaul and retirements schedule shall include the following statement and the associated table.

**"OVERHAUL AND RETIREMENT SCHEDULE**

Units of operating equipment that are to be overhauled or retired at the period specified are listed here. Unless otherwise specified in TM 1-1500-328-23, Aeronautical Equipment Maintenance Management Policies and Procedures, removal of equipment for overhaul may be accomplished at the inspection nearest the time when overhaul is due."

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The overhaul and retirement schedule shall be prepared as a table and shall consist of the following entries:

**a. Part name.** The name of the part shall be listed. An asterisk (*) shall precede the part name if the part is an indentured subassembly.

**b. Part number.** The official P/N of the part listed.

**c. Overhaul interval hours.** The maximum operating time allowed on the part before it is to be overhauled.

**d. Overhaul interval notes.** Any additional information required on the part's overhaul interval.

**e. Retirement interval hours.** Maximum operating time allowed on the part before it is removed and condemned.

**f. Retirement interval notes.** Any additional information required on the part's retirement interval.

## E.7.17 Preventive maintenance services / preventive maintenance daily inspection work package (aircraft preventive maintenance services / preventive maintenance daily only) <pms-inspecwp>.

A WP shall be developed for each specific inspection interval (for example, daily, intermediate, periodic, 10 hour/ 14 day, 30 hr/42 day, etc.), as applicable to the aircraft. Inspection checklists shall be divided by areas of the aircraft (for example, nose, fuselage, tail, etc.). All items requiring inspection shall be listed in the logical sequence of inspection that would require a minimum of time and motion on the part of the individual performing the inspection. The checklist data shall be formatted and delivered to support the inspection requirements in DA Pam 738-751. Refer to FIGURE E-13 for a diagram of modules and WPs in an aircraft Preventive maintenance services / preventive maintenance daily inspection manual. The following components shall be included these WPs.

**a.** WP identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** WP initial setup **<initial_setup>** is required for this WP (5.4.3).

**c.** Information described in E.7.17.1 through E.7.17.4.

### E.7.17.1 Actuation warning.

The following warning shall appear before the first step of instruction (italicized text within parentheses shall be replaced with the appropriate information):

**WARNING**

"Accidental actuation of the aircraft power plant or hydraulic system, or (*insert aircraft specific equipment as applicable, for example, firing of armament, jettison ballistics*) may cause severe injury or death. Before starting inspection, the aircraft safety check must be performed, if applicable IAW (*insert specific technical manual here*) if applicable the following statement may be inserted here "and all armament must be safetied, deactivated, and cleared (*insert technical manuals here*)."

### E.7.17.2 Mandatory safety-of-flight inspection items.

Mandatory safety-of-flight inspection items shall be highlighted. Mandatory safety-of-flight inspection items shall have WARNING on the WARNING SUMMARY page at the front of the manual. The WARNING shall be verbatim as follows: **" CSI**

**WARNING**

Certain inspections are mandatory Safety-of-Flight requirements, and the inspection intervals cannot be exceeded. In the event these inspections cannot be accomplished at the specified interval, the aircraft condition status symbol will be immediately changed to a red "X."

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### E.7.17.3 Area diagram.

An area diagram of the aircraft, showing sequences for inspection by area shall be included. The area identified shall include all surfaces, material, components and equipment pertaining to that specific location.

### E.7.17.4 Standard checklists.

If applicable, the standard inspection checklist shall be further divided into Power Off checks and Power On checks.

**a.** The following statement shall be the first item for each aircraft. It shall read: "Inspect aircraft forms and records for recorded discrepancies (DA Pam 738-751, Functional Users' Manual for the Army Maintenance Management System Aviation (TAMMS-A))."

**b.** The WPs shall be divided into the proper sequence of steps as outlined in the area diagrams. For PMD manuals, there shall be one WP for each inspection area.

**c.** The following statement shall be the final instruction of the checklist: "Inspect for foreign object damage and ensure all access panels or doors opened or removed for this inspection are closed or reinstalled."

## E.7.18 Phased maintenance checklist <pmi-cklistwp>.

A phased maintenance checklist WP shall be prepared. Refer to FIGURE E-14 for a diagram of modules and WPs in an aircraft preventive maintenance inspection manual. The following components shall be included these WPs.

**a.** WP identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** WP initial setup **<initial_setup>** is required for this WP (5.4.3).

**c.** Information described in E.7.18.1 and E.7.18.2.

### E.7.18.1 Inspection area diagrams <figure>.

Diagrams locating the inspection areas and the access doors and panels that require removal at various phased maintenance inspections of the aircraft shall be included.

### E.7.18.2 Phased maintenance checklist.

The phased maintenance checklist shall include all the inspection steps required to complete the given inspection. It may contain illustrations to aid in the performance of the inspection. Inspection steps shall be organized in a logical flow to minimize inspector movement. The inspection data shall be formatted and presented to support the inspection requirements in DA Pam 738-751. The WP shall begin with the following note:

**Note**

"Before start of the Phased Maintenance Inspection, it is recommended that a pre-inspection maintenance test flight (MTF) be conducted. Accomplishment of the MTF shall be determined by the unit maintenance officer. The pre-inspection MTF should be conducted by a maintenance test pilot following a review of the aircraft forms and records and a briefing from the crew of the aircraft. The MTF is recommended to assess the aircraft performance and identify deficiencies that should be corrected while the aircraft is undergoing phased maintenance inspections."


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## E.7.19 Preservation, packaging, and marking general information work package <ppmgeninfowp> (DMWR/NMWR only).

This WP shall be prepared and shall be the first WP in the first maintenance module. Preservation, packaging, and marking general information shall include the below preservation, packaging, and marking general information verbatim. Individual maintenance WPs shall be prepared for:

- Pack – (E.7.4.4.20)
- Mark – (E.7.4.4.16)
- Preserve – (E.7.4.4.27)

This WP shall refer to the individual WPs containing the instructions for packing, marking, and preserving, and shall not contain any tasks/ instructions. The following components shall be included these WPs.

**a.** WP identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** WP initial setup **<initial_setup>** is not required for this WP.

**c.** Preservation, packaging, and marking general information stated below shall be included verbatim.

**"PACKAGING**

Military preservation, Level A packing, and marking shall be accomplished in accordance with the specific packaging instructions contained in WP *insert WP number*.

**MARKING FOR SHIPMENT AND STORAGE**

Storage: In addition to any special markings called out on the special packaging instruction (SPI) or in the packaging requirements code, all unit packages, intermediate packs, exterior shipping containers, and as applicable, unitized loads shall be marked in accordance with MIL-STD-129 including bar coding. The repair facility is responsible for application of special markings as required by MIL-STD-129 regardless of whether specified in the contract/order or not. Special markings include, but are not limited to, shelf-life markings, structural markings, and transportation special handling markings. The marking of pilferable and sensitive materiel will not identify the nature of the materiel.

Shipment: The repair facility shall apply identification and address markings with bar codes in accordance with MIL-STD-129. A Military Shipment Label (MSL) is required for all shipments except contractor to contractor. The MSL will include both linear and 2D barcodes per the standard. Military Shipping Label: Military Shipment Labels may be created using the Computer Automated Transportation Tool Military Shipment Label/Issue Receipt Release Document (CATT MSL/IRRD).

**HEAT TREATMENT AND MARKING OF WOOD PACKAGING MATERIALS**

Wood Packaging Materials (WPM) (for example, boxes, crates, skids, pallets, and any wood used as inner packaging made of non-manufactured wood) shall be constructed of lumber that has been heat-treated in accordance with the requirements of International Standard for Phytosanitary Measures (ISPM) –15. The WPM manufacturer shall be affiliated with an inspection agency accredited by the board of review of the American Lumber Standard Committee. The WPM manufacturer shall ensure traceability to the original source of heat treatment. Each piece of WPM shall be marked to show the conformance to the International Plant Protection Convention Standard. Certification markings shall be indelible and permanent. They may be stamped, stenciled, or branded directly onto or into the WPM. Certification marks shall be applied in a visible location on at least two opposite sides of the wood packaging product but are not required on each individual component piece of a wood packaging product. On dunnage, the marking shall be applied every 2 feet to opposite surfaces of each piece. If possible, the mark shall be visible when the dunnage is placed in the load to enable inspectors to verify the WPM's compliance without unloading or unstuffing the container. Foreign manufacturers shall have the heat treatment of WPM verified in accordance with their National Plant Protection Organization's compliance program.

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**ALTERNATIVES**

The packaging requirements have been validated and the method of preservation/packing has proven successful in meeting the needs of the military distribution system, including undefined storage and shipment throughout the world. Tailoring of the packaging instructions may only be authorized by the packaging requirements developer. If tailored, prototype package is required to validate the sizes and fit requirements. Minor dimensional and size changes are acceptable provided email notification is provided to the packaging requirements developer. Any design changes or changes in the method of preservation that provide a cost savings without degrading the method of preservation or packing or affecting the serviceability of the item will be considered and responded to within 10 days of submission. The equipment MATDEV reserves the right to require testing to validate alternate preservation methods, materials, alternates, blocking, bracing, cushioning, and packing.

**RULES OF PACKAGING MATERIALS**

The cushioning material and the fiberboard boxes may be reused provided:

**a.** There is no visible damage to material.

**b.** The foam cushioning has not taken a permanent set.

**c.** The fiberboard has no punctures, delaminating, or crushed flutes.

The water vapor proof barrier bag shall never be reused. Always use new barrier material, evacuate air from the barrier bag, and conduct a snap test after 2 hours on each bag to ensure seal is holding. All components of the wood box/crate must be present, properly secured in position, and not broken. Splits are acceptable provided the boards remain secured and not loose. When reapplying the lid, fasteners shall be placed 1/2 inch away from the previous fastener hole. Strapping shall be applied per MIL-HDBK-147.

**CONTAINER REPAIR**

Each long-life metal reusable container will be inspected and reconditioned in accordance with TB 9-289, TM 1-8145-695-24&P, or SB 725-92-1 and the applicable container drawing package. Container drawings are available upon request from the packaging requirements developer. This reconditioning effort includes mandatory replacement of breather valves, humidity indicators, data plates, sealing gaskets, and desiccant, plus all shear mounts with an age factor of 5 years or older. It also includes a leak test after reconditioning, inspection and replacement of unserviceable wood skids, and touch up or total stripping and refinishing of the container surfaces with CARC paint."

## E.7.20 Facilities work package <facilwp>.

This WP shall be prepared as directed by the MATDEV. A description of all facilities (for example, test stands, test tracks, clean rooms, shielded rooms, or other facilities) that are required to do the maintenance work shall be included. Reference shall be provided for any specifications or standards that these facilities must meet. When approved by the MATDEV, data from these standards may be included in this WP. The following components shall be included in facilities WPs.

**a.** WP identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** WP initial setup **<initial_setup>** is required for this WP (5.4.3).

**c.** Facilities required to do maintenance work.

## E.7.21 Overhaul inspection procedures (OIPs) work package <oipwp>.

Unless otherwise specified by the MATDEV, OIPs shall be prepared for items that have parts with specific characteristics, wear limits, specified performance requirements, or fatigue characteristics or tolerances. A separate WP shall be provided for each item containing such parts. Within each WP, a separate OIP table or list shall be provided for each part of the item that requires a critical inspection. The OIP shall consist of the characteristics being

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inspected for, inspection methods, and the acceptance/reject criteria that must be met. Unless otherwise specified by the MATDEV, an illustration shall accompany the OIP. Illustrations for OIPs are strongly encouraged and shall only be omitted for very simple systems/parts. A reference letter may be included on the illustration to aid in locating the critical inspection characteristics of the parts. The following components shall be included in OIP WPs.

**a.** WP identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** WP initial setup **<initial_setup>** is required for this WP (5.4.3).

**c.** OIP information for specific characteristics, wear limits, specified performance requirements, or fatigue characteristics or tolerances in accordance with E.7.21.1.

### E.7.21.1 OIPs.

The OIP shall contain the characteristics being inspected for, the inspection methods being used, and the acceptance/reject criteria that must be met. Unless otherwise specified, an illustration shall accompany the OIP. Illustrations are strongly encouraged for OIPs and shall only be omitted for very simple systems/parts. A reference letter may be included in the OIP to locate the critical inspection characteristics of the parts on the illustrations. The OIPs may be contained in a table or a list. References to these OIP WPs shall be included within the applicable maintenance procedural step (for example, disassembly, reassembly, testing, etc.) or preshop analysis procedural step where they apply.

## E.7.22 Depot mobilization requirements work package <mobilwp>.

This WP shall be prepared and shall include the modifications, deletions, or additions to the preshop analysis or overhaul procedures required during mobilization shall be included in this WP. The following information shall be included:

**a.** WP identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** WP initial setup **<initial_setup>** is required for this WP (5.4.3).

**c.** Introduction in accordance with E.7.22.1.

**d.** Mobilization requirements in accordance with E.7.22.2.

### E.7.22.1 Introduction for depot mobilization requirements <intro>.

**"DEPOT MOBILIZATION REQUIREMENTS INTRODUCTION**

**Scope**

The purpose of this WP is to streamline and accelerate the overhaul process during the mobilization of the depot.

**Explanation of Mobilization Requirements**

The mobilization requirements include a list of instructions for modifying preshop analysis and/or overhaul procedures. The pertinent procedures to be modified are referred to by WP number followed by the action to be taken."

### E.7.22.2 Mobilization requirements <mobilreq>.

Mobilization requirements consist of a list of actions that shall be in effect during depot mobilization. The WPs that are modified by these actions shall be noted. These data shall be provided in a standard table **<mobiltab>**. Alternatively, if the actions are already listed in another WP or WPs, a statement shall be made that includes references to those actions.

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## E.7.23 QA requirements work package <qawp>.

This WP shall be prepared when equipment has special quality requirements. The following components shall be included in QA requirements WPs.

**a.** WP identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** WP initial setup **<initial_setup>** is required for this WP (5.4.3).

**c.** Information in E.7.23.1 thru E.7.23.8.

### E.7.23.1 Statement of responsibility <responsibility>.

**"STATEMENT OF RESPONSIBILITIES**

The depot/contractor is responsible for complying with the quality assurance requirements contained in this WP and in accordance with International Standards Organization (ISO) 9000 Series standards or equivalent. The commodity manager reserves the right to perform inspections or make changes that ensure the depot work being done meets the quality standards of the DMWR and preserves the inherent reliability of the item."

Definitions shall be prepared for all QA terms extensively used in the DMWR and NMWR. If the definitions are listed in another publication, that publication shall be referenced."

### E.7.23.2 Definitions <definitions>.

Definitions shall be prepared for all QA terms extensively used in the DMWR and NMWR. If the definitions are listed in another publication, that publication shall be referenced.

### E.7.23.3 Special requirements for inspection tools and equipment <specialreq>.

Any special requirements for the maintenance and calibration of tools and test equipment used for QA inspections shall be listed.

### E.7.23.4 Certification requirements <certreq>.

Any certification or licensing requirements for processes, instructions, materials, equipment, or personnel skills shall be listed. The list shall include appropriate standards, specifications, regulations, and/or laws that apply. The list shall reference the text in the DMWR/NMWR where a requirement exists for a soldering, welding, or magnetic particle inspection certification, radioactive substance, or test driver licenses.

### E.7.23.5 Quality program <quality-program>.

Any requirements for a quality program shall be listed.

### E.7.23.6 In-process inspections <inprocess>.

The following statement shall be included:

**"IN-PROCESS INSPECTIONS**

In-process Quality Assurance (QA) inspections are contained throughout the overhaul procedures of this DMWR. These inspections are immediately preceded by a statement such as "QA check" to identify them. They are the minimum inspections required. Additional QA inspections may be established by the depot or the commodity manager."

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### E.7.23.7 Acceptance inspections <acceptance>.

The following statement shall be included:

**"ACCEPTANCE INSPECTIONS**

Items overhauled in accordance with this DMWR will be accepted based on the following criteria:

1. Conformance to quality of material requirements.
2. Conformance to all in-process quality assurance inspections.
3. Conformance to all final assembly testing requirements.
4. Conformance to the preservation, packaging, and marking requirements."

### E.7.23.8 First article inspection <first>.

When applicable, reference to first article inspection/test prepared for the DMWR/NMWR in accordance with ISO 9000 Series standards or equivalent shall be included.

## E.7.24 Auxiliary equipment maintenance work package <auxeqpwp>.

When auxiliary equipment (for example, Modified Tables of Organization and Equipment (MTOE) items, etc.) maintenance TMs are not procured for peculiar equipment furnished by the contractor, separate maintenance WPs shall be prepared for each maintenance task. The following components shall be included in these WPs.

**a.** WP identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** WP initial setup **<initial_setup>** is required for this WP (5.4.3).

**c.** Information in E.7.24.1.

### E.7.24.1 Auxiliary equipment instructions <maintsk>/<proc>.

Concise step-by-step instructions shall be prepared for proper care of auxiliary equipment while in and out of service.

There shall be WPs for each of the following tasks:

**a.** Storage.

**b.** Preventive maintenance.

**c.** Lubrication.

**d.** Operating checks.

**e.** Adjustments.

**f.** Maintenance instructions (refer to E.7.4). For special tools/kits that have been fabricated, refer to E.7.23.8.

## E.7.25 Ammunition identification work package <ammoidentwp>.

This WP prepared as directed by the MATDEV. It shall provide applicable information on care and handling and identification of ammunition, components, and non-ammunition Class V items such as packaging. Individual paragraphs shall be prepared for each ammunition type/group/classification. The following components shall be included in these WPs.

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**a.** WP identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** WP initial setup **<initial_setup>** is required for this WP (5.4.3).

**c.** Information in E.7.25.1.

### E.7.25.1 Care and handling.

Concise step-by-step instructions required for the care and handling of ammunition shall be prepared. These shall include hazard distances, storage, special requirements, prevention of deterioration due to rough handling, exposure to adverse weather conditions, or any other hazards that may be encountered.

### E.7.26 Ammunition maintenance work package <ammowp>.

This WP shall be prepared as directed by the MATDEV and shall reference or contain (in separate WPs) any maintenance tasks identified in E.7.4.4, among others, including but not limited to "inspect," "unpack," "preparation for firing" (including tasks such as "clean," "set fuze," "adjust charges," "remove safety clips," etc.), "loading," "firing" (including unfired ammunition), and "pack." The following components shall be included in these WPs.

**a.** WP identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** WP initial setup **<initial_setup>** is required for this WP (5.4.3).

**c.** Maintenance tasks identified in E.7.4.4, among others (referenced and included in WP).

### E.7.27 Foreign ammunition work package <natowp>.

A WP to describe foreign ammunition shall be prepared when applicable and contain the following information.

**a.** WP identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** WP initial setup **<initial_setup>** is required for this WP (5.4.3).

**c.** Information describing foreign ammunition.

## E.8 NOTES.

The notes in Section 6 apply to this appendix.

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# APPENDIX F

# PARTS INFORMATION

## F.1 SCOPE.

This appendix establishes the technical content requirements for the preparation of parts information for major equipment systems, their subsystems, assemblies, components, Shop Replaceable Units (SRUs), and Line Replaceable Units (LRUs). This appendix also defines and details a subset of parts information called a Repair Parts and Special Tools List (RPSTL), which consists of an introduction work package (WP) and one or more accompanying WPs that authorize repair to system components. Parts information for an equipment system can stand alone as its own technical publication or be included with other operations and maintenance data as a module in a publication. This appendix is a mandatory part of this standard. The information contained herein is intended for compliance. Requirements in this appendix are applicable for all maintenance levels. For equipment with DMWRs or NMWRs, parts information shall be contained in those documents.

**Note**

To be compliant with this standard, publications shall be developed within **<production>** (opening) and **</production>** (closing) root tags.

When preparing information in accordance with this appendix, all references shall be linked using the appropriate tags (**<xref>**, **<extref>**, etc.). Links shall not be typed as text without tagging.

### F.1.1 Content tailoring using Appendix A matrices.

Applicability of some requirements contained in this standard are the responsibility of the materiel developer (MATDEV) (refer to AR 750–1, AR 700–127) and shall be accomplished using matrices referenced in TABLE A-I. The applicability of requirements are also designated by statements including, but not limited to, the following:

**a.** "...as specified by the MATDEV."

**b.** "...as required by the MATDEV."

**c.** "Refer to the MATDEV for..."

**d.** "Contact MATDEV if..."

**e.** "The MATDEV shall determine..."

**f.** "...as needed."

#### F.1.1.1 Front and rear matter.

Front and rear matter requirements for documents discussed in this appendix shall be determined in accordance with TABLE A-II.

## F.2 APPLICABLE DOCUMENTS.

The applicable documents in Section 2 apply to this appendix.

## F.3 ACRONYMS AND DEFINITIONS.

The acronyms and terms defined in Section 3 of this standard apply to this appendix.

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## F.4 PARTS INFORMATION APPLICABILITY.

The RPSTL provides authorized spares and repair parts; special tools; special Test, Measurement, and Diagnostic Equipment (TMDE); and other special support equipment required for performance of all levels of maintenance of the weapon system/equipment, subsystems, assemblies, and components. The RPSTL authorizes the requisitioning, issue and disposition of spares, repair parts and special tools in accordance with the Source Maintenance and Recoverability (SMR) codes.

Requirements contained in this appendix are applicable to all maintenance levels unless specifically labeled in bold and in parentheses. The labeled requirement may specify a specific maintenance level (for example, Field, Maintainer, or AMC) . The labeled requirement shall be applicable to all TMs containing that maintenance level. An explanation of applicable DA maintenance levels is provided in Section 3. All parts information for all levels of maintenance (except sustainment) may be in a single parts manual or may be in multiple parts manuals. When separate parts manuals are specified by the MATDEV, they shall be grouped by maintenance level. Duplication of the parts data should be avoided when separating by maintenance level.

## F.5 PARTS INFORMATION CONFIGURATION.

Parts shall be prepared for equipment systems and applicable support and interface equipment. This information shall be contained in one of the following:

**a.** Separate parts information technical publication (F.5.1).

**b.** Parts information included in a Depot Maintenance Work Requirement (DMWR) / National Maintenance Work Requirement (NMWR) (F.5.2).

**c.** Parts information included as a module in a technical publication (F.5.3).

### F.5.1 Separate parts information technical publication.

Separate parts information technical publications shall consist of front and rear matter and a parts information module **<pim>** containing the work packages (WPs) listed in TABLE F-I (as required; see individual WP sections for more details). Contrarily, for maintenance publications including parts information, the front and rear matter requirements shall be included in front and rear matter for the publication as a whole (see F.5.3). See FIGURE F-1 for a diagram of components for a parts information TM.

### F.5.2 Parts information included in a DMWR/NMWR.

If an item of equipment is programmed for depot overhaul and no repair parts (including modules, printed circuits, and components) are authorized for replacement below depot-level maintenance, authorized repair parts data shall be included in the applicable DMWR/NMWR. Depot-level repair part information shall be included in the DMWR/NMWR. The appropriate Source Maintenance and Recoverability (SMR) code shall identify the repair level.

### F.5.3 Parts information included as a module in a technical publication.

Parts information WPs developed in accordance with this appendix shall comprise the parts information module **<pim>** which shall immediately follow the last maintenance information module **<mim>** in a publication. Refer to APPENDIX A for content selection matrices that designate Required ("R"), Prohibited ("P"), and As Needed (shaded) sections of a parts module. A parts information module shall consist of a RPSTL and other tools and parts information WPs. Refer to TABLE F-I for a list of individual types of parts information WPs. See FIGURE F-1 and FIGURE F-2 for a diagram of components for a publication containing parts information.

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**TABLE F-I. Parts information WPs.**

| | WP | WP Tag | Appendix Section |
|---|---|---|---|
| **RPSTL** | RPSTL introduction WP | **<introwp>** | F.8.1 |
| | Repair parts list WP(s) [1] | **<plwp>** | F.8.2 |
| **Other Available Tools and Parts Information** | Special tools/kits WP | **<stl_partswp>** | F.8.3 |
| | Kits parts list WP | **<kitswp>** | F.8.4 |
| | Bulk items WP | **<bulk_itemswp>** | F.8.5 |
| | Special tools/kits WP | **<stlwp>** | F.8.6 |

[1] WP type frequently duplicated in a module for effective information display.

**Note**

Tags/output shown in figure(s) below may vary slightly based on version of DTD/stylesheets utilized (see applicable DTD or tag definition list for available tags).

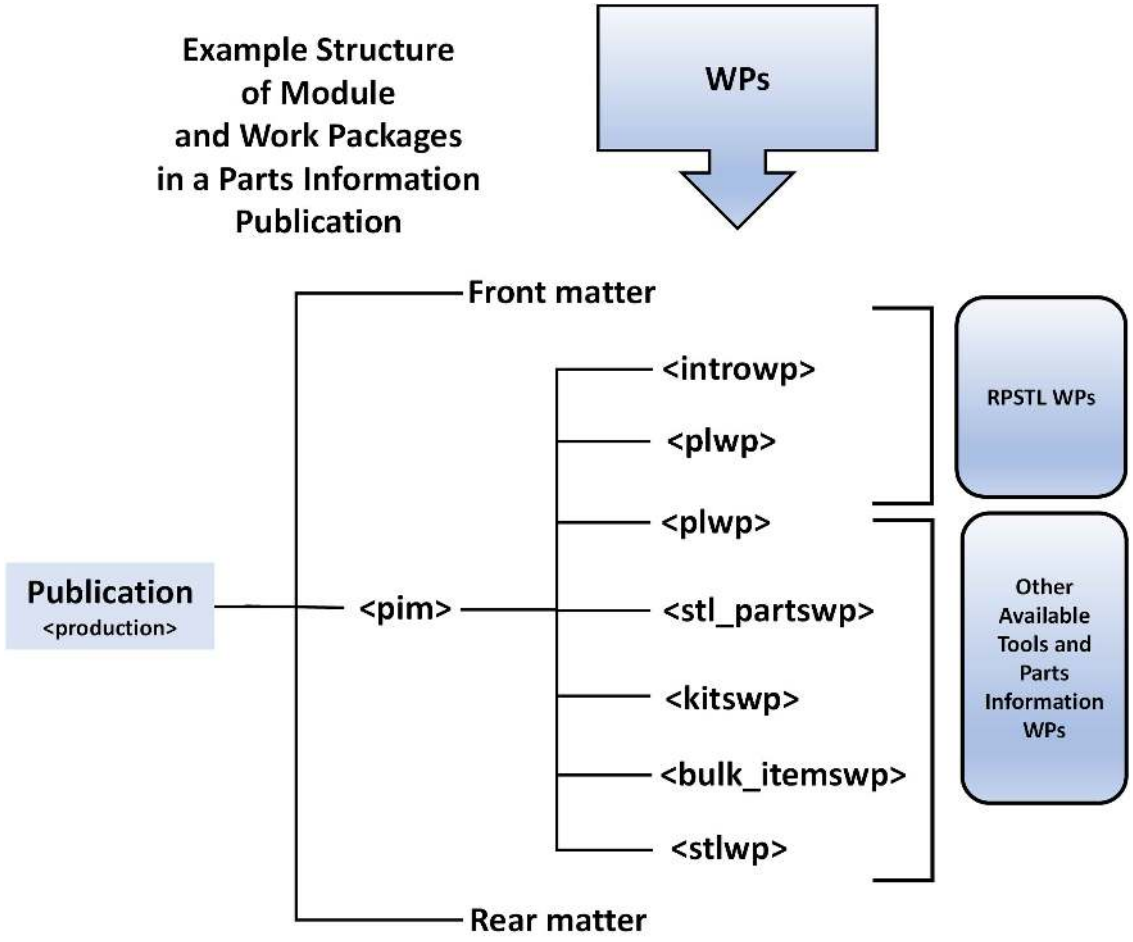


**FIGURE F-1. Stand-alone parts information publication.**

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**Note**

Tags/output shown in figure(s) below may vary slightly based on version of DTD/stylesheets utilized (see applicable DTD or tag definition list for available tags).

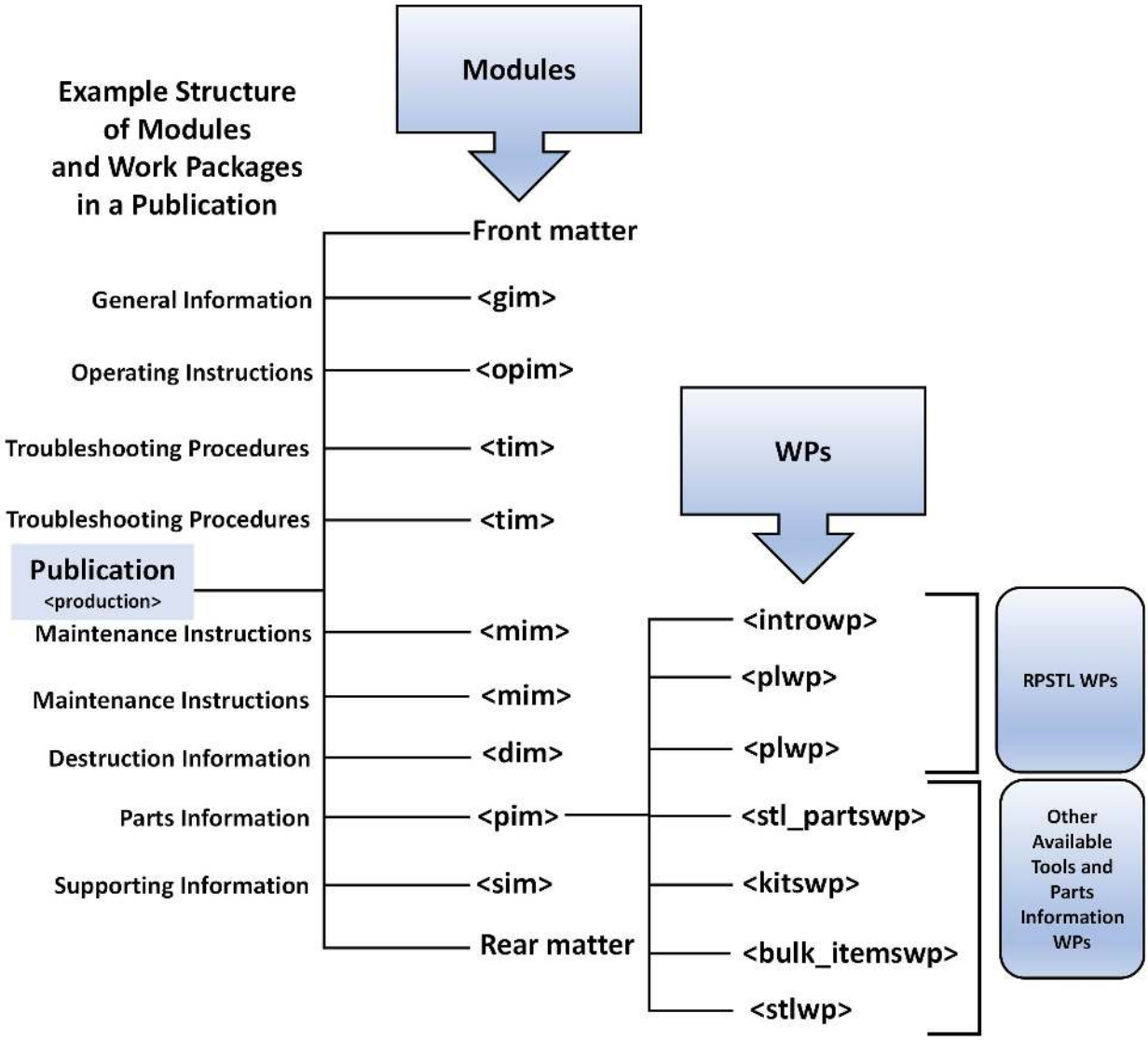


**FIGURE F-2. Parts information module and WPs in an equipment publication.**

## F.6 RPSTL METHODOLOGY.

As stated above, a RPSTL is a subset of parts information for an equipment system. This section discusses RPSTLs and how RPSTL data informs various publication components.

### F.6.1 RPSTL components.

RPSTLs shall always include the following components: (1) introductory information; (2) listings of all authorized spare and repair parts, special tools/kits, special TMDE and other support equipment required for maintenance; (3) illustrations to identify and locate the spare and repair parts; and (4) National Stock Numbers (NSNs).

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## F.6.2 RPSTL layout.

All RPSTL WPs shall start on the right-hand page as shown in FIGURE F-3. For parts list, special tools/kits, and kit WPs, the first page shall contain the WP identification **<wpidinfo>**. The figure **<figure>** (or first figure if multiple figures are included ) shall start on page 2. The parts list shall follow the figure(s). Each RPSTL WP (after the RPSTL introduction) ends with text stating "END OF FIGURE."

**Note**

Figure appearance may vary slightly based on version of DTD/stylesheets utilized.

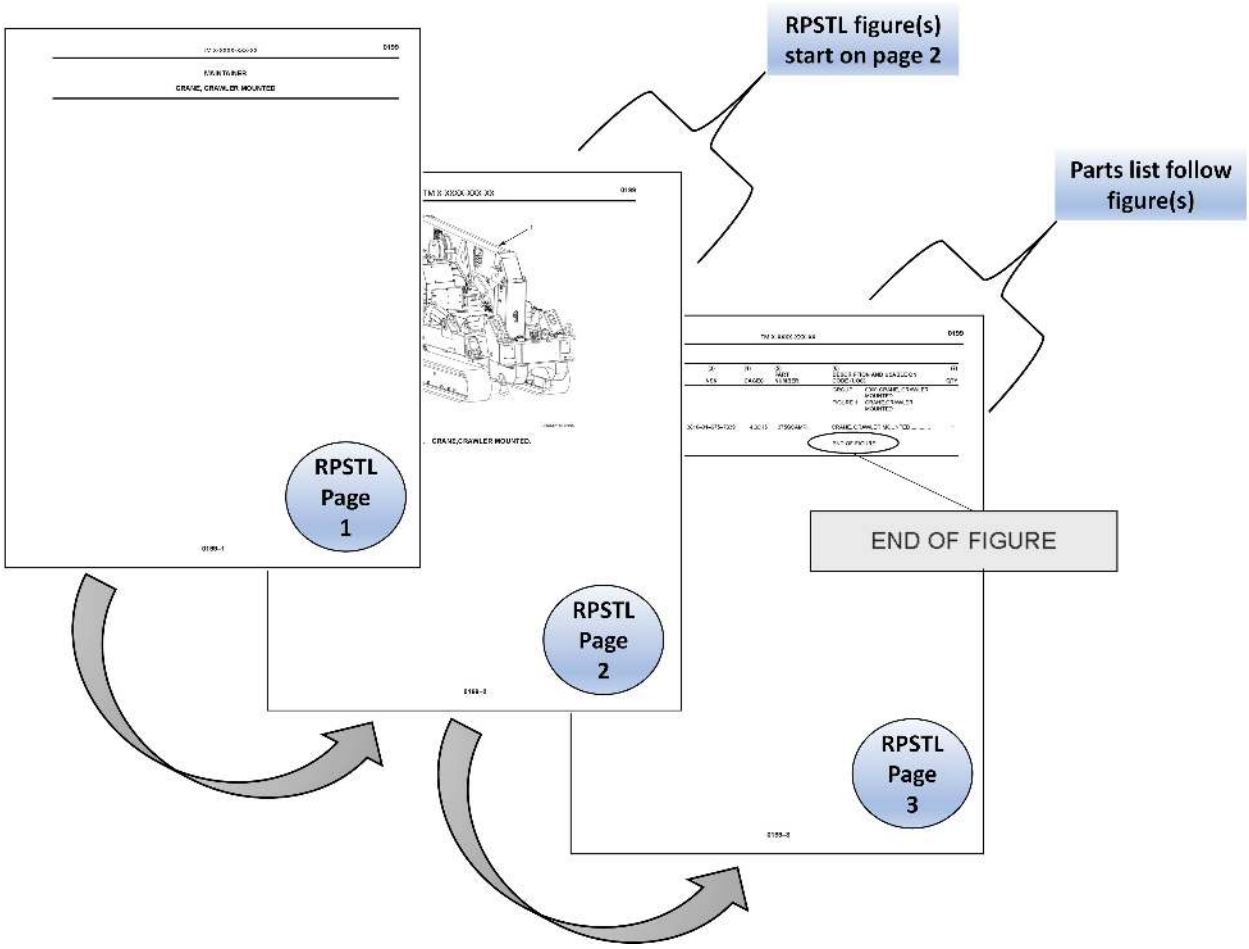


**FIGURE F-3. RPSTL layout.**

## F.6.3 RPSTL data and functionality within a technical publication.

The RPSTL authorizes the requisitioning, issue, and disposition of spares, repair parts, and special tools/kits in accordance with SMR codes listed for each part. SMR codes are a component of the Logistics Product Data (LPD) for a system. LPD is derived from reports such as the Logistics Support Analysis Report (LSAR) 004 that provides Maintenance Allocation Chart (MAC) data; the LSAR 080 (Bill of Materials (BoM) list) that provides RPSTL data; among other reports. Maintenance in a technical publication is developed in accordance with MAC-authorized repair. LPD (including SMR codes) contained in the RPSTL is used to produce the MAC; thus, data contained in the MAC and RPSTL shall align. Moreover, three components of a technical publication (maintenance WPs, the MAC, and the RPSTL) are tied to each other and must always agree (discussed further in F.6.4). FIGURE F-4 shows how logistics data affecting MAC entries and maintenance WPs are represented in the RPSTL. SMR codes and

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Functional Group Codes (FGCs) (along with corresponding equipment) are listed in columns (2) and (6), respectively, of a RPSTL table (FIGURE F-4, right-side page showing table). Item numbers are listed in column (1) and directly correspond to the graphical representations of parts (with item numbers shown as callouts) (FIGURE F-4, left-side page showing graphic). RPSTLs also provide the following tabular parts data: column (3) NSN (National Stock Number); column (4) CAGEC (Commercial and Government Entity Code); column (5) Description and UOC (Usable on Code); Part Number; column (6); and column (7) QTY (Quantity). These items are discussed in greater detail in F.6.4.

**Note**

Figure appearance may vary slightly based on version of DTD/stylesheets utilized.

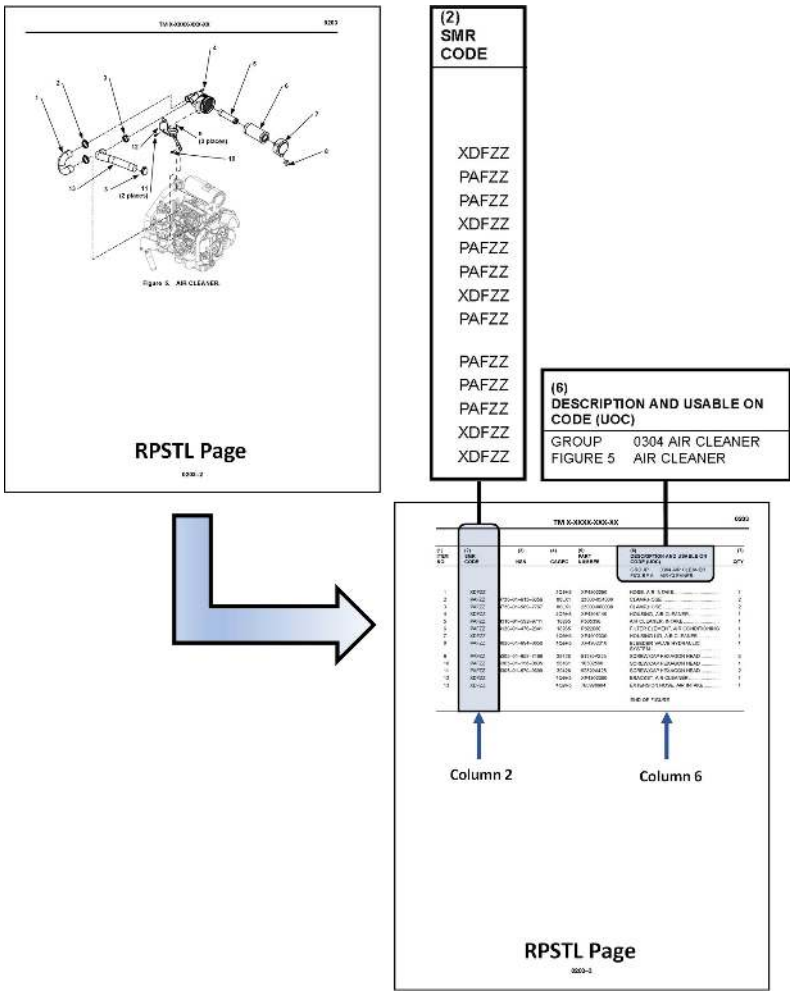


**FIGURE F-4. RPSTL page example.**

## F.6.4 RPSTL alignment with MAC and maintenance.

RPSTL data inform the development of MAC entries, which, in turn, guide the development of maintenance WPs. Every entry in a MAC shall have a corresponding WP containing the relevant maintenance procedure (unless stipulated otherwise in remarks codes), and these maintenance WPs shall be arranged to coincide with the FGC, or top-down breakdown sequence, followed in the MAC and RPSTL. FIGURE F-5 and FIGURE F-6 show an example FGC 0304, "AIR CLEANER" represented in a technical publication. The SMR codes for components and subcomponents of FGC 0304 are reflected in the RPSTL table shown in FIGURE F-5. As stated in Appendix E,

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maintenance is developed in accordance with the third and fourth SMR code positions that define levels of maintenance authorized to maintain and use an item and denote the lowest maintenance level with the capability and resources to perform a complete repair, respectively. The MAC entry for FGC 0304 shown in FIGURE F-5 designates the authorized level of repair for components of FGC 0304. FIGURE F-6 shows one of the maintenance WPs ("AIR INNER AND OUTER FILTERS") developed for FGC 0304. In this example, replacement tasks for the inner and outer air filter are designated as a crew (operator)-level task and, thus, integrated into an operator maintenance WP.

**Note**

Figure appearance may vary slightly based on version of DTD/stylesheets utilized.

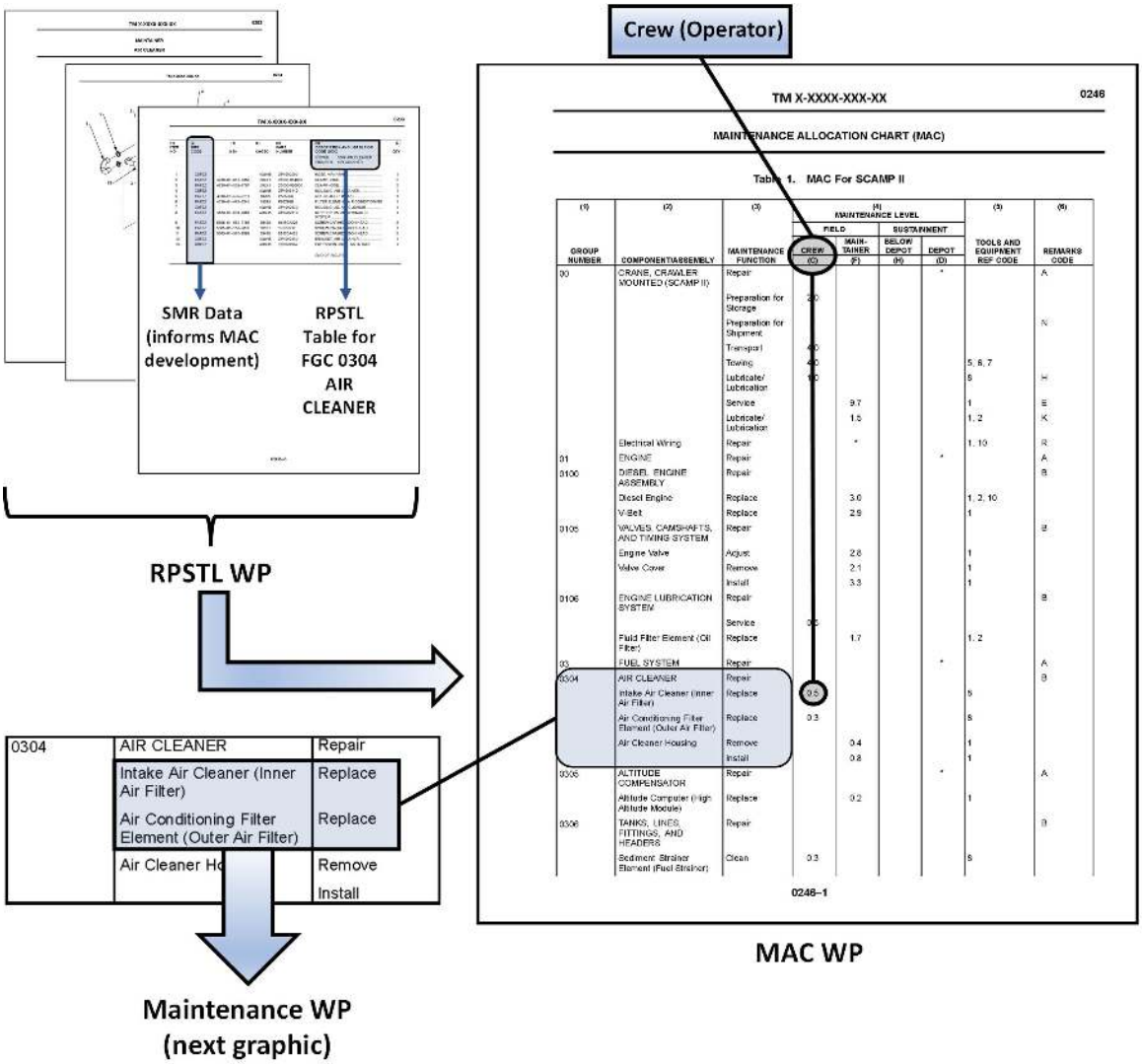


**FIGURE F-5. Maintenance alignment (crosswalk) in technical publications (sheet 1 of 2).**

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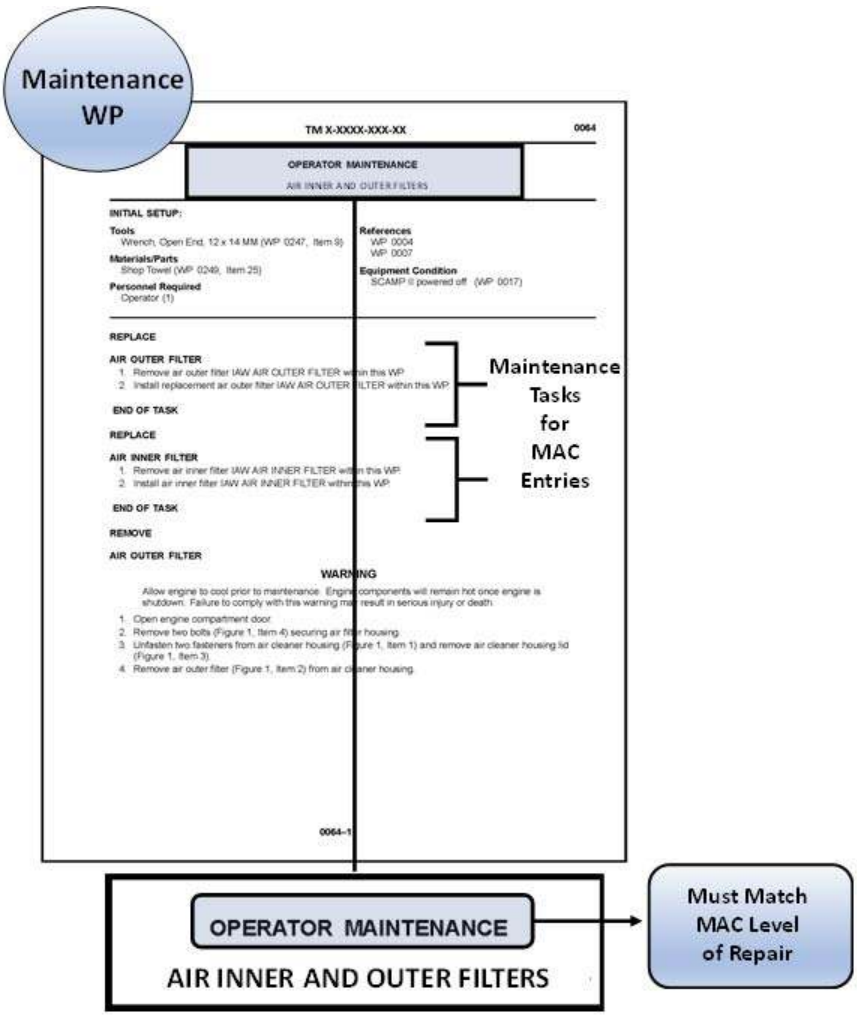


**FIGURE F-6. Maintenance alignment (crosswalk) in technical publications (sheet 2 of 2).**

### F.6.4.1 Component naming conventions in RPSTLs, MACs, and maintenance.

Component names listed in a MAC may not always exactly match component names listed in RPSTL for the same part/piece of equipment. The RPSTL lists H6 names for parts, which are derived from the item name directory used for National Stock Number (NSN) assignments. This directory defines the names and cross references the Item Name Codes (INC) and the Federal Supply Classes (FSC). MAC component names shall match maintenance tasks titles appearing in a technical publication. These titles are the H6 names, or when necessary for clarity, Logistics Control Number (LCN) nomenclature. The LCN nomenclature may be used in place of the nomenclature used in the FEDLOG H6 listing. Functional Group Codes (FGC) for a component shall always align across MAC entries and RPSTL entries and, thus, can serve to prevent potential cross-referencing confusion.

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## F.6.5 Sustainment maintenance and alignment.

**Note**

While this standard governs the development of field-level technical publications, the information in this section ensures a basic understanding of alignment requirements beyond field-level repair. Logistics data for an equipment system guides the organization, structure, and data provided within RPSTLs, MACs, and maintenance. Alignment consistency is key among technical publications – field-level and above – that support these equipment systems.

The alignment principles discussed in F.6.4 are also applicable to sustainment maintenance provided in depot-level technical publications such as Depot Maintenance Work Requirements (DMWRs), etc. Repair parts associated with FGCs for field-level maintenance shall directly correlate to FGCs for sustainment-level repair data. Depot-level publications, however, shall provide more detail about components and parts needed for maintenance intended to overhaul a system or portion of a system. See example DMWR graphic and table listing SMR and FGC codes.

**Note**

Figure appearance may vary slightly based on version of DTD/stylesheets utilized.

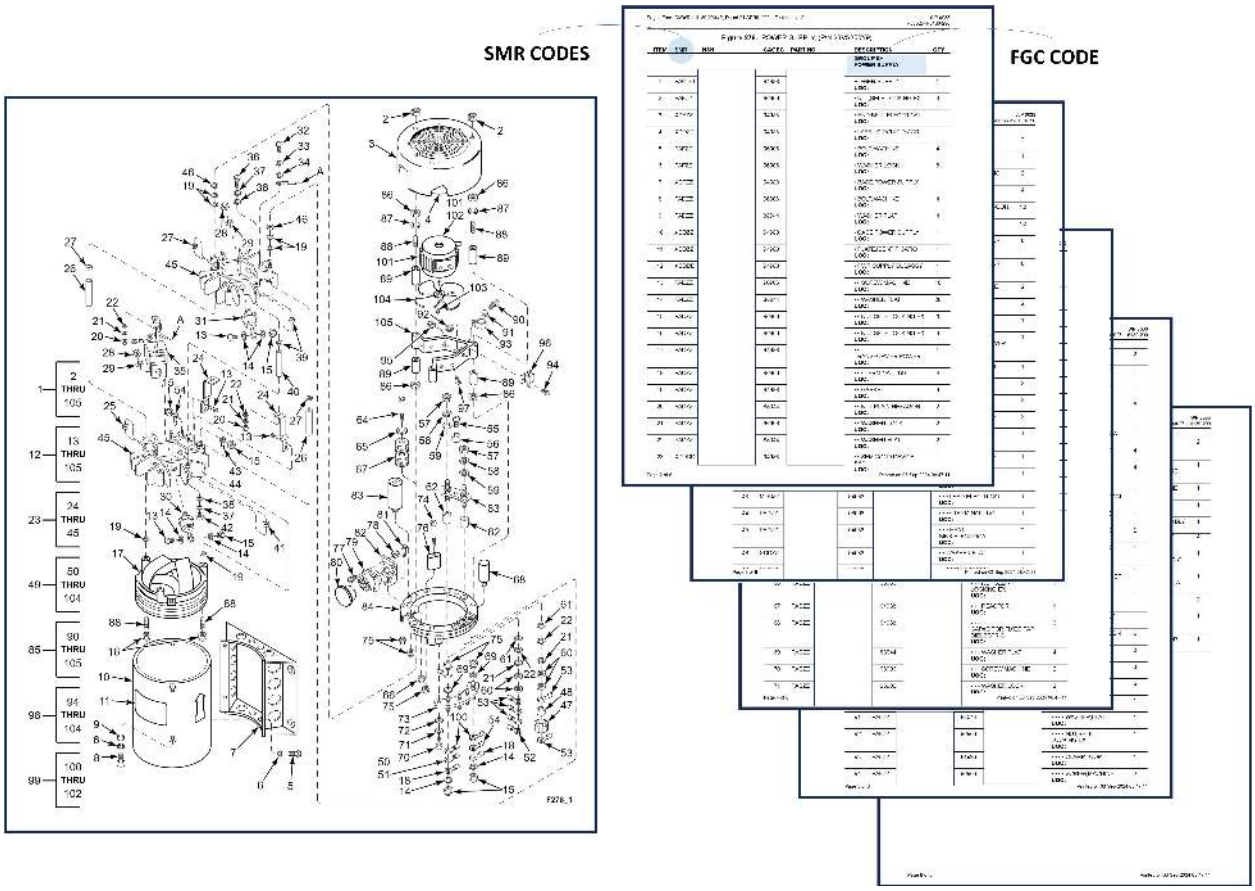


**FIGURE F-7. Example DMWR graphic and table.**

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## F.7 DEVELOPMENT OF PARTS INFORMATION MODULES.

The following modules (and respective WPs) shall be prepared for stand-alone parts technical publications (F.7.1) and parts information included in an equipment system publications as a RPSTL (F.7.2).

### F.7.1 Development of a stand-alone parts technical publication.

As stated in section F.5.1, when specified by the MATDEV, a stand-alone parts technical publication shall be prepared (see TABLE A-V). It shall include the following WPs in a parts information **<pim>** module as needed. See FIGURE F-1 for a graphical representation of WP organization and tags discussed in this section.

**a.** RPSTL introduction WP **<introwp>** (F.8.1).

**b.** RPSTL parts list WPs **<plwp>** (F.8.2).

**c.** Additional RPSTL parts list WP(s) as needed.

**d.** Repair parts for special tools/kits list WP **<stl_partswp>** (F.8.3).

**e.** Kit parts list WP **<kitswp>** (F.8.4).

**f.** Bulk items WP **<bulk_itemswp>** (F.8.5).

**g.** Repair parts for special tools/kits list WP **<stlwp>** (F.8.3).

### F.7.2 Development of a parts information module in an equipment system technical publication.

As stated in section F.5.3, when specified by the MATDEV, a parts information module **<pim>** shall be provided in an equipment system technical publication as a RPSTL. It shall consist of the WPs listed in step G. (below) as needed. All other modules in an equipment technical publication shall be prepared as required by publication type (see A.4.4). See FIGURE F-2 for a graphical representation of WP organization and tags discussed in this section.

**a.** General information module **<gim>** (B.6).

**b.** Operating instructions module(s) **<opim>** (C.6).

**c.** Troubleshooting procedures module(s) **<tim>** (D.7).

**d.** Maintenance instructions module(s) **<mim>** (E.6).

**e.** Destruction information module **<dim>** (H.5).

**f.** Software information module **<soim>** (M.6).

**g.** Parts information module **<pim>**:

**(1)** RPSTL introduction WP **<introwp>** (F.8.1).

**(2)** RPSTL parts list WP(s) **<plwp>** (F.8.2).

**(3)** Additional RPSTL parts list WP(s) as needed.

**(4)** Repair parts for special tools/kits list WP **<stl_partswp>** (F.8.3).

**(5)** Kit parts list WP **<kitswp>** (F.8.4).

**(6)** Bulk items WP **<bulk_itemswp>** (F.8.5).

**(7)** Special tools/kits list WP **<stlwp>** (F.8.6).

**h.** Supporting information module **<sim>** (G.6).

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## F.8 DEVELOPMENT OF PARTS INFORMATION WORK PACKAGES.

The inclusion of various WP types depends on configuration of the parts information (stand-alone, F.5.1 or integrated, F.5.3) and the required tools, kits, etc. necessary for maintenance of equipment. This section describes each type of parts information WP and their functions.

### F.8.1 RPSTL introduction work package <introwp>.

A RPSTL introduction WP shall be prepared and shall include the verbatim text (within the quotation marks). The italicized text shall be replaced with the required system specific information or select the corresponding phrase for the specific system. If there are any SMR codes in the RPSTL data that use the 6th position, information for the 6th position found in AR 700-82 shall be included in the RPSTL introduction after the explanation of the 5th position. The following components shall be included in a RPSTL introduction WP.

**a.** WP identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** Initial setup **<initial_setup>** is not required for this WP.

**c.** RPSTL introduction information (F.8.1.1 thru F.8.1.4) based on type of publication:

**(1)** Non-aviation (F.8.1.1).

**(2)** Aviation (F.8.1.2).

**(3)** Marine Corps (MC) (F.8.1.3).

#### F.8.1.1 Non-aviation RPSTL introduction.

**"INTRODUCTION**

**SCOPE**
This RPSTL lists the authorized spares and repair parts; special tools; special Test, Measurement, and Diagnostic Equipment (TMDE); and other special support equipment required for performance of maintenance. It authorizes the requisitioning, issue, and disposition of spares, repair parts, and special tools/kits as indicated by the source, maintenance, and recoverability (SMR) codes.

**GENERAL**
In addition to the Introduction work package, this RPSTL is divided into the following work packages.

1. Repair Parts List WPs. These WPs contain lists of spare and repair parts authorized for use in the performance of maintenance at the levels determined by the MAC/SMR code. These WPs also include parts that must be removed for replacement of the authorized parts. Parts lists are composed of functional groups in ascending numeric or alphanumeric sequence, with the parts in each group listed in ascending figure and item number sequence. Sending units, brackets, filters, and bolts are listed with the component upon which they mount. Bulk materials are listed by item name in the Bulk Items WP that follows either: (*select one of the two following items:(1) the last Parts List, Special Tools Repair Parts, or (2) Kits WP*). Repair parts kits are either (*select one of the two following items: (1) listed separately in their own functional group and WP* ***OR*** *(2) listed at the end of the individual WPs*). Repair parts for repairable special tools/kits are also listed in a separate WP. Items listed are shown on the associated illustrations.

**Note**

*Include the text in items 2 through 4 only if the described work package is included in the TM.*

2. Special Tools Repair Parts WP. This WP lists any spare parts required for the special tools/kits, TMDE, or other support equipment listed in the Special Tools WP that are not listed in any other publication.

3. Kits WP. This WP lists all repair kits and their component parts.

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4. Bulk Items WP. This WP lists all items identified as "bulk" in the parts lists. Due to the nature of bulk items, this WP does not include a figure.
5. Special Tools/Kits List WPs. This WP lists those special tools/kits, special TMDE, and special support equipment authorized by this RPSTL (as indicated by Basis of Issue (BOI) information in the DESCRIPTION AND USABLE ON CODE (UOC) column). Tools that are components of common tool sets and/or Class VII are not listed.

**EXPLANATION OF COLUMNS IN THE REPAIR PARTS LIST AND SPECIAL TOOLS/KITS LIST WPS**

ITEM NO. (Column 1). Indicates the number used to identify items called out in the illustration.

SMR CODE (Column 2). The SMR code contains supply/requisitioning information, maintenance level authorization criteria, and disposition instruction in accordance with AR 700-82, as shown in the following breakout. This entry may be subdivided into 4 subentries, one for each service.

**TABLE 1. SMR Code Explanation.**

| Source Code | Maintenance Code | | Recoverability Code |
|---|---|---|---|
| XX | XX | | XX |
| 1st two positions: How to get an item. | 3rd position: Who can install, replace, or use the item. | 4th position: Who can do complete repair on the item. | 5th position: Who determines disposition action on unserviceable items. |

**Note**

Complete Repair: Maintenance capacity, capability, and authority to perform all corrective maintenance tasks of the "Repair" function in a use/user environment in order to restore serviceability to a failed item.

Source Code. The source code tells you how you get an item needed for maintenance, repair, or overhaul of an end item/equipment. Explanations of source codes follow:

**TABLE 2. SMR Code Explanation.**

| Source Code | Application/Explanation |
|---|---|
| PA<br>PB<br>PC<br>PD<br>PE<br>PF<br>PG<br>PH | Stock items; use the applicable NSN to requisition/request items with these source codes. They are authorized to the level indicated by the code entered in the third position of the SMR code.<br>**Note**<br>Items coded PC are subject to deterioration. Items coded PR or PZ are obsolete and may not be able to be ordered like other P coded items. |
| PR<br>PZ<br>KD<br>KF<br>KB | Items with these codes are not to be requested/requisitioned individually. They are part of a kit that is authorized to the maintenance level indicated in the third position of the SMR code. The complete kit must be requisitioned and applied. |
| MF - Made at maintainer level | Items with these codes are not to be requisitioned/requested individually. They must be made from bulk material which is identified by the P/N in the DESCRIPTION AND USABLE ON |

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| | |
|---|---|
| MH - Made at below-depot sustainment level | CODE (UOC) entry and listed in the bulk material group work package of the RPSTL. If the item is authorized to you by the third position code of the SMR code, but the source code indicates it is made at higher level, order the item from the higher level of maintenance. |
| ML - Made at SRA | |
| MD - Made at depot | |
| MG - **Navy only** | |
| AF - Assembled by maintainer level | Items with these codes are not to be requested/requisitioned individually. The parts that make up the assembled item must be requisitioned or fabricated and assembled at the level of maintenance indicated by the source code. If the third position of the SMR code authorizes you to replace the item, but the source code indicates the item is assembled at a |
| AH - Assembled by below depot sustainment level | |
| AL - Assembled by depot | |
| AD - Assembled by depot | |
| AG - **Navy only** | |
| XA | Do not requisition an "XA" coded item. Order the next higher assembly. (Refer to NOTE below.) |
| XB | If an item is not available from salvage, order it using the CAGEC and P/N. |
| XC | Installation drawings, diagrams, instruction sheets, field service drawings; identified by manufacturer's P/N. |
| XD | Item is not stocked. Order an XD-coded item through local purchase or normal supply channels using the CAGEC and P/N given, if no NSN is available. |

**Note**

Cannibalization or controlled exchange, when authorized, may be used as a source of supply for items with the above source codes except for those items source coded "XA" or those aircraft support items restricted by requirements of AR 750-1.

Maintenance Code. Maintenance codes tell you the level(s) of maintenance authorized to use and repair support items. The maintenance codes are entered in the third and fourth positions of the SMR code as follows:

Third Position. The maintenance code entered in the third position tells you the lowest maintenance class authorized to remove, replace, and use an item. The maintenance code entered in the third position will indicate authorization to the following classes of maintenance:

| **Maintenance Code** | **Application/Explanation** |
|---|---|
| C - | Crew (Operator) |
| F - | Maintainer maintenance can remove, replace, and use the item. |
| H - | Below Depot Sustainment maintenance can remove, replace, and use the item. |
| L - | Specialized repair activity can remove, replace, and use the item. |

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G - Afloat and ashore intermediate maintenance can remove, replace, and use the item (Navy only).

K - Contractor facility can remove, replace, and use the item.

Z - Item is not authorized to be removed, replace, or used at any maintenance level.

D - Depot can remove, replace, and use the item.

**Note**

Army will use C in the third position. However, for multi-service publications, other services may use O.

Fourth Position. The maintenance code entered in the fourth position tells you whether or not the item is to be repaired and identifies the lowest maintenance class with the capability to do complete repair (perform all authorized repair functions).

C - Crew (operator) is the lowest class that can do complete repair.

F - Maintainer is the lowest class that can do complete repair of the item.

H - Below Depot Sustainment is the lowest class that can do complete repair of the item.

L - Specialized repair activity (*enter specialized repair activity designator)* is the lowest class that can do complete repair of the item.

D - Depot is the lowest class that can do complete repair of the item.

G - Both afloat and ashore intermediate levels are capable of complete repair of item. (**Navy only**).

K - Complete repair is done at contractor facility.

Z - Nonreparable.

B - No repair is authorized. No repair is authorized. No parts or special tools are authorized for maintenance of "B" coded item. However, the item may be reconditioned by adjusting, lubricating, etc., at the user level.

Recoverability Code. Recoverability codes are assigned to items to indicate the disposition action on unserviceable items. The recoverability code is shown in the fifth position of the SMR code as follows:

| **Recoverability Code** | **Application/Explanation** |
|---|---|
| C - | Reparable item. When uneconomically reparable, condemn and dispose of the item at the crew/operator level. |
| Z - | Nonreparable item. When unserviceable, condemn and dispose of the item at the level of maintenance shown in the third position of the SMR code. |
| F - | Reparable item. When uneconomically reparable, condemn and dispose of the item at the field level. |
| H - | Reparable item. When uneconomically reparable, condemn and dispose of the item at the below depot sustainment. |
| D - | Reparable item. When beyond lower level repair capability, return to depot. Condemnation and disposal of item are not authorized below depot. |
| L - | Reparable item. Condemnation and disposal not authorized below Specialized Repair Activity (SRA). |

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A - Item requires special handling or condemnation procedures because of specific reasons (such as precious metal content, high dollar value, critical material, or hazardous material). Refer to appropriate manuals/directives for specific instructions.

G - Field level reparable item. Condemn and dispose at either afloat or ashore intermediate levels. (Navy only).

K - Reparable item. Condemnation and disposal to be performed at contractor facility.

NSN (Column (3)). The NSN(s) for the item is listed in this column.

CAGEC (Column (4)). The Commercial and Government Entity Code (CAGEC) is a five-digit code which is used to identify the manufacturer, distributor, or Government agency/activity that supplies the item.

PART NUMBER (Column (5)). Indicates the primary number used by the manufacturer (individual, company, firm, corporation, or Government activity), which controls the design and characteristics of the item by means of its engineering drawings, specifications, standards, and inspection requirements to identify an item or range of items.

**Note**

When you use an NSN to requisition an item, the item you receive may have a different part number from the number listed.

DESCRIPTION AND USABLE ON CODE (UOC) (Column (6)). This column includes the following information:

1. The federal item name, and when required, a minimum description to identify the item.
2. Part numbers of any bulk materials required if the item is to be locally manufactured or fabricated.
3. Hardness Critical Item (HCI). Items that require special handling or procedures to ensure protection against electromagnetic pulse (EMP) damage are marked with the letters ‘HCI.’
4. Refer to Usable on Code details presented later in this work package under SPECIAL INFORMATION.
5. Dot indentions indicate the relationship of the part (or parts) to its next higher assembly (NHA) in the tabular listing. The NHA for this part (or parts) is listed right before the part (or parts) that it is the NHA for. If the item is connected directly to (can be disassembled from) the item identified in the functional group code title for that specific tabular listing, it shall have one dot indentation. Otherwise, that item in the tabular list will not have a dot indention.
6. The statement END OF FIGURE appears below the last item description in column (6) for each Figure in the repair parts list, special tools repair parts, kits, bulk items, and special tools list work packages.

QTY (Column (7)). The QTY (quantity per Figure) column indicates the quantity of the item used in the breakout shown on the illustration/Figure. A "V" appearing in this column instead of a quantity indicates that the quantity is variable and quantity may change from application to application.

*(MC)Include in multiservice manuals involving Marine Corps.*

USMC QTY per Equip (Column (8)). This column indicates the total quantity of the item used on the equipment.

**SPECIAL INFORMATION**

UOC. The UOC appears in the lower left corner of the Description Column heading. Usable on codes are shown as "UOC:" in the Description Column (justified left) on the first line under the applicable item/nomenclature. Un-coded items are applicable to all models. UOCs used in the RPSTL are:

| Code | Used On |
|---|---|
| PAA | Model M114 |

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PAB Model M114A

PAC Model M114B

*Include appropriate UOC content, as applicable.*

Fabrication Instructions. Bulk materials required to manufacture items are listed in the bulk material work package of this RPSTL. Part numbers for bulk material are also referenced in the Description Column of the line item entry for the item to be manufactured/fabricated. Detailed fabrication instructions for items source coded to be manufactured or fabricated are found in (*enter applicable TM number or work package sequence number*).

Index Numbers. Items which have the word BULK in the Figure column will have an index number shown in the item number column. This index number is a cross-reference between the NSN/Part Number (P/N) Index work packages and the bulk material list in the bulk items work package.

*For a combined narrative-RPSTL manual associated publications shall not be included.*

Associated Publications. The publication(s) listed below pertain to the (*enter item name*):

**Publication** **Short Title**

*The following paragraph shall appear only in the field maintenance RPSTL special instructions.*

Illustrations List. The illustrations in this RPSTL contain field authorized items.

Illustrations published in (*enter applicable TM number for the higher maintenance level RPSTL, e.g., for field, below depot sustainment, etc.*) that contain field authorized items also appear in this RPSTL. The tabular list in the repair parts list work package contains only those parts coded "F" in the third position of the SMR code; therefore, there may be a break in the item number sequence.

**HOW TO LOCATE REPAIR PARTS**

First. Using the table of contents, determine the assembly group to which the item belongs. This is necessary since Figures are prepared for assembly groups and subassembly groups, and lists are divided into the same groups.

Second. Find the Figure covering the functional group or the sub functional group to which the item belongs.

Third. Identify the item on the Figure and note the number(s).

Fourth. Look in the repair parts list work packages for the Figure and item numbers. The NSNs and part numbers are on the same line as the associated item numbers.

**ABBREVIATIONS**

**Abbreviation** **Explanation**

*Include all abbreviations used in the RPSTL. "*

## F.8.1.2 Aviation RPSTL introduction.

**"INTRODUCTION**

**SCOPE**

This RPSTL lists the authorized spares and repair parts; special tools; special Test, Measurement, and Diagnostic Equipment (TMDE); and other special support equipment required for performance of maintenance. It authorizes the requisitioning, issue, and disposition of spares, repair parts, and special tools/kits as indicated by the source, maintenance, and recoverability (SMR) codes.

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## GENERAL

In addition to the Introduction work package, this RPSTL is divided into the following work packages.

1. Repair Parts List WPs. These WPs contain lists of spare and repair parts authorized for use in the performance of maintenance at the levels determined by the MAC/SMR code. These WPs also include parts that must be removed for replacement of the authorized parts. Parts lists are composed of functional groups in ascending numeric or alphanumeric sequence, with the parts in each group listed in ascending figure and item number sequence. Sending units, brackets, filters, and bolts are listed with the component upon which they mount. Bulk materials are listed by item name in the Bulk Items WP that follows either: (*select one of the two following items:(1) the last Parts List, Special Tools Repair Parts, or (2) Kits WP*). Repair parts kits are either (*select one of the two following items: (1) listed separately in their own functional group and WP* ***OR*** *(2) listed at the end of the individual WPs*). Repair parts for repairable special tools/kits are also listed in a separate WP. Items listed are shown on the associated illustrations.

**Note**

*Include the text in items 2 through 4 only if the described work package is included in the TM.*

2. Special Tools Repair Parts WP. This WP lists any spare parts required for the special tools/kits, TMDE, or other support equipment listed in the Special Tools WP that are not listed in any other publication.
3. Kits WP. This WP lists all repair kits and their component parts.
4. Bulk Items WP. This WP lists all items identified as "bulk" in the parts lists. Due to the nature of bulk items, this WP does not include a figure.
5. Special Tools/Kits List WPs. This WP lists those special tools/kits, special TMDE, and special support equipment authorized by this RPSTL (as indicated by Basis of Issue (BOI) information in the DESCRIPTION AND USABLE ON CODE (UOC) column). Tools that are components of common tool sets and/or Class VII are not listed.

## EXPLANATION OF COLUMNS IN THE REPAIR PARTS LIST AND SPECIAL TOOLS/KITS LIST WPS

ITEM NO. (Column 1). Indicates the number used to identify items called out in the illustration.

SMR CODE (Column 2). The SMR code contains supply/requisitioning information, maintenance level authorization criteria, and disposition instruction in accordance with AR 700-82, as shown in the following breakout. This entry may be subdivided into 4 subentries, one for each service.

**TABLE 1. SMR Code Explanation.**

| Source Code | Maintenance Code | | Recoverability Code |
|---|---|---|---|
| XX | XX | | XX |
| 1st two positions: How to get an item. | 3rd position: Who can install, replace, or use the item. | 4th position: Who can do complete repair on the item. | 5th position: Who determines disposition action on unserviceable items. |

**Note**

Complete Repair: Maintenance capacity, capability, and authority to perform all corrective maintenance tasks of the "Repair" function in a use/user environment in order to restore serviceability to a failed item.

Source Code. The source code tells you how you get an item needed for maintenance, repair, or overhaul of an end item/equipment. Explanations of source codes follow:


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**TABLE 1. Source Code Explanation.**

| Source Code | Application/Explanation |
|---|---|
| PA<br>PB<br>PC<br>PD<br>PE<br>PF<br>PG<br>PH<br>PR<br>PZ | Stock items; use the applicable NSN to requisition/request items with these source codes. They are authorized to the level indicated by the code entered in the third position of the SMR code.<br>**Note**<br>Items coded "PC" are subject to deterioration. Items coded "PR" or "PZ" are obsolete and may not be able to be ordered like other P-coded items. |
| KD<br>KF<br>KB | Items with these codes are not to be requisitioned/requested individually. They are part of a kit that is authorized to the maintenance level indicated in the third position of the SMR code. The complete kit must be requisitioned and applied. |
| MO - Made at AMC level<br>MF - Made at ASB level<br>ML - Made at TASMG<br>MD - Made at depot<br>MG - **Navy only** | Items with these codes are not to be requisitioned/requested individually. They must be made from bulk material that is identified by the P/N in the DESCRIPTION AND USABLE ON CODE (UOC) entry and listed in the bulk material group work package of the RPSTL. If the item is authorized to you by the third position code of the SMR code, but the source code indicates it is made at higher level, order the item from the higher level of maintenance. |
| AO - Assembled at AMC level<br>AF - Assembled at ASB level<br>AL - Assembled at TASMG<br>AD - Assembled at depot<br>AG - Navy only | Items with these codes are not to be requested/requisitioned individually. The parts that make up the assembled item must be requisitioned or fabricated and assembled at the level of maintenance indicated by the source code. If the third position of the SMR code authorizes you to replace the item, but the source code indicates the item is assembled at a higher level, order the item from the higher level of maintenance. |
| XA | Do not requisition an XA-coded item. Order the next higher assembly. (Refer to NOTE below.) |
| XB | If an item is not available from salvage, order it using the CAGEC and P/N. |
| XC | Installation drawings, diagrams, instruction sheets, field service drawings; identified by manufacturer's P/N. |
| XD | Item is not stocked. Order an XD-coded item through local purchase or normal supply channels using the CAGEC and P/N given if no NSN is available. |


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**Note**

Cannibalization or controlled exchange, when authorized, may be used as a source of supply for items with the above source codes except for those items source coded "XA" or those aircraft support items restricted by requirements of AR 750-1.

Maintenance Code. Maintenance codes tell you the level(s) of maintenance authorized to use and repair support items. The maintenance codes are entered in the third and fourth positions of the SMR code as follows:

Third Position. The maintenance code entered in the third position tells you the lowest maintenance level authorized to remove, replace, and use an item. The maintenance code entered in the third position will indicate authorization to the following levels of maintenance:

| Maintenance Code | Application/Explanation |
|---|---|
| O - | AMC maintenance can remove, replace, and use the item. |
| F - | ASB maintenance can remove, replace, and use the item. |
| L - | TASMG can remove, replace, and use the item. |
| G - | |
| K - | Afloat and ashore intermediate maintenance can remove, replace, and use the item (Navy only). |
| Z - | Contractor facility can remove, replace, and use the item. Item is not authorized to be removed, replaced, or used at any maintenance level. |
| D - | Depot can remove, replace, and use the item. |

Fourth Position. The maintenance code entered in the fourth position tells you whether or not the item is to be repaired and identifies the lowest maintenance level with the capability to do complete repair (perform all authorized repair functions).

| | |
|---|---|
| O - | AMC is the lowest class that can do complete repair of item. |
| F - | ASB is the lowest class that can do complete repair of the item. |
| L - | TASMG is the lowest class that can do complete repair of the item. |
| D - | Depot is the lowest class that can do complete repair of the item. |
| G - | Both afloat and ashore intermediate levels are capable of complete repair of item. (Navy only.) |
| K - | Complete repair is done at contractor facility. |
| Z - | Non-repairable. No repair is authorized. |
| B - | No repair is authorized. No parts or special tools/kits are authorized for maintenance of B-coded item. However, the item may be reconditioned by adjusting, lubricating, etc., at the user level. |

Recoverability Code. Recoverability codes are assigned to items to indicate the disposition action on unserviceable items. The recoverability code is shown in the fifth position of the SMR code as follows:

| Recoverability Code | Application/Explanation |
|---|---|
| Z - | Non-repairable item. When unserviceable, condemn and dispose of the item at the level of maintenance shown in the third position of the SMR code. |

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O - Repairable item. When uneconomically repairable, condemn and dispose of the item at the AMC level.

F - Repairable item. When uneconomically repairable, condemn and dispose of the item at the ASB level.

D - Repairable item. When beyond lower level repair capability, return to depot. Condemnation and disposal of item are not authorized below depot-level.

L - Repairable item. Condemnation and disposal not authorized below TASMG.

A - Item requires special handling or condemnation procedures because of specific reasons (such as precious metal content, high dollar value, critical material, or hazardous material). Refer to appropriate manuals/directives for specific instructions.

G - Field-level repairable item. Condemn and dispose at either afloat or ashore intermediate levels. (Navy only.)

K - Repairable item. Condemnation and disposal to be performed at contractor facility.

NSN (Column 3). The NSN for the item is listed in this column.

CAGEC (Column 4). The Commercial and Government Entity Code (CAGEC) is a five-digit code that is used to identify the manufacturer, distributor, or Government agency/activity that supplies the item.

PART NUMBER (Column 5). Indicates the primary number used by the manufacturer (individual, company, firm, corporation, or Government activity) that controls the design and characteristics of the item by means of its engineering drawings, specifications, standards, and inspection requirements to identify an item or range of items.

**Note**

When you use an NSN to requisition an item, the item you receive may have a different P/N from the number listed.

DESCRIPTION AND USABLE ON CODE (UOC) (Column (6)). This column includes the following information:

**1.** The federal item name, and when required, a minimum description to identify the item.

**2.** P/Ns of bulk materials are referenced in this column in the line entry to be manufactured or fabricated.

**3.** Hardness Critical Item (HCI). A support item that provides the equipment with special protection from Electromagnetic Pulse (EMP) damage during a nuclear attack.

**4.** Refer to Usable on Code details presented later in this work package under SPECIAL INFORMATION.

**5.** Dot indentions indicate the relationship of the part (or parts) to its next higher assembly (NHA) in the tabular listing. The NHA for this part (or parts) is listed right before the part (or parts) for which it is the NHA. If the item is connected directly to (can be disassembled from) the item identified in the functional group code title for that specific tabular listing, it shall have no dot indentation.

**6.** The statement "END OF FIGURE" appears just below the last item description in column (6) for a given figure in both the repair parts list and special tools/kits list work packages.

QTY (Column (7)). The QTY column indicates the quantity of the item used in the breakout shown on the illustration/ figure, which is prepared for a functional group, sub-functional group, or an assembly. A "V" appearing in this column instead of a quantity indicates that the quantity is variable and quantity may change from application to application.

*(MC) Include for multiservice manuals involving the Marine Corps.*

USMC QTY per Equip (Column 8). This column accommodates the Marine Corps quantity per equipment requirement.

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**SPECIAL INFORMATION**

UOC. The UOC appears in the lower left corner of the Description Column heading.

Usable on codes are shown as "UOC" in the Description Entry (justified left) on the first line under the applicable item/nomenclature. Uncoded items are applicable to all models. Identification of the UOCs used in the RPSTL are:

| Code | Used On |
|---|---|
| PAA | Model M114 |
| PAB | Model M114A |
| PAC | Model M114B |

*Include appropriate UOC content, as applicable.*

Fabrication Instructions. Bulk materials required to manufacture items are listed in the bulk material functional group of this RPSTL. Part numbers for bulk material are also referenced in the Description Column of the line item entry for the item to be manufactured/fabricated. Detailed fabrication instructions for items source coded to be manufactured or fabricated are found in (*enter applicable TM number or work package sequence number*).

Index Numbers. Items that have the word "BULK" in the figure column will have an index number shown in the item number column. This index number is a cross-reference between the NSN / P/N index work packages and the bulk material list in the repair parts list work package.

*For a combined narrative-RPSTL manual, associated publications shall not be included.*

Associated Publications. The publication or publications listed below pertains to the (*enter item name*):

| Publication | Short Title |
|---|---|

*The following paragraph shall appear only in the unit maintenance RPSTL special instructions.*

Illustrations List. The illustrations in this RPSTL contain field authorized items. Illustrations published in (*enter applicable TM number for the higher maintenance level RPSTL, e.g., for field, below depot sustainment, etc.*) that contain field authorized items also appear in this RPSTL. The tabular list in the repair parts list work package contains only those parts coded "O" in the third position of the SMR code; therefore, there may be a break in the item number sequence.

**HOW TO LOCATE REPAIR PARTS**

First. Using the table of contents, determine the assembly group to which the item belongs. This is necessary since figures are prepared for assembly groups and subassembly groups, and lists are divided into the same groups.

Second. Find the figure covering the functional group or the sub-functional group to which the item belongs.

Third. Identify the item on the figure and note the number(s).

Fourth. Look in the repair parts list work packages for the figure and item numbers. The NSNs and part numbers are on the same line as the associated item numbers.

**ABBREVIATIONS**

| Abbreviations | Explanation |
|---|---|

*Include all abbreviations used in the RPSTL."*

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## F.8.1.3 Marine Corps (MC) RPSTL introduction.

The following introduction shall be used for RPSTL manuals for Marine Corps:

**"INTRODUCTION**

**SCOPE**
This RPSTL lists the authorized spares and repair parts; special tools; special Test, Measurement, and Diagnostic Equipment (TMDE); and other special support equipment required for performance of maintenance. It authorizes the requisitioning, issue, and disposition of spares, repair parts, and special tools/kits as indicated by the source, maintenance, and recoverability (SMR) codes.

**GENERAL**
In addition to the Introduction work package, this RPSTL is divided into the following work packages.

1. Repair Parts List WPs. These WPs contain lists of spare and repair parts authorized for use in the performance of maintenance at the levels determined by the MAC/SMR code. These WPs also include parts that must be removed for replacement of the authorized parts. Parts lists are composed of functional groups in ascending numeric or alphanumeric sequence, with the parts in each group listed in ascending figure and item number sequence. Sending units, brackets, filters, and bolts are listed with the component upon which they mount. Bulk materials are listed by item name in the Bulk Items WP that follows either: (*select one of the two following items:(1) the last Parts List, Special Tools Repair Parts, or (2) Kits WP*). Repair parts kits are either (*select one of the two following items: (1) listed separately in their own functional group and WP* ***OR*** *(2) listed at the end of the individual WPs*). Repair parts for repairable special tools/kits are also listed in a separate WP. Items listed are shown on the associated illustrations.

**Note**

*Include the text in items 2 through 4 only if the described work package is included in the TM.*

2. Special Tools Repair Parts WP. This WP lists any spare parts required for the special tools/kits, TMDE, or other support equipment listed in the Special Tools WP that are not listed in any other publication.
3. Kits WP. This WP lists all repair kits and their component parts.
4. Bulk Items WP. This WP lists all items identified as "bulk" in the parts lists. Due to the nature of bulk items, this WP does not include a figure.
5. Special Tools/Kits List WPs. This WP lists those special tools/kits, special TMDE, and special support equipment authorized by this RPSTL (as indicated by Basis of Issue (BOI) information in the DESCRIPTION AND USABLE ON CODE (UOC) column). Tools that are components of common tool sets and/or Class VII are not listed.

**EXPLANATION OF COLUMNS IN THE REPAIR PARTS LIST AND SPECIAL TOOLS/KITS LIST WPS**

ITEM NO. (Column 1). Indicates the number used to identify items called out in the illustration.

SMR CODE (Column 2). The SMR code contains supply/requisitioning information, maintenance level authorization criteria, and disposition instruction in accordance with MCO 4400.120, as shown in the following breakout. This entry may be subdivided into 4 subentries, one for each service.

**TABLE 1. SMR Code Explanation.**

| Source Code | Maintenance Code | Recoverability Code |
|---|---|---|
| XX | XX | XX |

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| 1st two positions: How to get an item. | 3rd position: Who can install, replace, or use the item. | 4th position: Who can do complete repair on the item. | 5th position: Who determines disposition action on unserviceable items. |
|---|---|---|---|

**Note**

Complete Repair: Maintenance capacity, capability, and authority to perform all corrective maintenance tasks of the "Repair" function in a use/user environment in order to restore serviceability to a failed item.

Source Code. The source code tells you how you get an item needed for maintenance, repair, or overhaul of an end item/equipment. Explanations of source codes follow:

**TABLE 1. SMR Code Explanation.**

| Source Code | Application/Explanation |
|---|---|
| PA<br>PB<br>PC<br>PD<br>PE<br>PF<br>PG<br>PH<br>PR<br>PZ | Stock items; use the applicable NSN to requisition/request items with these source codes. They are authorized to the level indicated by the code entered in the third position of the SMR code.<br>**Note**<br>Items coded "PC" are subject to deterioration. Items coded "PR" or "PZ" are obsolete and may not be able to be ordered like other P-coded items. |
| KD<br>KF<br>KB | Items with these codes are not to be requested/requisitioned individually. They are part of a kit that is authorized to the maintenance level indicated in the third position of the SMR code. The complete kit must be requisitioned and applied. |
| MO-Made at Field/Organizational level<br>MF-Made at Field/Intermediate level<br>MH- Field/Intermediate level<br>ML-Made at Specialized Maintenance Facility<br>MD-Made at depot MG-Navy only | Items with these codes are not to be requisitioned/requested individually. They must be made from bulk material that is identified by the P/N in the DESCRIPTION AND USABLE ON CODE (UOC) entry and listed in the bulk material group work package of the RPSTL. If the item is authorized to you by the third position code of the SMR code, but the source code indicates it is made at higher level, order the item from the higher level of maintenance. |
| AO-Assembled at Field/Organizational level | Items with these codes are not to be requested/requisitioned individually. The parts that make up the assembled item must be requisitioned or fabricated and assembled at the level of maintenance indicated by the source code. If the third position of the SMR code authorizes you to replace the |


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| | |
|---|---|
| AF-Assembled at Field/Intermediate level | item, but the source code indicates the item is assembled at a higher level, order the item from the higher level of maintenance. |
| AH -Assembled at Field/Intermediate level | |
| AL-Assembled at Specialized Maintenance Facility | |
| AD-Assembled at depot | |
| AG-Navy only | |
| XA | Do not requisition an XA-coded item. Order the next higher assembly. (Refer to NOTE below.) |
| XB | If an item is not available from salvage, order it using the CAGEC and P/N. |
| XC | Installation drawings, diagrams, instruction sheets, field service drawings; identified by manufacturer's P/N. |
| XD | Item is not stocked. Order an XD-coded item through local purchase or normal supply channels using the CAGEC and P/N given, if no NSN is available. |

**Note**

Cannibalization or controlled exchange, when authorized, may be used as a source of supply for items with the above source codes except for those items source coded "XA" or those aircraft support items restricted by requirements of AR 750-1.

Maintenance Code. Maintenance codes tell you the level(s) of maintenance authorized to use and repair support items. The maintenance codes are entered in the third and fourth positions of the SMR code as follows:

Third Position. The maintenance code entered in the third position tells you the lowest maintenance level authorized to remove, replace, and use an item. The maintenance code entered in the third position will indicate authorization to the following levels of maintenance:

| **Maintenance Code** | **Application/Explanation** |
|---|---|
| O - | Field/Organizational level maintenance can remove, replace, and use the item. |
| F - | Field/Intermediate maintenance can remove, replace, and use the item. |
| H - | Field/Intermediate maintenance can remove, replace, and use the item. |
| L - | Specialized Maintenance Facility can remove, replace, and use the item. |
| G - | Afloat and ashore intermediate maintenance can remove, replace, and use the item (Navy only). |
| K - | Contractor facility can remove, replace, and use the item. |
| Z - | Item is not authorized to be removed, replace, or used at any maintenance level. |
| D - | Depot can remove, replace, and use the item. |

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Fourth Position. The maintenance code entered in the fourth position tells you whether or not the item is to be repaired and identifies the lowest maintenance level with the capability to do complete repair (perform all authorized repair functions).

| Maintenance Code | Application/Explanation |
|---|---|
| O - | Field/Organizational level is the lowest class that can do complete repair of item. |
| F - | Field/Intermediate is the lowest class that can do complete repair of the item. |
| H - | Field/Intermediate is the lowest class that can do complete repair of the item. |
| L - | Specialized Maintenance Facility is the lowest class that can do complete repair of the item. |
| D - | Depot is the lowest class that can do complete repair of the item. |
| G - | Both afloat and ashore intermediate levels are capable of complete repair of item. (Navy only.) |
| K - | Complete repair is done at contractor facility. |
| Z - | Non-repairable. No repair is authorized. |
| B - | No repair is authorized. No parts or special tools/kits are authorized for maintenance of a B-coded item. However, the item may be reconditioned by adjusting, lubricating, etc., at the user level. |

Recoverability Code. Recoverability codes are assigned to items to indicate the disposition action on unserviceable items. The recoverability code is shown in the fifth position of the SMR code as follows:

| Recoverability Code | Application/Explanation |
|---|---|
| Z - | Non-repairable item. When unserviceable, condemn and dispose of the item at the level of maintenance shown in the third position of the SMR code. |
| O - | Repairable item. When uneconomically repairable, condemn and dispose of the item at the Field/Organizational level. |
| F - | Repairable item. When uneconomically repairable, condemn and dispose of the item at the Field/Intermediate level. |
| H - | Repairable item. When uneconomically repairable, condemn and dispose of the item at the Field/Intermediate level. |
| D - | Repairable item. When beyond lower level repair capability, return to depot. Condemnation and disposal of item are not authorized below depot-level. |
| L - | Repairable item. Condemnation and disposal not authorized below Specialized Maintenance Facility. |
| A - | Item requires special handling or condemnation procedures because of specific reasons (such as precious metal content, high dollar value, critical material, or hazardous material). Refer to appropriate manuals/directives for specific instructions. |
| G - | Field-level repairable item. Condemn and dispose at either afloat or ashore intermediate levels. (Navy only.) |
| K - | Repairable item. Condemnation and disposal to be performed at contractor facility. |

NSN (Column 3). The NSN for the item is listed in this column.

CAGEC (Column 4). The Commercial and Government Entity Code (CAGEC) is a five-digit code that is used to identify the manufacturer, distributor, or Government agency/activity that supplies the item.

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PART NUMBER (Column 5). Indicates the primary number used by the manufacturer (individual, company, firm, corporation, or Government activity) that controls the design and characteristics of the item by means of its engineering drawings, specifications, standards, and inspection requirements to identify an item or range of items.

**Note**

When you use an NSN to requisition an item, the item you receive may have a different P/N from the number listed.

DESCRIPTION AND USABLE ON CODE (UOC) (Column (6)). This column includes the following information:

1. The federal item name, and when required, a minimum description to identify the item.
2. P/Ns of bulk materials are referenced in this column in the line entry to be manufactured or fabricated.
3. Hardness Critical Item (HCI). A support item that provides the equipment with special protection from electromagnetic pulse (EMP) damage during a nuclear attack.
4. Refer to Usable on Code details presented later in this work package under SPECIAL INFORMATION.
5. Dot indentions indicate the relationship of the part (or parts) to its next higher assembly (NHA) in the tabular listing. The NHA for this part (or parts) is listed right before the part (or parts) for which it is the NHA. If the item is connected directly to (can be disassembled from) the item identified in the functional group code title for that specific tabular listing, it shall have one dot indentation. Otherwise, that item in the tabular list will not have a dot indention.
6. The statement "END OF FIGURE" appears just below the last item description in column (6) for a given figure in both the repair parts list and special tools/kits list work packages.

QTY (Column (7)). The QTY (quantity per figure) column indicates the quantity of the item used in the breakout shown on the illustration/figure, which is prepared for a functional group, sub-functional group, or an assembly. A "V" appearing in this column instead of a quantity indicates that the quantity is variable and quantity may change from application to application.

**SPECIAL INFORMATION**

UOC. The UOC appears in the lower left corner of the Description Column heading.

Usable on codes are shown as "UOC" in the Description Entry (justified left) on the first line under the applicable item/nomenclature. Uncoded items are applicable to all models. Identification of the UOCs used in the RPSTL are:

| Code | Used On |
|---|---|
| PAA | Model M114 |
| PAB | Model M114A |
| PAC | Model M114B |

*Include appropriate UOC content, as applicable.*

Fabrication Instructions. Bulk materials required to manufacture items are listed in the bulk material functional group of this RPSTL. Part numbers for bulk material are also referenced in the Description Column of the line item entry for the item to be manufactured/fabricated. Detailed fabrication instructions for items source coded to be manufactured or fabricated are found in (*enter applicable TM number or work package sequence number*).

Index Numbers. Items that have the word "BULK" in the figure column will have an index number shown in the item number column. This index number is a cross-reference between the NSN / P/N index work packages and the bulk material list in the repair parts list work package.

*For a combined narrative-RPSTL manual, associated publications shall not be included.*

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Associated Publications. The publication or publications listed below pertains to the (*enter item name*):

| Publication | Short Title |
|---|---|

*The following paragraph shall appear only in the unit maintenance RPSTL special instructions.*

Illustrations List. The illustrations in this RPSTL contain field authorized items. Illustrations published in (enter applicable TM number for the higher maintenance level RPSTL, e.g., for field, below depot sustainment, etc.) that contain field authorized items also appear in this RPSTL. The tabular list in the repair parts list work package contains only those parts coded "O" in the third position of the SMR code; therefore, there may be a break in the item number sequence.

**HOW TO LOCATE REPAIR PARTS**

First. Using the table of contents, determine the assembly group to which the item belongs. This is necessary since figures are prepared for assembly groups and subassembly groups, and lists are divided into the same groups.

Second. Find the figure covering the functional group or the sub-functional group to which the item belongs.

Third. Identify the item on the figure and note the number(s).

Fourth. Look in the repair parts list work packages for the figure and item numbers. The NSNs and part numbers are on the same line as the associated item numbers.

**ABBREVIATIONS**

| Abbreviation | Explanation |
|---|---|

*Include all abbreviations used in the RPSTL.".*

## F.8.1.4 Indexed RPSTL illustration and legend <figure>.

When specified by the MATDEV, an indexed RPSTL illustration and legend shall be added to the end of the introduction WP. These additions help the user understand major system components organized in the RPSTL under Functional Group Codes (FGCs). The illustration shall have a legend that defines the item number, major functional group figure title, and the respective figure number. FIGURE F-8 shows example illustration and legend. The whole-system illustration is shown in this example, and contains callouts pointing to significant subsystems. The legend lists item numbers that directly correlate to callouts in the illustration. Each item number has a corresponding figure title (with FGC in parentheses) and figure number identifying subsequent illustrations in this introduction WP. RPSTL illustrations may contain one or multiple graphics depending on system size and complexity.



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**Note**

Figure appearance may vary slightly based on version of DTD/stylesheets utilized.

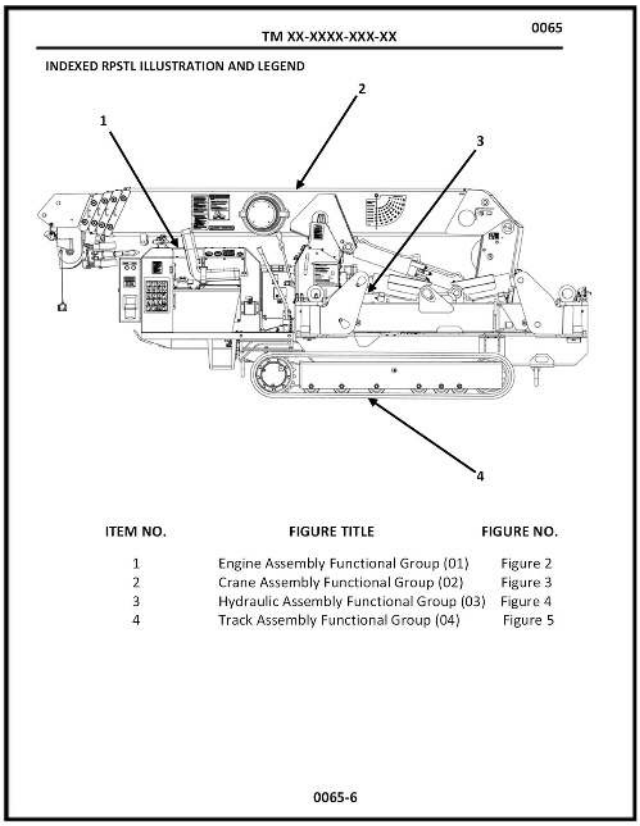


**FIGURE F-8. Indexed RPSTL illustration and legend example.**

## F.8.2 RPSTL parts list work package <plwp>.

A stand-alone parts information technical publication or a parts information module in a combined manual shall contain at least one repair parts list WP. For less complex equipment, the RPSTL may be contained in a single WP **<plwp>** or a few WPs **<plwp>**. Each RPSTL WP shall have at least one figure (illustration) and one parts list; however, the figure for complex equipment may have multiple sheets. See FIGURE F-9 through FIGURE F-13 for RPSTL WP components and associated tags. The following components shall be included in a RPSTL WP.

**a.** Work package identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** Initial setup **<initial_setup>** is not required for this WP.

**c.** Parts information as described in F.8.2.1 through F.8.2.1.10.



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**Note**

Figure appearance and tags/output may vary slightly based on version of DTD/stylesheets utilized (see applicable DTD or tag definition list for available tags).

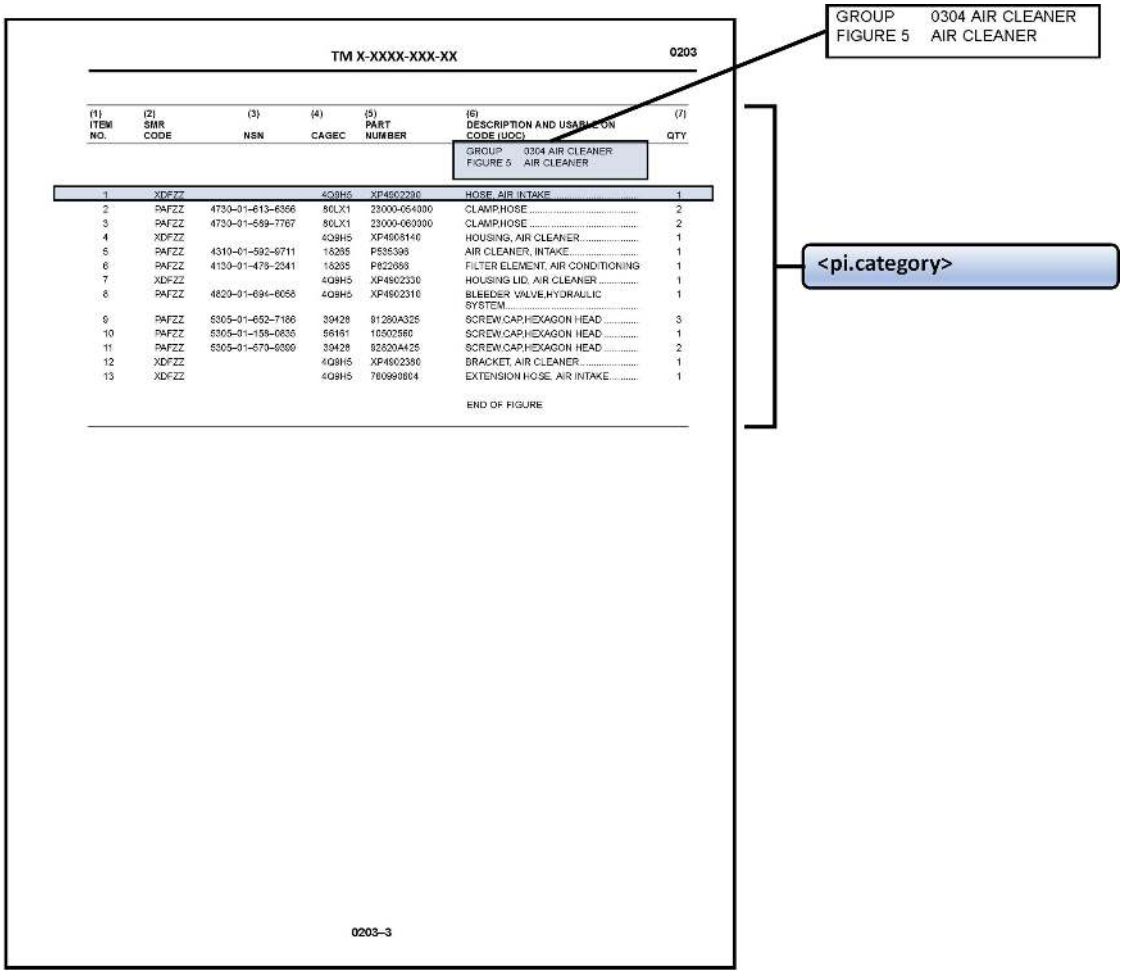


**FIGURE F-9. RPSTL WP example.**

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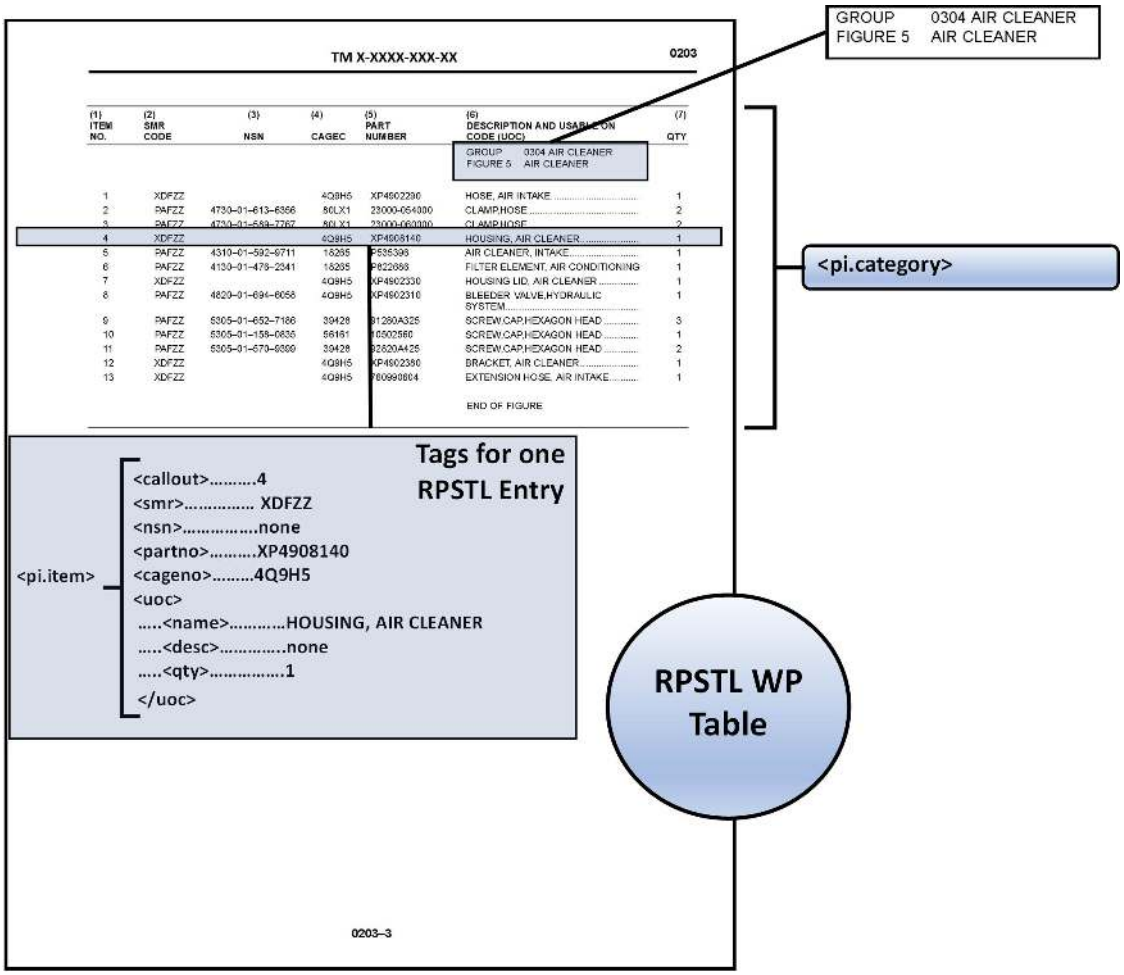


**FIGURE F-10. RPSTL WP example (with tagging details).**



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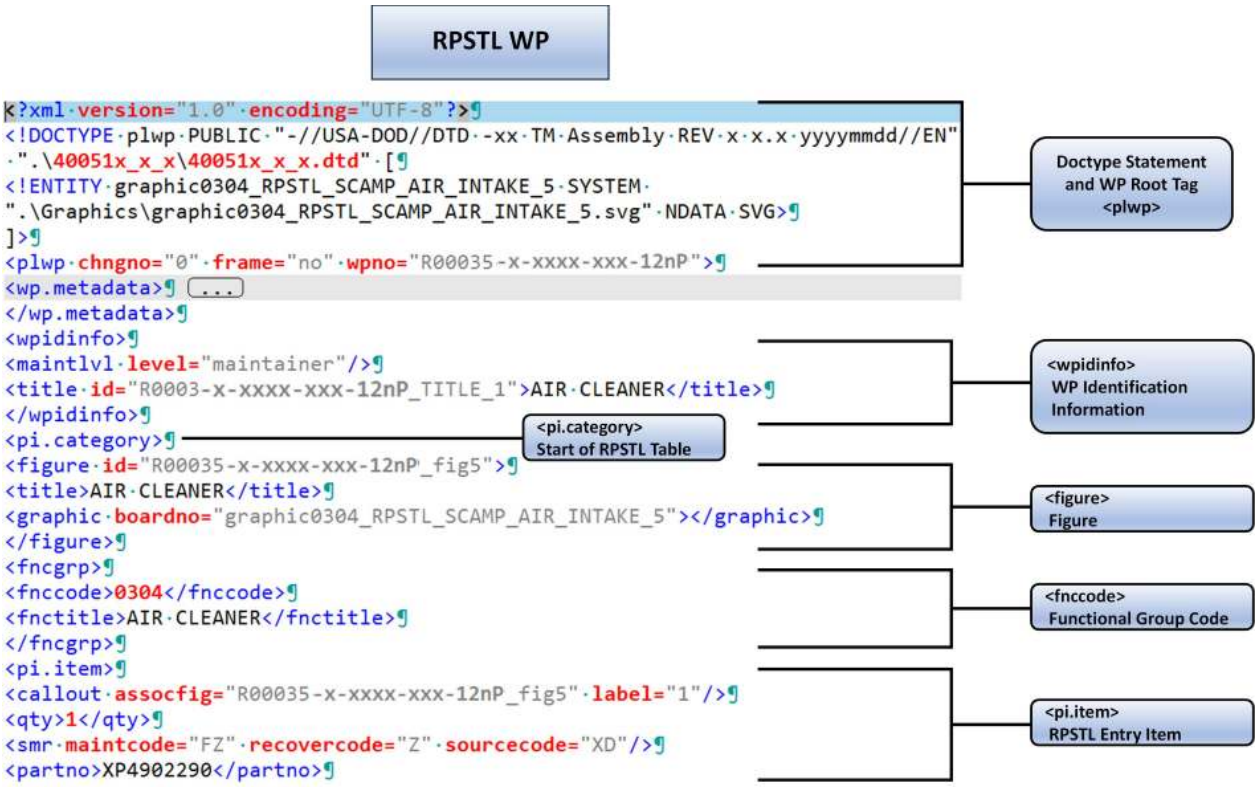


**FIGURE F-11. RPSTL WP tags.**

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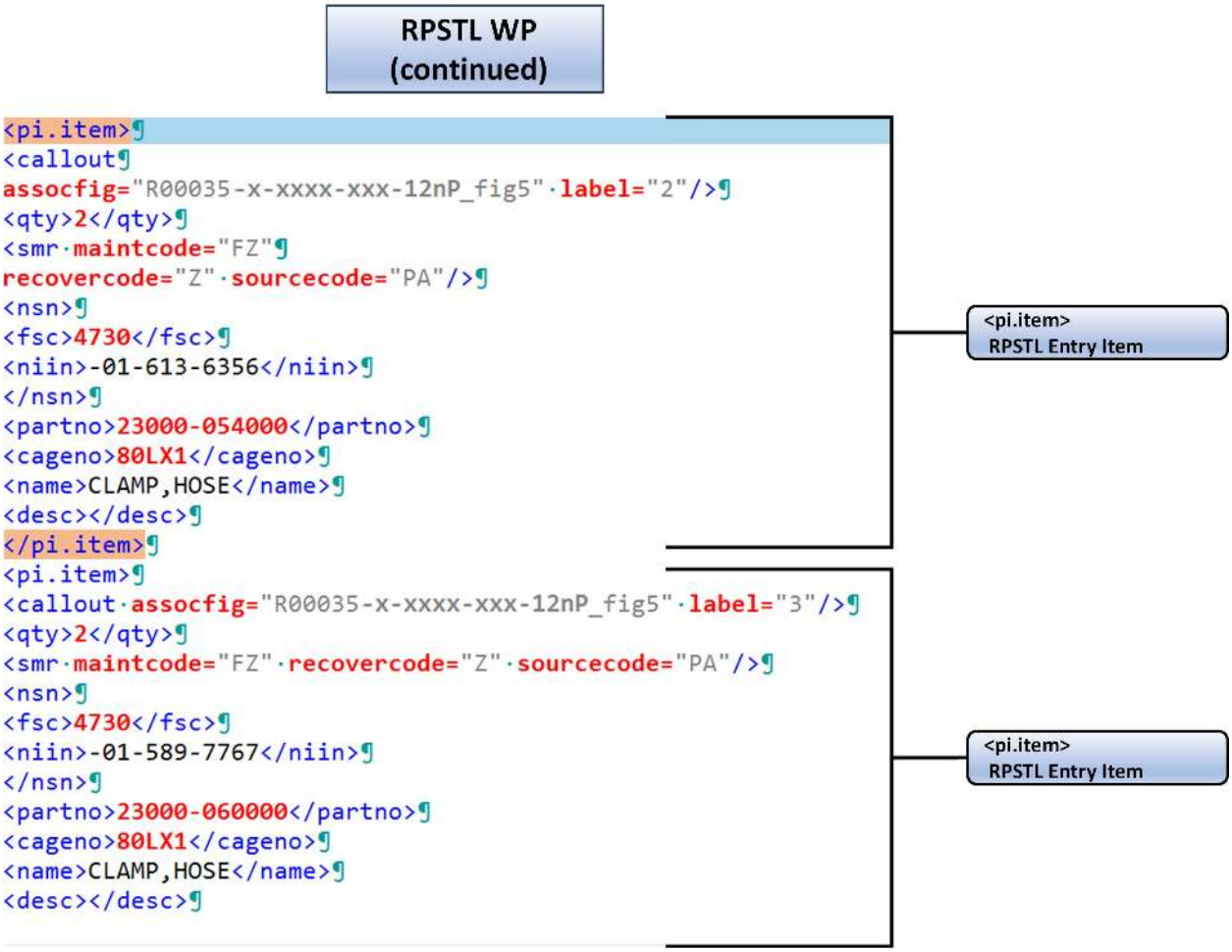


**FIGURE F-12. RPSTL WP tags (sheet 1 of 2).**

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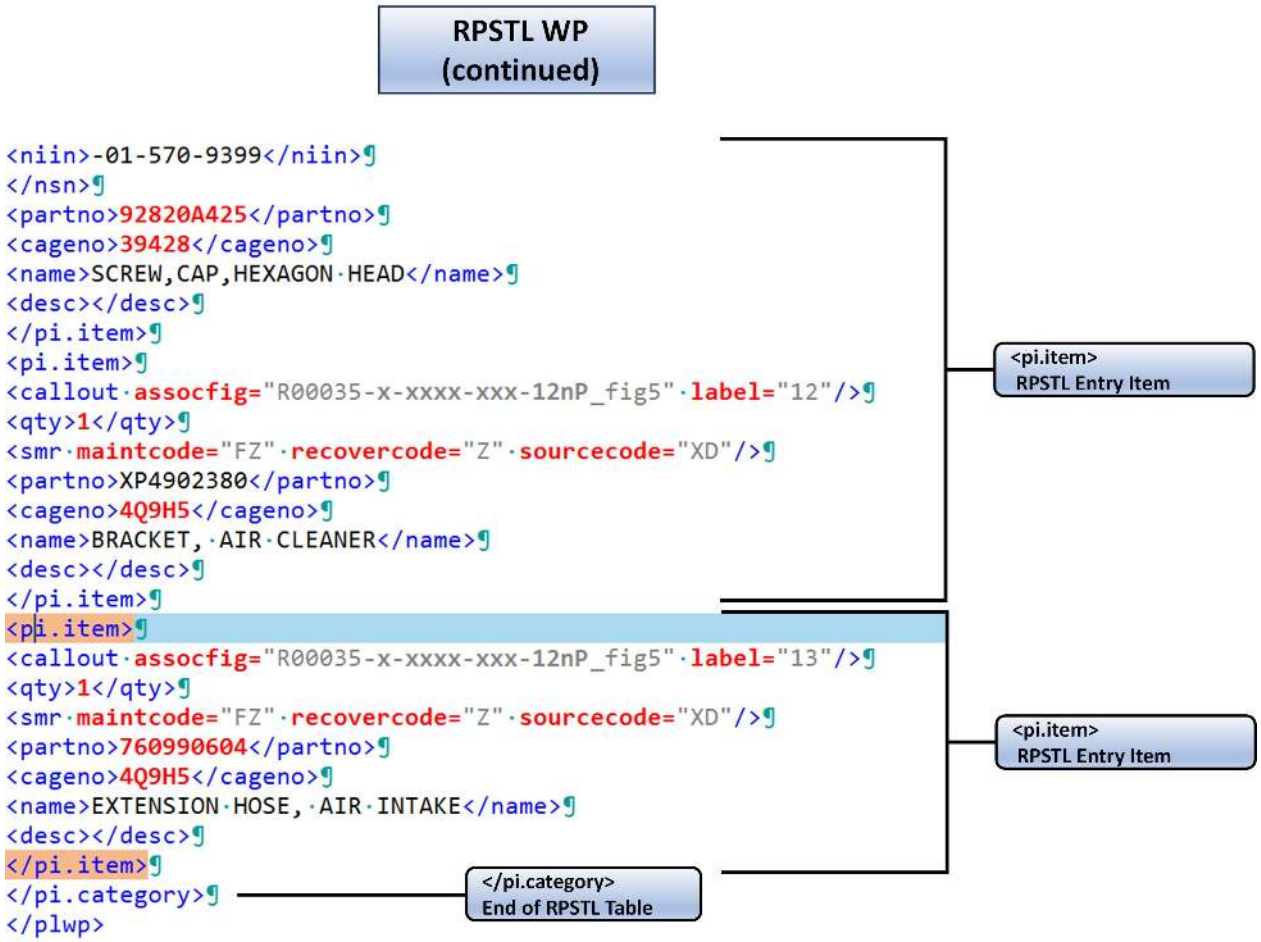


**FIGURE F-13. RPSTL WP tags (sheet 2 of 2).**

### F.8.2.1 Repair parts list <pi.category>.

The repair parts list (tags shown in FIGURE F-10 through FIGURE F-13) shall have a figure **<figure>** and a list of repair part items **<pi.item>** as specified in F.8.2.1.1 through F.8.2.1.12. The repair parts list has standard format information per 4.8. Only one repair parts list **<pi.category>** shall appear in a RPSTL WP **<plwp>**.

### F.8.2.1.1 Repair parts figure title <title>.

When available, figure titles (tags shown in FIGURE F-10 through FIGURE F-13) shall be taken from provisioning documentation. The RPSTL figure title, the functional group title, and the applicable MAC title shall be the same. When no provisioning documentation is provided, the MATDEV or contractor shall develop a title. This title shall be used consistently throughout the publication.

### F.8.2.1.2 Functional group header <fncgrp>.

The functional group header (tags shown in FIGURE F-10 through ) shall precede the first repair part item in the DESCRIPTION AND USABLE ON CODE (UOC) column (FIGURE F-15, Item 6). The header shall consist of the functional group number and title **<fnccode>** appearing on the top line(s). The line(s) below the functional group header shall include the figure number and the figure title **<fnctitle>**. The functional group codes shall not exceed 11 characters. The FGCs in the RPSTL shall match the FCGs in the MAC.

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### F.8.2.1.3 Repair part item <pi.item>.

Each repair part (tags shown in FIGURE F-10 through FIGURE F-13) shall include the column requirements in F.8.2.1.4 thru F.8.2.1.11. Each repair part may also include the optional items in F.8.1.2.

### F.8.2.1.4 Item number column <callout>.

Callouts (tags shown in FIGURE F-10 through FIGURE F-13) shall be listed on the repair parts list in the ITEM NO. column (FIGURE F-15, Item 1) by the same callout number shown on the associated figure. The items shall be listed in ascending sequence by item number.

### F.8.2.1.5 SMR code column <smr>.

The SMR CO. column (tags shown in FIGURE F-10 through FIGURE F-13) shall include SMR codes assigned to the applicable items. For multiple service technical publications, the SMR CO. column (FIGURE F-15, Item 2) shall be divided into subcolumns, one for each service involved (see FIGURE F-14). Each service shall identify the appropriate SMR code subentry. When services share the same SMR code for an item, the SMR code shall be listed for each service.


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**Note**

Figure appearance may vary slightly based on version of DTD/stylesheets utilized (see applicable DTD or tag definition list for available tags).

| (1) ITEM | (2) SMR CODE | | | | (3) NSN | (4) CAGEC | (5) PART | (6) DESCRIPTION AND | (7) QTY | (8) USMC |
|---|---|---|---|---|---|---|---|---|---|---|
| NO | a. ARMY | b. AIR FORCE | c. NAVY | d. USMC | | | NUMBER | USABLE ON CODE (UOC) | | QTY PER EQUIP |
| 1 | PACZZ | PAOZZ | PAOZZ | PAOZZ | 5310-01-012-3595 | 81205 | 2740-0003 | . NUT, PLAIN, ASSEMBLED | 7 | 374 |
| 2 | PACZZ | PAOZZ | PAOZZ | PAOZZ | XXXX-01-XXX-XXXX | 19207 | 12325869 | . BOLT, MACHINE | 6 | 300 |
| 3 | PACZZ | PAOZZ | PAOZZ | PAOZZ | XXXX-01-XXX-XXXX | 96906 | MS35207-269 | . SCREW, MACHINE | 1 | 1 |
| 4 | XDCZZ | XB | XBOZZ | XBOZZ | | 30554 | 88-20036 | . SPACER, TRAY, OUTPUT | 1 | 1 |
| 5 | PACZZ | PAOZZ | PAOZZ | PAOZZ | XXXX-01-XXX-XXXX | 96906 | MS27183-42 | . WASHER, FLAT | 1 | 1 |
| 6 | XDCCC | XB | XBOZZ | XBOZZ | | 30554 | 88-20036 | . OUTPUT BOX ASSEMBLY SEE FIGURE 13 FOR BREAKDOWN | 1 | 1 |
| 7 | PACZZ | PAOZZ | PAOZZ | PAOZZ | XXXX-01-XXX-XXXX | 45722 | P15121-64 | . . SCREW, ASSEMBLED, WAS | 2 | 25 |
| 8 | PACZZ | PAOZZ | PAOZZ | PAOZZ | XXXX-01-XXX-XXXX | 81205 | 2740-0003 | . . NUT, PLAIN, ASSEMBLED | 10 | 374 |
| 9 | PACZZ | PAOZZ | PAOZZ | PAOZZ | XXXX-01-XXX-XXXX | 96906 | 88-20314-4 | . . CLAMP, LOOP | 2 | 7 |
| 10 | PACZZ | PAOZZ | PAOZZ | PAOZZ | XXXX-01-XXX-XXXX | 45722 | P15121-67 | . . SCREW, ASSEMBLED, WAS | 9 | 9 |
| 11 | XDCZZ | XB | XBOZZ | XBOZZ | | 9R803 | 4300-12-XP-74 | . . MARKER STRIP, TERMINAL | 1 | 1 |
| 12 | XDCZZ | XB | XBOZZ | XBOZZ | | 9R803 | 3300-14-XP-74 | . . MARKER STRIP, TERMINAL | 1 | 1 |
| 13 | PACZZ | PAOZZ | PAOZZ | PAOZZ | XXXX-01-XXX-XXXX | 78189 | 511-081800-00 | . . NUT, PLAIN, ASSEMBLED | 2 | 8 |
| 14 | PACZZ | PAOZZ | PAOZZ | PAOZZ | XXXX-01-XXX-XXXX | 45722 | P15121-37 | . . SCREW, ASSEMBLED, WAS | 2 | 6 |
| 15 | PACZZ | PAOZZ | PAOZZ | PAOZZ | XXXX-01-XXX-XXXX | 96906 | MS27183-42 | . . WASHER, FLAT | 6 | 87 |

**FIGURE F-14. Example of a multi-service RPSTL.**

### F.8.2.1.6 NSN column <nsn>.

The NSN column (tags shown in FIGURE F-10 through FIGURE F-13) shall include the NSN (FIGURE F-15, Item 3) assigned to the applicable item. All P-coded items shall have NSNs.

### F.8.2.1.7 Commercial and Government Entity Code (CAGEC) column <cageno>.

The applicable five-digit CAGEC number (tags shown in FIGURE F-10 through FIGURE F-13), found at https://cage.dla.mil/, shall appear in the CAGEC column (FIGURE F-15, Item 4).

### F.8.2.1.8 Part number column <partno>.

Each assigned part number (tags shown in FIGURE F-10 through FIGURE F-13) shall be listed in the PART NUMBER column (FIGURE F-15, Item 5). When multiple part numbers exist for a single item (for example, an

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end-item design number and a subsidiary suppliers number), the part number column shall list the manufacturer's number. The subsidiary identification information shall be included in the description column.

### F.8.2.1.9 Description and Usable On Code (UOC) column <uoc>.

The DESCRIPTION AND USABLE ON CODE column (FIGURE F-15, Item 6) shall include the following information:

**a.** Item name **<name>**. The item name (tags shown in FIGURE F-10 through FIGURE F-13) shall consist of the official nomenclature (refer to 4.16.2). If the item is an HCI or ESD item, the symbol HCI and/or ESD shall precede the item name.

**b.** Description **<desc>**. The description (tags shown in FIGURE F-10 through FIGURE F-13) shall consist of the data from the provisioning document. The description may also contain other information to assist in identifying the item. This includes, but is not limited to, original manufacturer's part number, Military Specification part numbers, or specific physical information about the item.

**c.** Indentions. The item name listed in the DESCRIPTION AND USABLE ON CODE (UOC) column shall be indented using dots (FIGURE F-15, right side) to show the parts relationship within the figure. No more than 5 indentures (dots) shall be used. For tagging purposes, indentions may be added by assigning an indent attribute to a **<pi.item>** tag.

**Note**

Figure appearance may vary slightly based on version of DTD/stylesheets utilized (see applicable DTD or tag definition list for available tags).

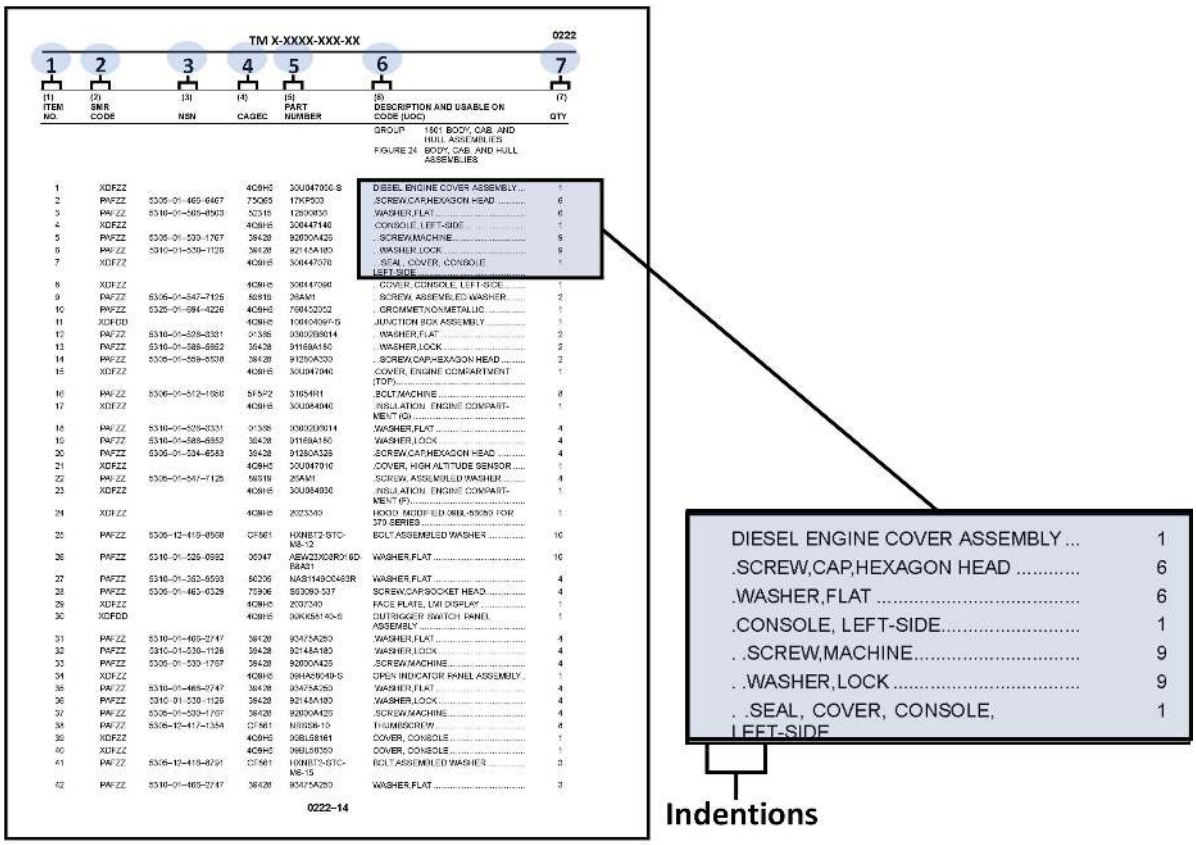


**FIGURE F-15. RPSTL parts list example.**

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**d.** UOC **<uoc>**. When an item has multiple configurations or multiple models, the three-position alphanumeric UOC, representing the applicable configuration in which the item is used, shall be placed on the last line under the item description. The letters "UOC" followed by the applicable UOC shall be indented. When an item is used on all configurations or when only one configuration is covered by the RPSTL, UOCs shall not be shown.

**e.** Serial number application **<usbefserno>**. When part numbers of spare/repair items are not the same for all serial numbered equipment of the same model, a statement identifying the Usable Effective (USBL EFF) serial numbers shall be placed on the last line under the item description. The letters "USBL EFF" followed by the applicable serial numbers shall be indented (for example, USBL EFF SER NOS 1719-1941). When an item is used on all models or when only one configuration is covered by the RPSTL, serial numbers shall not be shown.

**f.** Assembled items. Spare and repair parts that are part of a non-stocked assembled item (source coded "AO," "AF," "AH," "AL," or "AD") shall be assigned item numbers on illustrations and shall be listed in item number sequence on the repair parts list. These items/parts shall be listed immediately below the item to be assembled on the repair parts list. When a particular illustration does not show the parts breakdown of the non-stocked assembly, reference shall be made to the breakdown illustration in the RPSTL. Instructions, drawings, charts, and tables showing how to assemble assemblies source coded "(A)" shall not appear in the RPSTL, but shall appear in the list of manufactured items (APPENDIX E) or by reference to the applicable assembled items maintenance publication if one is available.

**g.** Manufactured items. All items source coded "MO," "MF," "MH," "ML," or "MD" shall have the statement in the DESCRIPTION AND USABLE ON CODE (UOC) column **<desc>** as follows: "MAKE FROM (*enter applicable bulk material or other replaceable item name, CAGEC, and part number*)." Material that is used to make items shall also be shown in a separate bulk items WP **<bulk_itemswp>** (F.8.5). Instructions, drawings, charts, and tables required to show how items are made shall not be contained in the RPSTL but shall appear in the illustrated list of manufactured items WP **<manuwp>** (E.7.8).

**h.** Kits and kit repair parts. Kits and repair parts shall conform to the format of either option 1 or option 2, as specified by the MATDEV. Only one option is to be used in a weapons systems RPSTL listing:

**(1)** Option 1 (kits):

**(a)** Kits shall appear at the end of the associated parts list (see FIGURE F-16). As specified by the MATDEV, the ITEM NO. column **<callout>** for kits shall be either left blank or contain alphabetical character(s). The QTY column **<qty>** for kits shall be a "V" (variable) when the exact quantity may vary.

**(b)** Parts **<kititem>**. Option 1 kit repair parts shall be listed with their applicable figure and appear in item number sequence. The statement "part of Kit P/N (*enter kit part number*)" shall follow item name **<name>**. Kit repair parts shall also be listed under the kit list at the end of the parts list. Parts of the kit list shall be indented and listed alphabetically by item name or in item number sequence immediately below the kit item name. The quantity **<qty>** (in parentheses), figure number, and item number **<callout>** shall follow the repair part item name.

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**Note**

Figure appearance may vary slightly based on version of DTD/stylesheets utilized (see applicable DTD or tag definition list for available tags).

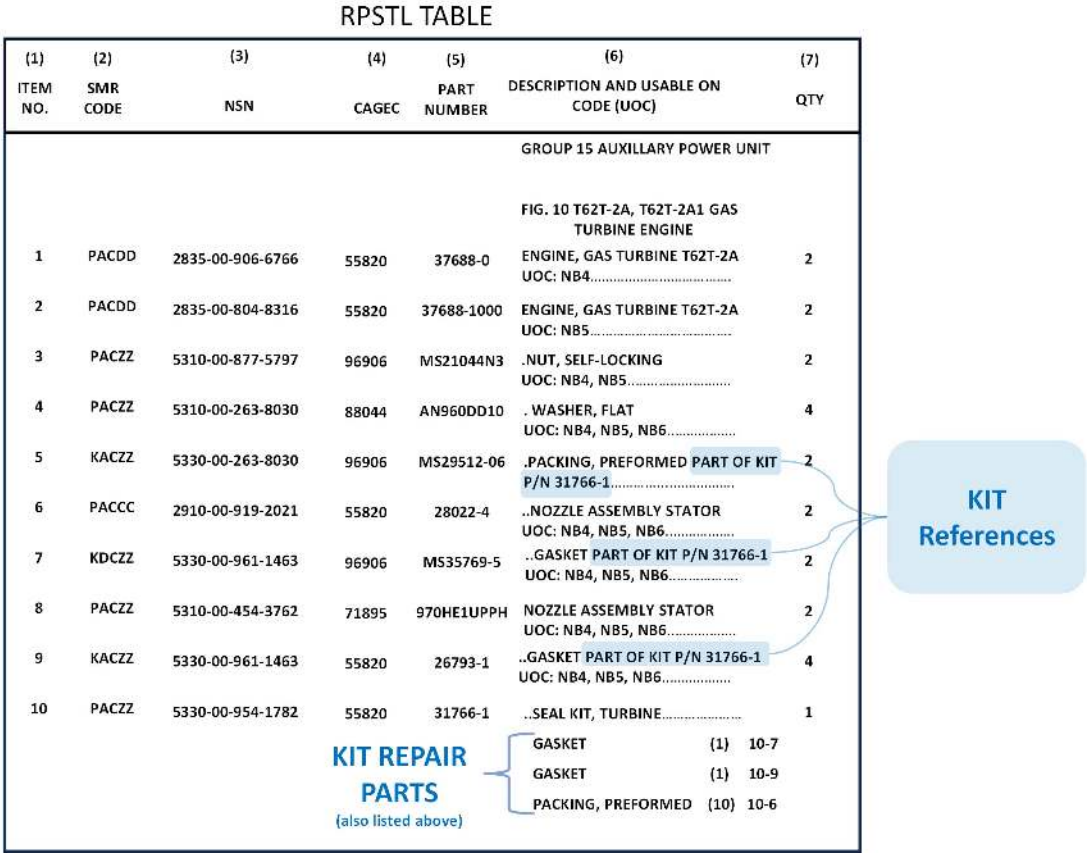


**FIGURE F-16. Example of kits breakdown (option 1).**

**(2)** Option 2 (kits):

**(a)** Kits shall be listed in the kit parts list WP **<kitswp>** (F.8.4).

**(b)** Parts **<pi.item>**. Option 2 kit repair parts shall appear in the parts list by item number as shown on the associated figure (see FIGURE F-17). They shall be listed in item number sequence. The statement "PART OF KIT P/N (*enter kit part number*)" shall follow the item name.



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**Note**

Figure appearance may vary slightly based on version of DTD/stylesheets utilized (see applicable DTD or tag definition list for available tags).

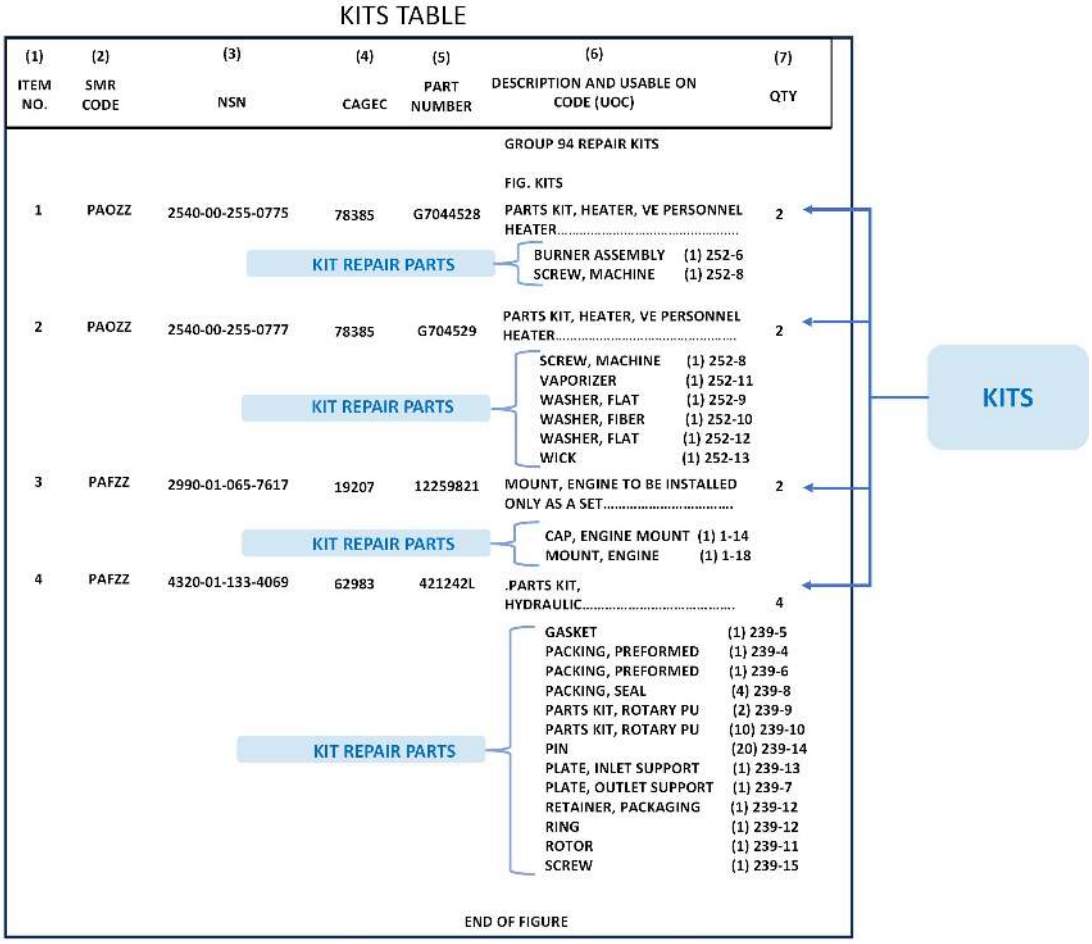


**FIGURE F-17. Example of kits breakdown (option 2).**

## F.8.2.1.10 Quantity column <qty>.

The number in the QTY column (FIGURE F-15, Item 7) (tags shown in FIGURE F-10 through ) shall be the total number of the item used on the part or component depicted in the illustration/figure. For clarity of illustration, the total number of occurrences of the item used may not be shown in the figure. The letter "V" (variable) shall be placed in the quantity column when a definite quantity cannot be determined because the number of uses or the size/ length of the item is not definite and may vary.

## F.8.2.1.11 (MC) United States Marine Corp. (USMC) quantity per equipment column <qty_per_end_item>.

The number in the USMC QTY per Equip column shall represent the total quantity for all the occurrences of that part in all the repair parts lists. This column shall be included in multi-service manuals involving the Marine Corps.

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### F.8.2.1.12 End of Figure.

The statement "END OF FIGURE" shall appear below the last item described in the column for each of the tabular lists in the repair parts list and the special tools/kits list WPs.

### F.8.2.2 Optional RPSTL components.

The following components may be included in the RPSTL:

**a.** Unit of issue **<ui>** for an item.

**b.** Unit of measure **<um>** for the item.

**c.** Reference designator for the item.

**d.** Next higher assembly **<nha_item>**.

**e.** Parts breakdown reference **<part.breakdown.ref>**.

**f.** Mandatory replacement parts (also included in the mandatory replacement parts list WP **<mrplwp>** (G.7.9).

**g.** Critical safety items (also included in the critical safety items WP **<csi.wp>** (APPENDIX G).

**h.** Basic Issue Items (BII). Repair parts for repairable BII that do not have separate operator publications, but are authorized in the RPSTL. If included, BII shall be supported by illustrations and listed in a functional group titled **<fnctitle>** "BASIC ISSUE ITEMS (REPAIR PARTS)." Items listed in functional and subfunctional groups shall be listed and identified with the same basic columnar data required for the end item repair parts.

### F.8.2.3 Prohibited in RPSTL.

Expendable and durable items shall not be listed in the RPSTL. These items shall appear in the expendable and durable items WP **<explistwp>** (APPENDIX G).

### F.8.2.4 Other RPSTL WPs.

Some equipment systems may require the development of additional RPSTL WPs to accompany **<plwp>** WPs. F.8.3 guides their development.

## F.8.3 Repair parts for special tools/kits list work package <stl_partswp>.

The special tools/kits repair parts list WP shall be prepared when all the following conditions are met:

**a.** A RPSTL identifies the special tool/kit in the special tools/kits list WP.

**b.** A special tool/kit has repair parts that may be replaced at any maintenance level covered in the technical publication.

**c.** A special tool/kit does not have repair instructions and/or parts listed in a technical publication for the special tool/kit.

When included, this WP shall follow the last repair parts list WP **<plwp>** and shall precede the kit parts list WP **<kitswp>** , bulk items WP **<bulk_itemswp>**, or special tools/ kits list WP **<stlwp>**. The following components shall be included in a repair parts for special tools/kits list WP.

**a.** Work package identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** Initial setup **<initial_setup>** is not required for this WP.

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**c.** Special tools/kits repair parts items list **<pi.category>** shall include components listed in section F.8.6 except items listed below. Only one special tools/kits repair parts items list **<pi.category>** shall appear in a special tools/kits list WP **<stl_partswp>**.

**(1)** The functional group header **<fncgrp>** shall precede the first special tools/kits repair part item in the DESCRIPTION AND USABLE ON CODE (UOC) column (tags shown in FIGURE F-10 through FIGURE F-13).

**(2)** The functional group number **<fnccode>** and title **<fnctitle>** shall be "SPECIAL TOOLS (REPAIR PARTS)" appearing on the top line(s). The next line(s) below shall be the figure number and the figure title.

## F.8.4 Kit parts list work package <kitswp>.

A kits parts WP shall be prepared when kit parts are listed separately in accordance with (2) (Option 2 (kits)) in section F.8.2.1.9. When included, this WP shall follow the last repair parts list WP **<plwp>** or repair parts for special tools/kits list WP **<stl_partswp>** and shall precede the bulk items list WP **<bulk_itemswp>** (if included) or the special tools/kits list WP **<stlwp>**.

The WP consists of one or more kits part item lists **<pi.category>** organized by functional group. This WP shall not have any illustrations. The following components shall be included in a kit parts list WP.

**a.** Work package identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** Initial setup **<initial_setup>** is not required for this WP.

**c.** Special tools/kits repair parts items list **<pi.category>** shall be listed alphanumerically by part number in the part number column and shall include components listed in section except the following:

**(1)** The functional group header **<fncgrp>** shall precede the first bulk item in the description column. The functional group number and title **<fnccode>** shall be "REPAIR KITS" appearing on the top line(s). The next line(s) below shall be the figure number and the figure title **<fnctitle>**.

**(2)** Kit part item group **<kititem>**. Parts in the kit group, in the DESCRIPTION AND USABLE ON CODE (UOC) column, shall be indented two positions and listed alphabetically by item name or in item number sequence under their kit name. Kit parts shall be listed by item names **<name>**, the quantity (in parentheses) **<qty>**, the figure number, and the item numbers **<callout>** that appear in the basic parts list.

**(3)** Kits part item quantity **<qty>**. The quantity column entry for kits part shall contain a "V" (variable) when the exact quantity may vary.

## F.8.5 Bulk items work package <bulk_itemswp>.

A bulk items WP shall be prepared when bulk items are required in the repair of any parts listed in a parts list, special tool/kit list or repair kit. This WP shall not have any illustrations. The following components shall be included in a bulk items WP.

**a.** Work package identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** Initial setup **<initial_setup>** is not required for this WP.

**c.** Bulk items list **<pi.item>** (see FIGURE F-18) shall be listed alphabetically by item name in the DESCRIPTION AND USABLE ON CODE (UOC) column and shall include components listed in F.8.2.1 except the following components:

**(1)** Functional group header **<fncgrp>**. The functional group header shall precede the first bulk item in the DESCRIPTION AND USABLE ON CODE (UOC) column. The functional group number and title **<fnccode>** shall be "BULK MATERIAL" appearing on the top line(s). The next line(s) below shall be the figure number and the figure title **<fnctitle>** and titled "FIG. BULK."

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**(2)** Item column **<callout>**. Numbers in the item number column of bulk material list apply to the FIG. BULK only and shall not be associated with item numbers (callouts appearing on the illustrations/figures).

**Note**

Figure appearance may vary slightly based on version of DTD/stylesheets utilized (see applicable DTD or tag definition list for available tags).

BULK TABLE

| (1) ITEM NO. | (2) SMR CODE | (3) NSN | (4) CAGEC | (5) PART NUMBER | (6) DESCRIPTION AND USABLE ON CODE (UOC) | (7) QTY |
|---|---|---|---|---|---|---|
| | | | | | GROUP 9501 BULK MATERIAL | |
| | | | | | FIG. BULK | |
| 1 | PACDD | 5330-00-982-5130 | 81349 | MILC7637TYP2 | ASBESTOS SHEET, WOVE | 1 |
| 2 | XBCZZ | 5975-00-285-0907 | 19207 | RRC271 | CHAIN, WELDLESS | 1 |
| 3 | PACZZ | 9340-00-142-6860 | 97030 | LOOM3/8D | CONDUIT, NONMETALIC | 1 |
| 4 | PACZZ | 9340-00-285-6775 | 19207 | 11633348 | GLASS, LAMINATED | 1 |
| 5 | PACZZ | 9340-00285-6775 | 19200 | 8635931 | GLASS, LAMINATED | 1 |
| 6 | PFCZZ | 4720-00-809-2429 | 30299 | FT3548-5 | HOSE, AIRDUCT | 1 |
| 7 | PAFZZ | 4720-00-001-0093 | 81349 | MIL-H-13531 | NOSE, ASSEMBLY, NONMETALIC | 1 |
| 8 | XBCZZ | | 85757 | 3250-0610 | HOSE, NONMETALIC | 1 |
| 9 | PAFZZ | 4720-00-999-8994 | 01276 | 303-8 | HOSE, NONMETALIC | 1 |
| 10 | PACZZ | 4720-00-951-2433 | 96909 | MS521301A229R | HOSE, NONMETALIC | 1 |
| 11 | PACZZ | 4720-01-009-9058 | 85757 | 3250-1010 | HOSE, NONMETALIC | 1 |
| 12 | PACZZ | 4720-00-683-8830 | 81349 | MIL-H-8788-4 | HOSE, NONMETALIC | 1 |
| 13 | PACZZ | 4720-00-999-4044 | 11083 | 3R7752 | HOSE, PREFORMED | 1 |
| 14 | XBOZZ | | 81349 | MIL-I-14511 | INSULATION BOARD TH | 1 |
| 15 | PACZZ | 9390-00488-2106 | 19207 | CPR102201 | NONMETALLIC SPECIAL | 1 |
| 16 | PACZZ | 5330-01-040-8923 | 81349 | HHP151 | RUBBER SHEET SOLID | 1 |
| 17 | XBCZZ | | 19207 | 10287823-7 | RUBBER STRIP | 1 |
| 18 | PAFZZ | 5330-01-040-8923 | 19207 | CPR104394 | SEAL, RUBBER CHANNEL | 1 |
| 19 | PAFZZ | 5330-01-082-3792 | 19207 | CPR102235 | SEAL, RUBBER SPECIAL | 1 |
| 20 | PACZZ | 5330-01-082-3793 | 19207 | CPR102232 | SEAL, RUBBER SPECIAL | 1 |

**FIGURE F-18. Example of a bulk material list.**

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## F.8.6 Special tools/kits list work package <stlwp>.

A special tools/kits list WP shall be prepared for special tools/kits, special TMDE, and other special support equipment authorized for maintenance of the end item/assembly. Repair parts for special tools/kits listed in this WP that have their own technical publication shall not be listed in the repair parts for special tools/kits list WP (F.8.3). The following components shall be included in a kit parts list WP.

**a.** Work package identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** Initial setup **<initial_setup>** is not required for this WP.

**c.** Special tools/kits list **<pi.category>** (FIGURE F-19) including components listed in section F.8.2.1 with the following amendments (listed below). Only one special tools/kits list **<pi.category>** shall appear in a special tools/kits list WP **<stlwp>**.

**(1)** Items shall be listed on the special tools/kits list in the item number column by the same callout number shown on the associated figure. The items shall be listed in ascending alphanumeric sequence.

**(2)** Functional group header **<fncgrp>**. The functional group header shall precede the first bulk item in the description column. The functional group number and title **<fnccode>** shall be "SPECIAL TOOLS" appearing on the top line(s). The next line(s) below shall be the figure number and the figure title **<fnctitle>**.

**(3)** D-coded items. When a depot-level RPSTL does not exist and items are maintained at depot-level, they shall be identified with a "D" in the third position of the SMR code in the highest level RPSTL prepared.

**(4)** The Basis of Issue (BOI) **<boi>**. The BOI **<boi>** shall be placed on the last line under the item description, in the DESCRIPTION AND USABLE ON CODE (UOC) column, for individual items, sets, or kits. The BOI shall indicate the quantity of the items, for example, sets, or kits authorized to support a quantity of end items/assemblies or a specific military unit. For example, BOI: 1 auth for 1-12 equip or BOI: 1 per BN HQ when BN has SVC CO.

**(5)** Quantity column. The quantity column shall be left blank.

**(6)** Components list **<kititem>**. Components of special tool sets and kits, in the DESCRIPTION AND USABLE ON CODE (UOC) column, shall be listed in figure and item number sequence **<callout>**. The component shall be indented two positions and listed by item name **<name>**, the figure number, and the item numbers **<callout>**. Quantities of components **<qty>** shall be included in BOI statement (see item (4) above).

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**Note**

Figure appearance may vary slightly based on version of DTD/stylesheets utilized (see applicable DTD or tag definition list for available tags).

Special Tools / Kit List

| (1) ITEM NO. | (2) SMR CODE | (3) NSN | (4) CAGEC | (5) PART NUMBER | (6) DESCRIPTION AND USABLE ON CODE (UOC) | (7) QTY |
|---|---|---|---|---|---|---|
| | | | | | GROUP 30 SPECIAL TOOLS | |
| | | | | | FIG. 254 | |
| 1 | PECDD | 6625-01-169-5333 | 80058 | TS-3920A-ASM | TEST SET, STABILIZATION (BOI: 1 AUTH PER 15 AIRCRAFT) | 1 |
| 1 | PECDD | 6625-01-266-1636 | 80058 | TS-3920B/ASM | TEST SET, STABILIZATION (BOI: 1 AUTH PER 15 AIRCRAFT) | 1 |
| 2 | XBCZZ | | 80063 | A3012556 | WEDGE, 30/60/90 DEG. (BOI: 1 AUTH PER TEST SET) | 1 |
| 3 | XBCZZ | | 80063 | A3012558 | WEDGE, 05-85-90 DEG. (BOI: 1 AUTH TEST SET) | 1 |
| 4 | XBCZZ | | 80063 | A3012559 | PIN, ALIGNMENT (BOI: 1 AUTH PER TEST SET) | 1 |
| 5 | XBCZZ | | 80063 | A3012559 | FIXTURE, ALIGNMENT (BOI: 1 AUTH PER TEST SET) | 1 |
| | | | | | END OF FIGURE | |

**FIGURE F-19. Example of a special tools/kit list WP.**

## F.8.7 Illustrations.

Additional RPSTL specific illustration requirements are described below:

**a.** Arrangement of illustrations. All illustrations prepared for spares, repair parts, special tools/kits, special TMDE, and other special support equipment shall be arranged in figure number sequence. They shall precede their companion parts list (on the left- hand page preceding the parts list or at the top of the same page of the parts list in ETMs). Illustrations shall not be duplicated to provide facing page illustrations for the second and subsequent pages of the companion parts list. Illustrations shall not be duplicated to show different models or configurations of an assembly when UOCs can be assigned to indicate differences in configurations. Refer to F.6.2 for RPSTL WP layout requirements to achieve facing page layout.

**b.** Use of illustrations. Foldout and foldout-foldup illustrations shall not be used in RPSTLs. References to illustrations in other technical publications or to illustrations in the narrative portion of a combined maintenance publication with a RPSTL shall not be made. Landscape pages shall not be prepared except for RPSTLs supporting nuclear weapons (regulated by the Department of Energy/ Defense Nuclear Agency). For clarity, multi-sheet illustrations may be used.

**c.** Identical parts/item numbers. Identical parts (same part number) appearing in a figure (illustration) having only one FGC shall have the same item number. If a figure has two or more FGCs/assemblies, only the identical parts with identical SMR codes within each FGC/assembly shall have the same item number.

**d.** Identical assemblies. When two or more identical assemblies (same part number) exist in different places, i.e., in the equipment, a breakdown of the parts shall be illustrated only once, i.e., the first time the assembly appears in the RPSTL. For subsequent times that the identical assembly appears, the assembly item name shall

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appear in the DESCRIPTION AND USABLE ON CODE (UOC) column and be followed by the statement **<part.breakdown.ref>**, "SEE FIG ## FOR BREAKDOWN."

## F.9 NOTES.

The notes in Section 6 apply to this appendix.

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# APPENDIX G

# SUPPORTING INFORMATION

## G.1 SCOPE.

This appendix establishes the technical content requirements for the preparation of supporting information for major systems and their related subsystems, equipment assemblies, components, Shop Replaceable Units (SRUs), and Line Replaceable Units (LRUs). This appendix is a mandatory part of this standard. The information contained herein is intended for compliance. Requirements in this appendix are applicable for all maintenance levels through sustainment including DMWRs and NMWRs.

**Note**

To be compliant with this standard, publications shall be developed within **<production>** (opening) and **</production>** (closing) root tags.

When preparing information in accordance with this appendix, all references shall be linked using the appropriate tags (**<xref>**, **<extref>**, etc.). Links shall not be typed as text without tagging.

### G.1.1 Content tailoring using Appendix A matrices.

Applicability of some requirements contained in this standard are the responsibility of the materiel developer (MATDEV) (refer to AR 750–1, AR 700–127) and shall be accomplished using matrices referenced in TABLE A-I. The applicability of requirements are also designated by statements including, but not limited to, the following:

**a.** "...as specified by the MATDEV."

**b.** "...as required by the MATDEV."

**c.** "Refer to the MATDEV for..."

**d.** "Contact MATDEV if..."

**e.** "The MATDEV shall determine..."

**f.** "...as needed."

#### G.1.1.1 Front and rear matter.

Front and rear matter requirements for documents discussed in this appendix shall be determined in accordance with TABLE A-II.

## G.2 APPLICABLE DOCUMENTS.

The applicable documents in Section 2 apply to this appendix.

## G.3 ACRONYMS AND DEFINITIONS.

The acronyms and terms defined in Section 3 of this standard apply to this appendix.

## G.4 SUPPORTING INFORMATION APPLICABILITY.

Requirements contained in this appendix are applicable to all maintenance levels unless specifically labeled in bold and in parentheses. An explanation of applicable DA maintenance levels is provided in Section 3.

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## G.5 SUPPORTING INFORMATION FUNCTIONALITY.

A supporting information module **<sim>** consists of a sequence of work packages (WPs) that provide lists, tables, illustrations, parts, and maintenance data for system components that are described, utilized, and referenced throughout a technical publication. The supporting information module shall appear as the last module in a publication and is broken into the WP types listed in TABLE G-I.

**TABLE G-I. Supporting information WPs.**

| WP | WP Tags | Appendix Section |
|---|---|---|
| References WP | **<refwp>** | G.7.1 |
| Maintenance Allocation Chart (MAC) Introduction WP | **<macintrowp>** | G.7.2 |
| MAC WP | **<macwp>** | G.7.3 |
| Components of End Item (COEI) / Basic Issue Items (BII) WP | **<coeibiiwp>** | G.7.4 |
| Additional Authorization List (AAL) WP | **<aalwp>** | G.7.5 |
| Collateral Material (CM) WP (Marine Corps only) | | G.7.6 |
| Expendable and durable items list WP | **<explistwp>** | G.7.7 |
| Tool identification list WP | **<toolidwp>** | G.7.8 |
| Mandatory Replacement Parts (MRP) list WP | **<mrplwp>** | G.7.9 |
| Critical Safety Items (CSIs) WP | **<csi.wp>** | G.7.10 |
| Support items WP | **<supitemwp>** | G.7.11 |
| Additional WP(s) | **<genwp>** | G.7.12 |

## G.6 DEVELOPMENT OF SUPPORTING INFORMATION MODULES.

A supporting information module **<sim>** shall be prepared as required by publication type (see A.4.4) and shall contain the WPs listed below as applicable. Supporting information modules and the individual WPs contained in this module shall not be divided based on maintenance level. Some WP types in a supporting information module can contain information for multiple maintenance levels (for example, the Expendable and Durable Items List WP). Therefore, there shall be one combined WP of each type (for example, one Expendable and Durable Items List WP covering all maintenance levels, etc.). See FIGURE G-1 for a diagram of components for a technical publication containing a supporting information module along with WPs available in the module.

**a.** References WP **<refwp>** (G.7.1).

**b.** MAC introduction WP **<macintrowp>** (G.7.2).

**c.** MAC WP **<macwp>** (G.7.3).

**d.** COEI / BII WP **<coeibiiwp>** (G.7.4.)

**e.** AAL WP **<aalwp>** (G.7.5).

**f.** Expendable and durable items list WP **<explistwp>** (G.7.7).

**g.** Tool identification (ID) list WP **<toolidwp>** (G.7.8).

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**h.** Mandatory replacement parts WP **<mrplwp>** (G.7.9).

**i.** Critical safety items (CSI) WP **<csi.wp>** (G.7.10).

**j.** Support items WP **<supitemwp>** (G.7.11).

**k.** General WP **<genwp>** (G.7.12).

**Note**

Tags/output shown in figure(s) below may vary slightly based on version of DTD/stylesheets utilized (see applicable DTD or tag definition list for available tags).

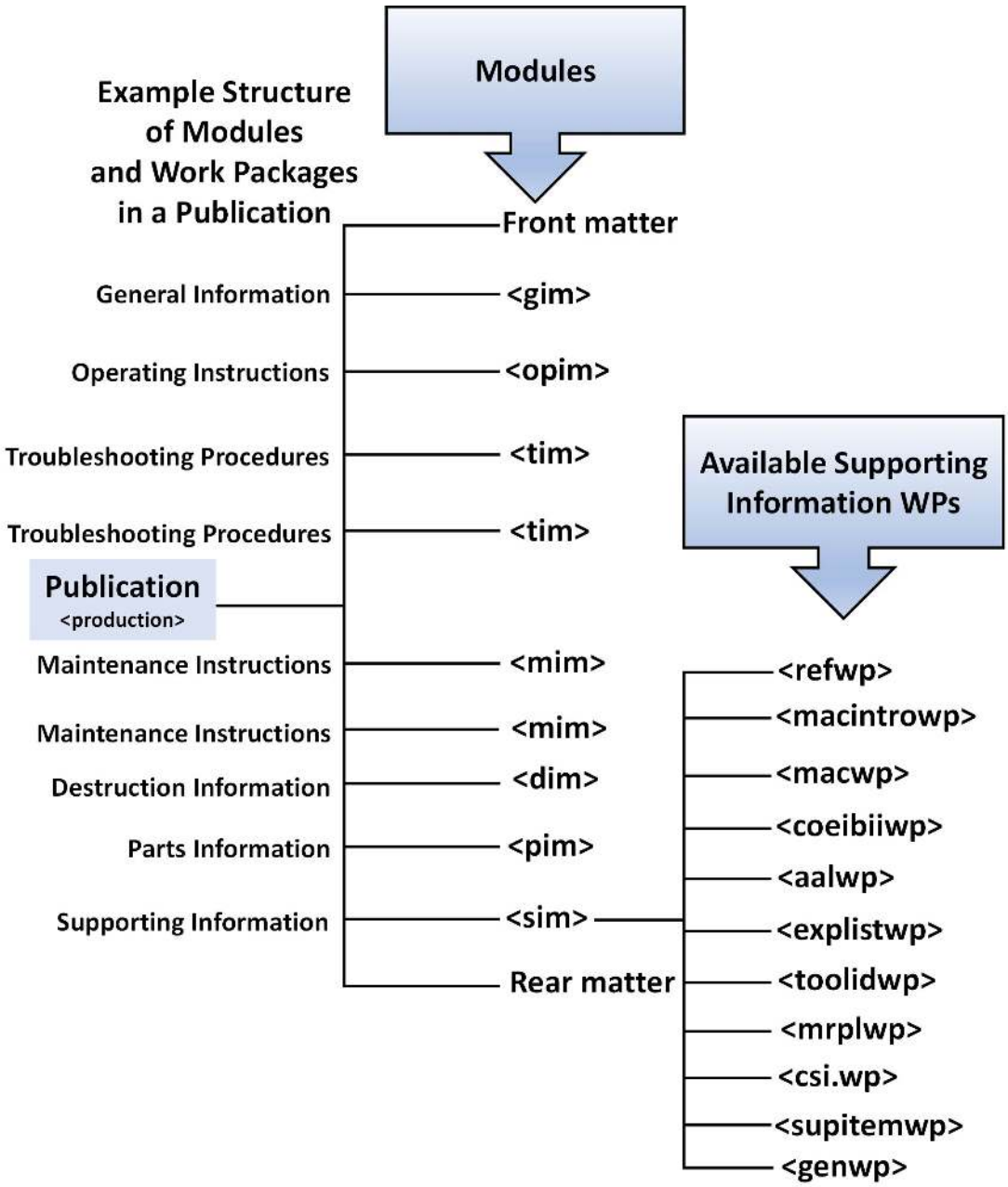


**FIGURE G-1. Supporting information modules and WPs.**

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## G.6.1 Multiple MACs <mac> in one WP <mac-alt>.

At times, MAC WPs may need to include multiple MACs in one WP. This may occur, for instance, when technical publications are combined for System of Systems (SoS) publications. SoS consist of dedicated equipment systems that pool resources and capabilities together to create a new, more complex system which offers greater functionality and performance than the sum of their individual parts. Similarly, the technical publications of these individual "parts" (or equipment systems) with their own independent technical publications (and MACs) are combined into one publication. To accommodate SoS needs, additional MACs **<mac-alt>** may be included in one WP with just one set of introductory and ending tags for the WP. FIGURE G-2 shows the structure of an example technical publication containing a MAC WP with multiple MACs. As needed, contact the Equipment Publications Control Office (EPCO) for assistance with integration of multiple MACs into one WP.

**Note**

Tags/output shown in figure(s) below may vary slightly based on version of DTD/stylesheets utilized (see applicable DTD or tag definition list for available tags).

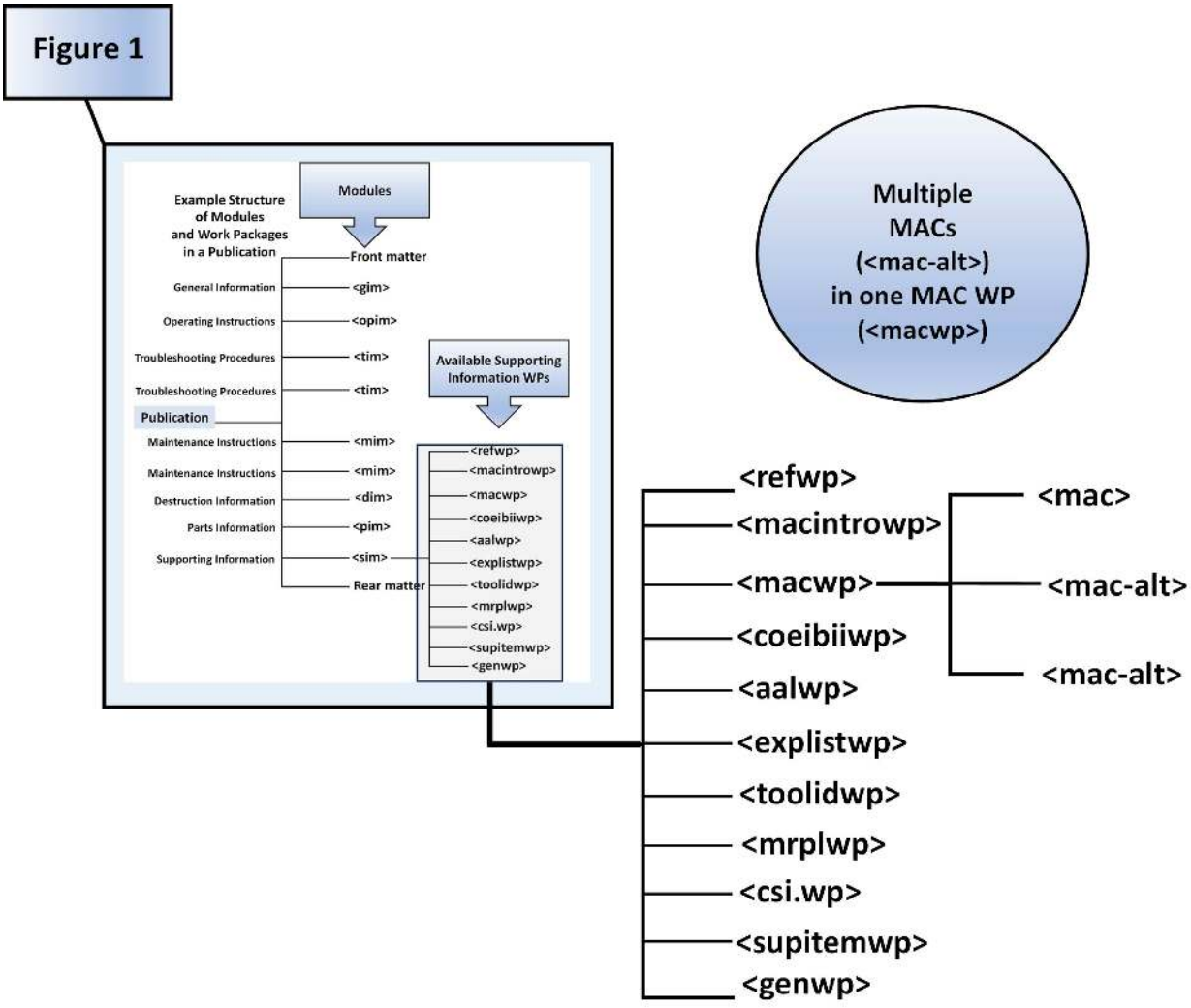


**FIGURE G-2. Multiple MACs example.**

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## G.7 DEVELOPMENT OF SUPPORTING INFORMATION WORK PACKAGES.

The inclusion of some supporting information WPs depends on the type/maintenance level of a technical publication as well as the required tools, kits, etc. necessary for maintenance of equipment. This section describes each type of supporting WP and their functions.

### G.7.1 References work package <refwp>.

The references WP shall be prepared and shall list all publications referenced in the technical publication that are required by the user to operate and/or maintain the equipment. This WP also may list any other helpful publications not listed in the publication. See FIGURE F-2 through FIGURE F-6 for references WP components and tags. The following components shall be included in a references WP.

**a.** Work package identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** Initial setup **<initial_setup>** is not required for this WP.

**c.** Scope **<scope>** is required for this WP and shall explain document types listed in the WP. This section may also include any other explanatory information such as where users can find full publication titles (as needed).

**d.** Publication list **<publist>** (required) as described in G.7.1.1 through G.7.1.1.2.

First Publication list. Individual paragraphs shall be prepared for each publication type. All referenced publications shall be listed; unpublished/unauthenticated publications shall not be listed in the references WP or referenced in the technical publication. This list shall identify the publication number **<name>** / **<extref>** / **<link>** in alphanumeric sequence and shall also include the title **<title>**. If a publication is non-government, the source shall be given and all such publications shall be listed alphabetically by title.

**e.** Second Publication list **<publist>** (optional).

If included, this list shall be created in accordance with the first publication list; however, it shall include other related documents (for example, technical publications for system-related documents, etc.) that are not referenced in the manual.

#### G.7.1.1 Publication titling <title>.

Many publications have long titles that may include part numbers, national stock numbers, etc. Publications titles shall be listed as simply as possible. See examples shown in FIGURE G-3; the titles are limited to document subject matter only (without part numbers, maintenance levels, etc.). Publication numbers listed on the left side of the work package identify maintenance levels, and title details are available if users research each publication. Therefore, this WP shall be utilized as a starting point for relevant reference research. Relevant tagging structure for this WP is shown in FIGURE G-4 through FIGURE G-5.

##### G.7.1.1.1 Publication titling of volumes <title>.

Publications that include volumes shall include the text "(all volumes)" rather than specifying number of volumes. This ensures that a reference is current regardless of updates to the referenced document.

##### G.7.1.1.2 Publication list entries.

The references WP shall only list publications in their entirety. References to sections of publications (paragraphs, chapters (or modules), specific volumes of multi-volume publications, figures, etc.) are prohibited.

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## Note

Figure appearance and tags/output may vary slightly based on version of DTD/stylesheets utilized (see applicable DTD or tag definition list for available tags).

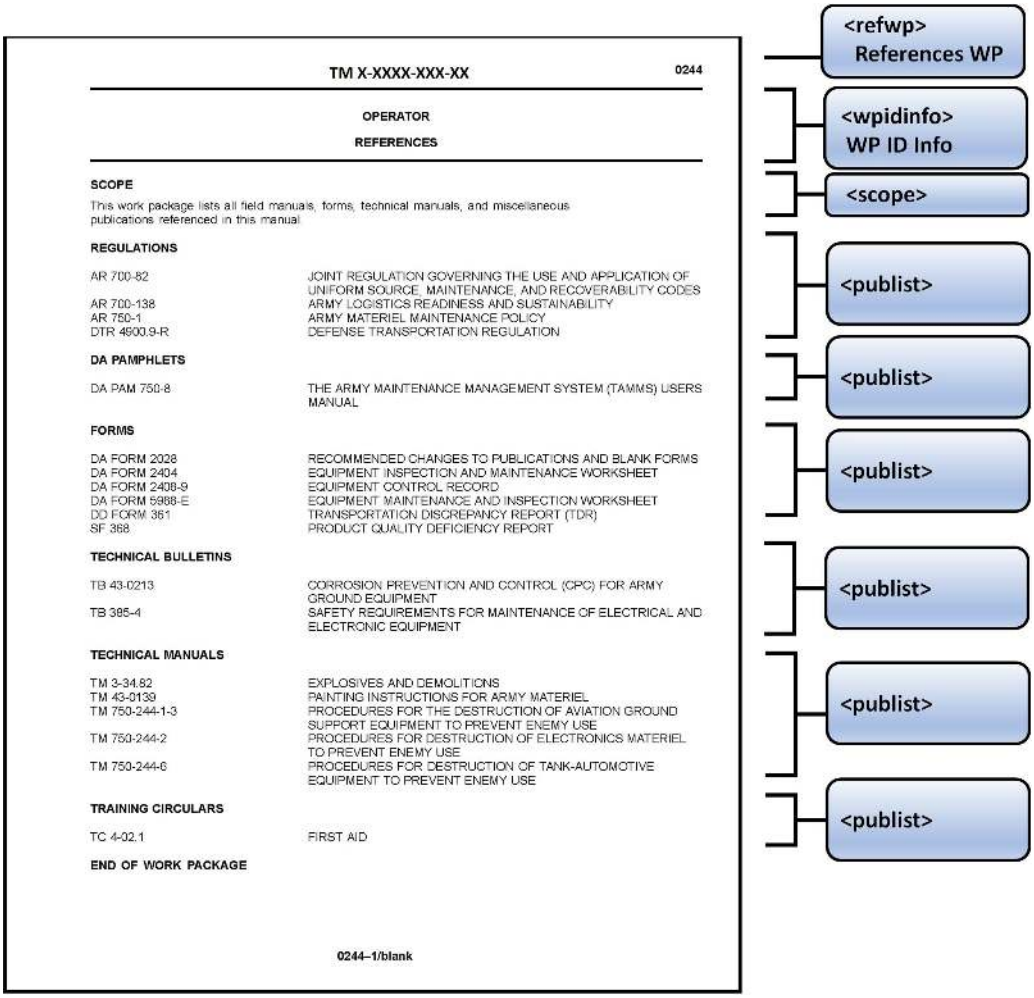


**FIGURE G-3. References WP example.**

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REFERENCES WP

```
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE refwp PUBLIC "-//USA-DOD//DTD xx TM Assembly REV x x.x yyyymmdd//EN"
 ".\40051x_x_x\40051x_x_x.dtd">
<refwp chngno="0" wpno="S00001-x-xxxx-xxx-12nP">
<wpidinfo>
<maintlvl level="operator"/>
<title id="S00001-x-xxxx-xxx-12nP_TITLE_1">REFERENCES</title>
</wpidinfo>
<scope>
<title>SCOPE</title>
<para>This work package lists all field manuals, forms, technical manuals, and
miscellaneous publications referenced in this manual.</para>
</scope>
<publist>
<title>REGULATIONS</title>
<pubident>
<extref docno="AR 700-82"/>
<title>Joint Regulation Governing The Use And Application Of Uniform Source,
Maintenance, And Recoverability Codes</title>
</pubident>
<pubident>
<extref docno="AR 700-138"/>
<title>Army Logistics Readiness And Sustainability</title>
</pubident>
<pubident>
<extref docno="AR 750-1"/>
<title>Army Materiel Maintenance Policy</title>
```

Doctype Statement and WP Root Tag <refwp>

<wpidinfo> WP Identification Information

<scope> Scope

<pubident> Publication Entry

<pubident> Publication Entry

<pubident> Publication Entry

<publist> Publication List "REGULATIONS"

**FIGURE G-4. References WP tags (1 of 2).**

REFERENCES WP (continued)

```
<pubident>
<extref docno="TM 43-0139"/>
<title>Painting Instructions For Army Materiel</title>
</pubident>
<pubident>
<extref docno="TM 750-244-1-3"/>
<title>Procedures For The Destruction Of Aviation Ground Support Equipment To Prevent
Enemy Use</title>
</pubident>
<pubident>
<extref docno="TM 750-244-2"/>
<title>Procedures For Destruction Of Electronics Materiel To Prevent Enemy Use</title>
</pubident>
<pubident>
<extref docno="TM 750-244-6"/>
<title>Procedures For Destruction Of Tank-Automotive Equipment To Prevent Enemy Use
</title>
</pubident>
</publist>
<publist>
<title>TRAINING CIRCULARS</title>
<pubident>
<extref docno="TC 4-02.1"/>
<title>First Aid</title>
</pubident>
</publist>
</refwp>
```

<publist> Publication List "REGULATIONS" (continued)

<publist> Publication List "TRAINING CIRCULARS"

**FIGURE G-5. References WP tags (2 of 2).**

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## G.7.2 MAC introduction work package <macintrowp>.

The MAC introduction shall be included when a MAC is in a technical publication and shall include the components listed below.

**a.** Work package identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** Initial setup **<initial_setup>** is not required for this WP.

**c.** Introduction data **<intro>**:

- Non-Aviation MAC (G.7.2.1).
- Aviation MAC (G.7.2.2).
- Marine Corps MAC (G.7.2.3).

## G.7.2.1 Non-Aviation MAC introduction WP <macintrowp>.

A non-aviation MAC introduction WP shall include the following text verbatim in between introduction tags**<intro>**.

"**INTRODUCTION**

**The Army Maintenance System MAC**

This introduction provides a general explanation of the maintenance levels, tasks, and other information contained in the MAC. Note that any included links (references) from MAC entries lead to their respective maintenance tasks within maintenance WPs (rather than to the top of maintenance WPs). These MAC links exist solely to support alignment between MAC entries and maintenance tasks in a publication. When performing maintenance procedures, refer to the top of a WP for necessary information such as initial setup data as well as any warnings, cautions, and notes, etc. that may pertain to equipment maintenance.

The MAC designates overall authority and responsibility for the performance of all maintenance tasks on the identified end item or component. The application of the maintenance tasks to the end item or component shall be consistent with the capacities and capabilities of the designated maintenance levels that are shown in the MAC in column (4). Column (4) is divided into two secondary columns. These columns indicate the maintenance levels of "Field" and "Sustainment." Each maintenance level column is further divided into two subcolumns. These subcolumns identify the maintenance levels and are as follows:

1. Field-level maintenance:
    **a.** Crew (operator) maintenance. This is the responsibility of a using organization to perform maintenance on its assigned equipment. It normally consists of inspecting, servicing, lubricating, adjusting, and replacing parts, minor assemblies, and subassemblies. Items with a "C" ("O" for multi-service reporting) in the third position of the Source, Maintenance, and Recoverability (SMR) code may be replaced at the crew (operator) level. A code of "C" ("O" for multi-service) in the fourth position of the SMR code indicates complete repair is authorized at the crew (operator) level.
    **b.** Maintainer maintenance. This is maintenance accomplished on a component, accessory, assembly, subassembly, plug-in unit, or other portion by field-level units. This maintenance is performed either on the system or after it is removed. An "F" in the third position of the SMR code indicates replacement of assemblies, subassemblies, or other components is authorized at this level. An "F" in the fourth position of the SMR code indicates complete repair of the identified item is allowed at the Maintainer level. Items repaired at this level are normally returned to the user after maintenance is performed.
2. Sustainment-level maintenance:
    **a.** Below depot sustainment. Maintenance accomplished on a component, accessory, assembly, subassembly, plug-in unit, or other portion either on the system or after it is removed. The replace function for this level of maintenance is indicated by the letter "H" appearing in the third position of the SMR code. An "H"


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appearing in the fourth position of the SMR code indicates complete repair is possible at the below depot sustainment maintenance level. Items are returned to the supply system after maintenance is performed at this level.

**b.** Depot. This is maintenance accomplished on a component, accessory, assembly, subassembly, plug-in unit, or other portion either on the system or after it is removed. Assets to be repaired at this level are normally returned to an Army Depot or authorized contractor facility. The "replace" task for this level of maintenance is indicated by the letter "D" or "K" appearing in the third position of the SMR code. A "D" or "K" appearing in the fourth position of the SMR code indicates complete repair is possible at the depot sustainment maintenance level. Items are returned to the supply system after maintenance is performed at this level.

The tools and test equipment requirements table (immediately following the MAC) lists the tools and test equipment (both special tools/kits and common tool sets) required for each maintenance task as referenced from the MAC. The remarks table (immediately following the tools and test equipment requirements) contains supplemental instructions and explanatory notes for a particular maintenance task.

**Maintenance Tasks**

Maintenance tasks are limited to and defined as follows (*Functions/tasks may be removed from introduction if not used*):

1. Inspect. Step-by-step instructions to determine the serviceability of an item by comparing its physical, mechanical, and/or electrical characteristics with established standards through examination (for example, by sight, sound, or feel).
2. Test. Step-by-step instructions to verify serviceability by measuring the mechanical, pneumatic, hydraulic, or electrical characteristics of an item and comparing those characteristics with prescribed standards, for example, load testing of lift devices or hydrostatic testing of pressure hoses. For software, to verify usability/ operability/functionality of the software.
3. Service. Step-by-step instructions to be performed periodically to keep an item in proper operating condition, such as replenishing fuel, lubricants, chemical fluids, or gases.
4. Adjust. Step-by-step instructions to maintain or regulate, within prescribed limits, by bringing into proper position, or by setting the operating characteristics to specified parameters.
5. Align. Step-by-step instructions to adjust specified variable elements of an item to bring about optimum or desired performance.
6. Calibrate. Step-by-step instructions to determine and cause corrections to be made or to be adjusted on instruments of test, measuring, and diagnostic equipment used in precision measurement. It consists of comparisons of two instruments, one of which is a certified standard of known accuracy, to detect and adjust any discrepancy in the accuracy of the instrument being compared.
7. Remove. Step-by-step instructions for taking a component off an asset to facilitate other maintenance on a different component or on the same component (except for "Replace" and "Repair." For software, it is step-by-step instructions for uninstalling/removing the software from a workstation or other viewing hardware.
8. Install. Step-by-step instructions for placing, positioning, or otherwise locating a component to make it part of a higher-level end item. The "install" task is authorized by the LPD/MAC and the assigned maintenance level is shown as the third position code of the SMR code. For software, it is step-by-step instructions putting the software on a workstation or other viewing hardware.
9. Replace. Step-by-step instructions for taking off an unserviceable component and putting a serviceable component in its place. The "replace" task is authorized by the LPD/MAC and the assigned maintenance level is shown as the third position code of the SMR code.
10. Repair. Step-by-step instructions for restoring an item or software to a completely serviceable or fully mission capable status. The "repair" task is authorized by the LPD/MAC and the assigned maintenance level is shown

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as the fourth position code of the SMR code. The following definitions are applicable to the "repair" maintenance task: welding, grinding, riveting, straightening, facing, machining, and/or resurfacing.

**11.** Paint. Step-by-step instructions to prepare and apply coats of paint. When used for ammunition, the paint is applied to the ammunition and its packaging so they can be identified and protected.

**12.** Overhaul. Step-by-step instructions to restore an item to a completely serviceable/operational condition as required by maintenance standards in the appropriate technical publications. "Overhaul" is normally the highest degree of maintenance performed by the Army. "Overhaul" does not normally return an item to a like new condition.

**13.** Rebuild. Step-by-step instructions required for the restoration of unserviceable equipment to a like new condition in accordance with original manufacturing standards. "Rebuild" is the highest degree of materiel maintenance applied to Army equipment. The "rebuild" operation includes the act of returning to zero those age measurements (for example, hours/miles) considered in classifying Army equipment/ components.

**14.** Lubricate. Step-by-step instructions for applying a material (for example, oil or grease) to reduce friction and allow a component to operate in a more efficient manner.

**15.** Mark. Step-by-step instructions for restoring obliterated identification on an asset.

**16.** Pack. Step-by-step instructions to place an item into a container for either storage or shipment after service and other maintenance operations have been completed.

**17.** Unpack. Step-by-step instructions for removing an asset from a storage or shipping container in preparation to perform further maintenance (for example, "repair" or "install").

**18.** Preserve. Step-by-step instructions for treating systems and equipment, whether installed or stored, to ensure a serviceable condition.

**19.** Prepare for use. Step-by-step instructions required to make an asset ready for other maintenance (for example, "remove preservatives," "lubricate," etc.).

**20.** Assemble. Step-by-step instructions to join the component pieces of an asset together to make a complete serviceable asset.

**21.** Disassemble. Step-by-step instructions to break down (take apart) a spare/functional group coded item to the level of its least component, which is assigned an SMR code for the level of maintenance under consideration (i.e., identified as maintenance significant).

**22.** Clean. Step-by-step instructions on how to remove dirt, corrosion, or other contaminants from equipment. Refer to appropriate painting, lubrication, and preservation methods to restore original corrosion prevention and control methods when removed as a result of cleaning and/or when using cleaning to remove corrosion from the item.

**23.** Nondestructive inspection. Step-by-step instructions on preparation and accomplishment inspections that do not destroy or damage the equipment.

**24.** Radio interference suppression. Step-by-step instructions to ensure installed equipment, either communication or other electronics, does not interfere with installed communication equipment

**25.** Place in service. Step-by-step instructions required to place an item into service that are not covered in the service upon receipt work package (WP).

**26.** Towing. Step-by-step instructions to connect one vehicle to another for the purpose of having one vehicle moved through the motive power of the other vehicle.

**27.** Jacking. Step-by-step instructions to mechanically raise or lift a vehicle to facilitate maintenance on the vehicle.

**28.** Parking. Step-by-step instructions to safely place a vehicle in a lot, ramp area or other designated location.

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**29.** Mooring. Step-by-step instructions to secure a vehicle by chains, ropes or other means to protect the vehicle from environmental conditions or secure for transportation.

**30.** Covering. Step-by-step instructions to place a protective wrapping over a vehicle to protect it from environmental conditions or to hide (for example, camouflage) it.

**31.** Hoisting. Step-by-step instructions to allow a vehicle to be raised by cables or ropes through attaching points.

**32.** Sling loading. Step-by-step instructions to place a sling around a vehicle to allow it to be raised.

**33.** External power. Step-by-step instructions on how to apply electrical power from any authorized power source (for example, external generator or facility power).

**34.** Preparation for storage. Step-by-step instructions for preparing the equipment for placement into administrative, short term, and/or long-term storage.

**35.** Preparation for shipment. Step-by-step instructions for preparing the equipment to be shipped or transported.

**36.** Transport. Step-by-step instructions and guidance for transporting/shipping the equipment.

**37.** Arm. Step-by-step instructions on activating ammunition prior to use.

**38.** Load. Step-by-step instructions for one of two tasks:

**a.** For transportation, the act of placing assets onto a transportation medium (for example, pallet, truck, container).

**b.** For weapons/weapons systems, the act of placing ammunition into the weapon/weapons system.

**39.** Unload. Step-by-step instructions for one of two tasks:

**a.** For transportation, the act of removing assets from a transportation medium (for example, pallet, truck, container).

**b.** For weapons/weapons systems, the act of removing ammunition from the weapon/weapons system.

**40.** Install peripheral device. Step-by-step instructions for installing peripheral devices such as printers, scanners, modems, etc.

**41.** Uninstall peripheral device. Step-by-step instructions for uninstalling peripheral devices such as printers, scanners, modems, etc.

**42.** Upgrade/patch. Step-by-step instructions for performing an upgrade to software or installing a patch to software.

**43.** Configure. Step-by-step instructions for configuring software for different uses/purposes and/or different users.

**44.** Debug. Step-by-step instructions for debugging software/correcting errors in the software.

**Explanation of Columns in the MAC**

Column (1) Group Number. Column (1) lists Functional Group Code (FGC) numbers, the purpose of which is to identify maintenance significant components, assemblies, subassemblies, and modules with the Next Higher Assembly (NHA).

Column (2) Component/Assembly. Column (2) contains the item names of components, assemblies, subassemblies, and modules for which maintenance is authorized.

Column (3) Maintenance task. Column (3) lists the tasks to be performed on the item listed in column (2). (For a detailed explanation of these tasks, refer to maintenance tasks outlined above.)

Column (4) Maintenance Level. Column (4) specifies each level of maintenance authorized to perform each task listed in column (3), by indicating man-hours required in the appropriate subcolumn. The man-hour figure is the task time multiplied by the number of maintainers required to perform that maintenance task. This time includes preparation (equipment conditions, inspections), task performance, follow-on maintenance, and quality assurance

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(inspections) time. Crew maintenance time will be entered as task (clock) time only. If different maintenance levels perform the same maintenance tasks due to the number or complexity of the tasks, appropriate man-hour figures are to be shown for each level. The symbol designations for the various maintenance levels are as follows:

- Field:

  C — Crew maintenance

  F — Maintainer maintenance

- Sustainment:

  L — Specialized Repair Activity (SRA)

  H — Below depot maintenance

  D — Depot maintenance

**Note**

The "L" maintenance level is not included in column (4) of the MAC. Functions to this level of maintenance are identified by work time figure in the "H" column of column (4), and an associated reference code is used in the REMARKS column (6). This code is keyed to the remarks and the SRA complete repair application is explained there.

Column (5) Tools and Equipment Reference Code. Column (5) specifies, by a number code, those common tool sets, kits, or outfits (not individual tools), common Test, Measurement and Diagnostic Equipment (TMDE), common tools that are not part of a set, kit, or outfit, special tools/kits, special TMDE, and special support equipment required to perform the designated function. Codes are keyed to the entries in the tools and test equipment table.

Column (6) Remarks Code. When applicable, this Column (6) contains a letter code, in alphabetical order, that is keyed to the remarks table entries.

**Explanation of Columns in the Tools and Test Equipment Requirements**

Column (1) Tool or Test Equipment Reference Code. The tool or test equipment reference code correlates with a code used in column (5) of the MAC.

Column (2) Maintenance Level. The lowest level of maintenance authorized to use the tool or test equipment.

Column (3) Nomenclature. Name or identification of the tool or test equipment.

Column (4) National Stock Number (NSN). The NSN of the tool or test equipment. Column (5) Tool Number. The manufacturer's part number and the CAGE code.

**Explanation of Columns in the Remarks**

Column (1) Remarks Code. The code recorded in column (6) of the MAC.

Column (2) Remarks. This column lists information pertinent to the maintenance task being performed as indicated in the MAC."

## G.7.2.2 Aviation MAC introduction WP <macintrowp>.

An aviation MAC introduction WP shall include the following text verbatim in between introduction tags**<intro>**.

"**INTRODUCTION**

**Aviation Maintenance Allocation Chart**

The MAC (immediately following the introduction) designates overall authority and responsibility for the performance of maintenance tasks on the identified end item or component. Note that any included links (references) from MAC entries lead to their respective maintenance tasks within maintenance WPs (rather than to the top of maintenance WPs). These MAC links exist solely to support alignment between MAC entries and maintenance tasks

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in a publication. When performing maintenance procedures, refer to the top of a WP for necessary information such as initial setup data as well as any warnings, cautions, and notes, etc. that may pertain to equipment maintenance. The application of the maintenance tasks to the end item or component shall be consistent with the capacities and capabilities of the designated maintenance level that is shown on the MAC as:

- Field:

  "O" — corresponds to Aviation Maintenance Company (AMC)

  "F’ — which corresponds to Aviation Support Battalion (ASB)

- Sustainment:

  "L"— corresponds to Theatre Aviation Sustainment Maintenance Group (TASMG) and other organizations that have National Maintenance Program certification and

  "D" — corresponds to Depot

The maintenance to be performed is described as follows:

1. Field-level maintenance activities:

   a. Aviation Maintenance Company (AMC). The aviation maintenance company is the lowest level of aviation field maintenance. The AMC provides direct support to aircraft operations, performing functions of aircraft servicing (daily, preflight, postflight inspections, refuel, arming), Battle Damage Assessment and Repair (BDAR), and repair or replacement actions as specified in the MAC.

   b. Aviation Support Company (ASC) in the Aviation Support Battalion (ASB). The ASB performs the following types of maintenance:

      (1) Off equipment repair of LRUs or other components within the limits prescribed in the MAC.

      (2) Inspections beyond the capability of the AMC.

      (3) BDAR as required.

      (4) Provide support to AMC personnel during peak workload periods as determined by local policy.

2. Sustainment maintenance:

   a. Theater Aviation Sustainment Maintenance Group (TASMG) (deployed). The TASMG performs the following:

      (1) Provides support to CONUS deploying forces.

      (2) Provides support to OCONUS deployed forces (as the Theater Aviation Support Maintenance Group (TASMG)).

      (3) Expands aviation maintenance capabilities of CONUS depots.

      (4) Classifies and inspects aviation stocks and components.

      (5) Performs maintenance actions beyond the scope of the AMC or ASB within the limits prescribed in the MAC.

      (6) Augments ASB and AMC maintenance tasks.

   b. Depot. This is maintenance accomplished on a component, accessory, assembly, subassembly, plug- in unit, or other portion either on the system or after it is removed. Assets to be repaired at this level are normally returned to an Army Depot or authorized contractor facility. The "replace" task for this level of maintenance is indicated by the letter "D" or "K" appearing in the third position of the Source, Maintenance, and Recoverability (SMR) code. A "D" or "K" appearing in the fourth position of the SMR code indicates complete repair is possible at the depot sustainment maintenance level. Items are returned to the supply system after maintenance is performed at this level.

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**Use of the MAC**

**Note**

Approved item names are used throughout this MAC. Generic terms/nomenclature (if any) are expressed in parentheses and are not to be considered as official terminology.

The MAC assigns maintenance tasks to the lowest level of maintenance.

**Maintenance tasks**

Maintenance tasks are limited to and defined as follows (Functions/tasks may be removed from introduction if not used):

1. Inspect. Step-by-step instructions to determine the serviceability of an item by comparing its physical, mechanical, and/or electrical characteristics with established standards through examination (for example, by sight, sound, or feel).
2. Test. Step-by-step instructions to verify serviceability by measuring the mechanical, pneumatic, hydraulic, or electrical characteristics of an item and comparing those characteristics with prescribed standards, for example, load testing of lift devices or hydrostatic testing of pressure hoses. For software, to verify usability/ operability/functionality of the software.
3. Service. Step-by-step instructions to be performed periodically to keep an item in proper operating condition such as replenishing fuel, lubricants, chemical fluids, or gases.
4. Adjust. Step-by-step instructions to maintain or regulate, within prescribed limits, by bringing into proper position, or by setting the operating characteristics to specified parameters.
5. Align. Step-by-step instructions to adjust specified variable elements of an item to bring about optimum or desired performance.
6. Calibrate. Step-by-step instructions to determine and cause corrections to be made or to be adjusted on instruments of test, measuring, and diagnostic equipment used in precision measurement. It consists of comparisons of two instruments, one of which is a certified standard of known accuracy, to detect and adjust any discrepancy in the accuracy of the instrument being compared.
7. Remove. Step-by-step instructions for taking a component off an asset to facilitate other maintenance on a different component or on the same component (except for replace and repair.) For software, it is step-by- step instructions for uninstalling/removing the software from a workstation or other viewing hardware.
8. Install. Step-by-step instructions for placing, positioning, or otherwise locating a component to make it part of a higher-level end item. The "install" task is authorized by the LPD/MAC and the assigned maintenance level is shown as the third position code of the SMR code. For software, it is step-by-step instructions putting the software on a workstation or other viewing hardware.
9. Replace. Step-by-step instructions for taking off an unserviceable component and putting a serviceable component in its place. The "replace" task is authorized by the LPD/MAC and the assigned maintenance level is shown as the fourth position code of the SMR code.
10. Repair. Step-by-step instructions for restoring an item or software to a completely serviceable or fully mission capable status. The "repair" task is authorized by the LPD/MAC and the assigned maintenance level is shown as the fourth position code of the SMR code. The following definitions are applicable to the "repair" maintenance task: welding, grinding, riveting, straightening, facing, machining, and/or resurfacing.
11. Paint. Step-by-step instructions to prepare and apply coats of paint. When used with ammunition, the paint is applied to the ammunition and its packaging so they can be identified and protected.
12. Overhaul. Step-by-step instructions to restore an item to a completely serviceable/operational condition as required by maintenance standards in the appropriate technical publications. "Overhaul" is normally the highest degree of maintenance performed by the Army. "Overhaul" does not normally return an item to a like new condition.

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**13.** Rebuild. Step-by-step instructions required for the restoration of unserviceable equipment to a like new condition in accordance with original manufacturing standards. "Rebuild" is the highest degree of materiel maintenance applied to Army equipment. The "rebuild" operation includes the act of returning to zero those age measurements (for example, hours/miles) considered in classifying Army equipment/components.

**14.** Lubricate. Step-by-step instructions for applying a material (for example, oil or grease) to reduce friction and allow a component to operate in a more efficient manner.

**15.** Mark. Step-by-step instructions for restoring obliterated identification on an asset.

**16.** Pack. Step-by-step instructions to place an item into a container for either storage or shipment after service and other maintenance operations have been completed.

**17.** Unpack. Step-by-step instructions for removing an asset from a storage or shipping container in preparation to perform further maintenance (for example, repair or install).

**18.** Preserve. Step-by-step instructions for treating systems and equipment, whether installed or stored, to en- sure a serviceable condition.

**19.** Prepare for use. Step-by-step instructions required to make an asset ready for other maintenance (for example, "re- move preservatives," "lubricate," etc.)

**20.** Assemble. Step-by-step instructions to join the component pieces of an asset together to make a complete serviceable asset.

**21.** Disassemble. Step-by-step instructions to break down (take apart) a spare/functional group coded item to the level of its least component, which is assigned an SMR code for the level of maintenance under consideration (i.e., identified as maintenance significant).

**22.** Clean. Step-by-step instructions on how to remove dirt, corrosion, or other contaminants from equipment. Refer to appropriate painting, lubrication, and preservation methods to restore original corrosion prevention and control methods when removed as a result of cleaning and/or when using cleaning to remove corrosion from the item.

**23.** Nondestructive inspection. Step-by-step instructions on preparation and accomplishment of inspections that do not destroy or damage the equipment.

**24.** Radio interference suppression. Step-by-step instructions to ensure installed equipment, either communication or other electronics, does not interfere with installed communication equipment.

**25.** Place in service. Step-by-step instructions required to place an item into service that are not covered in the service upon receipt work package (WP).

**26.** Towing. The step-by-step instructions to connect one vehicle to another for the purpose of having one vehicle moved through the motive power of the other vehicle.

**27.** Jacking. The step-by-step instructions to mechanically raise or lift a vehicle to facilitate maintenance on the vehicle.

**28.** Parking. Step-by-step instructions to safely place a vehicle in a lot, ramp area, or other designated location.

**29.** Mooring. Step-by-step instructions to secure a vehicle by chains, ropes, or other means to protect the vehicle from environmental conditions or secure for transportation.

**30.** Covering. Step-by-step instructions to place a protective wrapping over a vehicle to protect it from environ- mental conditions or to hide (for example, camouflage) it.

**31.** Hoisting. Step-by-step instructions to allow a vehicle to be raised by cables or ropes through attaching points.

**32.** Sling loading. Step-by-step instructions to place a sling around a vehicle to allow it to be raised.

**33.** External power. Step-by-step instructions on how to apply electrical power from any authorized power source (for example, external generator or facility power).

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**34.** Preparation for storage. Step-by-step instructions for preparing the equipment for placement into administrative, short term, and/or long-term storage.

**35.** Preparation for shipment. Step-by-step instructions for preparing the equipment to be shipped or transported.

**36.** Transport. Step-by-step instructions and guidance for transporting/shipping the equipment.

**37.** Arm. Step-by-step instructions on activating prior to use.

**38.** Load. Step-by-step instructions for one of two tasks:

**a.** For transportation, the act of placing assets onto a transportation medium (for example, pallet, truck, container).

**b.** For weapons/weapon systems, the act of placing ammunition into the weapon/weapon system.

**39.** Unload. Step-by-step instructions for one of two tasks:

**a.** a. For transportation, the act of removing assets from a transportation medium (for example, pallet, truck, container).

**b.** b. For weapons/weapon systems, the act of removing ammunition from the weapon/weapon system.

**40.** Install peripheral device. Step-by-step instructions for installing peripheral devices such as printers, scanners, modems, etc.

**41.** Uninstall peripheral device. Step-by-step instructions for uninstalling peripheral devices such as printers, scanners, modems, etc.

**42.** Upgrade/patch. Step-by-step instructions for performing an upgrade to software or installing a patch to software.

**43.** Configure. Step-by-step instructions for configuring software for different uses/purposes and/or different users.

**44.** Debug. Step-by-step instructions for debugging software/correcting errors in the software.

**Explanation of Columns in the MAC**

Column (1) Group Number. Column (1) lists Functional Group Code (FGC) numbers, the purpose of which is to identify maintenance significant components, assemblies, subassemblies, and modules with the Next Higher Assembly (NHA).

Column (2) Component/Assembly. Column (2) contains the item names of components, assemblies, subassemblies, and modules for which maintenance is authorized.

Column (3) Maintenance task. Column (3) lists the tasks to be performed on the item listed in column (2). (For a detailed explanation of these tasks, refer to maintenance tasks outlined above.)

Column (4) Maintenance Level. Column (4) specifies each level of maintenance authorized to perform each task listed in column (3), by indicating man-hours required in the appropriate subcolumn. The man-hour figure is the task time multiplied by the number of maintainers required to perform that maintenance task. This time includes preparation (equipment conditions, inspections), task performance, follow-on maintenance and quality assurance (inspections) time. Crew maintenance time will be entered as task (clock) time only. If different maintenance levels perform the same maintenance tasks due to the number or complexity of the tasks, appropriate man-hour figures are to be shown for each level. The symbol designations for the various maintenance levels are as follows:

- Field:

  O — Aviation Maintenance Company

  F — Aviation Support Battalion

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- Sustainment:

  L — Theater Aviation Support Maintenance Group

  D — Depot

Column (5) Tools and Equipment Reference Code. Column (5) specifies, by a number code, those common tool sets, kits, or outfits (not individual tools), common Test, Measurement and Diagnostic Equipment (TMDE), common tools that are not part of a set, kit, or outfit, special tools/kits, special TMDE, and special support equipment required to perform the designated task. Codes are keyed to the entries in the tools and test equipment table.

Column (6) Remarks Code. When applicable, this Column (6) contains a letter code, in alphabetical order, that is keyed to the remarks table entries.

**Explanation of Columns in the Tools and Test Equipment Requirements**

Column (1) Tool or Test Equipment Reference Code. The tool or test equipment reference code correlates with a code used in column (5) of the MAC.

Column (2) Maintenance Level. The lowest level of maintenance authorized to use the tool or test equipment.

Column (3) Nomenclature. Name or identification of the tool or test equipment.

Column (4) National Stock Number (NSN). The NSN of the tool or test equipment.

Column (5) Tool Number. The manufacturer's part number.

**Explanation of Columns in the Remarks**

Column (1) Remarks Code. The code recorded in column (6) of the MAC.

Column (2) Remarks. This column lists information pertinent to the maintenance task being performed as indicated in the MAC."

### G.7.2.3 Marine Corps MAC introduction <macintrowp>.

Marine Corps Military Equipment (ME) will determine Source Maintenance Recoverability (SMR) code through the use of SECNAVINST 4410.23A which is used to communicate maintenance and supply instructions to the various logistic support levels and using commands for support. Refer to section G.7.3.3 for an example Marine Corps MAC.

A marine corps MAC introduction WP shall include the following text verbatim in between introduction tags**<intro>**.

"**INTRODUCTION**

The MAC designates overall authority and responsibility for the performance of maintenance tasks on the identified Military Equipment or component. The application of the maintenance tasks to the Military Equipment or component shall be consistent with the capacities and capabilities of the designated maintenance levels, which are shown in the MAC in the Maintenance Levels column which is divided into five sub-columns. These columns indicate the maintenance levels of Field and Depot.

**Levels of Maintenance**

Field and Depot. The distinction between field and depot Level Of Maintenance (LOM) is based on the maintenance tasks performed, within each LOM and as communicated through the SMR code.

1. Field Level of Maintenance. Field maintenance is any maintenance that does not require depot maintenance capability and is performed by equipment operator(s) and mechanics/technicians within Marine Corps organizations and activities, and/or by approved commercial/contract sources. Maintenance tasks performed within the field LOM are categorized as organizational or intermediate. A unit may perform any field maintenance tasks for which it is manned, trained, and equipped. Units are not authorized to conduct

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maintenance outside of their assigned capabilities. However, supported and supporting unit commanders are authorized to coordinate in order to ensure the most effective use of maintenance resources within their respective capabilities, and will incorporate local standard operating procedures to ensure the same. Additionally, commanders must consider their impacts not only to their own operations and resources, but also on those of their supporting/supported units.

2. Depot Level of Maintenance. Maintenance actions taken on material or software involving the inspection, repair, overhaul, or the modification or reclamation (as necessary) of weapons systems, equipment end items, parts, components, assemblies, and subassemblies that are beyond field maintenance capabilities, and/or are authorized and directed by HQMC.
3. Maintenance tasks performed within the field level of maintenance are categorized as:
   a. Organizational
      1. Field - Operator/Crew (C). Provides instructions for operation, theory of operation, operator maintenance, checks and adjustments, inventory, cleaning, and inspecting. Includes minor repairs and parts replacement with limited training and can be completed with equipment available to the operator/crew.
      2. Field - Organizational Maintenance Activity (O). Provides instructions to, allow specially trained maintenance technicians at the unit level, to assess and maintain equipment under battlefield conditions. This includes performing scheduled Preventive Maintenance Checks and Services (PMCS), diagnosis and isolation of equipment malfunction and replacement of minor parts, assemblies, and subassemblies.
   b. Intermediate
      1. Field - Intermediate Maintenance Activity (IMA) (F). Provides instructions that include inspection/indepth diagnosis, modification, modular parts replacement, adjustment, and limited repair or evacuation/disposal of PEIs. Intermediate manuals also include calibration and repair of Test, Measurement and Diagnostic Equipment as well as item fabrication, precision machining, and various welding methods.
      2. Field – Intermediate Maintenance Activity - Component Repair (H). Provides instructions for diagnosis, isolation, adjustment, alignment and in-depth repair of internal component parts and pieces.
3. Depot (D). Provides instructions for performing major repair, overhaul, or complete rebuild of parts, subassemblies, assemblies, or principal end items. Includes manufacturing parts and conducting required modifications, testing, calibration, and reclamation.

**Maintenance Tasks**

Introduction. The following series of steps depict the logical sequence necessary to complete the various types of maintenance tasks.

Maintenance tasks are limited to and defined as follows *(Functions/tasks may be removed from introduction if not used):*

1. Inspect. Step-by-step instructions to determine the serviceability of an item by comparing its physical, mechanical, and/or electrical characteristics with established standards through examination (for example, by sight, sound, or feel).
2. Test. Step-by-step instructions to verify serviceability by measuring the mechanical, pneumatic, hydraulic, or electrical characteristics of an item and comparing those characteristics with prescribed standards, for example, load testing of lift devices or hydrostatic testing of pressure hoses. For software, to verify usability/operability/functionality of the software.
3. Service. Step-by-step instructions to be performed periodically to keep an item in proper operating condition such as replenishing fuel, lubricants, chemical fluids, or gases.

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4. Adjust. Step-by-step instructions to maintain or regulate, within prescribed limits, by bringing into proper position, or by setting the operating characteristics to specified parameters.
5. Align. Step-by-step instructions to adjust specified variable elements of an item to bring about optimum or desired performance.
6. Calibrate. Step-by-step instructions to determine and cause corrections to be made or to be adjusted on instruments of test, measuring, and diagnostic equipment used in precision measurement. It consists of comparisons of two instruments, one of which is a certified standard of known accuracy, to detect and adjust any discrepancy in the accuracy of the instrument being compared.
7. Remove. Step-by-step instructions for taking a component off an asset to facilitate other maintenance on a different component or on the same component (except for replace and repair.) For software, it is step-by-step instructions for uninstalling/removing the software from a workstation or other viewing hardware.
8. Install. Step-by-step instructions for placing, positioning, or otherwise locating a component to make it part of a higher-level end item. The install task is authorized by the LPD/MAC and the assigned maintenance level is shown as the third position code of the SMR code. For software, it is step-by-step instructions putting the software on a workstation or other viewing hardware.
9. Replace. Step-by-step instructions for taking off an unserviceable component and putting a serviceable component in its place. The replace task is authorized by the LPD/MAC and the assigned maintenance level is shown as the third position code of the SMR code.
10. Repair. Step-by-step instructions for restoring an item or software to a completely serviceable or fully mission capable status. The repair task is authorized by the LPD/MAC and the assigned maintenance level is shown as the fourth position code of the SMR code. The following definitions are applicable to the "repair" maintenance task: welding, grinding, riveting, straightening, facing, machining, and/or resurfacing.
11. Paint. Step-by-step instructions to prepare and apply coats of paint. When used with munitions, the paint is applied so the ammunition can be identified and protected.
12. Overhaul. Step-by-step instructions to restore an item to a completely serviceable/operational condition as required by maintenance standards in the appropriate technical publications. Overhaul is normally the highest degree of maintenance performed by the Army. Overhaul does not normally return an item to a like new condition.
13. Rebuild. Step-by-step instructions required for the restoration of unserviceable equipment to a like new condition in accordance with original manufacturing standards. Rebuild is the highest degree of materiel maintenance applied to Army equipment. The rebuild operation includes the act of returning to zero those age measurements (for example, hours/miles) considered in classifying Army equipment/components.
14. Lubricate. Step-by-step instructions for applying a material (for example, oil or grease) to reduce friction and allow a component to operate in a more efficient manner.
15. Mark. Step-by-step instructions for restoring obliterated identification on an asset.
16. Pack. Step-by-step instructions to place an item into a container for either storage or shipment after service and other maintenance operations have been completed.
17. Unpack. Step-by-step instructions for removing an asset from a storage or shipping container in preparation to perform further maintenance (for example, repair or install).
18. Preserve. Step-by-step instructions for treating systems and equipment whether installed or stored, to ensure a serviceable condition.
19. Prepare for use. Step-by-step instructions required to make an asset ready for other maintenance (for example, remove preservatives, lubricate, etc.).
20. Assemble. Step-by-step instructions to join the component pieces of an asset together to make a complete serviceable asset.

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**21.** Disassemble. Step-by-step instructions to break down (take apart) a spare/functional group coded item to the level of its least component, that is assigned an SMR code for the level of maintenance under consideration (i. e., identified as maintenance significant).

**22.** Clean. Step-by-step instructions on how to remove dirt, corrosion or other contaminants from equipment. Refer to appropriate painting, lubrication, and preservation methods to restore original corrosion prevention and control methods when removed as a result of cleaning and/or when using cleaning to remove corrosion from the item.

**23.** Nondestructive inspection. Step-by-step instructions on preparation and accomplishment inspections which do not destroy or damage the equipment.

**24.** Radio interference suppression. Step-by-step instructions to ensure installed equipment, either communication or other electronics, does not interfere with installed communication equipment.

**25.** Place in service. Step-by-step instructions required to place an item into service that are not covered in the service upon receipt work package.

**26.** Towing. Step-by-step instructions to connect one vehicle to another for the purpose of having one vehicle moved through the motive power of the other vehicle.

**27.** Jacking. Step-by-step instructions to mechanically raise or lift a vehicle to facilitate maintenance on the vehicle.

**28.** Parking. Step-by-step instructions to safely place a vehicle in a lot, ramp area or other designated location.

**29.** Mooring. Step-by-step instructions to secure a vehicle by chains, ropes or other means to protect the vehicle from environmental conditions or secure for transportation.

**30.** Covering. Step-by-step instructions to place a protective wrapping over a vehicle to protect it from environmental conditions or to hide (for example, camouflage) it.

**31.** Hoisting. Step-by-step instructions to allow a vehicle to be raised by cables or ropes through attaching points.

**32.** Sling loading. Step-by-step instructions to place a sling around a vehicle to allow it to be raised.

**33.** External power. Step-by-step instructions on how to apply electrical power from any authorized power source (for example, external generator or facility power).

**34.** Preparation for storage. Step-by-step instructions for preparing the equipment for placement into administrative, short term, and/or long-term storage.

**35.** Preparation for shipment. Step-by-step instructions for preparing the equipment to be shipped or transported.

**36.** Transport. Step-by-step instructions and guidance for transporting/shipping the equipment.

**37.** Arm. Step-by-step instructions on activating munitions prior to use.

**38.** Load. Step-by-step instructions for one of three tasks:

- **a.** For transportation, the act of placing assets onto a transportation medium (for example, pallet, truck, container).
- **b.** For weapons/weapons systems, the act of placing munitions into the weapon/weapons system.

**39.** Unload. Step-by-step instructions for one of three tasks:

- **a.** For transportation, the act of removing assets from a transportation medium (for example, pallet, truck, container).
- **b.** For weapons/weapons systems, the act of removing munitions from the weapon/weapons system.

**40.** Install peripheral device. Step-by-step instructions for installing peripheral devices such as printers, scanners, modems, etc.

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**41.** Uninstall peripheral device. Step-by-step instructions for uninstalling peripheral devices such as printers, scanners, modems, etc.

**42.** Upgrade/patch. Step-by-step instructions for performing an upgrade to software or installing a patch to software.

**43.** Configure. Step-by-step instructions for configuring software for different uses/purposes and/or different users.

**44.** Debug. Step-by-step instructions for debugging software/correcting errors in the software.

**Explanation of Columns in the MAC**

Column (1) Group Number. Column (1) lists Functional Group Code (FGC) numbers, the purpose of which is to identify maintenance significant components, assemblies, subassemblies, and modules with the Next Higher Assembly (NHA).

Column (2) Component/Assembly. Column (2) contains the item names of components, assemblies, subassemblies, and modules for which maintenance is authorized.

Column (3) Maintenance Task. Column (3) lists the functions to be performed on the item listed in column (2). (For a detailed explanation of these functions, refer to maintenance tasks outlined previously.)

Column (4) Maintenance Level. Column (4) specifies each level of maintenance (1st through 5th) authorized to perform each function listed in column (3), by indicating man-hours required in the appropriate sub-column. The number of maintainers required to perform that maintenance task the task time multiplies the man-hour figure. This time includes preparation (equipment conditions, inspections), task performance, follow-on maintenance and quality assurance (inspections) time. Maintenance time will be entered as task (clock) time only. If different maintenance levels perform the same maintenance tasks due to the number or complexity of the tasks, appropriate man-hour figures are to be shown for each class. The symbol designations for the various maintenance levels as follows:

**1.** Field

  **a.** Organizational

    **(1)** 1st Operator Crew (C)

    **(2)** 2nd Mech/ Tech (O)

**2.** Intermediate

  **a.** 3rd Mech/Tech (F)

  **b.** 4th Mech/Tech (H)

**3.** Depot

    5th Mech/Tech (D)

Column (5) Tools and Equipment Reference Code. Column (5) specifies, by a number code, those common tool sets, kits, or outfits (not individual tools), common Test, Measurement and Diagnostic Equipment (TMDE), common tools that are not part of a set, kit, or outfit, special tools, special TMDE, and special support equipment required to perform the designated function. Codes are keyed to the entries in the tools and test equipment table. Column (6) Remarks Code. When applicable, this Column (6) contains a letter code, in alphabetical order, which is keyed to the remarks table entries.

**Explanation of Columns in the Tool and Test Equipment Requirements**

Column (1) Tool or Test Equipment Reference Code. The tool or test equipment reference code correlates with a code used in column (5) of the MAC.

Column (2) Maintenance Level. The lowest class of maintenance authorized to use the tool or test equipment.

Column (3) Nomenclature. Name or identification of the tool or test equipment.

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Column (4) National Stock Number (NSN). The NSN of the tool or test equipment. Column (5) Tool Number. The manufacturer's part number

**Explanation of Columns in the Remarks**

Column (1) Remarks Code. The code recorded in column (6) of the MAC.

Column (2) Remarks. This column lists information pertinent to the maintenance task being performed as indicated in the MAC."

## G.7.3 MAC work package <macwp>.

A MAC WP shall be prepared as required by publication type (see A.4.4) and guides the creation of maintenance in a publication (refer to G.7.3.3 for an example Marine Corps MAC). MAC entries shall consist of a list of functional groups applicable to the end item that require maintenance. Parts that are not subject to maintenance shall not be listed in the MAC. The MAC shall be in a 20-level maintenance manual or a combined -12 manual and shall contain entries for all levels of maintenance through depot for both hardware and software maintenance tasks. A maintenance WP may be in a lower or higher-level publication than its relevant MAC provided that these publications are for the same equipment system. The MAC WP shall include the components listed below.

**a.** Work package identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** Initial setup **<initial_setup>** is not required for this WP.

**c.** MAC table **<mac>** (non-aviation and joint-service non-aviation) and **<avmac>** ) (aviation) as discussed in G.7.3.1 through G.7.3.2.3.

**d.** Tools and test equipment requirements **<tereqtab>** (G.7.3.2.3).

**e.** Remarks **<remarktab>** (G.7.3.2.4).

### G.7.3.1 MAC development methodology.

The MAC drives (or informs) the creation of all maintenance tasks in a technical publication by identifying and authorizing specific maintenance tasks to be performed at each maintenance level. The MAC shall be prepared with Functional Group Codes (FGC) or Logistics Control Numbers (LCN) according to Logistics Product Data (LPD) provided by the MATDEV and listed in the RPSTL. Every entry in the MAC shall correspond to a maintenance task or maintenance tasks within a system/equipment publication, and the MAC shall not contain items, maintenance tasks, or procedures included in other systems/equipment publications. The MAC guides the development of field and sustainment maintenance.

#### G.7.3.1.1 MAC task organization.

**Note**

The MAC development methodology in this section shall serve as the preferred method of organizing the MAC and its alignment with accompanying maintenance tasks and RPSTL entries. Publications for large equipment systems may depart from the methodology presented in this section. Contact the Equipment Publications Control Office (EPCO) for any MAC-related inquiries.

This section and G.7.3.1.2 shall serve as a guide for MAC development. Since the purpose of a MAC is to serve as a maintenance roadmap, all task entries that appear in a MAC shall correspond to tasks in a publication unless that task is accomplished through maintenance to one or more subcomponents. For example, if the MAC lists a repair task for an assembly and that repair is accomplished through replacement of a subcomponent of that assembly, only the task for that subassembly would be represented in the publication.

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Remarks codes directing the procedure of any major task shall not serve as (or take the place of) a task entry for that task in the MAC. For example, in FIGURE G-6, "repair" is authorized for FGC 1303 "TRACK IDLERS AND BRACKETS," and remarks code B explains that this repair is accomplished through "maintenance to one or more subcomponents." This "repair" for FGC 1303 is to replace the following subcomponents: "Track Idler Assembly" and "Track Tensioner Grease Cylinder." These items are explicitly listed below FGC 1303 in the MAC because their authorized maintenance ("replace") are tasks represented in the publication and, therefore, shall not be lumped in with a "repair" task for the greater assembly (as text in a remarks code). Grouping various maintenance tasks under the "umbrella" concept of "repair maintenance" often negatively affects technical publication usability because alignment between the MAC and maintenance tasks represented in the publication is not readily apparent. Traceability between these two components (and RPSTL WPs) is mandatory for usability.

As the example in FIGURE G-7 demonstrates, remarks codes stating that a "repair" task is accomplished through the replacement (or any other authorized major maintenance) of subcomponents without identification of these specific subcomponents is prohibited. FIGURE G-7 through FIGURE G-11 shall serve as an organizational template for all MAC development. MAC and maintenance task organization is described in greater detail in G.7.3.1.2.

**Note**

LPD shall determine the grouping of system components in a MAC, the FGCs of these components, and which MAC components shall have FGCs. Any requests to depart from this MAC organizational structure shall be directed to the EPCO.

Figure appearance may vary slightly based on version of DTD/stylesheets utilized (see applicable DTD or tag definition list for available tags).

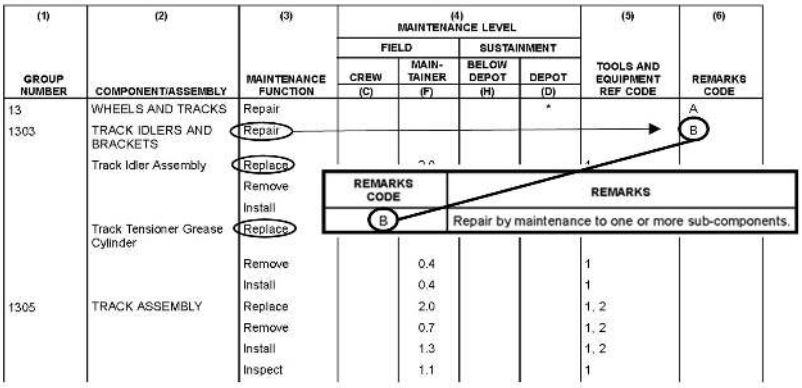


**FIGURE G-6. MAC task organization.**

## G.7.3.1.2 MACs and maintenance task organization (example).

**Note**

The MAC development methodology in this section shall serve as the preferred method of organizing the MAC and its alignment with accompanying maintenance tasks and RPSTL entries. Publications for large equipment systems may depart from the methodology presented in this section. Contact the EPCO for any MAC-related inquiries.

The MAC entries for FGC 13 shown in FIGURE G-7 through FIGURE G-11 illustrate available maintenance for this major assembly titled, "WHEELS AND TRACKS" as well as two subassemblies titled, "TRACK IDLERS AND BRACKETS" (FGC 1303) and "TRACK ASSEMBLY" (FGC 1305). Depot-level repair only is authorized for FGC 13 (remarks code A); however, field-level repair is authorized for both subassemblies of this major assembly. For FGC 1303, a "repair" task is authorized as field-level and is accomplished by replacement of the "Track Idler Assembly" and replacement of the "Track Tensioner Grease Cylinder" (remarks code B). In other words,

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replacement of each of these subassemblies serve to restore the "TRACK IDLERS AND BRACKETS" greater assembly to a serviceable or mission capable status. Thus, a "repair" task accompanies the greater assembly in the MAC, and "replace" tasks accompany each subassembly in the MAC.

For FGC 1305, a "replace" task is authorized as field-level and is accomplished by replacement of the entire track assembly. An "inspect" task is also authorized for FGC 1305. For "remove" and "install" tasks listed with both of the subassemblies, LPD dictates inclusion of these tasks based on the need to facilitate other maintenance. Accordingly, these maintenance tasks are listed in column F and shall be represented in maintenance WPs.

**Note**

If LPD does not dictate the inclusion of "remove" and "install" tasks, they shall not be listed in the MAC and shall not be included in maintenance WPs. Also, any "replace" task that is accompanied by "remove" and "install" tasks for that same component shall reference the "remove" / "install" tasks as much as possible to reduce procedural redundancy.

Figure appearance may vary slightly based on version of DTD/stylesheets utilized (see applicable DTD or tag definition list for available tags).



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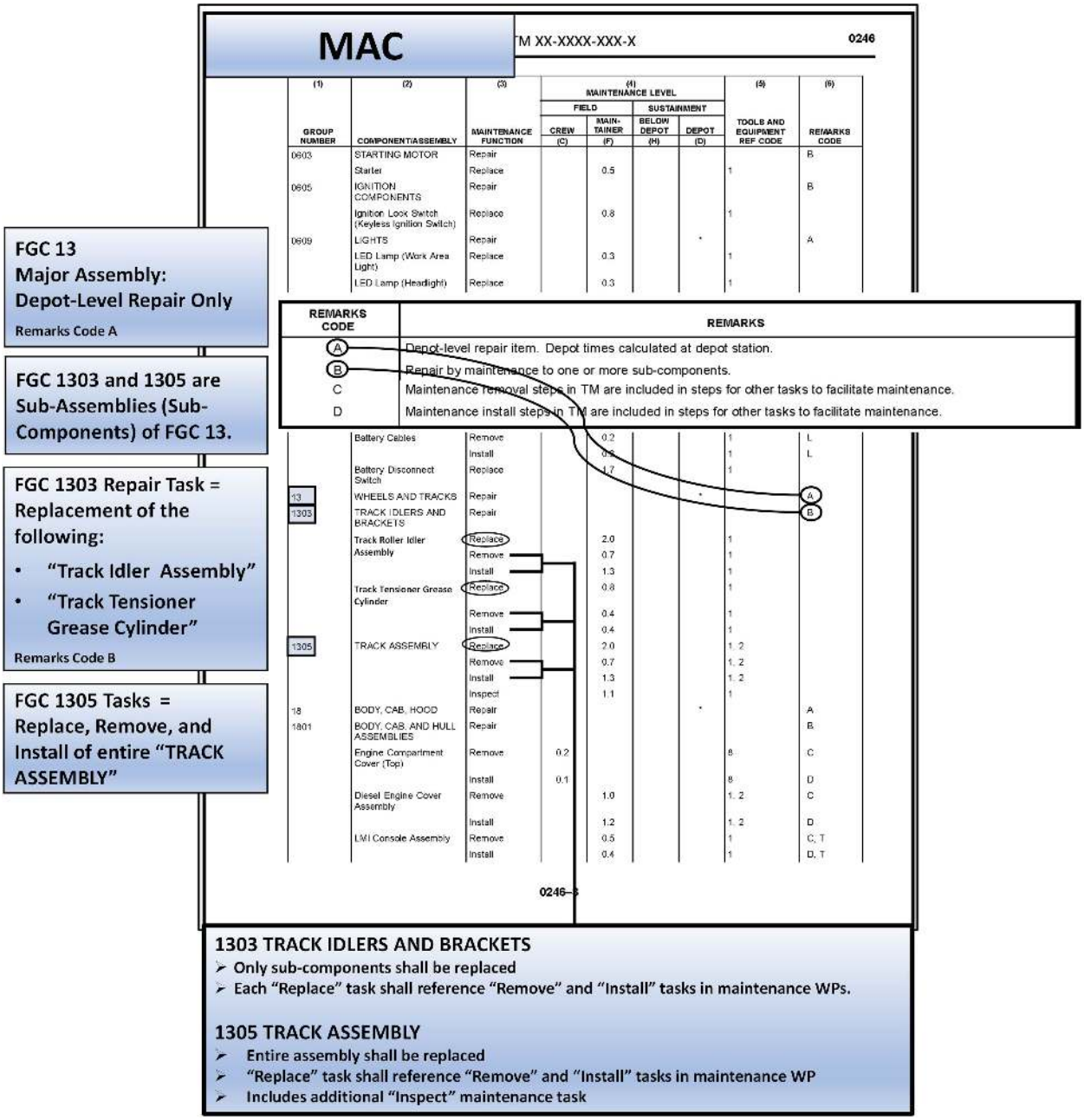


**FIGURE G-7. MAC and maintenance tasks (sheet 1 of 5).**

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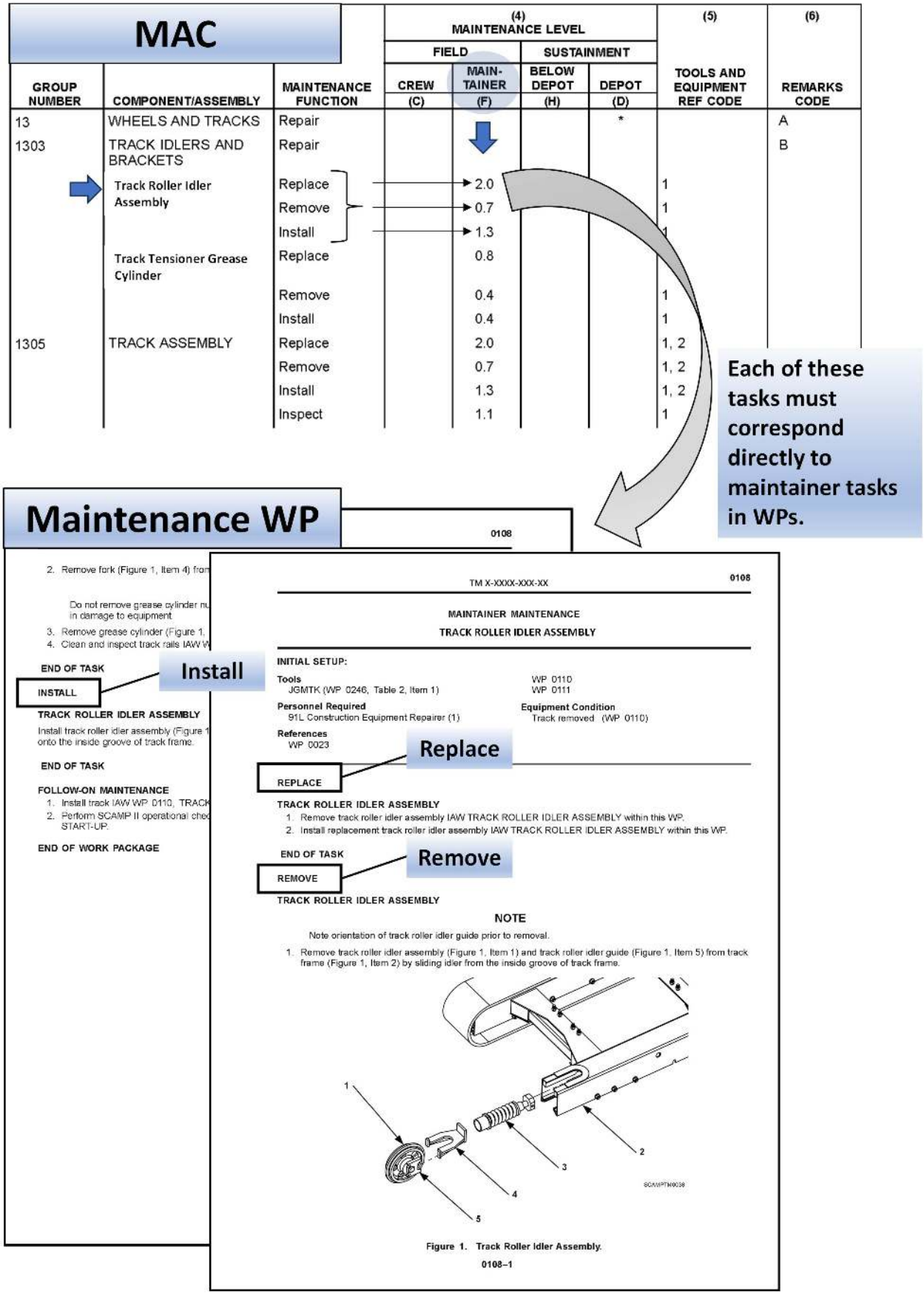


**FIGURE G-8. MAC and maintenance tasks (sheet 2 of 5).**

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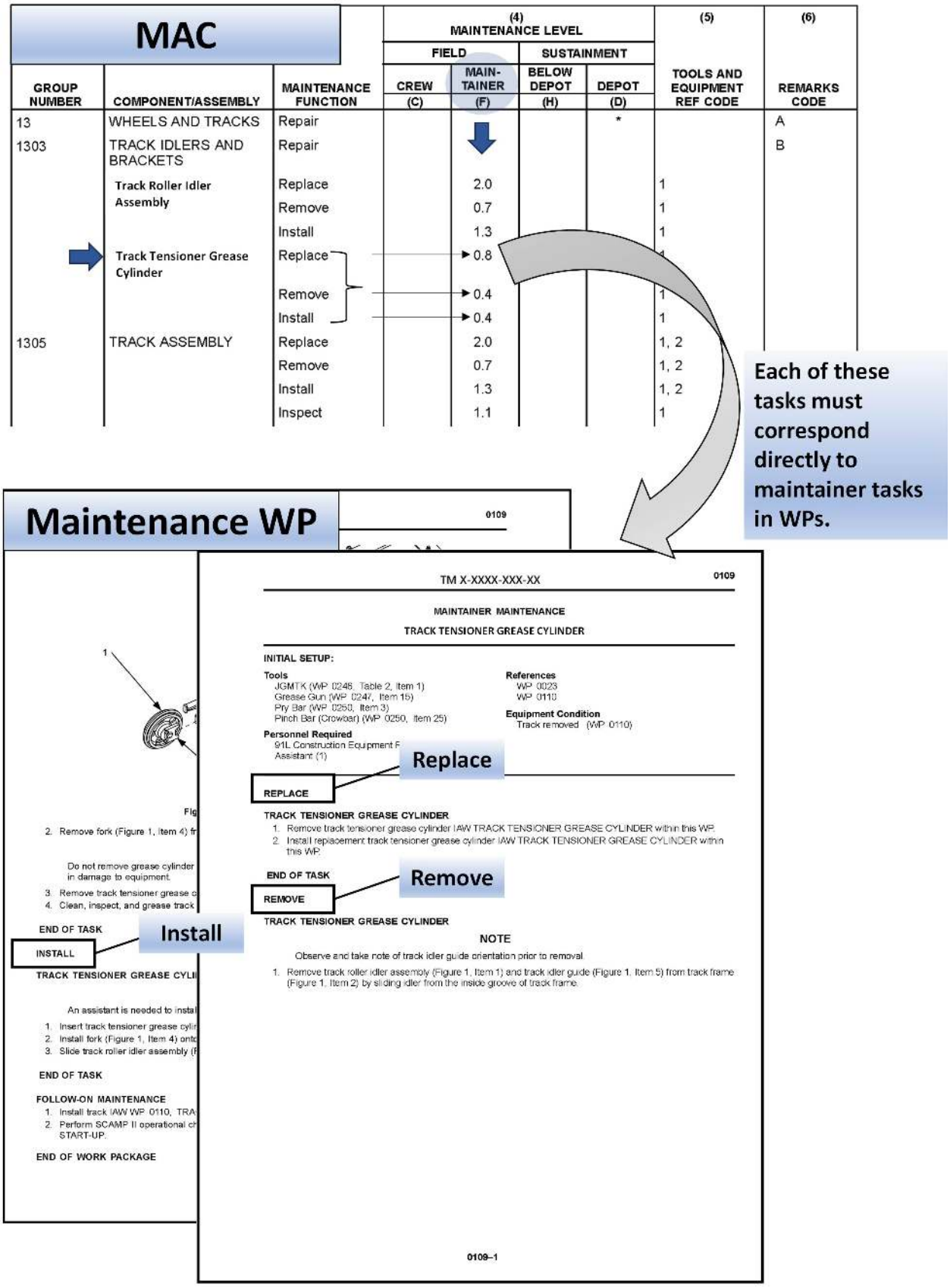


**FIGURE G-9. MAC and maintenance tasks (sheet 3 of 5).**



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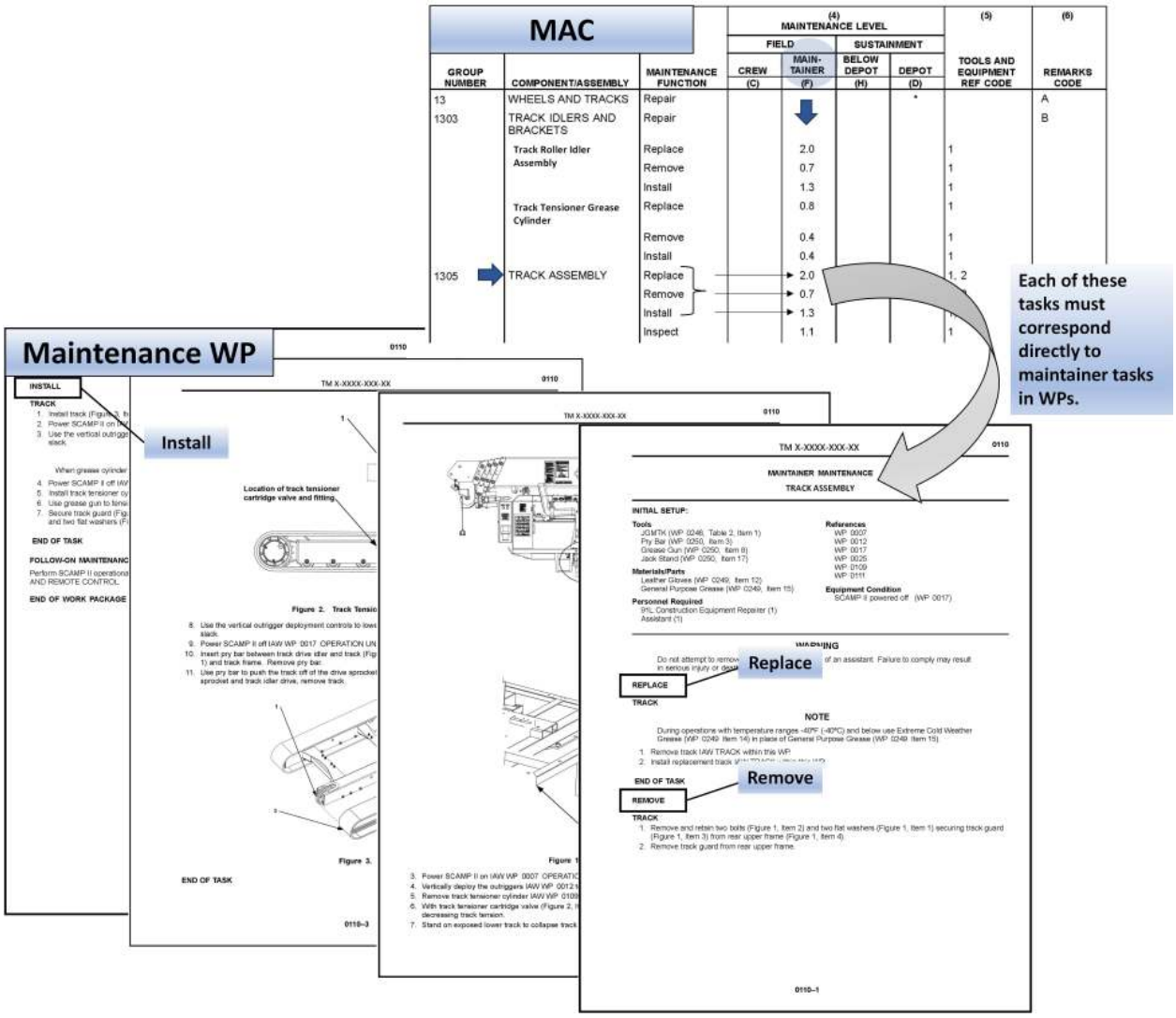


**FIGURE G-10. MAC and maintenance tasks (sheet 4 of 5).**

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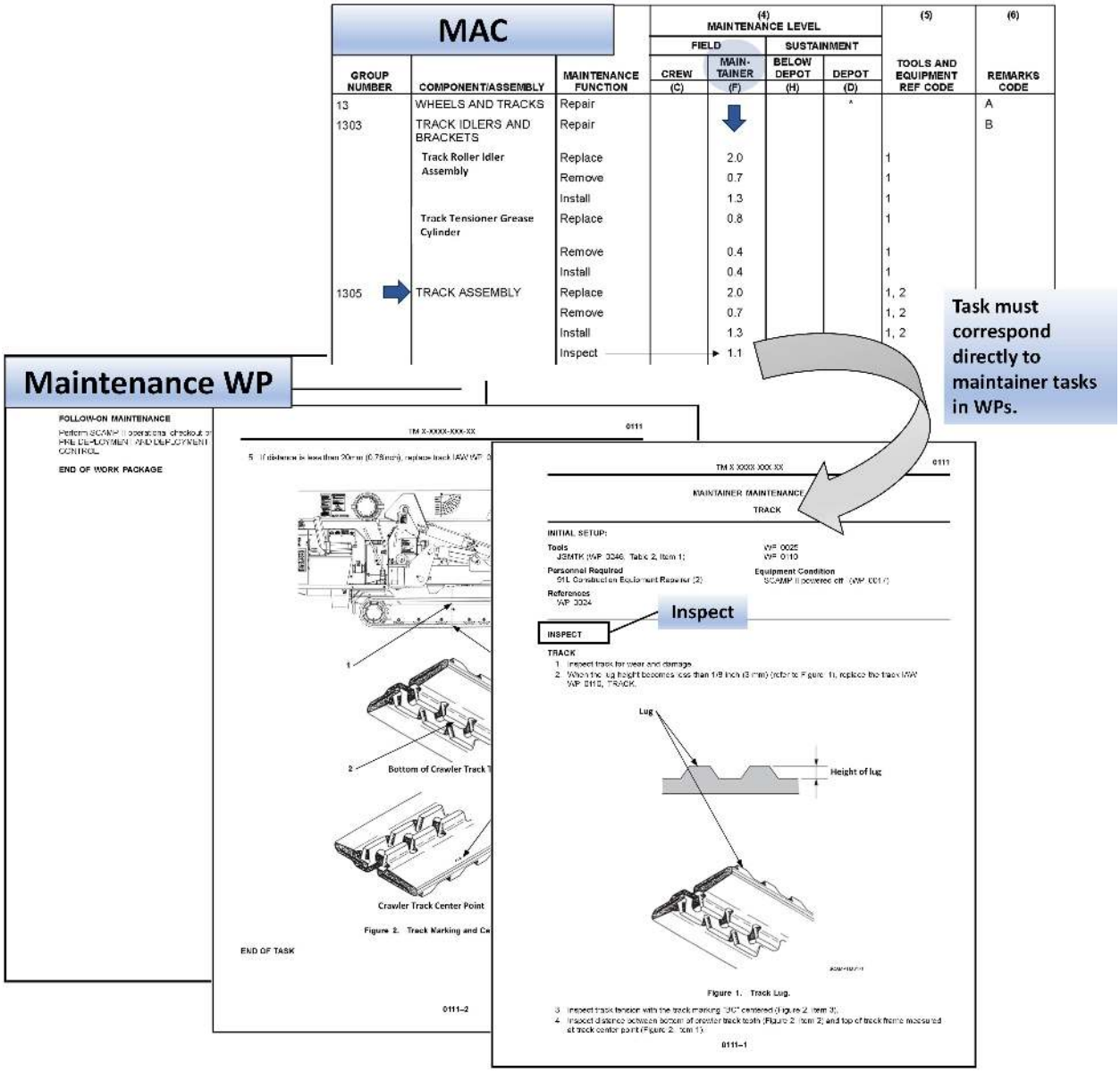


**FIGURE G-11. MAC and maintenance tasks (sheet 5 of 5).**

## G.7.3.1.3 MACs and RPSTLs.

RPSTL WPs shall align with MAC entries and maintenance WPs. FIGURE G-12 shows the RPSTL WP illustrating the repair parts associated with authorized repair for FGC 1303. FIGURE G-13 shows the RPSTL WP illustrating the repair parts associated with authorized repair for FGC 1305. These RPSTL WPs align with MAC entries and maintenance WPs discussed in sections G.7.3.1.1 and G.7.3.1.2.

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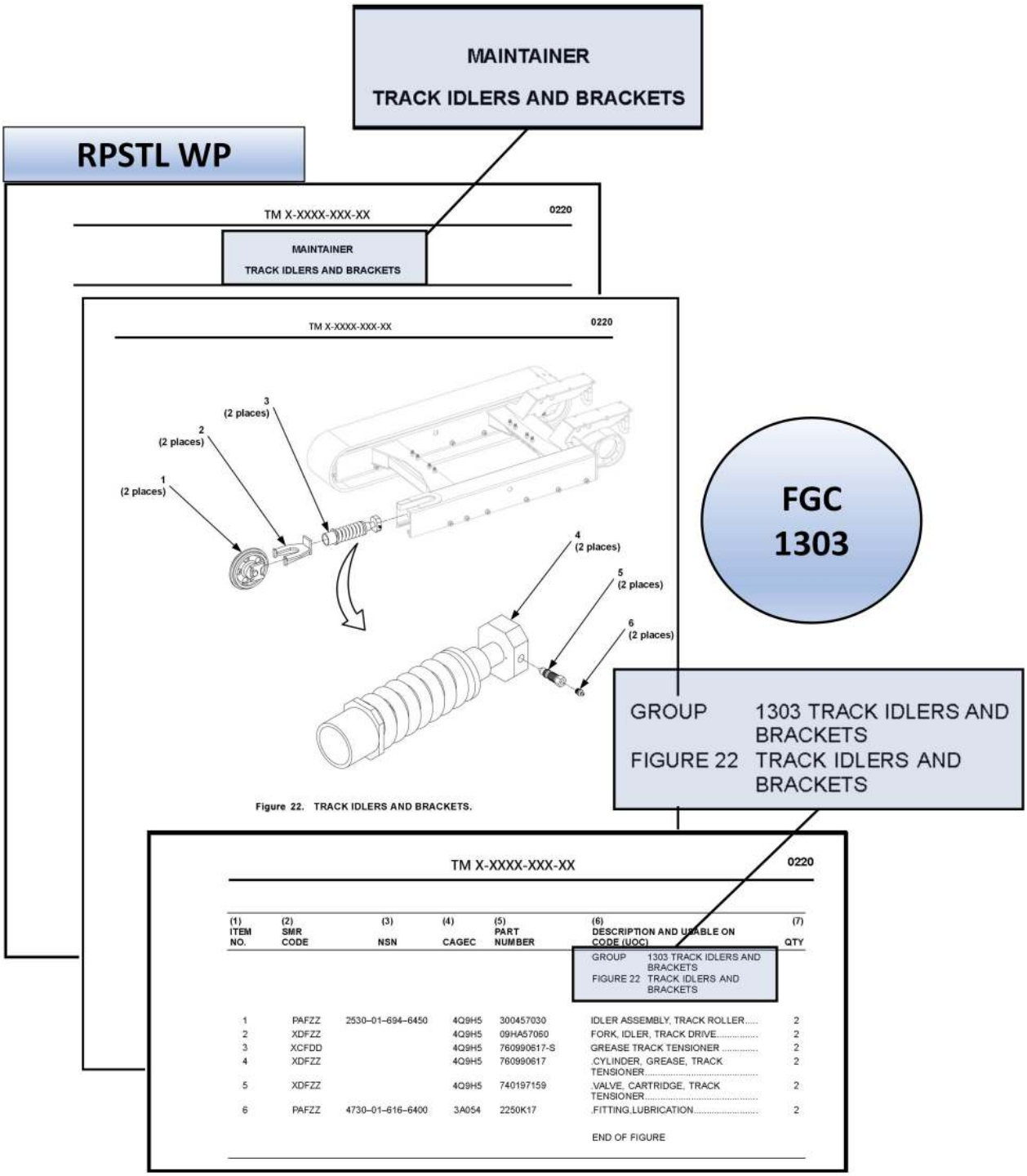


**FIGURE G-12. RPSTL WP (sheet 1 of 2).**

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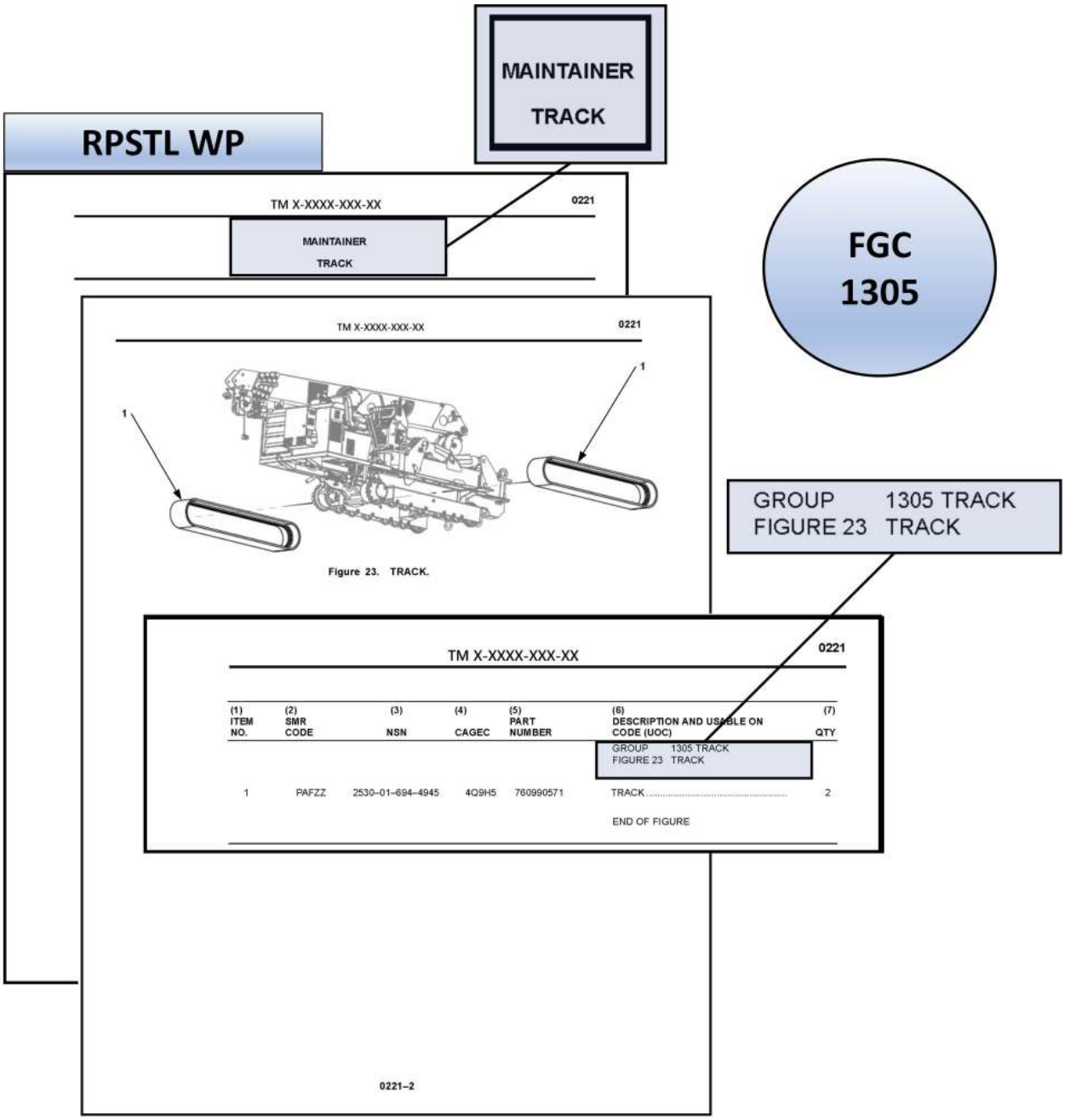


**FIGURE G-13. RPSTL WP (sheet 2 of 2).**

### G.7.3.1.4 MACs and maintenance tasks to facilitate other maintenance.

As mentioned in section G.7.3.1.1, the sole purpose of some MAC entries is to facilitate other maintenance. For example, FIGURE G-14 shows that LPD-authorized field-level repair for FGC 1801, "BODY, CAB, AND HULL ASSEMBLIES," is limited to "remove" and "install" tasks for a multitude of consoles and covers that are simply "in the way" of other components and, thus, preventing maintenance to internal system components. Remarks codes C and D explain that for these entries "steps in TM are included in steps for other tasks to facilitate maintenance." These types of MAC entries shall be included when a component would be removed/installed many times during the course of other maintenance tasks. Within maintenance procedures, these tasks can then be referenced as needed to

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reduce procedural redundancy. For instance, FIGURE G-14 shows "remove" and "install" tasks for three covers associated with FGC 1801: "Engine Compartment Cover (Top), "Diesel Engine Cover Assembly," and "LMI Console Assembly." FIGURE G-14 shows that the corresponding maintenance task for the diesel engine cover assembly only contains a "REMOVE" task and an "INSTALL" task. This WP does not include a "REPLACE" task for this cover because LPD does not authorize its replacement. This maintenance WP (and, more specifically, these individual tasks) shall be referenced as needed by other maintenance tasks (as shown in FIGURE G-15).

**Note**

Figure appearance and tags/output may vary slightly based on version of DTD/stylesheets utilized (see applicable DTD or tag definition list for available tags).

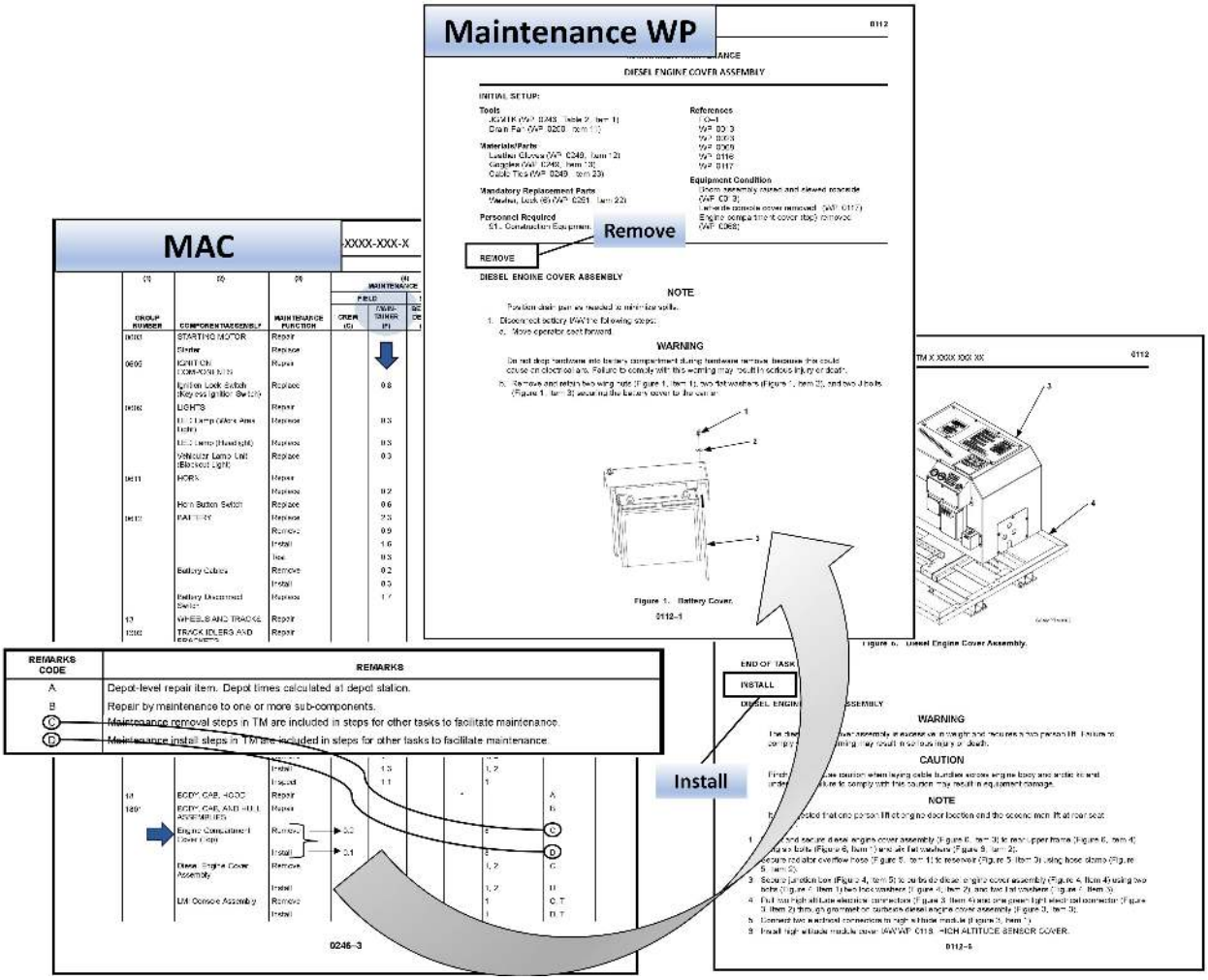


**FIGURE G-14. MAC and remove/install maintenance tasks.**

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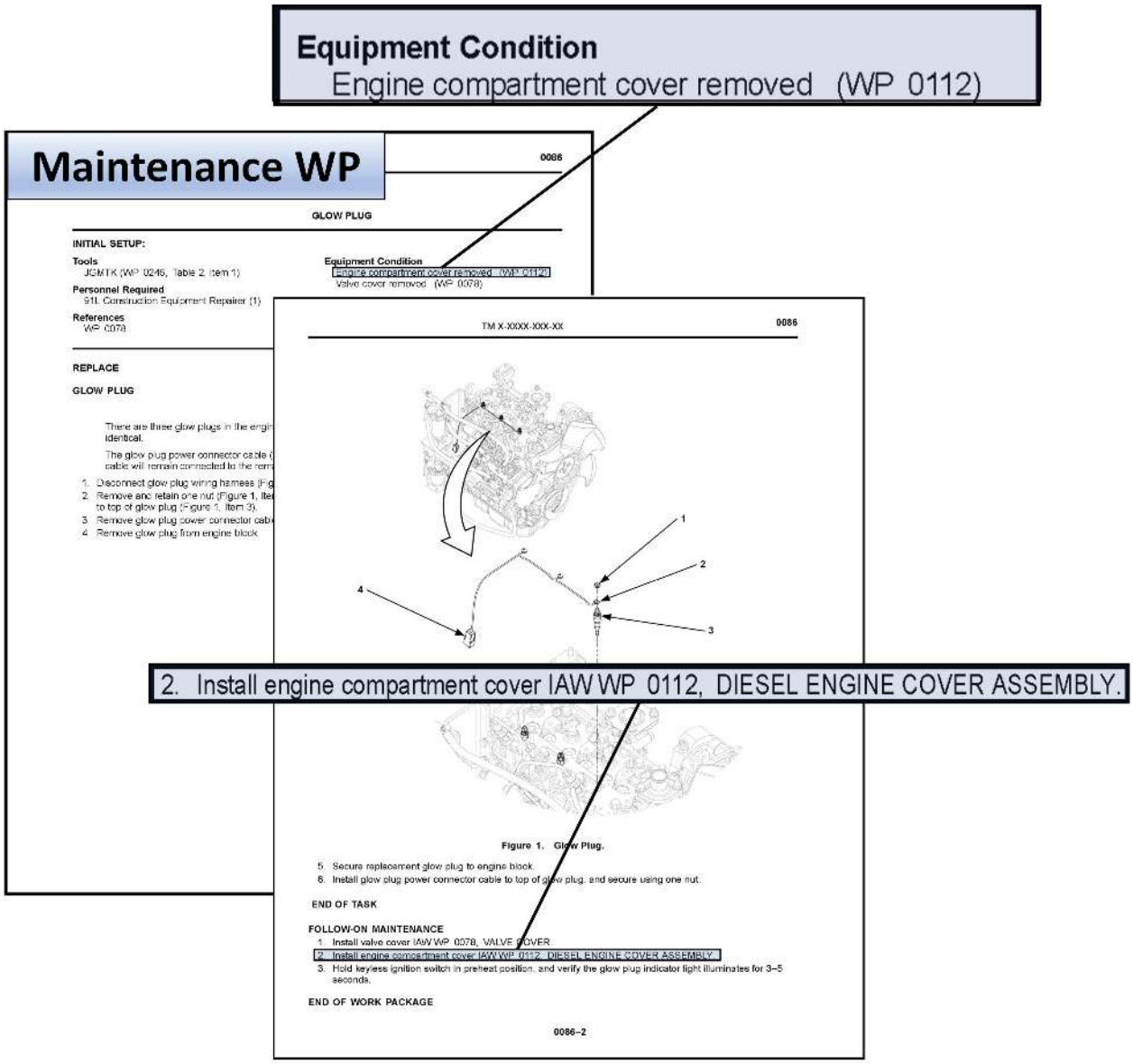


**FIGURE G-15. References in maintenance WP.**

## G.7.3.2 MAC table features, tags, and linking.

As stated above, the MAC drives (or informs) the creation of all maintenance tasks in a technical publication. Various MAC table features mentioned in this section serve to guide the creation of maintenance tasks and ensure the alignment across a publication. MAC tagging structure and linking further serve to facilitate alignment. The sections below discuss these concepts.

### G.7.3.2.1 MAC table features.

TABLE G-II lists general MAC table features (Column 1) and all associated MAC requirements (Column 2). FIGURE G-16 illustrates how these elements are presented in a MAC table and the tagging structure for an example FGC 00 ("CRANE, CRAWLER, MOUNTED, SCAMP II").

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**TABLE G-II. General MAC requirements.**

| MAC Feature | General Requirement |
|---|---|
| Functional Groups | A "functional group" is a repairable assembly or subassembly (for example, spares). |
| | The end item group shall be numbered "00." |
| | FGCs shall not exceed 11 characters in length. |
| | FGCs shall match those used in the RPSTL (except the FGCs used for software). |
| | Software shall be considered a major component and assigned a FGC accordingly. Software FGCs shall be listed in the MAC. The FGCs for software do not need to have a corresponding RPSTL WP (detailed software information shall be contained in COEI/BII or AAL WPs as necessary). |
| Nomenclature | When possible, all item names of MAC functional groups shall be the official nomenclature used in the FEDLOG H6 listing (https://public.logisticsinformationservice.dla.mil/H6/search.aspx) in reverse word order except for software.<br>**Note**<br>For ammunition TMs, "AA" may be used if needed (tactical entries). |
| | When necessary for clarity, the LCN nomenclature may be included in the MAC beneath the official nomenclature in the component/assembly column |
| | Where applicable, type designators may be used, without stock or part numbers (P/Ns) if possible, in order to minimize need for subsequent change; however, entries shall contain positive identification. |
| Maintenance Tasks | Individual components of a FGC requiring maintenance shall have their own entries beneath the FGC nomenclature in column (2). Level of maintenance, required task time, required tools, and remarks (if necessary) shall be added for each component maintenance task.<br>**Note**<br>Grouping tasks via individual maintenance tasks shall only be utilized if the maintenance task, level of maintenance, required task time, and required tools are identical for all components within the same FGC. If any of these attributes are different, a separate component entry shall be listed. A remark to clarify the grouped component tasks shall be created and referenced. |
| Maintenance Levels | All items in the MAC shall specify the maintenance levels to which a function is authorized. Authorizations are based on SMR source data in RPSTL (refer to APPENDIX F). |
| | Maintenance levels shall match maintenance WPs. The MAC shall be updated during change/revision cycle to reflect any changes made to maintenance tasks in that change/revision such as adding new tasks, deleting tasks, or changing times. |
| Man Hours | Man-hours equals the task time multiplied by the number of personnel required for a task. |
| | Operator maintenance time shall be entered as a task (clock) time only. If different levels of maintenance perform the same maintenance tasks due to the number or complexity of the tasks, appropriate man-hour figures shall be shown for each level. |
| | Task time includes task itself, follow-on maintenance, and inspection times. |
| | For ammunition, an "X" may be used in place of man-hour figure. |
| | For sustainment maintenance tasks where the man-hour figures are not yet known, an asterisk "*" or a dash "-" may be used in place of a time until such time as a time can be determined. |

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**TABLE G-II. General MAC requirements. (continued**

| MAC Feature | General Requirement |
|---|---|
| Tool Code References | Lists reference to table with tools required for each maintenance task listed in the MAC. Discussed further in Section G.7.3.2.4. |
| Remarks Code References | Lists reference to table with any information pertinent to maintenance task listed in the MAC. Discussed further in G.7.3.2.5. |

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**Note**

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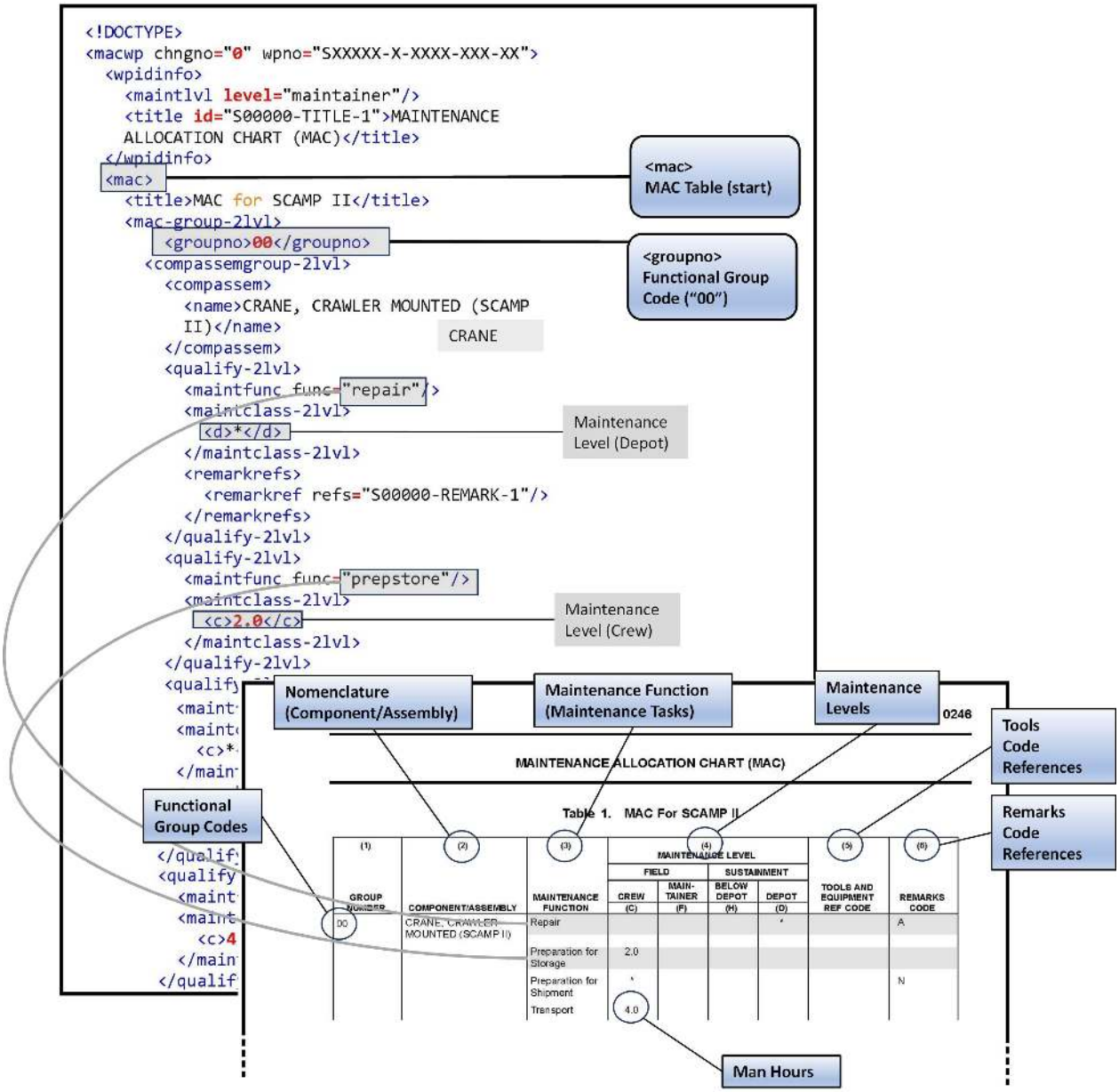


**FIGURE G-16. MAC entry components.**

## G.7.3.2.2 MAC tagging.

TABLE G-III lists MAC columns and their relevant functions while FIGURE G-17 through FIGURE G-18 show example associated tagging structures for a non-aviation MAC. The tags shown in these figures create the MAC columns listed in TABLE G-III.


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**TABLE G-III. MAC columns and functions.**

<table>
<tr><th>MAC Column</th><th>MAC Compo-nent</th><th colspan="2">Tags</th><th>Function</th></tr>
<tr><td>Column 1</td><td>FGC</td><td colspan="2"><groupno></td><td>Numbers which identify maintenance significant compo-nents, assemblies, subassemblies, and modules associate them with the Next Higher Assembly (NHA).</td></tr>
<tr><td>Column 2</td><td>Component/ Assembly</td><td colspan="2"><compassem></td><td>Item names of components, assemblies, subassemblies, and modules for which maintenance is authorized.</td></tr>
<tr><td>Column 3</td><td>Maintenance Task</td><td colspan="2"><maintfunc></td><td>Tasks to be performed on the item listed in column 2.</td></tr>
<tr><td rowspan="4">Column 4 (Non-Aviation MAC)</td><td rowspan="8">Maintenance Level</td><td rowspan="4"><main-tclass-2lvl></td><td>Crew</td><td rowspan="4">Specifies levels of maintenance authorized to perform each function listed in column 3.<br>Indicated by man-hours required listed in the appropriate subcolumn.<br>For multi-service manuals, an asterisk shall be placed next to the "C" and the following note shall follow the table to explain the asterisk:<br>Note<br>"This is a multi-service manual. While Army uses a "C." other services may use an "O" in this column."</td></tr>
<tr><td><c> <f><br>Maintainer</td></tr>
<tr><td><h><br>Below Depot</td></tr>
<tr><td><d><br>Depot</td></tr>
<tr><td rowspan="4">Column 4 (Aviation MAC)</td><td rowspan="4"><main-tclass-2lvl></td><td><o><br>Aviation Maintenance Company</td><td rowspan="4">Specifies levels of maintenance authorized to perform each function listed in column 3.<br>Indicated by man-hours required listed in the appropriate subcolumn.</td></tr>
<tr><td><f><br>Aviation Support Battalion</td></tr>
<tr><td><l><br>Theater Aviation Sustainment Maintenance Group</td></tr>
<tr><td><d><br>Sustainment</td></tr>
<tr><td>Column 5</td><td>Tools and Equipment Reference Code</td><td colspan="2"><teref></td><td>Number code that specifies common tool sets, kits, or out-fits (not individual tools), common Test, Measurement and Diagnostic Equipment (TMDE), common tools that are not part of a set, or outfit, special tools/kits, special TMDE, and special support equipment required to per-form the designated function.<br>Codes are linked to tools and test equipment listed in MAC Table 2.<br>Not required for all MAC entries (as required).</td></tr>
</table>

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**TABLE G-III. MAC columns and functions. (continued**

| MAC Column | MAC Compo-nent | Tags | Function | Function |
|---|---|---|---|---|
| Column 6 | Remarks Code | **<remarkref>** | | Letter code linked to remarks listed in MAC Table 3.<br>Listed in alphabetical order at each MAC entry.<br>Not required for all MAC entries (as required). |

**Note**

Tags/output shown in figure (s) below may vary slightly based on version of DTD/stylesheets utilized (see applicable DTD or tag definition list for available tags).

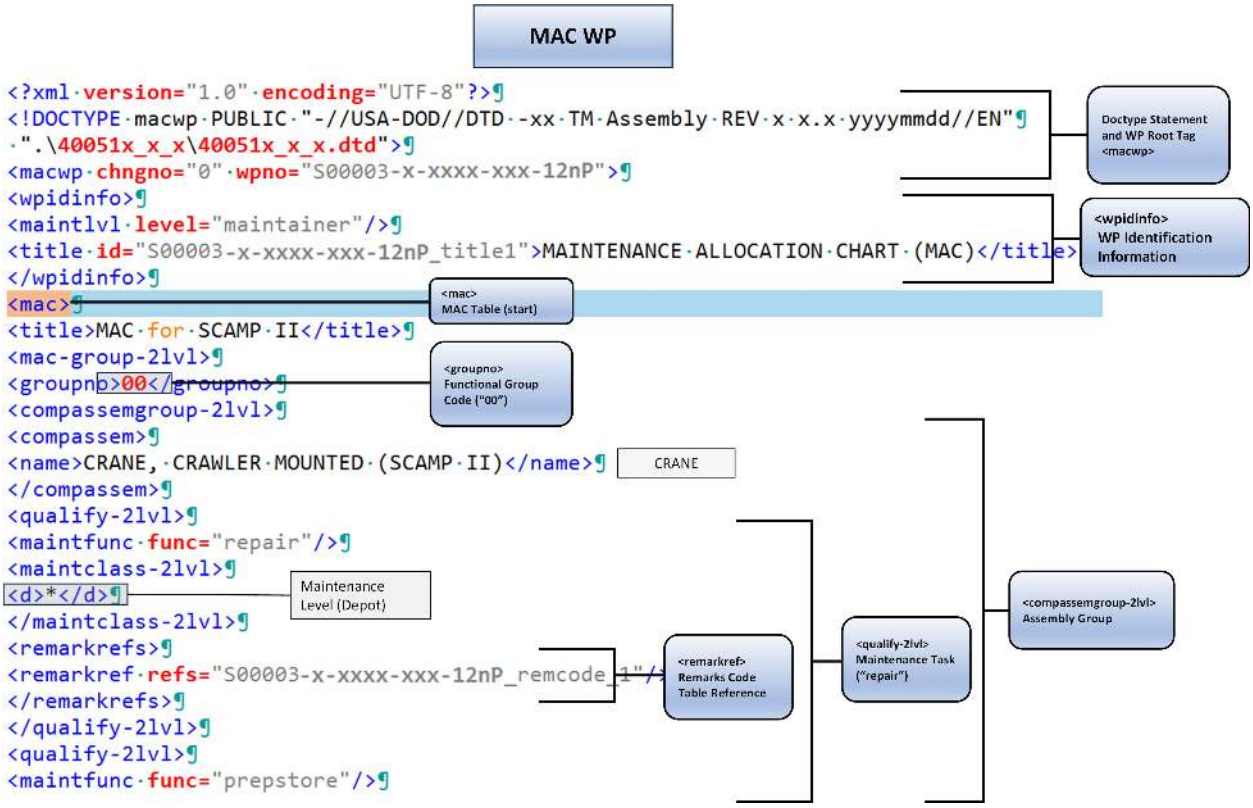


**FIGURE G-17. MAC WP tags (sheet 1 of 2).**



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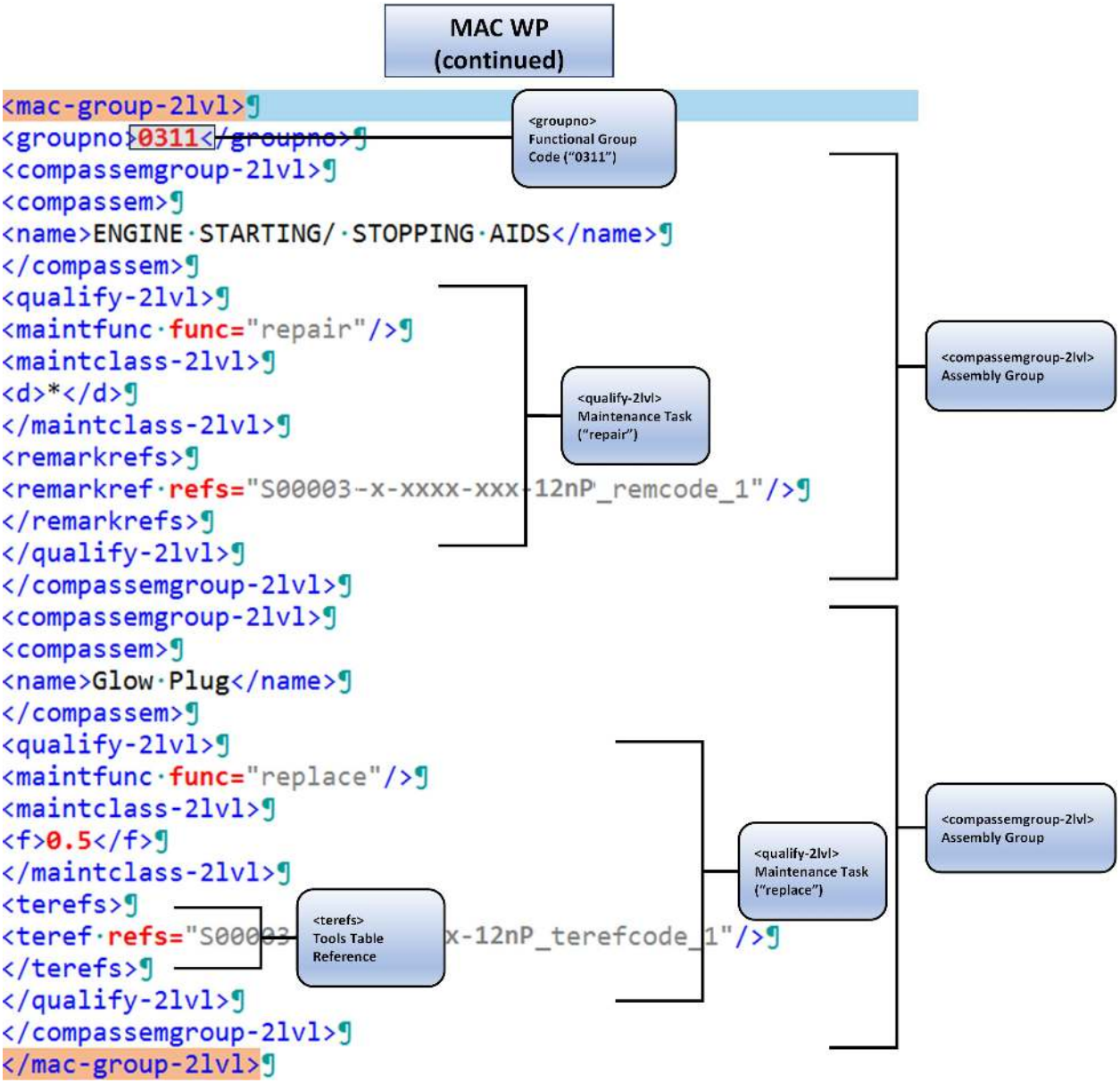


**FIGURE G-18. MAC WP tags (sheet 2 of 2).**

### G.7.3.2.3 MAC referencing to maintenance tasks.

MAC entries shall be represented in a technical publication in some manner. This representation is usually accomplished with maintenance tasks in maintenance modules; however, it may also include PMCS entries (as necessary). At times, MAC entries may be represented as operator instructions that include maintenance procedures.

Conversely, not all maintenance tasks that appear in a technical publication shall necessarily be represented in the MAC. Smaller tasks like cleaning, for example, may not be listed in a MAC. Most notably, LPD specifies which tasks **<maintfunc>** shall be listed in a MAC, and these corresponding MAC entries shall be linked (referenced) to maintenance tasks within maintenance WPs as frequently as possible.

FIGURE G-20 and FIGURE G-22 depict how MAC entries shall reference (and, thus, align) with maintenance tasks. These graphics include tagging examples that successfully link specific MAC entries to their corresponding

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maintenance tasks in maintenance WPs. Referencing MAC entries to maintenance tasks is also discussed in section E.5.3.2. Troubleshooting tasks **<tswp>** shall not be included nor linked in maintenance line items in the MAC.

FIGURE G-19 and FIGURE G-21 depict the same MAC entries and corresponding maintenance WPs shown in FIGURE G-20 and FIGURE G-22. These graphics, however, provide additional information in blue sections explaining how maintenance tasks are represented within context of each other as well as the MAC based on system LPD. Ensuring MAC entries reference their respective maintenance tasks as much as possible accomplishes two things:

**a.** Ensures alignment between the MAC and maintenance tasks included in a technical publication.

**b.** For ETMs and IETMs, allows users to easily navigate between the MAC (where equipment hierarchy is outlined) and maintenance tasks (where maintenance on specific parts is provided).

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**Note**

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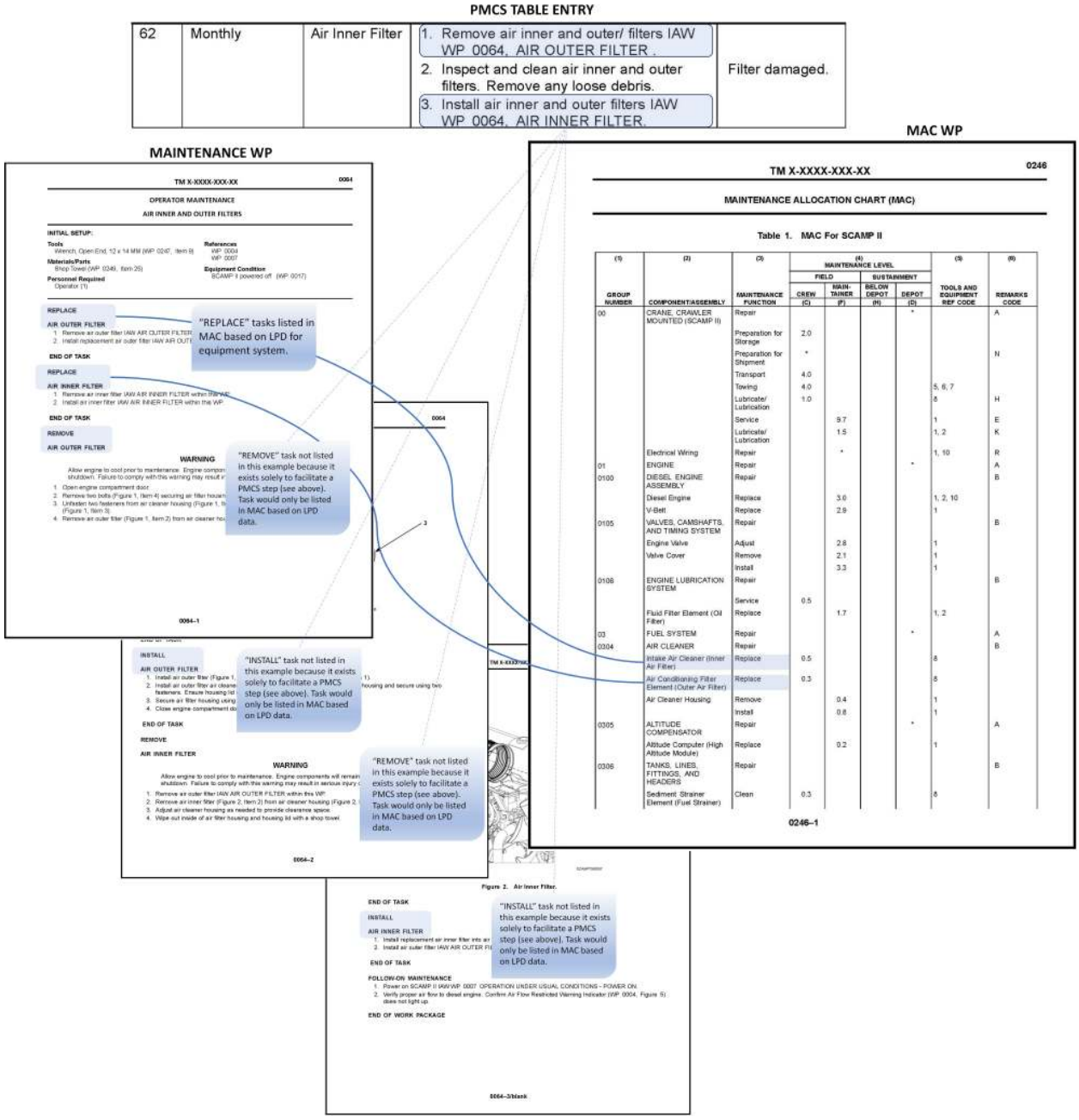


**FIGURE G-19. MAC and maintenance task linking (sheet 1 of 4).**

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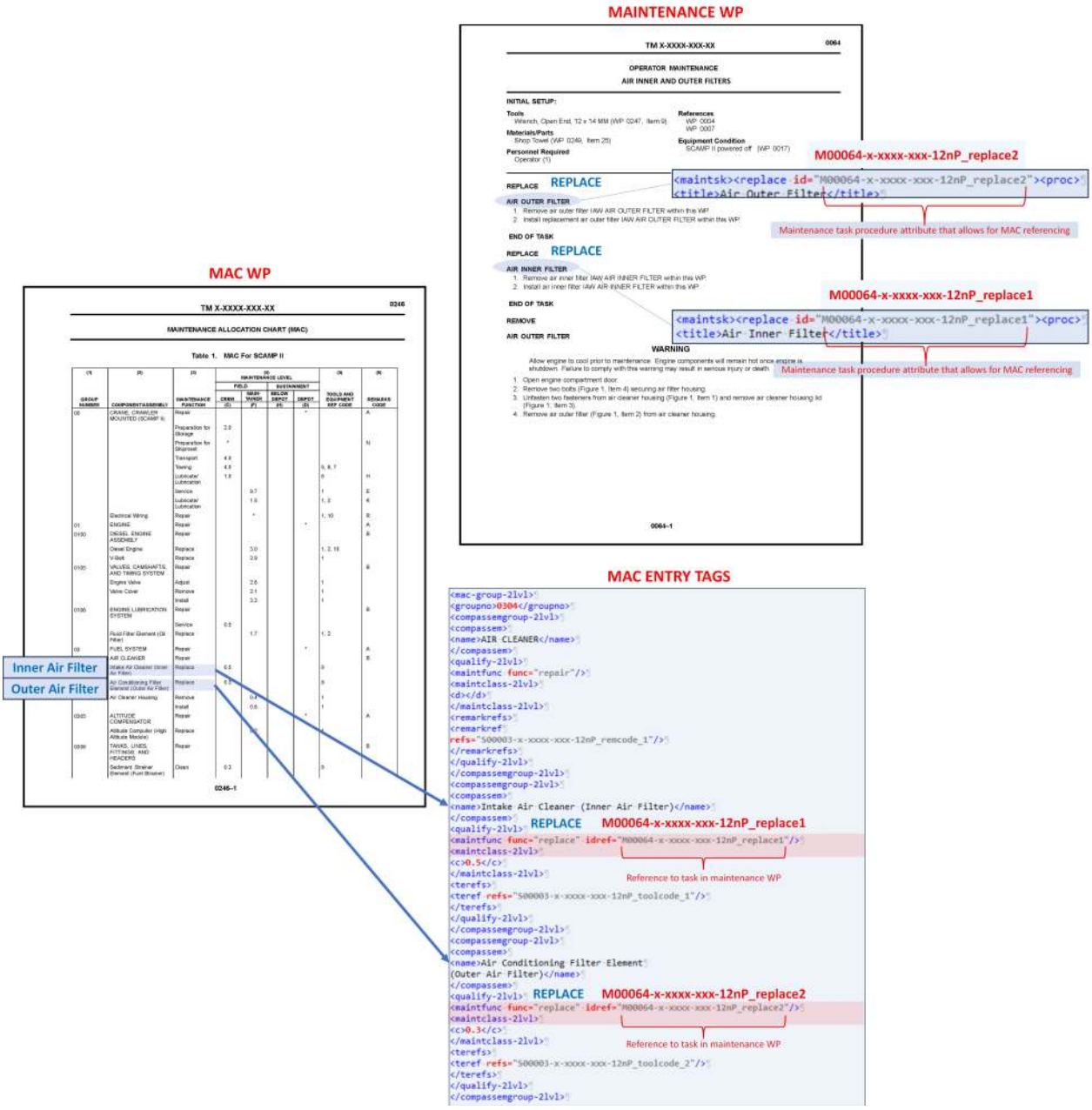


**FIGURE G-20. MAC and maintenance task linking (sheet 2 of 4).**

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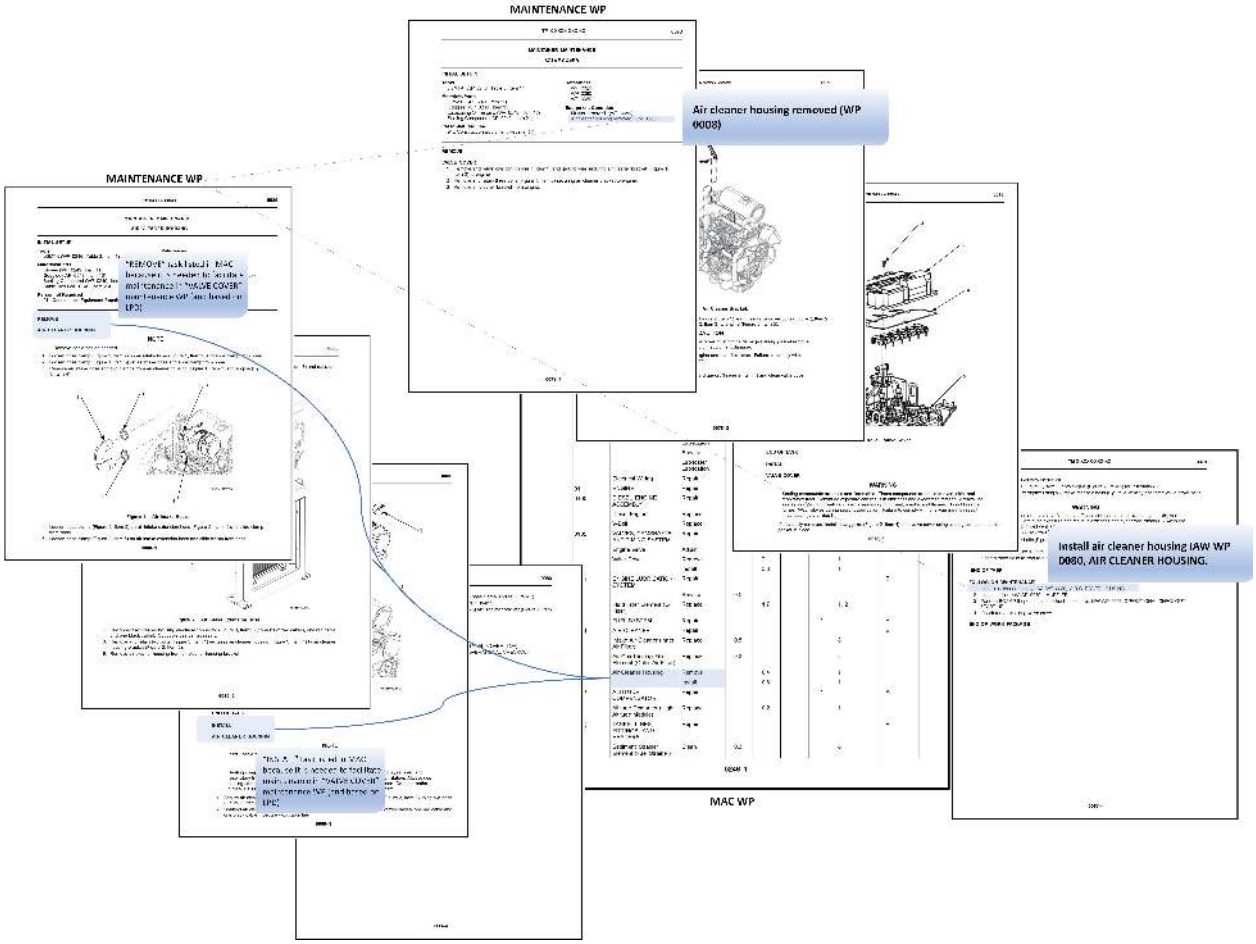


**FIGURE G-21. MAC and maintenance task linking (sheet 3 of 4).**

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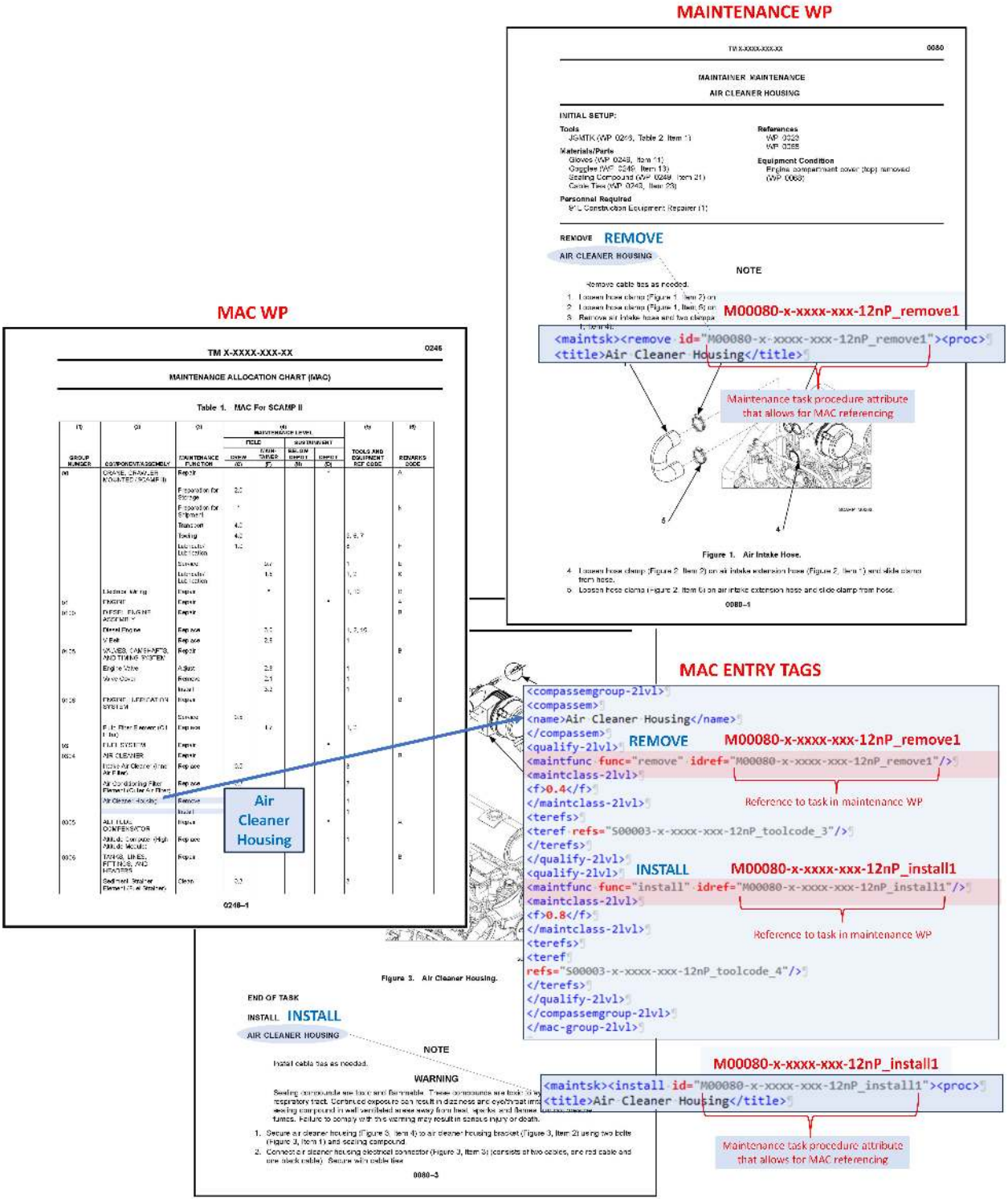


**FIGURE G-22. MAC and maintenance task linking (sheet 4 of 4).**



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## G.7.3.2.4 Tools and test equipment requirements <tereqtab>.

A tools and test equipment list (standard format information per 4.8) shall be prepared in accordance with the format shown in FIGURE G-23 through FIGURE G-24, as applicable. This list shall be in alphabetical order by name and serves to provide any tool and test equipment data necessary to support maintenance listed in the MAC. Every item listed in this table shall be referenced in the MAC, Column 5 at least once. The following items may be included in this list:

**a.** Common tool sets, kits, or outfits (not individual tools).

**b.** Common Test, Measurement and Diagnostic Equipment (TMDE).

**c.** Special tools/kits.

**d.** Special TMDE.

**e.** Special support equipment required to maintain the equipment.

**Note**

As necessary, this table may duplicate tool sets/kits/outfits listed in the tool ID WP.

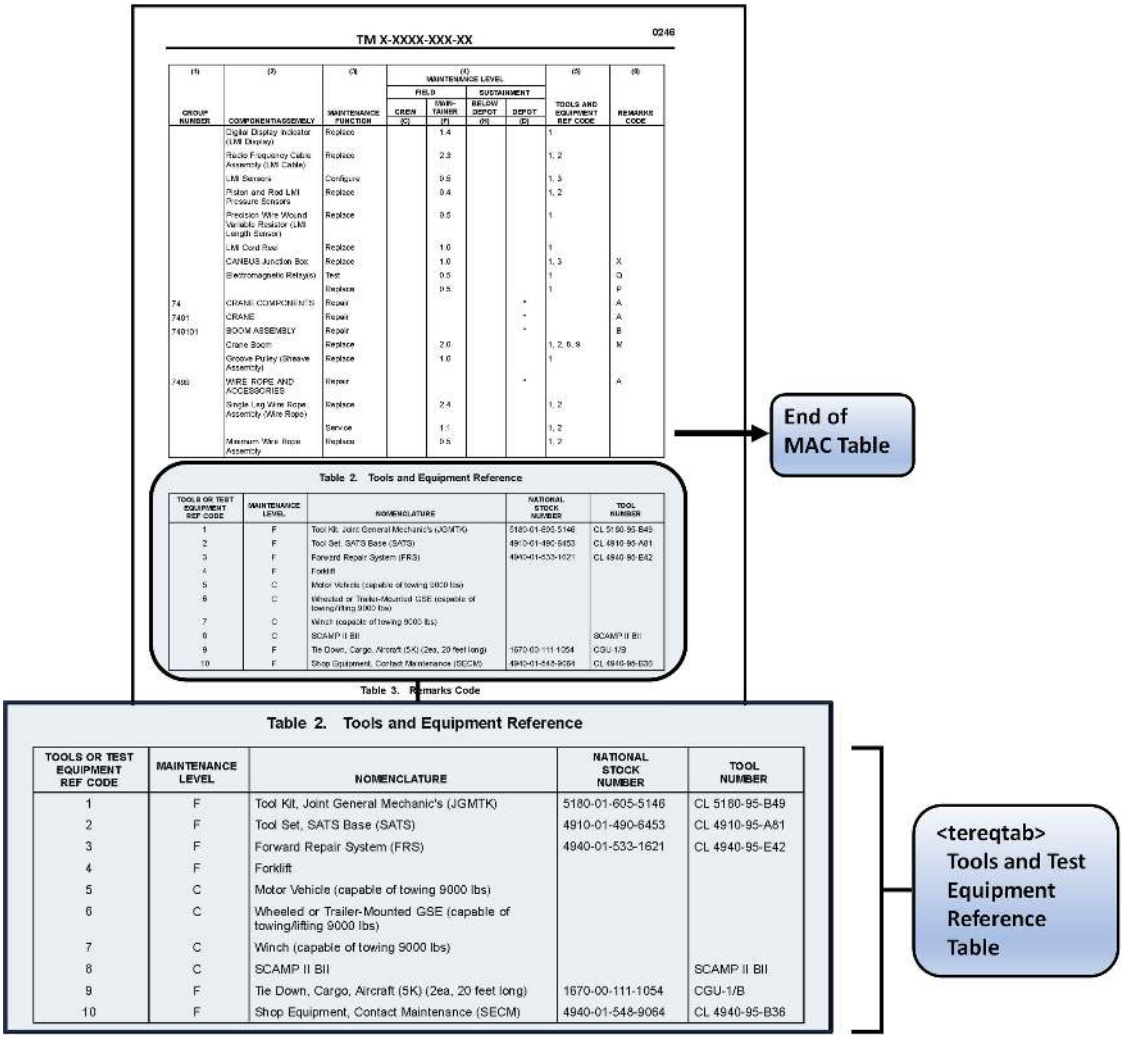


**FIGURE G-23. Tool and test equipment reference table.**

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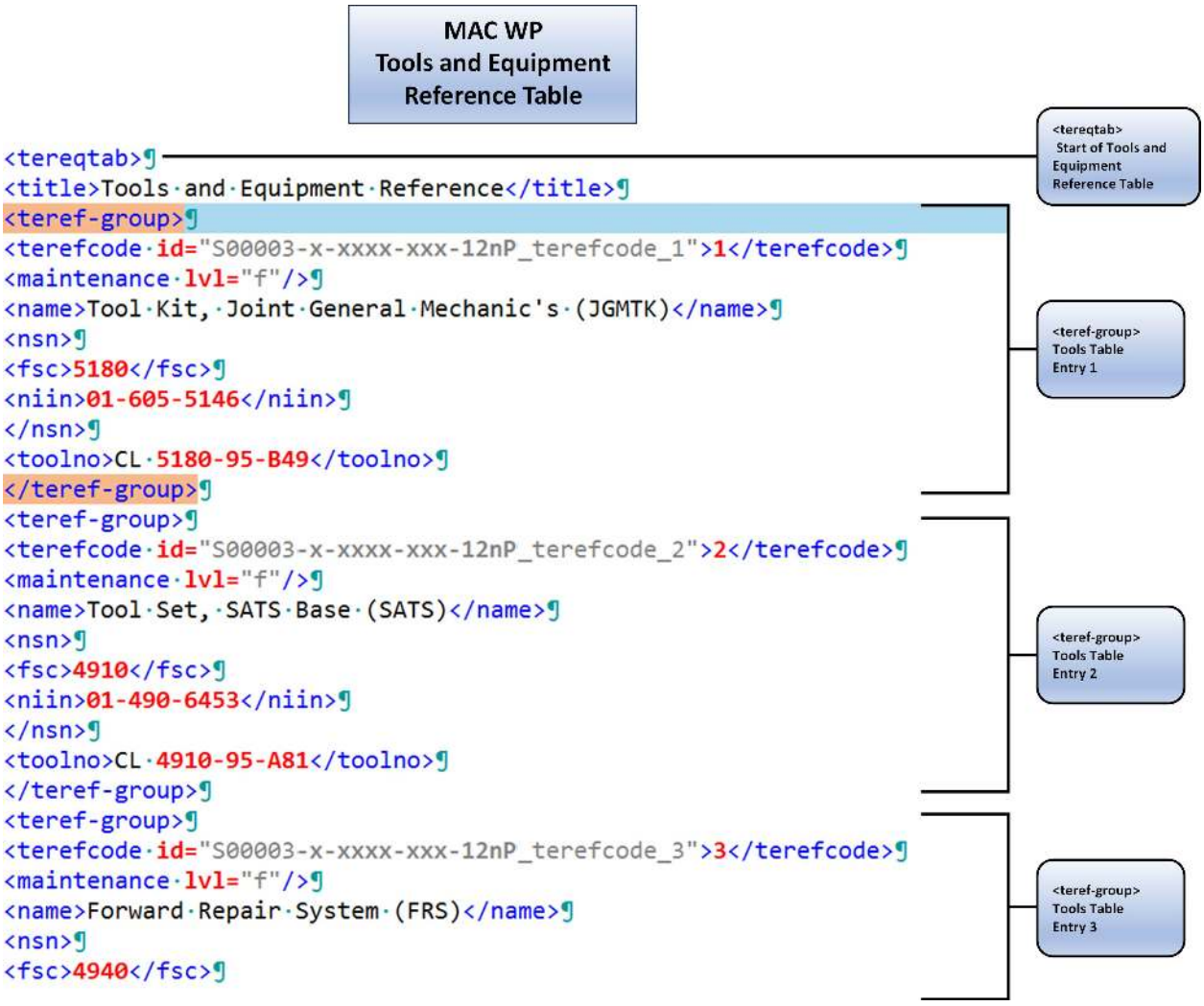


**FIGURE G-24. Tool and test equipment reference table tags.**

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FIGURE G-25 emphasizes that all tools and special tools listed in the initial setup sections of WPs shall link directly to the tool ID WP (not MAC, Table 2). FIGURE G-26 provides greater detail about tool classifications and whether they shall be listed in the tool and test equipment table (MAC, Table 2) or the tool ID WP (G.7.8).

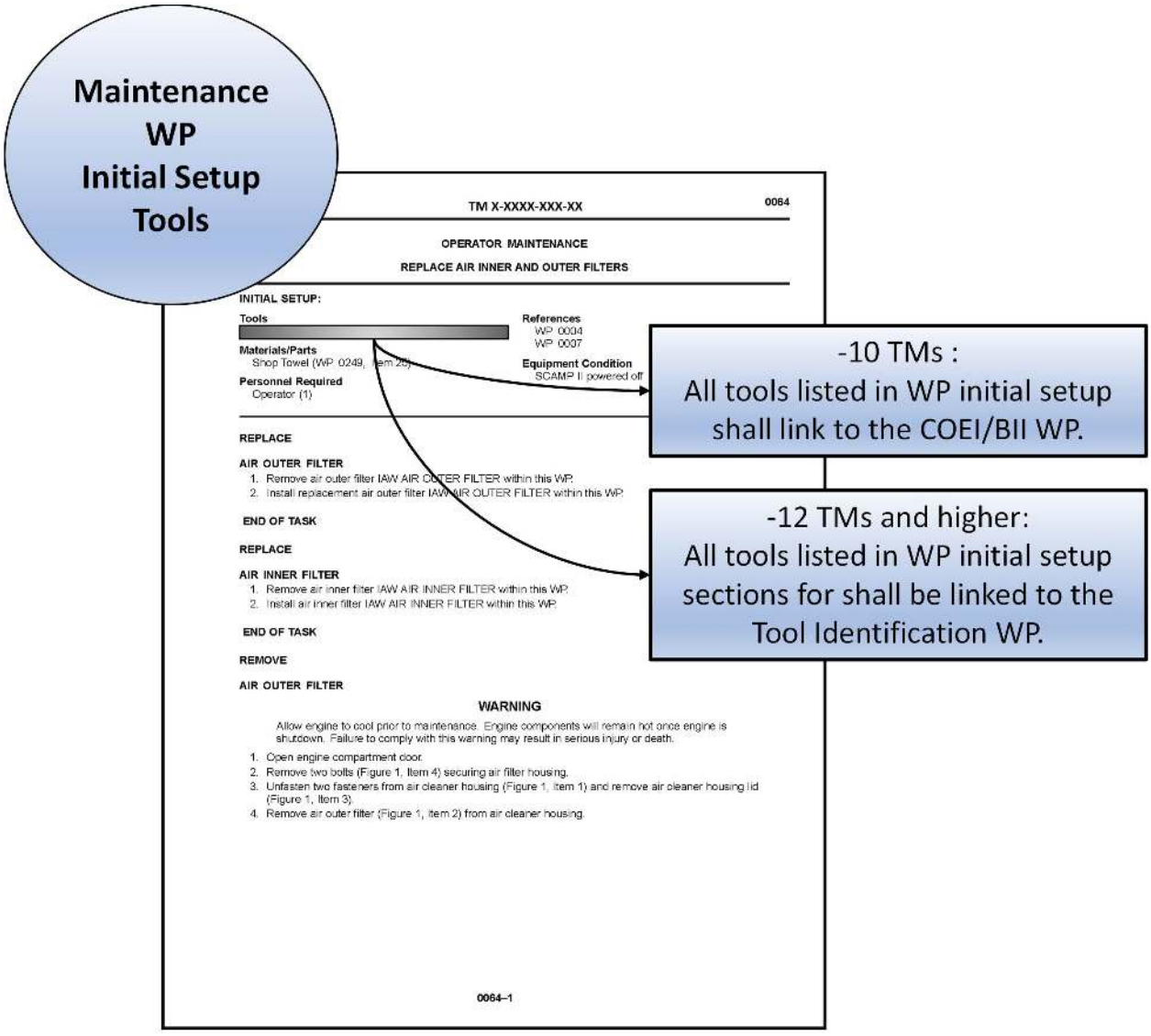


**FIGURE G-25. Tools classifications and listings.**

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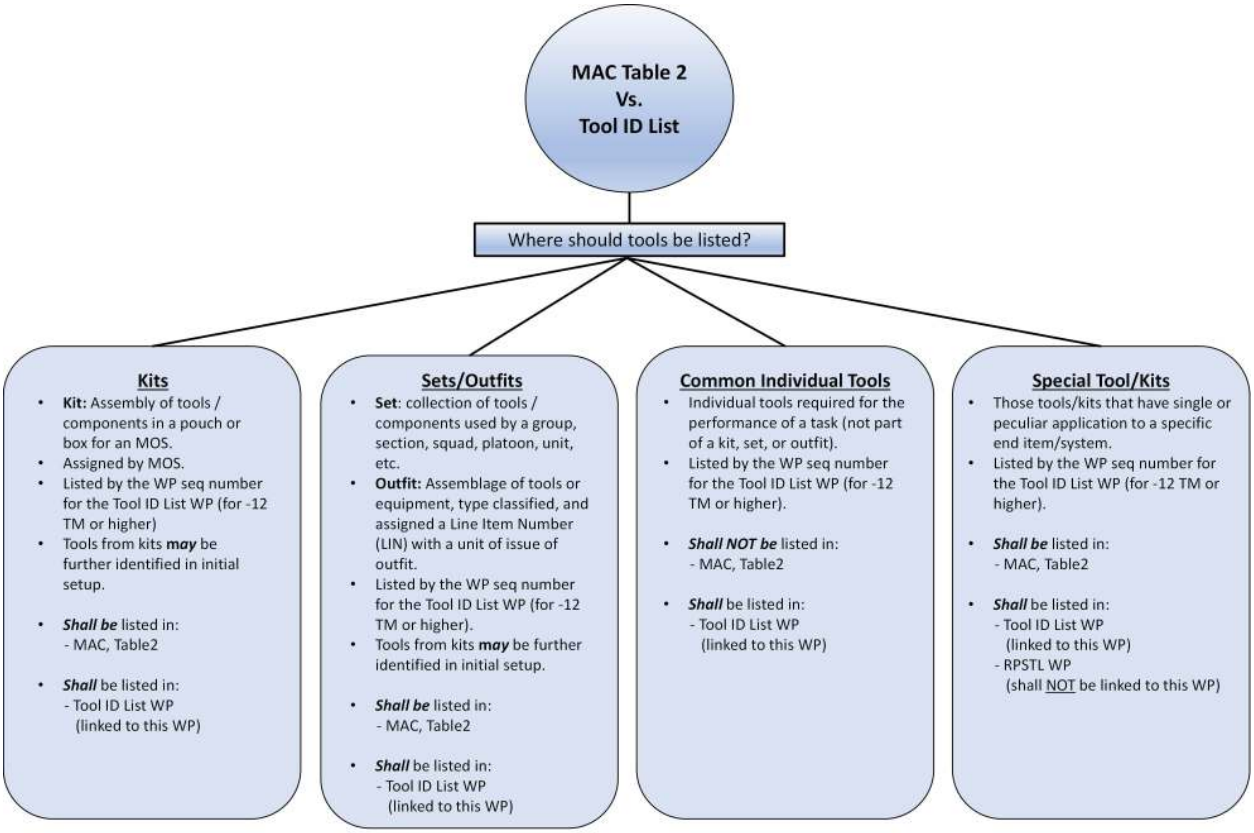


**FIGURE G-26. Tool and test equipment reference table (MAC table 2) versus the tool identification list.**

## G.7.3.2.5 Remarks <remarktab>.

Remarks (standard format information per 4.8) pertinent to maintenance tasks shall be prepared as applicable and shall be listed in this table alphabetically by remarks code. Remarks table format shall be as shown in FIGURE G-27 and FIGURE G-28, as applicable. Remarks codes serve to provide additional information related to maintenance tasks including, but not limited to, the following:

**a.** Depot-level repair designations.

**b.** Below-depot repairs (sustainment) included in technical publication as field-level maintenance.

**c.** Specific WP numbers where tasks are located in a technical publication.

**d.** Maintenance tasks included as part of maintenance to a greater assembly listed in MAC column 3 (i.e., "Repair by maintenance to one or more subcomponents.").

**Note**

Maintenance tasks can be linked within remarks codes for easy user navigation.

**e.** Identification of any below-depot tasks (H) that are included in the technical publication and considered "field maintenance" because a soldier may perform them. See also APPENDIX E for information about sustainment maintenance structure in a technical publication.

**f.** Equipment details such as "battery cables removed and installed to facilitate other maintenance."

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**Note**

Figure appearance and tags/output may vary slightly based on version of DTD/stylesheets utilized (see applicable DTD or tag definition list for available tags).

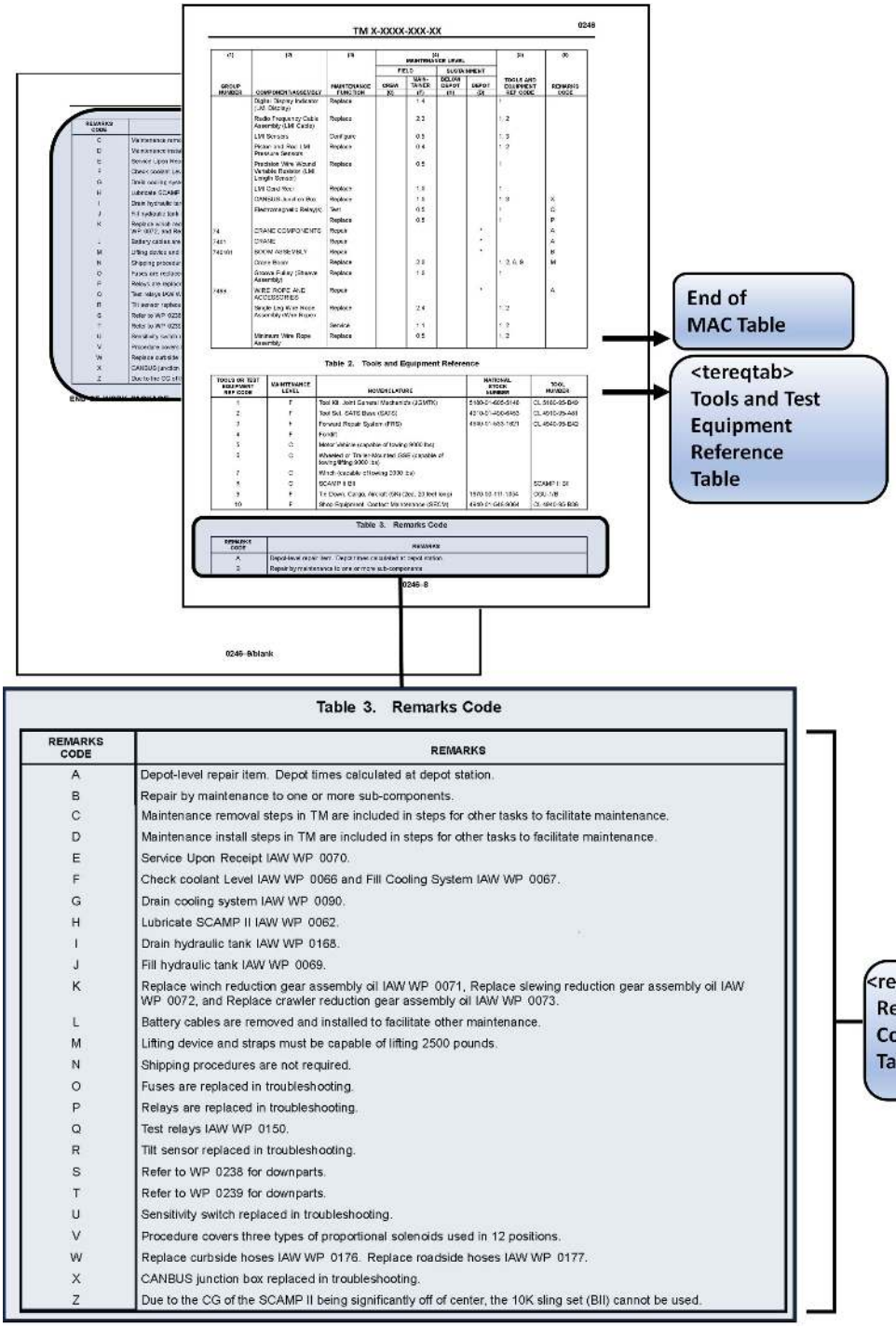


Table 3. Remarks Code

| REMARKS CODE | REMARKS |
|---|---|
| A | Depot-level repair item. Depot times calculated at depot station. |
| B | Repair by maintenance to one or more sub-components. |
| C | Maintenance removal steps in TM are included in steps for other tasks to facilitate maintenance. |
| D | Maintenance install steps in TM are included in steps for other tasks to facilitate maintenance. |
| E | Service Upon Receipt IAW WP 0070. |
| F | Check coolant Level IAW WP 0066 and Fill Cooling System IAW WP 0067. |
| G | Drain cooling system IAW WP 0090. |
| H | Lubricate SCAMP II IAW WP 0062. |
| I | Drain hydraulic tank IAW WP 0168. |
| J | Fill hydraulic tank IAW WP 0069. |
| K | Replace winch reduction gear assembly oil IAW WP 0071, Replace slewing reduction gear assembly oil IAW WP 0072, and Replace crawler reduction gear assembly oil IAW WP 0073. |
| L | Battery cables are removed and installed to facilitate other maintenance. |
| M | Lifting device and straps must be capable of lifting 2500 pounds. |
| N | Shipping procedures are not required. |
| O | Fuses are replaced in troubleshooting. |
| P | Relays are replaced in troubleshooting. |
| Q | Test relays IAW WP 0150. |
| R | Tilt sensor replaced in troubleshooting. |
| S | Refer to WP 0238 for downparts. |
| T | Refer to WP 0239 for downparts. |
| U | Sensitivity switch replaced in troubleshooting. |
| V | Procedure covers three types of proportional solenoids used in 12 positions. |
| W | Replace curbside hoses IAW WP 0176. Replace roadside hoses IAW WP 0177. |
| X | CANBUS junction box replaced in troubleshooting. |
| Z | Due to the CG of the SCAMP II being significantly off of center, the 10K sling set (BII) cannot be used. |

**FIGURE G-27. Remarks code table.**

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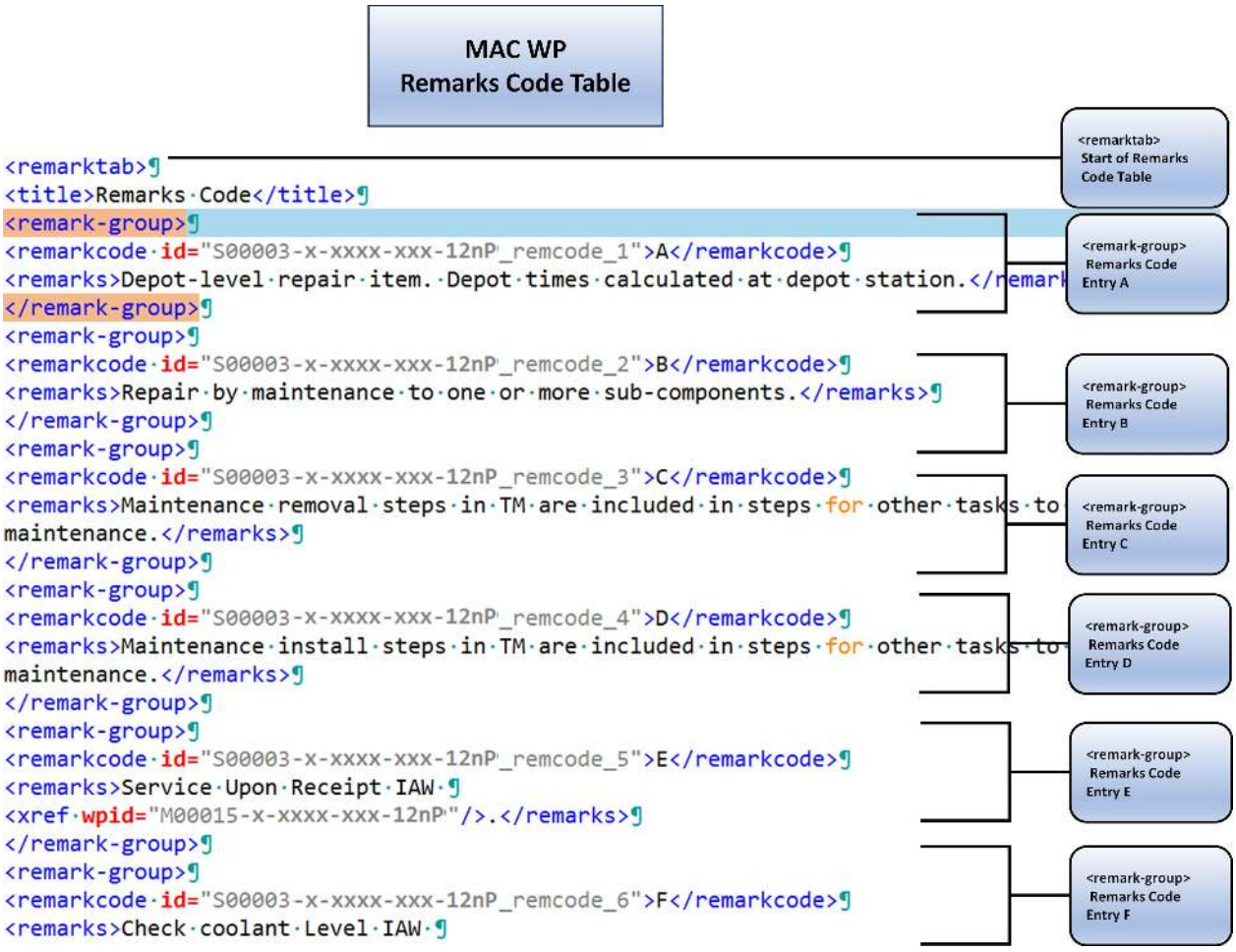


**FIGURE G-28. Remarks code table tags.**

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### G.7.3.3 Marine Corps MAC WP <macwp>.

FIGURE G-29 shows an example Marine Corps MAC WP.

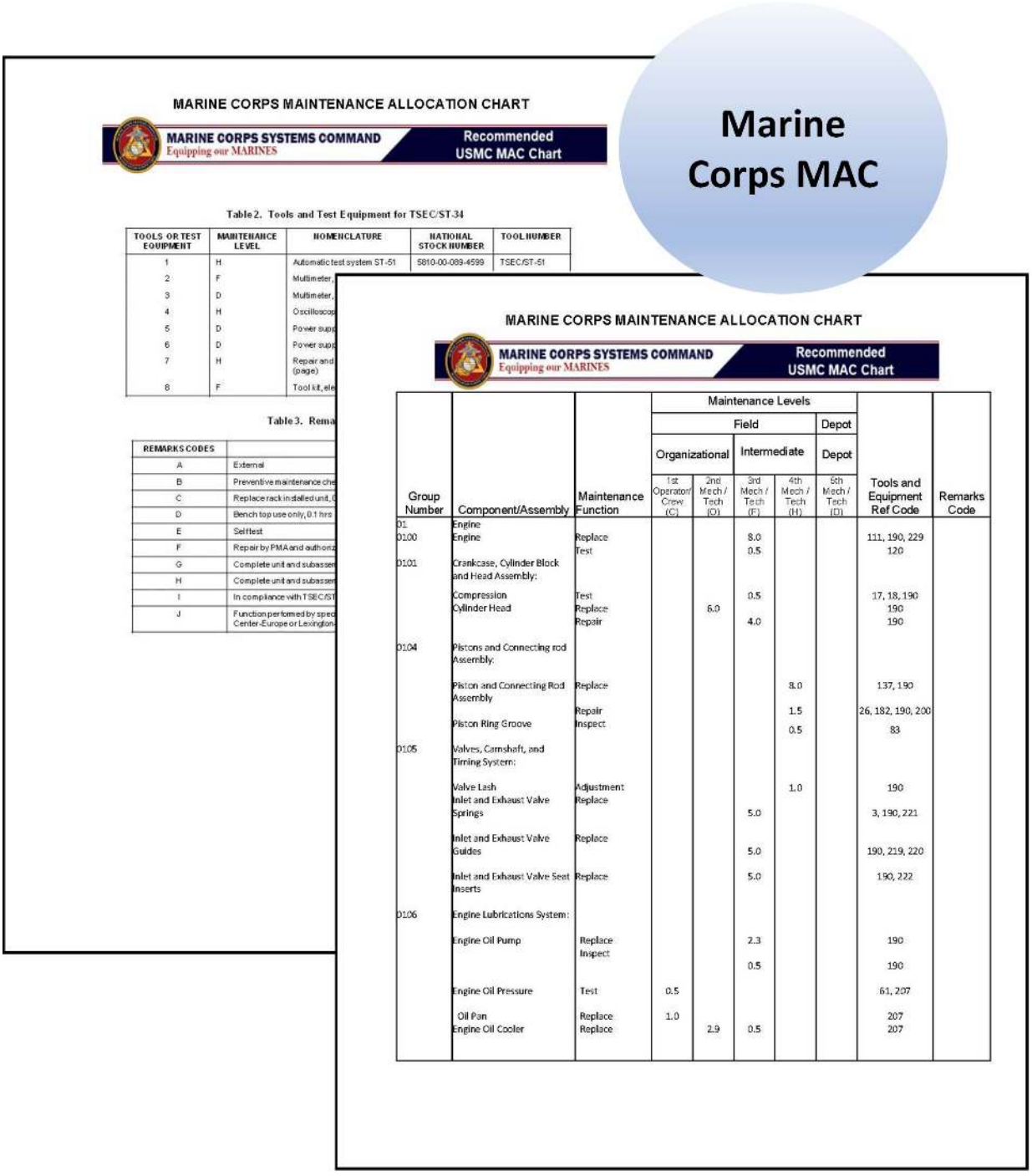


**FIGURE G-29. Marine Corps MAC.**

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## G.7.4 COEI / BII (Army or Supply System Responsibility (SSR (Marine Corps lists work package <coeibiiwp>.

The COEI / BII or SSR WP shall be prepared as required by publication type (see A.4.4). This WP is an inventory for equipment to ensure safe and efficient operation. Army only manuals and multi-service manuals shall use term "COEI/BII," while Marine Corps only manuals shall use the term "Supply System Responsibility (SSR)." For multiservice manuals involving the Marine Corps, the term "COEI/BII" shall be used regardless of the lead service. WP format shall be based on the number of items and usability. Two formats are available. Method A incorporates graphics above a tabular equipment listing (see example in FIGURE G-30 and FIGURE G-31). Also, for method A, larger quantities of items may be broken into several smaller tables for usability. Method B incorporates graphics within the tabular listing for better usability (see example in FIGURE G-32). Method B is typically used for large quantities of items. WP components are listed below:

**a.** Work package identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** Initial setup **<initial_setup>** is not required for this WP.

**c.** Introduction **<intro>** (G.7.4.1 through G.7.4.2 (MC)).

**d.** COEI **<coei>** (G.7.4.3 through G.7.4.5).

**e.** BII **<bii>** (G.7.4.6).



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**Note**

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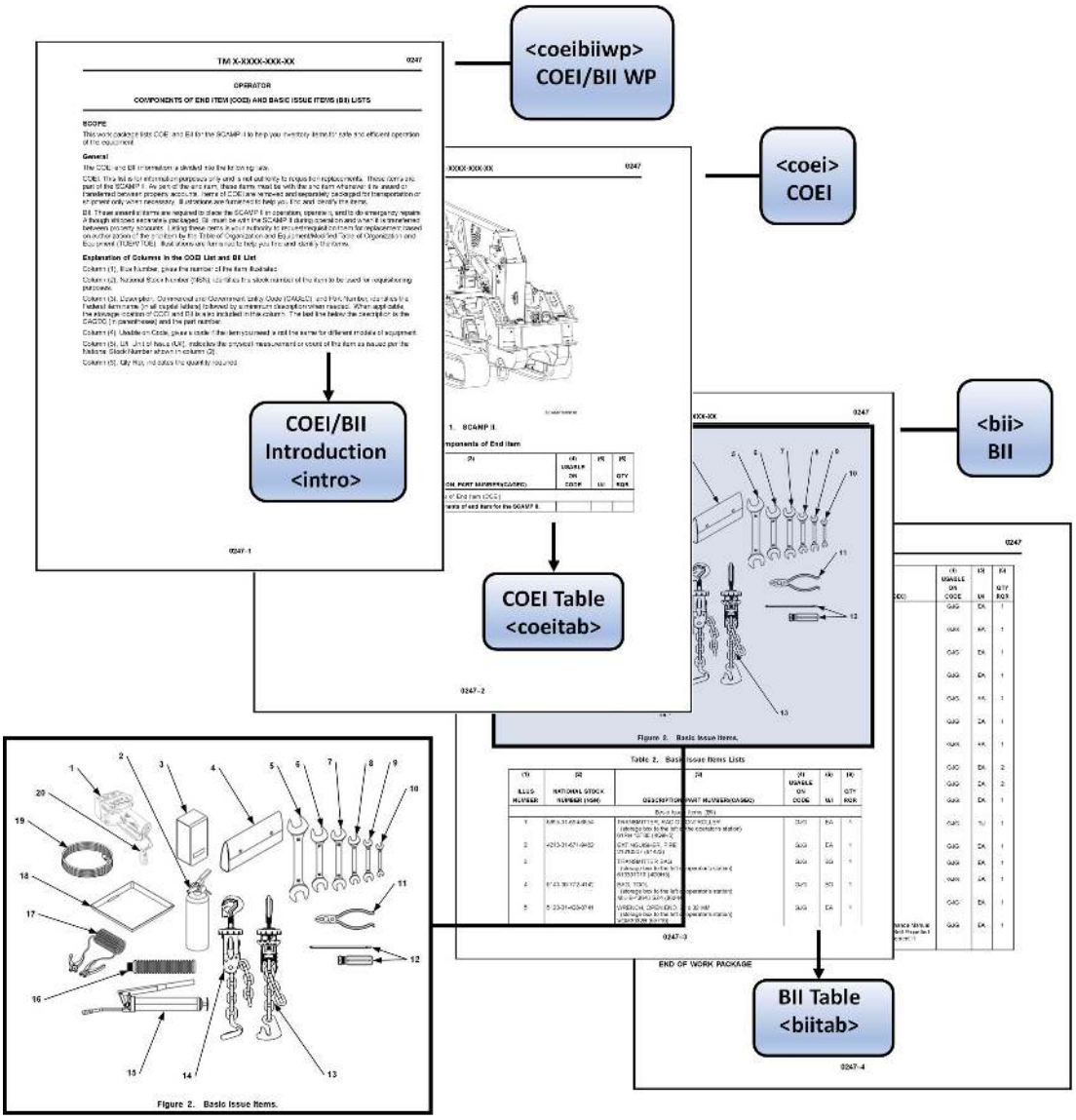


**FIGURE G-30. COEI/BII method A example (sheet 1 of 2).**

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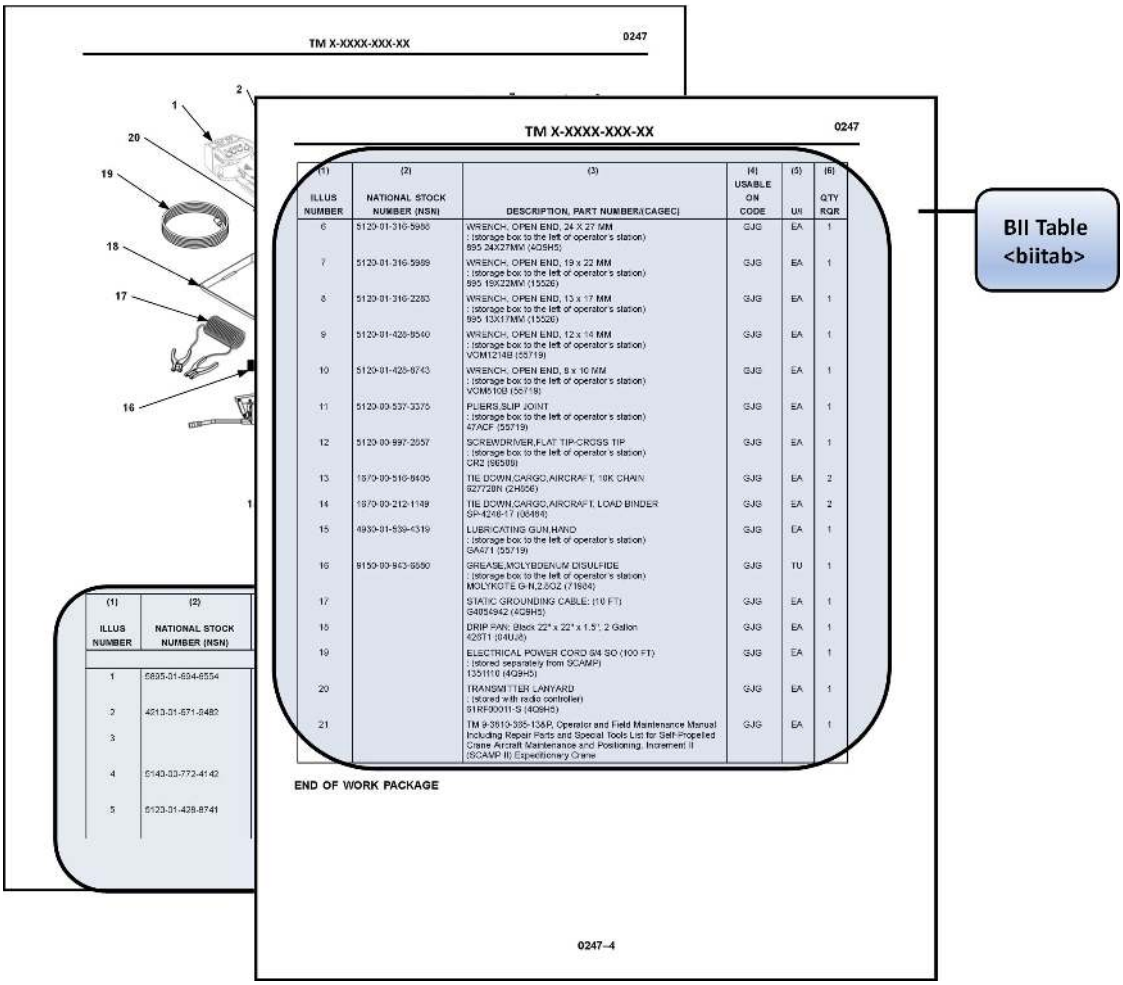


**FIGURE G-31. COEI/BII method A example (sheet 2 of 2).**

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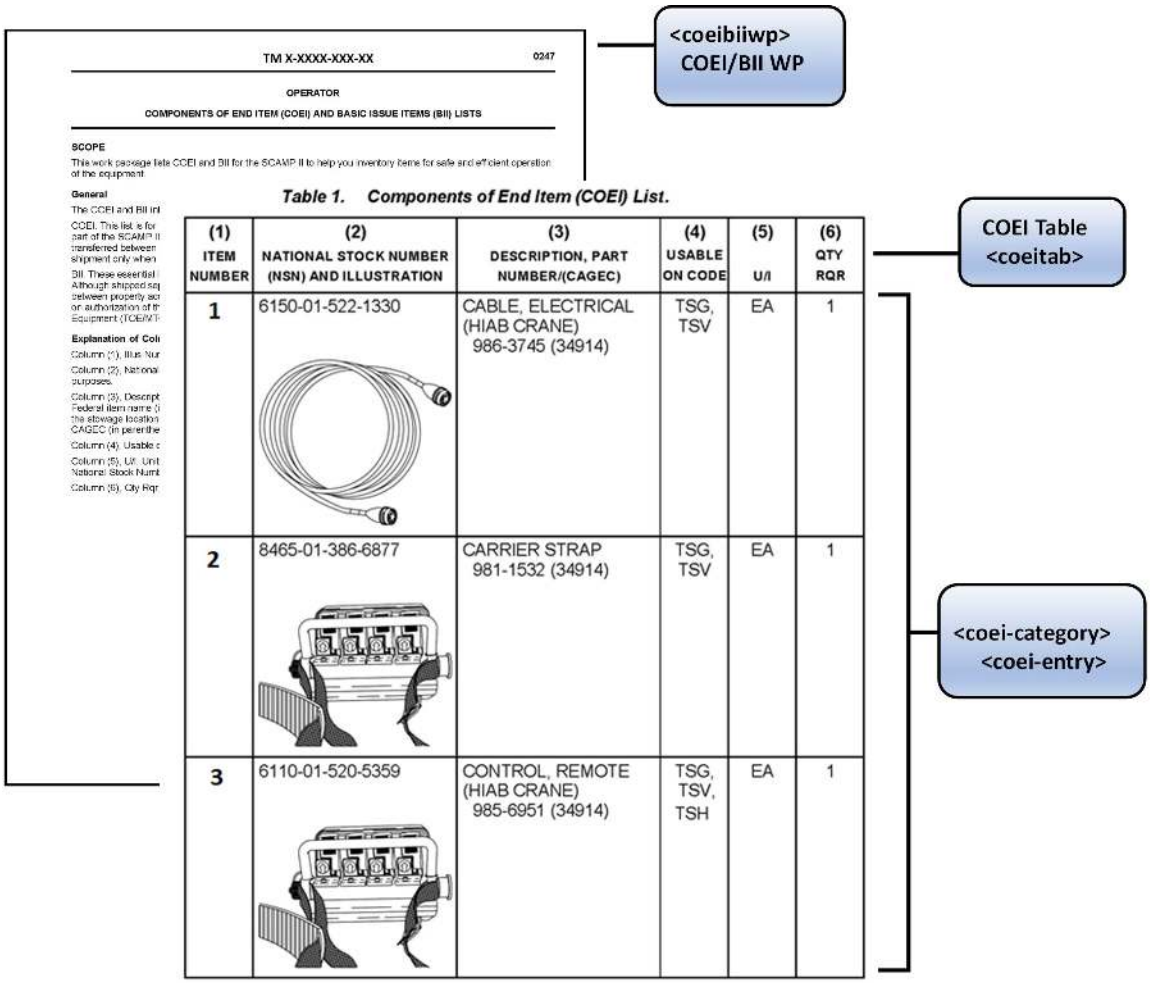


| (1) ITEM NUMBER | (2) NATIONAL STOCK NUMBER (NSN) AND ILLUSTRATION | (3) DESCRIPTION, PART NUMBER/(CAGEC) | (4) USABLE ON CODE | (5) U/I | (6) QTY RQR |
|---|---|---|---|---|---|
| 1 | 6150-01-522-1330 | CABLE, ELECTRICAL (HIAB CRANE) 986-3745 (34914) | TSG, TSV | EA | 1 |
| 2 | 8465-01-386-6877 | CARRIER STRAP 981-1532 (34914) | TSG, TSV | EA | 1 |
| 3 | 6110-01-520-5359 | CONTROL, REMOTE (HIAB CRANE) 985-6951 (34914) | TSG, TSV, TSH | EA | 1 |

**FIGURE G-32. COEI/BII method B example.**

## G.7.4.1 COEI / BII Introduction.

Include the following introduction verbatim. If the COEI/BII list is long and there are multiple UOCs, additional information may be provided in the UOC portion of the introduction.

**"COMPONENTS OF END ITEM (COEI) AND BASIC ISSUE ITEMS (BII) INTRODUCTION**

**Scope**

This work package (WP) lists COEI and BII for the *insert the end item name; Armaments Center: insert text "end item"* to help you inventory items for safe and efficient operation of the equipment.

**General**

The COEI and BII information is divided into the following lists:

Components of End Item (COEI). This list is for information purposes only and is not authority to requisition replacements. These items are part of the *insert the end item name; Armaments Center: insert text "end item"*. As part of the end item, these items must be with the end item whenever it is issued or transferred between property accounts. Items of COEI are removed and separately packaged for transportation or shipment only when necessary. Illustrations are furnished to help you find and identify the items.

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Basic Issue Items (BII). These essential items are required to place the *insert the end item name; Armaments Center: insert text "end item"* ) in operation, operate it, and to do emergency repairs. Although shipped separately packaged, BII must be with the *insert the end item name; Armaments Center: insert text "end item"*) during operation and when it is transferred between property accounts. Listing these items is your authority to request/requisition them for replacement based on authorization of the end item by the Table of Organization and Equipment/Modified Table of Organization and Equipment (TOE/MTOE). Illustrations are furnished to help you find and identify the items.

**Explanation of Columns in the COEI List and BII List**

*Select Method A text.*

"Column (1) Illus Number. Gives you the number of the item illustrated.

Column (2) National Stock Number (NSN). Identifies the stock number of the item to be used for requisitioning purposes.

Column (3) Description, Part Number/Commercial and Government Entity Code (CAGEC). Identifies the Federal item name (in all capital letters) followed by a minimum description when needed. When applicable, the stowage location of COEI and BII is also included in this column. The last line below the description is the CAGEC (in parentheses) and the part number.

Column (4) Usable On Code. When applicable, gives you a code if the item you need is not the same for different models of equipment. (*Add the following only as applicable. Replace As, Bs, or Cs with appropriate codes and model numbers.*) These codes are identified below:

**Code Used on**

AA *Model* XXX

BB *Model* XXXX

CC *Model* XXXXX

Add if applicable: Model XXX uses COEI items (*insert item numbers*) and BII items (*insert item numbers*), Model XXXX use COEI items (*insert item numbers*) and BII items (*insert item numbers*), and Model XXXXX use COEI items (*insert item numbers*) and BII items (*insert item numbers*).

Column (5) U/I. Unit of Issue (U/I) indicates the physical measurement or count of the item as issued per the National Stock Number shown in column (2).

Column (6) Qty Rqr. Indicates the quantity required."

OR

*Select Method B text.*

"Column (1) Item Number. Gives you the reference number of the item listed.

Column (2) National Stock Number (NSN) and Illustration. Identifies the stock number of the item to be used for requisitioning purposes and provides an illustration of the item.

Column (3) Description, Part Number/(CAGEC). Identifies the Federal item name (in all capital letters) followed by a minimum description when needed. When applicable, the stowage location of COEI and BII is also included in this column. The last line below the description is the CAGEC (Commercial and Government Entity Code) (in parentheses) and the part number.

Column (4) Usable On Code. When applicable, gives you a code if the item you need is not the same for different models of equipment. (Add the following only as applicable. Replace As, Bs, or Cs with appropriate codes and model numbers.) These codes are identified below:

**Code Used on**

AA *Model* XXX

BB *Model* XXXX


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:03Z
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CC *Model* XXXXX

Add if applicable: Model XXX uses COEI items (*insert item numbers*) and BII items (*insert item numbers*), Model XXXX use COEI items (*insert item numbers*) and BII items (*insert item numbers*), and Model XXXXX use COEI items (*insert item numbers*) and BII items (*insert item numbers*).

Column (5) U/I. Unit of Issue (U/I) indicates the physical measurement or count of the item as issued per the National Stock Number shown in column (2).

Column (6) Qty Rqr. Indicates the quantity required."

## G.7.4.2 (MC Introduction for SSR

Include the following introduction verbatim for SSR. If the SSR list is long and there are multiple UOCs, additional information may be provided in the UOC portion of the introduction.

**"SUPPLY SYSTEM RESPONSIBILITY (SSR LIST INTRODUCTION**

**Scope**

This work package (WP) lists SSR for the *insert the end item name; Armaments Center: insert text "end item"* to help you inventory items for safe and efficient operation of the equipment.

**General**

Supply System Responsibility. This list is for information purposes only and is not authority to requisition replacements. These items are part of the *insert the end item name; Armaments Center: insert text "end item"*. As part of the end item, these items must be with the end item whenever it is issued or transferred between property accounts. Illustrations are furnished to help you find and identify the items.

**Explanation of Columns in the SSR List**

*Select Method A text.*

Illus Number. Gives you the number of the item illustrated.

National Stock Number (NSN). Identifies the stock number of the item to be used for requisitioning purposes.

Description, Part Number/Commercial and Government Entity Code (CAGEC). Identifies the Federal item name (in all capital letters) followed by a minimum description when needed. When applicable, the stowage location of COEI and BII is also included in this column. The last line below the description is the CAGEC (in parentheses) and the part number.

Usable On Code. When applicable, gives you a code if the item you need is not the same for different models of equipment. *Add the following only as applicable. Replace As, Bs, or Cs with appropriate codes and model numbers.*) These codes are identified below:

**Code Used on**

AA *Model* XXX

BB *Model* XXXX

CC *Model* XXXXX

Add if applicable: Model XXX uses COEI items (*insert item numbers*) and BII items (*insert item numbers*), Model XXXX use COEI items (*insert item numbers*) and BII items (*insert item numbers*), and Model XXXXX use COEI items (*insert item numbers*) and BII items (*insert item numbers*).

U/I. Unit of Issue (U/I) indicates the physical measurement or count of the item as issued per the National Stock Number shown in column (2).

Qty Rqr. Indicates the quantity required."

OR

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*Select Method B text.*

Item Number. Gives you the reference number of the item listed.

National Stock Number (NSN) and Illustration. Identifies the stock number of the item to be used for requisitioning purposes and provides an illustration of the item.

Description, Part Number/(CAGEC). Identifies the Federal item name (in all capital letters) followed by a minimum description when needed. When applicable, the stowage location of COEI and BII is also included in this column. The last line below the description is the CAGEC (Commercial and Government Entity Code) (in parentheses) and the part number.

Usable On Code. When applicable, gives you a code if the item you need is not the same for different models of equipment. (Add the following only as applicable. Replace As, Bs, or Cs with appropriate codes and model numbers.) These codes are identified below:

**Code Used on**

AA *Model* XXX

BB *Model* XXXX

CC *Model* XXXXX

Add if applicable: Model XXX uses COEI items (*insert item numbers*) and BII items (*insert item numbers*), Model XXXX use COEI items (*insert item numbers*) and BII items (*insert item numbers*), and Model XXXXX use COEI items (*insert item numbers*) and BII items (*insert item numbers*).

U/I. Unit of Issue (U/I) indicates the physical measurement or count of the item as issued per the National Stock Number shown in column (2).

Qty Rqr. Indicates the quantity required."

### G.7.4.3 COEI/SSR list <coei>.

As the introduction (G.7.4.1) states, COEI consists of all parts or components of a system that must be with the system whenever it is issued or transferred, and these parts are officially identified as part of the system. The COEI shall be prepared as an illustrated tabular list, and the illustrations shall be placed above the list (Method A) or within the list (Method B). The arrangement of the illustrations and list shall be as shown in FIGURE G-30 and FIGURE G-31 (Method A) or FIGURE G-32 (Method B). For Method B, illustrations shall be placed in the National Stock Number column.

### G.7.4.4 List <coeitab>.

The COEI/SSR list (standard format information per 4.8) shall include the headings and basic content shown in FIGURE G-33, as applicable to the specific equipment. The description **<dcpno>** of each item shall consist of the approved Federal item name <name>, followed by a short description **<desc>** when needed. The part number **<partno?** shall be located below the item. The CAGEC **<cageno>** shall follow the part number and in parentheses. Items shall be listed in alphabetical order by description. When applicable, the stowage location of COEI/SSR shall also be included with the description in the table. When more than one model or configuration is applicable and Usable On Codes (UOCs) **<uoc>** are assigned, the UOC shall appear in a separate column adjacent to the description column. See FIGURE G-33 for example COEI table entry tags.



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**Note**

Tags/output shown in figure(s) below may vary slightly based on version of DTD/stylesheets utilized (see applicable DTD or tag definition list for available tags).

*Table 1. Components of End Item (COEI) List.*

| (1) ITEM NUMBER | (2) NATIONAL STOCK NUMBER (NSN) AND ILLUSTRATION | (3) DESCRIPTION, PART NUMBER/(CAGEC) | (4) USABLE ON CODE | (5) U/I | (6) QTY RQR |
|---|---|---|---|---|---|
| 1 | 6150-01-522-1330 | CABLE, ELECTRICAL (HIAB CRANE) 986-3745 (34914) | TSG, TSV | EA | 1 |
| 2 | 8465-01-386-6877 | CARRIER STRAP 981-1532 (34914) | TSG, TSV | EA | 1 |
| 3 | 6110-01-520-5359 | CONTROL, REMOTE (HIAB CRANE) 985-6951 (34914) | TSG, TSV, TSH | EA | 1 |

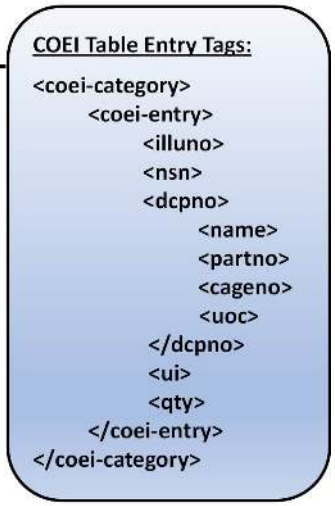


**FIGURE G-33. COEI list example (method B).**

## G.7.4.5 On-board spares <on-board-spares>.

When on-board spares **<on-board-spares>** apply, they shall be listed in one of three ways, as specified by the MATDEV.

**a.** The on-board spares may be listed at the bottom of the COEI table. There shall be a break in the test of the COEI list and a new heading "ON-BOARD SPARES" shall be used. A list of the on-board spares shall appear in the same format as required for the basic COEI/SSR list.

**b.** A separate table within the COEI WP may be prepared to list the on-board spares.

**c.** If necessary to aid in usability with large number of on-board spares, a separate supporting information WP may be prepared using the additional supporting information WP (G.7.11).

## G.7.4.6 Basic Issue Items (BII) <bii>.

As the introduction (G.7.4.1) states, BII must be with the system during operation and when it is issued or transferred; however, these items are not identified as part of the system. Any tools required for operator maintenance as well as operator equipment publications shall also be included in the BII. All BII items shall be listed

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in alphabetical order by description; and, when applicable, the stowage location of BII shall also be included in the description column in the table. See FIGURE G-34 through FIGURE G-37 for example BII table entry tags.

**Note**

Tags/output shown in figure(s) below may vary slightly based on version of DTD/stylesheets utilized (see applicable DTD or tag definition list for available tags).

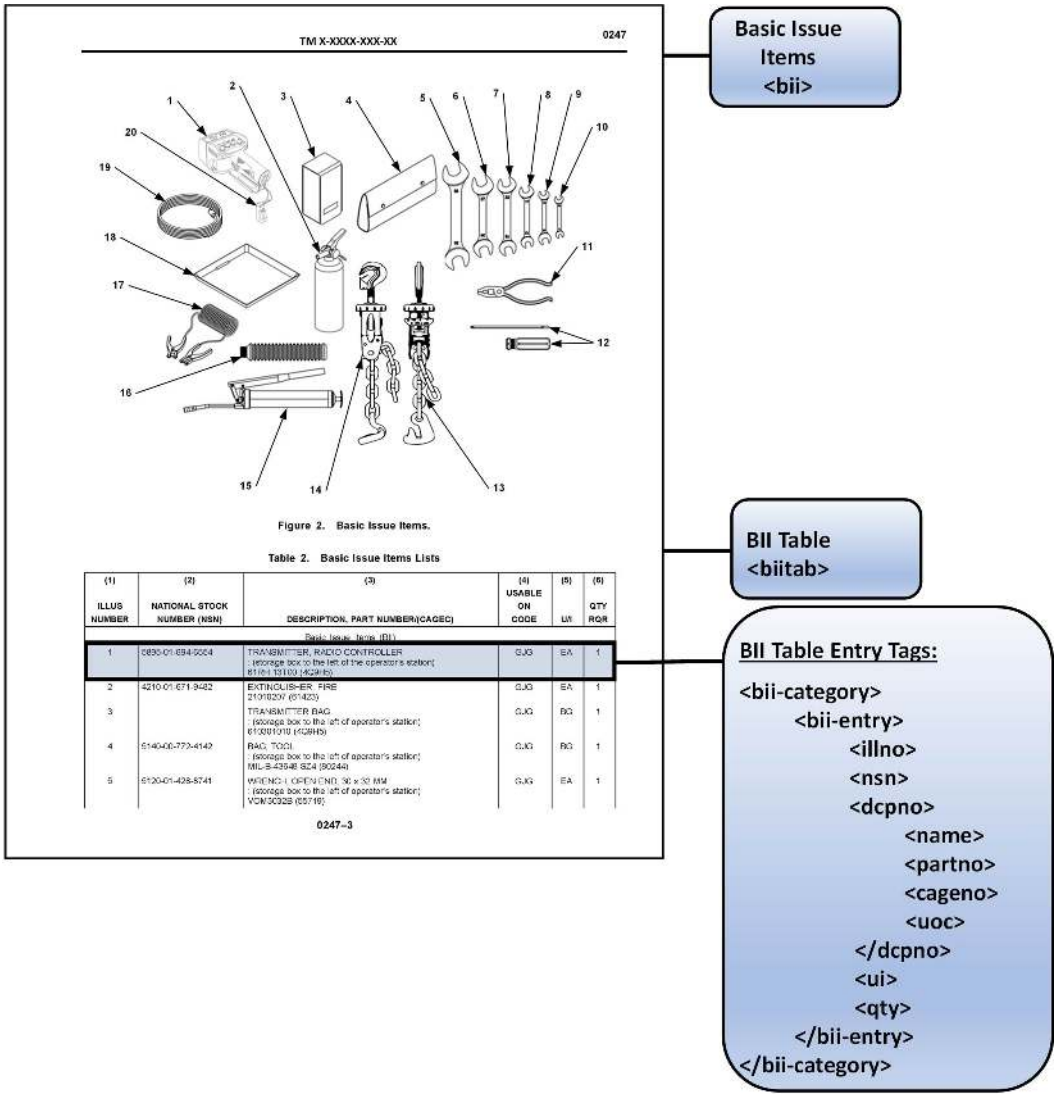


**FIGURE G-34. BII list example (method A).**

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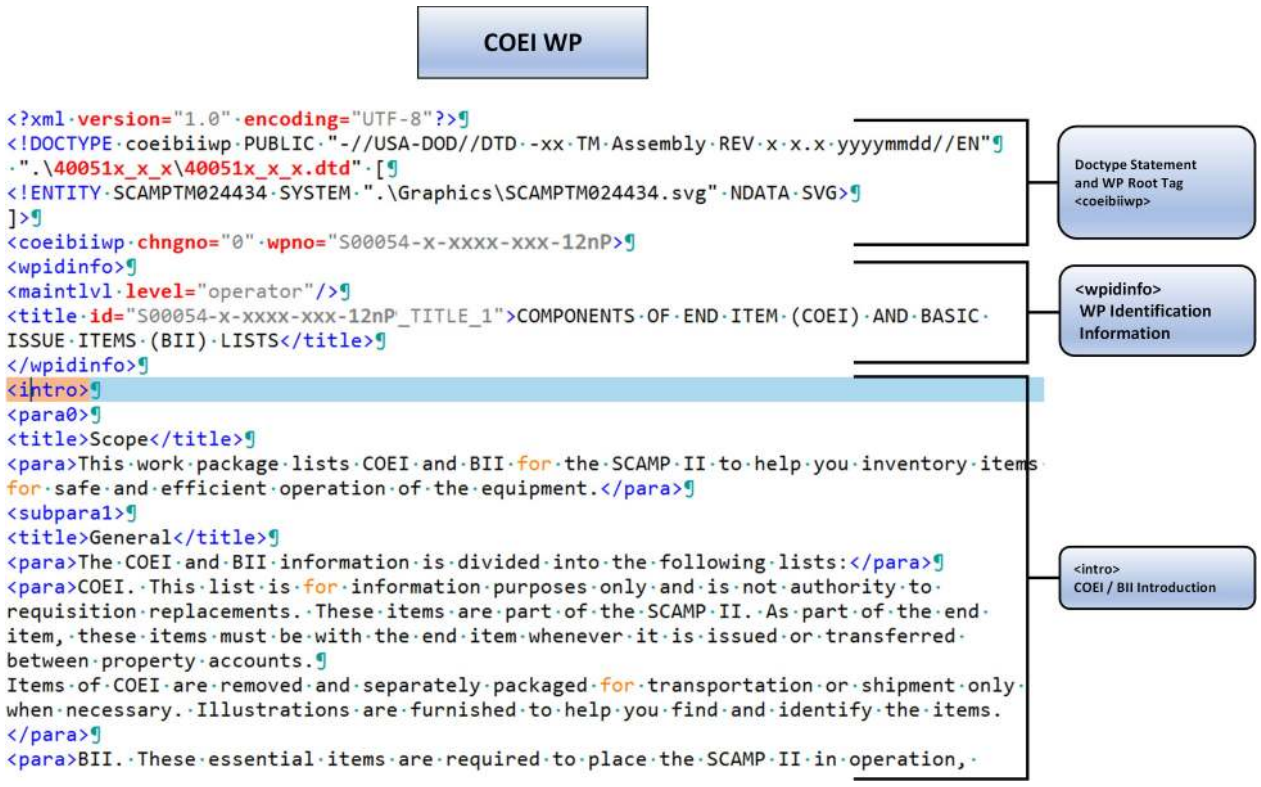


**FIGURE G-35. COEI WP tags (method A) (sheet 1 of 2).**



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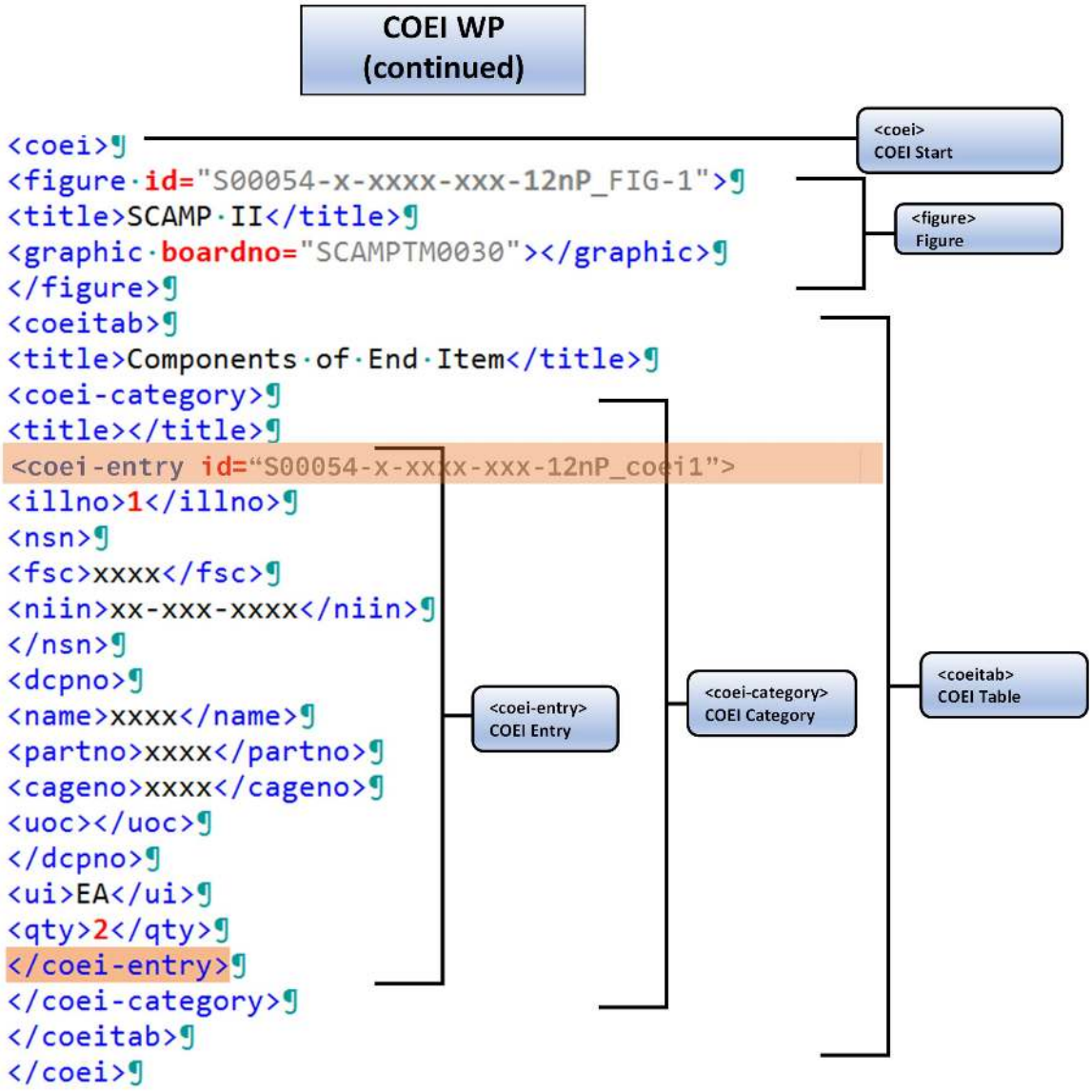


**FIGURE G-36. COEI WP tags (method A) (sheet 2 of 2).**

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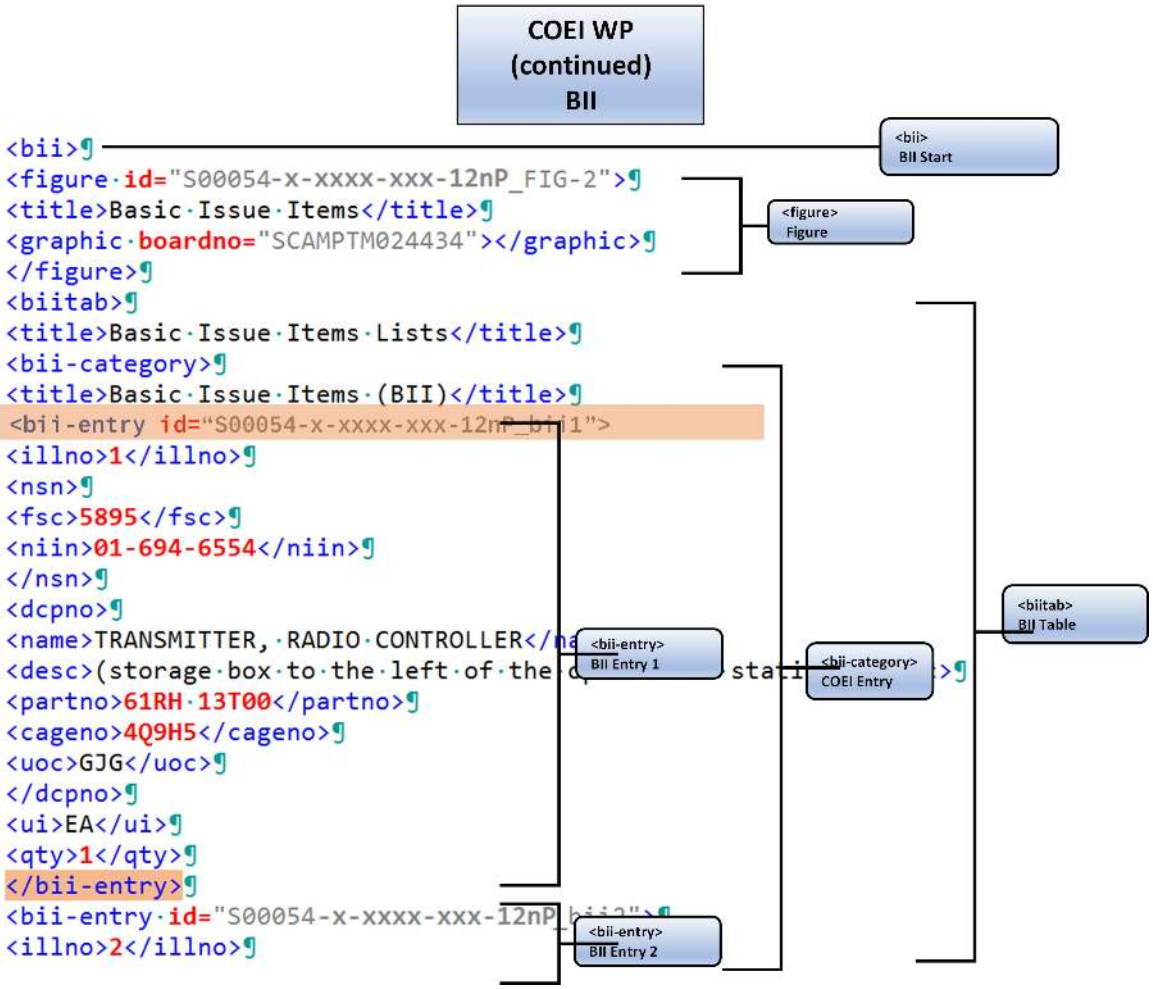


**FIGURE G-37. COEI WP (BII) tags (method A).**

## G.7.5 Additional Authorization List (AAL) (Army) or Using Unit Responsibility Items (UURI) (Marine Corps) list work package <aalwp>.

An AAL WP shall be prepared as applicable for an equipment system (as directed by MATDEV). AAL items do not have to be with the system when it is issued or transferred, and these items are not identified as part of the system. AAL items, for example, may include equipment covers, mats, or padding, as well as extra lighting, etc. depending on equipment type. AAL items may include tools and test equipment only in the cases described in 5.4.3.6.1. The AAL WP shall include the components listed below.

**a.** Work package identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** Initial setup **<initial_setup>** is not required for this WP.

**c.** AAL/UURI introduction **<intro>** (G.7.5.1).

**d.** AAL list **<aal>** (G.7.5.2).

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**Terminology** :

- "Using Unit Responsibility Items (UURI)" – Marine Corps only manuals (WP shall list the items to be requisitioned separately by the using unit).
- "AAL" – Multiservice manuals involving the Marine Corps (term shall be used regardless of the lead service).
- "AAL" – Army only manuals and multi-service manuals (term shall be used regardless of lead service).

**Army only** :

Army AAL WPs shall include the components listed below.

**a.** Items not issued with the end item.

**b.** Items not listed on the end item engineering drawing as part of the end item.

**c.** Items not part of the end item NSN configuration.

**d.** Items not required to be turned in with the end item.

**e.** Items that are separately authorized by MTOE, TDA, CTA, or JTA.

**f.** Items provided for information only.

### G.7.5.1 AAL introduction <intro>.

As applicable, the following introductions (text within the quotation marks) shall be prepared and included verbatim.

**Note**

If an AAL list is long and there are multiple UOCs, additional information may be provided in the UOC portion of the introduction.

**"ADDITIONAL AUTHORIZATION LIST (AAL) INTRODUCTION**

**Scope**

This work package (WP) lists additional items you are authorized for the support of the *enter item name*.

**General**

This list identifies items that do not have to accompany the *enter short item name* ) and that do not have to be turned in with it. These items are all authorized to you by CTA, MTOE, TDA, or JTA.

**Explanation of Columns in the AAL**

Column (1) Item Number. Identifies table entry that may be referenced in initial setup sections of work packages as applicable.

Column (2) National Stock Number (NSN). Identifies the stock number of the item to be used for requisitioning purposes.

Column (3) Description, Part Number/(CAGEC). Identifies the Federal item name (in all capital letters) followed by a minimum description when needed. The last line below the description is the part number and the Commercial and Government Entity Code (CAGEC) (in parentheses).

Column (4) Usable On Code. When applicable, gives you a code if the item you need is not the same for different models of equipment. (Add the following only as applicable. *Add the following only as applicable. Replace Xs with appropriate codes and model numbers*.) These codes are identified below:

**Code Used on**

AA *Model* XXX

BB *Model* XXXX


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CC *Model* XXXXX

Add if applicable: Model XXX uses AAL items (*insert item numbers*), Model XXXX use AAL items (insert item numbers, and Model XXXXX use AAL items (*insert item numbers*).

Column (5) U/I. Unit of Issue (U/I) indicates the physical measurement or count of the item as issued per the National Stock Number shown in column (1).

Column (6) Qty Recm. Indicates the quantity recommended."

**"USING UNIT RESPONSIBILITY ITEMS (UURI LIST INTRODUCTION**

**Scope**

This work package (WP) lists using unit responsibility items you are authorized for the support of the *enter item name*.

**General**

This list identifies items that are to be requisitioned by the using unit.

**Explanation of Entries in the UURI**

National Stock Number (NSN). Identifies the stock number of the item to be used for requisitioning purposes.

Description, Part Number/Commercial and Government Entity Code (CAGEC). Identifies the Federal item name (in all capital letters) followed by a minimum description when needed. The last line below the description is the part number and the CAGEC (in parentheses).

Usable On Code. When applicable, gives you a code if the item you need is not the same for different models of equipment. *Add the following only as applicable. Replace Xs with appropriate codes and model numbers* .) These codes are identified below:

**Code Used on**

AA *Model* XXX

BB *Model* XXXX

CC *Model* XXXXX

Add if applicable: Model XXX uses UURI items (*insert item numbers*), Model XXXX uses UURI items (*insert item numbers*), and Model XXXXX use UURI items (*insert item numbers*).

U/I. Unit of Issue (U/I) indicates the physical measurement or count of the item as issued per the National Stock Number.

Qty Recm. Indicates the quantity recommended."



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## G.7.5.2 AAL /UURI list <aal>.

A tabular list (standard format information per 4.8) of all AAL items/UURI shall be prepared. The headings and subsequent information for this list shall be the same as the COEI list (G.7.4.3) except the ILLUS NUMBER column required for the COEI/BII/SSR lists shall not apply since no illustrations are used, and the QTY column shall be QTY RECM (quantity recommended). The items shall be listed alphabetically. For Army operator manuals, any parts (except for mandatory replacement parts) required for maintenance shall be listed in the AAL. See FIGURE G-38 through FIGURE G-40 for example AAL table entry tags.

**Note**

Figure appearance and tags/output may vary slightly based on version of DTD/stylesheets utilized (see applicable DTD or tag definition list for available tags).

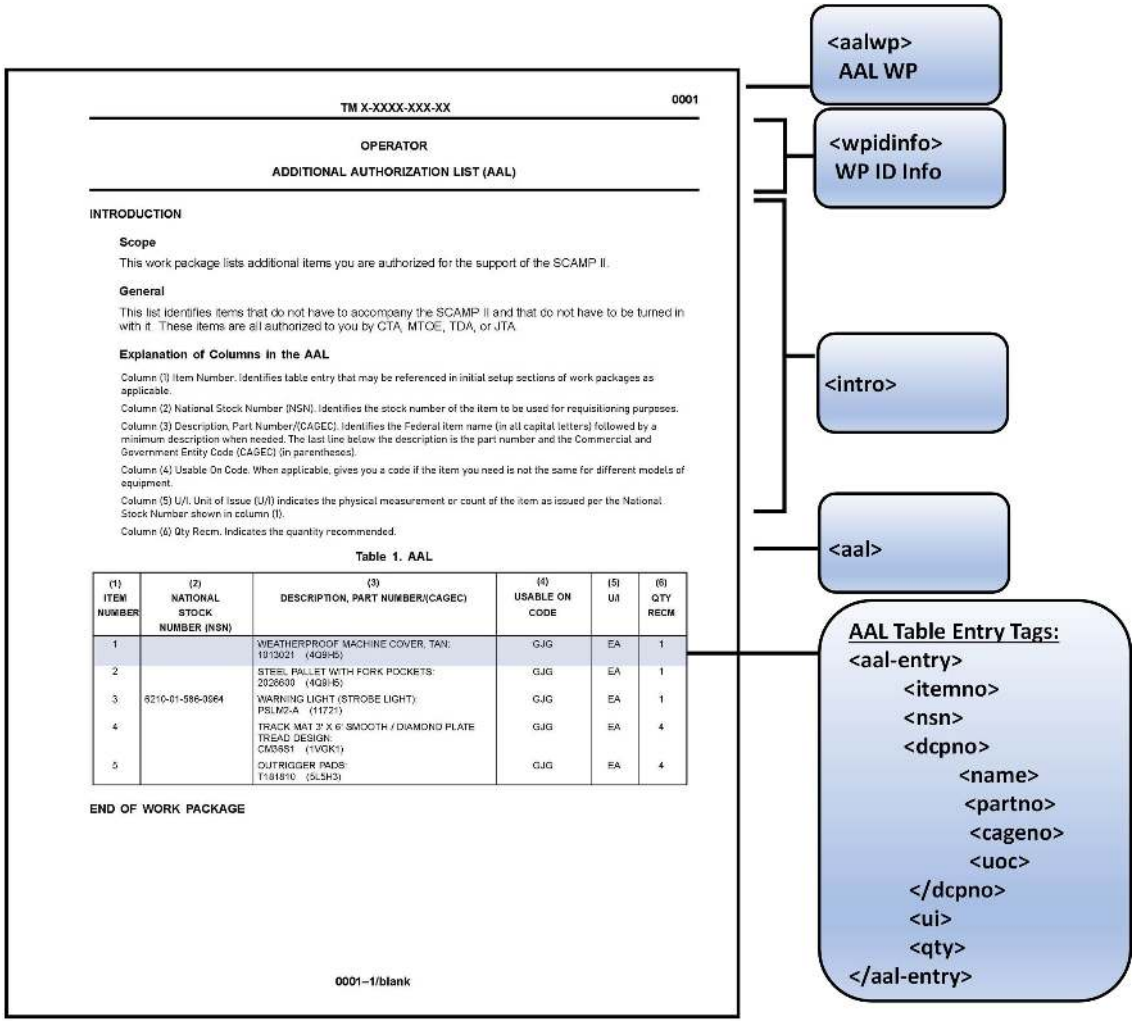


**FIGURE G-38. AAL WP example.**

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AAL WP

```
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE aalwp PUBLIC ""-//USA-DOD//DTD xx TM Assembly REV x x.x yyyymmdd//EN"
".\40051x_x_x\40051x_x_x.dtd">
<aalwp chngno="0" wpno="S00055-x-xxxx-xxx-12nP">
<wpidinfo>
<maintlvl level="operator"/>
<title id="S00055-x-xxxx-xxx-12nP_TITLE-1">ADDITIONAL AUTHORIZATION LIST (AAL)</title>
</wpidinfo>
<intro>
<para0>
<title>INTRODUCTION</title>
<subpara1>
<title>Scope</title>
<para>This work package lists additional items you are authorized for the support of the SCAMP
II.</para>
</subpara1>
<subpara1>
<title>General</title>
<para>This list identifies items that do not have to accompany the SCAMP II and that do not
have to be turned in with it. These items are all authorized to you by CTA, MTOE, TDA, or JTA.
</para>
</subpara1>
<subpara1>
<title>Explanation of Columns In the AAL</title>
<para>Column (1) National Stock Number (NSN). Identifies the stock number of the item to be
used for requisitioning purposes.</para>
<para>Column (2) Description, Part Number/(CAGEC). Identifies the
Federal item name (in all capital letters) followed by a minimum description when needed. The
```

Doctype Statement and WP Root Tag <aalwp>

<wpidinfo> WP Identification Information

<intro> AAL Introduction

FIGURE G-39. AAL WP tags (sheet 1 of 2).

AAL WP (continued)

```
<aal>
<title>AAL</title>
<aal-entry id="S00055-x-xxxx-xxx-12nP_aal1"><itemno>1</itemno><nsn><fsc></fsc>
<niin></niin></nsn><dcpno>
<name>WEATHERPROOF MACHINE COVER, TAN</name>
<partno>1013021</partno><cageno>4Q9H5</cageno><uoc>GJG</uoc></dcpno><ui>EA</ui>
<qty>1</qty></aal-entry>
<aal-entry id="S00055-x-xxxx-xxx-12nP_aal2"><itemno>2</itemno><nsn><fsc></fsc>
<niin></niin></nsn><dcpno>
<name>STEEL PALLET WITH FORK POCKETS</name>
<partno>2028600</partno><cageno>4Q9H5</cageno><uoc>GJG</uoc></dcpno><ui>EA</ui>
<qty>1</qty></aal-entry>
<aal-entry id="S00055-x-xxxx-xxx-12nP_aal3"><itemno>3</itemno><nsn><fsc>6210
</fsc><niin>01-586-0964</niin></nsn><dcpno>
<name>WARNING LIGHT (STROBE LIGHT)</name>
<partno>PSLM2-A</partno><cageno>11721</cageno><uoc>GJG</uoc></dcpno><ui>EA</ui>
<qty>1</qty></aal-entry>
<aal-entry id="S00055-x-xxxx-xxx-12nP_aal4"><itemno>4</itemno><nsn><fsc></fsc>
<niin></niin></nsn><dcpno>
<name>TRACK MAT 3' X 6' SMOOTH / DIAMOND PLATE TREAD DESIGN</name>
<partno>CM36S1</partno><cageno>1VGK1</cageno><uoc>GJG</uoc></dcpno><ui>EA</ui>
```

<aal> AAL Start

<aal-entry> AAL Entry

FIGURE G-40. AAL WP tags (sheet 2 of 2).

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## G.7.6 Collateral Material (CM work package (Marine Corps only.

This WP shall be prepared for Marine Corps technical publications only, and shall contain a list of items furnished with the end items upon initial issue. These items typically remain with the using unit during redistribution/rebuild or other change of custody of the end item unless otherwise directed by MARCORLOGCOM. These items are required to be maintained on hand, on order, or identified as an unfunded deficiency unless otherwise specifically directed. CM will be maintained and replaced by the using unit, except for materiel with 9999 series NSNs. Using units are not authorized to requisition items using the assigned 9999 series NSNs. The 9999 series NSN shown under the heading of "Collateral Materiel" is for control within the distribution system only, and is not authorized for requisitioning purposes. Items under this category will be requisitioned by individual NSN/NIIN, and/or part number and CAGE Codes. CM MC WPs shall include the components listed below.

**a.** Work package identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** Initial setup **<initial_setup>** is not required for this WP.

**c.** Introduction data **<intro>**. The introduction text below shall be included verbatim.

**d.** CM list **<intro>** (G.7.6.1).

**"COLLATERAL MATERIAL(CM) ITEMS**

**LIST INTRODUCTION**

**Scope**

This work package (WP) lists collateral material items you are authorized for the support of the *enter item name*.

**General**

This list identifies items that are to be requisitioned by the using unit except those with 9999 NSN.

**Explanation of Entries in the UURI**

National Stock Number (NSN). Identifies the stock number of the item to be used for requisitioning purposes.

Description, Part Number/Commercial and Government Entity Code (CAGEC). Identifies the Federal item name (in all capital letters) followed by a minimum description when needed. The last line below the description is the part number and the CAGEC (in parentheses).

Usable On Code. When applicable, gives you a code if the item you need is not the same for different models of equipment. *Add the following only as applicable. Replace Xs with appropriate codes and model numbers.* These codes are identified below:

**Code Used on**

AA *Model* XXX

BB *Model* XXXX

CC *Model* XXXXX

Add if applicable: Model XXX uses CM items (*insert item numbers*), Model XXXX uses CM items (*insert item numbers*), and Model XXXXX use CM items (*insert item numbers*).

U/I. Unit of Issue (U/I) indicates the physical measurement or count of the item as issued per the National Stock Number.

Qty Recm. Indicates the quantity recommended."

## G.7.6.1 CM list (Marine Corps only).

Collateral material list. A tabular list (standard format information per 4.8) of all CM items shall be prepared. The entries and subsequent information for this list shall be the same as the COEI/BII/SSR lists except the ILLUS


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NUMBER entry required for the COEI/BII/SSR lists shall not apply since no illustrations are used, and the QTY entry shall be QTY RECM (quantity recommended). The items shall be listed alphabetically.

## G.7.7 Expendable and durable items list work package <explistwp>.

This WP shall be prepared as required by publication type (see A.4.4). This WP shall provide a technical publication user with a list of all expendable and durable items called out in the publication text that are necessary to operate and/or maintain the equipment. These items typically get "used up" during equipment operation or maintenance and could include adhesives, fuses, antifreeze, lubricants, sealants, tape, fuel, and tags, etc., for example. The expendable and durable items list WP components are listed below.

**a.** Work package identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** Initial setup **<initial_setup>** is not required for this WP.

**c.** Introduction data **<intro>**. The introduction text below shall be included verbatim.

**d.** Expendable and durable items list **<explist>** (G.7.7.1).

**EXPENDABLE AND DURABLE ITEMS**

**LIST INTRODUCTION**

**Scope**

This WP lists expendable and durable items that you will need to operate and/or maintain the (*enter equipment/end item name; Armaments Center: insert text "end item"*). This list is for information only and is not authority to requisition the listed items. These items are authorized to you by CTA 50-970, Expendable/Durable Items (Except Medical, Class V Repair Parts, and Heraldic Items), CTA 50-909, Field and Garrison Furnishings and Equipment, or CTA 8-100, Army Medical Department Expendable/Durable Items.

**Explanation of Columns in the Expendable/Durable Items List**

Column (1) Item No. This number is assigned to the entry in the list and is referenced in the narrative instructions to identify the item (for example, "Use brake fluid (WP 0098, item 5)").

Column (2) Level. This column identifies the lowest class of maintenance that requires the listed item (*include as applicable: C = Crew, O = AMC, F = Maintainer or ASB, H = TASMG, D = Depot*).

Column (3) National Stock Number (NSN). This is the NSN assigned to the item that you can use to requisition it.

Column (4) Item Name, Description, Part Number/(CAGEC). This column provides the other information you need to identify the item. The last line below the description is the part number and the Commercial and Government Entity Code (CAGEC) (in parentheses).

Column (5) U/I. Unit of Issue (U/I) code shows the physical measurement or count of an item, such as gallon, dozen, gross, etc."

## G.7.7.1 Expendable and durable items list <explist>.

This list (standard format information per 4.8) shall be prepared in tabular format, listed alphabetically by item name, and contain no illustrations. Items to be listed shall be those approved by the MATDEV. Entries shall include the information listed below. See FIGURE G-41 through FIGURE G-44 for example expendable and durable table entry tags.

**a.** Item number.

**b.** Lowest maintenance class.

**c.** National Stock Number (NSN).

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**d.** Item name or nomenclature.

**e.** If applicable, a description.

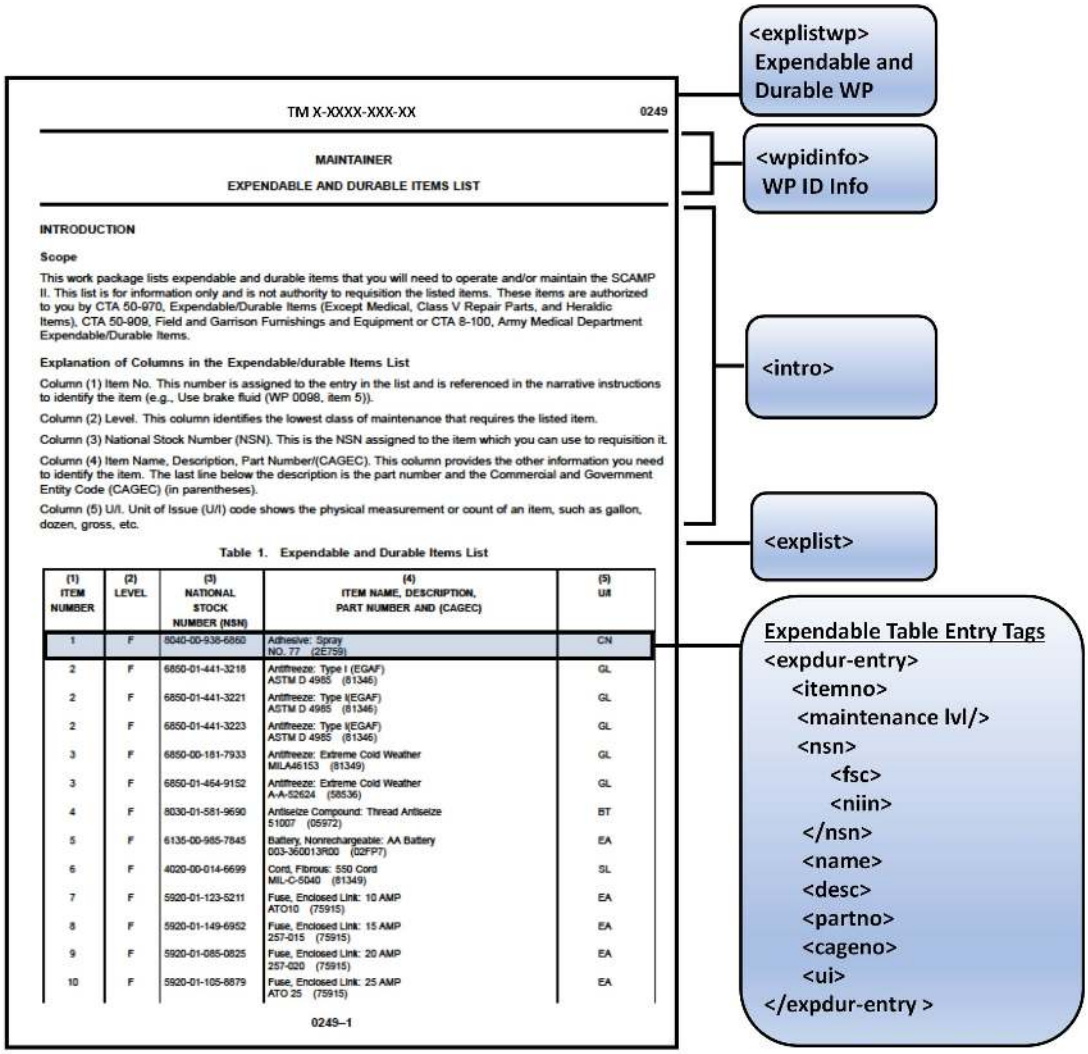
**f.** Part number.

**g.** Commercial and Government Entity Code (CAGEC).

**h.** Unit of Issue (U/I).

**Note**

Figure appearance and tags/output may vary slightly based on version of DTD/stylesheets utilized (see applicable DTD or tag definition list for available tags).




TM X-XXXX-XXX-XX 0249

MAINTAINER

EXPENDABLE AND DURABLE ITEMS LIST

INTRODUCTION

Scope

This work package lists expendable and durable items that you will need to operate and/or maintain the SCAMP II. This list is for information only and is not authority to requisition the listed items. These items are authorized to you by CTA 50-970, Expendable/Durable Items (Except Medical, Class V Repair Parts, and Heraldic Items), CTA 50-909, Field and Garrison Furnishings and Equipment or CTA 8-100, Army Medical Department Expendable/Durable Items.

Explanation of Columns in the Expendable/durable Items List

Column (1) Item No. This number is assigned to the entry in the list and is referenced in the narrative instructions to identify the item (e.g., Use brake fluid (WP 0098, item 5)).

Column (2) Level. This column identifies the lowest class of maintenance that requires the listed item.

Column (3) National Stock Number (NSN). This is the NSN assigned to the item which you can use to requisition it.

Column (4) Item Name, Description, Part Number/(CAGEC). This column provides the other information you need to identify the item. The last line below the description is the part number and the Commercial and Government Entity Code (CAGEC) (in parentheses).

Column (5) U/I. Unit of Issue (U/I) code shows the physical measurement or count of an item, such as gallon, dozen, gross, etc.

Table 1. Expendable and Durable Items List

| (1) ITEM NUMBER | (2) LEVEL | (3) NATIONAL STOCK NUMBER (NSN) | (4) ITEM NAME, DESCRIPTION, PART NUMBER AND (CAGEC) | (5) U/I |
|---|---|---|---|---|
| 1 | F | 8040-00-938-6860 | Adhesive: Spray NO. 77 (2E759) | CN |
| 2 | F | 6850-01-441-3218 | Antifreeze: Type I (EGAF) ASTM D 4985 (81346) | GL |
| 2 | F | 6850-01-441-3221 | Antifreeze: Type I(EGAF) ASTM D 4985 (81346) | GL |
| 2 | F | 6850-01-441-3223 | Antifreeze: Type I(EGAF) ASTM D 4985 (81346) | GL |
| 3 | F | 6850-00-181-7933 | Antifreeze: Extreme Cold Weather MILA46153 (81349) | GL |
| 3 | F | 6850-01-464-9152 | Antifreeze: Extreme Cold Weather A-A-52624 (58536) | GL |
| 4 | F | 8030-01-581-9690 | Antiseize Compound: Thread Antiseize 51007 (05972) | BT |
| 5 | F | 6135-00-985-7845 | Battery, Nonrechargeable: AA Battery 003-360013R00 (02FP7) | EA |
| 6 | F | 4020-00-014-6699 | Cord, Fibrous: 550 Cord MIL-C-5040 (81349) | SL |
| 7 | F | 5920-01-123-5211 | Fuse, Enclosed Link: 10 AMP ATO10 (75915) | EA |
| 8 | F | 5920-01-149-6952 | Fuse, Enclosed Link: 15 AMP 257-015 (75915) | EA |
| 9 | F | 5920-01-085-0825 | Fuse, Enclosed Link: 20 AMP 257-020 (75915) | EA |
| 10 | F | 5920-01-105-8879 | Fuse, Enclosed Link: 25 AMP ATO 25 (75915) | EA |

0249–1

**FIGURE G-41. Expendable and durable WP example.**



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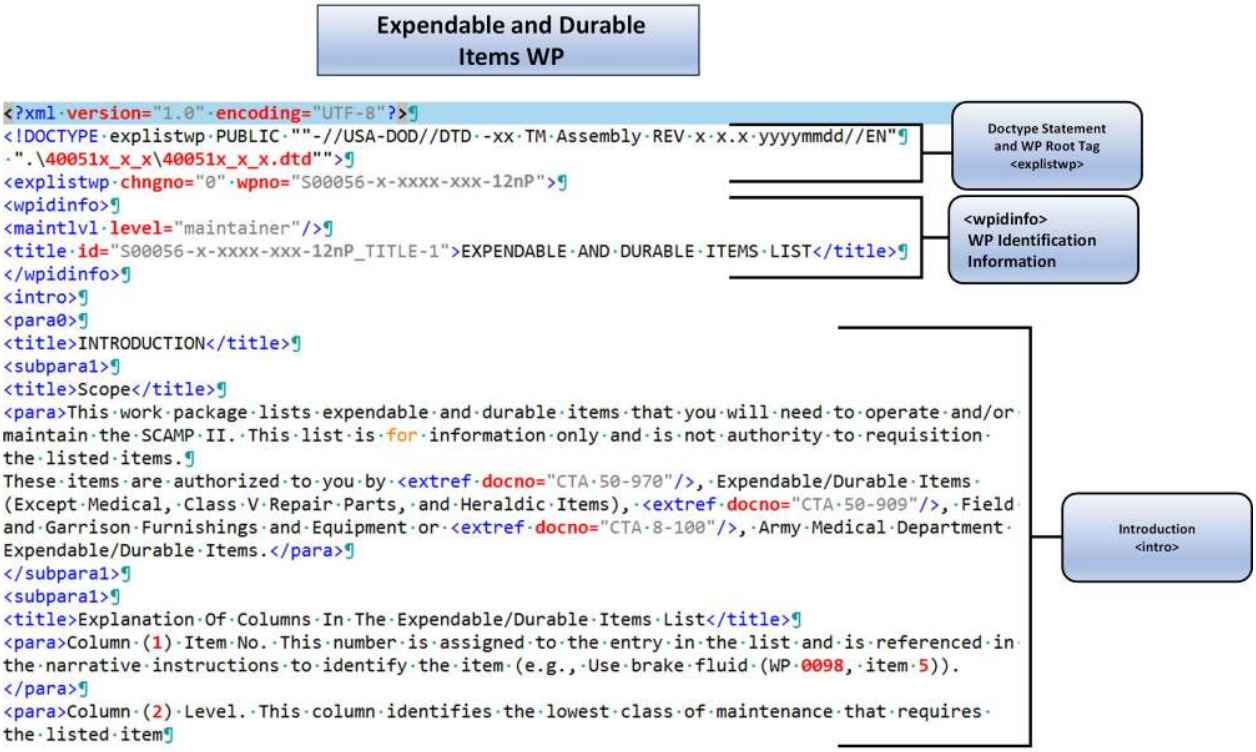


**FIGURE G-42. Expendable and durable WP tags (1 of 3).**

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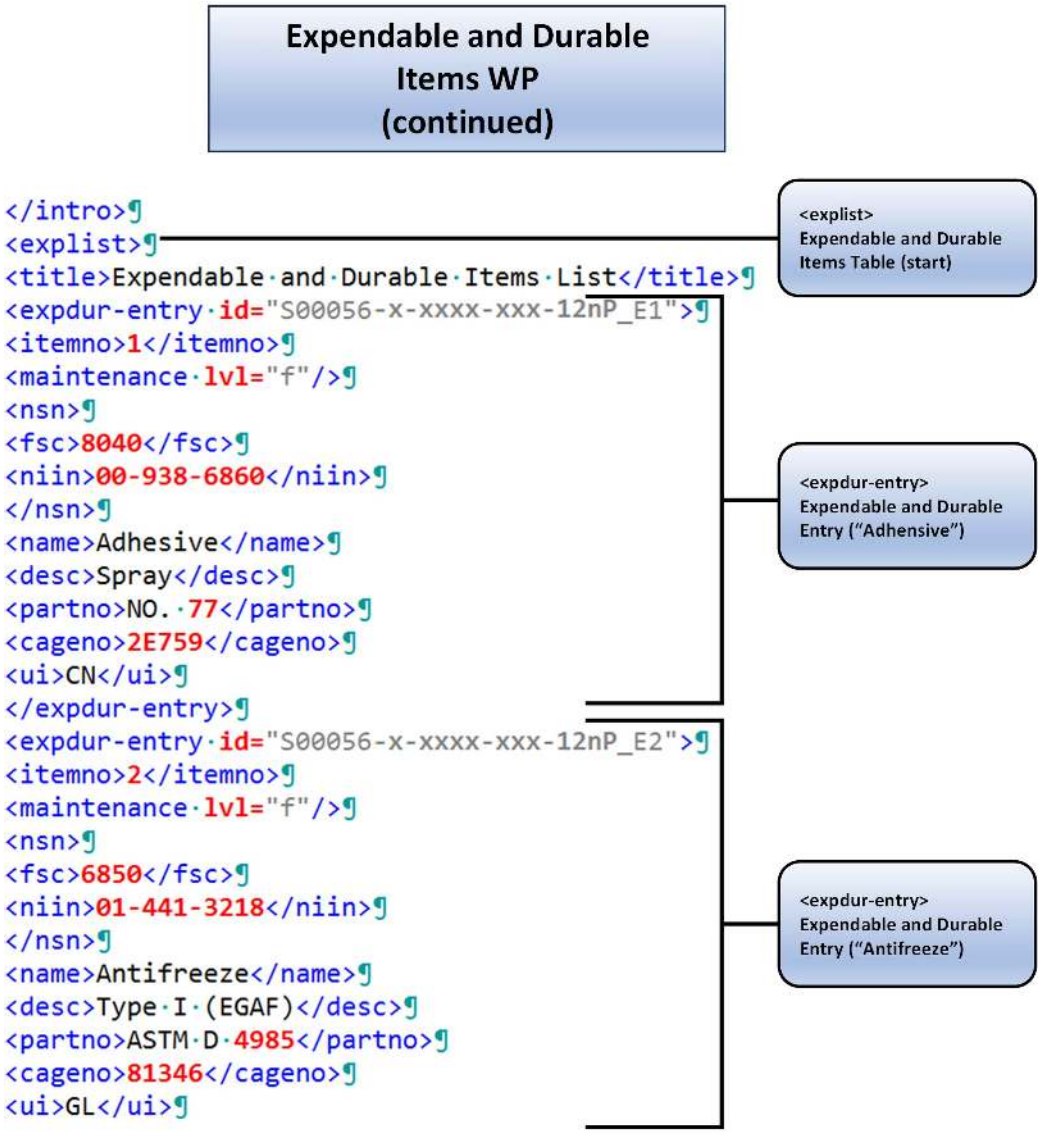


**FIGURE G-43. Expendable and durable WP tags (2 of 3).**



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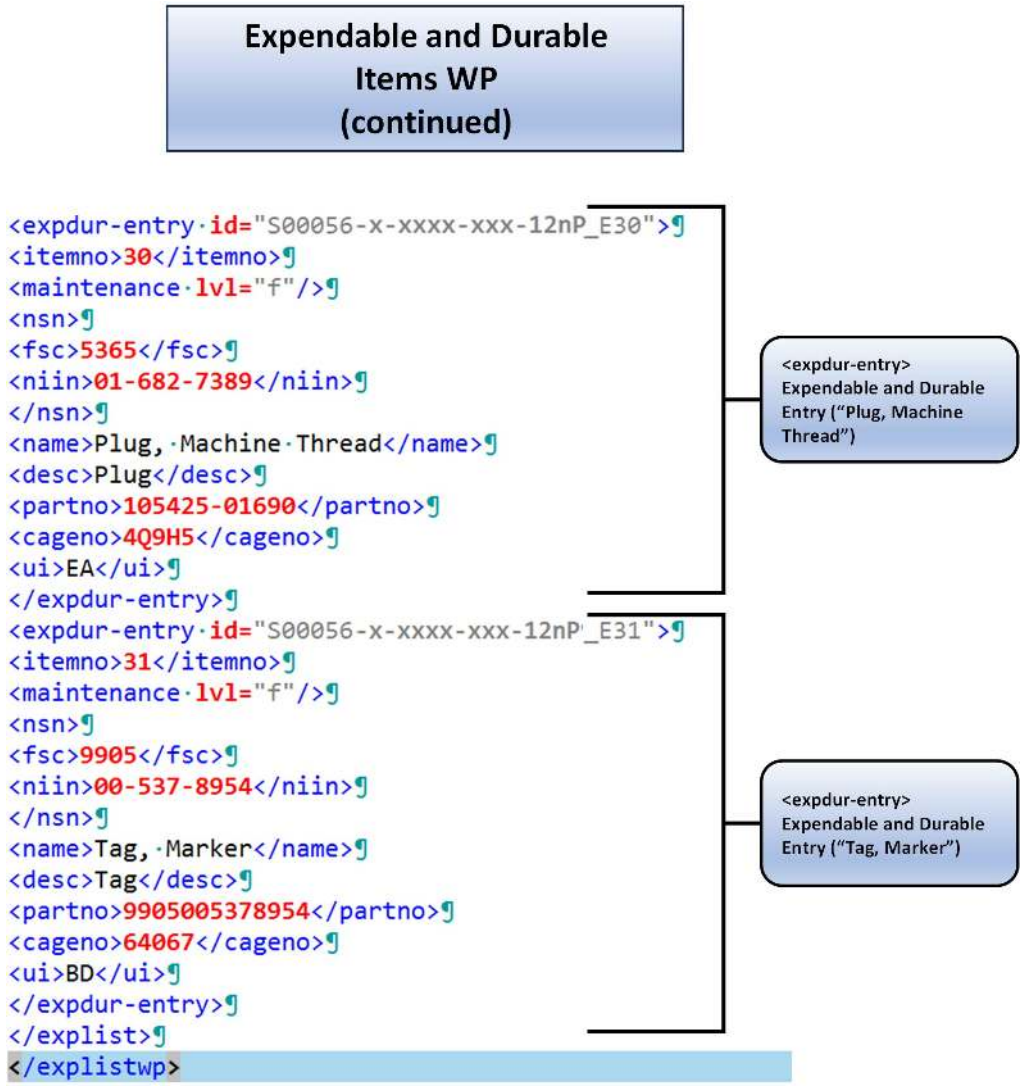


**FIGURE G-44. Expendable and durable WP tags (3 of 3).**

## G.7.8 Tool identification (ID) list work package <toolidwp>.

This WP shall be prepared if tools are listed in the initial setup sections of troubleshooting and/or maintenance WPs. This WP serves as the location where tools, kits, sets/outfits are that referenced in troubleshooting and maintenance WP initial setup sections (and used in the narrative portion of these WPs) are listed; thus, these items shall be linked to the tool ID WP. The tool ID WP ensures users have a comprehensive accounting of tools needed to perform troubleshooting and maintenance tasks by listing the following information: national stock numbers (table column 3); part numbers and CAGE codes (table column 4); and relevant sets/ kits/outfits, etc. these tools belong to (table column 5). Troubleshooting and maintenance WPs shall strive to list individual tools (that are part of a kit, set, or outfit) as much as possible in initial setup sections to aid technical publication users (see FIGURE G-25 and FIGURE G-26).


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**Note**

The TIL does not authorize the requisitioning of tools. The Modified Table of Organization and Equipment (MTOE) and commands authorize tool requisition.

FIGURE G-45 emphasizes that all tools and special tools listed in the initial setup sections of WPs shall link directly to the tool ID WP (not MAC, Table 2). FIGURE G-46 provides greater detail about tool classifications and whether they shall be listed in the tool and test equipment table (MAC, Table 2) or the tool ID WP (G.7.8).

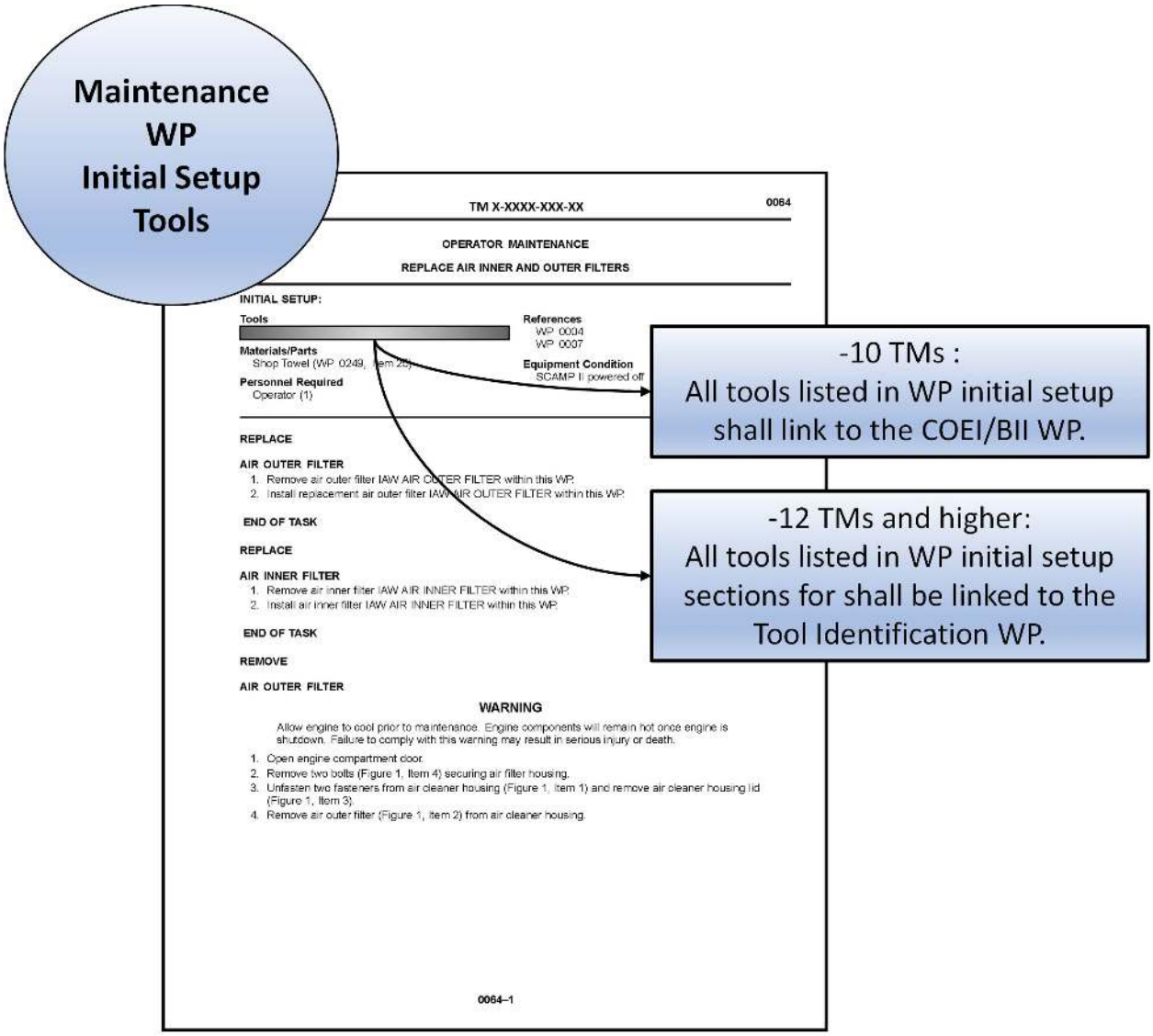


**FIGURE G-45. Tools classifications and listings.**



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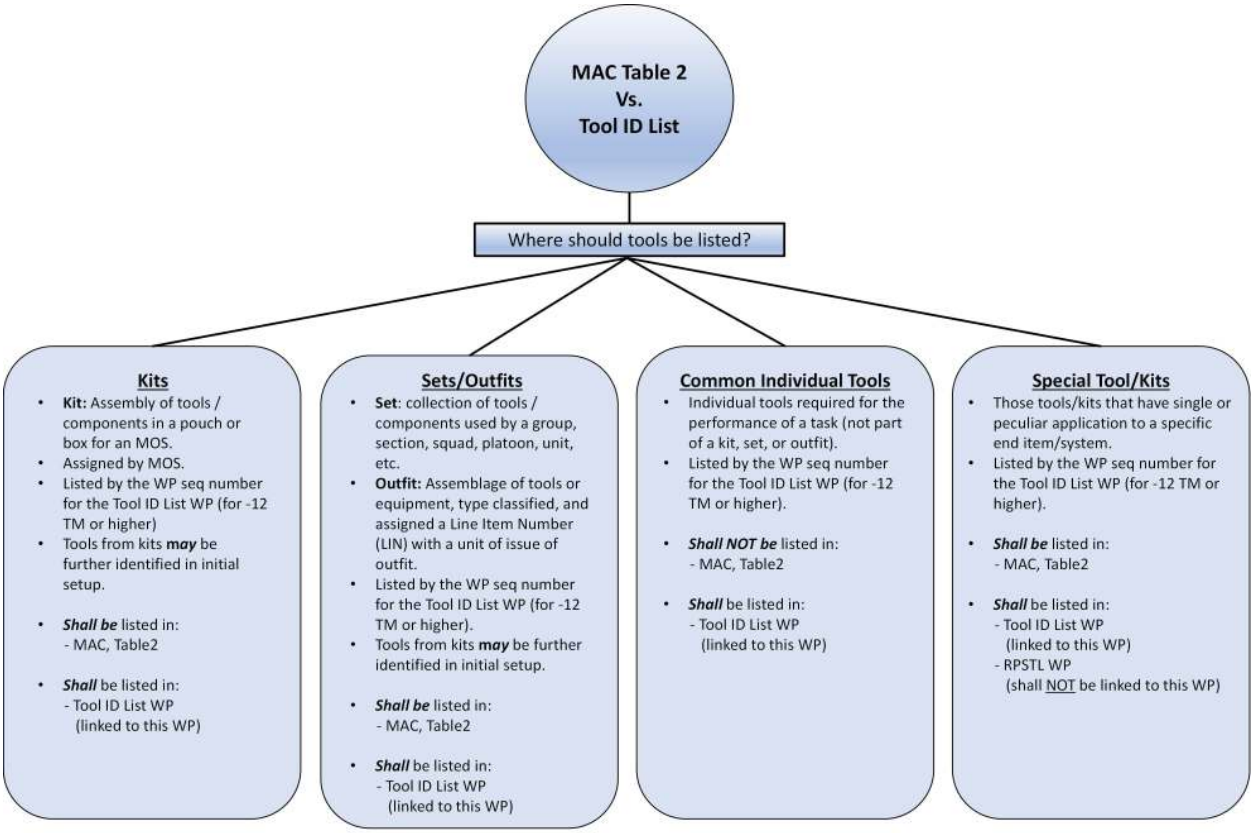


**FIGURE G-46. Tool and test equipment reference table (MAC table 2) versus the tool identification list.**

FIGURE G-47 demonstrates that the maintenance WP for replacing a hydraulic inline filter references directly to the tool ID list WP item 14 ("WRENCH, BOX AND OPEN-END COMBINATION"), which is contained in the tool set "CL 4910-95-A81SATS."

**Note**

For DMWRs/NMWRs, a list of all common and special tools/kits and TMDE not contained in lower-level technical publications or in the RPSTL and required to perform the procedures in the Depot Maintenance Work Requirements (DMWRs) / National Maintenance Work Requirements (NMWRs) shall be included. This list shall include any special inspection equipment used only for the item that the DMWR/NMWR covers.


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**Note**

Figure appearance may vary slightly based on version of DTD/stylesheets utilized.

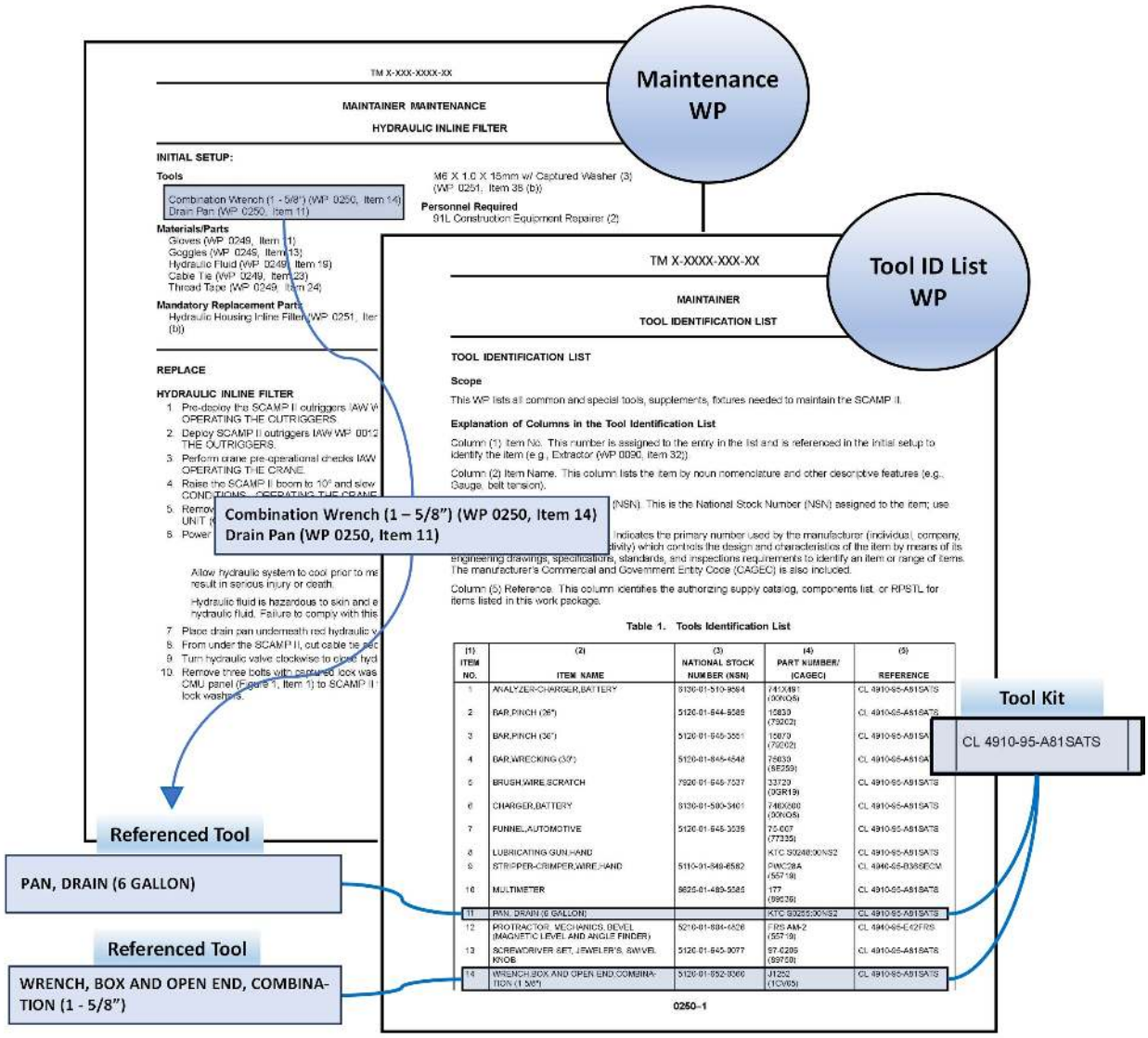


**FIGURE G-47. Tool identification list entry example.**



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### G.7.8.1 Tool identification list WP components <toolidwp>.

The tool identification list WP components are listed below. See FIGURE G-48 through for example tool identification list table entry tags. If there are no tools required for maintenance of the equipment, the following statement shall be included in the tool identification list WP in lieu of the tool list table:

"No tools are required for the maintenance of (*insert the system name*)."

**a.** Work package identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** Initial setup **<initial_setup>** is not required for this WP.

**c.** Introduction data **<intro>**. The following text shall be prepared and included verbatim.

**"TOOL IDENTIFICATION LIST INTRODUCTION**

**Scope**

"This WP lists all common and special tools/kits, supplements, fixtures needed to maintain the *insert equipment name*."

OR

"This work package (WP) lists all common and special tools/kits and equipment not listed in the lower-level manuals for this system and that are needed to maintain the *insert equipment name*." **DMWRs/NMWRs only**

**"Explanation of Columns in the Tool Identification List**

Column (1) Item No. This number is assigned to the entry in the list and is referenced in the initial setup to identify the item (for example, "Extractor (WP 0090, item 32)").

Column (2) Item Name. This column lists the item by noun nomenclature and other descriptive features (for example, Gauge, belt tension).

Column (3) National Stock Number (NSN). This is the National Stock Number (NSN) assigned to the item; use it to requisition the item.

Column (4) Part Number/(CAGEC). Indicates the primary number used by the manufacturer (individual, company, firm, corporation, or Government activity) that controls the design and characteristics of the item by means of its engineering drawings, specifications, standards, and inspection requirements to identify an item or range of items. The manufacturer's Commercial and Government Entity Code (CAGEC) is also included.

Column (5) Reference. This column identifies the authorizing supply catalog, components list, or RPSTL for items listed in this WP." (Not required for DMWRs/NMWRs)."

### G.7.8.2 Tool identification list <toolidlist>.

The tool list (standard format information per 4.8) shall be prepared in tabular format, listed alphabetically by item name, and contain no illustrations. A lead-in paragraph to the tool identification list may be included, and table entries shall include the information listed below. See FIGURE G-48 through FIGURE G-51 for example tool identification list table entry tags.

**a.** Item number.

**b.** Item name or nomenclature.

**c.** National Stock Number (NSN).

**d.** Part Number.

**e.** Commercial and Government Entity Code (CAGEC).

**f.** Reference.



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**Note**

Figure appearance and tags/output may vary slightly based on version of DTD/stylesheets utilized (see applicable DTD or tag definition list for available tags).

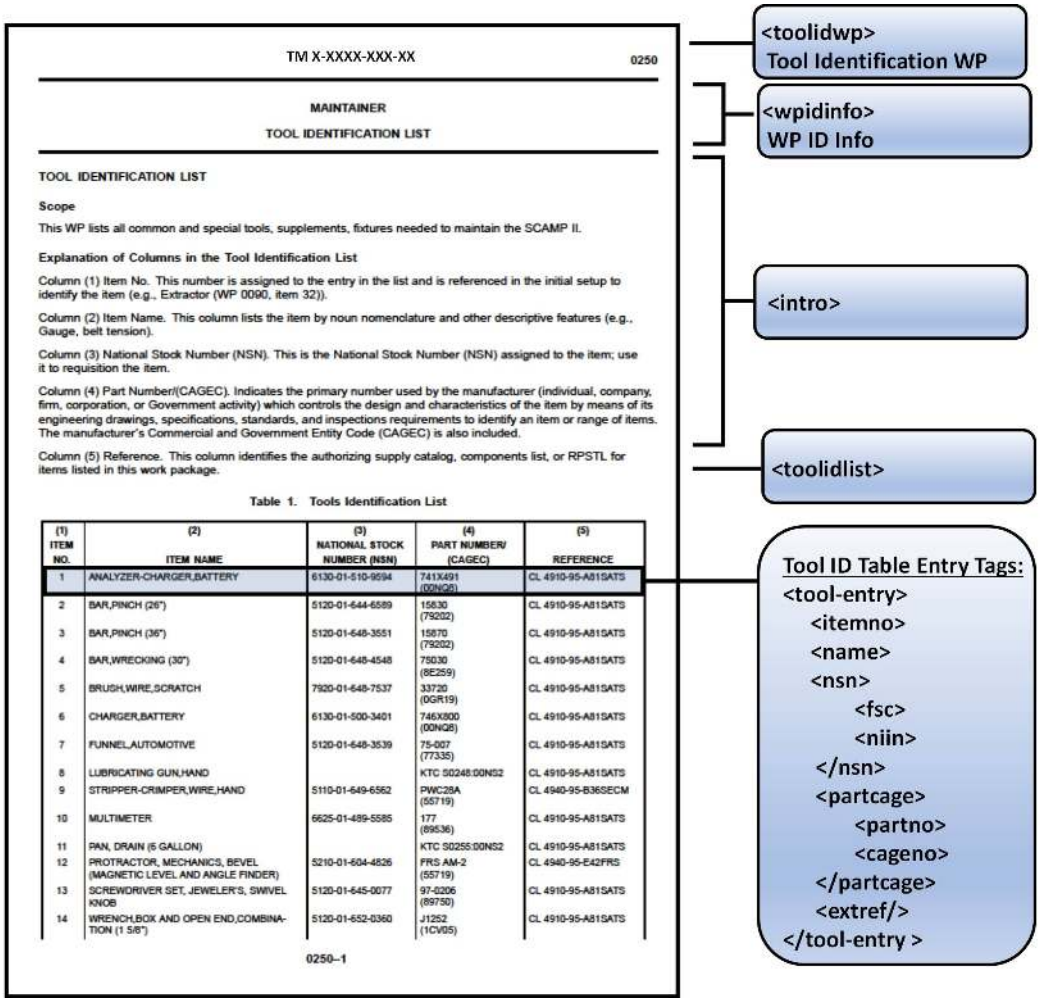


**FIGURE G-48. Tool identification WP example.**

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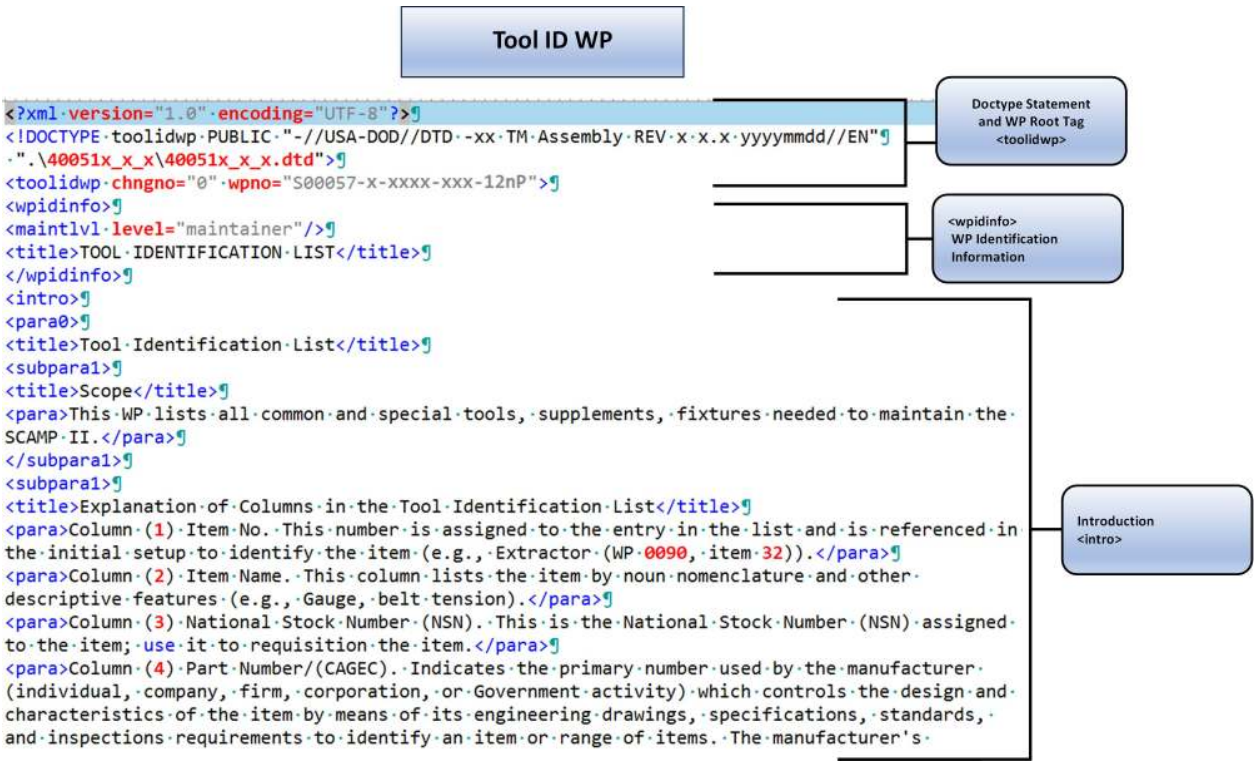


**FIGURE G-49. Tool identification WP tags (sheet 1 of 3).**

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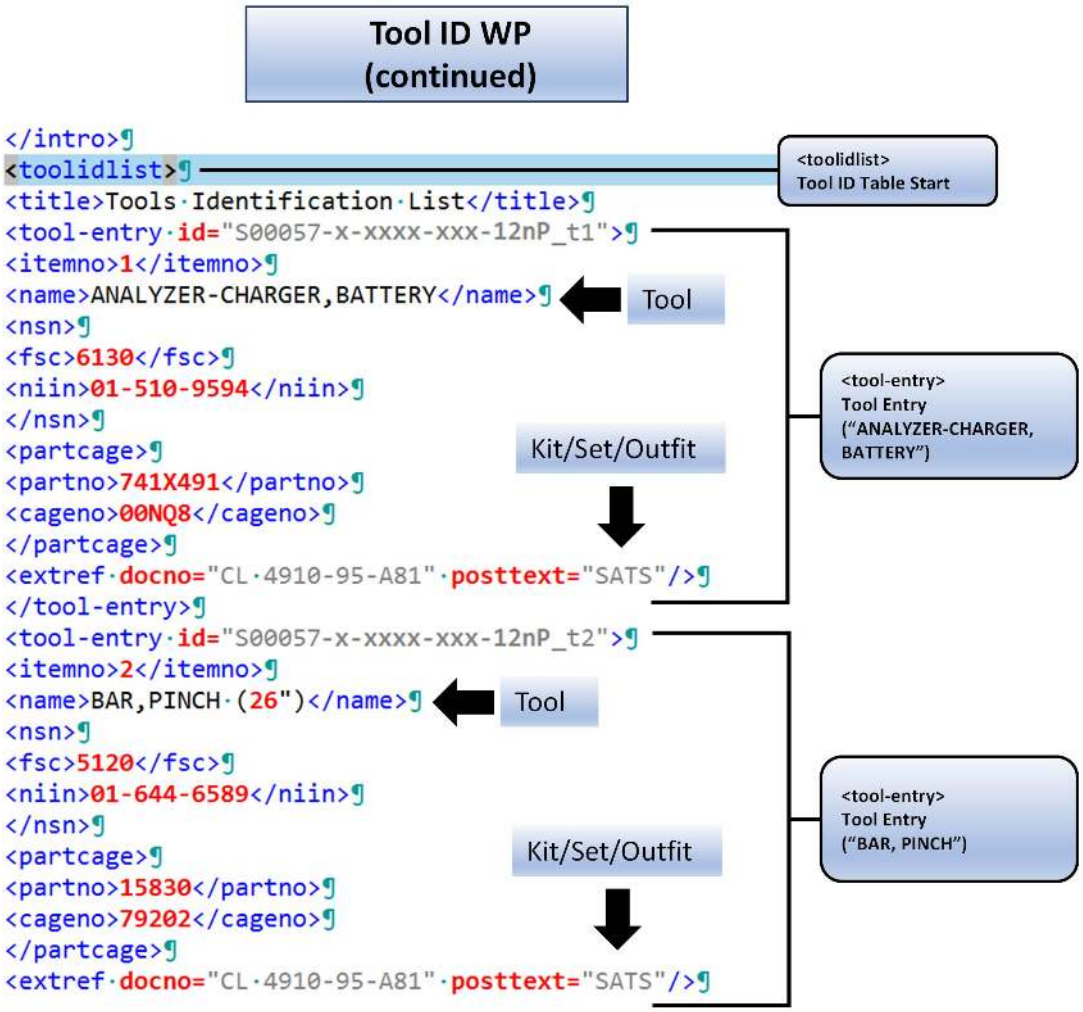


**FIGURE G-50. Tool identification WP tags (sheet 2 of 3).**

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Check the source to verify that this is the current version before use.

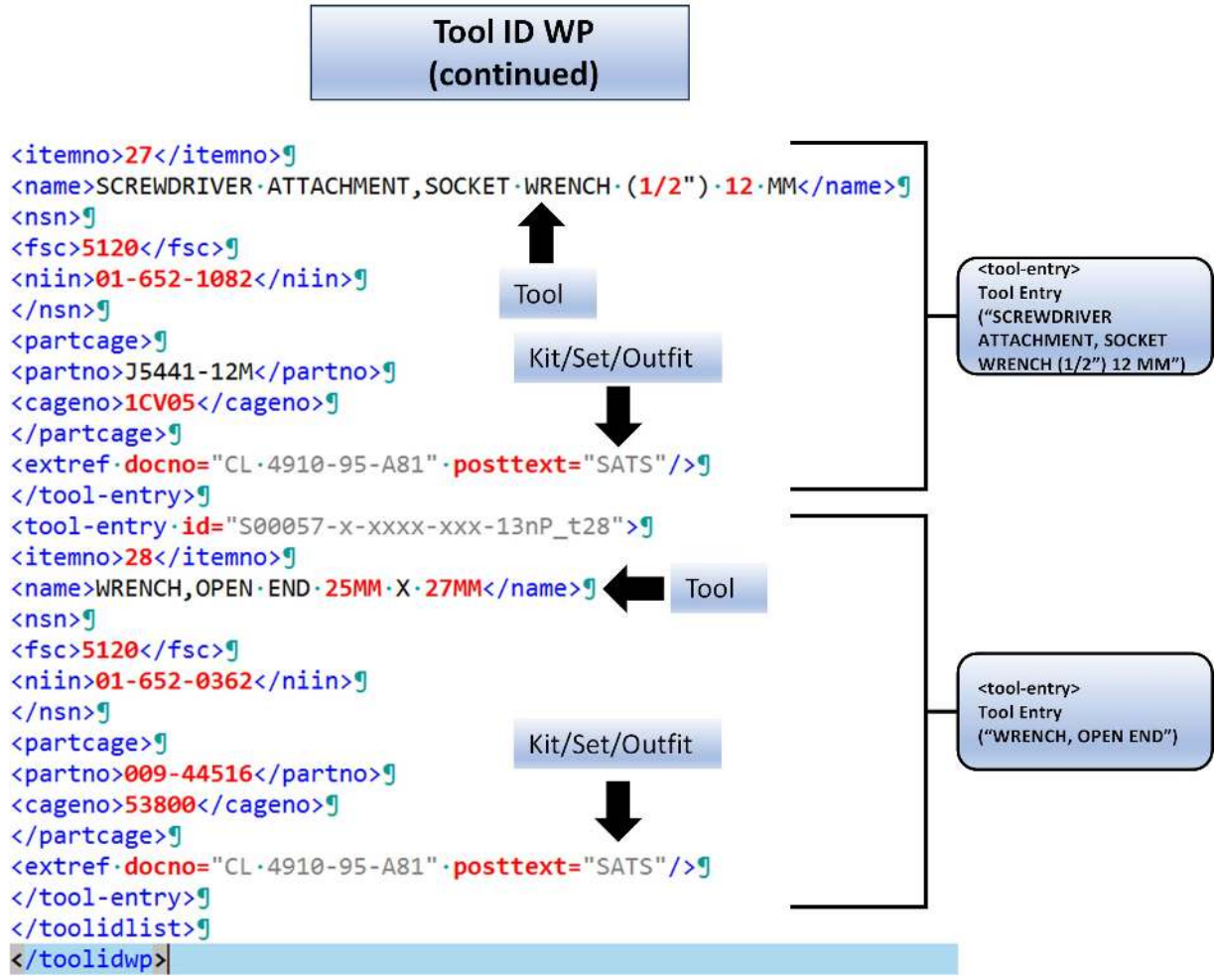


**FIGURE G-51. Tool identification WP tags (sheet 3 of 3).**

## G.7.9 Mandatory replacement parts work package <mrplwp>.

This WP shall list all mandatory replacement parts referenced in the task initial setups for all maintenance tasks in the manual including PMCS tasks. Mandatory replacement parts are items that must be replaced during maintenance whether they have failed or not. This includes items based on usage intervals such as miles, time, rounds, fired, etc. For all technical publications that include repair maintenance, as well as DMWRs/NMWRs, a mandatory replacement parts list consisting of all items that must be replaced during the repair and overhaul of the equipment, whether or not they have been disturbed, shall be developed. When an item or component is not disassembled based on Preshop Analysis (PSA), the item will not be disassembled for the sole purpose to add a mandatory part. All items that must be replaced during overhaul or repair procedures (based on usage intervals such as miles, time, or rounds fired, or replaced on a Time Between Overhaul (TBO) interval) shall be included in the parts list table. A reference shall be made to the TM that covers the equipment. The mandatory replacement parts WP components are listed below. See FIGURE G-52 through FIGURE G-55 for example mandatory replacement parts table entry tags. If there are no mandatory replacement parts for the equipment, a statement shall be added to the general information WP (refer to B.7.1).

**a.** Work package identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** Initial setup **<initial_setup>** is not required for this WP.

**c.** Introduction data **<intro>**. The following text shall be prepared and included verbatim.

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**"INTRODUCTION**

**Scope**

This work package WP includes a list of all the mandatory replacement parts referenced in the task initial setups and procedures including those referenced in Preventive Maintenance Checks and Services. These are items that must be replaced during maintenance whether they have failed or not. This includes items based on usage intervals such as miles, time, rounds fired, etc.

**Explanation of Columns in the Mandatory Replacement Parts List**

Column (1) Item No. This number is assigned to the entry in the list and is referenced in the narrative instructions to identify the item (for example, "Use O-ring (WP 0098, Item 5)").

Column (2) Part Number (CAGEC). Identifies the part number and CAGEC of the item to be used for requisitioning purposes.

Column (3) National Stock Number (NSN) Identifies the stock number of the item to be used for requisitioning purposes.

Column (4) Description This column lists the item by noun nomenclature and other descriptive features (for example, Gauge, belt tension).

Column (5) Qty. Indicates the quantity required."

## G.7.9.1 Mandatory replacement parts list <mrpl>.

Mandatory replacement parts shall be listed (standard column headings in quotes) by item number **<itemno>** "Item No.," part number **<partno>** and CAGEC **<cageno>** "Part Number/(CAGEC)," NSN **<nsn>** "National Stock Number (NSN)," nomenclature **<name>** "Nomenclature," and quantity **<qty>** "Qty." Items shall be listed in alphanumeric order by part number.


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**Note**

Figure appearance and tags/output may vary slightly based on version of DTD/stylesheets utilized (see applicable DTD or tag definition list for available tags).

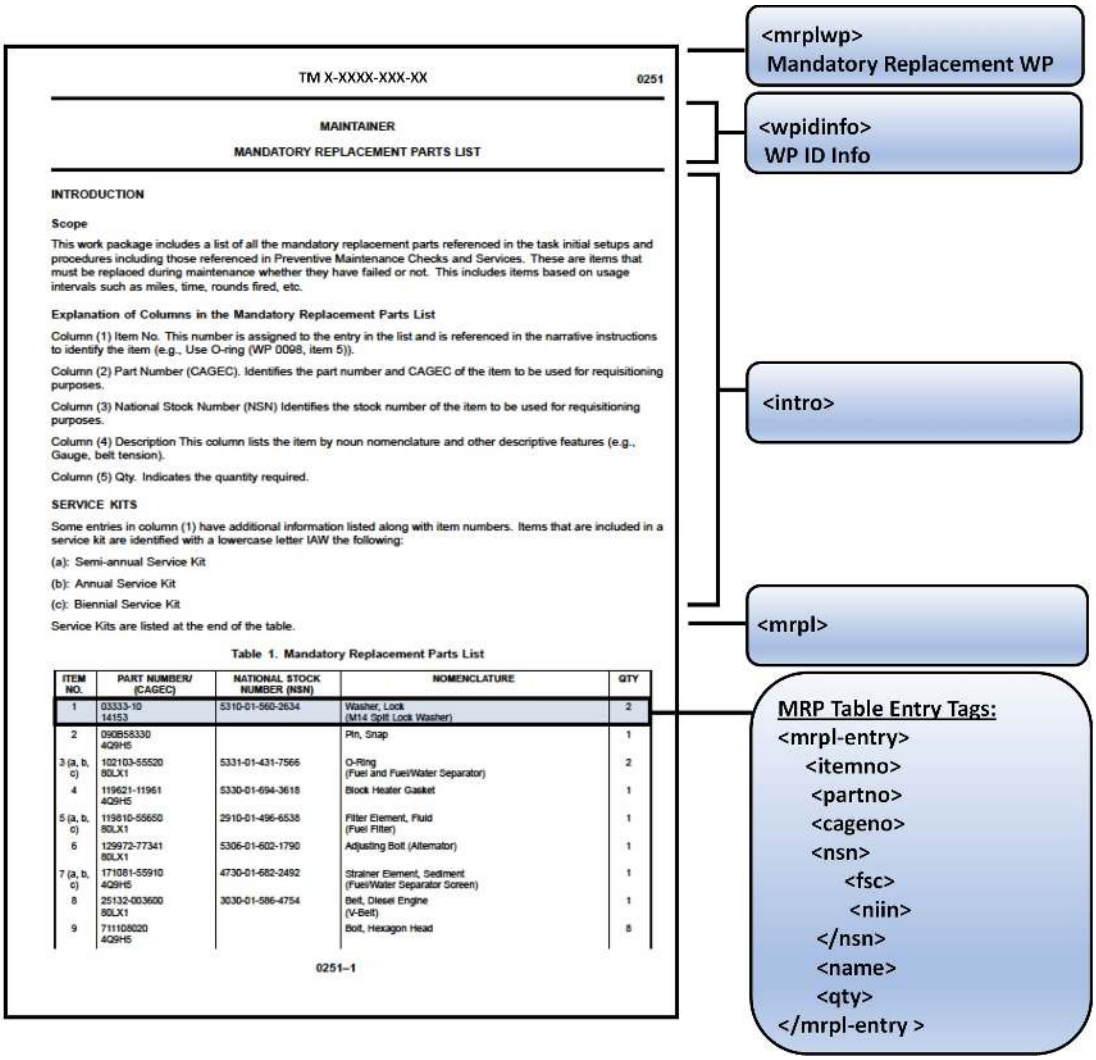


**FIGURE G-52. Mandatory replacement parts WP example.**


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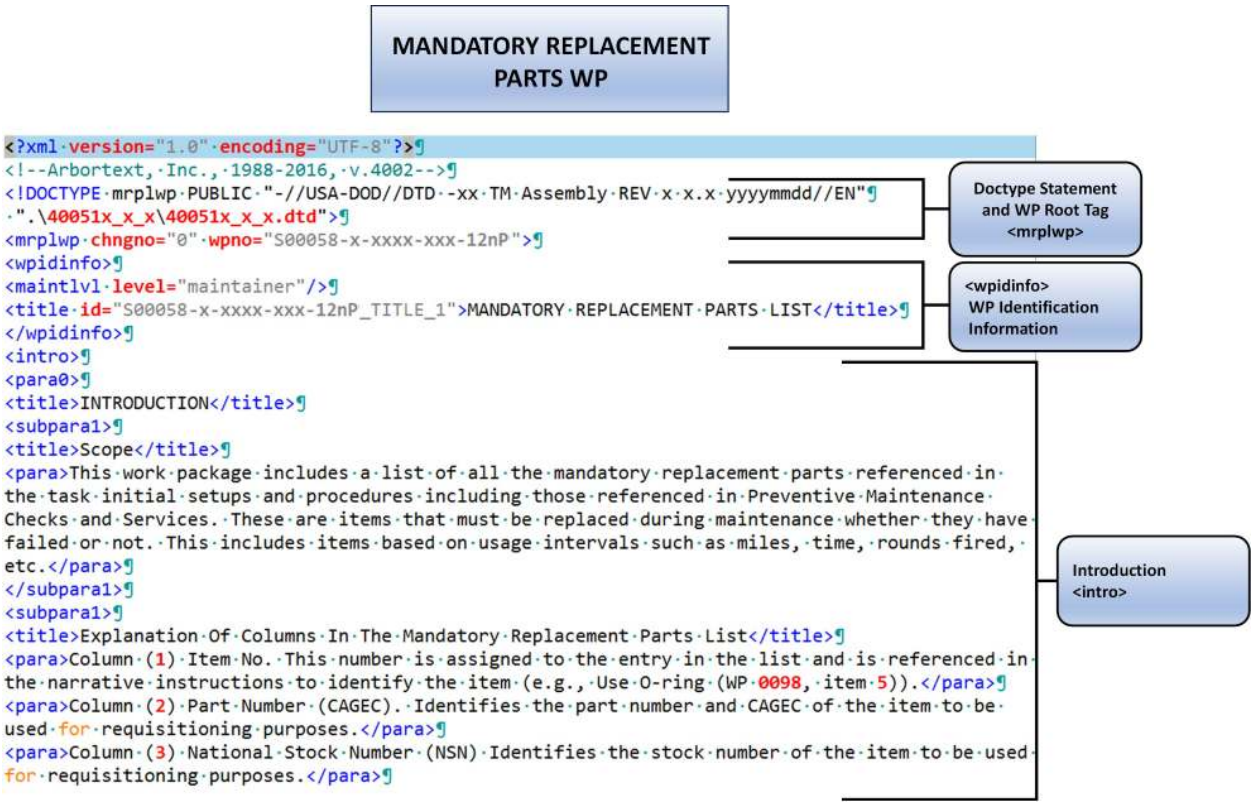


**FIGURE G-53. Mandatory replacement parts WP tags (sheet 1 of 3).**



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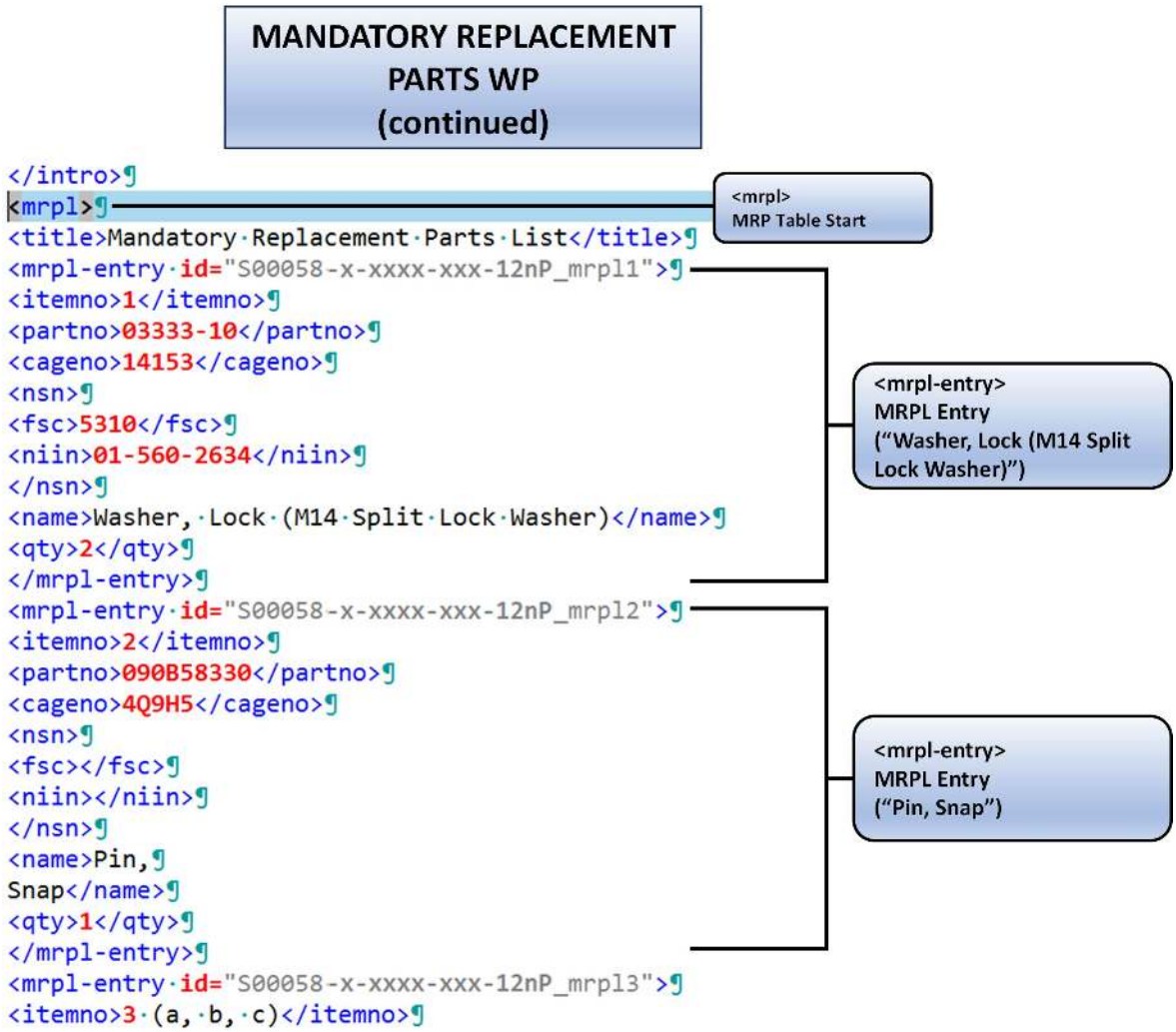


**FIGURE G-54. Mandatory replacement parts WP tags (sheet 2 of 3).**

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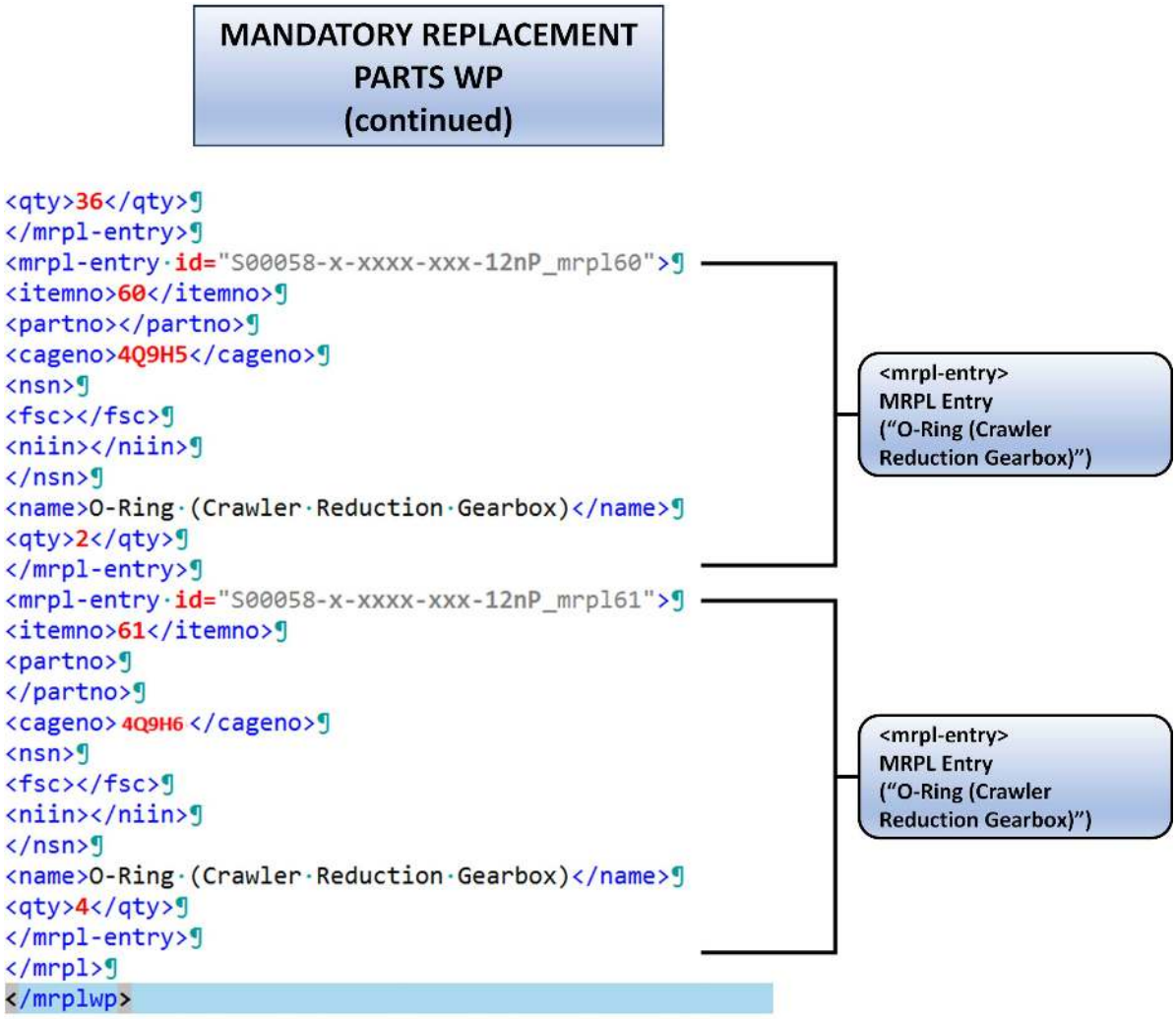


**FIGURE G-55. Mandatory replacement parts WP tags (sheet 3 of 3).**

## G.7.10 Critical safety items (CSI) work package <csi.wp>.

This WP shall be prepared on any system that contains a CSI as specified by the MATDEV. All CSIs shall be listed by their nomenclature **<name>**, part number **<partno>**, CAGE code **<cageno>**, and critical characteristic **<desc>** . If there are no critical safety items for the equipment, the following statement shall be entered in the CSI WP in lieu of the table: "There are no critical safety items for the *system name*.

**a.** Work package identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** Initial setup **<initial_setup>** is not required for this WP.

**c.** Critical safety items list **<csi>** is required for this WP.



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### G.7.11 Support items work package <supitemwp>.

This WP shall be prepared as directed by the MATDEV and shall combine any of the supporting lists described in through , as applicable. This WP shall be developed when the data contained in these supporting lists are minimal and creating a separate WP for each list is unnecessary.

**a.** Work package identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** Initial setup **<initial_setup>** is not required for this WP.

**c.** Introduction data **<intro>**. The WP may include an introduction to the information.

**d.** The WP shall include the applicable support item lists.

### G.7.12 General work package <genwp>.

When specified by the MATDEV additional WPs shall be prepared when the WPs previously described herein do not support the data/information to be presented.

**a.** Work package identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** Initial setup **<initial_setup>** is not required for this WP.

**c.** c. Introduction data **<intro>**. The WP may include an introduction to the information.

**d.** d. The WP shall include the applicable additional information.

### G.7.13 Notes.

The notes in Section 6 apply to this appendix.



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# APPENDIX H

# DESTRUCTION OF EQUIPMENT TO PREVENT ENEMY USE

## H.1 SCOPE.

This appendix establishes the technical content requirements for developing generic information and/or specific procedures regarding the destruction of equipment to prevent enemy use for major weapon systems and their related systems, subsystems, equipment, assemblies, components, Shop Replaceable Units (SRUs), and Line Replaceable Units (LRUs). This appendix is a mandatory part of this standard. The information contained herein is intended for compliance. The requirements are applicable for all maintenance levels/classes through "overhaul" (depot), including Depot Maintenance Work Requirements (DMWRs) and National Maintenance Work Requirements (NMWRs).

**Note**

To be compliant with this standard, publications shall be developed within **<production>** (opening) and **</production>** (closing) root tags.

When preparing information in accordance with this appendix, all references shall be linked using the appropriate tags (**<xref>**, **<extref>**, etc.). Links shall not be typed as text without tagging.

### H.1.1 Content tailoring using Appendix A matrices.

Applicability of some requirements contained in this standard are the responsibility of the materiel developer (MATDEV) (refer to AR 750–1, AR 700–127) and shall be accomplished using matrices referenced in TABLE A-I. The applicability of requirements are also designated by statements including, but not limited to, the following:

**a.** "...as specified by the MATDEV."

**b.** "...as required by the MATDEV."

**c.** "Refer to the MATDEV for..."

**d.** "Contact MATDEV if..."

**e.** "The MATDEV shall determine..."

**f.** "...as needed."

#### H.1.1.1 Front and rear matter.

Front and rear matter requirements for documents discussed in this appendix shall be determined in accordance with TABLE A-II.

## H.2 APPLICABLE DOCUMENTS.

The applicable documents in Section 2 of the basic standard apply to this appendix.

## H.3 ACRONYMS AND DEFINITIONS.

The acronyms and terms defined in Section 3 of this standard apply to this appendix.

## H.4 DESTRUCTION OF EQUIPMENT CONFIGURATION.

Several approaches are available for preparing manuals for destruction of equipment. These include, but are not limited to:

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**a.** Instructions or procedures developed for a particular stock class of materiel, as identified by its Federal Supply Classification (FSC).

**b.** Procedures that provide detailed destruction instructions for specific equipment system(s) or equipment and any installed subsystems.

**c.** Simple standardized destruction methods based on the assumption that time and demolition materials may not always be available for carrying out complicated demolition or other destruction procedures.

### H.4.1 Types of destruction manuals.

Each weapon system or major item of equipment shall have destruction procedures prepared that cover the approaches in B. and C. mentioned previously. Equipment managers may direct that a generic destruction manual be developed for assets they control in approach A. above that are not covered in an equipment system-specific manual. Equipment managers and weapons system program managers should work together to ensure that destruction procedures do not provide conflicting destruction requirements or overly duplicated destruction procedures. Some duplication of destruction procedures is allowed for components in an equipment system, but only those specific procedures (refer to H.6.3.2) for the component shall be duplicated. Duplication of this information is preferred to having users in a combat situation looking for destruction information in multiple technical publiccations.

### H.4.2 General destruction rules.

When preparing any destruction manual, the following priority guidelines shall be followed. These are provided to ensure a common approach to destruction of materiel:

**a.** Any cryptographic equipment or materiel shall be destroyed first.

**b.** Classified equipment or materiel is to be destroyed after any cryptographic assets. A statement to this effect shall be included in the introductory materiel. The destruction of classified materiel statement is required regardless of the classification of the materiel covered in the current technical publication.

**c.** Essential material shall be destroyed when time precludes the destruction of the entire system. In this case, essential material consists of such material identified for the system or stock class in the manual being prepared. The system manual shall include a list of essential material. A statement shall be included stating that essential material be destroyed in the order provided and that the same material be destroyed on each system (refer to H.6.2.5).

**d.** Any repair parts that may be on the verge of capture shall be destroyed in the same order as the essential material.

## H.5 DEVELOPMENT OF DESTRUCTION OF EQUIPMENT MODULES.

The following modules (and respective WPs) shall be prepared for stand-alone destruction of equipment technical publications (H.5.1) and destruction of equipment information included in an equipment system publication (H.5.2).

### H.5.1 Development of a stand-alone destruction of equipment technical publication.

When directed by an AMC supply class custodian or manager, a separate destruction technical publication **<destruction_ manual>** shall be prepared (see TABLE A-XIV). The manual shall contain generic destruction procedures and, when possible, should include specific procedures for each item in the stock class. If a separate manual is used, these destruction procedures shall be contained in a minimum of three work packages organized into two modules (listed below). See FIGURE H-1 for a graphical representation of WP organization and tags discussed in this section.

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### H.5.1.1 Destruction general information module <gim>.

A general information module **<gim>** shall be prepared and shall contain the following WPs.

**a.** General information work package (WP) **<ginfowp>** (H.6.1).

**b.** Introduction WP **<destruct-introwp>** (H.6.2).

### H.5.1.2 Destruction procedures module <dim>

A destruction procedures module **<dim>** shall be prepared shall contain at least one of the WP type listed below.

**a.** Destruction procedures WP **<destruct-materialwp>** (H.6.3).

**Note**

Tags/output shown in figure(s) below may vary slightly based on version of DTD/stylesheets utilized (see applicable DTD or tag definition list for available tags).

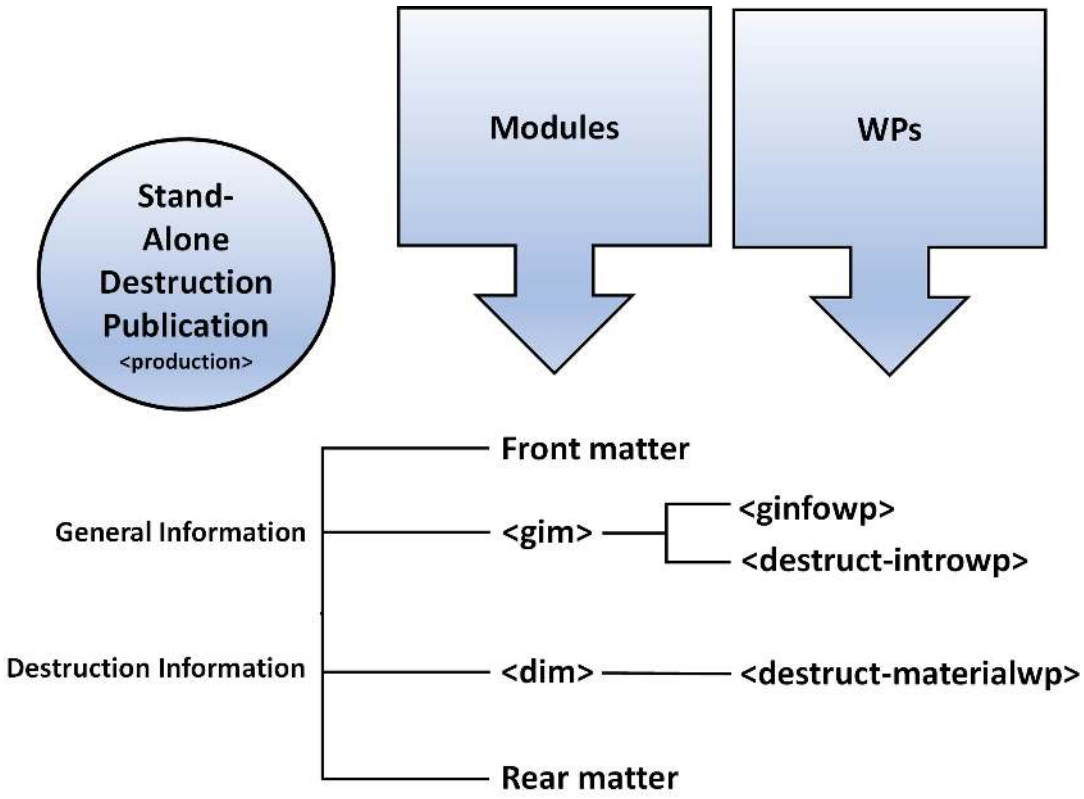


**FIGURE H-1. Modules and WPs in a stand-alone destruction technical publication.**

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## H.5.2 Development of a destruction of equipment module in an equipment system technical publication.

If destruction of equipment information is included as a module in an equipment system manual, these procedures shall be developed in a destruction of equipment module **<dim>**. It shall consist of the WPs listed in step E. (below) as needed. All other modules in an equipment technical publication shall be developed as required by publication type (see A.4.4). See FIGURE H-2 for a graphical representation of WP organization and tags discussed in this section.

**a.** General information module **<gim>** (B.6).

**b.** Operating instructions module(s) **<opim>** (C.6).

**c.** Troubleshooting procedures module(s) **<tim>** (D.8).

**d.** Maintenance instructions module(s) **<mim>** (E.6).

**e.** Destruction of equipment information module **<dim>** (H.5.1.2):

**(1)** Destruction of equipment introduction WP **<destruct-introwp>** (F.8.1).

**(2)** Destruction procedures WPs **<destruct-materialwp>** (F.8.2).

**f.** Software information module **<soim>** (M.6).

**g.** Parts information module **<pim>** (F.7.2).

**h.** Supporting information module **<sim>** (M.6).



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**Note**

Tags/output shown in figure(s) below may vary slightly based on version of DTD/stylesheets utilized (see applicable DTD or tag definition list for available tags).

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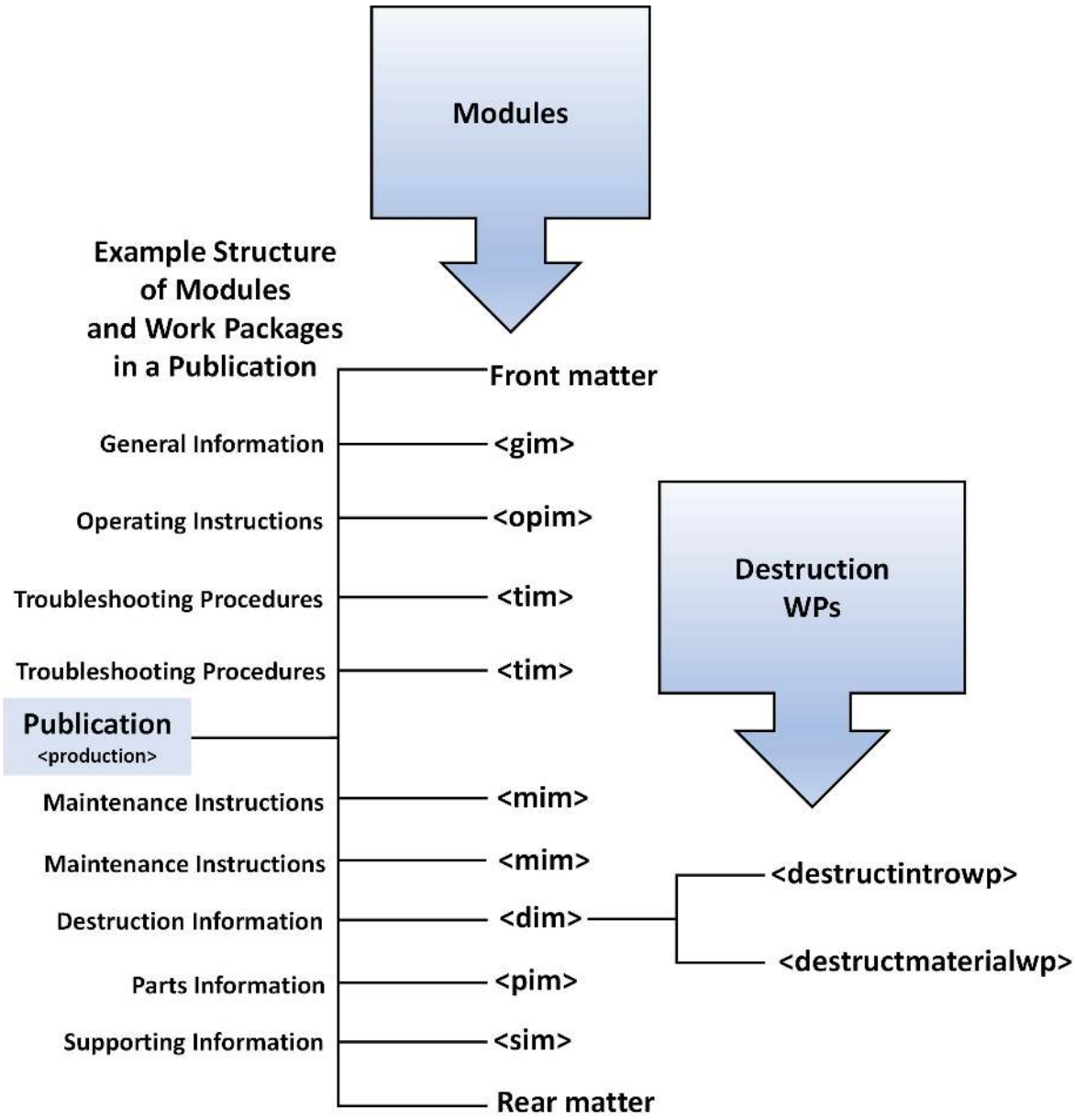


**FIGURE H-2. Destruction modules and WPs in an equipment system technical publication.**

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## H.6 DEVELOPMENT OF DESTRUCTION OF EQUIPMENT WORK PACKAGES.

### H.6.1 General information work package <ginfowp>.

A general information WP shall be prepared (see APPENDIX B). At a minimum, it shall contain a scope statement containing the following verbatim text:
"This manual is for the guidance of those whose duty it is to render inoperable or destroy equipment that is in imminent danger of capture by an enemy."

For destruction procedures that will implement any international standards, the following text shall be included. For a stand-alone destruction manual, the statement shall be in the **<ginfowp>** scope paragraph. For destruction procedures included in a weapon system manual, this statement shall be included in the "About this Manual" (italicized text within parentheses shall be replaced with the appropriate information).

"Certain provisions of this technical manual (identify by module, WP, paragraph, or similar manner, if appropriate) are the subject of international standardization agreement (*insert the ABCA or ASCC standard number; the NATO, STANAG, NETR, or NEPR number; or appropriate documentary reference*). When revision or cancellation of this technical manual is proposed that will modify the international agreement concerned, the technical manual management activity will take appropriate action through international standardization channels, including departmental standardization offices, to change the agreement or make other appropriate accommodations."

### H.6.2 Destruction procedures introduction work package <destruct-introwp>.

The destruction procedures introduction WP shall be prepared as required by publication type (see A.4.4) and shall contain the information described in H.6.2.1 through H.6.2.5. FIGURE H-3 depicts some of these sections and their associated tags. Destruction procedures introduction WP components are listed below:

**a.** Work package identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** Initial setup **<initial_setup>** is not required for this WP.

#### H.6.2.1 Authority to destroy materiel <authorize_to_destroy>.

The following paragraph shall be included verbatim:
"Authorization. Only division or higher commanders have the authority to order destruction of equipment. They may, however, delegate this authority to subordinate commanders when the situation demands it."

#### H.6.2.2 Reporting destruction <report_destruct>.

A paragraph shall be included that requires any destruction activity be reported through command channels.

#### H.6.2.3 General destruction information <general_destruct_info>.

Text shall be included that provides the user with information that is generic to most destruction processes. These data shall include, but are not limited to, the following types of information:

**a.** Information on types of destructive processes such as burning, use of explosives, burying, or self destruction devices/techniques. This explanation shall include the advantages and disadvantages of each process.

**b.** For complex weapons systems, the reason to perform any subordinate destruction procedures in conjunction with those for the weapons system.


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**c.** Any considerations relative to physical location or weather related (wind, rain, temperature) that users should consider when destroying materiel.

**d.** Explanations on the priority for materiel destruction (refer to TABLE A-XV).

### H.6.2.4 Degree of destruction <degree_of_destruct>.

The following information shall be included verbatim:
"Methods of Destruction. Choose methods of destruction that will cause such damage that it will be impossible to restore the equipment to a usable condition within the combat zone.
Classified Equipment. Classified equipment must be destroyed to such a degree as to prevent duplication by, or revealing means of operation or function to, the enemy.
Associated Classified Documents. Any classified documents, notes, instructions, or other written material pertaining to function, operation, maintenance, or employment, including drawings or parts lists, must be destroyed in a manner to render them useless to the enemy."

### H.6.2.5 Essential components and spare parts <component_spares>.

When specified by the MATDEV, the destruction procedures may identify essential components whose destruction will incapacitate the weapons system. In certain conditions, the destruction of essential components may be used. If destruction of essential components is allowed, statements shall be included that for each weapons system, the same components will be destroyed. A similar statement shall be included that for any spare parts requiring destruction, the same essential spare parts shall be destroyed. If a weapons system operator determines the component parts to be essential, they should notify the components item manager so that they may identify those items for higher priority destruction in any item-level destruction procedures manual.

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**Note**

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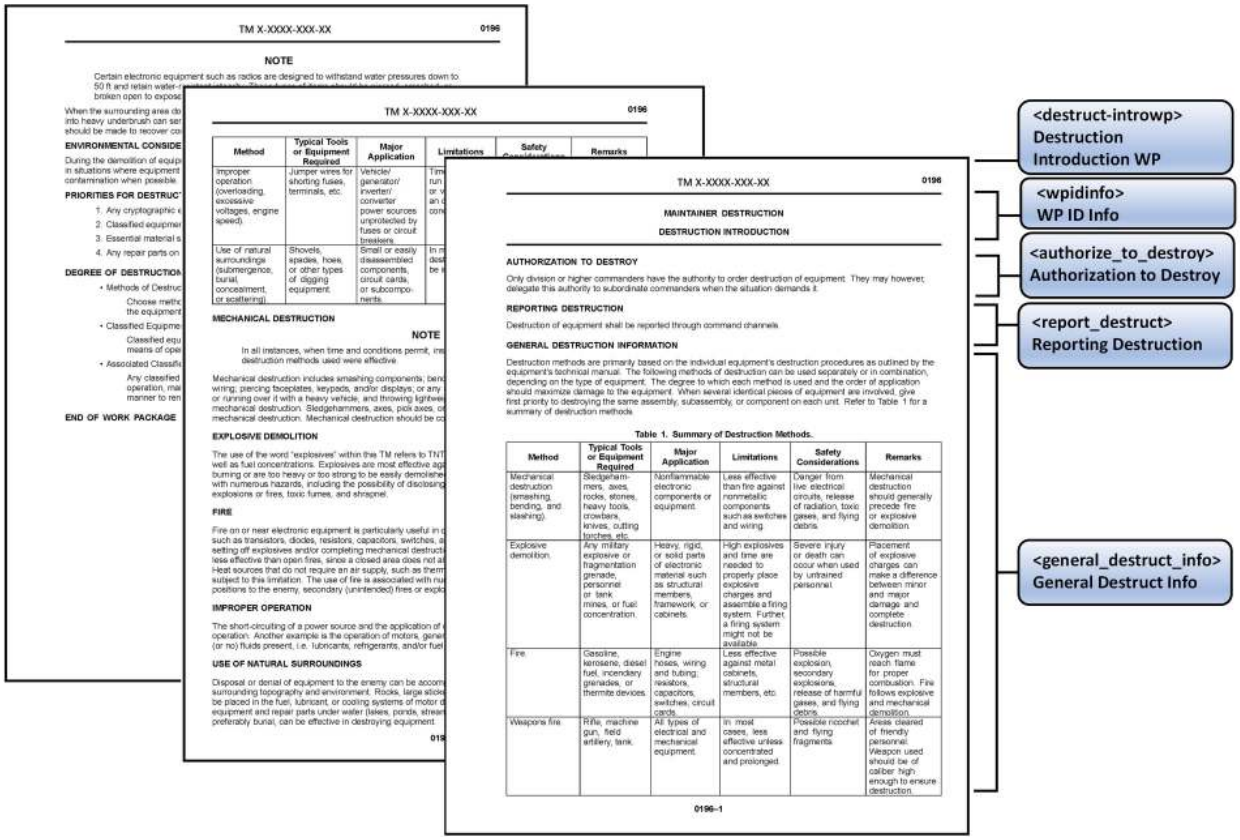


**FIGURE H-3. Destruction procedures introduction WP.**

## H.6.3 Destruction procedures work package <destruct-materialwp>.

The destruction procedures WP shall be prepared as required by publication type (see A.4.4) and shall contain the following information as described in H.6.3.1 through H.6.3.3. FIGURE H-4 depicts some of these sections and their associated tags. The destruction procedures WP shall contain only destruction procedures. All general or explanatory information shall be contained in the destruction introduction WP (H.5.2). Destruction procedures introduction WP components are listed below:

**a.** Work package identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** Initial setup **<initial_setup>** shall be included as needed (5.4.3). Any procedure, **<proc>**, required by this WP shall include initial setup **<initial_setup>**.

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### H.6.3.1 Parts list <essential_spares>.

When an equipment system technical publication contains a requirement to allow destruction of essential or spare parts (H.6.2.5), a list of essential components and spares shall be developed and included in the WP.

### H.6.3.2 Specific destruction procedures <proc>.

The destruction procedures WP shall include specific destruction procedures for the weapons system or items (for item-level technical publications). When required, specific procedures to destroy subordinate components shall be included. Specific destruction procedures for subordinate components shall not be referenced. As applicable, the order in which the procedures should be applied and the results of applying in the wrong order shall be included in this WP. When destruction procedures are developed, authors shall ensure the procedures use resources a soldier in the field would have readily accessible. The following methods shall be included as applicable:

**a.** Self-destruction options.

**b.** Explosive devices.

**c.** Improper operation.

**d.** Fire.

**e.** Mechanical devices (for example, sledgehammers, crowbars, cranes, etc.).

**f.** Natural surroundings (for example, rivers, lakes, caves, burying, hiding in vegetation, etc.).

As applicable, the procedures shall identify the points on the equipment that would be most advantageous to apply the previously described methods (for example, where to place explosives or where to apply force with a mechanical device).

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**Note**

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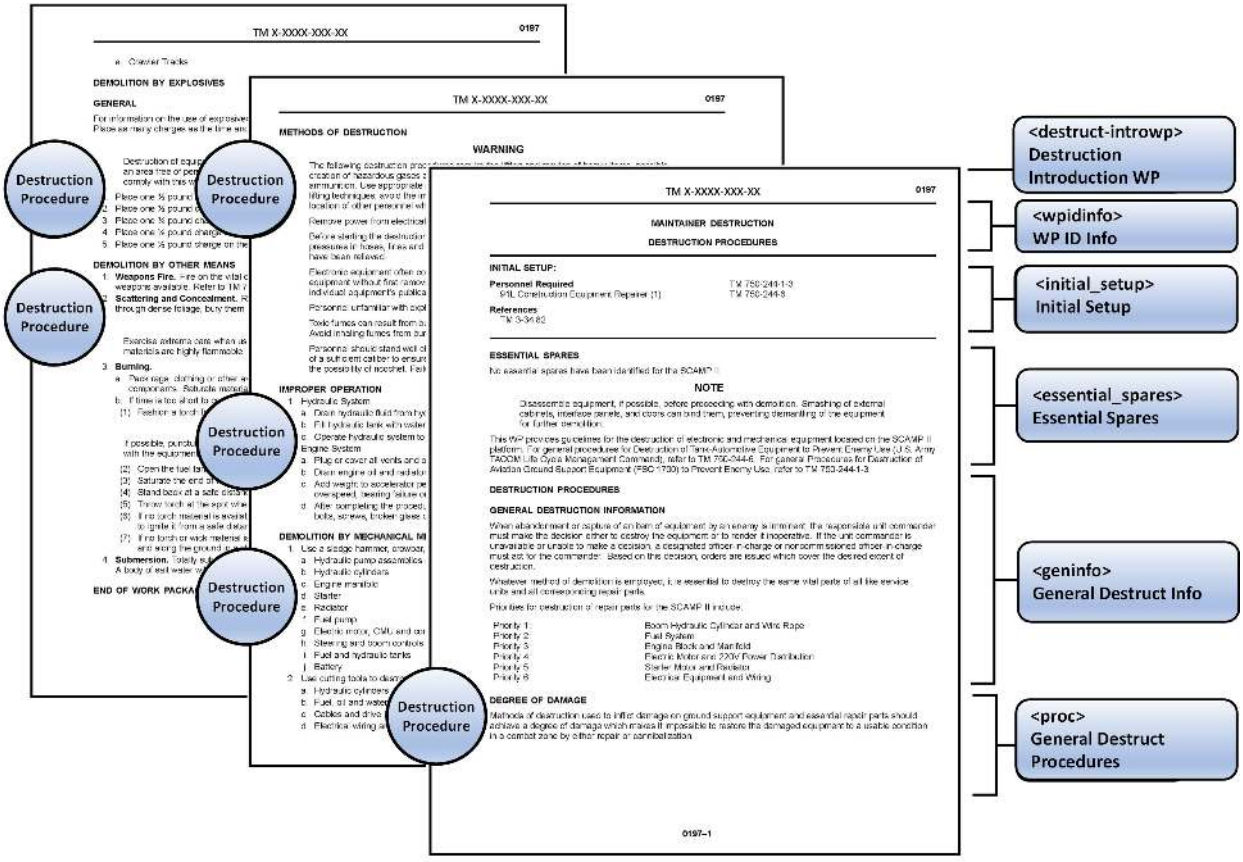


**FIGURE H-4. Destruction procedures WP.**

## H.6.3.3 Classified equipment and documents.

Special instructions for destruction of classified equipment and documents shall be provided.

## H.7 NOTES.

The notes in Section 6 apply to this appendix.

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# APPENDIX I

# BATTLE DAMAGE ASSESSMENT AND REPAIR

## I.1 SCOPE.

The requirements provided in this appendix provide the technical content requirements for the preparation of Battle Damage Assessment and Repair (BDAR) procedures. This appendix covers only assessment and repair of equipment failures occurring on the battlefield. This repair is sometimes limited to such means of fixing as bypassing, patching, or jury-rigging components, or the use of alternative procedures to restore the equipment/system performance to a minimum operating condition. Fix procedures in BDAR information are for use in combat only. Standard maintenance procedures are used as soon as practicable. This appendix is a mandatory part of this standard. The information contained herein is intended for compliance.

**Note**

To be compliant with this standard, publications shall be developed within **<production>** (opening) and **</production>** (closing) root tags.

When preparing information in accordance with this appendix, all references shall be linked using the appropriate tags (**<xref>**, **<extref>**, etc.). Links shall not be typed as text without tagging.

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### I.1.1 Content tailoring using Appendix A matrices.

Applicability of some requirements contained in this standard are the responsibility of the materiel developer (MATDEV) (refer to AR 750–1, AR 700–127) and shall be accomplished using matrices referenced in TABLE A-I. The applicability of requirements are also designated by statements including, but not limited to, the following:

**a.** "...as specified by the MATDEV."

**b.** "...as required by the MATDEV."

**c.** "Refer to the MATDEV for..."

**d.** "Contact MATDEV if..."

**e.** "The MATDEV shall determine..."

**f.** "...as needed."

#### I.1.1.1 Publication suffix.

BDAR publications shall contain "-BD" (for Battle Damage) as a suffix to the publication number (rather than "-BDAR"). See example publication suffix at the top of the pages shown in FIGURE I-3. However, these publications shall still be formally known as BDARs.

#### I.1.1.2 Front and rear matter.

Front and rear matter requirements for documents discussed in this appendix shall be determined in accordance with TABLE A-II.

## I.2 APPLICABLE DOCUMENTS.

The applicable documents in Section 2 apply to this appendix.

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## I.3 ACRONYMS AND DEFINITIONS.

The acronyms and terms defined in Section 3 of this standard apply to this appendix.

## I.4 BDAR MAINTENANCE LEVEL.

Unless otherwise specified, BDAR repair functions shall be accomplished by the following maintenance levels/ classes:

**a.** Field (Crew (operator)/AMC). Performed by crew (operator) or by a forward organizational maintenance team.

**b.** Field (Maintainer/ASB). Performed by maintainer or ASB, when damage exceeds service repair capability. When required repair time or tactical conditions dictate, the damaged/failed item will be recovered or evacuated as appropriate.

## I.5 BDAR INFORMATION INTEGRATION METHODOLOGY.

BDAR information shall be developed in accordance with this appendix. BDAR information shall be divided into work packages (WPs) that stand alone and are broken into the WP types listed below. Prior to starting BDAR information development, refer to the sections listed below in TABLE I-I for WP requirements.

**TABLE I-I. BDAR information WPs.**

| WP | WP Tag | Appendix Section |
|---|---|---|
| General information WP | **<ginfowp>** | I.7.1 |
| BDAR unique general information WP | **<bdar-geninfowp>** | I.7.2 |
| Battle damage assessment WP(s) | **<damage-assesswp>** | I.7.3 |
| Battle damage repair WP(s) | **<genrepairwp>** | I.7.4 |
| References WP | **<refwp>** | I.7.5 |
| Special or fabricated tools WP | **<bdartoolswp>** | I.7.6 |
| Expendable and durable items list WP | **<explistwp>** | I.7.7 |
| Substitute materials/parts WP | **<substitute-matwp>** | I.7.8 |

### I.5.1 BDAR procedure development format options.

There are two format options available for the development of BDAR procedures (listed below). The option chosen for an equipment system shall be specified by the MATDEV.

1. BDAR procedures may be included as a section in an equipment system technical publication (I.5.1.1).
2. BDARs may be developed as separate manual (I.5.1.2).

#### I.5.1.1 BDAR procedures included in an equipment system technical publication.

When specified by the MATDEV, various WPs that appear in a stand-alone BDAR **<bdar>** shall be included in an equipment system technical publication module **<bim>**.

**Note**

BDAR modules (**<bim>**) included in IETMs utilize system **<systemhierarchy>** or functional **<functionhierarchy>** hierarchy elements.

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Technical publications that allow BDAR information are identified within the content selection matrices in APPENDIX A. Content shall be limited to information and procedures related to battle damage assessment and repair procedures; however, this content can be referenced from (and referenced out to) other WPs in the equipment system publication. FIGURE I-1 shows an equipment system technical publication that includes BDAR as a module.

**Note**

Tags/output shown in figure(s) below may vary slightly based on version of DTD/stylesheets utilized (see applicable DTD or tag definition list for available tags).

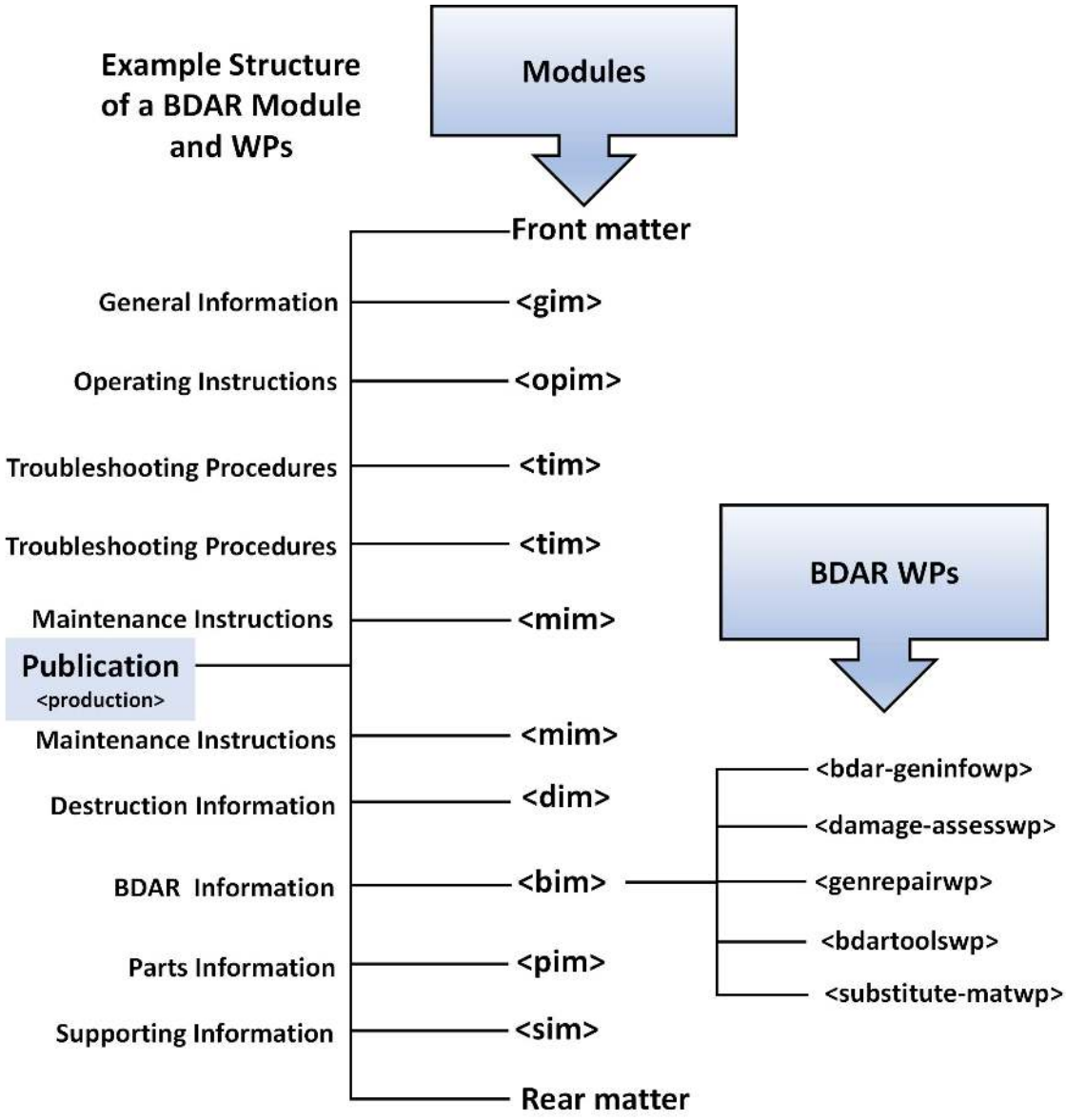


**FIGURE I-1. BDAR module in an equipment system technical publication.**

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### I.5.1.2 BDAR procedures as a stand-alone manual.

When specified by the MATDEV, a stand-alone battle damage manual **<bdar>** shall be prepared (see TABLE A-XIV). Content shall be directed to fix-forward battlefield conditions; for example, repairs must be made as quickly as possible and to the extent necessary to restore or maintain the applicable equipment/system. The modules and WPs listed below shall be prepared as required (see A.6.5). Refer to FIGURE I-2 for a graphical representation of module and WP organization as well as tags discussed in this section.

**Note**

Tags/output shown in figure (s) below may vary slightly based on version of DTD/stylesheets utilized (see applicable DTD or tag definition list for available tags).

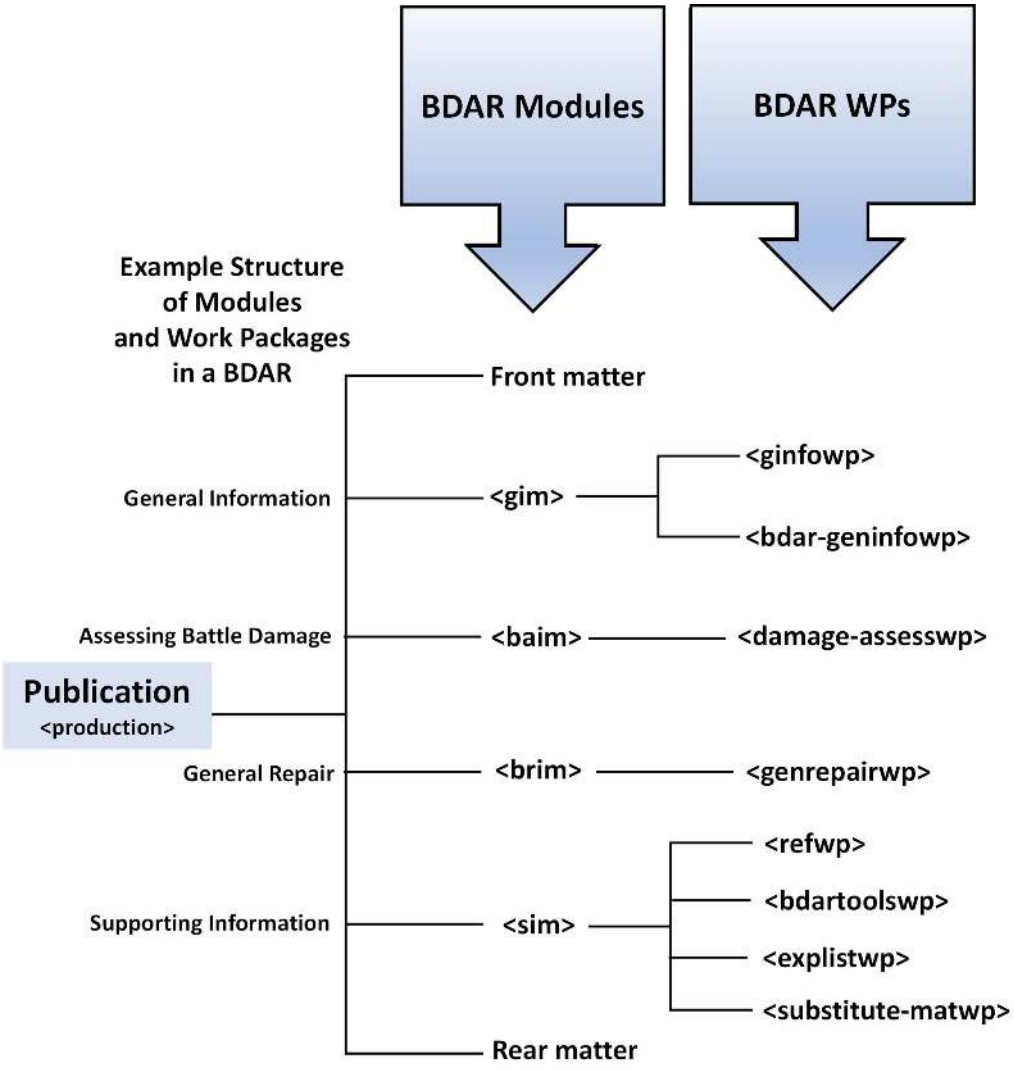


**FIGURE I-2. BDAR modules and WPs.**



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## I.6 DEVELOPMENT OF BDAR MODULES.

### I.6.1 BDAR module <bim> (module included in technical publication only).

A BDAR module **<bim>** shall be included in an equipment system technical publication as specified by the MATDEV. It shall consist of the following WPs.

**a.** BDAR unique general information WP **<bdar-geninfowp>** (I.7.2).

**b.** Battle damage assessment WP(s) **<damage-assesswp>** (I.7.3).

**c.** Battle damage repair WP(s) **<genrepairwp>** (I.7.4).

**d.** Special or fabricated tools WP **<bdartoolswp>** (I.7.6).

**e.** Substitute matrials/parts WP **<substitute-matwp>** (I.7.8).

### I.6.2 BDAR general information module <gim> (stand-alone BDAR).

A BDAR general information module **<gim>** shall be prepared. It shall consist of the following WPs.

**a.** BDAR general information WP **<ginfowp>** (I.7.1).

**b.** BDAR unique general information WP **<bdar-geninfowp>** (I.7.2).

### I.6.3 Battle damage assessment instructions module <baim> (stand-alone BDAR).

A battle damage assessment instructions module **<baim>** shall be prepared. It shall consist of the following WPs.

**a.** Battle damage assessment WP(s) **<damage-assesswp>** (I.7.3).

**b.** Additional battle damage assessment WP(s) as needed.

### I.6.4 Battle damage repair instructions module <brim> (stand-alone BDAR).

A battle damage repair instructions module **<brim>** shall be prepared. It shall consist of the following WPs.

**a.** Battle damage repair WP(s) **<genrepairwp>** (I.7.4).

**b.** Additional battle damage repair WP(s) as needed.

### I.6.5 BDAR supporting information module <sim> (stand-alone BDAR).

A supporting information module **<sim>** with the following WPs shall be prepared as required.

**a.** References WP **<refwp>** (I.7.5).

**b.** Special or fabricated tools WP **<bdartoolswp>** (I.7.6).

**c.** Expendable and durable items list WP **<explistwp>** (I.7.7).

**d.** Substitute materials/parts WP **<substitute-matwp>** (I.7.8).

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## I.7 DEVELOPMENT OF BDAR WORK PACKAGES.

The following types of BDAR WPs shall be prepared as required (see ). The following statement (in block below) shall appear at the beginning of each WP in the BDAR manual.

> BDAR FIXES SHALL BE USED ONLY IN COMBAT OR FOR TRAINING AT THE DISCRETION OF THE COMMANDER. (AUTHORIZED TRAINING FIXES ARE LISTED IN THE BDAR TRAINING PROCEDURES WORK PACKAGE.) IN ANY CASE, DAMAGE SHALL BE REPAIRED BY STANDARD MAINTENANCE PROCEDURES AS SOON AS PRACTICABLE.

### I.7.1 General information work package <ginfowp>.

A general information WP shall be prepared. It shall contain those elements required to support all Army publications (see B.7.1). General information WP components are listed below:

**a.** WP identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** Initial setup **<initial_setup>** is not required for this WP.

**c.** Scope **<scope>** is required for this WP.

**d.** Equipment Improvement Reporting (EIR) **<eir>** and other general information WP elements required to support Army publications (B.7.1.3) as needed.

### I.7.2 BDAR unique general information work package <bdar-geninfowp>.

This WP shall contain information that is general in nature, but unique to a BDAR manual (see FIGURE I-3). It shall inform the user/reader of the purpose of the BDAR information and its relationship to user personnel, other publications, and the end item/system it supports. BDAR unique general information WP components are listed below:

**a.** WP identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** Initial setup **<initial_setup>** is not required for this WP.

**c.** BDAR fixes statement is required for this WP (see blocked text in I.7).

**d.** Standards and practices **<bdar-std-practices>** are required for this WP (I.7.2.1).

**e.** Tasks and responsibilities**<bdar-task-resp>** are required for this WP (I.7.2.2).

#### I.7.2.1 Standards and practices <bdar-std-practices>.

This paragraph shall contain information pertaining to standards and practices peculiar to combat conditions. It shall include, as a minimum, the following subparagraph headings and data (expanded as applicable):

**a.** BDAR Characteristics. An explanation of the expediency of repair, reason for deviation from standard maintenance practices, need to take greater risks, and other characteristics specific to repair under combat conditions shall be included.

**b.** Waiver of precautions. A reference to deviations from normal peacetime precautions shall be included. If such deviations are summarized in another portion of the BDAR information, reference shall be made to that portion.

**c.** Operating characteristics. The minimum functional combat capability criteria for the applicable end item/system shall be listed.


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**d.** Training. The explanation/rationale concerning the use of BDAR fixes for training shall be addressed. It shall list all BDAR procedures that are authorized for training. The fix (training) procedures shall be grouped by major system(s) or components(s) as they appear in the BDAR information. Each procedure shall be cross-referenced to the WP where it appears. The following statement shall be included:

"After completion of training, the end item/system shall be returned to full serviceable condition using regular repair procedures as applicable."

### I.7.2.2 Tasks and responsibilities <bdar-task-resp>.

This paragraph shall consist of tasks that may be required as a result of battlefield damage. The person/group responsible for each task shall be identified. The tasks shall appear in the order in which they should be performed. This information shall be presented in narrative form. This shall include the following subparagraphs: I.7.2.2.1 through I.7.2.2.3.

### I.7.2.2.1 Tagging/identifying BDAR repairs.

Instructions for identifying components affected by BDAR fixes shall be included.

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**Note**

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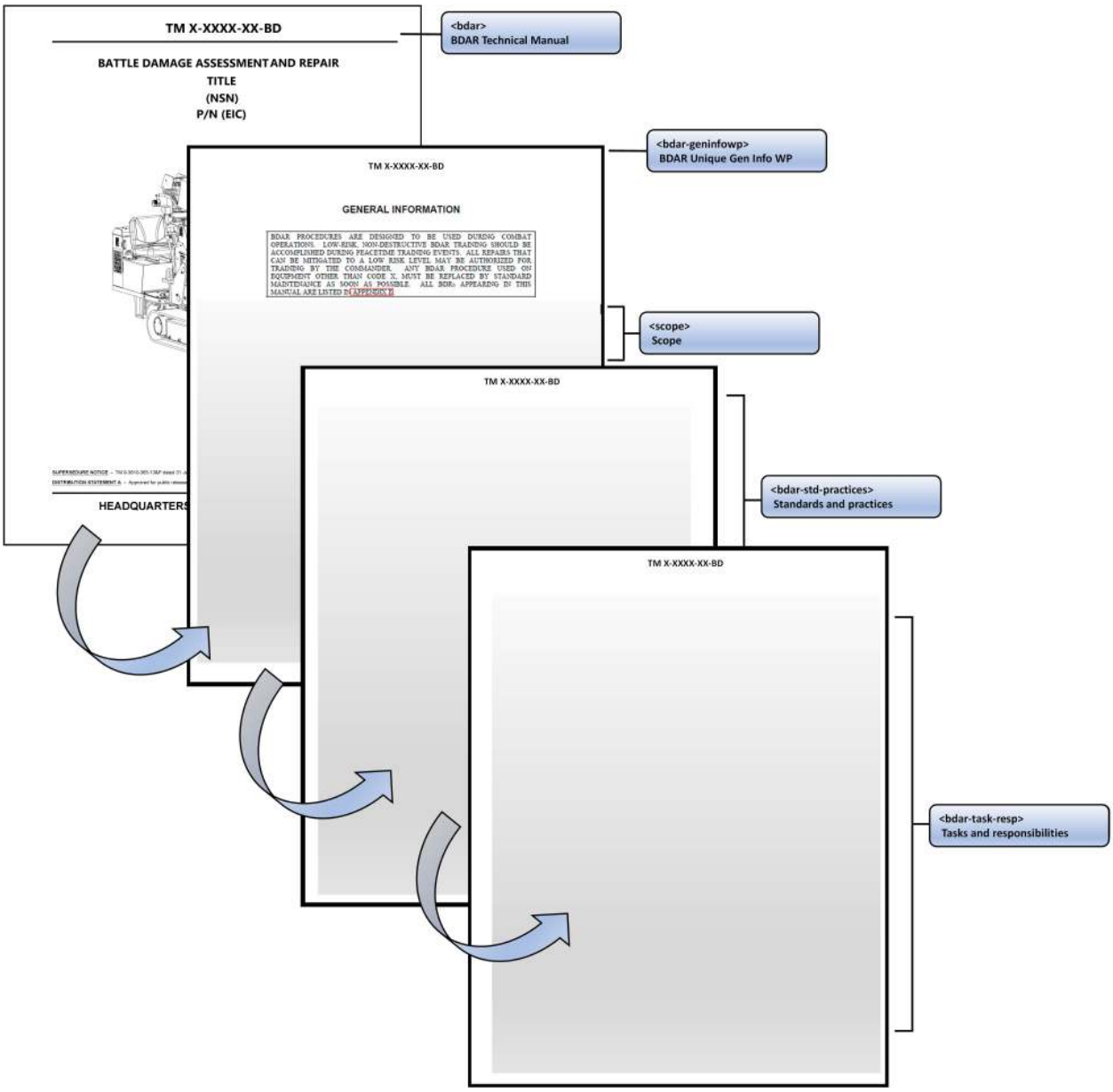


**FIGURE I-3. BDAR cover and unique general information WP example.**

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#### I.7.2.2.2 Tagging/identifying BDAR repairs.

Instructions for identifying components affected by BDAR fixes shall be included.

#### I.7.2.2.3 Reports.

Instructions for completing reports resulting from BDAR fixes shall be addressed.

### I.7.2.3 Combat Threats <bdar-combat-threat> (Aviation Only.

This paragraph shall be prepared as needed. It shall consist of the description of damage from threats confronting aircraft while on combat missions from armor-piercing, armor-piercing incendiary projectiles, and high-explosive incendiary projectiles. It shall also describe damage from exposure to bombs, mortars, and artillery fragments and blasts when on the ground. The resulting effects to the metal airframe structure and follow-on effects should the mission be continued shall be given. The effects of secondary damage such as cracks, crippling, or buckling and loss or damage to mechanical fasteners shall also be given. Structure damage modes shall be defined for the type of materials and structure affected.

## I.7.3 BDAR assessment work package <damage-assesswp>.

One or more battle damage assessment work packages shall be prepared. Each of these work packages shall contain an introduction and fault assessment tables and shall be organized as follows:

**a.** End item. These shall be a battle damage assessment WP pertaining to the overall end item or major subsystems and shall discuss the capability of the end item/subsystem to perform its mission essential functions.

**b.** Major functional group. Unless otherwise specified by the MATDEV, these work packages shall be titled, arranged, and shall correspond to the functional groups as they appear in the MAC and the parts information. Each major functional group shall have at least one and may have more than one.

**c.** Auxiliary Equipment. As required, battle damage assessment work packages shall be prepared for any auxiliary equipment.

Battle damage assessment WP components are listed below:

**a.** WP identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** Initial setup **<initial_setup>** is not required for this WP.

**c.** BDAR fixes statement is required for this WP (see graphic in I.5).

**d.** Introduction **<intro>**. This paragraph shall be prepared and shall contain paragraphs that cover the scope of the WP and application of assessment tables.

**e.** Fault assessment tables (I.7.3.1) are required for this WP.

### I.7.3.1 Fault assessment tables.

This paragraph shall contain assessment tables that lead the user to a repair procedure or another chart/table that will further aid in analyzing/assessing damage. As specified by the MATDEV, the format of the assessment tables shall be either a troubleshooting procedure or a table (see to FIGURE I-4 and FIGURE I-5 for examples.) The assessment procedures shall be developed and arranged so that logical and expedient methods are used to locate trouble.

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**Note**

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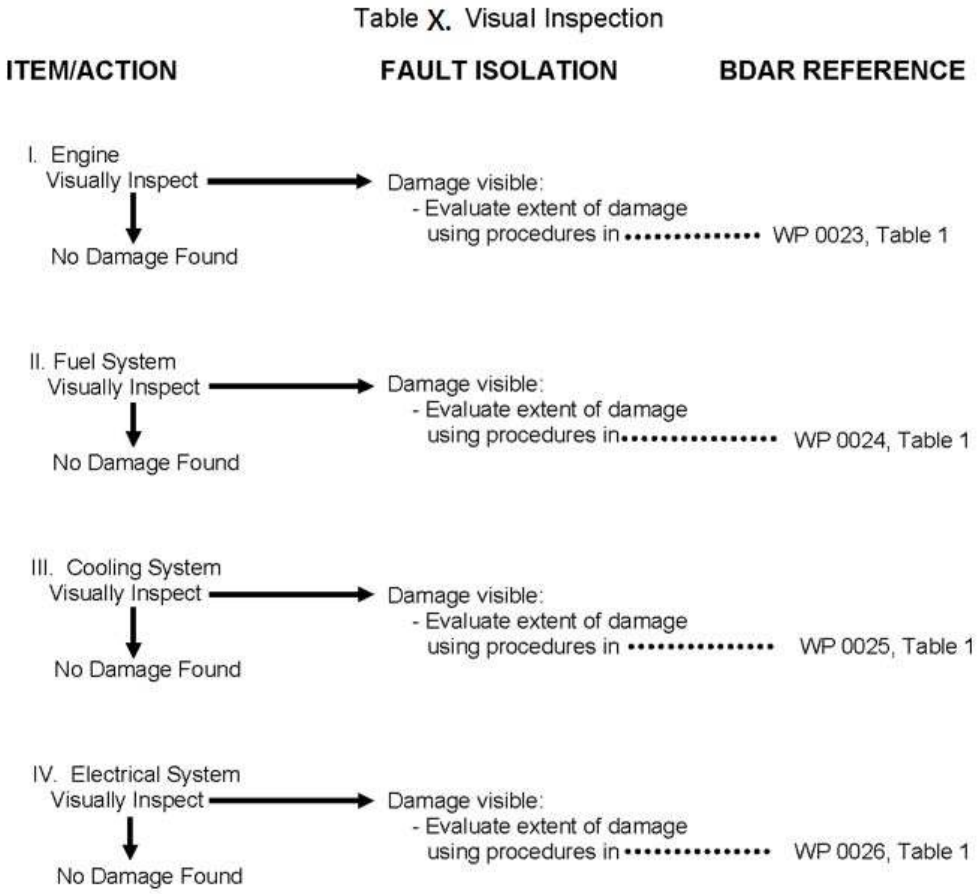


**FIGURE I-4. BDAR assessment flowchart diagram.**

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How to use the fault assessment tables:

a. A fault assessment table is organized so the user can quickly assess a particular system or capability by asking a series of questions.

b. First, ask a question. Your response will be either a "yes" or "no". If it is "yes", then you have no problem so go to the next question.

c. If it is "no", then proceed to the work package listed.

Table 1. Mobility

| | |
|---|---|
| Does engine start/run? | If no go to WP 0032 |
| Does tank move in "D" and "R"? | If no go to WP 0033 |
| Are the track and suspension intact? | If no go to WP 0034 for track repair Or go to WP 0035 for suspension repair |
| Does tank steer/pivot? | If no go to WP 0036 |
| Does tank brake? | If no go to WP 0037 |
| Does tank have full power | If no go to WP 0038 |

**FIGURE I-5. Example BDAR assessment table.**

## I.7.4 Battle damage repair work package <genrepairwp>.

Unless otherwise specified by the MATDEV, these work packages shall provide information for battlefield repair of end items, components, etc. The following types of repair work packages shall be included in the BDAR information module:

**a.** General repair. As required, procedures shall be provided for items that are not necessarily associated with a specific component or subsystem of the end item.

**b.** End item repair. Procedures for repair of the overall end item shall be provided in the WP(s).

**c.** Major functional group repair. Unless otherwise specified by the MATDEV, these WP(s) shall be titled, arranged, and shall correspond to the functional groups as they appear in the MAC and the parts information. Each major functional group shall have at least one and may have more than one.

**d.** Auxiliary equipment. As required, procedures for repair of battle damage to auxiliary equipment shall be provided.

Battle damage repair WP components are listed below:

**a.** WP identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** Initial setup **<initial_setup>** is required for this WP (5.4.3). For a stand-alone BDAR manual, the initial shall not be referred or link but shall contain the details.

**c.** BDAR fixes statement is required for this WP (see graphic in I.6).



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**d.** Introduction **<geninfo>**. This paragraph shall contain the following subparagraphs:

**(1)** Scope. A brief statement that describes the purpose and application of the overall coverage of the WP shall be included.

**(2)** Repair procedure index. A list of all procedures shall be contained in this WP. The procedures shall be listed in the order in which they appear. Procedures authorized for training and listed in the training fixes WP shall be boxed in.

**e.** Repair procedure(s) **<bdar-repair-proc>** are required for this WP (see I.7.4.1).

### I.7.4.1 BDAR repair procedures <bdar-repair-proc>.

This paragraph shall contain the repair procedure for the item(s) covered in the WP. The format and content of these paragraphs shall be as follows.

**General.** Remarks concerning the general nature and causes related to the damage and repair of the item shall be included. These remarks shall be brief.

**Item name, trouble.** The item name and the trouble shall be used as the subparagraph side head. The side head shall be followed with a general statement or statements concerning the particular type of trouble and repair to be made. Statements shall be brief and as concise as possible. Subparagraphs shall be as follows.

**Limitations <bdar-limitation>.** This statement or statements shall identify, in relation to operational capability, the limits that would be imposed on the equipment/end item if the fix that follows is performed.

**Procedural steps <proc>.** Each step shall be listed numerically and placed in the sequential order in which it will be performed. Steps shall be as prescribed in 4.12.8.1. The last procedural step for every BDAR fix shall be: "Record BDAR action taken. When mission is complete, as soon as practical, repair the equipment/system using standard maintenance procedures."

**Options.** When more than one method of making the same repair/fix exists, multiple options shall be included. Options shall be listed in order of effectiveness and listed consecutively as option 1, option 2, etc. Each option provided under the item name/trouble paragraph side head shall contain the subparagraphs required by . Alternatives that do not include fixes shall also be listed as options.

**Item name, category.** When the basic item, identified in the section title, is divided into categories or types, each specific item shall be titled and covered within a separate paragraph. Each of these paragraphs shall contain only the information that applies to that specific item. For example: information or procedures under a heading "high pressure" shall pertain to high pressure; low pressure information/ procedures (if applicable) shall appear under the heading "low pressure."

## I.7.5 References work package <refwp>.

This WP shall be prepared and shall be included in the references for the BDAR information. This WP shall be prepared in accordance with APPENDIX G.

## I.7.6 Special or fabricated tools work package <bdartoolswp>.

This WP shall be prepared and shall contain a list of all tools and test equipment that are required for BDAR procedures and that are not common. This list shall be prepared in accordance with the requirements for a tool identification list in APPENDIX G. When fabrication of tools is required for BDAR, this WP shall also contain fabrication instructions for those tools. The fabrication instructions shall be prepared in accordance with the requirements for an illustrated list of manufactured items contained in APPENDIX E. The special or fabricated tools WPe shall consist of the following components:

**a.** WP identification information **<wpidinfo>** is required for this WP (5.4.1).

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**b.** Initial setup **<initial_setup>** is not required for this WP.

**c.** BDAR fixes statement is required for this WP (see graphic in I.5).

**d.** Introduction **<intro>**. This paragraph shall introduce the assessment table(s) in the WP. It shall contain paragraphs that will cover the scope of the WP and application of assessment tables.

**e.** List of all tools and test equipment that are required for BDAR procedures and that are not common.

**f.** Fabrication instructions for any fabrication tools as needed.

### I.7.7 Expendable and durable items work package <explistwp>.

This WP shall be prepared and shall include the expendable and durable items required for BDAR. This WP shall be prepared in accordance with APPENDIX G.

### I.7.8 Substitute materials/parts work package <substitute-matwp>.

This WP shall list materials and parts that may be used for BDAR fixes. Lists or tables shall include the primary material/part, the substitute/alternate material/part, and remarks (when applicable) that identify the limitations or degradation effected by use of the substitutes. The WP shall be divided into paragraphs by material type. For an example of alternate/substitute material listing, refer to FIGURE I-6. Substitute materials/parts WP components are listed below:

**a.** WP identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** Initial setup **<initial_setup>** is not required for this WP.

**c.** BDAR fixes statement is required for this WP (see graphic in I.5).

**d.** Introduction **<intro>**. As needed when paragraphs are required, the first paragraph shall be titled introduction and shall provide a general explanation of the purpose and content of the other paragraphs.

**e.** POL substitutes. As needed, paragraph(s)/table(s) shall be dedicated to POL substitutes. For examples of Petroleum, Oil, and Lubricant (POL) substitutes, refer to FIGURE I-7 and FIGURE I-8.

**f.** Other substitutes. As needed, paragraph(s)/table(s) shall be dedicated to any other substitute materials/parts information.

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**Note**

Figure appearance may vary slightly based on version of DTD/stylesheets utilized (see applicable DTD or tag definition list for available tags).

Table X. Hull Spares and Repair Parts

| Parts | | Applies To | | From Weapons System | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| NSN | Description | M1 (PM) | M1A1 | M2 M3 Tank | M34 A2 Track | M48 A5 Famil | M60 Famil | M88 Famil | M109 Veh |
| 5935-00-001-7325 | Connector Plug | X | X | X | | | | | |
| 5315-00-014-1283 | Pin, Straight, Headless | X | X | | | X | X | | |
| 2530-00-015-2774 | Spacer, Hub Track | X | X | X | | X | X | X | X |
| 4730-00-018-9566 | Plug, Pipe | X | X | X | | X | X | X | X |
| 4730-00-050-4203 | Fitting, Lubrication | X | X | | | X | X | | |
| 4730-00-050-4208 | Fitting, Lubrication | X | X | | | X | X | | |
| 5340-00-057-3537 | Clevis, Road End | X | X | | | X | X | | |
| 2530-01-201-4816 | Roadwheel Assembly | X | X | | | | X | X | |
| 4730-00-080-9847 | Adaptor, Straight | X | X | | | X | X | X | X |
| 5340-00-088-4254 | Clamp, Loop | X | X | | | X | X | | |
| 5340-00-088-6655 | Clamp, Loop | X | X | | | X | X | | |
| 2920-00-088-8613 | | X | X | | | | X | | |

**FIGURE I-6. Example of substitute materials/parts table.**



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**Note**

Figure appearance may vary slightly based on version of DTD/stylesheets utilized (see applicable DTD or tag definition list for available tags).

Table X. Substitute Lubricants and Hydraulic Fluids

| Primary | | | | Alternate | | Expedient | Notes |
|---|---|---|---|---|---|---|---|
| Lubrication Point | Temperature Range | Military Specifications | NATO Product | US or NATO Equivalent | Soviet | | |
| Gun Bore | Above 32°F<br>+40°F/65°F | (PL-M) MIL-L-3150<br>PL-S VV-L-800 | 02-192<br>0-190 | | | | Not BDAR critical |
| Bore Evacuator | Above 32°F<br>+40°F/65°F | (PL-M) MIL-L-3150<br>PL-S VV-L-800 | 0-192<br>0-190 | OE/HDO-10 MIL-L-2104<br>OEA, MIL-G-46167 | | | |
| Breech Block | Above 32°F<br>+40°F/65°F | (PL-M) MIL-L-3150<br>PL-S VV-L-800 | 0-192<br>0-190 | Any MIL-L-2104<br>OEA, MIL-G-46167 | | | |
| Grenade Dischargers | Above 32°F<br>+40°F/65°F | (PL-M) MIL-L-3150<br>PL-S VV-L-800 | 0-192<br>0-190 | Any MIL-L-2104<br>OEA, MIL-G-46167 | | | Not BDAR critical |

**FIGURE I-7. Example of substitute lubricants and hydraulic fluids table.**

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**Note**

Figure appearance may vary slightly based on version of DTD/stylesheets utilized (see applicable DTD or tag definition list for available tags).

TableX. Substitute Fuels for Diesel Fuel W-F-800, DF-1, and NATO-F-54

| Primary Fuel | Alternate Fuel | Expedient Fuel | Military Specification | Commercial Specification |
|---|---|---|---|---|
| Diesel Fuel<br>VV-F-800<br>DF-1<br>NATO-F-54 | See Below | See Below | X | |
| | *Automotive<br>Diesel:<br>ASTM-D-975<br>(1-D and 2-D) | | | X |
| | Kerosene:<br>ASTM-D-3699 | | | X |
| | Fuel Oil:<br>ASTM-D-396<br>(Numbers 1 and 2) | | | X |
| | Distillate:<br>NATO-F-75<br>(Low pour point) | | X | |
| | Kerosene:<br>NATO-F-5B | | X | |
| | Aviation Turbine:<br>MIL-T-5624<br>(JP4 and JP5)<br>NATO-F-40 | | X | X |

**FIGURE I-8. Example of substitute fuels table.**

## I.8 NOTES.

The notes in Section 6 apply to this appendix.

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# APPENDIX J

# PREVENTIVE MAINTENANCE CHECKLIST

## J.1 SCOPE.

This appendix contains detailed requirements for the preparation of an operator's pocket-size Preventive Maintenance Checklist (PMC) for major weapon systems and their related systems, subsystems, equipment, assemblies, components, Shop Replaceable Units (SRUs), and Line Replaceable Units (LRUs). This appendix is a mandatory part of the standard. The information contained herein is intended for compliance. The requirements are applicable for all maintenance levels/classes through "overhaul" (depot), including Depot Maintenance Work Requirements (DMWRs) and National Maintenance Work Requirements (NMWRs). A PMC **<pmc>** shall be prepared when specified by the MATDEV. The MATDEV shall specify those inspection intervals for the PMC using those intervals as stated in APPENDIX E (refer to J.3.1).

**Note**

When preparing information in accordance with this appendix, all references shall be linked using the appropriate tags (**<xref>**, **<extref>**, etc.). Links shall not be typed as text without tagging.

### J.1.1 Content tailoring using Appendix A matrices.

Applicability of some requirements contained in this standard are the responsibility of the materiel developer (MATDEV) (refer to AR 750–1, AR 700–127) and shall be accomplished using matrices referenced in TABLE A-I. The applicability of requirements are also designated by statements including, but not limited to, the following:

**a.** "...as specified by the MATDEV."

**b.** "...as required by the MATDEV."

**c.** "Refer to the MATDEV for..."

**d.** "Contact MATDEV if..."

**e.** "The MATDEV shall determine..."

**f.** "...as needed."

#### J.1.1.1 Front and rear matter.

Front and rear matter requirements for documents discussed in this appendix shall be determined in accordance with TABLE A-II.

## J.2 ACRONYMS AND DEFINITIONS.

The acronyms and terms defined in Section 3 of this standard apply to this appendix.

## J.3 DEVELOPMENT OF PMC MANUALS.

The PMC shall consist of front matter, a PMCS introduction work package (WP) (E.7.1), one or more PMCS WPs (E.7.2), and rear matter. FIGURE J-1 shows an example PMC manual front cover.

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**Note**

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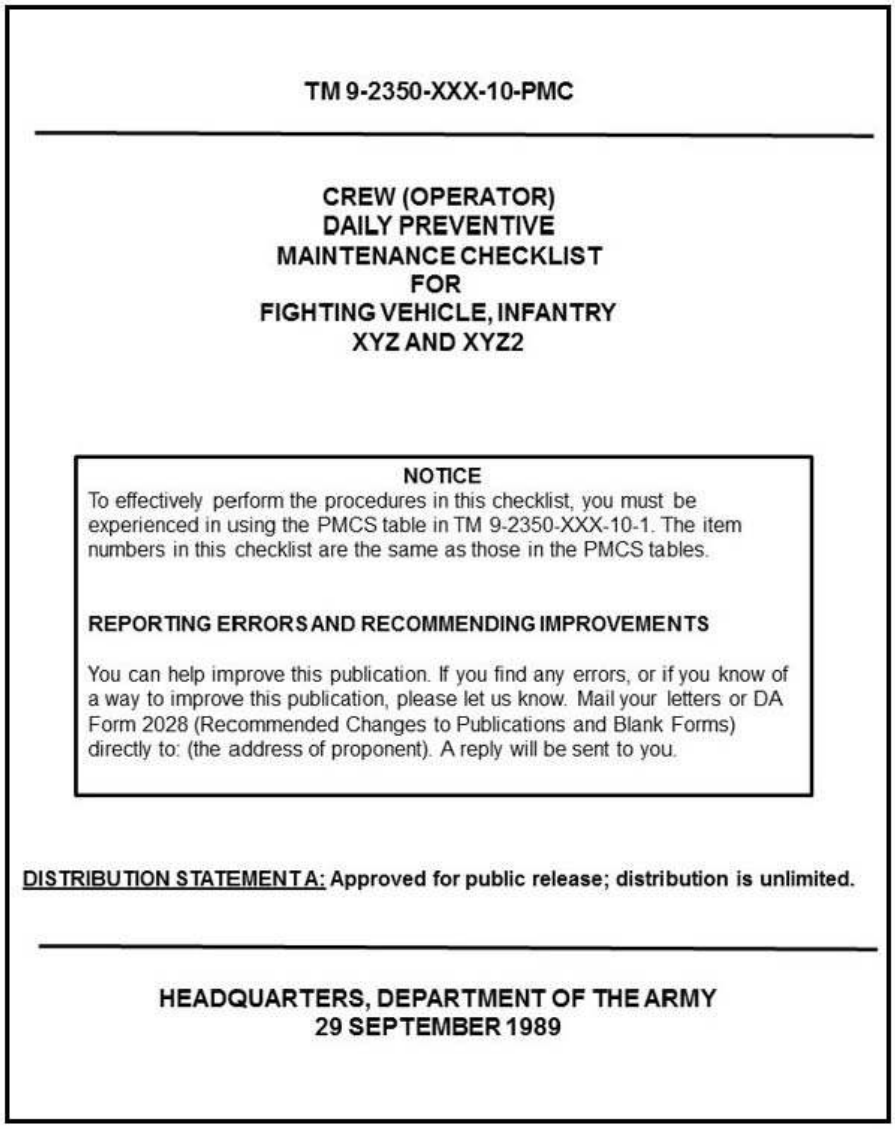

TM 9-2350-XXX-10-PMC

CREW (OPERATOR)
DAILY PREVENTIVE
MAINTENANCE CHECKLIST
FOR
FIGHTING VEHICLE, INFANTRY
XYZ AND XYZ2

NOTICE
To effectively perform the procedures in this checklist, you must be experienced in using the PMCS table in TM 9-2350-XXX-10-1. The item numbers in this checklist are the same as those in the PMCS tables.

REPORTING ERRORS AND RECOMMENDING IMPROVEMENTS

You can help improve this publication. If you find any errors, or if you know of a way to improve this publication, please let us know. Mail your letters or DA Form 2028 (Recommended Changes to Publications and Blank Forms) directly to: (the address of proponent). A reply will be sent to you.

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HEADQUARTERS, DEPARTMENT OF THE ARMY
29 SEPTEMBER 1989

**FIGURE J-1. Example PMC cover page.**

## J.3.1 PMCS introduction work package <pmcsintrowp>.

A PMCS introduction WP shall be prepared in accordance with APPENDIX E.

## J.3.2 PMCS work package <pmcswp>.

The PMC shall contain one or more PMCS WP(s) containing all items and intervals for the maintenance level(s) as specified by the MATDEV. The specified inspections shall be taken directly from the applicable PMCS table (refer to APPENDIX E) in the operator-level or field-level manual containing the inspection. Item numbers in the checklist shall be the same as those assigned to the procedures in the operator or field-level maintenance PMCS table. For stand-alone PMC manuals, the initial setups shall contain detail for items and shall not be linked or referenced.

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## J.4 NOTES.

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

### J.4.1 Acquisition requirements.

The MATDEV should specify the inspection intervals to be included in the PMC (refer to E.7.2.2).

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# APPENDIX K

# LUBRICATION ORDERS

## K.1 SCOPE.

This appendix establishes the technical content requirements for the preparation of stand-alone Lubrication Orders (LOs) **<lubeorder>** for equipment systems and their related systems, subsystems, equipment, assemblies, components, Shop Replaceable Units (SRUs), and Line Replaceable Units (LRUs). This appendix is a mandatory part of this standard. The information contained herein is intended for compliance. The requirements are applicable for all maintenance levels/classes through "overhaul" (depot) including Depot Maintenance Work Requirements (DMWRs) and National Maintenance Work Requirements (NMWRs).

**Note**

To be compliant with this standard, publications shall be developed within **<production>** (opening) and **</production>** (closing) root tags.

When preparing information in accordance with this appendix, all references shall be linked using the appropriate tags (**<xref>**, **<extref>**, etc.). Links shall not be typed as text without tagging.

### K.1.1 Content tailoring using Appendix A matrices.

Applicability of some requirements contained in this standard are the responsibility of the materiel developer (MATDEV) (refer to AR 750–1, AR 700–127) and shall be accomplished using matrices referenced in TABLE A-I. The applicability of requirements are also designated by statements including, but not limited to, the following:

**a.** "...as specified by the MATDEV."

**b.** "...as required by the MATDEV."

**c.** "Refer to the MATDEV for..."

**d.** "Contact MATDEV if..."

**e.** "The MATDEV shall determine..."

**f.** "...as needed."

#### K.1.1.1 Front and rear matter.

Front and rear matter requirements for documents discussed in this appendix shall be determined in accordance with TABLE A-II. Refer to K.5.1.3 for LO front cover details.

## K.2 APPLICABLE DOCUMENTS.

The applicable documents in Section 2 apply to this appendix.

## K.3 ACRONYMS AND DEFINITIONS.

The acronyms and terms defined in Section 3 of this standard apply to this appendix.

## K.4 INTEGRATION OF LUBRICATION INSTRUCTIONS.

Lubrication instructions may be integrated into publications in two possible ways:

**a.** PMCS entries or additional work packages (WPs) in an equipment technical publication.

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**b.** Stand-alone LO technical publication.

### K.4.1 Lubrication instructions in an equipment technical publication.

Lubrication instructions shall be prepared for all equipment, except aircraft, that require lubrication. Lubrication instructions may be integrated in an equipment technical publication in the following two ways.

**a.** When specified by the MATDEV, the lubrication instructions may be included in the PMCS WP (E.7.2) or as a lubrication WP (refer to K.6.2).

**b.** When the lubrication instructions are classified, they shall be included in the PMCS or a lubrication WP that is classified to at least the classification level of the instructions or higher. Classified instructions shall be marked and handled as specified in the current security regulations.

### K.4.2 Stand-alone LO technical publication.

Data developed in accordance with this appendix shall be divided into modules and WPs. See TABLE K-I for a list of and FIGURE K-1 for a diagram of components of a technical publication containing this module along with WPs available in the module.

**TABLE K-I. LO WPs types.**

| WP | WP Tag | Appendix Section |
|---|---|---|
| LO introduction WP | **<lointrowp>** | K.7.1 |
| Lubrication instructions WP | **<lubewp>** | |
| LO references WP | **<refwp>** | K.7.3 |
| LO tool identification WP | **<toolidwp>** | K.7.4 |
| Lubricants and military symbols WP | **<lubricantsymbolswp>** | K.7.5 |
| Lubricant types WP | **<lubetypeswp>** | K.7.6 |
| Special notes WP | **<lospecnoteswp>** | K.7.7 |


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**Note**

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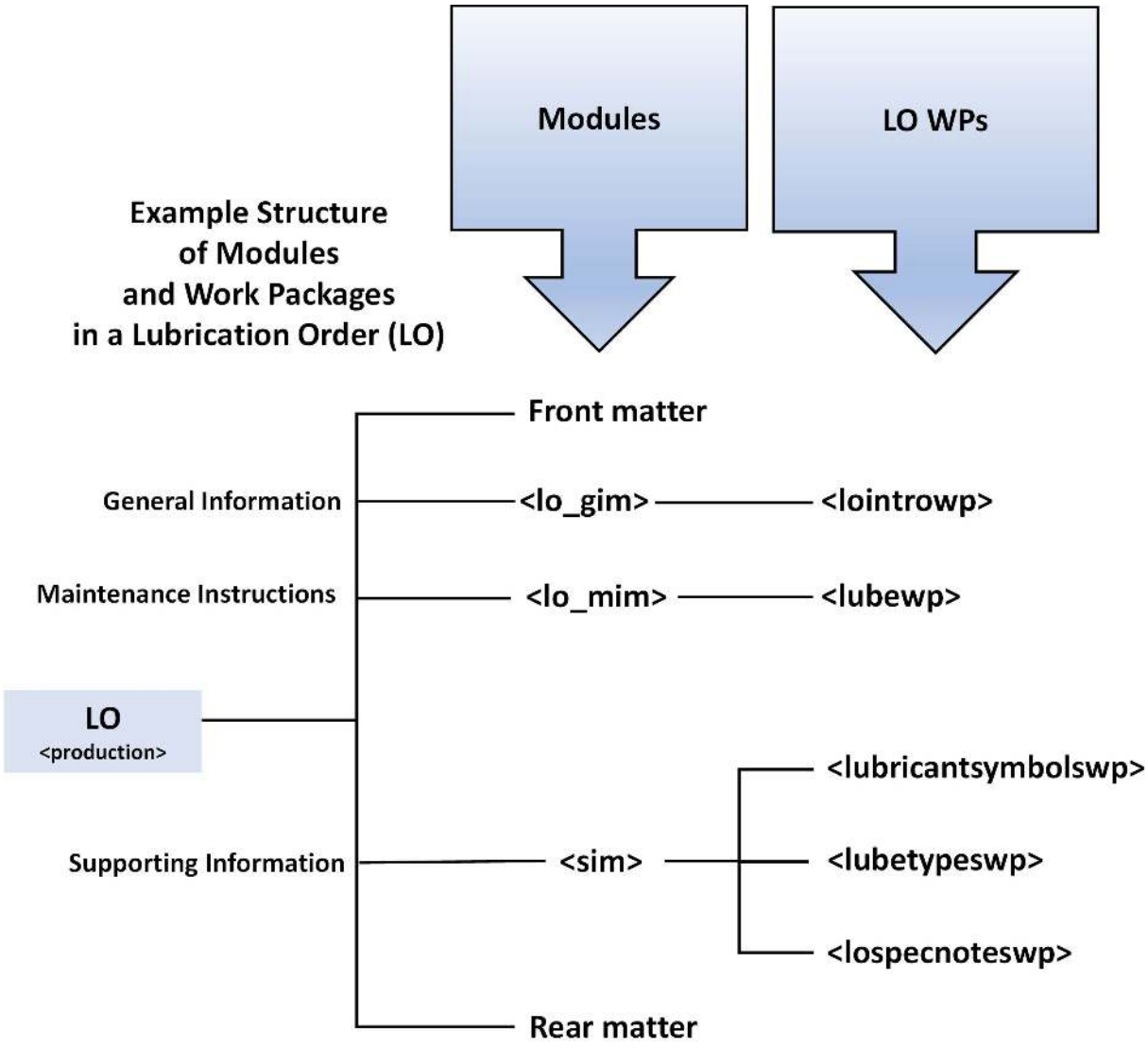


**FIGURE K-1. LO manual modules and WPs.**

## K.5 LUBRICATION ORDER KEY ELEMENTS.

Sections K.5.1 through K.5.1.3 discuss some key elements of lubrication orders. Section K.6 provides in-depth guidelines for the development of individual LO WPs.

### K.5.1 Page size

LOs shall be prepared in either log book or standard page size (4.15.3).



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### K.5.1.1 Lubrication interval symbols.

Unless otherwise specified by the MATDEV, the lubrication interval symbols in TABLE K-I shall be used.

**TABLE K-II. LO intervals.**

| Symbol | Definition |
|---|---|
| D | Daily |
| W | Weekly |
| M | Monthly |
| Q | Quarterly |
| S | Semiannually |
| A | Annually |
| B | Biennially |
| H | Hours (operated) |
| MI | Miles (operated) |
| KM | Kilometers (operated) |
| RDS | Rounds (fired) |
| OC | On Condition |
| MRA | Maintenance Repair Action |

### K.5.1.2 LO measurements.

Unless otherwise specified by the MATDEV, all measurements expressed in the text, in tables, or in illustrations shall be expressed in both U.S. standard units and metric units. The order shall be in accordance with equipment markings.

### K.5.1.3 LO front cover.

The LO cover shall be formatted in accordance with 5.2.1. It shall contain the components listed below (some components are shown in FIGURE K-2).

**a.** Number.

**b.** Heading.

**c.** Title **<tmtitle>**.

**d.** Reference Line **<lube-refs>**: shall be placed below the title within the applicable area; shall consist of the publication number(s) of the related technical publications.

**e.** Reporting Errors **<reporting>**.

**f.** Supersession notice **<super>**.

**g.** Distribution statement, export control warning, and destruction notice **<notices>**.

**h.** Lubrication Order (LO) statement **<general_purpose_notices>**: the statement listed below shall be included on the title page of the LO.

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"A copy of this lubrication order will remain with the equipment at all times; instructions contained herein are mandatory."

**i.** Service nomenclature **<servnomen>**.

**j.** Date **<date>**.

**Note**

Figure appearance may vary slightly based on version of DTD/stylesheets utilized.

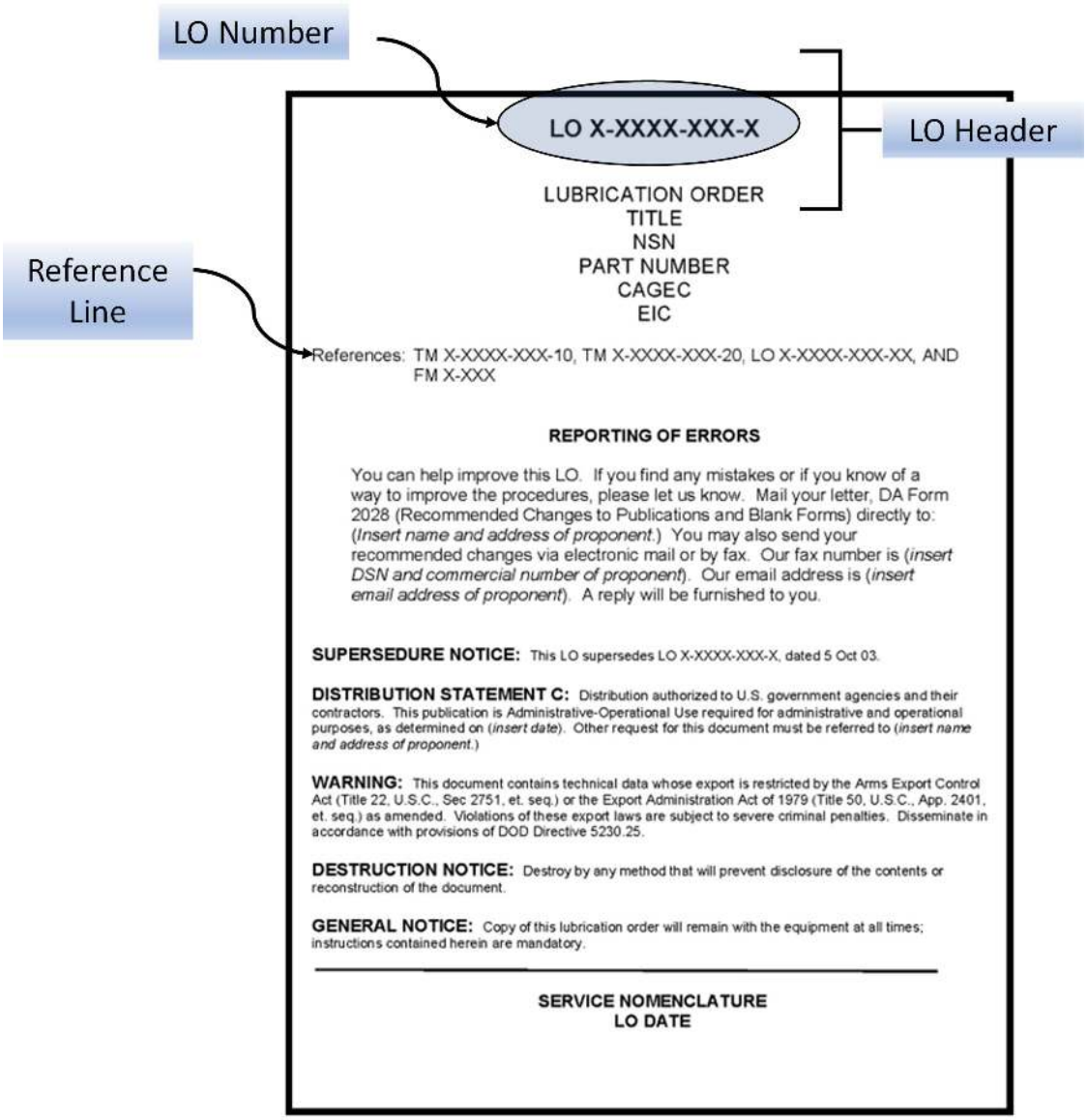


**FIGURE K-2. LO cover example.**



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## K.6 DEVELOPMENT OF LUBRICATION ORDER MODULES.

When specified by the MATDEV, a lubrication order manual **<lubeorder>** shall be prepared. The modules and WPs listed below shall be developed as required (see TABLE A-XVI). Refer to FIGURE K-1 for a graphical representation of module and WP organization as well as tags discussed in this section.

### K.6.1 LO general information module <lo_gim>.

A LO general information module **<lo_gim>** shall be prepared. It shall consist of the following WP.

**a.** LO general information WP **<lointrowp>** (K.7.1).

### K.6.2 LO maintenance instructions module <lo_mim>.

A LO maintenance instructions module **<lo_mim>** shall be prepared. It shall consist of the following WP(s).

**a.** LO lubrication instructions WP **<lubewp>** ().

**b.** Additional lubrication procedures WP(s) as needed.

### K.6.3 LO supporting information module <sim>.

A LO supporting information module **<sim>** shall be prepared. It shall consist of the following WP(s) as applicable for an equipment system.

**a.** References WP **<refwp>** (K.7.3).

**b.** Tool identification WP **<toolidwp>** (K.7.4).

**c.** Lubricants and military symbols WP **<lubricantsymbolswp>** (K.7.5).

**d.** Lubricant types WP **<lubetypeswp>** (K.7.6).

**e.** Special notes WP **<lospecnoteswp>** (K.7.7).

## K.7 DEVELOPMENT OF LUBRICATION ORDER WORK PACKAGES.

When lubrication instructions are integrated into a manual (when specified by the MATDEV), WPs shall be divided into WP types listed in TABLE K-I. These work packages should stand alone and contain all information and references required to support the WP type.

### K.7.1 LO introduction work package <lointrowp>.

The LO introduction WP shall contain the following information (shown in FIGURE K-3 and FIGURE K-4):

**a.** WP identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** Initial setup **<initial_setup>** is not required (but allowable) for this WP.

**c.** LO introduction contents. The LO introduction shall contain the statements/information as prescribed in K.7.1.1 through K.7.1.4.

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### K.7.1.1 General statement(s/notes.

General statement(s)/notes shall be placed in the LO introduction WP and are applicable to the overall understanding of requirements of the LO instructions (shown in FIGURE K-3). The statement(s) shall include information such as:

**a.** Adherence to lubrication intervals.

**b.** Explanation of interval symbols.

**c.** Maintenance levels/classes.

**d.** Exceptional operational requirements.

**e.** Abbreviations.

**f.** Fittings.

**g.** Parts cleaning.

**h.** Statement concerning corrosion control (as applicable). The statement shall provide instructions or reference corrosion control requirements provided in the applicable narrative technical publications.

### K.7.1.2 Oil filter statement.

As applicable, an oil statement (shown in FIGURE K-3) similar to the following shall be included:

"Oil filters shall be serviced/cleaned/changed as applicable, when:

**a.** they are known to be contaminated or clogged;

**b.** service is recommended by Army Oil Analysis Program (AOAP) laboratory analysis; or

**c.** at prescribed hardtime maintenance intervals."

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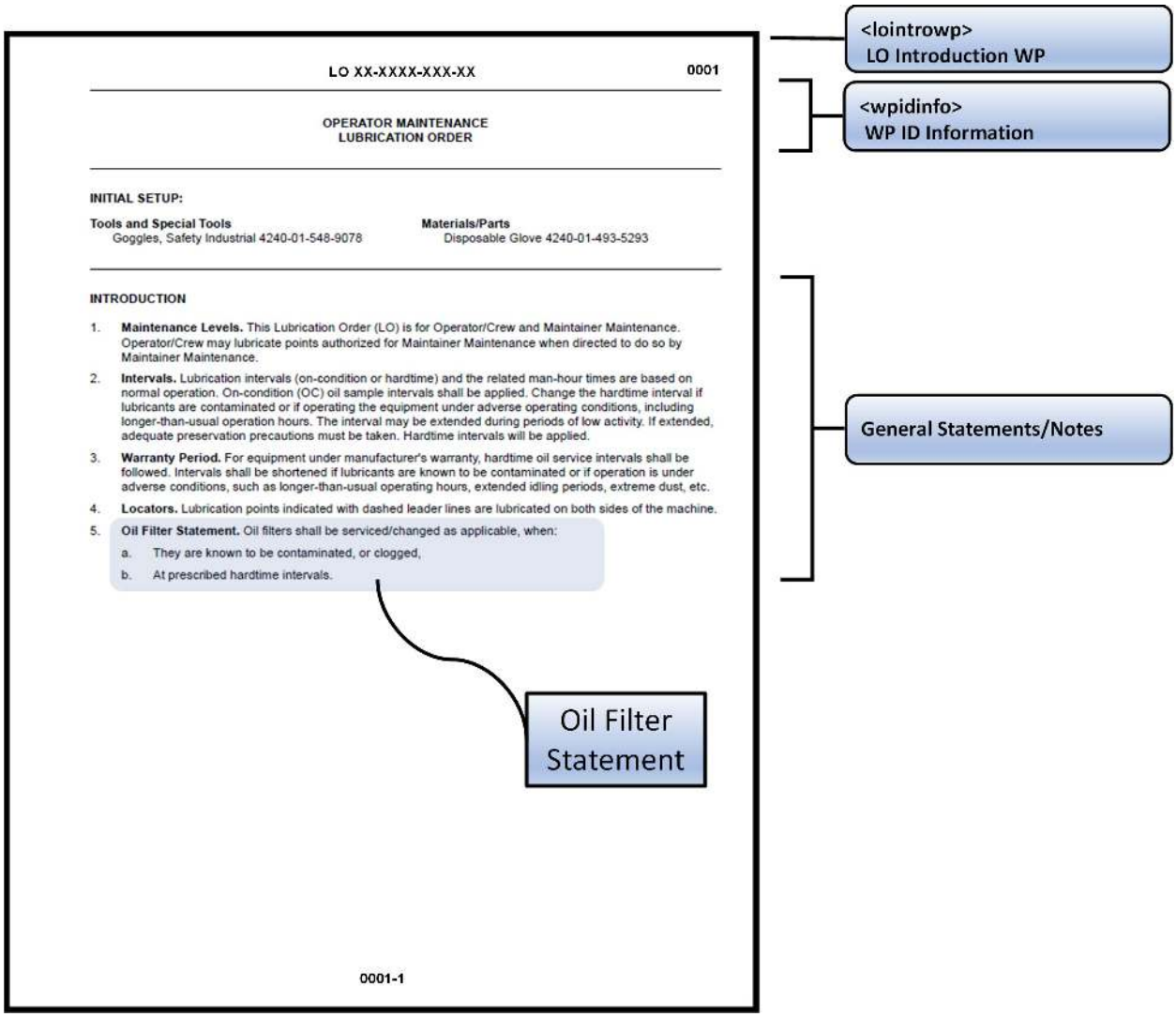


**FIGURE K-3. LO introduction WP (sheet 1 of 2).**

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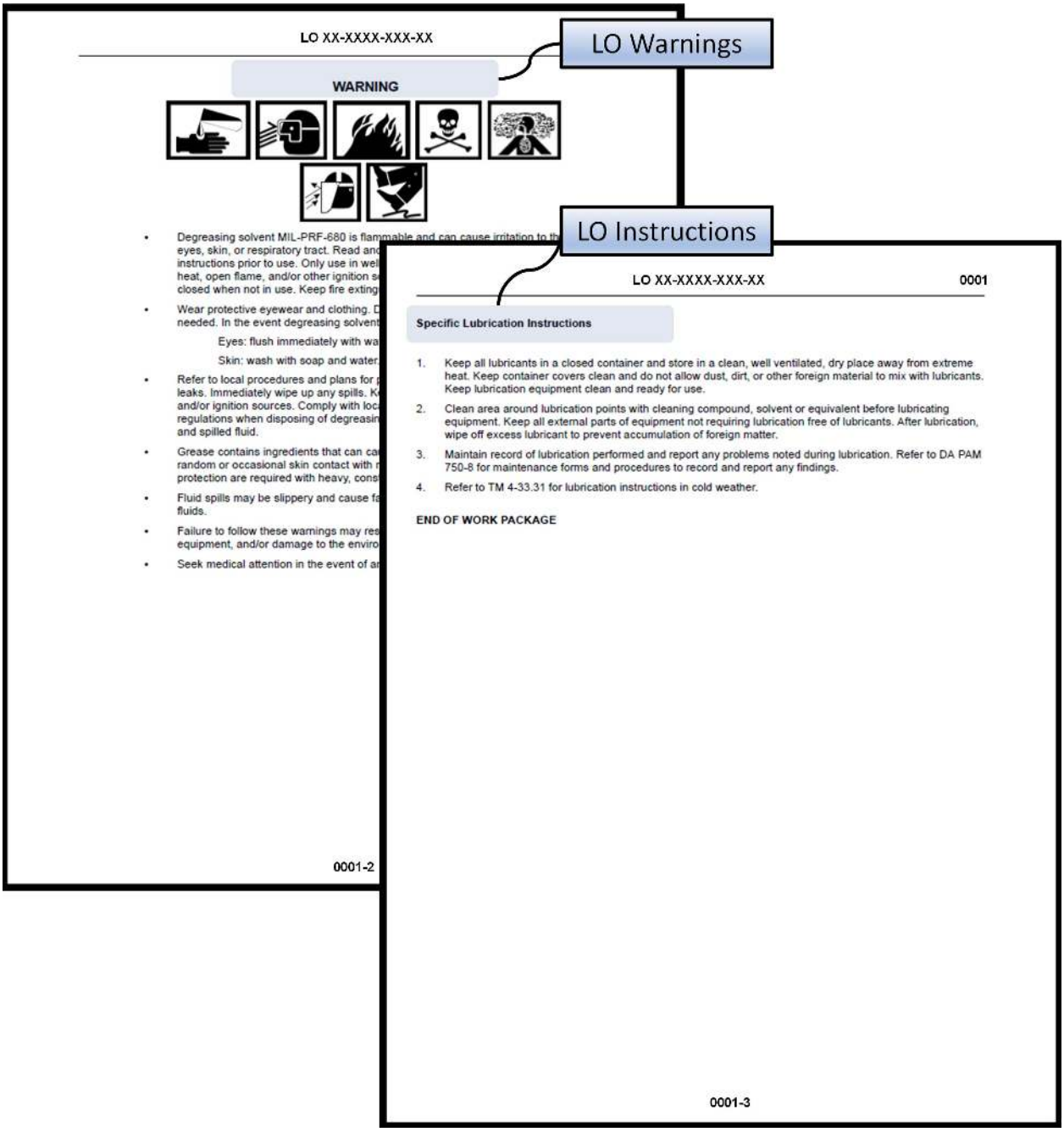


**FIGURE K-4. LO introduction WP (sheet 2 of 2).**



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### K.7.1.3 Army Oil Analysis Program (AOAP) statements.

One of the following statements shall be included for all equipment falling under the AOAP.

**a. Army Oil Analysis Program (AOAP) sampling interval statement.** A statement similar to the following shall be included: "Engine oil/transmission oil/hydraulic fluids must be sampled at *insert applicable hour/ mileage time frame*) as prescribed by *insert DA Pam 750-8 or DA Pam 738-751*."

**b. Army Oil Analysis Program (AOAP) not available/non-enrolled statement.** When a component/equipment is not enrolled in the AOAP, or oil analysis support is not available, a statement similar to the following shall be used: "This *enter name of component/equipment* is not enrolled in the Army Oil Analysis Program. HARDTIME MAINTENANCE INTERVALS APPLY."

### K.7.1.4 Warranty hardtime statement.

When applicable, the following statement shall be used:

"For equipment under manufacturer's warranty, hardtime oil service intervals shall be followed. Intervals shall be shortened if lubricants are known to be contaminated or if operation is under adverse conditions such as longer than usual operating hours, extended idling periods, extreme dust, etc."

## K.7.2 Lubrication instructions work package <lubewp>.

One or more lubrication instructions WPs shall be prepared (refer to APPENDIX E) and shall include the following:

**a.** Applications.

**b.** Instructions.

**c.** Authorized lubricants.

**d.** Intervals.

**e.** Man-hour requirements.

**f.** Lubrication points.

**g.** AOAP requirements.

Unless otherwise specified by the contracting activity, the lubrication instructions shall be presented in grouped sequence by interval to enable the user to receive, lubricate, and return to an acceptable performance standard all components of the equipment in a minimum amount of time with the skills, tools, test equipment, and spare parts authorized by the Logistics Product Data (LPD) or the Maintenance Allocation Chart (MAC). Unless otherwise specified by the contracting activity, lubrication instructions shall be based upon the principles of Reliability Centered Maintenance (RCM) logic. Any other maintenance procedures from the related maintenance technical publication that are required for the LO may be included in the LO only if the information is less than two pages. If the information is more than two pages, a reference to the maintenance technical publication shall be included in the LO. The lubrication instructions WP shall contain the following information:

**a.** WP identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** Initial setup **<initial_setup>** is required for this WP (5.4.3).

**c.** Lubrication instructions as prescribed in K.7.2.1 through K.7.2.3.


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### K.7.2.1 Grouped lubrication points.

When grouped lubrication points require the same lubricant at the same interval, the type and number of points shall be identified and described by one of the following methods:

**a.** Multi-headed, solid-shafted arrows shall point to each of the lubrication points (refer to FIGURE K-6).

**b.** Lubrication point notes shall provide instructions for applying lubricants, taking into account the following factors:

**(1)** Type, grade, availability, and properties of prescribed lubricant.

**(2)** Expected temperature.

**(3)** Lubrication gun and tools available to authorized maintenance level.

**(4)** Types of lubrication fittings.

**(5)** Possible ill effects of excessive or insufficient lubrication.

Caution shall be stressed where over or under lubrication of a part will damage that part or closely associated parts. Such cautionary notes shall either be included as a portion of the point note or as a special note (refer to K.7.5).


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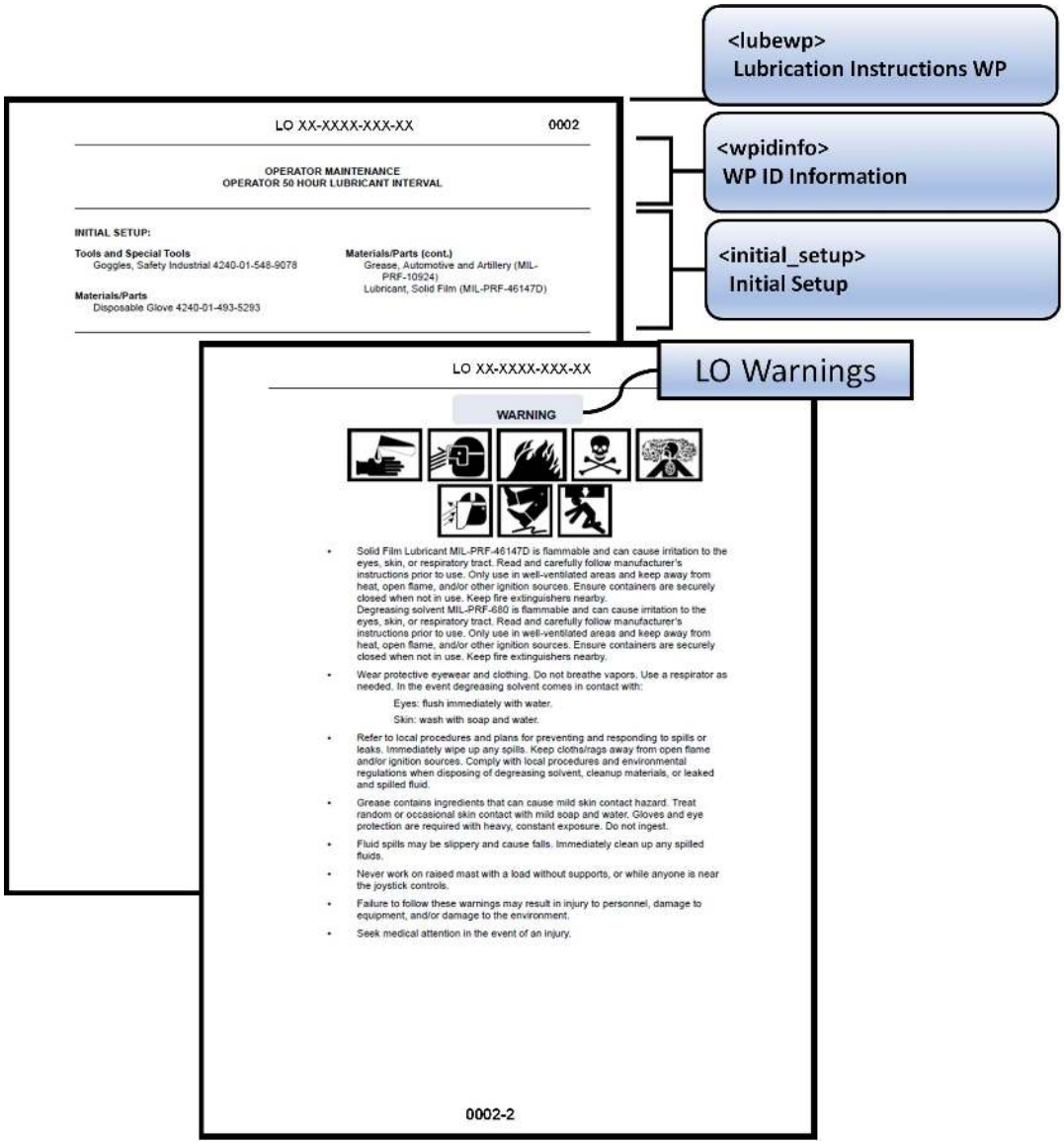


**FIGURE K-5. Lubrication instructions WP (sheet 1 of 2).**

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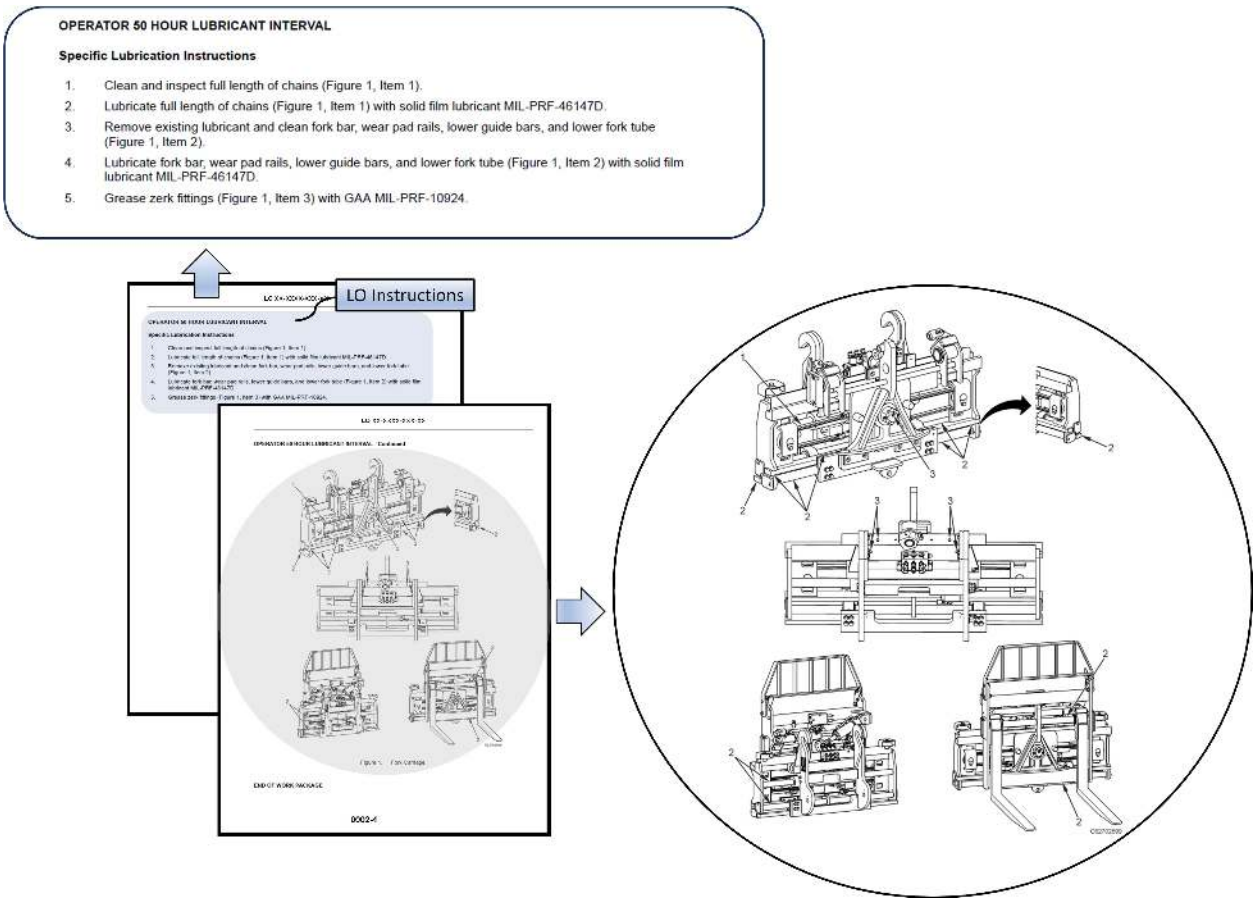


**FIGURE K-6. Lubrication instructions WP (sheet 2 of 2).**

### K.7.2.2 Washing and natural drying.

If applicable, instructions shall be given for washing and natural drying of finely machined and dirt-sensitive parts before re-lubricating. Use of compressed air jets or temperatures above 212 degrees Fahrenheit shall not be prescribed.

### K.7.2.3 Preservation information.

If preservation procedures are required for the LO, reference shall be made to the related maintenance technical publication containing preservation procedures. Any special preservation procedures needed for the LO that are not covered in the related maintenance technical publication may be included in the LO.

## K.7.3 LO references work package <refwp>.

An LO references WP shall be developed, as needed, in accordance with G.7.1 if the LO contains references to other publications.

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### K.7.4 LO tool identification work package <toolidwp>.

An LO tool identification WP may be developed, as needed, in accordance with G.7.8 if the LO utilizes tools in the narrative instructions.

### K.7.5 Lubricants and military symbols work package <lubricantsymbolswp>.

Unless otherwise specified by the MATDEV, lubricants shall be identified by standard military symbols, in accordance with MIL-HDBK-113 and MIL-HDBK-275. The lubricants and military symbols WP shall contain the following information:

**a.** WP identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** Initial setup **<initial_setup>** is not required for this WP.

**c.** Lubricant and military symbols information. The lubricant symbols and interval symbols shall be contained in a table. These columns shall be headed by the words "Lubricant" and "Interval." Those lubrication points that are serviced or lubricated by checking the level, replenishing the lubricant, or draining and refilling shall be indicated by the lubricant's symbol at the point on the illustration that is designated for replenishing or refilling. The amount of lubricant required shall be given either in the point note or in the "Capacity" column of the table, if applicable.

### K.7.6 Lubricant types work package <lubetypeswp>.

As application, this WP shall provide information needed to select the proper lubricant for various temperature ranges and uses. The lubricant types WP shall contain the following information:

**a.** WP identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** Initial setup **<initial_setup>** is not required for this WP.

**c.** Lubricant types information as prescribed in K.7.6.1 and K.7.6.2.

### K.7.6.1 Lubricant table.

As applicable, this WP shall contain a table or tables to provide information needed to select the proper lubricant for various temperature ranges and uses. The size and location of the table(s) shall be tailored to meet layout requirements and shall include as applicable, information on temperature range, lubricant, military symbol, NATO code, specification, NSN, capacity, interval between service, and man-hours required to complete all service by type stated to the nearest tenth for all lubricants prescribed by the LO (refer to FIGURE K-7).

### K.7.6.2 Notes to lubrication table.

As necessary, when specific restrictions, preferred grades, and other conditions exist, notes shall be annotated on tables in accordance with 4.15.8. For example: "When MIL-PRF-2104 lubricant is authorized, use 15W-40 (OE/HDO-15/40) when available and the applicable temperature range exists," or "15W-40 oil is not authorized in this particular *enter component name*)." Where applicable, the statement "For Arctic Operation, refer to TM 4-33.31" shall be included as a note (refer to FIGURE K-5).

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**Note**

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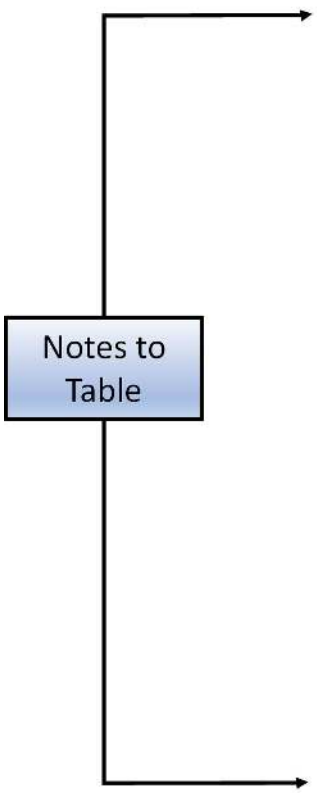


*Table 1. Lubrication Table.*

| System | Temperature Range | Lubricant Specification | Capacity | Interval | Man-hour |
|---|---|---|---|---|---|
| Engine Oil w/ Filters | 0°F to +120°F (-18°C to +49°C)<br><br>-70°F to +40°F (-57°C to +4°C) | Lubricating Oil, Engine 15W-40 (MIL-PRF-2104)<br><br>Lubricating Oil, Internal Combustion Engine, Synthetic Base SMPL (MIL-PRF-32626) * | 15.85 qt (15 L) | First 100 hours then 500 hours | 1 |
| * If SMPL lubricant is required to meet the low expected temperature range, SMPL lubricant is to be used in lieu of 15W-40 lubricant for all expected temperatures where 15W-40 is specified. | | | | | |
| Hydrostatic Drive Motor Oil | All Temperatures | Lubricating Oil, Gear 80W-90 (SAE J2360) | 3.7 qt (3.5 L) | 500 hours | 2 |
| Hydraulic Oil | All Temperatures | Lubricating Oil, Internal Combustion Engine, Synthetic Base SMPL (MIL-PRF-32626) | 13.4 gal. (51 L) | 2000 hours | 2 |
| Front Axle Differential Housing | -40°F to +120°F (-40°C to +49°C) | Lubricating Oil, Gear 80W-90 (SAE J2360) | 3.4 gal. (13 L) | 500 hours | 1 |
| Front Axle Wheel End | -40°F to +120°F (-40°C to +49°C) | Lubricating Oil, Gear 80W-90 (SAE J2360) | 0.5 gal. (2 L) | 500 hours | 1 |
| Rear Axle Differential Housing | -40°F to +120°F (-40°C to +49°C) | Lubricating Oil, Gear 80W-90 (SAE J2360) | 2.75 gal. (10.4 L) | 500 hours | 1 |
| Rear Axle Wheel End | -40°F to +120°F (-40°C to +49°C) | Lubricating Oil, Gear 80W-90 (SAE J2360) | 0.5 gal. (2 L) | 500 hours | 1 |
| Brake Fluid Reservoir | All Temperatures | Hydraulic Fluid, Fire Resistant (MIL-PRF-46170) | | 2000 hours | |
| Engine Radiator, Coolant Expansion Tank | -60°F to +90°F (-51°C to +32°C)<br><br>-90°F to -40°F (-68°C to -40°C) | Antifreeze, Ethylene Glycol, 50/50 Mix (A-A-52624 Type I C)<br><br>Antifreeze, Ethylene Glycol, 60/40 Mix (A-A52624 Type I B) | 5.8 gal. (22 L) | 2000 hours | |
| Grease Fittings | All Temperatures | Grease, Automotive and Artillery (MIL-PRF-10924) | | | |
| *For Arctic operation, refer to TM 4-33.31. | | | | | |

**FIGURE K-7. Notes to lubrication table.**

## K.7.7 Special notes work package <lospecnoteswp>.

As applicable, additional pertinent lubrication point information shall be incorporated into the LO. When applicable, the LO shall contain a special note referencing, but not repeating, instructions in technical publications. The special notes WP shall contain the following information:

**a.** WP identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** Initial setup **<initial_setup>** is not required for this WP.

**c.** Special notes information as prescribed in K.7.7.1 through K.7.7.2.



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### K.7.7.1 Pertinent lubrication point information.

As applicable, additional pertinent lubrication point information shall be incorporated into the LO. When applicable, the LO shall contain a special note referencing, but not repeating, instructions in technical publications.

### K.7.7.2 Effect of extreme conditions.

If applicable, pertinent instructions relevant to the effect of extreme conditions such as temperature, humidity, or altitude on lubrication requirements for the equipment shall be included as a special note.

## K.8 NOTES.

The notes in Section 6 apply to this appendix.

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# APPENDIX L

# AMMUNITION DEPOT MAINTENANCE WORK REQUIREMENTS

## L.1 SCOPE.

This appendix establishes the technical content requirements for the preparation of Depot Maintenance Work Requirements (DMWRs) for the demilitarization or maintenance of ammunition, referred to herein as ammunition DMWR. This appendix is a mandatory part of this standard. The information contained herein is intended for compliance.

**Note**

To be compliant with this standard, publications shall be developed within **<production>** (opening) and **</production>** (closing) root tags.

When preparing information in accordance with this appendix, all references shall be linked using the appropriate tags (**<xref>**, **<extref>**, etc.). Links shall not be typed as text without tagging.

### L.1.1 Content tailoring using Appendix A matrices.

Applicability of some requirements contained in this standard are the responsibility of the materiel developer (MATDEV) (refer to AR 750–1, AR 700–127) and shall be accomplished using matrices referenced in TABLE A-I. The applicability of requirements are also designated by statements including, but not limited to, the following:

**a.** "...as specified by the MATDEV."

**b.** "...as required by the MATDEV."

**c.** "Refer to the MATDEV for..."

**d.** "Contact MATDEV if..."

**e.** "The MATDEV shall determine..."

**f.** "...as needed."

#### L.1.1.1 Front and rear matter.

Front and rear matter requirements for documents discussed in this appendix shall be determined in accordance with TABLE A-II.

## L.2 APPLICABLE DOCUMENTS.

The applicable documents in Section 2 apply to this appendix.

## L.3 ACRONYMS AND DEFINITIONS.

The acronyms and terms defined in Section 3 of this standard apply to this appendix.

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## L.4 AMMUNITION DMWR FUNCTIONALITY.

An ammunition DMWR **<dmwr_ammo>** consists of a sequence of modules and work packages (WPs) that provide formation for the demilitarization or maintenance of ammunition. Data developed in accordance with this appendix shall be divided into modules and WPs. See TABLE L-I for a list of work packages for an ammunition DMWR. See FIGURE L-1 for a diagram of modules and WPs for an ammunition DMWR.

**TABLE L-I. Ammunition DMWR WPs.**

| WP | WP Tags | Appendix Section |
|---|---|---|
| General information WP [1] | **<ginfowp>** | L.6.1 |
| DMWR introduction WP | **<dmwr_introwp>** | L.6.2 |
| Maintenance requirements WP | **<dmwr_operationalreqwp>** | L.6.3 |
| Quality acceptance requirements WP | **<dmwr_qarwp>** | L.6.4 |
| References WP | **<refwp>** | L.6.5 |
| Expendable and durable items list WP | **<explistwp>** | L.6.6 |
| Tool identification list WP | **<toolidwp>** | L.6.7 |
| Ammunition Peculiar Equipment (APE) WP | **<genwp>** | L.6.8 |
| Special equipment, locally available materials, and special facilities WP | **<genwp>** | L.6.9 |
| Approved intraplant transfer equipment WP | **<genwp>** | L.6.10 |
| Chemically treated packing materials WP | **<genwp>** | L.6.11 |
| Environmental requirements WP | **<genwp>** | L.6.12 |
| Hazard analysis WP | **<genwp>** | L.6.13 |
| Hazardous Component Safety Data Statement (HCSDS) WP | **<genwp>** | L.6.14 |
| Other supporting information WP | **<genwp>** | L.6.15 |

[1] WP type duplication is prohibited.


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**Note**

Tags/output shown in figure(s) below may vary slightly based on version of DTD/stylesheets utilized (see applicable DTD or tag definition list for available tags).

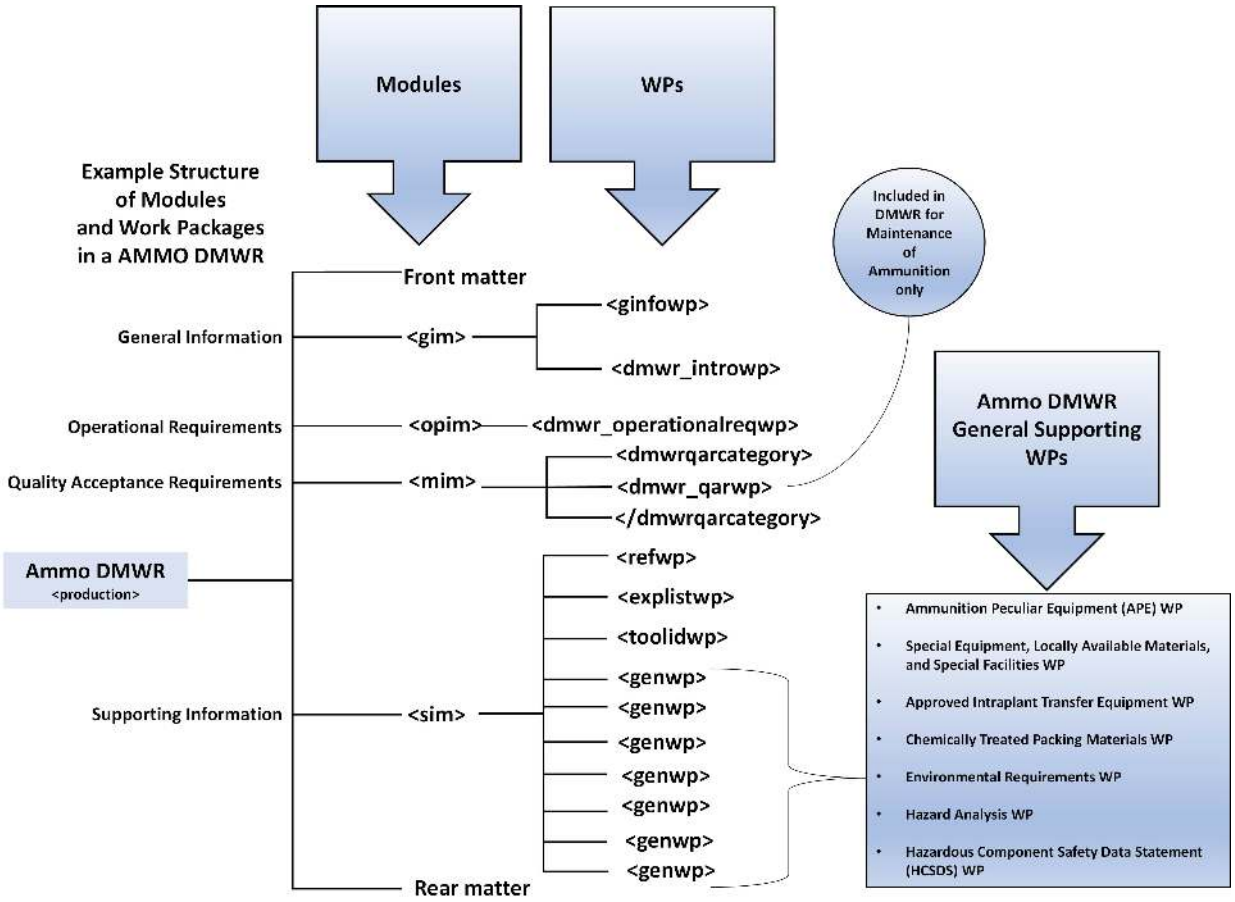


**FIGURE L-1. Ammunition DMWR modules and WPs.**

## L.5 DEVELOPMENT OF AMMUNITION DMWR MODULES.

When specified by the material developer (MATDEV), an ammunition DMWR **<dmwr_ammo>** shall be prepared (see TABLE A-IX). Sections L.5.1 through L.5.4 outline module requirements for this publication type. FIGURE L-1 is a graphical representation of WP organization and tags discussed in this section.

### L.5.1 General information and DMWR introduction module <gim>.

A general information module **<gim>** shall be provided. It shall consist of the following WPs.

**a.** General information WP **<ginfowp>** (L.6.1).

**b.** DMWR introduction WP **<dmwr_introwp>** (L.6.2).



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### L.5.2 Maintenance requirements module <mim>.

A maintenance requirements module **<mim>** shall be provided. It shall consist of the following WPs.

**a.** DMWR maintenance requirements WP(s) **<dmwr_operationalreqwp>** (L.6.3).

**b.** Additional maintenance requirements WP(s) as needed.

### L.5.3 Quality acceptance requirements module <mim> (maintenance of ammunition DMWR only).

A quality acceptance requirements module **<mim>** shall be provided in a maintenance of ammunition DMWR. It shall consist of the following WP between **<dmwrqarcategory.>** tags.

**a.** Quality acceptance requirements WP **<dmwr_qarwp>** (L.6.4).

### L.5.4 DMWR supporting information module <sim>.

A DMWR supporting information module **<sim>** shall be provided. It shall consist of the following WP(s) as applicable.

**a.** References WP **<refwp>** (L.6.5).

**b.** Expendable and durable items list WP **<explistwp>** (L.6.6).

**c.** Tool identification list WP **<toolidwp>** (L.6.7).

**d.** Ammunition Peculiar Equipment (APE) WP **<genwp>** (L.6.8).

**e.** Special equipment, locally available materials, and special facilities WP **<genwp>** (L.6.9).

**f.** Approved intraplant transfer equipment WP **<genwp>** (L.6.10).

**g.** Chemically treated packing materials WP **<genwp>** (L.6.11).

**h.** Environmental requirements WP **<genwp>** (L.6.12).

**i.** Hazard analysis WP **<genwp>** (L.6.13).

**j.** Hazardous Component Safety Data Statement (HCSDS) WP **<genwp>** (L.6.14).

**k.** Other supporting information WP **<genwp>** (L.6.15).

## L.6 DEVELOPMENT OF AMMUNITION DMWR WORK PACKAGES.

### L.6.1 General information work package <ginfowp>.

A general information WP shall be prepared in accordance with B.7.1. and shall contain the following information:

**a.** Work package (WP) identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** Initial setup **<initial_setup>** is not required for this WP.

**c.** Scope **<scope>** (B.7.1.1).

**d.** Maintenance forms, records, and reports **<mfrr>** (B.7.1.2).

**e.** Corrosion prevention and controls (CPC) **<cpcdata>** (B.7.1.5) (maintenance of ammunition DMWRs only).

**f.** List of abbreviations / acronyms **<loa>** (B.7.1.11).

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## L.6.2 DMWR introduction work package <dmwr_introwp>.

A DMWR introduction WP shall be prepared and contain the following information:

**a.** Work package identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** Initial setup **<initial_setup>** is not required for this WP.

**c.** Work planning **<work_planning>**. Accumulation of excess ammunition items, removal of line rejects or explosive waste/hazardous waste, and removal of items containing precious metals shall be addressed.

**d.** Disposition **<disposition>**. Disposition guidelines for serviceable and unserviceable components and materials shall be included as a part of each procedure and shall address removal of hazardous materials or components and inspection of salvaged materials prior to transfer to Locally Approved Disposition Services (LADS). Reference may be made to publications for information on packing, marking, and shipping generated assemblies, components, and materials.

**e.** Equipment **<equipment>**. The equipment information provided shall contain, but not be limited to, the following paragraphs:

"Equipment cited herein for the various procedures has been approved for the procedures specified. Activities intending to use other equipment for these procedures must obtain approval from the proponent agency of this DMWR by filing a deviation, waiver, or exception in accordance with AR 700-20.

Transfer and materials handling equipment must conform to requirements set forth in AR 385-10. The Approved Intraplant Transfer Equipment Work Package lists preferred approved Ammunition Peculiar Equipment (APE) for moving and handling ammunition and components.

Use of APE or nonstandard APE is governed by AR 700-20. All modifications to existing APE and locally fabricated nonstandard APE must have prior approval in accordance with AR 700-20. Locally designed and fabricated equipment, other than APE or nonstandard APE, must be approved in accordance with AR 700-20."

**f.** Safety requirements **<sfty_req>**. The safety requirements information provided shall contain, but not be limited to, the following paragraph:

"Guidance for safety requirements as prescribed by current safety directives and regulations must be addressed."

**g.** Protection against general hazards **<gen_hazards>**.

Guidance for general hazards shall be addressed for the ammunition and materials requiring protection against the general hazards. Additionally, requirements for handling of ammunition, requirements for wearing of suitable protective clothing, and precautions when handling chemically treated packing materials and pallets shall be included. Reference shall be made to Chemically Treated Packing Materials Work Package for additional data on personal hygiene requirements, working with chemically treated wood, and the disposition of contaminated clothing.

**h.** Protection against specific hazards **<spec_hazards>** ). Specific hazards shall be listed in each applicable procedure for the ammunition and materials requiring protection against the specific hazards.

**i.** Hazard analysis **<haz_analysis>**. As a minimum, the hazard analysis information provided shall contain the following statement and shall reference the Hazard Analysis Work Package.

"A hazard analysis identifies potential hazards associated with these procedures. It provides countermeasures to mitigate these hazards and assesses the probability and effect of occurrence."

**j.** Environmental regulation compliance **<erc>**. Environmental regulations implemented by Federal, State, and local governments shall be addressed (refer to L.6.12).

**k.** Resource conservation and recovery regulations **<rcrr>**. Pertinent resource conservation and recovery regulations, as contained in the Resource Conservation and Recovery Act, 42 U.S.C. 6901 et seq., shall be addressed.

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**l.** Resource recovery **<resource_recovery>**. Resource recovery shall contain a paragraph similar to the following:

"All items of salvageable value will be salvaged as scrap or reusable material. All propellent, explosives, and hazardous materials that can be successfully recovered and reused must be recovered; otherwise, the materials must be disposed of by an environmentally safe and State approved method."

**m.** Reporting requirements **<reporting_req>**. Guidance for reporting work accomplishments shall be addressed.

**n.** Tabulated data **<tabdata>**. Reference shall be made to the applicable Army Ammunition Data Sheets technical publication.

**o.** Flowchart **<flowchart>**. A flowchart providing an overview of all procedures may be included but is not mandatory.

### L.6.3 Maintenance requirements work package <dmwr_operationalreqwp>.

Maintenance requirements WPs shall be prepared and shall include information and procedures necessary to demilitarize ammunition or perform maintenance on ammunition. This WP type may be repeated for each procedure as necessary to meet all the maintenance requirements. FIGURE L-2 and FIGURE L-3 show an example of a maintenance requirements WP. Components of the maintenance requirements WP are listed below.

**a.** Work package identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** Initial setup **<initial_setup>** is required for this WP (5.4.3).

**c.** Safety requirements **<special_sfty>**. Safety requirements for equipment, material, facilities, environmental conditions, etc. shall be prepared.

**d.** Procedural steps **<op_steps>**. Procedural steps, which are to include warnings, cautions, and notes, shall be prepared.

**e.** Flowchart **<flowchart>**. A flowchart providing an overview of all procedural steps may be included but is not mandatory.

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**Note**

Figure appearance and tags/output may vary slightly based on version of DTD/stylesheets utilized (see applicable DTD or tag definition list for available tags).

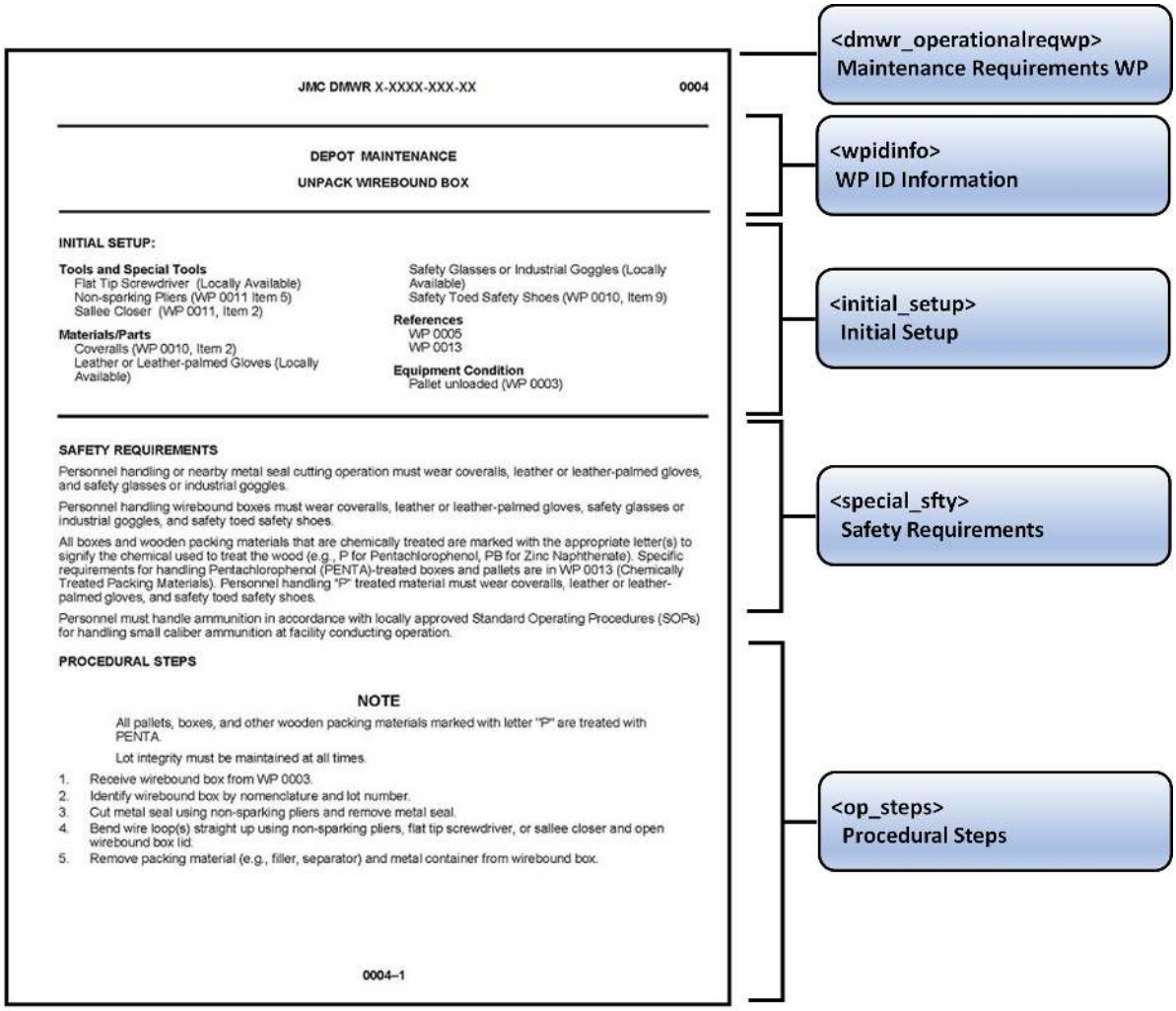


**FIGURE L-2. Maintenance requirements WP example (sheet 1 of 2).**

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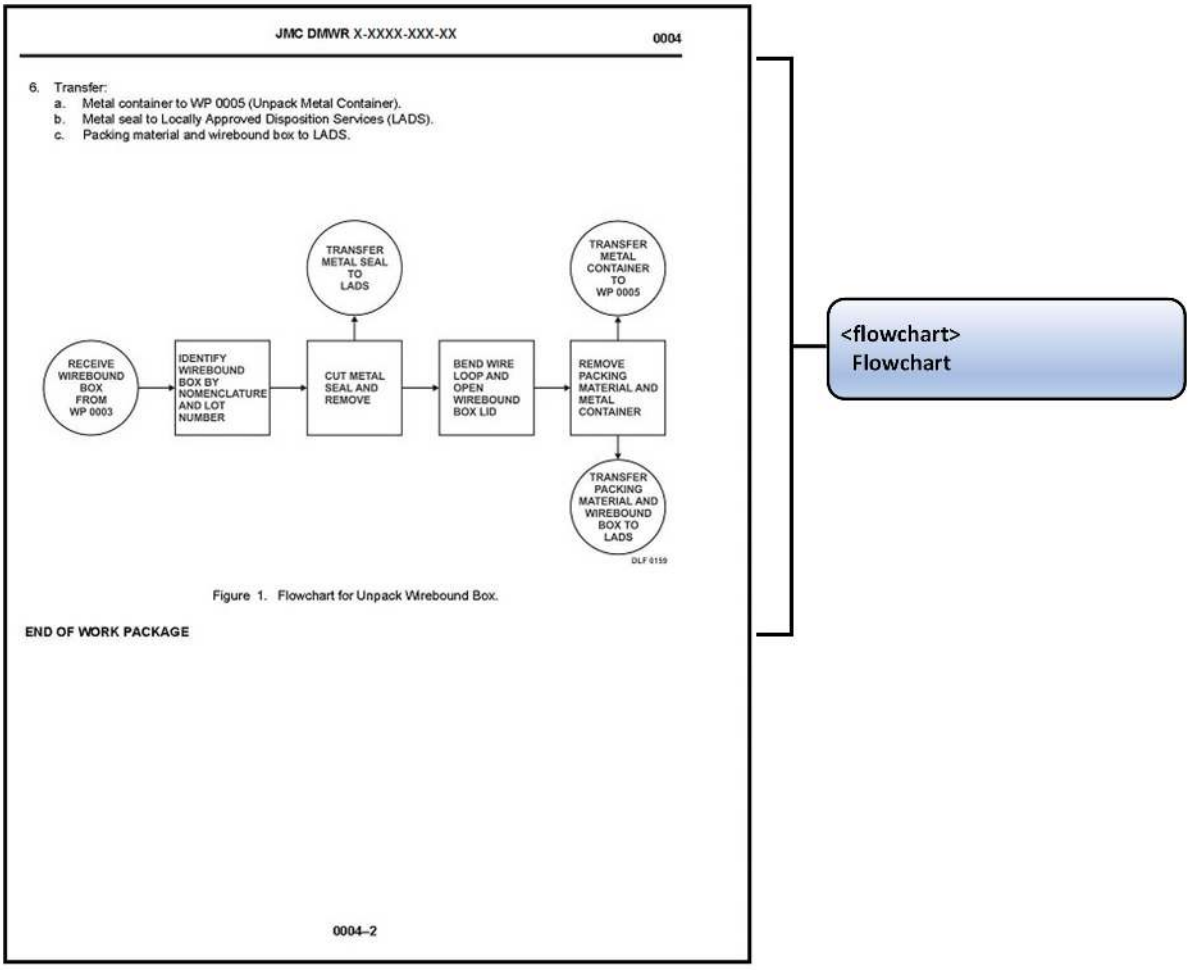


**FIGURE L-3. Maintenance requirements WP example (sheet 2 of 2).**

## L.6.4 Quality acceptance requirements work package <dmwr_qarwp> (maintenance of ammunition DMWRs only).

A quality acceptance requirements WP, if included in a DMWR, is integrated into the second maintenance module **<mim>**, as shown in FIGURE L-1.

**Note**

An ammunition DMWR for demilitarization of ammunition shall **not** contain a quality acceptance requirements module.

Components of the quality acceptance requirements work package are listed below.

**a.** Work package identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** Initial setup **<initial_setup>** is not required for this WP.

**c.** Ammunition maintenance **<maintenance_qar>**. The quality acceptance requirements for ammunition subject to maintenance shall address inspection methods and procedures, workmanship, lot formation, ammunition data card requirements, testing (for example, ballistic, pre-renovation), and/or product defect criteria, including defect classification. Incorporation of appropriate Statistical Process Control (SPC) statements for performing activities will be included as required.

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### L.6.5 References work package <refwp>.

This WP shall be prepared in accordance with G.7.1.

### L.6.6 Expendable and durable items list work package <explistwp>.

This WP shall be prepared in accordance with G.7.7.

### L.6.7 Tool identification list work package <toolidwp>.

This WP shall be prepared in accordance with G.7.8.

### L.6.8 Ammunition Peculiar Equipment (APE work package <genwp>.

An APE WP shall be prepared in accordance with G.7.12. This WP shall consist of a list of APE required to perform the procedures described in the DMWR.

### L.6.9 Special equipment, locally available materials, and special facilities work package <genwp>.

A special equipment, locally available materials, and special facilities WP shall be prepared in accordance with G.7.12. This WP shall consist of a list of special equipment, locally available materials, and special facilities required to perform the procedures described in the DMWR.

### L.6.10 Approved intraplant transfer equipment work package <genwp>.

An approved intraplant transfer equipment WP shall be prepared in accordance with G.7.12. This WP shall list suggested or commonly available intraplant transfer equipment.

### L.6.11 Chemically treated packing materials work package <genwp>.

A chemically treated packing materials WP shall be prepared in accordance with G.7.12. This WP shall be used to include the latest chemically treated packing materials requirements.

### L.6.12 Environmental requirements work package <genwp>.

An environmental requirements WP shall be prepared in accordance with G.7.12. This WP shall be used to include the latest environment requirements. At a minimum, this WP shall include air, noise, emission problems and controls, as applicable.

### L.6.13 Hazard analysis work package <genwp>.

A hazard analysis WP shall be prepared in accordance with G.7.12. This WP shall contain a hazard analysis updated to include the latest requirements. Potential hazards that may result in injury or death shall be identified. Appropriate countermeasures shall be provided.

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### L.6.14 Hazardous component safety data statement (HCSDS) work package <genwp>.

A HCSDC WP shall be prepared in accordance with G.7.12. This WP shall list all HCSDS documents and appropriate material, component, or assembly for the end item.

### L.6.15 Other supporting information work package <genwp>.

When specified by the MATDEV, other supporting information WPs may be included in an ammunition DMWR. This WP shall be prepared in accordance with G.7.12.

## L.7 NOTES.

The notes in Section 6 apply to this appendix.

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# APPENDIX M

# SOFTWARE USERS MANUALS AND SOFTWARE ADMINISTRATORS MANUALS

## M.1 SCOPE.

This appendix establishes the technical content requirements for equipment systems with software components requiring specific and detailed user instruction. Software refers to the computer programming provided as part of an equipment system. Software does not refer to the viewing software for an interactive electronic technical manual (IETM) or the viewing hardware/workstation used to view/use an equipment system. The requirements provided in this appendix provide the technical content requirements for the preparation of Software Users Manuals (SUMs) and Software Administrators Manuals (SAMs). This appendix is a mandatory part of this standard. The information herein is intended for compliance.

**Note**

To be compliant with this standard, publications shall be developed within **<production>** (opening) and **</production>** (closing) root tags.

When preparing information in accordance with this appendix, all references shall be linked using the appropriate tags (**<xref>, <extref>**, etc.). Links shall not be typed as text without tagging.

Also, the terms "software" and "application(s)" may be used synonymously.

### M.1.1 Content tailoring using Appendix A matrices.

Applicability of some requirements contained in this standard are the responsibility of the materiel developer (MATDEV) (refer to AR 750–1, AR 700–127) and shall be accomplished using matrices referenced in TABLE A-I. The applicability of requirements are also designated by statements including, but not limited to, the following:

**a.** "...as specified by the MATDEV."

**b.** "...as required by the MATDEV."

**c.** "Refer to the MATDEV for..."

**d.** "Contact MATDEV if..."

**e.** "The MATDEV shall determine..."

**f.** "...as needed."

#### M.1.1.1 Front and rear matter.

Front and rear matter requirements for documents discussed in this appendix shall be determined in accordance with TABLE A-II.

## M.2 APPLICABLE DOCUMENTS.

The applicable documents in Section 2 apply to this appendix.

## M.3 ACRONYMS AND DEFINITIONS.

The acronyms in Section 3 of this standard apply to this appendix. The definitions listed in TABLE G-I , in addition to the definitions in Section 3 of this standard, apply to this appendix.

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**TABLE M-I. Definitions.**

| Term | Definition |
|---|---|
| Software | Software refers to the computer programming provided as part of a weapon system. In this appendix, software does not refer to the viewing software for an IETM. It does not include the viewing hardware/workstation used to view/ use the weapon system software. |
| Software Administrator | A person who has administrative rights and can make changes to software configurations and/or changes to code. |
| Software Users Manual (SUM) | A SUM is a manual that contains information and procedures for the user of the software. The user usually does not have administrator rights and cannot make changes to the software code. |
| Software Administrators Manual (SAM) | A SAM provides information and procedures to the individuals who have the responsibilities and administrator rights to make changes to the software code, do repairs, updates, patches, etc. |
| Supervisor | User of software that has special privileges above and beyond a regular user but is not an administrator. |

## M.4 SELECTIVE APPLICATION AND TAILORING OF CONTENT.

This standard contains some requirements that may not be applicable to the preparation of all technical manuals. Selective application and tailoring of requirements contained in this standard are the responsibility of the MATDEV and shall be accomplished using matrices referenced in TABLE A-I matrixes. The applicability of some requirements is also designated by one of the following statements: unless specified otherwise by the MATDEV; as specified by the MATDEV; or when specified by the MATDEV.

## M.5 SOFTWARE INFORMATION INTEGRATION METHODOLOGY.

Software information data shall be developed in accordance with this appendix. Software information shall be divided into WPs (WPs) that stand alone and are broken into the WP types listed below. Prior to starting software information development, refer to the sections listed below in TABLE M-II for WP requirements.

**TABLE M-II. Software information WPs.**

| WP | WP Tag | Appendix Section |
|---|---|---|
| Software general information WP | **<softginfowp>** | M.7.1 |
| Software summary WP | **<softsumwp>** | M.7.2 |
| Software effectivity WP | **<softeffectwp>** | M.7.3 |
| Differences between versions WP | **<softdiffversionwp>** | M.7.4 |
| Features and capabilities WP | **<softfeaturescapwp>** | M.7.5 |
| Functional display WP | **<functionaldisplaywp>** | M.7.6 |
| Security and privacy procedures WP | **<softsecprivwp>** | M.7.7.1 |
| Supervisory controls WP | **<softsuperctrlswp>** | M.7.7.2 |
| Power up and access software procedures WP | **<softpwraccesswp>** | M.7.7.3 |
| Software operator procedures WP(s) [1] | **<softoperwp >** | M.7.7.4 |

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**TABLE M-II. Software information WPs. (continued**

| WP | WP Tag | Appendix Section |
|---|---|---|
| Power down and exit software procedures WP | **<softpwrexitwp>** | M.7.7.5 |
| Troubleshooting introduction WP | **<tsintrowp>** | M.7.8.1 |
| Troubleshooting index WP (Required for SAM only) | **<tsindxwp>** | M.7.8.2 |
| Messages WP(s) (Required for SAM only) | **<softmessageswp>** | M.7.8.3 |
| Recovery from errors, malfunctions, and emergencies WP | **<softerrorswp>** | M.7.8.4 |
| Troubleshooting WP [1] | **<tswp>** | M.7.8.5 |
| Software maintenance WP(s) [1] | **<maintwp>** | M.7.9 |
| Software PMCS WPs [1] | **<pmcswp>** | M.7.9.3 |
| Software destruction WP(s) | **<destruct-materialwp>** | M.7.10 |
| Reference WP | **<refwp>** | M.7.11.1 |
| Maintenance allocation chart introduction WP | **<macintrowp>** | M.7.11.2 |
| Maintenance allocation chart WP | **<macwp>** | M.7.11.3 |
| Software basic issue items (BII) WP | **<softbiiwp>** | M.7.11.4 |
| Additional authorization list WP | **<aalwp>** | M.7.11.5 |
| Expendable and durable items list WP | **<explistwp>** | M.7.11.6 |
| Additional supporting information WP | **<genwp>** | M.7.11.7 |

[1] WP type frequently duplicated in a module for effective information display.

## M.5.1 Hardware related procedures.

Hardware related procedures in SUM/SAM manuals shall be restricted to powerup/startup procedures required to get to the software and to explanation/use of hardware buttons/controls that affect the software operation. Reference shall be made to the hardware manual for any other hardware related procedures. If such a manual does not exist or is not authorized to the user, contact your Equipment Publications Control Office (EPCO) office to discuss how to handle documenting required hardware-related information.

## M.5.2 Updates to SUMs/SAMs.

As specified by the MATDEV, a change or revision shall be used to update the SUM or SAM. Generally, changes update information related to the same version of the software and a revision is used for a new software version. If required, the information for the earlier version(s) may be retained within the revised SUM or SAM. If information for older versions is retained in a revision, the software effectivity WP and differences between versions WP shall be prepared.


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## M.5.3 Bolding in software information WPs.

All software information WPs shall bold the following items in procedural steps: menu options, check boxes, radio buttons, text boxes, icons, drop-down list names, etc. For these items, bolding shall be used in lieu of quotation marks and italics. This practice is to ensure these items stand out among other text. FIGURE M-1 shows examples of text that shall be bolded.

**Note**

Figure appearance may vary slightly based on version of DTD/stylesheets utilized.

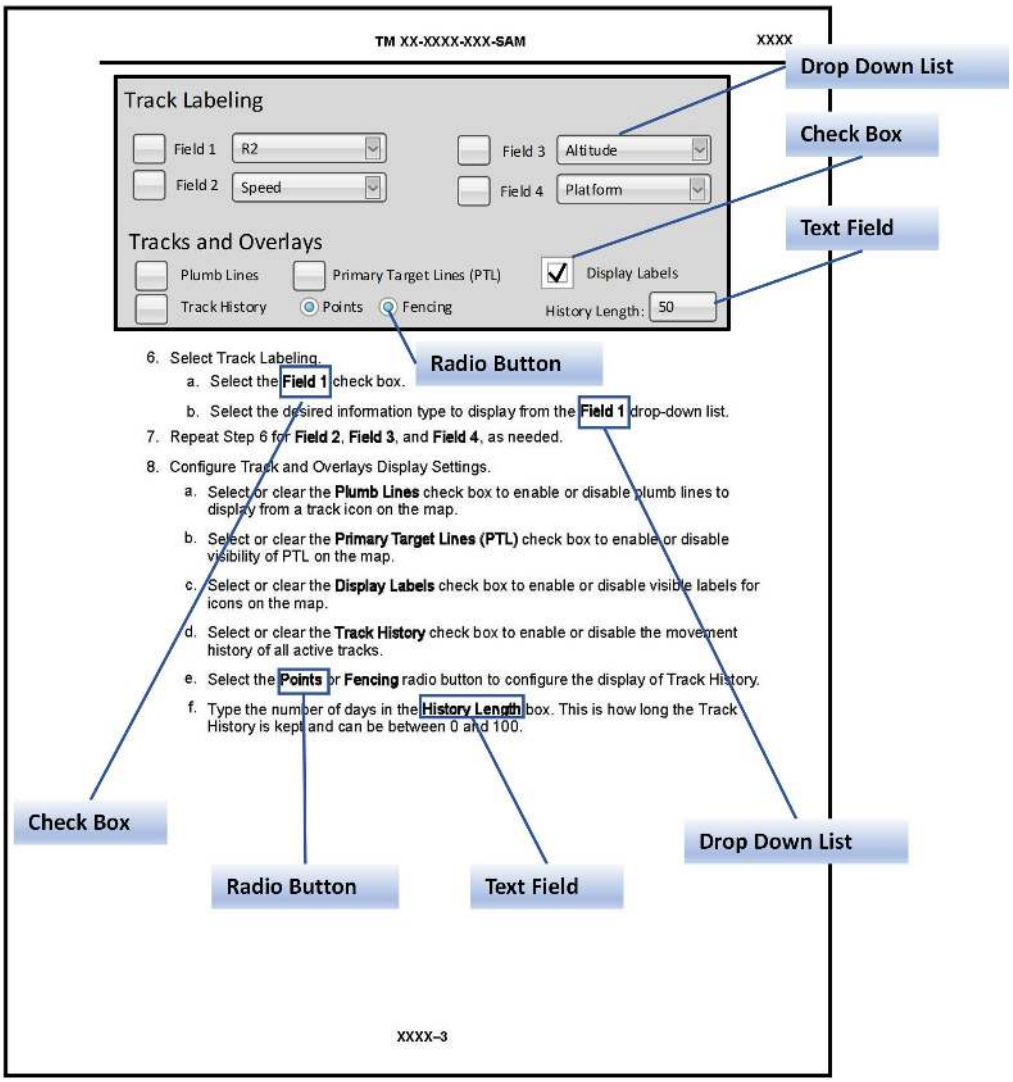


**FIGURE M-1. SUM/SAM included in an equipment system technical publication.**

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### M.5.4 Use of graphics in software WPs.

Graphics in software WPs shall be used to show various screen displays to enhance users' understanding of concepts and procedures. Graphics shall not be overused in software operation, troubleshooting, and maintenance WPs. Overuse results in excessive WP length and informational redundancy. For example, multiple graphics showing the same screen with different buttons, toggles etc. called out while performing a software operation shall not be used; callouts shall all appear in one graphic provided all elements of the screen are still easily viewable. Software description WPs **<softginfowp>**, **<softsumwp>**, **<softeffectwp>**, etc.) may require greater use of similar graphics; however, attempts shall still be made to minimize graphical overuse. Contact the EPCO if unsure about graphic use.

### M.5.5 Software information development format options.

There are three format options available for the development of software information (listed below). Frequency of changes to software versus hardware, volume of software information, and degree of integration of software and hardware, etc. should be considered when deciding whether to combine software information within the hardware operator and/or maintenance manuals. If a technical publication is a software-only publication, options 2 or 3 below are the only available choices. The option chosen for an equipment system shall be specified by the MATDEV. Any procedures requiring administrative privileges to perform shall be put in a SAM and shall not be put in a SUM.

1. Software information may be included as a section in an equipment system technical publication (M.5.5.1)
2. SUMs and SAMs may be developed as separate manuals for each system (M.5.5.2)
3. SUMs and SAMs may be combined as one manual (M.5.5.3)

### M.5.5.1 Software information included in an equipment system technical publication.

When specified by the MATDEV, various WPs that appear in a stand-alone SUM **<sum>** and/or SAM **<sam>** shall be included in an equipment system technical publication module **<soim>**. Content shall be limited to information and procedures related to the software; however, this content can be referenced from (and reference out to) other WPs in the equipment system publication. FIGURE M-2 shows an equipment system technical publication that includes a SUM/SAM as a module. See detailed requirements in section M.6.1.


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**Note**

Tags/output shown in figure(s) below may vary slightly based on version of DTD/stylesheets utilized (see applicable DTD or tag definition list for available tags).

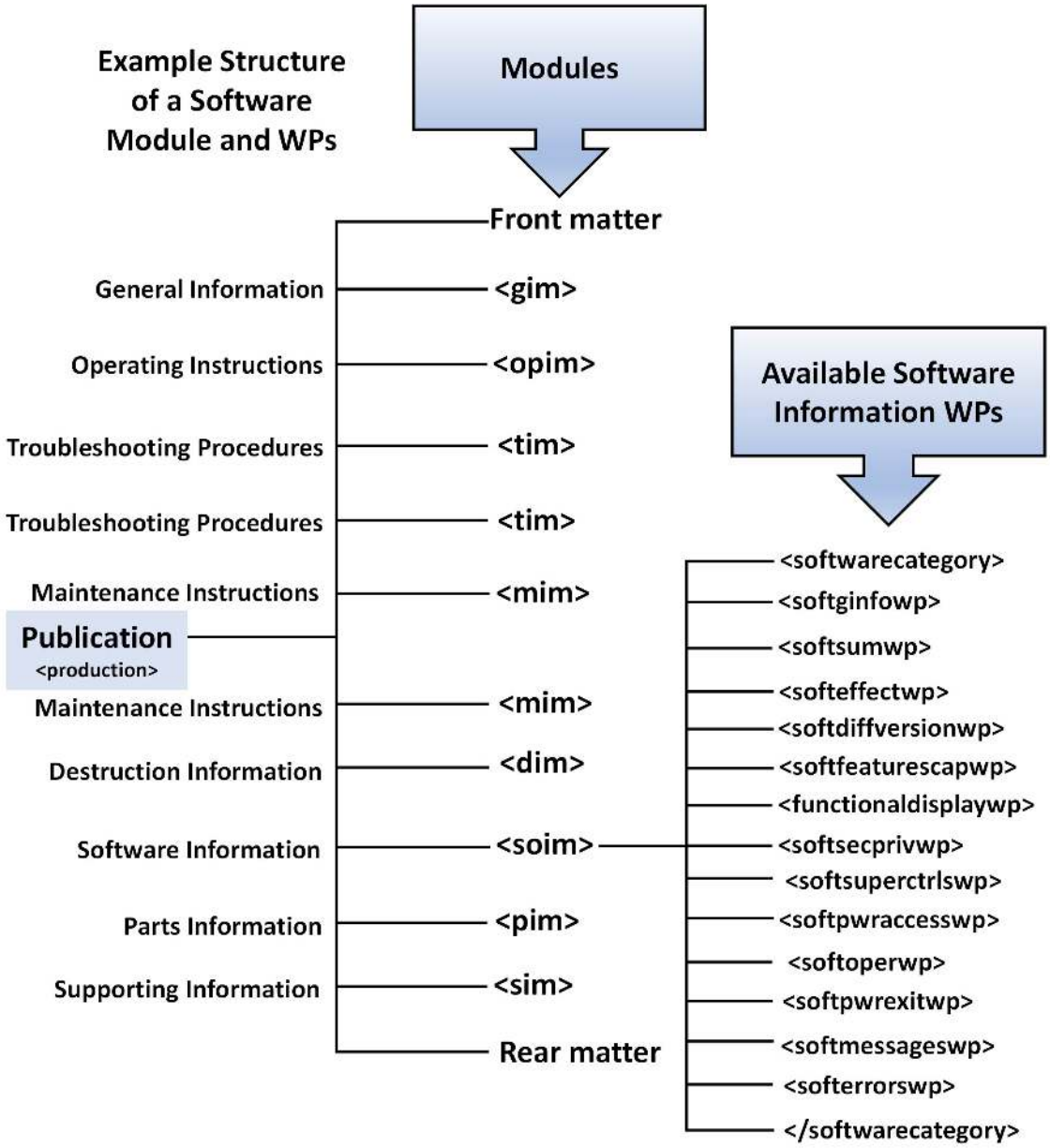


**FIGURE M-2. SUM/SAM modules and WPs.**

### M.5.5.2 SUMs and SAMs developed as separate manuals.

When specified by the MATDEV, a SUM **<sum>** and/or SAM **<sam>** shall be prepared. Content shall be limited to information/procedures related to the software. In accordance with DA Pam 25-40, individual SUMs and SAMs shall have a -SUM or -SAM suffix. The differentiation between an administrator manual and a combined user and administrator manual shall be made in the publication title. FIGURE M-3 shows components of SUM/SAM manuals



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as stand-alone documents. FIGURE M-4 shows WP identification numbering for SUM/SAM manuals as stand-alone documents. See detailed requirements in section M.6.2.

**Note**

Tags/output shown in figure(s) below may vary slightly based on version of DTD/stylesheets utilized (see applicable DTD or tag definition list for available tags).

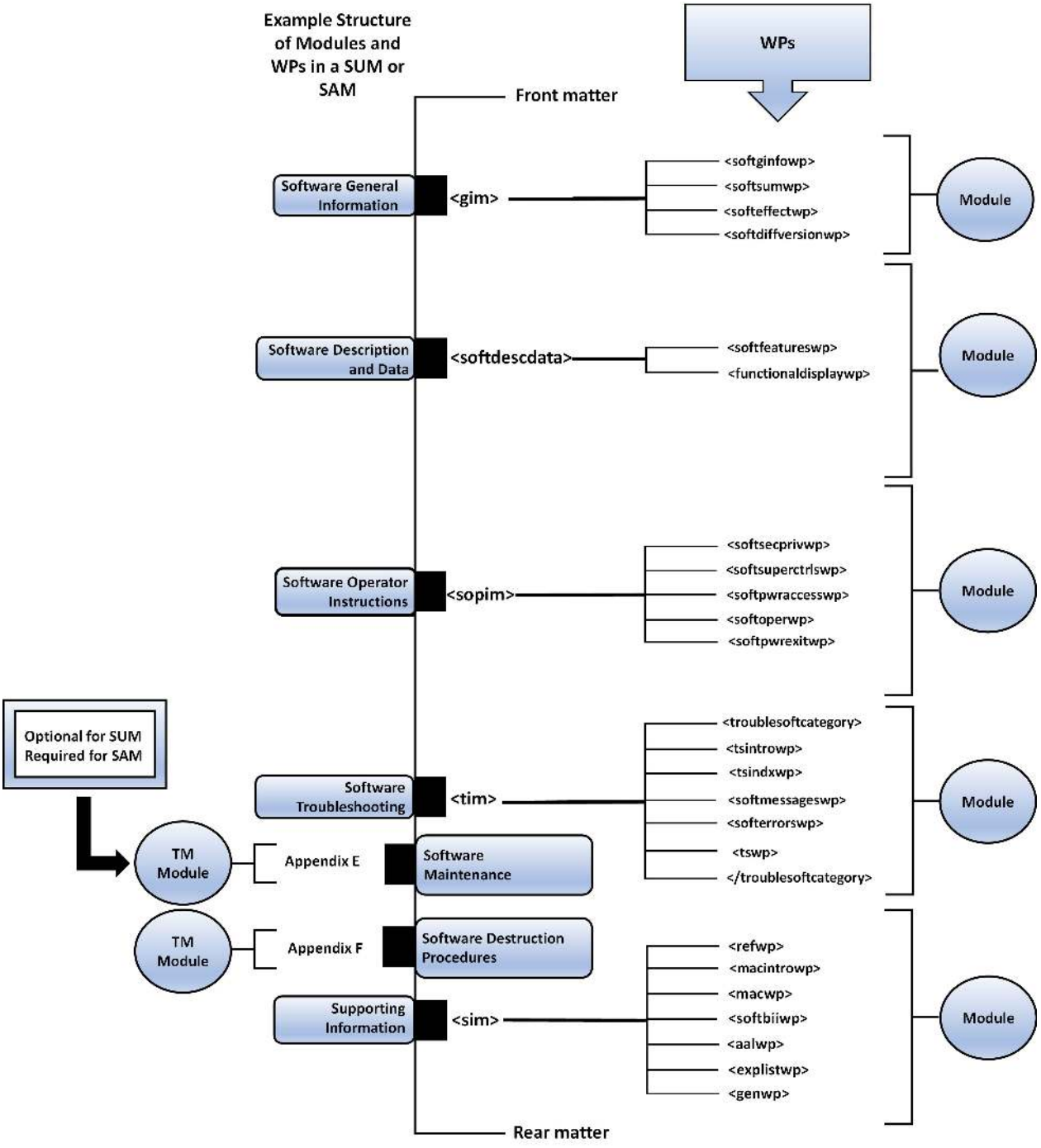


**FIGURE M-3. SUM/SAM stand-alone technical publication.**

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### M.5.5.3 SUMs and SAMs combined in one manual.

When specified by the MATDEV, a combined SUM **<sum>** and SAM **<sam>** manual shall be prepared. A combined SUM and SAM manual shall have a -SAM suffix. The differentiation between an administrator manual and a combined user and administrator manual shall be made in the publication title. See detailed requirements in section M.6.2.

### M.5.5.4 Configuration tagging.

For configuration-dependent software, the intended display format (printed vs. electronic) may influence desired planning and development of software technical publications (as discussed in M.5.5) as well as functional display WPs (M.7.6) within software publications. Contact the EPCO as needed for details concerning intended display format for a system. For printed technical publications, it is likely preferable to develop separate publications for each configuration. This is due to confusion that may result from significant configuration dependent text-and graphics that cannot be hidden (as can be done with electronic TMs). For electronic technical publications, this confusion is less of a concern, and various configurations may be combined into one publication. When one software configuration is needed, it is easily clickable and displayed while other configuration text is not on the screen. FIGURE M-4 displays one example of configuration tagging at the WP level. Refer to M.7.4 for more information about differences between versions of software.

**Note**

Tags/output shown in figure(s) below may vary slightly based on version of DTD/stylesheets utilized (see applicable DTD or tag definition list for available tags).

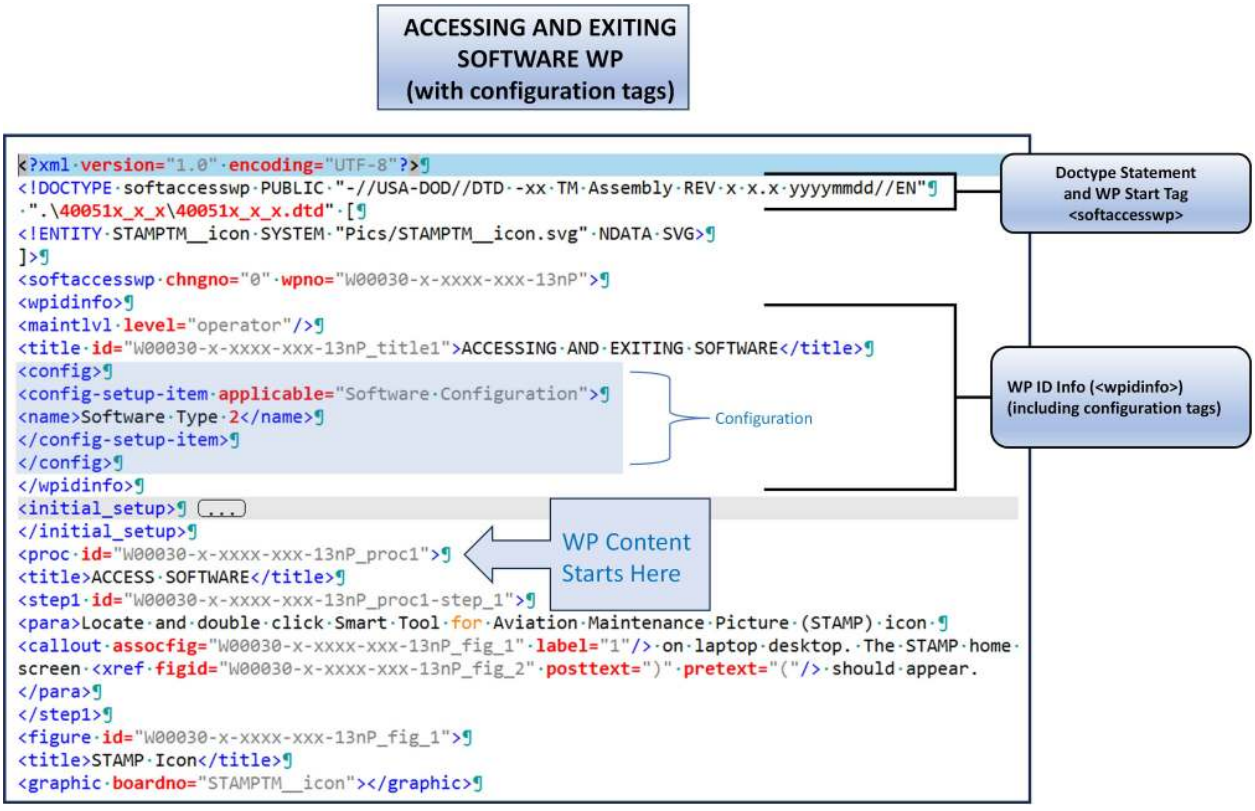


**FIGURE M-4. Configuration tagging example (WP level).**

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### M.5.5.5 WP identification numbering and alpha designators for stand-alone and combined SUMs and SAMs.

WP identification numbering for stand-alone and combined SUMs and SAMs shall follow guidance described in General Requirements (1). WP identification numbers shall also contain a dash and then a -SUM or -SAM suffix. A combined SUM and SAM manual shall have a -SAM suffix. FIGURE M-5 shows a sample publication with example WP identification numbering for each module.

**Note**

Tags/output shown in figure(s) below may vary slightly based on version of DTD/stylesheets utilized (see applicable DTD or tag definition list for available tags).

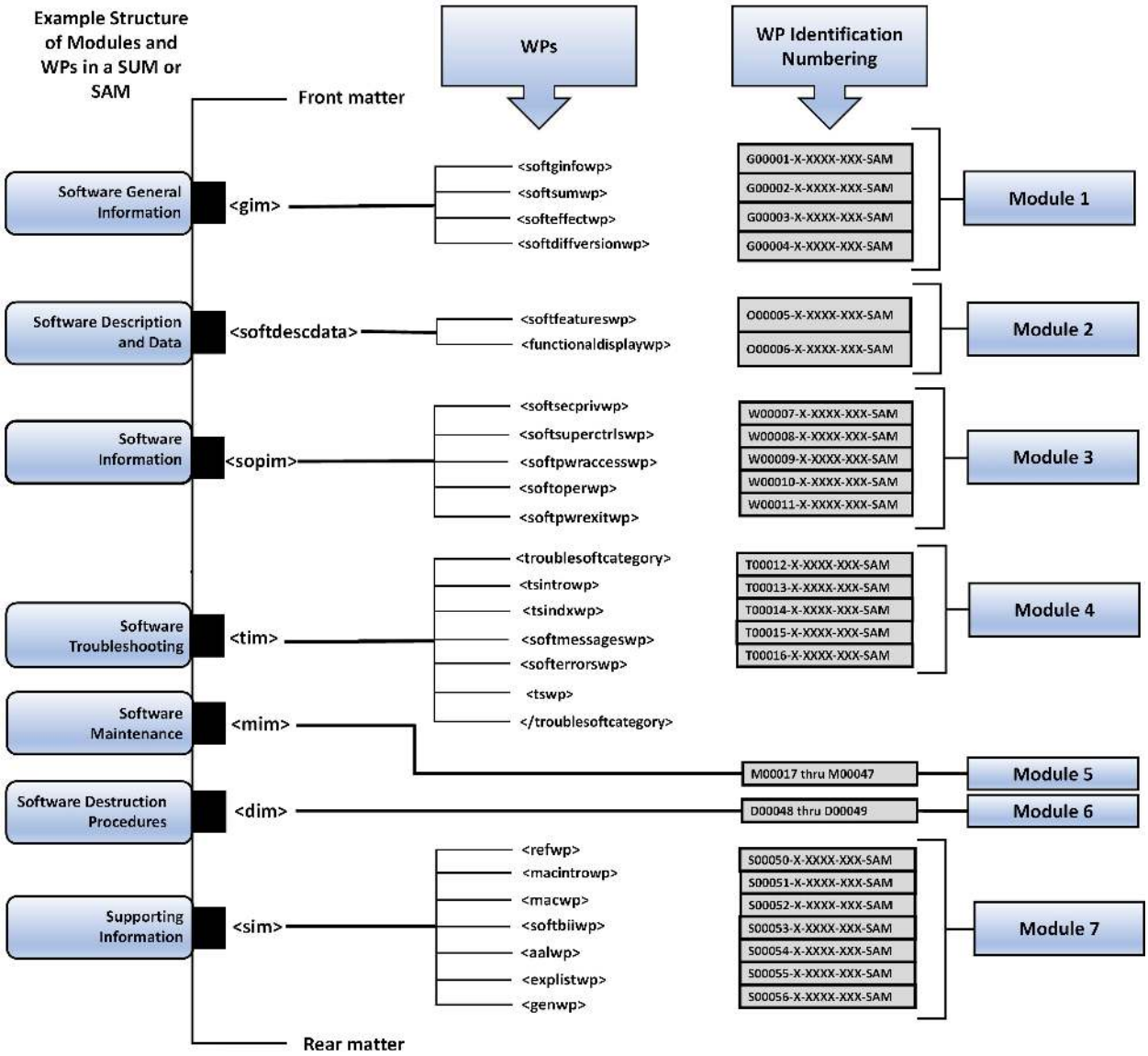


**FIGURE M-5. Example WP identification numbering SUM/SAM stand-alone technical publication.**



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## M.6 DEVELOPMENT OF SOFTWARE INFORMATION MODULES.

The following modules (and respective WPs) shall be developed for software information included in an equipment system technical publication (M.6.1) and stand-alone software publications (M.6.2).

### M.6.1 Development of software module(s included in an equipment system technical publication.

As stated in section M.5.5.1, when specified by the MATDEV, a SUM **<sum>** and/or SAM **<sam>** shall be included in an equipment system technical publication as a software information module **<soim>**. Section M.6.1 outline requirements of the **<soim>** module. All other modules in an equipment publication shall be developed in accordance with appendices in this standard. See FIGURE M-3 for a graphical representation of WP organization and tags discussed in this section.

#### M.6.1.1 Software information module <soim> in an equipment system technical publication.

A software information module **<soim>** in an equipment system technical publication shall be provided. It shall consist of the following WPs within **<softwarecategory>** tags.

**a.** Software general information WP **<softginfowp>** (M.7.1).

**b.** Software summary WP **<softsumwp>** (M.7.2).

**c.** Software effectivity WP **<softeffectwp>** (M.7.3) (Revisions only).

**d.** Differences between versions WP **<softdiffversionwp>** (M.7.4) (Revisions only).

**e.** Features and capabilities WP **<softfeaturescapwp>** (M.7.5).

**f.** Functional display WP **<functionaldisplaywp>** (M.7.6).

**g.** Security and privacy WP **<softsecprivwp>** (M.7.7.1).

**h.** Supervisory controls WP **<softsuperctrlswp>** (M.7.7.2).

**i.** Power up and access software procedures WP **<softpwraccesswp>** (M.7.7.3).

**j.** Software operator procedures WPs **<softoperwp>** (M.7.7.4).

**k.** Power down and exit software procedures WP **<softpwrexitwp>** (M.7.7.5).

**l.** Messages WP **<softmessageswp>** (M.7.8.3) (Required for SAM only).

**m.** Recovery from errors, malfunctions, and emergencies WP **<softerrorswp>** (M.7.8.4).

### M.6.2 Development of software modules for stand-alone software technical publications (SUMs, SAMs, and SUM/SAMs).

As stated in sections M.5.5.2 and M.5.5.3, when specified by the MATDEV, a stand-alone SUM **<sum>**, a stand-alone SAM **<sam>** a combined SUM/SAM **<sam>** manual shall be prepared (see TABLE A-VIII). Sections M.6.2.1 through M.6.2.7 outline module requirements for these publication types. See FIGURE M-4 for a graphical representation of WP organization and tags discussed in this section.

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### M.6.2.1 Software general information module <gim>.

A software general information module **<gim>** shall be provided. It shall consist of the following WPs.

**a.** Software general information WP **<softginfowp>** (M.7.1).

**b.** Software summary WP **<softsumwp>** (M.7.2).

**c.** Software effectivity WP **<softeffectwp>** (M.7.3) (Revisions only).

**d.** Differences between versions WP **<softdiffversionwp>** (M.7.4) (Revisions only).

### M.6.2.2 Software description and data module <softdescdata>.

A software description and data module **<softdescdata>** shall be provided. It shall consist of the following WPs.

**a.** Features and capabilities WP **<softfeaturescapwp>** (M.7.5).

**b.** Functional display WP **<functionaldisplaywp>** (M.7.6).

### M.6.2.3 Software operator instructions module(s) <sopim>.

The software operator instructions module **<sopim>** shall be prepared and shall contain, as applicable, the following work packages in the order provided.

**a.** Security and privacy WP **<softsecprivwp>** (M.7.7.1).

**b.** Supervisory controls WP **<softsuperctrlswp>** (M.7.7.2).

**c.** Power up and access software procedures WP **<softpwraccesswp>** (M.7.7.3).

**d.** Software operator procedures WPs **<softoperwp>** (M.7.7.4).

**e.** Power down and exit software procedures WP **<softpwrexitwp>** (M.7.7.5).

### M.6.2.4 Software troubleshooting module <tim>.

This troubleshooting module shall be prepared for SUM/SAM and SAM (optional for SUM) and shall contain the following WPs in the order provided, as applicable, between **<troublesoftcategory>** category tags.

**a.** Introduction WP **tsintrowp** (M.7.8.1)

**b.** Troubleshooting index WP **tsindx** (M.7.8.2) (Required for SUM/SAM and SAM only).

**c.** Messages WP **<softmessageswp>** (M.7.8.3) (Required for SUM/SAM and SAM only).

**d.** Recovery from errors, malfunctions, and emergencies WP **<softerrorswp>** (M.7.8.4).

**e.** Troubleshooting WP(s) **<tswp>** (M.7.8.5).

**f.** Additional troubleshooting WP(s) as needed.

### M.6.2.5 Software maintenance instructions module <mim>.

This module shall be prepared in accordance with M.7.9.

### M.6.2.6 Software destruction to prevent enemy use module <dim>.

This module shall be prepared in accordance with M.7.10.

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### M.6.2.7 Supporting information module <sim> (Required.

This supporting information module shall be prepared and shall contain the following WPs in the order provided, as applicable.

**a.** References WP **<refwp>** (M.7.11.1) (Required).

**b.** Maintenance Allocation Chart Introduction **<macintrowp>** (M.7.11.2).

**c.** Maintenance Allocation Chart **<macwp>** (M.7.11.3).

**d.** Basic issue items WP **<softbiiwp>** (M.7.11.4) (Required).

**e.** Additional authorization list WP **<aalwp>** (M.7.11.5).

**f.** Expendable and durable items list WP **<explist>** (M.7.11.6).

**g.** Additional supporting information WP(s) **<genwp>** (M.7.11.7).

## M.7 DEVELOPMENT OF SOFTWARE INFORMATION WORK PACKAGES.

The following types of software information WPs shall be developed as applicable.

### M.7.1 Software general information work package <softginfowp>.

A software general information WP shall be prepared and shall contain the following information in the order presented and as applicable. This WP provides an explanation of the scope and functionality of the software as well as its features and capabilities. This WP shall not contain specific software operation procedures. Refer to FIGURE M-6 for an example of a software general information WP tagging structure.

**a.** WP identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** Initial setup **<initial_setup>** is not required for this WP.

**c.** Scope **<scope>** (B.7.1.1) is required for this WP and shall contain purpose of the software.

**d.** Maintenance forms, records, and reports **<mfrr>** is required for this WP (B.7.1.2).

**e.** Software improvement recommendations **<eir>** is required for this WP (B.7.1.3). This paragraph shall provide the software user/administrator with instructions for submitting recommendations for improvements related to the software.

**f.** System overview **<softsysover>** is required for this WP and shall contain a description of the software and describe its capabilities. More detailed information shall be provided in the features and capabilities WP described in .

**g.** Document overview **<softdocover>** is required for this WP. This paragraph may contain brief introduction to the kinds of software WPs included from a high-level viewpoint. This section shall also reference any additionally available documentation for the software. If this section is not included, reference additional software in the system overview **<softsysover>**) section.

**h.** Warranty information **<wrntyref>** included as applicable. Any warranty information related to the software shall be provided in this paragraph. Hardware related warranty information shall not be included here unless it also applies to the software.

**i.** Destruction of software to prevent enemy use **<destructmat>** is required for this WP (B.7.1.7).

**j.** Nomenclature cross reference list **<nomenreflist>** (B.7.1.10) included as applicable. If included, this list shall only contain nomenclature related to the software. Hardware nomenclature shall be contained in the operator and/or maintenance manuals for the hardware.

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**k.** List of abbreviations **<loa>** shall be included in this WP. For equipment system technical publications containing a software information module, this section shall contain a list of the software-related abbreviations and acronyms used within the module. For stand-alone SUMs/SAMs, this section shall contain a list of all abbreviation/acronyms in the publication.

**l.** Copyright credit line **<copyrt>** may be included in this WP for stand-alone SUMs/SAMs only as needed. Technical publications should not contain copyrighted material except as specified in the Federal Acquisition Regulations and Defense Federal Acquisition Regulation Supplement. When copyrighted material is included in a technical publication, the publication author shall obtain prior written permission from the copyright owner or authorized agent for its use. The written permission shall contain a statement declaring whether or not a copyright credit line is required. When a copyright credit line is required, the information shall appear as the last paragraph of the general information.

**m.** Endorsement notice **<endorse>** may be included as applicable. If included, this section shall state that the Army is not endorsing specific software regardless of software type specificity in text or graphic representations of software screens.

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**Note**

Figure appearance and tags/output may vary slightly based on version of DTD/stylesheets utilized (see applicable DTD or tag definition list for available tags).

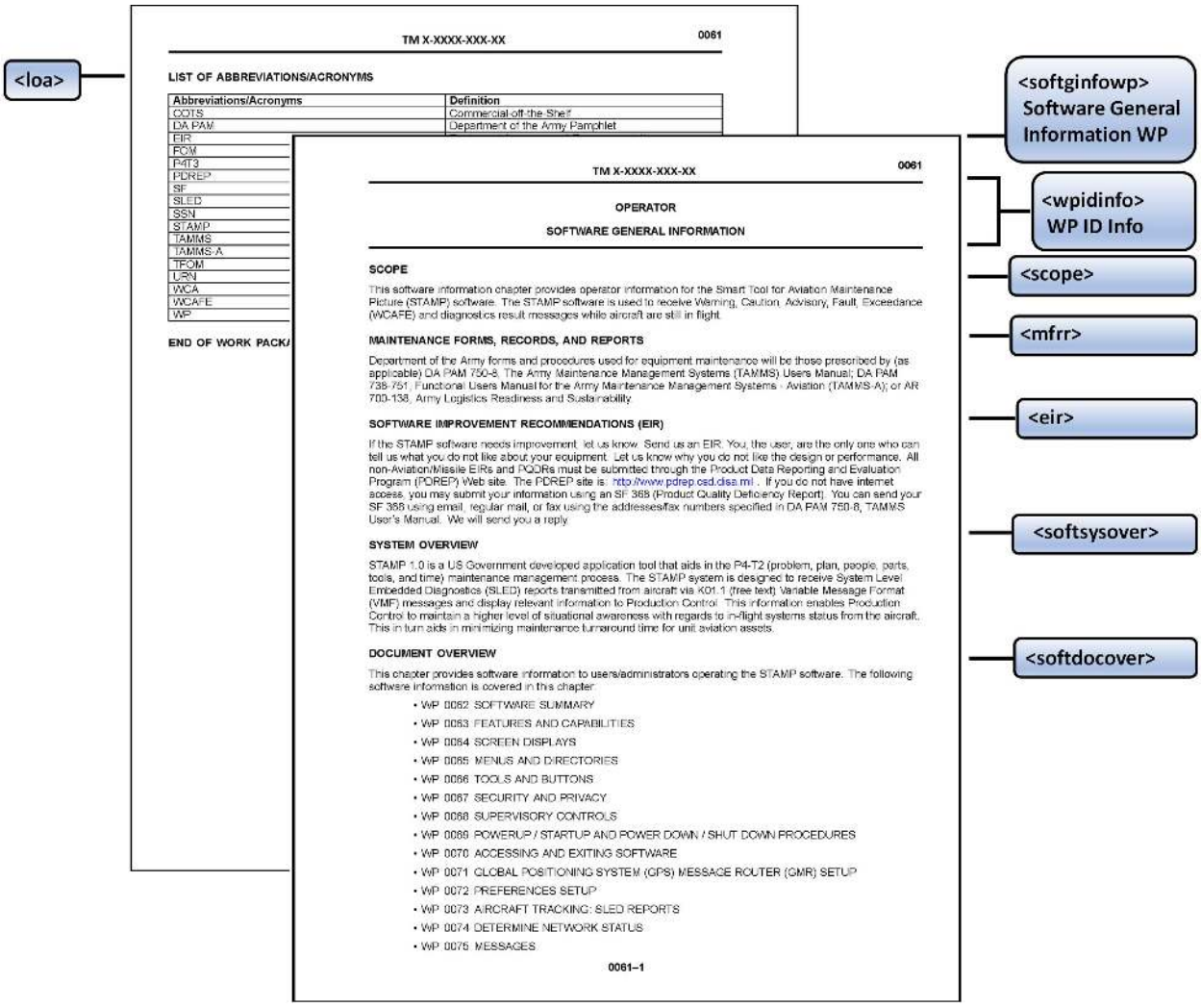


**FIGURE M-6. Software general information WP.**

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## M.7.2 Software summary work package <softsumwp>.

A software summary WP shall contain the following information in the order presented and as applicable. This WP provides a description of software environment(s) as well as security and privacy measures and supervisor controls. This WP shall not contain specific software operation procedures. Refer to FIGURE M-7 for an example of a software summary WP tagging structure.

**a.** WP identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** Initial setup **<initial_setup>** is not required for this WP.

**c.** Software application **<soft_app>** is required for this WP and shall contain a description of how the software applies at various levels of command. Additionally, this section shall describe how the software applies to different organizations/activities and military units (System Battalion, Brigade, Squadron, etc.).

**d.** Software inventory **<soft_inventory>** is required for this WP and shall contain a list/description of the software components provided.

**e.** Software environment **<soft_environment>** is optional for SUMs and required for SAMs. This paragraph shall contain information about software run environments such as the operating system, the database system, specific developmental tools, or compiler. Any relevant environment requirements shall also be included.

**f.** Security and privacy **<soft_secpriv>** is required for this WP and shall contain a brief description of the security and privacy measures provided with the software.

**Note**

Detailed security and privacy procedures shall be provided in the operating procedures.

**g.** Supervisory controls **<soft_superctrls>** shall be included as needed and contain a brief description of any supervisory controls provided with the software.

**h.** Assistance and problem reporting **<soft_assistreport>** is required for this WP. This paragraph shall contain information for obtaining assistance with the software and how/where to report problems with the software.



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**Note**

Figure appearance and tags/output may vary slightly based on version of DTD/stylesheets utilized (see applicable DTD or tag definition list for available tags).

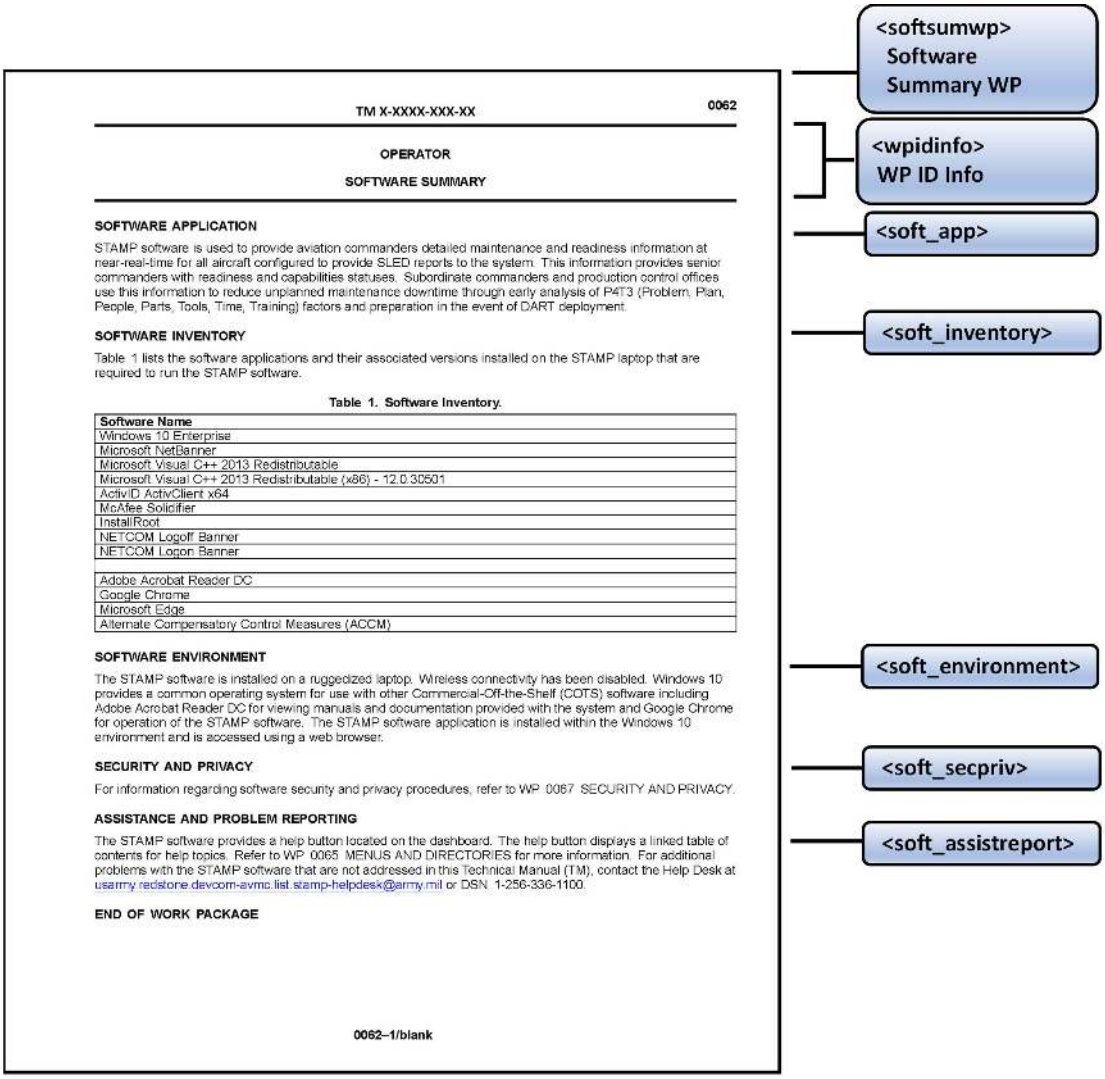


**FIGURE M-7. Software summary WP.**

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### M.7.3 Software effectivity work package <softeffectwp> (Revisions only).

A software effectivity WP shall be prepared as needed for revisions and shall contain information for all software versions available on a system. This WP shall not contain specific software operation procedures. This WP shall contain the following information in the order presented and as applicable.

**a.** WP identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** Initial setup **<initial_setup>** is not required for this WP.

**c.** Software effectivity information **<geninfo>** shall contain software version details. More specifically, when multiple software versions are included a SUM or SAM, this WP shall identify operating systems relevant to each software version.

### M.7.4 Differences between versions work package <softdiffversionwp> (Revisions only).

A software differences WP shall be prepared as needed for revisions and shall contain information about the differences between software versions available on a system. This WP shall not contain specific software operation procedures. This WP shall contain the following information in the order presented and as applicable.

**a.** WP identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** Initial setup **<initial_setup>** is not required for this WP.

**c.** Differences between software versions information **<eqpdiff>**) shall contain software version details. More specifically, this WP shall be included when more than one version of the software must be covered in a SUM or SAM. If more than 25 percent of the SUM/SAM data are affected by the differences between versions, configuration tagging shall be used or a separate technical publication shall be developed. Refer to M.5.5.4 for more information regarding configuration tagging.

### M.7.5 Features and capabilities work package <softfeaturescapwp>.

A software features and capabilities WP shall be prepared as needed and shall contain information about the features and capabilities of the software. This WP shall not contain specific software operation procedures. This WP shall contain the following information in the order presented and as applicable. Refer to FIGURE M-8 for an example of a software features and capabilities WP tagging structure.

**a.** WP identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** Initial setup **<initial_setup>** is not required for this WP.

**c.** Features and capabilities **<para0>** shall not only include descriptions of the features and capabilities of the software and shall but also provide general information for how to use these features and/or capabilities. This WP shall be used for features/capabilities other than those covered in other WPs (for example, key commands, security, etc.).

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**Note**

Figure appearance and tags/output may vary slightly based on version of DTD/stylesheets utilized (see applicable DTD or tag definition list for available tags).

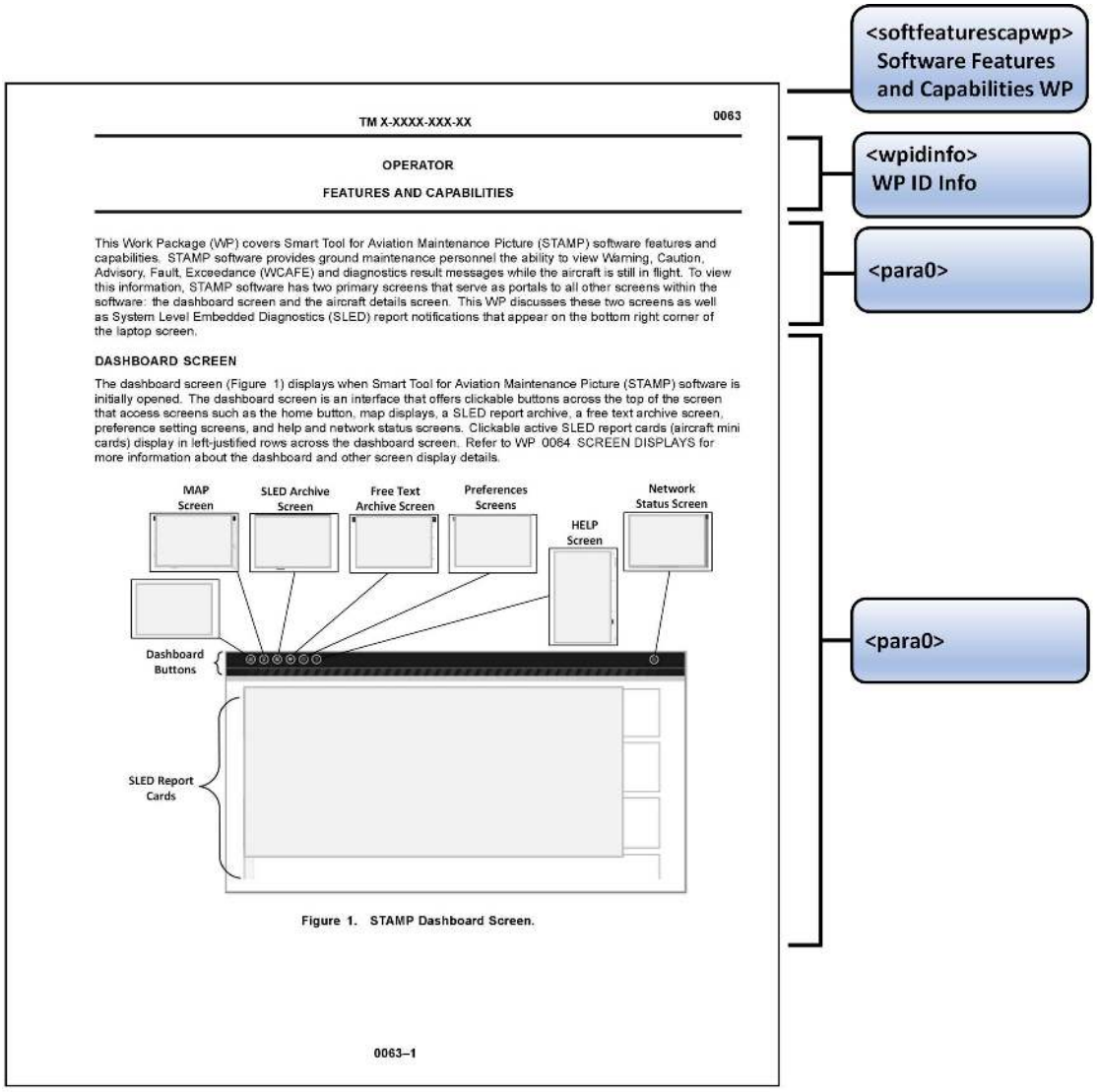


**FIGURE M-8. Software features and capabilities package.**

## M.7.6 Functional display work package <functionaldisplaywp>.

This WP shall be prepared and shall contain any descriptive (non-procedural) data a user may need to be familiar with and understand to navigate through the software during software procedures. The purpose of this WP is to provide a writer the freedom to develop WPs for simple or complex software and provide context and familiarity with screens, menus, and buttons, etc. This WP shall not contain specific software operation procedures. Technical publications may contain multiple functional display WPs as needed. This WP shall contain the following information in the order presented and as applicable.

**a.** WP identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** Initial setup **<initial_setup>** is not required for this WP.



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**c.** Screen displays **<softscreendisplay>** (as needed) (M.7.6.1).

**d.** Menus and directories information **<para0>** (as needed) (M.7.6.2).

**e.** Tools and buttons information **<para0>** (as needed) (M.7.6.3).

## M.7.6.1 Screen displays <softscreendisplay>.

This content may be prepared as needed in a functional display WP **<functionaldisplaywp>** as discussed in M.7.6. Screen display text may contain information about and descriptions of the screens that display to the user/ administrator while using the software. This text shall not contain specific software operation procedures. FIGURE M-9 shows example information for screen displays.

**Note**

Figure appearance may vary slightly based on version of DTD/stylesheets utilized.

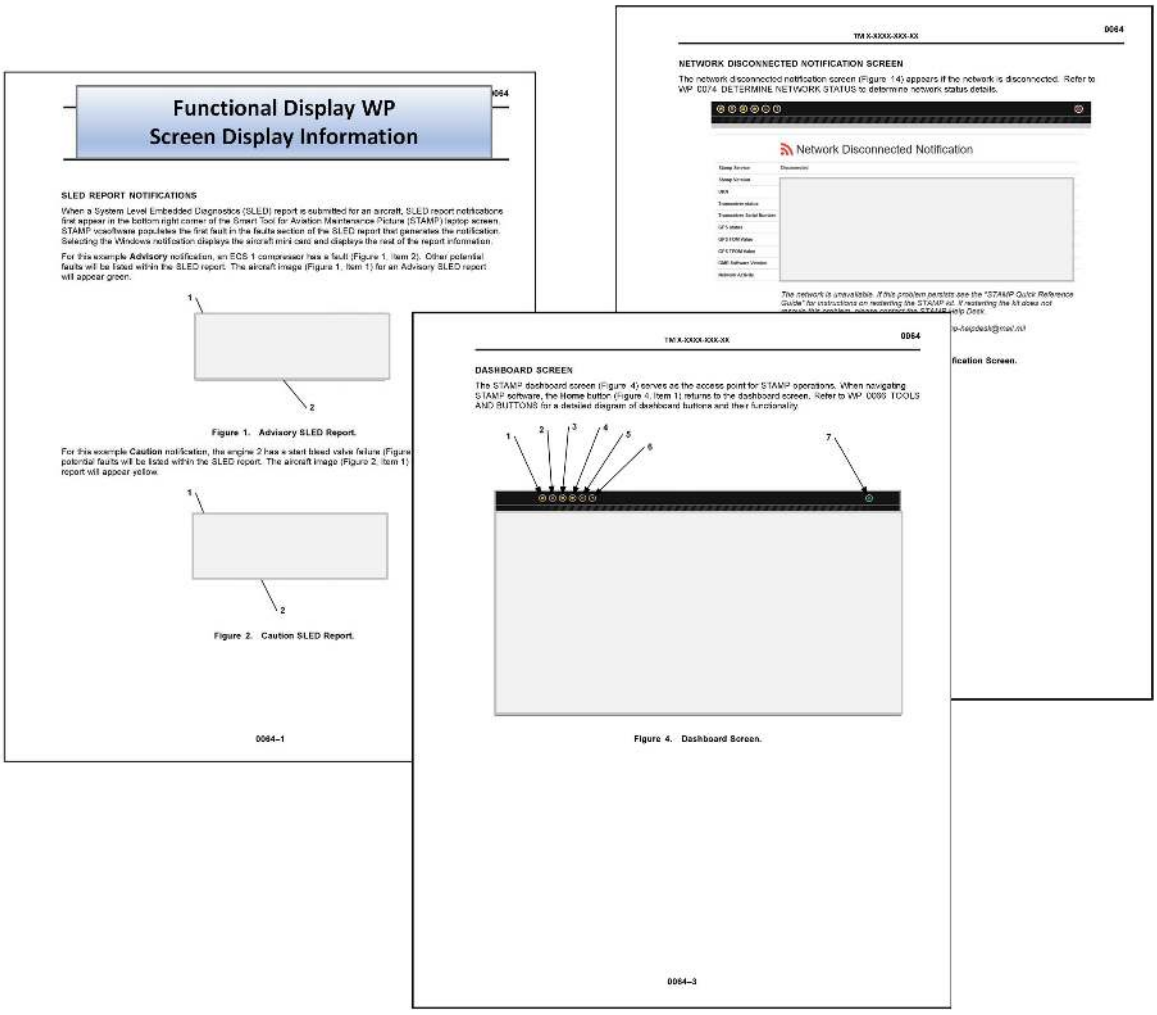


**FIGURE M-9. Software screen displays information.**



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### M.7.6.2 Menus and directories information.

This content may be prepared as needed in a functional display WP **<functionaldisplaywp>** as discussed in M.7.6. Menus and directories text may contain descriptions of the menus and directory structure for the software. This text may also contain general information for how to use the menus/submenus, file structure, and file management tools, etc. This text shall not contain specific software operation procedures. FIGURE M-10 through shows example information for screen displays.

**Note**

Figure appearance may vary slightly based on version of DTD/stylesheets utilized.

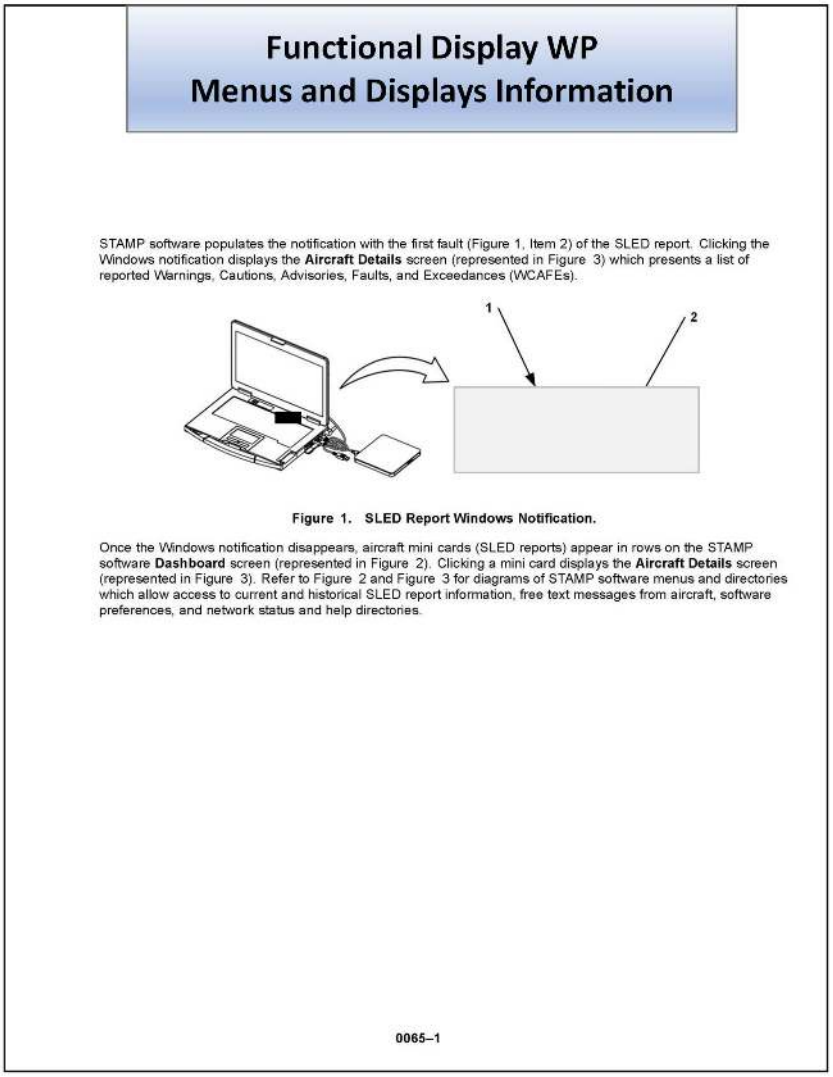


**FIGURE M-10. Software menus and directories information (sheet 1 of 3).**



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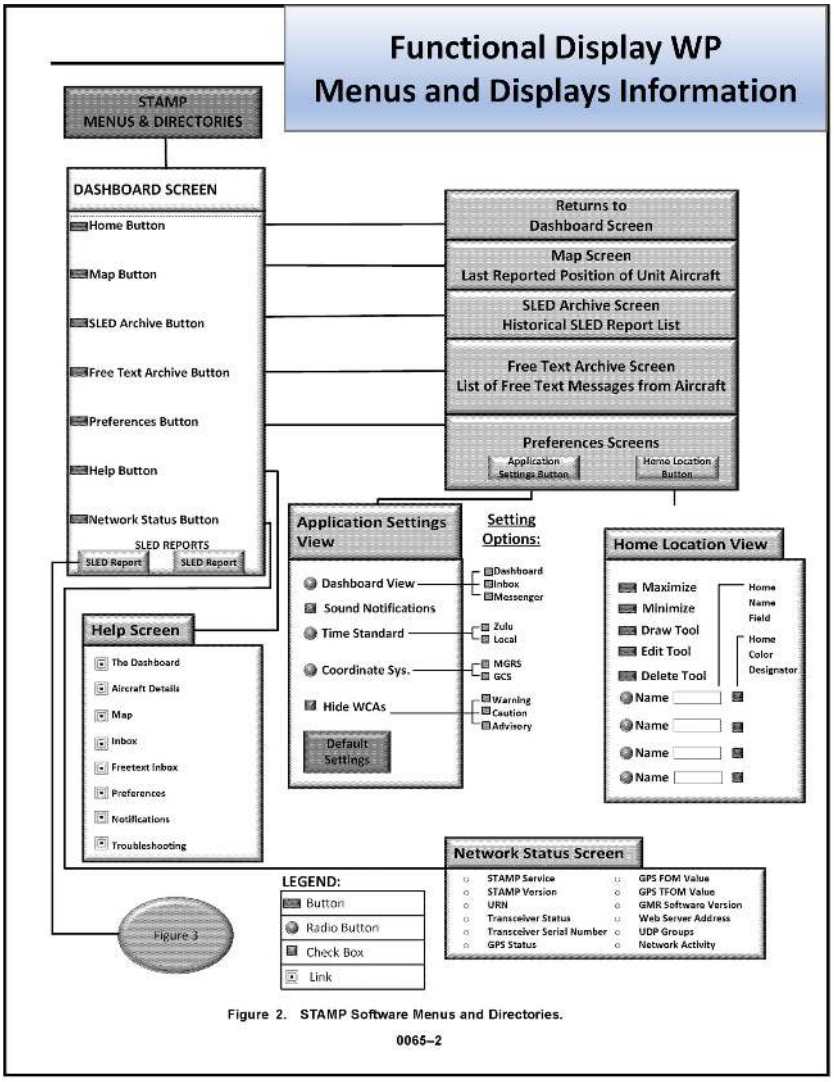


Figure 2. STAMP Software Menus and Directories.

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**FIGURE M-11. Software menus and directories information (sheet 2 of 3).**



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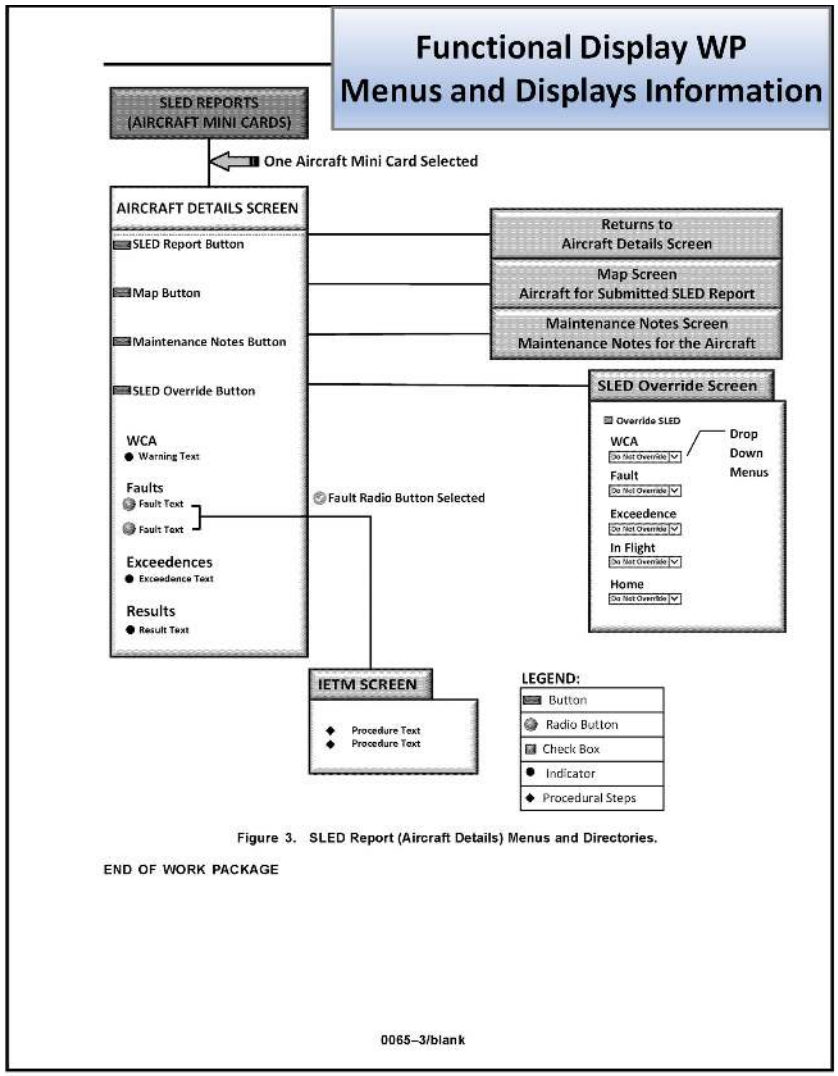


Figure 3. SLED Report (Aircraft Details) Menus and Directories.

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**FIGURE M-12. Software menus and directories information (sheet 3 of 3).**



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### M.7.6.3 Tools and buttons information.

This content may be prepared as needed in a functional display WP **<functionaldisplaywp>**) as discussed in M.7.6. Tools and buttons text may contain descriptions of buttons, tools, and/or toolbars provided by the software including general descriptive information for accessing these buttons and tools/toolbars. As applicable, this section may also contain general descriptive information for accessing hardware buttons/controls that are used in the operation of the software. This text shall not contain specific software operation procedures. FIGURE M-13 shows example information for tools and buttons.

**Note**

Figure appearance may vary slightly based on version of DTD/stylesheets utilized.

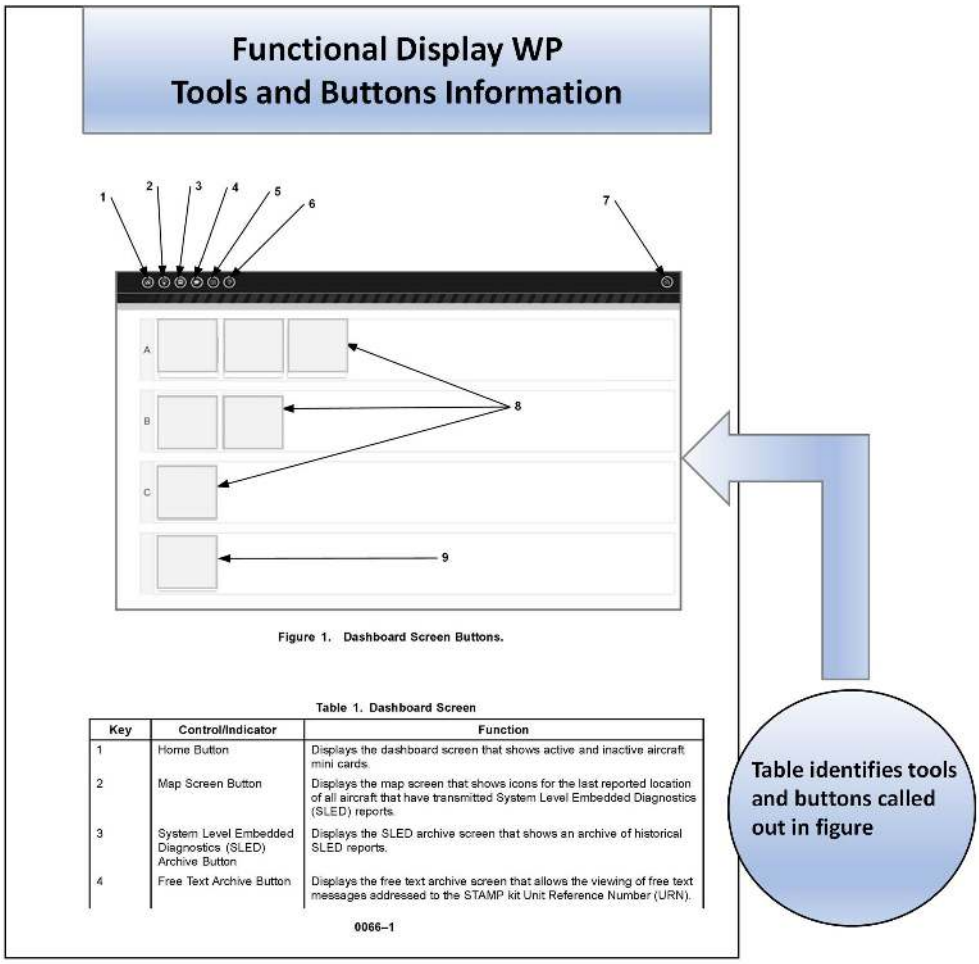


**FIGURE M-13. Tools and buttons WP.**

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## M.7.7 Software operator instructions modules <soim> and <sopim>.

Software operator instructions shall appear in either a **<soim>** module (software module within an equipment technical publication) or a **<sopim>** module (stand-alone SUM/SAM). These modules shall contain, as applicable, the following WPs shown in FIGURE M-14. The circles in this figure highlight where the same WPs appear in each module type.

**Note**

Tags/output shown in figure(s) below may vary slightly based on version of DTD/stylesheets utilized (see applicable DTD or tag definition list for available tags).

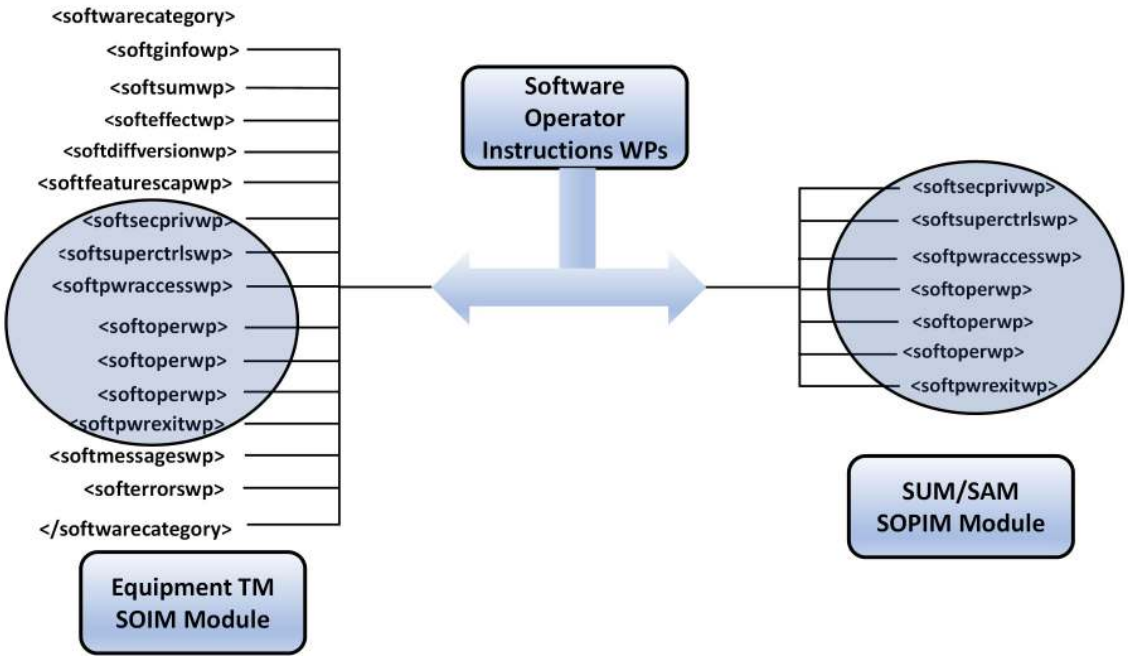


**FIGURE M-14. Software operator instructions WPs.**



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### M.7.7.1 Security and privacy procedures WP <softsecprivwp>.

A software security and privacy procedures WP shall be prepared and shall contain procedures for the security and privacy features of the software. This WP shall contain the following information in the order presented and as applicable. Refer to FIGURE M-15 for an example of a software security and privacy WP tagging structure.

**a.** WP identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** Initial setup **<initial_setup>** is required for this WP (5.4.3).

**c.** Security and privacy procedures **<proc>** shall include tasks such as setting passwords, changing passwords, setting file access restrictions, and account management (for example, setting up new accounts and removing accounts, etc.). Bolding in procedures shall be done in accordance with M.5.3.

**Note**

Figure appearance and tags/output may vary slightly based on version of DTD/stylesheets utilized (see applicable DTD or tag definition list for available tags).

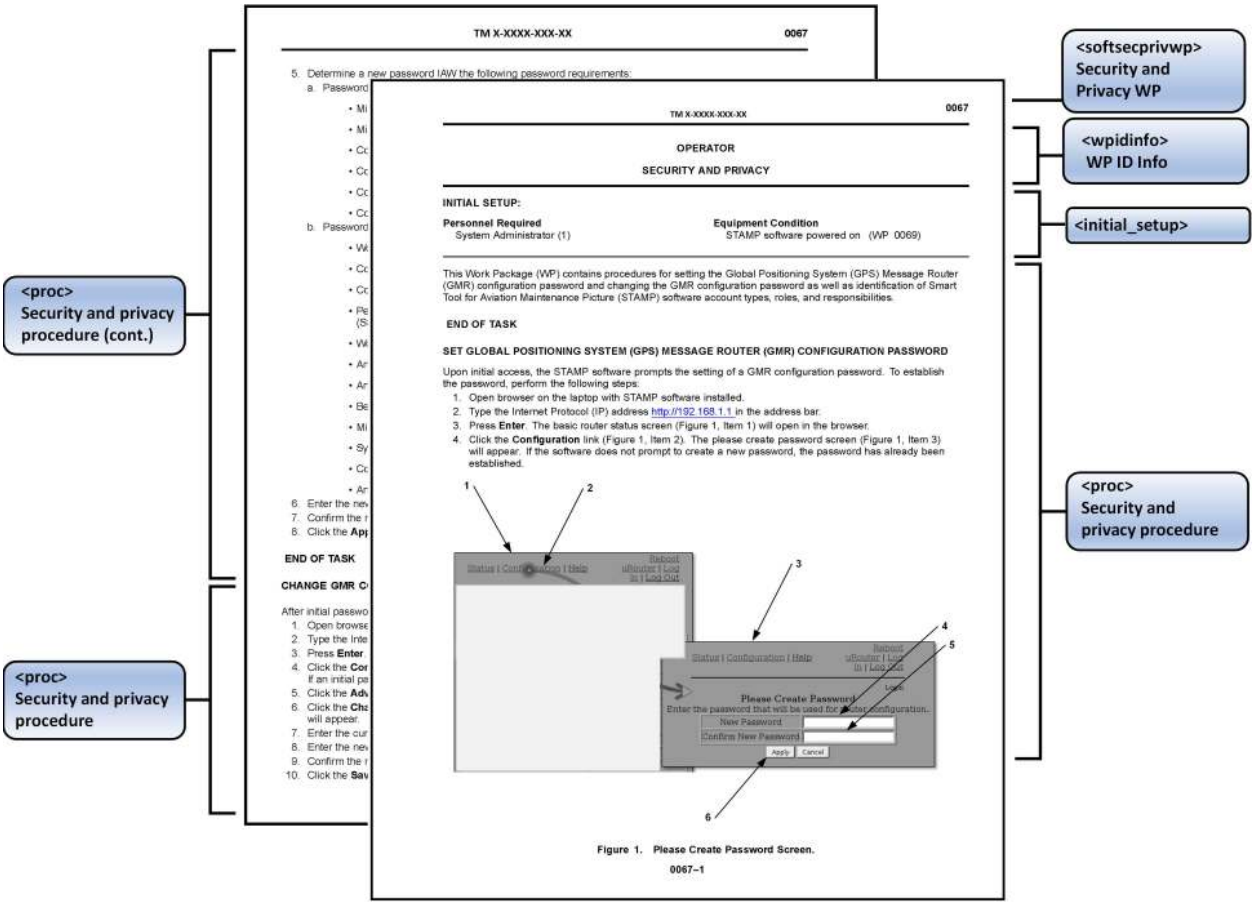


**FIGURE M-15. Security and privacy WP.**



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## M.7.7.2 Supervisory controls WP <softsuperctrlswp> (Required for SAM only.

A software supervisory controls WP shall be prepared for a SUM/SAM and SAM only (optional for SUM). The software supervisory controls WP shall contain information about controls only a supervisor/administrator would use that are available within the software. This WP shall contain the following information in the order presented and as applicable. Refer to FIGURE M-16 for an example of a software supervisory controls WP tagging structure.

**a.** WP identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** Initial setup **<initial_setup>** is required for this WP (5.4.3).

**c.** Supervisor controls procedures **<proc>** shall include tasks for how to use supervisory controls (for example, setting the controls, turning the controls on/off, etc.). Bolding in procedures shall be done in accordance with M.5.3.

**Note**

Figure appearance and tags/output may vary slightly based on version of DTD/stylesheets utilized (see applicable DTD or tag definition list for available tags).

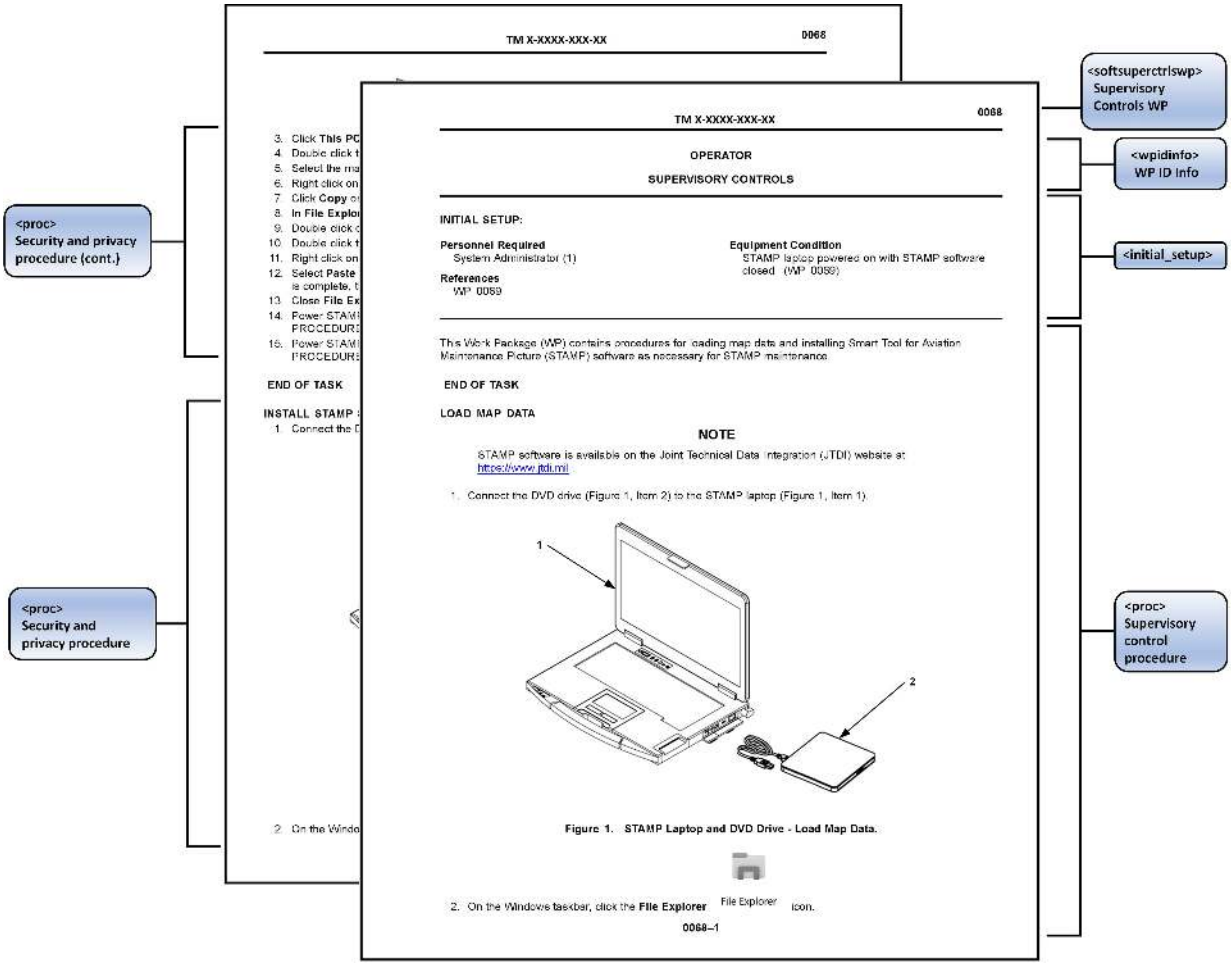


**FIGURE M-16. Supervisory controls WP.**

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### M.7.7.3 Power up and access software procedures WP <softpwraccesswp>.

This software WP shall be prepared as needed and shall contain information about procedures for powering or starting up viewing equipment that enables access to the software. This WP shall also contain procedures for accessing or logging onto software. This WP shall contain the following information in the order presented and as applicable. Refer to FIGURE M-17 for an example of a software power up and access software procedures WP tagging structure.

**a.** WP identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** Initial setup **<initial_setup>** is required for this WP (5.4.3).

**c.** Power up procedures **<proc>** shall include tasks for how to power up a viewing station. Bolding in procedures shall be done in accordance with M.5.3.

**d.** Access procedures **<proc>** shall include tasks for how to access a software application. Bolding in procedures shall be done in accordance with M.5.3.


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**Note**

Figure appearance and tags/output may vary slightly based on version of DTD/stylesheets utilized (see applicable DTD or tag definition list for available tags).

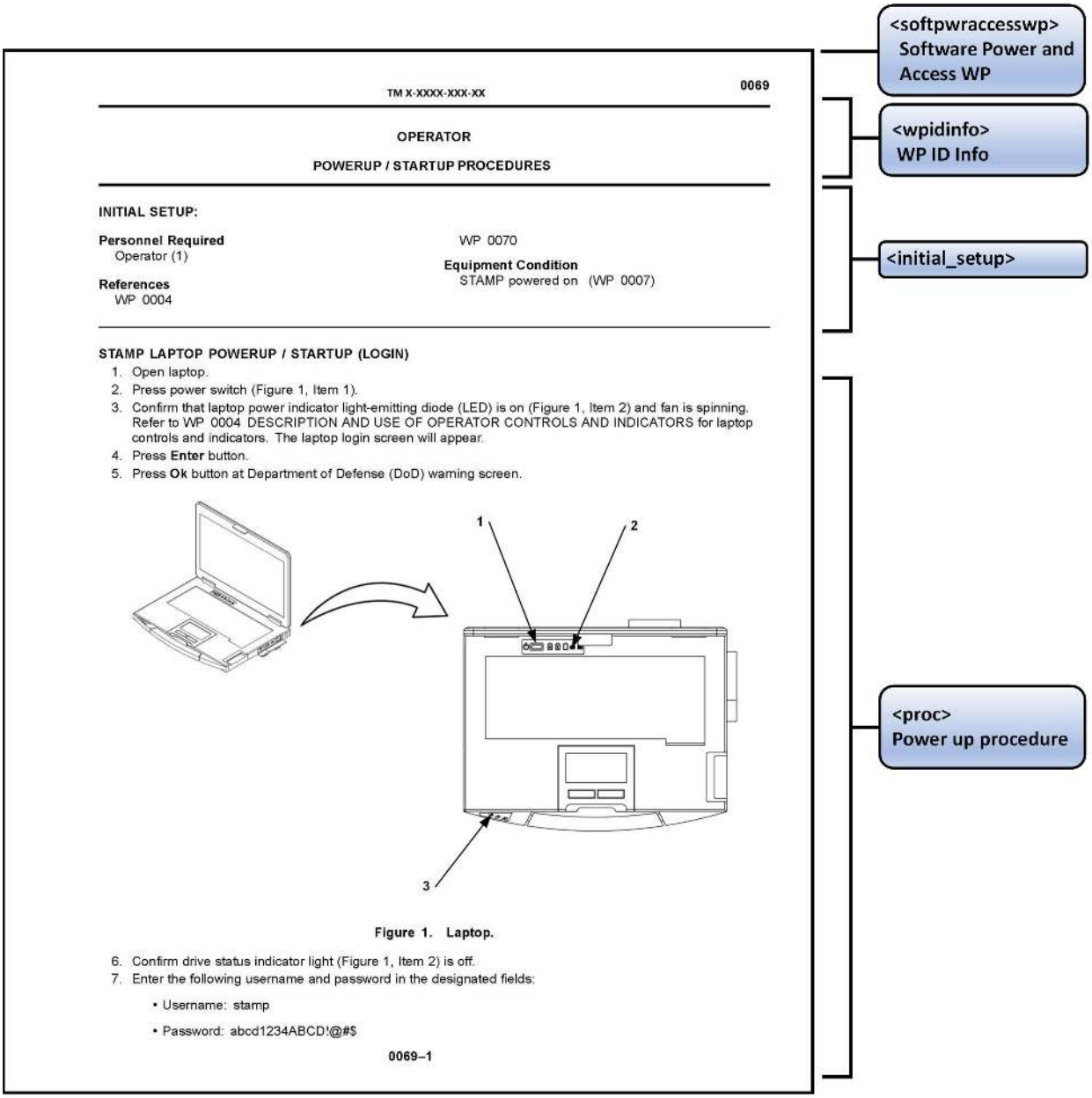


**FIGURE M-17. Power up / access software WP.**

## M.7.7.4 Software operator procedures WP(s) <softoperwp>.

A software operator procedures WP shall include operational procedures for using the software as intended by the MATDEV. Information provided in this WP and the way data is presented (order of operations) shall vary from equipment system to equipment system. Sections M.7.7.4.1 through M.7.7.4.4 present some possible software concepts that may be included in a technical publication; however, these WPs are not limited to these topics and the **<softoperwp>** tag can be used for all software operation procedural content as needed. A technical publication may


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contain one or more software operator procedure WPs. These WPs shall be developed to ensure clear and useful transmission of important software operations concepts and procedures. This WP shall contain WP identification information **<wpidinfo>**and initial setup **<initial_setup>** information listed below and in the order presented below. The rest of the information in these WPs may be classified as one of the concepts listed below (key commands, user interface, conventions, and operating procedures). The associated tags may be utilized in any order and as needed to accurately convey software operator procedures.

**a.** WP identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** Initial setup **<initial_setup>** is required for this WP (5.4.3).

**c.** Key commands **<softkeycmd>** is optional for this WP (M.7.7.4.1).

**d.** User interface **<softgui>** is optional for this WP (M.7.7.4.2).

**e.** Software operating conventions **<softopconventions>** is optional for this WP (M.7.7.4.3).

**f.** Software operating procedures **<softoperproc>** is optional for this WP (M.7.7.4.4).

### M.7.7.4.1 Key commands <softkeycmd>.

Software key command procedure(s) may contain descriptions for any relevant software key commands as well as steps for how to use key commands contained in the software. Key commands may include tasks for running any processes and/or executing any commands contained in the software (for example, any common processes the software typically runs and any unique sequence or combination of keystrokes to which the software responds). Bolding in procedures shall be done in accordance with M.5.3.

### M.7.7.4.2 User interface <softgui>.

Software user interface procedure(s) may contain descriptions for any Graphical User Interfaces (GUIs) of the software as well as methods for the user to interact with a computer system using the software. Details how a user can control a computer system using software may be included (for example, input controls as well as navigational and informational components of the software). Bolding in procedures shall be done in accordance with M.5.3.

### M.7.7.4.3 Software operating conventions <softopconventions>.

Software operating conventions procedure(s) may contain descriptions for any operating conventions that are unique to the software as well as tasks for degraded operation. Bolding in procedures shall be done in accordance with M.5.3.

### M.7.7.4.4 Software operating procedures <softoperproc>.

Software operating procedure(s) may contain procedural steps for use of the software during unit movement and any other software-related tasks in all modes of unit operation. Bolding in procedures shall be done in accordance with M.5.3.

## M.7.7.5 Power down and exit software procedures WP <softpwrexitwp>.

This software WP shall be prepared and shall contain information about procedures for powering down viewing equipment that enables access to the software. This WP shall also contain procedures for logging off software. This WP shall contain the following information in the order presented and as applicable.

**a.** WP identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** Initial setup **<initial_setup>** is required for this WP (5.4.3).

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**c.** Power down procedures **<proc>** shall include tasks for how to power down a viewing station. Bolding in procedures shall be done in accordance with M.5.3.

**d.** Exit procedures **<proc>** shall include tasks for how to exit a software application. Bolding in procedures shall be done in accordance with M.5.3.

## M.7.8 Software troubleshooting procedures (Required for SUM/SAM and SAM only).

Software troubleshooting procedures shall include WPs covered in sections M.7.8.1 through M.7.8.5 as needed.

### M.7.8.1 Troubleshooting introduction work package <tsintrowp> (Required for SUM/SAM and SAM only).

This WP shall be prepared in accordance with APPENDIX D.

### M.7.8.2 Troubleshooting index work package <tsindxwp> (Required for SUM/SAM and SAM only).

This WP shall be prepared in accordance with APPENDIX D.

### M.7.8.3 Messages WP <softmessageswp> work package (Required for SUM/SAM and SAM only).

This software messages WP shall be prepared as needed and shall describe possible error messages the user is likely to encounter. This WP provides the user with a centralized area to research error messages and their meanings. This WP shall contain the following information in the order presented and as applicable. When possible, ensure this WP contains references to relevant troubleshooting and maintenance WPs to provide user context for when these screens would appear. Refer to FIGURE M-18 for an example of a messages WP tagging structure.

**a.** WP identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** Initial setup **<initial_setup>** is required for this WP (5.4.3).

**c.** Messages **<message>** shall include the wording for error messages and explanations of message meanings. Error messages may also be repeated in the troubleshooting index WP to refer the user to the troubleshooting and/or correction action needed for an error message.

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**Note**

Figure appearance and tags/output may vary slightly based on version of DTD/stylesheets utilized (see applicable DTD or tag definition list for available tags).

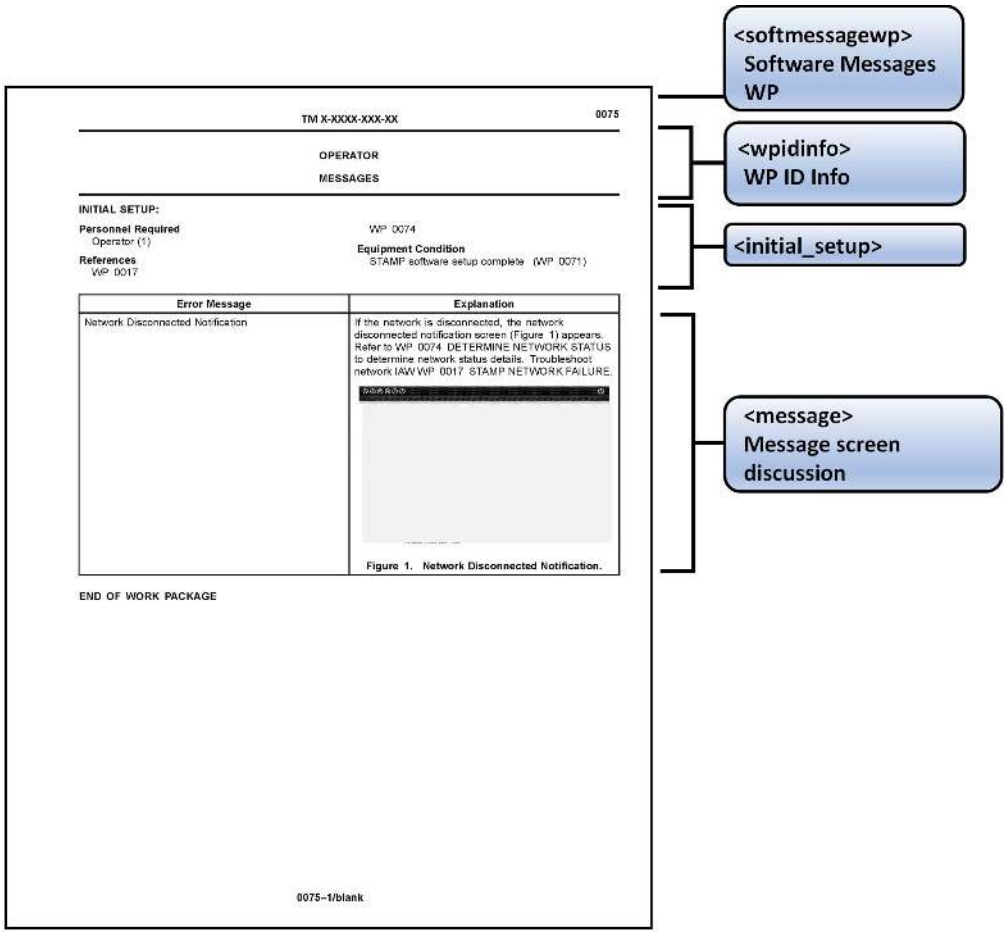


**FIGURE M-18. Software messages WP.**

## M.7.8.4 Recovery from errors, malfunctions, and emergencies work package <softerrorswp>.

This software WP shall be prepared as needed and shall provide procedures for recovering from errors, correcting malfunctions, and handling emergencies. This WP shall contain the following information in the order presented and as applicable.

**a.** WP identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** Initial setup **<initial_setup>** is required for this WP (5.4.3).

**c.** Recovery from errors, malfunctions, and emergencies procedures **<proc>**) shall include any steps to correct any software malfunctions or handle any emergencies using the software.

## M.7.8.5 Troubleshooting work package <tswp>.

This module shall be prepared in accordance with APPENDIX D. As applicable, these WP shall contain any information/procedures needed to cover interfaces between software troubleshooting and hardware troubleshooting.


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## M.7.9 Software maintenance instructions module <mim> (Required for SUM/SAM and SAM).

Software maintenance shall be prepared as necessary and in accordance with the methods listed below.

**a.** Software maintenance incorporated into an equipment technical publication (M.7.9.1):

- Software maintenance tasks incorporated into the maintenance WPs and/or modules in an equipment technical publication.
- Separate software maintenance instruction modules.

**b.** Software maintenance in a stand-alone SUM/SAM (M.7.9.2).

## M.7.9.1 Software maintenance incorporated into in an equipment technical publication.

Software maintenance may be incorporated into an equipment system technical publication as individual software maintenance tasks (E.7.4.4.39) developed within WPs including other non-software tasks. Software maintenance may also be incorporated into an equipment system technical publication with separate software maintenance WPs included in the same module with other non-software maintenance WPs. In these cases, maintenance modules **<mim>** shall be prepared in accordance with E.6, and WPs shall appear between **<maintenancecategory>** category tags (example shown in FIGURE M-19).

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**Note**

Tags/output shown in figure(s) below may vary slightly based on version of DTD/stylesheets utilized (see applicable DTD or tag definition list for available tags).

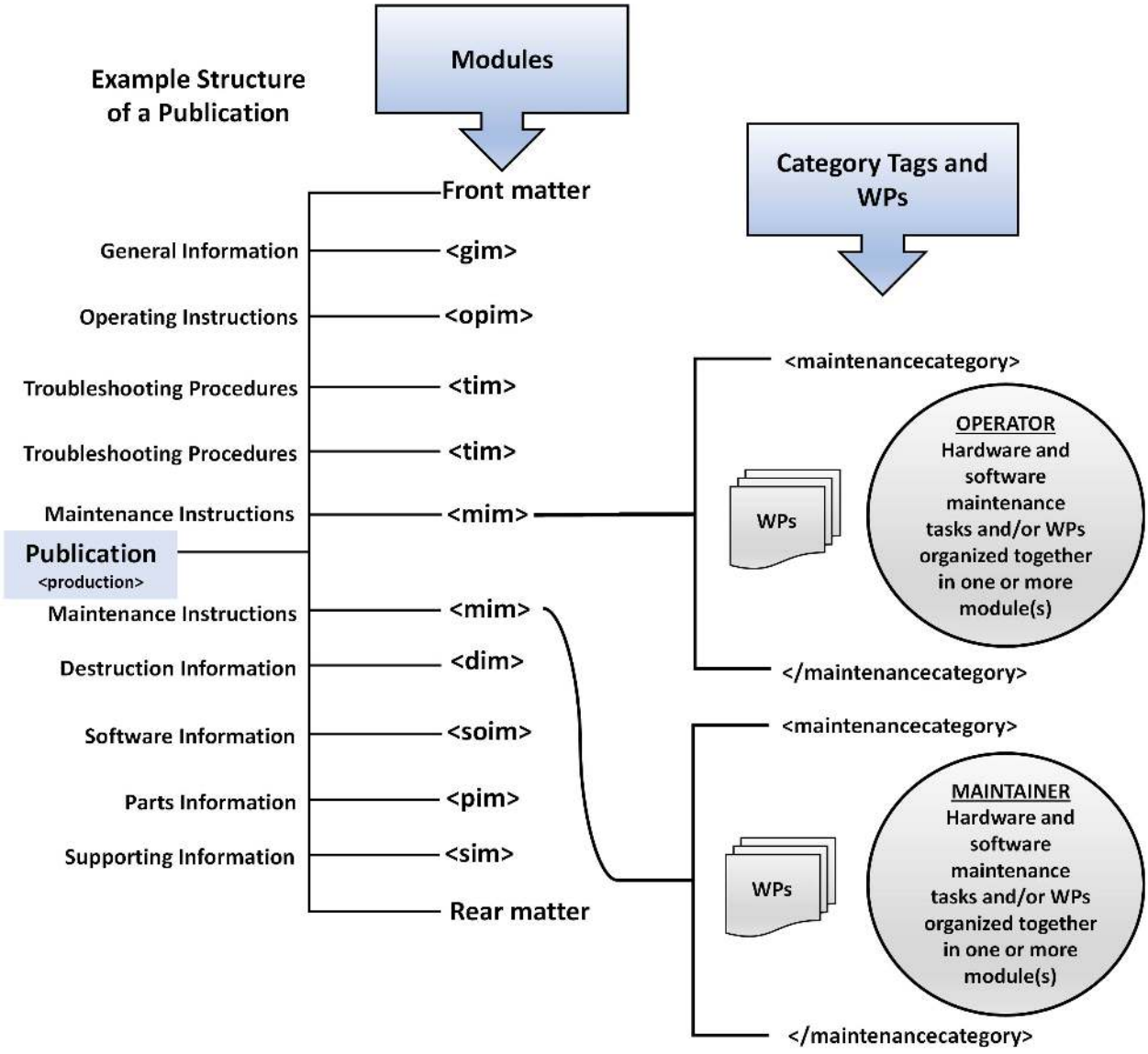


**FIGURE M-19. Example publication with hardware and software maintenance together in a module.**



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Software maintenance may also be incorporated into an equipment system technical publication as a separate software maintenance instructions module (separate from non-software maintenance WPs). In these cases, the maintenance module **<mim>** shall be prepared in accordance with E.6, and WPs shall appear between **<softmaintcategory>** category tags in accordance with E.6.10 (example shown in FIGURE M-20).

**Note**

Tags/output shown in figure(s) below may vary slightly based on version of DTD/stylesheets utilized (see applicable DTD or tag definition list for available tags).

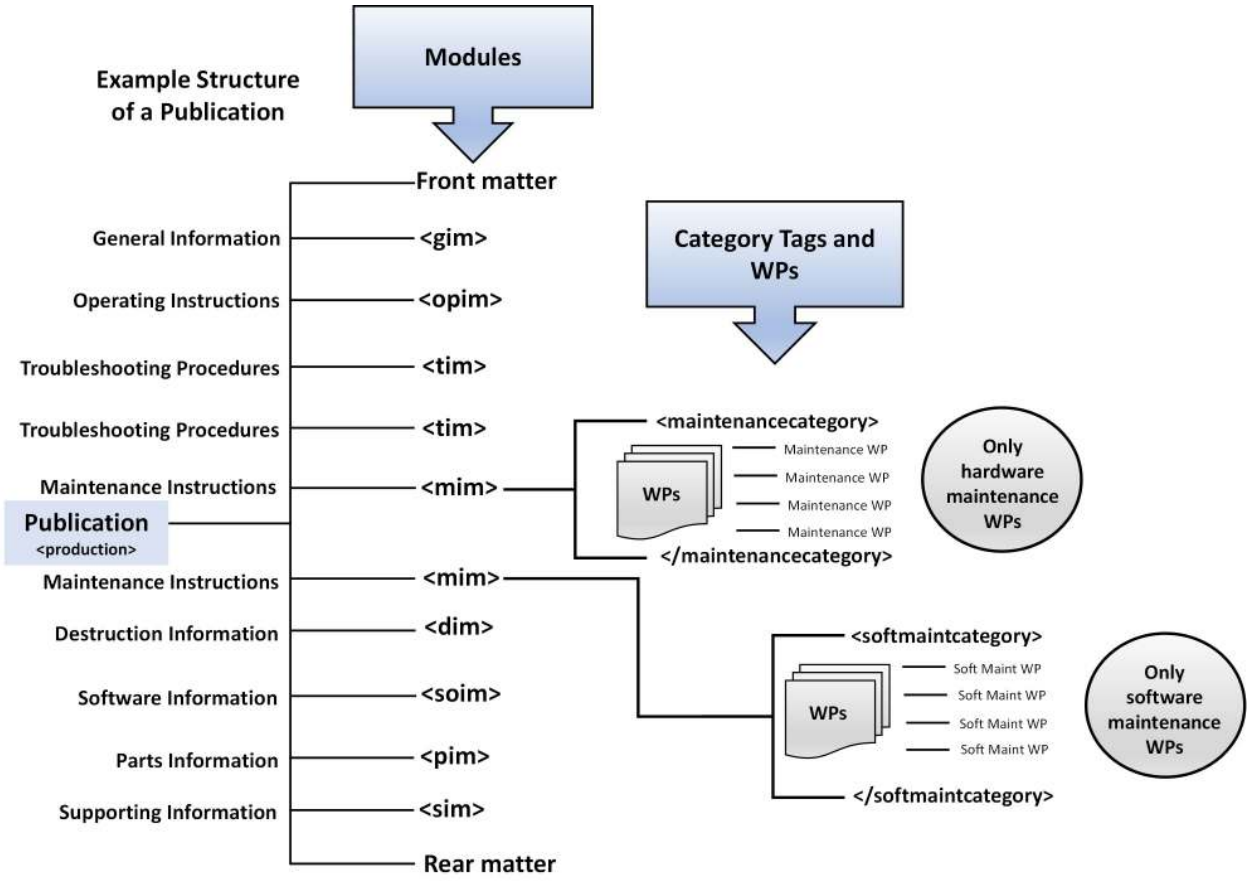


**FIGURE M-20. Example publication with hardware and software maintenance in separate modules.**

## M.7.9.2 Software maintenance in a SUM/SAM and SAM.

For stand-alone SUM/SAM and SAM publications, software maintenance shall conform to requirements shown in the matrices in APPENDIX A (TABLE A-VIII).

## M.7.9.3 Software PMCS.

Software PMCS includes an introduction WP (M.7.9.3.1) and PMCS data organized in accordance with the methods described in M.7.9.3.2.



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### M.7.9.3.1 Software PMCS introduction <pmcsintrowp>.

A software PMCS introduction shall be prepared as necessary and in accordance with E.7.1.

### M.7.9.3.2 Software PMCS WPs <pmcswp>.

Software PMCS data shall be prepared for equipment systems as necessary and in accordance with the options discussed below.

- ***Option 1***: Software PMCS incorporated into an equipment technical publication (M.7.9.3.2.1).

  The structure of this integration can take three different forms listed below.

  – PMCS entries for software incorporated into existing PMCS data for an equipment technical publication.

  – Separate WPs for equipment PMCS and software PMCS together in a module.

  – Separate modules for equipment PMCS WPs and software PMCS WPs.

- Software PMCS in a stand-alone SUM/SAM (M.7.9.3.2.2).

#### M.7.9.3.2.1 Software PMCS incorporated into in an equipment technical publication.

Software PMCS data may be incorporated into an equipment technical publication in various ways. This data may be blended in with PMCS information for an equipment system in a WP or it may be separated into dedicated software PMCS WPs and/or modules. Sections M.7.9.3.2.1.1 and M.7.9.3.2.1.2 outline these options.

##### M.7.9.3.2.1.1 Software PMCS data incorporated into equipment PMCS.

When necessary, software PMCS data may be included with regular PMCS information for the larger equipment system in a WP. This data may simply be integrated into the same PMCS table with other equipment PMCS entries as shown in FIGURE M-21 and structured as shown in FIGURE M-22.


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**Note**

Figure appearance may vary slightly based on version of DTD/stylesheets utilized.

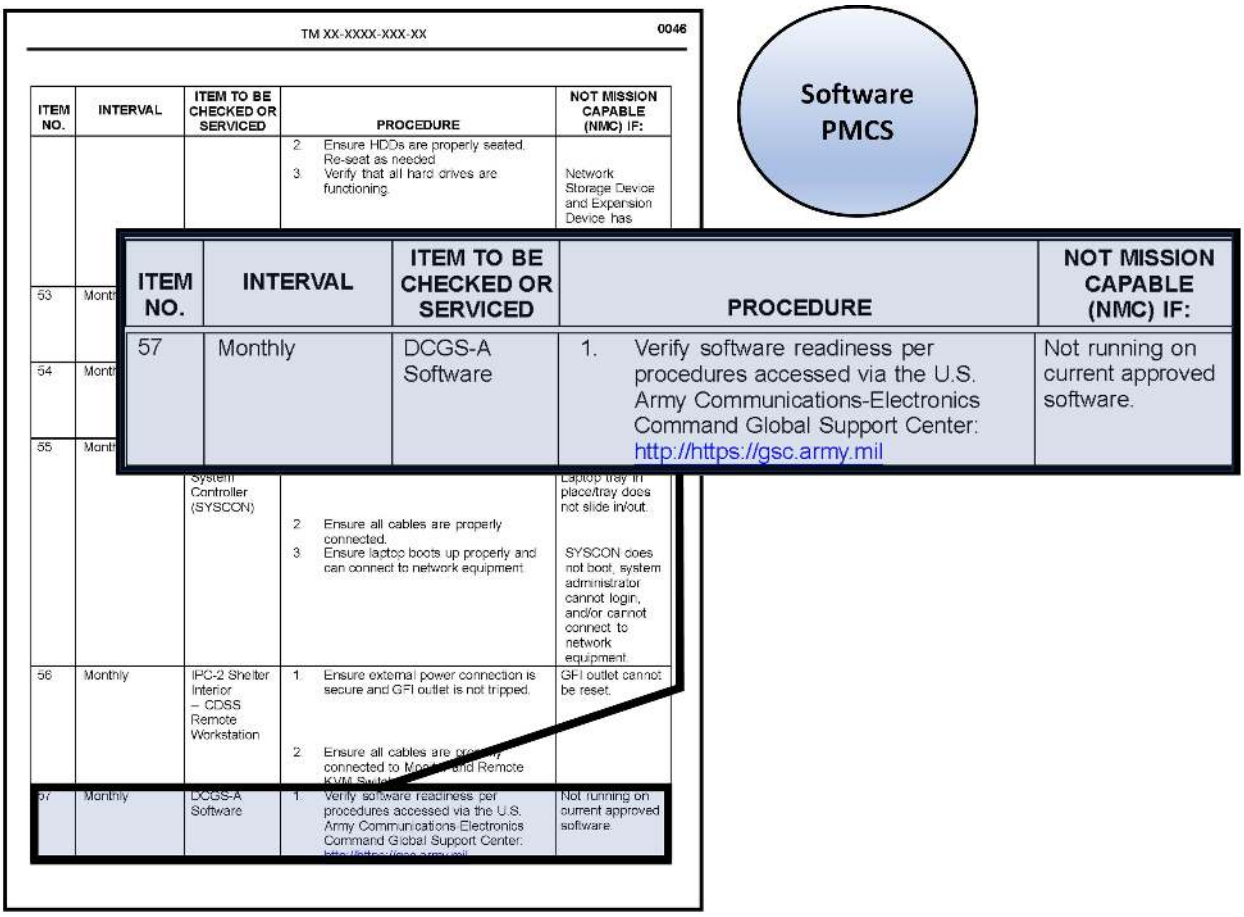


**FIGURE M-21. Software PMCS.**

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**Note**

Tags/output shown in figure (s) below may vary slightly based on version of DTD/stylesheets utilized (see applicable DTD or tag definition list for available tags).

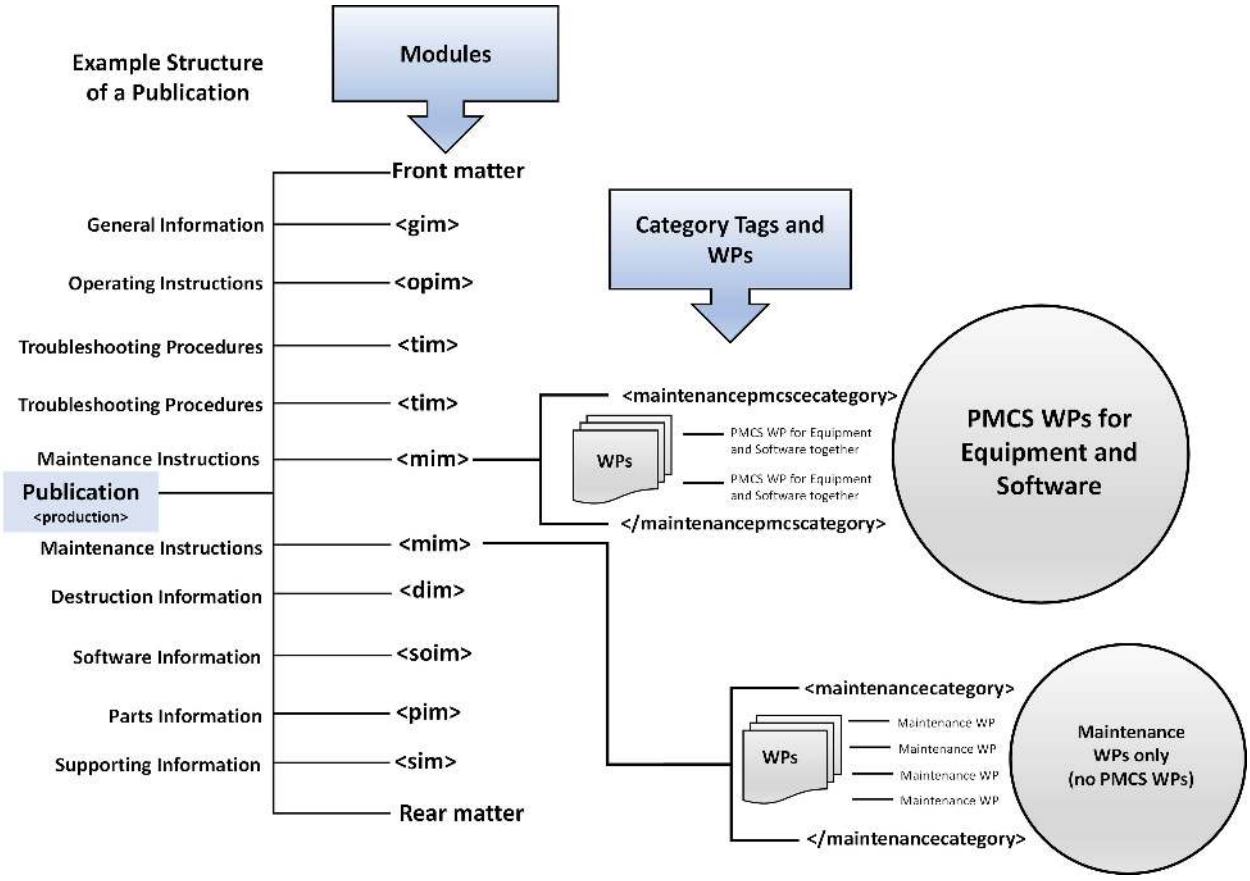


**FIGURE M-22. Example publication including PMCS for equipment and software together in WPs.**

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### M.7.9.3.2.1.2 Software PMCS data provided in separate WPs and/or modules.

Software PMCS data may also be separated into dedicated software PMCS WPs and/or modules depending on complexity of software and number of PMCS entries. As sections E.6.2 and E.6.3 explain, when PMCS WPs stand alone or are combined with other maintenance WPs in a single maintenance module, this module shall be comprised of WPs between **<maintenancepmcscategory>** category tags (examples shown in FIGURE M-23, FIGURE M-24, and FIGURE M-25).

**Note**

Tags/output shown in figure (s) below may vary slightly based on version of DTD/stylesheets utilized (see applicable DTD or tag definition list for available tags).

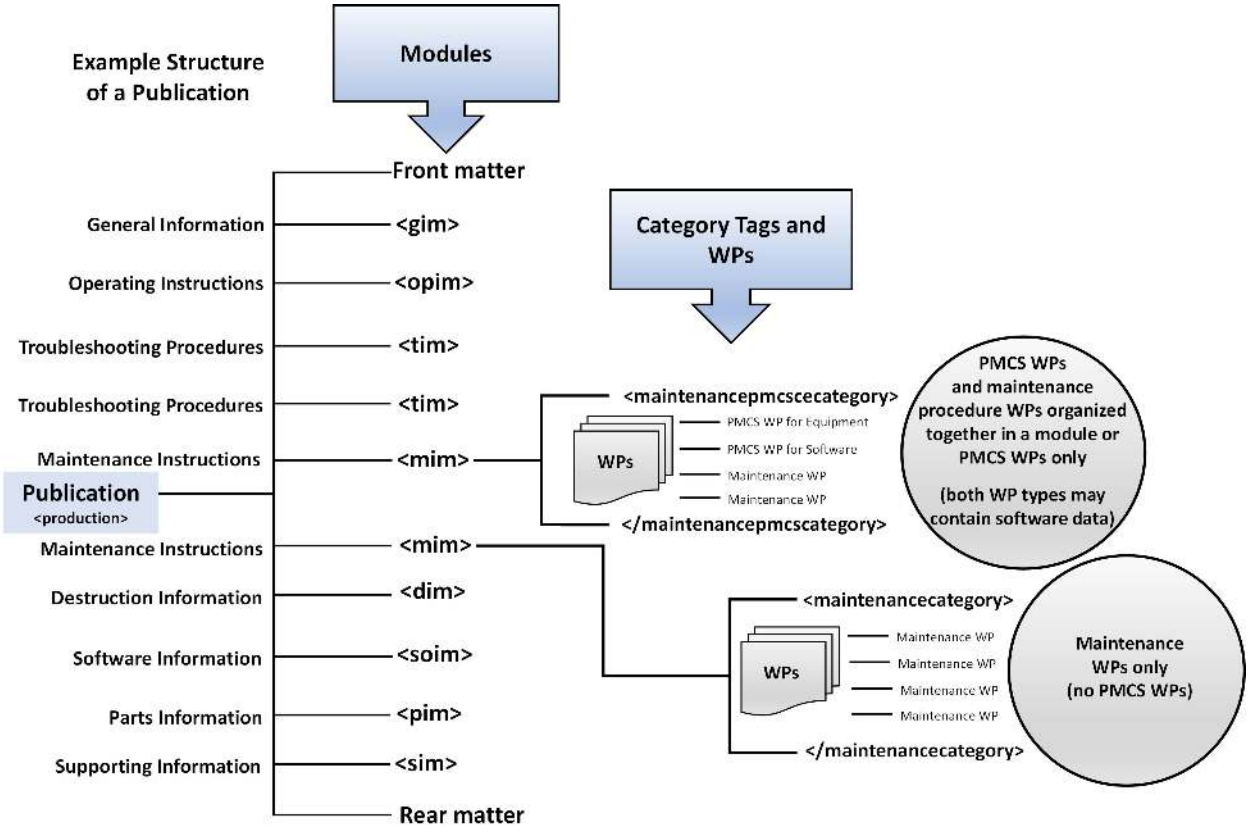


**FIGURE M-23. Example publication including PMCS WPs and maintenance procedure WPs organized together in a module.**



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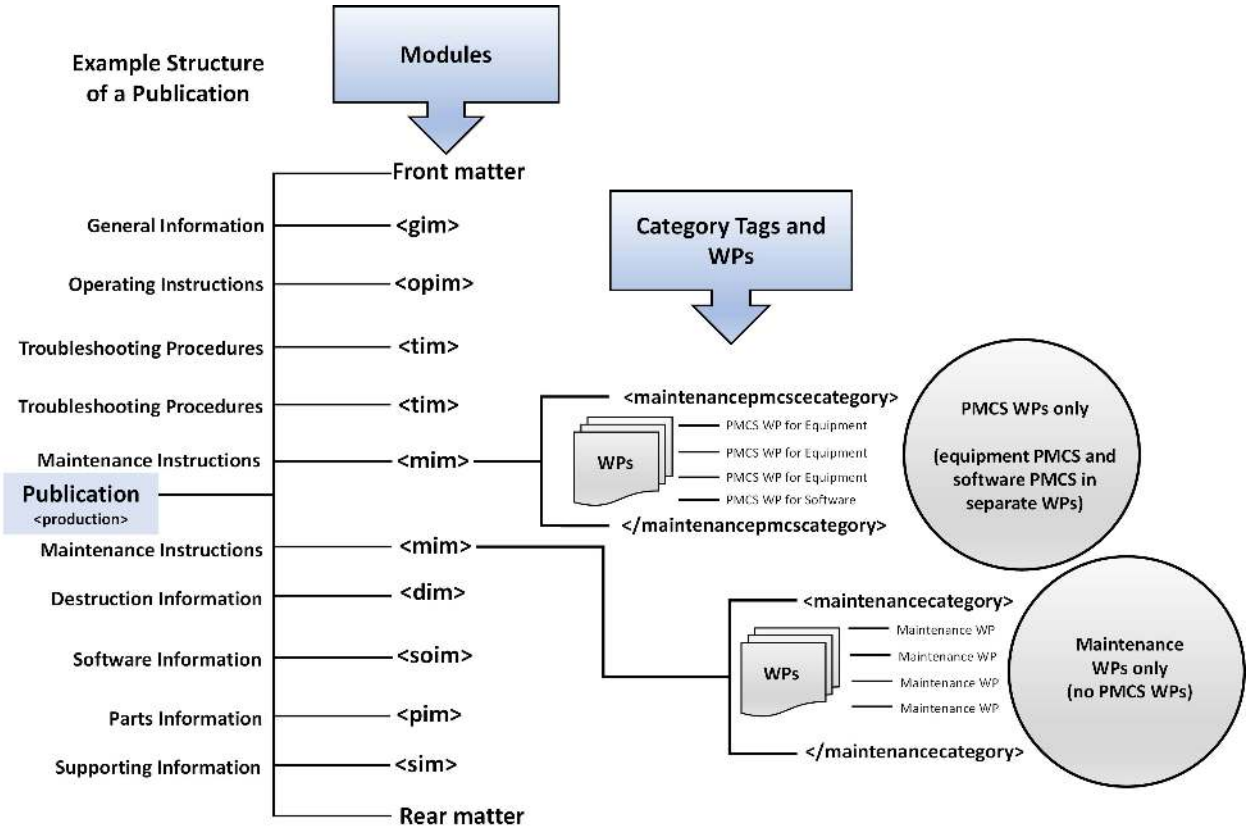


**FIGURE M-24. Example publication including separate PMCS WPs for equipment and software together in a module.**

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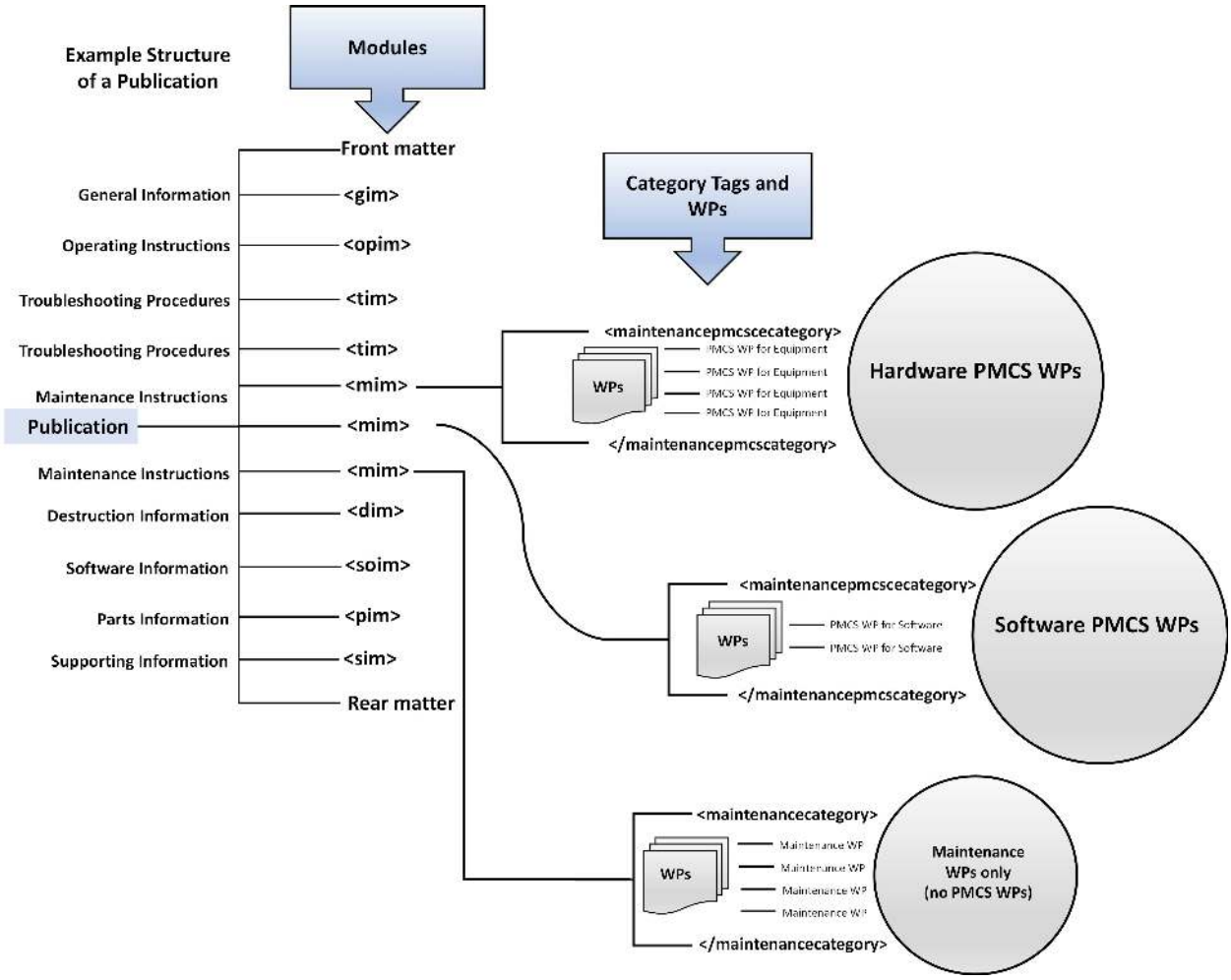


**FIGURE M-25. Example publication including PMCS for hardware and software in separate modules.**

### M.7.9.3.2.2 Software PMCS in a stand-alone SUM/SAM.

For stand-alone SUM/SAMs, software PMCS data shall conform to requirements shown in the matrices in APPENDIX A (TABLE A-VIII).

## M.7.10 Software destruction to prevent enemy use module <dim>.

This module shall be prepared in accordance with APPENDIX H.

## M.7.11 Supporting information module <sim> (Required).

Software supporting information WPs shall include WPs covered in sections through as needed.

### M.7.11.1 References WP <refwp> (Required).

References for the SUM or SAM information shall be included in this WP. This WP shall be prepared in accordance with APPENDIX G.



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### M.7.11.2 Maintenance allocation chart introduction WP <macintrowp>.

This WP shall be prepared for software only systems and shall be prepared in accordance with APPENDIX G.

### M.7.11.3 Maintenance allocation chart WP <macwp>

This WP shall be prepared for software only systems and shall be prepared in accordance with APPENDIX G.

### M.7.11.4 Software basic issue items (BII) WP <softbiiwp> (Required).

This WP shall be prepared and shall contain the following information in the order presented and as applicable.

**a.** WP identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** Initial setup **<initial_setup>** is not required for this WP.

**c.** Introduction for BII. Include the below introduction verbatim.

**Note**

Software-only programs may alter boilerplate as needed. Contact EPCO to discuss appropriate boilerplate alterations.

**"BASIC ISSUE ITEMS (BII) LIST**

**INTRODUCTION**

**Scope**

This WP lists BII for the *insert the end item name* to help you inventory items for safe and efficient operation of the equipment.

**General**

Basic Issue Items (BII). These essential items are required to place the *enter name of the software* in operation, operate it, and to do emergency repairs. Although shipped separately packaged, BII must be with the *enter name of end item*) during operation and when it is transferred between property accounts. Listing these items is your authority to request/requisition them for replacement based on authorization of the end item by the Table of Organization and Equipment/Modified Table of Organization and Equipment (TOE/MTOE). Illustrations are furnished to help you find and identify the items.

**Explanation of Columns in the BII List**

*Select Method A text.*

"Column (1) Illus Number. Gives you the number of the item illustrated.

Column (2) National Stock Number (NSN). Identifies the stock number of the item to be used for requisitioning purposes.

Column (3) Description, Part Number/Commercial and Government Entity Code (CAGEC). Identifies the Federal item name (in all capital letters) followed by a minimum description when needed. When applicable, the stowage location of BII is also included in this column. The last line below the description is the CAGEC (in parentheses) and the part number.

Column (4) Usable On Code. When applicable, gives you a code if the item you need is not the same for different models of equipment. *Add the following only as applicable. Replace As, Bs, or Cs with appropriate codes and model numbers*.) These codes are identified below:

**Code Used on**
AA *Model* XXX

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BB *Model* XXXX

CC *Model* XXXXX

Add if applicable: Model XXX uses BII items (*insert item numbers*), Model XXXX uses BII items (*insert item numbers*), and Model XXXXX uses BII items (*insert item numbers*).

Column (5) U/I. Unit of Issue (U/I) indicates the physical measurement or count of the item as issued per the National Stock Number shown in column (2).

Column (6) Qty Rqr. Indicates the quantity required."

OR

*Select Method B text.*

"Column (1) Item Number. Gives you the reference number of the item listed.

Column (2) National Stock Number (NSN) and Illustration. Identifies the stock number of the item to be used for requisitioning purposes and provides an illustration of the item.

Column (3) Description, Part Number/(CAGEC). Identifies the Federal item name (in all capital letters) followed by a minimum description when needed. When applicable, the stowage location of BII is also included in this column. The last line below the description is the CAGEC (Commercial and Government Entity Code) (in parentheses) and the part number.

Column (4) Usable On Code. When applicable, gives you a code if the item you need is not the same for different models of equipment. (Add the following only as applicable. Replace As, Bs, or Cs with appropriate codes and model numbers.) These codes are identified below:

**Code Used on**

AA *Model* XXX

BB *Model* XXXX

CC *Model* XXXXX

Add if applicable: Model XXX uses BII items (*insert item numbers*), Model XXXX uses BII items (*insert item numbers*), and Model XXXXX use BII items (*insert item numbers*).

Column (5) U/I. Unit of Issue (U/I) indicates the physical measurement or count of the item as issued per the National Stock Number shown in column (2).

Column (6) Qty Rqr. Indicates the quantity required."

**d.** Bii <softbii>.

This list shall be prepared in accordance with APPENDIX G and shall contain as a minimum the SUM and SAM. This list shall follow the introduction. This list shall also include any other software related BII items. This list shall not be a duplication of the BII contained in the hardware operator manual and it shall not include any hardware related BII items. If applicable, the software shall be listed in the BII. Instructions for how to obtain the software/disc shall be included.

### M.7.11.5 Additional authorization list (AAL) WP <aalwp>.

If applicable, this WP may be included in the SUM. If prepared, this WP shall be prepared in accordance with APPENDIX G and shall include only applicable software related AAL items. This WP shall not be a duplication of the AAL contained in the hardware operator manual and it shall not include hardware related AAL items.

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### M.7.11.6 Expendable and durable items list WP <explistwp>.

If applicable, this WP may be included in the SUM and/or SAM. If prepared, this WP shall be prepared in accordance with APPENDIX G and shall include only software related expendable items. This WP shall not be a duplication of the expendable list from the operator and/or maintenance manuals for the hardware or contain any hardware related expendables.

### M.7.11.7 Additional supporting information WP <genwp>.

This WP shall be prepared as required for information not covered in M.7.11.1 through M.7.11.6. This WP shall be prepared in accordance with APPENDIX G.

### M.7.12 Rear Matter.

Rear matter for SUMs and SAMs shall be prepared in accordance with 5.5.

## M.8 NOTES.

The notes in Section 6 apply to this appendix.

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# APPENDIX N

# GENERAL MAINTENANCE MANUALS

## N.1 SCOPE.

The requirements provided in this appendix provide the technical content requirements for the preparation of General Maintenance Manuals. This appendix is a mandatory part of this standard. The information contained herein is intended for compliance.

**Note**

To be compliant with this standard, publications shall be developed within **<production>** (opening) and **</production>** (closing) root tags.

When preparing information in accordance with this appendix, all references shall be linked using the appropriate tags (**<xref>**, **<extref>**, etc.). Links shall not be typed as text without tagging.

### N.1.1 Content tailoring using Appendix A matrices.

Applicability of some requirements contained in this standard are the responsibility of the materiel developer (MATDEV) (refer to AR 750–1, AR 700–127) and shall be accomplished using matrices referenced in TABLE A-I. The applicability of requirements are also designated by statements including, but not limited to, the following:

**a.** "...as specified by the MATDEV."

**b.** "...as required by the MATDEV."

**c.** "Refer to the MATDEV for..."

**d.** "Contact MATDEV if..."

**e.** "The MATDEV shall determine..."

**f.** "...as needed."

#### N.1.1.1 Front and rear matter.

Front and rear matter requirements for documents discussed in this appendix shall be determined in accordance with TABLE A-II.

## N.2 APPLICABLE DOCUMENTS.

The applicable documents in Section 2 apply to this appendix.

## N.3 ACRONYMS AND DEFINITIONS.

The acronyms and terms defined in Section 3 of this standard apply to this appendix.

## N.4 DEVELOPMENT OF GENERAL MAINTENANCE MANUAL MODULES.

When specified by the MATDEV, a general maintenance manual **<genmaintman>** shall be prepared, and content shall be limited to information/procedures related to general maintenance (see TABLE A-XVIII). FIGURE N-1 shows a diagram of modules and work packages in a general maintenance manual. The following modules (and respective WPs) shall be prepared for a general maintenance manual.

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Check the source to verify that this is the current version before use.

**Note**

Tags/output shown in figure(s) below may vary slightly based on version of DTD/stylesheets utilized (see applicable DTD or tag definition list for available tags).

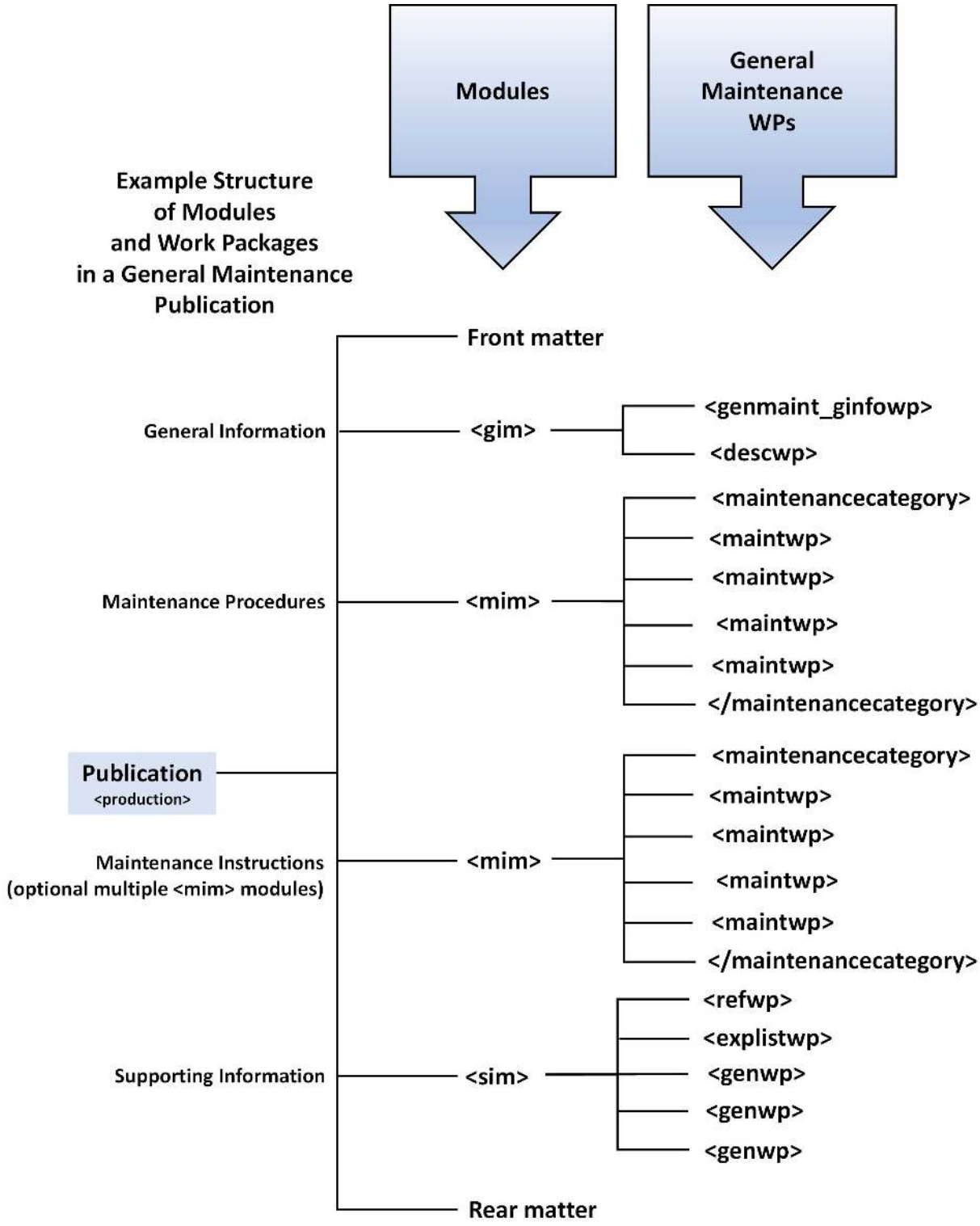


**FIGURE N-1. General maintenance technical publication modules and WPs.**



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Check the source to verify that this is the current version before use.

### N.4.1 General information module <gim>.

A general information module **<gim>** for a general information manual shall be prepared and shall contain the following WPs as required (see ).

**a.** General information WP **<genmaint_ginfowp>** (N.5.1).

**b.** Equipment description and data WP **<descwp>** (N.5.1).

### N.4.2 Maintenance module <mim>.

A general maintenance module shall be prepared and shall contain one or more maintenance modules as needed. These modules shall contain WPs with maintenance procedures and/or general maintenance procedures as listed below.

**a.** Maintenance WP(s) **<maintwp>** (N.5.1).

**b.** General maintenance WP(s) **<gen.maintwp>** (N.5.1).

### N.4.3 Supporting information module <sim>.

A supporting information module shall be prepared and shall contain the following WPs in the order provided, as applicable:

**a.** Reference WPs **<refwp>** (N.5.4).

**b.** Expendable and durable items list WP **<explistwp>** (N.5.5).

**c.** Additional supporting information WP **<genwp>** (N.5.6).

## N.5 DEVELOPMENT OF GENERAL MAINTENANCE MANUAL WORK PACKAGES.

### N.5.1 General information work package <genmaint_ginfowp>.

A general information module **<gim>** for a general information manual shall be prepared and shall contain the following WPs as required (see ). FIGURE N-2 shows an example general information WP. This WP shall contain the following components (in the order presented and as applicable):

**a.** WP identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** Initial setup **<initial_setup>** is not required for this WP.

**c.** Scope **<scope>** is required for this WP (B.7.1.1).

**d.** Maintenance forms, records, and reports **<mfrr>** is required for this WP (B.7.1.2).

**e.** Equipment improvement recommendations **<eir>** is required for this WP (B.7.1.3).

**f.** Policy **<policy>**. As applicable, this paragraph shall contain policy information related to general maintenance.

**g.** Safety **<safety>**. As applicable, this paragraph shall contain safety information related to general maintenance.

**h.** Warranty information **<wrntyref>**. As applicable, warranty information related to the equipment and/or general maintenance shall be included.

**i.** Nomenclature cross-reference list **<nomenreflist>**. As applicable, a nomenclature cross-reference list shall be prepared in accordance with B.7.1.10. This list shall only contain nomenclature related to general maintenance.

**j.** List of abbreviations/acronyms **<loa>** is required for this WP. This paragraph shall contain a list of the general maintenance related abbreviations and acronyms used within the general maintenance manual.

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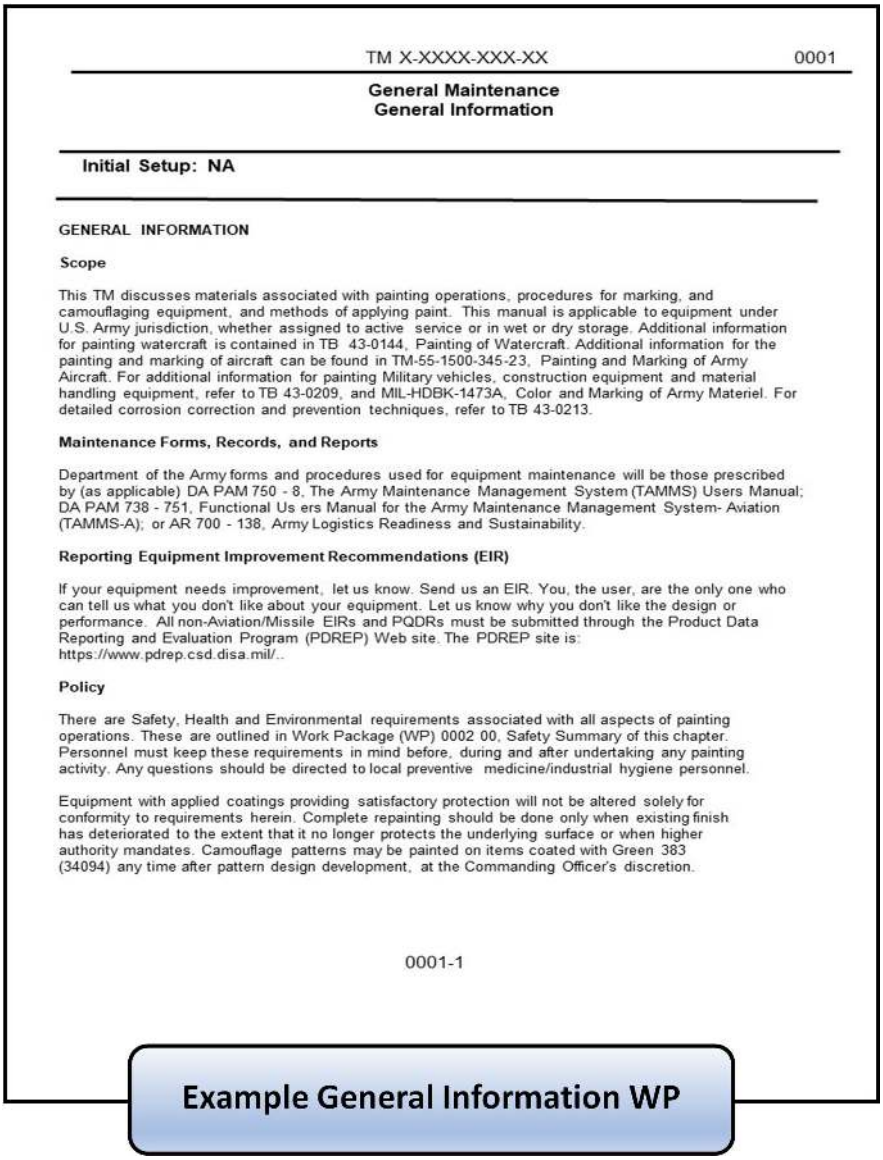
Check the source to verify that this is the current version before use.

**Note**

Figure appearance may vary slightly based on version of DTD/stylesheets utilized.



TM X-XXXX-XXX-XX 0001

**General Maintenance**
**General Information**

**Initial Setup: NA**

**GENERAL INFORMATION**

**Scope**

This TM discusses materials associated with painting operations, procedures for marking, and camouflaging equipment, and methods of applying paint. This manual is applicable to equipment under U.S. Army jurisdiction, whether assigned to active service or in wet or dry storage. Additional information for painting watercraft is contained in TB 43-0144, Painting of Watercraft. Additional information for the painting and marking of aircraft can be found in TM-55-1500-345-23, Painting and Marking of Army Aircraft. For additional information for painting Military vehicles, construction equipment and material handling equipment, refer to TB 43-0209, and MIL-HDBK-1473A, Color and Marking of Army Materiel. For detailed corrosion correction and prevention techniques, refer to TB 43-0213.

**Maintenance Forms, Records, and Reports**

Department of the Army forms and procedures used for equipment maintenance will be those prescribed by (as applicable) DA PAM 750 - 8, The Army Maintenance Management System (TAMMS) Users Manual; DA PAM 738 - 751, Functional Us ers Manual for the Army Maintenance Management System- Aviation (TAMMS-A); or AR 700 - 138, Army Logistics Readiness and Sustainability.

**Reporting Equipment Improvement Recommendations (EIR)**

If your equipment needs improvement, let us know. Send us an EIR. You, the user, are the only one who can tell us what you don't like about your equipment. Let us know why you don't like the design or performance. All non-Aviation/Missile EIRs and PQDRs must be submitted through the Product Data Reporting and Evaluation Program (PDREP) Web site. The PDREP site is: https://www.pdrep.csd.disa.mil/..

**Policy**

There are Safety, Health and Environmental requirements associated with all aspects of painting operations. These are outlined in Work Package (WP) 0002 00, Safety Summary of this chapter. Personnel must keep these requirements in mind before, during and after undertaking any painting activity. Any questions should be directed to local preventive medicine/industrial hygiene personnel.

Equipment with applied coatings providing satisfactory protection will not be altered solely for conformity to requirements herein. Complete repainting should be done only when existing finish has deteriorated to the extent that it no longer protects the underlying surface or when higher authority mandates. Camouflage patterns may be painted on items coated with Green 383 (34094) any time after pattern design development, at the Commanding Officer's discretion.

0001-1



**FIGURE N-2. General information WP.**

## N.5.2 Equipment description and data work package <descwp>.

As applicable, an equipment description and data work package may be included and shall be prepared in accordance with B.7.2.

## N.5.3 Maintenance instruction work package <maintwp> and general maintenance WP(s) <gen.maintwp>.

Maintenance WPs shall be prepared and shall contain the following components as applicable. Refer to FIGURE N-3 shows an example general maintenance WP.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:03Z
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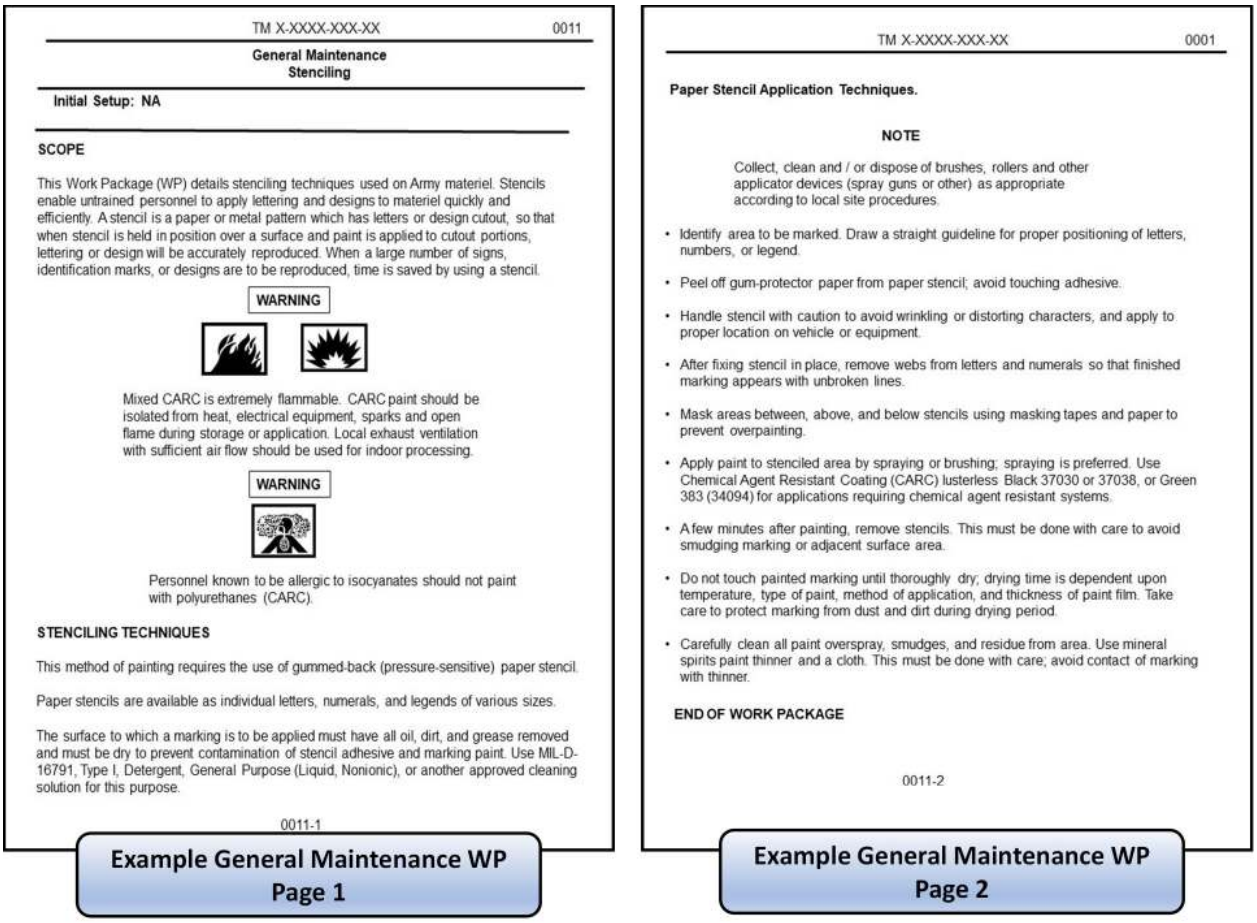
**a.** Maintenance instruction WPs **<maintwp>** shall be developed in accordance with E.7.4.

**b.** General maintenance WPs **<gen.maintwp>**shall be developed in accordance with E.7.5.

.

**Note**

Figure appearance may vary slightly based on version of DTD/stylesheets utilized.



TM X-XXXX-XXX-XX 0011

General Maintenance
Stenciling

Initial Setup: NA

SCOPE

This Work Package (WP) details stenciling techniques used on Army materiel. Stencils enable untrained personnel to apply lettering and designs to materiel quickly and efficiently. A stencil is a paper or metal pattern which has letters or design cutout, so that when stencil is held in position over a surface and paint is applied to cutout portions, lettering or design will be accurately reproduced. When a large number of signs, identification marks, or designs are to be reproduced, time is saved by using a stencil.

WARNING

Mixed CARC is extremely flammable. CARC paint should be isolated from heat, electrical equipment, sparks and open flame during storage or application. Local exhaust ventilation with sufficient air flow should be used for indoor processing.

WARNING

Personnel known to be allergic to isocyanates should not paint with polyurethanes (CARC).

STENCILING TECHNIQUES

This method of painting requires the use of gummed-back (pressure-sensitive) paper stencil.

Paper stencils are available as individual letters, numerals, and legends of various sizes.

The surface to which a marking is to be applied must have all oil, dirt, and grease removed and must be dry to prevent contamination of stencil adhesive and marking paint. Use MIL-D-16791, Type I, Detergent, General Purpose (Liquid, Nonionic), or another approved cleaning solution for this purpose.

0011-1

Example General Maintenance WP Page 1

TM X-XXXX-XXX-XX 0001

Paper Stencil Application Techniques.

NOTE

Collect, clean and / or dispose of brushes, rollers and other applicator devices (spray guns or other) as appropriate according to local site procedures.

- Identify area to be marked. Draw a straight guideline for proper positioning of letters, numbers, or legend.
- Peel off gum-protector paper from paper stencil; avoid touching adhesive.
- Handle stencil with caution to avoid wrinkling or distorting characters, and apply to proper location on vehicle or equipment.
- After fixing stencil in place, remove webs from letters and numerals so that finished marking appears with unbroken lines.
- Mask areas between, above, and below stencils using masking tapes and paper to prevent overpainting.
- Apply paint to stenciled area by spraying or brushing; spraying is preferred. Use Chemical Agent Resistant Coating (CARC) lusterless Black 37030 or 37038, or Green 383 (34094) for applications requiring chemical agent resistant systems.
- A few minutes after painting, remove stencils. This must be done with care to avoid smudging marking or adjacent surface area.
- Do not touch painted marking until thoroughly dry; drying time is dependent upon temperature, type of paint, method of application, and thickness of paint film. Take care to protect marking from dust and dirt during drying period.
- Carefully clean all paint overspray, smudges, and residue from area. Use mineral spirits paint thinner and a cloth. This must be done with care; avoid contact of marking with thinner.

END OF WORK PACKAGE

0011-2

Example General Maintenance WP Page 2

**FIGURE N-3. General maintenance WP example.**

## N.5.4 Reference work package <refwp>.

References for the general maintenance manual shall be included in this WP. This WP shall be prepared in accordance with G.6.1.

## N.5.5 Expendable and durable items list work package <explistwp>.

Expendable and durable items required for general maintenance shall be included in this WP. This WP shall be prepared in accordance with G.7.7.

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### N.5.6 Additional supporting information work package <genwp>.

This WP shall be prepared as required for information not covered in N.5.4 through N.5.5. This WP shall be prepared in accordance with G.7.12.

## N.6 NOTES.

The notes in Section 6 apply to this appendix.

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# APPENDIX O

# HAND RECEIPTS

## O.1 SCOPE.

A hand receipt (HR) is a signed document acknowledging acceptance of and responsibility for items of property. This appendix establishes the technical content requirements for the preparation of an operator's HR for major equipment systems and their related systems, subsystems, equipment, assemblies, components, shop replaceable units (SRUs), and line replaceable units (LRUs). This appendix is a mandatory part of the standard. The information contained herein is intended for compliance. The requirements are applicable for all maintenance levels through depot, including Depot Maintenance Work requirements (DMWRs) and National Maintenance Work Requirements (NMWRs).

**Note**

To be compliant with this standard, publications shall be developed within **<production>** (opening) and **</production>** (closing) root tags.

When preparing information in accordance with this appendix, all references shall be linked using the appropriate tags (**<xref>**, **<extref>**, etc.). Links shall not be typed as text without tagging.

### O.1.1 Content tailoring using Appendix A matrices.

Applicability of some requirements contained in this standard are the responsibility of the materiel developer (MATDEV) (refer to AR 750–1, AR 700–127) and shall be accomplished using matrices referenced in TABLE A-I. The applicability of requirements are also designated by statements including, but not limited to, the following:

**a.** "...as specified by the MATDEV."

**b.** "...as required by the MATDEV."

**c.** "Refer to the MATDEV for..."

**d.** "Contact MATDEV if..."

**e.** "The MATDEV shall determine..."

**f.** "...as needed."

#### O.1.1.1 Front and rear matter.

Front and rear matter requirements for documents discussed in this appendix shall be determined in accordance with TABLE A-II.

## O.2 APPLICABLE DOCUMENTS.

The applicable documents in Section 2 apply to this appendix.

## O.3 ACRONYMS AND DEFINITIONS.

The acronyms and terms defined in Section 3 of this standard apply to this appendix.

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## O.4 DEVELOPMENT OF HAND RECEIPT MODULES.

When specified by the MATDEV, an HR **<hr>** shall be prepared. The modules and WPs listed below shall be developed as required (see TABLE A-XIX). Refer to FIGURE O-1 for a graphical representation of module and WP organization as well as tags discussed in this section.

**Note**

Tags/output shown in figure(s) below may vary slightly based on version of DTD/stylesheets utilized (see applicable DTD or tag definition list for available tags).

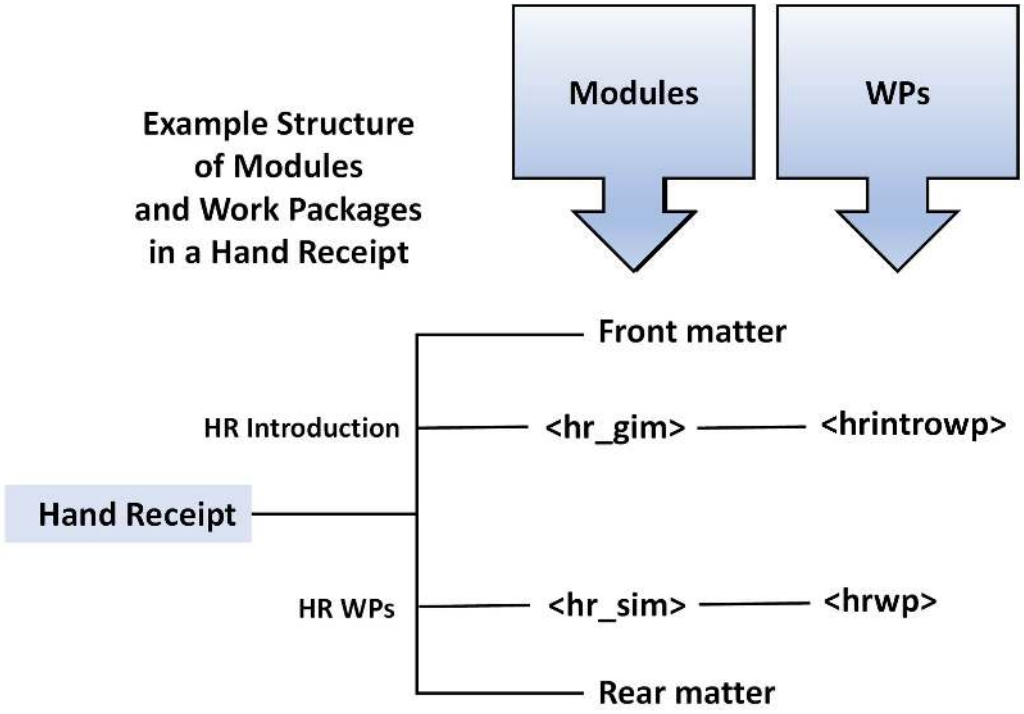


**FIGURE O-1. Hand receipt modules and WPs.**

### O.4.1 HR introduction module <hr_gim>.

An HR introduction module **<hr_gim>** shall be prepared. It shall consist of the following WP.

**a.** HR introduction WP **<hrintrowp>** (O.5.1).

### O.4.2 HR module <hr_sim>.

An HR module **<hr_sim>** shall be prepared. It shall consist of the following WP.

**a.** HR WP **<hrwp>** (O.5.2).

## O.5 DEVELOPMENT OF HAND RECEIPT WORK PACKAGES.

The following types of hand receipt WPs shall be developed.


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## O.5.1 HR introduction work package <hrintrowp>.

An HR introduction WP shall be prepared and shall contain the following paragraphs:ne

**a.** Scope **<scope>**. This paragraph shall describe the scope and purpose of the HR.

**b.** General. This paragraph shall explain the overprinted DA Form 2062 and its purpose, give local reproduction authorization, and provide an address for requisitioning additional copies of HRs.

**c.** Explanation of Blocks and Columns (DA Form 2062). This paragraph shall include explanations of all applicable codes used on the DA Form 2062 (for example, controlled inventory item code (formerly SEC) and Accounting Requirements Code (ARC)).

**d.** Authorization documents. This paragraph shall provide the authorization documents for COEI, BII, and AAL.

**e.** Additional introductory information **<para0>**. Additional paragraphs addressing other introductory information may be added as appropriate (refer to FIGURE O-2 through FIGURE O-4 for an example HR introduction.

**Note**

Figure appearance may vary slightly based on version of DTD/stylesheets utilized.

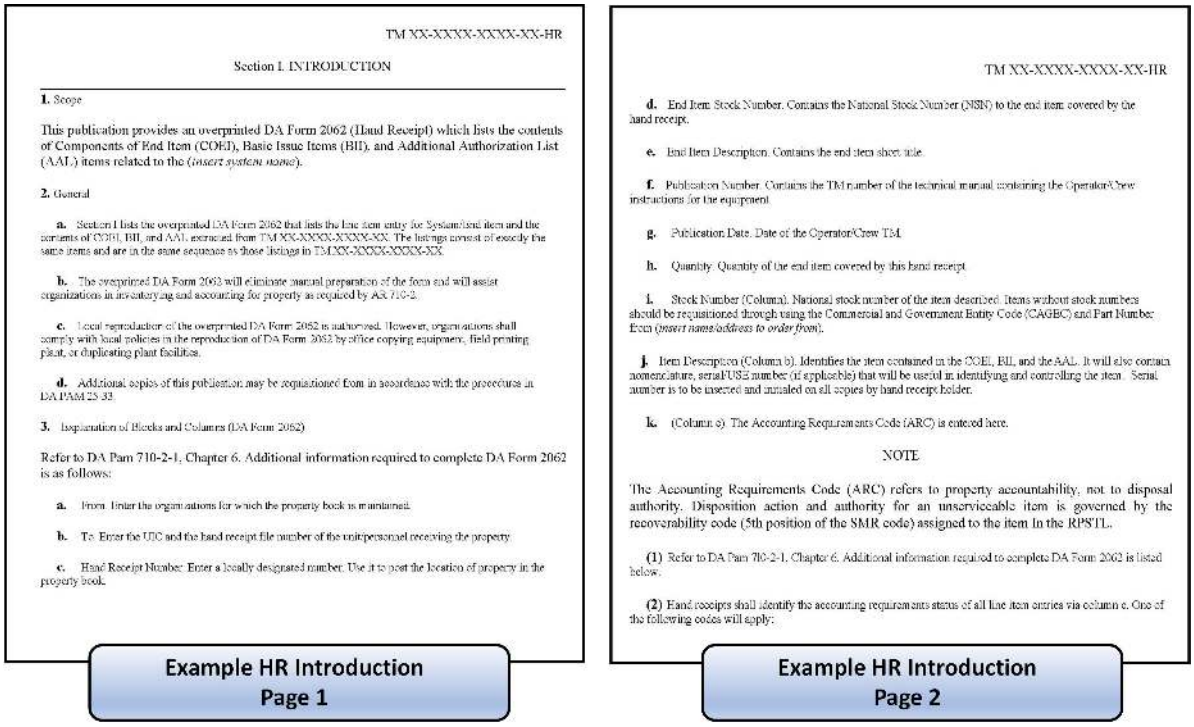


**FIGURE O-2. Example HR introduction (pages 1 and 2).**

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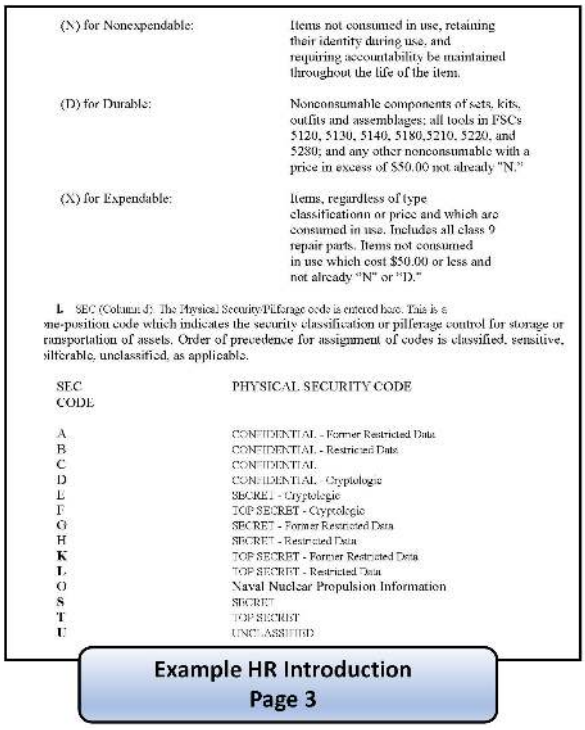


(N) for Nonexpendable: Items not consumed in use, retaining their identity during use, and requiring accountability be maintained throughout the life of the item.

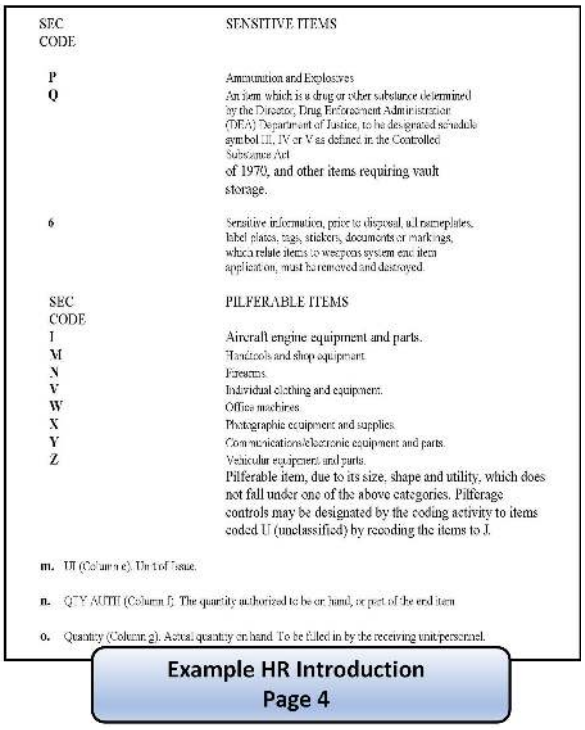
(D) for Durable: Nonconsumable components of sets, kits, outfits and assemblages; all tools in FSCs 5120, 5130, 5140, 5180,5210, 5220, and 5280; and any other nonconsumable with a price in excess of $50.00 not already "N."

(X) for Expendable: Items, regardless of type classificationn or price and which are consumed in use. Includes all class 9 repair parts. Items not consumed in use which cost $50.00 or less and not already "N" or "D."

**L** SEC (Column d). The Physical Security/Pilferage code is entered here. This is a one-position code which indicates the security classification or pilferage control for storage or transportation of assets. Order of precedence for assignment of codes is classified, sensitive, pilferable, unclassified, as applicable.

| SEC CODE | PHYSICAL SECURITY CODE |
|---|---|
| A | CONFIDENTIAL - Former Restricted Data |
| B | CONFIDENTIAL - Restricted Data |
| C | CONFIDENTIAL |
| D | CONFIDENTIAL - Cryptologic |
| E | SECRET - Cryptologic |
| F | TOP SECRET - Cryptologic |
| G | SECRET - Former Restricted Data |
| H | SECRET - Restricted Data |
| K | TOP SECRET - Former Restricted Data |
| L | TOP SECRET - Restricted Data |
| O | Naval Nuclear Propulsion Information |
| S | SECRET |
| T | TOP SECRET |
| U | UNCLASSIFIED |

Example HR Introduction
Page 3



| SEC CODE | SENSITIVE ITEMS |
|---|---|
| P | Ammunition and Explosives |
| Q | An item which is a drug or other substance determined by the Director, Drug Enforcement Administration (DEA) Department of Justice, to be designated schedule symbol III, IV or V as defined in the Controlled Substance Act of 1970, and other items requiring vault storage. |
| 6 | Sensitive information, prior to disposal, all nameplates, label plates, tags, stickers, documents or markings, which relate items to weapons system end item application, must be removed and destroyed. |

| SEC CODE | PILFERABLE ITEMS |
|---|---|
| I | Aircraft engine equipment and parts. |
| M | Handtools and shop equipment. |
| N | Firearms. |
| V | Individual clothing and equipment. |
| W | Office machines. |
| X | Photographic equipment and supplies. |
| Y | Communications/electronic equipment and parts. |
| Z | Vehicular equipment and parts. |
| | Pilferable item, due to its size, shape and utility, which does not fall under one of the above categories. Pilferage controls may be designated by the coding activity to items coded U (unclassified) by recoding the items to J. |

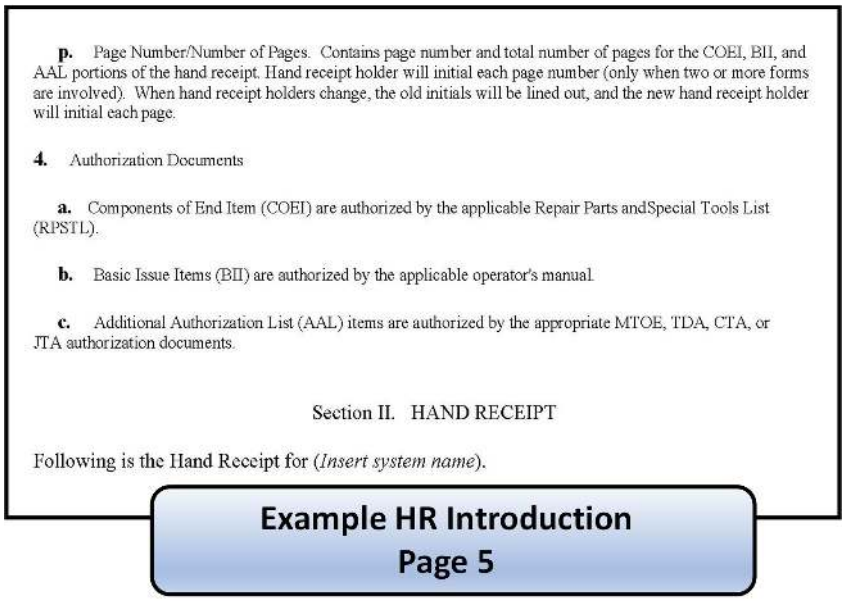
**m.** UI (Column e). Unit of Issue.

**n.** QTY AUTH (Column f). The quantity authorized to be on hand, or part of the end item

**o.** Quantity (Column g). Actual quantity on hand. To be filled in by the receiving unit/personnel.

Example HR Introduction
Page 4

**FIGURE O-3. Example HR introduction (pages 3 and 4).**



**p.** Page Number/Number of Pages. Contains page number and total number of pages for the COEI, BII, and AAL portions of the hand receipt. Hand receipt holder will initial each page number (only when two or more forms are involved). When hand receipt holders change, the old initials will be lined out, and the new hand receipt holder will initial each page.

**4.** Authorization Documents

**a.** Components of End Item (COEI) are authorized by the applicable Repair Parts andSpecial Tools List (RPSTL).

**b.** Basic Issue Items (BII) are authorized by the applicable operator's manual.

**c.** Additional Authorization List (AAL) items are authorized by the appropriate MTOE, TDA, CTA, or JTA authorization documents.

Section II. HAND RECEIPT

Following is the Hand Receipt for (*Insert system name*).

Example HR Introduction
Page 5

**FIGURE O-4. Example HR introduction (page 5).**

## O.5.2 HR work packages <hrwp>.

The HR shall contain one or more HR work packages containing:

**a. Lead-in paragraph.** This paragraph shall contain the following lead-in sentence:

"Following is the Hand Receipt for *insert system name* )."



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Check the source to verify that this is the current version before use.

**b. Technical content.** The hand receipt work package shall contain DA Form 2062 in the form of a graphic. The DA Form 2062 shall consist of COEI, BII, and AAL contents extracted from the applicable validated and verified operator's manual. Refer to O.5.2.1 and O.5.2.2 for technical content requirement details.

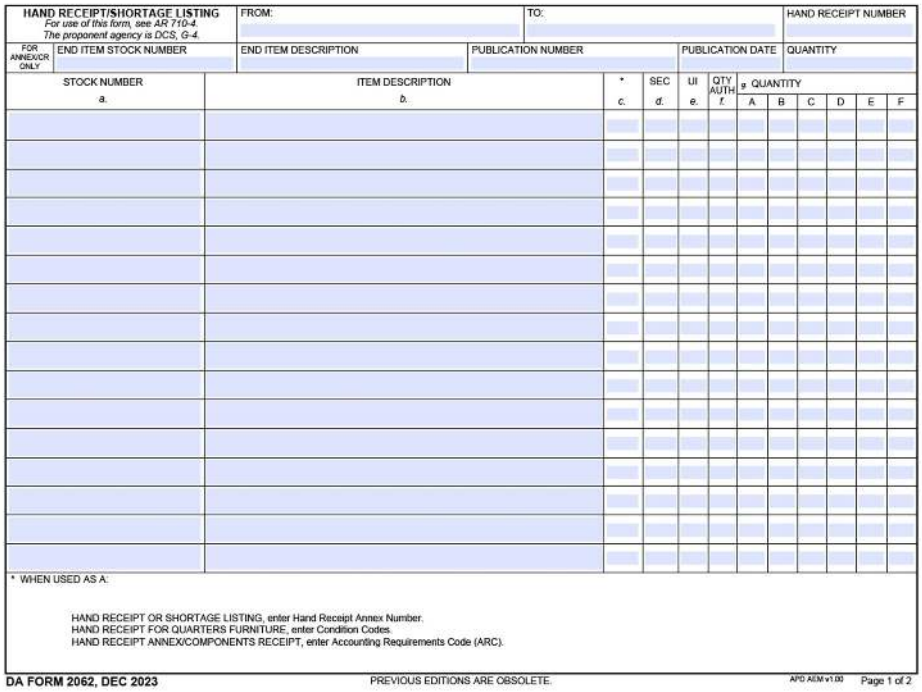
## O.5.2.1 Overprinted DA Form 2062.

The hand receipt work packages shall include an overprinted DA Form 2062 for line item entry for system/end item and the contents of the applicable COEI, BII, and AAL. The hand receipt work package COEI, BII, and AAL title headers and listings shall be in the same sequence as used in the related operator technical manual (i.e., (1) system/ end item line item entry; (2) COEI; (3) BII; and (4) AAL). Hand receipt contents shall be as follows:

**a.** The related technical publication number and date of publication, end item stock number, and the end item description and quantity shall be shown in the applicable blocks in the heading of DA Form 2062. The line item entry for the end item and contents of COEI, BII and AAL lists, with applicable headers, shall comprise one DA Form 2062 (front side with continuation sheet(s), see FIGURE O-5 and FIGURE O-6).

**Note**

Figure appearance may vary slightly based on version of DTD/stylesheets utilized.



HAND RECEIPT/SHORTAGE LISTING
*For use of this form, see AR 710-4.*
*The proponent agency is DCS, G-4.*

FROM: | TO: | HAND RECEIPT NUMBER

FOR ANNEX/OR ONLY | END ITEM STOCK NUMBER | END ITEM DESCRIPTION | PUBLICATION NUMBER | PUBLICATION DATE | QUANTITY

| STOCK NUMBER a. | ITEM DESCRIPTION b. | * c. | SEC d. | UI e. | QTY AUTH f. | g. QUANTITY A | B | C | D | E | F |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | |

* WHEN USED AS A:

HAND RECEIPT OR SHORTAGE LISTING, enter Hand Receipt Annex Number.
HAND RECEIPT FOR QUARTERS FURNITURE, enter Condition Codes.
HAND RECEIPT ANNEX/COMPONENTS RECEIPT, enter Accounting Requirements Code (ARC).

DA FORM 2062, DEC 2023 | PREVIOUS EDITIONS ARE OBSOLETE. | APD AEM v1.00 | Page 1 of 2



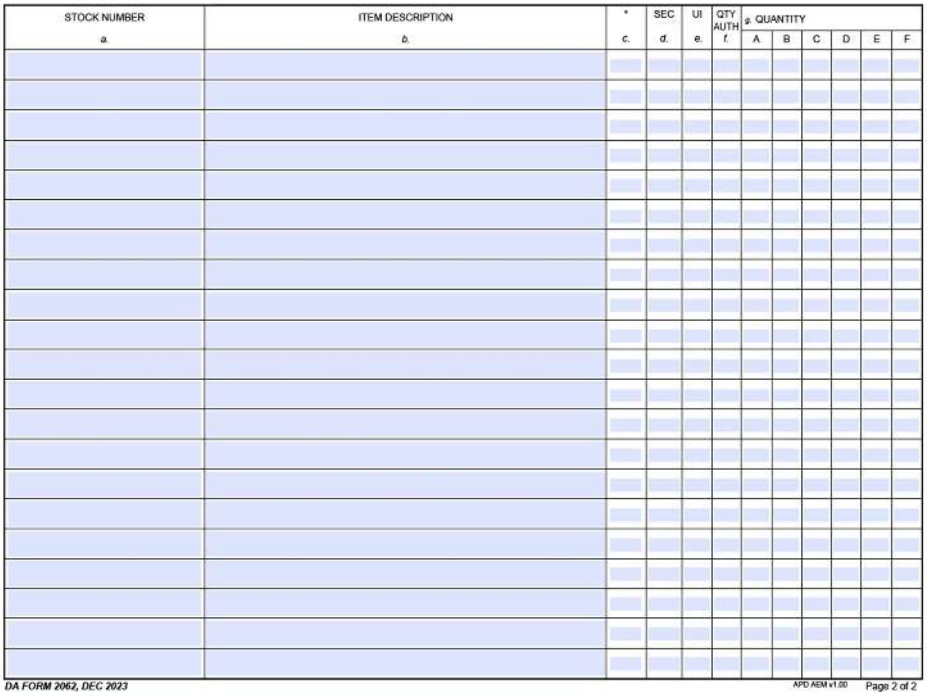
**FIGURE O-5. Example DA form 2062 (front side).**



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:03Z
Check the source to verify that this is the current version before use.

**Note**

Figure appearance may vary slightly based on version of DTD/stylesheets utilized.



| STOCK NUMBER<br>a. | ITEM DESCRIPTION<br>b. | *<br>c. | SEC<br>d. | UI<br>e. | QTY AUTH<br>f. | g. QUANTITY<br>A | B | C | D | E | F |
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DA FORM 2062, DEC 2023 APD AEM v1.00 Page 2 of 2

Example DA Form 2062
(back side)

**FIGURE O-6. Example DA form 2062 (back side).**

**b.** The continuation sheet(s) shall contain the technical publication number and short end item title placed at the top left (outside the margin of the DA Form 2062). When COEI, BII, and/or AAL listings require a continuation sheet, the first line of the description column shall include the applicable title (i.e., COEI, BII, or AAL) followed by a dash and the word "Continued" (see FIGURE O-5).

**c.** Sufficient space (8 lines minimum) shall be left at the end of a listing for signature of recipient of COEI, BII, and AAL items. If a list of components extends too far on the page to allow for signature in the balance columns, a blank continuation sheet shall be added (see FIGURE O-6).

**d.** When a title header (for example, COEI, BII, or AAL) has an applicable listing in the operator technical manual, the technical publication hand receipt shall consist of the title header followed by the line item entry for NSN; brief item description (which shall include Commercial and Government Entity Code (CAGEC) and Part Number (PN), in that order); the accounting requirements code; the physical security/pilferage code; unit of issue information; and quantity authorized.

**e.** When a title header (for example, COEI, BII, or AAL) has no applicable listing in the operator technical manual, the title header shall appear on the technical publication hand receipt with the line entry "NOT APPLICABLE" (see FIGURE O-7).



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**Note**

Figure appearance may vary slightly based on version of DTD/stylesheets utilized.

TM 10-9925-100-10-HR Containerized Chapel

| STOCK NUMBER a. | ITEM DESCRIPTION b. | * c. | SEC d. | UI e. | QTY AUTH f. | g. QUANTITY A | B | C | D | E | F |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 6110-01-413-6474 | Convenience Outlet Assembly, with Cable | N | U | EA | 4 | | | | | | |
| 6110-01-251-0402 | TEMPER Electrical Box Type III, 120V | N | U | EA | 2 | | | | | | |
| | BASIC ISSUE ITEMS LIST | | | | | | | | | | |
| | TM 10-9925-100-12&P | N | U | EA | 1 | | | | | | |
| 7520-00-559-9618 | Pouch, Documentation | N | C | EA | 1 | | | | | | |
| | Tie Down Strap | N | U | PR | 5 | | | | | | |
| 3390-01-204-3009 | Tie Down Strap, Sling Gear | N | U | EA | 7 | | | | | | |
| 5975-00-878-3791 | Grounding Rod, Sectional, Type III Class B, with attachments | N | U | EA | 1 | | | | | | |
| | ADDITIONAL AUTHORIZATIONS LIST | | | | | | | | | | |
| | NOT APPLICABLE | | | | | | | | | | |

Reverse of DA Form 2062 CURRENT AS OF 30 NOVEMBER 2003 PAGE 7 OF 7 PAGES

TM 10-9925-100-10-HR

Example DA Form 2062 (Continuation Sheet With "NOT APPLICABLE" entry)

**FIGURE O-7. Example DA form 2062 (continuation sheet with "NOT APPLICABLE" entry).**

## O.5.2.2 "Current as of" date.

The statement "Current as of" shall be shown at the bottom of each DA Form 2062 as shown in FIGURE O-5. The cited date shall be the publication date of the current operator's manual/change from which the technical publication hand receipt data was extracted.

## O.6 NOTES.

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

## O.6.1 Acquisition requirements.

Acquisition requirements. The MATDEV should specify the inspection intervals to be included in the PMC (refer to APPENDIX E).



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# APPENDIX P

# AMMUNITION AND MUNITIONS EQUIPMENT DATA SHEETS TECHNICAL MANUALS

## P.1 SCOPE.

This appendix establishes the technical content requirements for the preparation of an ammunition or munitions equipment data sheets technical publication, referred to herein as data sheets publication. This appendix is a mandatory part of the standard. The information in this appendix is intended for compliance.

**Note**

To be compliant with this standard, publications shall be developed within **<production>** (opening) and **</production>** (closing) root tags.

When preparing information in accordance with this appendix, all references shall be linked using the appropriate tags (**<xref>**, **<extref>**, etc.).Links shall not be typed as text without tagging.

### P.1.1 Content tailoring using Appendix A matrices.

Applicability of some requirements contained in this standard are the responsibility of the materiel developer (MATDEV) (refer to AR 750–1, AR 700–127) and shall be accomplished using matrices referenced in TABLE A-I. The applicability of requirements are also designated by statements including, but not limited to, the following:

**a.** "...as specified by the MATDEV."

**b.** "...as required by the MATDEV."

**c.** "Refer to the MATDEV for..."

**d.** "Contact MATDEV if..."

**e.** "The MATDEV shall determine..."

**f.** "...as needed."

#### P.1.1.1 Front and rear matter.

Front and rear matter requirements for documents discussed in this appendix shall be determined in accordance with TABLE A-II.

## P.2 APPLICABLE DOCUMENTS.

The applicable documents in Section 2 apply to this appendix.

## P.3 ACRONYMS AND DEFINITIONS.

The acronyms and terms defined in Section 3 of this standard apply to this appendix.

## P.4 AMMUNITION AND MUNITIONS EQUIPMENT DATA SHEETS PUBLICATION FUNCTIONALITY.

TABLE P-I lists work packages (WPs) in an ammunition data sheets publication **<ammo_ds>**. TABLE P-II lists WPs in a munitions equipment data sheets publication **<ammo_ds>**. FIGURE P-1 and FIGURE P-2 are diagrams of modules and work packages for these publications.

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**TABLE P-I. Ammunition data sheets publication WPs.**

| WP | WP Tag | Appendix Section |
|---|---|---|
| General information WP [1] | **<ginfowp>** | P.6.1 |
| Data sheets introduction WP | **<ammods_introwp>** | P.6.2 |
| Data sheets WP [2] | **<ammo_dswp>** | P.6.3 |
| References WP | **<refwp>** | P.6.4 |
| Other supporting information WP | **<genwp>** | P.6.8 |

[1] WP type duplication is prohibited.

[2] WP type frequently duplicated in a module for effective information display.

**TABLE P-II. Munitions equipment data sheets publication WPs.**

| WP | WP Tag | Appendix Section |
|---|---|---|
| General information WP [1] | **<ginfowp>** | P.6.1 |
| Data sheets introduction WP | **<ammods_introwp>** | P.6.2 |
| Data sheets WP [2] | **<ammo_dswp>** | P.6.3 |
| References WP | **<refwp>** | P.6.4 |
| Deleted items WP | **<genwp>** | P.6.5 |
| Operational index WP | **<genwp>** | P.6.6 |
| Preparation and handling of ammunition peculiar equipment (APE) for shipment and storage WP | **<genwp>** | P.6.7 |
| Other supporting information WP | **<genwp>** | P.6.8 |

[1] WP type duplication is prohibited.

[2] WP type frequently duplicated in a module for effective information display.



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**Note**

Tags/output shown in figure(s) below may vary slightly based on version of DTD/stylesheets utilized (see applicable DTD or tag definition list for available tags).

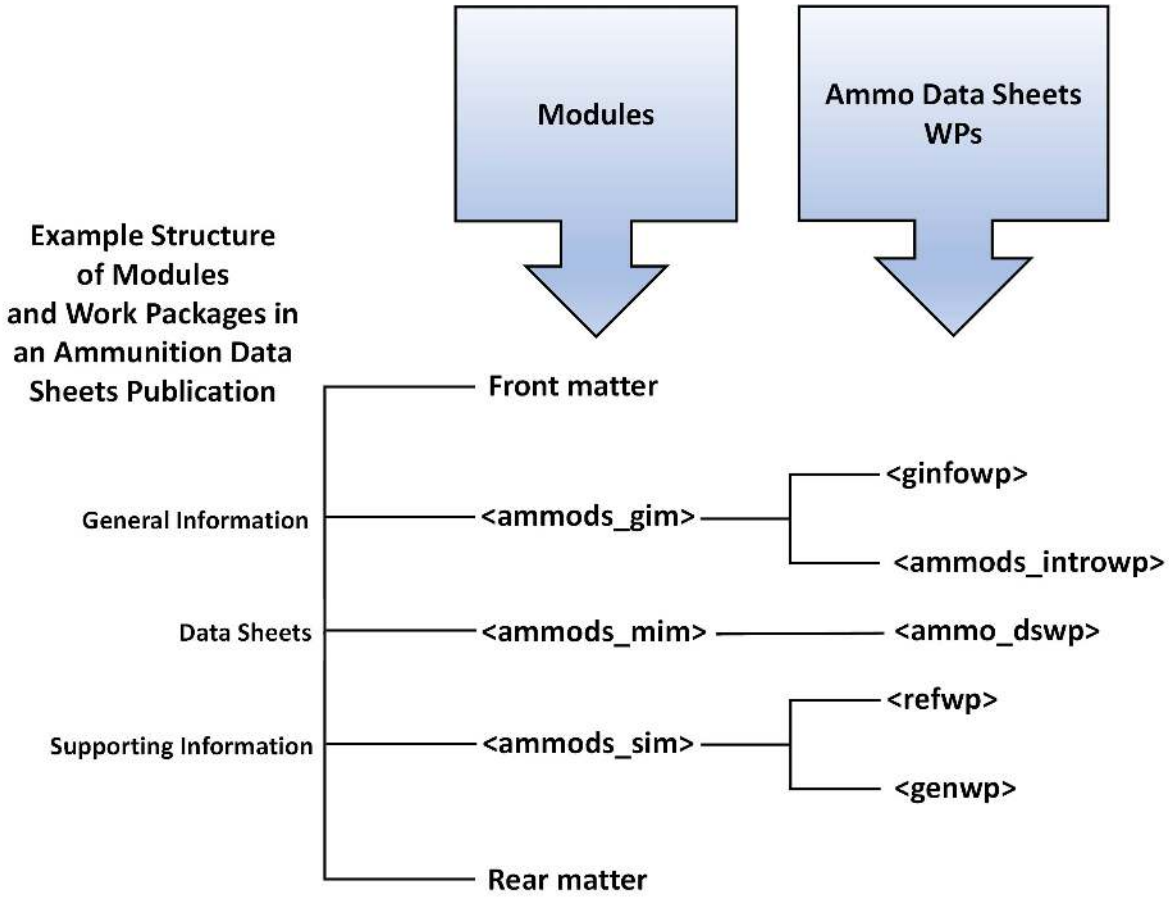


**FIGURE P-1. Ammunition data sheets publication modules and WPs.**

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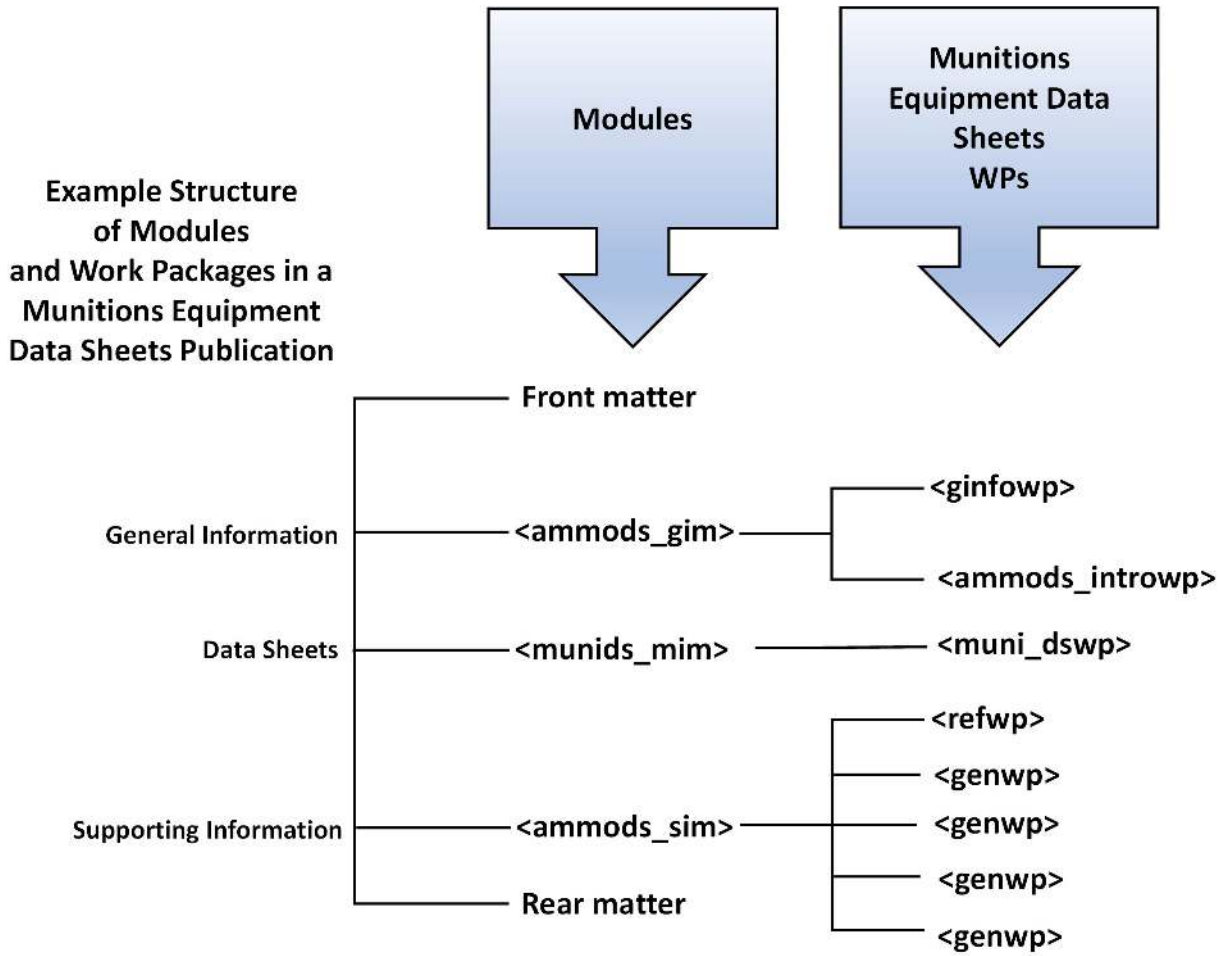


**FIGURE P-2. Munitions equipment data sheets publication modules and WPs.**

## P.5 DEVELOPMENT OF AMMUNITION OR MUNITIONS EQUIPMENT DATA SHEETS MODULES.

When specified by the materiel developer (MATDEV), an ammunition data sheets publication **<ammo_ds>** or a munitions equipment data sheets publication **<ammo_ds>** shall be prepared (see TABLE A-X). Refer to P.5.1 for ammunition data sheet module requirements. Refer to P.5.2 for munitions equipment data sheets module requirements.

### P.5.1 Ammunition data sheet modules.

The ammunition data sheets modules and WPs listed below shall be developed as required (see TABLE A-X). Refer to FIGURE P-1 for a graphical representation of module and WP organization as well as tags in a ammunition data sheets publication.

#### P.5.1.1 General information and data sheets introduction module <ammods_gim>.

A general information and data sheets introduction module **<ammods_gim>** shall be prepared. It shall consist of the following WPs.

**a.** General information WP **<ginfowp>** (P.6.1).


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**b.** Data sheets introduction WP **<ammods_introwp>** (P.6.2).

### P.5.1.2 Army ammunition data sheets module <ammods_mim>.

A ammunition data sheets module **<ammods_mim>** shall be prepared. It shall consist of the following WP(s).

**a.** Data sheets WP **<ammo_dswp>** (P.6.3).

**b.** Additional data sheets WP(s) as needed.

### P.5.1.3 Supporting information module <ammods_sim>.

A ammunition data sheets supporting information module **<ammods_sim>** shall be prepared. It shall consist of the following WP(s) as applicable.

**a.** References WP **<refwp>** (P.6.4).

**b.** Other supporting information WP **<genwp>** (P.6.8).

## P.5.2 Munitions equipment data sheet modules.

The munitions equipment data sheets modules and WPs listed below shall be developed as required (see TABLE A-X). Refer to FIGURE P-2 for a graphical representation of module and WP organization as well as tags in a munitions equipment data sheets publication.

### P.5.2.1 General information and data sheets introduction module <ammods_gim>.

A general information and data sheets introduction module **<ammods_gim>** shall be prepared. It shall consist of the following WPs.

**a.** General information WP **<ginfowp>** (P.6.1).

**b.** Data sheets introduction WP **<ammods_introwp>** (P.6.2).

### P.5.2.2 Army munition equipment data sheets module <munids_mim>.

A munitions equipment data sheets module **<munids_mim>** shall be prepared. It shall consist of the following WP (s).

**a.** Data sheets WP **<muni_dswp>** (P.6.3).

**b.** Additional data sheets WP(s) as needed.

### P.5.2.3 Supporting information module <ammods_sim>.

A supporting information module **<ammods_sim>** shall be prepared. It shall consist of the following WP as applicable.

**a.** References WP **<refwp>** (P.6.4).

**b.** Deleted items WP **<genwp>** (P.6.5).

**c.** Operational index WP **<genwp>** (P.6.6).

**d.** Preparation and handling of ammunition peculiar equipment (APE) for shipment and storage WP **<genwp>** (P.6.7).

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**e.** Other supporting information WP **<genwp>** (P.6.8).

## P.6 DEVELOPMENT OF AMMUNITION OR MUNITIONS EQUIPMENT DATA SHEETS WORK PACKAGES.

### P.6.1 General information work package <ginfowp>.

A general information WP shall be prepared in accordance with B.7.1. and shall contain the following information (at a minimum):

**a.** Work package identification information **<wpidinfo>** is required for this WP (P.6.1).

**b.** Initial setup **<initial_setup>** is not required for this WP.

**c.** Scope **<scope>** (B.7.1.1).

**d.** Maintenance forms, records, and reports **<mfrr>** (B.7.1.2).

**e.** List of abbreviations / acronyms **<loa>** (B.7.1.11).

### P.6.2 Data sheets introduction work package <ammods_introwp>.

A data sheets introduction WP shall be prepared and contain the following information:

**a.** Work package identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** Initial setup **<initial_setup>** is not required for this WP.

**c.** Purpose and use of data sheets. The purpose and use of data sheets shall be explained. For ammunition data sheets publications, the work package shall contain a statement similar to the following:

"This manual is a reference document published as an aid in training, familiarization, and identification of (*insert commodity item and FSC*). This manual is not authorization for requisitioning, stockage, maintenance, or issue of the material described herein."

### P.6.3 Data sheets work package <ammo_dswp>.

A data sheets work package shall be prepared. FIGURE P-3 and FIGURE P-4 show an example ammunition data sheet work package. This WP shall contain the following (as applicable):

**a.** Work package identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** Initial setup **<initial_setup>** is not required for this WP.

**c.** Figure of ammunition or munitions equipment item **<ammods.fig>**. As required by the MATDEV, a figure of the item shall be included on the data sheet.

**d.** Use **<ammods.use>**. This section shall describe the use of the item (for example, purpose, intended audience, applicable weapons from which the item is fired).

**e.** Description **<ammods.desc>**. This section shall describe the general description, characteristics, capabilities, and features of the item. Color coding information for ammunition items shall be included, as applicable.

**f.** Functioning **<ammods.func>**. This section shall describe how the item functions. The description shall explain what functional affect it has on other components (for example, from trigger pull to target).

**g.** Differences between models **<ammods.diff>**. This section shall identify the physical or functioning differences in configurations or models when more than one configuration or model is described.

**h.** Tabulated data **<ammods.tabdata>**. This section shall list descriptive data of the item and its components.

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Ammunition data sheets publications shall describe the following (at a minimum):

**(1)** Department of Defense Ammunition Code (DODAC)

**(2)** National Stock Number (NSN)

**(3)** Type

**(4)** Weight

**(5)** Dimensions

**(6)** Explosive/energetic information

**(7)** Color

Munitions equipment data sheet publications shall describe the following (at a minimum):

**(1)** Ammunition Peculiar Equipment (APE) number

**(2)** Unit of issue

**(3)** Installation data

**(4)** Utilities required

**(5)** Production capacity

**i.** Performance **<ammods.perf>** (ammunition data sheets publication only). This section shall list information describing the normal use or operation of the item (for example, chamber pressure, velocity, maximum range, trace).

**j.** Temperature limits **<ammods.temp>** (ammunition data sheets publication only). This section shall list the operational and storage temperature limits of the item. Dummy/inert items do not require firing temperature limits.

**k.** Drawings/specifications **<ammods.draw>** (ammunition data sheets publication only). This section shall list applicable ammunition drawings for the item (for example, Combination of Adopted Items (COAI), top drawing, packing and marking drawings, specification, unitization).

**l.** Packing configuration **<ammods.packcfg>** (ammunition data sheets publication only). This section shall describe how the item is packed, including all levels of packaging from item to pallet, for each NSN.

**m.** Packing data **<ammods.pack>** (ammunition data sheets publication only). This section shall list the dimensions (for example, length, width, height, cube) and weight of a full pack configuration of the item (for example, pallet, shippable container).

**n.** Shipping and storage data **<ammods.shipstor>**. For ammunition data sheets publications, this section shall list, at a minimum, DoD hazard class/division, Department of Transportation (DOT) reference number, compatibility group, proper shipping name, United Nations (UN) number, and net explosive weight. This section is not applicable for dummy/inert items. For munitions equipment data sheets publications, this section shall list the physical dimensions, weight, and cube of the equipment.

**o.** Logistics data **<ammods.logdata>** (ammunition data sheets publication only). This section shall list logistics data for each NSN. The logistics data may contain information such as class of supply, controlled inventory item code, NSN established date, official nomenclature, standard study number, shelf life, type classification, and unit of issue.

**p.** Limitations **<ammods.limit>** (ammunition data sheets publication only). This section shall provide any restrictions or weaknesses associated with the item.

**q.** Remarks **<ammods.rem>** (ammunition data sheets publication only). This section shall provide any other information deemed pertinent.

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**r.** References **<ammods.refs>** (ammunition data sheets publication only). This section shall list, citing only publication numbers, related publications (i.e., publications that use the item).

**s.** Associated equipment **<ammods.eqp>** (munitions equipment data sheets publications only). This section shall list any additional APE that is used in conjunction with the described munitions equipment.

**t.** Kits **<ammods.kits>** (munitions equipment data sheets publications only). This section shall list any APE accessory equipment not normally supplied with the end item.

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**Note**

Figure appearance and tags/output may vary slightly based on version of DTD/stylesheets utilized (see applicable DTD or tag definition list for available tags).

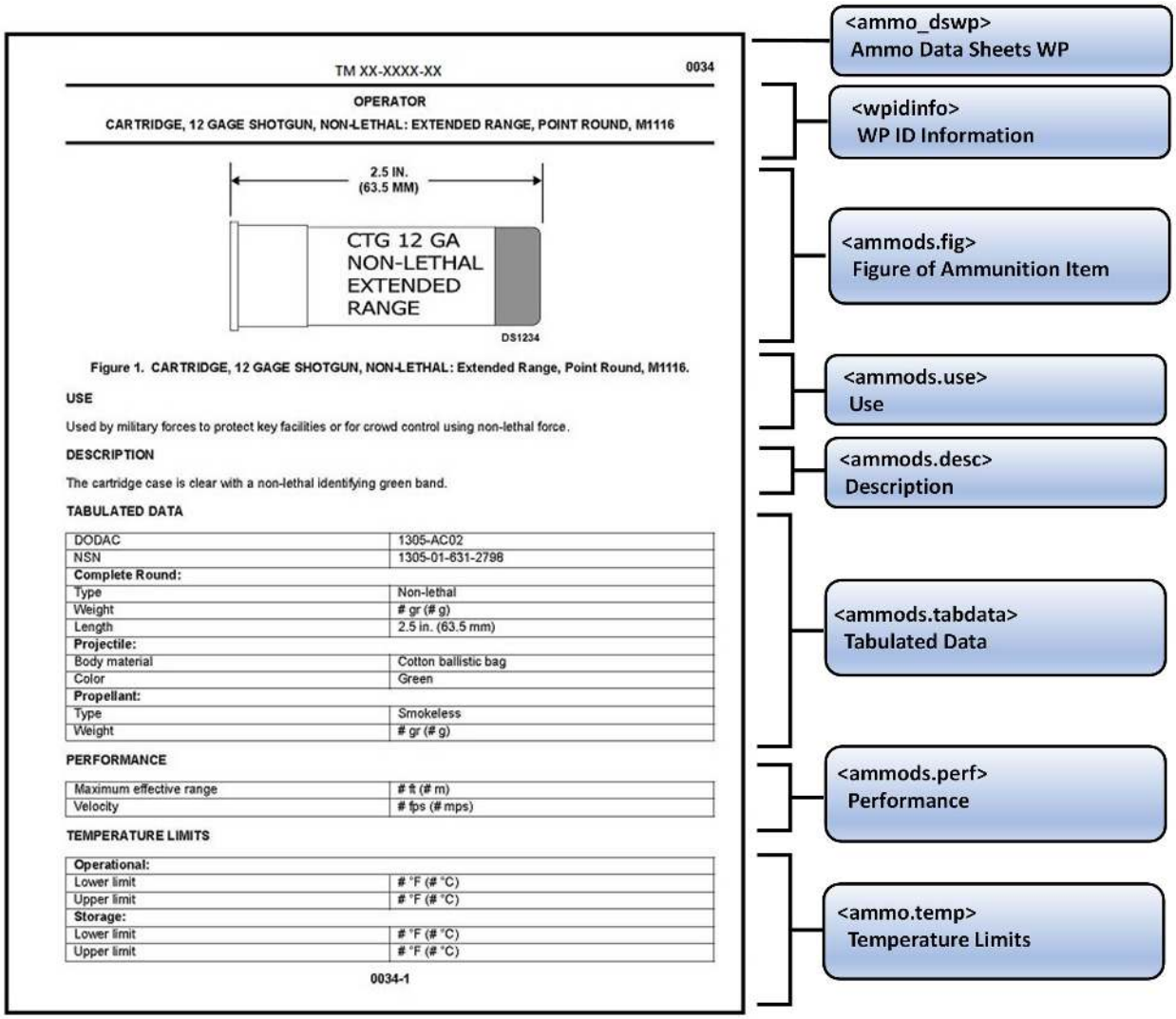


**FIGURE P-3. Example of an ammunition data sheet WP (sheet 1 of 2).**



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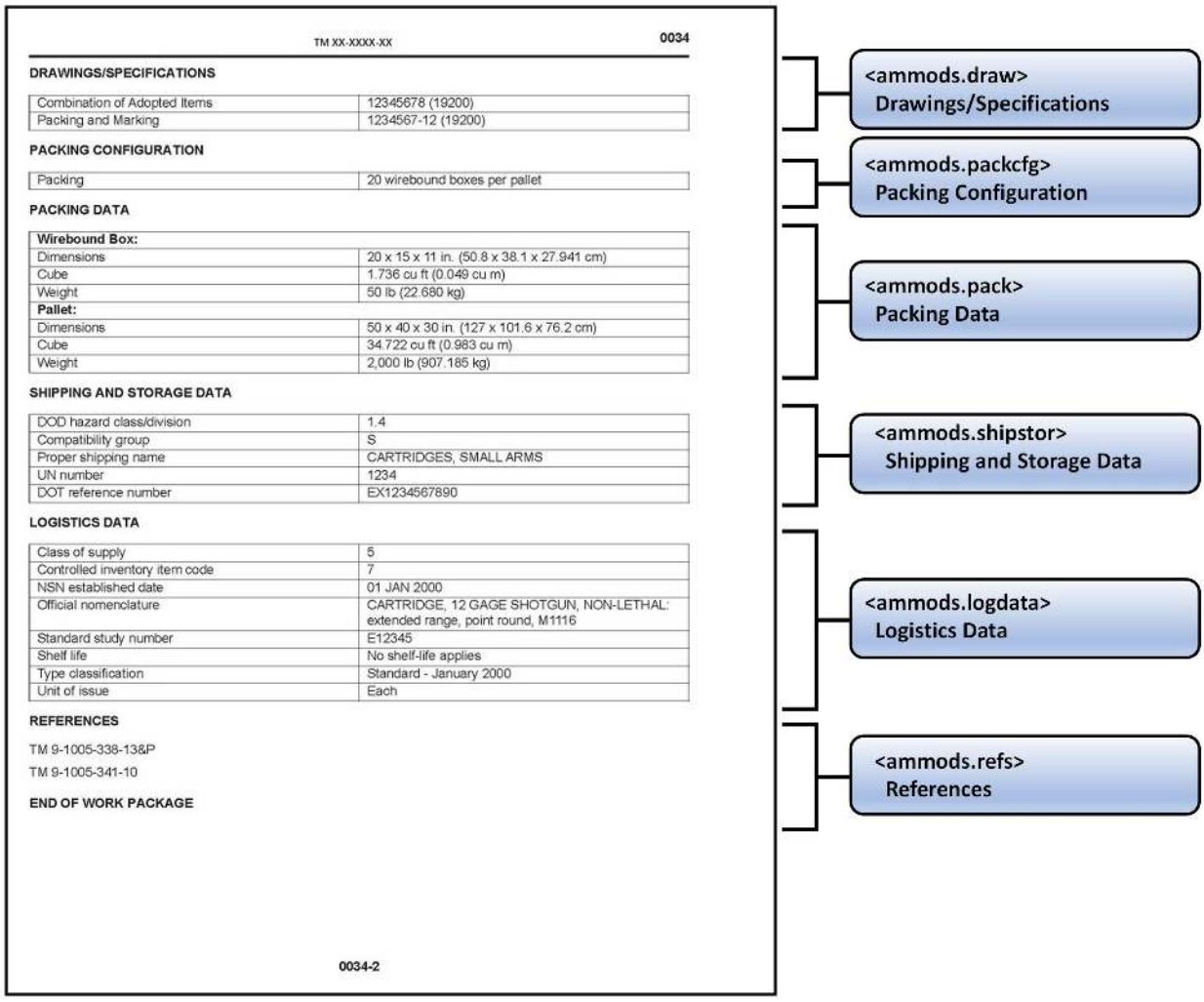


**FIGURE P-4. Example of an ammunition data sheet WP (sheet 2 of 2).**

## P.6.4 References work package <refwp>.

This WP shall be prepared in accordance with G.7.1.

## P.6.5 Deleted items work package <genwp> (munitions equipment data sheets TM only).

A deleted items WP shall appear in all editions of a munitions equipment data sheets publications subsequent to the first edition. The deleted items WP shall contain the following components:

**a.** Work package identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** Initial setup **<initial_setup>** is not required for this WP.

**c.** An alphabetical list of all items deleted from this data sheets publication as a result of a technical committee action or Materiel Status Record (MSR). On first editions that contain no deleted items, this WP shall contain the statement: "Since this is the first edition, there are no deleted items."

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### P.6.6 Operational index work package <genwp> (munitions equipment data sheets publication only.

When specified by the MATDEV, munitions equipment data sheets publication shall contain an operational index WP. This WP shall provide a cross-reference between ammunition items and the APE items needed for function testing, inspecting, maintenance, renovation, and demilitarization of the ammunition item. This WP shall contain the following components:

**a.** Work package identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** Initial setup **<initial_setup>** is not required for this WP.

**c.** Cross-reference between ammunition items and the APE items.

### P.6.7 Preparation and handling of ammunition peculiar equipment (APE) for shipment and storage work package <genwp> (munitions equipment data sheets TM only).

When specified by the MATDEV, the munitions equipment data sheets publication shall contain a preparation and handling of APE for shipment and storage WP. This WP shall contain the following components:

**a.** Work package identification information **<wpidinfo>** is required for this WP (5.4.1).

**b.** Initial setup **<initial_setup>** is not required for this WP.

**c.** Introduction (scope and definitions).

**d.** General requirements (program requirements, levels of protection, and basic requirements), as applicable.

**e.** Detailed requirements, as applicable.

**f.** Requirements for checking for and removal of explosive contamination.

### P.6.8 Other supporting information work package <genwp>.

When specified by the MATDEV, other supporting information WPs may be included in a data sheets publication. These WPs shall be prepared in accordance with G.7.12.

## P.7 NOTES.

The notes in Section 6 apply to this appendix.

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# APPENDIX Q

# TECHNICAL BULLETINS

## Q.1 SCOPE.

A technical bulletin (TB) provides supplementary information to technical publications. TBs may contain such things as maintenance expenditure limits, equipment improvement recommendation material, functional group code information, general information, procedures for kit installation or special mission changes to materiel, warranty information, schematics/diagrams to supplement a technical publication, ammunition information, etc. TBs do not contain operator or maintenance instructions, are not used for quick reference guides or other similarly named publications, are not used in lieu of operator/maintenance manuals or to change them, and do not contain parts information. The requirements provided in this appendix provide the technical content requirements for the preparation of Technical Bulletins (TBs) other than warranty TBs, medical TBs, or calibration TBs. This appendix is a mandatory part of this standard. The information contained herein is intended for compliance.

**Note**

To be compliant with this standard, publications shall be developed within **<production>** (opening) and **</production>** (closing) root tags.

When preparing information in accordance with this appendix, all references shall be linked using the appropriate tags (**<xref>**, **<extref>**, etc.). Links shall not be typed as text without tagging.

### Q.1.1 Content tailoring using Appendix A matrices.

Applicability of some requirements contained in this standard are the responsibility of the materiel developer (MATDEV) (refer to AR 750–1, AR 700–127) and shall be accomplished using matrices referenced in TABLE A-I. The applicability of requirements are also designated by statements including, but not limited to, the following:

**a.** "...as specified by the MATDEV."

**b.** "...as required by the MATDEV."

**c.** "Refer to the MATDEV for..."

**d.** "Contact MATDEV if..."

**e.** "The MATDEV shall determine..."

**f.** "...as needed."

#### Q.1.1.1 Front and rear matter.

Front and rear matter requirements for documents discussed in this appendix shall be determined in accordance with TABLE A-II.

## Q.2 APPLICABLE DOCUMENTS.

The applicable documents in Section 2 apply to this appendix.

## Q.3 ACRONYMS AND DEFINITIONS.

The acronyms and terms defined in Section 3 of this standard apply to this appendix.

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## Q.4 DEVELOPMENT OF TECHNICAL BULLETIN MODULES.

When specified by the MATDEV, a TB **<tech_bltn>** shall be prepared. The modules listed in Q.4.1 through Q.4.3 and WPs listed in TABLE Q-I shall be developed as required (see TABLE A-XX). Refer to FIGURE Q-1 for a graphical representation of module and WP organization as well as tags discussed in this section.

**TABLE Q-I. Technical bulletin WPs.**

| WP | WP Tag | Appendix Section |
|---|---|---|
| General information WP [1] | **<ginfowp>** | Q.5.1 |
| Equipment description and data WP (s) | **<descwp>** | Q.5.2 |
| Maintenance WP(s) | **<maintwp>** | Q.5.3 |
| Wiring diagrams and schematics WP(s) | **<wiringwp>** | Q.5.4 |
| Additional maintenance WP(s) | **<gen.maintwp>** | Q.5.5 |
| Reference WP [1] | **<refwp>** | Q.5.6 |
| Expendable and durable items list WP [1] | **<explistwp>** | Q.5.7 |
| Additional supporting WP | **<genwp>** | |

[1] WP type duplication is prohibited.


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**Note**

Tags/output shown in figure(s) below may vary slightly based on version of DTD/stylesheets utilized (see applicable DTD or tag definition list for available tags).

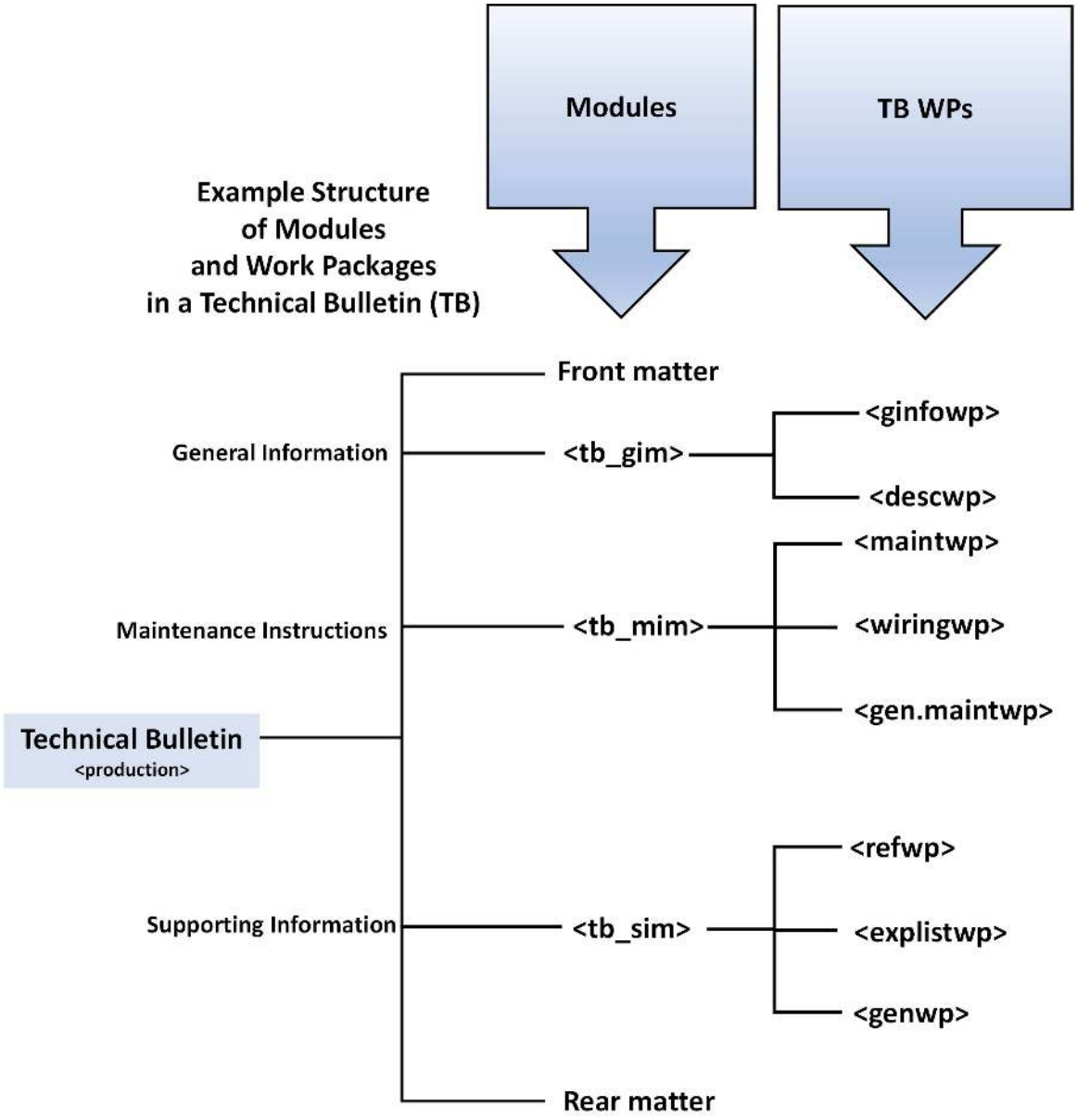


**FIGURE Q-1. TB manual modules and WPs.**

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### Q.4.1 TB general information and equipment description module <tb_gim>.

An TB general information and equipment description module **<tb_gim>** shall be prepared. It shall consist of the following WP(s).

**a.** General information WP **<ginfowp>** (Q.5.1) (Required).

**b.** Equipment description and data WP(s) **<descwp>** (Q.5.2) (As required).

**c.** Additional equipment description and data WP(s) as needed.

### Q.4.2 TB maintenance instructions module <tb_mim>.

A TB maintenance instructions module **<tb_mim>** shall be prepared as needed. It shall consist of the following WP (s).

**a.** Maintenance WP(s) **<maintwp>** (Q.5.3).

**b.** Wiring diagrams and schematics WP(s) **<wiringwp>** (Q.5.4).

**c.** Additional maintenance WP(s) **<gen.maintwp>** (Q.5.5) as needed.

### Q.4.3 TB supporting information module <tb_sim>.

A TB supporting information module **<tb_sim>** shall be prepared as needed.

**a.** Reference WP **<refwp>** (Q.5.6).

**b.** Expendable and durable items list WP **<explistwp>** (Q.5.7).

**c.** Additional supporting WP(s) **<genwp>** ().

## Q.5 DEVELOPMENT OF TECHNICAL BULLETIN WORK PACKAGES.

The following types WPs shall be developed for a TB.

### Q.5.1 General information work package <ginfowp>.

A general information WP shall be prepared in accordance with B.7.1

### Q.5.2 Equipment description and data work package <descwp>.

An equipment description and data WP shall be prepared in accordance with B.7.2.

### Q.5.3 Maintenance work package <maintwp>.

A TB may contain maintenance WPs in a maintenance module limited to the following tasks:

**a.** Remove **<remove>** (E.7.4.4.31) (for kits only).

**b.** Install **<install>** (E.7.4.4.31.1) (for kits only).

**c.** Inspect **<inspect>** (E.7.4.4.11).

**d.** Pack **<pack>** (E.7.4.4.20).

**e.** Unpack **<unpack>** (E.7.4.4.20.1).


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**f.** Load **<load>** (E.7.4.4.14).

**g.** Unload **<unload>** (E.7.4.4.14.1).

**h.** Preparation for Shipment **<prepship>** (E.7.4.4.25).

**i.** Transport **<transport>** (E.7.4.4.38).

### Q.5.4 Wiring diagrams and schematics work package <wiringwp>.

For a TB containing schematic diagrams for technical publications, the maintenance module may contain wiring diagram WP(s). This WP shall be prepared in accordance with E.7.10.

### Q.5.5 Additional maintenance work package <gen.maintwp>.

This WP shall be used for general supplementary information that does not fit in the above WPs. This WP shall be prepared in accordance with E.7.5.

### Q.5.6 References work package <refwp>.

As specified by the MATDEV, references for the TB shall be included in this WP. This WP shall be prepared in accordance with G.7.1.

### Q.5.7 Expendable and durable items list work package <explistwp>.

As specified by the MATDEV, expendable and durable items required for the TB shall be included in this WP. This WP shall be prepared in accordance with G.7.7.

### Q.5.8 Additional supporting information work package <genwp>.

The TB may contain one or more supporting information WPs for supplementary information needed in the TB as specified by the MATDEV. These WPs shall be prepared in accordance with G.7.12.

## Q.6 NOTES.

The notes in Section 6 apply to this appendix.

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# MIL-STD-40051E

# CONCLUDING MATERIAL

Custodians:

Army – TM

Navy – MC

Preparing Activity:

Army – TM

Review Activities:

Army – APD, AR, AT, AV,

CR, EA, MI, PT

Project Number:

TMSS-2025-007

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.



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